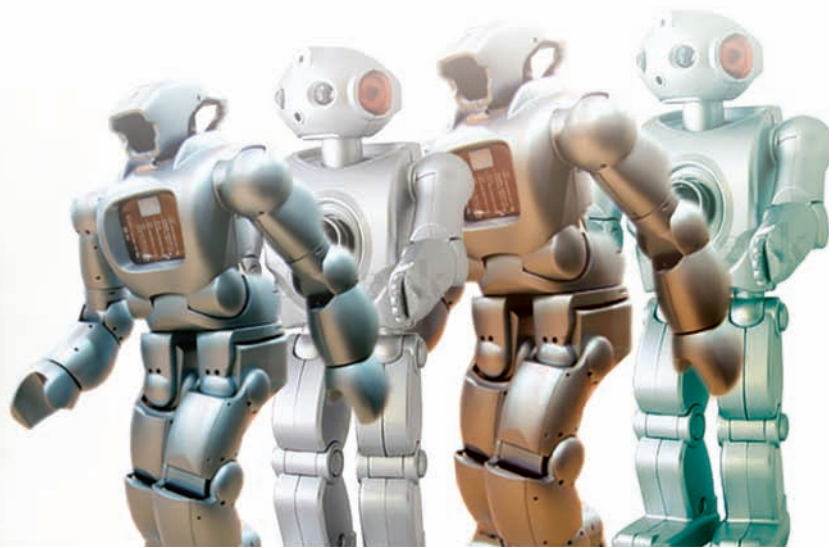


Probability, Statistics and Queueing Theory

V. Sundarapandian



PROBABILITY, STATISTICS AND QUEUEING THEORY

V. Sundarapandian

Professor, Department of Mathematics
Vel Tech Dr. RR & Dr. SR Technical University
Chennai

Formerly, Professor and Academic Convenor
Indian Institute of Information Technology and Management (IIITM-K)
Thiruvananthapuram, Kerala

PHI Learning Private Limited

New Delhi - 110 001
2009

PROBABILITY, STATISTICS AND QUEUEING THEORY

V. Sundarapandian

© 2009 by PHI Learning Private Limited, New Delhi. All rights reserved. No part of this book may be reproduced in any form, by mimeograph or any other means, without permission in writing from the publisher.

ISBN-978-81-203-3844-9

The export rights of this book are vested solely with the publisher.

Published by Asoke K. Ghosh, PHI Learning Private Limited, M-97, Connaught Circus, New Delhi-110001 and Printed by Mohan Makhijani at Rekha Printers Private Limited, New Delhi-110020.

To
My Daughters
Laxmi
and
Shree

Contents

Preface

ix

1. PROBABILITY	1–91
1.1 Brief History of Probability	1
1.2 Sample Space and Events	2
1.3 Classical and Empirical Probability	6
<i>Problem Set 1.1</i>	18
1.4 Axiomatic Definition of Probability	19
<i>Problem Set 1.2</i>	38
1.5 Conditional Probability	39
<i>Problem Set 1.3</i>	71
1.6 Total Probability	73
<i>Problem Set 1.4</i>	80
1.7 Bayes' Theorem	81
<i>Problem Set 1.5</i>	90
2. RANDOM VARIABLE	92–192
2.1 Definition of a Random Variable	92
<i>Problem Set 2.1</i>	95
2.2 Distribution Function of a Random Variable	95
<i>Problem Set 2.2</i>	102
2.3 Discrete Random Variable	103
<i>Problem Set 2.3</i>	110
2.4 Continuous Random Variable	111
<i>Problem Set 2.4</i>	124
2.5 Mathematical Expectation	126
<i>Problem Set 2.5</i>	153
2.6 Chebyshev's Inequality	155
<i>Problem Set 2.6</i>	163
2.7 Moments of a Random Variable	163
<i>Problem Set 2.7</i>	172

2.8	Moment Generating Function	174
	<i>Problem Set 2.8</i>	183
2.9	Characteristic Function	184
	<i>Problem Set 2.9</i>	192
3.	STANDARD PROBABILITY DISTRIBUTIONS	193–318
3.1	Degenerate Distribution	193
3.2	Bernoulli Distribution	194
3.3	Binomial Distribution	196
	<i>Problem Set 3.1</i>	215
3.4	Poisson Distribution	216
	<i>Problem Set 3.2</i>	233
3.5	Geometric Distribution	234
	<i>Problem Set 3.3</i>	242
3.6	Negative Binomial Distribution	242
	<i>Problem Set 3.4</i>	246
3.7	Uniform Distribution	247
	<i>Problem Set 3.5</i>	255
3.8	Exponential Distribution	255
	<i>Problem Set 3.6</i>	264
3.9	Gamma Distribution	265
	<i>Problem Set 3.7</i>	271
3.10	Weibull Distribution	271
	<i>Problem Set 3.8</i>	275
3.11	Normal Distribution	276
	<i>Problem Set 3.9</i>	296
3.12	Functions of a Random Variable	298
	<i>Problem Set 3.10</i>	316
4.	TWO-DIMENSIONAL RANDOM VARIABLES	319–520
4.1	Joint Distribution Functions	319
	<i>Problem Set 4.1</i>	341
4.2	Marginal Distributions	342
	<i>Problem Set 4.2</i>	367
4.3	Conditional Distributions	369
	<i>Problem Set 4.3</i>	397
4.4	Expectation	400
	<i>Problem Set 4.4</i>	408
4.5	Sums of Independent Random Variables	409
	<i>Problem Set 4.5</i>	417
4.6	Functions of Random Variables	418
	<i>Problem Set 4.6</i>	440
4.7	Covariance	441
	<i>Problem Set 4.7</i>	454
4.8	Conditional Expectation	455
	<i>Problem Set 4.8</i>	462

4.9	Correlation and Regression	463
	<i>Problem Set 4.9</i>	510
4.10	Central Limit Theorem	512
	<i>Problem Set 4.10</i>	520
5	RANDOM PROCESSES	521–635
5.1	Definition and Description of Random Processes	521
5.1.1	Classification of Random Processes	522
5.1.2	Description of Random Processes	523
5.1.3	Mean, Correlation and Covariance Functions	524
	<i>Problem Set 5.1</i>	531
5.2	Stationary Random Processes	532
5.2.1	First-Order Stationary Processes	532
5.2.2	Second-Order Stationary Processes	534
5.2.3	Order n and Strict-Sense Stationary Processes	536
	<i>Problem Set 5.2</i>	548
5.3	Autocorrelation and Cross-correlation Functions	549
	<i>Problem Set 5.3</i>	557
5.4	Ergodic Process	558
	<i>Problem Set 5.4</i>	572
5.5	Markov Process	573
5.5.1	Markov Chain	574
5.5.2	Probability Distribution of a Markov Chain	576
5.5.3	Chapman-Kolmogorov Theorem	577
5.5.4	Stationary Distribution for a Markov Chain	581
5.5.5	Classification of States of a Markov Chain	593
	<i>Problem Set 5.5</i>	602
5.6	Binomial, Poisson and Normal Processes	604
	<i>Problem Set 5.6</i>	624
5.7	Sine Wave Process	625
5.8	Birth and Death Process	626
5.8.1	Pure Birth Process	627
5.8.2	Poisson Process	628
5.8.3	Birth and Death Process	632
6.	SPECTRAL ANALYSIS OF RANDOM PROCESSES	636–685
6.1	Autocorrelation and Cross-correlation Functions	636
	<i>Problem Set 6.1</i>	640
6.2	Power Spectral Density and Cross-spectral Density Functions	641
	<i>Problem Set 6.2</i>	667
6.3	Linear Systems with Random Inputs	668
	<i>Problem Set 6.3</i>	683
7.	QUEUEING THEORY	686–749
7.1	Basic Characteristics of Queueing Models	686
7.1.1	Transient and Steady-States	687

7.1.2	Kendall's Notation of a Queueing System	688
7.1.3	Transient State Probabilities for Poisson Queue Systems	689
7.1.4	Steady State Probabilities for Poisson Queue Systems	690
7.2	Model I—($M/M/1$): (∞ /FIFO) Single Server with Infinite Capacity	692
7.2.1	Characteristics of Model I	692
	<i>Problem Set 7.1</i>	707
7.3	Model II—($M/M/s$): (∞ /FIFO), Multiple Server with Infinite Capacity	709
7.3.1	Characteristics of Model II	711
	<i>Problem Set 7.2</i>	721
7.4	Model III—($M/M/1$): (k /FIFO) Single Server with Finite Capacity	722
7.4.1	Characteristics of Model III	725
	<i>Problem Set 7.3</i>	730
7.5	Model IV—($M/M/s$): (k /FIFO) Multiple Server with Finite Capacity	732
7.5.1	Characteristics of Model IV	734
	<i>Problem Set 7.4</i>	739
7.6	The ($M/G/1$) Queueing System	740
7.6.1	Pollaczek-Khinchine Formula	742
	<i>Problem Set 7.5</i>	748
<i>Bibliography</i>		751–752
<i>Answers to Problems</i>		753–806
<i>Index</i>		807–809

Preface

The notions of Probability, Statistics and Queueing Theory are fundamental to learning many advanced concepts in various branches of Engineering, Mathematics, biological sciences, and many others. This book is the result of my teaching of Statistics at B.Tech./M.Sc. levels at Indian Institute of Technology Kanpur, Vellore Institute of Technology (VIT), and SRM University at Chennai. All basic theories have been presented to make the students clearly understand the subject.

The book is organized into seven chapters. Chapter 1 gives a detailed review of probability (with classical, empirical and axiomatic definitions), conditional probability, total probability, and Bayes' theorem. Chapter 2 elaborates the concept of distribution function, types, and the moment-generating and characteristic functions of a random variable. It also discusses mathematical expectation and Chebyshev's inequality.

Chapter 3 provides a detailed description of the standard probability distributions. Chapter 4 discusses the important concepts of two-dimensional random variables such as joint distribution functions, marginal and conditional density functions, sums of independent random variables, conditional expectation, covariance, correlation, and regression. The chapter concludes with a study on Central Limit Theorem, which is important in engineering and physical applications.

Chapter 5 deals with random processes and their types, autocorrelation and cross-correlation functions, and the ergodic, Markov, binomial, Poisson, normal, sine wave, and birth and death processes. Chapter 6 analyzes the spectral properties of random processes such as power spectral density and cross-spectral density functions of random processes, and their applications to linear systems with random inputs.

Chapter 7 discusses the basics of queueing theory with a detailed study on the five important queueing models, viz., $(M/M/1):(\infty/\text{FIFO})$, $(M/M/s):(\infty/\text{FIFO})$, $M/M/1:(k/\text{FIFO})$, $M/M/s:(k/\text{FIFO})$, and $(M/G/1)$. Besides, the book contains numerous definitions, examples, theorems, and proofs to help the students understand the concepts well.

This book is mainly intended as a text for the undergraduate courses in engineering (B.Tech./BE) and postgraduate courses in science (M.Sc.) offered by engineering institutes like IITs and by various Indian universities.

I thank my mentor, Professor Christopher I. Byrnes, Electrical and Systems Engineering, Washington University, who has been a great source of inspiration for me in both teaching

and research. I also wish to thank Professor P.C. Joshi, Indian Institute of Technology Kanpur, for assigning me the task of teaching Statistics for the M.Sc. Mathematics students at IIT Kanpur. I also thank my colleagues, Professors M.S. Gopinathan and C.S. Padmanabha Iyer at IIITM-K, Thiruvananthapuram, Kerala, and Professors E.D. Jemmis and V.M. Nandakumaran at Indian Institute of Science Education and Research (IISER), Thiruvananthapuram, for their advice and encouragement. I thank my Publishers, PHI Learning, New Delhi, especially their editorial and production team, for their careful processing of the manuscript.

I hope that the students, faculty and other readers will find this book very useful for meeting their needs.

I welcome any valuable suggestions and constructive comments for improving the contents of the book.

V. Sundarapandian

E-mail: sundar@iiitmk.ac.in

Chapter 1

Probability

1.1 BRIEF HISTORY OF PROBABILITY

It is well-known that the theory of probability had its origin in gambling and games of chance. The foundation of probability theory, as a precise mathematical science, was laid in the mid-seventeenth century by two great French mathematicians Blaise Pascal (1623–1662) and Pierre de Fermat (1601–1665). In 1654, the French gambler and noble man Chavalier de Mere (1607–1648) asked many prominent mathematicians including Pascal several questions related to problems of chance, the best known of which was the *problem of points*. The problems posed by de Mere paved the way for a fruitful correspondence of letters on the games of chance between Pascal and Fermat. The famous Pascal-Fermat correspondence not only solved de Mere's problems but laid the foundation for more general results on the games of chance also. To tackle these problems, Fermat used combinatorial analysis (finding the number of possible and that of favourable outcomes in ideal games of chance using permutations and combinations), while Pascal reasoned by recursion (an iterative process in which the result of the next case is determined by the present case).

EXAMPLE 1.1. (*Problem of Points*) Two players, A and B , play a series of fair games until one person has won six games. Both A and B have wagered the same amount of money with the understanding that the winner takes all. But, suppose that the series of fair games is prematurely stopped for whatever reasons at which point A has won five games and B three games. How is the stake to be divided among the players?

Solution. (Pascal and Fermat) As A has won five games and B three games, A needs to win just one more game, while B needs to win three more games. Therefore, the play will be over after at most three further games. It is clear that there are a total of $2^3 = 8$ possible future outcomes, and all these outcomes are equally likely. The exhaustive possible outcomes of the play are AAA , AAB , ABA , ABB , BAA , BAB , BBA and BBB .

$$\text{Probability}(A \text{ winning}) = \frac{\text{No. of outcomes in which } A \text{ wins}}{\text{Total no. of outcomes}} = \frac{7}{8}$$

$$\text{Probability}(B \text{ winning}) = \frac{\text{No. of outcomes in which } B \text{ wins}}{\text{Total no. of outcomes}} = \frac{1}{8}$$



Remark 1.1. Pascal and Fermat, in fact, presented a solution to a more general problem in which player A needs m more wins to take the whole stake and player B needs n more wins to take the whole stake. The game will be over after at most $m + n - 1$ further plays. There are a total of 2^{m+n-1} possible outcomes. Pascal and Fermat formulated that

$$P(\text{A winning the game}) = \sum_{i=0}^{n-1} \binom{m+n-i}{i} \frac{1}{2^{m+n-1}}$$

where

$$\binom{n}{i} = {}^nC_i = \frac{n!}{i!(n-i)!}$$

Next, the great mathematician Jacob Bernoulli (1654–1705) gave a sound theoretical base to probability. His work *Ars conjectandi* (The Art of Conjecture) was the first substantial treatise on the theory of probability. Thomas Bayes (1702–1761) proved a special case of what is now known as the *Bayes' Theorem*. Pierre-Simon Laplace (1749–1827) proved a more general version of Bayes' Theorem and used it for various applications in Celestial Mechanics and Medical Statistics. In his monumental work titled *Théorie Analytique des Probabilités* (Analytic Theory of Probability) first published in 1812, Laplace described many of the tools he invented for calculating the mathematical probability of events. Paul Levy (1887–1971) published his major results in the work *Calcul des Probabilités* (1925). Richard von Mises (1883–1953) contributed to the empirical or frequency interpretation of probability.

Russian mathematicians also have made very valuable contributions to the theory of probability. The chief Russian contributor is Andrei Nikolaevich Kolmogorov (1903–1987), one of the most important mathematicians of the twentieth century. His work *Grundbegriffe der Wahrscheinlichkeit-srechnung* (1933) contains his most famous contributions to probability theory, namely, an axiomatic definition of probability. The modern probability theory is based on this definition of probability. Other major contributors from Russia are P.L. Chebychev (1821–1894), A.A. Markov (1856–1922), A.Y. Khintchine (1894–1959), A.M. Lyapunov (1857–1918), etc. Chebychev proved a generalized form of the *Law of Large Numbers*, which used *Chebychev's inequality*. Markov and Lyapunov were students of Chebychev in the University of St. Petersburg at Moscow. Both had contributed to established topics like *Central Limit Theorem* and the *Law of Large Numbers*. Markov introduced the concept of *Markov chain* in the theory of modern probability. In the 1930s, Khintchine developed the theory of random processes and, in particular, that of stationary processes.

1.2 SAMPLE SPACE AND EVENTS

Probability theory is based on the paradigm of a *random experiment*. So, we start this section with a formal definition of a random experiment.

Definition 1.1. A *random experiment* is an experiment in which the following hold:

- (1) The experiment can be repeated under identical conditions.
- (2) We have knowledge of all possible outcomes in advance.
- (3) Any trial of the experiment results in an outcome that is not known in advance.

Next, we define the *sample space* associated with a random experiment.

Definition 1.2. If E is a random experiment, then the set of all possible outcomes of E , denoted by S , is called the *sample space* associated with the random experiment E . The elements of the sample space are called as *sample points*.

Definition 1.3. A sample space S is called

- (a) *finite* if it contains a finite number of sample points;
- (b) *discrete* if it contains at most a countable number of sample points; and
- (c) *uncountable* if it contains an uncountable number of sample points.

EXAMPLE 1.2. Let E be the random experiment of tossing a coin. It has two outcomes: *Head* (H) and *Tail* (T), where we assume that the coin does not stand on its edge when it falls. Thus, the sample space $S = \{H, T\}$. S is an example for a finite sample space.

EXAMPLE 1.3. Let E be the random experiment of tossing two coins simultaneously. It has four outcomes, viz. HH, HT, TH , and TT . Thus, the sample space $S = \{HH, HT, TH, TT\}$. S is an example for a finite sample space. In general, if we consider the random experiment E of tossing n coins simultaneously, then E would have a total of 2^n exhaustive outcomes. The associated sample space S also is finite.

EXAMPLE 1.4. Let E be the random experiment of throwing a die. It has six outcomes: faces with dots numbered from one to six, which may be simply represented by the numbers on them. Thus, the sample space S is given by $S = \{1, 2, 3, 4, 5, 6\}$.

Similarly, if we consider the random experiment E of throwing a pair of dice simultaneously, then it is clear that E will have a total of $6^2 = 36$ outcomes. The associated sample space S is given by

$$S = \{(1, 1), \dots, (1, 6), (2, 1), \dots, (2, 6), \dots, (6, 1), \dots, (6, 6)\}$$

In general, if we consider the random experiment E of throwing n dice simultaneously, then it is clear that E will have a total of 6^n outcomes. The associated sample space S also is finite.

EXAMPLE 1.5. Consider the random experiment E of drawing a card from a well-shuffled pack of playing cards. It has a total of 52 outcomes. It has a total of four suits called *Spades*, *Hearts*, *Diamonds* and *Clubs*. Each suit has a total of 13 cards of which nine are numbered from 2 to 10, an ace, a king, a queen and a jack. Spades and clubs are *black-faced* cards, while hearts and diamonds are *red-faced* cards. Similarly, if we consider the random experiment E of drawing two cards from the well-shuffled pack of cards simultaneously, then the associated sample space S would have a total of ${}^{52}C_2$ outcomes. It also is a finite sample space.

Next, we will present an example for a discrete sample space having a countably infinite number of sample points.

EXAMPLE 1.6. Consider the random experiment E of choosing a positive integer at random. Then the associated sample space is given by $S = \mathcal{Z}_+ = \{1, 2, 3, \dots\}$. The sample space S is discrete.

Next, we present an example for an uncountable sample space.

EXAMPLE 1.7. Let B be unit sphere given by $B = \{x \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$. The sphere is tossed and a point is selected at random on the sphere. Then this random experiment E will have an uncountable number of sample points. It is clear that the sample space equals the unit sphere B . Such a sample space is also called as *continuous sample space*.

Definition 1.4. Let S be the sample space associated with a random experiment E . Any set $A \subset S$ is called an *event*. If $A \subset S$ is a singleton, then A is called a *simple event*; otherwise, it is called a *compound event*.

EXAMPLE 1.8. Let E be the random experiment of throwing a die. Consider the event A defined by “getting the number 5”. Then, $A = \{5\}$ so that A is a simple event. Consider the event B defined by “getting an odd number”. Then $B = \{1, 3, 5\}$ so that B is a compound event.

EXAMPLE 1.9. Let E be the random experiment of throwing a pair of dice. The sample space S will contain 36 sample points as detailed in Example 1.4. The numbers on the faces of the dice are recorded and summed up. Event A represents the claim that the sum is equal to 12 and event B represents the claim that the sum is equal to 8. Then, it is easy to see that $A = \{(6, 6)\}$ and $B = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$. Thus, A and B are, respectively simple and compound events.

Definition 1.5. A collection of events $\mathcal{A} = \{A_i : i \in I\}$, where I is an index set, is called *exhaustive* if the union of the member sets A_i equals the whole sample sample, i.e.

$$\bigcup_{i \in I} A_i = S$$

EXAMPLE 1.10. Let E be the random experiment of throwing a die. Then $S = \{1, 2, \dots, 6\}$. Consider the events A_1 and A_2 defined by “getting an odd number”, and “getting an even number”, respectively. Then, $A_1 = \{1, 3, 5\}$ and $A_2 = \{2, 4, 6\}$. Here, A_1 and A_2 are exhaustive events because $A_1 \cup A_2 = S$.

EXAMPLE 1.11. Let E be the random experiment of drawing a card from a pack of well-shuffled cards. Let A_1 and A_2 denote the events defined by “drawing a black-faced card” and “drawing a red-faced” respectively. Then, A_1 will have a total of 26 sample points consisting of all the cards in spade and club suits, while A_2 will have a total of 26 sample points consisting of all the cards in diamond and heart suits. Here, A_1 and A_2 are exhaustive events because $A_1 \cup A_2 = S$.

Definition 1.6. If A is any event, then the *complementary event* of A , denoted by $\bar{A}$, is defined as

$$\bar{A} = S - A$$

(see Figure 1.1)

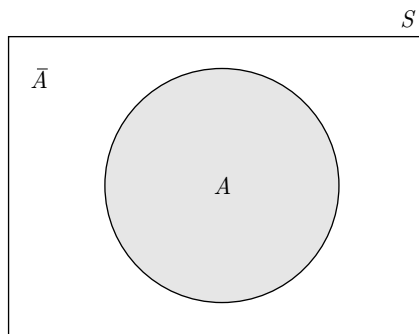


Figure 1.1: A complementary event.

EXAMPLE 1.12. Let E be the random experiment of tossing a coin. Then, $S = \{H, T\}$. Let $A = \{H\}$. Then, $\bar{A} = S - A = \{T\}$.

EXAMPLE 1.13. Let E be the random experiment of throwing a die. Then, $S = \{1, 2, \dots, 6\}$. Let $A = \{1, 3, 5\}$. Then, $\bar{A} = S - A = \{2, 4, 6\}$.

Definition 1.7. Two events A and B are called *mutually exclusive* or *disjoint* if they have empty intersection, i.e. if $A \cap B = \emptyset$ (see Figure 1.2).

A collection of events $\mathcal{A} = \{A_1, A_2, \dots, A_k, \dots\}$ is called mutually exclusive if they are pairwise disjoint, i.e. if $A_i \cap A_j = \emptyset$, whenever $i \neq j : i, j = 1, 2, 3, \dots$

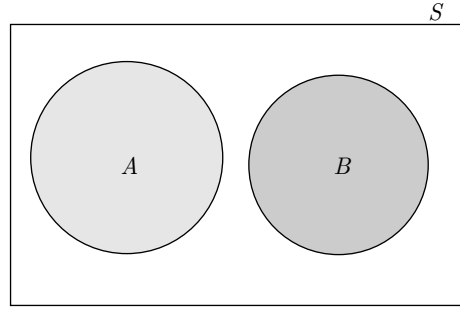


Figure 1.2: Mutually exclusive events.

EXAMPLE 1.14. Let E be the random experiment of tossing a coin. Then, $S = \{H, T\}$. The events $A = \{H\}$ and $B = \{T\}$ are clearly mutually exclusive because $A \cap B = \emptyset$.

EXAMPLE 1.15. Let E be the random experiment of drawing a card from a pack of well-shuffled playing cards. Consider the events A_1, A_2, A_3 and A_4 are defined by “drawing a spade”, “drawing a club”, “drawing a diamond” and “drawing a heart”, respectively. The events A_i are mutually exclusive as they are pairwise disjoint and because

$$A_1 \cup A_2 \cup A_3 \cup A_4 = S \quad (1.1)$$

In this case, as the events A_i are both mutually exclusive and exhaustive, Eq. (1.1) may also be represented as

$$S = A_1 + A_2 + A_3 + A_4$$

Remark 1.2. If A is any event, then it is easy to see that A and $\bar{A}$ are mutually exclusive and exhaustive events. Thus, we can express the sample space S as

$$S = A + \bar{A}$$

Definition 1.8. Let S be a finite sample space. Then its sample points are called *equally likely* if there is no reason to expect any sample point to occur in preference to the other sample points.

Remark 1.3. In applications, the phrases *chosen at random*, *selected at random*, etc. simply signify that the outcomes of the random experiment in question are equally likely to occur. Some examples are given as follows.

- (a) For the random experiment of tossing a fair coin, the outcomes H and T are equally likely.
- (b) For the random experiment of throwing an unbiased die, the outcomes 1 to 6 are equally likely.
- (c) For the random experiment of drawing a card from a well-shuffled pack of cards, all the 52 outcomes are equally likely.

Definition 1.9. A finite collection of events $\{A_1, A_2, \dots, A_n\}$ is called *independent* if the occurrence (or non-occurrence) of any event A_i is not affected by the occurrence or non-occurrence of the remaining events.

EXAMPLE 1.16. For the experiment of tossing a fair coin twice, we consider the events A and B defined by “getting a head in the first toss” and “getting a head in the second toss”, respectively. Then, it is easy to see that A and B are independent events.

EXAMPLE 1.17. For the experiment of throwing a pair of dice, consider the events A and B defined by “getting a 6 in the first die” and “getting a 6 in the second die”, respectively. Then, it is clear that A and B are independent events.

1.3 CLASSICAL AND EMPIRICAL PROBABILITY

The classical definition of probability is due to the French mathematician P.S. Laplace, and it is stated in his classic work *Théorie analytique des probabilités* (Paris, 1812). The underlying assumption in Laplace’s definition is that the sample space S of the random experiment E is finite, and that all the sample points are equally likely to occur. The *classical* or *mathematical* or ‘*a priori*’ definition of probability is stated as follows.

Definition 1.10. (Laplace, 1812) Let E be a random experiment such that its sample space S contains a finite number $n(S)$ of sample points, all of which are equally likely. Let A be any event, i.e. $A \subset S$. Then, the *probability* of A , denoted by $P(A)$, is defined by

$$P(A) = \frac{n(A)}{n(S)}$$

where $n(A)$ is the number of sample points in the set A .

Remark 1.4. A simple consequence of Definition 1.10 is the following:

$$P(\bar{A}) = 1 - P(A)$$

To see this, note that $S = A + \bar{A}$ and hence $n(\bar{A}) = n(S) - n(A)$. Therefore,

$$P(\bar{A}) = \frac{n(\bar{A})}{n(S)} = \frac{n(S) - n(A)}{n(S)} = 1 - \frac{n(A)}{n(S)} = 1 - P(A)$$

Definition 1.11. If A is any event, then the *odds in favour* of A are defined as a to b if

$$P(A) = \frac{a}{a+b}$$

In this case, the *odds against* A are defined as b to a , because

$$P(\bar{A}) = 1 - P(A) = \frac{b}{a+b}$$

Remark 1.5. The probability of an event A , as defined in Definition 1.10 has many interesting properties. Before enumerating them, we fix some notations. Let

$$p = P(A) \text{ and } q = P(\bar{A})$$

In the literature, the probabilities p and q represent the probabilities of the *success* and *failure*, respectively for the occurrence of the event A .

1. $p \geq 0$, $q \geq 0$, and $p + q = 1$.
2. $0 \leq p \leq 1$ (The inequality $p \leq 1$ follows because $n(A) \leq n(S)$).
3. $P(\emptyset) = 0$ and $P(S) = 1$ (Note that $n(\emptyset) = 0$).
4. If A and B are mutually exclusive events, then

$$P(A \cup B) = P(A) + P(B)$$

This follows because $n(A \cup B) = n(A) + n(B)$ if A and B are mutually exclusive. Using the principle of mathematical induction, we can generalize the above result to conclude thus, if $A_1, A_2, \dots, A_n$ are mutually exclusive events (note that, by definition, they are pairwise disjoint), then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

Laplace's definition of probability was criticized by the 19th century mathematicians such as John Venn (1834–1923) and George Boole (1815–1864). While the classical definition of probability is very simple to postulate, it has some serious limitations, which are listed Remark 1.6.

Remark 1.6. (*Limitations of Classical Probability*)

1. Laplace's definition assumes that the sample space has only finitely many outcomes. This has many serious practical limitations. The definition is not applicable in the cases when $n(S)$ is countably infinite or uncountable.
2. It also assumes that the outcomes of the random experiment are all equally likely. This has many serious practical limitations as many of the events in life are not equally likely. Some examples are given below.
 - (i) If a student takes an examination, the outcomes, *pass* and *failure*, are not equally likely. A well-prepared student has a high chance of getting passed, while a poorly prepared student has a high chance of getting failed in the examination.
 - (ii) If a person jumps out of an aircraft without a parachute, the chances of his survival are very low, and so the two outcomes, survival and death, are not equally likely.

The criticisms placed on Laplace's definition of probability stressed the need for a new definition of probability that is applicable for more general situations. Richard von Mises (1883–1953) formulated the *frequency definition* of probability. John Venn and R.A. Fisher (1890–1962) also developed the theory of probability based on the frequency definition. In the frequency definition, the probability of an event A is the *limit* of the relative frequency of A in a large number of values. (Here, the "limit" is not taken strictly in the sense of limits as in Calculus.) Formally, the *frequency interpretation* or the *empirical probability* can be defined as follows.

Definition 1.12. (*von Mises*) Let E be a random experiment which is repeated n times under homogeneous conditions, in which an event A is found to occur, say, $n(A)$ times. Then the *relative frequency* or the *frequency ratio* of the event A is defined as

$$f_n(A) = \frac{n(A)}{n}$$

Then the *probability* of the event A is defined as the limiting value of the frequency ratio of A as n gets indefinitely large, i.e.

$$P(A) = \lim_{n \rightarrow \infty} f_n(A) \quad (1.2)$$

As in the case of classical definition of probability (Definition 1.10), the frequency definition (Definition 1.12) also is found to satisfy many interesting properties.

Remark 1.7. Let $p = P(A)$ and $q = P(\bar{A})$. Recall that the probabilities p and q represent the probabilities, respectively of the *success* and *failure* for the occurrence of the event A .

1. $p \geq 0$, $q \geq 0$, and $p + q = 1$. It should be clear from the definition that $p \geq 0$ and $q \geq 0$. The relation $p + q = 1$ essentially follows by noting that $n(\bar{A}) = n - n(A)$. Hence,

$$f_n(\bar{A}) = \frac{n(\bar{A})}{n} = \frac{n - n(A)}{n} = 1 - \frac{n(A)}{n} = 1 - f_n(A)$$

Taking limits (or as n gets infinitely large), we see that

$$q = P(\bar{A}) = \lim_{n \rightarrow \infty} f_n(\bar{A}) = 1 - \lim_{n \rightarrow \infty} f_n(A) = 1 - P(A) = 1 - p.$$

2. $0 \leq p \leq 1$. (The inequality $p \leq 1$ follows because $n(A) \leq n$.)
3. $P(\emptyset) = 0$ and $P(S) = 1$. (Note that $n(\emptyset) = 0$ and $n(S) = n$.)
4. If A and B are mutually exclusive events, then

$$P(A \cup B) = P(A) + P(B)$$

This follows because $n(A \cup B) = n(A) + n(B)$ if A and B are mutually exclusive. Using the principle of mathematical induction, we can generalize the above result to conclude thus, if $A_1, A_2, \dots, A_n$ are mutually exclusive events (note that, by definition, they are pairwise disjoint), then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

Remark 1.8. (*Limitations of Frequency Interpretation of Probability*)

1. If an experiment is repeated a very large number of times or indefinitely, then the conditions of the experiment need not remain homogeneous. As a consequence, the frequency ratio of A is subject to change.
2. The frequency ratio of A , $f_n(A)$ as $n \rightarrow \infty$ need not converge to a unique value. Hence, $P(A)$ is not well-defined in Eq. (1.2).

EXAMPLE 1.18. From 21 ticket marked with 20 to 40 numerals, one is drawn at random. Find the chance that it is a multiple of 5.

(Anna, Nov. 2003)

Solution. Clearly, $n(S) = 21$. If A denotes the event that the drawn ticket is a multiple of 5, then $n(A) = 5$ because $A = \{20, 25, 30, 35, 40\}$. Hence, the required probability is

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{21}$$

◆

EXAMPLE 1.19. If you twice flip a balanced coin, what is the probability of getting at least one head?
(Anna, April 2004)

Solution. The sample space $S = \{HH, HT, TH, TT\}$. Let A denote the event of getting at least one head. Then $A = \{HT, TH, HH\}$. Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{4}$$

◆

EXAMPLE 1.20. A coin is tossed four times. Find the probability of getting two heads and two tails.

Solution. The sample space S has $2^4 = 16$ outcomes. The event A is defined by “getting two heads and two tails”. It is easy to see that

$$A = \{HHTT, HTHT, HTTH, TTHH, THTH, THHT\}$$

Therefore,

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{16} = \frac{3}{8}$$

◆

EXAMPLE 1.21. If two dice are rolled, find the probability that the sum of the uppermost faces will (i) equal 6 and (ii) equal 8.

Solution. The sample space S has $6^2 = 36$ outcomes. Let A and B be the events defined by “getting the sum equal to 6” and “getting the sum equal to 8”, respectively. We are asked to find the probabilities of the events A and B .

(i) Note that $A = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$, hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{36} = \frac{5}{36}$$

(ii) Note that $B = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$, hence,

$$P(B) = \frac{n(B)}{n(S)} = \frac{5}{36}$$

◆

EXAMPLE 1.22. Two dice are thrown. What is the probability that the product of the numbers on them lies between 7 and 13?

Solution. Here, S contains $6^2 = 36$ outcomes. Let A be the event defined by “getting the product of the numbers on the uppermost faces lying between 7 and 13”. Equivalently, A is the event where the product of the numbers on the uppermost faces equal 8, 9, 10 or 12. Thus, it is easy to see that

$$A = \{(2, 4), (4, 2), (3, 3), (2, 5), (5, 2), (2, 6), (3, 4), (4, 3)\}$$

Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{9}{36} = \frac{1}{4}$$

◆

EXAMPLE 1.23. A number is drawn from the first 20 natural numbers. What is the probability that it is a multiple of 3 or 5?

(Madras, Oct. 2000)

Solution. Clearly, $n(S) = 20$. Let A denote the event that the number drawn is a multiple of 3 or 5. Then $A = \{3, 5, 6, 9, 10, 12, 15, 18\}$. Therefore, $n(A) = 8$. Hence, the required probability is

$$P(A) = \frac{n(A)}{n(S)} = \frac{8}{20} = \frac{2}{5}$$

◆

EXAMPLE 1.24. Two dice are thrown together. Find the probability that (i) the total of the numbers on the top face is 9 and (ii) the top faces are the same.

(Anna, May 2006)

Solution. Clearly, $n(S) = 6^2 = 36$. Let A and B denote the events of getting a total of 9 and of having the same top faces, respectively.

(i) Clearly, $A = \{(3, 6), (4, 5), (5, 4), (6, 3)\}$. Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

(ii) Clearly, $B = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$. Hence,

$$P(B) = \frac{n(B)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

◆

EXAMPLE 1.25. From a group of 15 chess players, 8 are selected by lot to represent a group at a convention. What is the probability that the selected include 3 of the 4 best players in the group?

(Anna, April 2000)

Solution. Since 8 players are selected from 15, $n(S) = {}^{15}C_8 = 6435$. Let A denote the event that the 8 selected include 3 of the 4 best players in the group. Then

$$n(A) = {}^4C_3 \times {}^{11}C_5 = 4 \times 462 = 1848$$

since 3 best players can be chosen from 4 best players in 4C_3 ways and the remaining players can be chosen from among the $15 - 4 = 11$ ordinary players in ${}^{11}C_5$ ways.

Thus, the required probability is

$$P(A) = \frac{n(A)}{n(S)} = \frac{1848}{6435} = \frac{56}{195} = 0.2872$$

◆

EXAMPLE 1.26. A lot of integrated circuit chips consist of 10 good, 4 with minor defects and 2 with major defects. Two chips are randomly chosen from the lot. What is the probability that at least one chip is good? (Anna, May 2007)

Solution. Define the events A and B as follows: A represents the event that exactly one chip is good and B represents the event that both chips are good. Clearly, the events A and B are mutually exclusive. Since there are totally 16 chips, it is clear that two chips can be drawn out of the lot in $n(S) = {}^{16}C_2 = 120$ ways.

Next, note that out of 10 good chips, one chip can be drawn in $n(A) = {}^{10}C_1 = 10$ ways and out of the 6 defective chips, one chip can be drawn in ${}^6C_1 = 6$ ways. Thus, we have

$$P(A) = \frac{n(A)}{n(S)} = \frac{10 \times 6}{120} = \frac{1}{2}$$

Next, note also that out of 10 good chips, two chips can be drawn in ${}^{10}C_2 = 45$ ways. Thus, we have

$$P(B) = \frac{n(B)}{n(S)} = \frac{45}{120} = \frac{3}{8}$$

Hence, the required probability is obtained as

$$P(A \cup B) = P(A) + P(B) = \frac{1}{2} + \frac{3}{8} = \frac{7}{8} \quad \blacklozenge$$

EXAMPLE 1.27. Two cards are drawn at random from a well-shuffled pack of playing cards. What is the probability that the drawn cards are (i) both clubs, (ii) King and Queen?

Solution. First, note that two cards can be drawn out of 52 cards in ${}^{52}C_2$ different ways. Hence,

$$n(S) = {}^{52}C_2 = \frac{52 \times 51}{2 \times 1} = 26 \times 51$$

- (i) Let A denote the event that both the drawn cards are clubs. Since there are a total of 13 clubs in the pack, it is clear that

$$n(A) = {}^{13}C_2 = \frac{13 \times 12}{2 \times 1} = 13 \times 6$$

Hence, the required probability is

$$P(A) = \frac{n(A)}{n(S)} = \frac{13 \times 6}{26 \times 51} = \frac{1}{17}$$

- (ii) Let B denote the event that the drawn cards are King and Queen. Since there are 4 Kings and 4 Queens in the pack, one King can be drawn from 4 Kings in 4C_1 different ways; similarly, one Queen can be drawn from 4 Queens in 4C_1 different ways, and also these two drawings are independent of each other. Hence, it follows that

$$n(B) = {}^4C_1 \times {}^4C_1 = 4 \times 4$$

So, the required probability is

$$P(B) = \frac{n(B)}{n(S)} = \frac{4 \times 4}{26 \times 51} = \frac{8}{663} \quad \blacklozenge$$

EXAMPLE 1.28. What is the probability that a leap year, selected at random, will contain 53 Sundays?

(Madras, May 2002)

Solution. Let A be the event that a randomly chosen leap year will contain 53 Sundays. Since a leap year has a total of 366 days, it will contain 52 complete weeks and two days over. These two days may be any of the following seven combinations: (i) Monday and Tuesday, (ii) Tuesday and Wednesday, (iii) Wednesday and Thursday, (iv) Thursday and Friday, (v) Friday and Saturday, (vi) Saturday and Sunday and (vii) Sunday and Monday.

Of the 7 possible, equally likely, outcomes, the last two outcomes contain Sunday. Hence, the number of favourable outcomes for the given event A are two. Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{7} \quad \blacklozenge$$

EXAMPLE 1.29. What is the probability that a non-leap year, selected at random, will contain 53 Sundays?

Solution. Let A denote the event that a randomly chosen non-leap year will contain 53 Sundays. Since a non-leap year consists of 365 days, it will contain 52 complete weeks and one day over. This day can be any of the 7 days out of which one day is Monday. Therefore, $n(S) = 7$ and $n(A) = 1$. Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{7} \quad \blacklozenge$$

EXAMPLE 1.30. An urn contains 4 white and 6 red balls. Two balls are drawn out together. What is the probability that both are red balls?

Solution. Since the urn contains a total of $4 + 6 = 10$ balls, two balls can be drawn together at random in

$${}^{10}C_2 = \frac{10 \times 9}{2 \times 1} = 45$$

different ways. Thus, $n(S) = 45$. Let A denote the event that the two balls, drawn together at random, are red balls. The number of favourable cases for A is obtained by noting that two red balls can be drawn out of the 6 red balls in

$${}^6C_2 = \frac{6 \times 5}{2 \times 1} = 15$$

different ways. Thus, $n(A) = 15$. Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{15}{45} = \frac{1}{3} \quad \blacklozenge$$

EXAMPLE 1.31. A bag contains 3 red and 2 green balls. Two balls are drawn at random from the bag. Find the probability that both balls have the same colour.

Solution. Since the bag contains a total of $3 + 2 = 5$ balls, 2 balls can be drawn in 5C_2 ways. Hence,

$$n(S) = {}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$$

Let E denote the event that both balls drawn have the same colour. Then E can materialize in the following mutually exclusive ways:

- (i) Both balls are red in colour.
- (ii) Both balls are green in colour.

Note that $n(A) = {}^3C_2 = 3$ and $n(B) = {}^2C_2 = 1$. Hence, the required probability is given by

$$P(E) = P(A \cup B) = P(A) + P(B) = \frac{n(A) + n(B)}{n(S)} = \frac{3 + 1}{10} = \frac{2}{5}$$

◆

EXAMPLE 1.32. A box contains 5 white, 4 red and 6 green balls. Three balls are drawn at random. Find the probability that a white, a red and a green ball are drawn.

Solution. The box contains a total of $5 + 4 + 6 = 15$ balls, out of which 3 balls are drawn. Hence,

$$n(S) = {}^{15}C_3 = \frac{15 \times 14 \times 13}{3 \times 2 \times 1} = 5 \times 7 \times 13$$

Let A be the event of drawing a white, a red and a green ball. Note that a white ball can be drawn out of 5 white balls in 5C_1 different ways, a red ball can be drawn out of 4 red balls in 4C_1 different ways, and a green ball can be drawn out of 6 green balls in 6C_1 different ways and also that these drawings are independent of each other. Hence,

$$n(A) = 5 \times 4 \times 6$$

Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{5 \times 4 \times 6}{5 \times 7 \times 13} = \frac{24}{91}$$

◆

EXAMPLE 1.33. Four persons are chosen at random from a group containing 3 men, 2 women and 4 children. Show that the chance that exactly two of them will be children is $\frac{10}{21}$.

(Anna, Nov. 2006)

Solution. The total number of persons in the group is $3 + 2 + 4 = 9$. Out of the 9 people, 4 are chosen at random. Hence,

$$n(S) = {}^9C_4 = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 9 \times 7 \times 2$$

Let A be the event that of the four people chosen at random, exactly two of them will be children. To calculate $n(A)$, note that 2 children can be selected from 4 children in 4C_2 different ways, 2 adults can be selected from $3 + 2 = 5$ adults in 5C_2 different ways, and that these selections are independent of each other. Hence,

$$n(A) = {}^4C_2 \times {}^5C_2 = \frac{4 \times 3}{2 \times 1} \times \frac{5 \times 4}{2 \times 1} = 5 \times 4 \times 3.$$

Hence,

$$P(A) = \frac{n(A)}{n(S)} = \frac{5 \times 4 \times 3}{9 \times 7 \times 2} = \frac{10}{21}$$

◆

EXAMPLE 1.34. Out of $(2n + 1)$ tickets consecutively numbered, three are drawn at random. Find the probability that the numbers on them are in Arithmetical Progression (A.P.).

(Anna, May 2006)

Solution. Since we are only interested in checking whether the numbers on the consecutively numbered tickets are in A.P., there is no loss of generality in assuming that the given tickets are numbered as $1, 2, \dots, 2n+1$. Since 3 tickets are drawn at random from the given $2n+1$ tickets, it is clear that

$$n(S) = {}^{2n+1}C_3 = \frac{(2n+1)(2n)(2n-1)}{3 \times 2 \times 1} = \frac{n(4n^2-1)}{3}$$

Let E denote the event that the three tickets chosen at random from the given set of tickets have numbers, which are in A.P.

If 1 is the smallest number of the three selected tickets, then there will be n possible sets of triplets in A.P., viz.

$$\begin{array}{ccc} (1 & 2 & 3) \\ (1 & 3 & 5) \\ (1 & 4 & 7) \\ \vdots & \vdots & \vdots \\ (1 & n+1 & 2n+1) \end{array}$$

If 2 is the smallest number of the three selected tickets, then there will be $n-1$ possible sets of triplets in A.P., viz.

$$\begin{array}{ccc} (2 & 3 & 4) \\ (2 & 4 & 6) \\ (2 & 5 & 8) \\ \vdots & \vdots & \vdots \\ (2 & n+1 & 2n) \end{array}$$

If 3 is the smallest number of the three selected tickets, then there will be $n-1$ possible sets of triplets in A.P., viz.

$$\begin{array}{ccc} (3 & 4 & 5) \\ (3 & 5 & 7) \\ (3 & 6 & 9) \\ \vdots & \vdots & \vdots \\ (3 & n+2 & 2n+1) \end{array}$$

Proceeding like this successively, if $2n-2$ is the smallest number of the three selected tickets, then the only possible set is

$$(2n-2 \quad 2n-1 \quad 2n).$$

Finally, if $2n-1$ is the smallest number of the three selected tickets, then the only possible set is

$$(2n-1 \quad 2n \quad 2n+1).$$

Hence, the total number of favourable cases for the event E is given by

$$n(E) = 1 + 2[(n-1) + (n-2) + \dots + 1] = n + 2 \frac{(n-1)n}{2} = n^2$$

Hence, the required probability is given by

$$P(E) = \frac{n(E)}{n(S)} = \frac{n^2}{\frac{n(4n^2-1)}{3}} = \frac{3n}{4n^2-1}$$



EXAMPLE 1.35. Out of $3n$ consecutive integers, three are selected at random. Find the probability that their sum is divisible by 3.

Solution. Let m be any integer. Then the total $3n$ consecutive integers, starting from m and up to $m + 3n - 1$, can be written in the following matrix structure:

$$\begin{array}{ccc} m & m+1 & m+2 \\ m+3 & m+4 & m+5 \\ m+6 & m+7 & m+8 \\ \vdots & \vdots & \vdots \\ m+3n-3 & m+3n-2 & m+3n-1 \end{array} \quad (1.3)$$

Since 3 integers can be chosen from $3n$ consecutive integers in ${}^{3n}C_3$ different ways, $n(S) = {}^{3n}C_3$.

If E denotes the event that the three numbers selected at random from the given $3n$ consecutive integers is divisible by 3, then E can be expressed as the union of the mutually exclusive events A and B , where A denotes the event that all the three chosen numbers are taken from any of the three columns of the array in Eq. (1.3), and B denotes the event that one integer is taken from each of the three column in Eq. (1.3).

Note that $n(A) = 3 \times {}^nC_3$ since each of the three columns contains n integers. Note also that $n(B) = n^3$ since one integer is taken from each of the three columns. Hence, the required probability

$$P(E) = P(A \cup B) = P(A) + P(B) = \frac{n(A) + n(B)}{n(S)}$$

Substituting, we get

$$P(E) = \frac{3 \times {}^nC_3 + n^3}{{}^{3n}C_3}$$

Simplifying, we get

$$P(E) = \frac{(n-1)(n-2) + 2n^2}{(3n-1)(3n-2)} = \frac{3n^2 - 3n + 2}{(3n-1)(3n-2)} \quad \blacklozenge$$

EXAMPLE 1.36. If $6n$ tickets numbered $0, 1, 2, \dots, 6n-1$ are placed in a bag, and three are drawn at random, show that the probability that the sum of the numbers on them is equal to $6n$ is equal to

$$\frac{3n}{(6n-1)(6n-2)}$$

Solution. First, note that 3 tickets can be drawn out of $6n$ tickets in ${}^{6n}C_3$ different ways. Hence,

$$n(S) = {}^{6n}C_3 = \frac{6n(6n-1)(6n-2)}{3 \times 2 \times 1} = n(6n-1)(6n-2)$$

Next, let E denote the event that the sum of the numbers on the three randomly chosen tickets is equal to $6n$.

If 0 is the smallest of the three selected numbers, then there are $3n - 1$ possible sets of triplets each having total equal to $6n$, viz.

$$\begin{pmatrix} 0 & 1 & 6n-1 \\ 0 & 2 & 6n-2 \\ \vdots & \vdots & \vdots \\ 0 & 3n-1 & 3n+1 \end{pmatrix}$$

If 1 is the smallest of the three selected numbers, then there are $3n - 2$ possible sets of triplets each having total equal to $6n$, viz.

$$\begin{pmatrix} 1 & 2 & 6n-3 \\ 1 & 3 & 6n-4 \\ \vdots & \vdots & \vdots \\ 1 & 3n-1 & 3n \end{pmatrix}$$

If 2 is the smallest of the three selected numbers, then there are $3n - 4$ possible sets of triplets each having total equal to $6n$, viz.

$$\begin{pmatrix} 2 & 3 & 6n-5 \\ 2 & 4 & 6n-6 \\ \vdots & \vdots & \vdots \\ 2 & 3n-2 & 3n \end{pmatrix}$$

If 3 is the smallest of the three selected numbers, then there are $3n - 5$ possible sets of triplets each having total equal to $6n$, viz.

$$\begin{pmatrix} 3 & 4 & 6n-7 \\ 3 & 5 & 6n-8 \\ \vdots & \vdots & \vdots \\ 3 & 3n-2 & 3n-1 \end{pmatrix}$$

This process can be continued successively.

If $2n - 4$ is the smallest of the three selected numbers, then there are 5 possible sets of triplets each having total equal to $6n$, viz.

$$\begin{pmatrix} 2n-4 & 2n-3 & 2n+7 \\ 2n-4 & 2n-2 & 2n+6 \\ 2n-4 & 2n-1 & 2n+5 \\ 2n-4 & 2n & 2n+4 \\ 2n-4 & 2n+1 & 2n+3 \end{pmatrix}$$

If $2n - 3$ is the smallest of the three selected numbers, then there are 4 possible sets of triplets each having total equal to $6n$, viz.

$$\begin{pmatrix} 2n-3 & 2n-2 & 2n+5 \\ 2n-3 & 2n-1 & 2n+4 \\ 2n-3 & 2n & 2n+3 \\ 2n-3 & 2n+1 & 2n+2 \end{pmatrix}$$

If $2n - 2$ is the smallest of the three selected numbers, then there are 2 possible sets of triplets each having total equal to $6n$, viz.

$$\begin{pmatrix} 2n-2 & 2n-1 & 2n+3 \\ 2n-2 & 2n & 2n+2 \end{pmatrix}$$

If $2n - 1$ is the smallest of the three selected numbers, then there is only 1 possible set of triplets each having total equal to $6n$, viz.

$$(2n - 1, 2n, 2n + 1)$$

Since the cases listed above are mutually exclusive, it is immediate that

$$\begin{aligned} n(E) &= 1 + 2 + 4 + 5 + \cdots + (3n - 5) + (3n - 4) + (3n - 2) + (3n - 1) \\ &= [1 + 4 + \cdots + (3n - 5) + (3n - 2)] + [2 + 5 + \cdots + (3n - 4) + (3n - 1)] \\ &= \frac{n}{2} [2 + 3(n - 1)] + \frac{n}{2} [4 + 3(n - 1)] \\ &= \frac{n}{2} [6 + 6(n - 1)] = 3n^2 \end{aligned}$$

where we have used the following formula that the sum S of the first n terms of the A.P.

$$a, a + d, a + 2d, \dots, a + (n - 1)d$$

is given by

$$S = \frac{n}{2} [2a + d(n - 1)]$$

Substituting, the required probability of the event E is given by

$$P(E) = \frac{n(E)}{n(S)} = \frac{3n^2}{n(6n - 1)(6n - 2)} = \frac{3n}{(6n - 1)(6n - 2)}$$

◆

EXAMPLE 1.37. Each coefficient in the equation $ax^2 + bx + c = 0$ is determined by throwing an ordinary die. Find the probability that the equation will have real roots.

Solution. First, note that each of the coefficients a, b and c of the given quadratic equation can take the values from 1 to 6 since its value is determined by throwing an ordinary die. Hence, the total number of possible outcomes is

$$n(S) = 6^3 = 216$$

Next, let A denote the event that the given quadratic equation will have real roots. The number of favourable outcomes for A is obtained by noting that the quadratic will have real roots if its discriminant is non-negative or equivalently

$$b^2 \geq 4ac \tag{1.4}$$

Since the maximum value that b can take is 6, the maximum value that the product ac can take to satisfy the condition (1.4) is clearly equal to $\frac{b^2}{4}$ or 9. Thus, one way to obtain $n(A)$ is to enumerate the cases for the product ac from $ac = 1$ up to $ac = 9$ as in Table 1.1 (note that ac cannot take the value 7).

Thus, we have

$$n(A) = 5 + 8 + 6 + 9 + 4 + 8 + 2 + 1 = 43$$

Hence, the required probability is

$$P(A) = \frac{n(A)}{n(S)} = \frac{43}{216}$$

◆

Table 1.1 No. of Favourable Cases

ac	a	c	$4ac$	$b(b^2 \geq 4ac)$	No. of Cases
1	1	1	4	2, 3, 4, 5, 6	5
2	1	2	8	3, 4, 5, 6	$2 \times 4 = 8$
3	1	3	12	4, 5, 6	$2 \times 3 = 6$
4	1	4	16	4, 5, 6	$3 \times 3 = 9$
5	1	5	20	5, 6	$2 \times 2 = 4$
6	1	6	24	5, 6	$4 \times 2 = 8$
8	2	4	32	6	$2 \times 1 = 2$
9	3	3	36	6	$1 \times 1 = 1$

PROBLEM SET 1.1

1. From 30 tickets, marked with 1 to 30 numerals, one is drawn at random. Find the probability that it is a multiple of 3.
2. From 20 tickets, marked with 1 to 20 numerals, two tickets are drawn at random. Find the chance that their product is an odd number.
3. An urn contains 50 tickets numbered 1 to 50 from which two tickets are drawn simultaneously. Find the probability that both the tickets drawn are prime numbers.
4. A coin is tossed three times. Find the probability of getting 1 head and 2 tails.
5. If two dice are rolled, what is the probability that the product of the numbers on them lies between 5 and 15?
6. If two dice are rolled, find the probability that the sum of the top faces will equal (i) 7, (ii) 10.
7. A number is drawn from the first 30 natural numbers. What is the probability that it is a multiple of 4 or 7?
8. Two dice are thrown together. Find the probability that the total of the numbers on the top faces is (i) even, (ii) divisible by 3.
9. Two cards are drawn at random from a well-shuffled pack of playing cards. What is the probability that both cards (i) are from the same suit, (ii) are red, (iii) are aces?
10. An urn contains 4 blue and 6 red balls. Two balls are drawn together. Find the probability that both balls have the same colour.
11. A bag contains 8 red balls and 4 white balls. Four balls are drawn at random. What is the probability that (i) all of them are red, (ii) two of them are red and two white?

12. An urn contains 3 white, 5 blue and 4 red balls. If 3 balls are drawn at random, find the probability that (i) two of the balls drawn are white, (ii) one is of each colour, (iii) none is blue, (iv) at least one is red.
13. What is the probability that a leap year selected at random will contain 53 Thursdays or 53 Fridays?
14. What is the probability of getting 8 cards of the same suit in one hand at a game of bridge?
15. There are four letters and four addressed envelopes. Find the probability that all letters are not dispatched in the right envelopes.

1.4 AXIOMATIC DEFINITION OF PROBABILITY

In this section, we present the *axiomatic definition of probability* due to A.N. Kolmogorov (1903–1987). We saw the limitations of the classical and frequency definitions of probability in Section 1.3. Kolmogorov, through his axiomatic definition of probability, helped in putting the theory of probability and random variables on a firm mathematical basis. The modern probability theory is based on the axioms of probability.

We need to introduce the concept of σ -field for a formal definition of axiomatic probability.

Definition 1.13. Let S be the sample space associated with a random experiment E . Then, a σ -field is a non-empty class $\mathcal{B}$ of subsets of S that is closed under the formation of countable unions and complementations, i.e. $\mathcal{B}$ satisfies the following conditions:

- (i) $A_1, A_2, \dots, A_n, \dots \in \mathcal{B} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{B}$;
- (ii) $A \in \mathcal{B} \Rightarrow \bar{A} \in \mathcal{B}$.

We state the De Morgan's laws from Set Theory, which will be useful for proving Lemma 1.1 and a few other results in this section.

Theorem 1.1. (*De Morgan's Laws*) If $\{A_i : i = 1, 2, \dots\}$ is any countable collection of sets, then the following results hold:

- (a) $\overline{\bigcup_{i=1}^{\infty} A_i} = \bigcap_{i=1}^{\infty} \bar{A}_i$
- (b) $\overline{\bigcap_{i=1}^{\infty} A_i} = \bigcup_{i=1}^{\infty} \bar{A}_i$

The basic properties of the σ -field are stated as follows.

Lemma 1.1. If $\mathcal{B}$ is a σ -field, then the following properties hold:

- (a) $\emptyset, S \in \mathcal{B}$
- (b) $A_1, A_2, \dots, A_n, \dots \in \mathcal{B} \Rightarrow \bigcap_{n=1}^{\infty} A_n \in \mathcal{B}$
- (c) $A, B \in \mathcal{B} \Rightarrow A - B \in \mathcal{B}$

Proof. (a) By definition, the class $\mathcal{B}$ is non-empty. Hence, there exists a set A contained in $\mathcal{B}$. By condition (ii), $\bar{A} \in \mathcal{B}$. By condition (i), it follows that

$$A \cup \bar{A} = S \in \mathcal{B}$$

By condition (ii) again, it follows that $\emptyset = \bar{S} \in \mathcal{B}$

- (b) Let $A_1, A_2, \dots, A_n, \dots \in \mathcal{B}$. By (ii), it follows that $\bar{A}_1, \bar{A}_2, \dots, \bar{A}_n, \dots \in \mathcal{B}$. By condition (i) and De Morgan's law (Theorem 1.1, (b)), we have

$$\bigcup_{n=1}^{\infty} \bar{A}_n = \overline{\bigcap_{n=1}^{\infty} A_n} \in \mathcal{B}.$$

Applying condition (ii) again, we see that $\bigcap_{n=1}^{\infty} A_n \in \mathcal{B}$.

- (c) Let $A, B \in \mathcal{B}$. By condition (ii), $\bar{B} \in \mathcal{B}$. Therefore, we have $A - B = A \cap \bar{B} \in \mathcal{B}$. ♦

Remark 1.9. The choice of the σ -field $\mathcal{B}$ is important in the axiomatic definition of probability. We remark how to choose $\mathcal{B}$ for discrete and uncountable sample spaces.

1. If the sample space S is discrete, i.e. if S contains only a finite number of sample points or at most a countable number of points, we can always take the σ -field $\mathcal{B}$ to be the power set of S , i.e. the set of all the subsets of S . In this σ -field, every one-point set, i.e. sets of the form $\{x\}$, is a member of the σ -field $\mathcal{B}$ and it is a special case of interest for assigning probabilities.
2. If the sample space S is uncountable, then the class of all subsets of S is still a σ -field, but it will be too large a class of sets to be of practical interest. For important special cases like $S = \mathbb{R}$ or an interval I of $\mathbb{R}$, i.e. if S is a continuous sample space, then we would like all intervals (open, closed, semi-open, semi-closed) to be members of the σ -field for assigning probabilities. Thus, a simple σ -algebra would be the class of all semi-closed intervals in $\mathbb{R}$ or contained in the interval I as $S = \mathbb{R}$ and I , respectively.

Definition 1.14. (*Axiomatic Definition of Probability*) Let S be a sample space and $\mathcal{B}$ be a σ -field of subsets of S . A set function P defined on $\mathcal{B}$ is called a *probability function* if it satisfies the following axioms:

- (i) For each $A \in \mathcal{B}$, $P(A) \geq 0$. (*Axiom of Non-Negativity*)
- (ii) $P(S) = 1$. (*Axiom of Certainty*)
- (iii) Let $\{A_i\}, A_i \in \mathcal{B}, i = 1, 2, \dots$ be a sequence of mutually exclusive events, i.e. $A_i \cap A_j = \emptyset$ whenever $i \neq j$. Then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad (\text{Axiom of Countable Additivity})$$

Remark 1.10. If the sample space S is discrete and has a finite number of points, say

$$S = \{x_1, x_2, \dots, x_n\}$$

then it suffices to assign probability to each simple event $\{x_i\}$ for $i = 1, 2, \dots, n$, where $P(x_i) \geq 0$ for each i and also that the total probability is equal to 1, i.e.

$$\sum_{i=1}^n P(x_i) = 1$$

In this probability space, for any event A , the probability of A is defined by

$$P(A) = \sum_{x \in A} P(x)$$

It is important to note that *equally likely* assignment is just one such assignment of probabilities. If the assignment is equally likely, then

$$P(x_i) = \frac{1}{n} \quad \text{for } i = 1, 2, \dots, n$$

so that for any event A , the probability of A is defined by

$$P(A) = \sum_{x \in A} P(x) = n(A) \cdot \frac{1}{n} = \frac{n(A)}{n(S)}$$

Thus, the classical definition of probability may be regarded as a special case of the axiomatic definition of probability.

Remark 1.11. If the sample space S is discrete and contains a countably infinite number of points, then we cannot make an equally likely assignment of probabilities. Rather, we perform probability assignments for all simple events of the form

$$\{x_i\}, \quad \text{where } i = 1, 2, \dots, \infty$$

If A is any event, then the probability of A is defined by

$$P(A) = \sum_{x \in A} P(x)$$

Remark 1.12. If the sample space S has uncountably many points, then we cannot assign even positive probability to simple events of the form $\{x_i\}$ without violating Axiom (ii), viz. $P(S) = 1$. In this case, we assign probability to compound events containing intervals. For example, if S is the sample space given by $S = [0, 1]$ (the unit interval of the real line $\mathbb{R}$) and $\mathcal{B}$ is the σ -field of all intervals contained in S , then the assignment

$$P(I) = \text{Length of } I \quad (I \text{ is an interval})$$

is a valid definition of probability. Some important points need to be understood in this context. We know that $P(\emptyset) = 0$ and $P(S) = 1$. This example illustrates that the converse of these statements are not always true. In other words, $P(A) = 0$ does not imply that $A = \emptyset$. Similarly, $P(B) = 1$ does not imply that $B = S$. To see this, consider the events $A = \{0\}$ and $B = (0, 1)$ contained in the given sample space S . Then $P(A) = 0$ (a single-point set does not have any length) and $P(B) = 1$. Note also that $A \neq \emptyset$ and $B \neq S$.

EXAMPLE 1.38. Let $S = \{1, 2, 3, \dots\}$ be a set of all positive integers. Then, S is a discrete sample space. We can define *probability* on the sample space S by assigning probabilities to all simple events $\{i\}$ for $i = 1, 2, \dots$. For example, we may define

$$P(i) = \frac{1}{2^i} \quad \text{for } i = 1, 2, \dots$$

Since the σ -field $\mathcal{B}$ is the set of all subsets of S , any event $A \in \mathcal{B}$ is the union of the simple events $\{i\}$. Hence,

$$P(A) = \sum_{x \in A} P(x)$$

So, $P(A)$ is defined for each $A \in \mathcal{B}$ and it is clear that $P(A) \geq 0$. Thus, Axiom (i) holds.

Note also that

$$P(S) = \sum_{i=1}^{\infty} P(i) = \sum_{i=1}^{\infty} \frac{1}{2^i} = \frac{0.5}{1-0.5} = 1$$

where

$$a + ax + ax^2 + ax^3 + \cdots = \frac{a}{1-x} \text{ where } |x| < 1$$

Thus, Axiom (ii) also holds. Since S is a discrete sample space, Axiom (iii) also holds trivially. Hence, P is a probability function on the sample space S .

Next, we derive many important properties of the axiomatic probability in the following theorems. In all the subsequent discussions, we consider the probability space $(S, \mathcal{B}, P)$. Also, whenever we speak of events, we always mean sets which are members of the σ -field $\mathcal{B}$.

Theorem 1.2. If A is any event, then $P(\bar{A}) = 1 - P(A)$.

Proof. A and $\bar{A}$ are mutually exclusive events and $A \cup \bar{A} = S$. Hence, by Axiom (iii), we have

$$P(S) = P(A \cup \bar{A}) = P(A) + P(\bar{A})$$

By Axiom (ii), $P(S) = 1$. Hence, it is immediate that

$$P(\bar{A}) = 1 - P(A)$$

◆

Corollary 1.1. $P(\emptyset) = 0$.

Proof. Since $\emptyset = \bar{S}$, it is immediate from Theorem 1.2 that

$$P(\emptyset) = 1 - P(S)$$

By Axiom (ii), $P(S) = 1$. Hence, $P(\emptyset) = 1 - 1 = 0$

◆

Corollary 1.2. If A is any event, then $P(A) \leq 1$.

Proof. By Axiom (i), $P(B) \geq 0$ for any event B . Taking $B = \bar{A}$, it follows that $P(\bar{A}) \geq 0$. By Theorem 1.2, we have

$$P(\bar{A}) = 1 - P(A) \geq 0$$

Thus, $P(A) \leq 1$.

◆

Remark 1.13. In numerical problems, sometimes it is easier to compute the probability of the complementary event $\bar{A}$ than that of A . In such cases, we first determine $P(\bar{A})$ and then compute $P(A)$ using the formula $P(A) = 1 - P(\bar{A})$.

The following examples illustrate the usefulness of Theorem 1.2.

EXAMPLE 1.39. A bag contains the numbers $1, 2, \dots, 100$ and two numbers are drawn at random. Find the probability that their product is even.

Solution. Since two numbers are drawn at random from a bag of 100 numbers,

$$n(S) = {}^{100}C_2 = \frac{100 \times 99}{2 \times 1} = 99 \times 50$$

Let A denote the event that the product of the two numbers is even. Then $\bar{A}$ represents the event that the product of the two numbers is odd. We know that the product of the two numbers can be odd only if each of them is odd. Thus,

$$n(\bar{A}) = {}^{50}C_2 = \frac{50 \times 49}{2 \times 1} = 25 \times 49$$

Hence

$$P(\bar{A}) = \frac{n(\bar{A})}{n(S)} = \frac{25 \times 49}{99 \times 50} = \frac{49}{198} = 0.2475$$

So, by Theorem 1.2, it follows that

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{49}{198} = \frac{149}{198} = 0.7525 \quad \blacklozenge$$

EXAMPLE 1.40. A room has 3 electric lamps. From a collection of 10 electric bulbs of which 6 are good, 3 are selected at random and put in the lamps. Find the probability that the room is lighted.

(Anna, Nov. 2007)

Solution. Let A denote the event that the room is lighted. It is clear that the room is lighted if at least one good bulb is put in the three lamps. Then $\bar{A}$ denotes the event that the room is not lighted, i.e. the three lamps carry only bad bulbs. Hence,

$$P(\bar{A}) = \frac{{}^4C_3}{{}^{10}C_3} = \frac{4}{120} = \frac{1}{30}$$

Thus, by Theorem 1.2, it follows that

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{1}{30} = \frac{29}{30} \approx 0.9667 \quad \blacklozenge$$

Theorem 1.3. The probability function P is *monotone* and *subtracive*. That is, if A and B are any two events with $A \subset B$, then the following hold:

- (a) $P(A) \leq P(B)$
- (b) $P(B - A) = P(B) - P(A)$, where $B - A = B \cap \bar{A}$.

Proof. (a) As seen in Figure 1.3, the events A and $B - A$ are mutually exclusive and their union is B . Hence, by Axiom (iii), we have

$$P(B) = P(A \cup B - A) = P(A) + P(B - A)$$

Thus, it follows that

$$P(B - A) = P(B) - P(A)$$

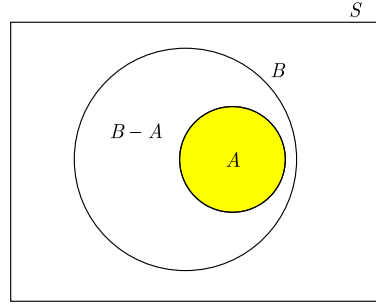


Figure 1.3: P is monotone and subtractive.

(b) By Axiom (i), we know that $P(B - A) \geq 0$. By (i), we know that

$$P(B - A) = P(B) - P(A)$$

Hence, it is immediate that $P(B) \geq P(A)$. ◆

Next, we establish the *Addition Rule*, which gives the relationship between the probability of $A \cup B$ in terms of the individual probabilities, viz. $P(A)$ and $P(B)$, and the probability of the intersection, viz. $P(A \cap B)$.

Theorem 1.4. (*Addition Rule*) If A and B are any two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

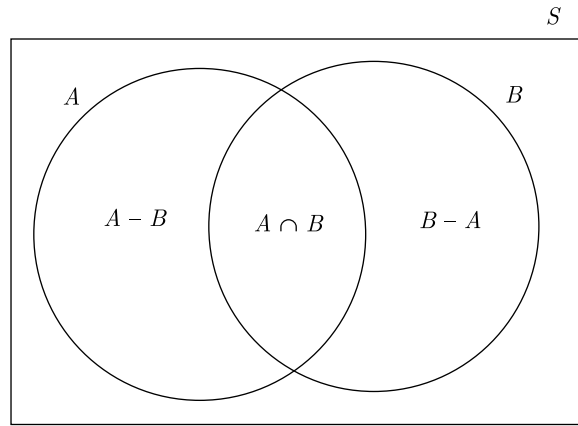


Figure 1.4: Addition rule for probability.

Proof. As illustrated in Figure 1.4, the events $A - B$, $A \cap B$ and $B - A$ are mutually exclusive, and their union is $A \cup B$. Hence, by Axiom (iii), we have

$$P(A \cup B) = P(A - B) + P(A \cap B) + P(B - A) \quad (1.5)$$

Since A is the union of mutually exclusive events $A - B$ and $A \cap B$, it follows from Axiom (iii) that

$$P(A) = P(A - B) + P(A \cap B)$$

or

$$P(A - B) = P(A) - P(A \cap B) \quad (1.6)$$

Since B is the union of mutually exclusive events $B - A$ and $A \cap B$, it follows from Axiom (iii) that

$$P(B) = P(B - A) + P(A \cap B)$$

or

$$P(B - A) = P(B) - P(A \cap B) \quad (1.7)$$

Substituting the results from Eq. (1.6) and Eq. (1.7) into Eq. (1.5), we get

$$\begin{aligned} P(A \cup B) &= [P(A) - P(A \cap B)] + P(A \cap B) + [P(B) - P(A \cap B)] \\ &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

◆

As a consequence of Theorem 1.4, we have the following result, which gives a simple formula for computing the probability of three events.

Corollary 1.3. If A, B and C are any three events, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - [P(A \cap B) + P(B \cap C) + P(A \cap C)] + P(A \cap B \cap C)$$

Proof. Let $E = B \cup C$. By Theorem 1.4, it follows that

$$P(A \cup B \cup C) = P(A \cup E) = P(A) + P(E) - P(A \cap E) \quad (1.8)$$

Since $E = B \cup C$, it follows from Theorem 1.4 that

$$P(E) = P(B \cup C) = P(B) + P(C) - P(B \cap C) \quad (1.9)$$

Since

$$A \cap E = A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

it follows again from Theorem 1.4 that

$$P(A \cap E) = P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \quad (1.10)$$

Substituting the results from Eqs. (1.9) and (1.10) into Eq. (1.8), it follows that

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$$

◆

As an extension of Theorem 1.4, we prove the following result, which provides a formula for computing the probability of the union of any n events.

Theorem 1.5. (*Addition Rule for n Events*) Let $A_1, A_2, \dots, A_n$ be any n events. Then

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) \\ &\quad - \dots + (-1)^{n-1} P\left(\bigcap_{i=1}^n A_i\right) \end{aligned} \quad (1.11)$$

Proof. We prove this result by the principle of mathematical induction.

For $n = 2$, the result is true by Theorem 1.4.

Induction Hypothesis: Assume that the result is true for $n = m$, i.e.

$$\begin{aligned} P\left(\bigcup_{i=1}^m A_i\right) &= \sum_{i=1}^m P(A_i) - \sum_{1 \leq i < j \leq m} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq m} P(A_i \cap A_j \cap A_k) \\ &\quad - \dots + (-1)^{m-1} P\left(\bigcap_{i=1}^m A_i\right) \end{aligned} \quad (1.12)$$

We shall show that the result is also true for $n = m + 1$. By Addition Rule for two events (Theorem 1.4), we have

$$P\left(\bigcup_{i=1}^{m+1} A_i\right) = P\left(\left(\bigcup_{i=1}^m A_i\right) \cup A_{m+1}\right) = P\left(\bigcup_{i=1}^m A_i\right) + P(A_{m+1}) - P\left(\left(\bigcup_{i=1}^m A_i\right) \cap A_{m+1}\right)$$

or

$$P\left(\bigcup_{i=1}^{m+1} A_i\right) = P\left(\bigcup_{i=1}^m A_i\right) + P(A_{m+1}) - P\left(\bigcup_{i=1}^m (A_i \cap A_{m+1})\right) \quad (1.13)$$

The result Eq. (1.12) in the induction hypothesis gives a formula for the probability for the union of any n events. Applying this result, we can simplify Eq. (1.13) as

$$\begin{aligned} P\left(\bigcup_{i=1}^{m+1} A_i\right) &= \sum_{i=1}^m P(A_i) - \sum_{1 \leq i < j \leq m} P(A_i \cap A_j) + \dots + (-1)^{m-1} P\left(\bigcap_{i=1}^m A_i\right) + P(A_{m+1}) \\ &\quad - \sum_{i=1}^m P(A_i \cap A_{m+1}) + \sum_{1 \leq i < j \leq m} P(A_i \cap A_j \cap A_{m+1}) - \dots + (-1)^m P\left(\bigcap_{i=1}^m A_i \cap A_{m+1}\right) \end{aligned}$$

Rearranging terms, it follows that

$$P\left(\bigcup_{i=1}^{m+1} A_i\right) = \sum_{i=1}^{m+1} P(A_i) - \sum_{1 \leq i < j \leq m+1} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq m+1} P(A_i \cap A_j \cap A_k) - \dots + (-1)^m P\left(\bigcap_{i=1}^{m+1} A_i\right)$$

Thus, we have shown that the result (1.11) is also true for $n = m + 1$. The proof is complete by the principle of mathematical induction. $\blacklozenge$

Remark 1.14. Let $A_1, A_2, \dots, A_n$ be any n events. We can define

$$\begin{aligned} S_1 &= \sum_{i=1}^n P(A_i) \\ S_2 &= \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \\ S_3 &= \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) \\ &\vdots \quad \vdots \quad \vdots \\ S_n &= P(A_1 \cap A_2 \cap \dots \cap A_n). \end{aligned}$$

Then we can express the Addition Rule for n events A_i equivalently as

$$P\left(\bigcup_{i=1}^n A_i\right) = S_1 - S_2 + S_3 - \dots + (-1)^{n-1} S_n \quad (1.14)$$

The result in Eq. (1.14) is also known as the *Inclusion-Exclusion Principle*.

EXAMPLE 1.41. Given an experiment such that $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}, P(A \cap B) = \frac{1}{4}$. Compute (i) $P(\overline{A \cap B})$, (ii) $P(\overline{A} \cap B)$.

(Anna, May 2007)

Solution. By the Addition Rule for 2 events, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Hence, it follows that

$$P(A \cup B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$$

(i) By De Morgan's law (Theorem 1.1), $\overline{A \cap B} = \overline{A} \cap \overline{B}$. Hence, it follows that

$$P(\overline{A \cap B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - \frac{7}{12} = \frac{5}{12}$$

(ii) Note that

$$P(\overline{A} \cap B) = P(B) - P(A \cap B) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

◆

EXAMPLE 1.42. If A and B are events with $P(A) = \frac{3}{8}, P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$, find $P(\overline{A} \cap \overline{B})$.

(Anna, April 2003)

Solution. By the Addition Rule for 2 events, we have

$$P(\bar{A} \cup \bar{B}) = P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cap \bar{B})$$

or

$$P(\bar{A} \cap \bar{B}) = P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cup \bar{B})$$

By the given data, we have

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{3}{8} = \frac{5}{8}, \quad P(\bar{B}) = 1 - P(B) = 1 - \frac{1}{2} = \frac{1}{2}$$

and

$$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - \frac{1}{4} = \frac{3}{4}$$

Hence, the required probability is

$$P(\bar{A} \cap \bar{B}) = P(\bar{A}) + P(\bar{B}) - P(\bar{A} \cup \bar{B}) = \frac{5}{8} + \frac{1}{2} - \frac{3}{4} = \frac{3}{8}$$

◆

EXAMPLE 1.43. The events A and B are such that

$$P(A+B) = \frac{3}{4}, \quad P(AB) = \frac{1}{4} \quad \text{and} \quad P(\bar{A}) = \frac{2}{3}$$

Find $P(B)$.

(Anna, Nov. 2004)

Solution. We have $P(\bar{A}) = 1 - P(A) = 1 - \frac{2}{3} = \frac{1}{3}$. Also,

$$P(A \cup B) = \frac{3}{4} \quad \text{and} \quad P(A \cap B) = \frac{1}{4}$$

From the Addition Rule for 2 events, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

or

$$P(B) = P(A \cup B) - P(A) + P(A \cap B) = \frac{3}{4} - \frac{1}{3} + \frac{1}{4} = \frac{2}{3}$$

◆

EXAMPLE 1.44. If $P(A) = P(B) = P(AB)$, prove that $P(A\bar{B} + \bar{A}B) = 0$.

(Anna, Nov. 2007)

Solution. Note that $A\bar{B} = A \cap \bar{B} = A - B$ and $\bar{A}B = B \cap \bar{A} = B - A$ are mutually exclusive events and also that

$$P(A - B) = P(A) - P(A \cap B) \quad \text{and} \quad P(B - A) = P(B) - P(A \cap B)$$

Hence, it follows that

$$P(A \cap \bar{B} + \bar{A} \cap B) = P(A - B) + P(B - A) = P(A) + P(B) - 2P(A \cap B) = 0$$

since it is given that $P(A) = P(B) = P(A \cap B)$.

◆

EXAMPLE 1.45. Three students A, B and C are in a swimming race. A and B have the same probability of winning and each is twice as likely to win as C . Find the probability of winning of each of the students.

(Anna, Nov. 2006)

Solution. Let A, B and C denote the events of the students A, B and C winning the swimming race, respectively. From the given data, we have

$$P(A) = P(B) = 2P(C)$$

Since the total probability is equal to 1, we have

$$P(A) + P(B) + P(C) = 1 \Rightarrow 2P(C) + 2P(C) + P(C) = 1,$$

i.e.

$$P(C) = \frac{1}{5}$$

Hence, it follows that

$$P(A) = P(B) = 2P(C) = \frac{2}{5}$$

◆

EXAMPLE 1.46. Show that for any two events A and B ,

$$P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$$

(Madras, Oct. 1997)

Solution. Since $A \cap B \subset A$ and P is subtractive, it is immediate that

$$P(A \cap B) \leq P(A)$$

Similarly, since $A \subset A \cup B$ and P is subtractive, it is immediate that

$$P(A) \leq P(A \cup B)$$

Next, by the Addition Rule for 2 events, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \geq P(A) + P(B) \quad (\text{since } P(A \cap B) \geq 0)$$

◆

EXAMPLE 1.47. If A and B are two mutually exclusive events, why the following assignment of probabilities is not permissible? $P(A) = 0.6$; $P(A \cap \bar{B}) = 0.5$

(Madras, Oct. 1996)

Solution. If A and B are mutually exclusive events, then $A \cap B = \emptyset$. Note that

$$P(A \cap \bar{B}) = P(A - B) = P(A) - P(A \cap B) = P(A) - 0 = P(A)$$

In the given assignment of probabilities, $P(A) \neq P(A \cap \bar{B})$ and hence it is not permissible.

◆

EXAMPLE 1.48. A card is drawn at random from a pack of well-shuffled playing cards. Find the probability that it is a Spade or a King.

Solution. Since the pack has a total of 52 cards, $n(S) = {}^{52}C_1 = 51$. Let A denote the event that a Spade is drawn and let B denote the event that a King is drawn. We are asked to compute the probability of drawing a Spade or a King, i.e. $P(A \cup B)$.

Since there are a total of 13 Spades, it is clear that $n(A) = {}^{13}C_1 = 13$. Since there are a total of 4 Kings, $n(B) = {}^4C_1 = 4$. Since there is just one card which is both a Spade and a King, $n(A \cap B) = 1$. Therefore,

$$P(A) = \frac{n(A)}{n(S)} = \frac{13}{52}, P(B) = \frac{n(B)}{n(S)} = \frac{4}{52} \text{ and } P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{52}$$

Hence, by the Addition Rule, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{13+4-1}{52} = \frac{16}{52} = \frac{4}{13}$$

◆

EXAMPLE 1.49. In a city, three English dailies A, B and C are published, and a recent survey of readers indicates the following: 30% read A , 20% read B , 15% read C , 10% read both A and B , 8% read both B and C , 4% read both A and C and 2% read A, B and C . Compute the probability that at least one paper among A, B and C is read by a randomly chosen person in the city.

Solution. Let A, B and C denote the events that a person reads the papers A, B and C , respectively. Then, we are given that

$$P(A) = \frac{30}{100} = \frac{3}{10}, P(B) = \frac{20}{100} = \frac{1}{5}, P(C) = \frac{15}{100} = \frac{3}{20}$$

$$P(A \cap B) = \frac{10}{100} = \frac{1}{10}, P(B \cap C) = \frac{8}{100} = \frac{2}{25}, P(A \cap C) = \frac{4}{100} = \frac{1}{25}$$

and

$$P(A \cap B \cap C) = \frac{2}{100} = \frac{1}{50}$$

The required probability is $P(A \cup B \cup C)$. By Addition Rule for 3 events,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - [P(A \cap B) + P(B \cap C) + P(A \cap C)] + P(A \cap B \cap C)$$

A simple calculation yields

$$P(A \cup B \cup C) = \frac{13}{20} - \frac{11}{50} + \frac{1}{50} = \frac{9}{20}$$

Thus, the probability that at least one of the 3 papers is read is $\frac{9}{20}$ or 45%. ◆

EXAMPLE 1.50. Find $P(A \text{ XOR } B)$, where $A \text{ XOR } B$ is the event that either A or B but not both will happen. (In Logic, the operation XOR is referred to as *exclusive OR* function.)

Solution. Note that $A \text{ XOR } B$ is the union of the two mutually exclusive events $A - B$ and $B - A$, i.e.

$$A \text{ XOR } B = (A - B) + (B - A)$$

(See Figure 1.4.) Hence, by Axiom (iii), we have

$$P(A \text{ XOR } B) = P(A - B) + P(B - A).$$

From the proof of Addition Rule for 2 events, we know that

$$P(A - B) = P(A) - P(A \cap B) \text{ and } P(B - A) = P(B) - P(A \cap B)$$

Hence,

$$P(A \text{ XOR } B) = P(A - B) + P(B - A) = P(A) + P(B) - 2P(A \cap B) \quad (1.15)$$

◆

EXAMPLE 1.51. The probability that a new airport will get an award for its design, an award for its efficient use of materials and both the awards is, respectively, 0.16, 0.24 and 0.11. What is the probability that it will get only one of the two awards?

(Anna, April 2005)

Solution. Define the events

- (i) The airport will get an award for its design.
- (ii) It will get an award for its efficient use of materials.

From the given data,

$$P(A) = 0.16, \quad P(B) = 0.24 \text{ and } P(A \cap B) = 0.11$$

The required probability is

$$P(A \text{ XOR } B) = P(A) + P(B) - 2P(A \cap B) = 0.16 + 0.24 - 0.22 = 0.18$$

◆

EXAMPLE 1.52. The probability that a contractor will get a plumbing contract is $\frac{2}{3}$ and the probability that he will not get an electric contract is $\frac{5}{9}$. If the probability of getting one contract is $\frac{4}{5}$, what is the probability that he will get both the contracts?

(Anna, Nov. 2005)

Solution. Define the events

- (i) The contractor will get a plumbing contract.
- (ii) He will get an electric contract.

From the given data,

$$P(A) = \frac{2}{3}, \quad P(\bar{B}) = \frac{5}{9} \text{ and } P(A \cup B) = \frac{4}{5}$$

Note that

$$P(B) = 1 - P(\bar{B}) = 1 - \frac{5}{9} = \frac{4}{9}$$

Hence, the probability that the contractor will get both the contracts is

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = \frac{2}{3} + \frac{4}{9} - \frac{4}{5} = \frac{14}{45} = 0.3111$$

◆

EXAMPLE 1.53. If A and B are any two events, show that

- (i) $P(A) + P(B) \geq 2P(A \cap B)$
- (ii) $P(A \cap B) \geq \frac{1}{2}[P(A) + P(B) - 1]$

Solution. The assertions follow from the formula, Eq. (1.15), derived in Example 1.50.

- (i) Since $P(A \text{ XOR } B) \geq 0$, it is immediate that

$$P(A \text{ XOR } B) = P(A - B) + P(B - A) = P(A) + P(B) - 2P(A \cap B) \geq 0$$

or

$$P(A) + P(B) \geq 2P(A \cap B)$$

- (ii) Since $P(A \text{ XOR } B) \leq 1$, it is immediate that

$$P(A \text{ XOR } B) = P(A - B) + P(B - A) = P(A) + P(B) - 2P(A \cap B) \leq 1$$

or

$$2P(A \cap B) \geq P(A) + P(B) - 1$$

or

$$P(A \cap B) \geq \frac{1}{2}[P(A) + P(B) - 1]$$

◆

Next, we establish the *Boole's inequality*, which is named after the British mathematician George Boole (1815–1864). The important contribution of this inequality is that it provides an *upper bound* for the probability of the union of n events $A_1, A_2, \dots, A_n$ in terms of the individual probabilities, $P(A_i)$.

Theorem 1.6. (*Boole's Inequality*) Let $A_1, A_2, \dots, A_n$ be any n events. Then

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i) \quad (1.16)$$

Proof. We prove this result by the principle of mathematical induction.

The result, Eq. (1.16), holds trivially for $n = 1$.

Next, we establish it for the case $n = 2$. By the Addition Rule for two events, we have

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$

By Axiom (i), $P(A_1 \cap A_2) \geq 0$. Hence, it follows that

$$P(A_1 \cup A_2) \leq P(A_1) + P(A_2) \quad (1.17)$$

Thus, the result, Eq. (1.16), holds for $n = 2$.

Induction Hypothesis: Assume that Eq. (1.16) holds for $n = m$, i.e. for any m events $A_1, A_2, \dots, A_m$, we have

$$P\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m P(A_i) \quad (1.18)$$

We shall show that the result, Eq. (1.16) holds for $n = m + 1$.

Consider any $m + 1$ events, say, $A_1, A_2, \dots, A_m, A_{m+1}$.

Define $E = A_1 \cup A_2 \cup \dots \cup A_m$.

By Eq. (1.17), we have

$$P\left(\bigcup_{i=1}^{m+1} A_i\right) = P(E \cup A_{m+1}) \leq P(E) + P(A_{m+1}) \quad (1.19)$$

By our induction hypothesis, Eq. (1.18), it follows that

$$P(E) = P\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m P(A_i) \quad (1.20)$$

Combining the inequalities, Eqs. (1.19) and (1.20), it follows that

$$P\left(\bigcup_{i=1}^{m+1} A_i\right) \leq \sum_{i=1}^m P(A_i) + P(A_{m+1}) = \sum_{i=1}^{m+1} P(A_i)$$

Hence, Eq. (1.16) also holds for $n = m + 1$.

The proof is complete by the principle of mathematical induction. ♦

Next, we prove the *Bonferroni's inequality*, which is named after the Italian mathematician Carlo Emilio Bonferroni (1892–1960). The importance of this inequality is that it gives a *lower bound* for the probability of the union of n events $A_1, A_2, \dots, A_n$ in terms of the individual probabilities, $P(A_i)$.

Theorem 1.7. (*Bonferroni's Inequality*) Let $A_1, A_2, \dots, A_n$ be any n events. Then

$$P\left(\bigcup_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \quad (1.21)$$

Proof. We prove this result by the principle of mathematical induction.

The result, Eq. (1.21), holds trivially for $n = 1$.

For $n = 2$, Eq. (1.21) holds by the Addition Rule for two events.

For $n = 3$, by the Addition Rule for three events (see Corollary 1.3), we have

$$P(A_1 \cup A_2 \cup A_3) = \sum_{i=1}^3 P(A_i) - \sum_{1 \leq i < j \leq 3} P(A_i \cap A_j) + P(A_1 \cap A_2 \cap A_3)$$

Hence, it is immediate that

$$P(A_1 \cup A_2 \cup A_3) \geq \sum_{i=1}^3 P(A_i) - \sum_{1 \leq i < j \leq 3} P(A_i \cap A_j)$$

where we have used the fact that $P(A_1 \cap A_2 \cap A_3) \geq 0$. Thus, the result also holds for $n = 3$.

Induction Hypothesis: Assume that Eq. (1.21) holds for $n = m$, i.e.

$$P\left(\bigcup_{i=1}^m A_i\right) \geq \sum_{i=1}^m P(A_i) - \sum_{1 \leq i < j \leq m} P(A_i \cap A_j) \quad (1.22)$$

We shall show that the result, Eq. (1.21), also holds for $n = m + 1$.

Let $A_1, A_2, \dots, A_m, A_{m+1}$ be any $m + 1$ events.

By the Addition Rule for two events, we have

$$\begin{aligned} P\left(\bigcup_{i=1}^{m+1} A_i\right) &= P\left(\left(\bigcup_{i=1}^m A_i\right) \cup A_{m+1}\right) \\ &= P\left(\bigcup_{i=1}^m A_i\right) + P(A_{m+1}) - P\left(\left(\bigcup_{i=1}^m A_i\right) \cap A_{m+1}\right) \end{aligned}$$

Hence,

$$P\left(\bigcup_{i=1}^{m+1} A_i\right) = P\left(\bigcup_{i=1}^m A_i\right) + P(A_{m+1}) - P\left(\bigcup_{i=1}^m (A_i \cap A_{m+1})\right) \quad (1.23)$$

By induction hypothesis, Eq. (1.22), we have

$$P\left(\bigcup_{i=1}^m A_i\right) \geq \sum_{i=1}^m P(A_i) - \sum_{1 \leq i < j \leq m} P(A_i \cap A_j) \quad (1.24)$$

By Boole's inequality for the m events $A_1 \cap A_{m+1}, \dots, A_m \cap A_{m+1}$, we also have

$$P\left(\bigcup_{i=1}^m (A_i \cap A_{m+1})\right) \leq \sum_{i=1}^m P(A_i \cap A_{m+1}) \quad (1.25)$$

Substituting the inequalities, Eqs. (1.24) and (1.25), into Eq. (1.23), we have

$$P\left(\bigcup_{i=1}^{m+1} A_i\right) \geq \sum_{i=1}^{m+1} P(A_i) - \sum_{1 \leq i < j \leq m+1} P(A_i \cap A_j)$$

Hence, the result Eq. (1.21), also holds for $n = m + 1$.

The proof is complete by the principle of mathematical induction. ◆

Remark 1.15. Combining the two inequalities from Theorems 1.6 and 1.7, we have

$$\sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \leq P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i) \quad (1.26)$$

These lower and upper bounds for the probability of the union of the n events A_i are also known as Bonferroni's inequalities. The importance of these bounds is that they help us provide an *estimate* of $P(\bigcup A_i)$ whenever the exact value of $P(\bigcup A_i)$ is not available.

Theorem 1.8. Let $\{A_n\}$ be a non-decreasing sequence of events of S , i.e.

$$A_n \subset A_{n+1} \text{ for } i = 1, 2, 3, \dots$$

Then

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\lim_{n \rightarrow \infty} A_n\right) = P\left(\bigcup_{n=1}^{\infty} A_n\right) \quad (1.27)$$

Proof. Let

$$A = \bigcup_{n=1}^{\infty} A_n$$

That is,

$$A = \lim_{n \rightarrow \infty} A_n$$

We can express A as

$$A = A_n \cup \left(\bigcup_{i=n}^{\infty} (A_{i+1} - A_i) \right)$$

By Axiom (iii), we have

$$P(A) = P(A_n) + \sum_{i=n}^{\infty} P(A_{i+1} - A_i)$$

Taking limits as $n \rightarrow \infty$, we have

$$P(A) = \lim_{n \rightarrow \infty} P(A_n) + \lim_{n \rightarrow \infty} \sum_{i=n}^{\infty} P(A_{i+1} - A_i) \quad (1.28)$$

The second term on the R.H.S. of Eq. (1.28) tends to zero as $n \rightarrow \infty$ because

$$\sum_{i=n}^{\infty} P(A_{i+1} - A_i) \leq 1 \text{ and } P(A_{i+1} - A_i) \geq 0 \text{ for each } i.$$

Hence, Eq. (1.28) simplifies to

$$P(A) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = P\left(\lim_{n \rightarrow \infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n) \quad \blacklozenge$$

Theorem 1.9. Let $\{A_n\}$ be a non-increasing sequence of events of S , i.e.

$$A_n \supset A_{n+1} \text{ for } i = 1, 2, 3, \dots$$

Then

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\lim_{n \rightarrow \infty} A_n\right) = P\left(\bigcap_{n=1}^{\infty} A_n\right) \quad (1.29)$$

Proof. This result is a simple consequence of Theorem 1.8. Define

$$B_n = \bar{A}_n \text{ for } n = 1, 2, \dots$$

Then $\{B_n\}$ is a non-decreasing sequence of events of S . By Theorem 1.8, we have

$$\lim_{n \rightarrow \infty} P(B_n) = P\left(\lim_{n \rightarrow \infty} B_n\right) = P\left(\bigcup_{n=1}^{\infty} B_n\right)$$

Since $B_n = \bar{A}_n$, we have

$$\lim_{n \rightarrow \infty} P(\bar{A}_n) = P\left(\lim_{n \rightarrow \infty} \bar{A}_n\right) = P\left(\bigcup_{n=1}^{\infty} \bar{A}_n\right)$$

Hence,

$$\lim_{n \rightarrow \infty} [1 - P(A_n)] = P\left(\overline{\bigcap_{n=1}^{\infty} A_n}\right) = 1 - P\left(\bigcap_{n=1}^{\infty} A_n\right)$$

Thus, it follows that

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\lim_{n \rightarrow \infty} A_n\right) = P\left(\bigcap_{n=1}^{\infty} A_n\right)$$

◆

Remark 1.16. Theorems 1.8 and 1.9 have important applications in later chapters. The property, Eq. (1.27), is often expressed as saying that the set function P is *continuous from below*, and the property Eq. (1.29), is often expressed as saying that P is *continuous from above*. Together, the two theorems assure us that the probability function P is *continuous from above and below*.

EXAMPLE 1.54. A die is loaded in such a way that each odd number is twice as likely to occur as each even number. Find $P(G)$, where G is the event that a number greater than 3 occurs on a single roll of the die.

(Anna, Nov. 2006)

Solution. Let p denote the probability of getting an even number. Given that

$$P(\text{odd number}) = 2p = 2P(\text{even number})$$

Since the total probability is unity, it follows that

$$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$$

which implies that

$$2p + p + 2p + p + 2p + p = 1 \text{ or } p = \frac{1}{9}$$

Therefore,

$$P(\text{even number}) = \frac{1}{9} \text{ and } P(\text{odd number}) = \frac{2}{9}$$

The given event $G = \{4, 5, 6\}$. Therefore, by Axiom (iii), we have

$$P(G) = P(4) + P(5) + P(6) = \frac{1}{9} + \frac{2}{9} + \frac{1}{9} = \frac{4}{9}$$

◆

EXAMPLE 1.55. Events A and B are such that

$$P(A \cup B) = \frac{3}{4}, P(A \cap B) = \frac{1}{4} \text{ and } P(\bar{A}) = \frac{2}{3}$$

Find $P(B)$.

(Anna, Nov. 2004)

Solution. First, note that $P(A) = 1 - P(\bar{A}) = 1 - \frac{2}{3} = \frac{1}{3}$.

Given also that $P(A \cup B) = \frac{3}{4}$ and $P(A \cap B) = \frac{1}{4}$.

By the Addition Rule for two events, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Substituting the known values in the above, we have

$$\frac{3}{4} = \frac{1}{3} + P(B) - \frac{1}{4}$$

Therefore,

$$P(B) = \frac{3}{4} - \frac{1}{3} + \frac{1}{4} = 1 - \frac{1}{3} = \frac{2}{3}$$

◆

EXAMPLE 1.56. Let $A_1, A_2, \dots, A_n$ be any n events. Show that

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n-1) \quad (1.30)$$

(Madras, Oct. 1999, Oct. 2000)

Solution. Let $B_i = \bar{A}_i$ for $i = 1, 2, \dots, n$. By Boole's inequality for n events (see Theorem 1.6), it follows that

$$P\left(\bigcup_{i=1}^n B_i\right) \leq \sum_{i=1}^n P(B_i)$$

or

$$P\left(\bigcup_{i=1}^n \bar{A}_i\right) \leq \sum_{i=1}^n P(\bar{A}_i)$$

or

$$P\left(\overline{\bigcap_{i=1}^n A_i}\right) = 1 - P\left(\bigcap_{i=1}^n A_i\right) \leq \sum_{i=1}^n [1 - P(A_i)]$$

Simplifying, we have

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - (n-1)$$

◆

PROBLEM SET 1.2

1. A and B are any two events. Let $P(A) = p_1$, $P(B) = p_2$ and $P(C) = p_3$. Determine the following:

- (a) $P(A \cup B)$
- (b) $P(\bar{A} \cup \bar{B})$
- (c) $P(\bar{A} \cap \bar{B})$
- (d) $P(A \cap \bar{B})$
- (e) $P(\bar{A} \cap B)$
- (f) $P(\bar{A} \cup B)$
- (g) $P(A \cup \bar{B})$

2. If A and B are events with $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{7}{12}$, find

- (a) $P(\bar{A} \cap B)$
- (b) $P(A \cap \bar{B})$
- (c) $P(\bar{A} \cap \bar{B})$
- (d) $P(\bar{A} \cup \bar{B})$

3. Events A and B are such that

$$P(A \cup B) = \frac{1}{2}, P(\bar{A} \cap B) = \frac{1}{5} \text{ and } P(\bar{B}) = \frac{3}{4}$$

Find (a) $P(A)$ and (b) $P(A \cap \bar{B})$.

4. If A and B are mutually exclusive events such that $P(A) = \frac{1}{4}$ and $P(B) = \frac{3}{5}$, find $P(\bar{A} \cap \bar{B})$.
5. Three newspapers A , B and C are published in a metropolitan city and a recent survey of readers in the city states the following: 50% read A , 30% read B , 20% read C , 20% read both A and B , 15% read both A and C , 10% read both B and C , and 5% read all the three papers. If a person is chosen at random from the city, compute the probability that he reads at least one of the papers A , B and C .
6. For any events A and B , show that $P[(A - B) \cup (B - A)] = P(A) + P(B) - 2P(A \cap B)$.
7. Three horses A , B and C are in a derby race. The chance of A winning the race is twice that of B and the chance of B winning the race is twice that of C . Find the respective chances of A , B and C winning the race.
8. From a faculty group of 5 mathematicians, 3 physicists and 2 chemists, a selection committee of 3 people is selected randomly. Find the probability that the selection committee will consist of (a) all mathematicians, (b) 2 mathematicians, 1 physicist, and (c) 1 mathematician, 1 physicist and 1 chemist.
9. A box contains 5 red, 6 black and 4 green balls. Three balls are drawn at random. Find the probability that the balls are (a) 1 red, 1 black and 1 green, (b) all red, (c) 2 red, 1 green, and (d) 1 black, 2 green.
10. Three cards are drawn from a well-shuffled pack of playing cards. Find the probability that they are of (a) 2 Spades, 1 Club, (b) all face cards, and (c) all different suits.

1.5 CONDITIONAL PROBABILITY

So far, we have calculated probabilities of various events on the assumption that no information was available about the random experiment other than the sample space S . If, however, it is known that an event B has occurred, then the probability of the event A has to be calculated with the condition that the event B has already occurred. In defining this (conditional) probability of A given that B has occurred, we take B to be the new sample space, and relative to this, we define the probability of A . Note that in the classical interpretation of probability, the definition for the conditional probability of A , given that B has occurred, will be

$$P(A|B) = \frac{n(A \cap B)}{n(B)} \text{ where } n(B) \neq 0 \quad (1.31)$$

(See Figure 1.5).

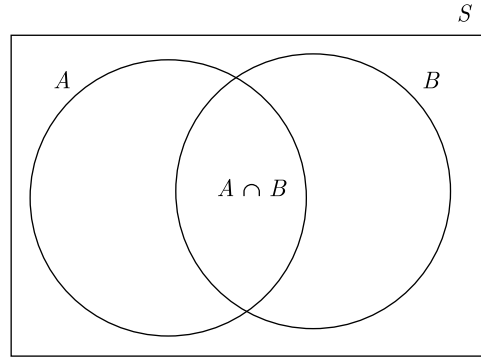


Figure 1.5: Conditional probability.

EXAMPLE 1.57. Consider the experiment of tossing two fair coins. Then, the sample space S is

$$S = \{HH, HT, TH, TT\}$$

Consider the two events A and B defined by

A : Both coins show the same face.

B : At least one coin shows T .

Then, we know that the favourable cases for A are HH and TT so that $n(A) = 2$. In this case,

$$P(A) = \frac{2}{4} = \frac{1}{2}$$

Suppose that the event B has already occurred, i.e. we are given the information “at least one coin shows T ”. This information *changes* the favourable number of cases for A . In fact, the favourable case for A , given the condition that B has occurred, is just one point, viz. TT . Note that the sample space has also changed. Since B has already occurred, the sample space for A is the event B . Note that B contains only the outcomes TH, HT, TT so that $n(B) = 3$. Thus, the (conditional) probability of A (given that B has occurred) is obtained as

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{1}{3}$$

EXAMPLE 1.58. Consider the experiment of tossing two fair dice. Then the sample space S consists of $6^2 = 36$ outcomes.

Consider the two events A and B defined by

A : The sum of the numbers on the dice is 6 or 8.

B : At least one die shows 4.

If we represent the sum of the numbers by the variable T , then the event A may be expressed as $T = 6$ or $T = 8$. Hence, the favourable cases for A are

$$(1, 5), (2, 4), (3, 3), (4, 2), (5, 1), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$$

so that $n(A) = 10$. In this case,

$$P(A) = \frac{10}{36} = \frac{5}{18}$$

It is clear that $n(B) = 6 + 6 = 12$ as we enumerate all sample points of the form $(\star, 4)$ and $(4, \star)$.

Suppose that the event B has already occurred, i.e. we are given the information “at least one die shows 4”. Now, the sample space S reduces to the event B so that $n(S) = n(B) = 12$. Of these total outcomes, the favourable outcomes for A are:

$$(2, 4), (4, 2), (4, 4)$$

so that $n(A \cap B) = 3$. Thus, the (conditional) probability of A , given that B has occurred, is obtained as

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{3}{12} = \frac{1}{4}$$

The above two examples are in the framework of classical interpretation of probability, namely, the number of outcomes of the random experiment are finite and equally likely. To derive a general definition for conditional probability of the event A , given that B has happened, we rewrite Eq. (1.31) as

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{n(A \cap B)/n(S)}{n(B)/n(S)} = \frac{P(A \cap B)}{P(B)} \text{ where } P(B) \neq 0$$

Interchanging the roles of A and B , we may also define the conditional probability of the event B , given that A has happened, as

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \text{ where } P(A) \neq 0$$

These observations lead to the following formal definition of *conditional probability* of an event given that another event has happened.

Definition 1.15. Let $(S, \mathcal{B}, P)$ be a probability space, and the events $A, B \in \mathcal{B}$. The *conditional probability of A given B* , denoted by $P(A|B)$, is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ if } P(B) \neq 0$$

Similarly, the *conditional probability of B given A* , denoted by $P(B|A)$, is defined as

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \text{ if } P(A) \neq 0$$

Remark 1.17. It is important to note that the conditional probability function $P(\cdot|B)$, where $P(B) > 0$, is well-defined, i.e. it satisfies the three axioms of probability. The relevant calculations are as follows:

1. For any event A , $A \cap B \in \mathcal{B}$ since $\mathcal{B}$ is a σ -algebra. Hence, $P(A \cap B) \geq 0$ since P is a probability function. Therefore, it follows that

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \geq 0$$

Hence, $P(\cdot|B)$ satisfies Axiom (i).

2. Since P is a probability function, $P(S) = 1$. Therefore, it follows that

$$P(S|B) = \frac{P(S \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

Hence, $P(\cdot|B)$ satisfies Axiom (ii).

3. If $\{A_n\}$ is any pairwise disjoint collection of events, then it is obvious that $\{A_n \cap B\}$ will be also a pairwise disjoint collection of events. Thus, we have

$$P\left(\bigcup_{n=1}^{\infty} A_n|B\right) = \frac{P\left(\left(\bigcup_{n=1}^{\infty} A_n\right) \cap B\right)}{P(B)} = \frac{P\left(\bigcup_{n=1}^{\infty} (A_n \cap B)\right)}{P(B)} = \frac{\sum_{n=1}^{\infty} P(A_n \cap B)}{P(B)}$$

Hence, it follows that

$$P\left(\bigcup_{n=1}^{\infty} A_n|B\right) = \sum_{n=1}^{\infty} \frac{P(A_n \cap B)}{P(B)} = \sum_{n=1}^{\infty} P(A_n|B)$$

Hence, $P(\cdot|B)$ satisfies Axiom (iii). Thus, the conditional probability function satisfies all the three axioms of probability.

An immediate consequence of Definition 1.15 is the following result, known as the *multiplication rule for 2 events on probability*.

Theorem 1.10. (*Multiplication Rule for 2 Events*) Let A and B be any two events. Then

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A) \text{ where } P(A) \neq 0, P(B) \neq 0 \quad (1.32)$$

Proof. The assertion is an immediate consequence of Definition 1.15. ♦

Next, we generalize this to the following result, known as the *multiplication rule for n events on probability*.

Theorem 1.11. (*Multiplication Rule for n Events*) Let $A_1, A_2, \dots, A_n$ be any n events. Then

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap A_2 \cap \cdots \cap A_{n-1}) \quad (1.33)$$

Proof. The proof follows from the principle of mathematical induction.

For $n = 2$, the result, Eq. (1.33), holds by Theorem 1.10.

Induction Hypothesis: Assume that the result holds for $n = m$, i.e. assume that

$$P\left(\bigcap_{i=1}^m A_i\right) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_m|A_1 \cap A_2 \cap \cdots \cap A_{m-1}) \quad (1.34)$$

We shall show that the result also holds for $n = m + 1$.

Let $A_1, A_2, \dots, A_m, A_{m+1}$ be any $m + 1$ events. By Theorem 1.10, we have

$$P\left(\bigcap_{i=1}^{m+1} A_i\right) = P\left(\left(\bigcap_{i=1}^m A_i\right) \cap A_{m+1}\right) = P\left(\bigcap_{i=1}^m A_i\right) P\left(A_{m+1} \mid \bigcap_{i=1}^m A_i\right)$$

Using the induction hypothesis, Eq. (1.34), it follows that

$$P\left(\bigcap_{i=1}^{m+1} A_i\right) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P\left(A_{m+1} \mid \bigcap_{i=1}^m A_i\right)$$

Thus, the result, Eq. (1.33), also holds for $n = m + 1$.

The proof is complete by the principle of mathematical induction. ◆

EXAMPLE 1.59. A box contains 4 bad and 6 good tubes. Two are drawn out from the box at a time. One of them is tested and found to be good. What is the probability that the other one also is good?

(Anna, May 2007)

Solution. Let A and B denote the events that the *first tube is good* and the *second tube is good*, respectively. From the given data, it follows that

$$P(A) = \frac{6}{10} = \frac{3}{5}$$

and

$$P(A \cap B) = \frac{{}^6C_2}{{}^{10}C_2} = \frac{6 \times 5}{10 \times 9} = \frac{1}{3}$$

Hence, the required probability is

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1/3}{3/5} = \frac{5}{9} \quad \text{◆}$$

EXAMPLE 1.60. If $P(A) = 0.5$, $P(B) = 0.3$ and $P(A \cap B) = 0.15$, find $P(A|\bar{B})$. (Madras, April 1997)

Solution. By definition,

$$P(A|\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})} = \frac{P(A) - P(A \cap B)}{1 - P(B)} = \frac{0.5 - 0.15}{1 - 0.3} = \frac{0.35}{0.7} = 0.5 \quad \text{◆}$$

EXAMPLE 1.61. If two cards are drawn successively without replacement from a well-shuffled pack of playing cards, find the probability that they are both aces.

Solution. This problem can be solved by Multiplication Rule for 2 events. Let A and B denote the events that the *first card drawn is ace* and the *second card drawn is ace*, respectively. The required probability is $P(A \cap B)$. By Multiplication Rule, we have

$$P(A \cap B) = P(A)P(B|A) = \frac{4C_1}{52C_1} \times \frac{3C_1}{51C_1} = \frac{1}{221}$$

◆

EXAMPLE 1.62. Cards are dealt one by one from a well-shuffled pack of playing cards until an ace appears. Show that the probability that exactly n cards are dealt before the first ace appears is

$$\frac{4(51-n)(50-n)(49-n)}{52 \times 51 \times 50 \times 49}$$

Solution. This problem can be solved using the Multiplication Rule for 2 events. We define the events

A : No ace is drawn in the first n draws.

B : An ace appears in the $(n+1)^{\text{st}}$ draw.

Hence, the probability that exactly n cards are dealt before the first ace appears is equal to $P(A \cap B)$. Note that

$$P(A) = \frac{48C_n}{52C_n} \text{ and } P(B|A) = \frac{4}{52-n}$$

By Multiplication Rule for 2 events, we have

$$P(A \cap B) = P(A)P(B|A) = \frac{48C_n}{52C_n} \times \frac{4}{52-n}$$

Thus,

$$P(A \cap B) = \frac{48!}{(48-n)!n!} \times \frac{(52-n)!n!}{52!} \times \frac{4}{52-n}$$

Simplifying, we get

$$P(A \cap B) = \frac{4(51-n)(50-n)(49-n)}{52 \times 51 \times 50 \times 49}$$

◆

EXAMPLE 1.63. Cards are drawn one by one at random from a well-shuffled pack of 52 cards until 2 aces appear for the first time. If n is the number of cards required to be drawn, then show that

$$P(N = n) = \frac{(n-1)(52-n)(51-n)}{50 \times 49 \times 17 \times 13}, \quad (2 \leq n \leq 50)$$

Solution. This problem can be solved using the Multiplication Rule for 2 events. We define the events

A : One ace is drawn in the first $n-1$ draws.

B : An ace is drawn in the n^{th} draw.

The required probability, $P(N = n)$, is equal to $P(A \cap B)$. We note that

$$P(A) = \frac{{}^4C_1 \times {}^{48}C_{n-2}}{{}^{52}C_{n-1}} \quad \text{and} \quad P(B|A) = \frac{3}{52 - (n-1)} = \frac{3}{53-n}$$

By Multiplication Rule for 2 events, we have

$$P(A \cap B) = P(A)P(B|A) = \frac{{}^4C_1 \times {}^{48}C_{n-2}}{{}^{52}C_{n-1}} \times \frac{3}{53-n}$$

Thus, we have

$$P(A \cap B) = \frac{4 \times 48!}{(n-2)!(50-n)!} \times \frac{(53-n)!(n-1)!}{52!} \times \frac{3}{53-n}$$

Simplifying, we get

$$P(A \cap B) = \frac{(n-1)(52-n)(51-n)}{50 \times 49 \times 17 \times 13} \quad \blacklozenge$$

EXAMPLE 1.64. A coin is tossed an infinite number of times. If the probability of a head in a single toss is p , find the probability that the k th head is obtained at the n th tossing but not earlier, with $q = 1 - p$.

(Anna, May 2006)

Solution. Define the events A and B as

A : $k-1$ heads are obtained in the first $n-1$ tosses.

B : A head is obtained in the n^{th} toss.

Clearly, A and B are independent events. Note that

$$P(A) = \binom{n-1}{k-1} p^{k-1} q^{(n-1)-(k-1)} = \binom{n-1}{k-1} p^{k-1} q^{n-k}$$

and

$$P(B|A) = P(B) = p$$

The required probability is given by

$$P(A \cap B) = P(A)P(B) = \binom{n-1}{k-1} p^k q^{n-k} \quad \blacklozenge$$

EXAMPLE 1.65. A box of light bulbs contains 30 bulbs, of which 5 bulbs are defective. If 3 of the bulbs are selected at random and taken out from the box in succession without replacement, find the probability that all three bulbs are defective.

Solution. Define the events

A : The first bulb selected is defective.

B : The second bulb selected is defective.

C : The third bulb selected is defective.

The required probability is $P(A \cap B \cap C)$. By Multiplication Rule for 3 events,

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|B \cap C) = \frac{5}{30} \times \frac{4}{29} \times \frac{3}{28} = \frac{1}{406}$$

◆

EXAMPLE 1.66. Four cards are to be dealt successively, at random and without replacement, from an ordinary pack of playing cards. Find the probability of receiving a Spade, a Heart, a Diamond and a Club in that order.

(Anna, Nov. 2006)

Solution. This problem can be solved using the Multiplication Rule for 4 events. Define the events

A: The first card is a Spade.

B: The second card is a Heart.

C: The third card is a Diamond.

D: The fourth card is a Club.

The events A, B, C and D are not independent since the cards drawn are not replaced. Note that

$$P(A) = \frac{13}{52}, P(B|A) = \frac{13}{51}, P(C|A \cap B) = \frac{13}{50}, P(D|A \cap B \cap C) = \frac{13}{49}$$

The required probability is

$$\begin{aligned} P(A \cap B \cap C \cap D) &= P(A)P(B|A)P(C|A \cap B)P(D|A \cap B \cap C) \\ &= \frac{13}{52} \times \frac{13}{51} \times \frac{13}{50} \times \frac{13}{49} = \frac{63}{14332} = 0.0044 \end{aligned}$$

◆

EXAMPLE 1.67. A manufacturer of airplane parts knows that the probability is 0.8 that an order will be ready for shipment on time, and it is 0.7 that an order will be ready for shipment and will be delivered on time. What is the probability that such an order will be delivered on time given that it was also ready for shipment on time?

(Anna, April 2004)

Solution. Define the events

A: The order will be ready for shipment on time.

B: The order will be delivered on time.

From the given data, we have

$$P(A) = 0.8 \text{ and } P(A \cap B) = 0.7$$

The required probability is

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.7}{0.8} = \frac{7}{8}$$

◆

EXAMPLE 1.68. A card is drawn from a 52-card pack, and without replacing the first card, a second card is drawn. The first and second cards are not replaced and a third card is drawn.

(i) If the first card is Heart, what is the probability of the second card being a Heart?

- (ii) If the first and second cards are Hearts, what is the probability that the third card is the King of Clubs?

(Bharathiar, Nov. 1996)

Solution. The events A, B and C are defined as follows:

A : The first card drawn is a Heart.

B : The second card drawn is a Heart.

C : The third card drawn is the King of Clubs.

- (i) The required probability is $P(A \cap B)$. By Multiplication Rule for 2 events,

$$P(A \cap B) = P(A)P(B|A) = \frac{13}{52} \times \frac{12}{51} = \frac{1}{17}$$

- (ii) The required probability is $P(A \cap B \cap C)$. By Multiplication Rule for 3 events,

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B) = \frac{13}{52} \times \frac{12}{51} \times \frac{1}{50} = \frac{1}{850}$$

◆

Next, we introduce the concept of *independence of events*. In many experiments, the knowledge that the event B has occurred does not affect the probability of the event A . That is, $P(A|B) = P(A)$. In these cases, it is natural to say that A and B are independent events. Example 1.69 illustrates this concept.

EXAMPLE 1.69. Consider the experiment of tossing two coins together. The sample space is

$$S = \{HH, HT, TH, TT\}$$

Solution. Define the two events:

A : The first coin shows H .

B : The second coin shows T .

We compute $P(A|B)$ by the definition

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Note that

$$A = \{HH, HT\}, B = \{TT, HT\} \text{ and } A \cap B = \{HT\}$$

Hence,

$$n(S) = 4, n(A) = 2, n(B) = 2 \text{ and } n(A \cap B) = 1$$

Thus, we see that

$$P(A \cap B) = \frac{1}{4}, P(B) = \frac{2}{4} \text{ and so } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{2}$$

We also see that

$$P(A) = \frac{2}{4} = \frac{1}{2}$$

Thus, $P(A|B) = P(A)$ so that the occurrence of B has not affected the probability of A .

◆

EXAMPLE 1.70. In a certain group of computer personnel, 65% have insufficient knowledge of hardware, 45% have inadequate idea of software and 70% are in either one or both of the two categories. What is the percentage of people who know software among those who have a sufficient knowledge of hardware?

(Anna, Nov. 2006)

Solution. The two events are defined as follows:

A: The computer personnel who have sufficient knowledge of hardware.

B: The computer personnel who have sufficient knowledge of software.

Given that

$$P(\bar{A}) = \frac{65}{100}, P(\bar{B}) = \frac{45}{100} \text{ and } P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = \frac{70}{100}$$

Thus, it follows that

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{65}{100} = \frac{35}{100}, P(B) = 1 - P(\bar{B}) = 1 - \frac{45}{100} = \frac{55}{100}$$

and

$$P(A \cap B) = 1 - P(\overline{A \cap B}) = 1 - \frac{70}{100} = \frac{30}{100}$$

The required probability is equal to

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{30/100}{35/100} = \frac{6}{7} = 0.8571$$

Hence, the people who know software among those who have a sufficient knowledge of hardware is 85.71 %.



EXAMPLE 1.71. Given a binary communication channel, where A is the input and B is the output, let

$$P(A) = 0.4, P(B|A) = 0.9 \text{ and } P(\bar{B}|\bar{A}) = 0.6$$

Find (i) $P(A|B)$, (ii) $P(A|\bar{B})$.

(Anna, Nov. 2006)

Solution. By the definition of conditional probability, we have

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \text{ and } P(\bar{B}|\bar{A}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{A})} = \frac{1 - P(A \cup B)}{1 - P(A)}$$

Hence, we have

$$P(A \cap B) = P(A)P(B|A) = 0.4 \times 0.9 = 0.36$$

Since $P(\bar{A}|\bar{B}) = 0.6$, we have

$$0.6 = \frac{1 - P(A \cup B)}{1 - 0.4} \Rightarrow P(A \cup B) = 1 - 0.36 = 0.64$$

By Addition Rule for 2 events, we know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

from which it follows that

$$P(B) = P(A \cup B) - P(A) + P(A \cap B) = 0.64 - 0.4 + 0.36 = 1 - 0.4 = 0.6$$

(i) We find that

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.36}{0.6} = 0.6$$

(ii) We find that

$$P(A|\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})} = \frac{P(A \cap \bar{B})}{1 - P(B)}$$

We know that the events $A \cap \bar{B}$ and $A \cap B$ are mutually exclusive and that their union is A (see Figure 1.4). Hence, by Axiom (iii), we have

$$P(A) = P(A \cap \bar{B}) + P(A \cap B)$$

so that

$$P(A \cap \bar{B}) = P(A) - P(A \cap B) = 0.4 - 0.36 = 0.04$$

Thus,

$$P(A|\bar{B}) = \frac{0.04}{0.4} = 0.1$$

◆

EXAMPLE 1.72. For any events A , B and C , prove that

$$P(A \cup B|C) = P(A|C) + P(B|C) - P(A \cap B|C)$$

Solution. By definition, we have

$$P(A \cup B|C) = \frac{P((A \cup B) \cap C)}{P(C)} = \frac{P((A \cap C) \cup (B \cap C))}{P(C)} \quad (1.35)$$

By Addition Rule for 2 events, we know that

$$P((A \cap C) \cup (B \cap C)) = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$$

Hence, Eq. (1.35) becomes

$$P(A \cup B|C) = \frac{P(A \cap C)}{P(C)} + \frac{P(B \cap C)}{P(C)} - \frac{P(A \cap B \cap C)}{P(C)}$$

i.e.

$$P(A \cup B|C) = P(A|C) + P(B|C) - P(A \cap B|C)$$

◆

EXAMPLE 1.73. For any three events A, B and C , prove that

$$P(A \cap \bar{B} | C) + P(A \cap B | C) = P(A | C)$$

Solution. By definition, we have

$$P(A \cap \bar{B} | C) = \frac{P(A \cap \bar{B} \cap C)}{P(C)}$$

and

$$P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)}$$

Hence, it follows that

$$P(A \cap \bar{B} | C) + P(A \cap B | C) = \frac{P(A \cap \bar{B} \cap C) + P(A \cap B \cap C)}{P(C)} \quad (1.36)$$

Note that the events $A \cap \bar{B} \cap C$ and $A \cap B \cap C$ are mutually exclusive and their union is the event $A \cap C$ because $B \cup \bar{B} = S$. By Axiom (iii), we have

$$P(A \cap C) = P(A \cap \bar{B} \cap C) + P(A \cap B \cap C)$$

Substituting the above into Eq. (1.36), we have

$$P(A \cap \bar{B} | C) + P(A \cap B | C) = \frac{P(A \cap C)}{P(C)} = P(A | C)$$

◆

EXAMPLE 1.74. If A and B are two events and $P(B) \neq 1$, prove that

$$P(A | \bar{B}) = \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

and deduce that $P(A \cap B) \geq P(A) + P(B) - 1$.

Solution. By definition, we have

$$P(A | \bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})} = \frac{P(A - B)}{1 - P(B)} = \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

(where we have used the fact that $P(A - B) = P(A) - P(A \cap B)$; see Proof of Theorem 1.4).

Next, for any event E , $P(E) \leq 1$. In particular, $P(A | B) \leq 1$. Therefore,

$$\frac{P(A) - P(A \cap B)}{1 - P(B)} \leq 1$$

or

$$P(A) - P(A \cap B) \leq 1 - P(B)$$

or

$$P(A \cap B) \geq P(A) + P(B) - 1$$

◆

Definition 1.16. (*Independent Events*) Two events A and B are called *independent* if

$$P(A \cap B) = P(A)P(B)$$

Remark 1.18. Whereas the conditional probability $P(A|B)$ is defined only when $P(B) > 0$, Definition 1.16 has not made any restriction on $P(A)$ or $P(B)$. This is because if A is any event such that $P(A) = 0$, then A is independent of any event B . Note that $A \cap B \subset A$ so that $P(A \cap B) \leq P(A)$ by the subtractive property of P . Thus, if $P(A) = 0$, then it automatically holds that $P(A \cap B) = 0$. Hence, in this case $P(A \cap B) = P(A)P(B) = 0$ so that A is independent of any event B .

Remark 1.19. Any event A is always independent of S and $\emptyset$. This can be easily seen as follows:

1. $A \cap S = A$ and $P(S) = 1$. Hence, it is immediate that

$$P(A \cap S) = P(A) = P(A)P(S)$$

2. $A \cap \emptyset = \emptyset$ and $P(\emptyset) = 0$. Hence, it is immediate that

$$P(A \cap \emptyset) = 0 = P(A)P(\emptyset)$$

Remark 1.20. We emphasize here that *independent events* and *mutually exclusive events* are different concepts. By definition, two events A and B are *mutually exclusive* if $A \cap B = \emptyset$ and they are *independent* if

$$P(A \cap B) = P(A)P(B)$$

If A and B are mutually exclusive events such that $P(A) > 0$ and $P(B) > 0$, then they cannot be independent because the occurrence of any of them automatically eliminates the occurrence of another. Likewise, if A and B are independent events with $P(A) > 0$ and $P(B) > 0$, then they cannot be mutually exclusive because $P(A \cap B) = P(A)P(B) > 0$.

EXAMPLE 1.75. Differentiate between *mutually exclusive events* and *independent events* with an example for each.

Solution. See Remark 1.20 for the difference between mutually exclusive and independent events. Here, we present examples only.

- (i) Consider the experiment of tossing a single coin. Here, $S = \{H, T\}$. The events A and B of getting a head and a tail, respectively, as the outcome are mutually exclusive. Note that $A = \{H\}$ and $B = \{T\}$ so that $A \cap B = \emptyset$. They cannot be independent because the occurrence of A or B automatically precludes the occurrence of other. Mathematically, $P(A) = P(B) = \frac{1}{2}$, but $A \cap B = \emptyset$ so that

$$P(A \cap B) = 0 \neq \frac{1}{4} = P(A)P(B)$$

- (ii) Consider the experiment of tossing a coin twice. Here, $S = \{HH, HT, TH, TT\}$. The events A and B of getting a head in the first and second tosses, respectively, are independent. Note that

$$A = \{HH, HT\}, \quad B = \{TH, HH\} \quad \text{and} \quad A \cap B = \{HH\}$$

so that

$$P(A \cap B) = \frac{1}{4} = P(A)P(B)$$

However, the events A and B are not mutually exclusive as $A \cap B \neq \emptyset$.



EXAMPLE 1.76. A card is chosen at random from a well-shuffled pack of playing cards. Define the events

A : A Spade is drawn.

B : A King is drawn.

Solution. We check whether A and B are independent. Note that

$$n(S) = 52, n(A) = 13, n(B) = 4, \text{ and } n(A \cap B) = 1$$

Hence, it follows that

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{52}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{13}{52} = \frac{1}{4}$$

and

$$P(B) = \frac{n(B)}{n(S)} = \frac{4}{52} = \frac{1}{13}$$

Note that

$$P(A \cap B) = P(A)P(B) = \frac{1}{52}$$

Hence, A and B are independent events. ◆

EXAMPLE 1.77. If $P(A) = 0.65$, $P(B) = 0.4$ and $P(A \cap B) = 0.24$, can A and B be independent events?

Solution. If A and B are independent events, then we must have

$$P(A \cap B) = P(A)P(B)$$

For the given events A and B , $P(A)P(B) = 0.65 \times 0.4 = 0.26 \neq P(A \cap B)$. Hence, the given events A and B cannot be independent. ◆

EXAMPLE 1.78. If A and B are independent events, prove that $\bar{A}$ and B also are independent. (Anna, April 2005)

Solution. Let A and B be independent events, then

$$P(A \cap B) = P(A)P(B) \tag{1.37}$$

We wish to show that $\bar{A}$ and B are independent events. Note that B is the union of the mutually exclusive events, $A \cap B$ and $B - A = \bar{A} \cap B$ (see Figure 1.4). By Axiom (ii), we have

$$P(B) = P(A \cap B) + P(\bar{A} \cap B) = P(A)P(B) + P(\bar{A} \cap B) \quad (\text{using Eq. (1.37)})$$

Hence,

$$P(\bar{A} \cap B) = P(B) - P(A)P(B) = [1 - P(A)]P(B) = P(\bar{A})P(B)$$

which shows that $\bar{A}$ and B are independent events. ◆

EXAMPLE 1.79. If A and B are independent events, prove that $\bar{A}$ and $\bar{B}$ also are independent.

(Madras, Oct. 2001, May 2002)

Solution. Let A and B be independent events, then

$$P(A \cap B) = P(A)P(B) \quad (1.38)$$

We wish to show that $\bar{A}$ and $\bar{B}$ are independent events. Note that

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$$

By Addition Rule for two events, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) \quad [\text{using Eq. (1.38)}]$$

Hence,

$$P(\bar{A} \cap \bar{B}) = 1 - P(A) - P(B) + P(A)P(B) = [1 - P(A)][1 - P(B)] = P(\bar{A})P(\bar{B})$$

which shows that $\bar{A}$ and $\bar{B}$ are independent events. ◆

EXAMPLE 1.80. If A and B are independent events with $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, find $P(\bar{A} \cap \bar{B})$.

(Anna, Nov. 2006)

Solution. If A and B are independent events, then we know that $\bar{A}$ and $\bar{B}$ also are independent. Hence, we have

$$P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B}) = \left[1 - \frac{1}{2}\right] \times \left[1 - \frac{1}{3}\right] = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} \quad \text{◆}$$

EXAMPLE 1.81. If A and B are two independent events with $P(A) = 0.4$ and $P(B) = 0.5$, find $P(A \cup B)$.

(Anna, Nov. 1996)

Solution. Since A and B are independent, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$.

From the Addition Rule for 2 events, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.7 \quad \text{◆}$$

EXAMPLE 1.82. Let events A and B be independent with $P(A) = 0.5$ and $P(B) = 0.8$. Find the probability that neither of the events A and B occurs.

(Anna, April 2000, May 2007)

Solution. Since A and B are independent events, it follows that $\bar{A}$ and $\bar{B}$ also are independent events. We also note that

$$P(\bar{A}) = 1 - P(A) = 1 - 0.5 = 0.5 \quad \text{and} \quad P(\bar{B}) = 1 - P(B) = 1 - 0.8 = 0.2$$

Hence, the required probability $P(\bar{A} \cap \bar{B})$ can be readily calculated as

$$P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B}) = 0.5 \times 0.2 = 0.1 \quad \text{◆}$$

EXAMPLE 1.83. Let $P(A \cup B) = \frac{5}{6}$, $P(A \cap B) = \frac{1}{3}$ and $P(\bar{B}) = \frac{1}{2}$. Are the events A and B independent? Explain.

(Anna, Nov. 2005)

Solution. First, note that $P(B) = 1 - P(\bar{B}) = 1 - \frac{1}{2} = \frac{1}{2}$
By the Addition Rule for two events, we know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

It follows that

$$\frac{5}{6} = P(A) + \frac{1}{2} - \frac{1}{3} = P(A) + \frac{1}{6}$$

So, we obtain

$$P(A) = \frac{5}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

Thus, we note that

$$P(A)P(B) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3} = P(A \cap B)$$

Hence, the events A and B are independent. ♦

Definition 1.17. (*Pairwise Independent Events*) Let $\mathcal{A}$ be a family of events from the σ -field $\mathcal{B}$. The events in $\mathcal{A}$ are called *pairwise independent* if for every pair of events $A, B \in \mathcal{A}$, we have

$$P(A \cap B) = P(A)P(B)$$

A stronger notion than the above is called *mutual or complete independence* of events.

Definition 1.18. (*Mutual or Complete Independence*) Let $\mathcal{A}$ be a family of events from the σ -field $\mathcal{B}$. The events in $\mathcal{A}$ are called *mutually independent* or *completely independent* if for every finite collection of events $A_1, A_2, \dots, A_n$ from $\mathcal{A}$ (where n is any positive integer), we have

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \dots P(A_n)$$

The next two examples support our claim that mutual independence of events is a stronger notion than pairwise independence of events.

EXAMPLE 1.84. Consider the experiment of tossing two coins simultaneously. Then the sample space S is

$$S = \{HH, HT, TH, TT\}$$

Define three events A, B and C as follows:

- A : The first coin shows H .
- B : The second coin shows H .
- C : Only one coin shows H .

Solution. Note that

$$A = \{HH, HT\}, B = \{HH, TH\} \text{ and } C = \{HT, TH\}$$

Therefore, it is immediate that

$$P(A) = P(B) = P(C) = \frac{2}{4} = \frac{1}{2}$$

Note also that

$$A \cap B = \{HH\}, B \cap C = \{TH\} \text{ and } A \cap C = \{HT\}$$

Therefore, it is immediate that

$$P(A \cap B) = P(B \cap C) = P(A \cap C) = \frac{1}{4}$$

Hence, it follows that

$$\begin{aligned} P(A \cap B) &= P(A)P(B) \\ P(B \cap C) &= P(B)P(C) \\ P(A \cap C) &= P(A)P(C) \end{aligned}$$

Thus, the events A , B and C are pairwise independent. However, they are not mutually independent. Note that $A \cap B \cap C = \emptyset$, and so

$$P(A \cap B \cap C) = 0, \text{ while } P(A)P(B)P(C) = \frac{1}{8}$$

So, $P(A \cap B \cap C) \neq P(A)P(B)P(C)$.

◆

EXAMPLE 1.85. An urn contains four tickets with numbers 112, 121, 211, 222 and one ticket is drawn at random. Let $A_i (i = 1, 2, 3)$ denote the event that the i^{th} digit of the number on the ticket drawn is 1. Discuss the independence of the events A_1, A_2, A_3 .

Solution. Note that the events are

$$A_1 = \{112, 121\}, A_2 = \{112, 211\} \text{ and } A_3 = \{121, 211\}.$$

Hence, it is immediate that

$$P(A_1) = P(A_2) = P(A_3) = \frac{2}{4} = \frac{1}{2}$$

Note also that

$$A_1 \cap A_2 = \{112\}, A_2 \cap A_3 = \{211\} \text{ and } A_1 \cap A_3 = \{121\}$$

so that

$$P(A_1 \cap A_2) = P(A_2 \cap A_3) = P(A_1 \cap A_3) = \frac{1}{4}$$

Hence, it follows that

$$P(A_1 \cap A_2) = P(A_1)P(A_2), P(A_2 \cap A_3) = P(A_2)P(A_3), P(A_1 \cap A_3) = P(A_1)P(A_3)$$

so that the events A_1, A_2, A_3 are pairwise independent.

However,

$$A_1 \cap A_2 \cap A_3 = \emptyset \Rightarrow P(A_1 \cap A_2 \cap A_3) = 0$$

whereas

$$P(A_1)P(A_2)P(A_3) = \frac{1}{8}$$

showing that $P(A_1 \cap A_2 \cap A_3) \neq P(A_1)P(A_2)P(A_3)$. Thus, the events A_1, A_2, A_3 are not mutually independent. ♦

EXAMPLE 1.86. A pair of dice are rolled once. Let A be the event that the first die has a 1 on it, B the event that the second die has a 6 on it and C the event that the sum is 7. Are A, B and C independent? (Bharatidasan, Nov. 1996)

Solution. Clearly, $n(S) = 6^2 = 36$. Note that

$$A = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)\},$$

$$B = \{(1, 6), (2, 6), (3, 6), (4, 6), (5, 6), (6, 6)\},$$

and

$$C = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

It is clear that

$$P(A) = P(B) = P(C) = \frac{6}{36} = \frac{1}{6}$$

Note also that

$$A \cap B = B \cap C = A \cap C = A \cap B \cap C = \{(1, 6)\}$$

Hence, it follows that

$$P(A \cap B) = P(B \cap C) = P(A \cap C) = P(A \cap B \cap C) = \frac{1}{36}$$

Since

$$P(A \cap B) = P(A)P(B), P(B \cap C) = P(B)P(C) \text{ and } P(A \cap C) = P(A)P(C)$$

it follows that A, B and C are pairwise independent.

However,

$$P(A \cap B \cap C) = \frac{1}{36} \neq P(A)P(B)P(C) = \frac{1}{216}$$

Hence, A, B and C are not mutually independent. ♦

EXAMPLE 1.87. If A, B, C are mutually independent events, prove that (i) A and $B \cup C$, (ii) A and $B \cap C$ are independent.

(Madurai, April 1996)

Solution. By hypothesis, A , B and C are mutually independent events. Hence, we know that

$$P(A \cap B) = P(A)P(B), P(B \cap C) = P(B)P(C), P(A \cap C) = P(A)P(C)$$

and

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

(i) To prove that A and $B \cup C$ are independent events, we have to show that

$$P(A \cap (B \cup C)) = P(A)P(B \cup C)$$

Note that

$$\begin{aligned} \text{L.H.S.} &= P((A \cap B) \cup (A \cap C)) = P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) \\ &= P(A)P(B) + P(A)P(C) - P(A)P(B)P(C) \\ &= P(A)[P(B) + P(C) - P(B)P(C)] \\ &= P(A)[P(B) + P(C) - P(B \cap C)] \\ &= P(A)P(B \cup C) = \text{R.H.S.} \end{aligned}$$

Hence, A and $B \cup C$ are independent events.

(ii) To prove that A and $B \cap C$ are independent events, we have to show that

$$P(A \cap (B \cap C)) = P(A)P(B \cap C)$$

Note that

$$\text{L.H.S.} = P(A \cap B \cap C) = P(A)P(B)P(C) = P(A)P(B \cap C) = \text{R.H.S.}$$

Hence, A and $B \cap C$ are independent events.

◆

EXAMPLE 1.88. Show that if A, B, C are pairwise independent events and also if C is independent of $A \cup B$, show that A, B, C are mutually independent events.

Solution. If A, B, C are pairwise independent events, we have

- (i) $P(A \cap B) = P(A)P(B)$
- (ii) $P(B \cap C) = P(B)P(C)$
- (iii) $P(A \cap C) = P(A)P(C)$

If C is independent of $A \cup B$, we have

$$P((A \cup B) \cap C) = P(A \cup B)P(C)$$

or

$$P[(A \cap C) \cup (B \cap C)] = P(A \cup B)P(C)$$

Using the Addition Rule for 2 events, we have

$$P(A \cap C) + P(B \cap C) - P(A \cap B \cap C) = (P(A) + P(B) - P(A \cap B))P(C)$$

Using (i), (ii) and (iii), we have

$$P(A)P(C) + P(B)P(C) - P(A \cap B \cap C) = P(A)P(C) + P(B)P(C) - P(A)P(B)P(C)$$

Simplifying, we get

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

Hence, A, B, C are mutually independent events. ♦

EXAMPLE 1.89. If the probability of success is 0.09, how many trials are needed to have a probability of at least one success as $\frac{1}{3}$ or more?

(Anna, April 2003)

Solution. Let p and q denote the probabilities of *success* and *failure*, respectively. Since $p = 0.09$, $q = 1 - p = 0.81$.

Let x be the number of trials needed to have a probability of at least one success.
The probability of failures in all the x trials is

$$q^x = (0.81)^x$$

So, the probability of at least one success in the x trials is

$$1 - q^x = 1 - (0.81)^x$$

We are asked to find x so that this probability is greater than $\frac{1}{3}$ or more. Hence,

$$1 - (0.81)^x \geq \frac{1}{3} \quad \text{or} \quad (0.81)^x \leq \frac{2}{3}$$

Thus,

$$x \log 0.81 \leq \log \left(\frac{2}{3} \right) \quad \text{or} \quad -0.2107x \leq -0.4055$$

Hence,

$$x \geq \left(\frac{0.4055}{0.2107} \right) = 1.9245$$

Since x is an integer, the required number of trials is $x = 2$. ♦

EXAMPLE 1.90. The probability of a man hitting a target is $\frac{1}{3}$. How many times must he fire so that the probability of hitting the target at least once is more than 90%?

(Anna, Nov. 2003)

Solution. Let

$$p = \text{probability of hitting the target} = \frac{1}{3}$$

$$q = \text{probability of missing the target} = 1 - \frac{1}{3} = \frac{2}{3}$$

Let x be the number of trials needed to have a probability of hitting the target at least once.

The probability of missing the target in all the x trials is

$$q^x = \left(\frac{2}{3}\right)^x$$

So, the probability of hitting the target at least once in all the x trials is

$$1 - q^x = 1 - \left(\frac{2}{3}\right)^x$$

We are asked to find x so that this probability is 90% or more. Hence,

$$1 - \left(\frac{2}{3}\right)^x \geq 0.9 \quad \text{or} \quad \left(\frac{2}{3}\right)^x \leq 1 - 0.9 = 0.1$$

i.e.

$$x \log\left(\frac{2}{3}\right) \leq \log 0.1 \quad \text{or} \quad -0.4055x \leq -2.3026$$

Thus, we have

$$x \geq \left(\frac{2.3026}{0.4055}\right) = 5.6784$$

Since x is an integer, the required number of trials is $x = 6$. ♦

EXAMPLE 1.91. A is known to hit the target in 2 out of 5 shots, whereas B is known to hit the target in 3 out of 4 shots. Find the probability of the target being hit when they both try?

(Anna, Nov. 2003)

Solution. Let A and B denote the events of the persons A and B hitting the target, respectively. From the given data, $P(A) = \frac{2}{5}$ and $P(B) = \frac{3}{4}$.

Since A and B are independent, it follows that

$$P(A \cap B) = P(A)P(B) = \frac{2}{5} \times \frac{3}{4} = \frac{6}{20}$$

The required probability is $P(A \cup B)$. By the Addition Rule for 2 events, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{2}{5} + \frac{3}{4} - \frac{6}{20} = \frac{17}{20} [-15pt]$$
♦

EXAMPLE 1.92. The odds in favour of A solving a mathematical problem are 3 to 4 and the odds against B solving the problems are 5 to 7. Find the probability that the problem will be solved by at least one of them.

(Anna, Nov. 2006)

Solution. The events A and B are defined as follows:

A : Solves the mathematical problem.

B : Solves the mathematical problem.

From the given data, we have

$$P(A) = \frac{3}{3+4} = \frac{3}{7} \text{ and } P(B) = \frac{7}{5+7} = \frac{7}{12}$$

The events A and B are clearly independent. Hence, it follows that

$$P(A \cap B) = P(A)P(B) = \frac{3}{7} \times \frac{7}{12} = \frac{1}{4}$$

The required probability is $P(A \cup B)$. By Addition Rule for 2 events, we have

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{3}{7} + \frac{7}{12} - \frac{1}{4} = \frac{16}{21} \end{aligned}$$

Hence, the probability that the problem will be solved by A or B is equal to $\frac{16}{21}$ or 0.7619. ♦

Remark 1.21. In the above problems, we calculated $P(A \cup B)$, where A and B are independent events, by means of Addition Rule for 2 events. In general, when $A_1, A_2, \dots, A_n$ are mutually independent events, there is a quick way of determining

$$P\left(\bigcup_{i=1}^n A_i\right) = P(A_1 \cup A_2 \cup \dots \cup A_n)$$

Let

$$p_i = P(A_i) \text{ for } i = 1, 2, \dots, n$$

It can be easily shown that if $A_1, A_2, \dots, A_n$ are mutually independent events, then $\bar{A}_1, \bar{A}_2, \dots, \bar{A}_n$ are also mutually independent events. Thus,

$$P\left(\bigcup_{i=1}^n A_i\right) = 1 - P\left(\bigcap_{i=1}^n \bar{A}_i\right) = 1 - P(\bar{A}_1)P(\bar{A}_2) \dots P(\bar{A}_n)$$

i.e.

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = 1 - (1 - p_1)(1 - p_2) \dots (1 - p_n)$$

EXAMPLE 1.93. A problem in Statistics is given to three students A , B and C whose chances of solving it are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$, respectively. What is the probability that the problem will be solved if all of them try independently?

(Madras, Oct. 2001)

Solution. Define the events

A : Solves the problem in Statistics.

B : Solves the problem in Statistics.

C : Solves the problem in Statistics.

Given that

$$p_1 = P(A) = \frac{1}{2}, p_2 = P(B) = \frac{3}{4} \text{ and } p_3 = P(C) = \frac{1}{4}$$

The required probability is

$$P(A \cup B \cup C) = 1 - (1 - p_1)(1 - p_2)(1 - p_3) = 1 - \frac{1}{2} \times \frac{1}{4} \times \frac{3}{4} = \frac{29}{32}$$

◆

EXAMPLE 1.94. $A_1, A_2, \dots, A_n$ are independent events with respective probabilities

$$P(A_i) = 1 - \frac{1}{2^i} \text{ for } i = 1, 2, \dots, n$$

Find $P(A_1 \cup A_2 \cup \dots \cup A_n)$.

Solution. By Remark 1.21, we know that

$$P\left(\bigcup_{i=1}^n A_i\right) = 1 - (1 - p_1)(1 - p_2) \cdots (1 - p_n)$$

Substituting the individual probabilities, we have

$$P\left(\bigcup_{i=1}^n A_i\right) = 1 - \left[\frac{1}{2}\right] \left[\frac{1}{2}\right]^2 \cdots \left[\frac{1}{2}\right]^n = 1 - \left[\frac{1}{2}\right]^{1+2+\dots+n} = 1 - \left[\frac{1}{2}\right]^{\frac{n(n+1)}{2}}$$

◆

EXAMPLE 1.95. A problem in Statistics is given to three students A, B and C , whose chances of solving it are p_1, p_2 and p_3 respectively. Find the probabilities of the following:

- (i) No one will solve the problem.
- (ii) Only one will solve the problem.
- (iii) At least one will solve the problem.

Solution. Let A_1, A_2 and A_3 denote the events of the students A, B and C solving the problem correctly. Note that these are mutually independent events, and their individual probabilities are given in the problem as

$$P(A_1) = p_1, P(A_2) = p_2, P(A_3) = p_3$$

- (i) The required probability is $P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3)$. Since the events A_i are mutually independent, the events $\bar{A}_i$ also will be mutually independent. Hence, we have

$$P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3) = P(\bar{A}_1)P(\bar{A}_2)P(\bar{A}_3) = (1 - p_1)(1 - p_2)(1 - p_3)$$

- (ii) Only one student can solve the problem in one of the following mutually exclusive ways:

- (a) Student A solves it, but students B and C fail.
- (b) Student B solves it, but students A and C fail.
- (c) Student C solves it, but students A and B fail.

Thus, the required probability is obtained as

$$P(A_1 \cap \bar{A}_2 \cap \bar{A}_3) + P(\bar{A}_1 \cap A_2 \cap \bar{A}_3) + P(\bar{A}_1 \cap \bar{A}_2 \cap A_3)$$

which is equal to

$$p_1(1-p_2)(1-p_3) + (1-p_1)p_2(1-p_3) + (1-p_1)(1-p_2)p_3$$

(iii) The required probability is $P(A_1 \cup A_2 \cup A_3)$, which can be easily calculated as

$$P(A_1 \cup A_2 \cup A_3) = 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3) = 1 - (1-p_1)(1-p_2)(1-p_3)$$

◆

EXAMPLE 1.96. A and B are two independent witnesses in a case. The probabilities that A and B will speak the truth are x and y , respectively. A and B agree in a certain statement. Show that the probability that this statement is true is

$$\frac{xy}{1-x-y+2xy}.$$

Solution. We are given that $P(A) = x$, $P(B) = y$, and that A, B are independent events. Now, A and B can agree (i) if they both speak the truth, the probability of which is given by xy or (ii) if they both lie, the probability of which is $(1-x)(1-y)$. Hence, the probability for A and B to agree in a certain statement is

$$xy + (1-x)(1-y)$$

So, the required probability that the agreed statement is true is given by

$$\frac{xy}{xy + (1-x)(1-y)}$$

◆

EXAMPLE 1.97. If A tells the truth 4 out of 5 times and B tells the truth 3 out of 4 times, what is the probability that both expressing the same fact contradict each other.

(Madras, April 2000)

Solution. From the given data,

$$P(A \text{ tells the truth}) = \frac{4}{5} \text{ and } P(B \text{ tells the truth}) = \frac{3}{4}$$

A and B will contradict each other if one speaks the truth and the other does not. Hence, there are two mutually exclusive possibilities:

- (i) A speaks the truth, and B tells a lie.
- (ii) B speaks the truth, and A tells a lie.

Note that

$$P(\text{i}) = \frac{4}{5} \times \frac{1}{4} = \frac{1}{5}$$

and

$$P(\text{ii}) = \frac{1}{5} \times \frac{3}{4} = \frac{3}{20}$$

Hence, the required probability is equal to

$$P(\text{i}) + P(\text{ii}) = \frac{1}{5} + \frac{3}{20} = \frac{7}{20}$$

◆

EXAMPLE 1.98. Three groups of children contain 3 girls and 1 boy, 2 girls and 2 boys, and 1 girl and 3 boys. One child is selected at random from each group. Show that the chance that the three selected consist of 1 girl and 2 boys is $\frac{13}{32}$.

Solution. We tabulate the given data as follows:

	Boys	Girls
Group 1	3	1
Group 2	2	2
Group 3	1	3

Out of each group, 1 child is selected. The three selected can consist of 1 girl and 2 boys in the following three mutually exclusive ways:

- (i) 1 boy from Group 1, 1 boy from Group 2, 1 girl from Group 3.
- (ii) 1 boy from Group 1, 1 girl from Group 2, 1 boy from Group 3.
- (iii) 1 girl from Group 1, 1 boy from Group 2, 1 boy from Group 3.

In calculating probabilities below, it is noted that choosing a child from any of the three groups is independent of choosing a child in the other groups. Hence,

$$P(\text{i}) = \frac{3}{4} \times \frac{2}{4} \times \frac{3}{4} = \frac{9}{32}$$

$$P(\text{ii}) = \frac{3}{4} \times \frac{2}{4} \times \frac{1}{4} = \frac{3}{32}$$

$$P(\text{iii}) = \frac{1}{4} \times \frac{2}{4} \times \frac{1}{4} = \frac{1}{32}$$

So, the required probability is equal to

$$P(\text{i}) + P(\text{ii}) + P(\text{iii}) = \frac{9}{32} + \frac{3}{32} + \frac{1}{32} = \frac{13}{32}$$



EXAMPLE 1.99. The odds that a book will be reviewed favourably by three independent critics are 5 to 1, 4 to 2 and 2 to 4, respectively. What is the probability that of the three reviews, a majority will be favourable?

Solution. Let A , B and C denote the events that the book will get a favourable review from the three critics in order. It is clear that A , B and C are mutually independent events and that

$$P(A) = \frac{5}{6}, P(B) = \frac{2}{3} \text{ and } P(C) = \frac{1}{3}$$

Of the three reviews, a majority will be favourable in the following mutually exclusive ways:

- (i) Critic 1 = Favourable; Critic 2 = Favourable; Critic 3 = Unfavourable
- (ii) Critic 1 = Favourable; Critic 2 = Unfavourable; Critic 3 = Favourable
- (iii) Critic 1 = Unfavourable; Critic 2 = Favourable; Critic 3 = Favourable
- (iv) Critic 1 = Favourable; Critic 2 = Favourable; Critic 3 = Favourable.

Note also that

$$P(\text{i}) = P(A \cap B \cap \bar{C}) = P(A)P(B)P(\bar{C}) = \frac{5}{6} \times \frac{2}{3} \times \frac{2}{3} = \frac{10}{27}$$

$$P(\text{ii}) = P(A \cap \bar{B} \cap C) = P(A)P(\bar{B})P(C) = \frac{5}{6} \times \frac{1}{3} \times \frac{1}{3} = \frac{5}{54}$$

$$P(\text{iii}) = P(\bar{A} \cap B \cap C) = P(\bar{A})P(B)P(C) = \frac{1}{6} \times \frac{2}{3} \times \frac{1}{3} = \frac{1}{27}$$

$$P(\text{iv}) = P(A \cap B \cap C) = P(A)P(B)P(C) = \frac{5}{6} \times \frac{2}{3} \times \frac{1}{3} = \frac{5}{27}$$

Hence, the required probability is equal to

$$P(\text{i}) + P(\text{ii}) + P(\text{iii}) + P(\text{iv}) = \frac{10}{27} + \frac{5}{54} + \frac{1}{27} + \frac{5}{27} = \frac{37}{54}$$

◆

EXAMPLE 1.100. A coin is tossed $m + n$ times. Show that the probability of getting m consecutive heads is

$$\frac{2+n}{2^{m+2}}$$

Solution. Let H = head, T = tail and X = any (head or tail). Note that the event E of getting m consecutive heads is the union of the following mutually exclusive events A_i , where $i = 0, 1, 2, \dots, n$.

A_0 : ($HHH \dots m$ times) T ($XXX \dots n-1$ times)

A_1 : T ($HHH \dots m$ times) T ($XXX \dots n-2$ times)

A_2 : $X T$ ($HHH \dots m$ times) T ($XXX \dots n-3$ times)

$\vdots$

A_{n-1} : ($XXX \dots n-2$ times) T ($HHH \dots m$ times) T

A_n : ($XXX \dots n-1$ times) T ($HHH \dots m$ times)

It is easy to see that

$$P(A_0) = P(A_n) = \left[\frac{1}{2}\right]^m \frac{1}{2}$$

and

$$P(A_1) = P(A_2) = \dots = P(A_{n-1}) = \frac{1}{2} \left[\frac{1}{2}\right]^m \frac{1}{2} = \left[\frac{1}{2}\right]^m \left[\frac{1}{4}\right]$$

Hence, the required probability is equal to

$$P(E) = P(A_0) + P(A_1) + \dots + P(A_{n-1}) + P(A_n)$$

i.e.

$$P(E) = \left[\frac{1}{2}\right]^m \frac{1}{2} + \left[\frac{1}{2}\right]^m \left[\frac{1}{4} + \frac{1}{4} + (n-2) \text{ times}\right] + \left[\frac{1}{2}\right]^m \frac{1}{2}$$

i.e.

$$P(E) = \left[\frac{1}{2}\right]^m \left(\frac{1}{2} + \frac{1}{2}\right) + \left[\frac{1}{2}\right]^m \left(\frac{n-2}{4}\right) = \left[\frac{1}{2}\right]^m \left(1 + \frac{n-2}{4}\right) = \frac{2+n}{2^{m+2}}$$

◆

EXAMPLE 1.101. A, B and C , in order, toss a coin. The first one to throw a head wins. If A starts, find their respective chances of winning.

(Anna, April 2003)

Solution. Let A_i, B_i and C_i denote the events that A, B and C , respectively, win the game in their i th attempt where $i = 1, 2, 3, \dots$. Then it is obvious that the collection of events

$$\mathcal{A} = \{A_i, B_i, C_i : i = 1, 2, 3, \dots\}$$

contains mutually independent events. Note that

$$P(A_i) = P(B_i) = P(C_i) = \frac{1}{2} \text{ for each } i$$

Therefore,

$$P(\bar{A}_i) = P(\bar{B}_i) = P(\bar{C}_i) = \frac{1}{2} \text{ for each } i$$

Since A starts the game, it is clear that A can win in the following mutually exclusive ways:

- (i) A wins the game in the first attempt, i.e. A_1 happens.
- (ii) None of A, B and C wins in the first attempt and A wins in the second attempt, i.e. the event

$$\bar{A}_1 \cap \bar{B}_1 \cap \bar{C}_1 \cap A_2 \text{ happens}$$

- (iii) None of A, B and C wins in the first and second attempts and A wins in the third attempt, i.e.

$$\bar{A}_1 \cap \bar{B}_1 \cap \bar{C}_1 \cap \bar{A}_2 \cap \bar{B}_2 \cap \bar{C}_2 \cap A_3 \text{ happens}$$

and so on.

By Axiom (iii),

$$P(A \text{ wins the game}) = P(\text{i}) + P(\text{ii}) + P(\text{iii}) + \dots$$

where

$$P(\text{i}) = \frac{1}{2}$$

$$P(\text{ii}) = P(\bar{A}_1)P(\bar{B}_1)P(\bar{C}_1)P(A_2) = \left[\frac{1}{2}\right]^4$$

$$P(\text{iii}) = P(\bar{A}_1)P(\bar{B}_1)P(\bar{C}_1)P(\bar{A}_2)P(\bar{B}_2)P(\bar{C}_2)P(A_3) = \left[\frac{1}{2}\right]^7$$

and so on.

Therefore,

$$P(A \text{ wins the game}) = \frac{1}{2} + \left[\frac{1}{2}\right]^4 + \left[\frac{1}{2}\right]^7 + \dots$$

which may be simplified as

$$P(A \text{ wins the game}) = \frac{1}{2} \left[1 + \left(\frac{1}{8}\right) + \left(\frac{1}{8}\right)^2 + \left(\frac{1}{8}\right)^3 + \dots \right] = \frac{1}{2} \times \frac{1}{1 - \frac{1}{8}} = \frac{4}{7}$$

Similarly, as A starts the game, B can win in the following mutually exclusive ways:

- (i) A does not win in the first attempt, but B wins the game in the first attempt, i.e. $\bar{A}_1 \cap B_1$ happens.
(ii) None of A , B and C wins the game in the first attempt, A does not win the game in the second attempt, but B wins the game in the second attempt, i.e.

$$\bar{A}_1 \cap \bar{B}_1 \cap \bar{C}_1 \cap \bar{A}_2 \cap B_2 \text{ happens}$$

- (iii) The event

$$\bar{A}_1 \cap \bar{B}_1 \cap \bar{C}_1 \cap \bar{A}_2 \cap \bar{B}_2 \cap \bar{C}_2 \cap \bar{A}_3 \cap B_3 \text{ happens}$$

and so on.

Hence, by Axiom (iii), we have

$$P(B \text{ wins the game}) = P(\text{i}) + P(\text{ii}) + P(\text{iii}) + \dots$$

A simple calculation shows that

$$\begin{aligned} P(B \text{ wins the game}) &= \left[\frac{1}{2}\right]^2 + \left[\frac{1}{2}\right]^5 + \left[\frac{1}{2}\right]^8 + \dots \\ &= \left[\frac{1}{4}\right] \left[1 + \left(\frac{1}{8}\right) + \left(\frac{1}{8}\right)^2 + \left(\frac{1}{8}\right)^3 + \dots\right] \\ &= \frac{1/4}{1 - 1/8} = \frac{1/4}{7/8} = \frac{2}{7} \end{aligned}$$

Note that the probability of C winning the game can be easily obtained as

$$P(C \text{ wins}) = 1 - P(A \text{ wins}) - P(B \text{ wins}) = 1 - \frac{4}{7} - \frac{2}{7} = \frac{1}{7}$$

◆

EXAMPLE 1.102. A and B throw alternately with a pair of dice. A will win if he throws 6 before B throws 7 and B will win if he throws 7 before A throws 6. If A begins, show that his chance of winning is $\frac{30}{61}$.

(Huyghen's Problem, Madras, 1999)

Solution. In a single throw with a pair of dice, there are $6^2 = 36$ possible outcomes. Of these, the outcomes favouring the event " $Total = 6$ " are $(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)$ and the outcomes favouring the event " $Total = 7$ " are $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$

Define the events A_i and B_i as

A_i : A wins the game in his i th throw ($i = 1, 2, \dots$)

B_i : B wins the game in his i th throw ($i = 1, 2, \dots$)

It is clear that the events A_i and B_i form a family of mutually independent events. Note also that for any value of i , we have

$$P(A_i) = \frac{5}{36}, \quad P(\bar{A}_i) = 1 - \frac{5}{36} = \frac{31}{36}$$

$$P(B_i) = \frac{6}{36} = \frac{1}{6}, \quad P(\bar{B}_i) = 1 - \frac{1}{6} = \frac{5}{6}$$

Given that the player A starts the game. Then he can win provided any of the following infinitely many, mutually exclusive, possibilities hold: (i) A_1 happens; (ii) $\bar{A}_1 \cap \bar{B}_1 \cap A_2$ happens; (iii) $\bar{A}_1 \cap \bar{B}_1 \cap \bar{A}_2 \cap B_2 \cap A_3$ happens, and so on.

We calculate the probabilities of (i), (ii), (iii), ... as follows.

$$P(\text{i}) = \frac{5}{36}$$

$$P(\text{ii}) = \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} = \frac{5}{6} \times \left[\frac{155}{216} \right]$$

$$P(\text{iii}) = \frac{31}{36} \times \frac{5}{6} \times \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} = \frac{5}{6} \times \left[\frac{155}{216} \right]^2$$

and so on.

By Axiom (iii), we have

$$P(A \text{ wins the game}) = P(\text{i}) + P(\text{ii}) + P(\text{iii}) + \dots = \frac{5}{36} \times \left[1 + \left[\frac{155}{216} \right] + \left[\frac{155}{216} \right]^2 + \dots \right]$$

Simplifying, we have

$$P(A \text{ wins the game}) = \frac{5}{36} \times \frac{1}{1 - \frac{155}{216}} = \frac{5}{36} \times \frac{1}{\frac{61}{216}} = \frac{30}{61}$$

◆

EXAMPLE 1.103. A coin is tossed until the first head appears. Assuming that the tosses are independent and the probability of a head occurring is p , find the value of p so that the probability that an odd number of tosses required is 0.6. Can you find a value of p so that the probability is 0.5 that an odd number of tosses required?

(Anna, April 2004)

Solution. Let p denote the probability of a head occurring. Let q denote the probability of a tail occurring. Then $p + q = 1$. Let A denote the event that an odd number of tosses are required for getting a head. Then A can materialize in the following mutually exclusive ways: (i) Head appears first in the first toss (probability = p), (ii) Head appears first in the third toss (probability = q^2p), (iii) Head appears first in the fifth toss (probability = q^4p), and so on.

By Axiom (iii), we have $P(A) = P(\text{i}) + P(\text{ii}) + P(\text{iii}) + \dots$, i.e.

$$P(A) = p + q^2p + q^4p + \dots = p [1 + q^2 + q^4 + \dots] = \frac{p}{1 - q^2}$$

i.e.

$$P(A) = \frac{p}{1 - (1 - p)^2} = \frac{p}{1 - (1 - 2p + p^2)} = \frac{1}{2 - p}$$

First, we find a value of p so that $P(A) = 0.6 = \frac{3}{5}$. Note that

$$P(A) = \frac{1}{2 - p} = \frac{3}{5} \Rightarrow 6 - 3p = 1 \Rightarrow p = \frac{1}{3}$$

The value of p is

$$p = \frac{1}{3}$$

Next, we see whether we can find a value of $p \neq 0$ satisfying $P(A) = 0.5$. Note that

$$P(A) = \frac{1}{2-p} = \frac{1}{2} \Rightarrow 2-p = 2 \Rightarrow p = 0$$

In this case, we cannot find a value of $p \neq 0$ satisfying $P(A) = 0.5$. ♦

EXAMPLE 1.104. A die is tossed until 6 appears. What is the probability that it must be tossed more than 4 times?

(Anna, April 2005)

Solution. Let p denote the probability that a 6 appears. Let q denote the probability that a 6 does not appear. Then $p = \frac{1}{6}$ and $q = \frac{5}{6}$. Let X denote the number of times the die must be tossed until 6 appears. Let A denote the event that $X > 4$.

It is easy to see that

$$P(X = 5) = q^4 p, \quad P(X = 6) = q^5 p, \quad P(X = 7) = q^6 p, \dots$$

Thus, the required probability is

$$\begin{aligned} P(A) &= P(X = 5) + P(X = 6) + P(X = 7) + \dots = q^4 p + q^5 p + q^6 p + \dots \\ &= q^4 p (1 + q + q^2 + \dots) = \frac{q^4 p}{1 - q} = \frac{q^4 p}{p} = q^4 \\ &= \left(\frac{5}{6}\right)^4 = \frac{625}{1296} = 0.4823 \end{aligned}$$
♦

EXAMPLE 1.105. (*Birthday Problem*) Find the probability that in a group of n people, at least two share the same birthday, where $n \leq 365$.

Solution. Let A denote the event that at least two people in the given group share the same birthday. Then $\bar{A}$ denotes the event that no two people in the given group share the same birthday. Since the birthday of any person is independent of the birthday of the other, we can compute $P(\bar{A})$ as

$$P(\bar{A}) = 1 \times \frac{364}{365} \times \frac{363}{365} \times \dots \times \frac{365 - (n-1)}{365} = \frac{365 \times 364 \times \dots \times [365 - (n-1)]}{365^n}$$

using the reasoning that the second person cannot have the same birthday as the first person (with probability $\frac{364}{365}$), the third person cannot have the same birthday as the first two (with probability $\frac{363}{365}$), etc., and finally, that the n^{th} person cannot have the same birthday as the first $n-1$ persons (with probability $\frac{365-(n-1)}{365}$).

Simplifying, we have

$$P(\bar{A}) = \frac{365!}{365^n (365 - n)!}$$

Hence, it follows that the required probability is

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{365!}{365^n(365-n)!}$$

◆

Definition 1.19. (*Bernoulli Trials*) Trials are called *independent* if the outcome of each trial is independent of the outcomes of the other trials. Repeated independent trials are called *Bernoulli trials* if the following conditions hold:

1. There are only two possible outcomes for each trial, which we may term as *success* and *failure* with probabilities p and q , respectively, where $p + q = 1$.
2. The probability p for success remains a constant throughout all the trials.

EXAMPLE 1.106. Consider the experiment of tossing a fair coin repeatedly. Each trial has only two outcomes H and T , where $p = P(H) = \frac{1}{2}$ and $q = 1 - p = \frac{1}{2}$. Since the trials are independent, and the probability p for getting H remains a constant throughout all the trials, the given trials constitute Bernoulli trials.

EXAMPLE 1.107. Consider the experiment of tossing a fair die repeatedly. We define *success* as *getting an even number*. Then each trial has only two outcomes with $p = \frac{3}{6} = \frac{1}{2}$ and $q = 1 - p = \frac{1}{2}$. Since the trials are independent and the probability p for getting an even number remains a constant throughout all the trials, the given trials constitute Bernoulli trials.

EXAMPLE 1.108. (*Bernoulli's problem*) If n Bernoulli trials are performed, find the probability that there will be exactly r successes in the n trials.

Solution. Let A denote the event of getting exactly r successes in n Bernoulli trials. Let p denote the probability of success in a trial and q denote the probability of failure in a trial, where $p + q = 1$. It is clear that A has a total of nC_r outcomes each having a probability of $p^r q^{n-r}$, where p^r corresponds to the probability of r successes and q^{n-r} corresponds to the probability of $n - r$ failures. Hence, it is immediate that

$$P(A) = {}^nC_r p^r q^{n-r}, \text{ where } r = 0, 1, 2, \dots, n$$

(Note that $P(A)$ is the $(r + 1)$ th term in the binomial expansion of $(q + p)^n$.)

◆

EXAMPLE 1.109. A coin is biased so that a head is twice as likely to occur as a tail. If the coin is tossed 4 times, what is the probability of getting exactly 2 tails?

(Madras, April 1999)

Solution. Let p and q denote the probabilities of getting a head and a tail in a toss, respectively. Given that $p = 2q$. Since $p + q = 1$, it follows that

$$2q + q = 1 \Rightarrow 3q = 1 \Rightarrow q = \frac{1}{3} \text{ and } p = \frac{2}{3}$$

Let A be the event of getting exactly two tails. Since the coin is tossed 4 times, A is also the same as the event of getting exactly 2 heads.

The required probability is given by

$$P(X = 2) = {}^4C_2 p^2 q^{4-2} = {}^4C_2 p^2 q^2 = 6 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2 = 6 \times \frac{4}{9} \times \frac{1}{9} = \frac{8}{27}$$

◆

EXAMPLE 1.110. Four coins are tossed simultaneously. What is the probability of getting (i) exactly 2 heads, (ii) at least 2 heads?

(Anna, Nov. 2006)

Solution. Let p and q denote the probabilities of getting a head and a tail, respectively, in a toss. Then $p = q = \frac{1}{2}$. Here, $n = 4$. Let X denote the number of heads in the four tosses. Then we know that

$$P(X = k) = {}^4C_k p^k q^{4-k} = {}^4C_k \left[\frac{1}{2}\right]^4 = {}^4C_k \times \frac{1}{16} \quad \text{for } k = 0, 1, 2, 3, 4$$

(since $p = q = \frac{1}{2}$).

$$(i) \quad P(X = 2) = {}^4C_2 \times \frac{1}{16} = \frac{6}{16} = \frac{3}{8}$$

(ii) The required probability is

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - ({}^4C_0 + {}^4C_1) \times \frac{1}{16} = 1 - \frac{5}{16} = \frac{11}{16}$$

◆

EXAMPLE 1.111. Ten coins are thrown simultaneously. Find the probability of getting at least 7 heads.

(Anna, Nov. 2005)

Solution. Let p and q denote the probabilities of getting a head and a tail, respectively. Then it is clear that $p = q = \frac{1}{2}$. Given that 10 coins are thrown simultaneously. Let X denote the number of heads. Then

$$P(X = x) = {}^{10}C_x p^x q^{10-x} = {}^{10}C_x \left(\frac{1}{2}\right)^{10} \quad \left(\text{since } p = q = \frac{1}{2}\right)$$

Hence, the probability of getting at least 7 heads is given by

$$P(X \geq 7) = P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10) = \left(\frac{1}{2}\right)^{10} [{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}].$$

Simplifying, we have

$$P(X \geq 7) = \left(\frac{1}{2}\right)^{10} [120 + 45 + 10 + 1] = \frac{176}{2^{10}} = \frac{11}{64}$$

◆

EXAMPLE 1.112. If m things are distributed among a men and b women, find the probability that the number of things received by men is odd.

(Anna, May 2006)

Solution. Let p and q denote the probabilities that a thing is received by men and women, respectively. Then it is easy to see that

$$p = \frac{a}{a+b} \quad \text{and} \quad q = \frac{b}{a+b}$$

By binomial expansion, it follows that

$$P(r \text{ successes in } m \text{ independent trials}) = {}^m C_r p^r q^{n-r}, \text{ where } r = 0, 1, 2, \dots, m$$

Since the events that men receive exactly 1 thing, exactly 3 things, exactly 5 things, etc. are mutually exclusive, the required probability that the number of things received by men are odd

$$\begin{aligned} &= {}^m C_1 p q^{m-1} + {}^m C_3 p^3 q^{m-3} + {}^m C_5 p^5 q^{m-5} + \dots \\ &= \frac{1}{2} [(q+p)^m - (q-p)^m] \\ &= \frac{1}{2} \left[\frac{(b+a)^m - (b-a)^m}{(b+a)^m} \right] \end{aligned}$$

◆

EXAMPLE 1.113. (*De Moivre's Problem*) If m dice are thrown, find the probability that the sum of the numbers on the m dice is equal to r .

Solution. First, note that $n(S) = 6^m$, because each die has 6 faces and there are m dice. Let A be the event that the sum of the numbers on the m dice is equal to r .

Then, A contains all points of the form

$$(x_1, x_2, \dots, x_m), \text{ where } x_1 + x_2 + \dots + x_m = r$$

Hence, $n(A)$ is equal to the coefficient of x^r in the expansion of

$$(x + x^2 + x^3 + x^4 + x^5 + x^6)^m$$

So, we have

$$\begin{aligned} n(A) &= \text{Coefficient of } x^r \text{ in } x^m (1 + x + x^2 + \dots + x^5)^m \\ &= \text{Coefficient of } x^{r-m} \text{ in } (1 + x + x^2 + \dots + x^5)^m. \\ &= \text{Coefficient of } x^{r-m} \text{ in } \left[\frac{1-x^6}{1-x} \right]^m \end{aligned}$$

Hence, the required probability is

$$P(A) = \frac{\text{Coefficient of } x^{r-m} \text{ in } (1-x^6)^m (1-x)^{-m}}{6^m}$$

◆

EXAMPLE 1.114. Find the probability of scoring 12 in a single throw with 6 dice.

Solution. In the notation of De Moivre's problem, we have $r = 12$ and $m = 6$. Here, $n(S) = 6^m = 6^6$. Also, we have

$$\begin{aligned} n(A) &= \text{Coefficient of } x^{r-m} \text{ in } (1-x^6)^m (1-x)^{-m} \\ &= \text{Coefficient of } x^6 \text{ in } (1-x^6)^6 (1-x)^{-6} \end{aligned}$$

Note that

$$(1-x^6)^6 = 1 - 6x^6 + 15x^{12} - \dots$$

and

$$(1-x)^{-6} = 1 + 6x + \cdots + {}^{5+r}C_r x^r + \cdots$$

Hence,

$$n(A) = \text{Coeff. of } x^6 \text{ in } (1 - 6x^6 + \cdots) (1 + 6x + \cdots + 462x^6 + \cdots) = 462 - 6 = 456$$

Hence, the required probability is equal to

$$P(A) = \frac{456}{6^6} = \frac{19}{1944}$$

◆

EXAMPLE 1.115. What is the probability of throwing a total of 15 from the toss of three ordinary dice together?

Solution. In the notation of De Moivre's problem, we have $r = 15$ and $m = 3$. Here, $n(S) = 6^m = 6^3$. Also, we have

$$\begin{aligned} n(A) &= \text{Coefficient of } x^{r-m} \text{ in } (1-x^6)^m (1-x)^{-m} \\ &= \text{Coefficient of } x^{12} \text{ in } (1-x^6)^3 (1-x)^{-3} \end{aligned}$$

Note that

$$(1-x^6)^3 = 1 - 3x^6 + 3x^{12} - x^{18}$$

and

$$(1-x)^{-3} = 1 + 3x + 6x^2 + \cdots + 28x^6 + \cdots + 91x^{12} + {}^{2+r}C_r x^r + \cdots$$

Hence,

$$\begin{aligned} n(A) &= \text{Coeff. of } x^6 \text{ in } (1 - 3x^6 + 3x^{12} - x^{18}) (1 + 3x + 6x^2 + \cdots + 28x^6 + \cdots + 91x^{12} + \cdots) \\ &= (1 \times 91) - (3 \times 28) + (3 \times 1) = 91 - 84 + 3 = 10 \end{aligned}$$

Hence, the required probability is equal to

$$P(A) = \frac{10}{6^3} = \frac{10}{216} = \frac{5}{108}$$

◆

PROBLEM SET 1.3

1. If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{5}$, find $P(\bar{B}|\bar{A})$.
2. If two cards are drawn successively without replacement from a well-shuffled pack of playing cards, find the probability that they are both face cards.
3. If A and B are events such that

$$P(A \cup B) = \frac{1}{2}, P(\bar{A} \cup \bar{B}) = \frac{3}{4} \text{ and } P(\bar{B}) = \frac{2}{3}$$

find (a) $P(\bar{A}|B)$, (b) $P(A|\bar{B})$, and (c) $P(\bar{A}|\bar{B})$.

4. Given a binary communication channel, where A is the input and B is the output, let

$$P(A) = 0.5, P(A|B) = 0.7 \text{ and } P(\bar{A}|\bar{B}) = 0.8$$

find (a) $P(\bar{A}|B)$ and (b) $P(A|\bar{B})$.

5. If A and B are two events such that $P(A) = \frac{1}{2}$, $P(B) = \frac{2}{3}$ and $P(A \cup B) = \frac{3}{4}$, find (a) $P(A|B)$, (b) $P(B|A)$, (c) $P(A \cap \bar{B})$, (d) $P(\bar{A} \cap B)$, (e) $P(A|\bar{B})$, (f) $P(B|\bar{A})$, and (g) $P(\bar{A} \cap \bar{B})$.
6. The probability that a boy will pass an examination is $\frac{2}{3}$ and that for a girl is $\frac{3}{4}$. What is the probability that at least one of them will pass the examination?
7. The odds against a certain event are 3 to 2 and the odds in favour of another event, independent of the former, are 3 to 5. Find the probability that at least one of the two events will happen.
8. If the probability that a communication system has high selectivity is 0.72 and the probability that it will have high fidelity is 0.63 and the probability that it will have both is 0.23. Find the probability that a system with high selectivity will have high fidelity.
9. In a mid-semester examination, 25% of the students have failed in mathematics, 20% have failed in physics and 10% have failed in both mathematics and physics. If a student is selected at random from the class, find the probability that
- the student has failed in physics if it is known that he has failed in mathematics, and
 - the student has failed in either mathematics or physics.
10. If A and B are independent events with $P(A) = 0.4$ and $P(B) = 0.6$, find (a) $P(A \cap \bar{B})$, (b) $P(\bar{A} \cap B)$ and (c) $P(\bar{A} \cap \bar{B})$.
11. If A and B are independent events with $P(A) = 0.6$ and $P(B) = 0.5$, find the probability that
- at least one of the two events, A and B , occurs;
 - exactly one of the two events, A and B , occurs; and
 - none of the two events, A and B , occurs.
12. If A and B are independent events such that $P(A) = \frac{2}{3}$ and $P(A \cup B) = \frac{3}{4}$, find $P(B)$.
13. The probability of a man hitting a target is 0.6. How many times must he fire so that the probability of hitting the target at least once is more than 80%?
14. Ram and his wife Geeta appear in an interview for two vacancies in the same Department. The probabilities for Ram and Geeta of getting selected equal 0.7 and 0.5, respectively. Find the probability that (a) at least one of them will be selected, (b) exactly one of them will be selected, and (c) both will be selected.
15. A is known to hit the target in 2 out of 3 shots, whereas B is known to hit the target in 3 out of 5 shots. What is the probability of the target being hit when both of them try?
16. The odds that A speaks the truth are 2 to 3 and the odds that B speaks the truth are 5 to 3. Find the probability that they contradict each other in a statement.
17. The probabilities of 3 students A, B, C solving a problem in Statistics are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{3}{4}$, respectively. If a problem is given to all the 3 students, find the probability that (a) no one will solve the problem, (b) at least one will solve the problem, and (c) only one will solve the problem.
18. The odds for 3 independent critics A, B and C favourably reviewing a book are 4 to 3, 2 to 3 and 5 to 2. What is the probability that a majority of review will be favourable?

19. If an event A is independent of the events B , $B \cup C$ and $B \cap C$, show that A is also independent of C .
20. Let p be the probability that a man aged x years dies in a year. Find the probability that out of n men $A_1, A_2, \dots, A_n$ each aged x , A will die and be the first to die.
21. The probability of n independent events $A_1, A_2, \dots, A_n$ are $p_1, p_2, \dots, p_n$ respectively. Find the probability that at least one of the n events A_i will happen.
22. Six coins are tossed simultaneously. Find the probability of getting (a) exactly 3 heads, (b) at least 3 heads.
23. Twelve coins are thrown simultaneously. Find the probability of getting at least 8 heads.
24. Find the probability of scoring 12 in a single throw with 4 dice.
25. Find the probability of scoring 15 in a single throw with 5 dice.

1.6 TOTAL PROBABILITY

We begin this section with a definition of events that *partition* the sample space S (see Figure 1.6).

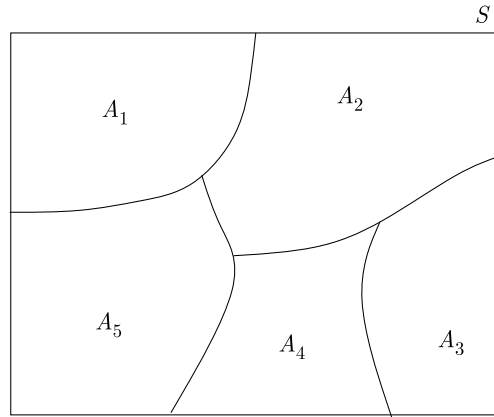


Figure 1.6: Partition of a sample space.

Definition 1.20. Let $\mathcal{A} = \{A_1, A_2, \dots, A_n\}$ be a finite collection of events. We say that $\mathcal{A}$ is a *partition* of the sample space S if the following three conditions hold:

1. $P(A_i) > 0$ for $i = 1, 2, \dots, n$
2. The events A_i are pairwise disjoint, i.e.

$$A_i \cap A_j = \emptyset \text{ whenever } i \neq j$$

3. The union of the events A_i equals the sample space S , i.e.

$$A_1 \cup A_2 \cup \dots \cup A_n = S$$

We illustrate the above definition with a few examples.

EXAMPLE 1.116. Consider the experiment of throwing a coin. The sample space $S = \{H, T\}$. Define the events $A_1 = \{H\}$ and $A_2 = \{T\}$. Then it is easy to see that A_1 and A_2 partition the sample space.

EXAMPLE 1.117. Consider the experiment of throwing a fair die. The sample space S is

$$S = \{1, 2, 3, 4, 5, 6\}$$

Note that there are many *partitions* for the sample space S .

- (i) Define the events A_i ($i = 1, 2, \dots, 6$) by $A_i = \{i\}$. Then it is clear that the events $A_1, A_2, \dots, A_6$ form a partition of the sample space S .
- (ii) Define the events B_i ($i = 1, 2, 3$) by

$$B_1 = \{1, 2\}, B_2 = \{3, 4\} \text{ and } B_3 = \{5, 6\}$$

Then it is clear that B_1, B_2 and B_3 form a partition of the sample space S .

- (iii) Define the events C_i ($i = 1, 2$) by

$$C_1 = \{1, 3, 5\} \text{ and } C_2 = \{2, 4, 6\}$$

Then it is clear that C_1 and C_2 form a partition of the sample space S .

Next, we prove the *total probability theorem*.

Theorem 1.12. (*Total Probability Theorem*) Let $\mathcal{A} = \{A_1, A_2, \dots, A_n\}$ be a partition of the sample space S , i.e. $\mathcal{A}$ has a finite collection of pairwise disjoint events A_i with $P(A_i) > 0$ for each i and whose union is the whole sample space S . If B is any event, then

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_n)P(B|A_n) \quad (1.39)$$

(The probability of B , as expressed in Eq. (1.39), is called the *total probability* of B .)

Proof. By hypothesis, the events A_i are pairwise disjoint and

$$\bigcup_{i=1}^n A_i = S$$

Thus, it follows that the events $A_i \cap B$ are also pairwise disjoint and

$$B = B \cap S = B \cap \bigcup_{i=1}^n A_i = \bigcup_{i=1}^n (B \cap A_i)$$

By Axiom (iii), we have

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_n)$$

By the multiplication rule for 2 events, the above equation may be simplified as

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_n)P(B|A_n)$$

◆

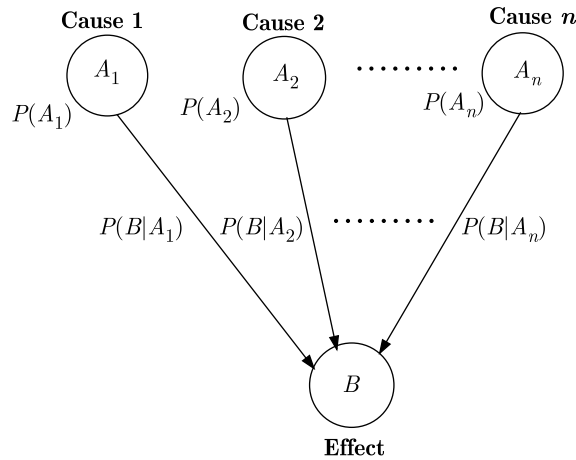


Figure 1.7: Total probability theorem.

Remark 1.22. The events $A_1, A_2, \dots, A_n$ that form a partition of the sample space S may be viewed as “causes” and the event B may be viewed as an “effect”. Thus, the total probability theorem calculates the *net effect* of all the n causes $A_1, A_2, \dots, A_n$ in determining the probability of the event B . The net effect is expressed in the formula Eq. (1.39). Figure 1.7 illustrates the total probability theorem.

EXAMPLE 1.118. Two-thirds of the students in a class are boys and the rest are girls. It is known that the probability of a girl getting a first class is 0.25 and that of a boy is 0.28. Find the probability that a student chosen at random will get first class.

(Madras, May 2002)

Solution. Define the events

A_1 : The student is a boy. (Cause 1)

A_2 : The student is a girl. (Cause 2)

B : The student will get first class. (Effect)

We are asked to compute the total probability, $P(B)$, given the causes A_1 and A_2 for the sample space (a classroom). From the data given in the problem,

$$P(A_1) = \frac{2}{3}, P(A_2) = \frac{1}{3}, P(B|A_1) = 0.28 \text{ and } P(B|A_2) = 0.25$$

By the total probability theorem (Theorem 1.12), the required probability is determined as

$$\begin{aligned} P(B) &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) = \frac{2}{3} \times 0.28 + \frac{1}{3} \times 0.25 \\ &= \frac{1}{3} [0.56 + 0.25] = \frac{1}{3} [0.8100] = 0.27 \end{aligned}$$

◆

EXAMPLE 1.119. An urn contains 10 white and 3 black balls. Another urn contains 3 white and 5 black balls. Two balls are drawn at random from the first urn and placed in the second urn, and then 1 ball is taken at random from the latter. What is the probability that it is a white ball?

(Anna, Nov. 2007)

Solution. Define the events:

A_1 : The two balls transferred are both white. (*Cause 1*)

A_2 : The two balls transferred are both black. (*Cause 2*)

A_3 : The two balls transferred are a white and a black. (*Cause 3*)

B : A white ball is selected from the second urn. (*Effect*)

First, we calculate the probabilities of the events A_i . Clearly,

$$P(A_1) = \frac{{}^{10}C_2}{{}^{13}C_2} = \frac{15}{26}, P(A_2) = \frac{{}^3C_2}{{}^{13}C_2} = \frac{1}{26} \text{ and } P(A_3) = \frac{{}^{10}C_1 {}^3C_1}{{}^{10}C_2} = \frac{10}{26}$$

If A_1 occurs, then Urn 2 will contain 5 white and 5 black balls. If A_2 occurs, then Urn 2 will contain 3 white and 7 black balls. If A_3 occurs, then Urn 2 will contain 4 white and 6 black balls. Hence, it follows that

$$P(B|A_1) = \frac{{}^5C_1}{{}^{10}C_1} = \frac{5}{10}, P(B|A_2) = \frac{{}^3C_1}{{}^{10}C_1} = \frac{3}{10} \text{ and } P(B|A_3) = \frac{{}^4C_1}{{}^{10}C_1} = \frac{4}{10}$$

The required probability, $P(B)$, can be easily determined using total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(B) &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) \\ &= \left[\frac{15}{26} \times \frac{5}{10} \right] + \left[\frac{1}{26} \times \frac{3}{10} \right] + \left[\frac{10}{26} \times \frac{4}{10} \right] \\ &= \frac{59}{130} = 0.4538 \end{aligned}$$



EXAMPLE 1.120. A bolt is manufactured by 3 machines A , B and C . A turns out twice as many items as B , and machines B and C produce equal number of items. 2% of bolts produced by A and B and 4% of bolts produced by C are defective. All bolts are put into 1 stock pile and 1 is chosen from this pile. What is the probability that it is defective?

(Anna, Nov. 2007)

Solution. Define the events

A : The bolt is made by machine A . (*Cause 1*)

B : The bolt is made by machine B . (*Cause 2*)

C : The bolt is made by machine C . (*Cause 3*)

D : The bolt is defective. (*Effect*)

From the given data,

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{4} \text{ and } P(C) = \frac{1}{4}$$

We also have

$$P(D|A) = \frac{2}{100}, P(D|B) = \frac{2}{100} \text{ and } P(D|C) = \frac{4}{100}$$

The probability that the chosen bolt is defective is given by the total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(D) &= P(A)P(D|A) + P(B)P(D|B) + P(C)P(D|C) \\ &= \left[\frac{1}{2} \times \frac{2}{100} \right] + \left[\frac{1}{4} \times \frac{2}{100} \right] + \left[\frac{1}{4} \times \frac{4}{100} \right] \\ &= \frac{1}{40} \end{aligned}$$

Thus, the required probability is $\frac{1}{40}$ or 2.5%. ♦

EXAMPLE 1.121. In a coin-tossing experiment, if the coin shows head, 1 die is thrown and the result is recorded. But if the coin shows tail, 2 dice are thrown and their sum is recorded. What is the probability that the recorded number will be 2?

(Anna, Nov. 2007)

Solution. Define the events

- A: The coin shows head. (Cause 1)
- B: The coin shows tail. (Cause 2)
- E: The recorded number is 2. (Effect)

It is clear that $P(A) = P(B) = \frac{1}{2}$. From the given data, we find that

$$P(E|A) = \frac{1}{6} \text{ and } P(E|B) = \frac{1}{36}$$

The required probability is given by total probability theorem (Theorem 1.12) as

$$P(E) = P(A)P(E|A) + P(B)P(E|B) = \frac{1}{2} \left[\frac{1}{6} + \frac{1}{36} \right] = \frac{7}{72}$$
♦

EXAMPLE 1.122. For a certain binary communication channel, the probability that a transmitted '0' is received as a '0' is 0.95 and the probability that a transmitted '1' is received as '1' is 0.90. If the probability that a '0' is transmitted is 0.4, find the probability that a '1' is received.

(Anna, May 2006)

Solution. Define the events

- A: '1' is transmitted. (Cause 1))
- $\bar{A}$: '0' is transmitted. (Cause 2)
- B: '1' is received. (Effect)

From the given data,

$$\begin{aligned} P(\bar{A}) &= 0.4 \Rightarrow P(A) = 1 - P(\bar{A}) = 0.6 \\ P(\bar{B}|\bar{A}) &= 0.95 \Rightarrow P(B|\bar{A}) = 1 - 0.95 = 0.05 \end{aligned}$$

and

$$P(B|A) = 0.90$$

The required probability is $P(B)$, which can be calculated using the total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(B) &= P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) \\ &= (0.6 \times 0.90) + (0.4 \times 0.05) = 0.56 \end{aligned}$$

◆

EXAMPLE 1.123. A binary communication channel carries data as one of 2 types of signals denoted by 0 and 1. Due to noise, a transmitted 0 is sometimes received as a 1 and a transmitted 1 is sometimes received as a 0. For a given channel, assume a probability of 0.94 that a transmitted 0 is correctly received as a 0 and a probability of 0.91 that a transmitted 1 is received as a 1. Further assume a probability of 0.45 of transmitted as a 0. If a signal is sent, determine the probability that

- (i) a 1 is received.
- (ii) a 1 was transmitted given that a 1 was received.
- (iii) a 0 was transmitted given that a 0 was received.
- (iv) an error occurs.

(Anna, May 2007)

Solution. Define the events

A : '0' is transmitted. (Cause 1))

$\bar{A}$: '1' is transmitted. (Cause 2)

B : '0' is received. (Effect)

From the given data,

$$P(A) = 0.45, P(B|A) = 0.94 \text{ and } P(\bar{B}|\bar{A}) = 0.91$$

- (i) The required probability is $P(\bar{B})$, which can be calculated using the total probability theorem. We have

$$\begin{aligned} P(\bar{B}) &= P(A)P(\bar{B}|A) + P(\bar{A})P(\bar{B}|\bar{A}) \\ &= (0.45 \times 0.06) + (0.55 \times 0.91) = 0.5275 \end{aligned}$$

- (ii) The required probability is $P(\bar{A}|\bar{B})$, which can be calculated as

$$P(\bar{A}|\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(\bar{A})P(\bar{B}|\bar{A})}{P(\bar{B})}$$

On substituting the probabilities, we get

$$P(\bar{A}|\bar{B}) = \frac{0.55 \times 0.91}{0.5275} = 0.9488$$

- (iii) The required probability is $P(A|B)$, which can be calculated as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B|A)}{P(B)}$$

On substituting the probabilities, we get

$$P(A|B) = \frac{0.45 \times 0.94}{0.4725} = 0.9895$$

(iv) The probability that an error occurs is given by

$$P(\text{error}) = P(A \cap \bar{B}) + P(\bar{A} \cap B) = P(A)P(\bar{B}|A) + P(B)P(\bar{A}|B)$$

On substituting the probabilities, we get

$$P(\text{error}) = [0.45 \times 0.06] + [0.4725 \times 0.0105] = 0.0320$$

◆

EXAMPLE 1.124. A box contains a white and b black balls and a second box contains c white and d black balls. One ball is transferred from the first box into the second and one ball is then drawn from the second box. Find the probability that it is a white ball.

Solution. Define the events

A_1 : A white ball is transferred into box 2 from box 1. (*Cause 1*)

A_2 : A black ball is transferred into box 2 from box 1. (*Cause 2*)

W : Event that a white ball is drawn from box 2. (*Effect*)

The probabilities of the causes are easily obtained as

$$P(A_1) = \frac{a}{a+b} \quad \text{and} \quad P(A_2) = \frac{b}{a+b}$$

Note also that

$$P(W|A_1) = \frac{c+1}{c+d+1} \quad \text{and} \quad P(W|A_2) = \frac{c}{c+d+1}$$

The required probability, $P(W)$, can be easily evaluated using the total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(W) &= P(A_1)P(W|A_1) + P(A_2)P(W|A_2) \\ &= \left[\frac{a}{a+b} \times \frac{c+1}{c+d+1} \right] + \left[\frac{b}{a+b} \times \frac{c}{c+d+1} \right] = \frac{a(c+1) + bc}{(a+b)(c+d+1)} \end{aligned}$$

◆

EXAMPLE 1.125. Each of the n urns contains a white and b black balls. One ball is transferred from the first urn to the second urn, then one ball is transferred from the second urn to the third urn and so on. Finally, one ball is drawn from the last urn. Find the probability of it being white.

Solution. Define the probability

$$p_k = \text{Probability that the ball drawn the } k\text{th urn is white.}$$

Then, it follows that

$$1 - p_k = \text{Probability that the ball drawn the } k\text{th urn is black.}$$

When $k = 1$, it is immediate that

$$p_1 = \frac{a}{a+b} \quad \text{and} \quad 1 - p_1 = \frac{b}{a+b}$$

To calculate p_{k+1} , the probability that the ball drawn from the $(k+1)$ th urn is white, we define the events

A_k : Ball transferred from the k th urn is white. (Cause 1)

$\bar{A}_k$: Ball transferred from the k th urn is black. (Cause 2)

W : White ball is drawn from the $(k+1)$ th urn is white. (Effect)

Note that

$$P(A_k) = p_k \text{ and } P(\bar{A}_k) = 1 - p_k$$

and also that

$$P(W|A_k) = \frac{a+1}{a+b+1} \text{ and } P(W|\bar{A}_k) = \frac{a}{a+b+1}$$

By the total probability theorem (Theorem 1.12), we have

$$p_{k+1} = P(A_k)P(W|A_k) + P(\bar{A}_k)P(W|\bar{A}_k) = p_k \frac{a+1}{a+b+1} + (1-p_k) \frac{a}{a+b+1}$$

Simplifying, we have

$$p_{k+1} = \frac{1}{a+b+1} [p_k + a] \text{ where } k = 1, 2, \dots, n-1$$

Taking $k = 1$, we have

$$p_2 = \frac{1}{a+b+1} [p_1 + a] = \frac{1}{a+b+1} \left[\frac{a}{a+b} + a \right] = \frac{a}{a+b}$$

Assume that

$$p_m = \frac{a}{a+b} \text{ (Induction Hypothesis)}$$

Then we have

$$p_{m+1} = \frac{1}{a+b+1} [p_m + a] = \frac{1}{a+b+1} \left[\frac{a}{a+b} + a \right] = \frac{a}{a+b}$$

Hence, by induction, it follows that

$$p_k = \frac{a}{a+b}, \text{ where } k \text{ is any positive integer}$$

In particular, we have

$$p_n = \frac{a}{a+b}, \text{ which is independent of } n.$$

◆

PROBLEM SET 1.4

1. An urn contains 6 white and 4 red balls, and another urn contains 3 white and 7 red balls. If one of the urns is chosen at random, and a ball is taken out of it, find the probability that it is white.
2. In a certain recruitment test, there are multiple choice questions. There are 4 possible answers to each question and of which one is correct. An intelligent student knows 80% of the answers. What is the probability that the student answers a given question correctly?

3. The probability that a doctor will diagnose a disease X correctly is 70%. The probability that a patient will die by his treatment after correct diagnosis is 30% and the probability of death by faulty diagnosis is 80%. What is the probability that a patient of this doctor, who has disease X , will die?
4. An industry has two plants to manufacture cars. Plants A and B produce 70% and 30% of the total output of cars. At plant A , 70% of the cars are of a new model X , and at plant B , only 40% of the cars are of the new model X . What is the probability that a car randomly selected from the total output of the company is of the new model X ?
5. Suppose that there are three boxes, A , B and C containing 3 white and 5 red balls, 5 white and 3 red balls, and 2 white and 4 red balls. If a box is chosen at random, and a ball is taken out of it, what is the probability that it is white?
6. Three persons, X , Y and Z , are short-listed for the post of Principal of an Engineering College. Their chances of getting selected are in the proportion 5 : 3 : 2, respectively. If X gets the post, the probability of introducing co-education in the college is 0.8. The probabilities of Y and Z doing the same are 0.6 and 0.4, respectively. What is the probability that co-education will be introduced in the College?
7. In a bolt factory, machines X , Y and Z produce 50%, 30% and 20% of the total. Suppose that of their output, 20%, 10% and 5% are defective. If a bolt is drawn at random, what is the probability of it being defective?

1.7 BAYES' THEOREM

In this section, we derive the *Bayes' Theorem*, named after Reverend Thomas Bayes (1702–1761). Thomas Bayes was an English clergyman and a mathematician whose main contribution to probability theory was his solution to a problem on *inverse probability*, which was published posthumously by his friend Richard Price in the *Philosophical Transactions of the Royal Society of London* (1764).

Theorem 1.13. (*Bayes' Theorem*) Let $\mathcal{A} = \{A_1, A_2, \dots, A_n\}$ be a partition of the sample space S , i.e. $\mathcal{A}$ has a finite collection of pairwise disjoint events A_i with $P(A_i) > 0$ for each i and whose union is the whole sample space S . If B is any event, then

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_n)P(B|A_n)} \quad (1.40)$$

for $i = 1, 2, \dots, n$. (The probability of $P(A_i|B)$, as expressed in Eq. (1.40), is called the *inverse probability* of the “cause” A_i given that the “effect” B has happened.)

Proof. The assertion is a simple consequence of the total probability theorem (see Theorem 1.12). Fix any i , where $i = 1, 2, \dots, n$.

By the definition of conditional probability, we have

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} \quad (1.41)$$

By multiplication rule for 2 events, we have

$$P(A_i \cap B) = P(A_i)P(B|A_i) \quad (1.42)$$

By total probability theorem (Theorem 1.12), we have

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n) \quad (1.43)$$

Substituting Eqs. (1.42) and (1.43) into the Eq. (1.41), we have

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n)} \quad \blacklozenge$$

Remark 1.23. (i) The probabilities

$$P(A_1), P(A_2), \dots, P(A_n)$$

of the *causes* are called *a priori probabilities* because they exist before the results of the experiment are known.

(ii) The inverse probabilities

$$P(B|A_1), P(B|A_2), \dots, P(B|A_n)$$

are called *a posteriori probabilities* because they are determined after the results of the experiment are known.

EXAMPLE 1.126. Given that a student studied, the probability of passing a certain quiz is 0.99. Given that a student did not study, the probability of passing the quiz is 0.05. Assume that the probability of studying is 0.7. A student flunked the quiz. What is the probability that he or she did not study? (Anna, Nov. 2006)

Solution. Define the events A and B as follows:

A : The student studied for the quiz. (*Cause 1*)

$\bar{A}$: The student did not study for the quiz. (*Cause 2*)

B : The student flunked the quiz. (*Effect*)

Given the two causes, viz. the student studied or did not study for the quiz, and the effect that the student flunked the quiz, we are asked to calculate the inverse probability $P(\bar{A}|B)$. This can be easily determined using the Bayes' theorem (Theorem 1.13).

From the given data, we have

$$P(A) = 0.7, \quad P(B|A) = 1 - 0.99 = 0.01 \quad \text{and} \quad P(B|\bar{A}) = 1 - 0.05 = 0.95$$

The total probability is

$$P(B) = P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) = (0.7 \times 0.01) + (0.3 \times 0.95) = 0.2920$$

Hence, by Bayes' theorem, we calculate the required inverse probability as

$$P(\bar{A}|B) = \frac{P(\bar{A})P(B|\bar{A})}{P(B)} = \frac{0.3 \times 0.95}{0.2920} = 0.9760 \quad \blacklozenge$$

EXAMPLE 1.127. A student knew only 60% of the questions in a test each with 5 answers. He simply guessed while answering the rest. What is the probability that he knew the answer to a question given that he answered it correctly?

(Madras, Oct. 1999)

Solution. Define the events

A : The student knew the answer. (*Cause 1*)

$\bar{A}$: The student guessed the answer. (*Cause 2*)

E : The student answered it correctly. (*Effect*)

From the given data,

$$P(A) = \frac{60}{100} = \frac{3}{5} \text{ and } P(\bar{A}) = 1 - \frac{3}{5} = \frac{2}{5}$$

Also,

$$P(E|A) = 1 \text{ and } P(E|\bar{A}) = \frac{1}{5}$$

Next, we find $P(E)$ using total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(E) &= P(A)P(E|A) + P(\bar{A})P(E|\bar{A}) \\ &= \left[\frac{3}{5} \times 1 \right] + \left[\frac{2}{5} \times \frac{1}{5} \right] = \frac{17}{25} \end{aligned}$$

The required probability is the inverse probability, $P(A|E)$, which can be easily evaluated using the Bayes' theorem (Theorem 1.13) as

$$P(A|E) = \frac{P(A)P(E|A)}{P(E)} = \frac{\frac{3}{5} \times 1}{\frac{17}{25}} = \frac{15}{17} = 0.8824$$

◆

EXAMPLE 1.128. In a certain recruitment test, there are multiple choice questions. There are 4 possible answers to each question and of which one is correct. An intelligent student knows 90% of the answer. If the intelligent student gets the correct answer, what is the probability that he is guessing?
(Anna, May 2006)

Solution. Define the events

A : The intelligent student knows the answer. (*Cause 1*)

$\bar{A}$: The intelligent student guesses the answer. (*Cause 2*)

E : The student answers it correctly. (*Effect*)

From the given data,

$$P(A) = \frac{90}{100} = \frac{9}{10} \text{ and } P(\bar{A}) = 1 - \frac{9}{10} = \frac{1}{10}$$

Also,

$$P(E|A) = 1 \text{ and } P(E|\bar{A}) = \frac{1}{4}$$

Next, we find $P(E)$ using total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(E) &= P(A)P(E|A) + P(\bar{A})P(E|\bar{A}) \\ &= \left[\frac{9}{10} \times 1 \right] + \left[\frac{1}{10} \times \frac{1}{4} \right] = \frac{37}{40} \end{aligned}$$

The required probability is the inverse probability, $P(\bar{A}|E)$, which can be easily evaluated using the Bayes' theorem (Theorem 1.13) as

$$P(\bar{A}|E) = \frac{P(\bar{A})P(E|\bar{A})}{P(E)} = \frac{\frac{1}{10} \times \frac{1}{4}}{\frac{37}{40}} = \frac{1}{37} = 0.0270$$

◆

EXAMPLE 1.129. A given lot of IC chips contains 2% defective chips. Each chip is tested before delivery. The tester itself is not totally reliable. Probability of tester says the chip is good when it really good is 0.95 and the chip is defective when it actually defective is 0.94. If a tested device is indicated to be defective, what is the probability that it is actually defective?

(Anna, Nov. 2006)

Solution. Define the events A and B as follows:

- $\bar{A}$: The IC chip is actually good. (Cause 1)
- A : The IC chip is actually defective. (Cause 2)
- B : The tested device is indicated to be defective. (Effect)

Given the two causes, viz. the IC chip is actually good or defective, and the effect that the tested device is found to be defective, we are asked to calculate the inverse probability $P(\bar{A}|B)$. This can be easily determined using the Bayes' theorem (Theorem 1.13).

From the given data, we have

$$P(\bar{A}) = 0.02, \quad P(B|A) = 1 - 0.95 = 0.05 \quad \text{and} \quad P(B|\bar{A}) = 0.94$$

The total probability is

$$P(B) = P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) = (0.98 \times 0.05) + (0.02 \times 0.94) = 0.0678$$

Hence, by Bayes' theorem, we calculate the required inverse probability as

$$P(\bar{A}|B) = \frac{P(\bar{A})P(B|\bar{A})}{P(B)} = \frac{0.02 \times 0.94}{0.0678} = 0.2773$$

◆

EXAMPLE 1.130. A bin contains three different types of disposable flashlights. The probability that a type 1 flashlight will give over 100 hours of use is 0.7 with the corresponding probabilities for type 2 and type 3 flashlights being 0.4 and 0.3, respectively. Suppose that 20% of the flashlights in the bin are type 1, 30% are type 2 and 50% are type 3.

- (i) Find the probability that a randomly chosen flashlight will give more than 100 hours of use.
- (ii) Given the flashlight lasted over 100 hours, what is the conditional probability that it was a type j flashlight, $j = 1, 2, 3$?

(Anna, Nov. 2007)

Solution. Define the events

- A : The flashlight is of type 1. (Cause 1)
- B : The flashlight is of type 2. (Cause 2)
- C : The flashlight is of type 3. (Cause 3)
- E : The flashlight will last over 100 hours. (Effect)

We are asked to find

- (i) $P(E)$, the probability that a randomly chosen flashlight will last over 100 hours.
- (ii) $P(A|E)$, $P(B|E)$ and $P(C|E)$, the conditional probabilities that the flashlight was of type 1, 2 or 3, respectively, given that it lasted over 100 hours.

By the data given in the problem,

$$P(A) = 0.2, P(B) = 0.3 \text{ and } P(C) = 0.5$$

$$P(E|A) = 0.7, P(E|B) = 0.4 \text{ and } P(E|C) = 0.3$$

- (i) We calculate $P(E)$ by total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(E) &= P(A)P(E|A) + P(B)P(E|B) + P(C)P(E|C) \\ &= (0.2 \times 0.7) + (0.3 \times 0.4) + (0.5 \times 0.3) = 0.41 \end{aligned}$$

Hence, $P(E) = 0.41$

- (ii) We calculate the conditional probabilities, $P(A|E)$, $P(B|E)$ and $P(C|E)$ by Bayes' theorem (Theorem 1.13).

First, we find that

$$P(A|E) = \frac{P(A)P(E|A)}{P(E)} = \frac{0.2 \times 0.7}{0.41} = 0.3415$$

Next, we find that

$$P(B|E) = \frac{P(B)P(E|B)}{P(E)} = \frac{0.3 \times 0.4}{0.41} = 0.2927$$

and

$$P(C|E) = \frac{P(C)P(E|C)}{P(E)} = \frac{0.5 \times 0.3}{0.41} = 0.3658$$



EXAMPLE 1.131. An urn contains 5 balls. Two balls are drawn at random and are found to be white. What is the probability of all the balls being white?

(Anna, April 2003, May 2007)

Solution. Define the events

- A_1 : The urn contains 2 white balls in total. (Cause 1)
- A_2 : The urn contains 3 white balls in total. (Cause 2)
- A_3 : The urn contains 4 white balls in total. (Cause 3)
- A_4 : The urn contains 5 white balls in total. (Cause 4)
- B : The two balls drawn are white. (Effect)

Since the 2 balls drawn are found to be white, A_1, A_2, A_3 and A_4 are the only four possibilities available. Since these 4 possibilities are equally likely, it follows that

$$P(A_1) = P(A_2) = P(A_3) = P(A_4) = \frac{1}{4}$$

Note also that

$$P(B|A_1) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}$$

$$P(B|A_2) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10}$$

$$P(B|A_3) = \frac{{}^4C_2}{{}^5C_2} = \frac{6}{10}$$

$$P(B|A_4) = \frac{{}^5C_2}{{}^5C_2} = \frac{10}{10}$$

The probability of the event B is obtained using total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(B) &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) + P(A_4)P(B|A_4) \\ &= \left(\frac{1}{4} \times \frac{1}{10}\right) + \left(\frac{1}{4} \times \frac{3}{10}\right) + \left(\frac{1}{4} \times \frac{6}{10}\right) + \left(\frac{1}{4} \times \frac{10}{10}\right) = \frac{1}{4} \times \frac{20}{10} = \frac{1}{2} \end{aligned}$$

We are asked to calculate the inverse probability, $P(A_4|B)$, which can be easily evaluated using the Bayes' theorem (Theorem 1.13) as

$$P(A_4|B) = \frac{P(A_4)P(B|A_4)}{P(B)} = \frac{\frac{1}{4} \times 1}{\frac{1}{2}} = \frac{1}{2}.$$

The required probability is $P(A_4|B) = \frac{1}{2}$. ◆

EXAMPLE 1.132. A box contains 5 red and 4 white balls. Two balls are drawn successively from the box without replacement and it is noted that the second one is white. What is the probability that the first is also white?

(Anna, Model)

Solution. Define the events

A : The first ball drawn is white. (Cause 1)

$\bar{A}$: The first ball drawn is red. (Cause 2)

B : The second ball drawn is white. (Effect)

First, we calculate the probability of the event A . Since the box contains 9 balls in total and A is the event that the ball drawn is white, it is immediate that

$$P(A) = \frac{4}{9} \Rightarrow P(\bar{A}) = 1 - \frac{4}{9} = \frac{5}{9}$$

Since the balls are drawn from the box without replacement and B represents the event that the second ball drawn is white, it follows that

$$P(B|A) = \frac{3}{8} \text{ and } P(B|\bar{A}) = \frac{4}{8}$$

Next, we calculate $P(B)$ by the total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(B) &= P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) \\ &= \left(\frac{4}{9} \times \frac{3}{8}\right) + \left(\frac{5}{9} \times \frac{4}{8}\right) = \frac{4}{9} \end{aligned}$$

The required probability is the inverse probability, $P(A|B)$, which can be easily evaluated using the Bayes' theorem (Theorem 1.13) as

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} = \frac{\frac{4}{9} \times \frac{3}{8}}{\frac{4}{9}} = \frac{3}{8}$$

◆

EXAMPLE 1.133. There are three boxes containing, respectively, 1 white, 2 red, 3 black balls; 2 white, 3 red, 1 black balls; 3 white, 1 red, 2 black balls. A box is chosen at random and from it, two balls are drawn at random. The two balls are 1 red and 1 white. What is the probability that they came from the second box?

(Anna, May 2007)

Solution. Define the events

B_1 : Box 1 is chosen. (Cause 1)

B_2 : Box 2 is chosen. (Cause 2)

B_3 : Box 3 is chosen. (Cause 3)

E : The two balls selected are 1 red and 1 white. (Effect)

Since one box is chosen at random, it is immediate that

$$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$$

Note that

$$P(E|B_1) = \frac{{}^2C_1 \times {}^1C_1}{{}^6C_2} = \frac{2}{15}$$

$$P(E|B_2) = \frac{{}^3C_1 \times {}^2C_1}{{}^6C_2} = \frac{6}{15}$$

$$P(E|B_3) = \frac{{}^3C_1 \times {}^1C_1}{{}^{12}C_2} = \frac{3}{15}$$

We can calculate $P(E)$ by total probability theorem (theorem 1.12) as

$$\begin{aligned} P(E) &= P(B_1)P(E|B_1) + P(B_2)P(E|B_2) + P(B_3)P(E|B_3) \\ &= \left(\frac{1}{3} \times \frac{2}{15}\right) + \left(\frac{1}{3} \times \frac{6}{15}\right) + \left(\frac{1}{3} \times \frac{3}{15}\right) = \frac{11}{45} \end{aligned}$$

In this problem, we are asked to calculate the inverse probability, $P(B_2|E)$. By Bayes' theorem (Theorem 1.13), we have

$$P(B_2|E) = \frac{P(B_2)P(E|B_2)}{P(E)} = \frac{\frac{1}{3} \times \frac{6}{15}}{\frac{11}{45}} = \frac{6}{11} = 0.5455$$

◆

EXAMPLE 1.134. There are three identical coins, one of which is ideal and the other two are biased with probabilities $\frac{1}{4}$ and $\frac{3}{4}$, respectively, for getting a head. One coin is taken at random and tossed twice. If a head appears both the times, show that the probability that the ideal coin was chosen is equal to $\frac{2}{7}$.

Solution. Denote the events

- A_1 : Ideal coin was chosen. (*Cause 1*)
- A_2 : Biased coin 1 was chosen. (*Cause 2*)
- A_3 : Biased coin 2 was chosen. (*Cause 3*)
- B : A head appears in both the tosses. (*Effect*)

Since the coins are identical, it is immediate that

$$P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}$$

Next, note that if the ideal coin was chosen, then the probability of getting heads in two tosses is equal to

$$P(B|A_1) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Similarly, we find that

$$P(B|A_2) = \left(\frac{1}{4}\right)^2 = \frac{1}{16} \text{ and } P(B|A_3) = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

Next, we calculate $P(B)$ by total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(B) &= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) \\ &= \left[\frac{1}{3} \times \frac{1}{4}\right] + \left[\frac{1}{3} \times \frac{1}{16}\right] + \left[\frac{1}{3} \times \frac{9}{16}\right] = \frac{7}{24}. \end{aligned}$$

We are asked to calculate the inverse probability, $P(A_1|B)$, which can be easily evaluated using the Bayes' theorem (Theorem 1.13) as

$$P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(B)} = \frac{\frac{1}{3} \times \frac{1}{4}}{\frac{7}{24}} = \frac{2}{7}$$

◆

EXAMPLE 1.135. The members of a consulting firm rent cars from rental agencies A , B and C as 60%, 30% and 10%, respectively. If 9%, 20% and 6% of cars from A , B and C , respectively, agencies need tune up and if a rental car delivered to the firm does not need tune up, what is the probability that it came from B agency?

(Anna, April 2004)

Solution. Define the events

A : The car came from rental agency A . (Cause 1)

B : The car came from rental agency B . (Cause 2)

C : The car came from rental agency C . (Cause 3)

G : The car does not need tune up. (Effect)

From the given data, we have

$$P(A) = \frac{60}{100} = \frac{3}{5}, \quad P(B) = \frac{30}{100} = \frac{3}{10}, \quad P(C) = \frac{10}{100} = \frac{1}{10}$$

and also that

$$P(G|A) = 1 - \frac{9}{100} = \frac{91}{100}, \quad P(G|B) = 1 - \frac{20}{100} = \frac{4}{5} \quad \text{and} \quad P(G|C) = 1 - \frac{6}{100} = \frac{47}{50}$$

First, we calculate $P(G)$, the probability that the car does not need tune up. By total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(G) &= P(A)P(G|A) + P(B)P(G|B) + P(C)P(G|C) \\ &= \left[\frac{3}{5} \times \frac{91}{100} \right] + \left[\frac{3}{10} \times \frac{4}{5} \right] + \left[\frac{1}{10} \times \frac{47}{50} \right] = \frac{22}{25} \end{aligned}$$

The required probability is $P(B|G)$, which can be easily evaluated using the Bayes' Theorem (Theorem 1.13) as

$$P(B|G) = \frac{P(B)P(G|B)}{P(G)} = \frac{\frac{3}{10} \times \frac{4}{5}}{\frac{22}{25}} = \frac{3}{11}$$

Thus, the required probability is $\frac{3}{11}$ or 27.27%. ◆

EXAMPLE 1.136. There are three unbiased coins and one biased coin with head on both sides. A coin is chosen at random and tossed 4 times. If head occurs all the 4 times, what is the probability that the biased coin has been chosen?

(Anna, April 2004)

Solution. Define the events

A_1 : Unbiased coin 1 is chosen. (Cause 1)

A_2 : Unbiased coin 2 is chosen. (Cause 2)

A_3 : Unbiased coin 3 is chosen. (Cause 3)

B : Biased coin is chosen. (Cause 4)

E : Head occurs all the 4 times. (Effect)

Since the coin is chosen at random, it follows that

$$P(A_1) = P(A_2) = P(A_3) = P(B) = \frac{1}{4}$$

Since coins 1, 2 and 3 are unbiased, it follows that

$$P(E|A_1) = P(E|A_2) = P(E|A_3) = \frac{1}{2^4} = \frac{1}{16}$$

(Note that of the 16 possible outcomes, there is only one outcome $HHHH$ having all the four heads.)

Since the biased coin has head on both sides, it follows that

$$P(E|B) = 1$$

First, we calculate $P(E)$ by total probability theorem (Theorem 1.12) as

$$\begin{aligned} P(E) &= P(A_1)P(E|A_1) + P(A_2)P(E|A_2) + P(A_3)P(E|A_3) + P(B)P(E|B) \\ &= 3 \times \left[\frac{1}{4} \times \frac{1}{16} \right] + \left[\frac{1}{4} \times 1 \right] = \frac{19}{64} \end{aligned}$$

The required probability is $P(B|E)$, which can be easily evaluated using Bayes' theorem (Theorem 1.13) as

$$P(B|E) = \frac{P(B)P(E|B)}{P(E)} = \frac{\frac{1}{4} \times 1}{\frac{19}{64}} = \frac{1}{4} \times \frac{64}{19} = \frac{16}{19}$$

Hence, the required probability is $\frac{16}{19}$ or 84.21%. ◆

PROBLEM SET 1.5

1. A Statistics quiz is given in a class. Given that a student studied, the probability of him passing the quiz is $\frac{9}{10}$. Given that a student did not study, the probability of him passing the quiz is $\frac{1}{8}$. Assume that 75% of the students in the class have studied for the quiz. A student, chosen at random, fails in the quiz. What is the probability that he or she did not study?
2. A student knows only 70% of the questions in an objective test each with four multiple-choice answers. He simply guesses while answering the test. What is the probability that he knew the answer to a question given that he answered it correctly?
3. In a new residence colony, there are three types of houses, A , B , C which constitute 20%, 30% and 50%, respectively, of the total houses in the colony. In these houses, 3%, 5% and 10% are occupied by engineers. A house was chosen at random and is occupied by an engineer. Find the probability that it is of (i) type A , (ii) type B and (iii) type C .
4. The contents of Boxes A , B and C are as follows:
 - Box A : 3 white, 2 black and 3 red balls.
 - Box B : 4 white, 3 black and 2 red balls.
 - Box C : 2 white, 4 black and 2 red balls.

One box is chosen at random and 2 balls are drawn. They happen to be white and black. Find the probability that they come from (i) Box A, (ii) Box B and (iii) Box C.

5. A factory has two machines A and B, which produce 70% and 30% of the total output, respectively. It has been found that 20% and 10%, respectively of the items produced by machines A and B are defective respectively. An item from the output is drawn at random and found to be defective. What is the probability that it was made by (i) machine A, (ii) machine B?
6. There are three machines A, B and C in a factory. They produce 40%, 35% and 25% of the products, respectively. Out of their output, 8%, 10% and 5% are defective, respectively. An item is chosen at random from the total output and tested. What is the probability that it is good? If it is good, find the probability that it was made by (i) machine A, (ii) machine B and (iii) machine C.
7. In a class of 60 students, 10 are considered to be very intelligent, 30 as medium and the rest below average. The probabilities of a very intelligent, a medium and a below average student passing the final examination are 95%, 50% and 10%, respectively. If a student is known to have passed the final examination, find the probability that he is (i) very intelligent, (ii) medium and (iii) below average.
8. One shot is fired from one of the three guns A, B and C. The probabilities of the target being hit by guns A, B and C are 0.6, 0.2 and 0.1, respectively. A shot is found to have hit the target. Find the probability that gun B fired the shot.
9. An urn contains 4 balls. Two balls are drawn at random and are found to be white. What is the probability of all the balls in the urn being white?
10. A box contains 6 white and 4 red balls. Two balls are drawn successively from the box without replacement and it is noted that the second ball is white. What is the probability that the first ball also is white?
11. There are two identical boxes. The first box contains 8 white and 7 red balls and the second box contains 4 white and 6 red balls. One box is chosen at random and one ball is picked randomly. It is found to be white. If so, what is the probability that the selected box is the first one?
12. There are three identical coins, one of which is ideal, while the other two coins are biased with probabilities 0.3 and 0.7, respectively, for getting a head. One coin is chosen at random and tossed twice. If a head appears both the times, find the probability that the ideal coin is chosen.
13. In a colony, there are 60% men and 40% women. According to a survey taken, 8 men out of 100 and 10 women out of 100 have high blood pressure. A high B.P. person is chosen at random. What is the probability that this person is male?
14. A medical shop in a city has three employees X, Y and Z, who fill 50%, 30% and 20% of the total orders respectively. While filling, the employees X, Y and Z make mistakes in 5%, 4% and 2% of the orders. Calculate the following probabilities:
 - (i) A mistake is made in filling an order.
 - (ii) If a mistake is made in an order, the order was filled by X.
 - (iii) If a mistake is made in an order, the order was filled by Y.
 - (iv) If a mistake is made in an order, the order was filled by Z.

Chapter 2

Random Variable

2.1 DEFINITION OF A RANDOM VARIABLE

In Chapter 1, we studied the notions of random experiment, sample space, and the various definitions of probability and applications. We studied the notions of finite, discrete and continuous sample spaces. We also studied the assignment and computation of probabilities of various events. In practice, when an experiment is performed, we are mostly interested in some function of the outcome rather than on the actual outcome itself. For instance, when a coin is tossed n times, we are often interested in knowing the number of heads in n tosses as opposed to knowing which n -tuple of the 2^n possible outcomes has occurred. As another example, when we play board games with two dice, we are often interested in the sum of numbers on the two dice and are not really concerned with the separate numbers on each die. In a board game like Monopoly, if throwing a six with two dice fetches us some attractive house like Rail Road or Electric Company, we are not bothered whether the six is obtained with $(5, 1)$ or $(3, 3)$ or $(2, 4)$. Such real-valued functions defined on a sample space S is called a *random variable*.

Definition 2.1. Let S be a sample space and $\mathcal{B}$ a σ -field of subsets of S . A function $X : S \rightarrow \mathbb{R}$ is called a *random variable* if the inverse images under X of all semi-closed intervals of the form $(-\infty, x]$, where $x \in \mathbb{R}$ are events in $\mathcal{B}$, i.e.

$$X^{-1}(-\infty, x] = \{\omega \in S : X(\omega) \leq x\} \in \mathcal{B}$$

Remark 2.1. Note that Definition 2.1 uses only the notions of sample space S and the σ -field $\mathcal{B}$ associated with the sample space S . It has not used the notion of probability. We also recall that if S is a discrete sample space having finite or countable number of sample points, then $\mathcal{B}$ can be taken as the set of all subsets of S . If S is a continuous sample space like $I = [a, b]$ or the whole real line $\mathbb{R}$, then the associated σ -field $\mathcal{B}$ is generated by the collection of all semi-closed intervals of the form $(a, b]$ contained in it.

Remark 2.2. The random variable X , as defined in Definition 2.1, is a well-defined, finite-valued real function defined on the sample space S . X is not a *variable*, and it is not *random* either. Note that X assigns values to the outcomes ω of the sample space S , which are random.

EXAMPLE 2.1. If A is any event and $A \in \mathcal{B}$, then the function $I_A : S \rightarrow \mathbb{R}$ defined by

$$I_A(\omega) = \begin{cases} 0, & \omega \notin A \\ 1, & \omega \in A \end{cases}$$

is called the *indicator function* of the set A . Note that

$$I_A^{-1}(-\infty, x] = \begin{cases} \emptyset & \text{if } x < 0 \\ \bar{A} & \text{if } 0 \leq x < 1 \\ A & \text{if } x \geq 1 \end{cases}$$

Thus, I_A is a random variable.

EXAMPLE 2.2. Consider the experiment of tossing a fair coin. Then $S = \{H, T\}$. Let X be the function defined as the “number of heads” in the outcome ω . Then

$$X(H) = 1 \text{ and } X(T) = 0$$

Note that

$$X^{-1}(-\infty, x] = \begin{cases} \emptyset & \text{if } x < 0 \\ \{T\} & \text{if } 0 \leq x < 1 \\ \{H, T\} & \text{if } x \geq 1 \end{cases}$$

Thus, it follows that X is a random variable.

EXAMPLE 2.3. Consider the experiment of tossing 2 fair coins. Then $S = \{HH, HT, TH, TT\}$. Let X be the function defined as the “number of heads” in the outcome ω . Then

$$X(HH) = 2, X(HT) = 1, X(TH) = 1 \text{ and } X(TT) = 0$$

Note that

$$X^{-1}(-\infty, x] = \begin{cases} \emptyset & \text{if } x < 0 \\ \{TT\} & \text{if } 0 \leq x < 1 \\ \{TT, TH, HT\} & \text{if } 1 \leq x < 2 \\ S & \text{if } x \geq 2 \end{cases}$$

Thus, it follows that X is a random variable.

EXAMPLE 2.4. Consider the experiment of throwing a fair die. Then $S = \{1, 2, 3, 4, 5, 6\}$. We can define a function X on S as

$$X(\omega) = \begin{cases} 1 & \text{if } \omega \text{ is even} \\ -1 & \text{if } \omega \text{ is odd} \end{cases}$$

Then, we have

$$X^{-1}(-\infty, x] = \begin{cases} \emptyset & \text{if } x < -1 \\ \{1, 3, 5\} & \text{if } -1 \leq x < 1 \\ S & \text{if } x \geq 1 \end{cases}$$

Thus, it follows that X is a random variable.

Remark 2.3. If S is a discrete sample space and $\mathcal{B}$ is the class of all subsets of S , then it can be easily shown that every real-valued function defined on S is a random variable.

EXAMPLE 2.5. Two fair dice are tossed. Let X be the sum of the numbers on the dice. Determine the following events:

- (i) $A = \{X = 6\}$
- (ii) $B = \{X = 10\}$
- (iii) $C = \{X \leq 5.4\}$
- (iv) $D = \{4.2 \leq X \leq 6.5\}$
- (v) $E = \{X \geq 10.2\}$

Solution. Note that $\{X = a\} = \{\omega \in S : X(\omega) = a\}$

- (i) $A = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$
- (ii) $B = \{(4, 6), (5, 5), (6, 4)\}$
- (iii) $C = \{X = 2\} \cup \{X = 3\} \cup \{X = 4\} \cup \{X = 5\}$. Hence,

$$C = \{(1, 1), (1, 2), (2, 1), (1, 3), (2, 2), (3, 1), (1, 4), (2, 3), (3, 2), (4, 1)\}$$

- (iv) $D = \{X = 5\} \cup \{X = 6\}$. Hence,

$$D = \{(1, 4), (2, 3), (3, 2), (4, 1), (1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$$

- (v) $E = \{X = 11\} \cup \{X = 12\}$. Hence,

$$E = \{(5, 6), (6, 5), (6, 6)\}$$

◆

Next, we shall state (without proof) some basic results on random variables.

Theorem 2.1. Let S be a sample space and let $\mathcal{B}$ be a σ -field defined on S . Then, the following properties hold:

- (i) X is a random variable on S if and only if for each real x ,

$$X^{-1}(-\infty, x) = \{\omega \in S : X(\omega) < x\} \in \mathcal{B}$$

- (ii) If X_1 and X_2 are random variables and α is a real constant, then $X_1 + X_2, \alpha X_1, X_1 X_2$ also are random variables.
- (iii) If X is a random variable, then X^2 and $|X|$ also are random variables.
- (iv) If X is a random variable taking non-negative values, then $\sqrt{X}$ also is a random variable.
- (v) If X is a random variable, then $\frac{1}{X}$ also is a random variable provided that $X^{-1}\{0\} = \emptyset$, i.e. there is no $\omega \in S$ for which $X(\omega) = 0$.
- (vi) If X_1 and X_2 are random variables, then $\max(X_1, X_2)$ and $\min(X_1, X_2)$ also are random variables.

- (vii) If X is a random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function, then $g(X)$ also is a random variable.

PROBLEM SET 2.1

1. Let A be any event of the sample space S . Show that the function $1 - I_A$ is the indicator function of A and deduce that it is a random variable.
2. If 3 fair coins are tossed and if X is the total number of heads in the outcome, show that X is a random variable.
3. If two fair dice are tossed and X is the sum of the numbers on the faces, determine the following events:
 - (a) $A = \{X = 4\}$
 - (b) $B = \{X = 7\}$
 - (c) $C = \{X \leq 3.2\}$
 - (d) $D = \{4.5 \leq X < 7\}$
 - (e) $E = \{X \geq 9.8\}$

2.2 DISTRIBUTION FUNCTION OF A RANDOM VARIABLE

In the previous section, we saw the definition of random variables and also how to associate events with random variables. However, random variables will be of real interest only when they are defined on a probability space, say, $(S, \mathcal{B}, P)$. The distribution function of a random variable X tells us how the probabilities of events in the σ -field $\mathcal{B}$ are distributed.

Definition 2.2. Let $(S, \mathcal{B}, P)$ be a probability space and X be a random variable defined on S . The function $F : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x) = P(-\infty, x] = P\{\omega \in S : X(\omega) \leq x\} \quad \text{for all } x \in \mathbb{R} \quad (2.1)$$

is called the *distribution function* or the *cumulative distribution function (CDF)* of the random variable X .

Remark 2.4. It is a common practice to express Definition (2.1) in an abbreviated form as

$$F(x) = P\{X \leq x\}$$

Sometimes, we use the notation F_X to emphasize that the distribution function is associated with the random variable X . For instance, if we are discussing the distribution functions of two random variables X and Y , we indicate them as F_X and F_Y , respectively, to indicate their associations clearly. Note also that for any distribution function F , the domain is the whole real line $\mathbb{R}$ and its range is $[0, 1]$.

EXAMPLE 2.6. If X is the random variable representing the number of head in the single toss of a fair coin, find the distribution function of X and plot its graph.

Solution. Here, $S = \{H, T\}$. Note that

$$P\{X = 1\} = P\{H\} = \frac{1}{2} \quad \text{and} \quad P\{X = 0\} = P\{T\} = \frac{1}{2}$$

Hence, the distribution function for X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{2} & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

and is plotted in Figure 2.1. It can be noted that F is non-decreasing, right-continuous, $F(-\infty) = 0$ and $F(+\infty) = 1$. ◆

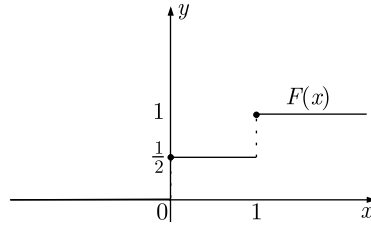


Figure 2.1: Distribution function for Example 2.6

EXAMPLE 2.7. If X is the random variable representing the number of heads in two tosses of a fair coin, find the distribution function of X .

Solution. Here, $S = \{HH, HT, TH, TT\}$. Note that

$$\begin{aligned} P\{X = 2\} &= P\{HH\} = \frac{1}{4} \\ P\{X = 1\} &= P\{HT, TH\} = \frac{1}{2} \\ P\{X = 0\} &= P\{TT\} = \frac{1}{4} \end{aligned}$$

Hence, the distribution function for X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{4} & \text{if } 0 \leq x < 1 \\ \frac{3}{4} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

and is plotted in Figure 2.2. It can be noted that F is non-decreasing, right-continuous, $F(-\infty) = 0$ and $F(+\infty) = 1$. ◆

Next, we state and prove some important properties of the distribution functions.

Theorem 2.2. Let F be the distribution function of a random variable X defined on a probability space $(S, \mathcal{B}, P)$. Then the following properties hold:

- (i) F is non-decreasing, i.e. $F(x) \leq F(y)$ whenever $x < y$.

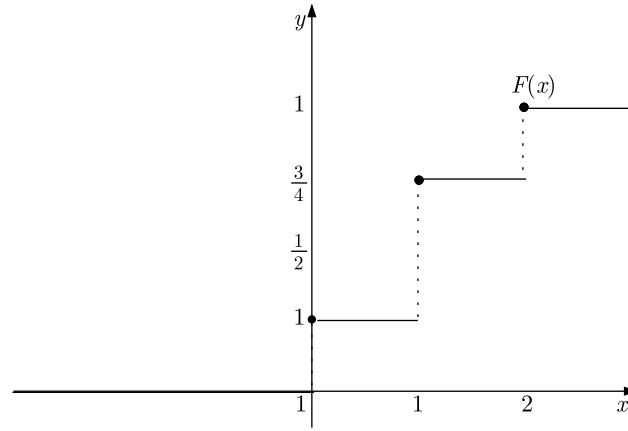


Figure 2.2: Distribution function for Example 2.7.

(ii) F is right-continuous, i.e.

$$F(x+0) = \lim_{h \rightarrow 0} F(x+h) = F(x) \text{ for all } x \in \mathbb{R}.$$

(iii) $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$ and $F(+\infty) = \lim_{x \rightarrow +\infty} F(x) = 1$

Proof. (i) Let $x, y \in \mathbb{R}$ with $x < y$. We assert that $F(x) \leq F(y)$. By definition,

$$F(x) = P\{X \leq x\} = P\{\omega \in S : X(\omega) \leq x\}$$

and

$$F(y) = P\{X \leq y\} = P\{\omega \in S : X(\omega) \leq y\}$$

Since $x < y$, it is clear that

$$\{X \leq x\} \subset \{X \leq y\}$$

Since P is subtractive (see Theorem 1.3), it follows that

$$P\{X \leq x\} \leq P\{X \leq y\}$$

i.e. $F(x) \leq F(y)$. Hence, F is non-decreasing.

(ii) We assert that F is right-continuous at each point $x \in \mathbb{R}$. For this, we have to show that

$$F(x+0) = F(x) \text{ for each } x \in \mathbb{R}$$

where

$$F(x+0) = \lim_{h \rightarrow 0} F(x+h)$$

By definition, we have

$$F(x+h) = P\{X \leq x+h\} = P\{\omega : X(\omega) \leq x+h\}$$

and

$$F(x) = P\{X \leq h\} = P\{\omega : X(\omega) \leq x\}$$

If we define the events

$$B = \{X \leq x+h\} \text{ and } A = \{X \leq x\}$$

then, $F(x+h) = P(B)$ and $F(x) = P(A)$.

Since $A \subset B$ and P is monotone (see Theorem 1.3), it follows that

$$P(B) - P(A) = P(B - A) = P\{x < X \leq x+h\} = P\{\omega : x < X(\omega) \leq x+h\}$$

Hence, it follows that

$$F(x+h) - F(x) = P(B) - P(A) = P\{x < X \leq x+h\} = P\{\omega : x < X(\omega) \leq x+h\}$$

Taking limits as $h \rightarrow 0$, we have

$$F(x+0) - F(x) = P(\emptyset) = 0$$

so that

$$F(x+0) = F(x) \text{ for all } x \in \mathbb{R}$$

Hence, F is right-continuous.

(iii) First, we show that $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$

Consider the events A_n defined by

$$A_n = \{X \leq -n\} = \{\omega : X(\omega) \leq -n\}, \text{ where } n = 1, 2, \dots$$

Then we see that $\{A_n\}$ is a non-increasing sequence of events in S , i.e.

$$A_n \supset A_{n+1} \text{ for } n = 1, 2, 3, \dots$$

Since P is continuous from above (see Theorem 1.9), it follows that

$$F(-\infty) = \lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = P(\emptyset) = 0$$

Next, consider the events B_n defined by

$$B_n = \{X \leq n\} = \{\omega : X(\omega) \leq n\}, \text{ where } n = 1, 2, \dots$$

Then we see that $\{B_n\}$ is a non-decreasing sequence of events in S , i.e.

$$B_n \supset B_{n+1} \text{ for } n = 1, 2, 3, \dots$$

Since P is continuous from below (see Theorem 1.8), it follows that

$$F(\infty) = \lim_{n \rightarrow \infty} P(B_n) = P\left(\bigcup_{n=1}^{\infty} B_n\right) = P(S) = 1$$

◆

The converse of Theorem 2.2 also is true. We state the following result without proof.

Theorem 2.3. If $F : \mathbb{R} \rightarrow \mathbb{R}$ is a function which is non-decreasing, right-continuous, $F(-\infty) = 0$ and $F(+\infty) = 1$, then there exists a unique probability function P defined on the Borel σ -field $\mathcal{B}$ generated by the semi-closed intervals of $\mathbb{R}$ such that

$$F(x) = P\{X \leq x\} = P\{\omega : X(\omega) \leq x\}.$$

Combining Theorems 2.2 and 2.3, we have the following characterizing result for distribution functions (CDF).

Theorem 2.4. A function $F : \mathbb{R} \rightarrow \mathbb{R}$ is a distribution function for some probability space $(S, \mathcal{B}, P)$ if and only if F satisfies the following properties:

- (i) F is non-decreasing.
- (ii) F is right-continuous.
- (iii) $F(-\infty) = 0$ and $F(+\infty) = 1$.

As Examples 2.6 and 2.7 illustrate, the distribution function of a random variable need not be left-continuous (see Figures 2.1 and 2.2). From the following result, we can easily deduce the condition under which the distribution function can be continuous from the left also.

Theorem 2.5. Let F be the distribution function of a random variable X defined on a probability space $(S, \mathcal{B}, P)$. Then,

$$F(x) - F(x-0) = P\{X = x\} \quad (2.2)$$

where $F(x-0) = \lim_{h \rightarrow 0} F(x-h)$

Proof. By definition, we have

$$F(x-h) = P\{X \leq x-h\} = P\{\omega : X(\omega) \leq x-h\}$$

and

$$F(x) = P\{X \leq x\} = P\{\omega : X(\omega) \leq x\}$$

Define the events $B = \{X \leq x\}$ and $A = \{X \leq x-h\}$. Then,

$$F(x-h) = P(B) \text{ and } F(x) = P(A)$$

Since $A \subset B$ and P is monotone (see Theorem 1.3), we have

$$P(B) - P(A) = P(B-A) = P\{x-h < X \leq x\} = P\{\omega : x-h < X(\omega) \leq x\}$$

Hence, it follows that

$$F(x) - F(x-h) = P(B) - P(A) = P\{x-h < X \leq x\}$$

Taking limits as $h \rightarrow 0$, we have

$$F(x) - F(x-0) = P\{X = x\}$$

◆

Theorem 2.6. A necessary and sufficient condition for a distribution function F of a random variable X to be continuous is that

$$P\{X = x\} = 0 \quad \text{for all } x \in \mathbb{R}$$

Proof. By Theorem 2.2, the distribution function F is right-continuous. Hence, F is continuous on $\mathbb{R}$ if and only if F is left-continuous on $\mathbb{R}$. By definition, the distribution function F is left-continuous on $\mathbb{R}$ if and only if

$$F(x) = F(x-0) \quad \text{for all } x \in \mathbb{R}$$

or if and only if

$$P\{X = x\} = 0 \quad \text{for all } x \in \mathbb{R}$$

(using Eq. (2.2)).

◆

Theorem 2.7. The set of points of discontinuity of a distribution function F is countable.

Proof. By Theorem 2.2, the distribution function F is a non-decreasing function on the real line. It is a standard result in Real Analysis that the set of points of discontinuity of every monotone function (non-decreasing or non-increasing) is countable.

◆

The following examples illustrate the distribution function of a random variable, which is continuous.

EXAMPLE 2.8. Show that the function $F : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-x} & \text{if } x \geq 0 \end{cases}$$

is a distribution function. Is it continuous?

Solution. We verify the properties (i), (ii) and (iii) stated in Theorem 2.4 for a distribution function.

(i) Since the exponential function is strictly non-decreasing, it follows that

$$x < y \Rightarrow e^x < e^y \Rightarrow e^{-x} > e^{-y} \Rightarrow 1 - e^{-x} < 1 - e^{-y}$$

Thus, it is immediate that

$$x < y \Rightarrow F(x) < F(y)$$

Hence, F is (strictly) non-decreasing on $\mathbb{R}$.

(ii) It is easy to see that F is continuous for each $x \neq 0$. A simple calculation also gives that

$$F(0+0) = F(0) = F(0-0)$$

so that F is continuous at the origin as well. Hence, F is continuous everywhere on $\mathbb{R}$. In particular, F is right-continuous.

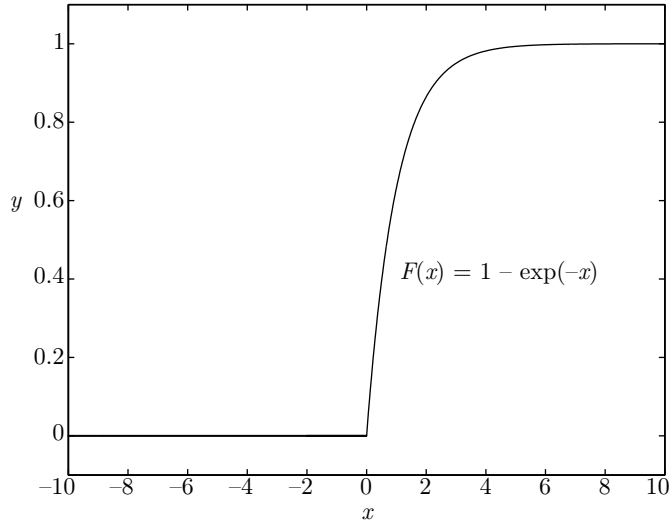


Figure 2.3: Distribution Function for Example 2.8

(iii) From the definition of F , it is immediate that $F(-\infty) = 0$. Note also that

$$F(+\infty) = \lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} [1 - e^{-x}] = 1 - 0 = 1$$

Hence, by Theorem 2.4, F is a distribution function for some probability function defined on $\mathbb{R}$. In fact, it will be shown later (see Section 3.8) that F is the cumulative distribution function for the *exponential density function* on $\mathbb{R}$. The plot of F is given in Figure 2.3. ◆

EXAMPLE 2.9. If F is the distribution function of a random variable X and if $a < b$, show that

$$F(b) - F(a) = P\{a < X \leq b\} = P\{\omega : a < X(\omega) \leq b\}$$

Solution. Define the events $A = \{X \leq a\}$ and $B = \{X \leq b\}$. By definition of the distribution function F , we have

$$F(a) = P(A) \text{ and } F(b)$$

Since $a < b$, it follows that $A \subset B$. Also, note that B is the union of the disjoint events

$$A = \{X \leq a\} \text{ and } C = \{a < X \leq b\}$$

By Axiom (iii) of the probability function P , we have

$$P(B) = P(A) + P\{a < X \leq b\}$$

from which it readily follows that

$$F(b) - F(a) = P(B) - P(A) = P\{a < X \leq b\}$$
◆

EXAMPLE 2.10. If the cumulative distribution function of X is $F(x)$, find the cumulative distribution function of $Y = X + a$, where a is a constant.

Solution. Let $G(y)$ be the cumulative distribution function of Y . Then

$$G(y) = P(Y \leq y) = P(X + a \leq y) = P(X \leq y - a) = F(y - a)$$

◆

PROBLEM SET 2.2

1. A coin is biased so that the head is twice as likely to appear as the tail. The coin is tossed twice. Find the distribution function of the number of heads.
2. A coin is biased so that the head is twice as likely to appear as the tail. The coin is tossed 3 times. Find the distribution function of the number of heads.
3. Two balls are selected at random and picked successively from a box containing 6 white and 4 red balls. Determine the sample space S , the probabilities of the sample points, and the corresponding values x of the random variable X , where X is the number of white balls drawn.
4. If X is a random variable taking on the values $x = 1, 2, 3, 4$ and

$$P(X = x) = \frac{x}{10} \text{ for } x = 1, 2, 3, 4$$

find the distribution function of X .

5. Let X be a random variable taking the values $x = 0, 1, 2, 3, 4$. For each of the following cases, determine whether the given values can serve as the values of a distribution function F of X . Explain your answer.
 - (a) $F(0) = 0, F(1) = 0.3, F(2) = 0.25, F(3) = 0.5, F(4) = 1$
 - (b) $F(0) = -0.1, F(1) = 0.2, F(2) = 0.3, F(3) = 0.8, F(4) = 1$
 - (c) $F(0) = 0, F(1) = 0.35, F(2) = 0.36, F(3) = 0.8, F(4) = 1$
 - (d) $F(0) = 0.2, F(1) = 0.4, F(2) = 0.5, F(3) = 0.8, F(4) = 1.1$
 - (e) $F(0) = 0, F(1) = 0.2, F(2) = 0.6, F(3) = 0.7, F(4) = 0.8$
6. If the distribution function of the random variable X is F_X , find the distribution functions of the following random variables:
 - (a) $Y = X - \varepsilon$, where ε is a constant.
 - (b) $Z = \alpha X$, where $\alpha > 0$ is a constant.
 - (c) $W = \alpha X + \beta$, where $\alpha > 0$ and β are constants.
7. If X has the distribution function F given by

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{4} & \text{for } 0 \leq x < 1 \\ \frac{1}{2} & \text{for } 1 \leq x < 3 \\ \frac{3}{4} & \text{for } 3 \leq x < 5 \\ 1 & \text{for } x \geq 5 \end{cases}$$

find the following probabilities: (a) $P(X = 3)$, (b) $P(X < 3)$, (c) $P(0 < X < 3)$, (d) $P(0 < X \leq 3)$, and (e) $P(1 < X < 5)$.

2.3 DISCRETE RANDOM VARIABLE

Definition 2.3. A random variable (RV) X defined on a sample space S is called *discrete* if X can take on at most a countable number of possible values, say, $x_1, x_2, \dots, x_n, \dots$ with $p_i = P(X = x_i) > 0$ for each i and satisfying

$$\sum_i p_i = \sum_i P\{X = x_i\} = 1$$

The points x_i of S having positive probability (*mass*) are said to be *mass points* of the discrete random variable X .

EXAMPLE 2.11. Let $S = \{H, T\}$. Consider the random variable X defined on S by

$$X(\omega) = \text{Number of heads in } \omega$$

Then X is a discrete random variable because it can take only two values, viz. $X = 0$ and $X = 1$. Note that

$$p_1 = P(X = 0) = P\{T\} = \frac{1}{2} \quad \text{and} \quad p_2 = P(X = 1) = P\{H\} = \frac{1}{2}$$

Since $p_1 > 0$ and $p_2 > 0$ and $p_1 + p_2 = 1$, X is a well-defined discrete random variable. The points $X = 0$ and $X = 1$ are the *mass points* for the discrete RV X .

EXAMPLE 2.12. Let $S = \{HH, HT, TH, TT\}$. Consider the random variable X defined on S by

$$X(\omega) = \text{Number of heads in } \omega$$

Then, X is a discrete random variable because it can take only three values, viz. $X = 0$, $X = 1$ and $X = 2$. Note that

$$\begin{aligned} p_1 &= P(X = 0) = P\{TT\} = \frac{1}{4}, \\ p_2 &= P(X = 1) = P\{TH, HT\} = \frac{1}{2} \\ p_3 &= P(X = 2) = P\{HH\} = \frac{1}{4} \end{aligned}$$

Since $p_1 > 0$, $p_2 > 0$ and $p_3 > 0$ and $p_1 + p_2 + p_3 = 1$, X is a well-defined discrete random variable. The points $X = 0$, $X = 1$ and $X = 2$ are the *mass points* for X .

Next, we define the *probability mass function* of a discrete random variable X .

Definition 2.4. Let X be a discrete RV defined on a sample space S with mass points $x_1, x_2, \dots, x_n, \dots$ and $p_i = P(X = x_i)$ for $i = 1, 2, 3, \dots$. Then the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} P(X = x_i) = p_i & \text{if } x = x_i \text{ for some } i \\ 0 & \text{if } x \neq x_i \text{ for any } i \end{cases}$$

is called the *probability mass function (PMF)* of the discrete RV X . (Sometimes, we also express f as f_X when it is necessary to emphasize that f is the PMF of the discrete RV X .)

Definition 2.5. If X is a discrete RV defined on a sample space S with mass points $x_1, x_2, \dots, x_n, \dots$ and $p_i = P(X = x_i)$ for $i = 1, 2, 3, \dots$, then the set of ordered pairs

$$\{(x_1, p_1), (x_2, p_2), \dots, (x_n, p_n)\}$$

is called the *probability distribution* of the discrete RV X .

EXAMPLE 2.13. For the discrete RV described in Example 2.11, the probability distribution is given by

$$\{(x_1, p_1), (x_2, p_2)\} = \left\{ \left(0, \frac{1}{2}\right), \left(1, \frac{1}{2}\right) \right\}$$

which can also be represented in a tabular form as below:

x	0	1
$f(x)$	$\frac{1}{2}$	$\frac{1}{2}$

Also, for the discrete RV described in Example 2.12, the probability distribution is given by

$$\{(x_1, p_1), (x_2, p_2), (x_3, p_3)\} = \left\{ \left(0, \frac{1}{4}\right), \left(1, \frac{1}{2}\right), \left(2, \frac{1}{4}\right) \right\}$$

which also can be represented in a tabular form as below:

x	0	1	2
$f(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

EXAMPLE 2.14. A fair coin is tossed 3 times. If X is the number of heads, give the probability distribution of X .

(Madras, Oct. 1996)

Solution. Let p and q denote the probabilities of getting a head and a tail in a single toss, respectively. Then $p = q = \frac{1}{2}$. Here, the sample space S consists of $2^3 = 8$ outcomes, namely

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Note that $\{X = 0\} = \{HTT, THT, TTH\}$ and hence $P(X = 0) = \frac{3}{8}$. Similarly, the probabilities $P(X = 1)$, $P(X = 2)$ and $P(X = 3)$ can be computed.

Thus, the probability distribution of X is given in the following table:

x	0	1	2	3
$f(x)$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$



EXAMPLE 2.15. A lot of 10 items contains 3 defectives from which a sample of 4 items is drawn without replacement. Let X be the random variable being the number of defective items in the sample. Find (i) the probability distribution of X , (ii) $P(X < 1)$, and (iii) $P(0 < X < 2)$.

(Madras, Oct. 1997)

Solution. (i) Clearly, X takes the values 0, 1, 2 and 3. We find the probabilities $P(X = i)$, where $i = 0, 1, 2, 3$.

We find that

$$P(X = 0) = P(\text{no defective}) = \frac{{}^7C_4}{{}^{10}C_4} = \frac{35}{210} = \frac{1}{6}$$

$$P(X = 1) = P(1 \text{ defective}) = \frac{{}^3C_1 {}^7C_3}{{}^{10}C_4} = \frac{3 \times 35}{210} = \frac{1}{2}$$

$$P(X = 2) = P(2 \text{ defectives}) = \frac{{}^3C_2 {}^7C_2}{{}^{10}C_4} = \frac{3 \times 21}{210} = \frac{3}{10}$$

$$P(X = 3) = P(3 \text{ defectives}) = \frac{{}^3C_3 {}^7C_1}{{}^{10}C_4} = \frac{1 \times 7}{210} = \frac{1}{30}$$

Hence, the probability distribution of X is given by

x	0	1	2	3
$f(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

$$(ii) \quad P(X < 1) = P(X = 0) = \frac{1}{6}.$$

$$(iii) \quad P(0 < X < 2) = P(X = 1) = \frac{1}{2}.$$

◆

Remark 2.5. If X is any discrete random variable, then the probability mass function f for X is also known as the *discrete density function* of X .

Theorem 2.8. Let X be a discrete random variable defined on a sample space S . Then the distribution function F of X can be obtained from the probability mass function f of X , and conversely.

Proof. Let X be a discrete random variable with mass points $x_1, x_2, \dots, x_n, \dots$

If f is the probability mass function of X , then

$$P(X = x_i) = f(x_i) \quad \text{for } i = 1, 2, \dots$$

Hence, the distribution function of X is obtained as

$$F(x) = P\{X \leq x\} = \sum_{x_i \leq x} f(x_i)$$

Conversely, if F is the distribution function of X , then the probability mass function f of X is obtained as

$$f(x) = \begin{cases} F(x_i) - F(x_i - 0) & \text{if } x = x_i \text{ for some } i \\ 0 & \text{if } x \neq x_i \text{ for any } i \end{cases}$$

where, by Theorem 2.5, it follows that

$$P(X = x) = F(x) - F(x - 0) \quad \text{for any } x \in \mathbb{R}$$

◆

EXAMPLE 2.16. If X is the discrete random variable representing the number of heads in two tosses of a single coin, find the probability mass function of X from the distribution function of X .

Solution. The mass points of X are $X = 0, 1, 2$. In Example 2.7, we obtained the distribution function of X as

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{4} & \text{if } 0 \leq x < 1 \\ \frac{3}{4} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

To describe the probability mass function f of X , it suffices to describe the values of f for the mass points $X = 0, 1, 2$. By Theorem 2.8, we have

$$f(x=0) = F(0) - F(0-0) = \frac{1}{4} - 0 = \frac{1}{4}$$

$$f(x=1) = F(1) - F(1-0) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$f(x=2) = F(2) - F(2-0) = 1 - \frac{3}{4} = \frac{1}{4}$$

◆

EXAMPLE 2.17. Let X be a discrete RV whose cumulative distribution function is

$$F(x) = \begin{cases} 0 & \text{if } x < -3 \\ \frac{1}{6} & \text{if } -3 \leq x < 6 \\ \frac{1}{2} & \text{if } 6 \leq x < 10 \\ 1 & \text{if } x \geq 10 \end{cases}$$

Find

- (i) $P(X \leq 4), P(-5 < X \leq 4), P(X = -3), P(X = 6), P(X = 10)$.
- (ii) The probability mass function of X .

(Anna, May 2000)

Solution. (i) $P(X \leq 4) = F(4) = \frac{1}{6}$. Note that

$$P(-5 < X \leq 4) = F(4) - F(-5) = \frac{1}{6} - 0 = \frac{1}{6}$$

(Recall the formula that $F(b) - F(a) = P(a < X \leq b)$.)

Using the formula that $P(X = x) = F(x) - F(x-0)$, we also have

$$P(X = -3) = F(-3) - F(-3-0) = \frac{1}{6} - 0 = \frac{1}{6}$$

$$P(X = 6) = F(6) - F(6-0) = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

and

$$P(X = 10) = F(10) - F(10-0) = 1 - \frac{1}{2} = \frac{1}{2}.$$

- (ii) The probability mass function of X is given by the following table:

x	-3	6	10
$f(x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$

◆

We state the following result without proof.

Theorem 2.9. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function with range $[0, 1]$ having the properties:

- (i) $f(x) = 0$ for all x except for a countable collection of points $x_1, x_2, \dots, x_n, \dots$ for which $f(x) > 0$
- (ii) $\sum_i f(x_i) = 1$

then f will be a probability mass function for some discrete RV X .

EXAMPLE 2.18. If a random variable X takes the values 1, 2, 3, 4 such that

$$2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$$

find the probability distribution of X .

(Anna, April 2003)

Solution. Let $P(X = 3) = k$. From the given data, it follows that

$$P(X = 1) = \frac{k}{2}, P(X = 2) = \frac{k}{3} \text{ and } P(X = 4) = \frac{k}{5}$$

Since the total probability is equal to 1, we must have

$$P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) = 1$$

or

$$\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1 \Rightarrow \frac{61k}{30} = 1 \Rightarrow k = \frac{30}{61}.$$

Hence, the probability distribution of X is obtained as:

x	1	2	3	4
$f(x)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{5}{61}$

◆

EXAMPLE 2.19. A random variable X has the following probability distribution

x :	0	1	2	3	4	5	6	7
$p(x)$:	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find

- (i) The value of k .
- (ii) $P(1.5 < X < 4.5 | X > 2)$.
- (iii) The smallest value of λ for which $P(X \leq \lambda) > \frac{1}{2}$.

Solution. (i) Since the total probability is equal to 1, we have

$$\sum_{x=0}^7 P(X = x) = 1$$

or

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1 \quad \text{or} \quad 10k^2 + 9k - 1 = 0$$

or

$$(10k - 1)(k + 1) = 0 \Rightarrow k = \frac{1}{10} \quad \text{or} \quad k = -1$$

The value $k = -1$ is inadmissible since probabilities cannot be negative. Hence,

$$k = \frac{1}{10}$$

(ii) By the definition of conditional probability, we have

$$P(1.5 < X < 4.5 | X > 2) = \frac{P[(1.5 < X < 4.5) \cap (X > 2)]}{P(X > 2)} = \frac{P(2 < X < 4.5)}{1 - P(X \leq 2)}$$

Hence, it follows that

$$P(1.5 < X < 4.5 | X > 2) = \frac{P(X = 3) + P(X = 4)}{1 - [P(X = 0) + P(X = 1) + P(X = 2)]} = \frac{5k}{1 - 3k} = \frac{\frac{5}{10}}{1 - \frac{3}{10}} = \frac{5}{7}$$

(iii) The distribution function of X is easily obtained as

$$F(x) = \begin{cases} 0 & \text{if } x < 1 \\ \frac{1}{10} & \text{if } 1 \leq x < 2 \\ \frac{3}{10} & \text{if } 2 \leq x < 3 \\ \frac{1}{2} & \text{if } 3 \leq x < 4 \\ \frac{4}{5} & \text{if } 4 \leq x < 5 \\ \frac{81}{100} & \text{if } 5 \leq x < 6 \\ \frac{83}{100} & \text{if } 6 \leq x < 7 \\ 1 & \text{if } x \geq 7 \end{cases}$$

Hence, it is clear that the smallest value of λ for which $P(X \leq \lambda) > \frac{1}{2}$ is given by $\lambda = 4$. ◆

EXAMPLE 2.20. If X is a discrete RV taking the values $1, 2, 3, \dots$ with the probability mass function f defined by

$$f(i) = P(X = i) = \frac{c^i}{i!} \quad \text{for } i = 1, 2, 3, \dots$$

find the value of c .

Solution. Since the total probability is equal to 1, we have

$$\sum_{i=1}^{\infty} P(X = i) = 1$$

i.e.

$$\sum_{i=1}^{\infty} \frac{c^i}{i!} = 1 \Rightarrow e^c - 1 = 1 \Rightarrow e^c = 2$$

Thus, $c = \log_e 2$.



EXAMPLE 2.21. Let X be a random variable such that

$$P(X = -2) = P(X = -1) = P(X = 1) = P(X = 2)$$

and

$$P(X < 0) = P(X = 0) = P(X > 0)$$

Determine the probability mass function and the distribution of X .

(Anna, Nov. 2006)

Solution. Let $P(X = 0) = \alpha$. Then, it follows that

$$P(X < 0) = P(X = 0) = P(X > 0) = \alpha$$

Since the total probability is equal to 1, we must have

$$P(X < 0) + P(X = 0) + P(X > 0) = 1 \Rightarrow 3\alpha = 1 \Rightarrow \alpha = \frac{1}{3}$$

Next, we note that

$$P(X < 0) = \frac{1}{3} = P(X = -2) + P(X = -1) = 2P(X = -2)$$

Hence, we have

$$P(X = -1) = \frac{1}{6} \text{ and } P(X = -2) = \frac{1}{6}$$

Similarly, we note that

$$P(X > 0) = \frac{1}{3} = P(X = 1) + P(X = 2) = 2P(X = 2)$$

Hence, we have

$$P(X = 1) = \frac{1}{6} \text{ and } P(X = 2) = \frac{1}{6}$$

Thus, the probability distribution of X is given by

x	-2	-1	0	1	2
$f(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{6}$

Also, the distribution function of X is given by

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < -2 \\ \frac{1}{6} & \text{if } -2 \leq x < -1 \\ \frac{1}{3} & \text{if } -1 \leq x < 0 \\ \frac{2}{3} & \text{if } 0 \leq x < 1 \\ \frac{5}{6} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$



PROBLEM SET 2.3

1. A coin is biased so that the head is twice as likely to appear as the tail. The coin is tossed twice. Find the probability distribution of the number of heads.
2. A coin is biased so that the head is twice as likely to appear as the tail. The coin is tossed 3 times. Find the probability distribution of the number of heads.
3. A box contains 3 white, 5 red and 2 green balls. Suppose that 2 balls are drawn successively from the box without replacement. If X represents the number of red balls, find the probability distribution of X .
4. A shipment of 10 cellphones contains 3 defective phones. A cellphone retail dealer makes a random purchase of 4 of the cellphones. If X is the random variable representing the number of defective cellphones purchased, find (a) the probability distribution of X , (b) $P(X < 2)$, and (c) $P(0 < X < 3)$.
5. Let X be a discrete RV whose cumulative distribution function is

$$F(x) = \begin{cases} 0 & \text{if } x < -2 \\ \frac{1}{8} & \text{if } -2 \leq x < 0 \\ \frac{1}{4} & \text{if } 0 \leq x < 2 \\ \frac{1}{2} & \text{if } 2 \leq x < 4 \\ 1 & \text{if } x \geq 4 \end{cases}$$

- (a) Find the probability mass function of X .
 - (b) Find $P(X \leq 1)$, $P(-1 < X \leq 1)$, $P(0 < X < 3)$, $P(1 < X \leq 4)$.
6. The probability mass function of a random variable X is given below:

x	0	1	2	3
$f(x)$	c	$2c^2$	c^2	$3c^2$

- (a) Determine the value of c .
- (b) Evaluate $P(X < 2)$, $P(X \geq 1)$, $P(1 < X \leq 3)$.

- (c) Find the distribution function of X .
 (d) Find the largest value of x such that $F(x) < \frac{1}{2}$.

7. X is a discrete RV with the probability mass function

$$f(x) = \frac{x}{10} \text{ for } x = 1, 2, 3, 4$$

- (a) Find $P(X = 1 \text{ or } 2)$.
 (b) Calculate $P\left[\frac{1}{2} < X < \frac{5}{2} | X > 1\right]$.

8. A random variable X has the following probability distribution:

x	0	1	2	3	4	5	6
$f(x)$	c	$2c$	$3c$	$4c$	$3c$	$2c$	c

- (a) Determine the value of c .
 (b) Find $P(X < 2), P(X \geq 2), P(0 < X < 4)$.
 (c) Find $P(0 < X < 4 | X \geq 2)$.
 (d) Find the smallest value of x for which $P(X \leq x) > \frac{1}{2}$.
 (e) Find the distribution function of X .
9. Show that $f(x) = \frac{2x}{n(n+1)}$ can serve as the probability mass function of a random variable X taking values $x = 0, 1, 2, \dots, n$.
10. Determine the values of α for which $f(x) = (1 - \alpha)\alpha^x$ can serve as the probability mass function of a random variable X taking values $x = 0, 1, 2, \dots$.

2.4 CONTINUOUS RANDOM VARIABLE

Definition 2.6. Let X be a random variable with distribution function (CDF) F . Then X is called a *continuous random variable* if F is a continuous function on $\mathbb{R}$.

Remark 2.6. By Theorem 2.6, a necessary and sufficient condition for the distribution function F to be continuous on $\mathbb{R}$ is that

$$P\{X = x\} = 0 \text{ for each } x \in \mathbb{R} \quad (2.3)$$

This shows that $P\{X = x\} = 0$ whenever X is a continuous random variable. Thus, for a continuous random variable, probability is not concentrated at any “mass points” as in the case of a discrete random variable. Another consequence of Eq. (2.3) is that for a continuous random variable X , the probabilities $P\{a < X \leq b\}, P\{a \leq X < b\}, P\{a \leq X \leq b\}, P\{a < X < b\}$ are all equal.

Definition 2.7. Let X be a continuous random variable with the distribution function F . We say that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is a *probability density function* (PDF) for F (or X) if $f(x) \geq 0$ for all $x \in \mathbb{R}$ such that

$$F(x) = \int_{-\infty}^x f(t) dt \quad (2.4)$$

Remark 2.7. We note the following:

1. F is continuously differentiable at x , then we can recover the probability density function at x . By the Fundamental Theorem of Calculus, it follows from Eq. (2.4) that

$$F'(x) = \frac{dF}{dx} = f(x)$$

2. If X is a continuous random variable with PDF f , then

$$P\{a < X < b\} = P\{a \leq X \leq b\} = \int_a^b f(t)dt$$

EXAMPLE 2.22. Consider the continuous distribution described in Example 2.8, viz.

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-x} & \text{if } x \geq 0 \end{cases}$$

By the Fundamental Theorem of Calculus, we derive the density function f of F as

$$f(x) = F'(x) = \begin{cases} 0 & \text{if } x < 0 \\ e^{-x} & \text{if } x \geq 0 \end{cases}$$

This density function is known as *exponential density function* and X is called an *exponential random variable*.

Next, we prove the following result.

Theorem 2.10. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is the probability density function of a continuous random variable if and only if it satisfies the following two conditions:

- (i) $f(x) \geq 0$ for all $x \in \mathbb{R}$
- (ii) $\int_{-\infty}^{\infty} f(t)dt = 1$

Proof. Necessity. Let f be a probability density function of a continuous random variable. By definition, $f(x) \geq 0$ for all $x \in \mathbb{R}$. It also follows that the associated distribution function F given by

$$F(x) = \int_{-\infty}^x f(t)dt$$

is continuous. Taking limits as $x \rightarrow \infty$, we have

$$F(\infty) = 1 = \int_{-\infty}^{\infty} f(t)dt$$

(by Theorem 2.2). Thus, Conditions (i) and (ii) are satisfied.

Sufficiency. Consider the distribution function F induced by the function f , namely,

$$F(x) = \int_{-\infty}^x f(t) dt$$

Clearly, F is continuous. We shall show that F is a distribution function for some random variable. Hence, it suffices to verify that F satisfies the three properties of Theorem 2.4, viz. (a) F is non-decreasing, (b) F is right-continuous, and (c) $F(-\infty) = 0$ and $F(+\infty) = 1$.

Since F is continuous, (b) is immediately satisfied, i.e. F is right-continuous. To prove that F is non-decreasing, let $x < y$. Then, we see that

$$F(y) - F(x) = \int_y^{\infty} f(t) dt - \int_x^{\infty} f(t) dt = \int_x^y f(t) dt \geq 0$$

since, by Condition (i), $f(t) \geq 0$ on $[a, b]$. Thus, $F(x) \leq F(y)$, i.e. F is non-decreasing.

It is clear from the definition that $F(-\infty) = 0$. By Condition (ii), we also see that

$$F(+\infty) = \int_{-\infty}^{\infty} f(t) dt = 1$$

◆

EXAMPLE 2.23. Given the PDF of a continuous random variable X as follows:

$$f(x) = \begin{cases} 12.5x - 1.25 & \text{for } 0.1 \leq x \leq 0.5 \\ 0 & \text{elsewhere} \end{cases}$$

Find $P(0.2 \leq X \leq 0.3)$

(Anna, Nov. 2007)

Solution. We find that

$$P(0.2 \leq X \leq 0.3) = \int_{x=0.2}^{0.3} f(x) dx = \int_{x=0.2}^{0.3} (12.5x - 1.25) dx = 1.25 \int_{x=0.2}^{0.3} (10x - 1) dx$$

Integrating, we get

$$P(0.2 \leq X \leq 0.3) = 1.25 [5x^2 - x]_{x=0.2}^{0.3} = 1.25[0.15 - 0] = 0.1875$$

◆

EXAMPLE 2.24. The diameter of an electric cable X is a continuous random variable with PDF

$$f(x) = \begin{cases} kx(1-x) & \text{if } 0 \leq x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find

(i) The value of k .

- (ii) The cumulative distribution function (CDF) of X .
 (iii) $P(X \leq \frac{1}{2} | \frac{1}{3} < X < \frac{2}{3})$.

(Anna, May 2006)

Solution. (i) We find the value of k by noting that

$$\int_{-\infty}^{\infty} f(t) dt = \int_0^1 kt(1-t) dt = k \int_0^1 (t-t^2) dt = 1$$

which implies that

$$k \left[\frac{t^2}{2} - \frac{t^3}{3} \right]_0^1 = 1$$

i.e.

$$k \left[\frac{1}{2} - \frac{1}{3} \right] = 1 \Rightarrow k \left[\frac{1}{6} \right] = 1$$

i.e.

$$k = 6$$

Thus, PDF of X is given by $f(x) = 6x(1-x)$ for $0 < x < 1$ and is zero elsewhere.

- (ii) Clearly, $F(x) = 0$ for $x < 0$. For $0 < x < 1$, we have

$$F(x) = \int_{-\infty}^x f(t) dt = \int_0^x 6t(1-t) dt = [3t^2 - 2t^3]_0^x = 3x^2 - 2x^3.$$

Also, for $x \geq 1$, $F(x) = 1$. Hence, we have

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 3x^2 - 2x^3 & \text{if } 0 < x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

- (iii) Define the events

$$A = \left\{ X \leq \frac{1}{2} \right\} \quad \text{and} \quad B = \left\{ \frac{1}{3} < X < \frac{2}{3} \right\}$$

The required probability is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Note that

$$P(B) = \int_{\frac{1}{3}}^{\frac{2}{3}} 6t(1-t) dt = [3t - 2t^2]_{\frac{1}{3}}^{\frac{2}{3}}$$

i.e.

$$P(B) = \left[\left(2 - \frac{8}{9} \right) - \left(1 - \frac{2}{9} \right) \right] = 1 - \frac{6}{9} = 1 - \frac{2}{3} = \frac{1}{3}$$

and

$$P(A \cap B) = P\left(\frac{1}{3} < X < \frac{1}{2}\right) = \int_{\frac{1}{3}}^{\frac{1}{2}} 6t(1-t) dt = [3t - 2t^2]_{\frac{1}{3}}^{\frac{1}{2}}$$

i.e.

$$P(A \cap B) = \left[\left(\frac{3}{2} - \frac{1}{2}\right) - \left(1 - \frac{2}{9}\right)\right] = \frac{2}{9}$$

Hence, the required probability is

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/9}{1/3} = \frac{2}{3}$$

◆

EXAMPLE 2.25. A continuous random variable X has a probability density function

$$f(x) = k(1+x), \quad 2 \leq x \leq 5$$

Find $P(X \leq 4)$.

(Anna, May 2007)

Solution. First, we find the value of the constant k . Since the total probability is unity, we have

$$\int_2^5 f(x) dx = 1 \Rightarrow \int_2^5 k(1+x) dx = 1$$

Integrating, we have

$$k \left[x + \frac{x^2}{2} \right]_2^5 = 1$$

Simplifying, we have

$$k \left[\left(5 + \frac{25}{2} \right) - (2 + 2) \right] = 1 \Rightarrow k \left[\frac{27}{2} \right] = 1 \Rightarrow k = \frac{2}{27}$$

Hence, the probability density function of X is given by

$$f(x) = \frac{2}{27} (1+x), \quad 2 \leq x \leq 5$$

The required probability is, thus, obtained as

$$P(X < 4) = \int_2^4 f(x) dx = \frac{2}{27} \int_2^4 (1+x) dx$$

Integrating, we have

$$P(X < 4) = \frac{2}{27} \left[x + \frac{x^2}{2} \right]_2^4 = \frac{2}{27} [(4+8) - (2+2)] = \frac{16}{27}$$

◆

EXAMPLE 2.26. A continuous random variable X has PDF $f(x) = 3x^2; 0 \leq x \leq 1$. Find a such that $P(X \leq a) = P(X > a)$.

(Anna, May 2006)

Solution. Note that $P(X > a) = 1 - P(X \leq a)$. Hence,

$$P(X \leq a) = P(X > a) \Rightarrow P(X \leq a) = 1 - P(X \leq a) \Rightarrow P(X \leq a) = \frac{1}{2}.$$

i.e.

$$\int_{-\infty}^a f(t) dt = \int_0^a 3t^2 dt = [t^3]_0^a = \frac{1}{2}$$

i.e.

$$a^3 = \frac{1}{2} \Rightarrow a = \left[\frac{1}{2} \right]^{\frac{1}{3}} = 0.7937$$

◆

EXAMPLE 2.27. A continuous random variable X has the PDF

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the cumulative distribution function (CDF) for X . Also find $P\left[\frac{1}{4} < X \leq \frac{3}{4}\right]$ and $P\left[X < \frac{1}{2}\right]$
(Anna, May 2007)

Solution. The given density function is known as a *triangular density function* (see Figure 2.4).

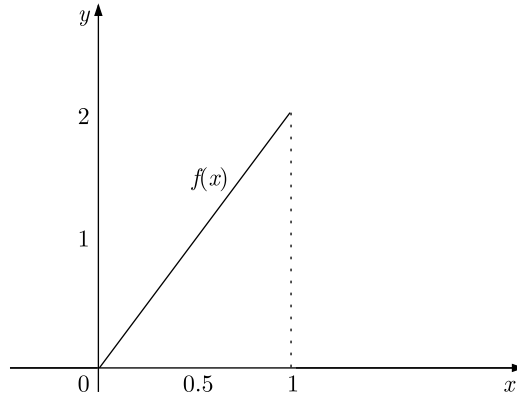


Figure 2.4: Triangular density function.

The CDF, denoted by F , for the random variable X , is easily obtained as:

$$F(x) = 0 \text{ if } x < 0$$

which follows because $f(x) = 0$ if $x < 0$.

Now, if $0 \leq x \leq 1$, then we have

$$F(x) = \int_{-\infty}^x f(t) dt = \int_0^x 2t dt = [t^2]_0^x = x^2$$

and if $x \geq 1$, then we have

$$F(x) = \int_{-\infty}^x f(t) dt = \int_0^1 2t dt = [t^2]_0^1 = 1$$

Thus, the cumulative distribution function F of the triangular density is obtained as

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x^2 & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

Next, note that

$$P\left[\frac{1}{4} < X \leq \frac{3}{4}\right] = F\left(\frac{3}{4}\right) - F\left(\frac{1}{4}\right) = \left[\frac{3}{4}\right]^2 - \left[\frac{1}{4}\right]^2 = \frac{1}{2}$$

We also find that

$$P\left(X < \frac{1}{2}\right) = F\left(\frac{1}{2}\right) = \left[\frac{1}{2}\right]^2 = \frac{1}{4}$$

◆

EXAMPLE 2.28. A random variable X has the density function

$$f(x) = K \frac{1}{1+x^2} \quad \text{where } -\infty < x < \infty$$

Determine K and the distribution function. Evaluate the probability $P(X \geq 0)$.

(Anna, Nov. 2003)

Solution. Since $f(x) \geq 0$ for all x , it is clear that $K \geq 0$. Also,

$$\int_{-\infty}^{\infty} f(t) dt = 1$$

i.e.

$$\int_{-\infty}^{\infty} K \frac{1}{1+t^2} dt = 1 \quad \text{or} \quad K [\tan^{-1} t]_{-\infty}^{\infty} = 1$$

i.e.

$$K \left[\frac{\pi}{2} - \frac{-\pi}{2} \right] = 1 \quad \text{or} \quad K[\pi] = 1 \quad \text{or} \quad K = \frac{1}{\pi}$$

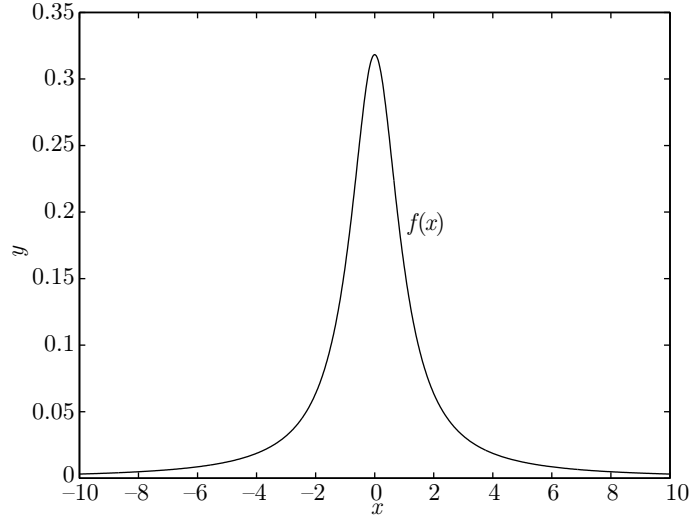


Figure 2.5: Standard Cauchy distribution.

Thus, $f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$ for $-\infty < x < \infty$.

This is known as the *standard Cauchy distribution* (see Figure 2.5).

Next, we find the distribution function of X . We see that

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt = \frac{1}{\pi} \int_{-\infty}^x \frac{1}{1+t^2} dt.$$

Integrating, we get

$$F(x) = \frac{1}{\pi} [\tan^{-1}(t)]_{-\infty}^x = \frac{1}{\pi} \left[x - \left(-\frac{\pi}{2} \right) \right] = \frac{1}{\pi} \left[x + \frac{\pi}{2} \right] \text{ for all } x$$

Next, we find that

$$P(X \geq 0) = 1 - P(X < 0) = 1 - F(0) = 1 - \frac{1}{\pi} \times \frac{\pi}{2} = 1 - \frac{1}{2} = \frac{1}{2}$$

◆

EXAMPLE 2.29. The cumulative distribution of X is

$$F(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{x^3+1}{9} & \text{if } -1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

Find $P(0 < X < 1)$.

(Madras, April 2000)

Solution. Clearly, $P(0 < X < 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}$.

◆

EXAMPLE 2.30. The CDF of a random variable X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x^2 & \text{if } 0 \leq x < \frac{1}{2} \\ 1 - \frac{3}{25}(3-x)^2 & \text{if } \frac{1}{2} \leq x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$$

Find the PDF of X and evaluate $P(|X| \leq 1)$ using both the PDF and the CDF.

(Anna, May 2007)

Solution. The PDF of f is given by

$$f(x) = F'(x) = \begin{cases} 0 & \text{if } x < 0 \text{ or } x \geq 3 \\ 2x & \text{if } 0 \leq x < \frac{1}{2} \\ \frac{6}{25}(3-x) & \text{if } \frac{1}{2} \leq x < 3 \end{cases}$$

Next, we find that

$$P(|X| \leq 1) = P(-1 \leq X \leq 1) = P(0 \leq X \leq 1) = \int_{x=0}^{\frac{1}{2}} f(x) dx + \int_{\frac{1}{2}}^1 f(x) dx$$

By the definition the PDF f , we have

$$P(|X| \leq 1) = \int_{x=0}^{\frac{1}{2}} 2x dx + \int_{\frac{1}{2}}^1 \frac{6}{25}(3-x) dx$$

Integrating, we have

$$P(|X| \leq 1) = [x^2]_{x=0}^{\frac{1}{2}} + \frac{6}{25} \left[3x - \frac{x^2}{2} \right]_{\frac{1}{2}}^1$$

Simplifying, we have

$$P(|X| \leq 1) = \frac{1}{4} + \frac{6}{25} \left[\left(3 - \frac{1}{2} \right) - \left(\frac{3}{2} - \frac{1}{8} \right) \right] = \frac{13}{25}$$

The same probability can also be evaluated using the CDF of X . Note that

$$P(|X| \leq 1) = P(0 < X < 1) = F(1) - F(0) = 1 - \left[\frac{3}{25} \times 4 \right] = 1 - \frac{12}{25} = \frac{13}{25}$$

◆

EXAMPLE 2.31. Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} |x| & \text{if } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is the PDF of a continuous random variable. Find the distribution function of X .

Solution. By definition, $f(x) \geq 0$ for all $x \in \mathbb{R}$. Thus, Condition (i) is satisfied. Next, note that

$$\int_{-\infty}^{\infty} f(t) dt = \int_{-1}^1 |t| dt = \int_0^1 2t dt = [t^2]_0^1 = 1$$

Thus, Condition (ii) also is satisfied.

Hence, f is the PDF of a continuous random variable X .

Let F be the distribution function of the random variable X . By the definition of f , it is immediate that $F(x) = 0$ for $x < -1$.

For $-1 \leq x \leq 0$, we have

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-1}^x (-t) dt = \left[-\frac{t^2}{2} \right]_{-1}^x = \frac{1-x^2}{2}$$

For $0 \leq x \leq 1$, we have

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt = F(0) + \int_0^x (t) dt$$

or

$$F(x) = \frac{1}{2} + \left[\frac{t^2}{2} \right]_0^x = \frac{1+x^2}{2}$$

Also, for $x > 1$, we have

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-1}^1 f(t) dt = 1$$

Thus, the distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{1-x^2}{2} & \text{if } -1 \leq x \leq 0 \\ \frac{1+x^2}{2} & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

The distribution function of X is plotted in Figure 2.6. ◆

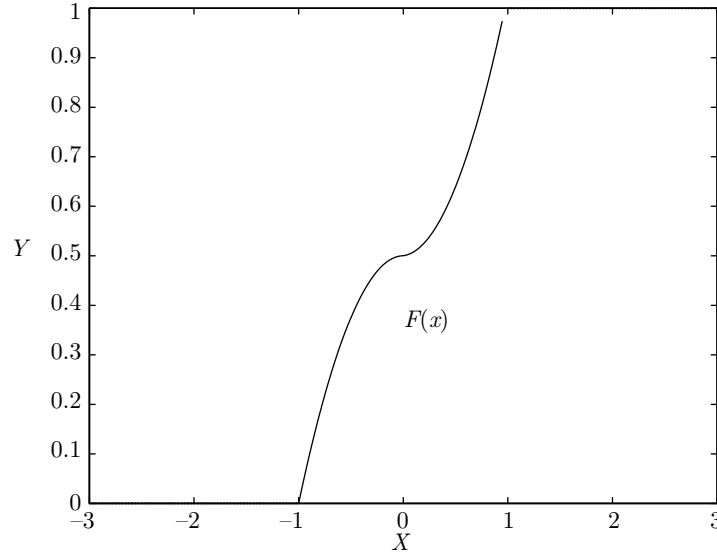


Figure 2.6: Distribution function.

EXAMPLE 2.32. If

$$p(x) = \begin{cases} xe^{-\frac{x^2}{2}} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

- (i) Show that $p(x)$ is a PDF of a continuous RV X .
(ii) Find its distribution function.

(Bharathiar, Nov. 1996)

Solution. (i) By definition, $p(x) \geq 0$ for all $x \in \mathbb{R}$. Thus, Condition (i) is satisfied. Also, note that

$$\int_{-\infty}^{\infty} p(t) dt = \int_0^{\infty} te^{-\frac{t^2}{2}} dt = \int_0^{\infty} e^{-u} du = [-e^{-u}]_0^{\infty} = 1$$

Thus, Condition (ii) also is satisfied. Hence, $p(x)$ is a PDF of a continuous random variable X .

- (ii) Let F be the distribution function of X . By the definition of f , it follows that $F(x) = 0$ for $x < 0$. Next, if $x \geq 0$, then

$$F(x) = \int_{-\infty}^x p(t) dt = \int_0^x te^{-\frac{t^2}{2}} dt = \int_0^{\frac{x^2}{2}} e^{-u} du = [e^{-u}]_0^{\frac{x^2}{2}} = 1 - e^{-\frac{x^2}{2}}$$

Hence, the distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-\frac{x^2}{2}} & \text{if } x \geq 0 \end{cases}$$

◆

EXAMPLE 2.33. A continuous random variable X has the PDF $f(x)$ given by

$$f(x) = Ce^{-|x|}, \quad -\infty < x < \infty$$

Find the value of C and the CDF of X .

(Anna, Nov. 2005)

Solution. Since f is the PDF of X , it follows that

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Thus, we have

$$\int_{-\infty}^{\infty} Ce^{-|x|} dx = 1 \Rightarrow 2 \int_0^{\infty} Ce^{-x} dx = 1$$

(since the integrand is an even function of x).

Integrating, we get

$$2C [-e^{-x}]_0^{\infty} = 1 \Rightarrow C = \frac{1}{2}$$

Hence, the PDF of X is given by

$$f(x) = \frac{1}{2}e^{-|x|} \quad \text{for } -\infty < x < \infty$$

This density function is known as *double-exponential* or *Laplace distribution* and is illustrated in Figure 2.7.

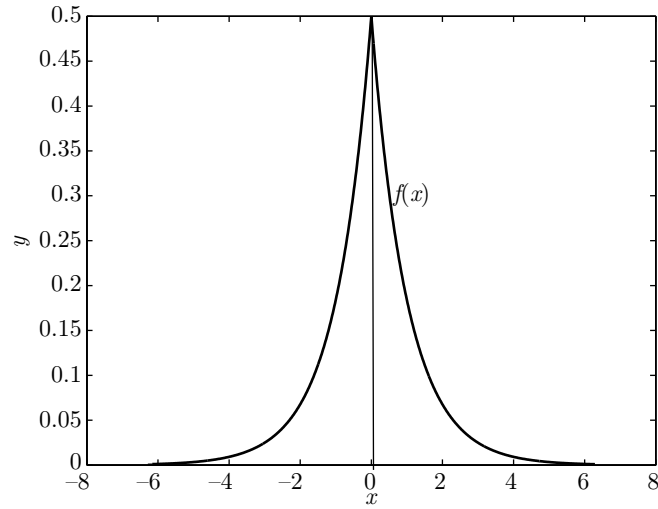


Figure 2.7: Laplace distribution.

Next, we calculate the CDF for Laplace distribution.

First, let $x \leq 0$. Then

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx = \int_{-\infty}^x \frac{1}{2} e^{-|x|} dx$$

Hence, we have

$$F(x) = \frac{1}{2} \int_{-\infty}^x e^x dx \quad (\text{note that } |x| = -x, \text{ when } x < 0)$$

Integrating we have

$$F(x) = \frac{1}{2} [e^x]_{-\infty}^x = \frac{1}{2} e^x$$

Next, let $x > 0$. Then

$$F(x) = P(X \leq x) = P(X \leq 0) + P(0 \leq X \leq x) = F(0) + P(0 \leq X \leq x) = \frac{1}{2} + P(0 \leq X \leq x)$$

Hence, it follows that

$$F(x) = \frac{1}{2} + \int_0^x \frac{1}{2} e^{-|x|} dx = \frac{1}{2} + \frac{1}{2} \int_0^x e^{-x} dx$$

Integrating, we have

$$F(x) = \frac{1}{2} + \frac{1}{2} [-e^{-x}]_0^x = \frac{1}{2} + \frac{1}{2} [1 - e^{-x}]$$

Simplifying, we have

$$F(x) = 1 - \frac{1}{2} e^{-x}$$

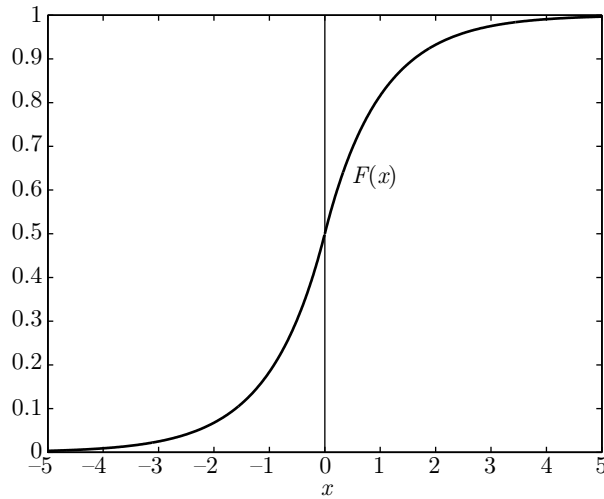


Figure 2.8: Distribution function for the Laplace distribution.

Combining the two cases, the distribution function of X (see Figure 2.8) is given by

$$F(x) = \begin{cases} \frac{1}{2}e^x & \text{if } x \leq 0 \\ 1 - \frac{1}{2}e^{-x} & \text{if } x > 0 \end{cases}$$



PROBLEM SET 2.4

1. A continuous random variable X has the PDF

$$f(x) = \begin{cases} kx(1+x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find

- (a) The value of k .
- (b) The cumulative distribution of X .
- (c) $P\left(X \leq \frac{1}{2} \mid \frac{1}{4} < X < \frac{3}{4}\right)$.

2. A continuous random variable X has the PDF

$$f(x) = \begin{cases} k(2-x) & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find

- (a) The value of k .
- (b) The cumulative distribution of X .
- (c) $P\left(\frac{1}{2} < X < \frac{3}{2} \mid X \geq 1\right)$.

3. A continuous random variable X has the PDF

$$f(x) = \begin{cases} 4x^3 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find a such that $P(X \leq a) = P(X > a)$.

4. (a) Verify whether the function

$$f(x) = \begin{cases} \frac{x}{4} & \text{for } 1 < x < 3 \\ 0 & \text{elsewhere} \end{cases}$$

is a probability density function.

- (b) Find the probability that a random variable X having this density will take values in the interval $1 < x < 2$.
- (c) Find $P\left(X \geq \frac{3}{2} \mid 1 < X < 2\right)$.

5. A continuous random variable X has the PDF

$$f(x) = \begin{cases} \frac{3x^2}{8} & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find

- (a) The CDF of X .
- (b) $P\left(\frac{1}{2} < X < \frac{3}{2}\right)$.
- (c) $P(X < 1)$.

6. A random variable X has the PDF $f(x)$ given by

$$f(x) = \begin{cases} Cx^2e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The value of C .
- (b) The CDF of X .

7. A continuous random variable X that can assume values between $X = 1$ and $X = 4$ has a density function given by

$$f(x) = \frac{2(1+x)}{21}.$$

- (a) The distribution function of X .
- (b) $P(2 < X < 3)$.

8. The cumulative distribution of X is

$$F(x) = \begin{cases} 0 & \text{if } x < -2 \\ \frac{x^3+8}{16} & \text{if } -2 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

Find

- (a) The PDF of X .
- (b) $P(0 < X < 1)$.

9. A continuous random variable X has the PDF

$$f(x) = \begin{cases} 6x(1-x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the distribution function of X .
- (b) Determine a number α such that $P(X < \alpha) = P(X > \alpha)$.

10. A continuous random variable X has the PDF

$$f(x) = \begin{cases} \frac{k}{\sqrt{x}} & \text{if } 0 < x < 9 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The value of k .
- (b) The distribution function of X .

- (c) $P(2 < X < 4)$.
 (d) $P(0 < X < 4|X > 2)$.

11. A continuous random variable X has the distribution function

$$F(x) = \begin{cases} 1 - \frac{4}{x^2} & \text{if } x > 2 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The PDF of X .
 (b) $P(3 < X < 5)$.
 (c) $P(X > 8)$.

12. A continuous random variable X has the distribution function

$$F(x) = \begin{cases} 0 & \text{if } x \leq -1 \\ \frac{x+1}{2} & \text{if } -1 < x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

Find

- (a) The PDF of X .
 (b) $P(-\frac{1}{2} < X < \frac{1}{2})$.
 (c) $P(X > \frac{2}{3})$.

2.5 MATHEMATICAL EXPECTATION

Mathematical expectation is a concept, which is commonly used to refer to the value expected as the outcome of any game or strategy. The expectation had its origin in gambling and the games of chance. In the first textbook on probability titled *Calculating in Games of Chance* (1657), Christian Huygens introduced the concept of expectation by analyzing a fair lottery.

Definition 2.8. If X is a discrete random variable with probability mass function $f(x)$ and with mass points $x_1, x_2, \dots, x_n, \dots$, then the *mathematical expectation* or *mean* of X , denoted by $E(X)$ or μ_X , is defined as

$$E(X) = \mu_X = \sum_i x_i f(x_i)$$

provided that the series converges absolutely, i.e.

$$\sum_i |x_i| f(x_i) < \infty$$

If X is a continuous random variable with PDF $f(x)$, then the *mathematical expectation* or *mean* of X , denoted by $E(X)$ or μ_X , is defined as

$$E(x) = \mu_X = \int_{-\infty}^{\infty} x f(x) dx$$

where the infinite integral converges absolutely, i.e.

$$\int_{-\infty}^{\infty} |x| f(x) dx < \infty$$

The following examples illustrate that expectation need not always exist for a random variable (discrete or continuous).

EXAMPLE 2.34. Let X be a discrete random variable with mass points

$$x_n = (-1)^n \frac{2^n}{n} \text{ for } n = 1, 2, 3, \dots$$

and the probability mass function

$$f(x_n) = \frac{1}{2^n} \text{ for } n = 1, 2, 3, \dots$$

Note that f is well-defined because $f(x_n) \geq 0$ for all n and also that

$$\sum_{n=1}^{\infty} f(x_n) = \frac{1}{2} + \left[\frac{1}{2}\right]^2 + \left[\frac{1}{2}\right]^3 + \dots = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1$$

We have

$$E(X) = \sum_{n=1}^{\infty} x_n f(x_n) = \sum_{n=1}^{\infty} (-1)^n \frac{2^n}{n} \times \frac{1}{2^n} = - \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = -\log_e 2$$

However,

$$\sum_{n=1}^{\infty} |x_n| f(x_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Thus, the series $\sum_{i=n}^{\infty} x_n f(x_n)$ does not converge absolutely. Hence, $E(X)$ does not exist for the given random variable X .

Remark 2.8. It is clear from the definition of expectation that $E(X)$ exists if and only if $E|X|$ exists.

EXAMPLE 2.35. Let X be a continuous random variable with the PDF

$$f(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2} \text{ for } -\infty < x < \infty$$

which is the density function of the *standard Cauchy distribution*.

We have,

$$E(X) = \int_{-\infty}^{\infty} x \cdot \frac{1}{\pi} \cdot \frac{1}{1+x^2} dx = 0$$

since the integrand is an odd function of x .

We find that

$$\int_{-\infty}^{\infty} |x| f(x) dx = \int_{-\infty}^{\infty} |x| \cdot \frac{1}{\pi} \cdot \frac{1}{1+x^2} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx$$

since the integrand is an even function of x . Integrating, we get

$$\int_{-\infty}^{\infty} |x| f(x) dx = \frac{1}{\pi} [\log_e(1+x^2)]_0^{\infty} = \infty$$

Since the integral $\int_{-\infty}^{\infty} x f(x) dx$ does not converge absolutely, $E(X)$ does not exist for the Cauchy random variable X .

EXAMPLE 2.36. A coin is biased so that the head is twice as likely to appear as the tail. The coin is tossed twice. Find the expected value of the number of heads.

Solution. Let p and q denote the probabilities of getting the head and tail respectively, where $p+q=1$. Given that $p=2q$. Hence, it follows that

$$p = \frac{2}{3} \text{ and } q = \frac{1}{3}$$

Let X denote the number of heads in the two tosses. Then X takes the values 0, 1 and 2. Note that

$$\{X=0\} = \{TT\}, \{X=1\} = \{HT, TH\} \text{ and } \{X=2\} = \{HH\}.$$

and

$$P(X=0) = q^2 = \frac{1}{9}, P(X=1) = 2pq = \frac{4}{9} \text{ and } P(X=2) = p^2 = \frac{4}{9}$$

Thus, the probability distribution of X is given by

x	0	1	2
$f(x)$	$\frac{1}{9}$	$\frac{4}{9}$	$\frac{4}{9}$

Hence, the expected value of X is

$$E(X) = \left(0 \times \frac{1}{9}\right) + \left(1 \times \frac{4}{9}\right) + \left(2 \times \frac{4}{9}\right) = \frac{12}{9} = \frac{4}{3} = 1\frac{1}{3}$$

Note that the expected value $1\frac{1}{3}$ is not one of the actual values taken by X . The result $E(X) = 1\frac{1}{3}$ indicates the *average* number of heads that we are expected to get if the experiment is performed repeatedly. ◆

EXAMPLE 2.37. Let A be any event in the sample space and $X = I_A$ be the *indicator variable* of A , i.e.

$$X(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

Note that $E(X)$ exists, and is equal to

$$E(X) = (1 \times P(A)) + (0 \times P(\bar{A})) = P(A)$$

EXAMPLE 2.38. A gambler plays a game of rolling a die with the following rules. He will win Rs. 200 if he throws a 6, but will lose Rs. 40 if throws 4 or 5 and lose Rs. 20 if throws 1, 2 or 3. Find the expected value that the gambler may gain.

Solution. Let X denote the value that the gambler may win or lose. Then X takes the values -40 , -20 and 200 .

Note that

$$\{X = -40\} = \{4, 5\}, \{X = -20\} = \{1, 2, 3\} \text{ and } \{X = 200\} = \{6\}$$

Hence, the probability distribution of X is given by

x	-40	-20	200
$f(x)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

By definition, the expected value of X is

$$E(X) = \left(-40 \times \frac{1}{3}\right) + \left(-20 \times \frac{1}{2}\right) + \left(200 \times \frac{1}{6}\right) = 10$$

Thus, the expected value that the gambler may gain is Rs. 10. ♦

We next state an important property of expectation of a random variable without proof.

Theorem 2.11. If X is a discrete random variable with mass points $x_1, x_2, \dots$ and with probability mass function f and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function, then

$$E[g(X)] = \sum_{i=1}^{\infty} g(x_i) f(x_i) = \sum_{i=1}^{\infty} g(x_i) P(X = x_i)$$

where it is assumed that the above series converges absolutely.

If X is a continuous random variable with PDF f and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function, then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

where it is assumed that the above integral converges absolutely.

Theorem 2.12. Let X be any random variable. Then the following properties hold:

- (i) $E(c) = c$, where c is any constant.
- (ii) $E(\alpha X) = \alpha E(X)$, where α is any constant.
- (iii) $E(aX + b) = aE(X) + b$, where a and b are any constants.

Proof. We prove this theorem only for the continuous case. The corresponding proof for the discrete case is left as an exercise for you.

(i) By definition, we have

$$E(c) = \int_{-\infty}^{\infty} c f(x) dx = c \int_{-\infty}^{\infty} f(x) dx = c[1] = c$$

(ii) By definition, we have

$$E(\alpha X) = \int_{-\infty}^{\infty} \alpha x f(x) dx = \alpha \int_{-\infty}^{\infty} x f(x) dx = \alpha E(X)$$

(iii) By definition, we have

$$E(aX + b) = \int_{-\infty}^{\infty} (ax + b) f(x) dx = a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx = aE(X) + b.$$

◆

Corollary 2.1. If X is a random variable and a is any constant, then

$$E(aX) = aE(X).$$

Proof. This follows by setting $b = 0$ in Property (iii).

◆

Theorem 2.13. If $X \geq 0$, then $E(X) \geq 0$.

Proof. We prove this result only for the continuous case, and the corresponding proof for the discrete case is left as an exercise for you.

By definition, we have

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx \geq 0$$

since the integrand is non-negative for all values of x in $(-\infty, \infty)$.

◆

EXAMPLE 2.39. Let X be a continuous random variable with set of possible values $\{x : 0 < x < \alpha\}$, (where $\alpha < \infty$) distribution function F and density function f . Prove the following:

$$E(X) = \int_0^{\alpha} [1 - F(t)] dt$$

(Anna, Nov. 2004)

Solution. By definition, we have

$$E(X) = \int_{t=0}^{\alpha} t f(t) dt = \int_{t=0}^{\alpha} t d(F(t))$$

Using integration by parts, we have

$$E(X) = [tF(t)]_0^\alpha - \int_{t=0}^\alpha F(t) dt$$

Since $F(0) = 0$ and $F(\alpha) = 1$, we have

$$E(X) = \alpha - \int_{t=0}^\alpha F(t) dt = \int_{t=0}^\alpha [1 - F(t)] dt$$

where we have used the fact that f is a PDF on $[0, \alpha]$, i.e.

$$\int_{t=0}^\alpha f(t) dt = 1$$

◆

Next, we define the *variance* and *standard deviation* of a random variable X . The motivation is as follows: We expect the random variable X to take on values about its mean $\mu = E(X)$. Thus, a reasonable way of measuring the possible variation of X is to see how far apart X lies from its mean value μ . A mathematically convenient way of analyzing this is to consider the quantity $E[(X - \mu)^2]$, called the *variance* of X . Formally, we have the following definition.

Definition 2.9. Let X be a random variable with the expected value (or mean) μ . Then the *variance* of X , denoted by $\text{Var}(X)$, is defined as

$$\text{Var}(X) = E[(X - \mu)^2]$$

and the *standard deviation* of X , denoted by σ_X or σ , is defined as

$$\sigma_X = +\sqrt{\text{Var}(X)}.$$

Remark 2.9. Let X be a random variable with probability mass function $f(x)$ (in discrete case) or with probability density function $f(x)$ (in continuous case). Both the variance and standard deviation are measures of the spread of the distribution of X about its mean. Physically, the mean can be interpreted as the *centre of gravity* of the density function $f(x)$ and the variance can be interpreted as the *moment of inertia* of $f(x)$ with respect to a perpendicular axis through the centre of gravity.

Theorem 2.14. Let X be a random variable with mean μ . Then the following properties hold:

- (i) $\text{Var}(X) = E(X^2) - \mu^2$.
- (ii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$, where a, b are any constants.
- (iii) $\text{Var}(b) = 0$, where b is any constant.
- (iv) $\text{Var}(X) = 0 \Rightarrow X = \mu$ almost everywhere.

Proof. (i) By definition, $\text{Var}(X) = E[X - \mu]^2$. Thus, we have

$$\text{Var}(X) = E[X^2 - 2\mu X + \mu^2] = E(X^2) - 2\mu E(X) + \mu^2 E(1)$$

Noting that $\mu = E(X)$ and that $E(c) = c$ for any constant c , we have

$$\text{Var}(X) = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2$$

(ii) Let $Y = aX + b$, where a and b are any constants. Then we have

$$\mu_Y = E(Y) = aE(X) + b = a\mu + b$$

By definition, we have

$$\text{Var}(Y) = E[Y - \mu_Y]^2 = E[aX - a\mu]^2 = E[a^2(X - \mu)^2] = a^2 \text{Var}(X)$$

(iii) Taking $a = 0$ in (ii), we have $\text{Var}(b) = 0$ for any constant b .

(iv) By definition, $\text{Var}(X) = 0$ implies that $E[X - \mu]^2 = 0$ or

$$\int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = 0$$

Note that the integrand $(x - \mu)^2 f(x)$ is non-negative for the whole range over $\mathbb{R}$, and also the probability density function $f(x) > 0$ in the range of X . Hence, we must have

$$X = \mu \text{ almost everywhere in } \mathbb{R}$$

◆

EXAMPLE 2.40. If $\text{Var}(X) = 4$, find $\text{Var}(3X + 8)$, where X is a random variable.

(Anna, Model, 2003)

Solution. We know that $\text{Var}(3X + 8) = 3^2 \text{Var}(X) = 9 \times 4 = 36$.

◆

Theorem 2.15. If X is any random variable, then $E[X - c]^2$, c being any constant, takes a minimum value when $c = E(X)$.

Proof. Let $\mu = E(X)$, the mean of X . We have

$$\begin{aligned} E[X - c]^2 &= E[(X - \mu) + (\mu - c)]^2 = E[(X - \mu)^2 + 2(\mu - c)(X - \mu) + (\mu - c)^2] \\ &= E[X - \mu]^2 + 2(\mu - c)E[X - \mu] + E[\mu - c]^2 \\ &= E[X - \mu]^2 + (\mu - c)^2 \text{ (since } E[X - \mu] = E(X) - \mu = 0) \end{aligned}$$

Hence, it is immediate that $E[X - c]^2 \geq E[X - \mu]^2$ for any constant c and that equality holds if and only if $c = \mu = E(X)$.

◆

EXAMPLE 2.41. Let X be a random variable with $E(X) = 1$ and $E[X(X - 1)] = 4$. Find $\text{Var}(X)$ and $\text{Var}(2 - 3X)$

(Anna, Model, 2003)

Solution. Note that

$$E[X(X-1)] = 4 \Rightarrow E[X^2 - X] = 4 \Rightarrow E(X^2) - E(X) = 4$$

Therefore,

$$E(X^2) = E(X) + 4 = 1 + 4 = 5$$

$$\text{Hence, } \text{Var}(X) = E(X^2) - [E(X)]^2 = 5 - 1 = 4.$$

$$\text{Also, } \text{Var}(2 - 3X) = (-3)^2 \text{Var}(X) = 9 \text{Var}(X) = 36. \quad \blacklozenge$$

Definition 2.10. Let X be any random variable with mean μ and standard deviation σ . We define the *standard random variable* associated with X by

$$Z = \frac{X - \mu}{\sigma}$$

EXAMPLE 2.42. If X is any random variable, show that the standard random variable has mean 0 and variance 1.

Solution. Let X be any random variable with mean μ and standard deviation σ . The standard random variable associated with X is defined by $Z = \frac{X - \mu}{\sigma}$. Note that

$$\mu_Z = E(Z) = \frac{1}{\sigma} E[X - \mu] = \frac{1}{\sigma} [E(X) - \mu] = 0 \quad (\text{since } \mu = E(X))$$

Next, we note that

$$\text{Var}(Z) = E[Z - \mu_Z]^2 = E[Z^2] = \frac{1}{\sigma^2} E[X - \mu]^2 = \frac{1}{\sigma^2} \times \sigma^2 = 1$$

Hence, Z has mean 0 and variance 1. $\blacklozenge$

At the beginning of this section, we showed examples of random variables (standard Cauchy variate, for example) for which even the mean does not exist. Next, we introduce some *order parameters* like *percentiles*, *quartiles*, and *median* which always exist.

Definition 2.11. Let X be a random variable (discrete or continuous). Consider any integer p , where $p = 1, 2, \dots, 100$. Then the *p th percentile* of X is defined as the value(s) x such that

$$P\{X \leq x\} \geq \frac{p}{100} \quad \text{and} \quad P\{X \geq x\} \geq 1 - \frac{p}{100}$$

(See Figure 2.9.)

The *quartiles* of the random variable X are defined as Q_1 , Q_2 , Q_3 and Q_4 , where Q_1 is the 25th, Q_2 is the 50th, Q_3 is the 75th and Q_4 is the 100th percentiles.

The second quartile, Q_2 , is also called the *median* of the random variable X and is denoted by M . Thus, the median of X is the value(s) M such that

$$P\{X \leq M\} \geq \frac{1}{2} \quad \text{and} \quad P\{X \geq M\} \geq \frac{1}{2}$$

(See Figure 2.10.)

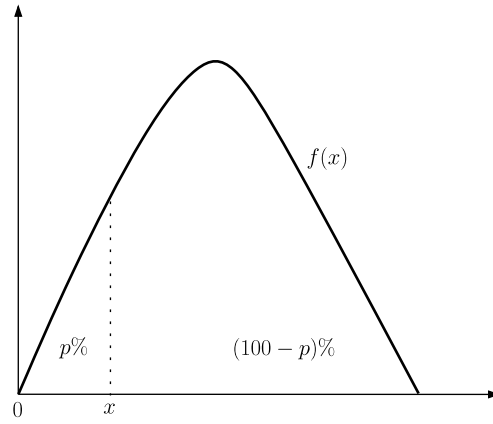


Figure 2.9: p th Percentile of a Random Variable.

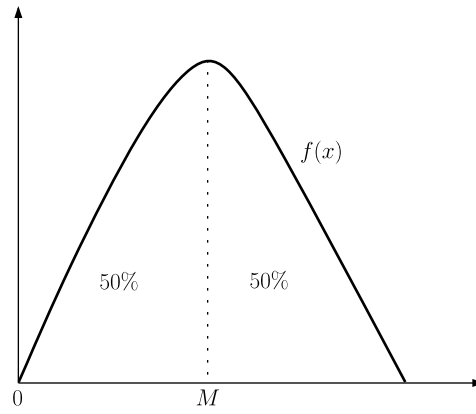


Figure 2.10: Median of a random variable.

Remark 2.10. If X is a continuous random variable with PDF $f(x)$, then the definitions for the various order parameters given in Definition 2.11 take the following simple form:

The p th percentile of X is the value x satisfying

$$\int_{-\infty}^x f(t)dt = \frac{p}{100}$$

The first and third quartiles of X , viz. Q_1 and Q_3 , are defined by the relations

$$\int_{-\infty}^{Q_1} f(t)dt = \frac{1}{4} \quad \text{and} \quad \int_{-\infty}^{Q_3} f(t)dt = \frac{3}{4}$$

Finally, the second quartile of X , also known as the median of X , is defined by the relation

$$\int_{-\infty}^M f(t)dt = \frac{1}{2}$$

Next, we define the *mode* of a random variable.

Definition 2.12. If X is a discrete random variable, then the *mode* of X is defined as the value(s) of X at which the probability mass function $f(x)$ takes its maximum value.

If X is a continuous random variable, then the *mode* of X is defined as the value of X at which the PDF $f(x)$ takes its maximum value.

Next, we establish an important property for the median of a continuous random variable.

Theorem 2.16. If X is a continuous random variable with PDF f , then the mean deviation of X about any point a takes the minimum value when $a = M$, where M is the median of X , i.e.

$$E |X - a| \geq E |X - M|$$

where a is any real number.

Proof. Let $K(a) \triangleq E |X - a|$. Then $K(a)$ is the mean deviation of X about any arbitrary real number a .

We claim that $K(a)$ takes its minimum value at $a = M$. The proof is via Calculus.

By definition, we have

$$K(a) = E |X - a| = \int_{-\infty}^{\infty} |x - a| f(x) dx = \int_{-\infty}^a |x - a| f(x) dx + \int_a^{\infty} |x - a| f(x) dx$$

Thus, we have

$$K(a) = \int_{-\infty}^a (a - x) f(x) dx + \int_a^{\infty} (x - a) f(x) dx$$

Differentiating K under the integration sign with respect to a , we get

$$K'(a) = \int_{-\infty}^a f(x) dx - \int_a^{\infty} f(x) dx \quad (2.5)$$

Differentiating $K'(a)$ under the integration sign with respect to a , we get

$$K''(a) = f(a) + f(a) = 2f(a) > 0$$

Since $K''(a) > 0$ for all values of a , we see that K has a global minimum at its turning point which is given by the root of the equation $K'(a) = 0$. From Eq. (2.5), it follows that the turning point c , of $K(a)$, is given by

$$\int_{-\infty}^c f(x) dx - \int_c^{\infty} f(x) dx = 0$$

or

$$\int_{-\infty}^c f(x) dx = \int_c^{\infty} f(x) dx \quad (2.6)$$

Since the total probability is unity, we also have

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx = 1 \quad (2.7)$$

Combining Eqs. (2.6) and (2.7), it is immediate that

$$\int_{-\infty}^c f(x) dx = \int_c^{\infty} f(x) dx = \frac{1}{2}$$

which implies that $c = M = \text{Median}(X)$. Hence, $K(a)$ takes its global minimum at the median M , of X . $\blacklozenge$

EXAMPLE 2.43. When a die is thrown, X denotes the number that turns up. Find $E(X)$, $E(X^2)$ and $\text{Var}(X)$.

(Anna, May 2006)

Solution. The probability distribution of X is given by

x	1	2	3	4	5	6
$f(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

We find that the mean of X is

$$\mu = E(X) = \sum x f(x) = \frac{1}{6} [1 + 2 + \cdots + 6] = \frac{1}{6} \times \frac{6 \times 7}{2} = \frac{7}{2}$$

$$E(X^2) = \sum x^2 f(x) = \frac{1}{6} [1^2 + 2^2 + \cdots + 6^2] = \frac{1}{6} \times \frac{6 \times 7 \times 13}{6} = \frac{91}{6}$$

Thus,

$$\text{Var}(X) = E(X^2) - \mu^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

$\blacklozenge$

EXAMPLE 2.44. A pair of dice is rolled. If X is the sum of the numbers, find the probability distribution of X . Find also the mean and variance.

(Madras, April 1998)

Solution. The probability distribution of X is given in the following table:

x	2	3	4	5	6	7	8	9	10	11	12
$f(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

A simple calculation gives

$$\mu = E(X) = \sum x P(X = x) = 7 \quad \text{and} \quad E(X^2) = \sum x^2 P(X = x) = \frac{329}{6}$$

Hence, the variance of X is given by

$$\text{Var}(X) = E(X^2) - \mu^2 = \frac{329}{6} - 49 = \frac{35}{6} = 5.8333$$

$\blacklozenge$

EXAMPLE 2.45. A random variable X has the following probability distribution:

$x:$	-2	-1	0	1	2	3
$f(x):$	0.1	k	0.2	$2k$	0.3	$3k$

Find

- The value of k .
- Evaluate $P(X < 2)$ and $P(-2 < X < 2)$.
- Find the cumulative distribution of X .
- Evaluate the mean of X .

(Madras, April 1997; Anna, Nov. 2007)

Solution. (i) Since the total probability is equal to 1, we must have

$$0.1 + k + 0.2 + 2k + 0.3 + 3k = 1 \Rightarrow 6k = 1 - 0.6 = 0.4 \Rightarrow k = \frac{0.4}{6} = \frac{1}{15}$$

Hence, the probability function is given by

$x:$	-2	-1	0	1	2	3
$f(x):$	$\frac{1}{10}$	$\frac{1}{15}$	$\frac{1}{5}$	$\frac{2}{15}$	$\frac{3}{10}$	$\frac{1}{5}$

(ii) We find that

$$P(X < 2) = 1 - P(X \geq 2) = 1 - \left(\frac{3}{10} + \frac{1}{5} \right) = 1 - \frac{5}{10} = \frac{1}{2}$$

and

$$P(-2 < X < 2) = f(-1) + f(0) + f(1) = \frac{1}{15} + \frac{1}{5} + \frac{2}{15} = \frac{2}{5}$$

(iii) The cumulative distribution of X is given by

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < -2 \\ \frac{1}{10} & \text{if } -2 \leq x < -1 \\ \frac{1}{6} & \text{if } -1 \leq x < 0 \\ \frac{11}{30} & \text{if } 0 \leq x < 1 \\ \frac{1}{2} & \text{if } 1 \leq x < 2 \\ \frac{4}{5} & \text{if } 2 \leq x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$$

(iv) The mean of X is given by

$$\mu = \sum_x xf(x) = -\frac{1}{5} - \frac{1}{15} + 0 + \frac{2}{15} + \frac{3}{5} + \frac{3}{5} = \frac{16}{15} = 1.0667$$



EXAMPLE 2.46. A man draws 3 balls from an urn containing 5 white and 7 black balls. He gets Rs. 10 for each white ball and Rs. 5 for each black ball. Find his expectation. (Anna, April 2003)

Solution. Let X denote the amount the man gets out of his drawing. Then X takes four values according to the following four possibilities:

(i) All balls are black $\Rightarrow X = 15$. Thus,

$$P(X = 15) = \frac{{}^7C_3}{{}^{12}C_3} = \frac{7}{44}$$

(ii) 1 white and 2 black balls $\Rightarrow X = 20$. Thus,

$$P(X = 20) = \frac{{}^5C_1 \times {}^7C_2}{{}^{12}C_3} = \frac{21}{44}$$

(iii) 2 white and 1 black balls $\Rightarrow X = 25$. Thus,

$$P(X = 25) = \frac{{}^5C_2 \times {}^7C_1}{{}^{12}C_3} = \frac{7}{22}$$

(iv) All balls are white $\Rightarrow X = 30$. Thus,

$$P(X = 30) = \frac{{}^5C_3}{{}^{12}C_3} = \frac{1}{22}$$

Hence, the probability distribution of X is given by

x	15	20	25	30
$f(x)$	$\frac{7}{44}$	$\frac{21}{44}$	$\frac{7}{22}$	$\frac{1}{22}$

Thus, the expectation of X is

$$E(X) = \left(15 \times \frac{7}{44}\right) + \left(20 \times \frac{21}{44}\right) + \left(25 \times \frac{7}{22}\right) + \left(30 \times \frac{1}{22}\right) = \frac{85}{4}$$

Hence, the expected value of X is Rs. 21.25. ◆

EXAMPLE 2.47. Let X be a random variable taking values $-1, 0$ and 1 such that

$$P(X = -1) = 2P(X = 0) = P(X = 1)$$

Find the mean of $2X - 5$.

(Anna, Model, 2003)

Solution. If $P(X = 1) = k$, then $P(X = -1) = k$ and $P(X = 0) = \frac{k}{2}$. Since the total probability is equal to 1, we must have

$$P(X = -1) + P(X = 0) + P(X = 1) = 1 \text{ or } k + \frac{k}{2} + k = 1 \text{ or } \frac{5k}{2} = 1$$

Thus, $k = \frac{2}{5}$. Next, if $Y = 2X - 5$, then Y takes on the values $-7, -5, -3$. Therefore, the probability distribution of Y is given by

y	-7	-5	-3
f(y)	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{2}{5}$

Hence, the mean of Y is

$$E(Y) = \left(-7 \times \frac{2}{5}\right) + \left(-5 \times \frac{1}{5}\right) + \left(-3 \times \frac{2}{5}\right) = -\frac{25}{5} = -5$$

◆

EXAMPLE 2.48. The monthly demand for Allwyn watches is known to have the following probability distribution:

Demand	1	2	3	4	5	6	7	8
Probability	0.08	0.12	0.19	0.24	0.16	0.10	0.07	0.04

Determine the expected demand for watches. Also compute the variance.

(Anna, Nov. 2006)

Solution. Let X be the demand for watches. The expectation of X is

$$\mu = \sum_{i=1}^8 x_i f(x_i) = 0.08 + 0.24 + 0.57 + 0.96 + 0.8 + 0.6 + 0.49 + 0.32 = 4.06$$

Hence, the expected demand for watches is 4.06.

Next, we compute the variance. We use the formula

$$\text{Var}(X) = E[X^2] - \mu^2$$

We find that

$$E[X^2] = \sum_{i=1}^8 x_i^2 f(x_i) = 0.08 + 0.48 + 1.71 + 3.84 + 4 + 3.6 + 3.43 + 2.56 = 19.7$$

Hence, we have

$$\text{Var}(X) = E[X^2] - \mu^2 = 19.7 - 16.4836 = 3.2164$$

◆

EXAMPLE 2.49. The probability function of an infinite discrete distribution is given by

$$P(X = j) = \frac{1}{2^j}, \quad j = 1, 2, \dots, \infty$$

Find (i) the mean and variance of the distribution, (ii) $P(X \text{ is even})$, (iii) $P(X \geq 5)$ and (iv) $P(X \text{ is divisible by } 3)$.

(Anna, May 2006)

Solution. (i) First, we find the mean of X . By definition, we have

$$\mu = E(X) = \sum_{j=1}^{\infty} jP(X=j) = \sum_{j=1}^{\infty} \frac{j}{2^j} = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \cdots$$

i.e.

$$\mu = \frac{1}{2} [1 + 2\theta + 3\theta^2 + 4\theta^3 + \cdots], \text{ where } \theta = \frac{1}{2}$$

Hence,

$$\mu = \frac{1}{2} \cdot (1 - \theta)^{-2} = \frac{1}{2} \left[\frac{1}{2} \right]^{-2} = 2$$

Next, we find the variance of X by the formula $\text{Var}(X) = E[X^2] - \mu^2$

Note that

$$E[X^2] = \sum_{j=1}^{\infty} j^2 P(X=j) = \sum_{j=1}^{\infty} \frac{j^2}{2^j}$$

Since we can write $j^2 = j(j+1) - j$, we have

$$E[X^2] = \sum_{j=1}^{\infty} \frac{j(j+1)}{2^j} - \sum_{j=1}^{\infty} \frac{j}{2^j} = \sum_{j=1}^{\infty} \frac{j(j+1)}{2^j} - 2 \quad (2.8)$$

Now, note that

$$\sum_{j=1}^{\infty} \frac{j(j+1)}{2^j} = 1 + \left(\frac{1}{2} \times 3 \right) + \left(\frac{1}{2^2} \times 6 \right) + \left(\frac{1}{2^3} \times 10 \right) + \cdots = \left[1 - \frac{1}{2} \right]^{-3}$$

i.e.

$$\sum_{j=1}^{\infty} \frac{j(j+1)}{2^j} = \left[\frac{1}{2} \right]^{-3} = 8 \quad (2.9)$$

Substituting Eq. (2.9) into Eq. (2.8), we have

$$E[X^2] = 8 - 2 = 6$$

Hence, it follows that $\text{Var}(X) = E[X^2] - \mu^2 = 6 - 4 = 2$.

(ii) Next, we find that

$$P(X \text{ is even}) = P(X=2) + P(X=4) + P(X=6) + \cdots = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \cdots$$

i.e.

$$P(X \text{ is even}) = \frac{\frac{1}{2^2}}{1 - \frac{1}{2^2}} = \frac{1}{3}$$

(iii) Also, we see that

$$P(X \geq 5) = P(X = 5) + P(X = 6) + P(X = 7) + \cdots = \frac{1}{2^5} + \frac{1}{2^6} + \frac{1}{2^7} + \cdots$$

i.e.

$$P(X \geq 5) = \frac{\frac{1}{2^5}}{1 - \frac{1}{2}} = \frac{1}{16}$$

(iv) Finally, we see that

$$P(X \text{ is divisible by } 3) = P(X = 3) + P(X = 6) + P(X = 9) + \cdots = \frac{1}{2^3} + \frac{1}{2^6} + \frac{1}{2^9} + \cdots$$

i.e.

$$P(X \text{ is divisible by } 3) = \frac{\frac{1}{2^3}}{1 - \frac{1}{2^3}} = \frac{1}{7}$$

◆

EXAMPLE 2.50. A bag contains a white and b black balls. c balls are drawn. Find the expected value of the number of white balls drawn.

(Madras, Oct. 2001)

Solution. We define the random variable X_i as

$$X_i = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ ball drawn is white} \\ 0 & \text{if } i^{\text{th}} \text{ ball drawn is black} \end{cases}$$

for $i = 1, 2, 3, \dots, c$. The probability distribution of X_i is obtained as

x	0	1
$f(x)$	$\frac{b}{a+b}$	$\frac{a}{a+b}$

Thus, it follows that

$$E(X_i) = \left(0 \times \frac{b}{a+b}\right) + \left(1 \times \frac{a}{a+b}\right) = \frac{a}{a+b}$$

If X is the number of white balls among the c balls drawn from the bag, then we have

$$X = X_1 + X_2 + \cdots + X_c$$

Hence, it follows that

$$\begin{aligned} E(X) &= E(X_1 + X_2 + \cdots + X_c) = E(X_1) + E(X_2) + \cdots + E(X_c) \\ &= \frac{a}{a+b} + \frac{a}{a+b} + \cdots (c \text{ times}) = \frac{ac}{a+b} \end{aligned}$$

Hence, the required expectation is $\frac{ac}{a+b}$.

◆

EXAMPLE 2.51. A coin is tossed until a head appears. What is the expected number of tosses?

(Madras, Oct. 2000)

Solution. Let p and q denote the probabilities of getting head and tail in a toss, respectively. Then $p = q = \frac{1}{2}$. If X denotes the number of tosses required until a head appears, then X takes the values $1, 2, 3, \dots$ with probabilities, respectively, $p, qp, q^2p, \dots$. Hence, the expected number of tosses is given by

$$\mu = E(X) = \sum_x xf(x) = p + 2qp + 3q^2p + 4q^3p + \dots$$

i.e.

$$\mu = p[1 + 2q + 3q^2 + 4q^3 + \dots] = p[1 - q]^{-2}$$

$$\mu = p[p]^{-2} = [p]^{-1} = 2$$

◆

EXAMPLE 2.52. The number of hardware failures of a computer system in a week of operations has the following PMF:

No. of failures	0	1	2	3	4	5	6
Probability	.18	.28	.25	.18	.06	.04	.01

Find the mean of the number of failures in a week.

(Anna, May 2006)

Solution. By definition, the mean value of X is given by

$$\mu = E(X) = \sum_x xf(x) = 1.82$$

◆

EXAMPLE 2.53. In a continuous distribution, the probability density is given by

$$f(x) = kx(2 - x), \quad 0 < x < 2$$

Find k , mean, variance and the distribution function.

(Anna, May 2007)

Solution. Since f is a PDF, we have

$$\int_{x=0}^2 f(x) dx = 1 \Rightarrow \int_{x=0}^2 kx(2 - x) dx = k \int_{x=0}^2 [2x - x^2] dx = 1$$

Integrating, we have

$$k \left[x^2 - \frac{x^3}{3} \right]_{x=0}^2 = 1 \Rightarrow k \left[\frac{4}{3} \right] = 1$$

Hence, $k = \frac{3}{4}$. Thus, the probability density of X is given by

$$f(x) = \frac{3x}{4} (2 - x), \quad 0 < x < 2$$

For finding the mean and variance of X , note that

$$\mu'_1 = E(X) = \int_{x=0}^2 x f(x) dx = \int_{x=0}^2 x \frac{3x}{4} (2-x) dx$$

Simplifying, we have

$$\mu'_1 = \frac{3}{4} \int_{x=0}^2 [2x^2 - x^3] dx = \frac{3}{4} \left[2\frac{x^3}{3} - \frac{x^4}{4} \right]_0^2$$

Evaluating, we have

$$\mu'_1 = \frac{3}{4} \left[\frac{16}{3} - 4 \right] = 1$$

Hence, the mean of X is $E(X) = 1$. Next, we find that

$$\mu'_2 = E(X^2) = \int_{x=0}^2 x^2 f(x) dx = \int_{x=0}^2 x^2 \frac{3x}{4} (2-x) dx$$

Simplifying, we have

$$\mu'_2 = \frac{3}{4} \int_{x=0}^2 [2x^3 - x^4] dx = \frac{3}{4} \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2$$

Evaluating, we have

$$\mu'_2 = \frac{3}{4} \left[8 - \frac{32}{5} \right] = \frac{6}{5}$$

Hence, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{6}{5} - 1 = \frac{1}{5}$$

Finally, we find the cumulative distribution function F of X .

Clearly, if $x \leq 0$, then $F(x) = P(X \leq x) = 0$. Also, if $x \geq 2$, then $F(x) = P(X \leq x) = P(X \leq 2) = 1$.

So, we let $0 < x < 2$. Then, by definition, we have

$$F(x) = P(X < x) = \int_{t=0}^x f(t) dt = \int_{t=0}^x \frac{3t}{4} (2-t) dt = \frac{3}{4} \int_{t=0}^x [2t - t^2] dt$$

Integrating, we have

$$F(x) = \frac{3}{4} \left[t^2 - \frac{t^3}{3} \right]_{t=0}^x = \frac{3}{4} \left[x^2 - \frac{x^3}{3} \right]$$

Hence, the CDF of X is given by

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ \frac{3}{4} \left[x^2 - \frac{x^3}{3} \right] & \text{for } 0 < x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$

◆

EXAMPLE 2.54. Find the value of (i) C and (ii) mean of the following distribution:

$$f(x) = \begin{cases} C(x - x^2) & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(Anna, Nov. 2006)

Solution. (i) Since f is the PDF, we have

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^1 C(x - x^2) dx = 1 \Rightarrow C \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

i.e.

$$C \left[\frac{1}{2} - \frac{1}{3} \right] = 1 \Rightarrow C \left[\frac{1}{6} \right] = 1 \Rightarrow C = 6$$

Thus, the probability density of X is given by

$$f(x) = \begin{cases} 6(x - x^2) & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(ii) The mean of X is given by

$$\mu = \int_{-\infty}^{\infty} xf(x) dx = \int_0^1 x \cdot 6(x - x^2) dx = 6 \int_0^1 (x^2 - x^3) dx$$

i.e.

$$\mu = 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 6 \left[\frac{1}{3} - \frac{1}{4} \right] = 6 \times \frac{1}{12} = \frac{1}{2}$$

◆

EXAMPLE 2.55. Suppose the duration 'X' in minutes of long distance calls from your home, follows exponential law with PDF $f(x) = \frac{1}{5}e^{-\frac{x}{5}}$ for $x > 0$ and 0 otherwise. Find $P[X > 5]$, $P[3 \leq X \leq 6]$, mean of X and variance of X .

(Anna, Nov. 2005)

Solution. First, we find that

$$P[X > 5] = \int_5^{\infty} f(x) dx = \int_5^{\infty} \frac{1}{5} e^{-\frac{x}{5}} dx$$

Integrating, we get

$$P[X > 5] = \left[-e^{-\frac{x}{5}} \right]_5^{\infty} = e^{-1} = \frac{1}{e} = 0.3679$$

Next, we find that

$$P[3 \leq X \leq 6] = \int_3^6 f(x) dx = \int_3^6 \frac{1}{5} e^{-\frac{x}{5}} dx$$

Integrating, we get

$$P[3 \leq X \leq 6] = \left[-e^{-\frac{x}{5}} \right]_3^6 = -e^{-\frac{6}{5}} + e^{-\frac{3}{5}} = e^{-0.6} - e^{-1.2} = 0.2476$$

Next, we find the mean of X .

$$\mu'_1 = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \frac{1}{5} e^{-\frac{x}{5}} dx = \int_0^{\infty} 5y e^{-y} dy \quad (\text{Putting } y = \frac{x}{5})$$

Integrating, we get

$$\mu'_1 = 5 \Gamma(2) = 5 \times 1 = 5$$

Hence, the mean of X is $\mu = \mu'_1 = 5$

Next, we find the variance of X .

Note that

$$\mu'_2 = E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \frac{1}{5} e^{-\frac{x}{5}} dx = \int_0^{\infty} 25y^2 e^{-y} dy$$

(The last integral is got by substituting $y = \frac{x}{5}$.)

Integrating, we get

$$\mu'_2 = 25 \Gamma(3) = 25 \times 2 = 50$$

Hence, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = 50 - 25 = 25$$

◆

EXAMPLE 2.56. The distribution function of a random variable X is given by

$$F(x) = 1 - (1+x)e^{-x}, x \geq 0$$

Find the density function, mean and variance.

(Anna, Nov. 2006; Nov. 2007)

Solution. The probability density function f of X is obtained by performing piecewise differentiation of the distribution function F . Hence, if $x \leq 0$, then $f(x) = F'(x) = 0$ and if $x > 0$, then

$$f(x) = F'(x) = 0 - (1)e^{-x} - (1+x)(-e^{-x}) = -e^{-x} + (1+x)e^{-x} = xe^{-x}$$

Thus, the probability density function of X is given by

$$f(x) = \begin{cases} xe^{-x} & \text{if } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the mean of X . By definition, we have

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_0^{\infty} x^2 e^{-x} dx = \Gamma(3) = 2! = 2$$

We also find the variance of X by using the formula

$$\text{Var}(X) = E[X^2] - \mu^2$$

Hence

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^3 e^{-x} dx = \Gamma(4) = 3! = 6$$

Thus, we have

$$\text{Var}(X) = E[X^2] - \mu^2 = 6 - 4 = 2$$

i.e., $\mu = E(X) = 2$ and $\text{Var}(X) = 2$. ◆

EXAMPLE 2.57. A random variable X has the probability density function given by

$$f(x) = \begin{cases} k \sin \frac{\pi}{6}x & \text{for } 0 \leq x \leq 6 \\ 0 & \text{elsewhere} \end{cases}$$

Find the constant k and obtain the median and quartiles of the distribution.

Solution. Since f is a probability density function, we have

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^6 k \sin \frac{\pi}{6} x dx = 1$$

i.e.

$$k \left[-\frac{\cos \frac{\pi}{6}x}{\frac{\pi}{6}} \right]_0^6 = 1 \Rightarrow \frac{6}{\pi} k [-\cos \pi + \cos 0] = 1 \Rightarrow \frac{6}{\pi} k [2] = 1$$

Hence, $k = \frac{\pi}{12}$.

The median, M , is given by

$$\int_{-\infty}^M f(x) dx = \frac{1}{2} \Rightarrow \int_0^M \frac{\pi}{12} \sin \frac{\pi}{6} x dx = \frac{1}{2}$$

i.e.

$$\left[-\frac{1}{2} \cos \frac{\pi}{6} x \right]_0^M = \frac{1}{2}$$

i.e.

$$-\frac{1}{2} \cos \frac{\pi}{6} M + \frac{1}{2} = \frac{1}{2} \Rightarrow \cos \frac{\pi}{6} M = 0 \Rightarrow \frac{\pi}{6} M = \frac{\pi}{2}$$

Hence, $M = 3$.

The first quartile, Q_1 , is given by

$$\int_{-\infty}^{Q_1} f(x) dx = \frac{1}{4} \Rightarrow \int_0^{Q_1} \frac{\pi}{12} \sin \frac{\pi}{6} x dx = \frac{1}{4}$$

i.e.

$$\left[-\frac{1}{2} \cos \frac{\pi}{6} x \right]_0^{Q_1} = \frac{1}{4}$$

i.e.

$$-\frac{1}{2} \cos \frac{\pi}{6} Q_1 + \frac{1}{2} = \frac{1}{4} \Rightarrow \cos \frac{\pi}{6} Q_1 = \frac{1}{2} \Rightarrow \frac{\pi}{6} Q_1 = \frac{\pi}{3}$$

Hence, $Q_1 = 2$.

The third quartile, Q_3 , is given by

$$\int_{-\infty}^{Q_3} f(x) dx = \frac{3}{4} \Rightarrow \int_0^{Q_3} \frac{\pi}{12} \sin \frac{\pi}{6} x dx = \frac{3}{4}$$

i.e.

$$\left[-\frac{1}{2} \cos \frac{\pi}{6} x \right]_0^{Q_3} = \frac{3}{4}$$

i.e.

$$-\frac{1}{2} \cos \frac{\pi}{6} Q_3 + \frac{1}{2} = \frac{3}{4} \Rightarrow \cos \frac{\pi}{6} Q_3 = -\frac{1}{4} \Rightarrow \frac{\pi}{6} Q_3 = \frac{2\pi}{3}$$

Hence, $Q_3 = 4$. ♦

EXAMPLE 2.58. A random variable X has the probability density function given by

$$f(x) = \frac{x}{b^2} e^{-\frac{x^2}{2b^2}} \text{ for } x \geq 0$$

Find the distance between the quartiles and show that the ratio of this distance to the standard deviation of X is independent of the parameter b .

Solution. Let Q_1 and Q_3 be the first and third quartiles of the variable X .

By definition, we have

$$\int_{-\infty}^{Q_1} f(x) dx = \frac{1}{4} \Rightarrow \int_0^{Q_1} \frac{x}{b^2} e^{-\frac{x^2}{2b^2}} dx = \frac{1}{4}$$

Substituting $u = \frac{x^2}{2b^2}$ and noting that $du = \frac{x}{b^2}$, we have

$$\int_0^{\frac{Q_1^2}{2b^2}} e^{-u} du = \frac{1}{4} \Rightarrow [-e^{-u}]_0^{Q_1^2/2b^2} = \frac{1}{4}$$

i.e.

$$1 - e^{-Q_1^2/2b^2} = \frac{1}{4} \text{ or } e^{-Q_1^2/2b^2} = \frac{3}{4}$$

Simplifying, we have

$$Q_1^2 = 2b^2 \log_e \left(\frac{4}{3} \right) \text{ or } Q_1 = b \sqrt{2} \sqrt{\log_e \left(\frac{4}{3} \right)}$$

To find Q_3 , we note that by definition

$$\int_{-\infty}^{Q_3} f(x) dx = \frac{3}{4} \Rightarrow \int_0^{Q_3} \frac{x}{b^2} e^{-\frac{x^2}{2b^2}} dx = \frac{3}{4}$$

Substituting $u = \frac{x^2}{2b^2}$ and noting that $du = \frac{x}{b^2}$, we have

$$\int_0^{\frac{Q_3^2}{2b^2}} e^{-u} du = \frac{3}{4} \Rightarrow [-e^{-u}]_0^{Q_3^2/2b^2} = \frac{3}{4}$$

i.e.

$$1 - e^{-Q_3^2/2b^2} = \frac{3}{4} \text{ or } e^{-Q_3^2/2b^2} = \frac{1}{4}$$

Simplifying, we have

$$Q_3^2 = 2b^2 \log_e (4) \text{ or } Q_3 = b \sqrt{2} \sqrt{\log_e (4)}$$

The distance between the quartiles (*inter-quartile range*), $Q_3 - Q_1$, is

$$Q_3 - Q_1 = b\sqrt{2} \left[\sqrt{\log_e (4)} - \sqrt{\log_e \left(\frac{4}{3} \right)} \right].$$

Next, we find the variance σ^2 using $\sigma^2 = E(X^2) - \mu^2$

Note that

$$\mu = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \cdot \frac{x}{b^2} e^{-\frac{x^2}{2b^2}} dx$$

Putting $u = \frac{x^2}{2b^2}$ and noting that $du = \frac{x}{b^2}$, we have

$$\mu = \int_0^{\infty} b\sqrt{2}u^{\frac{1}{2}}e^{-u}du = b\sqrt{2}\Gamma\left(\frac{3}{2}\right) = b\sqrt{2}\frac{1}{2}\sqrt{\pi}.$$

i.e.

$$\mu = b\sqrt{\frac{\pi}{2}}$$

Note also that

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \cdot \frac{x}{b^2} e^{-\frac{x^2}{2b^2}} dx$$

Putting $u = \frac{x^2}{2b^2}$ and noting that $du = \frac{x}{b^2}$, we have

$$E(X^2) = \int_0^{\infty} 2b^2 u e^{-u} du = 2b^2 \Gamma(2) = 2b^2$$

Therefore,

$$\sigma^2 = E(X^2) - \mu^2 = 2b^2 - b^2 \left(\frac{\pi}{2}\right) = b^2 \left[2 - \frac{\pi}{2}\right]$$

Hence,

$$\sigma = b\sqrt{2 - \frac{\pi}{2}}$$

Therefore,

$$\frac{Q_3 - Q_1}{\sigma} = \frac{\sqrt{2} \left[\sqrt{\log_e(4)} - \sqrt{\log_e\left(\frac{4}{3}\right)} \right]}{\sqrt{2 - \frac{\pi}{2}}}$$

which is independent of the parameter b . ♦

Next, we define the notions of *arithmetic*, *geometric* and *harmonic means* for a random variable X .

Definition 2.13. Let X be a continuous random variable with probability density function f .

(a) The *arithmetic mean* (AM) or *mean* of X is denoted by μ and is defined as

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

provided the integral converges absolutely.

(b) The *geometric mean* (GM) of X is denoted by G and is defined by the relation

$$\log_e G = \int_{-\infty}^{\infty} \log x f(x) dx$$

provided the integral converges absolutely.

(c) The *harmonic mean* (HM) of X is denoted by H and is defined by the relation

$$\frac{1}{H} = \int_{-\infty}^{\infty} \frac{1}{x} f(x) dx$$

provided the integral converges absolutely.

The definition is similar for the case of a discrete random variable.

EXAMPLE 2.59. If X is a continuous random variable with probability density function

$$f(x) = \begin{cases} 6x(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the arithmetic mean, geometric mean, harmonic mean, median and mode.

Solution. The arithmetic mean, μ , is defined by

$$\mu = E(X) = \int_0^1 x f(x) dx = 6 \int_0^1 (x^2 - x^3) dx$$

Integrating, we have

$$\mu = 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 6 \left[\frac{1}{3} - \frac{1}{4} \right]$$

Simplifying, we have

$$\mu = 6 \times \frac{1}{12} = \frac{1}{2}$$

Thus, AM = $\mu = \frac{1}{2}$.

For the given density function, geometric mean does not exist.

The harmonic mean, H , is defined by the relation

$$\frac{1}{H} = \int_0^1 \frac{1}{x} f(x) dx = 6 \int_0^1 (1-x) dx$$

Integrating, we have

$$\frac{1}{H} = 6 \left[x - \frac{x^2}{2} \right]_0^1 = 6 \left[1 - \frac{1}{2} \right] = 3$$

Hence, HM = $H = \frac{1}{3}$.

Note that AM > HM

The median, M , is defined by the relation

$$\int_{-\infty}^M f(x) dx = \frac{1}{2} \Rightarrow \int_0^M f(x) dx = \frac{1}{2}$$

since $f(x) = 0$ for $x < 0$.

Therefore, we have

$$6 \int_0^M (x - x^2) dx = \frac{1}{2}$$

Integrating, we have

$$6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^M = \frac{1}{2} \quad \text{or} \quad 6 \left[\frac{M^2}{2} - \frac{M^3}{3} \right] = \frac{1}{2}$$

Simplifying, we have

$$3M^2 - 2M^3 = \frac{1}{2} \quad \text{or} \quad 4M^3 - 6M^2 + 1 = 0$$

Factorizing the cubic in M , we have

$$(2M - 1)(2M^2 - 2M - 1) = 0$$

Hence, it follows that $M = \frac{1}{2}$ or $M = \frac{1}{2}(1 \pm \sqrt{3})$.

Since $0 < M < 1$, the only admissible value of the median is $M = \frac{1}{2}$.

The mode, c , is the point in $0 < x < 1$, where f attains its maximum value. Differentiating f successively, we have

$$f'(x) = 6 - 12x \quad \text{and} \quad f''(x) = -12$$

Note that $f''(x) < 0$ for all x . Hence, if f has a turning point, then it is a global maximum for f . The turning points of f are given by the roots of $f'(x) = 0$. Solving, we get

$$f'(x) = 6 - 12x = 0 \Rightarrow x = \frac{1}{2}$$

Hence, mode $c = \frac{1}{2}$.

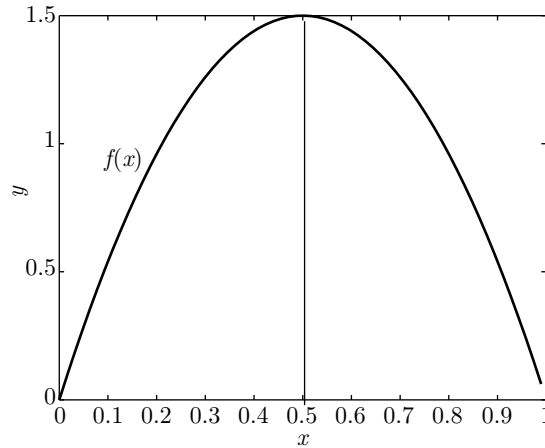


Figure 2.11: Symmetric distribution.

We note that for the given PDF f , mean, median and mode coincide. This should not be surprising because the given random variable X has a symmetric distribution. The probability density f is plotted in Figure 2.11, and it is easy to see that f is symmetric about the line $x = \frac{1}{2}$. ◆

EXAMPLE 2.60. Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} 6(2-x)(x-1) & \text{for } 1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Show that the geometric mean, G , of X is given by $16G = e^{\frac{19}{6}}$.

Solution. By definition, the geometric mean, G , is given by

$$\log G = \int_{-\infty}^{\infty} \log x f(x) dx = \int_1^2 \log x f(x) dx$$

since $f(x) = 0$ for $x < 1$ and $f(x) = 0$ for $x > 2$. Hence, we have

$$\log G = 6 \int_1^2 \log x ((2-x)(x-1)) dx = 6 \int_1^2 \log x (3x - x^2 - 2) dx$$

Therefore,

$$\log G = 6 \int_1^2 \log x d \left[\frac{3x^2}{2} - \frac{x^3}{3} - 2x \right]$$

Using integration by parts, we have

$$\log G = 6 \left[\left\{ \log x \left[\frac{3x^2}{2} - \frac{x^3}{3} - 2x \right] \right\}_1^2 - \int_1^2 \left[\frac{3x^2}{2} - \frac{x^3}{3} - 2x \right] \left(\frac{1}{x} \right) dx \right]$$

$$\log G = 6 \left[\left\{ \log 2 \left[6 - \frac{8}{3} - 4 \right] \right\} - \int_1^2 \left[\frac{3x}{2} - \frac{x^2}{3} - 2 \right] dx \right]$$

Integrating, we have

$$\log G = 6 \left[-\frac{2}{3} \log 2 - \left[\frac{3x^2}{4} - \frac{x^3}{9} - 2x \right]_1^2 \right]$$

Therefore,

$$\log G = -4 \log 2 - 6 \left[\left(3 - \frac{8}{9} - 4 \right) - \left(\frac{3}{4} - \frac{1}{9} - 2 \right) \right]$$

Simplifying, we have

$$\log G = -4 \log 2 + \frac{19}{6}$$

or

$$\log G + 4\log 2 = \frac{19}{6} \quad \text{or} \quad \log(16G) = \frac{19}{6}$$

Hence, G is given by

$$16G = e^{\frac{19}{6}}$$



PROBLEM SET 2.5

1. If X is a discrete random variable with the probability mass function

$$f(x) = \begin{cases} \frac{1}{x(x+1)} & \text{if } x = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

show that the mean does not exist.

2. If X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{(x+1)^2} & \text{if } 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

show that the mean does not exist.

3. A coin is biased so that the head is twice as likely to appear as the tail. The coin is tossed 3 times. Find the expected value of the number of heads.
4. A and B throw alternatively with one die for a stake of Rs. 5500. The player who first throws 6 wins the stake. If A has the first throw, find the expectations of A and B winning the stake.
5. (a) If X is any random variable, establish that

$$\text{Var}(X) = E[X(X-1)] - E(X)[E(X) - 1]$$

(b) If X is a random variable with $E(X) = 2$ and $E[X(X-1)] = 6$, find $\text{Var}(X)$ and $\text{Var}(4-5X)$.

6. If X is a random variable with mean 12 and variance 36, find the positive values of a and b so that $Y = aX - b$ has mean 0 and variance 1.
7. If X is a random variable that assumes the values -2 , 1 and 2 with probabilities $\frac{1}{4}$, $\frac{1}{8}$ and $\frac{5}{8}$, find the mean and variance of X .
8. A random variable X has the probability distribution

x	-3	-2	-1	0	1	2	3
$f(x)$	0.2	$2k$	0.1	k	0.4	k	0.1

Find

- (a) The value of k .
- (b) $P(X < 1)$.
- (c) The mean and variance of X .
9. The monthly demand for pocket radios is known to have the following probability distribution:

Demand	1	2	3	4	5	6	7	8
Probability	0.02	0.03	0.16	0.2	0.05	0.2	0.16	0.18

Find the expected demand for the pocket radios. Also compute the variance.

10. A man draws 3 balls from an urn containing 6 white and 4 red balls. He gets Rs. 10 for each white ball and Rs. 20 for each red ball. Find his expectation and variance.
11. If X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{\alpha}{x} & \text{if } 1 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The value of α .
- (b) The mean and variance of X .
- (c) The median of X .
12. The demand for a new product of a company is assumed to be a continuous random variable with the distribution function

$$F(x) = \begin{cases} 1 - e^{-\frac{x^2}{2a^2}} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Find

- (a) The probability density function of X .
- (b) The mean and variance of X .
- (c) The probability that the demand for the new product will exceed α .
13. If X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{3}{(x+1)^4} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

find the mean of X .

14. A continuous random variable X has the probability density function

$$f(x) = \begin{cases} k \sin \frac{\pi x}{4} & \text{if } 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The value of k .
- (b) The mean and variance of X .
- (c) The quartiles of X .
- (d) The median of X .
- (e) The mode of X .

15. A continuous random variable X has the probability density function

$$f(x) = \begin{cases} kx^3 e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The value of k .
- (b) The mean and variance of X .
- (c) The harmonic mean of X .

2.6 CHEBYSHEV'S INEQUALITY

We start this section with *Markov's inequality*, which is used to prove *Chebyshev's inequality*. Markov's inequality provides an upper bound for the probability that the absolute value of a random variable is greater than or equal to some positive constant. This inequality is named after the Russian mathematician, Andrey Markov (1856–1922).

Theorem 2.17. (*Markov's Inequality*) If X is any random variable and $K > 0$, then

$$P[|X| \geq K] \leq \frac{E|X|}{K} \quad (2.10)$$

Proof. Define the event A by

$$A = \{|X| \geq K\} = \{\omega \in S : |X(\omega)| \geq K\}$$

Let I_A be the indicator variable for the event A , i.e.

$$I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

We claim that

$$KI_A \leq |X| \quad (2.11)$$

To prove this, note that given any $\omega \in S$, either $\omega \in A$ or $\omega \notin A$. If $\omega \in A$, then $|X(\omega)| \geq K$, i.e. $|X(\omega)| \geq KI_A(\omega)$, which proves Eq. (2.11). If $\omega \notin A$, then $I_A(\omega) = 0$ so that Eq. (2.11) holds trivially. Thus, we have established Eq. (2.11).

From Eq. (2.11), it follows immediately that

$$E[KI_A] \leq E|X| \Rightarrow KE[I_A] \leq E|X|$$

By Example 2.37, we know that

$$E[I_A] = P(A) = P[|X| \geq K]$$

Hence, we have

$$KP[|X| \geq K] \leq E|X|$$

or

$$P[|X| \geq K] \leq \frac{E|X|}{K}$$

Alternative Proof. First, we prove this for the continuous case. Let X be a continuous random variable with probability density function f . By definition, we have

$$E|X| = \int_{-\infty}^{\infty} |x|f(x)dx = \int_{|x| \geq K} |x|f(x)dx + \int_{|x| < K} |x|f(x)dx$$

Thus, it follows that

$$E|X| \geq \int_{|x| \geq K} |x|f(x)dx \geq \int_{|x| \geq K} Kf(x)dx$$

i.e.

$$E|X| \geq K \int_{|x| \geq K} f(x)dx = KP[|X| \geq K]$$

i.e.

$$P[|X| \geq K] \leq \frac{E|X|}{K}$$

Next, we prove this for the discrete case. Let X be a discrete random variable with mass points x_i ($i = 1, 2, 3, \dots$) and probability mass function $f(x)$. By definition, we have

$$E|X| = \sum_i |x_i|f(x_i) = \sum_{|x_i| \geq K} |x_i|f(x_i) + \sum_{|x_i| < K} |x_i|f(x_i)$$

Thus, it follows that

$$E|X| \geq \sum_{|x_i| \geq K} |x_i|f(x_i) \geq \sum_{|x_i| \geq K} Kf(x_i)$$

i.e.

$$E|X| \geq K \sum_{|x_i| \geq K} f(x_i) = KP[|X| \geq K]$$

i.e.

$$P[|X| \geq K] \leq \frac{E|X|}{K}$$

◆

Next, we prove *Chebyshev's inequality* (also known as *Tchebysheff's inequality* or *Chebyshev's theorem*), which is named after the Russian mathematician Pafnuty Chebyshev (1821–1894). Essentially, Chebyshev's inequality asserts that nearly all the values of a probability distribution stay close to the mean value, and that the standard deviation serves as a good measure of the dispersion or the spread of the distribution about the mean. This inequality is useful in determining probabilities of events centred about the mean, when the actual probability distribution is not available, and when we know only the mean and variance of the distribution.

Theorem 2.18. (*Chebyshev's Inequality*) Let X be a random variable with mean μ and finite variance σ^2 . Then for any real number $K > 0$,

$$P[|X - \mu| \geq K\sigma] \leq \frac{1}{K^2} \quad (2.12)$$

Proof. Chebyshev's inequality follows immediately from Markov's inequality. By Markov's inequality, if Z is any random variable and $a > 0$ is a constant, then

$$P[|Z| \geq a] \leq \frac{E|Z|}{a}$$

Take $Z = (X - \mu)^2$ and $a = (K\sigma)^2$. Then it follows that

$$P[|(X - \mu)^2| \geq (K\sigma)^2] \leq \frac{E|X - \mu|^2}{(K\sigma)^2}$$

i.e.

$$P[|X - \mu| \geq K\sigma] \leq \frac{\sigma^2}{K^2\sigma^2} = \frac{1}{K^2}$$

This proves Eq. (2.12).

Alternative Proof. First, we prove Chebyshev's inequality for the continuous case. Let X be a continuous random variable with probability density function f . (The shaded area in Figure 2.12 describes the probability for the region $|X - \mu| \geq K\sigma$.) By definition, we have

$$\sigma^2 = E|X - \mu|^2 = \int_{-\infty}^{\infty} |x - \mu|^2 f(x) dx$$

i.e.

$$\sigma^2 = \int_{|X - \mu| \geq K\sigma} |x - \mu|^2 f(x) dx + \int_{|X - \mu| < K\sigma} |x - \mu|^2 f(x) dx$$

i.e.

$$\sigma^2 \geq \int_{|X - \mu| \geq K\sigma} |x - \mu|^2 f(x) dx \geq \int_{|X - \mu| \geq K\sigma} K^2\sigma^2 f(x) dx$$

i.e.

$$\frac{1}{K^2} \geq \int_{|X - \mu| \geq K\sigma} f(x) dx = P[|X - \mu| \geq K\sigma]$$

This proves Eq. (2.12).

Next, we prove Chebyshev's inequality for the discrete case. Let X be a discrete random variable with mass points x_i ($i = 1, 2, 3, \dots$) and probability mass function f . Then we have

$$\sigma^2 = E|X - \mu|^2 = \sum_i |x_i - \mu|^2 f(x_i)$$

i.e.

$$\sigma^2 = \sum_{|x_i - \mu| \geq K\sigma} |x_i - \mu|^2 f(x_i) + \sum_{|x_i - \mu| < K\sigma} |x_i - \mu|^2 f(x_i)$$

i.e.

$$\sigma^2 \geq \sum_{|x_i - \mu| \geq K\sigma} |x_i - \mu|^2 f(x_i) \geq \sum_{|x_i - \mu| \geq K\sigma} K^2\sigma^2 f(x_i)$$

i.e.

$$\frac{1}{K^2} \geq \sum_{|x_i - \mu| \geq K\sigma} f(x_i) = P[|X - \mu| \geq K\sigma]$$

◆

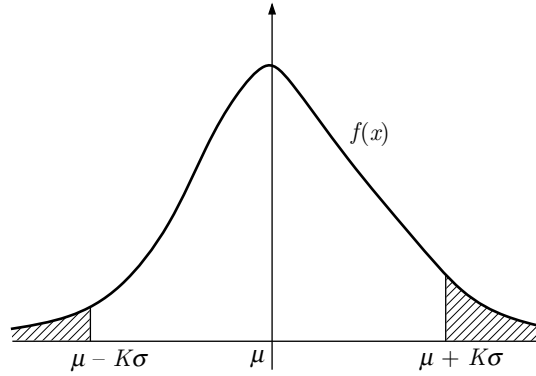


Figure 2.12: Chebyshev's inequality.

Equivalent forms of Chebyshev's inequality, Eq. (2.12), are given in the following results.

Corollary 2.2. Let X be a random variable with mean μ and finite variance σ^2 . Then for any real number $K > 0$,

$$P[|X - \mu| < K\sigma] \geq 1 - \frac{1}{K^2}$$

Proof. For any event A , we know that $P(A) + P(\bar{A}) = 1$. Hence, we have

$$P[|X - \mu| < K\sigma] = 1 - P[|X - \mu| \geq K\sigma] \geq 1 - \frac{1}{K^2}$$

using Chebyshev's inequality. ◆

Remark 2.11. Chebyshev's inequality stated in Corollary 2.2 describes how well the standard deviation serves as a measure of the dispersion or spread of the probability distribution about the mean. Taking $K = 2$, we have

$$P[|X - \mu| < 2\sigma] \geq 1 - \frac{1}{2^2} = 0.75 = 75\%$$

Taking $K = 3$, we have

$$P[|X - \mu| < 3\sigma] \geq 1 - \frac{1}{3^2} = 0.8889 = 88.9\%$$

Taking $K = 4$, we have

$$P[|X - \mu| < 4\sigma] \geq 1 - \frac{1}{4^2} = 0.9375 = 93.4\%$$

Taking $K = 5$, we have

$$P[|X - \mu| < 5\sigma] \geq 1 - \frac{1}{5^2} = 0.96 = 96\%$$

Corollary 2.3. Let X be a random variable with mean μ and finite variance σ^2 . Then for any real number $C > 0$,

$$P[|X - \mu| \geq C] \leq \frac{\sigma^2}{C^2}$$

Proof. Setting $K\sigma = C$ or $K = \frac{C}{\sigma}$ in Eq. (2.12), the result follows. $\blacklozenge$

Corollary 2.4. Let X be a random variable with mean μ and finite variance σ^2 . Then, for any real number $C > 0$,

$$P[|X - \mu| < C] \geq 1 - \frac{\sigma^2}{C^2}$$

Proof. For any event A , we know that $P(A) + P(\bar{A}) = 1$. Hence,

$$P[|X - \mu| < C] + P[|X - \mu| \geq C] = 1$$

By the previous Corollary, we know that

$$P[|X - \mu| \geq C] \leq \frac{\sigma^2}{C^2}$$

Hence, it follows that

$$P[|X - \mu| < C] = 1 - P[|X - \mu| \geq C] \geq 1 - \frac{\sigma^2}{C^2} \quad \blacklozenge$$

Remark 2.12. Chebyshev's inequality Eq. (2.12) provides an upper bound for the probability, $P[|X - \mu| \geq K\sigma]$. This information is very useful when the probability distribution of the random variable X is not available, and we are provided with only the mean μ and variance σ of X . Chebyshev's inequality is very simple and it can be used universally for random variables of all types.

EXAMPLE 2.61. Use Chebyshev's inequality to show that, if X is the number scored in a throw of a fair die, $P\{|X - 3.5| > 2.5\} < 0.47$.

(Anna, Model, 2003)

Solution. As calculated in Example 2.43, X has mean $\mu = \frac{7}{2} = 3.5$ and variance $\sigma^2 = \frac{35}{12} = 2.9167$.

By Chebyshev's inequality, we have

$$P[|X - \mu| \geq C] \leq \frac{\sigma^2}{C^2}$$

Taking $C = 2.5$, we have

$$P\{|X - 3.5| > 2.5\} < \frac{2.9167}{6.25} = 0.4667 \approx 0.47 \quad \blacklozenge$$

EXAMPLE 2.62. Two unbiased dice are thrown. Let X be a random variable that represents the sum of the numbers showing up. Prove by Chebyshev's inequality

$$P\{|X - 7| \geq 2\} \leq \frac{35}{24}$$

Compare this result with the actual probability.

(Anna, Model, 2003)

Solution. The probability distribution of X , mean and variance were determined in Example 2.44. Recall that $\mu = 7$ and $\sigma^2 = \frac{35}{6}$.

By Chebyshev's inequality, we have

$$P[|X - \mu| \geq C] \leq \frac{\sigma^2}{C^2}$$

Taking $C = 2$, we get

$$P\{|X - 7| \geq 2\} \leq \frac{35/6}{4} = \frac{35}{24}$$

The actual probability is equal to

$$P\{|X - 7| \geq 2\} = 1 - P\{|X - 7| < 2\} = 1 - P\{-2 < X - 7 < 2\} = 1 - P\{5 < X < 9\}$$

i.e.

$$P\{|X - 7| \geq 2\} = 1 - [P(X = 6) + P(X = 7) + P(X = 8)] = 1 - \left[\frac{5}{36} + \frac{6}{36} + \frac{5}{36} \right]$$

i.e.

$$P\{|X - 7| \geq 2\} = 1 - \frac{16}{36} = \frac{20}{36} = \frac{5}{9} = 0.5556$$

Thus, the actual probability is 0.5556, while the upper bound given by Chebyshev's inequality is $\frac{35}{24} = 1.4583$. ♦

EXAMPLE 2.63. A random variable X has the density function e^{-x} , $x \geq 0$. Show that Chebyshev's inequality gives $P[|X - 1| > 2] < \frac{1}{4}$ and show that actual probability is e^{-3} .

(Anna, April 2003)

Solution. First, we calculate the mean and variance of X . By definition,

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x e^{-x} dx = \Gamma(2) = 1$$

Note also that

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 e^{-x} dx = \Gamma(3) = 2! = 2$$

Hence, the variance of X is given by

$$\sigma^2 = \text{Var}(X) = E(X^2) - \mu^2 = 2 - 1 = 1$$

By Chebyshev's inequality, we have

$$P[|X - \mu| > C] < \frac{\sigma^2}{C^2}$$

Taking $C = 2$, we have

$$P(|X - 1| > 2) < \frac{\sigma^2}{C^2} = \frac{1}{4}$$

The actual probability is equal to

$$\begin{aligned} P(|X-1| > 2) &= 1 - P\{|X-1| \leq 2\} \\ &= 1 - P(-2 \leq X-1 \leq 2) \\ &= 1 - P(-1 \leq X \leq 3) \end{aligned}$$

i.e.

$$P(|X-1| > 2) = 1 - \int_0^3 e^{-x} dx = 1 - [-e^{-x}]_0^3 = 1 - [-e^{-3} + 1] = e^{-3}$$

Thus, the actual probability is e^{-3} , while the bound given by Chebyshev's inequality is $\frac{1}{4}$. $\blacklozenge$

EXAMPLE 2.64. A random variable X has a mean 10, a variance 4 and an unknown probability distribution. Find the value of C such that $P[|X-10| \geq C] \leq 0.04$.

(Madras, April 1997; April 1999)

Solution. Given that $\mu = 10$ and $\sigma^2 = 4$. By Chebyshev's inequality, we have

$$P[|X - \mu| \geq C] \leq \frac{\sigma^2}{C^2}$$

Substituting the given values, we have

$$P[|X - 10| \geq C] \leq \frac{4}{C^2}$$

We find C so that

$$\frac{4}{C^2} = 0.04 = \frac{4}{100} \Rightarrow C^2 = 100$$

Thus, $C = 10$. $\blacklozenge$

EXAMPLE 2.65. A random variable X has a mean $\mu = 8$, a variance $\sigma^2 = 9$ and an unknown probability distribution. Find (i) $P(-4 < X < 20)$ and (ii) $P(|X-8| \geq 6)$.

(Madras, Oct. 1997; Oct. 1998; April 2002)

Solution. (i) By Chebyshev's inequality, we have

$$P[|X - \mu| < C] \geq 1 - \frac{\sigma^2}{C^2}$$

Thus,

$$P(-4 < X < 20) = P(|X-8| < 12) \geq 1 - \frac{9}{12^2} = 1 - \frac{1}{16} = \frac{15}{16}$$

Hence, $P(-4 < X < 20) \geq \frac{15}{16} = 0.9375$.

(ii) By Chebyshev's inequality, we have

$$P[|X - \mu| \geq C] \leq \frac{\sigma^2}{C^2}$$

Thus,

$$P(|X - 8| \geq 6) \leq \frac{9}{6^2} = \frac{1}{4}$$

Hence, $P(|X - 8| \geq 6) \leq \frac{1}{4} = 0.25$. ◆

EXAMPLE 2.66. A random variable X has a mean 12, a variance 9 and an unknown probability distribution. Find $P(6 < X < 18)$ and $P(3 < X < 21)$.

(Madras, Oct. 1999; April 2000)

Solution. Given that $\mu = 12$ and $\sigma^2 = 9$. By Chebyshev's inequality, we have

$$P[|X - \mu| < C] \geq 1 - \frac{\sigma^2}{C^2}$$

First, we find that

$$P(6 < X < 18) = P(|X - 12| < 6) \geq 1 - \frac{9}{6^2} = 1 - \frac{1}{4} = \frac{3}{4}$$

Hence, $P(6 < X < 18) \geq \frac{3}{4} = 0.75$.

Next, we find that

$$P(3 < X < 21) = P(|X - 12| < 9) \geq 1 - \frac{9}{9^2} = 1 - \frac{1}{9} = \frac{8}{9}$$

Hence, $P(3 < X < 21) \geq \frac{8}{9} = 0.8889$. ◆

EXAMPLE 2.67. Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50. If the variance of a week's production is known to equal 25, then what can be said about the productivity that will be between 40 and 60?

(Anna, Nov. 2006)

Solution. Let X be the number of items produced in a factory during a week. Then it is given that X has mean $\mu = 50$ and variance $\sigma^2 = 25$. The probability that the productivity will be between 40 and 60 is given by

$$P(40 \leq X \leq 60) = P(|X - \mu| \leq 20)$$

By Chebyshev's inequality, we know that

$$P(|X - \mu| \leq C) \geq 1 - \frac{\sigma^2}{C^2}$$

Hence, it follows that

$$P(|X - \mu| \leq 20) \geq 1 - \frac{25}{20^2} = 0.9375$$
◆

PROBLEM SET 2.6

1. A random variable X has the probability distribution

x	0	1	2	4
$f(x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$

Find

(a) An upper bound for $P(|X - 1| \geq 2)$ by Chebychev's inequality.

(b) $P(|X - 1| \geq 2)$ by direct computation.

2. Two fair dice are thrown. If X is the sum of numbers showing up, prove that

$$P[|X - 7| \geq 4] \leq \frac{35}{96}$$

Compare this result with the actual probability.

3. X is a continuous random variable with the exponential density

$$f(x) = \begin{cases} \frac{1}{2}e^{-\frac{x}{2}} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Show that Chebychev's inequality gives $P[|X - 2| \geq 2] \leq \frac{1}{2}$ and that actual probability is e^{-2} .

4. A random variable X has a mean $\mu = 6$, a variance $\sigma^2 = 4$ and an unknown probability distribution. Find

(a) $P(-2 < X < 14)$

(b) $P(|X - 6| \geq 6)$

5. Suppose that on average, a call centre receives 8000 calls a day with a variance of 3000. What can be said about the probability that this call centre will receive between 6000 and 10000 calls tomorrow?
6. The number of plastic toys produced in a toy factory during a week is a random variable with mean 100. If the variance of a week's production in the toy factory is equal to 50, then what can be said about the statement that the productivity will be between 80 and 120 for the following week?
7. A random variable X has a mean μ , a variance σ^2 and an unknown probability distribution. Find the smallest value of K that guarantees that $P(|X - \mu| < K\sigma)$ is (a) at least 90%, (b) at least 95%, (c) at least 99%.

2.7 MOMENTS OF A RANDOM VARIABLE

Definition 2.14. Let X be a random variable. Let k be a positive integer and c be a constant. The *moment of order k* or the *k^{th} moment* of X about the point c is defined as $E[(X - c)^k]$.

First, we consider moments about the origin.

Definition 2.15. Let X be a random variable. The k^{th} moment of X about the origin or simply the k^{th} moment of X is denoted by μ'_k and is defined as $E[X^k]$, i.e.

$$\mu'_k = E[X^k], \quad k = 1, 2, 3, 4, \dots$$

The moments μ'_k are also known as *raw moments*.

If X is a continuous random variable with probability density function f , then

$$\mu'_k = \int_{-\infty}^{\infty} x^k f(x) dx$$

and if X is a discrete random variable with probability mass function f , then

$$\mu'_k = \sum_i x_i^k f(x_i)$$

Note that $\mu'_0 = 1$ and $\mu'_1 = E(X) = \mu$, the mean of X .

The moments of X defined about $c = \mu$ are known as the *central moments* and they convey important properties of the distribution.

Definition 2.16. Let X be a random variable. The k^{th} central moment of X is denoted by μ_k and is defined as $E[(X - \mu)^k]$, i.e.

$$\mu_k = E[(X - \mu)^k], \quad k = 1, 2, 3, 4, \dots$$

If X is a continuous random variable with probability density function f , then

$$\mu_k = \int_{-\infty}^{\infty} (x - \mu)^k f(x) dx$$

and if X is a discrete random variable with probability mass function f , then

$$\mu_k = \sum_i (x_i - \mu)^k f(x_i)$$

Remark 2.13. We note the following:

1. For any distribution, $\mu_0 = 1$ and $\mu_1 = 0$. Indeed,

$$\mu_1 = E[X - \mu] = E(X) - \mu = \mu - \mu = 0$$

2. The second central moment, μ_2 , is the *variance* of the distribution, i.e. $\mu_2 = \text{Var}(X) = \sigma^2$. The formula for computing $\text{Var}(X)$ can be rewritten as follows:

$$\mu_2 = \text{Var}(X) = E(X^2) - \mu^2 = \mu'_2 - \mu_1'^2$$

3. The significance of the second central moment, $\mu_2 = \sigma^2$, is that it describes the *spread* or *dispersion* of the distribution.

4. The significance of the third central moment, μ_3 , is that it gives a measure of the *lopsidedness* or *skewness* of the distribution. For a symmetric distribution, $\mu_3 = 0$. If $\mu_3 < 0$, then the distribution is skewed to the left (i.e. the tail of the distribution is heavier on the left). If $\mu_3 > 0$, then the distribution is skewed to the right (i.e. the tail of the distribution is heavier on the right).
5. Note that $\mu_4 = E[(X - \mu)^4] > 0$ for all probability distributions except for the degenerate point (constant) distribution. The significance of the fourth central moment, μ_4 , is that it gives a measure of the *peakedness* of the distribution compared to the normal distribution of the same variance.

Karl Pearson (1857–1936) had done significant work in classifying various probability distributions. Here, we give the definition of his β and γ coefficients (see Figure 2.13) which are useful in understanding the shape properties like symmetry and kurtosis of the distribution.

Definition 2.17. (Karl Pearson's β and γ Coefficients) Based on the first four central moments, Pearson's β and γ coefficients can be defined as follows:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \gamma_1 = +\sqrt{\beta_1}, \beta_2 = \frac{\mu_4}{\mu_2^2} \text{ and } \gamma_2 = \beta_2 - 3$$

The coefficients γ_1 and γ_2 are known as the *coefficients of skewness* and *kurtosis*, respectively.

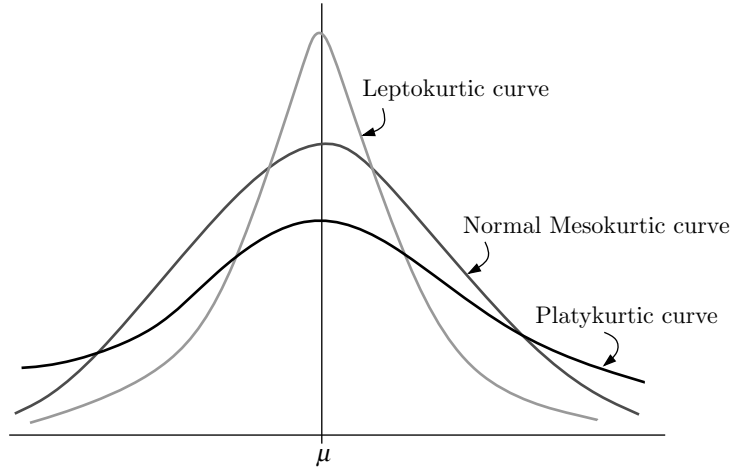


Figure 2.13: Karl Pearson's coefficients.

Remark 2.14. For a normal distribution, it can be shown that $\mu_4 = 3\mu_2^2$ and hence $\beta_2 = 3$ or $\gamma_2 = 0$. Distributions having $\gamma_2 = 0$ are said to be *mesokurtic*. Distributions having $\beta_2 < 3$ or $\gamma_2 < 0$ are flatter than the normal curve and they are called as *platykurtic*. Distributions having $\beta_2 > 3$ or $\gamma_2 > 0$ are more peaked than the normal curve and they are called as *leptokurtic*.

The following result describes the relation between the central and the raw moments.

Theorem 2.19.

$$\mu_k = \mu'_k - {}^k C_1 \mu'_{k-1} \mu'_1 + {}^k C_2 \mu'_{k-2} \mu_1'^2 - \cdots + (-1)^k \mu_1'^k \quad (2.13)$$

Proof. By binomial expansion, we have

$$(X - \mu)^k = X^k - {}^kC_1 X^{k-1} \mu + {}^kC_2 X^{k-2} \mu^2 - \dots + (-1)^k \mu^k$$

By taking expectation on both sides of the above equation, the result follows. $\blacklozenge$

As a consequence of Theorem 2.19, we can develop simple formulas for calculating the first four central moments from the first four raw moments.

For any distribution, we always have $\mu_0 = 1, \mu_1 = 0$.

Next, putting $k = 2$ in Eq. (2.13), we have

$$\begin{aligned} \mu_2 &= \mu'_2 - {}^2C_1 \mu'_1 \mu'_1 + \mu_1'^2 \\ &= \mu'_2 - 2\mu_1'^2 + \mu_1'^2 \\ &= \mu'_2 - \mu_1'^2 \end{aligned}$$

Putting $k = 3$ in Eq. (2.13), we have

$$\begin{aligned} \mu_3 &= \mu'_3 - {}^3C_1 \mu'_2 \mu'_1 + {}^3C_2 \mu'_1 \mu_1'^2 - \mu_1'^3 \\ &= \mu'_3 - 3\mu'_2 \mu'_1 + 3\mu'_1 \mu_1'^2 - \mu_1'^3 \\ &= \mu'_3 - 3\mu'_2 \mu'_1 + 2\mu_1'^3 \end{aligned}$$

Putting $k = 4$ in Eq. (2.13), we have

$$\begin{aligned} \mu_4 &= \mu'_4 - {}^4C_1 \mu'_3 \mu'_1 + {}^4C_2 \mu'_2 \mu_1'^2 - {}^4C_3 \mu'_1 \mu_1'^3 + \mu_1'^4 \\ &= \mu'_4 - 4\mu'_3 \mu'_1 + 6\mu'_2 \mu_1'^2 - 4\mu'_1 \mu_1'^3 + \mu_1'^4 \\ &= \mu'_4 - 4\mu'_3 \mu'_1 + 6\mu'_2 \mu_1'^2 - 3\mu_1'^4 \end{aligned}$$

EXAMPLE 2.68. Find the mean, variance and the coefficients β_1, β_2 of the random variable X with the probability density function given by

$$f(x) = \begin{cases} kx^2 e^{-x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Solution. First, we find the value of k . Since f is the probability density function, we have

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^{\infty} kx^2 e^{-x} dx = 1 \Rightarrow k\Gamma(3) = k(2!) = 1$$

Thus, $k = \frac{1}{2}$.

Next, we calculate the first four raw moments μ'_r . Note that

$$\mu'_r = \int_{-\infty}^{\infty} x^r f(x) dx = \int_0^{\infty} \frac{1}{2} x^{r+2} e^{-x} dx = \frac{1}{2} \Gamma(r+3) = \frac{(r+2)!}{2}$$

Hence, we have

$$\mu'_1 = \frac{3!}{2} = 3, \mu'_2 = \frac{4!}{2} = 12, \mu'_3 = \frac{5!}{2} = 60 \text{ and } \mu'_4 = \frac{6!}{2} = 360$$

Therefore,

$$\begin{aligned}\mu_2 &= \mu'_2 - \mu_1'^2 = 12 - 9 = 3 \\ \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3 = 60 - 108 + 54 = 6 \\ \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4 = 360 - 720 + 648 - 243 = 45\end{aligned}$$

Thus, the mean $\mu = \mu'_1 = 3$ and the variance $\sigma^2 = \mu_2 = 3$. Also,

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{36}{27} = \frac{4}{3} \text{ and } \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{45}{9} = 5$$

◆

EXAMPLE 2.69. If the probability density of X is given by

$$f(x) = \begin{cases} 2(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(i) Show that $E[X^r] = \frac{2}{(r+1)(r+2)}$.

(ii) Use this result to evaluate $E[(2X+1)^2]$.

(Anna, Nov. 2006)

Solution. (i) By definition, we have

$$E[X^r] = \int_{-\infty}^{\infty} x^r f(x) dx = \int_0^1 x^r \cdot 2(1-x) dx = 2 \int_0^1 (x^r - x^{r+1}) dx$$

Integrating, we get

$$E[X^r] = 2 \left[\frac{x^{r+1}}{r+1} - \frac{x^{r+2}}{r+2} \right]_0^1 = 2 \left[\frac{1}{r+1} - \frac{1}{r+2} \right] = \frac{2}{(r+1)(r+2)}$$

(ii) We find that

$$\begin{aligned}E[(2X+1)^2] &= E[4X^2 + 4X + 1] = 4E[X^2] + 4E[X] + 1 \\ &= 4 \left[\frac{2}{3 \times 4} \right] + 4 \left[\frac{2}{2 \times 3} \right] + 1 = \frac{2}{3} + \frac{4}{3} + 1 = 3\end{aligned}$$

◆

EXAMPLE 2.70. The density function of a random variable X is given by

$$f(x) = Kx(2-x), 0 \leq x \leq 2$$

Find K , mean, variance and r th moment.

(Anna, Nov. 2006)

Solution. Since f is the probability density function, we have

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow \int_0^2 Kx(2-x) dx = 1 \Rightarrow K \int_0^2 (2x - x^2) dx = 1$$

Integrating, we get

$$K \left[x^2 - \frac{x^3}{3} \right]_0^2 = 1 \Rightarrow K \left[\left(4 - \frac{8}{3} \right) - 0 \right] = 1 \Rightarrow K \left(\frac{4}{3} \right) = 1$$

Thus, $K = \frac{3}{4}$. Hence, the PDF of X is given by

$$f(x) = \begin{cases} \frac{3}{4}x(2-x) & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

The r th raw moment of X is given by

$$\mu'_r = E[X^r] = \int_{-\infty}^{\infty} x^r f(x) dx = \int_0^2 x^r \frac{3}{4}x(2-x) dx = \frac{3}{4} \int_0^2 (2x^{r+1} - x^{r+2}) dx$$

Integrating, we get

$$\mu'_r = \frac{3}{4} \left[2 \frac{x^{r+2}}{r+2} - \frac{x^{r+3}}{r+3} \right]_0^2 = \frac{3}{4} \left[2 \frac{2^{r+2}}{r+2} - \frac{2^{r+3}}{r+3} \right]$$

Simplifying, it follows that

$$\mu'_r = \frac{3 \cdot 2^{r+1}}{(r+2)(r+3)}.$$

In particular, we have

$$\mu'_1 = \frac{3 \times 4}{3 \times 4} = 1 \quad \text{and} \quad \mu'_2 = \frac{3 \times 8}{4 \times 5} = \frac{6}{5}$$

Hence, the mean and the variance of X are obtained as

$$\text{Mean}(X) = \mu'_1 = 1 \quad \text{and} \quad \text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{6}{5} - 1 = \frac{1}{5}$$

◆

EXAMPLE 2.71. The first four moments of a distribution about $X = 4$ are 1, 4, 10 and 45, respectively. Show that the mean is 5, variance is 3, $\mu_3 = 0$ and $\mu_4 = 26$.

(Anna, Nov. 2004)

Solution. From the given data, we have

$$(i) \quad E(X - 4) = 1$$

- (ii) $E(X - 4)^2 = 4$
- (iii) $E(X - 4)^3 = 10$
- (iv) $E(X - 4)^4 = 45$

Note that

$$\mu = E(X) = E[\overline{X - 4} + 4] = E[X - 4] + 4 = 1 + 4 = 5$$

Next, we note that

$$\text{Var}(X) = E[\overline{X - 4} - 1]^2 = E[X - 4]^2 - 2E[X - 4] + 1 = 4 - 2 + 1 = 3$$

Next, we find that

$$\mu_3 = E[\overline{X - 4} - 1]^3 = E[(X - 4)^3] - 3E[(X - 4)^2] + 3E[X - 4] - 1$$

Using the given data, we have

$$\mu_3 = 10 - 3(4) + 3(1) - 1 = 0$$

Finally, we find that

$$\mu_4 = E[\overline{X - 4} - 1]^4 = E[(X - 4)^4] - 4E[(X - 4)^3] + 6E[(X - 4)^2] - 4E[X - 4] + 1$$

Using the given data, we have

$$\mu_4 = 45 - 4(10) + 6(4) - 4(1) + 1 = 26$$

◆

EXAMPLE 2.72. Show that for the symmetrical density function

$$f(x) = \frac{2a}{\pi} \left[\frac{1}{a^2 + x^2} \right], \quad -a \leq x \leq a$$

the odd central moments are zero, and the second and fourth central moments are given by

$$\mu_2 = \frac{a^2(4 - \pi)}{\pi}, \quad \mu_4 = a^4 \left[1 - \frac{8}{3\pi} \right]$$

Solution. First, we compute μ'_1 , the mean of X . We have

$$\mu'_1 = E(X) = \int_{-a}^a x \frac{2a}{\pi} \left[\frac{1}{a^2 + x^2} \right] dx = 0$$

since the integrand is an odd function of x . Since $\mu = 0$, the central moments for the distribution are the same as the raw moments, i.e. moments about the origin.

Next, we show that the odd central moments are zero. For any non-negative integer k , we have

$$\mu_{2k+1} = \mu'_{2k+1} = E(X^{2k+1}) = \int_{-a}^a x^{2k+1} \frac{2a}{\pi} \left[\frac{1}{a^2 + x^2} \right] dx = 0$$

since the integrand is an odd function of x .

The second central moment is given by

$$\mu_2 = \mu_2' = E(X^2) = \int_{-a}^a x^2 \frac{2a}{\pi} \left[\frac{1}{a^2 + x^2} \right] dx = 2 \frac{2a}{\pi} \int_0^a \frac{x^2}{a^2 + x^2} dx$$

since the integrand is an even function of x .

Note that

$$\mu_2 = \frac{4a}{\pi} \int_0^a \left[1 - \frac{a^2}{a^2 + x^2} \right] dx = \frac{4a}{\pi} \int_0^a \left[1 - \frac{1}{1 + \left(\frac{x}{a}\right)^2} \right] dx$$

Integrating, we get

$$\begin{aligned} \mu_2 &= \frac{4a}{\pi} \left[x - \frac{\tan^{-1}\left(\frac{x}{a}\right)}{\frac{1}{a}} \right]_0^a \\ &= \frac{4a}{\pi} \left[a - a \frac{\pi}{4} \right] = \frac{4a^2}{\pi} \left[1 - \frac{\pi}{4} \right] \end{aligned}$$

Simplifying, we get

$$\mu_2 = \frac{4a^2}{\pi} \frac{4 - \pi}{4} = \frac{a^2(4 - \pi)}{\pi}$$

The fourth central moment is given by

$$\mu_4 = \mu_4' = E(X^4) = \int_{-a}^a x^4 \frac{2a}{\pi} \left[\frac{1}{a^2 + x^2} \right] dx = 2 \frac{2a}{\pi} \int_0^a \frac{x^4}{a^2 + x^2} dx$$

since the integrand is an even function of x .

Note that

$$\mu_4 = \frac{4a}{\pi} \int_0^a \left[x^2 - a^2 + \frac{a^4}{a^2 + x^2} \right] dx$$

Integrating, we get

$$\begin{aligned} \mu_4 &= \frac{4a}{\pi} \left[\frac{x^3}{3} - a^2 x + a^2 \frac{\tan^{-1} \frac{x}{a}}{\frac{1}{a}} \right]_0^a \\ &= \frac{4a}{\pi} \left[\frac{a^3}{3} - a^3 + a^3 \frac{\pi}{4} \right] = \frac{4a^4}{\pi} \left[\frac{\pi}{4} - \frac{2}{3} \right] \end{aligned}$$

Simplifying, we get

$$\mu_4 = a^4 \left[1 - \frac{8}{3\pi} \right]$$

◆

EXAMPLE 2.73. Let X be a random variable with the Laplace distribution

$$f(x) = \frac{1}{2} e^{-|x|}, \quad -\infty < x < \infty$$

Show that X has mean 0, variance 2 and the mean deviation about the mean is 1.

Solution. The mean of X is given by

$$\mu'_1 = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} x \frac{1}{2} e^{-|x|} dx = 0$$

since the integrand is an odd function of x . Hence, X has mean 0.

The second raw moment is given by

$$\mu'_2 = E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-\infty}^{\infty} x^2 \frac{1}{2} e^{-|x|} dx = 2 \int_0^{\infty} x^2 \frac{1}{2} e^{-x} dx$$

since the integrand is an even function of x . Hence,

$$\mu'_2 = \int_0^{\infty} x^2 e^{-x} dx = \Gamma(3) = 2$$

Thus, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = 2 - 0 = 2$$

Next, the mean deviation about the mean is given by

$$E|X - \mu| = E|X| = \int_{-\infty}^{\infty} |x| f(x) dx = \int_{-\infty}^{\infty} |x| \frac{1}{2} e^{-|x|} dx = 2 \int_0^{\infty} |x| \frac{1}{2} e^{-x} dx$$

since the integrand is an even function of x . Hence,

$$E|X - \mu| = \int_0^{\infty} x e^{-x} dx = \Gamma(2) = 1$$

◆

EXAMPLE 2.74. Let X be a random variable X with the probability density function

$$f(x) = \begin{cases} \frac{1}{2} \sin x & \text{if } 0 \leq x \leq \pi \\ 0 & \text{otherwise} \end{cases}$$

Show that the mean, median and mode of X coincide.

Solution. The mean of X is given by

$$\mu'_1 = E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\pi} x \frac{1}{2} \sin x dx = \frac{1}{2} \int_0^{\pi} x d(-\cos x)$$

Using integration by parts, we have

$$\mu'_1 = \frac{1}{2} \left[[x(-\cos x)]_0^{\pi} - \int_0^{\pi} (-\cos x)(1) dx \right] = \frac{1}{2} [\pi + [\sin x]_0^{\pi}] = \frac{1}{2} [\pi + 0]$$

Hence, the mean of X is given by $\mu = \mu'_1 = \frac{\pi}{2}$.

The median M of X is given by the relation

$$\int_{-\infty}^M f(x) dx = \frac{1}{2} \Rightarrow \int_0^M \frac{1}{2} \sin x dx = \frac{1}{2}$$

i.e.

$$\int_0^M \sin x dx = 1 \Rightarrow [-\cos x]_0^M = 1 \Rightarrow -\cos M + 1 = 1 \Rightarrow \cos M = 0$$

The only value of M satisfying $\cos M = 0$ and $0 \leq M \leq \pi$ is $M = \frac{\pi}{2}$. Hence, the median of X is $M = \frac{\pi}{2}$.

The mode of X is the point c in $0 \leq x \leq \pi$ where f attains its maximum. We find that

$$f'(x) = \frac{1}{2}(\cos x) \text{ and } f''(x) = \frac{1}{2}(-\sin x)$$

So, $f'(x) = 0 \Rightarrow x = \frac{\pi}{2}$ and note that $f''\left(\frac{\pi}{2}\right) = -\frac{1}{2} < 0$.

Thus, the mode of x is $c = \frac{\pi}{2}$. Since $\mu = M = c$, the mean, median and mode of X coincide. ♦

PROBLEM SET 2.7

1. The first four moments of a distribution about $X = 2$ are 1, 10, 28 and 60, respectively. Show that the mean is 3, variance is 9, $\mu_3 = 0$ and $\mu_4 = 5$.
2. X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} 1 - \frac{x}{2} & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $E[X^r] = \frac{2^{r+1}}{(r+1)(r+2)}$.
- (b) Use this result to evaluate $E[(3X + 2)^2]$.

3. X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} Kx(1-x) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The value of K .
 - (b) The mean and variance of X .
 - (c) The r^{th} moment of X .
4. X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} Kx^2 & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of K .
- (b) Find the mean and variance of X .
- (c) Show that all the odd order central moments vanish and the even order central moments are given by

$$\mu_{2n} = \frac{3}{2n+3} \text{ where } n = 0, 1, 2, \dots$$

- (d) Find β_1 and β_2 .

5. X is a continuous random variable with the uniform density function

$$f(x) = \begin{cases} K & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of K .
- (b) Find the mean and variance of X .
- (c) Show that all the odd order central moments vanish and the even order central moments are given by

$$\mu_{2n} = \frac{1}{2n+1} \text{ where } n = 0, 1, 2, \dots$$

- (d) Find β_1 and β_2 .

6. X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} xe^{-x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) The mean and variance of X .
- (b) The first four central moments of X .
- (c) β_1 and β_2 .
- (d) The mode of X .

7. X is a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{\ln 2} \cdot \frac{1}{x} & \text{for } 1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find

- (a) The mean and variance of X .
- (b) μ'_r .
- (c) The median of X .

2.8 MOMENT GENERATING FUNCTION

Definition 2.18. The *moment generating function* (MGF) of a random variable X is denoted by $M_X(t)$ and is defined as

$$M_X(t) = E(e^{tX}), \quad t \in \mathbb{R}$$

wherever this expectation exists. If X is a continuous random variable with probability density function f , then

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

If X is a discrete random variable with mass points $\{x_i\}$ and with probability mass function f , then

$$M_X(t) = \sum_i e^{tx_i} f(x_i)$$

As the name indicates, the moment generating function of a random variable X generates all its (raw) moments.

Theorem 2.20. Let X be a random variable with moment generating function $M_X(t)$. For any positive integer r , the r^{th} raw moment of X is given by

$$\mu'_r = M_X^{(r)}(0) = \frac{d^r}{dt^r} M_X(t)|_{t=0}$$

Proof. By definition, we have

$$\begin{aligned} M_X(t) &= E([e^{tX}]) = E\left[1 + tX + \frac{t^2}{2!}X^2 + \cdots + \frac{t^r}{r!}X^r + \cdots\right] \\ &= 1 + tE(X) + \frac{t^2}{2!}E(X^2) + \cdots + \frac{t^r}{r!}E(X^r) + \cdots \\ &= 1 + t\mu'_1 + \frac{t^2}{2!}\mu'_2 + \cdots + \frac{t^r}{r!}\mu'_r + \cdots \end{aligned}$$

Hence, it follows immediately that

$$\mu'_r = M_X^{(r)}(0) = \frac{d^r}{dt^r} M_X(t)|_{t=0}$$

◆

Remark 2.15. By Theorem 2.20, it follows that

$$\mu'_r = \text{Coefficient of } \frac{t^r}{r!} \text{ in the series expansion of } M_X(t)$$

EXAMPLE 2.75. Let X have the probability mass function

$$p(k) = \begin{cases} \frac{6}{\pi^2} \frac{1}{k^2} & \text{if } k = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Find the moment generating function of X

(Anna, April 2003)

Solution. By definition, we have

$$M_X(t) = E[e^{tX}] = \frac{6}{\pi^2} \sum_{k=1}^{\infty} \frac{e^{tk}}{k^2}$$

The series is not convergent (by D'Alembert's Ratio Test) for every $t > 0$. Hence, the moment generating function of X does not exist. ♦

EXAMPLE 2.76. Let X have the exponential density function

$$f(x) = \begin{cases} \theta e^{-\theta x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find $M_X(t)$. Does $M_X(t)$ exist for all values of t ?

Solution. By definition, we have

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \int_{x=0}^{\infty} e^{tx} \cdot \theta e^{-\theta x} dx \\ &= \theta \int_0^{\infty} e^{-(\theta-t)x} dx \end{aligned}$$

The indefinite integral on the right hand side of above converges only when $|t| < \theta$. Hence, $M_X(t)$ exists only for values of t such that $|t| < \theta$. In this case, we have ♦

$$M_X(t) = \theta \left[-\frac{e^{-(\theta-t)x}}{\theta-t} \right]_0^{\infty} = \frac{\theta}{\theta-t}$$

EXAMPLE 2.77. Let X have the Poisson distribution, i.e.

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!} \text{ for } k = 0, 1, 2, \dots$$

Find the MGF of X .

Solution. By definition, we have

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \sum_{k=0}^{\infty} e^{tk} P(X=k) = e^{-\lambda} \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{-\lambda(1-e^t)} \end{aligned}$$

which exists for all values of t . ◆

Next, we prove some properties of the moment generating functions.

Theorem 2.21. If $M_X(t)$ is the moment generating function of the random variable X , then $M_X(ct)$ is the moment generating function of the random variable cX , where c is any constant, i.e.

$$M_{cX}(t) = M_X(ct)$$

Proof. Let $Y = cX$. By definition, we have

$$M_{cX}(t) = M_Y(t) = E[e^{tY}] = E[e^{tcX}] = E[e^{(ct)X}] = M_X(ct)$$
◆

Theorem 2.22. (*Effect of Change of Origin and Scale*) If we transform the variable X to a new variable Y by changing both the origin and the scale in X as

$$Y = \frac{X - a}{h}$$

then

$$M_Y(t) = e^{-\frac{at}{h}} M_X\left(\frac{t}{h}\right)$$

Proof. By definition, we have

$$\begin{aligned} M_Y(t) &= E[e^{tY}] = E\left[e^{t \frac{X-a}{h}}\right] = E\left[e^{\frac{tX}{h}} \cdot e^{-\frac{ta}{h}}\right] = e^{-\frac{ta}{h}} \cdot E\left[e^{\frac{t}{h}X}\right] \\ &= e^{-\frac{ta}{h}} M_X\left(\frac{t}{h}\right) \end{aligned}$$
◆

We state the following uniqueness result without proof.

Theorem 2.23. The moment generating function uniquely determines the distribution function. In other words, if X and Y are any two random variables, then

$$M_X(t) = M_Y(t) \iff F_X = F_Y$$

Remark 2.16. By Theorem 2.23, it follows that corresponding to a given probability distribution, there is a unique moment generating function (provided it exists) and corresponding to a given moment generating function, there is a unique probability distribution. Hence, if $M_X(t) = M_Y(t)$, then X and Y are identically distributed, i.e. they have the same probability density function.

EXAMPLE 2.78. The moment generating function of a random variable X is given by $M_X(t) = e^{3(e^t-1)}$. Find $P(X = 1)$.

(Anna, Model, 2003)

Solution. We can rewrite the given moment generating function of X as

$$M_X(t) = e^{-3(1-e^t)}$$

By Example 2.77, we know that the moment generating function of the Poisson distribution with parameter λ is given by

$$M_Y(t) = e^{-\lambda(1-e^t)}$$

By the uniqueness theorem for moment generating functions (see Theorem 2.23), it follows that the given random variable X also follows the Poisson distribution with parameter $\lambda = 3$. Hence, we know that

$$P(X = k) = e^{-3} \frac{3^k}{k!} \text{ for } k = 0, 1, 2, \dots$$

In particular, we have $P(X = 1) = e^{-3} \frac{3}{1!} = 3e^{-3} = 0.1494$. ◆

EXAMPLE 2.79. A perfect coin is tossed twice. Find the moment generating function of the number of heads. Hence, find the mean and variance.

Solution. The probability distribution of X , the number of heads, is given as

x	0	1	2
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

By definition, the MGF of X is given by

$$M_X(t) = E[e^{tX}] = \sum_x e^{tx} P(X = x) = \left[e^{t0} \frac{1}{4} \right] + \left[e^{t1} \frac{1}{2} \right] + \left[e^{t2} \frac{1}{4} \right]$$

Simplifying, we get

$$M_X(t) = \frac{1}{4} [1 + 2e^t + e^{2t}] = \frac{1}{4} (1 + e^t)^2$$

Next, note that

$$M'_X(t) = \frac{1}{2} (1 + e^t) e^t = \frac{1}{2} (e^t + e^{2t}) \text{ and } M''_X(t) = \frac{1}{2} (e^t + 2e^{2t})$$

Hence, it follows that

$$\text{Mean}(X) = \mu'_1 = M'_X(0) = 1 \text{ and } \mu'_2 = M''_X(0) = \frac{3}{2}$$

The variance of X is given by

$$\text{Var}(X) = \mu_2 = \mu'_2 - \mu_1'^2 = \frac{3}{2} - 1 = \frac{1}{2}$$

◆

EXAMPLE 2.80. Find the moment generating function for the distribution where

$$f(x) = \begin{cases} \frac{2}{3} & \text{at } x = 1 \\ \frac{1}{3} & \text{at } x = 2 \\ 0 & \text{otherwise} \end{cases}$$

(Anna, Nov. 2006)

Solution. Note that X is a discrete random variable with mass points $x = 1, 2$. By definition, we have

$$M_X(t) = E[e^{tX}] = \sum_x e^{tx} f(x) = e^t P(X=1) + e^{2t} P(X=2) = \frac{2}{3}e^t + \frac{1}{3}e^{2t}$$

Thus, the moment generating function of X is given by

$$M_X(t) = \frac{1}{3} [2e^t + e^{2t}]$$

◆

EXAMPLE 2.81. Find the moment generating function of the discrete random variable X with the probability mass function

$$f(x) = \frac{1}{n} \quad \text{for } x = 1, 2, \dots, n$$

Solution. By definition, the MGF of X is given by

$$\begin{aligned} M_X(t) &= E[e^{tX}] = \sum_{x=1}^n e^{tx} \frac{1}{n} = \frac{1}{n} [e^t + e^{2t} + \dots + e^{nt}] \\ &= \frac{e^t}{n} [1 + e^t + (e^t)^2 + \dots + (e^t)^{n-1}] = \frac{e^t}{n} \frac{1 - e^{nt}}{1 - e^t} \end{aligned}$$

◆

EXAMPLE 2.82. Find the moment generating function of the random variable with the probability law $P(X=x) = q^{x-1}p, x=1, 2, 3, \dots$. Also find the mean and variance.

(Anna, Nov. 2006)

Solution. By definition, the MGF of X is given by

$$\begin{aligned} M_X(t) &= E[e^{tX}] = \sum_{x=1}^{\infty} e^{tx} q^{x-1} p \\ &= e^t p + e^{2t} qp + e^{3t} q^2 p + \dots + e^{kt} q^{k-1} p + \dots \\ &= pe^t [1 + (qe^t) + (qe^t)^2 + \dots] = pe^t \left[\frac{1}{1 - qe^t} \right] = \frac{pe^t}{1 - qe^t} \end{aligned}$$

which converges provided $qe^t < 1$ or equivalently that $t < \log_e(\frac{1}{q})$.

Hence, the MGF of X is given by $M_X(t) = \frac{pe^t}{1 - qe^t}$ for values of t satisfying $t < \log_e(\frac{1}{q})$.

Next, we find the mean and variance of X . A simple calculation shows that

$$M'_X(t) = \frac{pe^t}{(1-qe^t)^2} \quad \text{and} \quad M''_X(t) = \frac{pe^t + pqe^{2t}}{(1-qe^t)^3}$$

Hence, we find that

$$\text{Mean}(X) = \mu'_1 = M'_X(0) = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}$$

and

$$\mu'_2 = M''_X(0) = \frac{p + pq}{(1-q)^3} = \frac{p(1+q)}{p^3} = \frac{1+q}{p^2}$$

Thus, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{1+q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$$

◆

EXAMPLE 2.83. Show that for the uniform distribution $f(x) = \frac{1}{2a}$, $-a < x < a$, the moment generating function about the origin is $\frac{\sinh at}{at}$.

(Anna, Nov. 2006)

Solution. By definition, the MGF of X is given by

$$M_X(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-a}^a e^{tx} \frac{1}{2a} dx$$

Integrating, we get

$$M_X(t) = \frac{1}{2a} \left[\frac{e^{tx}}{t} \right]_{-a}^a = \frac{1}{2at} [e^{at} - e^{-at}] = \frac{1}{at} \left[\frac{e^{at} - e^{-at}}{2} \right] = \frac{1}{at} \sinh at$$

Thus, the MGF of X about the origin is $M_X(t) = \frac{\sinh at}{at}$

◆

EXAMPLE 2.84. A random variable X has the density function

$$f(x) = \begin{cases} \theta e^{-\theta x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the moment generating function and the mean and variance.

(Anna, Nov. 2006)

Solution. By Example 2.76, we know that the MGF of X is given by

$$M_X(t) = \frac{\theta}{\theta - t} \text{ which exists for } t < \theta$$

Note that we can write $M_X(t)$ as

$$M_X(t) = \frac{1}{1 - \frac{t}{\theta}} = 1 + \left(\frac{t}{\theta}\right) + \left(\frac{t}{\theta}\right)^2 + \cdots + \left(\frac{t}{\theta}\right)^r + \cdots$$

Thus, it follows that

$$\mu'_r = \text{Coefficient of } \frac{t^r}{r!} \text{ in } M_X(t) = \frac{r!}{\theta^r}$$

In particular, we have

$$\mu'_1 = \frac{1}{\theta} \text{ and } \mu'_2 = \frac{2}{\theta^2}$$

Hence,

$$\text{Mean}(X) = \mu = \mu'_1 = \frac{1}{\theta}$$

and

$$\text{Var}(X) = \sigma^2 = \mu'_2 - \mu_1'^2 = \frac{2}{\theta^2} - \frac{1}{\theta^2} = \frac{1}{\theta^2}$$

◆

EXAMPLE 2.85. Find the MGF of the random variable X , with probability density function

$$f(x) = \begin{cases} x & \text{for } 0 \leq x \leq 1 \\ 2-x & \text{for } 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Also find μ'_1, μ'_2 .

(Anna, Model; April 2003)

Solution. By definition, the MGF of X is given by

$$M_X(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_0^1 e^{tx} x dx + \int_1^2 e^{tx} (2-x) dx = I_1 + I_2 (\text{say})$$

We find that

$$I_1 = \int_0^1 e^{tx} x dx = \int_0^1 x d\left(\frac{e^{tx}}{t}\right)$$

Using integration by parts, we get

$$I_1 = \left[x \left(\frac{e^{tx}}{t} \right) \right]_0^1 - \int_0^1 \left(\frac{e^{tx}}{t} \right) (1) dx = \left[\frac{e^t}{t} \right] - \frac{1}{t} \left[\frac{e^{tx}}{t} \right]_0^1 = \frac{e^t}{t} - \frac{1}{t^2} [e^t - 1]$$

Simplifying, we get

$$I_1 = \frac{(t-1)e^t + 1}{t^2}$$

Next, we find that

$$I_2 = \int_1^2 e^{tx}(2-x) dx = \int_0^1 e^{t(2-y)} y dy = e^{2t} \int_0^1 e^{-ty} y dy = e^{2t} \int_0^1 y d\left(\frac{e^{-ty}}{-t}\right)$$

Using integration by parts, we get

$$I_2 = e^{2t} \left[\left[y \left(\frac{e^{-ty}}{-t} \right) \right]_0^1 - \int_0^1 \left(\frac{e^{-ty}}{-t} \right) (1) dy \right] = e^{2t} \left[\frac{e^{-t}}{-t} - \frac{1}{t} \left[\frac{e^{-ty}}{-t} \right]_0^1 \right]$$

Simplifying, we get

$$I_2 = \frac{e^{2t} - (t+1)e^t}{t^2}$$

Hence, the MGF of X is given by

$$M_X(t) = I_1 + I_2 = \frac{e^{2t} - 2e^t + 1}{t^2} = \left[\frac{e^t - 1}{t} \right]^2$$

which exists for all values of t .

To find μ'_1, μ'_2 , we note the following:

$$e^t - 1 = \left(1 + \frac{t}{1!} + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots \right) - 1 = t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots$$

which shows that

$$\frac{e^t - 1}{t} = 1 + \frac{t}{2} + \frac{t^2}{6} + \frac{t^3}{24} + \dots$$

Hence, it is immediate that

$$M_X(t) = \left[\frac{e^t - 1}{t} \right]^2 = 1 + t + \left(\frac{1}{4} + \frac{1}{3} \right) t^2 + \dots$$

Thus, it follows that

$$\mu'_1 = \text{Coefficient of } \frac{t}{1!} \text{ in } M_X(t) = 1$$

and

$$\mu'_2 = \text{Coefficient of } \frac{t^2}{2!} \text{ in } M_X(t) = \frac{7}{6}$$

◆

EXAMPLE 2.86. If X has the probability density function (*Laplace distribution*)

$$f(x) = \frac{1}{2} e^{-|x|} \quad \text{for } -\infty < x < \infty$$

show that the MGF of X is given by $M_X(t) = \frac{1}{1-t^2}$, which exists for $|t| < 1$. Hence, find the mean and variance of X .

Solution. By definition, the MGF of X is given by

$$M_X(t) = E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^0 e^{tx} \frac{1}{2} e^{-|x|} dx + \int_0^{\infty} e^{tx} \frac{1}{2} e^{-|x|} dx$$

Noting that $|x| = -x$ for $x < 0$ and $|x| = x$ for $x \geq 0$, we have

$$\begin{aligned} M_X(t) &= \frac{1}{2} \int_{-\infty}^0 e^x e^{tx} dx + \frac{1}{2} \int_0^{\infty} e^{-x} e^{tx} dx = \frac{1}{2} \int_0^{\infty} e^{-y} e^{-ty} dy + \frac{1}{2} \int_0^{\infty} e^{-x} e^{tx} dx \\ &= \frac{1}{2} \int_0^{\infty} e^{-(1+t)y} dy + \frac{1}{2} \int_0^{\infty} e^{-(1-t)x} dx \end{aligned}$$

which converges for $1+t > 0$ and $1-t > 0$, i.e. $-1 < t < 1$, or $|t| < 1$

Integrating, we get

$$\begin{aligned} M_X(t) &= \frac{1}{2} \left[\frac{-e^{-(1+t)y}}{1+t} \right]_0^{\infty} + \frac{1}{2} \left[\frac{-e^{-(1-t)x}}{1-t} \right]_0^{\infty} \\ &= \frac{1}{2} \left[0 + \frac{1}{1+t} \right] + \frac{1}{2} \left[0 + \frac{1}{1-t} \right] = \frac{1}{2} \left[\frac{1}{1+t} + \frac{1}{1-t} \right] = \frac{1}{1-t^2} \end{aligned}$$

which converges for $|t| < 1$.

When $|t| < 1$, we can express $M_X(t)$ as

$$M_X(t) = 1 + (t^2) + (t^2)^2 + (t^2)^3 + \dots = 1 + t^2 + t^4 + t^6 + \dots$$

Thus, it follows that

$$\mu'_1 = \text{Coefficient of } \frac{t}{1!} \text{ in } M_X(t) = 0$$

and

$$\mu'_2 = \text{Coefficient of } \frac{t^2}{2!} \text{ in } M_X(t) = 2$$

Thus, the mean of X is $\mu'_1 = 0$ and the variance of X is given by

$$\text{Var}(X) = \mu_2 = \mu'_2 - \mu'^2_1 = 2 - 0 = 2$$

◆

EXAMPLE 2.87. A continuous random variable X has the probability density function given by

$$f(x) = Ce^{-|x|}, \quad -\infty < x < \infty$$

Find the value of C and the moment generating function of X .

(Anna, May 2007)

Solution. The given random variable X follows the Laplace distribution. As worked out in Example 2.33, $C = \frac{1}{2}$. Next, as in Example 2.86, the MGF of X is given by

$$M_X(t) = \frac{1}{1-t^2}$$

which converges for $|t| < 1$. ♦

EXAMPLE 2.88. Find the MGF of the random variable whose moments are

$$\mu'_r = (r+1)!2^r, \quad r = 1, 2, 3, \dots$$

Solution. By definition, the MGF of X is given by

$$\begin{aligned} M_X(t) &= E[e^{tX}] = 1 + t\mu'_1 + \frac{t^2}{2!}\mu'_2 + \dots + \frac{t^r}{r!}\mu'_r + \dots \\ &= 1 + t(2!2) + \frac{t^2}{2!}3!2^2 + \frac{t^3}{3!}4!2^3 + \dots + \frac{t^r}{r!}(r+1)!2^r + \dots \\ &= 1 + 2\theta + 3\theta^2 + 4\theta^3 + \dots + (r+1)\theta^r + \dots \quad \text{where } \theta = 2t \end{aligned}$$

Hence, we have

$$M_X(t) = (1-\theta)^{-2} = (1-2t)^{-2}$$

which converges provided $\theta < 1$ or equivalently that $t < \frac{1}{2}$. ♦

PROBLEM SET 2.8

1. Let X be a discrete random variable with the probability mass function

$$f(x) = \frac{1}{16} {}^4C_x \quad \text{for } x = 0, 1, 2, 3, 4$$

Find the moment generating function of X and use it to determine the mean and variance of X .

2. Let X be a discrete random variable with the probability mass function

$$f(x) = 2 \left(\frac{1}{3} \right)^x \quad \text{for } x = 1, 2, 3, \dots$$

Find the moment generating function of X and use it to determine the mean and variance of X .

3. Let X be a continuous random variable with the uniform distribution

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

Find the moment generating function of X and use it to determine the mean and variance of X .

4. Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} xe^{-x} & \text{if } 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find the moment generating function of X and use it to determine the mean and variance of X .

5. Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{2}x^2e^{-x} & \text{if } 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find the moment generating function of X and use it to determine the mean and variance of X .

6. Let X be a continuous random variable with the Laplace distribution

$$f(x) = \frac{1}{2\alpha} e^{-\frac{|x-\alpha|}{\alpha}} \text{ for } -\infty < x < \infty$$

where $\alpha > 0$ is a constant. Find the moment generating function of X and use it to determine the mean and variance of X .

7. If the moment generating function of X is given by

$$M_X(t) = \frac{1}{1-t^2} \text{ for } -1 < t < 1$$

find the moment generating function of $Y = 3X$.

8. If the moment generating function of X is given by

$$M_X(t) = \frac{1}{4} (1 + e^t)^2$$

find the moment generating function of $Y = \frac{X-4}{2}$.

9. If the moment generating function of X is given by

$$M_X(t) = \frac{1}{2t} [e^{2t} - 1]$$

find the moment generating function of $Y = \frac{X-6}{2}$.

10. If the moment generating function of X is given by

$$M_X(t) = \frac{1}{1-t^2} \text{ where } |t| < 1$$

find the moment generating function of $Y = \frac{X-4}{4}$.

2.9 CHARACTERISTIC FUNCTION

In this section, we study a generating function called the *characteristic function*, which exists for all distributions.

Definition 2.19. If X is any random variable, the complex-valued function ϕ_X defined on $\mathbb{R}$ by

$$\phi_X(t) = E(e^{itX}) \quad t \in \mathbb{R}$$

is called the *characteristic function (CF)* of the random variable X .

If X is a continuous random variable with probability density function f , then

$$\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

If X is a discrete random variable with mass points $\{x_i\}$ and probability mass function f , then

$$\phi_X(t) = \sum_i e^{itx_i} f(x_i)$$

Theorem 2.24. The characteristic function is bounded by 1, i.e.

$$|\phi_X(t)| \leq 1 \quad \text{for all } t$$

Proof. If X is a continuous random variable, then we have

$$|\phi_X(t)| = \left| \int_{-\infty}^{\infty} e^{itx} f(x) dx \right| \leq \int_{-\infty}^{\infty} |e^{itx}| f(x) dx = \int_{-\infty}^{\infty} f(x) dx = 1$$

If X is a discrete random variable, then we have

$$|\phi_X(t)| = \left| \sum_i e^{itx_i} f(x_i) \right| \leq \sum_i |e^{itx_i}| f(x_i) = \sum_i f(x_i) = 1$$

◆

The following result follows immediately from the fact that $\phi_X(t)$ is absolutely convergent for all values of t .

Corollary 2.5. The characteristic function $\phi_X(t)$ exists for all values of t .

Theorem 2.25. Let $\phi_X(t)$ be the characteristic function of a random variable X . Then $\phi_X(0) = 1$.

Proof. By definition, $\phi_X(0) = E(e^{i \cdot 0 \cdot X}) = E(1) = 1$.

◆

Theorem 2.26. $\phi_X(t)$ and $\phi_X(-t)$ are conjugate functions, i.e. $\overline{\phi_X(t)} = \phi_X(-t)$.

Proof. By definition, we have

$$\phi_X(t) = E(e^{itX}) = E(\cos tX) + iE(\sin tX)$$

Hence, it follows that

$$\overline{\phi_X(t)} = (\cos tX) - iE(\sin tX)$$

Note also that

$$\phi_X(-t) = E(\cos -tX) + iE(\sin -tX) = E(\cos tX) - iE(\sin tX)$$

Hence, it follows that $\overline{\phi_X(t)} = \phi_X(-t)$.

◆

Theorem 2.27. If the random variable X has a symmetrical distribution about the origin, i.e.

$$f(-x) = f(x)$$

then $\phi_X(t)$ is real-valued and an even function of t .

Proof. We prove this result only for the continuous case. (The discrete case is left as an exercise for the reader.) By definition, we have

$$\begin{aligned}\phi_X(t) &= \int_{-\infty}^{\infty} e^{itx} f(x) dx = \int_{-\infty}^{\infty} e^{-ity} f(-y) dy \quad (\text{by putting } x = -y) \\ &= \int_{-\infty}^{\infty} e^{-ity} f(y) dy = \phi_X(-t)\end{aligned}$$

which shows that ϕ_X is an even function of t . From the preceding result, it follows that

$$\phi_X(t) = \phi_X(-t) = \overline{\phi_X(t)}$$

which shows that $\phi_X(t)$ is real-valued. ◆

The characteristic function generates all the raw moments, as stated in the next theorem.

Theorem 2.28. If X is a random variable with characteristic function $\phi_X(t)$ and if $\mu'_r = E(X^r)$, then

$$\mu'_r = \frac{\phi_X^{(r)}(0)}{i^r} = \frac{1}{i^r} \left[\frac{d^r}{dt^r} \phi_X(t) \right]_{t=0}$$

Proof. We prove this result for the continuous case. (The discrete case is left as an exercise for the reader.) By definition, we have

$$\phi_X(t) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$$

Differentiating under the integral sign r times with respect to t , we get

$$\frac{d^r}{dt^r} \phi_X(t) = \int_{-\infty}^{\infty} (ix)^r e^{itx} f(x) dx = i^r \int_{-\infty}^{\infty} x^r e^{itx} f(x) dx$$

Putting $t = 0$ on both sides of the above, we have

$$\phi_X^{(r)}(0) = i^r \int_{-\infty}^{\infty} x^r f(x) dx = E(X^r) = \mu'_r$$

Hence, it follows immediately that

$$\mu'_r = \frac{1}{i^r} \phi_X^{(r)}(0) \quad \text{◆}$$

Theorem 2.29. (*Effect of Change of Origin and Scale*) If we transform the variable X to a new variable Y by changing both the origin and the scale in X as

$$Y = \frac{X - a}{h}$$

then

$$\phi_Y(t) = e^{-\frac{iat}{h}} \phi_X\left(\frac{t}{h}\right)$$

Proof. By definition, we have

$$\begin{aligned} \phi_Y(t) &= E(e^{iYt}) = E\left(e^{\frac{i(X-a)t}{h}}\right) = E\left(e^{-\frac{iat}{h}} \cdot e^{\frac{iXt}{h}}\right) \\ &= e^{-\frac{iat}{h}} E\left(e^{\frac{iXt}{h}}\right) = e^{-\frac{iat}{h}} \phi_X\left(\frac{t}{h}\right) \end{aligned}$$

◆

We state the following results without proof.

Theorem 2.30. ϕ_X is uniformly continuous in $\mathbb{R}$.

Theorem 2.31. The characteristic function uniquely determines the distribution function. In other words, if X and Y are any random variables, then

$$\phi_X(t) = \phi_Y(t) \iff F_X = F_Y$$

Theorem 2.32. (*Inversion Theorem*) Let X be a continuous random variable with the distribution function F_X , probability density function f_X and characteristic function $\phi_X(t)$. If $a + h$ and $a - h$ ($h > 0$) are the continuity points of F_X , then

$$F_X(a + h) - F_X(a - h) = \lim_{T \rightarrow \infty} \int_{-\infty}^{\infty} \frac{\sinh t}{t} e^{-ita} \phi_X(t) dt \quad (2.14)$$

The integral in Eq. (2.14) can be inverted and the probability density function f_X can be expressed in terms of $\phi_X(t)$ by the integral

$$\frac{d}{dx} F(x) = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi_X(t) dt \quad (2.15)$$

The formula, Eq. (2.15), is known as the *inversion theorem*.

EXAMPLE 2.89. Find the characteristic function of the discrete random variable X whose probability function is given below.

x	-1	1
$f(x)$	$\frac{1}{2}$	$\frac{1}{2}$

Solution. By definition, the characteristic function of X is given by

$$\begin{aligned}\phi_X(t) &= E[e^{itX}] = \sum_x e^{itx} f(x) = e^{-it} \frac{1}{2} + e^{it} \frac{1}{2} \\ &= \frac{1}{2} [e^{-it} + e^{it}] = \frac{1}{2} [2 \cos t] = \cos t\end{aligned}$$

◆

EXAMPLE 2.90. Find the characteristic function of the uniform distribution

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

Hence, find the mean and variance.

Solution. By definition, the characteristic function of X is given by

$$\phi_X(t) = E[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f(x) dx = \int_a^b e^{itx} \frac{1}{b-a} dx$$

Integrating, we get

$$\phi_X(t) = \frac{1}{b-a} \left[\frac{e^{itx}}{it} \right]_a^b = \frac{1}{b-a} \left[\frac{e^{itb}}{it} - \frac{e^{ita}}{it} \right] = \frac{e^{itb} - e^{ita}}{(b-a)it}$$

It is easy to see that

$$e^{itb} - e^{ita} = it(b-a) + \frac{i^2 t^2 (b^2 - a^2)}{2} + \frac{i^3 t^3 (b^3 - a^3)}{3!} + \dots$$

from which it follows that

$$\phi_X(t) = \frac{e^{itb} - e^{ita}}{(b-a)it} = 1 + \frac{it(b+a)}{2} + \frac{i^2 t^2 (b^2 + ab + a^2)}{3!} + \dots$$

Note that

$$\mu'_1 = \text{Coefficient of } \frac{it}{1!} \text{ in } \phi_X(t) = \frac{b+a}{2}$$

and

$$\mu'_2 = \text{Coefficient of } \frac{(it)^2}{2!} \text{ in } \phi_X(t) = \frac{b^2 + ab + a^2}{3}$$

Thus, the mean of X is given by

$$\mu = \mu'_1 = \frac{b+a}{2}$$

and the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{b^2 + ab + a^2}{3} - \frac{(a+b)^2}{4}$$

Simplifying, we get

$$\text{Var}(X) = \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}$$

◆

EXAMPLE 2.91. Find the characteristic function of the exponential distribution

$$f(x) = \begin{cases} \theta e^{-\theta x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Hence, find the mean and variance.

Solution. By definition, the characteristic function of X is given by

$$\begin{aligned} \phi_X(t) &= E[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f(x) dx = \int_0^{\infty} e^{itx} \theta e^{-\theta x} dx \\ &= \theta \int_0^{\infty} e^{-(\theta - it)x} dx = \theta \left[\frac{-e^{-(\theta - it)x}}{\theta - it} \right]_0^{\infty} = \theta \left[0 + \frac{1}{\theta - it} \right] = \frac{\theta}{\theta - it} \end{aligned}$$

Note that we can also express $\phi_X(t)$ as

$$\phi_X(t) = \frac{1}{1 - \frac{it}{\theta}} = 1 + \left(\frac{it}{\theta} \right) + \left(\frac{it}{\theta} \right)^2 + \left(\frac{it}{\theta} \right)^3 + \dots$$

Hence, we find that

$$\mu'_1 = \text{Coefficient of } \frac{it}{1!} \text{ in } \phi_X(t) = \frac{1}{\theta}$$

and

$$\mu'_2 = \text{Coefficient of } \frac{(it)^2}{2!} \text{ in } \phi_X(t) = \frac{2}{\theta^2}$$

Hence, the mean of X is given by

$$\text{Mean}(X) = \mu = \mu'_1 = \frac{1}{\theta}$$

and the variance of X is given by

$$\text{Var}(X) = \mu_2 = \mu'_2 - \mu_1'^2 = \frac{2}{\theta^2} - \frac{1}{\theta^2} = \frac{1}{\theta^2}$$

◆

EXAMPLE 2.92. If X has the probability density function (Laplace distribution)

$$f(x) = \frac{1}{2} e^{-|x|} \quad \text{for } -\infty < x < \infty$$

show that the characteristic function of X is given by $\phi_X(t) = \frac{1}{1+t^2}$. Hence, find the mean and variance of X .

Solution. By definition, the characteristic function of X is given by

$$\begin{aligned}
\phi_X(t) &= E[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f(x) dx = \int_{-\infty}^0 e^{itx} \frac{1}{2} e^{-|x|} dx + \int_0^{\infty} e^{itx} \frac{1}{2} e^{-|x|} dx \\
&= \frac{1}{2} \int_{-\infty}^0 e^{itx} e^x dx + \frac{1}{2} \int_0^{\infty} e^{itx} e^{-x} dx = \frac{1}{2} \int_0^{\infty} e^{-ity} e^{-y} dy + \frac{1}{2} \int_0^{\infty} e^{itx} e^{-x} dx \\
&= \frac{1}{2} \int_0^{\infty} e^{-(1+it)y} dy + \frac{1}{2} \int_0^{\infty} e^{-(1-it)x} e^{-x} dx
\end{aligned}$$

Integrating, we get

$$\phi_X(t) = \frac{1}{2} \left[\frac{-e^{-(1+it)y}}{1+it} \right]_0^{\infty} + \frac{1}{2} \left[\frac{-e^{-(1-it)x}}{1-it} \right]_0^{\infty} = \frac{1}{2} \left[\frac{1}{1+it} \right] + \frac{1}{2} \left[\frac{1}{1-it} \right]$$

Simplifying, we get

$$\phi_X(t) = \frac{1}{2} \left[\frac{1}{1+it} + \frac{1}{1-it} \right] = \frac{1}{(1+it)(1-it)} = \frac{1}{1+t^2}$$

Next, note that

$$\phi_X(t) = 1 - t^2 + t^4 - t^6 + \dots = 1 + (it)^2 + (it)^4 + (it)^6 + \dots$$

Thus, it follows that

$$\mu'_1 = \text{Coefficient of } \frac{it}{1!} \text{ in } \phi_X(t) = 0$$

and

$$\mu'_2 = \text{Coefficient of } \frac{(it)^2}{2!} \text{ in } \phi_X(t) = 2$$

Hence, the mean of X is $\mu'_1 = 0$ and the variance of X is given by

$$\text{Var}(X) = \mu_2 = \mu'_2 - \mu_1'^2 = 2 - 0 = 2$$

◆

EXAMPLE 2.93. If the characteristic function of a random variable X is $e^{-|t|}$, show that X has the probability density function given by

$$f(x) = \frac{1}{\phi} \frac{1}{1+x^2}, \quad -\infty < x < \infty$$

Solution. Given that $\phi(t) = e^{-|t|}$. By inversion theorem, we know that the probability density function is given by

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} \phi(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} e^{-|t|} dt$$

$$\begin{aligned}
&= \frac{1}{2\pi} \int_{-\infty}^0 e^{itx} e^t dt + \frac{1}{2\pi} \int_0^{\infty} e^{itx} e^{-t} dt = \frac{1}{2\pi} \int_0^{\infty} e^{-ixu} e^{-u} du + \frac{1}{2\pi} \int_0^{\infty} e^{itx} e^{-t} dt \\
&= \frac{1}{2\pi} \int_0^{\infty} e^{-(1+ix)u} du + \frac{1}{2\pi} \int_0^{\infty} e^{-(1-ix)t} dt
\end{aligned}$$

Integrating, we get

$$f(x) = \frac{1}{2\pi} \left[\frac{-e^{-(1+ix)u}}{1+ix} \right]_0^{\infty} + \frac{1}{2\pi} \left[\frac{-e^{-(1-ix)t}}{1-ix} \right]_0^{\infty} = \frac{1}{2\pi} \left[\frac{1}{1+ix} \right] + \frac{1}{2\pi} \left[\frac{1}{1-ix} \right]$$

Simplifying, we get

$$f(x) = \frac{1}{2\pi} \left[\frac{1}{1+ix} + \frac{1}{1-ix} \right] = \frac{1}{2\pi} \frac{2}{1+x^2} = \frac{1}{\pi} \frac{1}{1+x^2}$$

◆

EXAMPLE 2.94. Show that the density function of the random variable, whose characteristic function is $e^{-\frac{t^2}{2}}$, is given by

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty$$

Solution. Given that $\phi(t) = e^{-\frac{t^2}{2}}$. By inversion theorem, the probability density function of X is given by

$$\begin{aligned}
f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} \phi(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{itx} e^{-\frac{t^2}{2}} dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{(ix+t)^2}{2}} e^{\frac{(ix)^2}{2}} dt = \frac{1}{2\pi} e^{\frac{(ix)^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{(ix+t)^2}{2}} dt \\
&= \frac{1}{2\pi} e^{-\frac{x^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} du = \frac{1}{2\pi} e^{-\frac{x^2}{2}} 2 \int_0^{\infty} e^{-\frac{u^2}{2}} du
\end{aligned}$$

Putting $v = \frac{u^2}{2}$, we get $dv = u du$. Hence,

$$\begin{aligned}
f(x) &= \frac{1}{2\pi} e^{-\frac{x^2}{2}} 2 \int_0^{\infty} \frac{1}{\sqrt{2v}} e^{-v} dv = \frac{1}{2\pi} e^{-\frac{x^2}{2}} \sqrt{2} \int_0^{\infty} v^{-\frac{1}{2}} e^{-v} dv \\
&= \frac{1}{2\pi} e^{-\frac{x^2}{2}} \sqrt{2} \Gamma\left(\frac{1}{2}\right) = \frac{1}{2\pi} e^{-\frac{x^2}{2}} \sqrt{2} \sqrt{\pi} = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}
\end{aligned}$$

◆

PROBLEM SET 2.9

1. Let X be a discrete random variable with the following probability distribution:

x	-1	0	1
$f(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Find the characteristic function of X and use it to determine the mean and variance of X .

2. Let X be a binomial random variable with the probability mass function

$$f(x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n \quad (p > 0, q > 0, p + q = 1)$$

Find the characteristic function of X and use it to determine the mean and variance of X .

3. Let X be a Poisson random variable with the probability mass function

$$f(x) = e^{-\lambda} \frac{\lambda^x}{x!}, \quad x = 0, 1, 2, \dots \quad (\text{where } \lambda > 0)$$

Find the characteristic function of X and use it to determine the mean and variance of X .

4. Let X be a gamma random variable with the probability density function

$$f(x) = \begin{cases} \frac{x^{\lambda-1}}{\Gamma(\lambda)} e^{-x} & \text{if } x > 0 \quad (\lambda > 0) \\ 0 & \text{otherwise} \end{cases}$$

Find the characteristic function of X and use it to determine the mean and variance of X .

5. If the characteristic function of a random variable X is given by

$$\phi(t) = \begin{cases} 1 - |t| & \text{for } |t| \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the probability density function of X .

Chapter 3

Standard Probability Distributions

We will first study some of the standard discrete probability distributions and then study some continuous probability distributions. These standard distributions serve as useful models for many real-life problems.

3.1 DEGENERATE DISTRIBUTION

In Statistics, a *degenerate distribution* is the probability distribution of a discrete random variable which assigns all the probability, i.e. probability 1, to a single point. In other words, degenerate random variables are those for which the whole mass of the variable is concentrated at a single point. Examples of degenerate variables are two-headed coins and a die having the same number in all the 6 faces (say, the number 6).

Definition 3.1. A random variable X is said to have a *degenerate distribution* if X is degenerated at a point c , i.e.

$$f(k; c) = P(X = k) = \begin{cases} 1 & \text{if } k = c \\ 0 & \text{if } k \neq c \end{cases}$$

The degenerate distribution has just one parameter c , where $-\infty < c < \infty$.

The degenerate distribution of a random variable with $c = 0$ is illustrated in Figure 3.1.

Clearly, the cumulative distribution of the degenerate random variable X with parameter c is given by

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < c \\ 1 & \text{if } x \geq c \end{cases}$$

The raw moments of the degenerate random variable X are given by

$$\mu'_r = E(X^r) = c^r P(X = c) = c^r \quad \text{for } r = 0, 1, 2, \dots$$

Hence, it follows that if X is a degenerate random variable with parameter c , then

$$\text{Mean}(X) = \mu'_1 = c \quad \text{and} \quad \text{Var}(X) = \mu'_2 - \mu'^2_1 = c^2 - c^2 = 0$$

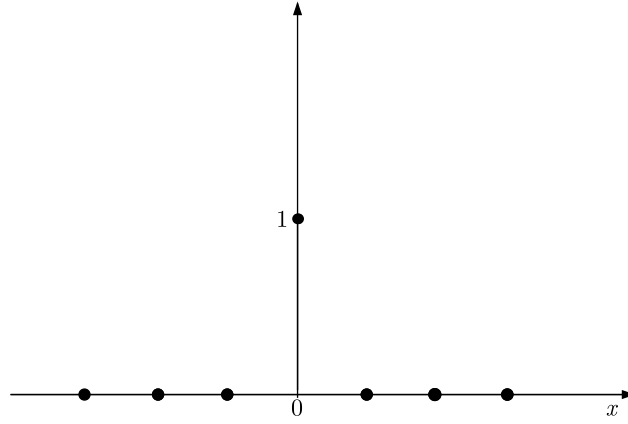


Figure 3.1: Degenerate distribution with $c = 0$.

As can be seen in Figure 3.1, a degenerate random variable is concentrated at the point $x = c$, and hence there is no spread or dispersion about its mean c . This illustrates the fact that $\text{Var}(X) = 0$ for a degenerate random variable.

The moment generating function (MGF) of the degenerate random variable X is given by

$$M_X(t) = E(e^{tX}) = e^{tc}P(X = c) = e^{tc}$$

3.2 BERNOULLI DISTRIBUTION

The *Bernoulli distribution* is named after the Swiss mathematician James Bernoulli (1654–1705). This distribution is associated with the *Bernoulli trial*. A Bernoulli trial is an experiment with two outcomes, which are random and which can be considered as *success* and *failure*, respectively. If we denote the probabilities of success and failure as p and q , respectively, then it follows that $p + q = 1$.

EXAMPLE 3.1. There are many real-life instances, which can be viewed as Bernoulli trials, like the following:

- (a) A coin is tossed. It has two outcomes, viz. H and T . Here, if we consider getting a head as a success and tail as a failure, then $p = \frac{1}{2}$ and $q = \frac{1}{2}$.
- (b) A fair die is tossed. If we consider getting the number six as a success and any other number as failure, then $p = \frac{1}{6}$ and $q = \frac{5}{6}$.
- (c) A card is drawn from a pack of cards. If we consider getting a card of Spade suit as a success and a card of any other suit as failure, then $p = \frac{1}{4}$ and $q = \frac{3}{4}$.
- (d) Two persons, A and B , are contesting in an election. A public opinion survey is conducted. A voter is selected at random, and he (or she) is asked whether he would vote for the candidate A .

As it can be noticed from the above example, Bernoulli trials feature questions with “yes” or “no” type of answers, where the answer “yes” corresponds to a *success*, while the answer “no” corresponds to a *failure*.

Mathematically, a Bernoulli trial can be modelled as follows.

Definition 3.2. A random variable X is said to have a *Bernoulli distribution* if it can take only two values 0 and 1, with 1 being considered as a *success* and 0 as a *failure*, with associated probabilities q and p , respectively, where $q = 1 - p$, i.e.

$$f(x; p) = P(X = x) = \begin{cases} p & \text{if } x = 1 \\ q = 1 - p & \text{if } x = 0 \end{cases}$$

If X is a Bernoulli random variable with parameter p , we write $X \sim B(1, p)$.

Theorem 3.1. Let X be a Bernoulli random variable with p as the probability of success, i.e. $X \sim B(1, p)$. Then the following properties hold:

- (a) $\text{Mean}(X) = p$ and $\text{Var}(X) = pq$.
- (b) The r^{th} raw moment is $\mu'_r = p$.
- (c) The moment generating function is $M_X(t) = q + pe^t$.
- (d) The characteristic function is $\phi_X(t) = q + pe^{it}$.

Proof. (a) We find that

$$\mu'_1 = E(X) = (0 \times f(0)) + (1 \times f(1)) = 0 + p = p$$

and

$$\mu'_2 = E(X^2) = (0^2 \times f(0)) + (1^2 \times f(1)) = p$$

Hence, the mean of X is

$$\text{Mean}(X) = \mu'_1 = p$$

and the variance of X is

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = p - p^2 = p(1 - p) = pq$$

(b) By definition, the r^{th} raw moment is

$$\mu'_r = E(X^r) = [0^r \times f(0)] + [1^r \times f(1)] = 0 + p = p$$

(c) By definition, the MGF of X is

$$M_X(t) = E(e^{tX}) = [e^{0t} \times f(0)] + [e^{1t} \times f(1)] = q + pe^t$$

(d) By definition, the characteristic function of X is

$$\phi_X(t) = E(e^{itX}) = [e^{0it} \times f(0)] + [e^{1it} \times f(1)] = q + pe^{it}$$

◆

3.3 BINOMIAL DISTRIBUTION

Binomial distribution was discovered by James Bernoulli, the inventor of Bernoulli distribution. It is the discrete probability distribution of the number of successes in a sequence of n independent Bernoulli trials with a constant probability of success p in all the n trials. It has two parameters, namely, (i) n , the number of independent Bernoulli trials and (ii) p , the constant probability of success in all the Bernoulli trials.

It is easy to see that the binomial random variable X takes on the values $0, 1, 2, \dots, n$ and satisfies the following probability law:

$$f(x; n, p) = P(X = x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

Remark 3.1. Consider a sequence of 3 Bernoulli trials with constant probability of success p in each trial. Then

$$P(X = 0) = P(\text{All failures}) = q \times q \times q = q^3$$

Note also that

$$\begin{aligned} P(X = 1) &= P(2 \text{ Failures and 1 Success}) = P(SFF) + P(FSF) + P(FFS) \\ &= pq^2 + pq^2 + pq^2 = 3pq^2 \end{aligned}$$

We also have

$$\begin{aligned} P(X = 2) &= P(1 \text{ Failure and 2 Successes}) = P(SSF) + P(SFS) + P(FSS) \\ &= p^2q + p^2q + p^2q = 3p^2q \end{aligned}$$

Next, we find that

$$P(X = 3) = P(3 \text{ Successes}) = P(SSS) = p \times p \times p = p^3$$

Generalizing this for the sequence of n independent Bernoulli trials, we have

$$P(X = x) = P(x \text{ Successes and } (n - x) \text{ Failures in any order})$$

Since there are nC_x different possibilities of x Successes and $(n - x)$ Failures and the probability in any particular instance is $p^x \times q^{n-x}$, it is clear that

$$P(X = x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

The distribution is named as “binomial distribution” because the probabilities $f(x; n, p)$ are successive terms in the binomial expansion of $(q + p)^n$. It also shows that the distribution is well-defined, i.e. the total probability is 1, because

$$\sum_{x=0}^n f(x; n, p) = \sum_{x=0}^n {}^nC_x q^{n-x} p^x = (q + p)^n = 1$$

Some instances of binomial distribution are as given below:

1. A coin is tossed 10 times and the number of times a head appears is counted. If we interpret getting a head as a success, then it is easy to see that the random variable (the number of heads) follows a binomial distribution with parameters $n = 10$ and $p = \frac{1}{2}$.

2. A die is rolled 100 times and the number of times a six appears is counted. If we interpret getting a six as a success, then it is easy to see that the random variable (the number of sixes) follows a binomial distribution with parameters $n = 100$ and $p = \frac{1}{36}$.

The binomial distribution is mathematically defined as follows.

Definition 3.3. A random variable X is said to have a *binomial distribution* with parameters n (a positive integer) and p ($0 < p < 1$) if it takes on the values $0, 1, 2, \dots, n$ and if its probability mass function is given by

$$f(x; n, p) = P(X = x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

where $q = 1 - p$.

If X is a binomial random variable with parameters n and p , we write $X \sim B(n, p)$.

Remark 3.2. Bernoulli distribution is a particular case of binomial distribution; indeed, we obtain it by taking $n = 1$ in the definition of binomial distribution.

Remark 3.3. The assumptions made or the physical conditions used in the derivation of binomial distributions are listed below:

1. There are totally n trials, where n is a positive integer.
2. Each trial is a Bernoulli trial, i.e. it has two only outcomes, which are random and which can be classified as *success* and *failure*, respectively.
3. The n Bernoulli trials are independent of each other.
4. The probability of success, p , is the same for all the n trials.

EXAMPLE 3.2. Find the formula for the probability distribution of the number of heads when a coin is tossed four times.

(Madras, Oct. 1996)

Solution. Clearly, X follows the binomial distribution with parameters $n = 4$ and $p = \frac{1}{2}$, i.e. $X \sim B(4, \frac{1}{2})$.

Then the probability mass function for X is given by

$$\begin{aligned} P(X = x) &= {}^4C_x p^x q^{4-x} = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}, \quad x = 0, 1, 2, 3, 4 \\ &= {}^4C_x \left(\frac{1}{2}\right)^4 = \frac{{}^4C_x}{16}, \quad x = 0, 1, 2, 3, 4 \end{aligned}$$

Thus, it is easy to see that the probability distribution of X is given by

x	0	1	2	3	4
$f(x)$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{16}$

◆

Theorem 3.2. If X is a binomial random variable with parameters n and p , i.e. $X \sim B(n, p)$, then X has mean np and variance npq , i.e. $\mu = np$ and $\sigma^2 = npq$.

Proof. Since $X \sim B(n, p)$, X has the binomial probability law given by

$$f(x; n, p) = P(X = x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

Since

$${}^nC_x = \frac{n \times (n-1) \times (n-2) \times \dots \times (n-x+1)}{x \times (x-1) \times (x-2) \times \dots \times 1}$$

it follows that

$${}^nC_x = \frac{n}{x} \times {}^{n-1}C_{x-1} = \frac{n}{x} \times \frac{n-1}{x-1} \times {}^{n-2}C_{x-2}, \text{ etc.} \quad (3.1)$$

Hence, we have

$$\mu'_1 = E(X) = \sum_{x=0}^n x {}^nC_x p^x q^{n-x} = \sum_{x=1}^n x {}^nC_x p^x q^{n-x}$$

Using the formula, Eq. (3.1), we have

$$\mu'_1 = \sum_{x=1}^n x \frac{n}{x} \times {}^{n-1}C_{x-1} p^x q^{n-x} = \sum_{x=1}^n n {}^{n-1}C_{x-1} p^x q^{n-x}$$

which can be rewritten as

$$\mu'_1 = np \sum_{x=1}^n {}^{n-1}C_{x-1} p^{x-1} q^{n-x} = \sum_{y=0}^{n-1} {}^{n-1}C_y p^y q^{n-1-y} \quad (\text{putting } y = x-1)$$

It follows that

$$\mu'_1 = np(q+p)^{n-1} = np(1) = np$$

Thus, X has mean $\mu = \mu'_1 = np$.

Next, we find the second raw moment, μ'_2 . For this, we use the formula

$$x^2 = x(x-1) + x$$

By definition, we have

$$\mu'_2 = E(X^2) = E(X(X-1) + X) = E(X(X-1)) + E(X)$$

Now,

$$E(X(X-1)) = \sum_{x=0}^n x(x-1) {}^nC_x p^x q^{n-x} = \sum_{x=2}^n x(x-1) {}^nC_x p^x q^{n-x}$$

Using formula, Eq. (3.1), we have

$$E(X(X-1)) = \sum_{x=2}^n x(x-1) \frac{n(n-1)}{x(x-1)} {}^{n-2}C_{x-2} p^x q^{n-x}$$

Simplifying, we get

$$E(X(X-1)) = n(n-1)p^2 \sum_{x=2}^n {}^{n-2}C_{x-2} p^{x-2} q^{n-x}$$

Putting $y = x - 2$ in the summation, we have

$$E(X(X-1)) = n(n-1)p^2 \sum_{y=0}^{n-2} {}^{n-2}C_y p^y q^{n-2-y}$$

Hence, it follows that

$$E(X(X-1)) = n(n-1)p^2 (q+p)^{n-2} = n(n-1)p^2$$

So, we have

$$\mu'_2 = E(X(X-1)) + E(X) = n(n-1)p^2 + np$$

Therefore, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = n(n-1)p^2 + np - n^2p^2$$

Simplifying, we get

$$\text{Var}(X) = -np^2 + np = np(1-p) = npq$$

◆

Remark 3.4. Let X be a binomial variable with parameters n and p . Then

$$\text{Mean}(X) = np \text{ and } \text{Var}(X) = npq$$

Since $0 < q < 1$, it follows that

$$npq < np \Rightarrow \text{Var}(X) < \text{Mean}(X)$$

Thus, for a binomial distribution, the variance is always less than the mean.

EXAMPLE 3.3. Criticize the following statement: “The mean of a binomial distribution is 5 and the standard deviation is 3”.

Solution. If $X \sim B(n, p)$, then $\mu = np = 5$ and $\sigma^2 = npq = 3^2 = 9$. Then

$$q = \frac{npq}{np} = \frac{9}{5} > 1$$

which is impossible since q is a probability. Hence, the given statement is not true.

◆

EXAMPLE 3.4. If X is a binomially distributed RV with $E(X) = 2$ and $\text{Var}(X) = \frac{4}{3}$, find $P(X = 5)$.
(Anna, May 2007)

Solution. If X is a binomial variate with parameters n and p , then we note that

$$E(X) = np = 2 \text{ and } \text{Var}(X) = npq = \frac{4}{3}$$

Hence, it is immediate that

$$q = \frac{\text{Var}(X)}{E(X)} = \frac{4/3}{2} = \frac{2}{3} \Rightarrow p = 1 - q = \frac{1}{3}$$

It also follows that

$$n = \frac{E(X)}{p} = \frac{2}{1/3} = 6$$

Hence, the probability mass function for X is given by

$$P(X = x) = {}^nC_x p^x q^{n-x} = {}^6C_x \left[\frac{1}{3}\right]^x \left[\frac{2}{3}\right]^{6-x} \text{ for } x = 0, 1, 2, \dots, 6$$

Hence, the required probability is obtained as

$$P(X = 5) = {}^6C_5 \left[\frac{1}{3}\right]^5 \left[\frac{2}{3}\right]^1 = \frac{4}{243} = 0.0165$$

◆

EXAMPLE 3.5. If the mean and variance of a binomial variable are 12 and 4, respectively, find the distribution.

(Madras, April 1999)

Solution. For a binomial variable X , $\text{Mean}(X) = np$ and $\text{Var}(X) = npq$.

Hence, we have $np = 12$ and $npq = 4$.

Thus, it follows that

$$q = \frac{npq}{np} = \frac{4}{12} = \frac{1}{3} \Rightarrow p = 1 - q = \frac{2}{3}$$

Also, we note that

$$n = \frac{12}{p} = \frac{12}{\frac{2}{3}} = 12 \times \frac{3}{2} = 18$$

Hence, X has the probability law

$$f(x) = P(X = x) = {}^{18}C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{18-x} \text{ for } x = 0, 1, 2, \dots, 18$$

◆

EXAMPLE 3.6. For a binomial distribution, the mean is 6 and standard deviation is $\sqrt{2}$. Find the first two terms of the distribution.

(Anna, April 2004)

Solution. For a binomial variable X , $\text{Mean}(X) = np$ and $\text{Var}(X) = \sigma^2 = npq$.

Hence, we have $np = 6$ and $npq = 2$.

Thus, it follows that

$$q = \frac{npq}{np} = \frac{2}{6} = \frac{1}{3} \Rightarrow p = 1 - q = \frac{2}{3}$$

Also, we note that

$$n = \frac{6}{p} = \frac{6}{\frac{2}{3}} = 6 \times \frac{3}{2} = 9$$

Hence, X has the probability law

$$f(x) = P(X = x) = {}^9C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{9-x} \text{ for } x = 0, 1, 2, \dots, 9$$

Thus, the first two terms of the distribution are

$$f(0) = P(X = 0) = \left(\frac{1}{3}\right)^9 = 5.0805 \times 10^{-5}$$

and

$$f(1) = P(X = 1) = 9 \times \frac{2}{3} \times \left(\frac{1}{3}\right)^8 = 9.1449 \times 10^{-4}$$

◆

Theorem 3.3. Let X be a binomial random variable with parameters n and p . Then the following properties hold:

(a) The central moments, μ_r , satisfy the recurrence relation

$$\mu_{r+1} = pq \left[nr\mu_{r-1} + \frac{d\mu_r}{dp} \right] \quad (\text{Renovsky Formula})$$

(Madras, April 2000)

(b) The second, third and fourth central moments are

$$\mu_2 = npq, \quad \mu_3 = npq(q-p) \quad \text{and} \quad \mu_4 = 3n^2p^2q^2 + npq(1-6pq).$$

(c) Karl Pearson's coefficients are

$$\text{Skewness} = \gamma_1 = \sqrt{\beta_1} = \frac{|q-p|}{\sqrt{npq}} \quad \text{and} \quad \text{Kurtosis} = \gamma_2 = \beta_2 - 3 = \frac{1-6pq}{npq}$$

Proof. (a) Since X has mean $\mu = np$, the r^{th} central moment μ_r is given by

$$\mu_r = E[(x - np)^r] = \sum_{x=0}^n (x - np)^r {}^nC_x p^x q^{n-x}$$

Hence, it follows that

$$\begin{aligned} \frac{d\mu_r}{dp} &= \sum_{x=0}^n {}^nC_x [r(x - np)^{r-1}(-n) p^x q^{n-x}] \\ &\quad + \sum_{x=0}^n {}^nC_x (x - np)^r [xp^{x-1}q^{n-x} + p^x(n-x)q^{n-x-1}(-1)] \end{aligned}$$

Simplifying, we have

$$\begin{aligned} \frac{d\mu_r}{dp} &= -nr\mu_{r-1} + \sum_{x=0}^n {}^nC_x (x - np)^r p^x q^{n-x} \left[\frac{x}{p} - \frac{n-x}{q} \right] \\ &= -nr\mu_{r-1} + \frac{1}{pq} \sum_{x=0}^n {}^nC_x (x - np)^r p^x q^{n-x} [x(p+q) - np] \end{aligned}$$

Since $p + q = 1$, it follows that

$$\begin{aligned}\frac{d\mu_r}{dp} &= -nr\mu_{r-1} + \frac{1}{pq} \sum_{x=0}^n {}^nC_x (x - np)^r p^x q^{n-x} (x - np) \\ &= -nr\mu_{r-1} + \frac{1}{pq} \sum_{x=0}^n {}^nC_x (x - np)^{r+1} p^x q^{n-x}\end{aligned}$$

Hence, we have

$$\frac{d\mu_r}{dp} = -nr\mu_{r-1} + \frac{1}{pq}\mu_{r+1}$$

Rearranging terms, we have

$$\mu_{r+1} = pq \left[nr\mu_{r-1} + \frac{d\mu_r}{dp} \right]$$

- (b) For any distribution, $\mu_0 = 1$ and $\mu_1 = 0$. Hence, taking $r = 1$ in the Renovsky formula (a), we have

$$\mu_2 = pq \left[n \cdot 1 \cdot \mu_0 + \frac{d\mu_1}{dp} \right] = pq[n + 0] = npq$$

Thus, $\mu_2 = npq$.

Taking $r = 2$ in the Renovsky formula (a), we have

$$\mu_3 = pq \left[n \cdot 2 \cdot \mu_1 + \frac{d\mu_2}{dp} \right] = pq \left[0 + \frac{d}{dp}(npq) \right] = pq[nq + np(-1)]$$

Simplifying, we have $\mu_3 = npq(q - p)$.

Next, taking $r = 3$ in the Renovsky formula (a), we have

$$\begin{aligned}\mu_4 &= pq \left[n \cdot 3 \cdot \mu_2 + \frac{d\mu_3}{dp} \right] = pq \left[n \cdot 3 \cdot (npq) + \frac{d}{dp}[npq(q - p)] \right] \\ &= 3n^2 p^2 q^2 + npq \frac{d}{dp}[pq(q - p)] \\ &= 3n^2 p^2 q^2 + npq [(1)q(q - p) + p(-1)(q - p) + pq(-1 - 1)] \\ &= 3n^2 p^2 q^2 + npq [(q - p)^2 - 2pq] = 3n^2 p^2 q^2 + npq [(q + p)^2 - 4pq - 2pq]\end{aligned}$$

Since $q + p = 1$, it follows that $\mu_4 = 3n^2 p^2 q^2 + npq[1 - 6pq]$.

- (c) First, we compute β_1 and β_2 . Note that

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{n^2 p^2 q^2 (q - p)^2}{n^3 p^3 q^3} = \frac{(q - p)^2}{npq}$$

and

$$\begin{aligned}\beta_2 &= \frac{\mu_4}{\mu_2^2} = \frac{3n^2 p^2 q^2 + npq[1 - 6pq]}{n^2 p^2 q^2} \\ &= 3 + \frac{[1 - 6pq]}{npq}\end{aligned}$$

Hence, the Karl Pearson's coefficients are

$$\text{Skewness} = \gamma_1 = \sqrt{\beta_1} = \frac{|q - p|}{\sqrt{npq}}$$

and

$$\text{Kurtosis} = \gamma_2 = \beta_2 - 3 = \frac{[1 - 6pq]}{npq}$$

◆

Theorem 3.4. Let X be a binomial random variable with parameters n and p .

(a) The moment generating function of X is given by

$$M_X(t) = (q + pe^t)^n$$

(Anna, May 2007; Nov. 2007)

(b) The characteristic function of X is given by

$$\phi_X(t) = (q + pe^{it})^n$$

Proof. (a) By definition, the MGF of X is given by

$$M_X(t) = E(e^{tX}) = \sum_{x=0}^n e^{tx} {}^nC_x p^x q^{n-x} = \sum_{x=0}^n {}^nC_x (pe^t)^x q^{n-x} = (q + pe^t)^n$$

(b) By definition, the characteristic function of X is given by

$$\phi_X(t) = E(e^{itX}) = \sum_{x=0}^n e^{itx} {}^nC_x p^x q^{n-x} = \sum_{x=0}^n {}^nC_x (pe^{it})^x q^{n-x} = (q + pe^{it})^n$$

◆

EXAMPLE 3.7. If the moment generating function of X is given by

$$M_X(t) = (0.6e^t + 0.4)^8$$

find the variance of X and the moment generating function of $Y = 3X + 4$.

(Anna, Model, 2003)

Solution. By Theorem 3.4, we know that the moment generating function of a binomial random variable with parameters n and p is given by

$$M_X(t) = (pe^t + q)^n$$

Comparing, we see that the given random variable X is binomial with parameters $n = 8$ and $p = 0.6$.

Hence, the variance of X is given by

$$\text{Var}(X) = npq = 8 \times 0.6 \times 0.4 = 1.92$$

Also, the moment generating function of $Y = 3X + 4$ is given by

$$M_Y(t) = E(e^{tY}) = E(e^{t(3X+4)}) = E(e^{3tX} \cdot e^{4t})$$

Hence, we have

$$M_Y(t) = e^{4t} E(e^{(3t)X}) = e^{4t} M_X(3t) = e^{4t} (0.6e^{3t} + 0.4)^8$$

◆

Theorem 3.5. Let X be a binomial random variable with parameters n and p .

(a) The probability law $f(x)$ satisfies

$$\frac{f(x)}{f(x-1)} = 1 + \frac{(n+1)p-x}{xq} \quad \text{for } i = 1, 2, \dots, n$$

(b) If $(n+1)p$ is not an integer, then the binomial distribution has a unique mode given by $m = [(n+1)p]$, i.e. the greatest integer less than $(n+1)p$.

(c) If $(n+1)p$ is an integer, say m , then the binomial distribution is bimodal and the two modal values are $x = m = (n+1)p$ and $x = m-1 = (n+1)p-1$.

Proof. (a) We know that

$$f(x) = {}^nC_x p^x q^{n-x} = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

Similarly,

$$f(x-1) = {}^nC_{x-1} p^{x-1} q^{n-x+1} = \frac{n!}{(x-1)!(n-x+1)!} p^{x-1} q^{n-x+1}$$

Thus, it follows that

$$\frac{f(x)}{f(x-1)} = \frac{n-x+1}{x} p \frac{1}{q} = \frac{(n-x+1)p}{xq}$$

Hence, we have

$$\frac{f(x)}{f(x-1)} = \frac{xq + (n-x+1)p - xq}{xq} = 1 + \frac{(n+1)p - x(p+q)}{xq}$$

Since $p+q=1$, it is immediate that

$$\frac{f(x)}{f(x-1)} = 1 + \frac{(n+1)p-x}{xq}$$

(b) If $(n+1)p$ is not an integer, let

$$(n+1)p = m + f$$

where m is the integral part of $(n+1)p$ and f is the fractional part of $(n+1)p$. Then, it is immediate that $0 < f < 1$. We claim that m is the unique mode of X .

By (a), we have

$$\frac{f(x)}{f(x-1)} = 1 + \frac{m+f-x}{xq} \quad (3.2)$$

From Eq. (3.2), it is obvious that

$$\frac{f(x)}{f(x-1)} > 1 \text{ for } x = 0, 1, 2, \dots, m$$

and

$$\frac{f(x)}{f(x-1)} < 1 \text{ for } x = m+1, m+2, \dots, n$$

Hence, it follows that

$$f(0) < f(1) < f(2) < \dots < f(m)$$

and

$$f(m) > f(m+1) > f(m+2) > \dots > f(n)$$

Hence, $f(x)$ is maximum when $x = m$ showing that $m = [(n+1)p]$ is the unique mode of X .

(c) Let $(n+1)p = m$ (an integer). By (a), we have

$$\frac{f(x)}{f(x-1)} = 1 + \frac{m-x}{xq} \quad (3.3)$$

From Eq. (3.3), it is obvious that

$$\frac{f(x)}{f(x-1)} > 1 \text{ for } x = 1, 2, \dots, m-1$$

i.e.

$$\frac{f(x)}{f(x-1)} = 1 \text{ for } x = m$$

and

$$\frac{f(x)}{f(x-1)} < 1 \text{ for } x = m+1, m+2, \dots, n$$

Hence, it follows that

$$f(x) > f(x-1) \text{ for } x = 1, 2, \dots, m-1$$

$$f(m) = f(m-1)$$

and

$$f(x) < f(x-1) \text{ for } x = m+1, m+2, \dots, n$$

Hence, it is immediate that X is bimodal and the two modal values are $x = m$ and $x = m-1$.

◆

EXAMPLE 3.8. Define binomial $B(n, p)$ distribution. Obtain its MGF, mean and variance.

(Anna, Nov. 2003)

Solution. See the proof of Theorem 3.4 for the result that the moment generating function of the binomial random variable X with parameters n and p is given by

$$M_X(t) = (q + pe^t)^n$$

Here, we illustrate a simple method of finding the mean and variance of the binomial distribution from its MGF

Differentiating the MGF successively with respect to t , we get

$$M'_X(t) = n(q + pe^t)^{n-1}(pe^t) = np(q + pe^t)^{n-1}e^t$$

and

$$M''_X(t) = np[(n-1)(q + pe^t)^{n-2}(pe^t)(e^t) + (q + pe^t)^{n-1}e^t]$$

Hence, it follows that

$$\mu'_1 = M'_X(0) = np(q + p)^{n-1} = np(1) = np \quad (\text{since } q + p = 1)$$

and

$$\begin{aligned} \mu'_2 = M''_X(0) &= np[(n-1)(q + p)^{n-2}(p) + (q + p)^{n-1}(1)] \\ &= np[(n-1)p + 1] = np[np + q] \quad (\text{since } q = 1 - p) \end{aligned}$$

Thus, $\mu'_1 = np$ and $\mu'_2 = n^2p^2 + npq$.

Hence, $\text{Mean}(X) = \mu_1 = np$ and $\text{Var}(X) = \mu'_2 - \mu_1^2 = npq$. ◆

EXAMPLE 3.9. Find the MGF of the standard binomial random variable

$$Z = \frac{X - \mu}{\sigma}$$

and obtain its limiting form as $n \rightarrow \infty$. Also, interpret the result.

Solution. First, note that $\mu = np$ and $\sigma = npq$ for a binomial random variable.

Next, by Theorem 2.22, we know that

$$M_Z(t) = e^{-\frac{\mu t}{\sigma}} M_X\left(\frac{t}{\sigma}\right)$$

Since $M_X(t) = (q + pe^t)^n$, it follows that

$$\begin{aligned} M_Z(t) &= e^{-\frac{np t}{\sqrt{npq}}} M_X\left(\frac{t}{\sqrt{npq}}\right) = \left(e^{-\frac{pt}{\sqrt{npq}}}\right)^n \left(q + pe^{\frac{t}{\sqrt{npq}}}\right)^n \\ M_Z(t) &= \left(qe^{-\frac{pt}{\sqrt{npq}}} + pe^{\frac{qt}{\sqrt{npq}}}\right)^n \quad (\text{since } 1 - p = q) \end{aligned} \tag{3.4}$$

Now, note that

$$qe^{-\frac{pt}{\sqrt{npq}}} = q \left[1 - \frac{pt}{\sqrt{npq}} + \frac{p^2 t^2}{2npq} + O\left(n^{-\frac{3}{2}}\right) \right] \quad (3.5)$$

where the expression $O\left(n^{-\frac{3}{2}}\right)$ denotes all terms involving powers $n^{\frac{3}{2}}$ or higher in the denominator. Similarly, we have

$$pe^{\frac{qt}{\sqrt{npq}}} = p \left[1 + \frac{qt}{\sqrt{npq}} + \frac{q^2 t^2}{2npq} + O\left(n^{-\frac{3}{2}}\right) \right] \quad (3.6)$$

Adding Eq. (3.5) and (3.6), we have

$$qe^{-\frac{pt}{\sqrt{npq}}} + pe^{\frac{qt}{\sqrt{npq}}} = (q+p) + \frac{t^2 pq}{2npq}(p+q) + O\left(n^{-\frac{3}{2}}\right) = 1 + \frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \quad (3.7)$$

Substituting Eq. (3.7) into Eq. (3.4), we have

$$M_Z(t) = \left[1 + \frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \right]^n$$

Taking natural logarithms on both sides, we have

$$\log M_Z(t) = n \log \left[1 + \frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \right]$$

Using the formula $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$, we have

$$\log M_Z(t) = n \left[\left(\frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \right) - \frac{1}{2} \left(\frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \right)^2 + \dots \right]$$

$$\log M_Z(t) = \frac{t^2}{2} + O\left(n^{-\frac{1}{2}}\right)$$

Taking limits as $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} \log M_Z(t) = \frac{t^2}{2}$$

or

$$\lim_{n \rightarrow \infty} M_Z(t) = e^{\frac{t^2}{2}} \quad (3.8)$$

Interpretation: The MGF on the RHS of (3.8) is the MGF of the standard normal variable. Hence, by the Uniqueness Theorem for MGFs (see Theorem 2.23), it follows that the standard binomial random variable approaches the standard normal variable as $n \rightarrow \infty$. In other words, the normal distribution is the limiting form of the binomial distribution as $n \rightarrow \infty$. ♦

EXAMPLE 3.10. For a random variable X , $M_X(t) = \frac{1}{81}(e^t + 2)^4$. Find $P(X \leq 2)$.

(Anna, Nov. 2004)

Solution. For a binomial random variable Y with parameters n and p , the moment generating function of X is given by

$$M_Y(t) = (q + pe^t)^n$$

Given that

$$M_X(t) = \frac{1}{81} (e^t + 2)^4 = \left(\frac{2}{3} + \frac{1}{3}e^t \right)^4$$

By the Uniqueness Theorem for MGFs, it follows that X follows a binomial distribution with parameters $n = 4$ and $p = \frac{1}{3}$. Thus, X has the probability mass function given by

$$P(X = x) = {}^4C_x p^x q^{4-x} = {}^4C_x \left(\frac{1}{3} \right)^x \left(\frac{2}{3} \right)^{4-x}, \quad x = 0, 1, 2, 3, 4$$

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = q^4 + 4pq^3 + 6p^2q^2$$

Simplifying, we have

$$P(X \leq 2) = \frac{16}{81} + \frac{32}{81} + \frac{8}{27} = \frac{8}{9}$$

◆

EXAMPLE 3.11. A coin is biased so that a head is twice as likely to appear as a tail. If the coin is tossed 6 times, find the probabilities of getting (i) exactly 2 heads, (ii) at least 3 heads, (iii) at most 4 heads.

Solution. If p and q denote the probabilities of getting a head and tail, respectively, then we are given that $p = 2q$. Since $p + q = 1$, it follows that

$$2q + q = 1 \Rightarrow 3q = 1 \Rightarrow q = \frac{1}{3}$$

Thus, $q = \frac{1}{3}$ and $p = 1 - q = \frac{2}{3}$.

If X denotes the number of heads, then X follows a binomial distribution with parameters $n = 6$ and $p = \frac{2}{3}$, i.e. $X \sim B(6, \frac{2}{3})$.

Then X follows the probability law

$$P(X = x) = {}^6C_x \left(\frac{2}{3} \right)^x \left(\frac{1}{3} \right)^{6-x}, \quad x = 0, 1, \dots, 6$$

$$P(X = x) = {}^6C_x \frac{2^x}{729}, \quad x = 0, 1, \dots, 6$$

x	0	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{729}$	$\frac{12}{729}$	$\frac{60}{729}$	$\frac{160}{729}$	$\frac{240}{729}$	$\frac{192}{729}$	$\frac{64}{729}$

(i) $P(\text{exactly 2 heads}) = P(X = 2) = \frac{60}{729} = \frac{20}{243} = 0.0823$.

(ii) $P(\text{at least 3 heads}) = P(X \geq 3) = 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$.

$$P(\text{at least 3 heads}) = 1 - \left[\frac{1}{729} + \frac{12}{729} + \frac{60}{729} \right] = 1 - \frac{73}{729} = \frac{656}{729} = 0.8999$$

(iii) $P(\text{at most four heads}) = P(X \leq 4) = P(X = 0) + P(X = 1) + \cdots + P(X = 4)$.

$$P(\text{at most four heads}) = \frac{1}{729} + \frac{12}{729} + \frac{60}{729} + \frac{160}{729} + \frac{240}{729} = \frac{473}{729} = 0.6488$$

◆

EXAMPLE 3.12. A box contains 100 cellphones, 20 of which are defective. 10 cellphones are selected for inspection. Find the probability that (i) at least one is defective, (ii) at the most three are defective, (iii) all the ten are defective, (iv) none of the ten is defective.

Solution. If X is the number of defective cellphones, then it is clear that X follows a binomial distribution with parameters $n = 10$ and $p = \frac{20}{100} = \frac{1}{5}$. Therefore, we have

$$\begin{aligned} P(X = x) &= {}^{10}C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{10-x}, \quad x = 0, 1, 2, \dots, 10 \\ &= {}^{10}C_x \frac{4^{10-x}}{5^{10}}, \quad x = 0, 1, 2, \dots, 10 \end{aligned}$$

(i) Probability that at least one cellphone is defective is given by

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{4}{5}\right)^{10} = 0.8926$$

(ii) Probability that at the most three cellphones are defective is given by

$$\begin{aligned} P(X \leq 3) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ &= \frac{1}{5^{10}} [4^{10} + (10 \times 4^9) + (45 \times 4^8) + (120 \times 4^7)] = 0.8791 \end{aligned}$$

(iii) Probability that all the ten cellphones are defective is given by

$$P(X = 10) = 1 \times \left(\frac{1}{5}\right)^{10} = 1.0240 \times 10^{-7}$$

(iv) Probability that none of the ten is defective is given by

$$P(X = 0) = 1 \times \left(\frac{4}{5}\right)^{10} = 0.1074$$

◆

EXAMPLE 3.13. It is known that 5% of the books bound at a certain bindery have defective bindings. Find the probability that 2 of 100 books bound by this bindery will have defective bindings.

(Anna, Nov. 2003)

Solution. Let X be the number of books with defective bindings. Then it is clear that X follows a binomial distribution with parameters $n = 100$ and $p = \frac{5}{100} = \frac{1}{20}$. Hence, X has the probability law

$$P(X = x) = {}^{100}C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{100-x}, \quad \text{for } x = 1, 2, \dots, 100$$

The required probability is

$$P(X = 2) = {}^{100}C_2 \left(\frac{1}{20}\right)^2 \left(\frac{19}{20}\right)^{98} = 0.0812$$

◆

EXAMPLE 3.14. A binomial distribution with parameter $n = 5$ satisfies the property $8P(X = 4) = P(X = 2)$. Find (i) p , (ii) $P(X \geq 1)$.

Solution. Since X follows a binomial distribution with parameter $n = 5$, we have

$$P(X = x) = {}^5C_x p^x q^{5-x}, x = 0, 1, \dots, 5$$

(i) Since $8P(X = 4) = P(X = 2)$, we have

$$8 \times {}^5C_4 p^4 q = {}^5C_2 p^2 q^3$$

$$8 \times 5 \times p^4 \times q = 10 \times p^2 \times q^3$$

Simplifying, we have

$$4p^2 = q^2 = (1 - p)^2 = 1 - 2p + p^2$$

$$3p^2 + 2p - 1 = (3p - 1)(p + 1) = 0$$

Thus,

$$p = \frac{1}{3} \text{ or } p = -1$$

Since p cannot be negative, $p = \frac{1}{3}$. Also, $q = 1 - p = \frac{2}{3}$.

(ii) We have

$$P(X \geq 1) = 1 - P(X = 0) = 1 - q^5 = 1 - \left(\frac{2}{3}\right)^5 = 0.8683$$

◆

EXAMPLE 3.15. Show that the r th raw moment μ'_r of the binomial distribution satisfies the property

$$\mu'_r = \left(p \frac{\partial}{\partial p}\right)^r (q + p)^n$$

Solution. We prove this result by the principle of induction. First, we show that the result holds for $n = 1$. By binomial theorem,

$$(q + p)^n = \sum_{x=0}^n {}^nC_x p^x q^{n-x}$$

Therefore, it follows that

$$\frac{\partial}{\partial p} (q + p)^n = \sum_{x=0}^n {}^nC_x x p^{x-1} q^{n-x}$$

Thus, we have

$$\left(p \frac{\partial}{\partial p}\right) (q+p)^n = \sum_{x=0}^n x {}^n C_x p^x q^{n-x} = \mu'_1$$

Hence, we have shown that the result holds for $r = 1$.

Next, suppose that the result holds for any $r = m$. (*Induction Hypothesis*)

We have

$$\mu'_m = \left(p \frac{\partial}{\partial p}\right)^m (q+p)^n$$

Therefore, it follows that

$$\left(p \frac{\partial}{\partial p}\right)^m (q+p)^n = \sum_{x=0}^n x^m {}^n C_x p^x q^{n-x}$$

Hence, we have

$$p \frac{\partial}{\partial p} \left(p \frac{\partial}{\partial p}\right)^m (q+p)^n = p \sum_{x=0}^n x^m {}^n C_x x p^{x-1} q^{n-x}$$

i.e.

$$\left(p \frac{\partial}{\partial p}\right)^{m+1} (q+p)^n = \sum_{x=0}^n x^{m+1} {}^n C_x p^x q^{n-x} = \mu'_{m+1}$$

Hence, the result also holds for $r = m + 1$. Thus, by the principle of induction, the result holds for all positive integral values of k . ♦

EXAMPLE 3.16. A pair of dice be rolled 900 times and X denotes the number of times a total of 9 occurs. Find $P(80 \leq X \leq 120)$ using Chebyshev inequality.

(Anna, Nov. 2003; May 2006)

Solution. Let p and q denote the probabilities of getting a 9 and not getting a 9 in a roll of a pair of dice. Then we know that

$$p = \frac{4}{36} = \frac{1}{9} \Rightarrow q = 1 - p = \frac{8}{9}$$

Given that $n = 900$, the number of times the dice are rolled. Since X follows a binomial distribution, it follows that

$$\mu = np = 900 \times \frac{1}{9} = 100 \text{ and } \sigma^2 = npq = 900 \times \frac{1}{9} \times \frac{8}{9} = \frac{800}{9}$$

By Chebyshev's inequality, we know that

$$P[|X - \mu| \leq C] \geq 1 - \frac{\sigma^2}{C^2}$$

i.e.

$$P(|X - 100| \leq 20) \geq 1 - \frac{800/9}{400} = 1 - \frac{2}{9} = \frac{7}{9}$$

Hence, we conclude that

$$P(80 \leq X \leq 100) \geq \frac{7}{9} = 0.7778$$
♦

EXAMPLE 3.17. A fair die is thrown 600 times. By Chebyshev's theorem, the lower bound for the probability of "getting a 6" 80 to 120 times.

Solution. Let p and q denote the probabilities of "getting a 6" and "not getting a 6" in a throw of a fair die. Then we know that

$$p = \frac{1}{6} \Rightarrow q = 1 - p = \frac{5}{6}$$

Given that $n = 600$, the number of times the dice are rolled. Since X follows a binomial distribution, it follows that

$$\mu = np = 600 \times \frac{1}{6} = 100 \text{ and } \sigma^2 = npq = 600 \times \frac{1}{6} \times \frac{5}{6} = \frac{250}{3}$$

By Chebyshev's inequality, we know that

$$P[|X - \mu| \leq C] \geq 1 - \frac{\sigma^2}{C^2}$$

i.e.

$$P(|X - 100| \leq 20) \geq 1 - \frac{250/3}{400} = 1 - \frac{5}{24} = \frac{19}{24}$$

Hence, a lower limit for $P(80 \leq X \leq 100)$ is given by

$$P(80 \leq X \leq 100) \geq \frac{19}{24} = 0.7917$$

◆

Remark 3.5. If there are N sets of n independent Bernoulli trials with constant parameter p of success, then the frequencies of $0, 1, 2, \dots, n$ successes are given by the successive terms of the expression

$$N(q + p)^n$$

i.e. the frequency of x successes is given by

$$N {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

EXAMPLE 3.18. Six dice are thrown 729 times. How many times do you expect at least 3 dice to show 5 or 6?

(Anna, Nov. 2003)

Solution. Let success denote the event getting a 5 or 6 in the throw of a die. Then

$$p = \frac{2}{6} = \frac{1}{3} \Rightarrow q = 1 - \frac{1}{3} = \frac{2}{3}$$

So, when 6 dice are thrown, the number of successes, X , follows a binomial distribution with parameters $n = 6$ and $p = \frac{1}{3}$. Thus,

$$P(X = x) = {}^6C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x}, \quad x = 0, 1, 2, \dots, 6$$

The probability that at least 3 of the 6 dice show a 5 or 6 is given by

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - \left[\left(\frac{2}{3}\right)^6 + 6 \cdot \left(\frac{1}{3}\right) \cdot \left(\frac{2}{3}\right)^5 + 15 \cdot \left(\frac{1}{3}\right)^2 \cdot \left(\frac{2}{3}\right)^4 \right] \\ &= 1 - \frac{1}{3^6} [2^6 + (6 \times 2^5) + (10 \times 2^4)] = 1 - \frac{496}{729} = \frac{233}{729} \end{aligned}$$

Thus, when 6 dice are thrown 729 times, the number of times we expect at least 3 dice to show 5 or 6 is given by

$$N \times P(X \geq 3) = 729 \times \frac{233}{729} = 233 \quad \blacklozenge$$

EXAMPLE 3.19. If X is a binomial distribution with parameters n and p , show that the probability mass function f of X satisfies the property

$$f(x+1) = \frac{n-x}{x+1} \cdot \frac{p}{q} \cdot f(x)$$

(Note: This formula will be very useful in fitting a binomial distribution to a given discrete data.)

Solution. Since $X \sim B(n, p)$, we have

$$f(x) = {}^nC_x p^x q^{n-x} = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

and

$$f(x+1) = {}^nC_{x+1} p^{x+1} q^{n-x-1} = \frac{n!}{(x+1)!(n-x-1)!} p^{x+1} q^{n-x-1}$$

Thus, it follows that

$$\frac{f(x+1)}{f(x)} = \frac{x!(n-x)!}{(x+1)!(n-x-1)!} \frac{p^{x+1}q^{n-x-1}}{p^xq^{n-x}} = \frac{n-x}{x+1} \cdot \frac{p}{q} \cdot f(x) \quad \blacklozenge$$

EXAMPLE 3.20. A biased coin is tossed 8 times and the number of heads are noted. The experiment is repeated 100 times and the following frequency distribution is obtained:

x	0	1	2	3	4	5	6	7	8
f	2	7	13	15	25	16	11	8	3

Fit a binomial distribution to these data. Calculate the theoretical frequencies.

Solution. First, we calculate the mean of the given data. We see that

$$\text{Mean, } \mu = \frac{\sum fx}{\sum f} = \frac{404}{100} = 4.04$$

Thus, we have

$$p = \frac{\mu}{n} = \frac{4.04}{8} = 0.505 \quad (\text{Recall that } \mu = E(X) = np)$$

Hence, $q = 1 - p = 0.495$.

Since $N = 100$, the theoretical frequencies for the given data are given by

$$Nf(x) = N^n C_x p^x q^{n-x} = 100^8 C_x (0.505)^x (0.495)^{8-x}, x = 0, 1, 2, \dots, 8$$

Using the recurrence relation

$$f(x+1) = \frac{n-x}{x+1} \cdot \frac{p}{q} \cdot f(x)$$

we have

$$Nf(x+1) = (100f(x)) \times 1.0202 \times \left(\frac{8-x}{x+1}\right), x = 0, 1, 2, \dots, 7$$

(where we have used the fact that $\frac{p}{q} = 1.0202$ for the given problem).

We find that

$$Nf(0) = 100 \times (0.495)^8 = 0.3604$$

Therefore,

$$Nf(1) = 0.3604 \times 1.0202 \times 8 = 2.9414$$

$$Nf(2) = 2.9414 \times 1.0202 \times \left(\frac{7}{2}\right) = 10.5030$$

$$Nf(3) = 10.5030 \times 1.0202 \times \left(\frac{6}{3}\right) = 21.4303$$

$$Nf(4) = 21.4303 \times 1.0202 \times \left(\frac{5}{4}\right) = 27.3290$$

$$Nf(5) = 27.3290 \times 1.0202 \times \left(\frac{4}{5}\right) = 22.3049$$

$$Nf(6) = 22.3049 \times 1.0202 \times \left(\frac{3}{6}\right) = 11.3777$$

$$Nf(7) = 11.3777 \times 1.0202 \times \left(\frac{2}{7}\right) = 3.3164$$

$$Nf(8) = 3.3164 \times 1.0202 \times \left(\frac{1}{8}\right) = 0.4229$$

We tabulate our results as follows:

No. of heads	Observed frequencies	Theoretical frequencies
0	2	0.3604
1	7	2.9414
2	13	10.5030
3	15	21.4303
4	25	27.3290
5	16	22.3049
6	11	11.3777
7	8	3.3164
8	3	0.4229

The frequency polygon for this problem is plotted in Figure 3.2.

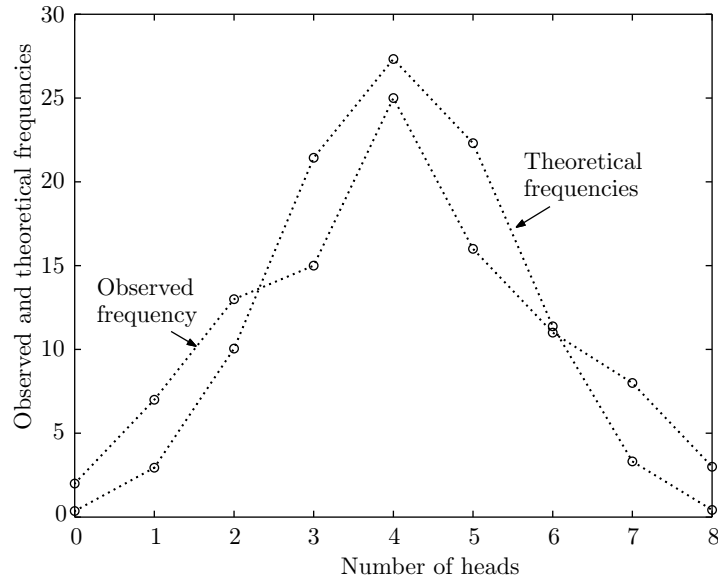


Figure 3.2: Fitting a binomial distribution.



PROBLEM SET 3.1

1. For a binomial distribution, the mean is 6 and the variance is 4. Find
 - (a) The first two terms of the distribution.
 - (b) $P(X = 4)$.

2. For a random variable X , the moment generating function is

$$M_X(t) = \frac{1}{64} (3 + e^t)^3$$

Find

- (a) $P(X \leq 2)$.
 - (b) The standard deviation of X .
 - (c) The moment generating function of $Y = 2X + 3$.
3. In a binomial distribution having 4 independent trials, it is known that the probabilities of 1 and 2 successes are $\frac{2}{3}$, and $\frac{1}{3}$ respectively. Find
 - (a) The parameter p of the distribution.
 - (b) The mean and variance of the distribution.

4. A coin is biased so that a head is thrice as likely to appear as a tail. If the coin is tossed 5 times, find the probability of getting (a) at least 3 heads, (b) at most 3 heads, and (c) exactly 3 tails.
5. A box contains 150 light bulbs, 30 of which are defective. A sample of 10 bulbs is selected for inspection. Find the probability that (a) at least 2 bulbs are defective, (b) at most 2 bulbs are defective, and (c) all the 10 bulbs are defective.
6. A box of 100 pens contains 10 defective pens. If 5 pens are selected at random from the box and sent to a retail store, find the probability that the store will receive at least one defective pen.
7. It is known that 5% of the screws manufactured by an automatic machine are defective. If a sample of 20 screws is selected at random, find the probability that the sample contains (a) exactly 2 defective screws, (b) at least 2 defective screws, (c) at most 2 defective screws, and (d) no defective screws.
8. A fair die is thrown, and an outcome of 4 or 5 is considered to be a success. If the die is thrown 9 times and X denotes the number of successes, find (a) the mean and variance of X , (b) $P(X = 2)$, (c) $P(X \leq 2)$ and (d) $P(X \geq 2)$.
9. A pair of fair dice is thrown 5 times. If getting a doublet is considered to be a success, find the probability of getting (a) at least 2 successes, (b) at most 2 successes, and (c) exactly 2 failures.
10. The probability that a bomb hits a target is given by 0.6. Assuming a binomial distribution, find the probability that out of 10 bombings, exactly 3 will hit the target.
11. Fit a binomial distribution to the following frequency distribution:

No. of successes	0	1	2	3	4
Frequency	4	11	21	9	5

12. Fit a binomial distribution to the following frequency distribution:

No. of successes	0	1	2	3	4	5
Frequency	7	18	26	24	17	8

3.4 POISSON DISTRIBUTION

Poisson distribution is a discrete probability distribution, which was discovered by Simon-Denis Poisson (1781–1840) and published in the year 1838.

Theorem 3.6. Poisson distribution is a limiting case of the binomial distribution under the following conditions:

- (a) n , the number of independent Bernoulli trials, increases indefinitely, i.e. $n \rightarrow \infty$.
- (b) p , the constant probability of success in each Bernoulli trial, decreases indefinitely, i.e. $p \rightarrow 0$.
- (c) $np = \mu$, which is the expected number of successes, remains constant.

(Anna, May 2007)

Proof. Suppose that X is a binomial random variable with parameters n and p satisfying the conditions (a), (b) and (c).

Note that

$$B(x; n, p) = {}^nC_x p^x q^{n-x} = {}^nC_x p^x (1-p)^{n-x} = {}^nC_x \frac{p^x}{(1-p)^x} (1-p)^n$$

Since $\mu = np$, we can write $p = \frac{\mu}{n}$ and note that μ is a constant.

$$B(x; n, p) = \frac{n(n-1)(n-2)\cdots(n-x+1)}{x!} \frac{\left(\frac{\mu}{n}\right)^x}{\left(1-\frac{\mu}{n}\right)^x} \left(1-\frac{\mu}{n}\right)^n$$

Simplifying, we get

$$B(x; n, p) = \frac{1\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{x-1}{n}\right)}{x! \left(1-\frac{\mu}{n}\right)^x} \mu^x \left(1-\frac{\mu}{n}\right)^n$$

Taking limits as $n \rightarrow \infty$, it follows that

$$\lim_{n \rightarrow \infty} B(x; n, p) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots \quad (3.9)$$

where we have used the results that

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\mu}{n}\right)^x = 1$$

and

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\mu}{n}\right)^n = e^{-\mu} \quad \blacklozenge$$

The probability law given in Eq. (3.9) is known as the *Poisson distribution*. It has just one parameter, namely, λ . Formally, the Poisson random variable is defined as follows.

Definition 3.4. A random variable X is said to have a Poisson distribution with parameter μ if its probability mass function is given by

$$f(x; \mu) = P(X = x) = \begin{cases} \frac{e^{-\mu} \mu^x}{x!} & \text{for } x = 0, 1, 2, \\ 0 & \text{elsewhere} \end{cases}$$

If X is a Poisson random variable with parameter μ , we write it as $X \sim P(\mu)$.

EXAMPLE 3.21. State some applications of Poisson distribution.

Solution. Some important applications of Poisson distribution are to find or calculate:

- (i) The load on Web servers.
- (ii) The load on telecommunication devices, for example, the number of telephone calls received per minute.
- (iii) The number of passengers entering a railway station on a given day.
- (iv) The number of air accidents in some unit of time.
- (v) The number of wrong telephone numbers dialled on a day.
- (vi) The number of typing mistakes in a book. \blacklozenge

Remark 3.6. Poisson distribution is well-defined because the probabilities $f(x; \lambda)$ are non-negative for all values of x and the total probability is 1, i.e.

$$\begin{aligned} \sum_{x=0}^{\infty} f(x; \mu) &= \sum_{x=0}^{\infty} \frac{e^{-\mu} \mu^x}{x!} \\ &= e^{-\mu} \sum_{x=0}^{\infty} \frac{\mu^x}{x!} = e^{-\mu} e^{\mu} = 1 \end{aligned}$$

Theorem 3.7. If X is a Poisson random variable with parameter μ , i.e. $X \sim P(\mu)$, then both the mean and variance of X are equal to μ , i.e. $E(X) = \mu$ and $\text{Var}(X) = \mu$.

(Anna, May 2007)

Proof. Let X be a Poisson random variable with parameter μ . Then, X follows the Poisson density given by

$$f(x; \mu) = P(X = \mu) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

By definition, we have

$$\begin{aligned} \mu'_1 &= E(X) = \sum_{x=0}^{\infty} x \frac{e^{-\mu} \mu^x}{x!} = \sum_{x=1}^{\infty} x \frac{e^{-\mu} \mu^x}{x!} \\ E(X) &= \sum_{x=1}^{\infty} x \frac{e^{-\mu} \mu^x}{x(x-1)!} = e^{-\mu} \sum_{x=1}^{\infty} \frac{\mu^x}{(x-1)!} \\ &= e^{-\mu} \sum_{y=0}^{\infty} \frac{\mu^{y+1}}{y!} \quad (\text{By putting } y = x - 1) \end{aligned}$$

Therefore,

$$E(X) = e^{-\mu} \mu \sum_{y=0}^{\infty} \frac{\mu^y}{y!} = e^{-\mu} \mu e^{\mu} = \mu$$

Thus, $\text{Mean}(X) = \mu$.

Next, we find $E(X^2)$ by noting that

$$\mu'_2 = E(X^2) = E[X(X-1)] + E(X) = E[X(X-1)] + \mu$$

Now,

$$\begin{aligned} E[X(X-1)] &= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\mu} \mu^x}{x!} = \sum_{x=2}^{\infty} x(x-1) \frac{e^{-\mu} \mu^x}{x!} \\ &= \sum_{x=2}^{\infty} x(x-1) \frac{e^{-\mu} \mu^x}{x(x-1)(x-2)!} = e^{-\mu} \sum_{x=2}^{\infty} \frac{e^{-\mu} \mu^x}{(x-2)!} \\ &= \mu^2 e^{-\mu} \sum_{x=2}^{\infty} \frac{\mu^{x-2}}{(x-2)!} = \mu^2 e^{-\mu} \sum_{y=0}^{\infty} \frac{\mu^y}{y!} = \mu^2 e^{-\mu} e^{\mu} = \mu^2 \end{aligned}$$

Thus,

$$\mu'_2 = E[X(X-1)] + \mu = \mu^2 + \mu$$

Therefore, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = \mu^2 + \mu - \mu^2 = \mu$$

Thus, for a Poisson distribution, both the mean and variance are equal to μ , the parameter of the distribution. ◆

Theorem 3.8. Let X be a Poisson random variable with parameter μ . Then the following properties hold:

- (a) The central moments, μ_r , satisfy the recurrence relation

$$\mu_{r+1} = \mu \left[r\mu_{r-1} + \frac{d\mu_r}{d\mu} \right]$$

- (b) The second, third and fourth central moments are

$$\mu_2 = \mu, \quad \mu_3 = \mu \quad \text{and} \quad \mu_4 = 3\mu^2 + \mu$$

- (c) Karl Pearson's coefficients are

$$\text{Skewness} = \gamma_1 = \frac{\sqrt{\beta_1}}{\sqrt{\mu}} \quad \text{and} \quad \text{Kurtosis} = \gamma_2 = \beta_2 - 3 = \frac{1}{\mu}$$

Proof. (a) Since X is a Poisson variate with parameter μ , the mean $E(X) = \mu$. Thus, the r th central moment, μ_r , is defined by

$$\mu_r = E[(X - \mu)^r] = \sum_{x=0}^{\infty} (x - \mu)^r \frac{e^{-\mu} \mu^x}{x!}$$

Hence, it follows that

$$\begin{aligned} \frac{d\mu_r}{d\mu} &= \sum_{x=0}^{\infty} r(x - \mu)^{r-1}(-1) \frac{e^{-\mu} \mu^x}{x!} + \sum_{x=0}^{\infty} (x - \mu)^r \frac{1}{x!} [-e^{-\mu} \mu^x + e^{-\mu} x \mu^{x-1}] \\ &= -r\mu_{r-1} + \sum_{x=0}^{\infty} (x - \mu)^r \frac{\mu^{x-1}(x - \mu)e^{-\mu}}{x!} \\ &= -r\mu_{r-1} + \frac{1}{\mu} \sum_{x=0}^{\infty} (x - \mu)^{r+1} \frac{\mu^x e^{-\mu}}{x!} \end{aligned}$$

Hence, we see that

$$\frac{d\mu_r}{d\mu} = -r\mu_{r-1} + \frac{1}{\mu} \mu_{r+1}$$

Rearranging terms, we have

$$\mu_{r+1} = \mu \left[r\mu_{r-1} + \frac{d\mu_r}{d\mu} \right]$$

(b) For any distribution, $\mu_0 = 1$ and $\mu_1 = 1$.

Taking $r = 1$ in the recurrence relation in (a), we have

$$\mu_2 = \mu \left[1\mu_0 + \frac{d\mu_1}{d\mu} \right] = \mu [1 + 0] = \mu$$

Thus, $\mu_2 = \mu$.

Taking $r = 2$ in the recurrence relation in (a), we have

$$\mu_3 = \mu \left[2\mu_1 + \frac{d\mu_2}{d\mu} \right] = \mu [0 + 1] = \mu$$

Thus, $\mu_3 = \mu$.

Taking $r = 3$ in the recurrence relation in (a), we have

$$\mu_4 = \mu \left[3\mu_2 + \frac{d\mu_3}{d\mu} \right] = \mu [3\mu + 1] = 3\mu^2 + \mu$$

Thus, $\mu_4 = 3\mu^2 + \mu$.

(c) Next, we calculate the Karl Pearson's coefficients of skewness and kurtosis. First, we find

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{\mu^2}{\mu^3} = \frac{1}{\mu} \quad \text{and} \quad \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\mu^2 + \mu}{\mu^2} = 3 + \frac{1}{\mu}$$

Therefore, we have

$$\text{Skewness} = \gamma_1 = \sqrt{\beta_1} = \frac{1}{\sqrt{\mu}} \quad \text{and} \quad \text{Kurtosis} = \gamma_2 = \beta_2 - 3 = \frac{1}{\mu}$$

◆

EXAMPLE 3.22. Show that for a Poisson distribution with parameter m , $\gamma_1 \gamma_2 \sigma m = 1$.

Solution. We know that the mean, $\mu = m$ and the standard deviation, $\sigma = \sqrt{m}$. It follows that

$$\gamma_1 = \frac{1}{\sqrt{m}} \quad \text{and} \quad \gamma_2 = \frac{1}{m}$$

Thus, it is immediate that $\gamma_1 \gamma_2 \sigma m = 1$.

◆

EXAMPLE 3.23. Deduce the first four central moments of the Poisson distribution from those of the binomial distribution.

Solution. For any distribution, $\mu_0 = 1$ and $\mu_1 = 0$. For the binomial distribution,

$$\mu_2 = npq, \quad \mu_3 = npq(q-p) \quad \text{and} \quad \mu_4 = 3n^2 p^2 q^2 + npq(1-6pq)$$

We know that Poisson distribution is a limiting case of the binomial distribution under the conditions, $n \rightarrow \infty$, $p \rightarrow 0$ (hence, $q \rightarrow 1$) and $np = \mu$, a constant. (Note that $\mu > 0$ as n and p are positive.)

Taking limits, the central moments of Poisson distribution are obtained as

$$\mu_2 = \lim[npq] = \lim[np] \lim[q] = \mu \times 1 = \mu$$

$$\mu_3 = \lim[npq(q-p)] = \lim[(np)q^2] - \lim[(np)(pq)] = \mu(1) - \mu(0) = \mu$$

$$\mu_4 = \lim[3n^2p^2q^2 + npq(1-6pq)] = 3\mu^2(1) + \mu(1)(1-0) = 3\mu^2 + \mu$$

◆

Theorem 3.9. Let X be a Poisson random variable with parameter μ .

(a) The moment generating function of X is given by

$$M_X(t) = e^{\mu(e^t - 1)}$$

(b) The characteristic function of X is given by

$$\phi_X(t) = e^{\mu(e^{it} - 1)}$$

Proof. (a) The MGF of the Poisson distribution is given by

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\mu} \mu^x}{x!} \\ &= e^{-\mu} \sum_{x=0}^{\infty} \frac{(\mu e^t)^x}{x!} = e^{-\mu} e^{\mu e^t} \end{aligned}$$

Simplifying, we get

$$M_X(t) = e^{-\mu + \mu e^t} = e^{\mu(e^t - 1)}$$

(b) The characteristic function of the Poisson distribution is given by

$$\begin{aligned} \phi_X(t) &= E(e^{itX}) = \sum_{x=0}^{\infty} e^{itx} \frac{e^{-\mu} \mu^x}{x!} \\ &= e^{-\mu} \sum_{x=0}^{\infty} \frac{(\mu e^{it})^x}{x!} = e^{-\mu} e^{\mu e^{it}} \end{aligned}$$

Simplifying, we get

$$\phi_X(t) = e^{-\mu + \mu e^{it}} = e^{\mu(e^{it} - 1)}$$

◆

EXAMPLE 3.24. Find the moment generating function, the mean and variance of a Poisson distribution.

(Anna, Nov. 2007)

Solution. If X is a Poisson random variable with parameter μ , then its MGF is given by

$$M_X(t) = e^{\mu(e^t - 1)}$$

Thus, we have

$$M'_X(t) = e^{\mu(e^t - 1)} [\mu e^t]$$

and

$$M''_X(t) = e^{\mu(e^t - 1)} [\mu e^t]^2 + e^{\mu(e^t - 1)} [\mu e^t]$$

Hence, $\mu'_1 = M'_X(0) = \mu$ and $\mu'_2 M''_X(0) = \mu^2 + \mu$.

Thus, $\text{Mean}(X) = \mu'_1 = \mu$ and $\text{Var}(X) = \mu'_2 - \mu'^2_1 = \mu$. ◆

EXAMPLE 3.25. The MGF of a random variable X is given by $M_X(t) = e^{3(e^t - 1)}$. Find $P(X = 1)$.
(Anna, Model 2003)

Solution. We know that the MGF of the Poisson random variable Y with parameter μ is given by

$$M_Y(t) = e^{\mu(e^t - 1)}$$

Given that $M_X(t) = e^{3(e^t - 1)}$. Hence, by the uniqueness theorem of moment generating functions, it follows that the given random variable X has a Poisson distribution with parameter $\mu = 3$. Thus, it follows that

$$P(X = x) = e^{-3} \frac{3^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

The required probability is

$$P(X = 1) = e^{-3} \frac{3^1}{1!} = 3 \times e^{-3} = 0.1494$$
◆

EXAMPLE 3.26. Find the moment generating function of the standard Poisson variable

$$Z = \frac{X - \mu}{\sigma}$$

where X is a Poisson variable with parameter p , and obtain its limiting form as $\mu \rightarrow \infty$. Also interpret the result.

Solution. If X is a Poisson random variable with parameter μ , then we know that X has mean and variance both equal to μ . Thus, $\sigma = \sqrt{\mu}$.

Next, by Theorem 2.22, we know that

$$M_Z(t) = e^{-\frac{\mu t}{\sigma}} M_X\left(\frac{t}{\sigma}\right)$$

Since $M_X(t) = e^{\mu(e^t - 1)}$, it follows that

$$M_Z(t) = e^{-t\sqrt{\mu}} e^{\mu\left(e^{\frac{t}{\sqrt{\mu}}} - 1\right)} = e^{-t\sqrt{\mu} + \mu\left(e^{\frac{t}{\sqrt{\mu}}} - 1\right)}$$

Note that

$$e^{\frac{t}{\sqrt{\mu}}} = 1 + \frac{t}{\sqrt{\mu}} + \frac{t^2}{2\mu} + \frac{t^3}{3\mu^{\frac{3}{2}}} + \dots$$

It follows that

$$\mu \left(e^{\frac{t}{\sqrt{\mu}}} - 1 \right) = t\sqrt{\mu} + \frac{t^2}{2} + \frac{t^3}{6\sqrt{\mu}} + \dots$$

Thus, we have

$$M_Z(t) = e^{-t\sqrt{\mu} + \mu \left(e^{\frac{t}{\sqrt{\mu}}} - 1 \right)} = e^{\frac{t^2}{2} + \frac{t^3}{6\sqrt{\mu}} + \dots} \quad (3.10)$$

Taking natural logarithms on both sides, we have

$$\log M_Z(t) = \frac{t^2}{2} + \frac{t^3}{6\sqrt{\mu}} + \dots$$

Taking limits as $\mu \rightarrow \infty$, we have

$$\lim_{\mu \rightarrow \infty} \log M_Z(t) = \frac{t^2}{2} + 0 = \frac{t^2}{2}$$

Hence, it follows that

$$\lim_{\mu \rightarrow \infty} M_Z(t) = e^{\frac{t^2}{2}} \quad (3.11)$$

Interpretation: The MGF on the RHS of Eq. (3.11) is the MGF of the standard normal variable. Hence, by the Uniqueness Theorem for MGFs (see Theorem 2.23), it follows that the standard Poisson random variable approaches the standard normal variable as $\mu \rightarrow \infty$. In other words, the normal distribution is the limiting form of the Poisson distribution as $\mu \rightarrow \infty$. $\blacklozenge$

Theorem 3.10. Let X be a Poisson random variable with parameter μ .

(a) The probability law $f(x)$ satisfies

$$\frac{f(x)}{f(x-1)} = \frac{\mu}{x} \quad \text{for } x = 1, 2, 3, \dots$$

(b) If μ is not an integer, then the Poisson distribution has a unique mode given by $m = [\mu]$, i.e. the greatest integer less than μ .

(c) If μ is an integer, say m , then the Poisson distribution is bimodal and the two modal values are $x = m = \mu$ and $x = m - 1 = \mu - 1$.

Proof. (a) By definition of Poisson probability law,

$$f(x) = e^{-\mu} \frac{\mu^x}{x!}$$

and

$$f(x-1) = e^{-\mu} \frac{\mu^{x-1}}{(x-1)!}$$

Thus, it follows that

$$\frac{f(x)}{f(x-1)} = \frac{\mu^x}{x!} \times \frac{(x-1)!}{\mu^{x-1}} = \frac{\mu}{x}$$

(b) If μ is not an integer, let $m = [\mu]$, the greatest integer less than μ . Then

$$\mu = m + f, \text{ where } 0 < f < 1$$

where m and f are the integral and fractional parts of μ , respectively. From (a) we have

$$\frac{f(x)}{f(x-1)} = \frac{m+f}{x} > 1 \text{ if } x = 0, 1, 2, \dots, m$$

and

$$\frac{f(x)}{f(x-1)} = \frac{m+f}{x} < 1 \text{ if } x = m+1, m+2, \dots$$

Hence, it is immediate that

$$f(m) > f(m-1) > f(m-2) > \dots > f(2) > f(1)$$

and

$$f(m) < f(m+1) < f(m+2) < \dots$$

Hence, $x = m$ is the unique point, where f attains its maximum value. Hence, $x = m = [\mu]$ is the unique mode for the Poisson distribution if μ is not an integer.

(c) If μ is an integer, let $m = \mu$.

From (a) we have

$$\begin{aligned} \frac{f(x)}{f(x-1)} &= \frac{m}{x} > 1 \text{ if } x = 0, 1, 2, \dots, m-1 \\ &= \frac{m}{m} = 1 \text{ if } x = m \end{aligned}$$

and

$$\frac{f(x)}{f(x-1)} = \frac{m}{x} < 1 \text{ if } x = m+1, m+2, \dots$$

Hence, it is immediate that

$$f(m) = f(m-1) > f(m-2) > \dots > f(2) > f(1)$$

and

$$f(m) < f(m+1) < f(m+2) < \dots$$

So, $x = m$ and $x = m-1$ are the two points, where f attains its maximum value. Hence, $x = \mu$ and $x = \mu - 1$ are the two modes for the Poisson distribution if μ is an integer.



EXAMPLE 3.27. A Poisson distribution has a double mode at $x = 2$ and $x = 3$. Find the probability that X will have either of these values.

Solution. By Theorem 3.10 (c), it is immediate that $\mu = 3$. (Alternatively, we may deduce the same by using $P(X = 2) = P(X = 3)$.)

Hence, X follows the Poisson probability law given by

$$f(x) = P(X = x) = e^{-3} \frac{3^x}{x!}, \quad x = 0, 1, 2, \dots$$

It follows that

$$P(X = 2) = e^{-3} \frac{3^2}{2!} = e^{-3}(4.5) = 0.2240$$

and

$$P(X = 3) = P(X = 2) = 0.2240$$

The required probability that X takes on any of the values 2 or 3 is given by

$$P(X = 2) + P(X = 3) = 0.2240 + 0.2240 = 0.4480$$

◆

EXAMPLE 3.28. Show that in a Poisson distribution with unit mean, the mean deviation (M.D.) about the mean is $\frac{2}{e}$ times the standard deviation.

Solution. For the Poisson distribution with parameter μ , both mean and variance are equal to μ . Since the given Poisson random variable X has unit mean, it is immediate that $\mu = 1$ and $\sigma = \sqrt{\mu} = 1$.

Note that X has the Poisson probability law

$$f(x) = P(X = x) = e^{-1} \frac{1^x}{x!} = e^{-1} \frac{1}{x!}, \quad x = 0, 1, 2, \dots$$

Hence, the mean deviation about the mean is given by

$$\begin{aligned} \text{M.D.} &= E|X - \mu| = \sum_{x=0}^{\infty} |x - 1| e^{-1} \frac{1}{x!} = \frac{1}{e} \sum_{x=0}^{\infty} \frac{|x - 1|}{x!} \\ &= \frac{1}{e} \left[1 + \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots \right] = \frac{1}{e} \left[1 + \frac{(2-1)}{2!} + \frac{(3-1)}{3!} + \frac{(4-1)}{4!} + \dots \right] \\ &= \frac{1}{e} \left[1 + \left(1 - \frac{1}{2!} \right) + \left(\frac{1}{2!} - \frac{1}{3!} \right) + \left(\frac{1}{3!} - \frac{1}{4!} \right) + \dots \right] = \frac{1}{e} [2] \end{aligned}$$

Hence,

$$\text{M.D.} = \frac{2}{e} = \frac{2}{e} \times \sigma$$

◆

EXAMPLE 3.29. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the variance.

(Anna, April 2003)

Solution. Let X be a Poisson variate with parameter μ . Then

$$P(X = x) = e^{-\mu} \frac{\mu^x}{x!} \quad \text{for } x = 0, 1, 2, 3, \dots$$

By hypothesis, we have

$$e^{-\mu} \frac{\mu^2}{2!} = 9 e^{-\mu} \frac{\mu^4}{4!} + 90 e^{-\mu} \frac{\mu^6}{6!}$$

Simplifying, we have

$$\frac{1}{2} = \frac{3}{8}\mu^2 + \frac{1}{8}\mu^4 \quad \text{or} \quad \mu^4 + 3\mu^2 - 4 = 0$$

Factorizing, we have

$$(\mu^2 + 4)(\mu^2 - 1) = 0 \Rightarrow \mu = \pm 2j, \pm 1$$

Since μ is a positive real number, we must have $\mu = 1$.

For a Poisson variate, $\text{Var}(X) = \mu$. Hence, the variance of X is 1. ◆

EXAMPLE 3.30. One per cent of jobs arriving at a computer system need to wait until weekends for scheduling, owing to core-size limitations. Find the probability that among a sample of 200 jobs there are no jobs that have to wait until weekends.

(Anna, May 2007)

Solution. Let p denote the probability that a job arriving at a computer system needs to wait until weekends for scheduling and n denote the sample size. Given that $p = 0.01$ and $n = 200$. Since p is very small, the given probability distribution can be assumed to be Poisson with the parameter

$$\mu = np = 200 \times 0.01 = 2$$

Hence, if X denotes the number of jobs arriving at the computer system that have to wait until weekends for scheduling, then the probability distribution of X is given by

$$P(X = x) = e^{-2} \frac{2^x}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

Hence, the required probability that there are no jobs that have to wait until weekends from the given sample is

$$P(X = 0) = e^{-2} = 0.1353 \quad \text{◆}$$

EXAMPLE 3.31. In a component manufacturing industry, there is a small probability of $\frac{1}{500}$ for any component to be defective. The components are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing (i) at least one defective, and (ii) at most one defective, in a consignment of 1000 packets.

Solution. Given that the probability for a component to be defective is

$$p = \frac{1}{500}$$

and that components are supplied in packets of 10, i.e. $n = 10$. Hence, the Poisson parameter is

$$\mu = np = 10 \times \frac{1}{500} = \frac{1}{50} = 0.02$$

Thus, X , the number of defective components in a packet, satisfies the Poisson probability law

$$P(X = x) = e^{-\mu} \frac{\mu^x}{x!} e^{-0.02} \frac{(0.02)^x}{x!}, \quad x = 0, 1, 2, \dots$$

(i) The probability that there is at least one defective component in a packet is equal to

$$P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0) = 1 - e^{-0.02} = 0.0198$$

Hence, the approximate number of packets containing at least one defective in $N = 1000$ packets is

$$N \times P(X \geq 1) = 1000 \times 0.0198 = 19.8 \approx 20$$

(ii) The probability that there is at most one defective component is equal to

$$P(X \leq 1) = P(X = 0) + P(X = 1) = e^{-0.02} [1 + 0.02] = e^{-0.02} \times 1.02 = 0.9998$$

Hence, the approximate number of packets containing at most one defective component in $N = 1000$ packets is

$$N \times P(X \leq 1) = 1000 \times 0.9998 = 999.8 \approx 1000$$

◆

EXAMPLE 3.32. The probability of an item produced by a certain machine will be defective is 0.05. If the produced items are sent to the market in packets of 20, find the number of packets containing (i) at least, (ii) exactly, and (iii) at most 2 defective items in a consignment of 1000 packets using Poisson approximation to a binomial distribution.

(Anna, April 2004)

Solution. Let p be the probability that an item produced is defective. Then it is given that $p = 0.05$. Also, the produced items are sent to the market in packets of 20. Therefore, $n = 20$. Hence, if X denotes the number of defective items in a packet, then X follows a Poisson distribution with mean

$$\mu = np = 20 \times 0.05 = 1$$

Thus, $\mu = 1$. It is also given that $N = 1000$ packets (in a consignment).

Thus, the probability that there will be x defective items in a packet is

$$P(X = x) = e^{-\mu} \frac{\mu^x}{x!} = e^{-1} \frac{1}{x!}, \quad x = 0, 1, 2, \dots$$

(i) The probability that a packet contains at least two defective items is equal to

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] \\ &= 1 - [e^{-1} + e^{-1}] = 1 - \frac{2}{e} = 0.2642 \end{aligned}$$

Hence, the number of packets containing at least two defective items in $N = 1000$ packets is

$$NP(X \geq 2) = 1000 \times 0.2642 = 264.2 \approx 264$$

(ii) The probability that a packet contains exactly two defective items is equal to

$$P(X = 2) = e^{-1} \frac{1}{2!} = 0.5e^{-1} = 0.1839$$

Hence, the number of packets containing exactly two defective items in $N = 1000$ packets is

$$NP(X = 2) = 1000 \times 0.1839 = 183.9 \approx 184$$

(iii) The probability that a packet contains at most two defective items is equal to

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = e^{-1} + e^{-1} + e^{-1} \frac{1}{2} = 2.5e^{-1} = 0.9197$$

Hence, the number of packets containing at most two defective items in $N = 1000$ packets is

$$NP(X \leq 2) = 1000 \times 0.9197 = 919.7 \approx 920$$

◆

EXAMPLE 3.33. If the probability that a person suffers from a disease is 0.001, find the probability that out of 3000 persons, (i) exactly 4, (ii) more than 2 persons will suffer from the disease.

(Anna, April 2005)

Solution. Let p be the probability that a person will suffer from a disease. Then $p = 0.001$. It is also given that $n = 3000$. If X denotes the number of persons suffering from the disease, then X can be assumed to follow a Poisson distribution since p is very small and n is large. Hence, X has the mean $\mu = np = 3$.

By Poisson law, the probability that there will be x persons suffering from the disease is

$$P(X = x) = e^{-\mu} \frac{\mu^x}{x!} = e^{-3} \frac{3^x}{x!}, \quad x = 0, 1, 2, \dots$$

(i) The probability that exactly 4 people will suffer from the disease is

$$P(X = 4) = e^{-3} \frac{3^4}{4!} = e^{-3} \frac{27}{8} = 0.1680$$

(ii) The probability that more than 2 persons will suffer from the disease is

$$P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)]$$

Thus,

$$P(X \geq 2) = 1 - [e^{-3} + e^{-3} \cdot 3] = 1 - 4e^{-3} = 0.8009$$

◆

EXAMPLE 3.34. In a city, 6% of all drivers get at least one parking ticket per year. Use Poisson distribution to determine the probability that among 80 drivers, 4 will get at least one parking ticket in any given year.

(Anna, Model 2003)

Solution. Let p denote the probability that a driver will get at least one parking ticket in the given year. It is given that $p = \frac{6}{100}$. Thus, if $n = 80$ drivers, then the mean μ is given by

$$\mu = np = 80 \times \frac{6}{100} = 4.8$$

If X denotes the number of drivers who will get at least one parking ticket in the given year, then X follows the Poisson distribution with parameter $\mu = 4.8$. Therefore,

$$P(X = x) = e^{-4.8} \frac{4.8^x}{x!}, \quad x = 0, 1, 2, \dots$$

Hence, the probability that 4 drivers will get at least one parking ticket in the given year is

$$P(X = 4) = e^{-4.8} \frac{4.8^4}{4!} = 0.1820$$

◆

EXAMPLE 3.35. Assuming that the probability of a fatal accident in a factory during the year is $\frac{1}{1200}$, calculate the probability that in a factory employing 300 workers, there will be at least two fatal accidents in a year.

(Anna, Nov. 2005)

Solution. Let p denote the probability of a fatal accident in a factory during the year. Given that $p = \frac{1}{2000}$. Let X denote the number of fatal accidents in a year. Since p is very small, X can be assumed to follow a Poisson distribution. Since $n = 300$ workers, it follows that the mean of the distribution is $\mu = np = \frac{1}{4} = 0.25$. Thus, X has the Poisson density given by

$$P(X = x) = \frac{e^{-0.25}(0.25)^x}{x!} \quad \text{for } x = 0, 1, 2, 3, \dots$$

Hence, the probability that there will be at least two fatal accidents in a year is

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] \\ &= 1 - e^{-0.25} [1 + 0.25] = 1 - e^{-0.25} \times 1.25 = 0.0265 \end{aligned}$$

◆

EXAMPLE 3.36. The number of planes landing at an airport in a 30-minutes interval obeys the Poisson law with mean 25. Use Chebyshev's inequality to find the least chance that the number of planes landing within the 30-minutes interval will be between 15 and 35.

(Madras, April 1997)

Solution. Let X be the number of planes landing at an airport in a 30-minutes interval. It is given that $X \sim P(25)$. Then X has mean $\mu = 25$ and variance $\sigma^2 = 25$.

By Chebyshev's inequality, we know that

$$\begin{aligned} P(|X - \mu| \leq C) &\geq \frac{\sigma^2}{C^2} \\ P(15 \leq X \leq 35) &= P(|X - \mu| \leq 10) \geq \frac{25}{10^2} = \frac{1}{4} \end{aligned}$$

Hence, the required least chance for the number of planes landing within in a given 30-minutes interval to lie between 15 and 35 is given by $\frac{1}{4}$ or 25%.

◆

EXAMPLE 3.37. The monthly breakdowns of a computer is a random variable having Poisson distribution with a mean equal to 1.8. Find the probability that this computer will function for a month (i) without a breakdown, (ii) with only one breakdown, and (iii) with at least one breakdown.

(Anna, May 2006; May 2007; Nov. 2007)

Solution. Let X be the number of monthly breakdowns of the computer. Given that X is a Poisson random variable with parameter $\mu = 1.8$. Hence, the probability mass function of X is given by

$$P(X = x) = \frac{e^{-1.8}(1.8)^x}{x!} \text{ for } x = 0, 1, 2, \dots$$

(i) The probability that the computer will function for a month without a breakdown is

$$P(X = 0) = e^{-1.8} = 0.1653$$

(ii) The probability that the computer will function for a month with only one breakdown is

$$P(X = 1) = e^{-1.8} \times 1.8 = 0.2975$$

(iii) The probability that the computer will function for a month with at least one breakdown is

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.1653 = 0.8347$$

◆

EXAMPLE 3.38. Let X be a Poisson random variable with parameter μ . Show that

(i) $P(X \text{ is even}) = \frac{1}{2} [1 + e^{-2\mu}]$.

(ii) $P(X \text{ is odd}) = \frac{1}{2} [1 - e^{-2\mu}]$.

Solution. By hypothesis, $P(X = x) = e^{-\mu} \frac{\mu^x}{x!}$ for $x = 0, 1, 2, \dots$

(i) We see that

$$\begin{aligned} P(X \text{ is even}) &= P(X = 0) + P(X = 2) + P(X = 4) + \dots + P(X = 2k) + \dots \\ &= e^{-\mu} + e^{-\mu} \frac{\mu^2}{2!} + e^{-\mu} \frac{\mu^4}{4!} + \dots + e^{-\mu} \frac{\mu^{2k}}{(2k)!} + \dots \\ &= e^{-\mu} \left[1 + \frac{\mu^2}{2!} + \frac{\mu^4}{4!} + \dots + \frac{\mu^{2k}}{(2k)!} + \dots \right] = e^{-\mu} \cosh \mu \end{aligned}$$

Therefore,

$$P(X \text{ is even}) = e^{-\mu} \frac{1}{2} [e^{\mu} + e^{-\mu}] = \frac{1}{2} [1 + e^{-2\mu}]$$

(ii) We see that

$$\begin{aligned} P(X \text{ is odd}) &= P(X = 1) + P(X = 3) + P(X = 5) + \dots + P(X = 2k+1) + \dots \\ &= e^{-\mu} \mu + e^{-\mu} \frac{\mu^3}{3!} + e^{-\mu} \frac{\mu^5}{5!} + \dots + e^{-\mu} \frac{\mu^{2k+1}}{(2k+1)!} + \dots \\ &= e^{-\mu} \left[\mu + \frac{\mu^3}{3!} + \frac{\mu^5}{5!} + \dots + \frac{\mu^{2k+1}}{(2k+1)!} + \dots \right] = e^{-\mu} \sinh \mu \end{aligned}$$

Therefore,

$$P(X \text{ is odd}) = e^{-\mu} \frac{1}{2} [e^{\mu} - e^{-\mu}] = \frac{1}{2} [1 - e^{-2\mu}]$$

◆

EXAMPLE 3.39. Let X be a Poisson random variable with parameter μ . Show that

- (i) $E(e^{-kX}) = e^{-\mu(1-e^{-k})}$ where k is any constant.
- (ii) $E(kXe^{-kX}) = \mu ke^{\mu(e^{-k}-1)-k}$ where k is any constant.

Solution. Given that X follows the Poisson probability law

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!} \quad \text{for } x = 0, 1, 2, \dots, \infty$$

(i) We find that

$$E(e^{-kX}) = \sum_{x=0}^{\infty} e^{-kx} \frac{e^{-\mu} \mu^x}{x!} = e^{-\mu} \sum_{x=0}^{\infty} \frac{(\mu e^{-k})^x}{x!} = e^{-\mu} e^{\mu e^{-k}}$$

Simplifying, we have

$$E(e^{-kX}) = e^{-\mu + \mu e^{-k}} = e^{-\mu(1-e^{-k})}$$

(ii) We find that

$$\begin{aligned} E(kXe^{-kX}) &= \sum_{x=0}^{\infty} kxe^{-kx} \frac{e^{-\mu} \mu^x}{x!} = ke^{-\mu} \sum_{x=1}^{\infty} xe^{-kx} \frac{\mu^x}{x!} \\ &= ke^{-\mu} \sum_{x=1}^{\infty} e^{-kx} \frac{\mu^x}{(x-1)!} = ke^{-\mu} \mu e^{-k} \sum_{x=1}^{\infty} \frac{(\mu e^{-k})^{x-1}}{(x-1)!} \end{aligned}$$

Putting $y = x - 1$ in the summation, we have

$$E(kXe^{-kX}) = \mu ke^{-\mu-k} \sum_{y=0}^{\infty} \frac{(\mu e^{-k})^y}{y!} = \mu ke^{-\mu-k} e^{\mu e^{-k}}$$

Simplifying, we have

$$E(kXe^{-kX}) = \mu ke^{\mu(e^{-k}-1)-k}$$

◆

Theorem 3.11. (*Fitting a Poisson Distribution*) If X is a Poisson random variable with parameter μ , then the probability mass function of X satisfies the following recurrence relation:

$$f(x+1) = \frac{\mu}{x+1} f(x), \quad x = 0, 1, 2, \dots$$

Proof. Since

$$f(x) = e^{-\mu} \frac{\mu^x}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

it follows that

$$f(x+1) = e^{-\mu} \frac{\mu^{x+1}}{(x+1)!}$$

Thus, it is immediate that

$$\frac{f(x+1)}{f(x)} = \frac{\mu^{x+1}}{(x+1)!} \times \frac{x!}{\mu^x} = \frac{\mu}{x+1}$$

Hence, we have

$$f(x+1) = \frac{\mu}{x+1} f(x), \quad x = 0, 1, 2, \dots$$

◆

Remark 3.7. The recurrence formula given by Theorem 3.11 provides a very simple method for fitting a given frequency distribution by a Poisson distribution. First, we find the mean μ of the frequency distribution. If N represents the total of all the frequencies, then the theoretical frequencies of the Poisson distribution can be easily calculated as follows:

$$Nf(0) = Ne^{-\mu}$$

$$Nf(1) = N \left[\frac{\mu}{x+1} \right]_{x=0} f(0) = \frac{\mu}{1} [Nf(0)]$$

$$Nf(2) = N \left[\frac{\mu}{x+1} \right]_{x=1} f(1) = \frac{\mu}{2} [Nf(1)]$$

$$Nf(3) = \left[\frac{\mu}{x+1} \right]_{x=2} f(2) = \frac{\mu}{3} [Nf(2)]$$

Similarly, $Nf(k)$ for $k > 3$ can be easily calculated in recursion.

EXAMPLE 3.40. Fit a Poisson distribution for the following distribution:

x	0	1	2	3	4	5	Total
f	142	156	69	27	5	1	400

(Anna, Nov. 2007)

Solution. First, we calculate the mean of the given data. We see that

$$\text{Mean, } \mu = \frac{\sum fx}{\sum f} = \frac{400}{400} = 1$$

Note that

$$Nf(0) = Ne^{-\mu} = 147.1518$$

Using the recurrence relation

$$f(x+1) = \frac{\mu}{x+1} \cdot f(x), \quad x = 0, 1, 2, \dots$$

we have

$$Nf(x+1) = [Nf(x)] \times \frac{\mu}{x+1}, \quad x = 0, 1, 2, 3, 4$$

Therefore,

$$Nf(1) = 147.1518 \times \frac{\mu}{1} = 147.1518$$

$$Nf(2) = 147.1518 \times \frac{\mu}{2} = 73.5759$$

$$Nf(3) = 73.5759 \times \frac{\mu}{3} = 24.5253$$

$$Nf(4) = 24.5253 \times \frac{\mu}{4} = 6.1313$$

$$Nf(5) = 6.1313 \times \frac{\mu}{5} = 1.2263$$

(Note also that $Nf(0) + Nf(1) + \dots + Nf(5) = 399.7623 \approx 400$)

We tabulate our results as follows:

<i>No. of accidents</i>	<i>Observed frequencies</i>	<i>Theoretical frequencies</i>
0	142	147.1518
1	156	147.1518
2	69	73.5759
3	27	24.5253
4	5	6.1313
5	1	1.2263



PROBLEM SET 3.2

- If X is a Poisson random variable such that $P(X = 1) = 0.4$ and $P(X = 2) = 0.2$, calculate
 - The variance of X .
 - $P(X = 0)$.
- If X is a Poisson random variable such that $P(X = 1) = P(X = 2)$, find the mean and variance of X .
- If X is a Poisson random variable such that $P(X = 2) = P(X = 3)$, find $P(X = 4)$.
- If X is a Poisson random variable such that $P(X = 0) = 0.2$, find $P(X = 1)$, $P(X = 2)$ and $P(X = 3)$.
- A Poisson random variable X has a double mode at $X = 4$ and $X = 5$. Find the probability that X will have either of these values.
- The probability of dialing a wrong number in a telephone booth is estimated as 3%. What is the probability of dialing not more than two wrong numbers in 50 dialings?
- If there are 200 misprints randomly distributed in a book of 500 pages, find the probability that a given page will contain at most 2 misprints?

8. Find the probability that at most 3 defective fuses will be found in a box of 400 fuses if experience shows that 2% of such fuses are defective.
9. The monthly breakdowns of a computer follow Poisson distribution with a mean equal to 1.2. Find the probability that this computer will function for a month (a) without a breakdown, (b) with only one breakdown, and (c) with at most two breakdowns.
10. Three per cent of the electric bulbs made by a certain company are known to be defective. If a sample of 200 bulbs is randomly chosen for inspection, find the probability that (a) there are no defective bulbs, (b) at most 2 bulbs are defective, and (c) at least 3 bulbs are defective.
11. In a certain factory producing razor blades, there is a 1% chance for any blade to be defective. The blades are in packets of 10. In a consignment of 1000 packets, calculate the approximate number of packets containing (a) no defective blade, (b) only one defective blade, (c) at most two defective blades, and (d) at least two defective blades.
12. The number of road accidents in a year due to taxi drivers in a city follows a Poisson distribution with mean equal to 2. Out of 500 taxi drivers randomly chosen, find approximately the number of drivers with (a) no accidents in a year, (b) at most 2 accidents in a year, and (c) at least 2 accidents in a year.
13. If the probability that an individual suffers a bad reaction from injection of a serum is 1%, find the probability that out of 800 individuals, (a) no individual will suffer a bad reaction, (b) exactly 2 individuals will suffer a bad reaction, and (c) more than 2 individuals will suffer a bad reaction.
14. Fit a Poisson distribution to the following set of observations and calculate the theoretical frequencies:

No. of deaths	0	1	2	3	4
Frequency	127	52	16	4	1

15. A skilled typist kept a record of mistakes made per day during 300 working days of a calendar year given as follows:

Mistakes per day	0	1	2	3	4	5	6
No. of days	167	70	35	17	7	3	1

Fit a Poisson distribution to this data and calculate the theoretical frequencies.

3.5 GEOMETRIC DISTRIBUTION

In probability theory, geometric distribution is the probability distribution of the number X of independent Bernoulli trials performed until a success occurs, where the Bernoulli trials have a constant probability of success, p . Then it is clear that X takes on the values 1, 2, 3, ... and

$$P(X = x) = P[x \text{ Bernoulli trials are performed until the first success is obtained}]$$

i.e.

$$P(X = x) = P[\text{failures in the first } x - 1 \text{ Bernoulli trials and a success in the } x\text{th trial}]$$

Using the independence of events, we get

$$P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

where $q = 1 - p$, the constant probability of failure in each Bernoulli trial.

There are many common examples of geometric distribution like tossing a coin repeatedly until the first time a head appears, throwing a die repeatedly until the first time a six appears, throwing a pair of dice repeatedly until the first time a 'double six' appears, etc.

Definition 3.5. A random variable X is said to have a *geometric distribution* with a parameter p ($0 < p < 1$) if it takes on the values $1, 2, 3, \dots$ and its probability mass function given by

$$f(x; p) = P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

where $q = 1 - p$.

Remark 3.8. If X is a geometric random variable, then the values of the probability mass function, viz. $p, qp, q^2p, q^3p, \dots, q^{x-1}p, \dots$ are the successive terms of a geometric progression series. This explains the name *geometric* in calling the random variable X .

Remark 3.9. We note that the geometric density function is well defined because $f(x) > 0$ for the mass points $x = 1, 2, 3, \dots$ and also that

$$\sum_{x=1}^{\infty} f(x) = \sum_{x=0}^{\infty} q^{x-1}p = p[1 + q + q^2 + q^3 + \dots] = p \frac{1}{1-q} = 1$$

since $1 - q = 1 - (1 - p) = p$.

Theorem 3.12. If X is a geometric random variable with parameter p , then

$$\text{Mean}(X) = \frac{1}{p} \quad \text{and} \quad \text{Var}(X) = \frac{q}{p^2}$$

(Anna, Nov. 2007)

Proof. If X is a geometric random variable with parameter p , then the probability mass function of X is given by

$$f(x) = P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

Hence, the first raw moment of X is given by

$$\begin{aligned} \mu'_1 &= \sum_{x=1}^{\infty} xq^{x-1}p = p[1 + 2q + 3q^2 + 4q^3 + \dots] \\ &= p[1 - q]^{-2} = p \times p^{-2} = \frac{1}{p} \end{aligned}$$

(We have used the formula $(1 - \theta)^{-r} = \sum_{x=r}^{\infty} {}^{x-1}C_{r-1} \theta^{x-r}$ if $0 < \theta < 1$.)

Thus, the mean of X is given by $\mu = \mu'_1 = \frac{1}{p}$.

Next, we find the second raw moment of X , i.e. $\mu'_2 = E(X^2)$.

Note that

$$\mu'_2 = E(X^2) = E[X(X-1)] + E(X) = E(X(X-1)) + \frac{1}{p}$$

and

$$\begin{aligned} E[X(X-1)] &= \sum_{x=1}^{\infty} x(x-1)q^{x-1}p = [2qp + 6q^2p + 12q^3p + 20q^4p + \dots] \\ &= 2qp[1 + 3q + 6q^2 + 10q^3 + \dots] = 2qp[1-q]^{-3} = 2qpp^{-3} = \frac{2q}{p^2} \end{aligned}$$

Hence, we have

$$\mu'_2 = E[X(X-1)] + \frac{1}{p} = \frac{2q}{p^2} + \frac{1}{p} = \frac{2q+p}{p^2} = \frac{1+q}{p^2}$$

since $p = 1 - q$

So, the variance of X is given by

$$\mu_2 = \mu'_2 - \mu_1'^2 = \frac{1+q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$$

Hence, $\text{Var}(X) = \sigma^2 = \frac{q}{p^2}$. ◆

EXAMPLE 3.41. Find the moment generating function of the geometric distribution and hence find its mean and variance.

(Anna, April 2003; May 2007)

Solution. If X is a geometric random variable with parameter p , then X has the probability mass function

$$f(x) = P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

Hence, the moment generating function of X is given by

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \sum_{x=1}^{\infty} e^{tx} q^{x-1} p = pe^t \sum_{x=1}^{\infty} (qe^t)^{x-1} \\ &= pe^t [1 + (qe^t) + (qe^t)^2 + (qe^t)^3 + \dots] \\ &= pe^t (1 - qe^t)^{-1} = \frac{pe^t}{1 - qe^t} \end{aligned}$$

Next, we note that

$$M'_X(t) = \frac{(1 - qe^t)(pe^t) - pe^t(-qe^t)}{(1 - qe^t)^2} = \frac{pe^t}{(1 - qe^t)^2}$$

Differentiating again with respect to t , we have

$$M''_X(t) = \frac{(1 - qe^t)^2(pe^t) - pe^t[2(1 - qe^t)(-qe^t)]}{(1 - qe^t)^4}$$

Simplifying, we have

$$M_X''(t) = \frac{pe^t + pqe^{2t}}{(1 - qe^t)^3}$$

Hence, the first two raw moments of X are given by

$$\mu_1' = M_X'(0) = \frac{p}{(1 - q)^2} = \frac{p}{p^2} = \frac{1}{p}$$

and

$$\mu_2' = M_X''(0) = \frac{p + pq}{(1 - q)^3} = \frac{p(1 + q)}{p^3} = \frac{1 + q}{p^2}$$

Hence, the mean of X is obtained as

$$\text{Mean}(X) = \mu = \mu_1' = \frac{1}{p}$$

and the variance of X is obtained as

$$\text{Var}(X) = \mu_2 = \mu_2' - \mu_1'^2 = \frac{1 + q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$$

◆

EXAMPLE 3.42. If X has a geometric distribution with parameter p , find

- (i) $P(X \text{ is even})$.
- (ii) $P(X \text{ is odd})$.

Solution. Since X is a geometric variable with parameter p , we have

$$f(x) = P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

where $q = 1 - p$.

- (i) The required probability is

$$\begin{aligned} P(X \text{ is even}) &= P(X = 2) + P(X = 4) + P(X = 6) + P(X = 8) + \dots = qp + q^3p + q^5p + q^7p + \dots \\ &= qp \left[1 + q^2 + q^4 + q^6 + \dots \right] = qp \frac{1}{1 - q^2} \\ &= \frac{qp}{(1 + q)(1 - q)} = \frac{q}{1 + q} \end{aligned}$$

since $1 - q = p$.

- (ii) The required probability is

$$\begin{aligned} P(X \text{ is odd}) &= P(X = 1) + P(X = 3) + P(X = 5) + P(X = 7) + \dots = p + q^2p + q^4p + q^6p + \dots \\ &= p \left[1 + q^2 + q^4 + q^6 + \dots \right] = p \frac{1}{1 - q^2} \\ &= \frac{p}{(1 + q)(1 - q)} = \frac{1}{1 + q} \end{aligned}$$

since $1 - q = p$.

Alternatively, we can also find $P(X \text{ is odd})$ by noting that

$$P(X \text{ is odd}) = 1 - P(X \text{ is even}) = 1 - \frac{q}{1+q} = \frac{1}{1+q}$$

◆

EXAMPLE 3.43. Suppose that a trainee soldier shoots a target in an independent fashion. If the probability that the target is shot in any one shot is 0.7, what is the

- (i) Probability that the target would be hit on 10th attempt?
- (ii) Probability that it takes him less than 4 shots?
- (iii) Probability that it takes him an even number of shots?
- (iv) Average number of shots needed to hit the target?

(Anna, Nov. 2006; Nov. 2007)

Solution. Let X denote the number of shots required until the first time the target is hit. Then it is given that X follows a geometric distribution with parameter $p = 0.7$. Note that the probability mass function of X is given by

$$f(x) = P(X = x) = q^{x-1} p = (0.3)^{x-1} (0.7), \quad x = 1, 2, 3, \dots$$

- (i) The probability that the target would be hit on 10th attempt is

$$P(X = 10) = q^9 p = (0.3)^9 0.7 = 1.3777 \times 10^{-5}$$

- (ii) The probability that it will take him less than 4 shots is

$$P(X = 1) + P(X = 2) + P(X = 3) = (1 + q + q^2)p = 1.3900 \times 0.7 = 0.9730$$

- (iii) The probability that it will take him an even number of shots is

$$\begin{aligned} P(X \text{ is even}) &= P(X = 2) + P(X = 4) + P(X = 6) + P(X = 8) + \dots \\ &= qp + q^3 p + q^5 p + q^7 p + \dots = qp \left[1 + q^2 + q^4 + q^6 + \dots \right] \\ &= qp \frac{1}{1 - q^2} = \frac{qp}{(1 + q)(1 - q)} = \frac{q}{1 + q} \end{aligned}$$

since $1 - q = p$.

Therefore, we have

$$P(X \text{ is even}) = \frac{q}{1 + q} = \frac{0.3}{1 + 0.3} = \frac{0.3}{1.3} = 0.2308$$

- (iv) The average number of shots needed to hit the target is given by

$$E(X) = \frac{1}{p} = 1.4286$$

◆

EXAMPLE 3.44. If X has geometric distribution, show that for any two positive integers s and t , $P(X > s+t|X > s) = P(X > t)$ (*Memoryless Property*).

(Anna, Model 2003)

Solution. Since X has geometric distribution, we have

$$f(x) = P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

Thus, for any positive integer k , we have

$$P(X > k) = \sum_{x=k+1}^{\infty} P(X = x) = \sum_{x=k+1}^{\infty} q^{x-1}p. \quad (3.12)$$

Simplifying, we have

$$P(X > k) = p [q^k + q^{k+1} + \dots] = \frac{pq^k}{1-q} = q^k$$

since $p + q = 1$.

If s and t are any two positive integers, then we have

$$P(X > s+t|X > s) = \frac{P(X > s+t \cap X > s)}{P(X > s)} = \frac{P(X > s+t)}{P(X > s)}.$$

Using Eq. (3.12), we have

$$P(X > s+t|X > s) = \frac{q^{s+t}}{q^s} = q^t = P(X > t).$$

◆

We note that the converse of the above result is also true. That is, if X is a discrete random variable taking positive integer values and having the memoryless property, then it can be shown (see Example 3.45) that X is geometric distribution. Thus, the *memoryless property* completely characterizes the geometric distribution among all discrete probability distributions.

EXAMPLE 3.45. Show that if X is a discrete random variable taking positive integer values with memoryless property, then X is the geometric distribution.

(Anna, Model 2003)

Solution. Let X be a discrete random variable taking positive integer values and with memoryless property, i.e. if s and t are any positive integers, then

$$P(X > s+t|X > s) = P(X > t)$$

i.e.

$$\frac{P(X > s+t \cap X > s)}{P(X > s)} = P(X > t)$$

i.e.

$$\frac{P(X > s+t)}{P(X > s)} = P(X > t)$$

i.e.

$$P(X > s+t) = P(X > s)P(X > t) \quad (3.13)$$

Since X takes the values $1, 2, 3, \dots$, it follows that $P(X \geq 1) = 1$.

Let $q \triangleq P(X > 1)$. Then $q > 0$. Let $p = 1 - q$.

For any positive integer s , we can write

$$P(X = s+1) = P(X > s) - P(X > s+1) \quad (3.14)$$

Thus, we have

$$\frac{P(X = s+1)}{P(X > s)} = 1 - \frac{P(X > s+1)}{P(X > s)} = 1 - \frac{P(X > s)P(X > 1)}{P(X > s)} \quad [\text{Using Eq. (3.13)}]$$

Simplifying, we have

$$\frac{P(X = s+1)}{P(X > s)} = 1 - P(X > 1) = 1 - q = p$$

Therefore,

$$P(X = s+1) = pP(X > s) \quad (3.15)$$

Replacing s by $s-1$ in Eq. (3.14), we have

$$P(X = s) = P(X > s-1) - P(X > s)$$

or

$$P(X > s) = P(X > s-1) - P(X = s) \quad (3.16)$$

Combining Eqs. (3.15) and (3.16), we have

$$P(X = s+1) = p[P(X > s-1) - P(X = s)] = p[P(X > s-1) - pP(X > s-1)]$$

[using Eq. (3.15)]

Simplifying, we have

$$P(X = s+1) = p(1-p)P(X > s-1) = pqP(X > s-1) \quad (3.17)$$

Replacing s by $s-2$ in Eq. (3.14), we have

$$P(X > s-1) = P(X > s-2) - P(X = s-1)$$

Thus, Eq. (3.16) becomes

$$\begin{aligned} P(X = s+1) &= pq[P(X > s-2) - P(X = s-1)] \\ &= pq[P(X > s-2) - qP(X > s-2)] \\ &= pq^2P(X > s-2) \end{aligned}$$

Repeating this process successively, we shall have

$$P(X = s+1) = pq^{s-1}P(X > 1) = pq^{s-1}q = pq^s$$

Hence, we have

$$P(X = s) = pq^{s-1}, \text{ where } s = 1, 2, 3, \dots$$

Thus, we have shown that X follows a geometric distribution with parameter p . ◆

EXAMPLE 3.46. Let X be the number of births in a hospital until the first girl is born. Determine the probability and the distribution functions of X . Assume the probability that the baby born is a girl is $\frac{1}{2}$.
(Anna, April 2003)

Solution. It is clear that X is a geometric random variable with parameter $p = \frac{1}{2}$, where p denotes the probability that the baby born is girl. Hence, the probability mass function of X is given by

$$f(x) = P(X = x) = q^{x-1}p = \left(\frac{1}{2}\right)^{x-1} \frac{1}{2} = \left(\frac{1}{2}\right)^x, \quad x = 1, 2, 3, \dots$$

Let F be the cumulative distribution function of X . Clearly, $F(x) = 0$ if $x < 1$.

If $1 \leq x < 2$, we have

$$F(x) = P(X \leq x) = P(X = 1) = p = \frac{1}{2}$$

If $2 \leq x < 3$, we have

$$F(x) = P(X \leq x) = P(X = 1) + P(X = 2) = p + qp = \frac{1}{2} + \left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

If $3 \leq x < 4$, we have

$$F(x) = P(X \leq x) = P(X = 1) + P(X = 2) + P(X = 3) = p + qp + q^2p = \frac{7}{8}$$

In general, if $r \leq x < r + 1$, where r is any positive integer, we have

$$F(x) = P(X \leq x) = P(X = 1) + P(X = 2) + \dots + P(X = r) = p + qp + \dots + q^{r-1}p$$

Thus,

$$F(x) = p[1 + q + q^2 + \dots + q^{r-1}] = p \left[\frac{1 - q^r}{1 - q} \right] = 1 - q^r = 1 - \frac{1}{2^r} = \frac{2^r - 1}{2^r}$$

since $p = q = \frac{1}{2}$. ◆

EXAMPLE 3.47. If X is a geometric random variable with parameter p , show that the probability mass function f of X satisfies the recurrence relation

$$f(x+1) = qf(x), \quad \text{where } x = 1, 2, 3, \dots$$

Solution. Since X has a geometric distribution with parameter p , it follows that

$$f(x) = P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

and so, we have

$$f(x+1) = P(X = x+1) = q^x p$$

Hence, it is immediate that

$$f(x+1) = q[q^{x-1}p] = qf(x), \quad x = 1, 2, 3, \dots$$
◆

PROBLEM SET 3.3

1. Find the mode of the geometric distribution

$$f(x) = \left(\frac{1}{2}\right)^x, \quad x = 1, 2, 3, \dots$$

2. If X is a geometric random variable, find the probability that X is divisible by 3.
3. Suppose that a trainee soldier shoots a target according to a geometric distribution. If the probability that a target is shot in any one shot is 0.8, find the probability that it takes him an odd number of shots.
4. Suppose that the probability for an applicant for a driver's license to pass the road test on any given attempt is $\frac{2}{3}$. What is the probability that the applicant will pass the road test on the third attempt?
5. For the geometric distribution with probability mass function

$$f(x) = \frac{2}{3} \left(\frac{1}{3}\right)^{x-1}, \quad x = 1, 2, 3, \dots$$

show that the Chebyshev's inequality gives

$$P(|X - 1.5| \leq 2) \geq \frac{13}{16}$$

while the actual probability is $\frac{26}{27}$.

3.6 NEGATIVE BINOMIAL DISTRIBUTION

In probability theory, negative binomial distribution is a discrete probability distribution, which is a generalization of the geometric distribution. It is also known as the *Pascal distribution*. Suppose that in an experiment, independent Bernoulli trials, each having constant probability of success p , $0 < p < 1$, are performed until a total of r successes are obtained. If X is the random variable representing the number of Bernoulli trials required, then it is clear that

$$P(X = x) = P[r - 1 \text{ successes in the first } x - 1 \text{ trials and a success in the } x^{\text{th}} \text{ trial}]$$

Using the independence of events, we get

$$\begin{aligned} P(X = x) &= \binom{x-1}{r-1} p^{r-1} q^{(x-1)-(r-1)} \times p \\ &= \binom{x-1}{r-1} p^r q^{x-r}, \quad x = r, r+1, \dots \end{aligned}$$

Definition 3.6. A random variable is said to have a *negative binomial distribution* or *Pascal distribution* with parameters r (a positive integer) and p (where $0 < p < 1$) if it takes on the values $r, r+1, r+2, \dots$ and if its probability mass function is

$$f(x; r, p) = P(X = x) = \binom{x-1}{r-1} p^r q^{x-r}, \quad x = r, r+1, \dots$$

In this case, we write $X \sim \text{NB}(r, p)$.

Remark 3.10. Taking $r = 1$ in the definition of negative binomial distribution, it follows that the probability mass function of X is given by

$$f(x) = P(X = x) = \binom{x-1}{0} p^1 q^{x-1} = q^{x-1} p, \quad x = 1, 2, \dots$$

which is the probability mass function of the geometric distribution with parameter p . Hence, the geometric distribution is a particular case of the negative binomial distribution by taking $r = 1$.

Remark 3.11. The negative binomial probability law is given by

$$f(x) = P(X = x) = \binom{x-1}{r-1} p^r q^{x-r}, \quad x = r, r+1, \dots \quad (3.18)$$

Note that for each value of $x = r, r+1, \dots$, $f(x) > 0$ and also that

$$\sum_{x=r}^{\infty} f(x) = \sum_{x=r}^{\infty} \binom{x-1}{r-1} p^r q^{x-r} = p^r \sum_{x=r}^{\infty} \binom{x-1}{r-1} q^{x-r} = p^r (1-q)^{-r} \quad (3.19)$$

Thus, the terms of the negative binomial probability law, Eq. (3.18), for $x = r, r+1, \dots$ are successive terms of the negative binomial expansion as given in Eq. (3.19). This explains the reason why the random variable X with density given in Eq. (3.18) is called a *negative random variable*. To see that the total probability law is unity, we see from Eq. (3.19),

$$\sum_{x=r}^{\infty} f(x) = p^r (1-q)^{-r} = p^r p^{-r} = p^{r-r} = p^0 = 1$$

which follows because $1 - q = p$.

EXAMPLE 3.48. If X has a negative binomial distribution with parameters r and p , obtain the moment generating function, mean and variance of X .

(Anna, April 2004, Nov. 2005)

Solution. Since $X \sim \text{NB}(r, p)$, it has the probability mass function

$$f(x) = P(X = x) = \binom{x-1}{r-1} p^r q^{x-r}, \quad x = r, r+1, \dots$$

Then, the moment generating function of X is given by

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \sum_{x=r}^{\infty} e^{tx} \binom{x-1}{r-1} p^r q^{x-r} \\ &= p^r \sum_{x=r}^{\infty} (e^t)^{x-r} (e^t)^r \binom{x-1}{r-1} q^{x-r} \\ &= (pe^t)^r \sum_{x=r}^{\infty} \binom{x-1}{r-1} (qe^t)^{x-r} \end{aligned}$$

Hence, it follows that

$$M_X(t) = (pe^t)^r (1 - qe^t)^{-r} = \left[\frac{pe^t}{1 - qe^t} \right]^r$$

which converges for values of t satisfying

$$0 < qe^t < 1 \quad \text{or} \quad 0 < t < \log_e \left(\frac{1}{q} \right)$$

Next, we calculate the first two raw moments of X . For that purpose, we calculate the first two derivatives of $M_X(t)$ with respect to t .

We find that

$$\begin{aligned} M'_X(t) &= r \left[\frac{pe^t}{1 - qe^t} \right]^{r-1} \left[\frac{(1 - qe^t)(pe^t) - (pe^t)(-qe^t)}{(1 - qe^t)^2} \right] \\ &= r \frac{(pe^t)^r}{(1 - qe^t)^{r+1}} \end{aligned}$$

Therefore, we have

$$\begin{aligned} M''_X(t) &= r \left[\frac{(1 - qe^t)^{r+1} r (pe^t)^{r-1} (pe^t) - (pe^t)^r (r+1) (1 - qe^t)^r (-qe^t)}{(1 - qe^t)^{2r+2}} \right] \\ &= r \frac{(pe^t)^r (r + qe^t)}{(1 - qe^t)^{r+2}} \end{aligned}$$

So, the first two raw moments of X are given by

$$\mu'_1 = M'_X(0) = r \frac{p^r}{(1 - q)^{r+1}} = r \frac{p^r}{p^{r+1}} = \frac{r}{p}$$

since $1 - q = p$.

Thus, the mean of X is $\mu = \frac{r}{p}$.

Next, we find that

$$\mu'_2 = M''_X(0) = r \frac{p^r (r + q)}{(1 - q)^{r+2}} = r \frac{p^r (r + q)}{p^{r+2}} = \frac{r(r + q)}{p^2}$$

Hence, we have

$$\mu_2 = \sigma^2 = \mu'_2 - \mu_1'^2 = \frac{r(r + q)}{p^2} - \frac{r^2}{p^2} = \frac{qr}{p^2}$$

Thus, the variance of X is given by $\text{Var}(X) = \frac{qr}{p^2}$. ♦

EXAMPLE 3.49. If X is a negative binomial random variable with parameters r and p , show that the probability mass function f of X satisfies the recurrence relation

$$f(x + 1) = \frac{x}{x - r + 1} q f(x), \quad x = r, r + 1, \dots$$

Solution. Since $X \sim \text{NB}(r, p)$, the PMF of X is given by

$$f(x) = P(X = x) = \binom{x-1}{r-1} p^r q^{x-r} = \frac{(x-1)!}{(r-1)!(x-r)!} p^r q^{x-r}$$

Hence, it follows that

$$f(x+1) = \binom{x}{r-1} p^r q^{x+1-r} = \frac{x!}{(r-1)!(x-r+1)!} p^r q^{x+1-r}$$

Thus, we have

$$\frac{f(x+1)}{f(x)} = \frac{x}{x-r+1} q$$

Hence, it is immediate that

$$f(x+1) = \frac{x}{x-r+1} q f(x), \quad x = r, r+1, \dots$$

◆

EXAMPLE 3.50. The probability that an experiment will succeed is 0.8. If the experiment is repeated until four successful outcomes have occurred, what is the expected number of repetitions required?

(Anna, Model 2003)

Solution. If X is the number of repetitions required until four successful outcomes have occurred, it is clear that X has a negative binomial distribution with parameters $r = 4$ and $p = 0.8$. Hence, the expectation of X is

$$\mu = E(X) = \frac{r}{p} = \frac{4}{0.8} = 5$$

◆

EXAMPLE 3.51. If the probability is 0.40 that a child exposed to a certain contagious will catch it, what is the probability that the tenth child exposed to the disease will be the third to catch it?

(Anna, Nov. 2006)

Solution. Let p denote the probability that a child exposed to a contagious disease will catch it, and X denote the number of children exposed to the disease till three children catch the disease. Then X follows a negative binomial distribution with parameters

$$p = 0.4 \text{ and } r = 3$$

So, X has the probability mass function given by

$$f(x) = P(X = x) = \binom{x-1}{r-1} p^r q^{x-r} = \binom{x-1}{2} p^3 q^{x-3}, \quad x = 3, 4, \dots$$

So, the required probability that the tenth child exposed to the disease will be the third to catch it is given by

$$P(X = 10) = \binom{9}{2} p^3 q^7 = 36 (0.4)^3 (0.6)^7 = 0.0645$$

◆

EXAMPLE 3.52. Let X be the number of births in a family until the second daughter is born. If the probability of having a male child is $\frac{1}{2}$, find the probability that the sixth child in the family is the second daughter.

Solution. Clearly, X follows a negative binomial distribution with parameters $r = 2$ and $p = \frac{1}{2}$. Hence, the probability mass function of X is given by

$$f(x) = P(X = x) = \binom{x-1}{r-1} p^r q^{x-r} = \binom{x-1}{1} \left(\frac{1}{2}\right)^x, \quad x = 2, 3, \dots$$

since $p = q = \frac{1}{2}$.

Hence, the probability that the sixth child in the family is the second daughter is given by

$$P(X = 6) = \binom{5}{1} \left(\frac{1}{2}\right)^6 = 0.0781$$

◆

EXAMPLE 3.53. In a company, 5% defective components are produced. What is the probability that at least 5 components are to be examined in order to get 3 defective?

(Anna, April 2004)

Solution. Let p be the probability that a defective component is produced. Let X be the number of components that are to be examined in order to get 3 defective components. Then X follows a negative binomial distribution with parameters $r = 3$ and $p = \frac{5}{100}$. Thus, X has the probability mass function given by

$$f(x) = P(X = x) = \binom{x-1}{r-1} p^r q^{x-r} = \binom{x-1}{2} (0.05)^3 (0.95)^{x-3}, \quad x = 3, 4, \dots$$

The probability that at least 5 components are to be examined in order to get 3 defective components is given by

$$\begin{aligned} P(X \geq 5) &= 1 - P(X < 5) = 1 - [P(X = 3) + P(X = 4)] \\ &= 1 - \left[\binom{2}{2} (0.05)^3 (0.95)^0 + \binom{3}{2} (0.05)^3 (0.95)^1 \right] \\ &= 1 - (0.05)^3 [1 + (3 \times 0.95)] = 0.9995 \end{aligned}$$

◆

PROBLEM SET 3.4

1. If the probability that a child exposed to certain viral fever will be infected is 0.3, find the probability that the eighth child exposed to the disease will be the fourth to be infected.
2. If the probability that a person will believe a rumour is 0.6, find the probability that the tenth person to hear the rumour will be the third person to believe.
3. A marksman is firing bullets at a target and the probability of hitting the target at any trial is 0.7. Find the probability that his seventh shot is his fourth hit.
4. If the probability of having a male child is 0.5, find the probability that in a family, the eighth child is the third boy.
5. In a company, 3% defective components are produced. What is the probability that at least 6 components are to be examined in order to get 3 defectives?

3.7 UNIFORM DISTRIBUTION

Definition 3.7. A random variable X is said to have a *uniform distribution* or *rectangular distribution* over a finite interval (a, b) , where $-\infty < a < b < \infty$ if its probability density function is given by

$$f(x; a, b) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{if otherwise} \end{cases}$$

If X has a uniform distribution on (a, b) , we write $X \sim U(a, b)$. The uniform distribution is plotted in Figure 3.3.

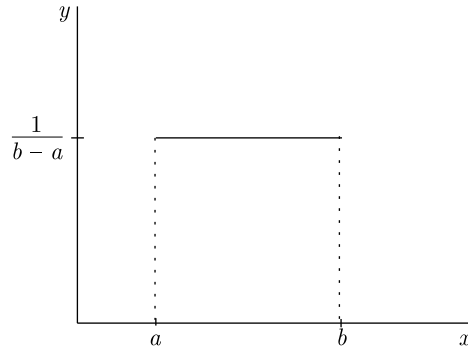


Figure 3.3: Uniform or rectangular distribution.

EXAMPLE 3.54. If X has a uniform distribution over the interval (a, b) , find the distribution function of X .

Solution. Let F be the distribution function of the uniform random variable X , i.e.

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

If $x \leq a$, then it is clear that $F(x) = 0$.

If $a < x < b$, then we have

$$F(x) = \int_a^x \frac{1}{b-a} dt = \frac{1}{b-a} [t]_a^x = \frac{x-a}{b-a}$$

If $x \geq b$, then we have

$$F(x) = \int_a^b \frac{1}{b-a} dt = \frac{1}{b-a} [b-a] = 1$$

Combining, we have

$$F(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a < x < b \\ 1 & \text{if } x \geq b \end{cases}$$

◆

EXAMPLE 3.55. If a random variable X has the probability density function

$$f(x) = \begin{cases} \frac{1}{4}, & |x| < 2 \\ 0, & \text{otherwise} \end{cases}$$

obtain (i) $P(X < 1)$, (ii) $P(|X| > 1)$, (iii) $P(2X + 3 > 5)$.

(Anna, May 2007)

Solution. The given random variable X has the uniform distribution over the interval $[-2, 2]$. Hence, we know that the cumulative distribution function of X is given by

$$F(x) = \frac{x-a}{b-a} = \frac{x+2}{4} \text{ for } -2 \leq x \leq 2$$

(i) We find that

$$P(X < 1) = F(1) = \frac{3}{4}$$

(ii) We find that

$$P(|X| > 1) = 1 - P(|X| \leq 1) = 1 - P(-1 < X < 1) = 1 - [F(1) - F(-1)].$$

Simplifying, we have

$$P(|X| > 1) = 1 - \left[\frac{3}{4} - \frac{1}{4} \right] = 1 - \frac{1}{2} = \frac{1}{2}$$

(iii) We find that

$$P(2X + 3 > 5) = P(2X > 2) = P(X > 1) = 1 - F(1) = 1 - \frac{3}{4} = \frac{1}{4}$$

◆

EXAMPLE 3.56. If X is uniformly distributed in $[-2, 2]$, find (i) $P(X < 0)$ and (ii) $P[|X - 1| \geq \frac{1}{2}]$.

(Anna, May 2007)

Solution. By hypothesis, the probability density function of X is

$$f(x) = \begin{cases} \frac{1}{4} & \text{if } -2 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(i) The required probability is

$$P(X < 0) = \int_{-2}^0 f(x) dx = \int_{-2}^0 \frac{1}{4} dx = \frac{1}{2}$$

(ii) The required probability is given by

$$\begin{aligned}
 P\left[|X-1| \geq \frac{1}{2}\right] &= 1 - P\left[|X-1| < \frac{1}{2}\right] \\
 &= 1 - P\left[-\frac{1}{2} < X-1 < \frac{1}{2}\right] \\
 &= 1 - P\left[\frac{1}{2} < X < \frac{3}{2}\right] \\
 &= 1 - \int_{\frac{1}{2}}^{\frac{3}{2}} f(x) dx = 1 - \int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{4} dx = \frac{3}{4}
 \end{aligned}$$

◆

Theorem 3.13. If X is a uniform random variable over the interval (a, b) , then the following properties hold:

- (i) $\text{Mean}(X) = \frac{a+b}{2}$ and $\text{Var}(X) = \frac{(b-a)^2}{12}$.
- (ii) $\text{Median}(X) = \frac{a+b}{2}$.
- (iii) $\text{Mode}(X)$ is any point x in the interval (a, b) .

Proof. (i) The mean of X is given by

$$\mu = E(X) = \int_a^b x f(x) dx = \int_a^b x \left[\frac{1}{b-a} \right] dx$$

Integrating, we have

$$\mu = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{1}{b-a} \left[\frac{b^2 - a^2}{2} \right]$$

Since $b^2 - a^2 = (b-a)(b+a)$, it follows immediately that

$$\mu = \frac{b+a}{2}$$

Thus, $\text{Mean}(X) = \frac{a+b}{2}$.

To find the variance of X , we first calculate the second raw moment of X . Note that

$$\mu'_2 = E(X^2) = \int_a^b x^2 f(x) dx = \int_a^b x^2 \left[\frac{1}{b-a} \right] dx$$

Integrating, we have

$$\mu'_2 = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{1}{b-a} \left[\frac{b^3 - a^3}{3} \right]$$

Since $b^3 - a^3 = (b - a)(b^2 + ab + a^2)$, it follows immediately that

$$\mu'_2 = \frac{b^2 + ab + a^2}{3}$$

Hence, the variance of X is given by

$$\mu_2 = \mu'_2 - \mu_1'^2 = \frac{b^2 + ab + a^2}{3} - \left[\frac{b+a}{2} \right]^2 = \frac{b^2 + ab + a^2}{3} - \frac{b^2 + 2ab + a^2}{4}$$

Simplifying, we have

$$\mu_2 = \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}$$

Thus, variance of X is given by $\text{Var}(X) = \frac{(b-a)^2}{12}$

(ii) The median, M , of X is obtained by solving the equation

$$\int_{-\infty}^M f(x) dx = \frac{1}{2} \Rightarrow \int_a^M f(x) dx = \frac{1}{2}$$

since $f(x) = 0$ for $x < a$. Therefore, we have

$$\int_a^M \left[\frac{1}{b-a} \right] dx = \frac{1}{2}$$

Integrating, we get

$$\frac{1}{b-a} [x]_a^M = \frac{1}{2} \Rightarrow \frac{M-a}{b-a} = \frac{1}{2}$$

Simplifying, we have

$$2M - 2a = b - a \Rightarrow 2M = a + b \Rightarrow M = \frac{a+b}{2}$$

Hence, $\text{Median}(X) = \frac{a+b}{2}$

(iii) Mode, c , is that value of x at which f attains its maximum value. From the definition of f , it is clear that any point c in the interval $a < x < b$ is a mode of X (see Figure 3.3).

◆

EXAMPLE 3.57. Find the moment generating function of a random variable uniformly distributed in the interval (a, b) and hence obtain its mean and variance.

(Anna, Nov. 2006)

Solution. Since X is uniformly distributed in the interval (a, b) , it has the probability density function given by

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a < x < b \\ 0 & \text{elsewhere} \end{cases}$$

Hence, the moment generating function of X is given by

$$M_X(t) = E(e^{tX}) = \int_a^b e^{tx} \left[\frac{1}{b-a} \right] dx = \frac{1}{b-a} \int_a^b e^{tx} dx$$

Integrating, we get

$$M_X(t) = \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_{x=a}^b = \frac{1}{b-a} \left[\frac{e^{bt}}{t} - \frac{e^{at}}{t} \right]$$

which exists for all values of t .

Simplifying,

$$M_X(t) = \frac{e^{bt} - e^{at}}{t(b-a)} \quad (3.20)$$

Next, we find the mean and variance of X . Note that

$$e^{bt} = 1 + \frac{(bt)}{1!} + \frac{(bt)^2}{2!} + \frac{(bt)^3}{3!} + \dots$$

and

$$e^{at} = 1 + \frac{(at)}{1!} + \frac{(at)^2}{2!} + \frac{(at)^3}{3!} + \dots$$

Therefore, it follows that

$$e^{bt} - e^{at} = (b-a)t + \frac{(b^2 - a^2)t^2}{2!} + \frac{(b^3 - a^3)t^3}{3!} + \dots$$

From Eq. (3.20), it is then immediate that

$$M_X(t) = 1 + \left(\frac{b+a}{2} \right) t + \left(\frac{b^2 + ab + a^2}{6} \right) t^2 + \dots$$

Hence, the first two raw moments of X are obtained as follows:

$$\mu'_1 = \text{Coefficient of } \frac{t}{1!} \text{ in } M_X(t) = \frac{b+a}{2}$$

and

$$\mu'_2 = \text{Coefficient of } \frac{t^2}{2!} \text{ in } M_X(t) = \frac{b^2 + ab + a^2}{3}$$

Hence, the mean of X is given by

$$\text{Mean}(X) = \mu = \mu'_1 = \frac{a+b}{2}$$

and the variance of X is given by

$$\text{Var}(X) = \mu_2 = \mu_2' - \mu_1'^2 = \left(\frac{b^2 + ab + a^2}{3} \right) - \left(\frac{a^2 + 2ab + b^2}{4} \right)$$

Simplifying, we have

$$\text{Var}(X) = \frac{b^2 - 2ab + a^2}{12} = \frac{(b - a)^2}{12}$$

◆

EXAMPLE 3.58. The number of personal computers (PCs) sold daily at CompuWorld is uniformly distributed with a minimum of 2000 PCs and a maximum of 5000 PCs.

- (i) Find the probability that the daily sales will fall between 2,500 and 3,000 PCs.
- (ii) What is the probability that CompuWorld will sell at least 4,000 PCs?
- (iii) What is the probability that CompuWorld will exactly sell 2,500 PCs? (Anna, Nov. 2006)

Solution. If X is the number of PCs sold daily at CompuWorld, then X is distributed uniformly over the interval $(2000, 5000)$. Hence, the probability density function of X is given by

$$f(x) = \begin{cases} \frac{1}{3000} & \text{for } 2000 < x < 5000 \\ 0 & \text{elsewhere} \end{cases}$$

- (i) The probability that the daily sales will fall between 2,500 and 3,000 PCs is

$$P(2500 < X < 3000) = \int_{2500}^{3000} f(x) dx = \int_{2500}^{3000} \left[\frac{1}{3000} \right] dx$$

Integrating, we have

$$P(2500 < X < 3000) = \frac{1}{3000} [x]_{2500}^{3000} = \frac{500}{3000} = \frac{1}{6} = 0.1667$$

- (ii) The probability that CompuWorld will sell at least 4,000 PCs is

$$P(X \geq 4000) = \int_{4000}^{5000} f(x) dx = \int_{4000}^{5000} \left[\frac{1}{3000} \right] dx$$

Integrating, we have

$$P(X \geq 4000) = \frac{1}{3000} [x]_{4000}^{5000} = \frac{1000}{3000} = \frac{1}{3} = 0.3333$$

- (iii) The probability that CompuWorld will exactly sell 2,500 PCs is

$$P(X = 2500) = 0$$

since X has a continuous probability distribution.

◆

EXAMPLE 3.59. Show that for the rectangular distribution

$$f(x) = \frac{1}{2a}, \quad -a < x < a$$

the moment generating function about the origin is $\frac{1}{at}(\sinh at)$. Also show that the central moments of even order are given by $\mu_{2n} = \frac{a^{2n}}{2n+1}$.

(Madras, Oct. 2001)

Solution. The moment generating function of X is given by

$$M_X(t) = E(e^{tX}) = \int_{-a}^a e^{tx} \left[\frac{1}{2a} \right] dx$$

Integrating, we get

$$M_X(t) = \frac{1}{2a} \left[\frac{e^{tx}}{t} \right]_{-a}^a = \frac{1}{2at} [e^{at} - e^{-at}]$$

Simplifying, we have

$$M_X(t) = \frac{1}{at} \left[\frac{e^{at} - e^{-at}}{2} \right] = \frac{1}{at} [\sinh at]$$

The mean of X is given by

$$\mu = E(X) = \int_{-a}^a x \left[\frac{1}{2a} \right] dx = 0$$

since the integrand is an odd function of X . Thus, all the central and raw moments coincide.

Hence, the central moments of even order are given by

$$\mu_{2n} = E(X^{2n}) = \int_{-a}^a x^{2n} \left[\frac{1}{2a} \right] dx = \frac{2}{2a} \int_0^a x^{2n} dx$$

since the integrand is an even function of x .

Integrating, we have

$$\mu_{2n} = \frac{1}{a} \left[\frac{x^{2n+1}}{2n+1} \right]_0^a = \frac{1}{a} \frac{a^{2n+1}}{2n+1}$$

Simplifying, we have

$$\mu_{2n} = \frac{a^{2n}}{2n+1}$$

◆

EXAMPLE 3.60. If the random variable X follows the rectangular distribution with probability density function

$$f(x) = \begin{cases} \frac{1}{\theta} & \text{for } 0 < x < \theta \\ 0 & \text{elsewhere} \end{cases}$$

show that

- (i) The r^{th} raw moment is $\mu'_r = \frac{\theta^r}{r+1}$;
- (ii) $\mu_1 = \mu_3 = 0$, $\mu_2 = \frac{\theta^2}{12}$, $\mu_4 = \frac{\theta^4}{80}$;
- (iii) $\beta_1 = 0$ and $\beta_2 = 1.8$; and
- (iv) Skewness = $\gamma_1 = 0$ and kurtosis = $\gamma_2 = -1.2$.

Solution. (i) We find that

$$\mu'_r = E(X^r) = \int_0^\theta x^r f(x) dx = \int_0^\theta x^r \left[\frac{1}{\theta} \right] dx$$

Integrating, we have

$$\mu'_r = \frac{1}{\theta} \left[\frac{x^{r+1}}{r+1} \right]_0^\theta = \frac{1}{\theta} \left[\frac{\theta^{r+1}}{r+1} \right]$$

Simplifying, we have

$$\mu'_r = \frac{\theta^r}{r+1} \text{ for } r = 1, 2, \dots$$

In particular, the first four raw moments are

$$\mu'_1 = \frac{\theta}{2}, \mu'_2 = \frac{\theta^2}{3}, \mu'_3 = \frac{\theta^3}{4} \text{ and } \mu'_4 = \frac{\theta^4}{5}$$

- (ii) For any distribution, $\mu_0 = 1, \mu_1 = 0$. Next, we find that

$$\mu_2 = \mu'_2 - \mu_1'^2 = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12}$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3 = \frac{\theta^3}{4} - \frac{\theta^3}{2} + \frac{\theta^3}{4} = 0$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4 = \frac{\theta^4}{5} - \frac{\theta^4}{2} + \frac{\theta^4}{2} - \frac{3\theta^4}{16} = \frac{\theta^4}{80}$$

We note that $\mu_3 = 0$ because the given distribution is symmetric about the mean $\mu = \frac{\theta}{2}$.

- (iii) We find that

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0 \text{ and } \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{\frac{\theta^4}{80}}{\frac{\theta^4}{144}} = \frac{144}{80} = 1.8$$

- (iv) We find that

$$\text{Skewness} = \gamma_1 = \sqrt{\beta_1} = 0 \text{ and Kurtosis} = \gamma_2 = \beta_2 - 3 = -1.2$$

◆

PROBLEM SET 3.5

1. The cumulative distribution function of a random variable X is given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq -\pi \\ \frac{x+\pi}{2\pi} & \text{if } -\pi < x < \pi \\ 1 & \text{if } x \geq \pi \end{cases}$$

Find

- The probability density function of X and show that X is uniformly distributed over $[-\pi, \pi]$.
 - $P(-\frac{\pi}{2} < X < \frac{\pi}{2})$.
2. If X is a uniform random variable in $[-1, 1]$, find the probability density of X , mean of X and $\text{Var}(X)$.
3. The number of cellphones sold daily at CellWorld is uniformly distributed with a minimum of 50 phones and a maximum of 100 phones. Find the probability that
- The daily sales will fall between 70 and 80 phones.
 - CellWorld will sell at least 75 phones for a day.
 - CellWorld will exactly sell 80 phones for a day.
4. Let X be a continuous random variable with the uniform density

$$f(x) = \begin{cases} \frac{1}{2a} & \text{if } 0 < x < 2a \\ 0 & \text{otherwise} \end{cases}$$

Find the

- Mean and variance of X .
 - Median of X .
 - Quartiles of X .
 - Mode of X .
5. Find the characteristic function of the random variable having the uniform density

$$f(x) = \begin{cases} \frac{1}{2h} & \text{if } \alpha - h < x < \alpha + h \\ 0 & \text{otherwise} \end{cases}$$

3.8 EXPONENTIAL DISTRIBUTION

In probability theory, an exponential distribution is a continuous probability distribution, which is commonly used to model the time between independent events that occur at a constant average rate.

Definition 3.8. A continuous random variable X is said to have an *exponential distribution* with parameter λ (known as the *rate parameter*) if X has the probability density function

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{elsewhere} \end{cases} \quad (3.21)$$

In this case, we write $X \sim \text{Exp}(\lambda)$.

Remark 3.12. It is easy to verify that the exponential distribution with rate parameter λ is well-defined. Indeed, $f(x) \geq 0$ for all x and

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_0^{\infty} = 1$$

The exponential probability density for $\lambda = 1$ is illustrated in Figure 3.4.

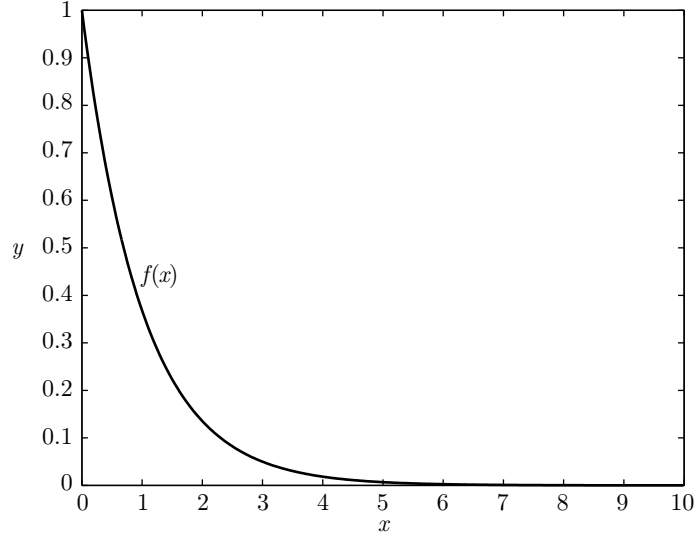


Figure 3.4: Exponential distribution for $\lambda = 1$.

EXAMPLE 3.61. If X is an exponential random variable with parameter λ , find the distribution function of X .

Solution. Let F be the distribution function of X , i.e. $F(x) = P(X \leq x)$.

If $x < 0$, then $F(x) = 0$. (This follows from the definition of f .)

If $x \geq 0$, then we have

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt = \int_0^x \lambda e^{-\lambda t} dt$$

Integrating, we have

$$F(x) = \left[-e^{-\lambda t} \right]_0^x = 1 - e^{-\lambda x}$$

Hence, the distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 - e^{-\lambda x} & \text{if } x \geq 0 \end{cases}$$

◆

Theorem 3.14. If X is an exponential random variable with parameter μ , then

- (i) $\mu'_r = \frac{r!}{\lambda^r}$.
- (ii) $\text{Mean}(X) = \frac{1}{\lambda}$.
- (iii) $\text{Var}(X) = \frac{1}{\lambda^2}$.

Proof. (i) By definition, we have

$$\mu'_r = E(X^r) = \int_0^{\infty} x^r f(x) dx = \int_0^{\infty} x^r \lambda e^{-\lambda x} dx = \int_{y=0}^{\infty} \frac{y^r}{\lambda^r} e^{-y} dy$$

(by substituting $y = \lambda x$ in the integral).

Hence, we have

$$\mu'_r = \frac{1}{\lambda^r} \int_0^{\infty} y^r e^{-y} dy = \frac{1}{\lambda^r} \Gamma(r+1) = \frac{r!}{\lambda^r}$$

(ii) Taking $r = 1$ in the Formula (i), we have

$$\text{Mean}(X) = \mu'_1 = \frac{1}{\lambda}$$

(iii) Taking $r = 2$ in the Formula (i), we have

$$\mu'_2 = \frac{2}{\lambda^2}$$

Hence, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

◆

Remark 3.13. We remark that it is also common in the literature to define the exponential distribution using an alternate parametrization as

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-\frac{1}{\theta}x} & \text{for } x \geq 0 \\ 0 & \text{elsewhere} \end{cases} \quad (3.22)$$

In Eq. (3.22), θ is known as the *survival parameter*. In this alternate specification, the random variable X is the *duration of time* that a biological or mechanical system manages to survive at the rate of θ . It can be easily seen that if X has the exponential density as given in Eq. (3.22), then $E(X) = \frac{1}{1/\theta} = \theta$ and $\text{Var}(X) = \frac{1}{1/\theta^2} = \theta^2$. In other words, the expected rate of survival is θ and the expected variance of survival is θ^2 . Because of these reasons, some authors use the alternate specification of the exponential distribution as given in Eq. (3.22). To avoid any ambiguity, we always use the definition of the exponential distribution given in Eq. (3.21) except when the need arises.

EXAMPLE 3.62. Find the moment generating function of an exponential random variable and hence find its mean and variance.

(Anna, Nov. 2004; Nov. 2006; Nov. 2007)

Solution. Let X be an exponential random variable with parameter λ . Then X has the probability density function given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

By definition, the moment generating function of X is

$$M_X(t) = E(e^{tX}) = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{-(\lambda-t)x} dx$$

Integrating, we have

$$M_X(t) = \lambda \left[\frac{e^{-(\lambda-t)x}}{-(\lambda-t)} \right]_0^{\infty}$$

which converges provided $\lambda - |t| > 0$ or $|t| < \lambda$. In this case, we have

$$M_X(t) = \lambda \left[0 + \frac{1}{\lambda-t} \right] = \frac{\lambda}{\lambda-t}$$

Hence, $M_X(t) = \frac{\lambda}{\lambda-t}$ for $|t| < \lambda$.

To find the mean and variance of X , we first calculate the first two raw moments, μ'_1, μ'_2 .

If $|t| < \lambda$, then we have

$$M_X(t) = \frac{1}{1 - \frac{t}{\lambda}} = 1 + \left(\frac{t}{\lambda}\right) + \left(\frac{t}{\lambda}\right)^2 + \left(\frac{t}{\lambda}\right)^3 + \dots$$

Hence, we find that

$$M_X(t) = 1 + \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \frac{t^3}{\lambda^3} + \dots$$

We have

$$\mu'_1 = \text{Coefficient of } \frac{t}{1!} \text{ in } M_X(t) = \frac{1}{\lambda}$$

and

$$\mu'_2 = \text{Coefficient of } \frac{t^2}{2!} \text{ in } M_X(t) = \frac{2}{\lambda^2}$$

Hence, the mean of X is $\text{Mean}(X) = \mu'_1 = \frac{1}{\lambda}$.

Also, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

Thus, the variance of X is $\text{Var}(X) = \frac{1}{\lambda^2}$. ◆

EXAMPLE 3.63. Let X be a random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{3} e^{-\frac{x}{3}} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (i) $P(X > 3)$.
- (ii) Moment generating function of X .
- (iii) $E(X)$ and $\text{Var}(X)$.

Solution. Given that X is an exponential random variable with parameter $\lambda = \frac{1}{3}$. Hence, we know that the cumulative distribution function of X is given by

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x} = 1 - e^{-\frac{x}{3}} \quad \text{for } x > 0$$

- (i) The required probability is given by

$$P(X > 3) = 1 - P(X \leq 3) = 1 - F(3) = e^{-1} = 0.3679$$

- (ii) As in Example 3.62, the MGF of X can be readily calculated as

$$M_X(t) = \frac{\lambda}{\lambda - t} = \frac{\frac{1}{3}}{\frac{1}{3} - t} = \frac{1}{1 - 3t}$$

which converges for all values of t such that $|t| < \lambda$ or $|t| < \frac{1}{3}$

- (iii) As in Example 3.62, the mean and variance of X can be readily calculated as

$$E(X) = \frac{1}{\lambda} = 3 \quad \text{and} \quad \text{Var}(X) = \frac{1}{\lambda^2} = 9$$

◆

EXAMPLE 3.64. The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{3}$. What is the probability that the repair time exceeds 3 hours?

(Anna, April 2003)

Solution. Let X be the time (in hours) required to repair a machine. Then it is given that X follows an exponential distribution with parameter $\lambda = \frac{1}{3}$. Hence, X has the probability density

$$f(x) = \begin{cases} \frac{1}{3}e^{-\frac{x}{3}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Hence, the probability that the repair time exceeds 3 hours is given by

$$P(X > 3) = \int_{x=3}^{\infty} f(x) dx = \int_{x=3}^{\infty} \frac{1}{3}e^{-\frac{x}{3}} dx$$

Integrating, we have

$$P(X > 3) = \left[-e^{-\frac{x}{3}} \right]_3^{\infty} = e^{-1} = \frac{1}{e} = 0.3679$$

◆

EXAMPLE 3.65. Establish the memoryless property of exponential distribution, i.e. if X is an exponential distribution, show that $P(X > s+t|X > s) = P(X > t)$, for any $s, t > 0$.

(Anna, May 2006; May 2007)

Solution. Let X be an exponential random variable with parameter λ . Then the probability density function of X is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

We have to show that X has the memoryless property, i.e. for any $s, t > 0$,

$$P(X > s+t|X > s) = P(X > t)$$

We note that for any $k > 0$,

$$\begin{aligned} P(X > k) &= \int_{x=k}^{\infty} f(x) dx = \int_{x=k}^{\infty} \lambda e^{-\lambda x} dx = \int_{x=k}^{\infty} d[-e^{-\lambda x}] \\ &= [-e^{-\lambda x}]_{x=k}^{\infty} = [0 + e^{-\lambda k}] = e^{-\lambda k} \end{aligned}$$

Thus,

$$P(X > s+t|X > s) = \frac{P(X > s+t \cap X > s)}{P(X > s)} = \frac{P(X > s+t)}{P(X > s)}$$

Hence, we have

$$P(X > s+t|X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$

Thus, X satisfies the memoryless property. ♦

We remark that the converse is also true. That is, if X is a continuous random variable taking only positive values and with memoryless property, then it can be shown (see the following example) that X has exponential distribution. Thus, the memoryless property completely characterizes the exponential distribution among all continuous probability distributions.

EXAMPLE 3.66. If X is a continuous random variable taking only positive values and with memoryless property, show that X has exponential distribution.

Solution. Given that $X \geq 0$ and X has memoryless property, i.e. if $s, t > 0$ then

$$P(X > s+t|X > s) = P(X > t)$$

i.e.

$$\frac{P(X > s+t \cap X > s)}{P(X > s)} = P(X > t) \quad \text{or} \quad \frac{P(X > s+t)}{P(X > s)} = P(X > t)$$

Thus, we have

$$P(X > s+t) = P(X > s)P(X > t) \tag{3.23}$$

If we define

$$G(x) = P(X > x), \quad \text{where } x > 0$$

then it follows from Eq. (3.23) that

$$G(s+t) = G(s)G(t) \text{ for any } s, t > 0$$

Hence, we have

$$G(x+h) = G(x)G(h) \text{ for any } h > 0$$

Therefore, we have

$$\frac{G(x+h) - G(x)}{h} = \frac{G(x)G(h) - G(x)}{h} = G(x) \frac{G(h) - 1}{h} \quad (3.24)$$

Note that $G(0) = P(X > 0) = 1$ since X takes only positive values. Hence, Eq. (3.24) can be written as

$$\frac{G(x+h) - G(x)}{h} = G(x) \frac{G(h) - G(0)}{h}$$

Taking limits on both sides as $h \rightarrow 0$, we have

$$G'(x) = G(x)G'(0) = -\lambda G(x)$$

where $\lambda \triangleq -G'(0)$.

Note that $G(x) = P(X > x)$ is a non-increasing function and so $G'(x) < 0$. This shows that $\lambda = -G'(0) > 0$.

Hence, we have

$$G'(x) = -\lambda G(x) \text{ with } G(0) = P(X > 0) = 1$$

Solving the differential equation, we have

$$G(x) = Ce^{-\lambda x}$$

The integration constant C is determined by using the condition that $G(0) = 1$.

Thus, $C = 1$. Hence, we have

$$G(x) = e^{-\lambda x} \text{ for } x > 0$$

Since $F(x) = P(X \leq x) = 1 - P(X > x) = 1 - G(x)$, we have

$$F(x) = 1 - e^{-\lambda x} \text{ for } x > 0$$

Hence, the probability density function, f , of X is obtained by

$$f(x) = F'(x) = \lambda e^{-\lambda x} \text{ for } x > 0$$

Thus, it follows that X has exponential distribution with parameter $\lambda > 0$. ◆

EXAMPLE 3.67. Show that if X has an exponential distribution, then for every constant $a \geq 0$, one has

$$P(Y \leq x | X \geq a) = P(X \leq x) \text{ for all } x$$

where $Y = X - a$.

(Madras, Oct. 2000)

Solution. Note that

$$P(Y \leq x | X \geq a) = P(X - a \leq x | X \geq a) = P(X \leq x + a | X \geq a)$$

Hence, we have

$$P(Y \leq x | X \geq a) = 1 - P(X > x + a | X \geq a) = 1 - P(X > x) = P(X \geq x)$$

by the memoryless property of X . ◆

EXAMPLE 3.68. A fast food chain finds that the average time customers have to wait for service is 45 seconds. If the waiting time can be treated as an exponential random variable, what is the probability that a customer will have to wait more than 5 minutes given that already he waited for 2 minutes? (Anna, May 2007)

Solution. If X denotes the waiting time at the fast food chain, then it is given that X is exponentially distributed with mean 45 seconds. Hence, the parameter of X is given by $\lambda = \frac{1}{45}$ and the probability density function of X is given by

$$f(x) = \lambda e^{-\lambda x} = \frac{1}{45} e^{-\frac{x}{45}} \text{ for } x \geq 0$$

Thus, the cumulative distribution function of X is given by

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x} = 1 - e^{-\frac{x}{45}} \text{ for } x \geq 0$$

By the memoryless probability of exponential distribution, the required probability that a customer will have to wait more than 5 minutes (or 300 seconds) given that already he waited for 2 minutes (or 120 seconds) is given by

$$P(X > 300 | X > 120) = P(X > 180) = 1 - P(X \leq 180) = 1 - F(180) = e^{-4} = 0.0183 \quad \text{◆}$$

EXAMPLE 3.69. The life length (in months) X of an electric component follows an exponential distribution with parameter $\lambda = \frac{1}{2}$. What is the probability that the component survives at least 10 months given that already it had survived for more than 9 months? (Anna, May 2006)

Solution. Since X is an exponential random variable with parameter $\lambda = \frac{1}{2}$, X has the probability density function given by

$$f(x) = \lambda e^{-\lambda x} = \frac{1}{2} e^{-\frac{x}{2}} \text{ for } x > 0$$

The required probability is given by

$$P(X \geq 10 | X \geq 9) = P(X \geq 1) = \int_{x=1}^{\infty} f(x) dx \quad (\text{by the memoryless property})$$

Hence, we have

$$P(X \geq 10 | X \geq 9) = \int_{x=1}^{\infty} \frac{1}{2} e^{-\frac{x}{2}} dx$$

Integrating, we have

$$P(X \geq 10 | X \geq 9) = \left[-e^{-\frac{x}{2}} \right]_1^{\infty} = e^{-\frac{1}{2}} = 0.6065 \quad \text{◆}$$

EXAMPLE 3.70. The time required to repair a machine is exponentially distributed with parameter $\frac{1}{2}$. What is the probability that a repair time exceeds 2 hours? What is the conditional probability that a repair time takes at least 10 hours given that its duration exceeds 9 hours?

(Anna, May 2006; Nov. 2006)

Solution. Since X is an exponential random variable with parameter $\lambda = \frac{1}{2}$, X has the probability density function given by

$$f(x) = \lambda e^{-\lambda x} = \frac{1}{2} e^{-\frac{x}{2}} \text{ for } x > 0$$

The probability that a repair time exceeds 2 hours is given by

$$\begin{aligned} P(X > 2) &= \int_{x=2}^{\infty} f(x) dx = \int_{x=2}^{\infty} \frac{1}{2} e^{-\frac{x}{2}} dx \\ &= \left[-e^{-\frac{x}{2}} \right]_{x=2}^{\infty} = e^{-1} = 0.3679 \end{aligned}$$

Next, the conditional probability that $X \geq 10$ given $X \geq 9$ is calculated as

$$P(X \geq 10 | X \geq 9) = P(X \geq 1) = \int_{x=1}^{\infty} f(x) dx \quad (\text{by the memoryless property})$$

As worked out in Example 3.69, it follows that

$$P(X \geq 10 | X \geq 9) = \left[-e^{-\frac{x}{2}} \right]_1^{\infty} = e^{-\frac{1}{2}} = 0.6065$$

◆

EXAMPLE 3.71. The length of time a person speaks over phone follows exponential distribution with mean 6. What is the probability that the person will talk for (i) more than 8 minutes and (ii) between 4 and 8 minutes?

(Anna, Nov. 2006)

Solution. If X is the random variable denoting the length of time a person speaks over phone, then X follows the exponential distribution with mean, $\mu = 6$. So, the probability density function of X

$$f(x) = \begin{cases} \frac{1}{6} e^{-\frac{x}{6}} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

So, the cumulative distribution function of X is

$$F(x) = P(X \leq x) = 1 - e^{-\frac{x}{6}} \text{ for } x > 0$$

(i) The probability that the person will talk for more than 8 minutes is

$$P(X > 8) = 1 - P(X \leq 8) = e^{-\frac{8}{6}} = e^{-\frac{4}{3}} = 0.2636$$

(ii) The probability that the person will talk between 4 and 8 minutes is

$$P(4 < X < 8) = F(8) - F(4) = e^{-\frac{4}{6}} - e^{-\frac{8}{6}} = e^{-\frac{2}{3}} - e^{-\frac{4}{3}} = 0.2498$$

◆

EXAMPLE 3.72. The daily consumption of milk in excess of 20,000 gallons is approximately exponentially distributed with $\theta = 3000$. The city has a daily stock of 35,000 gallons. What is the probability that of two days selected at random, the stock is insufficient for both days.

(Anna, April 2003)

Solution. Let X be the daily consumption of milk in the city, and $Y = X - 20,000$. By hypothesis, Y is approximately exponentially distributed with $\theta = 3000$ (mean). Hence, the probability density function of Y is given by

$$f(y) = \begin{cases} \frac{1}{3000} e^{-\frac{y}{3000}} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

The cumulative distribution function of Y is

$$F(y) = P(Y \leq y) = 1 - e^{-\frac{y}{3000}} \quad \text{for } y > 0$$

Since the city has a daily stock of 35,000 gallons, the probability that the city is insufficient for a day is given by

$$P(X > 35000) = P(Y > 15000) = 1 - P(Y \leq 15000) = 1 - F(y) = e^{-\frac{15000}{3000}} = e^{-5}$$

Using independence of events, the probability that of two days selected at random, the stock is insufficient for both days is given by

$$e^{-5} \times e^{-5} = [e^{-5}]^2 = e^{-10}$$



PROBLEM SET 3.6

1. Let X be a random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{2} e^{-\frac{x}{2}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find

- (a) $P(X > 2)$.
 - (b) The MGF of X and hence the mean and variance of X .
2. A fast food restaurant finds that the average time customers have to wait for service is 30 seconds. If the waiting time can be treated as an exponential random variable, what is the probability that a customer will have to wait more than 4 minutes given that already he waited for 2 minutes?
 3. The life length (in months) X of an electric bulb follows an exponential distribution with parameter $\lambda = 0.4$. What is the probability that the component survives at least 2 years given that it had already survived for more than 6 months?
 4. The time required (in hours) to repair a machine is exponentially distributed with parameter $\lambda = \frac{1}{3}$.
 - (a) What is the probability that a repair time exceeds 3 hours?

- (b) What is the conditional probability that a repair times takes at least 8 hours given that its duration exceeds 2 hours?
5. The failure pattern of an electronic system is known to follow an exponential distribution with mean time to failure of 500 hours. Find the probability that the system failure occurs within 300 hours.
6. The life (in years) of a certain type of electrical switch has an exponential distribution with a mean of 3. Find the probability that a switch installed fails within two years.
7. The length of time (in minutes) a person speaks over the phone follows an exponential distribution with mean 4. Find the probability that the person will talk for (a) more than 8 minutes, (b) between 3 and 6 minutes, and (c) less than 2 minutes.

3.9 GAMMA DISTRIBUTION

Definition 3.9. A continuous random variable is said to have a *gamma distribution* with parameters (α, λ) , where $\alpha > 0$ and $\lambda > 0$, if its probability density function is given by

$$f(x; \alpha, \lambda) = \begin{cases} \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

In this case, we write $X \sim \gamma(\alpha, \lambda)$.

Theorem 3.15. Let X be a gamma random variable with parameters α and λ . Then

- (i) $\text{Mean}(X) = \frac{\alpha}{\lambda}$.
(ii) $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.

(Madras, Sep. 1997; Anna, May 2007)

Proof. (i) The mean of X is given by

$$\mu'_1 = E(X) = \int_0^{\infty} x f(x) dx = \int_0^{\infty} x \left[\frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)} \right] dx$$

Simplifying, we have

$$\mu'_1 = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \lambda^\alpha x^\alpha e^{-\lambda x} dx$$

Putting $y = \lambda x$ in the integral and noting that $dx = \frac{1}{\lambda} dy$, we have

$$\begin{aligned} \mu'_1 &= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} y^\alpha e^{-y} \frac{dy}{\lambda} = \frac{1}{\lambda \Gamma(\alpha)} \int_0^{\infty} y^\alpha e^{-y} dy \\ &= \frac{1}{\lambda \Gamma(\alpha)} \Gamma(\alpha + 1) = \frac{\alpha \Gamma(\alpha)}{\lambda \Gamma(\alpha)} = \frac{\alpha}{\lambda} \end{aligned}$$

Thus, $\text{Mean}(X) = \frac{\alpha}{\lambda}$

(ii) First, we calculate the second raw moment, μ'_2 . We have

$$\mu'_2 = E(X^2) = \int_0^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \left[\frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)} \right] dx$$

Simplifying, we have

$$\mu'_2 = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \lambda^{\alpha} x^{\alpha+1} e^{-\lambda x} dx$$

Putting $y = \lambda x$ in the integral and noting that $dx = \frac{1}{\lambda} dy$, we have

$$\begin{aligned} \mu'_2 &= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \lambda^{\alpha} \frac{y^{\alpha+1}}{\lambda^{\alpha+1}} e^{-y} \frac{dy}{\lambda} = \frac{1}{\lambda^2 \Gamma(\alpha)} \int_0^{\infty} y^{\alpha+1} e^{-y} dy \\ &= \frac{1}{\lambda^2 \Gamma(\alpha)} \Gamma(\alpha+2) = \frac{(\alpha+1)\alpha \Gamma(\alpha)}{\lambda^2 \Gamma(\alpha)} = \frac{(\alpha+1)\alpha}{\lambda^2} \end{aligned}$$

Thus, the variance of X is given by

$$\text{Var}(X) = \mu_2 = \mu'_2 - \mu_1'^2 = \frac{(\alpha+1)\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}$$

Hence, $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.

◆

EXAMPLE 3.73. Derive the moment generating function of the gamma distribution, and hence obtain its mean and variance.

(Anna, Nov. 2004; May 2007)

Solution. By definition, the moment generating function of the gamma distribution is given by

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \int_{x=0}^{\infty} e^{tx} f(x) dx = \int_{x=0}^{\infty} e^{tx} \frac{1}{\Gamma(\alpha)} \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx \\ &= \frac{1}{\Gamma(\alpha)} \int_{x=0}^{\infty} e^{-(\lambda-t)x} (\lambda x)^{\alpha-1} (\alpha dx) \end{aligned}$$

Substituting $y = \alpha x$, we have

$$M_X(t) = \frac{1}{\Gamma(\alpha)} \int_{y=0}^{\infty} e^{-\frac{(\lambda-t)}{\lambda} y} y^{\alpha-1} dy$$

Next, we substitute $z = (\frac{\lambda-t}{\lambda})y = (1 - \frac{t}{\lambda})y$. Then, $dz = (1 - \frac{t}{\lambda})dy$ and the integral become

$$\begin{aligned} M_X(t) &= \frac{1}{\Gamma(\alpha)} \int_{z=0}^{\infty} e^{-z} \frac{z^{\alpha-1}}{(1 - \frac{t}{\lambda})^{\alpha-1}} \frac{dz}{(1 - \frac{t}{\lambda})} \\ &= \frac{1}{\Gamma(\alpha)} \left(1 - \frac{t}{\lambda}\right)^{-\alpha} \int_{z=0}^{\infty} e^{-z} z^{\alpha-1} dz \end{aligned}$$

where t satisfies the condition that $|\frac{t}{\lambda}| < 1$

Integrating, we have

$$M_X(t) = \frac{1}{\Gamma(\alpha)} \left(1 - \frac{t}{\lambda}\right)^{-\alpha} \Gamma(\alpha) = \left(1 - \frac{t}{\lambda}\right)^{-\alpha}$$

Hence, the MGF of X is given by

$$M_X(t) = \left(1 - \frac{t}{\lambda}\right)^{-\alpha}$$

where $|t| < \lambda$.

Next, we find the mean and variance of X . Expanding $M_X(t)$, we have

$$M_X(t) = 1 + \alpha \frac{t}{\lambda} + \frac{\alpha(\alpha+1)}{2} \left(\frac{t}{\lambda}\right)^2 + \dots$$

Thus, the first raw moment of X is given by

$$\mu'_1 = \text{Coefficient of } \frac{t}{1!} \text{ in } M_X(t) = \frac{\alpha}{\lambda}$$

and the second raw moment of X is given by

$$\mu'_2 = \text{Coefficient of } \frac{t^2}{2!} \text{ in } M_X(t) = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Hence, the mean of X is given by

$$\mu = \mu'_1 = \frac{\alpha}{\lambda}$$

and the variance of X is given by

$$\text{Var}(X) = \mu_2 = \mu'_2 - \mu_1'^2 = \frac{\alpha(\alpha+1)}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}$$

◆

EXAMPLE 3.74. Find the moment generating function for the gamma density function

$$f(x) = \begin{cases} \frac{1}{\Gamma(n)} x^{n-1} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Also obtain its mean, variance, β_1, β_2 , harmonic mean and mode. Prove that

$$2\mu_4\sigma^2 - 3\mu_3^2 - 6\sigma^6 = 0$$

Solution. From our calculations in Example 3.73, it follows that

$$M_X(t) = (1-t)^{-n}, \quad \text{where } |t| < 1$$

Expanding $M_X(t)$ as a Taylor series, we get

$$M_X(t) = 1 + t + t^2 + \cdots + t^n + \cdots$$

Thus, the r th raw moment of X is given by

$$\mu'_r = \text{Coefficient of } \frac{t^r}{r!} \text{ in } M_X(t) = r!$$

Thus, the mean of X is given by

$$\mu'_1 = 1! = 1$$

and the variance of X is given by

$$\text{Var}(X) = \sigma^2 = \mu_2 = \mu'_2 - \mu_1'^2 = 2! - 1^2 = 2 - 1 = 1$$

Next, note that

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3 = 6 - 6 + 2 = 2$$

and

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu_1'^2 - 3\mu_1'^4 = 24 - 24 + 12 - 3 = 9$$

Hence, we have

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{4}{1} = 4$$

and

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{9}{1} = 9$$

We also see that

$$2\mu_4\sigma^2 - 3\mu_3^2 - 6\sigma^6 = 18 - 12 - 6 = 0$$

The harmonic mean, H , of X is given by

$$\frac{1}{H} = \int_0^\infty \frac{1}{x} f(x) dx = \int_0^\infty \frac{1}{x} \frac{1}{\Gamma(n)} x^{n-1} e^{-x} dx = \frac{1}{\Gamma(n)} \int_0^\infty x^{n-2} e^{-x} dx$$

Integrating, we have

$$\frac{1}{H} = \frac{\Gamma(n-1)}{\Gamma(n)} = \frac{1}{n}$$

Hence,

$$H = n$$

Finally, to calculate the mode, c , of the gamma variate X , we note that

$$f'(x) = \frac{1}{\Gamma(n)} [x^{n-2}e^{-x} + x^{n-1}(-e^{-x})]$$

Hence, we have

$$f'(x) = \frac{e^{-x}}{\Gamma(n)} (x^{n-2} - x^{n-1})$$

Differentiating again, we get

$$f''(x) = \frac{e^{-x}}{\Gamma(n)} [(n-2)x^{n-3} - nx^{n-2} + x^{n-1}]$$

In the range $0 < x < \infty$, $f'(x) = 0 \Rightarrow x = 1$. We also note that

$$f''(1) = -\frac{e^{-1}}{\Gamma(n)} < 0$$

Hence, $x = 1$ is the mode of X . ♦

EXAMPLE 3.75. If X is a random variable and the gamma density function is given by

$$f(x) = \begin{cases} \frac{1}{\Gamma(n)} x^{n-1} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the limiting form of the standard gamma variate as $n \rightarrow \infty$. Interpret your result.

Solution. We know that the moment generating function of X is given by

$$M_X(t) = (1-t)^{-n} \quad \text{where } |t| < 1$$

The standard gamma variate, Z , is given by

$$Z = \frac{X - \mu}{\sigma} = \frac{X - n}{\sqrt{n}}$$

Hence, by Theorem 2.22, the moment generating function of Z is given by

$$M_Z(t) = e^{-\sqrt{n}t} \left(1 - \frac{t}{\sqrt{n}}\right)^{-n}$$

Taking natural logarithms on both sides, we have

$$K_Z(t) \triangleq \log M_Z(t) = -\sqrt{n}t - n \log \left[1 - \frac{t}{\sqrt{n}}\right]$$

Expanding the right hand side, we have

$$\begin{aligned} K_Z(t) &= -\sqrt{n}t + n \left[\frac{t}{\sqrt{n}} + \frac{t^2}{2n} + \frac{t^3}{3n^{\frac{3}{2}}} + \dots \right] \\ &= \frac{t^2}{2} + \frac{t^3}{3\sqrt{n}} + \frac{t^4}{4n} + \dots \end{aligned}$$

Taking limits as $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} K_Z(t) = \frac{t^2}{2}$$

Hence, it follows that

$$\lim_{n \rightarrow \infty} M_Z(t) = e^{\frac{t^2}{2}}$$

Interpretation. We know that $e^{\frac{t^2}{2}}$ is the moment generating function of the standard normal variate. By the uniqueness theorem for moment generating functions, it follows that the standard gamma variate approaches the standard normal variate as the parameter $n \rightarrow \infty$. ♦

EXAMPLE 3.76. The daily consumption of milk in a city, in excess of 20,000 gallons, is approximately distributed as a gamma variate with parameters $\alpha = 2$ and $\lambda = \frac{1}{10,000}$. The city has a daily stock of 30,000 gallons. What is the probability that the stock is insufficient on a particular day?

Solution. Let X be the daily consumption of milk in a city, and $Y = X - 20,000$. Given that Y follows a gamma distribution with parameters $\alpha = 2$ and $\lambda = \frac{1}{10,000}$. Thus, the probability density function of Y is given by

$$f(y) = \frac{1}{(10,000)^2} \frac{y^{2-1}}{\Gamma(2)} e^{-\frac{y}{10,000}} \quad \text{for } y > 0$$

Hence,

$$f(y) = \frac{y}{(10,000)^2} e^{-\frac{y}{10,000}} \quad \text{for } y > 0$$

Since the city has a daily stock of 30,000 gallons, the probability that the stock is insufficient on a particular day is given by

$$P(X > 30,000) = P(Y > 10,000) = \int_{10,000}^{\infty} \frac{y}{(10,000)^2} e^{-\frac{y}{10,000}} dy$$

Hence,

$$P(X > 30,000) = \frac{1}{10,000} \int_{10,000}^{\infty} y d \left[-e^{-\frac{y}{10,000}} \right]$$

Using integration by parts,

$$P(X > 30,000) = \frac{1}{10,000} \left[\left[y \left(-e^{-\frac{y}{10,000}} \right) \right]_{10,000}^{\infty} - \int_{10,000}^{\infty} \left(-e^{-\frac{y}{10,000}} \right) dy \right]$$

Simplifying,

$$P(X > 30,000) = e^{-1} - \frac{1}{10,000} \int_{10,000}^{\infty} \left(-e^{-\frac{y}{10,000}} \right) dy$$

Integrating, we have

$$P(X > 30,000) = e^{-1} - \left[e^{-\frac{y}{10,000}} \right]_{10,000}^{\infty} = e^{-1} - [0 - e^{-1}]$$

Hence, the required probability is

$$P(X > 30,000) = 2e^{-1} = \frac{2}{e} = 0.7358$$



PROBLEM SET 3.7

1. If X is a gamma random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{2!} x^2 e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the moment generating function of X and hence obtain its mean and variance.

2. If X is a random variable having its r th raw moment as

$$\mu'_r = \frac{(k+r)!}{k!}$$

where k is a positive integer, show that the probability density function of X is

$$f(x) = \begin{cases} \frac{x^k}{k!} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

3. Let X be a gamma random variable with parameters α and λ . When $\alpha > 1$, show that X has a unique mode at

$$x^* = \frac{\alpha - 1}{\lambda}$$

4. If X is a gamma random variable with parameters $\alpha = 2$ and $\lambda = 3$, find $P(X \geq 2)$.
5. The daily consumption of milk in a city, in excess of 30,000 gallons, is approximately distributed as a gamma variate with parameters $\alpha = 2$ and $\lambda = \frac{1}{20,000}$. The city has a daily stock of 40,000 gallons. What is the probability that the stock is insufficient on a particular day?

3.10 WEIBULL DISTRIBUTION

Weibull distribution is named after Waloddi Weibull (1887–1979), a Swedish engineer, scientist and a mathematician. He published his paper on Weibull distribution in 1939. Weibull distribution is often used in the field of data analysis due to its flexibility.

Definition 3.10. A continuous random variable X is said to have the *Weibull distribution* with parameters (α, β) , where $\alpha > 0$ and $\beta > 0$, if its probability density function is defined by

$$f(x; \alpha, \beta) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

EXAMPLE 3.77. If X is a Weibull random variable with parameters $\alpha > 0$ and $\beta > 0$, find the distribution function of X .

Solution. Let F be the distribution function of X , i.e. $F(x) = P(X \leq x)$.

If $x < 0$, then $F(x) = 0$.

Let $x \geq 0$. Then we have

$$F(x) = \int_0^x f(t)dt = \int_0^x \alpha \beta t^{\beta-1} e^{-\alpha t^\beta} dt = \int_0^x d[-e^{-\alpha t^\beta}]$$

Integrating, we have

$$F(x) = [-e^{-\alpha t^\beta}]_0^x = 1 - e^{-\alpha x^\beta}$$

Combining, we have

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - e^{-\alpha x^\beta} & \text{for } x \geq 0 \end{cases}$$

◆

EXAMPLE 3.78. Find the mean and variance of Weibull distribution.

(Anna, May 2006; May 2007)

Solution. The mean of X is given by

$$\mu = E(X) = \int_{x=0}^{\infty} x f(x) dx = \int_{x=0}^{\infty} x [\alpha \beta x^{\beta-1} e^{-\alpha x^\beta}] dx$$

Substitute $y = \alpha x^\beta$ in the integral. Then $dy = \alpha \beta x^{\beta-1} dx$. Hence,

$$\mu = \int_{y=0}^{\infty} \left(\frac{y}{\alpha}\right)^{\frac{1}{\beta}} e^{-y} dy = \alpha^{-\frac{1}{\beta}} \int_{y=0}^{\infty} y^{\frac{1}{\beta}} e^{-y} dy$$

It follows that

$$\mu = \alpha^{-\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right)$$

To find the variance of X , we first calculate the second raw moment of X . We have

$$\mu'_2 = E(X^2) = \int_{x=0}^{\infty} x^2 f(x) dx = \int_{x=0}^{\infty} x^2 [\alpha \beta x^{\beta-1} e^{-\alpha x^\beta}] dx$$

Substitute $y = \alpha x^\beta$ in the integral. Then $dy = \alpha \beta x^{\beta-1} dx$. Hence,

$$\mu'_2 = \int_{y=0}^{\infty} \left(\frac{y}{\alpha}\right)^{\frac{2}{\beta}} e^{-y} dy = \alpha^{-\frac{2}{\beta}} \int_{y=0}^{\infty} y^{\frac{2}{\beta}} e^{-y} dy$$

It follows that

$$\mu'_2 = \alpha^{-\frac{2}{\beta}} \Gamma\left(1 + \frac{2}{\beta}\right)$$

Hence, the variance of X is given by

$$\text{Var}(X) = \mu'_2 - \mu_1'^2 = \alpha^{-\frac{2}{\beta}} \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \Gamma^2\left(1 + \frac{1}{\beta}\right) \right]$$

◆

EXAMPLE 3.79. The life time of a component measured in hours is Weibull distribution with parameters $\alpha = 0.2, \beta = 0.5$. Find the mean life time of the component.

(Anna, Nov. 2003)

Solution. If X denotes the life time of the component measured in hours, then it is given that X follows Weibull distribution with parameters $\alpha = 0.2$ and $\beta = 0.5$. Hence, the mean life time of the component is

$$\mu = E(X) = \alpha^{-\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right) = (0.2)^{-\frac{1}{0.5}} \Gamma\left(1 + \frac{1}{0.5}\right)$$

Simplifying, we have

$$\mu = (0.2)^{-2} \Gamma(1 + 2) = 25 \times 2 = 50 \text{ hours}$$

◆

EXAMPLE 3.80. Suppose that the lifetime of a certain kind of an emergency back up battery (in hours) is a random variable X having the Weibull distribution with $\alpha = 0.1$ and $\beta = 0.5$. Find:

- The mean lifetime of these batteries.
- The probability that such a battery will last more than 300 hours.
- The probability that such a battery will not last 100 hours.

(Anna, April 2005)

Solution. Let X denote the lifetime of certain kind of an emergency back up battery (in hours). Then X is a Weibull random variable with parameters $\alpha = 0.1$ and $\beta = 0.5$. Then, the probability density function of X is given by

$$f(x) = \alpha \beta x^{\beta-1} e^{-\alpha x^\beta} \text{ for } x > 0$$

and $f(x) = 0$ elsewhere.

- The mean lifetime of the battery is given by

$$\mu = \alpha^{-\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right) = (0.1)^{-\frac{1}{0.5}} \Gamma\left(1 + \frac{1}{0.5}\right) = (0.1)^{-2} \Gamma(1 + 2)$$

Simplifying, we have

$$\mu = \frac{1}{0.01} \times 2 = 200 \text{ hours}$$

(b) The probability that the battery will last more than 300 hours is given by

$$P(X > 300) = \int_{300}^{\infty} f(x) dx = \int_{300}^{\infty} \alpha \beta x^{\beta-1} e^{-\alpha x^{\beta}} dx$$

Substitute $y = \alpha x^{\beta}$ in the integral. Then $dy = \alpha \beta x^{\beta-1} dx$. Note also that

$$x = 300 \Rightarrow y = 0.1 (300)^{0.5} = 0.1 \times 10 \times \sqrt{3} = \sqrt{3}$$

Hence, the integral becomes

$$P(X > 300) = \int_{y=\sqrt{3}}^{\infty} e^{-y} dy = [-e^{-y}]_{\sqrt{3}}^{\infty} = e^{-\sqrt{3}} = 0.1769$$

(c) The probability that the battery will not last 100 hours is given by

$$P(X < 100) = \int_0^{100} f(x) dx = \int_0^{100} \alpha \beta x^{\beta-1} e^{-\alpha x^{\beta}} dx$$

Substitute $y = \alpha x^{\beta}$ in the integral. Then $dy = \alpha \beta x^{\beta-1} dx$. Note also that

$$x = 100 \Rightarrow y = 0.1 (100)^{0.5} = 0.1 \times 10 = 1$$

Hence, the integral becomes

$$P(X < 100) = \int_{y=0}^1 e^{-y} dy = [-e^{-y}]_0^1 = [1 - e^{-1}] = 0.6321$$

◆

EXAMPLE 3.81. The life time X in hours of a component is modelled by a Weibull distribution with $\beta = 2$. Starting with a large number of components, it is observed that 15% of the components that have lasted 90 hours fail before 100 hours. Find the parameter α .

(Anna, May 2007)

Solution. Since $\beta = 2$, the given Weibull distribution is described by the density function

$$f(x) = 2\alpha x e^{-\alpha x^2} \quad \text{for } x > 0$$

Given that $P(X < 100 | X > 90) = 0.15$. Using the definition of conditional probability, we have

$$\frac{P(90 < X < 100)}{P(X > 90)} = 0.15$$

Note that

$$P(X > 90) = \int_{x=90}^{\infty} f(x) dx = \int_{x=90}^{\infty} 2\alpha x e^{-\alpha x^2} dx$$

Integrating, we have

$$P(X > 90) = \left[-e^{-\alpha x^2} \right]_{x=90}^{\infty} = e^{-90^2 \alpha}$$

Similarly, it can be seen that

$$P(90 < X < 100) = e^{-90^2 \alpha} - e^{-100^2 \alpha}$$

Hence, it follows that

$$\frac{P(90 < X < 100)}{P(X > 90)} = \frac{e^{-90^2 \alpha} - e^{-100^2 \alpha}}{e^{-90^2 \alpha}} = 0.15$$

i.e.

$$1 - e^{-(100^2 - 90^2) \alpha} = 0.15 \quad \text{or} \quad e^{-1900 \alpha} = 1 - 0.15 = 0.85$$

i.e.

$$\alpha = -\frac{\log 0.85}{1900} = 8.5536 \times 10^{-5}$$

◆

EXAMPLE 3.82. Each of the six tubes of a radio set has a life length (in years) which follows a Weibull distribution with parameters $\alpha = 25$ and $\beta = 2$. If these tubes function independently of one another, what is the probability that no tube will have to be replaced during the first two months of operation?

(Anna, May 2006)

Solution. Let X denote the life length (in years) of the tube of the radio set. Then, by hypotheses, X follows a Weibull distribution with parameters $\alpha = 25$ and $\beta = 2$. Hence, the probability density function of X is given by

$$f(x) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

and the cumulative distribution function of X is given by

$$F(x) = 1 - e^{-\alpha x^\beta} = 1 - e^{-25x^2} \quad \text{for } x > 0$$

and the probability that a tube will not be replaced during the first 2 months is given by

$$P\left(X > \frac{2}{12}\right) = P\left(X > \frac{1}{6}\right) = 1 - P\left(X \leq \frac{1}{6}\right) = 1 - F\left(\frac{1}{6}\right) = e^{-\frac{25}{36}} = 0.4994$$

So, the probability that none of the six tubes will be replaced during the first 2 months is given by

$$\left[e^{-\frac{25}{36}}\right]^6 = 0.0155 \quad (\text{using independence})$$

◆

PROBLEM SET 3.8

1. The life time of a component measured in hours is Weibull distribution with parameters $\alpha = 0.5$ and $\beta = 0.2$. Find the mean life time of the component.

2. Suppose that the lifetime of a certain kind of an emergency back-up battery (in hours) is a random variable X having the Weibull distribution with $\alpha = 0.4$ and $\beta = 0.25$. Find:
 - (a) The mean lifetime of these batteries.
 - (b) The probability that such a battery will last more than 200 hours.
 - (c) The probability that such a battery will not last 300 hours.
3. The life time X in hours of a component is modelled by a Weibull distribution with $\beta = 2$. Starting with a large number of components, it is observed that 20% of the components that have lasted 80 hours fail before 100 hours. Find the parameter α .
4. Each of the six tubes of a radio set has a life length (in years) which follows a Weibull distribution with parameters $\alpha = 10$ and $\beta = 3$. If these tubes function independently of one another, what is the probability that no tube will have to be replaced during the first three months of operation?

3.11 NORMAL DISTRIBUTION

The normal distribution, also called as *Gaussian distribution*, is a continuous probability distribution of great importance. It has numerous applications in diverse fields. Its importance stems from the fact that it can be used as a model of quantitative phenomena in the biological and behavioural sciences due to the *Central Limit Theorem* (see Section 4.10 for a detailed discussion on the Central Limit Theorem). In Statistics, normal distribution arises as a limiting case of several discrete and continuous probability distributions. It has applications in Sampling Theory. By Central Limit Theorem, it follows that the sampling distribution of the sample mean is approximately normal even if the distribution of the population from which the sample is drawn from is not normal. Normal distribution has large applications in Statistical Quality Control in industry for setting control limits.

The normal distribution was first introduced by Abraham de Moivre (1667–1754) in 1734, who deduced it as a limiting form of the binomial distribution. Pierre-Simon Laplace (1749–1827) used normal distributions in the analysis of errors of experiments. C.F. Gauss (1777–1855) used them in his work on the theory of celestial movement.

Definition 3.11. A continuous random variable X with parameters μ and σ , where $-\infty < \mu < \infty$ and $\sigma > 0$, is said to have a *normal distribution* or *Gaussian distribution* if its probability density function is given by

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \text{where } -\infty < x < \infty$$

If X is a normal random variable with parameters μ and σ , we denote it by $X \sim N(\mu, \sigma^2)$.

The normal probability density function is also called as *bell-shaped curve*. For the values of $\mu = 0$ and $\sigma = 2$, the normal probability density is illustrated in Figure 3.5.

Definition 3.12. If X is a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$, then the *standard normal variable* is defined by

$$Z = \frac{X - \mu}{\sigma}$$

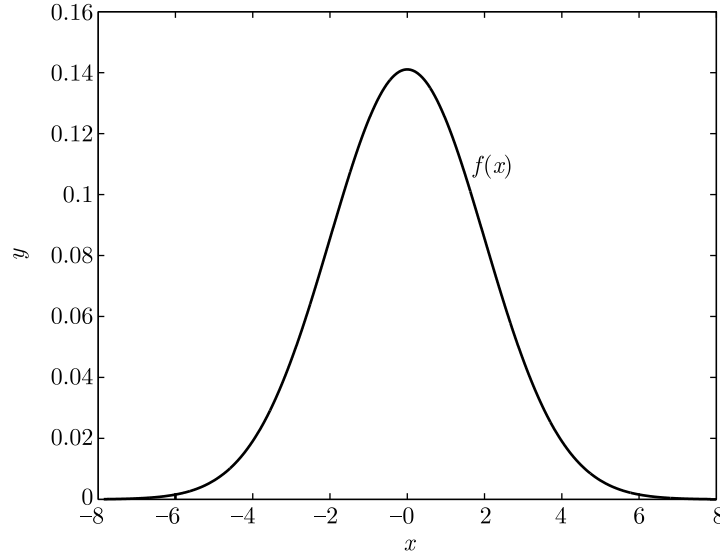


Figure 3.5: Normal probability density function for $\mu = 0, \sigma = 2$.

It can be easily shown that the standard normal variable Z has mean $\mu_Z = 0$ and variance $\sigma_Z^2 = 1$. Hence, the probability density function of Z , called as the *standard normal density*, is given by

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \text{ where } -\infty < z < \infty$$

EXAMPLE 3.83. List some important properties of the normal distribution.

Solution. Let $X \sim N(\mu, \sigma^2)$. Some important properties of the normal distribution are listed below:

- (i) The curve $y = f(x)$ is bell-shaped and symmetrical about the line $x = \mu$.
- (ii) The mean, median and mode of X coincide at $x = \mu$.
- (iii) Since $x = \mu$ is the mode of X , the maximum value of $f(x)$ occurs at the point $x = \mu$ and is equal to

$$f_{\max} = [f(x)]_{x=\mu} = \frac{1}{\sigma\sqrt{2\pi}}$$

- (iv) All the odd central moments are zero, i.e. $\mu_{2r+1} = 0$ for $r = 0, 1, 2, \dots$ and the even central moments are given by

$$\mu_{2r} = 1 \cdot 3 \cdot 5 \cdots (2r-1) \sigma^{2r} \text{ for } r = 0, 1, 2, \dots$$

- (v) $\beta_1 = 0$ and $\beta_2 = 3$. Thus,

$$\gamma_1 = \text{Skewness} = \sqrt{\beta_1} = 0 \text{ and } \gamma_2 = \text{Kurtosis} = \beta_2 - 3 = 0$$

- (vi) The moment generating function of the normal distribution is given by

$$M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

- (vii) The points of inflexion of the curve $y = f(x)$ are $x = \mu \pm \sigma$.
 (viii) The mean deviation (M.D.) about the mean is given by

$$\text{M.D.} = \sqrt{\frac{2}{\pi}} \sigma \approx \frac{4}{5} \sigma (\text{approx.})$$

- (ix) The quartiles are given by

$$Q_1 = \mu - 0.6745\sigma \quad \text{and} \quad Q_3 = \mu + 0.6745\sigma$$

Hence, the quartile deviation (Q.D.) is given by

$$\text{Q.D.} = \frac{Q_3 - Q_1}{2} \approx \frac{2}{3} \sigma$$

Thus,

$$\text{Q.D.} : \text{M.D.} : \text{S.D.} :: \frac{2}{3} \sigma : \frac{4}{5} \sigma : \sigma :: 10 : 12 : 15$$

- (x) *Area Property:*

$$P(\mu - \sigma < X < \mu + \sigma) = 0.6826$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9544$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$$



The first property follows immediately from the definition (see also Figure 3.5). The remaining properties will be established in the problems in this section.

EXAMPLE 3.84. Find the mean and variance of the normal distribution.

(Madras, Sept., 1997)

Solution. Let X be a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$. By definition, the mean of X is given by

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} x \left[\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx$$

Substitute $u = \frac{x-\mu}{\sqrt{2}\sigma}$ in the integral. Then $du = \frac{1}{\sqrt{2}\sigma} dx$. The integral becomes

$$\begin{aligned} E(X) &= \int_{u=-\infty}^{\infty} (\sqrt{2}\sigma u + \mu) \left[\frac{1}{\sigma \sqrt{2\pi}} e^{-u^2} \right] \sqrt{2}\sigma du \\ &= \frac{1}{\sqrt{\pi}} \int_{u=-\infty}^{\infty} (\sqrt{2}\sigma u + \mu) e^{-u^2} du \\ &= \frac{2}{\pi} \sigma \int_{u=-\infty}^{\infty} u e^{-u^2} du + \frac{\mu}{\sqrt{\pi}} \int_{u=-\infty}^{\infty} e^{-u^2} du \end{aligned}$$

Noting that the integrand in the first integral is an odd function of u and that the integrand in the second integral is an even function of u , we have

$$E(X) = 0 + \frac{2\mu}{\sqrt{\pi}} \int_{u=0}^{\infty} e^{-u^2} du = \frac{2\mu}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \mu$$

Hence, $\text{Mean}(X) = \mu$.

Next, we find the variance of X . By definition, we have

$$\text{Var}(X) = E[(X - \mu)^2] = \int_{x=-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Hence,

$$\text{Var}(X) = \int_{x=-\infty}^{\infty} (x - \mu)^2 \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx$$

Substitute $u = \frac{x-\mu}{\sqrt{2}\sigma}$ in the integral. Then $du = \frac{1}{\sqrt{2}\sigma} dx$. The integral becomes

$$\text{Var}(X) = \int_{u=-\infty}^{\infty} 2\sigma^2 u^2 \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-u^2} \right] \sqrt{2}\sigma du$$

Simplifying, we have

$$\text{Var}(X) = \frac{2\sigma^2}{\sqrt{\pi}} \int_{u=-\infty}^{\infty} u^2 e^{-u^2} du = \frac{2\sigma^2}{\sqrt{\pi}} 2 \int_{u=0}^{\infty} u^2 e^{-u^2} du = \frac{4\sigma^2}{\sqrt{\pi}} \int_{u=0}^{\infty} u^2 e^{-u^2} du$$

Substitute $w = u^2$ in the above integral. Then $dw = 2u du$. Hence,

$$\begin{aligned} \text{Var}(X) &= \frac{4\sigma^2}{\sqrt{\pi}} \int_{w=0}^{\infty} w e^{-w} \frac{dw}{2\sqrt{w}} = \frac{2\sigma^2}{\sqrt{\pi}} \int_{w=0}^{\infty} w^{\frac{1}{2}} e^{-w} dw \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \left[\frac{1}{2} \Gamma\left(\frac{1}{2}\right) \right] = \frac{2\sigma^2}{\sqrt{\pi}} \left[\frac{1}{2} \sqrt{\pi} \right] = \sigma^2 \end{aligned}$$

Hence, $\text{Var}(X) = \sigma^2$. ♦

EXAMPLE 3.85. (a) Obtain the moment generating function of the normal distribution.

(Madras, April 2000; Anna, Nov. 2006)

(b) Deduce the moment generating function of the standard normal distribution. Obtain its mean and variance.

Solution. (a) Let X be a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$. By definition, the moment generating function of X is given by

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx \\ &= \int_{-\infty}^{\infty} e^{tx} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx \end{aligned}$$

Substitute $u = \frac{x-\mu}{\sqrt{2}\sigma}$ in the integral. Then $du = \frac{1}{\sqrt{2}\sigma} dx$. The integral becomes

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty} e^{t(\sqrt{2}\sigma u + \mu)} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-u^2} \right] \sqrt{2}\sigma du = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{\sqrt{2}\sigma t u} \cdot e^{\mu t} e^{-u^2} du \\ &= \frac{e^{\mu t}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{\sqrt{2}\sigma t u - u^2} du = \frac{e^{\mu t}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\left(u - \frac{\sigma t}{\sqrt{2}}\right)^2 + \frac{\sigma^2 t^2}{2}} du \\ &= \frac{e^{\mu t}}{\sqrt{\pi}} e^{\frac{\sigma^2 t^2}{2}} \int_{-\infty}^{\infty} e^{-\left(u - \frac{\sigma t}{\sqrt{2}}\right)^2} du \end{aligned}$$

Substituting $v = u - \frac{\sigma t}{\sqrt{2}}$, the integral becomes

$$M_X(t) = \frac{e^{\mu t + \frac{\sigma^2 t^2}{2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-v^2} dv = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

since

$$\int_{-\infty}^{\infty} e^{-v^2} dv = 2 \int_0^{\infty} e^{-v^2} dv = 2 \left(\frac{\pi}{2} \right) = \pi$$

Thus, the MGF of X is $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$.

(b) The standard normal variate Z is defined by $Z = \frac{X-\mu}{\sigma}$. By Theorem 2.22, the MGF of Z is given by

$$M_Z(t) = e^{-\frac{\mu t}{\sigma}} M_X\left(\frac{t}{\sigma}\right) = e^{-\frac{\mu t}{\sigma}} e^{\frac{\mu t}{\sigma} + \frac{t^2}{2}}$$

Simplifying,

$$M_Z(t) = e^{\frac{t^2}{2}}$$

Expanding, $M_Z(t)$ as a Taylor series, we get

$$M_Z(t) = 1 + \frac{(t^2/2)}{1!} + \frac{(t^2/2)^2}{2!} + \cdots = 1 + \frac{t^2}{2} + \frac{t^4}{8} + \cdots$$

Hence, the first raw moment of Z is

$$\mu'_1 = E(Z) = \text{Coefficient of } \frac{t}{1!} \text{ in } M_Z(t) = 0$$

and the second raw moment of Z is

$$\mu'_2 = E(Z^2) = \text{Coefficient of } \frac{t^2}{2!} \text{ in } M_Z(t) = 1$$

the mean of Z is

$$\mu_Z = E(Z) = 0$$

and the variance of Z is

$$\text{Var}(Z) = E(Z^2) - \mu_Z^2 = 1 - 0 = 1$$

The standard normal variate Z has mean 0 and variance 1.



EXAMPLE 3.86. Find the median of a normal distribution.

(Anna, Nov. 2006)

Solution. Let X be a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$. If M is the median of X , then M satisfies the relation

$$\int_{-\infty}^M f(x) dx = \frac{1}{2}$$

Hence, we have

$$\int_{-\infty}^M \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^M e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \frac{1}{2}$$

and

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx + \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu}^M e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \triangleq I_1 + I_2 = \frac{1}{2} \quad (3.25)$$

Now, we evaluate

$$I_1 = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\mu} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Substituting $u = \frac{x-\mu}{\sqrt{2}\sigma}$, we have

$$I_1 = \frac{1}{\sigma\sqrt{2\pi}} \int_{u=-\infty}^0 e^{-u^2} \sqrt{2}\sigma du = \frac{1}{\sqrt{\pi}} \int_{u=-\infty}^0 e^{-u^2} du$$

Substituting $v = -u$, we have

$$I_1 = \frac{1}{\sqrt{\pi}} \int_{v=0}^{\infty} e^{-v^2} dv = \frac{1}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} = \frac{1}{2}$$

Substituting the value of I_1 in Eq. (3.25), we have

$$I_2 = \int_{\mu}^M e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 0$$

Since the integrand in the above integral takes only positive values, we must have

$$\mu = M$$

Hence, for a normal distribution, $\text{Median}(X) = \text{Mean}(X) = \mu$. ♦

EXAMPLE 3.87. Find the mode of normal distribution.

(Madras, Oct. 2000)

Solution. Let X be a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$. By definition, the probability density of X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for } -\infty < x < \infty$$

We find that

$$f(x) < \frac{1}{\sigma\sqrt{2\pi}} \quad \text{if } x \neq \mu$$

and

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \quad \text{if } x = \mu$$

Since the maximum value of f occurs at $x = \mu$, we find that

$$\text{Mode}(X) = \text{Mean}(X) = \mu$$

and the maximum value of f is given by

$$f_{\max} = \frac{1}{\sigma\sqrt{2\pi}}$$

We note that the same conclusion can also be drawn by using Calculus.

Differentiating the normal density f , we have

$$f'(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} (x-\mu)$$

Differentiating again, we have

$$f''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \left[1 - \frac{(x-\mu)^2}{\sigma^2} \right]$$

Setting $f'(x) = 0$, we obtain $x = \mu$.

Note that

$$f''(\mu) = -\frac{1}{\sigma^3\sqrt{2\pi}} < 0$$

Hence, $x = \mu$ is the mode of the distribution. ♦

EXAMPLE 3.88. Find the points of inflexion of the general normal curve and show that they occur at a distance equal to the standard deviation from the mean ordinate.

Solution. Let X be a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$. A point of inflexion, x^* , for a smooth curve $y = f(x)$ is the point where the curvature changes sign, i.e. the curve changes from concave upwards (positive curvature) to concave downwards (negative curvature) or vice versa. Equivalently, the point of inflexion x^* can be characterized by the conditions $f''(x^*) = 0$ and $f'''(x^*) \neq 0$.

From Example 3.87, we know that

$$f''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \left[1 - \frac{(x-\mu)^2}{\sigma^2} \right]$$

Setting $f''(x) = 0$, we obtain

$$1 - \frac{(x-\mu)^2}{\sigma^2} = 0 \Rightarrow (x-\mu)^2 = \sigma^2$$

Taking square roots, we have

$$x - \mu = \pm\sigma \Rightarrow x = \mu \pm \sigma$$

Note also that

$$f'''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \left[-\frac{2(x-\mu)}{\sigma^2} - \frac{(x-\mu)}{\sigma^2} \left[1 - \frac{(x-\mu)^2}{\sigma^2} \right] \right]$$

Hence, we find that when $x = \mu + \sigma$,

$$f'''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-0.5} \left[-\frac{2}{\sigma} \right] \neq 0$$

Similarly, when $x = \mu - \sigma$,

$$f'''(x) = -\frac{1}{\sigma^3\sqrt{2\pi}} e^{-0.5} \left[\frac{2}{\sigma} \right] \neq 0$$

Hence, the points of inflexion of the general normal distribution are given by

$$x^* = \mu \pm \sigma$$

The points of inflexion for the standard normal curve ($\mu = 0, \sigma = 1$) are illustrated in Figure 3.6. ♦

EXAMPLE 3.89. Show that the mean, median and mode of the normal distribution coincide.

(Madras, April 2000)

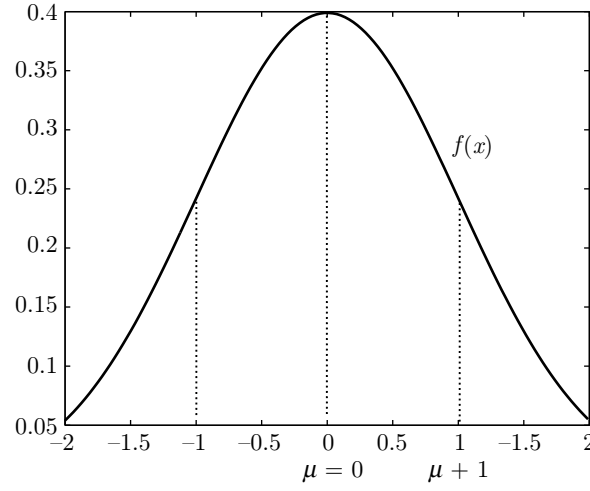


Figure 3.6: Points of inflexion for the standard normal distribution.

Solution. This is an immediate consequence of the results obtained in Examples 3.84, 3.86 and 3.87. ◆

EXAMPLE 3.90. Let X be a normal distribution with parameters μ and σ . Show that

(a) the odd order central moments are zero, i.e.

$$\mu_{2r+1} = E[(X - \mu)^{2r+1}] = 0 \quad \text{for } r = 0, 1, 2, \dots$$

(b) the even order central moments are

$$\mu_{2r} = E[(X - \mu)^{2r}] = 1 \cdot 3 \cdot 5 \cdots (2r - 1) \sigma^{2r} \quad \text{for } r = 1, 2, \dots$$

(Anna, May 2007)

(c) the first four central moments are

$$\mu_1 = 0, \mu_2 = \sigma^2, \mu_3 = 0 \quad \text{and} \quad \mu_4 = 3\sigma^4.$$

(Anna, Nov. 2007)

(d) $\beta_1 = 0, \beta_2 = 3, \gamma_1 = 0$ and $\gamma_2 = 0$.

Solution. First, note that the mean of X is $E(X) = \mu$. Hence, the k th order central moment of X is given by

$$\mu_k = E[(X - E(X))^k] = E[(X - \mu)^k] = \int_{-\infty}^{\infty} (x - \mu)^k f(x) dx$$

(a) The odd order central moments are given by

$$\mu_{2r+1} = \int_{x=-\infty}^{\infty} (x-\mu)^{2r+1} f(x) dx = \int_{x=-\infty}^{\infty} (x-\mu)^{2r+1} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx$$

Substitute $u = \frac{x-\mu}{\sqrt{2}\sigma}$ in the integral. Then $du = \frac{1}{\sqrt{2}\sigma} dx$. The integral becomes

$$\begin{aligned} \mu_{2r+1} &= \int_{u=-\infty}^{\infty} (\sqrt{2}\sigma)^{2r+1} u^{2r+1} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-u^2} \right] \sqrt{2}\sigma du \\ &= \frac{(\sqrt{2}\sigma)^{2r+1}}{\sqrt{\pi}} \int_{u=-\infty}^{\infty} u^{2r+1} e^{-u^2} du = 0 \end{aligned}$$

since the integrand is an odd function of u .

(b) The even order central moments are given by

$$\mu_{2r} = \int_{x=-\infty}^{\infty} (x-\mu)^{2r} f(x) dx = \int_{x=-\infty}^{\infty} (x-\mu)^{2r} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx$$

Substitute $u = \frac{x-\mu}{\sqrt{2}\sigma}$ in the integral. Then $du = \frac{1}{\sqrt{2}\sigma} dx$. The integral becomes

$$\begin{aligned} \mu_{2r} &= \int_{u=-\infty}^{\infty} (\sqrt{2}\sigma)^{2r} u^{2r} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-u^2} \right] \sqrt{2}\sigma du \\ &= \frac{(\sqrt{2}\sigma)^{2r}}{\sqrt{\pi}} \int_{u=-\infty}^{\infty} u^{2r} e^{-u^2} du = \frac{(\sqrt{2}\sigma)^{2r}}{\sqrt{\pi}} 2 \int_{u=0}^{\infty} u^{2r} e^{-u^2} du \end{aligned}$$

since the integrand is an even function of u .

Hence, we have

$$\mu_{2r} = \frac{2(\sqrt{2}\sigma)^{2r}}{\sqrt{\pi}} \int_{u=0}^{\infty} u^{2r} e^{-u^2} du$$

Substitute $v = u^2$ in the above integral. Then $dv = 2u du$ and the integral becomes

$$\mu_{2r} = \frac{2(\sqrt{2}\sigma)^{2r}}{\sqrt{\pi}} \int_{v=0}^{\infty} v^r e^{-v} \frac{dv}{2\sqrt{v}} = \frac{(\sqrt{2}\sigma)^{2r}}{\sqrt{\pi}} \int_{v=0}^{\infty} v^{r-\frac{1}{2}} e^{-v} dv$$

Hence, it is immediate that

$$\mu_{2r} = \frac{(\sqrt{2}\sigma)^{2r}}{\sqrt{\pi}} \Gamma\left(r + \frac{1}{2}\right)$$

Replacing r by $r - 1$ in the above equation, it is immediate that

$$\mu_{2(r-1)} = \frac{(\sqrt{2}\sigma)^{2r-2}}{\sqrt{\pi}} \Gamma\left(r - \frac{1}{2}\right)$$

Note that

$$\Gamma\left(r + \frac{1}{2}\right) = \Gamma\left(\left(r - \frac{1}{2}\right) + 1\right) = \left(r - \frac{1}{2}\right) \Gamma\left(r - \frac{1}{2}\right)$$

Hence, it is immediate that

$$\frac{\mu_{2r}}{\mu_{2(r-1)}} = \left(\sqrt{2}\sigma\right)^2 \left(r - \frac{1}{2}\right)$$

Simplifying, we have

$$\mu_{2r} = (2r - 1)\sigma^2 \mu_{2(r-1)}$$

Hence,

$$\begin{aligned} \mu_{2r} &= (2r - 1)\sigma^2 \cdot (2r - 3)\sigma^2 \cdot \mu_{2(r-2)} = (2r - 1)(2r - 3)(\sigma^2)^2 \mu_{2(r-2)} \\ &\vdots \\ &= (2r - 1)(2r - 3) \cdots 3 \cdot 1 (\sigma^2)^r \mu_0 \end{aligned}$$

Since $\mu_0 = 1$, it follows that

$$\mu_{2r} = 1 \cdot 3 \cdot 5 \cdots (2r - 1) \sigma^{2r} \quad \text{for } r = 0, 1, 2, \dots$$

(c) By (a), it follows that $\mu_1 = 0, \mu_3 = 0$. By (b), it follows that $\mu_2 = \sigma^2$ and $\mu_4 = 3\sigma^4$.

(d) By (c), we have

$$\beta_1 = \frac{\mu_3}{\mu_2^{3/2}} = 0 \quad \text{and} \quad \beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\sigma^4}{\sigma^4} = 3$$

Hence, it follows that

$$\gamma_1 = \text{Skewness} = \sqrt{\beta_1} = 0 \quad \text{and} \quad \gamma_2 = \text{Kurtosis} = \beta_2 - 3 = 0$$

◆

EXAMPLE 3.91. For a certain normal distribution, the first moment about 10 is 40 and the fourth moment about 50 is 48. Find the mean and S.D. of the distribution.

(Madras, April 2000).

Solution. Given that (a) $E(X - 10) = 40$ and (b) $E[(X - 50)^4] = 48$.

By Example 3.90, from (a), it is immediate that $\mu = E(X) = 10 + 40 = 50$. Hence, $\mu = 50$

From (b), we have

$$E[(X - 50)^4] = \mu_4 = 48$$

By Example 3.90, we know that

$$\mu_4 = 3\sigma^4$$

Hence, it follows that

$$3\sigma^4 = 48 \Rightarrow \sigma^4 = 16 \Rightarrow \sigma = 2$$

i.e., $\sigma = \text{S.D.} = 2$

◆

EXAMPLE 3.92. In a normal distribution $N(\mu, \sigma^2)$, prove that

$$\mu_{2r+2} = \sigma^2 \mu_{2r} + \sigma^3 \frac{d\mu_{2r}}{d\sigma} \quad (\text{Madras, May 2002})$$

Solution. By Example 3.90, we know that

$$\mu_{2r} = 1 \cdot 3 \cdots (2r-1) \sigma^{2r} \quad (3.26)$$

from which it follows that

$$\frac{d\mu_{2r}}{d\sigma} = 1 \cdot 3 \cdots (2r-1) \cdot (2r) \sigma^{2r-1}$$

Hence, we find that

$$\sigma^3 \frac{d\mu_{2r}}{d\sigma} = 1 \cdot 3 \cdots (2r-1) \cdot (2r) \sigma^{2(r+1)}$$

From Eq. (3.26), we also obtain

$$\sigma^2 \mu_{2r} = 1 \cdot 3 \cdots (2r-1) \sigma^{2(r+1)}$$

Hence, it is immediate that

$$\sigma^3 \frac{d\mu_{2r}}{d\sigma} + \sigma^2 \mu_{2r} = 1 \cdot 3 \cdots (2r-1) \cdot (2r+1) \sigma^{2(r+1)} = \mu_{2r+2}$$

◆

EXAMPLE 3.93. Find the mean deviation from the mean for the normal distribution.

(Madras, March 1997)

Solution. Let X be a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$. We know that the mean of X is $E(X) = \mu$. Hence, by definition, the mean deviation about the mean is given by

$$\begin{aligned} \text{M.D.} &= E|X - \mu| = \int_{x=-\infty}^{\infty} |x - \mu| f(x) dx \\ &= \int_{x=-\infty}^{\infty} |x - \mu| \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx \end{aligned}$$

Substitute $u = \frac{x-\mu}{\sqrt{2}\sigma}$ in the integral. Then $du = \frac{1}{\sqrt{2}\sigma} dx$. The integral becomes

$$\begin{aligned} \text{M.D.} &= \int_{u=-\infty}^{\infty} \sqrt{2}\sigma |u| \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-u^2} \right] \sqrt{2}\sigma du \\ &= \sqrt{\frac{2}{\pi}} \sigma \int_{u=-\infty}^{\infty} |u| e^{-u^2} du = \sqrt{\frac{2}{\pi}} \sigma \cdot 2 \int_{u=0}^{\infty} u e^{-u^2} du \end{aligned}$$

since the integrand is an even function of u .

Note that

$$\text{M.D.} = \sqrt{\frac{2}{\pi}} \sigma \int_{u=0}^{\infty} d(-e^{-u^2})$$

Integrating, we have

$$\text{M.D.} = \sqrt{\frac{2}{\pi}} \sigma [-e^{-u^2}]_0^{\infty} = \sqrt{\frac{2}{\pi}} \sigma$$

Hence, the mean deviation about the mean is given by

$$\text{M.D.} = \sqrt{\frac{2}{\pi}} \sigma = 0.7979\sigma \approx \frac{4}{5}\sigma$$

◆

EXAMPLE 3.94. Find the quartile deviation for the normal distribution.

Solution. Let X be a normal random variable with parameters μ and σ , i.e. $X \sim N(\mu, \sigma^2)$. Let Q_1 and Q_3 be the lower and upper quartiles of X , respectively. Then, by definition, we have

$$P(X < Q_1) = \frac{1}{4} \text{ and } P(X < Q_3) = \frac{3}{4}$$

The points $x = Q_1$ and $x = Q_3$ are located as shown in Figure 3.7.

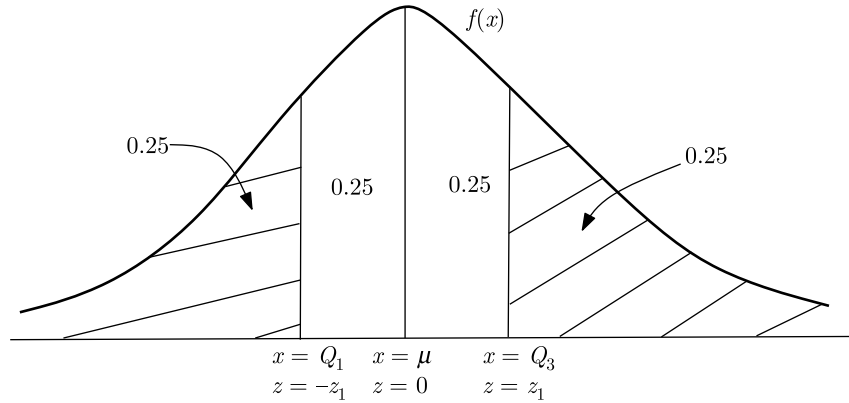


Figure 3.7: Quartiles of the normal distribution.

Define the standard normal variate $Z = \frac{X - \mu}{\sigma}$. Since $X = \mu$ is the median of X and the normal distribution is symmetrical about the line $X = \mu$, it is clear that Q_1 and Q_3 are symmetrically located to the left and right of the line $x = \mu$.

When $X = Q_3$, $Z = \frac{Q_3 - \mu}{\sigma} = z_1$ (say).

When $X = Q_1$, $Z = \frac{Q_1 - \mu}{\sigma} = -z_1$ (by symmetry).

Subtracting the second equation from the first, we get

$$\frac{Q_3 - Q_1}{\sigma} = 2z_1$$

Hence, the quartile deviation of the normal distribution is given by

$$\text{Q.D.} = \frac{Q_3 - Q_1}{2} = z_1 \sigma$$

From the Table 3.1, we find that

$$P(0 < Z < z_1) = 0.25 \Rightarrow z_1 = 0.675 \approx \frac{2}{3}$$

Hence, the quartiles are given by

$$Q_3 = \mu + z_1 \sigma = \mu + 0.675 \sigma$$

$$Q_1 = \mu - z_1 \sigma = \mu - 0.675 \sigma$$

and the quartile deviation is given by

$$\text{Q.D.} = 0.675 \sigma \approx \frac{2}{3} \sigma$$

◆

EXAMPLE 3.95. Prove that for the normal distribution, the quartile deviation, the mean and the standard deviations are approximately 10 : 12 : 15.

Solution. By Examples 3.84, 3.93 and 3.94, it follows that

$$\text{Q.D.} : \text{M.D.} : \text{S.D.} :: \frac{2}{3} \sigma : \frac{4}{5} \sigma : \sigma :: 10 : 12 : 15.$$

◆

EXAMPLE 3.96. A distribution is known to be normal. The quartiles are 8.64 and 14.32. Calculate the mean and the standard deviation of the distribution.

Solution. If Q_1 and Q_3 are the lower and upper quartiles of the normal distribution, then we have

$$Q_3 = \mu + 0.675 \sigma = 14.32$$

$$Q_1 = \mu - 0.675 \sigma = 8.64$$

Adding the two equations, we get

$$2\mu = 22.96 \Rightarrow \mu = 11.48$$

Substituting the value of μ in the equation for Q_3 , we have

$$\sigma = \frac{14.32 - \mu}{0.675} = \frac{2.84}{0.675} = 4.2074$$

Hence, the mean, $\mu = 11.48$ and the standard deviation, $\sigma = 4.2074$.

◆

EXAMPLE 3.97. An electrical firm manufactures light bulbs that have life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns

Table 3.1: Areas under the standard normal curve*

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2612	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.7	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.9	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
4.0	.49997									

*The table provides the area, $P(0 < Z < z)$, for the standard normal curve $\phi(z)$.

(i) more than 834 hours.

(ii) between 778 and 834 hours.

(Anna, Nov. 2006)

Solution. Let X be the life of the light bulb manufactured by the electrical firm. Then X is normally distributed with mean $\mu = 800$ hours and standard deviation $\sigma = 40$ hours.

The standard normal variate is defined by

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 800}{40}$$

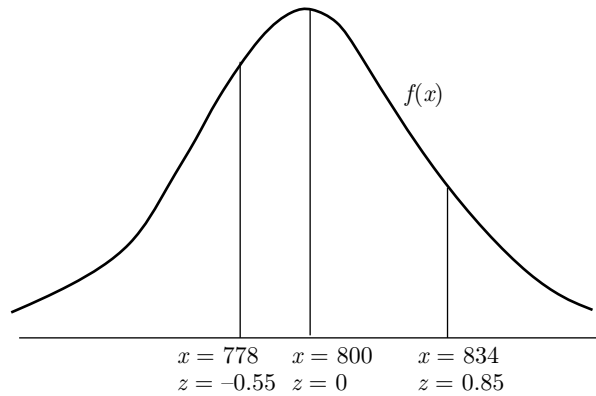


Figure 3.8: Normal distribution.

(i) The probability that a bulb burns more than 834 hours is given by $P(X > 834)$.

When $X = 834$, $Z = \frac{834-800}{40} = \frac{34}{40} = 0.85$. Hence, we have

$$\begin{aligned} P(X > 834) &= P(Z > 0.85) = 0.5 - P(0 < Z < 0.85) \\ &= 0.5 - 0.3023 = 0.1977 \end{aligned}$$

(ii) The probability that a bulb burns between 778 and 834 hours is given by $P(778 < X < 834)$.

When $X = 778$, $Z = \frac{778-800}{40} = -\frac{22}{40} = -0.55$. Hence, we have

$$\begin{aligned} P(778 < X < 834) &= P(-0.55 < Z < 0.85) \\ &= P(-0.55 < Z < 0) + P(0 < Z < 0.85) \end{aligned}$$

Since the standard normal curve is symmetrical about the line $z = 0$ (see Figure 3.8), it follows that

$$\begin{aligned} P(778 < X < 834) &= P(0 < Z < 0.55) + P(0 < Z < 0.85) \\ &= 0.2088 + 0.3023 = 0.5111 \end{aligned}$$

◆

EXAMPLE 3.98. The savings bank account of a customer showed an average balance of Rs. 150 and a standard deviation of Rs. 50. Assuming that the account balances are normally distributed, find what percentage of account is

- (i) over Rs. 200.
- (ii) between Rs. 120 and Rs. 170.
- (iii) less than Rs. 75.

(Anna, Nov. 2006)

Solution. Let X be the balance of the savings bank account of a customer. Given that X is normally distributed with mean $\mu = 150$ and standard deviation $\sigma = 50$.

The standard normal variate is defined by

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 150}{50}$$

- (i) The probability that the account balance is over Rs. 200 is given by $P(X > 200)$.

When $X = 200$, $Z = \frac{200-150}{50} = \frac{50}{50} = 1$. Hence, we have

$$\begin{aligned} P(X > 200) &= P(Z > 1) = 0.5 - P(0 < Z < 1) \\ &= 0.5 - 0.3413 = 0.1587 \end{aligned}$$

So, the percentage of account balance over Rs. 200 is 15.87%.

- (ii) The probability that the account balance is between Rs. 120 and Rs. 170 is given by $P(120 < X < 170)$.

When $X = 120$, $Z = \frac{120-150}{50} = -\frac{30}{50} = -0.6$.

When $X = 170$, $Z = \frac{170-150}{50} = \frac{20}{50} = 0.4$.

Hence, we have

$$\begin{aligned} P(120 < X < 170) &= P(-0.6 < Z < 0.4) \\ &= P(-0.6 < Z < 0) + P(0 < Z < 0.4) \end{aligned}$$

Since the standard normal distribution is symmetrical about the line $Z = 0$, we have

$$\begin{aligned} P(120 < X < 170) &= P(0 < Z < 0.6) + P(0 < Z < 0.4) \\ &= 0.2257 + 0.1554 = 0.3811 \end{aligned}$$

Hence, the percentage of account balance between Rs. 120 and Rs. 170 is 38.11%

- (iii) The probability that the account balance is less than Rs. 75 is given by $P(X < 75)$.

When $X = 75$, $Z = \frac{75-150}{50} = -\frac{75}{50} = -1.5$. Hence, we have

$$P(X < 75) = P(Z < -1.5) = 0.5 - P(-1.5 < Z < 0)$$

Since the standard normal distribution is symmetrical about the line $Z = 0$, we have

$$P(X < 75) = 0.5 - P(0 < Z < 1.5) = 0.5 - 0.4332 = 0.0668$$

Hence, the percentage of account balance less than Rs. 75 is 6.68%.



EXAMPLE 3.99. The marks obtained by a number of students in a certain subject are approximately normally distributed with mean 65 and standard deviation 5. If 3 students are selected at random from this group, what is the probability that at least one of them would have scored above 75?

(Anna, May 2007)

Solution. Let X be the marks obtained by a student in a subject. It is given that X is normally distributed with mean $\mu = 65$ and standard deviation 5. Define the standard normal variate by

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 65}{5}$$

Then, the probability that a student would have scored above 75 is given by

$$p = P(X > 75) = P(Z > 2) = 0.5 - P(0 < Z < 2) = 0.5 - 0.4772 = 0.0228$$

Since 3 students are selected at random from the given group of students, we have $n = 3$. If we consider the event of a student scoring above 75 as a "success", then it is clear that X follows a binomial distribution with parameters $n = 3$ and $p = 0.0228$, i.e.

$$P(X = x) = {}^3C_x p^x q^{3-x}, \text{ where } x = 0, 1, 2, 3$$

Hence, the required probability that out of the 3 students selected at random, at least one student would have scored above 75 is given by

$$P(X \geq 1) = 1 - P(X = 0) = 1 - q^3 = 1 - 0.9772^3 = 0.0669$$

◆

EXAMPLE 3.100. The weekly wages of 1000 workmen are normally distributed around a mean of Rs. 500 with a standard deviation of Rs. 50. Estimate the number of workers, whose weekly wages will be

- (i) between Rs. 400 and Rs. 600.
- (ii) less than Rs. 400.
- (iii) more than Rs. 600.

Solution. Let X be the weekly wage of worker. Given that $X \sim N(\mu, \sigma^2)$, where $\mu = 500$ and $\sigma = 50$.

The standard normal variate is defined by

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 500}{50}$$

- (i) First, we find $P(400 < X < 600)$.

$$\text{When } X = 400, Z = \frac{400 - 500}{50} = -\frac{100}{50} = -2.$$

$$\text{When } X = 600, Z = \frac{600 - 500}{50} = \frac{100}{50} = 2.$$

Hence, we have

$$P(400 < X < 600) = P(-2 < Z < 2) = 2P(0 < Z < 2) = 2(0.4772) = 0.9544$$

Out of 1000 workmen, the expected number of workers, whose weekly wages will be between Rs. 400 and Rs. 600, is given by

$$NP(400 < X < 600) = 1000 \times 0.9544 = 954.4 \approx 954$$

(ii) Here, we find $P(X < 400)$. Note that

$$\begin{aligned} P(X < 400) &= P(Z < -2) = P(Z > 2) \quad (\text{by symmetry}) \\ &= 0.5 - P(0 < Z < 2) \\ &= 0.5 - 0.4772 = 0.0228 \end{aligned}$$

Out of 1000 workmen, the expected number of workers, whose weekly wages will be less than Rs. 400 is given by

$$NP(X < 400) = 1000 \times 0.0228 = 22.8 \approx 23$$

(iii) Here, we find $P(X > 600)$. Note that

$$P(X > 600) = P(Z > 2) = 0.0228 \quad (\text{by (ii)})$$

Out of 1000 workmen, the expected number of workers, whose weekly wages will be more than Rs. 600 is given by

$$NP(X > 600) = 1000 \times 0.0228 = 22.8 \approx 23$$

◆

EXAMPLE 3.101. In an intelligence test administered on 1000 children, the average was 60 and the standard deviation was 20. Assuming that the marks obtained by the children follow a normal distribution, find the number of children who have scored (i) over 90 marks, (ii) below 40 marks, and (iii) between 50 and 80 marks.

Solution. Let X be the marks obtained by the children in the intelligence test.

Given that $X \sim N(\mu, \sigma^2)$ with $\mu = 60$ and $\sigma = 20$.

The standard normal variate is defined by

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 60}{20}$$

(i) First, we find $P(X > 90)$.

$$\text{When } X = 90, Z = \frac{90-60}{20} = \frac{30}{20} = 1.5.$$

Hence, we have

$$\begin{aligned} P(X > 90) &= P(Z > 1.5) = 0.5 - P(0 < Z < 1.5) \\ &= 0.5 - 0.4332 = 0.0668 \end{aligned}$$

Out of 1000 children, the expected number of children who had scored above 90 marks is given by

$$NP(X > 90) = 1000 \times 0.0668 = 66.8 \approx 67$$

(ii) Here, we find $P(X < 40)$.

$$\text{When } X = 40, Z = \frac{40-60}{20} = -\frac{20}{20} = -1.$$

Hence, we have

$$\begin{aligned} P(X < 40) &= P(Z < -1) = P(Z > 1) \quad (\text{by symmetry}) \\ &= 0.5 - P(0 < Z < 1) \\ &= 0.5 - 0.3413 = 0.1587 \end{aligned}$$

Out of 1000 children, the expected number of children who had scored below 40 marks is given by

$$NP(X < 40) = 1000 \times 0.1587 = 158.7 \approx 159$$

(iii) Here, we find $P(50 < X < 80)$.

$$\text{When } X = 50, Z = \frac{50-60}{20} = -\frac{10}{20} = -0.5.$$

$$\text{When } X = 80, Z = \frac{80-60}{20} = \frac{20}{20} = 1.$$

Thus, we have

$$\begin{aligned} P(50 < X < 80) &= P(-0.5 < Z < 1) \\ &= P(-0.5 < Z < 0) + P(0 < Z < 1) \\ &= P(0 < Z < 0.5) + P(0 < Z < 1) \quad (\text{by symmetry}) \\ &= 0.1915 + 0.3413 = 0.5328 \end{aligned}$$

Out of 1000 children, the expected number of children who had scored between 50 and 80 marks is given by

$$NP(50 < X < 80) = 1000 \times 0.5328 = 532.8 \approx 533$$

◆

EXAMPLE 3.102. In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and the standard deviation of the distribution.

Solution. Let $X \sim N(\mu, \sigma^2)$. We are given that

$$(i) \quad P(X > 64) = 0.08.$$

$$(ii) \quad P(X < 45) = 0.31.$$

We define the standard normal variate as

$$Z = \frac{X - \mu}{\sigma}$$

When $X = 64$, let $Z = z_1$. Then

$$\frac{64 - \mu}{\sigma} = z_1 \quad \text{or} \quad 64 - \mu = z_1 \sigma$$

When $X = 45$, let $Z = -z_2$ (say).

(Since $P(X < 45) = 0.31 < 0.5$, it is clear that the corresponding value of Z is negative.)

Then

$$\frac{45 - \mu}{\sigma} = -z_2 \text{ or } 45 - \mu = -z_2 \sigma$$

From Figure 3.9, it is clear that

$$P(0 < Z < z_1) = 0.42 \Rightarrow z_1 = 1.4$$

and

$$P(-z_2 < Z < 0) = 0.31 \Rightarrow P(0 < Z < z_2) = 0.31 \Rightarrow z_2 = 0.45$$

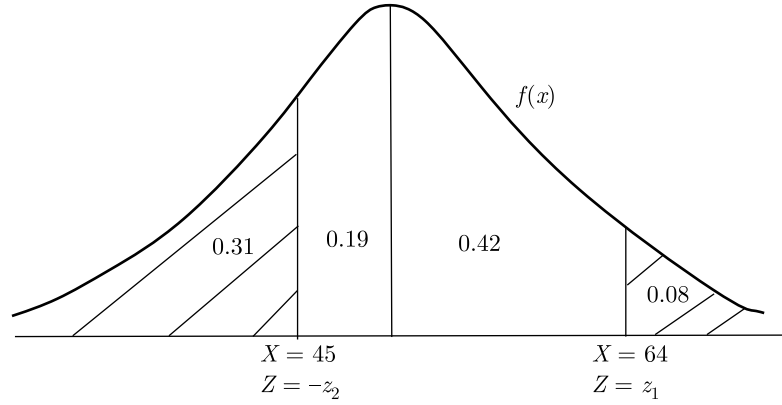


Figure 3.9: Normal distribution

Hence, we obtain the equations

$$64 - \mu = 1.4\sigma \quad (3.27)$$

and

$$45 - \mu = -0.5\sigma \quad (3.28)$$

Subtracting Eq. (3.28) from Eq. (3.27), we get

$$19 = 1.9\sigma \Rightarrow \sigma = 10$$

Substituting the value of σ in Eq. (3.27), we get

$$\mu = 64 - 14 = 50.$$

Hence, the mean, $\mu = 50$ and the standard deviation, $\sigma = 10$. ◆

PROBLEM SET 3.9

1. For a certain normal distribution, the first moment about 5 is 15 and the fourth moment about 20 is 243. Find the mean and the standard deviation of the distribution.
2. For a certain normal distribution, the lower and upper quartiles are 3.5 and 12.5, respectively. Find the mean and the standard deviation of the distribution.

3. If X is normally distributed with mean 20 and standard deviation 5, find the probabilities of the following: (a) $X \geq 35$, (b) $X \leq 30$, (c) $10 \leq X \leq 40$, and (d) $|X - 20| > 10$.
4. If X is normally distributed with mean 25 and variance 36, find
 - (a) $P(13 < X < 43)$.
 - (b) $P(X < 43|X > 13)$.
5. In an examination, 30% of the candidates obtained marks below 40 and 10% of the candidates got above 75 marks. Assuming that the marks are normally distributed, find the mean and the standard deviation of the distribution.
6. In a class quiz, 8% of the students obtained marks below 25 and 90% of the students got below 85 marks. Assuming that the marks are normally distributed, find the mean and the standard deviation of the distribution.
7. The life of a certain kind of electronic device is normally distributed with a mean of 400 hours and a standard deviation of 50 hours. Find the probability that the device will have a lifetime
 - (a) more than 500 hours.
 - (b) between 250 and 450 hours.
 - (c) less than 300 hours.
8. The savings bank account of a customer showed an average balance of Rs. 1000 and a standard deviation of Rs. 500. Assuming that the account balances are normally distributed, find what percentage of account is
 - (a) is over Rs. 2600.
 - (b) between Rs. 800 and Rs. 2000.
 - (c) less than Rs. 500.
9. The marks obtained by students in an intelligence test were found to be normally distributed with mean 45 and standard deviation 25. Find what percentage of students scored marks
 - (a) more than 80.
 - (b) between 30 and 70.
 - (c) below 35.
10. The heights of individuals of a college are normally distributed with mean 160 cms and standard deviation 24 cms. Determine the probability that a randomly selected person is (a) less than 150 cms, (b) over 6 feet, and (c) between 150 and 170 cms.
11. In a sample of 1000 items, the mean weight is 50 kgs with a standard deviation of 20 kgs. Assuming that the distribution is normal, find the number of items weighing (a) between 40 and 70 kgs, (b) over 80 kgs., and (c) under 30 kgs.
12. In a test on 2500 electric bulbs, it was found that the life of a particular make was normally distributed with a mean life of 2000 hours and a standard deviation of 400 hours. Estimate the number of bulbs likely to burn (a) for more than 3000 hours, (b) for less than 1500 hours and (c) between 1800 and 2300 hours.
13. In a science test administered on 1200 children, the average was 70 and the standard deviation was 15. Assuming that the marks obtained by the children follow a normal distribution, find the number of children who had scored (a) over 90 marks, (b) below 35 marks, and (c) between 40 and 80 marks.

14. The weekly wages of 1000 workmen are normally distributed with a mean of Rs. 500 and a standard deviation of Rs. 150. Estimate the number of workers whose weekly wages will lie (a) over Rs. 800, (b) less than Rs. 300 and (c) between Rs. 400 and Rs. 700.
15. The heights of individuals in a college are normally distributed with a mean height of 65 inches and standard deviation of 10 inches. In a group of 500 individuals randomly selected, find the number of persons with heights (a) over 6 feet, (b) under 5 feet, and (c) between 60 and 70 inches.

3.12 FUNCTIONS OF A RANDOM VARIABLE

If X is a random variable (discrete or continuous) with probability density function f , and if $v : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function, then

$$Y = v(X)$$

also is a random variable. In this section, we illustrate the methods for finding the probability density function g for Y . We discuss two important methods for finding the PDF g of Y , namely, *distribution function method*, and *transformation method*.

Method I: Distribution Function Method

Let X be a continuous random variable with probability density function f and cumulative distribution function F . Let $Y = v(X)$ be a random variable, where $v : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function. A fairly straightforward method of finding the probability function g for Y is to first find the cumulative distribution function

$$G(y) = P(Y \leq y) = P(v(x) \leq y)$$

and then find the probability density function g of Y by direct differentiation:

$$g(y) = \frac{d}{dy} G(y)$$

EXAMPLE 3.103. If X is a random variable with the uniform density

$$f(x) = \begin{cases} 1 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

and $Y = \sqrt{X}$, find (i) the distribution function of Y and (ii) the probability density of Y .

Solution. (i) First, we calculate the distribution function F of X , where $F(x) = P(X \leq x)$. If $x < 0$, then $F(x) = 0$. If $0 \leq x \leq 1$, then we see that

$$F(x) = \int_{-\infty}^x f(x) dx = \int_0^x 1 dx = x$$

Also, if $x \geq 1$, then

$$F(x) = \int_{-\infty}^x f(x) dx = \int_0^1 1 dx = 1$$

Hence, the distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

Next, let G be the distribution function of Y , i.e.

$$G(y) = P(Y \leq y) = P(\sqrt{X} \leq y)$$

If $y < 0$, then it is clear that $G(y) = 0$. If $0 \leq y \leq 1$, then

$$G(y) = P(\sqrt{X} \leq y) = P(X \leq y^2) = F(y^2) = y^2$$

Also, if $y \geq 1$, then

$$G(y) = P(\sqrt{X} \leq y) = P(X \leq y^2) = F(y^2) = 1$$

Hence, the distribution function of Y is given by

$$G(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ y^2 & \text{if } 0 \leq y \leq 1 \\ 1 & \text{if } y \geq 1 \end{cases}$$

(ii) The probability density g of Y is obtained by differentiating G , i.e.

$$g(y) = G'(y) = \begin{cases} 2y & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 3.104. If X is a random variable with the exponential density

$$f(x) = \begin{cases} \theta e^{-\theta x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

and $Y = X^2$, find (i) the distribution function of Y and (ii) the probability density of Y .

Solution. (i) First, we calculate the distribution function F of X , i.e. $F(x) = P(X \leq x)$. If $x \leq 0$, then $F(x) = 0$. If $x \geq 0$, then we have

$$F(x) = \int_{-\infty}^x f(x) dx = \int_0^x \theta e^{-\theta x} dx = \left[-e^{-\theta x} \right]_0^x = 1 - e^{-\theta x}$$

Hence, the distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-\theta x} & \text{if } x \geq 0 \end{cases}$$

Let G be the distribution function of Y , i.e.

$$G(y) = P(Y \leq y) = P(X^2 \leq y)$$

If $y \leq 0$, then it is clear that $G(y) = 0$. If $y > 0$, then

$$G(y) = P(X^2 \leq y) = P(|X| \leq \sqrt{y}) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = F(\sqrt{y}) - F(-\sqrt{y})$$

Since $F(-\sqrt{y}) = 0$, it is immediate that

$$G(y) = F(\sqrt{y}) = 1 - e^{-\theta\sqrt{y}}$$

Hence, the distribution function of Y is given by

$$G(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ 1 - e^{-\theta\sqrt{y}} & \text{if } y > 0 \end{cases}$$

(ii) The probability density function of Y is obtained by differentiating G , i.e.

$$g(y) = G'(y) = \begin{cases} \frac{\theta}{2\sqrt{y}} e^{-\theta\sqrt{y}} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 3.105. If X is uniformly distributed in $(-\frac{\pi}{2}, \frac{\pi}{2})$, find the probability density function of $y = \tan x$.

(Anna, Nov. 2003)

Solution. Since X has the uniform density in $(-\frac{\pi}{2}, \frac{\pi}{2})$, we have

$$f(x) = \begin{cases} \frac{1}{\pi} & \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$

It is easy to see that the distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq -\frac{\pi}{2} \\ \frac{1}{\pi} [x + \frac{\pi}{2}] & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ 1 & \text{if } x \geq \frac{\pi}{2} \end{cases}$$

Since $y = \tan x$ and x takes values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, it is clear that y takes all the values in $(-\infty, \infty)$.

Let G be the distribution function of $Y = \tan X$. Then we have

$$G(y) = P(Y \leq y) = P(\tan X \leq y) = P(X \leq \tan^{-1} y)$$

(since $y = \tan x$ is non-decreasing).

Hence, it follows that

$$G(y) = F(\tan^{-1} y) = \frac{1}{\pi} \left[\tan^{-1} y + \frac{\pi}{2} \right]$$

So, the probability density function of Y is obtained as

$$g(y) = G'(y) = \frac{1}{\pi} \frac{1}{1+y^2} \quad -\infty < y < \infty$$

i.e. Y follows the Cauchy distribution. ◆

EXAMPLE 3.106. If X is a continuous random variable with probability density function f and if $Y = |X|$, show that the probability density function g of Y is given by

$$g(y) = \begin{cases} f(y) + f(-y) & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Also, use this result to show that the probability density of $Y = |X|$ is exponentially distributed with parameter 1 when X has the Laplace distribution

$$f(x) = \frac{1}{2} e^{-|x|} \quad \text{for } -\infty < x < \infty$$

Solution. Let G be the distribution function of $Y = |X|$. Then, we have

$$G(y) = P(Y \leq y) = P(|X| \leq y)$$

If $y \leq 0$, then it is clear that $G(y) = 0$. If $y > 0$, then

$$G(y) = P(|X| \leq y) = P(-y \leq X \leq y) = F(y) - F(-y)$$

Thus, the distribution function of Y is given by

$$G(y) = \begin{cases} F(y) - F(-y) & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Hence, the probability density function of Y is obtained by differentiating G , i.e.

$$g(y) = G'(y) = \begin{cases} f(y) + f(-y) & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases} \quad (3.29)$$

Next, let X be a random variable with the Laplace transformation

$$f(x) = \frac{1}{2} e^{-|x|} \quad \text{for } -\infty < x < \infty$$

and let $Y = |X|$. Then the probability density function of Y is given by the formula (3.29). That is, $g(y) = 0$ for $y \leq 0$ and if $y > 0$, then

$$g(y) = f(y) + f(-y) = \frac{1}{2} e^{-|y|} + \frac{1}{2} e^{-|y|} = e^{-y}$$

Hence, the probability density of Y is given by

$$g(y) = \begin{cases} e^{-y} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

i.e. Y is exponentially distributed with parameter 1. ♦

EXAMPLE 3.107. If X is a Gaussian random variable with mean zero and variance σ^2 , find the probability density function of $Y = |X|$. (Anna, Nov. 2007)

Solution. Since X is a normal (Gaussian) random variable with mean zero and variance σ^2 , the probability density function of X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} \quad \text{for } -\infty < x < \infty$$

By Example 3.106, the probability density function of $Y = |X|$ is given by

$$g(y) = \begin{cases} f(y) + f(-y) & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Substituting and simplifying, we have

$$g(y) = \begin{cases} \frac{1}{\sigma} \sqrt{\frac{2}{\pi}} e^{-\frac{y^2}{2\sigma^2}} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases} \quad \diamond$$

EXAMPLE 3.108. (i) If X is a continuous random variable with probability density function f and $Y = aX + b$, where a and b are constants with $a \neq 0$, show that the probability density function g of Y can be expressed as

$$g(y) = \frac{1}{|a|} f\left(\frac{y-b}{a}\right)$$

(ii) Using (i), show that if X is uniformly distributed in $(-k, k)$ and $Y = aX + b$, then Y is also uniformly distributed. Find the PDF of Y .

Solution. (i) Let F and G be the distribution functions of X and Y , respectively. Since $a \neq 0$, we have two cases to consider, viz. $a > 0$ and $a < 0$.

First, let $a > 0$. Then for any y , we have

$$\begin{aligned} G(y) &= P(Y \leq y) = P(aX + b \leq y) = P(aX \leq y - b) = P\left(X \leq \frac{y-b}{a}\right) \\ &= F\left(\frac{y-b}{a}\right) \end{aligned}$$

Hence, the probability density function g of Y can be obtained as

$$g(y) = G'(y) = \frac{1}{a} f\left(\frac{y-b}{a}\right) \quad (\text{since } F'(x) = f(x))$$

Next, let $a < 0$. Then for any y , we have

$$\begin{aligned} G(y) &= P(Y \leq y) = P(aX + b \leq y) = P(aX \leq y - b) = P\left(X \geq \frac{y-b}{a}\right) \\ &= 1 - P\left(X \leq \frac{y-b}{a}\right) = 1 - F\left(\frac{y-b}{a}\right) \end{aligned}$$

Hence, the probability density function g of Y can be obtained as

$$g(y) = G'(y) = -\frac{1}{a} f\left(\frac{y-b}{a}\right) \quad (\text{since } F'(x) = f(x))$$

Combining the two cases, for any $a \neq 0$, the probability density function of Y is given by

$$g(y) = \frac{1}{|a|} f\left(\frac{y-b}{a}\right)$$

(ii) Suppose that X is uniformly distributed in $(-k, k)$, i.e. X has the uniform density

$$f(x) = \begin{cases} \frac{1}{2k} & \text{for } -k < x < k \\ 0 & \text{elsewhere} \end{cases}$$

By (i), the PDF of $Y = aX + b$ is given by

$$g(y) = \frac{1}{|a|} f\left(\frac{y-b}{a}\right)$$

Note that

$$-k < x < k \iff -k < \frac{y-b}{a} < k \iff b - ak < y < b + ak$$

Hence, it is immediate that

$$g(y) = \begin{cases} \frac{1}{|a|} \times \frac{1}{2k} & \text{for } b - ak < y < b + ak \\ 0 & \text{elsewhere} \end{cases}$$

Thus, we conclude that Y is uniformly distributed in $(b - ak, b + ak)$. ♦

EXAMPLE 3.109. Let X be a random variable with the PDF given by

$$f(x) = \begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the PDF of $Y = 3X + 1$.

(Anna, April 2000; Nov. 2007)

Solution. By Example 3.108, the probability density function of $Y = aX + b$ is given by

$$g(y) = \frac{1}{|a|} f\left(\frac{y-b}{a}\right)$$

Here, $a = 3$ and $b = 1$. Note also that

$$0 < x < 1 \iff 0 < \frac{y-1}{3} < 1 \iff 0 < y-1 < 3 \iff 1 < y < 4$$

Thus, for $1 < y < 4$, we have

$$g(y) = \frac{1}{3} f\left(\frac{y-1}{3}\right) = \frac{1}{3} 2^{\frac{y-1}{3}} = \frac{2}{9} (y-1)$$

Hence, the probability density function of Y is given by

$$g(y) = \begin{cases} \frac{2}{9} (y-1) & \text{for } 1 < y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

♦

EXAMPLE 3.110. If X is a continuous random variable with probability density function f and if $Y = X^2$, show that the probability density function g of Y is given by

$$g(y) = \begin{cases} \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Also, use this result to obtain the probability density of $Y = X^2$ if X has the probability density function

$$f(x) = \begin{cases} \frac{1}{8}(x+2) & \text{for } -2 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Solution. Let G be the distribution function of $Y = X^2$. Then we have

$$G(y) = P(Y \leq y) = P(X^2 \leq y)$$

Therefore, if $y \leq 0$, then it is clear that $G(y) = 0$. If $y > 0$, then

$$G(y) = P(X^2 \leq y) = P(|X| \leq \sqrt{y}) = P(-\sqrt{y} < X < \sqrt{y})$$

Thus, it follows that

$$G(y) = F(\sqrt{y}) - F(-\sqrt{y})$$

Hence, the distribution function of Y is obtained as

$$G(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ F(\sqrt{y}) - F(-\sqrt{y}) & \text{if } y > 0 \end{cases}$$

and, the probability density function g of Y is obtained as

$$g(y) = G'(y) = \begin{cases} \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases} \quad (3.30)$$

Next, let X be the random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{8}(x+2) & \text{for } -2 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

and $Y = X^2$. Then, the probability density function of Y is as given by Eq. (3.30).

A simple calculation gives

$$g(y) = \begin{cases} \frac{1}{4\sqrt{y}} & \text{for } 0 < y < 4 \\ 0 & \text{elsewhere} \end{cases}$$

(Note that $-2 < x < 2 \Rightarrow |x| < 2 \Rightarrow 0 < y = x^2 < 4$.)

◆

EXAMPLE 3.111. If $Y = X^2$, where X is a Gaussian random variable with zero mean and variance σ^2 , find the probability density function of the random variable Y .

(Anna, May 2007)

Solution. Given that X follows a normal distribution with zero mean and variance σ^2 . Hence, the probability density function of X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} \quad \text{for } -\infty < x < \infty$$

By Example 3.110, the probability density function of $Y = X^2$ is given by the formula

$$g(y) = \begin{cases} \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Substituting the probability density function of Y is given by

$$g(y) = \frac{1}{2\sqrt{y}} \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{y}{2\sigma^2}} + \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{y}{2\sigma^2}} \right] = \frac{1}{\sigma\sqrt{2\pi y}} e^{-\frac{y}{2\sigma^2}} \quad \text{for } y > 0$$

◆

EXAMPLE 3.112. Let $Y = X^2$. Find the PDF of Y if X is a uniform random variable over $(-1, 2)$.

(Anna, Model 2003)

Solution. Given that X has the uniform density over $(-1, 2)$, i.e.

$$f(x) = \begin{cases} \frac{1}{3} & \text{for } -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Since $Y = X^2$, by Example 3.110, the PDF of Y is given by the formula

$$g(y) = \begin{cases} \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Since $-1 < x < 4$, $y = x^2$ takes values from 0 to 4.

For values of y satisfying $0 < y < 1$, we have

$$g(y) = \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] = \frac{1}{2\sqrt{y}} \left[\frac{1}{3} + \frac{1}{3} \right] = \frac{1}{6\sqrt{y}}$$

For values of y satisfying $1 < y < 4$, we have

$$g(y) = \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] = \frac{1}{2\sqrt{y}} \left[\frac{1}{3} + 0 \right] = \frac{1}{3\sqrt{y}}$$

Hence, the probability density function of Y is given by

$$g(y) = \begin{cases} \frac{1}{6\sqrt{y}} & \text{if } 0 < y < 1 \\ \frac{1}{3\sqrt{y}} & \text{if } 1 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

◆

EXAMPLE 3.113. A random variable X has the probability density function

$$f(x) = e^{-2|x|}, \quad -\infty < x < \infty$$

If $Y = X^2$, find the probability density function of Y .

(Anna, Nov. 2006)

Solution. Since $Y = X^2$, by Example 3.110, the PDF of Y is given by the formula

$$g(y) = \begin{cases} \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Since $-\infty < X < \infty$, $Y = X^2$ takes values from 0 to ∞ .

For $y > 0$, we have

$$g(y) = \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] = \frac{1}{2\sqrt{y}} [e^{-2\sqrt{y}} + e^{-2\sqrt{y}}] = \frac{1}{\sqrt{y}} e^{-2\sqrt{y}}$$

Hence, the probability density function of Y is given by

$$g(y) = \begin{cases} \frac{1}{\sqrt{y}} e^{-2\sqrt{y}} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

◆

Method II: Transformation Method

Here, we describe how to obtain the density function g of the random variable $Y = v(X)$ without first getting the distribution function G of Y .

First, we consider the case of a discrete random variable. If $v : \mathbb{R} \rightarrow \mathbb{R}$ is a *one-to-one* or *injective* function and if the probability distribution of X is available, then the probability distribution of $Y = v(X)$ can be calculated by means of direct substitution. This is illustrated in the following examples.

EXAMPLE 3.114. Let X be a random variable with probability distribution

$$f(x) = \begin{cases} \frac{1}{3}, & x = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability distribution of the random variable $Y = 2X - 1$.

(Madras, April 1998)

Solution. The probability distribution of X can be tabulated as

x	1	2	3
$f(x)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

Note that the given function $Y = 2X - 1$ is one-to-one. Hence, the probability distribution of Y is obtained by directly substituting the values of Y for the values of X using the relationship $Y = 2X - 1$.

y	1	3	5
$g(y)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

◆

EXAMPLE 3.115. Let X be the number of heads when two coins are tossed simultaneously. Find the probability distribution of $Y = (1 + X)^3$.

Solution. The probability distribution of X can be tabulated as

x	0	1	2
$f(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Note that the given function $Y = (1 + X)^3$ is one-to-one. Hence, the probability distribution of Y is obtained by directly substituting the values of Y for the values of X using the relationship $Y = (1 + X)^3$.

y	1	8	27
$g(y)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

◆

EXAMPLE 3.116. Let X be a binomial random variable with the probability distribution

$$f(x) = {}^3C_x \left(\frac{2}{5}\right)^x \left(\frac{3}{5}\right)^{3-x}, \quad x = 0, 1, 2, 3$$

Find the probability distribution of the random variable $Y = X^2$.

(Madras, April 1997; Oct. 1998)

Solution. From the given data, the probability distribution of X can be tabulated as follows:

x	0	1	2	3
$f(x)$	$\frac{27}{125}$	$\frac{54}{152}$	$\frac{36}{125}$	$\frac{8}{125}$

Note that the given function $Y = X^2$ is one-to-one as X takes only non-negative values. Hence, the probability distribution of Y is obtained by directly substituting the values of Y for the values of X using the relationship $Y = X^2$.

y	0	1	4	9
$g(y)$	$\frac{27}{125}$	$\frac{54}{152}$	$\frac{36}{125}$	$\frac{8}{125}$

◆

EXAMPLE 3.117. Let X be a geometric random variable with the probability distribution

$$f(x) = \frac{3}{4} \left(\frac{1}{4} \right)^{x-1}, \quad x = 1, 2, 3, \dots$$

Find the probability distribution of the variable $Y = X^2$.

(Madras, Oct. 1998; April 1999; Oct. 1999; Oct. 2002)

Solution. Since X takes only positive values, the relationship $Y = X^2$ is one-to-one. Hence, the probability distribution of the variable Y is given by

$$g(y) = \frac{3}{4} \left(\frac{1}{4} \right)^{\sqrt{y}-1}, \quad y = 1, 4, 9, 16, 25, \dots$$

where $\sqrt{y}$ denotes the positive square root of each perfect square y .

◆

If the relationship $Y = v(X)$ is not one-to-one, then the probability distribution of Y can be calculated as illustrated in the following example.

EXAMPLE 3.118. Let X be the number of heads in two tosses of a fair coin. Find the probability distribution of $Y = (X - 1)^2$.

Solution. The probability distribution of X is tabulated as follows:

x	0	1	2
$f(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Note that the given relationship $Y = (X - 1)^2$ is not one-to-one because Y takes the same value $Y = 1$ for $X = 0$ and $X = 2$.

Note that Y takes the values 0 and 1, respectively. Hence,

$$g(0) = P(Y = 0) = P(X = 1) = \frac{1}{2}$$

and

$$g(1) = P(Y = 1) = P(X = 0) + P(X = 2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Hence, the probability distribution of Y can be tabulated as follows:

y	0	1
$g(y)$	$\frac{1}{2}$	$\frac{1}{2}$

◆

Next, we consider the case of the continuous random variable. We derive a formula for the probability density function g of the variable $Y = v(X)$ under the assumptions that v is continuously differentiable and either non-decreasing or non-increasing for all values within the range of X for which $f(x) \neq 0$.

The main result is given in Theorem 3.16.

Theorem 3.16. Let X be a continuous random variable with probability density function f . If the function given by $y = v(x)$ is continuously differentiable and either non-decreasing or non-increasing for all values within the range of X for which $f(x) \neq 0$, then, for these values of x , the equation $y = v(x)$ can be uniquely solved for x to yield $x = x(y)$, and for the corresponding values of y , the probability density of $Y = v(X)$ is given by

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right|$$

Elsewhere, $g(y) = 0$.

Proof. Let $Y = v(X)$, where v satisfies the assumptions of the theorem. Let G be the distribution function of Y and g the probability density function of Y . We have two cases to consider: (i) v is non-decreasing, and (ii) v is non-increasing.

(i) Suppose that v is non-decreasing.

The distribution function of Y is given by

$$G(y) = P(Y \leq y) = P(v(X) \leq y) = P(X \leq v^{-1}(y)) = F(v^{-1}(y)) = F(x(y)).$$

Hence, the probability density function of Y is obtained as

$$g(y) = G'(y) = f(x(y)) \cdot \frac{dx}{dy}.$$

(ii) Suppose that v is non-increasing.

The distribution function of Y is given by

$$\begin{aligned} G(y) &= P(Y \leq y) = P(v(X) \leq y) = P(X \geq v^{-1}(y)) = 1 - P(X \leq v^{-1}(y)) \\ &= 1 - F(v^{-1}(y)) = 1 - F(x(y)) \end{aligned}$$

Hence, the probability density function of Y is obtained as

$$g(y) = G'(y) = -f(x(y)) \cdot \frac{dx}{dy} = f(x(y)) \left[-\frac{dx}{dy} \right]$$

Combining the two cases, we conclude that

$$g(y) = f(x(y)) \left| \frac{dx}{dy} \right|$$

◆

Remark 3.14. When the equation $x = v^{-y}$ has a finite number of solutions, say $x_1, x_2, \dots, x_n$, we can use the following extended formula for finding the probability density function $g(y)$ of Y :

$$g(y) = f(x_1) \left| \frac{dx_1}{dy} \right| + f(x_2) \left| \frac{dx_2}{dy} \right| + \dots + f(x_n) \left| \frac{dx_n}{dy} \right|$$

EXAMPLE 3.119. Let $Y = e^X$. Find the PDF of Y if X is a uniform random variable over $(0, 1)$.

(Anna, Model 2003).

Solution. Note that the given function $y = e^x$ is continuously differentiable and non-decreasing. Hence, we can obtain the probability density g of Y by transformation method.

Given that X has a uniform density over $(0, 1)$, i.e.

$$f(x) = \begin{cases} 1 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Note that $y = e^x \Rightarrow x = \log_e y$

Hence, it follows that

$$\frac{dx}{dy} = \frac{1}{y}$$

Since $0 < x < 1$, it follows that $0 < \log_e y < 1$, i.e. $1 < y < e$.

Thus, for values of y satisfying $1 < y < e$, we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = 1 \cdot \frac{1}{y} = \frac{1}{y}$$

Hence, by transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} \frac{1}{y} & \text{for } 1 < y < e \\ 0 & \text{elsewhere} \end{cases}$$

◆

Remark 3.15. After calculating the probability density g of Y , it is recommended to verify that it is indeed a probability density function, i.e. $g(y) \geq 0$ for all y and that

$$\int_{-\infty}^{\infty} g(y) dy = 1$$

For instance, for the above example, it is easily verified that

$$\int_{-\infty}^{\infty} g(y) dy = \int_1^e \frac{1}{y} dy = [\log_e y]_1^e = 1 - 0 = 1$$

EXAMPLE 3.120. Given the random variable X with probability density function

$$f(x) = \begin{cases} 2x, & \text{if } 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the probability density function of $Y = 8X^3$.

(Anna, April 2003; May 2007)

Solution. Note that the given function $y = 8x^3$ is continuously differentiable and non-decreasing in the region $0 < x < 1$. Hence, we can obtain the probability density function g of Y by transformation method.

$$\text{Note that } y = 8x^3 \Rightarrow x = \frac{1}{2} y^{\frac{1}{3}}$$

Hence, it follows that

$$\frac{dx}{dy} = \frac{1}{6} y^{-\frac{2}{3}}$$

Since $0 < x < 1$ and $y = 8x^3$, it follows that $0 < y < 8$

Thus, for values of y satisfying $0 < y < 8$, we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = y^{\frac{1}{3}} \cdot \frac{1}{6} y^{-\frac{2}{3}} = \frac{1}{6} y^{-\frac{1}{3}}.$$

Hence, by transformation method, the probability density function of Y is given by

$$g(y) = \begin{cases} \frac{1}{6} y^{-\frac{1}{3}} & \text{for } 0 < y < 8 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 3.121. If X is a random variable with CDF as $F(x)$, show that the random variable $Y = F(X)$ is uniformly distributed in $(0, 1)$.

(Anna, May 2006)

Solution. We know that the cumulative distribution function $F(x)$ is differentiable and non-decreasing when X is a continuous random variable.

Let $Y = F(X)$. Then it is clear that Y takes values in $(0, 1)$. Also, note that

$$f(x) = F'(x) = \frac{dy}{dx} \Rightarrow \frac{dx}{dy} = \frac{1}{f(x)}$$

By the transformation method, for values of y in $(0, 1)$, the PDF of Y is given by

$$g(y) = f(x(y)) \left| \frac{dx}{dy} \right| = f(x) \cdot \frac{1}{f(x)} = 1 \quad (\text{note that } f(x) \geq 0 \text{ for all } x)$$

Hence, it follows that Y is uniformly distributed in $(0, 1)$.

◆

EXAMPLE 3.122. If X is a random variable uniformly distributed in $(0, 1)$, find the PDF of $Y = \sin X$. Also find the mean and variance of Y .

(Anna, May 2006)

Solution. Note that the given function $y = \sin x$ is continuously differentiable and non-decreasing in the interval $(0, 1)$. Hence, we can obtain the probability density g of Y by transformation method.

Given that X has a uniform density over $(0, 1)$, i.e.

$$f(x) = \begin{cases} 1 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Note that $y = \sin x \Rightarrow x = \sin^{-1} y$.

Hence, it follows that

$$\frac{dx}{dy} = \frac{1}{\sqrt{1-y^2}}$$

Since $0 < x < 1$, it follows that $0 < \sin^{-1} y < 1$, i.e. $0 < y < \sin 1$.

Thus, for values of y satisfying $0 < y < \sin 1$, we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = 1 \times \frac{1}{\sqrt{1-y^2}} = \frac{1}{\sqrt{1-y^2}}$$

Hence, by transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} \frac{1}{\sqrt{1-y^2}} & \text{for } 0 < y < \sin 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the mean and variance of $Y = \sin X$.

Note that the first two raw moments of Y are

$$E(Y) = E(\sin X) \quad \text{and} \quad E(Y^2) = E(\sin^2 X)$$

which can be easily evaluated using the probability density of X . Hence, we have

$$E(Y) = \int_0^1 \sin x \cdot 1 \, dx = [-\cos x]_0^1 = 1 - \cos 1$$

and

$$E(Y^2) = \int_0^1 \sin^2 x \cdot 1 \, dx = \frac{1}{2} \int_0^1 (1 - \cos 2x) \, dx$$

Integrating, we get

$$E(Y^2) = \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^1 = \frac{1}{2} \left[1 - \frac{\sin 2}{2} \right]$$

Hence, the mean of Y is

$$\mu_Y = E(Y) = 1 - \cos 1 = 0.4597$$

and the variance of Y is

$$\begin{aligned} \text{Var}(Y) &= E(Y^2) - \mu_Y^2 = \frac{1}{2} \left[1 - \frac{\sin 2}{2} \right] - (1 - \cos 1)^2 \\ &= 2 \cos 1 - \frac{1}{2} - \frac{\sin 2}{4} - \cos^2 1 = 0.0614 \end{aligned}$$

◆

EXAMPLE 3.123. Let X be a continuous random variable with PDF

$$f(x) = \begin{cases} \frac{x}{12} & \text{if } 1 < x < 5 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability density function of $Y = 2X - 3$.

(Madras, Oct. 2002; Anna, Nov. 2006)

Solution. Note that the given function $y = 2x - 3$ is continuously differentiable and non-decreasing. Hence, we can obtain the probability density g of Y by transformation method.

Note that $y = 2x - 3 \Rightarrow x = \frac{y+3}{2}$.

Hence, it follows that

$$\frac{dx}{dy} = \frac{1}{2}$$

Since $1 < x < 5$, it follows that

$$1 < \frac{y+3}{2} < 5 \Rightarrow 2 < y+3 < 10 \Rightarrow -1 < y < 7$$

Thus, for values of y satisfying $-1 < y < 7$, we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = \frac{y+3}{24} \times \frac{1}{2} = \frac{y+3}{48}$$

Hence, by transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} \frac{y+3}{48} & \text{for } -1 < y < 7 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 3.124. If the random variable X is uniformly distributed over $(-1, 1)$, find the density function of $Y = \sin\left(\frac{\pi X}{2}\right)$.

(Anna, April 2003)

Solution. Note that the given function $y = \sin\left(\frac{\pi x}{2}\right)$ is continuously differentiable and non-decreasing in the interval $(-1, 1)$. Hence, we can obtain the probability density g of Y by transformation method.

Given that X is uniformly distributed over $(-1, 1)$, i.e. X has the PDF

$$f(x) = \begin{cases} \frac{1}{2} & \text{for } -1 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Note that $y = \sin\left(\frac{\pi x}{2}\right) \Rightarrow x = \frac{2}{\pi} \sin^{-1}y$.

Hence, it follows that

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Since $-1 < x < 1$, it follows that

$$-1 < \frac{2}{\pi} \sin^{-1}y < 1 \Rightarrow -\frac{\pi}{2} < \sin^{-1}y < \frac{\pi}{2} \Rightarrow -1 < y < 1$$

Thus, for values of y satisfying $-1 < y < 1$, we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = \frac{1}{2} \times \frac{2}{\pi} \times \frac{1}{\sqrt{1-y^2}} = \frac{1}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

Hence, by transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} \frac{1}{\pi} \cdot \frac{1}{\sqrt{1-y^2}} & \text{for } -1 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 3.125. Let the density function of X be

$$f(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Using the method of transformation, find the density functions of $Y = X\sqrt{X}$, $Z = e^{-X}$.
(Anna, April 2003)

Solution. First, we find the probability density of $Y = X\sqrt{X} = X^{\frac{3}{2}}$. We note that $y = x^{\frac{3}{2}}$ is continuously differentiable and non-decreasing in the range $x > 0$. We note also that $y > 0$ when $x > 0$.

Since $y = x^{\frac{3}{2}}$, it follows that $x = y^{\frac{2}{3}}$.

Hence, we have

$$\frac{dx}{dy} = \frac{2}{3} y^{-\frac{1}{3}}$$

Thus, for $y > 0$, then the probability density of Y is given by

$$g(y) = f(x(y)) \left| \frac{dx}{dy} \right| = e^{-y^{\frac{2}{3}}} \times \frac{2}{3} y^{-\frac{1}{3}}$$

Hence, by the transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} \frac{2}{3} y^{-\frac{1}{3}} e^{-y^{\frac{2}{3}}} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the probability density of $Z = e^{-X}$. We note that $z = e^{-x}$ is continuously differentiable and non-increasing in the range $x > 0$. We note also that $z > 0$ when $x > 0$.

Since $z = e^{-x}$, it follows that $x = \frac{1}{\log z}$.

Hence, we have

$$\frac{dx}{dz} = -\frac{1}{(\log z)^2} \cdot \frac{1}{z}$$

Thus, for $z > 0$, then the probability density of Z is given by

$$h(z) = f(x(z)) \left| \frac{dx}{dz} \right| = e^{-\frac{1}{\log z}} \times \frac{1}{(\log z)^2} \cdot \frac{1}{z}$$

Hence, by the transformation method, the probability density of Z is given by

$$h(z) = \begin{cases} \frac{1}{z(\log z)^2} e^{-\frac{1}{\log z}} & \text{for } z > 0 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 3.126. If X is a normal random variable with mean zero and variance σ^2 , find the probability density function of $Y = e^X$.

(Anna, Nov. 2003; Nov. 2007)

Solution. Since $X \sim N(0, \sigma^2)$, the probability density function of X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} \quad \text{for } -\infty < x < \infty$$

Note that the given function $y = e^x$ is continuously differentiable and non-decreasing over the real line $\mathbb{R}$. Hence, we can obtain the probability density g of Y by transformation method.

Note that $y = e^x \Rightarrow x = \log_e y$.

Hence, it follows that

$$\frac{dx}{dy} = \frac{1}{y}$$

Since $-\infty < x < \infty$ and $y = \log_e(x)$, it is clear that y takes all values in the real line $\mathbb{R}$ except $y = 0$. Thus, for values of y satisfying $-\infty < y < \infty$ (except $y \neq 0$), we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(\log y)^2}{2\sigma^2}} \times \frac{1}{|y|}$$

Hence, by transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} \frac{1}{\sigma\sqrt{2\pi|y|}} e^{-\frac{(\log y)^2}{2\sigma^2}} & \text{for } y \neq 0, -\infty < y < \infty \\ 0 & \text{for } y = 0 \end{cases}$$

◆

EXAMPLE 3.127. If X has exponential distribution with parameter λ , find the distribution of $Y = e^{-\lambda X}$:

(Anna, Nov. 2004)

Solution. Note that the given function $y = e^{-\lambda x}$ is continuously differentiable and non-increasing. Hence, we can obtain the probability density g of Y by transformation method.

Note that $y = e^{-\lambda x} \Rightarrow x = -\frac{1}{\lambda} \log y$.

Hence, it follows that

$$\frac{dx}{dy} = -\frac{1}{\lambda} \frac{1}{y}$$

Since $x > 0$, it follows that

$$-\frac{1}{\lambda} \log y > 0 \Rightarrow \log y < 0 \Rightarrow 0 < y < 1$$

Thus, for values of y satisfying $0 < y < 1$, we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = \lambda \times y \times \frac{1}{\lambda} \times \frac{1}{y} = 1$$

So, by transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} 1 & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Thus, Y is uniformly distributed over $(0, 1)$. ◆

EXAMPLE 3.128. Let X be a random variable with probability density function

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If $Y = -2 \log X$, find the probability density function of the random variable Y and $E(Y)$.

(Anna, May 2007)

Solution. Note that the given function $y = -2 \log x$ is continuously differentiable and non-increasing. Hence, we can obtain the probability density g of Y by transformation method.

Note that $y = -2 \log x \Rightarrow x = e^{-\frac{y}{2}}$.

Hence, it follows that

$$\frac{dx}{dy} = -\frac{1}{2} e^{-\frac{y}{2}}$$

Since $0 < x < 1$, it follows that

$$0 < e^{-\frac{y}{2}} < 1 \Rightarrow 0 < \frac{y}{2} < \infty \Rightarrow 0 < y < \infty$$

Thus, if $y > 0$, then we have

$$g(y) = f(x(y)) \cdot \left| \frac{dx}{dy} \right| = 1 \cdot \frac{1}{2} e^{-\frac{y}{2}}$$

Hence, by transformation method, the probability density of Y is given by

$$g(y) = \begin{cases} \frac{1}{2} e^{-\frac{y}{2}} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Thus, Y is exponentially distributed with parameter $\lambda = \frac{1}{2}$. Hence, it is immediate that $E(Y) = \frac{1}{\lambda} = 2$. ◆

PROBLEM SET 3.10

1. If X is a random variable with the probability density function

$$f(x) = \begin{cases} 2x & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and $Y = \sqrt{X}$, find (a) the distribution function of Y and (b) the PDF of Y .

2. If X is a random variable uniformly distributed in $(-1, 1)$ and $Y = X^2$, find (a) the distribution function of Y and (b) the PDF of Y .
3. If X is a random variable with the exponential density function

$$f(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

find the probability density function of $Y = 2X + 3$.

4. If X is a random variable with the Laplace density function

$$f(x) = \frac{1}{2} e^{-|x|} \text{ for } -\infty < x < \infty$$

find the probability density function of $Y = X^2$.

5. If X is a random variable with the probability density function

$$f(x) = \begin{cases} \theta e^x e^{-\theta e^x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and $Y = e^X$, find (a) the distribution function of Y and (b) the PDF of Y .

6. If X is a random variable with the probability density function

$$f(x) = \begin{cases} 2xe^{-x^2} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and $Y = X^2$, find (a) the distribution function of Y and (b) the PDF of Y .

7. If X is a random variable with the probability distribution

x	0	1	2	3
$f(x)$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

find the probability distribution of the random variable $Y = 3X - 2$.

8. If X is a binomial random variable with the probability mass function

$$f(x) = {}^2C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{2-x}, \quad x = 0, 1, 2$$

find the probability distribution of the random variable $Y = X^2$.

9. If X is a geometric random variable with the probability mass function

$$f(x) = \frac{2}{3} \left(\frac{1}{3}\right)^{x-1}, \quad x = 1, 2, 3, \dots$$

find the probability mass function of the random variable $Y = X^2$.

10. If X is the number of heads in three tosses of a fair coin, find the probability distribution of $Y = (X - 1)^2$.
11. If X is a random variable uniformly distributed in $(0, 1)$, find the probability density function of $Y = \ln X$ by transformation method.
12. If X is a random variable with the exponential density function

$$f(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

find the probability density function of $Y = 8X^3$ by transformation method.

13. If X is a random variable uniformly distributed in $(0, 1)$, find the probability density function of $Y = \cos X$. Find also the mean and variance of Y .
14. If X is a random variable with the probability density function

$$f(x) = \begin{cases} \frac{1}{4\sqrt{x}} & \text{if } 0 < x < 4 \\ 0 & \text{otherwise} \end{cases}$$

find the probability density function of $Y = \sqrt{X}$ by transformation method.

15. If X is a random variable with the probability density function

$$f(x) = \begin{cases} \frac{x}{2} & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

find the probability density function of $Y = 2X - 3$ by transformation method.

Chapter 4

Two-Dimensional Random Variables

4.1 JOINT DISTRIBUTION FUNCTIONS

Let $(S, \mathcal{B}, P)$ be any probability space, where S is the sample space, $\mathcal{B}$ is the σ -algebra or Borel algebra defined on S and P is the probability function on $\mathcal{B}$ satisfying the axiomatic definition of probability given by Kolmogorov.

Definition 4.1. Let X and Y be two random variables defined on the sample space S . The function $X \times Y : S \rightarrow \mathbb{R}^2$ defined by

$$(X, Y)(\omega) = (X(\omega), Y(\omega)), \quad \omega \in S$$

is called a *two-dimensional random variable* on S .

Next, we define the distribution function of a two-dimensional random variable.

Definition 4.2. Let (X, Y) be a two-dimensional random variable defined on the sample space S . Then the *distribution function* of (X, Y) is denoted by F and is defined as

$$F(x, y) = P(X \leq x, Y \leq y), \quad \text{where } (x, y) \in \mathbb{R}^2$$

F is also called the *joint distribution function* of the random variables X and Y .

Next, we establish an important result for the distribution function of a two-dimensional random variable (X, Y) .

Theorem 4.1. Let F be the distribution function of a two-dimensional random variable (X, Y) . Then, if $a < b$ and $c < d$, we have

$$P(a < X \leq b, c < Y \leq d) = F(b, d) - F(b, c) - F(a, d) + F(a, c)$$

Proof. Let us define the events

$$A = \{X \leq a\}, B = \{X \leq b\}, C = \{Y \leq c\} \text{ and } D = \{Y \leq d\}$$

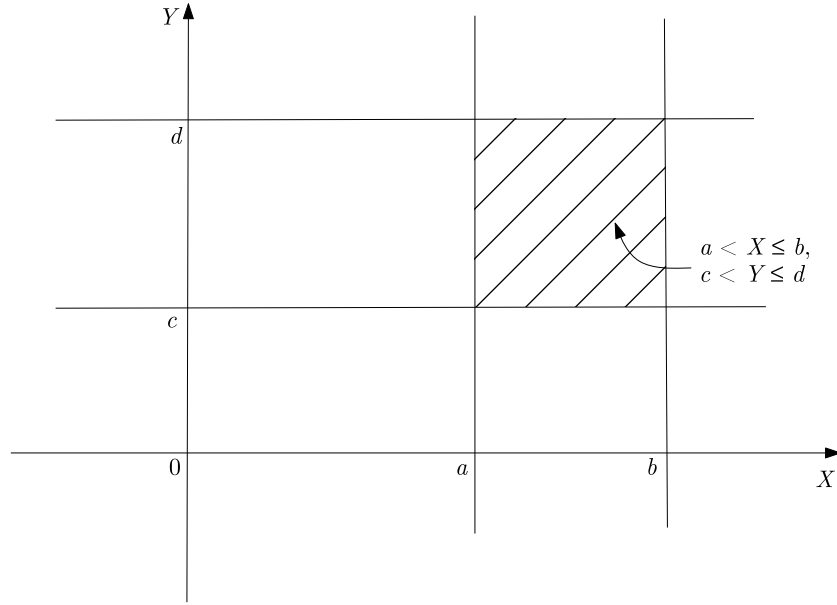


Figure 4.1: Property of distribution function.

It is clear that (see Figure 4.1)

$$P(a < X \leq b, c < Y \leq d) = P[(B-A) \cap (D-C)]$$

By the distributive property of sets, we have

$$P(a < X \leq b, c < Y \leq d) = P[B \cap (D-C) - A \cap (D-C)] \quad (4.1)$$

From the subtractive property of probability (see Theorem 1.3), we know that

$$E \subset F \Rightarrow P(F-E) = P(F) - P(E)$$

Since $A \subset B$, it follows that

$$A \cap (D-C) \subset B \cap (D-C)$$

Hence, by the subtractive property of probability, Eq. (4.1) simplifies to

$$\begin{aligned} P(a < X \leq b, c < Y \leq d) &= P[B \cap (D-C)] - P[A \cap (D-C)] \\ &= P[(B \cap D) - (B \cap C)] - P[(A \cap D) - (A \cap C)] \end{aligned}$$

By the subtractive property of probability, it again follows that

$$P(a < X \leq b, c < Y \leq d) = P(B \cap D) - P(B \cap C) - P(A \cap D) + P(A \cap C) \quad (4.2)$$

Note that

$$P(B \cap D) = P(X \leq b, Y \leq d) = F(b, d)$$

$$P(B \cap C) = P(X \leq b, Y \leq c) = F(b, c)$$

$$P(A \cap D) = P(X \leq a, Y \leq d) = F(a, d)$$

$$P(A \cap C) = P(X \leq a, Y \leq c) = F(a, c)$$

Hence, Eq. (4.2) simplifies to

$$P(a < X \leq b, c < Y \leq d) = F(b, d) - F(b, c) - F(a, d) + F(a, c)$$

◆

Remark 4.1. $P(a < X \leq b, c < Y \leq d)$ represents the probability that the two-dimensional random variable (X, Y) has a value in the rectangular region $(a, b] \times (c, d]$, i.e. it represents the total mass in $(-\infty, a] \times (-\infty, b]$. As in the one-dimensional case, here also there are two types of distributions, viz. *discrete* and *continuous*. For the discrete distributions, the entire mass is concentrated at a countable number of points, known as the *mass points*. For the continuous distributions the mass is continuously distributed with uniform or varying thickness all over the (x, y) plane.

The properties of the distribution function are summarized in the following theorem. We state this result without proof.

Theorem 4.2. Let F be the distribution function of a two-dimensional random variable (X, Y) . Then, the following properties hold:

- (i) $F(x, y)$ is non-decreasing with respect to both arguments, i.e.

$$F(x_1, y_1) \leq F(x_2, y_2) \text{ for } x_1 < x_2 \text{ and } y_1 < y_2$$

and is right-continuous with respect to both arguments, i.e.

$$F(x+0, y) = F(x, y+0) = F(x, y)$$

- (ii) We have

$$F(-\infty, y) = \lim_{x \rightarrow -\infty} F(x, y) = 0, \quad F(x, -\infty) = \lim_{y \rightarrow -\infty} F(x, y) = 0$$

and

$$F(+\infty, +\infty) = \lim_{\substack{x \rightarrow +\infty \\ y \rightarrow +\infty}} F(x, y) = 1$$

- (iii) For each (x_1, y_1) and (x_2, y_2) with $x_1 < x_2$ and $y_1 < y_2$, we have

$$P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) \geq 0$$

The following theorem gives a characterization for the distribution function of a two-dimensional random variable (X, Y) . We state this result without proof.

Theorem 4.3. A real-valued function F of two variables x, y is a distribution function of some two-dimensional random variable (X, Y) if and only if F satisfies the following conditions:

- (i) F is non-decreasing and right-continuous with respect to both arguments, x and y .
(ii) $F(-\infty, y) = F(x, -\infty) = 0$ and $F(+\infty, +\infty) = 1$.

(iii) For every $(x_1, y_1), (x_2, y_2)$ with $x_1 < x_2$ and $y_1 < y_2$, the inequality

$$F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) \geq 0$$

holds.

Next, we define a two-dimensional *discrete* random variable.

Definition 4.3. A two-dimensional random variable (X, Y) is said to be *discrete* if it takes on a countable number of values in $\mathbb{R}^2$. We call every pair (x_i, y_j) that is assumed with positive probability p_{ij} a *mass point* or *jump point* of the distribution function of (X, Y) , and call p_{ij} the *mass* or *jump* at (x_i, y_j) .

Since the total probability for a random variable is unity, it is clear that

$$\sum_i \sum_j p_{ij} = 1$$

The distribution function or cumulative distribution function (CDF) of a discrete random variable (X, Y) is given by

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{x \leq x_i} \sum_{y \leq y_j} f(x, y)$$

Next, we define the *probability mass function* (PMF) of a two-dimensional discrete random variable.

Definition 4.4. Let (X, Y) be a two-dimensional discrete random variable with mass points $(x_i, y_j), i = 1, 2, \dots$ and $j = 1, 2, \dots$. Then the *probability mass function* or *discrete density function* of the random variable (X, Y) is denoted by $f_{X,Y}$ and is defined by

$$f_{X,Y}(x, y) = \begin{cases} P(X = x_i, Y = y_j) = p_{ij} & \text{if } (x, y) = (x_i, y_j) \text{ for some } i \text{ and } j \\ 0 & \text{otherwise} \end{cases}$$

We state the following result without proof.

Theorem 4.4. A set of real numbers $\Lambda = \{p_{ij} : i = 1, 2, \dots, \infty, j = 1, 2, \dots, \infty\}$ is the probability mass function for some two-dimensional discrete random variable (X, Y) if the following conditions hold:

- (i) $p_{ij} \geq 0$ for all i and j .
- (ii) $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} p_{ij} = 1$.

EXAMPLE 4.1. Suppose that two fair coins are tossed simultaneously. We define two random variables, X and Y , as

X = The number of heads in the toss of the first coin.

Y = The number of heads in the toss of the second coin.

Since X and Y take on the values 0 and 1, it is clear that the mass points for (X, Y) are given by

$$(x_1, y_1) = (0, 0), (x_1, y_2) = (0, 1), (x_2, y_1) = (1, 0) \text{ and } (x_2, y_2) = (1, 1)$$

Hence, it is clear that the probability mass function for (X, Y) is given by

$$p_{ij} = P(X = x_i, Y = y_j) = \frac{1}{4} \text{ for } i = 1, 2, j = 1, 2$$

It can be also put in a tabular form as follows:

X \ Y	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$

EXAMPLE 4.2. Three balls are drawn at random without replacement from a box containing 2 white, 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn, find the joint probability distribution of (X, Y) .

(Anna, April 2004; May 2007)

Solution. Since there are totally 2 white balls and 3 red balls, it is clear that X takes the values 0, 1, 2 and Y takes the values 0, 1, 2, 3. We are asked to calculate the joint probability distribution of (X, Y) , i.e. to calculate the probabilities $P(X = x, Y = y)$, where $x = 0, 1, 2$ and $y = 0, 1, 2, 3$. The calculations are carried out as follows.

First, we note that there are totally $2 + 3 + 4 = 9$ balls and that 3 balls can be drawn at random without replacement from the box containing 9 balls in ${}^9C_3 = 84$ different ways.

$$P(X = 0, Y = 0) = P(\text{only black balls}) = \frac{{}^4C_3}{{}^9C_3} = \frac{4}{84} = \frac{1}{21}$$

$$P(X = 1, Y = 0) = P(1 \text{ white, } 2 \text{ black balls}) = \frac{{}^2C_1 \times {}^4C_2}{{}^9C_3} = \frac{2 \times 6}{84} = \frac{1}{7}$$

$$P(X = 2, Y = 0) = P(2 \text{ white, } 1 \text{ black ball}) = \frac{{}^2C_2 \times {}^4C_1}{{}^9C_3} = \frac{1 \times 4}{84} = \frac{1}{21}$$

$$P(X = 0, Y = 1) = P(1 \text{ red, } 2 \text{ black balls}) = \frac{{}^3C_1 \times {}^4C_2}{{}^9C_3} = \frac{3 \times 6}{84} = \frac{3}{14}$$

$$P(X = 1, Y = 1) = P(1 \text{ white, } 1 \text{ black, } 1 \text{ red}) = \frac{{}^2C_1 \times {}^3C_1 \times {}^4C_1}{{}^9C_3} = \frac{2 \times 3 \times 4}{84} = \frac{2}{7}$$

$$P(X = 2, Y = 1) = P(2 \text{ white, } 1 \text{ red ball}) = \frac{{}^2C_2 \times {}^3C_1}{{}^9C_3} = \frac{1 \times 3}{84} = \frac{1}{28}$$

$$P(X = 0, Y = 2) = P(2 \text{ red, } 1 \text{ black ball}) = \frac{{}^3C_2 \times {}^4C_1}{{}^9C_3} = \frac{3 \times 4}{84} = \frac{1}{7}$$

$$P(X = 1, Y = 2) = P(1 \text{ white, } 2 \text{ red balls}) = \frac{{}^2C_1 \times {}^3C_2}{{}^9C_3} = \frac{2 \times 3}{84} = \frac{1}{14}$$

$$P(X = 2, Y = 2) = 0 \quad (\text{since only three balls are drawn})$$

$$P(X = 0, Y = 3) = P(3 \text{ red balls}) = \frac{{}^3C_3}{{}^9C_3} = \frac{1}{84}$$

$$P(X = 1, Y = 3) = 0 \quad (\text{since only three balls are drawn})$$

$$P(X = 2, Y = 3) = 0 \quad (\text{since only three balls are drawn})$$

Thus, the joint probability distribution of (X, Y) can be tabulated as follows:

X \ Y	0	1	2	3
0	$\frac{1}{21}$	$\frac{3}{14}$	$\frac{1}{7}$	$\frac{1}{84}$
1	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{14}$	0
2	$\frac{1}{21}$	$\frac{1}{28}$	0	0

◆

Next, we define a two-dimensional *continuous* random variable.

Definition 4.5. A two-dimensional random variable (X, Y) is said to be *continuous* if there exists a non-negative function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that for every pair $(x, y) \in \mathbb{R}^2$, we have

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{u=-\infty}^x \int_{v=-\infty}^y f(u, v) dv du$$

where F is the distribution function of (X, Y) . The function f is called the *(joint) probability density function* of (X, Y) .

Theorem 4.5. The probability density function f of a continuous random variable (X, Y) satisfies the following properties:

- (i) $f(x, y) \geq 0$ for all x and y .
- (ii) $\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y) dy du = 1$.

Proof. (i) This is immediate from the definition of the continuous RV (X, Y) .

(ii) We know that $F(+\infty, +\infty) = 1$. Hence, it follows that

$$F(+\infty, +\infty) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y) dy du = 1$$

◆

If the joint density function f is continuous at (x, y) , then we have

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y)$$

EXAMPLE 4.3. Let (X, Y) be a continuous random variable with the joint probability density function given by

$$f(x, y) = \begin{cases} e^{-(x+y)} & \text{if } 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find the distribution function of (X, Y) .

Solution. If $x > 0$ and $y > 0$, then we have

$$\begin{aligned} F(x, y) &= \int_{u=0}^x \int_{v=0}^y f(u, v) \, dv \, du \\ &= \int_{u=0}^x \int_{v=0}^y e^{-(u+v)} \, dv \, du \end{aligned}$$

Since the integrals are separable, we have

$$\begin{aligned} F(x, y) &= \left[\int_{u=0}^x e^{-u} \, du \right] \left[\int_{v=0}^y e^{-v} \, dv \right] = [-e^{-u}]_0^x [-e^{-v}]_0^y \\ &= (1 - e^{-x})(1 - e^{-y}) \quad \text{for } x > 0, y > 0 \end{aligned}$$

Hence, the distribution function of (X, Y) is given by

$$F(x, y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}), & \text{for } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

♦

The converse of Theorem 4.5 also is true. We state the following result without proof.

Theorem 4.6. If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a non-negative function satisfying

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y) \, dy \, dx = 1$$

then f is the joint probability density function of some continuous random variable (X, Y) .

EXAMPLE 4.4. Find the value of K if $f(x, y) = K(1 - x)(1 - y)$ for $0 < x, y < 1$ is to be a joint density function of X and Y .

(Anna, April 2003)

Solution. Clearly, $K \geq 0$ since joint the density function f takes only non-negative values. Also, we use the property that

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y) \, dy \, dx = 1$$

Using the definition of f , we have

$$\int_{x=0}^1 \int_{y=0}^1 K(1 - x)(1 - y) \, dy \, dx = 1$$

Since the integrals are separable, we have

$$K \left[\int_{x=0}^1 (1-x) dx \right] \left[\int_{y=0}^1 (1-y) dy \right] = 1$$

Integrating, we have

$$K \left[-\frac{(1-x)^2}{2} \right]_0^1 \left[-\frac{(1-y)^2}{2} \right]_0^1 = 1$$

Simplifying, we have

$$K \left[\frac{1}{2} \right] \left[\frac{1}{2} \right] = 1 \quad \text{or} \quad K \left[\frac{1}{4} \right] = 1$$

Hence, $K = 4$ ◆

EXAMPLE 4.5. The joint probability density function of a bivariate random variable (X, Y) is given by

$$f(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find $P(X + Y < 1)$.

(Anna, Model 2003)

Solution. Define the region A by

$$A = \{(x, y) \in \mathbb{R}^2 : 0 < x < 1, 0 < y < 1, x + y < 1\}$$

Then the region A is as sketched in Figure 4.2. We are asked to find the probability that $X + Y < 1$. By the definition of f , this is equal to the probability that $(X, Y) \in A$.

The required probability is given by

$$\begin{aligned} P[(X, Y) \in A] &= \int_{x=0}^1 \int_{y=0}^{1-x} f(x, y) dy dx = \int_{x=0}^1 \int_{y=0}^{1-x} 4xy dy dx \\ &= \int_{x=0}^1 2x [y^2]_{y=0}^{1-x} dx = \int_{x=0}^1 2x(1-x)^2 dx \\ &= \int_{x=0}^1 2x(x^2 - 2x + 1) dx \\ P[(X, Y) \in A] &= \int_{x=0}^1 [2x^3 - 4x^2 + 2x] dx = \left[\frac{x^4}{2} - \frac{4x^3}{3} + x^2 \right]_0^1 \end{aligned}$$

Simplifying, we have

$$P[(X, Y) \in A] = \frac{1}{2} - \frac{4}{3} + 1 = \frac{3-8+6}{6} = \frac{1}{6} \quad \text{◆}$$

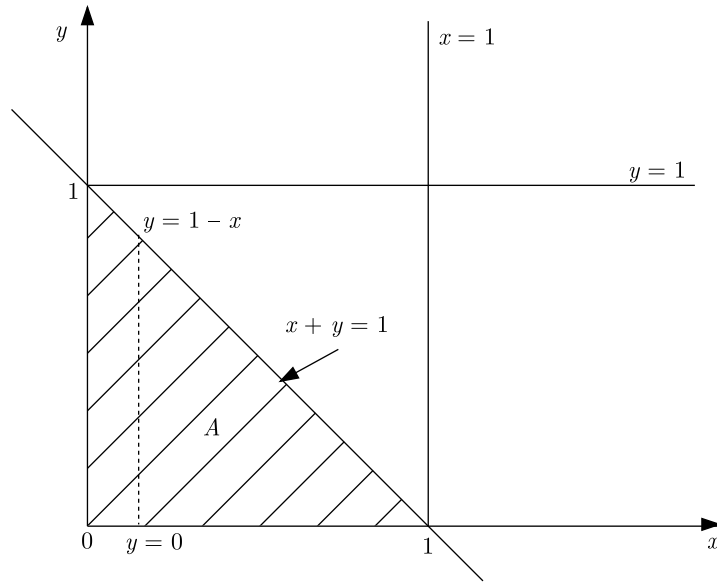


Figure 4.2: The region A.

EXAMPLE 4.6. If the joint probability density function of X and Y is given by

$$f(x,y) = \begin{cases} 24xy & \text{for } 0 < x < 1, 0 < y < 1, x+y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find $P(X+Y < \frac{1}{2})$.

Solution. Define the region A as

$$A = \{(x,y) \in \mathbb{R}^2 : 0 < x < 1, 0 < y < 1, x+y < \frac{1}{2}\}$$

Then the region A is as sketched in Figure 4.3.

By the definition of f , the probability that $X+Y < \frac{1}{2}$ is equal to the probability that $(X,Y) \in A$. Thus, the required probability is given by

$$P[(X,Y) \in A] = \int_{x=0}^{\frac{1}{2}} \int_{y=0}^{\frac{1}{2}-x} f(x,y) dy dx = \int_{x=0}^{\frac{1}{2}} \int_{y=0}^{\frac{1}{2}-x} 24xy dy dx$$

Integrating y , we have

$$\begin{aligned} P[(X,Y) \in A] &= 12 \int_{x=0}^{\frac{1}{2}} x [y^2]_0^{\frac{1}{2}-x} dx \\ &= 12 \int_{x=0}^{\frac{1}{2}} x \left[\left(\frac{1}{2} - x \right)^2 \right] dx = 1 \end{aligned}$$

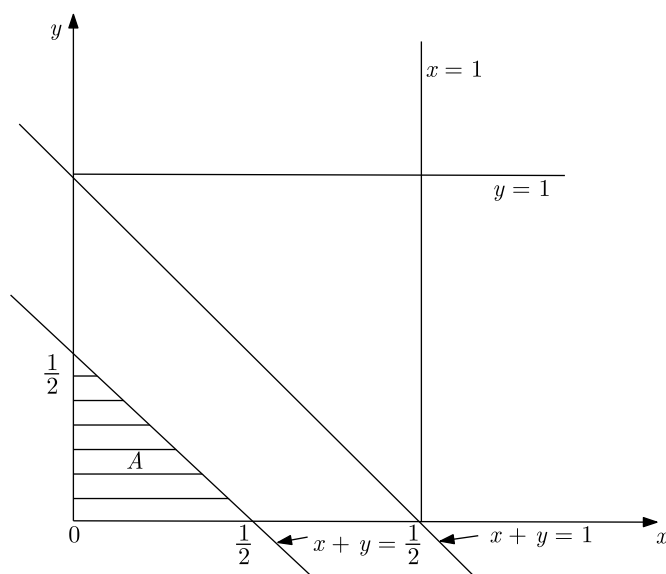


Figure 4.3: The region A, Example 4.6

$$= 12 \int_{x=0}^{\frac{1}{2}} x \left[x^2 - x + \frac{1}{4} \right] dx = 1$$

$$= 12 \int_{x=0}^{\frac{1}{2}} \left[x^3 - x^2 + \frac{x}{4} \right] dx = 1$$

Integrating x , we have

$$\begin{aligned} P[(X, Y) \in A] &= 12 \left[\frac{x^4}{4} - \frac{x^3}{3} + \frac{x^2}{8} \right]_0^{\frac{1}{2}} = 12 \left[\frac{1}{64} - \frac{1}{24} + \frac{1}{32} \right] \\ &= 12 \times \frac{1}{192} = \frac{1}{16} \end{aligned}$$

◆

EXAMPLE 4.7. If the joint density function of the two random variables X and Y be

$$f(x, y) = \begin{cases} e^{-(x+y)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

find (i) $P(X < 1)$, (ii) $P(X + Y < 1)$.

(Anna, Nov. 2003)

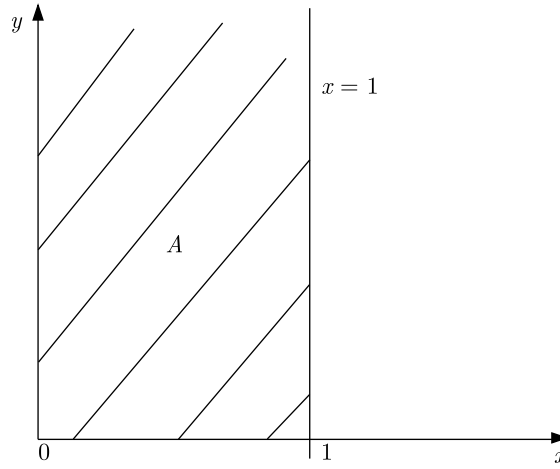


Figure 4.4: The region A.

Solution. (i) Define the region A by

$$A = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x < 1\}$$

The region A is as illustrated in Figure 4.4. We are asked to find $P(X < 1)$. By the definition of f , this is equal to $P[(X, Y) \in A]$.

The required probability is

$$P[(X, Y) \in A] = \int_{x=0}^1 \int_{y=0}^{\infty} f(x, y) dy dx = \int_{x=0}^1 \int_{y=0}^{\infty} e^{-(x+y)} dy dx$$

Since $e^{-(x+y)} = e^{-x-y} = e^{-x}e^{-y}$, the integrals are separable and hence

$$P[(X, Y) \in A] = \left[\int_{x=0}^1 e^{-x} dx \right] \left[\int_{y=0}^{\infty} e^{-y} dy \right]$$

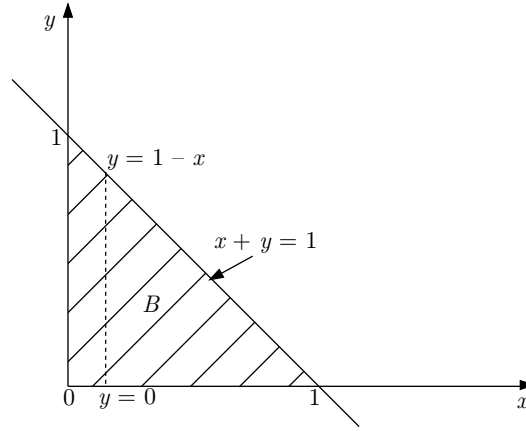
Integrating, we have

$$\begin{aligned} P[(X, Y) \in A] &= [-e^{-x}]_0^1 [-e^{-y}]_0^{\infty} = [-e^{-1} + 1] [0 + 1] \\ &= (1 - e^{-1}) = 1 - \frac{1}{e} = 0.6321 \end{aligned}$$

(ii) Define the region B as

$$B = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x + y < 1\}$$

The region B is as illustrated in Figure 4.5.

Figure 4.5: The region B .

We are asked to find the probability that $X + Y < 1$. By the definition of f , this is equal to $P[(x, y) \in B]$.

Hence, the required probability is

$$P[(X, Y) \in B] = \int_{x=0}^1 \int_{y=0}^{1-x} f(x, y) dy dx = \int_{x=0}^1 \int_{y=0}^{1-x} e^{-x} e^{-y} dy dx$$

Integrating, we have

$$\begin{aligned} P[(X, Y) \in B] &= \int_{x=0}^1 e^{-x} [-e^{-y}]_{y=0}^{1-x} dx = \int_{x=0}^1 e^{-x} [-e^{x-1} + 1] dx \\ &= \int_{x=0}^1 [-e^{-1} + e^{-x}] dx \end{aligned}$$

Integrating, we have

$$\begin{aligned} P[(X, Y) \in B] &= [-xe^{-1} - e^{-x}]_0^1 = [(-1e^{-1} - e^{-1}) - (0 - 1)] \\ &= [1 - 2e^{-1}] = 1 - \frac{2}{e} = 0.2642 \end{aligned}$$

◆

EXAMPLE 4.8. If two random variables have the joint density

$$f(x_1, x_2) = \begin{cases} x_1 x_2 & \text{for } 0 < x_1 < 1, 0 < x_2 < 2 \\ 0 & \text{elsewhere} \end{cases}$$

find the probability that both random variables will take on values less than 1.

(Anna, Nov. 2003)

Solution. Using the definition of f , the required probability is given by

$$P(X_1 < 1, X_2 < 1) = \int_{x_1=0}^1 \int_{x_2=0}^1 f(x_1, x_2) dx_2 dx_1 = \int_{x_1=0}^1 \int_{x_2=0}^1 x_1 x_2 dx_2 dx_1$$

Since the integrals can be separated, we have

$$P(X_1 < 1, X_2 < 1) = \left[\int_{x_1=0}^1 x_1 dx_1 \right] \left[\int_{x_2=0}^1 x_2 dx_2 \right]$$

Integrating, we have

$$P(X_1 < 1, X_2 < 1) = \left[\frac{x_1^2}{2} \right]_0^1 \left[\frac{x_2^2}{2} \right]_0^1 = \left[\frac{1}{2} - 0 \right] \left[\frac{1}{2} - 0 \right]$$

Simplifying, we have

$$P(X_1 < 1, X_2 < 1) = \left[\frac{1}{2} \right]^2 = \frac{1}{4}$$

◆

EXAMPLE 4.9. If the joint probability density of two random variables is given by

$$f(x_1, x_2) = \begin{cases} 6e^{-2x_1-3x_2} & \text{for } x_1 > 0, x_2 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the probability that the first random variable will take on a value between 1 and 2, and the second random variable will take on a value between 2 and 3. Also find the probability that the first random variable will take on a value less than 2 and the second random variable will take on a value greater than 2.

(Anna, Nov. 2003; April 2005)

Solution. First, we calculate the probability that $1 < X_1 < 2$ and $2 < X_2 < 3$, which is given by

$$P(1 < X_1 < 2, 2 < X_2 < 3) = \int_{x_1=1}^2 \int_{x_2=2}^3 f(x_1, x_2) dx_2 dx_1 = \int_{x_1=1}^2 \int_{x_2=2}^3 6e^{-2x_1-3x_2} dx_2 dx_1$$

Since $e^{-2x_1-3x_2} = e^{-2x_1} e^{-3x_2}$, the integrals can be separated and we have

$$P(1 < X_1 < 2, 2 < X_2 < 3) = \left[\int_{x_1=1}^2 2e^{-2x_1} dx_1 \right] \left[\int_{x_2=2}^3 3e^{-3x_2} dx_2 \right]$$

Integrating, we have

$$\begin{aligned} P(1 < X_1 < 2, 2 < X_2 < 3) &= [-e^{-2x_1}]_1^2 [-e^{-3x_2}]_2^3 = [-e^{-4} + e^{-2}] [-e^{-9} + e^{-6}] \\ &= \left[\frac{1}{e^2} - \frac{1}{e^4} \right] \left[\frac{1}{e^6} - \frac{1}{e^9} \right] = \left[\frac{e^2 - 1}{e^4} \right] \left[\frac{e^3 - 1}{e^9} \right] \\ &= \frac{(e^2 - 1)(e^3 - 1)}{e^{13}} \end{aligned}$$

Next, we calculate the probability that $0 < X_1 < 2$ and $X_2 > 2$, which is given by

$$P(0 < X_1 < 2, X_2 > 2) = \int_{x_1=0}^2 \int_{x_2=2}^{\infty} 6e^{-2x_1-3x_2} dx_2 dx_1 = \int_{x_1=0}^2 \int_{x_2=2}^{\infty} 6e^{-2x_1} e^{-3x_2} dx_2 dx_1$$

Since the integrals can be separated, we have

$$P(0 < X_1 < 2, X_2 > 2) = \left[\int_{x_1=0}^2 2e^{-2x_1} dx_1 \right] \left[\int_{x_2=2}^{\infty} 3e^{-3x_2} dx_2 \right]$$

Integrating, we have

$$\begin{aligned} P(0 < X_1 < 2, X_2 > 2) &= [-e^{-2x_1}]_0^2 [-e^{-3x_2}]_2^{\infty} = [-e^{-4} + 1] [0 + e^{-6}] \\ &= \left[1 - \frac{1}{e^4} \right] \left[\frac{1}{e^6} \right] = \frac{e^4 - 1}{e^{10}} \end{aligned}$$

◆

EXAMPLE 4.10. If the joint probability density function of X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{8}(6 - x - y), & 0 < x < 2, 2 < y < 4 \\ 0, & \text{otherwise} \end{cases}$$

find

- (i) $P(X < 1 \cap Y < 3)$.
- (ii) $P(X < 1 | Y < 3)$.
- (iii) $P(X + Y < 3)$.

(Madras, Oct. 1997; Anna, April 2003)

Solution. (i) The required probability is given by

$$P(X < 1, Y < 3) = \int_{x=-\infty}^1 \int_{y=-\infty}^3 f(x, y) dy dx$$

Using the definition of f , we have

$$P(X < 1, Y < 3) = \int_{x=0}^1 \int_{y=2}^3 \frac{1}{8}(6-x-y) dy dx$$

Integrating y , we have

$$P(X < 1, Y < 3) = \frac{1}{8} \int_{x=0}^1 \left[(6-x)y - \frac{y^2}{2} \right]_2^3 dx$$

Simplifying, we have

$$\begin{aligned} P(X < 1, Y < 3) &= \frac{1}{8} \int_{x=0}^1 \left\{ \left[3(6-x) - \frac{9}{2} \right] - \left[2(6-x) - \frac{4}{2} \right] \right\} dx \\ &= \frac{1}{8} \int_{x=0}^1 \left[\frac{7}{2} - x \right] dx = \frac{1}{8} \left[\frac{7x}{2} - \frac{x^2}{2} \right]_0^1 = \frac{1}{8} \left[\left(\frac{7}{2} - \frac{1}{2} \right) - (0-0) \right] \\ &= \frac{1}{8}[3] = \frac{3}{8} \end{aligned}$$

(ii) By definition, the required probability is given by

$$P(X < 1 | Y < 3) = \frac{P(X < 1 \cap Y < 3)}{P(Y < 3)}$$

Now, we have

$$P(Y < 3) = \int_{y=-\infty}^3 \int_{x=-\infty}^{\infty} f(x,y) dx dy$$

Using the definition of f ,

$$P(Y < 3) = \int_{y=2}^3 \int_{x=0}^2 \frac{1}{8}(6-x-y) dx dy$$

Integrating x ,

$$\begin{aligned} P(Y < 3) &= \frac{1}{8} \int_{y=2}^3 \left[(6-y)x - \frac{x^2}{2} \right]_{x=0}^2 dy \\ P(Y < 3) &= \frac{1}{8} \int_{y=2}^3 \left[\left(2(6-y) - \frac{2^2}{2} \right) - (0-0) \right] dy \end{aligned}$$

$$P(Y < 3) = \frac{1}{8} \int_{y=2}^3 [10 - 2y] dy$$

Integrating y ,

$$\begin{aligned} P(Y < 3) &= \frac{1}{8} [10y - y^2]_{y=2}^3 = \frac{1}{8} [(30 - 9) - (20 - 4)] \\ &= \frac{1}{8} [21 - 16] = \frac{5}{8} \end{aligned}$$

Hence, the required probability is given by

$$P(X < 1 | Y < 3) = \frac{P(X < 1 \cap Y < 3)}{P(Y < 3)} = \frac{3/8}{5/8} = \frac{3}{5}$$

(iii) Here, we have to calculate the probability $P(X + Y < 3)$. Define the region A as

$$A = \{(x, y) : 0 < x < 2, 2 < y < 4, x + y < 3\}$$

The region A is as illustrated in Figure 4.6.

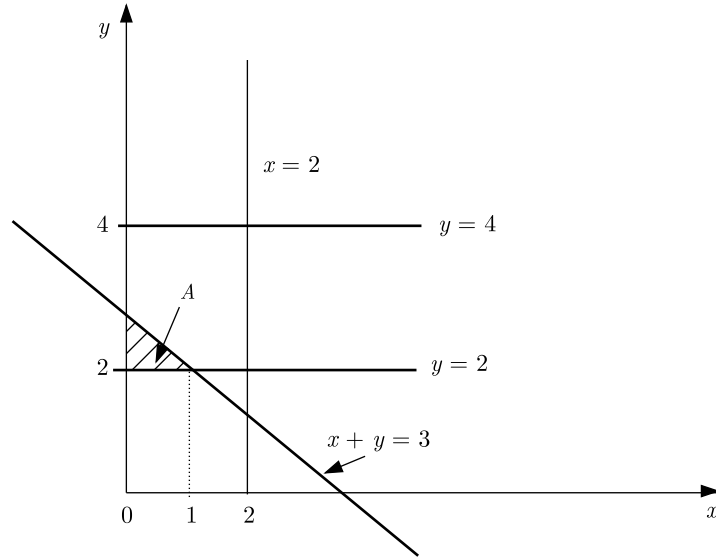


Figure 4.6: The region A .

By the definition of f , the probability that $X + Y < 3$ is equal to the probability that $(X, Y) \in A$. Hence, the required probability is given by

$$P[(X, Y) \in A] = \int_{x=0}^1 \int_{y=2}^{3-x} f(x, y) dy dx$$

$$= \int_{x=0}^1 \int_{y=2}^{3-x} \frac{1}{8} (6-x-y) dy dx$$

Integrating y , we have

$$\begin{aligned} P[(X, Y) \in A] &= \frac{1}{8} \int_{x=0}^1 \left[(6-x)y - \frac{y^2}{2} \right]_2^{3-x} dx \\ &= \frac{1}{8} \int_{x=0}^1 \left[\left((6-x)(3-x) - \frac{(3-x)^2}{2} \right) - \left(2(6-x) - \frac{2^2}{2} \right) \right] dx \\ P[(X, Y) \in A] &= \frac{1}{8} \int_{x=0}^1 \left[\frac{x^2}{2} - 4x + \frac{7}{2} \right] dx \end{aligned}$$

Integrating x , we have

$$\begin{aligned} P[(X, Y) \in A] &= \frac{1}{8} \left[\frac{x^3}{6} - 2x^2 + \frac{7x}{2} \right]_0^1 = \frac{1}{8} \left[\frac{1}{6} - 2 + \frac{7}{2} \right] \\ &= \frac{1}{8} \times \frac{10}{6} = \frac{5}{24} \end{aligned}$$

◆

EXAMPLE 4.11. If the joint PDF of the random variables (X, Y) is given by

$$f(x, y) = kxye^{-(x^2+y^2)}, x > 0, y > 0$$

find the value of k .

(Anna, Nov. 2004; Nov. 2007)

Solution. Since $f(x, y) \geq 0$ for all x and y , it is clear that $k \geq 0$. To find the value of k , we use the property that

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y) dy dx = 1$$

Using the definition of f , we have

$$\int_{x=0}^{\infty} \int_{y=0}^{\infty} kxye^{-(x^2+y^2)} dx dy = 1$$

Since $e^{-(x^2+y^2)} = e^{-x^2-y^2} = e^{-x^2} \cdot e^{-y^2}$, the integrals are separable and we have

$$k \left[\int_{x=0}^{\infty} xe^{-x^2} dx \right] \left[\int_{y=0}^{\infty} ye^{-y^2} dy \right] = 1$$

which can be rewritten as

$$\frac{k}{4} \left[\int_{x=0}^{\infty} 2xe^{-x^2} dx \right] \left[\int_{y=0}^{\infty} 2ye^{-y^2} dy \right] = 1$$

Integrating, we have

$$\frac{k}{4} \left[-e^{-x^2} \right]_0^{\infty} \left[-e^{-y^2} \right]_0^{\infty} = 1$$

Simplifying, we have

$$\frac{k}{4} [0 + 1] [0 + 1] = 1 \Rightarrow \frac{k}{4} = 1$$

Hence, $k = 4$. ◆

EXAMPLE 4.12. If $f(x, y) = Kye^{-x}$, $x > 0$, $0 < y < 2$ is the joint PDF of two random variables X and Y , what is the value of K ?

(Anna, Nov. 2007)

Solution. Since $f(x, y) \geq 0$ for all x and y , it is clear that $K \geq 0$. To find the value of k , we use the property that

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x, y) dy dx = 1$$

Using the definition of f , we have

$$\int_{x=0}^{\infty} \int_{y=0}^2 Kye^{-x} dy dx = 1$$

Since the integrals can be separated, it follows that

$$K \left[\int_{x=0}^{\infty} e^{-x} dx \right] \left[\int_{y=0}^2 y dy \right] = 1$$

Integrating, we get

$$K \left[-e^{-x} \right]_{x=0}^{\infty} \left[\frac{y^2}{2} \right]_{y=0}^2 = 1$$

Simplifying, we have

$$K [0 + 1] [2 - 0] = 1 \text{ or } 2K = 1$$

Hence, $K = \frac{1}{2}$. ◆

EXAMPLE 4.13. If two random variables, X and Y , have the PDF

$$f(x, y) = ke^{-(2x+y)} \text{ for } x, y > 0$$

evaluate k .

(Anna, Nov. 2005)

Solution. Since $f(x,y) \geq 0$ for all x and y , it is clear that $k \geq 0$. To find the value of k , we use the property that

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x,y) dy dx = 1$$

Using the definition of f , we have

$$\int_{x=0}^{\infty} \int_{y=0}^{\infty} k e^{-(2x+y)} dy dx = 1$$

Since $e^{-(2x+y)} = e^{-2x-y} = e^{-2x} \cdot e^{-y}$, it is clear that the integrals are separable and hence we have

$$k \left[\int_{x=0}^{\infty} e^{-2x} dx \right]_0^{\infty} \left[\int_{y=0}^{\infty} e^{-y} dy \right]_0^{\infty} = 1$$

Integrating, we have

$$k \left[-\frac{1}{2} e^{-2x} \right]_0^{\infty} \left[-e^{-y} \right]_0^{\infty} = 1$$

Simplifying, we have

$$k \left[0 + \frac{1}{2} \right] [0 + 1] = 1 \Rightarrow k \left[\frac{1}{2} \right] = 1$$

Hence, $k = 2$. ♦

EXAMPLE 4.14. The joint probability density function of two random variables X and Y is

$$f(x,y) = \begin{cases} cx(x-y) & \text{for } 0 < x < 2, -x < y < x \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate c .

(Anna, May 2007)

Solution. Since $f(x,y) \geq 0$ for all x and y , it is clear that $c \geq 0$. To find the value of c , we use the property that

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x,y) dy dx = 1$$

Using the definition of f , we have

$$\int_{x=0}^2 \int_{y=-x}^x cx(x-y) dy dx = 1$$

Thus,

$$c \int_{x=0}^2 x \int_{y=-x}^x (x-y) dy dx = 1$$

Using the property of definite integrals, we have

$$c \int_{x=0}^2 x \cdot 2 \int_{y=0}^x (x) dy dx = 1$$

Integrating y ,

$$c \int_{x=0}^2 2x^2 [y]_0^x dx = 1$$

Simplifying,

$$c \int_{x=0}^2 2x^3 dx = 1$$

Integrating x , we have

$$c \left[\frac{x^4}{2} \right]_0^2 = 1$$

Simplifying,

$$c[8 - 0] = 1 \Rightarrow c[8] = 1$$

Hence, $c = \frac{1}{8}$. ◆

EXAMPLE 4.15. Find the joint probability density function of the two random variables, X and Y , whose joint distribution function is given by

$$F(x,y) = \begin{cases} 1 - e^{-x} - e^{-y} + e^{-x-y} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Solution. The joint PDF of X and Y is given by

$$f(x,y) = \frac{\partial^2 F(x,y)}{\partial x \partial y} \text{ for all } x,y$$

For $x > 0, y > 0$, we note that

$$\frac{\partial F(x,y)}{\partial y} = e^{-y} - e^{-x-y}$$

and hence, it follows that

$$f(x,y) = \frac{\partial^2 F(x,y)}{\partial x \partial y} = e^{-x-y}$$

Hence, the joint PDF of X and Y is

$$f(x,y) = \begin{cases} e^{-x-y} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases} \quad \text{◆}$$

EXAMPLE 4.16. If the joint probability density of X and Y is given by

$$f(x,y) = \begin{cases} \frac{1}{2} & \text{for } x > 0, y > 0, x+y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

find:

(a) $P(X \leq 1, Y \leq 1)$.

(b) $P(X + Y < 1)$.

(c) $P(X > 2Y)$.

Solution. (a) The required probability is given by

$$P(X \leq 1, Y \leq 1) = \int_{x=0}^1 \int_{y=0}^1 f(x,y) dy dx = \int_{x=0}^1 \int_{y=0}^1 \frac{1}{2} dy dx$$

Integrating y , we have

$$P(X \leq 1, Y \leq 1) = \frac{1}{2} \int_{x=0}^1 [y]_0^1 dx = \frac{1}{2} \int_{x=0}^1 dx$$

Integrating x , we have

$$P(X \leq 1, Y \leq 1) = \frac{1}{2} [x]_0^1 = \frac{1}{2}$$

(b) The required probability is given by

$$P(X + Y < 1) = \int_{x=0}^1 \int_{y=0}^{1-x} f(x,y) dy dx = \int_{x=0}^1 \int_{y=0}^{1-x} \frac{1}{2} dy dx$$

Integrating y , we have

$$P(X + Y < 1) = \frac{1}{2} \int_{x=0}^1 [y]_0^{1-x} dx = \frac{1}{2} \int_{x=0}^1 (1-x) dx$$

Integrating x , we have

$$P(X + Y < 1) = \frac{1}{2} \left[-\frac{(1-x)^2}{2} \right]_0^1 = \frac{1}{2} \left[0 + \frac{1}{2} \right] = \frac{1}{4}$$

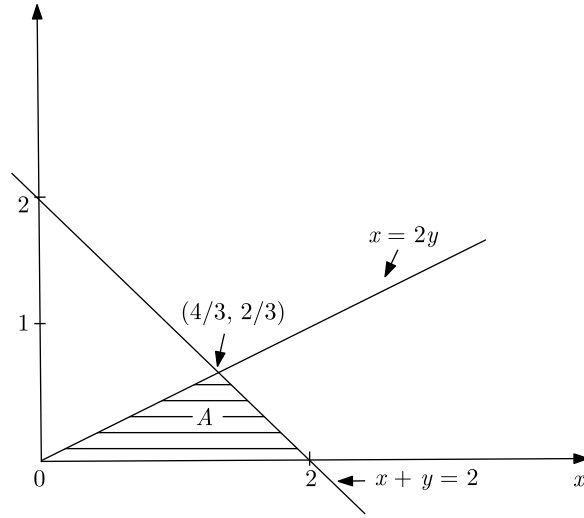


Figure 4.7: Region A.

(c) Define the region A as

$$A = \{(x, y) \in \mathbb{R}^2 : x > 0, y > 0, x + y < 2, x > 2y\}$$

Then the region A is as illustrated in Figure 4.7.

From the definition of f , it follows that the required probability that $X > 2Y$ is equal to the probability that $(X, Y) \in A$.

Hence, we have

$$P[(X, Y) \in A] = \int_{y=0}^{\frac{2}{3}} \int_{x=2y}^{2-y} f(x, y) dx dy = \int_{y=0}^{\frac{2}{3}} \int_{x=2y}^{2-y} \frac{1}{2} dx dy$$

Integrating x , we have

$$\begin{aligned} P[(X, Y) \in A] &= \frac{1}{2} \int_{y=0}^{\frac{2}{3}} [x]_{2y}^{2-y} dy = \frac{1}{2} \int_{y=0}^{\frac{2}{3}} [(2-y) - 2y] dy \\ &= \frac{1}{2} \int_{y=0}^{\frac{2}{3}} (2-3y) dy \end{aligned}$$

Integrating y , we have

$$\begin{aligned} P[(X, Y) \in A] &= \frac{1}{2} \left[-\frac{1}{3} \frac{(2-3y)^2}{2} \right]_0^{\frac{2}{3}} = \frac{1}{2} \left[0 + \frac{1}{3} \times 2 \right] \\ &= \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} \end{aligned}$$

◆

PROBLEM SET 4.1

- Three balls are drawn at random without replacement from a box containing 3 white, 2 red and 3 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn, find the joint probability distribution of (X, Y) .
- Let (X, Y) be a continuous random variable with the joint probability density function given by

$$f(x, y) = \begin{cases} 4xye^{-x^2-y^2} & \text{if } 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find the distribution function of (X, Y) .

- If X and Y have the joint probability density function

$$f(x, y) = \begin{cases} Ky(y-x) & \text{if } -y < x < y, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

find the value of K .

- If X and Y have the joint probability density function

$$f(x, y) = \begin{cases} K \frac{1+x+y}{(1+x)^4 (1+y)^4} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

find the value of K .

- If X and Y have the joint probability density function

$$f(x, y) = \begin{cases} K(x+2y) & \text{if } 0 < x < 2, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

find the value of K .

- If X and Y have the joint probability density function

$$f(x, y) = \begin{cases} K(x+y) & \text{if } 0 < x < 2, 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

find (a) the value of K and (b) $P(X+Y < 1)$.

- If the joint probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} 2e^{-x-2y} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

find $P(X+Y < 1)$.

- If the joint probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} 6x^2y & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

find $P(Y > X)$.

9. If the joint probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{y}{2} (1+x) & \text{if } 0 < x < 2, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

find $P(X + Y > 1)$.

10. If the joint probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} 12e^{-3x-4y} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

find (a) $P(1 < X < 3, 2 < Y < 4)$, (b) $P(X < 2, Y < 2)$ and (c) $P(X + Y < 1)$.

11. If the joint probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} 1 & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

find (a) $P\{X < \frac{1}{2}, Y > \frac{1}{2}\}$ and (b) $P(X + Y < \frac{1}{2})$.

12. If the joint probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} 2 & \text{if } x > 0, y > 0, x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

find (a) $P(X < \frac{1}{2}, Y < \frac{1}{2})$, (b) $P(X + Y < \frac{1}{2})$, and (c) $P(X > 2Y)$.

4.2 MARGINAL DISTRIBUTIONS

First, we define the marginal distributions for the discrete case.

Definition 4.6. If X and Y are discrete random variables and f is the joint probability mass function for the random variables X and Y , the function given by

$$f_X(x) = \sum_y f(x, y)$$

for each x within the range of X is called the *marginal density function* or the *marginal distribution* of X . Similarly, the function given by

$$f_Y(y) = \sum_x f(x, y)$$

for each y within the range of Y is called the *marginal density function* or the *marginal distribution* of Y .

Remark 4.2. Suppose that (X, Y) is a two-dimensional random variable with mass points (x_i, y_j) , where $i = 1, 2, \dots$ and $j = 1, 2, \dots$. Then the marginal distribution of X is defined by

$$f_X(x_i) = \sum_j f(x_i, y_j) \quad \text{for each } i = 1, 2, \dots$$

Note that f_X is a probability mass function, i.e.

- (a) $f_X(x_i) \geq 0$ for each i ; and
- (b) $\sum_i f_X(x_i) = 1$.

Similarly, the marginal distribution of Y is defined by

$$f_Y(y_j) = \sum_i f(x_i, y_j) \text{ for each } j = 1, 2, \dots$$

Note that f_Y is a probability mass function, i.e.

- (a) $f_Y(y_j) \geq 0$ for each j .
- (b) $\sum_j f_Y(y_j) = 1$.

Next, we define the marginal density functions for the continuous case. Here, the probability distributions are replaced by probability densities and the summations are replaced by integrals.

Definition 4.7. If X and Y are continuous random variables and f is the joint probability density function for the random variables, X and Y , the function given by

$$f_X(x) = \int_{y=-\infty}^{\infty} f(x, y) dy \text{ for } -\infty < x < \infty$$

is called the *marginal density* of X . Similarly, the function given by

$$f_Y(y) = \int_{x=-\infty}^{\infty} f(x, y) dx \text{ for } -\infty < y < \infty$$

is called the *marginal density* of Y .

Remark 4.3. If (X, Y) is a two-dimensional continuous random variable with joint probability density function f , then we remark that the marginal density functions, f_X and f_Y , are probability density functions, i.e. we have

- (a) $f_X(x) \geq 0$ for each x .
- (b) $\int_{x=-\infty}^{\infty} f_X(x) dx = 1$.

and also that

- (a) $f_Y(y) \geq 0$ for each y .
- (b) $\int_{y=-\infty}^{\infty} f_Y(y) dy = 1$.

Next, we define the marginal cumulative distribution functions for the two-dimensional random variable (X, Y) (discrete or continuous).

Definition 4.8. Let (X, Y) be a random variable with joint cumulative distribution function F . Then the *marginal cumulative distribution function* of X is denoted by F_X and is defined by

$$F_X(x) = F(x, \infty) = \lim_{y \rightarrow \infty} F(x, y)$$

If (X, Y) is of discrete type, then

$$F_X(x) = \sum_{x_i \leq x} P(X = x_i) = \sum_{x_i \leq x} f_X(x_i)$$

and if (X, Y) is of continuous type, then

$$F_X(x) = \int_{t=-\infty}^x f_X(t) dt$$

Similarly, the marginal cumulative distribution of Y is denoted by F_Y and is defined by

$$F_Y(y) = F(y, \infty) = \lim_{x \rightarrow \infty} F(x, y)$$

If (X, Y) is of discrete type, then

$$F_Y(y) = \sum_{y_j \leq y} P(X = y_j) = \sum_{y_j \leq y} f_Y(y_j)$$

and if (X, Y) is of continuous type, then

$$F_Y(y) = \int_{t=-\infty}^y f_Y(t) dt$$

Next, we define the *independence* of random variables.

Definition 4.9. Let (X, Y) be a two-dimensional random variable (discrete or continuous). Then X and Y are called *independent* if and only if

$$F(x, y) = F_X(x)F_Y(y)$$

where F is the joint cumulative distribution function of X and Y , and F_X and F_Y are the marginal cumulative distribution functions of X and Y , respectively.

Next, we establish a simple result for independent random variables.

Theorem 4.7. If X and Y are independent random variables and $a < b, c < d$ are real numbers, then

$$P(a < X \leq b, c < Y \leq d) = P(a < X \leq b) P(c < Y \leq d)$$

Proof. Let X and Y be independent random variables. Let $a < b, c < d$ be any real numbers. By Theorem 4.1, we have

$$P(a < X \leq b, c < Y \leq d) = F(b, d) - F(b, c) - F(a, d) + F(a, c) \quad (4.3)$$

Since X and Y are independent, Eq. (4.3) can be simplified as

$$P(a < X \leq b, c < Y \leq d) = F_X(b)F_Y(d) - F_X(b)F_Y(c) - F_X(a)F_Y(d) + F_X(a)F_Y(c)$$

Factorizing the R.H.S. of the above equation, we have

$$\begin{aligned} P(a < X \leq b, c < Y \leq d) &= [F_X(b) - F_X(a)][F_Y(d) - F_Y(c)] \\ &= P(a < X \leq b) P(c < Y \leq d) \end{aligned}$$

◆

Next, we establish an important characterization for independent random variables.

Theorem 4.8. (a) Two discrete random variables, X and Y , are independent if and only if

$$P(X = x_i, Y = y_j) = P(X = x_i)P(Y = y_j) \text{ for all pairs } (x_i, y_j)$$

or equivalently

$$f(x_i, y_j) = f_X(x_i) f_Y(y_j) \text{ for all pairs } (x_i, y_j)$$

where f is the joint probability mass function of X and Y , and f_X, f_Y are the marginal probability mass functions of X and Y , respectively.

(b) Two continuous random variables, X and Y , are independent if and only if

$$f(x, y) = f_X(x) f_Y(y) \text{ for all } (x, y) \in \mathbb{R}^2$$

where f is the joint probability density function of X and Y , and f_X, f_Y are the marginal probability density functions of X and Y , respectively.

Proof. (a) Let X, Y be independent discrete random variables. By Theorem 4.3, for any $\varepsilon > 0$ and mass point (x_i, y_j) for (X, Y) , we have

$$P(x_i - \varepsilon < X < x_i, y_j - \varepsilon < Y \leq y_j) = P(x_i - \varepsilon < X < x_i) P(y_j - \varepsilon < Y \leq y_j)$$

Taking limits as $\varepsilon \rightarrow 0$, we have

$$P(X = x_i, Y = y_j) = P(X = x_i) P(Y = y_j)$$

Conversely, suppose that

$$P(X = x_i, Y = y_j) = P(X = x_i)P(Y = y_j) \text{ for all pairs } (x_i, y_j) \quad (4.4)$$

We shall establish that (X, Y) are independent random variables.

Note that

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} P(X = x_i, Y = y_j)$$

Using Eq. (4.4), we have

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} P(X = x_i)P(Y = y_j)$$

It follows that

$$F(x, y) = \left[\sum_{x_i \leq x} P(X = x_i) \right] \left[\sum_{y_j \leq y} P(Y = y_j) \right] = F_X(x_i) F_Y(y_j)$$

Hence, X and Y are independent random variables.

(b) Let X and Y be independent continuous random variables.

By Theorem 4.7, it follows that if a, b, c, d are any real numbers with $a < b$ and $c < d$, then

$$\int_{x=a}^b \int_{y=c}^d f(x, y) dy dx = \left[\int_{x=a}^b f_X(x) dx \right] \left[\int_{y=c}^d f_Y(y) dy \right]$$

Hence, we have

$$\int_{x=a}^b \int_{y=c}^d f(x, y) dy dx = \int_{x=a}^b \int_{y=c}^d f_X(x) f_Y(y) dy dx$$

Since a, b, c and d are arbitrary real numbers with $a < b$ and $c < d$, we must have

$$f(x, y) = f_X(x) f_Y(y) \quad \text{for all } (x, y) \in \mathbb{R}^2 \quad (4.5)$$

Conversely, suppose that Eq. (4.5) holds. We shall show that X and Y are independent random variables.

Note that

$$F(x, y) = \int_{s=-\infty}^x \int_{t=-\infty}^y f(s, t) dt ds$$

Using Eq. (4.5), we have

$$F(x, y) = \int_{s=-\infty}^x \int_{t=-\infty}^y f_X(s) f_Y(t) dt ds$$

Since the integrals are separable, we have

$$F(x, y) = \left[\int_{s=-\infty}^x f_X(s) ds \right] \left[\int_{t=-\infty}^y f_Y(t) dt \right] = F_X(x) F_Y(y)$$

Thus, X and Y are independent random variables. ◆

Definition 4.10. Two random variables X and Y are called *identically distributed* if X and Y have the same distribution function, i.e.

$$F_X(x) = F_Y(y) \quad \text{for all } x \in \mathbf{R}$$

EXAMPLE 4.17. The joint distribution of X and Y is given by

$$f(x,y) = \frac{x+y}{21}, \quad x = 1, 2, 3; y = 1, 2$$

Find the marginal distributions of X and Y . Find also the means of X and Y .

(Madras, Oct. 1996; Oct. 2000)

Solution. By definition, the marginal distribution of X is

$$f_X(x) = \sum_{y=1}^2 f(x,y) \quad \text{for } x = 1, 2, 3$$

and the marginal distribution of Y is given by

$$f_Y(y) = \sum_{x=1}^3 f(x,y) \quad \text{for } y = 1, 2$$

The calculations are summarized in the following table:

X \ Y	Y		$f_X(x)$
	1	2	
1	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{5}{21}$
2	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{7}{21}$
3	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{9}{21}$
$f_Y(y)$	$\frac{9}{21}$	$\frac{12}{21}$	1

Hence, the marginal distribution of X is given by

x	1	2	3
$f_X(x)$	$\frac{5}{21}$	$\frac{7}{21}$	$\frac{9}{21}$

Hence, the mean of X is

$$\mu_X = \sum_{x=1}^3 x f_X(x) = \frac{5}{21} + \frac{2}{3} + \frac{9}{7} = \frac{46}{21} = 2.1905$$

Also, the marginal distribution of Y is

y	1	2
$f_Y(y)$	$\frac{3}{7}$	$\frac{4}{7}$

Hence, the mean of Y is

$$\mu_Y = \sum_{y=1}^2 y f_Y(y) = \frac{3}{7} + \frac{8}{7} = \frac{11}{7} = 1.5714$$

◆

EXAMPLE 4.18. The joint probability distribution of X and Y is given by

$$f(x, y) = \frac{1}{27}(2x + y), \quad x = 0, 1, 2; y = 0, 1, 2$$

- (a) Find the marginal distributions of X and Y .
 (b) Are X and Y independent random variables?

(Madras, April 2000)

Solution. (a) By definition, the marginal distribution of X is given by

$$f_X(x) = \sum_{y=0}^2 f(x, y) \quad \text{for } x = 0, 1, 2$$

and the marginal distribution of Y is given by

$$f_Y(y) = \sum_{x=0}^2 f(x, y) \quad \text{for } y = 0, 1, 2$$

The calculations are summarized in the following table:

$X \backslash Y$	0	1	2	$f_X(x)$
0	0	$\frac{1}{27}$	$\frac{2}{27}$	$\frac{3}{27}$
1	$\frac{2}{27}$	$\frac{3}{27}$	$\frac{4}{27}$	$\frac{9}{27}$
2	$\frac{4}{27}$	$\frac{5}{27}$	$\frac{6}{27}$	$\frac{15}{27}$
$f_Y(y)$	$\frac{6}{27}$	$\frac{9}{27}$	$\frac{12}{27}$	1

Hence, the marginal distribution of X is

x	0	1	2
$f_X(x)$	$\frac{1}{9}$	$\frac{1}{3}$	$\frac{5}{9}$

Also, the marginal distribution of Y is

y	0	1	2
$f_Y(y)$	$\frac{2}{9}$	$\frac{1}{3}$	$\frac{4}{9}$

(b) No, X and Y are not independent random variables because

$$f(x, y) \neq f_X(x)f_Y(y) \text{ for some values of } (x, y)$$

For example, $f(0, 0) = 0$ but $f_X(0)f_Y(0) = \frac{1}{9} \times \frac{2}{9} = \frac{2}{81}$. Hence,

$$f(0, 0) \neq f_X(0)f_Y(0)$$

Also, $f(1, 1) = \frac{1}{27}$ but $f_X(1)f_Y(1) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$. Hence,

$$f(1, 1) \neq f_X(1)f_Y(1)$$

These calculations show that X and Y are not independent random variables. ♦

EXAMPLE 4.19. The following table gives the probability distribution of X and Y :

$\begin{matrix} \text{Y} \backslash \text{X} \\ \text{Y} \end{matrix}$	1	2	3
1	0.1	0.1	0.2
2	0.2	0.3	0.1

Find (a) marginal density function of X , (b) marginal density function of Y . (*Anna, Nov. 2003*)

Solution. (a) By definition, the marginal density function of X is given by

$$f_X(x) = \sum_{y=1}^2 f(x, y) \text{ for } x = 1, 2, 3$$

Hence, we have

x	1	2	3
$f_X(x)$	0.3	0.4	0.3

(b) By definition, the marginal density function of Y is given by

$$f_Y(y) = \sum_{x=1}^3 f(x, y) \text{ for } y = 1, 2$$

Hence, we have

y	1	2
$f_Y(y)$	0.4	0.6

♦

EXAMPLE 4.20. For the bivariate probability distribution of (X, Y) given below, find $P(X \leq 1)$, $P(Y \leq 3)$ and $P(X + Y \leq 4)$.

X \ Y	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

(Anna, Nov. 2007)

Solution. The marginal distribution of X is given by

x	0	1	2
$f_X(x)$	$\frac{1}{4}$	$\frac{5}{8}$	$\frac{1}{8}$

and the marginal distribution of Y is given by

y	1	2	3	4	5	6
$f_Y(y)$	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{3}{16}$	$\frac{1}{4}$

Thus, it follows that

$$P(X \leq 1) = f_X(0) + f_X(1) = \frac{1}{4} + \frac{5}{8} = \frac{7}{8}$$

and

$$P(Y \leq 3) = f_Y(1) + f_Y(2) + f_Y(3) = \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}$$

Note also that the mass points corresponding to the event $X + Y \leq 4$ are given by

$$(0, 1), (0, 2), (0, 3), (0, 4), (1, 1), (1, 2), (1, 3), (2, 1), (2, 2).$$

Thus, from the given probability distribution, it is easy to calculate and obtain

$$P(X + Y \leq 4) = 0 + 0 + \frac{1}{32} + \frac{2}{32} + \frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{32} + \frac{1}{32} = \frac{13}{32}$$

◆

EXAMPLE 4.21. The joint density function of two continuous random variables X and Y is

$$f(x, y) = \begin{cases} Cxy, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find the value of the constant C .
- (ii) Find $P(X \geq 3, Y \leq 2)$.
- (iii) Find the marginal distribution function of X .

(Anna, May 2000)

Solution. (i) Since f is the probability density function, we must have

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} f(x,y) dy dx = 1$$

Using the definition of f , we have

$$\int_{x=0}^4 \int_{y=1}^5 Cxy dy dx = 1$$

Since the integrals are separable, we have

$$C \left[\int_{x=0}^4 x dx \right] \left[\int_{y=1}^5 y dy \right] = 1$$

Integrating, we have

$$C \left[\frac{x^2}{2} \right]_0^4 \left[\frac{y^2}{2} \right]_1^5 = 1$$

Simplifying, we have

$$C[8-0] \left[\frac{25}{2} - \frac{1}{2} \right] = 1$$

or

$$C[8][12] = 1 \Rightarrow C[96] = 1$$

Hence, $C = \frac{1}{96}$.

Thus, the joint density function of (X, Y) is given by

$$f(x,y) = \begin{cases} \frac{1}{96} xy, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

(ii) The required probability is given by

$$P(X \geq 3, Y \leq 2) = \int_{x=3}^4 \int_{y=1}^2 f(x,y) dy dx = \int_{x=3}^4 \int_{y=1}^2 \frac{1}{96} xy dy dx$$

Integrating y , we have

$$\begin{aligned} P(X \geq 3, Y \leq 2) &= \frac{1}{96} \int_{x=3}^4 x \left[\frac{y^2}{2} \right]_1^2 dx = \frac{1}{96} \int_{x=3}^4 x \left[2 - \frac{1}{2} \right] dx \\ &= \frac{1}{96} \int_{x=3}^4 x \left[\frac{3}{2} \right] dx = \frac{1}{64} \int_{x=3}^4 x dx \end{aligned}$$

Integrating x , we have

$$\begin{aligned} P(X \geq 3, Y \leq 2) &= \frac{1}{64} \left[\frac{x^2}{2} \right]_3^4 = \frac{1}{64} \left[8 - \frac{9}{2} \right] \\ &= \frac{1}{64} \times \frac{7}{2} = \frac{7}{128} \end{aligned}$$

(iii) Next, we find the marginal density function of X .

If $0 < x < 4$, then we have

$$f_X(x) = \int_{y=1}^5 f(x,y) dy = \int_{y=1}^5 \frac{1}{96} xy dy$$

Integrating y , we have

$$\begin{aligned} f_X(x) &= \frac{x}{96} \left[\frac{y^2}{2} \right]_1^5 = \frac{x}{96} \left[\frac{25}{2} - \frac{1}{2} \right] \\ &= \frac{x}{96} \times 12 = \frac{x}{8} \end{aligned}$$

Hence, the marginal density of X is given by

$$f_X(x) = \begin{cases} \frac{x}{8} & \text{for } 0 < x < 4 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 4.22. Given the joint probability density function of (X, Y) as

$$f(x,y) = \begin{cases} 8xy, & 0 < y < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(i) Show that X and Y are not independent and (ii) find $P(X + Y > 1)$.

(Anna, May 1999)

Solution. (i) First, we find the marginal density functions of X and Y . Define the region A as

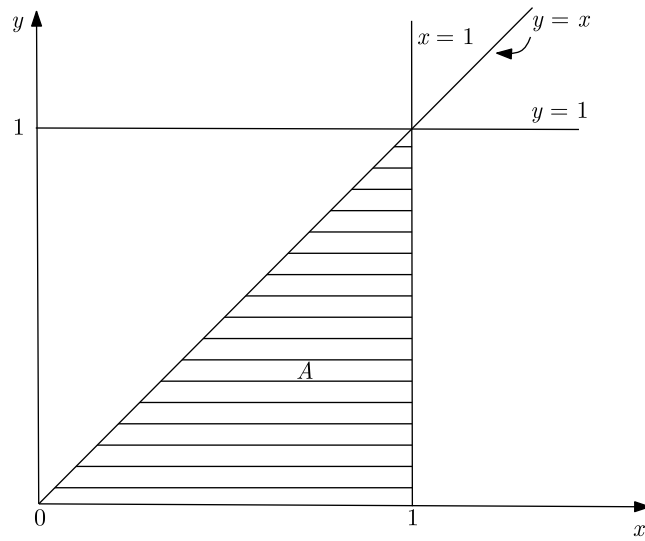
$$A = \{(x,y) \in \mathbb{R}^2 : 0 < y < x < 1\}$$

Then $f(x,y) = 8xy$ for $(x,y) \in A$ and zero elsewhere. The region is as illustrated in Figure 4.8.

We begin with finding the marginal density function of X .

If $0 < x < 1$, then we have

$$f_X(x) = \int_{y=0}^x f(x,y) dy \quad [\text{See Figure 4.8}]$$

**Figure 4.8:** Region A.

Hence,

$$f_X(x) = \int_{y=0}^x 8xy \, dy = 4x [y^2]_0^x = 4x^3$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} 4x^3 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the marginal density function of Y .

If $0 < y < 1$, then we have

$$f_Y(y) = \int_{x=y}^1 f(x,y) \, dx \quad [\text{See Figure 4.8}]$$

Hence,

$$f_Y(y) = \int_{x=y}^1 8xy \, dx = 4y [x^2]_{x=y}^1 = 4y [1 - y^2]$$

Hence, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 4y(1 - y^2) & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Note that

$$f_X(x)f_Y(y) = \begin{cases} 16x^3y(1-y^2) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent random variables.

(ii) Next, we compute $P(X+Y > 1)$. Define the region B as

$$B = \{(x,y) \in \mathbb{R}^2 : 0 < y < x < 1, x+y > 1\}$$

Then the region B is as illustrated in Figure 4.9. By the definition of f , the required probability is equal to the probability that $(X,Y) \in B$.

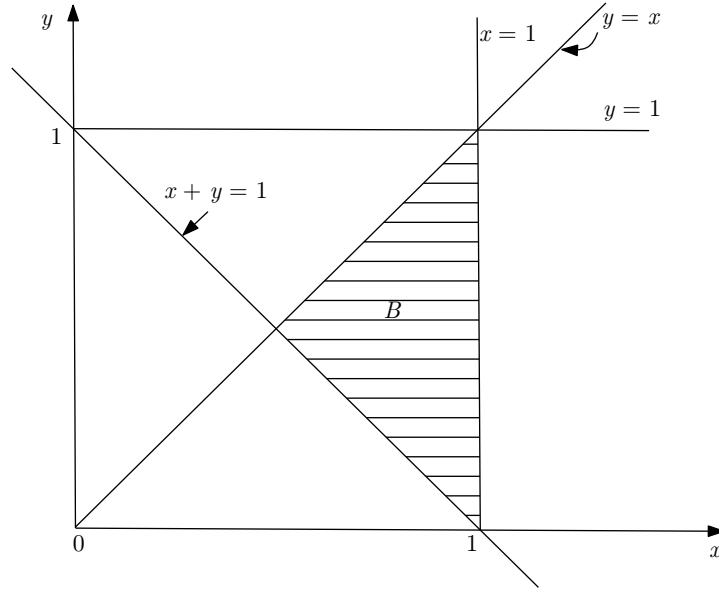


Figure 4.9: Region B .

Hence, the required probability is given by

$$P[(X,Y) \in B] = \int_{x=\frac{1}{2}}^1 \int_{y=1-x}^x f(x,y) dy dx = \int_{x=\frac{1}{2}}^1 \int_{y=1-x}^x 8xy dy dx$$

Integrating y , we have

$$\begin{aligned} P[(X,Y) \in B] &= \int_{x=\frac{1}{2}}^1 4x [y^2]_{y=1-x}^x dx \\ &= \int_{x=\frac{1}{2}}^1 4x [x^2 - (1-x)^2] dx \end{aligned}$$

$$= \int_{x=\frac{1}{2}}^1 4x(2x-1) dx = \int_{x=\frac{1}{2}}^1 (8x^2 - 4x) dx$$

Integrating x ,

$$\begin{aligned} P[(X, Y) \in B] &= \left[\frac{8x^3}{3} - 2x^2 \right]_{\frac{1}{2}}^1 \\ &= \left[\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - \frac{1}{2} \right) \right] = \left[\frac{2}{3} + \frac{1}{6} \right] \end{aligned}$$

Hence, we have

$$P(X + Y > 1) = P[(X, Y) \in B] = \frac{5}{6}$$

◆

EXAMPLE 4.23. If two random variables X and Y have the joint density

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

find the probabilities $P(0.2 < X < 0.5)$ and $P(0.4 < Y < 0.6)$.

(Anna, April 2004)

Solution. First, we find the marginal density functions of X and Y .

If $0 < x < 1$, then we have

$$f_X(x) = \int_{y=0}^1 f(x, y) dy = \int_{y=0}^1 \frac{6}{5} [x + y^2] dy$$

Integrating y ,

$$f_X(x) = \frac{6}{5} \left[xy + \frac{y^3}{3} \right]_{y=0}^1 = \frac{6}{5} \left[x + \frac{1}{3} \right]$$

Hence, the marginal density of X is given by

$$f_X(x) = \begin{cases} \frac{6}{5} \left(x + \frac{1}{3} \right) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

If $0 < y < 1$, then we have

$$f_Y(y) = \int_{x=0}^1 f(x, y) dx = \int_{x=0}^1 \frac{6}{5} (x + y^2) dx$$

Integrating x , we have

$$f_Y(y) = \frac{6}{5} \left[\frac{x^2}{2} + xy^2 \right]_{x=0}^1 = \frac{6}{5} \left[\frac{1}{2} + y^2 \right]$$

Hence, the marginal density of Y is given by

$$f_Y(y) = \begin{cases} \frac{6}{5} (y^2 + \frac{1}{2}) & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we compute the required probabilities.

We find that

$$P(0.2 < X < 0.5) = \int_{x=0.2}^{0.5} f_X(x) dx = \int_{x=0.2}^{0.5} \frac{6}{5} \left(x + \frac{1}{3} \right) dx$$

Integrating, we have

$$\begin{aligned} P(0.2 < X < 0.5) &= \frac{6}{5} \left[\frac{x^2}{2} + \frac{x}{3} \right]_{0.2}^{0.5} \\ &= \frac{6}{5} [0.2917 - 0.0867] = 0.2460 \end{aligned}$$

Next, we find that

$$P(0.4 < Y < 0.6) = \int_{y=0.4}^{0.6} f_Y(y) dy = \int_{y=0.4}^{0.6} \frac{6}{5} \left(y^2 + \frac{1}{2} \right) dy$$

Integrating, we have

$$\begin{aligned} P(0.4 < Y < 0.6) &= \frac{6}{5} \left[\frac{y^3}{3} + \frac{y}{2} \right]_{0.4}^{0.6} \\ &= \frac{6}{5} [0.3720 - 0.2213] = 0.1808 \end{aligned}$$

◆

EXAMPLE 4.24. The joint density function of X and Y is

$$f(x, y) = e^{-(x+y)} \text{ when } x > 0, y > 0$$

Are X and Y independent?

(Madras, Oct. 1997; April 2002)

Solution. The marginal density of X is given by

$$f_X(x) = e^{-x} \text{ when } x > 0$$

and the marginal density of Y is given by

$$f_Y(y) = e^{-y} \text{ when } y > 0$$

Since $f(x, y) = f_X(x) \cdot f_Y(y)$, X and Y are independent.

◆

EXAMPLE 4.25. If the random variables X and Y have joint probability density function given by

$$f(x,y) = \begin{cases} 4xye^{-(x^2+y^2)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

check whether they are independent.

(Anna, Model 2003)

Solution. The marginal density function of X is given by

$$f_X(x) = \begin{cases} 2xe^{-x^2} & \text{for } x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 2ye^{-y^2} & \text{for } y \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) = f_X(x)f_Y(y)$ for all $(x,y) \in \mathbb{R}^2$, it follows that X and Y are independent random variables. ♦

EXAMPLE 4.26. If the joint density of X and Y is given by

$$f(x,y) = \begin{cases} \frac{1}{2}xe^{-y}, & 0 < x < 2, y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

find the marginal density functions, and check whether X and Y are independent.

(Madras, Oct. 2002; Anna, April 2003)

Solution. The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{x}{2} & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} e^{-y} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) = f_X(x)f_Y(y)$ for all $(x,y) \in \mathbb{R}^2$, we conclude that X and Y are independent random variables. ♦

EXAMPLE 4.27. Given the joint density function of X and Y as

$$f(x,y) = \begin{cases} \frac{1}{4}(1+xy), & |x| < 1, |y| < 1 \\ 0, & \text{elsewhere} \end{cases}$$

find the marginal densities and check whether X and Y are independent.

(Madras, Oct. 2002)

Solution. The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{1}{2} & \text{for } |x| < 1 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{2} & \text{for } |y| < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Hence, it follows that

$$f_X(x)f_Y(y) = \begin{cases} \frac{1}{4} & \text{for } |x| < 1, |y| < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, it follows that X and Y are not independent random variables. $\blacklozenge$

EXAMPLE 4.28. The random variables X and Y have the following joint probability density function:

$$f(x,y) = \begin{cases} K(4-x-y), & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Find K , the marginal densities and test whether X and Y are independent.

(Madras, April 2002)

Solution. A simple calculation gives $K = \frac{1}{8}$.

The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{3-x}{4} & \text{for } 0 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{3-y}{4} & \text{for } 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Note that

$$f_X(x)f_Y(y) = \begin{cases} \frac{(3-x)(3-y)}{16} & \text{for } 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent random variables. $\blacklozenge$

EXAMPLE 4.29. A gun is aimed at a certain point (origin of the coordinate system). Because of the random factors, the actual hit point can be any point (X, Y) in a circle of radius R about the origin. Assume that the joint density of X and Y is constant in this circle given by

$$f(x, y) = \begin{cases} C & \text{for } x^2 + y^2 \leq R^2 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Compute C .
(ii) Show that the marginal density of X is given by

$$f_X(x) = \begin{cases} \frac{2}{\pi R} \sqrt{1 - \left[\frac{x}{R}\right]^2} & \text{for } -R \leq x \leq R \\ 0 & \text{otherwise} \end{cases}$$

(Bharathiar, Nov. 1996)

Solution. (i) Since f is the joint PDF of (X, Y) , $f(x, y) \geq 0$ for all (x, y) . Hence, $C \geq 0$. To find the value of C , we use the property that

$$\iint_{x^2+y^2 \leq R^2} f(x, y) dy dx = \iint_{x^2+y^2 \leq R^2} C dy dx = 1$$

In other words,

$$C [\text{Area of the circle } x^2 + y^2 = R^2] = 1$$

or

$$C [\pi R^2] = 1 \Rightarrow C = \frac{1}{\pi R^2}$$

Alternatively, we can use polar coordinates $x = R \cos \theta, y = R \sin \theta$ to note that

$$\int_{r=0}^R \int_{\theta=0}^{2\pi} Cr d\theta dr = 1$$

The integrals are separable and we have

$$C \left[\int_{r=0}^R r dr \right] \left[\int_{\theta=0}^{2\pi} d\theta \right] = 1$$

Integrating, we have

$$C \left[\frac{r^2}{2} \right]_0^R [\theta]_0^{2\pi} = 1$$

Simplifying, we have

$$C \left[\frac{R^2}{2} \right] [2\pi] = 1 \Rightarrow C [R^2 \pi] = 1$$

Hence, the value of C is given by $C = \frac{1}{\pi R^2}$.

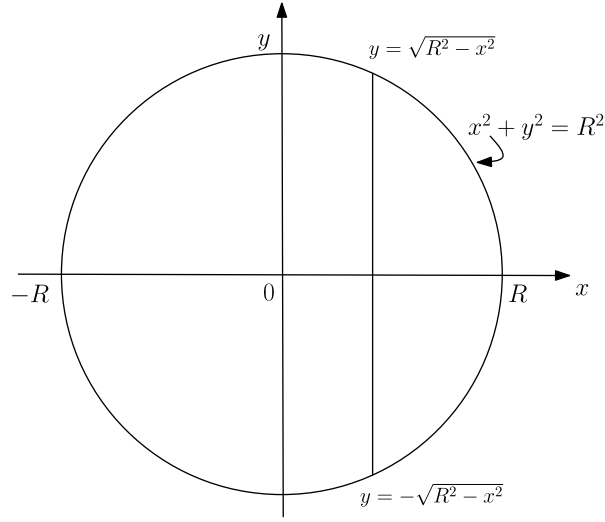


Figure 4.10: The circle $x^2 + y^2 = R^2$.

(ii) Next, we find the marginal density of X . If $|x| \leq R$, then we have

$$f_X(x) = \int_{y=-\sqrt{R^2-x^2}}^{y=\sqrt{R^2-x^2}} f(x,y) dy = \int_{y=-\sqrt{R^2-x^2}}^{y=\sqrt{R^2-x^2}} \frac{1}{\pi R^2} dy$$

Using the property of definite integrals, we have

$$\begin{aligned} f_X(x) &= \frac{2}{\pi R^2} \int_{y=0}^{\sqrt{R^2-x^2}} dy = \frac{2}{\pi R^2} [y]_0^{\sqrt{R^2-x^2}} \\ &= \frac{2}{\pi R^2} \sqrt{R^2-x^2} = \frac{2}{\pi R} \sqrt{\frac{R^2-x^2}{R^2}} = \frac{2}{\pi R} \sqrt{1-\frac{x^2}{R^2}} \end{aligned}$$

Thus, the marginal density of X is given by

$$f_X(x) = \begin{cases} \frac{2}{\pi R} \sqrt{1 - \left[\frac{x}{R}\right]^2} & \text{for } -R \leq x \leq R \\ 0 & \text{otherwise} \end{cases}$$

◆

EXAMPLE 4.30. The random variables X and Y have the joint probability density function given by

$$f(x,y) = \begin{cases} 6(1-x-y) & \text{for } x > 0, y > 0, x+y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal density functions of X and Y . Examine if X and Y are independent.

(Madras, April 2000).

Solution. Define the region A as

$$A = \{(x, y) \in \mathbb{R}^2 : x > 0, y > 0, x + y < 1\}$$

Then the region A is as illustrated in Figure 4.11.

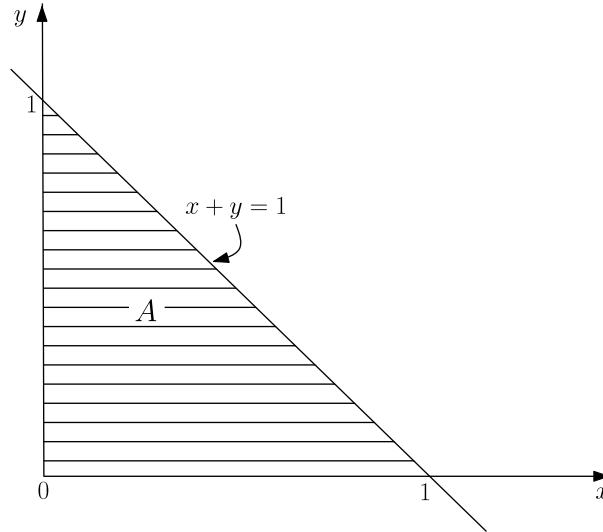


Figure 4.11: The region A .

First, we find the marginal density of X .

If $0 < x < 1$, then we have

$$f_X(x) = \int_{y=0}^{1-x} f(x, y) \, dy = \int_{y=0}^{1-x} 6(1-x-y) \, dy$$

Integrating, we have

$$\begin{aligned} f_X(x) &= 6 \left[(1-x)y - \frac{y^2}{2} \right]_0^{1-x} = 6 \left[(1-x)^2 - \frac{(1-x)^2}{2} \right] \\ &= 6 \times \frac{(1-x)^2}{2} = 3(1-x)^2 \end{aligned}$$

Thus, the marginal density of X is given by

$$f_X(x) = \begin{cases} 3(1-x)^2 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the marginal density of Y .

If $0 < y < 1$, then we have

$$f_Y(y) = \int_{x=0}^{1-y} f(x, y) \, dx = \int_{x=0}^{1-y} 6(1-x-y) \, dx$$

Integrating, we have

$$\begin{aligned} f_Y(y) &= 6 \left[(1-y)x - \frac{x^2}{2} \right]_0^{1-y} = 6 \left[(1-y)^2 - \frac{(1-y)^2}{2} \right] \\ &= 6 \times \frac{(1-y)^2}{2} = 3(1-y)^2 \end{aligned}$$

Thus, the marginal density of Y is given by

$$f_Y(y) = \begin{cases} 3(1-y)^2 & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Hence, we find that

$$f_X(x)f_Y(y) = \begin{cases} 9(1-x)^2(1-y)^2 & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent random variables. $\blacklozenge$

EXAMPLE 4.31. Examine whether the variables X and Y are independent, whose joint density is

$$f(x,y) = xe^{-xy-x}, \quad 0 < x, y < \infty$$

(Anna, May 2006)

Solution. The marginal density function of X is given by

$$f_X(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{(1+y)^2} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Note that

$$f_X(x)f_Y(y) = \begin{cases} \frac{e^{-x}}{(1+y)^2} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent random variables. $\blacklozenge$

EXAMPLE 4.32. The joint probability density function of a two-dimensional random variable (X, Y) is given by

$$f(x,y) = xy^2 + \frac{x^2}{8}, \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 1$$

Compute (i) $P(X > 1)$, (ii) $P(Y < \frac{1}{2})$, (iii) $P[X > 1 | Y < \frac{1}{2}]$, (iv) $P[Y < \frac{1}{2} | X > 1]$, (v) $P(X < Y)$ and (vi) $P(X + Y \leq 1)$.

(Anna, May 2007)

Solution. To solve this problem, we first need to compute the marginal density functions of X and Y .

(i) First, we find the marginal density function of X . If $0 \leq x \leq 2$, then we have

$$f_X(x) = \int_{y=0}^1 f(x,y) dy = \int_{y=0}^1 \left[xy^2 + \frac{x^2}{8} \right] dy$$

Integrating, we have

$$f_X(x) = \left[\frac{xy^3}{3} + \frac{x^2y}{8} \right]_{y=0}^1 = \frac{x}{3} + \frac{x^2}{8}$$

$$f_X(x) = \frac{1}{24} (3x^2 + 8x) \quad \text{for } 0 \leq x \leq 2$$

(ii) Next, we find the marginal density function of Y . If $0 \leq y \leq 1$, then we have

$$f_Y(y) = \int_{x=0}^2 f(x,y) dx = \int_{x=0}^2 \left[xy^2 + \frac{x^2}{8} \right] dx$$

Integrating, we have

$$\begin{aligned} f_Y(y) &= \left[\frac{x^2y^2}{2} + \frac{x^3}{24} \right]_{x=0}^2 = 2y^2 + \frac{1}{3} \\ &= \frac{1}{3} (6y^2 + 1) \quad \text{for } 0 \leq y \leq 1 \end{aligned}$$

(iii) Next, we find that

$$P(X > 1) = \int_{x=1}^2 f_X(x) dx = \int_{x=1}^2 \frac{1}{24} (3x^2 + 8x) dx$$

Integrating, we have

$$P(X > 1) = \frac{1}{24} [x^3 + 4x^2]_{x=1}^2 = \frac{1}{24} [(8 + 16) - (1 + 4)] = \frac{19}{24}$$

Next, we find that

$$P\left[Y < \frac{1}{2}\right] = \int_{y=0}^{\frac{1}{2}} f_Y(y) dy = \int_{y=0}^{\frac{1}{2}} \frac{1}{3} (6y^2 + 1) dy$$

Integrating, we have

$$P\left[Y < \frac{1}{2}\right] = \frac{1}{3} [2y^3 + y]_{y=0}^{\frac{1}{2}} = \frac{1}{3} \left[\frac{1}{4} + \frac{1}{2} \right] = \frac{1}{4}$$

(iv) Next, we compute $P[X > 1, Y < \frac{1}{2}]$. By definition, we have

$$P\left[X > 1, Y < \frac{1}{2}\right] = \int_{x=1}^2 \int_{y=0}^{\frac{1}{2}} \left[xy^2 + \frac{x^2}{8}\right] dy dx$$

Integrating y , we have

$$P\left[X > 1, Y < \frac{1}{2}\right] = \int_{x=1}^2 \left[\frac{xy^3}{3} + \frac{x^2y}{8}\right]_{y=0}^{\frac{1}{2}} dx = \int_{x=1}^2 \left[\frac{x}{24} + \frac{x^2}{16}\right] dx$$

Integrating x , we have

$$\begin{aligned} P\left[X > 1, Y < \frac{1}{2}\right] &= \left[\frac{x^2}{48} + \frac{x^3}{48}\right]_{x=1}^2 = \frac{5}{24} \\ &= \frac{P[X > 1, Y < \frac{1}{2}]}{P(Y < \frac{1}{2})} = \frac{5/24}{1/4} = \frac{5}{6} \end{aligned}$$

Similarly, we find that

$$P\left[Y < \frac{1}{2} | X > 1\right] = \frac{P[X > 1, Y < \frac{1}{2}]}{P(X > 1)} = \frac{5/24}{19/24} = \frac{5}{19}$$

(v) Next, we find that

$$P(X < Y) = \int_{x=0}^1 \int_{y=x}^1 f(x, y) dy dx = \int_{x=0}^1 \int_{y=x}^1 \left[xy^2 + \frac{x^2}{8}\right] dy dx$$

A simple calculation gives $P(X < Y) = \frac{37}{480}$.

(vi) Finally, we note that

$$P(X + Y \leq 1) = \int_{x=0}^1 \int_{y=0}^{1-x} f(x, y) dy dx = \int_{x=0}^1 \int_{y=0}^{1-x} \left[xy^2 + \frac{x^2}{8}\right] dy dx$$

A simple calculation gives $P(X + Y \leq 1) = \frac{13}{480}$.

◆

EXAMPLE 4.33. Given the joint probability density function

$$f(x,y) = \begin{cases} \frac{2}{\pi} (1 - x^2 - y^2), & 0 < x^2 + y^2 < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Determine the marginal probability density functions of X and Y .
- (ii) Compute $P(X^2 + Y^2 < b^2)$, $0 < b < 1$.

(Anna, Nov. 2006)

Solution. Define the region A as

$$A = \{(x,y) \in \mathbb{R}^2 : 0 < x^2 + y^2 < 1\}$$

Then A is the interior of the unit circle as illustrated in Figure 4.12. The given joint density function f takes values only in the region A .

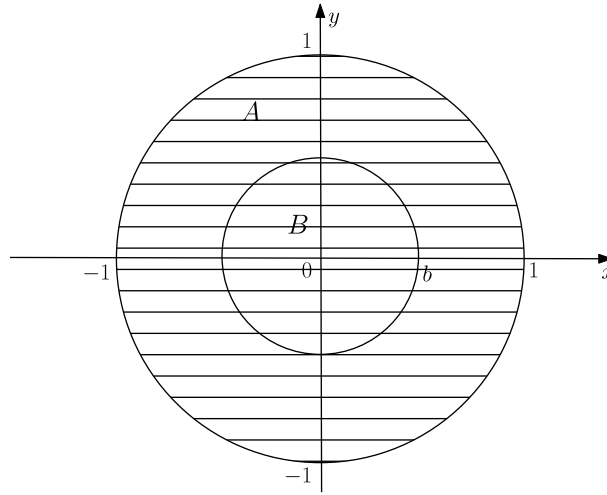


Figure 4.12: Region A .

- (i) First, we compute the marginal density function of X .

If $-1 < x < 1$, then we have

$$f_X(x) = \int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \sqrt{1-x^2} f(x,y) dy = \int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \sqrt{1-x^2} \frac{2}{\pi} (1-x^2-y^2) dy$$

Since the integrand is an even function of y , we have

$$f_X(x) = \frac{2}{\pi} 2 \int_{y=0}^{\sqrt{1-x^2}} (1-x^2-y^2) dy$$

Integrating,

$$\begin{aligned} f_X(x) &= \frac{4}{\pi} \left[(1-x^2)y - \frac{y^3}{3} \right]_0^{\sqrt{1-x^2}} \\ &= \frac{4}{\pi} (1-x^2)^{\frac{3}{2}} \left[1 - \frac{1}{3} \right] = \frac{8}{3\pi} (1-x^2)^{\frac{3}{2}} \end{aligned}$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{8}{3\pi} (1-x^2)^{\frac{3}{2}} & \text{for } -1 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Interchanging the roles of x and y , it is easy to see that the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{8}{3\pi} (1-y^2)^{\frac{3}{2}} & \text{for } -1 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

(ii) Next, we find that

$$P(X^2 + Y^2 < b^2) = P(-b \leq \sqrt{x^2 + y^2} \leq b) = \int \int_B f(x, y) dy dx$$

where the integral is taken over the circular region

$$B = \{(x, y) \in \mathbb{R}^2 : 0 < x^2 + y^2 < b^2\}$$

Using polar coordinates, the integral becomes

$$P(X^2 + Y^2 < b^2) = \int_{r=0}^b \int_{\theta=0}^{2\pi} \frac{2}{\pi} [1-r^2] r d\theta dr$$

Since the integrals are separable, we have

$$P(X^2 + Y^2 < b^2) = \frac{2}{\pi} \left[\int_{r=0}^b (r-r^3) dr \right] \left[\int_{\theta=0}^{2\pi} d\theta \right]$$

Integrating, we have

$$\begin{aligned} P(X^2 + Y^2 < b^2) &= \frac{2}{\pi} \left[\frac{r^2}{2} - \frac{r^4}{4} \right]_0^b [2\pi] \\ &= 4 \left[\frac{b^2}{2} - \frac{b^4}{4} \right] = b^2(2-b^2) \end{aligned}$$

◆

PROBLEM SET 4.2

1. The joint probability distribution of X and Y is given by

$$f(x,y) = \frac{1}{4} \quad \text{for } x = 0, 1; y = 0, 1$$

- (a) Find the marginal distributions of X and Y .
- (b) Find the means of X and Y .
- (c) Are X and Y independent?

2. The joint probability distribution of X and Y is given by

X \ Y	Y	
	0	1
0	$\frac{2}{9}$	$\frac{1}{9}$
1	$\frac{1}{9}$	$\frac{5}{9}$

- (a) Find the marginal distributions of X and Y .
- (b) Find the means of X and Y .
- (c) Are X and Y independent?

3. The joint probability distribution of X and Y is given by

$$f(x,y) = \frac{3x+4y}{63} \quad \text{for } x = 0, 1, 2; y = 0, 1, 2$$

- (a) Find the marginal distributions of X and Y .
- (b) Find the means of X and Y .
- (c) Are X and Y independent?

4. The joint probability distribution of X and Y is given by

$$f(x,y) = \frac{x+2y}{54} \quad \text{for } x = 1, 2, 3; y = 1, 2, 3$$

- (a) Find the marginal distributions of X and Y .
- (b) Find the means of X and Y .
- (c) Are X and Y independent?

5. The joint probability distribution of X and Y is given by

$$f(x,y) = K(3x+y) \quad \text{for } x = 0, 1, 2; y = 1, 2, 3$$

- (a) Find the value of K .
- (b) Find the marginal distributions of X and Y .
- (c) Check if X and Y are independent.
- (d) Find $P(X \leq 1, Y \leq 2)$.

6. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} Kxy & \text{if } 0 < x < 2, 0 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of the constant K .
- (b) Find $P(X \geq 1, Y \leq 2)$.
- (c) Find the marginal density functions of X and Y .
- (d) Check if X and Y are independent.

7. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} K(x^2 + y^2) & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of the constant K .
- (b) Find the marginal density functions of X and Y .
- (c) Check if X and Y are independent.
- (d) Find $P(0.2 < X < 0.4)$ and $P(0.3 < Y < 0.6)$.

8. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} 6x & \text{if } 0 < x < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginal density functions of X and Y .
- (b) Check if X and Y are independent.
- (c) Find $P(X + Y > 1)$.

9. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} 4e^{-2x}y & \text{if } x > 0, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginal density functions of X and Y .
- (b) Check if X and Y are independent.
- (c) Find the means of X and Y .

10. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} K(1+x^2)y & \text{if } 0 < x < 1, 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of the constant K .
- (b) Find the marginal density functions of X and Y .
- (c) Check if X and Y are independent.
- (d) Find $P(X + Y < 1)$

11. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} K & \text{if } 0 < y < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of the constant K .
- (b) Find the marginal density functions of X and Y .
- (c) Check if X and Y are independent.
- (d) Find the means of X and Y .

12. The joint density function of X and Y is given by

$$f(x,y) = \begin{cases} K(x+1)e^{-y} & \text{if } 0 < x < 1, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of the constant K .
- (b) Find the marginal density functions of X and Y .
- (c) Check if X and Y are independent.
- (d) Find $P(0 < x < \frac{1}{2}, y > 1)$.

4.3 CONDITIONAL DISTRIBUTIONS

First, we define the *conditional distributions* for the discrete case. In Probability Theory (Chapter 1), we defined the conditional probability of an event A given event B as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) \neq 0$$

Suppose that A and B are the events $X = x$ and $Y = y$, where X and Y are discrete random variables having the joint probability mass function $f(x,y)$. Then we have

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{f(x,y)}{f_Y(y)}$$

provided that $f_Y(y) = P(Y = y) \neq 0$, where f_Y is the marginal distribution of Y . If we denote the conditional probability by $f_{X|Y}(x|y)$ to indicate that x is a variable and y is fixed, we have the following definition.

Definition 4.11. If (X,Y) is a two-dimensional discrete random variable with joint probability mass function f and if f_X and f_Y are the marginal distributions of X and Y , respectively, then the *conditional distribution of X given $Y = y$* is denoted by $f_{X|Y}(x|y)$, and is defined as

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}, \quad \text{where } f_Y(y) \neq 0$$

for each x within the range of X . Similarly, the *conditional distribution of Y given $X = x$* is denoted by $f_{Y|X}(y|x)$, and is defined as

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}, \quad \text{where } f_X(x) \neq 0$$

for each y within the range of Y .

Remark 4.4. The conditional density functions $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$ are both probability density functions. This can be easily seen as follows:

(i) First, we note that

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} \geq 0$$

for all values of x and y within their range.

(ii) We also note that

$$\sum_x f_{X|Y}(x|y) = \frac{\sum_x f(x,y)}{f_Y(y)} = \frac{f_Y(y)}{f_Y(y)} = 1$$

from the definition of the marginal distribution $f_Y(y)$.

Hence, it follows that $f_{X|Y}(x|y)$ is a probability density function. In a similar manner, it can be established that $f_{Y|X}(y|x)$ is a probability density function.

Remark 4.5. If X and Y are independent discrete random variables, then we know that

$$f(x,y) = f_X(x) f_Y(y)$$

for all values of x and y within their range. In this case,

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{f_X(x) f_Y(y)}{f_Y(y)} = f_X(x)$$

Similarly, we also have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{f_X(x) f_Y(y)}{f_X(x)} = f_Y(y)$$

The converse also is true. If

$$f_{X|Y}(x|y) = f_X(x)$$

for all values of x and y within their range, then it is easy to see that

$$f(x,y) = f_{X|Y}(x|y) f_Y(y) = f_X(x) f_Y(y)$$

so that X and Y are independent random variables.

Similarly, if

$$f_{Y|X}(y|x) = f_Y(y)$$

for all values of x and y within their range, then

$$f(x,y) = f_{Y|X}(y|x) f_X(x) = f_Y(y) f_X(x)$$

so that X and Y are independent random variables.

The conditional distributions can be similarly defined for two-dimensional continuous random variables by replacing probability distributions with probability density functions.

Definition 4.12. If (X, Y) is a two-dimensional continuous random variable with joint probability density function f and if f_X and f_Y are the marginal density functions of X and Y , respectively, then the *conditional density function of X given $Y = y$* is denoted by $f_{X|Y}(x|y)$ and is defined as

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}, \text{ where } f_Y(y) \neq 0$$

for $-\infty < x < \infty$. Similarly, the *conditional density function of Y given $X = x$* is denoted by $f_{Y|X}(y|x)$, and is defined as

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}, \text{ where } f_X(x) \neq 0$$

for $-\infty < y < \infty$.

Remark 4.6. Both the conditional density functions $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$ are probability density functions. This can be easily seen as follows:

(i) First, we note that

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} \geq 0 \quad \forall (x, y) \in \mathbb{R}^2$$

(ii) We also note that

$$\int_{x=-\infty}^{\infty} f_{X|Y}(x|y) dx = \frac{\int_{x=-\infty}^{\infty} f(x, y) dx}{f_Y(y)} = \frac{f_Y(y)}{f_Y(y)} = 1$$

from the definition of the marginal density function $f_Y(y)$.

Hence, it follows that $f_{X|Y}(x|y)$ is a probability density function. In a similar manner, it can be established that $f_{Y|X}(y|x)$ is a probability density function.

Remark 4.7. If X and Y are independent continuous random variables, then we know that

$$f(x, y) = f_X(x) f_Y(y)$$

for all values of $(x, y) \in \mathbb{R}^2$. In this case, it is easy to see that

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{f_X(x) f_Y(y)}{f_Y(y)} = f_X(x)$$

Similarly, we also have

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{f_X(x) f_Y(y)}{f_X(x)} = f_Y(y)$$

The converse also is true. If

$$f_{X|Y}(x|y) = f_X(x)$$

for all values of $(x, y) \in \mathbb{R}^2$, then it is easy to see that

$$f(x, y) = f_{X|Y}(x|y) f_Y(y) = f_X(x) f_Y(y)$$

so that X and Y are independent random variables.

Similarly, if

$$f_{Y|X}(y|x) = f_Y(y)$$

for all values of $(x,y) \in \mathbb{R}^2$, then it is easy to see that

$$f(x,y) = f_{Y|X}(y|x)f_X(x) = f_Y(y)f_X(x)$$

so that X and Y are independent random variables.

EXAMPLE 4.34. The joint probability mass function of (X,Y) is given by

$$p(x,y) = K(2x + 3y), \quad x = 0, 1, 2; y = 1, 2, 3$$

Find all the marginal and conditional probability distributions.

(Anna, Nov. 2007)

Solution. The given joint probability distribution may be tabulated as follows:

X \ Y	Y		
	1	2	3
0	3K	6K	9K
1	5K	8K	11K
2	7K	10K	13K

First, we find the value of K .

Since the total probability is equal to 1, we must have

$$\sum_x \sum_y P(x,y) = 1$$

Hence, we have

$$72K = 1 \Rightarrow K = \frac{1}{72}$$

Thus, the jointly probability distribution of X and Y is given as follows:

X \ Y	Y		
	1	2	3
0	$\frac{1}{24}$	$\frac{1}{12}$	$\frac{1}{8}$
1	$\frac{5}{72}$	$\frac{1}{9}$	$\frac{11}{72}$
2	$\frac{7}{72}$	$\frac{5}{36}$	$\frac{13}{72}$

Next, we find the marginal probability distribution of X . By definition, we have

$$f_X(x) = \sum_{y=1}^3 p(x,y) \quad \text{for } x = 0, 1, 2$$

Hence, the marginal distribution of X is given by

x	0	1	2
$f_X(x)$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$

We also find the marginal probability distribution of Y . By definition, we have

$$f_Y(y) = \sum_{x=0}^2 p(x,y) \text{ for } y = 1, 2, 3$$

Hence, the marginal distribution of Y is given by

y	1	2	3
$f_Y(y)$	$\frac{5}{24}$	$\frac{1}{3}$	$\frac{11}{24}$

Next, we find the conditional distributions of X for various values of Y .

First, we find the conditional distribution of X given that $Y = 1$. By definition, we have

$$f_{X|Y}(x|y=1) = \frac{p(x,1)}{f_Y(1)} = \frac{p(x,1)}{\frac{5}{24}} = \frac{24}{5} p(x,1) \text{ for } x = 0, 1, 2$$

Thus, the conditional distribution of X given that $Y = 1$ is obtained as

x	0	1	2
$f_{X Y}(x y=1)$	$\frac{1}{5}$	$\frac{1}{3}$	$\frac{7}{15}$

Similarly, the conditional distribution of X given that $Y = 2$ is defined by

$$f_{X|Y}(x|y=2) = \frac{p(x,2)}{f_Y(2)} = \frac{p(x,2)}{\frac{1}{3}} = 3p(x,2) \text{ for } x = 0, 1, 2$$

Thus, the conditional distribution of X given that $Y = 2$ is obtained as

x	0	1	2
$f_{X Y}(x y=2)$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{5}{12}$

Similarly, the conditional distribution of X given that $Y = 3$ is defined by

$$f_{X|Y}(x|y=3) = \frac{p(x,3)}{f_Y(3)} = \frac{p(x,3)}{\frac{11}{24}} = \frac{24}{11} p(x,3) \text{ for } x = 0, 1, 2$$

Thus, the conditional distribution of X given that $Y = 3$ is obtained as

x	0	1	2
$f_{X Y}(x y=3)$	$\frac{3}{11}$	$\frac{1}{3}$	$\frac{13}{33}$

Next, we find the conditional distributions of Y for various values of X .

First, we find the conditional distribution of Y given that $X = 0$. By definition, we have

$$f_{Y|X}(y|x=0) = \frac{p(0,y)}{f_X(0)} = \frac{p(0,y)}{\frac{1}{4}} = 4p(0,y) \text{ for } y = 1, 2, 3$$

Thus, the conditional distribution of Y given that $X = 0$ is obtained as

y	1	2	3
$f_{Y X}(y x=0)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$

Similarly, the conditional distribution of Y given that $X = 1$ is defined by

$$f_{Y|X}(y|x=1) = \frac{p(1,y)}{f_X(1)} = \frac{p(1,y)}{\frac{1}{3}} = 3p(1,y) \text{ for } y = 1, 2, 3$$

Thus, the conditional distribution of Y given that $X = 1$ is obtained as

y	1	2	3
$f_{Y X}(y x=1)$	$\frac{5}{24}$	$\frac{1}{3}$	$\frac{11}{24}$

Similarly, the conditional distribution of Y given that $X = 2$ is defined by

$$f_{Y|X}(y|x=2) = \frac{p(2,y)}{f_X(2)} = \frac{p(2,y)}{\frac{5}{12}} = \frac{12}{5} p(2,y) \text{ for } y = 1, 2, 3$$

Thus, the conditional distribution of Y given that $X = 2$ is obtained as

y	1	2	3
$f_{Y X}(y x=2)$	$\frac{7}{30}$	$\frac{1}{3}$	$\frac{13}{30}$



EXAMPLE 4.35. The joint probability distribution of X and Y is given by the following table:

$X \backslash Y$	1	3	9
2	$\frac{1}{8}$	$\frac{1}{24}$	$\frac{1}{12}$
4	$\frac{1}{4}$	$\frac{1}{4}$	0
6	$\frac{1}{8}$	$\frac{1}{24}$	$\frac{1}{12}$

Find the marginal probability distribution of Y and the conditional distribution of Y given that $X = 2$. Are X and Y independent?

(Anna, May 1999; Madras, Oct. 2001)

Solution. By definition, the marginal distribution of Y is given by

$$f_Y(y) = \sum_x f(x,y)$$

It follows that

$$f_Y(1) = \frac{1}{8} + \frac{1}{4} + \frac{1}{8} = \frac{1}{2}$$

$$f_Y(3) = \frac{1}{24} + \frac{1}{4} + \frac{1}{24} = \frac{1}{3}$$

$$f_Y(9) = \frac{1}{12} + 0 + \frac{1}{12} = \frac{1}{6}$$

Hence, the marginal distribution of Y is given by

y	1	3	9
$f_Y(y)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

By definition, the conditional distribution of Y given that $X = 2$ is given by

$$f_{Y|X}(y|X=2) = \frac{f(2,y)}{f_X(2)}$$

Now, $f_X(2)$ is obtained as

$$f_X(2) = \sum_y f(2,y) = \frac{1}{8} + \frac{1}{24} + \frac{1}{12} = \frac{6}{24} = \frac{1}{4}$$

It follows that

$$f_{Y|X}(y|X=2) = \frac{f(2,y)}{f_X(2)} = 4f(2,y)$$

Hence, the conditional distribution of Y given $X = 2$ is obtained as follows:

y	1	3	9
$f_{Y X}(y X=2)$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{3}$

We note that $f_{Y|X}(3) \neq f_Y(3)$ and $f_{Y|X}(9) \neq f_Y(9)$. This shows that X and Y are not independent random variables. $\blacklozenge$

EXAMPLE 4.36. Given is the joint distribution of X and Y :

$Y \backslash X$	0	1	2
0	0.02	0.08	0.10
1	0.05	0.20	0.25
2	0.03	0.12	0.15

Obtain

- Marginal distributions of X and Y .
- The conditional distribution of X given $Y = 0$.

(Anna, Nov. 2003)

Solution. (i) By definition, the marginal distribution of X is given by

$$f_X(x) = \sum_{y=0}^2 f(x,y)$$

Hence, the marginal distribution of X is obtained as follows:

x	0	1	2
$f_X(x)$	0.1	0.4	0.5

By definition, the marginal distribution of Y is given by

$$f_Y(y) = \sum_{x=0}^2 f(x,y)$$

Hence, the marginal distribution of Y is obtained as follows:

y	0	1	2
$f_Y(y)$	0.2	0.5	0.3

(ii) By definition, the conditional distribution of X given $Y = 0$ is given by

$$f_{X|Y}(x|0) = \frac{f(x,0)}{f_Y(0)} = \frac{f(x,0)}{0.2} \text{ for } x = 0, 1, 2$$

Hence, we find that

$$f_{X|Y}(0|0) = \frac{0.02}{0.2} = 0.1$$

$$f_{X|Y}(1|0) = \frac{0.08}{0.2} = 0.4$$

$$f_{X|Y}(2|0) = \frac{0.10}{0.2} = 0.5$$

So, the conditional distribution of X given $Y = 0$ is obtained as follows:

x	0	1	2
$f_{X Y}(x 0)$	0.1	0.4	0.5

◆

EXAMPLE 4.37. The joint probability distribution of X and Y is given by

$$f(x,y) = \frac{x+y}{21}, \quad x = 1, 2, 3; y = 1, 2$$

Find the conditional distribution of Y for $X = x$.

(Madras, Oct. 1999)

Solution. The joint probability distribution of X and Y is given:

$X \backslash Y$	1	2	$f_X(x)$
1	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{5}{21}$
2	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{7}{21}$
3	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{9}{21}$
$f_Y(y)$	$\frac{9}{21}$	$\frac{12}{21}$	1

Next, we compute the marginal distribution of X .

By definition, we have

$$f_X(x) = \sum_{y=1}^2 f(x,y) \text{ for } x = 1, 2, 3$$

Hence, the marginal distribution of X is obtained as follows:

x	1	2	3
$f_X(x)$	$\frac{5}{21}$	$\frac{7}{21}$	$\frac{9}{21}$

Next, we compute the conditional distribution of Y for $X = x$, where $x = 1, 2, 3$.

First, we compute the conditional distribution of Y for $X = 1$.

By definition,

$$f_{Y|X}(y|1) = \frac{f(1,y)}{f_X(1)} = \frac{f(1,y)}{5/21} \text{ for } y = 1, 2$$

Hence, we have

$$f_{Y|X}(1|1) = \frac{f(1,1)}{f_X(1)} = \frac{2/21}{5/21} = \frac{2}{5}$$

and

$$f_{Y|X}(2|1) = \frac{f(1,2)}{f_X(1)} = \frac{3/21}{5/21} = \frac{3}{5}$$

Thus, the conditional distribution of Y for $X = 1$ is obtained as follows:

y	1	2
$f_{Y X}(y 1)$	$\frac{2}{5}$	$\frac{3}{5}$

Next, we compute the conditional distribution of Y for $X = 2$.

By definition, we have

$$f_{Y|X}(y|2) = \frac{f(2,y)}{f_X(2)} = \frac{f(2,y)}{7/21} \text{ for } y = 1, 2.$$

Hence,

$$f_{Y|X}(1|2) = \frac{f(2,1)}{f_X(2)} = \frac{3/21}{7/21} = \frac{3}{7}$$

and

$$f_{Y|X}(2|2) = \frac{f(2,2)}{f_X(2)} = \frac{4/21}{7/21} = \frac{4}{7}$$

Thus, the conditional distribution of Y for $X = 2$ is obtained as follows:

y	1	2
$f_{Y X}(y 2)$	$\frac{3}{7}$	$\frac{4}{7}$

Finally, we compute the conditional distribution of Y for $X = 3$.

By definition, we have

$$f_{Y|X}(y|3) = \frac{f(3,y)}{f_X(3)} = \frac{f(3,y)}{9/21} \text{ for } y = 1, 2$$

Hence

$$f_{Y|X}(1|3) = \frac{f(3,1)}{f_X(3)} = \frac{4/21}{9/21} = \frac{4}{9}$$

and

$$f_{Y|X}(2|3) = \frac{f(3,2)}{f_X(3)} = \frac{5/21}{9/21} = \frac{5}{9}$$

Thus, the conditional distribution of Y for $X = 3$ is obtained as follows:

y	1	2
$f_{Y X}(y 3)$	$\frac{4}{9}$	$\frac{5}{9}$

◆

EXAMPLE 4.38. Random variables X and Y have the joint distribution (when $p + q = 1$, $0 < p < 1$ and $\lambda > 0$):

$$f(x,y) = \frac{\lambda^x e^{-\lambda} p^y q^{x-y}}{y!(x-y)!}; \quad y = 0, 1, 2, \dots, x \text{ and } x = 0, 1, 2, \dots$$

Find the marginal and conditional distributions and evaluate $P(X < 1)$.

(Anna, April 2004; Nov. 2005)

Solution. The marginal distribution of X is given by

$$f_X(x) = \sum_{y=0}^x f(x,y) \text{ for } x = 0, 1, 2, \dots$$

Hence, we have

$$f_X(x) = \sum_{y=0}^x \frac{\lambda^x e^{-\lambda} p^y q^{x-y}}{y!(x-y)!} = \frac{e^{-\lambda} \lambda^x}{x!} \sum_{y=0}^x \frac{x!}{y!(x-y)!} p^y q^{x-y}$$

Simplifying, we have

$$f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!} \sum_{y=0}^x \binom{x}{y} p^y q^{x-y} = \frac{e^{-\lambda} \lambda^x}{x!} (p+q)^x = \frac{e^{-\lambda} \lambda^x}{x!}$$

since $p + q = 1$

Thus, the marginal distribution of X is given by

$$f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!} \text{ for } x = 0, 1, 2, \dots$$

which is Poisson distribution with parameter $\lambda > 0$.

Next, we calculate the marginal distribution of Y .

We have

$$f_Y(y) = \sum_{x=y}^{\infty} f(x,y) \quad \text{for } y = 0, 1, 2, \dots$$

Hence,

$$\begin{aligned} f_Y(y) &= \sum_{x=y}^{\infty} \frac{\lambda^x e^{-\lambda} p^y q^{x-y}}{y!(x-y)!} = e^{-\lambda} \frac{(\lambda p)^y}{y!} \sum_{x=y}^{\infty} \frac{(\lambda q)^{x-y}}{(x-y)!} \\ &= e^{-\lambda} \frac{(\lambda p)^y}{y!} \sum_{z=0}^{\infty} \frac{(\lambda q)^z}{z!} = e^{-\lambda} \frac{(\lambda p)^y}{y!} e^{\lambda q} \quad (\text{Putting } z = x - y) \\ &= e^{-\lambda(1-q)} \frac{(\lambda p)^y}{y!} = e^{-\lambda p} \frac{(\lambda p)^y}{y!} \end{aligned}$$

Thus, the marginal distribution of Y is given by

$$f_Y(y) = e^{-\lambda p} \frac{(\lambda p)^y}{y!} \quad \text{for } y = 0, 1, 2, \dots$$

which is a Poisson distribution with parameter $\lambda p > 0$.

Next, we find the conditional distribution of X given $Y = y$.

Fix any value of y in the range $y = 0, 1, 2, \dots$

Then, by definition, we have

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x,y)}{f_Y(y)} \quad \text{where } x = y, y+1, y+2, \dots \\ &= \frac{\lambda^x e^{-\lambda} p^y q^{x-y}}{y!(x-y)!} \times \frac{y!}{e^{-\lambda p} (\lambda p)^y} \end{aligned}$$

Simplifying, we have

$$f_{X|Y}(x|y) = e^{-\lambda q} \frac{(\lambda q)^{x-y}}{(x-y)!} \quad \text{for } x = y, y+1, y+2, \dots$$

Thus, for any fixed value of y in the range $y = 0, 1, 2, \dots$, the conditional distribution of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = e^{-\lambda q} \frac{(\lambda q)^{x-y}}{(x-y)!} \quad \text{for } x = y, y+1, y+2, \dots$$

Next, we find the conditional distribution of Y given $X = x$.

Fix any value of x in the range $x = 0, 1, 2, \dots$

Then, by definition, we have

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f(x,y)}{f_X(x)} = \frac{\lambda^x e^{-\lambda} p^y q^{x-y}}{y!(x-y)!} \times \frac{x!}{\lambda^x e^{-\lambda}} \\ &= \frac{x!}{y!(x-y)!} p^y q^{x-y} = \binom{x}{y} p^y q^{x-y} \end{aligned}$$

Thus, for any fixed value of x in the range $x = 0, 1, 2, \dots$, the conditional distribution of Y given $X = x$ is obtained as

$$f_{Y|X}(y|x) = \binom{x}{y} p^y q^{x-y} \quad \text{for } y = 0, 1, 2, \dots, x$$

Finally, we compute $P(X < 1)$.

We have

$$P(X < 1) = P(X = 0) = f_X(0) = e^{-\lambda}$$

◆

EXAMPLE 4.39. Suppose the random variables X and Y have the joint probability density function

$$f(x, y) = \begin{cases} xye^{-\frac{1}{2}(x^2+y^2)}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Find

- (i) the marginal PDF of X .
- (ii) The marginal PDF of Y .
- (iii) The conditional PDF of X given $Y = y$.
- (iv) If X and Y are independent.

(Anna, May 2007)

Solution. (i) It is easy to see that the marginal PDF of X is given by

$$f_X(x) = \begin{cases} xe^{-\frac{x^2}{2}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

(ii) Similarly, it can be shown that the marginal PDF of Y is given by

$$f_Y(y) = \begin{cases} ye^{-\frac{y^2}{2}} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(iii) Fix any value y of Y in the region $y > 0$. Then the conditional PDF of X given $Y = y$ is obtained by the formula

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} \quad \text{for } x > 0$$

Substituting the values of the density functions, we get

$$f_{X|Y}(x|y) = \frac{xye^{-\frac{1}{2}(x^2+y^2)}}{ye^{-\frac{y^2}{2}}}, \quad \text{where } x > 0$$

Simplifying, we get

$$f_{X|Y}(x|y) = xe^{-\frac{x^2}{2}}, \quad \text{where } x > 0$$

which is the same as the marginal density function of X .

(iv) By the answer obtained in (iii), it is immediate that X and Y are independent random variables. $\blacklozenge$

EXAMPLE 4.40. Given

$$f(x,y) = \begin{cases} Cx(x-y) & \text{if } 0 < x < 2, -x < y < x \\ 0 & \text{otherwise} \end{cases}$$

- (i) Evaluate C .
- (ii) Find $f_X(x)$.
- (iii) Find $f_{Y|X}(y|x)$.
- (iv) Find $f_Y(y)$.

(Anna, Model, May 2007)

Solution. Define the region A as

$$A = \{(x,y) \in \mathbb{R}^2 : 0 < x < 2, -x < y < x\}$$

Then the region A is as illustrated in Figure 4.13. The joint density function f takes values only in the region A .

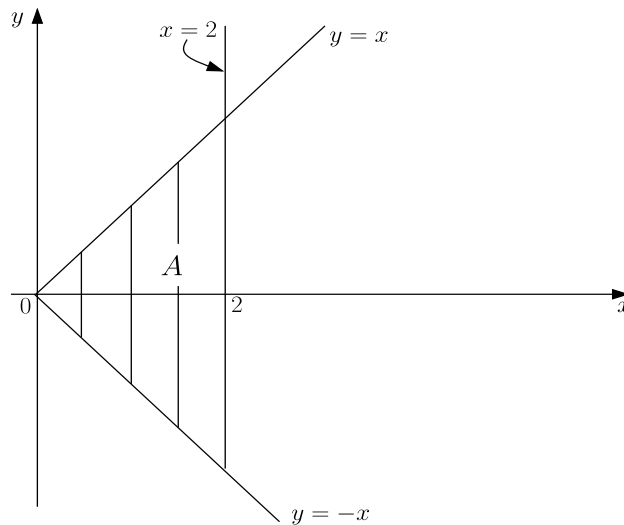


Figure 4.13: Region A .

- (i) First, we calculate the value of C . Note that

$$\int_{x=0}^2 \int_{y=-x}^x f(x,y) dy dx = 1 \Rightarrow \int_{x=0}^2 \int_{y=-x}^x Cx(x-y) dy dx = 1$$

Integrating y , we have

$$C \int_{x=0}^2 x \left[xy - \frac{y^2}{2} \right]_{y=-x}^x dx = 1$$

Simplifying, we have

$$C \int_{x=0}^2 x [2x^2] dx = 1 \Rightarrow 2C \int_{x=0}^2 x^3 dx = 1$$

Integrating x , we have

$$2C \left[\frac{x^4}{4} \right]_{x=0}^2 = 1 \Rightarrow 8C = 1$$

Hence, $C = \frac{1}{8}$ and the PDF of (X, Y) is given by

$$f(x, y) = \frac{1}{8} x(x-y) \text{ for } 0 < x < 2, -x < y < x$$

(ii) Here, we find the marginal density function of X . If $0 < x < 2$, then we have

$$f_X(x) = \int_{y=-x}^x f(x, y) dy = \int_{y=-x}^x \frac{1}{8} x(x-y) dy \quad (\text{See Figure 4.13})$$

Using the property of definite integrals, we have

$$f_X(x) = \frac{1}{8} \int_{y=-x}^x x^2 dy - \frac{1}{8} \int_{y=-x}^x xy dy = \frac{1}{8} 2 \int_{y=0}^x x^2 dy - 0 = \frac{x^2}{4} \int_{y=0}^x dy$$

Integrating, we have

$$f_X(x) = \frac{x^2}{4} [x] = \frac{x^3}{4}$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{x^3}{4} & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

(iii) Here, we compute the conditional density function, $f_{Y|X}(y|x)$.

Fix any value of x in the interval $0 < x < 2$.

Then, by definition, we have

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f(x, y)}{f_X(x)} = \frac{\frac{1}{8}x(x-y)}{\frac{1}{4}x^3}, \text{ for } -x < y < x \\ &= \frac{1}{2} \frac{x-y}{x^2} \text{ for } -x < y < x \end{aligned}$$

(iv) Here, we find the marginal density function of Y . First, note that y takes all values in the interval $(-2, 2)$. We have two cases to consider.

If $-2 < y < 0$, then we have

$$f_Y(y) = \int_{x=-y}^0 f(x, y) dx = \int_{x=-y}^0 \frac{1}{8} x(x-y) dx = \frac{1}{8} \int_{x=-y}^0 [x^2 - xy] dx$$

Integrating,

$$\begin{aligned} f_Y(y) &= \frac{1}{8} \left[\frac{x^3}{3} - \frac{x^2 y}{2} \right]_{x=-y}^0 = \frac{1}{8} \left[\left(\frac{8}{3} - 2y \right) - \left(-\frac{y^3}{3} - \frac{y^3}{2} \right) \right] \\ &= \frac{1}{3} - \frac{y}{4} + \frac{5}{48} y^3 \quad \text{if } -2 < y < 0 \end{aligned}$$

Similarly, if $0 < y < 2$, then we have

$$f_Y(y) = \int_{x=y}^0 f(x, y) dx = \int_{x=y}^0 \frac{1}{8} x(x-y) dx = \frac{1}{8} \int_{x=y}^0 [x^2 - xy] dx$$

Integrating, we have

$$\begin{aligned} f_Y(y) &= \frac{1}{8} \left[\frac{x^3}{3} - \frac{x^2 y}{2} \right]_{x=y}^0 = \frac{1}{8} \left[\left(\frac{8}{3} - 2y \right) - \left(\frac{y^3}{3} - \frac{y^3}{2} \right) \right] \\ &= \frac{1}{3} - \frac{y}{4} + \frac{1}{48} y^3 \quad \text{if } 0 < y < 2 \end{aligned}$$

Combining the two cases, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{3} - \frac{y}{4} + \frac{5}{48} y^3 & \text{if } -2 < y < 0 \\ \frac{1}{3} - \frac{y}{4} + \frac{1}{48} y^3 & \text{if } 0 < y < 2 \end{cases}$$

◆

EXAMPLE 4.41. The joint probability density function of the two-dimensional random variable (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{8}{9} xy & 1 \leq x \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

- (i) Find the marginal density functions of X and Y .
- (ii) Find also the conditional density function of Y given $X = x$ and the conditional density function of X given $Y = y$.

(Anna, April 2003)

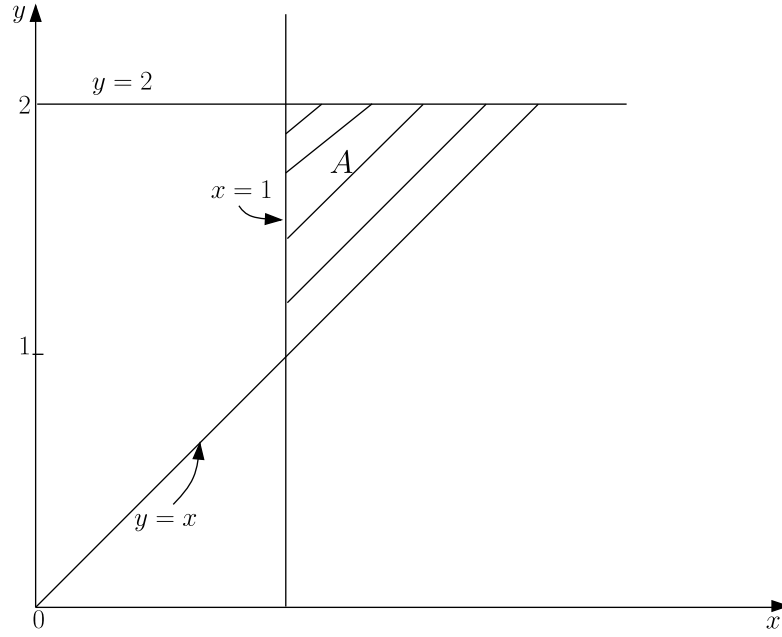


Figure 4.14: Region A.

Solution. Define the region A as

$$A = \{(x, y) \in \mathbb{R}^2 : 1 \leq x \leq y \leq 2\}$$

Then the region A is as illustrated in Figure 4.14. The joint PDF f takes values only in the region A .

(i) First, we find the marginal density function of X .

If $1 \leq x \leq 2$, then we have

$$f_X(x) = \int_{y=x}^2 f(x, y) dy = \int_{y=x}^2 \frac{8}{9} xy dy$$

Integrating y , we have

$$f_X(x) = \frac{4x}{9} [y^2]_{y=x}^2 = \frac{4x}{9} (4 - x^2)$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{4x}{9} (4 - x^2) & \text{for } 1 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the marginal density function of Y .

If $1 \leq y \leq 2$, then we have

$$f_Y(y) = \int_{x=1}^y f(x,y) dx = \int_{x=1}^y \frac{8}{9} xy dx$$

(See Figure 4.14.)

Integrating x , we have

$$f_Y(y) = \frac{4y}{9} [y^2]_{x=1}^y = \frac{4y}{9} (y^2 - 1)$$

Thus, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{4y}{9} (y^2 - 1) & \text{for } 1 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

(ii) Here, we find the conditional density of Y given $X = x$.

Fix any value of x in the interval $1 \leq x \leq 2$.

Then the conditional density of Y given $X = x$ is given by

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f(x,y)}{f_X(x)} = \frac{\frac{8}{9} xy}{\frac{4x}{9} (4-x^2)} \quad \text{for } x \leq y \leq 2 \\ &= \frac{2y}{4-x^2} \quad \text{for } x \leq y \leq 2 \end{aligned}$$

Thus, for any fixed value of x in $1 \leq x \leq 2$, the conditional density of Y given $X = x$ is given by

$$f_{Y|X}(y|x) = \begin{cases} \frac{2y}{4-x^2} & \text{for } x \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the conditional density of X given $Y = y$.

Fix any value of y in the interval $1 \leq y \leq 2$.

Then the conditional density of X given $Y = y$ is given by

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{\frac{8}{9} xy}{\frac{4y}{9} (y^2 - 1)} \quad \text{for } 1 \leq x \leq y$$

Simplifying, we have

$$f_{X|Y}(x|y) = \frac{2x}{y^2 - 1} \quad \text{for } 1 \leq x \leq y$$

Thus, for any fixed value of y in $1 \leq y \leq 2$, the conditional density of X given $Y = y$ is given by

$$f_{X|Y}(x|y) = \begin{cases} \frac{2x}{y^2 - 1} & \text{for } 1 \leq x \leq y \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 4.42. If two random variables have the joint probability density

$$f(x_1, x_2) = \begin{cases} \frac{2}{3}(x_1 + 2x_2) & \text{for } 0 < x_1 < 1, 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the conditional density of the first given that the second takes on the value x_2 . (Anna, Nov. 2003)

Solution. Clearly, the marginal density of X_2 is given by

$$f_{X_2}(x_2) = \begin{cases} \frac{1}{3}(1 + 4x_2) & \text{for } 0 < x_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the conditional density of X_1 given that $X_2 = x_2$.

Fix any value of x_2 in the interval $0 < x_2 < 1$.

Then the conditional density of X_1 given $X_2 = x_2$ is obtained as

$$f_{X_1|X_2}(x_1|x_2) = \frac{f(x_1, x_2)}{f_{X_2}(x_2)} = \frac{\frac{2}{3}(x_1 + 2x_2)}{\frac{1}{3}(1 + 4x_2)} = \frac{2(x_1 + 2x_2)}{1 + 4x_2} \quad \text{for } 0 < x_1 < 1$$

Thus, the conditional density of X_1 given $X_2 = x_2$ is given by

$$f_{X_1|X_2}(x_1|x_2) = \begin{cases} \frac{2(x_1 + 2x_2)}{1 + 4x_2} & \text{for } 0 < x_1 < 1 \\ 0 & \text{elsewhere} \end{cases} \quad \blacklozenge$$

EXAMPLE 4.43. Let X and Y be continuous random variables with joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{2}(x^2 + y^2) & 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Find $f_{X|Y}(x|y)$.

(Anna, Nov. 2004)

Solution. Clearly, the marginal density of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{2}(1 + 3y^2) & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we compute the marginal density of X given $Y = y$.

Fix any value of y in the interval $0 < y < 1$. Then, by definition, we have

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} = \frac{\frac{3}{2}(x^2 + y^2)}{\frac{1}{2}(1 + 3y^2)} \quad \text{for } 0 < x < 1 \\ &= \frac{3(x^2 + y^2)}{1 + 3y^2} \quad \text{for } 0 < x < 1 \end{aligned}$$

Thus, the conditional density of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = \begin{cases} \frac{3(x^2 + y^2)}{1 + 3y^2} & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases} \quad \blacklozenge$$

EXAMPLE 4.44. The joint probability density function of random variables X and Y is given by

$$f(x,y) = \begin{cases} \frac{1}{y} e^{-\frac{x}{y}} e^{-y}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Find $P(X > 1|Y = y)$.

(Anna, April 2005)

Solution. First, we find the marginal density function of Y .

If $y > 0$, then we have

$$f_Y(y) = \int_{x=0}^{\infty} f(x,y) dx = \int_{x=0}^{\infty} \frac{1}{y} e^{-\frac{x}{y}} e^{-y} dx = e^{-y} \int_{x=0}^{\infty} d \left[-e^{-\frac{x}{y}} \right]$$

Integrating, we have

$$f_Y(y) = e^{-y} \left[-e^{-\frac{x}{y}} \right]_{x=0}^{\infty} = e^{-y} [0 + 1] = e^{-y}$$

Thus, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} e^{-y} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the conditional density of X given $Y = y$.

Fix any value of y in the interval $y > 0$.

Then, by definition, we have

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x,y)}{f_Y(y)} = \frac{\frac{1}{y} e^{-\frac{x}{y}} e^{-y}}{e^{-y}} \quad \text{for } x > 0 \\ &= \frac{1}{y} e^{-\frac{x}{y}} \quad \text{for } x > 0 \end{aligned}$$

Thus, for any fixed $y > 0$, the conditional density of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = \begin{cases} \frac{1}{y} e^{-\frac{x}{y}} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

The required probability is given by

$$P(X > 1|Y = y) = \int_{x=1}^{\infty} f_{X|Y}(x|y) dx = \int_{x=1}^{\infty} \frac{1}{y} e^{-\frac{x}{y}} dx = \int_{x=1}^{\infty} d \left[-e^{-\frac{x}{y}} \right]$$

Integrating, we have

$$P(X > 1|Y = y) = \left[-e^{-\frac{x}{y}} \right]_{x=1}^{\infty} = \left[0 + e^{-\frac{1}{y}} \right] = e^{-\frac{1}{y}}$$

◆

EXAMPLE 4.45. The random variables X and Y have the joint probability density function

$$f(x,y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Are X and Y independent?
- (ii) Find the conditional probability density function of X given $Y = y$.
- (iii) Find the conditional probability density function of Y given $X = x$.

(Anna, April 2005; Nov. 2006)

Solution. (i) Clearly, the marginal density function of X is given by

$$f_X(x) = \begin{cases} 2x(x + \frac{1}{3}) & \text{for } 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{6}(y+2) & \text{for } 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Note that

$$f_X(x)f_Y(y) = \begin{cases} \frac{x}{3}(x + \frac{1}{3})(y+2) & \text{for } 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent random variables.

- (ii) Next, we compute the conditional density function of X given $Y = y$.

Fix any value of y in the interval $0 \leq y \leq 2$.

Then, by definition, we have

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{x(x + \frac{y}{3})}{\frac{1}{6}(y+2)} \quad \text{for } 0 \leq x \leq 1$$

Simplifying, we have

$$f_{X|Y}(x|y) = \frac{2x(3x+y)}{y+2} \quad \text{for } 0 \leq x \leq 1$$

Thus, for any fixed value of y in the interval $0 \leq y \leq 2$, the conditional density function of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = \begin{cases} \frac{2x(3x+y)}{y+2} & \text{for } 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

(iii) Next, we calculate the conditional density function of Y given $X = x$.

Fix any value of x in the interval $0 \leq x \leq 1$.

Then, by definition, we have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{x(x + \frac{y}{3})}{2x(x + \frac{1}{3})} \text{ for } 0 \leq y \leq 2$$

Simplifying, we have

$$f_{Y|X}(y|x) = \frac{3x+y}{2(3x+1)}$$

Thus, for any fixed x in $0 \leq x \leq 1$, the conditional density function of Y given $X = x$ is obtained as

$$f_{Y|X}(y|x) = \begin{cases} \frac{3x+y}{2(3x+1)} & \text{for } 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 4.46. Two random variables X and Y have the following joint probability density function

$$f(x,y) = \begin{cases} 2-x-y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find

- (i) marginal probability density functions of X and Y .
- (ii) conditional density functions.
- (iii) $\text{Var}(X)$ and $\text{Var}(Y)$.

(Anna, May 2006)

Solution. (i) Clearly, the marginal probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{3}{2} - x & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal probability density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{3}{2} - y & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

(ii) First, we find $f_{X|Y}(x|y)$, the conditional density function of X given $Y = y$.

Fix any value of y in the interval $0 \leq y \leq 1$.

Then, by definition, we have

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{2-x-y}{\frac{3}{2}-y} = \frac{2(2-x-y)}{3-2y} \text{ for } 0 \leq y \leq 1$$

Thus, for any fixed value of y in $0 \leq y \leq 1$, the conditional density function of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = \begin{cases} \frac{2(2-x-y)}{3-2y} & \text{for } 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find $f_{Y|X}(y|x)$, the conditional density function of Y given $X = x$.

Fix any value of x in the interval $0 \leq x \leq 1$.

Then, by definition, we have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{2-x-y}{\frac{3}{2}-x} = \frac{2(2-x-y)}{3-2x} \text{ for } 0 \leq x \leq 1$$

Thus, for any fixed value of x in $0 \leq x \leq 1$, the conditional density function of Y given $X = x$ is obtained as

$$f_{Y|X}(y|x) = \begin{cases} \frac{2(2-x-y)}{3-2x} & \text{for } 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

(iii) We start with finding the first two raw moments of X .

We have

$$\mu'_1(X) = E(X) = \int_{x=0}^1 x f_X(x) dx = \int_{x=0}^1 x \left[\frac{3}{2} - x \right] dx = \int_{x=0}^1 \left[\frac{3x}{2} - x^2 \right] dx$$

Integrating,

$$\mu'_1(X) = E(X) = \left[\frac{3x^2}{4} - \frac{x^3}{3} \right]_0^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$$

Next, we have

$$\mu'_2(X) = E(X^2) = \int_{x=0}^1 x^2 f_X(x) dx = \int_{x=0}^1 x^2 \left[\frac{3}{2} - x \right] dx = \int_{x=0}^1 \left[\frac{3x^2}{2} - x^3 \right] dx$$

Integrating,

$$\mu'_2(X) = E(X^2) = \left[\frac{x^3}{2} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

Hence, the variance of X is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{4} - \frac{25}{144} = \frac{11}{144}$$

Next we find the first two raw moments of Y .

We have

$$\mu'_1(Y) = E(Y) = \int_{y=0}^1 y f_Y(y) dy = \int_{y=0}^1 y \left[\frac{3}{2} - y \right] dy = \int_{y=0}^1 \left[\frac{3y}{2} - y^2 \right] dy$$

Integrating,

$$\mu'_1(Y) = E(Y) = \left[\frac{3y^2}{4} - \frac{y^3}{3} \right]_0^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$$

Next, we have

$$\mu'_2(Y) = E(Y^2) = \int_{y=0}^1 y^2 f_Y(y) dy = \int_{y=0}^1 y^2 \left[\frac{3}{2} - y \right] dy = \int_{y=0}^1 \left[\frac{3y^2}{2} - y^3 \right] dy$$

Integrating,

$$\mu'_2(Y) = E(Y^2) = \left[\frac{y^3}{2} - \frac{y^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

Hence, the variance of Y is given by

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{1}{4} - \frac{25}{144} = \frac{11}{144}$$

◆

EXAMPLE 4.47. Given the joint probability density function of (X, Y) as

$$f(x, y) = \begin{cases} 8xy, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the marginal and conditional probability density functions of X and Y . Are X and Y independent?

(Anna, May 2006)

Solution. Define the region A as

$$A = \{(x, y) \in \mathbb{R}^2 : 0 < x < y < 1\}$$

The region A is illustrated in Figure 4.15. The joint density function f takes values only in the region A .

Now, we find the marginal density function of X .

If $0 < x < 1$, then we have

$$f_X(x) = \int_{y=x}^1 f(x, y) dy = \int_{y=x}^1 8xy dy \quad [\text{See Figure 4.15}]$$

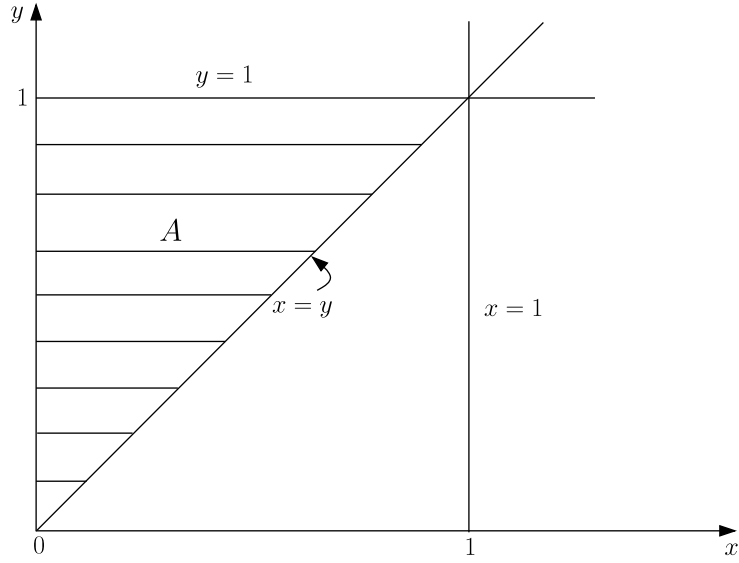


Figure 4.15: Region A.

Integrating,

$$f_X(x) = 4x [y^2]_{y=x}^1 = 4x [1 - x^2]$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} 4x(1 - x^2) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the marginal density function of Y .

If $0 < y < 1$, then we have

$$f_Y(y) = \int_{x=0}^y f(x, y) dx = \int_{x=0}^y 8xy dx \quad [\text{See Figure 4.15}]$$

Integrating, we have

$$f_Y(y) = 4y [x^2]_{x=0}^y = 4y^3$$

Thus, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 4y^3 & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the conditional density of X given $Y = y$.

Fix any value of y in the interval $0 < y < 1$.

Then, by definition, we have

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{8xy}{4y^3} = \frac{2x}{y^2} \quad \text{for } 0 < x < y$$

Thus, for any fixed value of y in $0 < y < 1$, the conditional density function of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = \begin{cases} \frac{2x}{y^2} & \text{for } 0 < x < y \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the conditional density of Y given $X = x$.

Fix any value of x in the interval $0 < x < 1$.

Then, by definition, we have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{8xy}{4x(1-x^2)} = \frac{2y}{1-x^2} \quad \text{for } x < y < 1$$

Thus, for any fixed value of x in $0 < x < 1$, the conditional density function of Y given $X = x$ is obtained as

$$f_{Y|X}(y|x) = \begin{cases} \frac{2y}{1-x^2} & \text{for } x < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Note that

$$f_X(x)f_Y(y) = \begin{cases} 16xy^3(1-x^2) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent random variables. ◆

EXAMPLE 4.48. The joint density function of the random variable (X, Y) is given by

$$f(x,y) = \begin{cases} 8xy & \text{for } 0 < x < 1, 0 < y < x \\ 0 & \text{elsewhere} \end{cases}$$

Find the

- (i) marginal density of Y ;
- (ii) conditional density of X given $Y = y$; and
- (iii) $P[X < \frac{1}{2}]$.

(Anna, May 2007)

Solution. (i) Similar to the calculations carried out in Example 4.47, it can be easily seen that the marginal density of Y is given by

$$f_Y(y) = \begin{cases} 4y(1-y^2) & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

and that the marginal density of X is given by

$$f_X(x) = \begin{cases} 4x^3 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- (ii) Fix any value of y in the interval $0 < y < 1$. Then the conditional density of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{8xy}{4y(1-y^2)} \text{ for } y < x < 1$$

Simplifying, we have

$$f_{X|Y}(x|y) = \frac{2x}{1-y^2} \text{ for } y < x < 1$$

- (iii) The required probability is calculated as

$$P\left[X < \frac{1}{2}\right] = \int_{x=0}^{\frac{1}{2}} f_X(x) dx = \int_{x=0}^{\frac{1}{2}} 4x^3 dx = [x^4]_0^{\frac{1}{2}} = \frac{1}{16}$$

◆

EXAMPLE 4.49. The joint probability density function of a bivariate random variable (X, Y) is

$$f_{XY}(x,y) = \begin{cases} k(x+y) & \text{if } 0 < x < 2, 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Find k .
- (ii) Find the marginal probability density functions of X and Y .
- (iii) Are X and Y independent?
- (iv) Find $f_{Y|X}(y|x)$ and $f_{X|Y}(x|y)$.

(Anna, Nov. 2006)

Solution. (i) It is easy to show that $k = \frac{1}{8}$

- (ii) Clearly, the marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{1}{4}(x+1) & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{4}(y+1) & \text{for } 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

- (iii) Note that

$$f_X(x)f_Y(y) = \begin{cases} \frac{1}{16}(x+1)(y+1) & \text{for } 0 < x < 2, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent random variables.

(iv) First, we find $f_{Y|X}(y|x)$, the conditional density of Y given $X = x$.

Fix any value of x in the interval $0 < x < 2$.

Then, by definition, we have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{\frac{1}{8}(x+y)}{\frac{1}{4}(x+1)} = \frac{x+y}{2(x+1)} \quad \text{for } 0 < x < 2$$

Thus, for any fixed y in $0 < y < 2$, the conditional density function of X given $Y = y$ is obtained as

$$f_{Y|X}(y|x) = \begin{cases} \frac{x+y}{2(x+1)} & \text{for } 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find $f_{X|Y}(x|y)$, the conditional density of X given $Y = y$.

Fix any value of y in the interval $0 < y < 2$.

Then, by definition, we have

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{\frac{1}{8}(x+y)}{\frac{1}{4}(y+1)} = \frac{x+y}{2(y+1)} \quad \text{for } 0 < x < 2$$

Thus, for any fixed x in $0 < x < 2$, the conditional density function of Y given $X = x$ is obtained as

$$f_{X|Y}(x|y) = \begin{cases} \frac{x+y}{2(y+1)} & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 4.50. The joint probability density function of the random variables X and Y is

$$f(x,y) = \begin{cases} 6x, & 0 < x < 1, 0 < y < 1-x \\ 0 & \text{otherwise} \end{cases}$$

(i) Show that X and Y are not independent.

(ii) Find $P(X > 0.3|Y = 0.5)$.

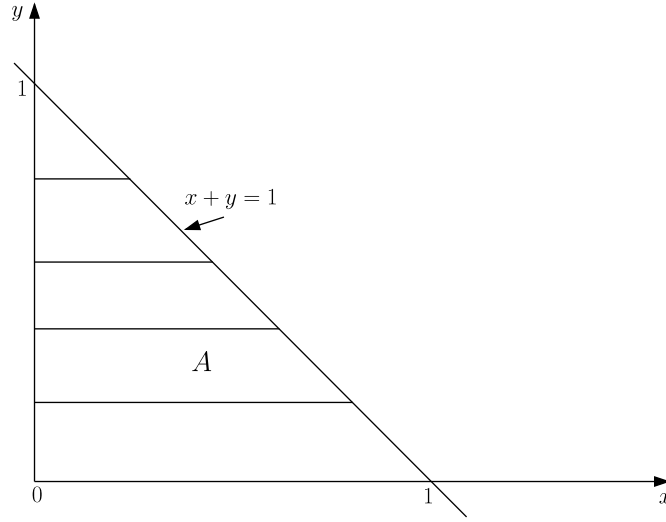
(Madras, April 1999)

Solution. Define the region A as

$$A = \{(x,y) \in \mathbb{R}^2 : 0 < x < 1, 0 < y < 1-x\}$$

Then the region A is as illustrated in Figure 4.16. The joint density function f takes values only in the region A .

(i) First, we show that X and Y are not independent. For this purpose, we compute the marginal density functions of X and Y .

**Figure 4.16:** Region A.

If $0 < x < 1$, then we have

$$f_X(x) = \int_{y=0}^{1-x} f(x,y) dy = \int_{y=0}^{1-x} 6x dy$$

Integrating, we have

$$f_X(x) = 6x[y]_{y=0}^{1-x} = 6x(1-x)$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} 6x(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we compute the marginal density function of Y .

If $0 < y < 1$, then we have

$$f_Y(y) = \int_{x=0}^{1-y} f(x,y) dx = \int_{x=0}^{1-y} 6x dx$$

Integrating, we have

$$f_Y(y) = [3x^2]_0^{1-y} = 3(1-y)^2$$

Thus, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 3(1-y)^2 & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Note that

$$f_X(x)f_Y(y) = \begin{cases} 18x(1-x)(1-y)^2 & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Thus, it follows that $f(x,y) \neq f_X(x)f_Y(y)$.

Hence, we conclude that X and Y are not independent random variables.

(ii) We find the conditional density function, $f_{X|Y}(x|0.5)$.

By definition, we have

$$f_{X|Y}(x|0.5) = \frac{f(x, 0.5)}{f_Y(0.5)} = \frac{6x}{3(0.5)^2} = \frac{2x}{0.25} = 8x \quad \text{for } 0 < x < 0.5$$

Thus,

$$f_{X|Y}(x|0.5) = \begin{cases} 8x & \text{for } 0 < x < 0.5 \\ 0 & \text{elsewhere} \end{cases}$$

Hence, the required probability can be readily computed as

$$P(X > 0.3|Y = 0.5) = \int_{x=0.3}^{0.5} f_{X|Y}(x|0.5) dx = \int_{x=0.3}^{0.5} 8x dx$$

Integrating, we have

$$P(X > 0.3|Y = 0.5) = [4x^2]_{x=0.3}^{0.5} = 4[0.25 - 0.09] = 4[0.16] = 0.64$$

Hence, we find that $P(X > 0.3|Y = 0.5) = 0.64$.

◆

PROBLEM SET 4.3

1. The joint probability distribution of X and Y is given by the following table.

$X \backslash Y$	0	1	2
0	0	$\frac{1}{10}$	$\frac{1}{5}$
1	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{10}$
2	$\frac{1}{10}$	$\frac{1}{5}$	0

- Find the marginal distributions of X and Y .
- Find the conditional distribution of Y given that $X = 2$.
- Are X and Y independent?

2. The joint probability distribution of X and Y is given by the following table:

$X \backslash Y$	1	2	3
1	$\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{12}$
2	$\frac{1}{24}$	$\frac{1}{6}$	$\frac{1}{12}$
3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{5}{24}$

- Find the marginal distributions of X and Y .
- Find the conditional distribution of X given that $Y = 1$.
- Are X and Y independent?

3. The joint probability distribution of X and Y is given by

$$f(x,y) = \frac{2x+3y}{54} \text{ for } x = 1, 2; y = 1, 2, 3$$

Find

- The marginal distributions of X and Y .
- The conditional distribution of X for $Y = y$.

4. The joint probability distribution of X and Y is given by

$$f(x,y) = \frac{x+4y}{45} \text{ for } x = 0, 1, 2; y = 0, 1, 2$$

Find:

- The marginal distributions of X and Y .
- The conditional distribution of Y for $X = x$.

5. The joint probability density function of X and Y is given by

$$f(x,y) = \begin{cases} \frac{2}{3} (2x+y), & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the marginal density functions of X and Y .
- Find the conditional density function of X given $Y = y$.
- Find the conditional density function of Y given $X = x$.
- Are X and Y independent?

6. The joint probability density function of X and Y is given by

$$f(x,y) = \begin{cases} \frac{9}{4} - x - y, & \text{if } 0 < x < 2, 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

- Find the marginal density functions of X and Y .
- Find the conditional density function of X given $Y = y$.
- Find the conditional density function of Y given $X = x$.
- Are X and Y independent?

7. The joint probability density function of X and Y is given by

$$f(x,y) = \begin{cases} \frac{K}{(1+x+y)^3} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find:

- The value of K .
 - The marginal density functions of X and Y .
 - The conditional density function of X given $Y = y$.
 - The conditional density function of Y given $X = x$.
 - $P(X < 4|Y = 1)$.
8. The joint probability density function of X and Y is given by

$$f(x,y) = \begin{cases} \frac{9(1+x+y)}{2(1+x)^4(1+y)^4} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Find the marginal density functions of X and Y .
 - Find the conditional density function of X given $Y = y$.
 - Find the conditional density function of Y given $X = x$.
 - Are X and Y independent?
9. The joint probability density function of X and Y is given by

$$f(x,y) = \begin{cases} K & \text{if } 0 < y < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the value of K .
 - Find the marginal density functions of X and Y .
 - Find the conditional density function of X given $Y = y$.
 - Find the conditional density function of Y given $X = x$.
 - Are X and Y independent?
10. The joint probability density function of X and Y is given by

$$f(x,y) = \begin{cases} Ke^{-x}(y+1) & \text{if } x > 0, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find the value of K .
- Find the marginal density functions of X and Y .
- Find the conditional density function of X given $Y = y$.
- Find the conditional density function of Y given $X = x$.
- Are X and Y independent?

4.4 EXPECTATION

First, we define the *expectation* or *expected value* for a two-dimensional random variable (X, Y) .

Definition 4.13. Let (X, Y) be a two-dimensional random variable. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function.

(a) If X and Y are discrete random variables and

$$\sum_i \sum_j |g(x_i, y_j)| P(X = x_i, Y = y_j) < \infty$$

then the *expectation* or *expected value* of $g(X, Y)$ is denoted by $Eg(X, Y)$ and is defined by

$$Eg(X, Y) = \sum_i \sum_j g(x_i, y_j) P(X = x_i, Y = y_j)$$

(b) If X and Y are continuous random variables and

$$\int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} |g(x, y)| f(x, y) dy dx < \infty$$

then the *expectation* or *expected value* of $g(X, Y)$ is denoted by $Eg(X, Y)$ and is defined by

$$Eg(X, Y) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} g(x, y) f(x, y) dy dx$$

Next, we establish some important properties of expectation.

Theorem 4.9. (*Addition Theorem of Expectation*) If X and Y are random variables of the same type, then

$$E(X + Y) = E(X) + E(Y)$$

provided all the expectations exist.

Proof. Let X and Y be continuous random variables with joint probability density function f . Let f_X and f_Y be the marginal density functions of X and Y , respectively. Then, by definition, we have

$$\begin{aligned} E(X + Y) &= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} (x + y) f(x, y) dy dx \\ &= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} x f(x, y) dy dx + \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} y f(x, y) dy dx \end{aligned}$$

Changing the order of integration in the second integral on the R.H.S. of the above equation, we have

$$\begin{aligned} E(X + Y) &= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} x f(x, y) dy dx + \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{\infty} y f(x, y) dx dy \\ &= \int_{x=-\infty}^{\infty} x f_X(x) dx + \int_{y=-\infty}^{\infty} y f_Y(y) dy = E(X) + E(Y) \end{aligned}$$

where we have used the definitions of marginal density functions, namely

$$f_X(x) = \int_{y=-\infty}^{\infty} f(x,y) dy \text{ and } f_Y(y) = \int_{x=-\infty}^{\infty} f(x,y) dx$$

The proof for the discrete case is very similar and hence, it is left as an exercise for the reader. $\blacklozenge$

Theorem 4.10. (*Expectation of a Linear Combination of RVs*) If X and Y are random variables of the same type and α, β are any real numbers, then

$$E(\alpha X + \beta Y) = \alpha E(X) + \beta E(Y)$$

Proof. By Theorem 4.9, we have

$$E(\alpha X + \beta Y) = E(\alpha X) + E(\beta Y) = \alpha E(X) + \beta E(Y)$$

$\blacklozenge$

Theorem 4.11. (*Multiplication Theorem of Expectation*) If X and Y are independent random variables of the same type, then

$$E(XY) = E(X)E(Y)$$

provided all the expectations exist.

Proof. Let X and Y be continuous, independent random variables; f be the joint probability density function of X and Y ; and f_X and f_Y be the marginal density functions of X and Y , respectively.

Then, by definition, we have

$$E(XY) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} xy f(x,y) dy dx$$

Since X and Y are independent, $f(x,y) = f_X(x)f_Y(y)$ for all $(x,y) \in \mathbb{R}^2$.
Then

$$E(XY) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} xy f_X(x) f_Y(y) dy dx$$

Since the integrals are separable, we have

$$E(XY) = \left[\int_{x=-\infty}^{\infty} f_X(x) dx \right] \left[\int_{y=-\infty}^{\infty} f_Y(y) dy \right] = E(X)E(Y)$$

The proof is similar for the discrete case and hence, it is left as an exercise for you. $\blacklozenge$

We note that Theorem 4.11 can be easily generalized to yield the following result. The proof is left as an exercise for you.

Theorem 4.12. If X and Y are independent random variables of the same type, and g, h are continuous real-valued functions, then

$$E[g(X)h(Y)] = E[g(X)] E[h(Y)]$$

As an application of Theorem 4.12, we have the following result.

Theorem 4.13. If X and Y are independent random variables, then the moment-generating function (MGF) of the sum of X and Y is equal to the product of the moment-generating functions of X and Y , i.e.

$$M_{X+Y}(t) = M_X(t) M_Y(t)$$

Proof. Suppose that X and Y are independent random variables. Then the MGF of $X + Y$ is given by

$$M_{X+Y}(t) = E e^{t(X+Y)} = E [e^{tX} e^{tY}]$$

By Theorem 4.12, we have

$$M_{X+Y}(t) = E [e^{tX}] E [e^{tY}] = M_X(t) M_Y(t)$$

◆

The above theorem can be easily generalized to n independent random variables as illustrated in the following example.

EXAMPLE 4.51. Prove that the MGF of the sum of a number of independent random variables is equal to the product of their respective moment generating functions.

(Anna, Nov. 2006)

Solution. Let $X_1, X_2, \dots, X_n$ be any n independent random variables. Let

$$Z \triangleq X_1 + X_2 + \dots + X_n$$

Then the MGF of Z is given by

$$M_Z(t) = E [e^{tZ}] = E [e^{t(X_1+X_2+\dots+X_n)}] = E [e^{tX_1} e^{tX_2} \dots e^{tX_n}]$$

Since $X_1, X_2, \dots, X_n$ are independent random variables, it follows that

$$M_Z(t) = E [e^{tX_1}] E [e^{tX_2}] \dots E [e^{tX_n}] = M_{X_1}(t) M_{X_2}(t) \dots M_{X_n}(t)$$

◆

Next, we prove the *Cauchy-Schwarz inequality* for expectation. Using this inequality, we can deduce an important property for the correlation coefficient of a bivariate distribution, which will be discussed later.

Theorem 4.14. (*Cauchy-Schwarz Inequality*) If X and Y are random variables of the same type, then

$$[E(XY)]^2 \leq E(X^2) E(Y^2)$$

Proof. Let Z be a real-valued function of $t \in \mathbb{R}$ defined by

$$\psi(t) = E[(X + tY)^2], \quad \text{where } t \in \mathbb{R}$$

Since $(X + tY)^2 \geq 0$ for all values of t , it follows that

$$\psi(t) = E[(X + tY)^2] \geq 0 \quad \text{for all } t \in \mathbb{R}$$

Thus, we have

$$\psi(t) = E[X^2 + 2tXY + t^2Y^2] = E[X^2] + 2tE[XY] + t^2E[Y^2] \geq 0$$

Define

$$A = E[Y^2], B = 2E[XY] \quad \text{and} \quad C = E[X^2]$$

Then, we have

$$\psi(t) = At^2 + Bt + C \geq 0 \quad \text{for all } t \in \mathbb{R}$$

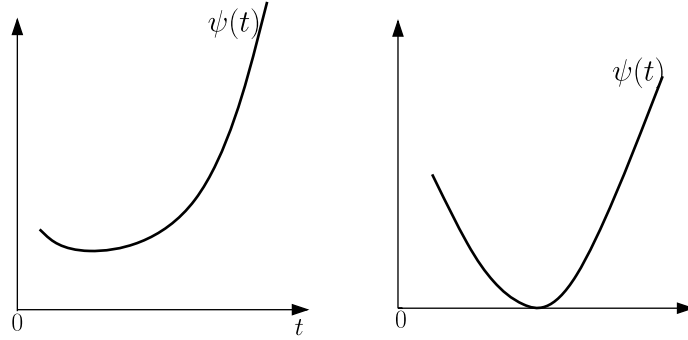


Figure 4.17: Graph of ψ , Cauchy-Schwarz Inequality.

From the graph of $\psi(t)$, as shown in Figure 4.17, it is clear that either ψ has no real roots (in which case $B^2 - 4AC < 0$) or ψ has a unique real root (in which case $B^2 - 4AC = 0$). Combining the two cases, we have $B^2 - 4AC \leq 0$.

Hence, we have

$$4E[XY]^2 - 4E[Y^2] E[X^2] \leq 0$$

Simplifying, we have

$$E[XY]^2 \leq E[X^2] E[Y^2]$$

◆

EXAMPLE 4.52. Let X and Y be random variables with joint density function

$$f(x, y) = \begin{cases} 4xy, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find $E(XY)$.

(Anna, Nov. 2006)

Solution. By definition, we have

$$E(XY) = \int_{x=0}^1 \int_{y=0}^1 xy f(x,y) dy dx = \int_{x=0}^1 \int_{y=0}^1 xy (4xy) dy dx$$

Since the integrals are separable, we have

$$E(XY) = 4 \left[\int_{x=0}^1 x^2 dx \right] \left[\int_{y=0}^1 y^2 dy \right]$$

Integrating,

$$\begin{aligned} E(XY) &= 4 \left[\frac{x^3}{3} \right]_{x=0}^1 \left[\frac{y^3}{3} \right]_{y=0}^1 \\ &= 4 \times \frac{1}{3} \times \frac{1}{3} = \frac{4}{9} \end{aligned}$$

◆

EXAMPLE 4.53. In producing gallium-arsenide microchips, it is known that the ratio between gallium and arsenide is independent of producing a high percentage of workable wafer, which are main components of microchips. Let X denote the ratio of gallium to arsenide and Y denote the percentage of workable microwafers retrieved during a 1-hour period. X and Y are independent random variables with the joint density being known as

$$f(x,y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Show that $E(XY) = E(X)E(Y)$.

(Anna, Nov. 2006)

Solution. We find that

$$E(XY) = \int_{x=0}^2 \int_{y=0}^1 xy f(x,y) dy dx = \int_{x=0}^2 \int_{y=0}^1 xy \left(\frac{x(1+3y^2)}{4} \right) dy dx$$

Since the integrals are separable, we have

$$E(XY) = \frac{1}{4} \left[\int_{x=0}^2 x^2 dx \right] \left[\int_{y=0}^1 y(1+3y^2) dy \right]$$

Integrating,

$$\begin{aligned} E(XY) &= \frac{1}{4} \left[\frac{x^3}{3} \right]_{x=0}^2 \left[\frac{y^2}{2} + \frac{3y^4}{4} \right]_{y=0}^1 \\ &= \frac{1}{4} \times \frac{8}{3} \times \frac{5}{4} = \frac{5}{6} \end{aligned}$$

To compute $E(X)$ and $E(Y)$, we need to find the marginal density functions of X and Y . If $0 < x < 2$, then the marginal density of X is given by

$$f_X(x) = \int_{y=0}^1 f(x,y) dy = \int_{y=0}^1 \frac{x(1+3y^2)}{4} dy$$

Integrating, we have

$$f_X(x) = \frac{x}{4} [y+y^3]_{y=0}^1 = \frac{x}{4} \times 2 = \frac{x}{2}$$

Thus, the marginal density of X is given by

$$f_X(x) = \begin{cases} \frac{x}{2} & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

If $0 < y < 1$, then the marginal density of Y is given by

$$f_Y(y) = \int_{x=0}^2 f(x,y) dx = \int_{x=0}^2 \frac{x(1+3y^2)}{4} dx$$

Integrating, we have

$$f_Y(y) = \frac{1+3y^2}{4} \left[\frac{x^2}{2} \right]_{x=0}^2 = \frac{1+3y^2}{4} \times 2 = \frac{1+3y^2}{2}$$

Thus, the marginal density of Y is given by

$$f_Y(y) = \begin{cases} \frac{1+3y^2}{2} & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we compute $E(X)$. By definition, we have

$$E(X) = \int_{x=0}^2 x f_X(x) dx = \int_{x=0}^2 x \left(\frac{x}{2} \right) dx$$

Integrating,

$$E(X) = \frac{1}{2} \left[\frac{x^3}{3} \right]_{x=0}^2 = \frac{1}{2} \times \frac{8}{3} = \frac{4}{3}$$

Next, we compute $E(Y)$. By definition,

$$E(Y) = \int_{y=0}^1 y f_Y(y) dy = \int_{y=0}^1 y \left(\frac{1+3y^2}{2} \right) dy$$

Integrating, we have

$$E(Y) = \frac{1}{2} \left[\frac{y^2}{2} + \frac{3y^4}{4} \right]_{y=0}^1 = \frac{1}{2} \times \frac{5}{4} = \frac{5}{8}$$

Thus, it follows that

$$E(X)E(Y) = \frac{4}{3} \times \frac{5}{8} = \frac{5}{6} = E(XY)$$

Hence, we conclude that $E(XY) = E(X)E(Y)$. ◆

EXAMPLE 4.54. Two random variables X and Y have the joint probability density function

$$f(x,y) = \begin{cases} \frac{xy}{96} & \text{if } 0 < x < 4, 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Find $E(XY)$.

(Anna, Nov. 2006)

Solution. By definition, we have

$$E(XY) = \int_{x=0}^4 \int_{y=1}^5 xy f(x,y) dy dx = \int_{x=0}^4 \int_{y=1}^5 xy \left(\frac{xy}{96} \right) dy dx$$

Since the integrals are separable, we have

$$E(XY) = \frac{1}{96} \left[\int_{x=0}^4 x^2 dx \right] \left[\int_{y=1}^5 y^2 dy \right]$$

Integrating, we have

$$\begin{aligned} E(XY) &= \frac{1}{96} \left[\frac{x^3}{3} \right]_{x=0}^4 \left[\frac{y^3}{3} \right]_{y=1}^5 \\ &= \frac{1}{96} \times \frac{8}{3} \times \frac{124}{3} = \frac{248}{27} = 9.1852 \end{aligned}$$

◆

EXAMPLE 4.55. The joint probability density function of the random variables, X and Y , is given by

$$f(x,y) = \begin{cases} 25, & 0.95 < x < 1.15, 0.95 < y < 1.15 \\ 0, & \text{otherwise} \end{cases}$$

Find $P(XY < 1), E(XY)$.

(Anna, May 2000, Model, 2003)

Solution. We find that

$$P(XY < 1) = \int_{x=0.95}^{1.15} \int_{y=0.95}^{1/x} f(x,y) dy dx = \int_{x=0.95}^{1.15} \int_{y=0.95}^{\frac{1}{x}} 25 dy dx$$

Integrating y , we have

$$P(XY < 1) = 25 \int_{x=0.95}^{1.15} [y]_{y=0.95}^{\frac{1}{x}} dx = 25 \int_{x=0.95}^{1.15} \left[\frac{1}{x} - 0.95 \right] dx$$

Integrating x , we have

$$\begin{aligned} P(XY < 1) &= 25 [\log x - 0.95x]_{x=0.95}^{1.15} = 25 [(0.1398 - 1.0925) - (-0.0513 - 0.9025)] \\ &= 25 \times 0.0011 = 0.0275 \end{aligned}$$

Next, we find $E(XY)$. We have

$$E(XY) = \int_{x=0.95}^{1.15} \int_{y=0.95}^{1/x} xy f(x,y) dy dx = \int_{x=0.95}^{1.15} \int_{y=0.95}^{1/x} 25xy dy dx$$

Since the integrals are separable, we have

$$E(XY) = 25 \left[\int_{x=0.95}^{1.15} x dx \right] \left[\int_{y=0.95}^{1/x} y dy \right]$$

Integrating, we have

$$\begin{aligned} E(XY) &= 25 \left[\frac{x^2}{2} \right]_{x=0.95}^{1.15} \left[\frac{y^2}{2} \right]_{y=0.95}^{1/x} = 25 [0.6612 - 0.4512] [0.6612 - 0.4512] \\ &= 25 \times 0.21 \times 0.21 = 1.1025 \end{aligned}$$

◆

EXAMPLE 4.56. If the joint probability density function of (X, Y) is given by

$$f(x,y) = e^{-(x+y)}, \quad x \geq 0, y \geq 0$$

find $E(XY)$.

(Anna, April 2005)

Solution. By definition, we have

$$E(XY) = \int_{x=0}^{\infty} \int_{y=0}^{\infty} xy f(x,y) dy dx = \int_{x=0}^{\infty} \int_{y=0}^{\infty} xye^{-x}e^{-y} dy dx$$

Since the integrals are separable, we have

$$E(XY) = \left[\int_{x=0}^{\infty} xe^{-x} dx \right] \left[\int_{y=0}^{\infty} ye^{-y} dy \right] = \left[\int_{x=0}^{\infty} xe^{-x} dx \right]^2$$

Now,

$$\begin{aligned} \int_{x=0}^{\infty} xe^{-x} dx &= \int_{x=0}^{\infty} x d[-e^{-x}] = [x(-e^{-x})]_{x=0}^{\infty} + \int_{x=0}^{\infty} e^{-x} dx \\ &= 0 + [-e^{-x}]_{x=0}^{\infty} = 0 + 1 = 1 \end{aligned}$$

Hence, it follows that

$$E(XY) = \left[\int_{x=0}^{\infty} xe^{-x} dx \right]^2 = 1^2 = 1$$

◆

PROBLEM SET 4.4

1. The joint probability distribution of X and Y is given by

$$f(x, y) = \frac{2x+y}{12} \text{ for } x = 0, 1; y = 0, 1, 2$$

- Find the marginal distributions of X and Y .
- Verify that $E(X+Y) = E(X) + E(Y)$.
- Compute $E(XY)$.

2. The joint probability distribution of X and Y is given by

X \ Y	-1	0	1
-1	$\frac{3}{20}$	$\frac{1}{10}$	$\frac{1}{10}$
0	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{5}$
1	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{1}{20}$

- Find the marginal distributions of X and Y .
- Find the means of X and Y .
- Verify that $E(X+Y) = E(X) + E(Y)$.
- Compute $E(XY)$.

3. The joint probability distribution of X and Y is given by

X \ Y	0	1	2
0	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{20}$
1	$\frac{1}{10}$	$\frac{1}{20}$	$\frac{1}{10}$
2	$\frac{1}{4}$	$\frac{1}{20}$	$\frac{1}{10}$

- (a) Find the marginal distributions of X and Y .
 (b) Find $E(X)$, $E(X^2)$, $E(Y)$ and $E(Y^2)$.
 (c) Verify that $E(X+Y) = E(X) + E(Y)$.
 (d) Verify Cauchy-Schwartz inequality for (X, Y) .
4. The joint probability density of random variables X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{8}(x+1)y & \text{if } 0 < x < 2, 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(XY < 1)$ and $E(XY)$.

5. The joint probability density of random variables X and Y is given by

$$f(x, y) = \begin{cases} \frac{1}{2} & \text{if } 0 < x < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginal density functions of X and Y .
 (b) Verify that $E(X+Y) = E(X) + E(Y)$.
 (c) Find $E(XY)$.
6. The joint probability density of random variables X and Y is given by

$$f(x, y) = \begin{cases} xye^{-\frac{1}{2}(x^2+y^2)} & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the marginal density functions of X and Y .
 (b) Find $E(X)$, $E(X^2)$, $E(Y)$ and $E(Y^2)$.
 (c) Verify that $E(X+Y) = E(X) + E(Y)$.
 (d) Compute $E(XY)$.
 (e) Verify Cauchy-Schwartz inequality for X and Y .

4.5 SUMS OF INDEPENDENT RANDOM VARIABLES

EXAMPLE 4.57. Show that the sum of two independent binomial random variables X and Y with parameters (n_1, p_1) and (n_2, p_2) , respectively, is also a binomial random variable provided $p_1 = p_2 = p$.
(Additive Property of Binomial Distribution)

Solution. Let X and Y be independent binomial random variables with parameters (n_1, p_1) and (n_2, p_2) respectively. We shall show that $X + Y$ also is a binomial random variable provided $p_1 = p_2 = p$.

By Theorem 3.4, we know that the MGF of X is given by

$$M_X(t) = (q_1 + p_1 e^t)^{n_1}, \text{ where } q_1 = 1 - p_1$$

Similarly, the MGF of Y is given by

$$M_Y(t) = (q_2 + p_2 e^t)^{n_2}, \text{ where } q_2 = 1 - p_2$$

Since X and Y are independent, by Theorem 4.13, we have

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$$

Hence

$$M_{X+Y}(t) = (q_1 + p_1 e^t)^{n_1} \cdot (q_2 + p_2 e^t)^{n_2} \quad (4.6)$$

From Eq. (4.6), it is evident that $M_{X+Y}(t)$ can assume the standard form of a binomial moment generating function only when $p_1 = p_2 = p$ or when $q_1 = q_2 = q$. Under this condition ($p_1 = p_2 = p$), Eq. (4.6) becomes

$$M_{X+Y}(t) = (q + p e^t)^{n_1} \cdot (q + p e^t)^{n_2} = (q + p e^t)^{n_1 + n_2}$$

Hence, when $p_1 = p_2 = p$, $X + Y$ is a binomial random variable with parameters $(n_1 + n_2, p)$. $\blacklozenge$

EXAMPLE 4.58. Consider $Z = X + Y$. Show that if X and Y are independent Poisson random variables with parameters λ_1 and λ_2 , respectively, then Z is also a Poisson random variable with parameter $\lambda_1 + \lambda_2$. (*Additive Property of Poisson Distribution*)

(Anna, Model 2003)

Solution. Let X and Y be two independent Poisson random variables with parameters λ_1 and λ_2 , respectively. We shall show that $X + Y$ is also a Poisson random variable with parameter $\lambda_1 + \lambda_2$.

By Theorem 3.9, we know that the MGF of X is given by

$$M_X(t) = e^{\lambda_1(e^t - 1)}$$

Similarly, the MGF of Y is given by

$$M_Y(t) = e^{\lambda_2(e^t - 1)}$$

Since X and Y are independent, by Theorem 4.13, we have

$$\begin{aligned} M_{X+Y}(t) &= M_X(t) \cdot M_Y(t) \\ &= e^{\lambda_1(e^t - 1)} \cdot e^{\lambda_2(e^t - 1)} \\ &= e^{(\lambda_1 + \lambda_2)(e^t - 1)} \end{aligned}$$

which is the standard form of the MGF of the Poisson random variable with parameter $\lambda_1 + \lambda_2$. By the uniqueness theorem of moment generating functions, it follows that $Z = X + Y$ is a Poisson random variable with parameter $\lambda_1 + \lambda_2$. $\blacklozenge$

EXAMPLE 4.59. Find the MGF of a Poisson variate. Deduce that the sum of two independent Poisson variates is a Poisson variate, while the difference is not a Poisson variate.

(Anna, May 2006)

Solution. Suppose that X and Y are independent Poisson variates with parameters λ_1 and λ_2 , respectively. By Theorem 3.9, we know that the MGF of X is given by

$$M_X(t) = e^{\lambda_1(e^t - 1)}$$

Similarly, the MGF of Y is given by

$$M_Y(t) = e^{\lambda_2(e^t - 1)}$$

As showed in Example 4.58, the sum of X and Y is a Poisson variate with parameter $\lambda_1 + \lambda_2$, which follows from the observation that

$$M_{X+Y}(t) = e^{(\lambda_1 + \lambda_2)(e^t - 1)}$$

which is the standard form of the MGF of the Poisson variate with parameter $\lambda_1 + \lambda_2$.

However, we remark that $X - Y$ is not a Poisson variate. Note that

$$M_{X-Y}(t) = M_X(t) \cdot M_{-Y}(t) = M_X(t) \cdot M_Y(-t)$$

since $M_{-Y}(t) = E(e^{-tY}) = M_Y(-t)$.

Hence, we have

$$M_{X-Y}(t) = e^{\lambda_1(e^t - 1)} e^{\lambda_2(e^{-t} - 1)} = e^{\lambda_1(e^t - 1) + \lambda_2(e^{-t} - 1)}$$

which is not the standard form of the moment generating function of a Poisson variate. By the uniqueness theorem of moment generating functions of random variables, we conclude that the difference of independent Poisson variates is not a Poisson variate. $\blacklozenge$

EXAMPLE 4.60. If X_1 and X_2 are independent Poisson variables with parameters λ_1 and λ_2 , show that the conditional distribution of X_1 given $X_1 + X_2$ follows binomial distribution.

(Anna, May 2006)

Solution. Suppose that X_1 and X_2 are independent Poisson variables with parameters λ_1 and λ_2 , respectively. By the additive property of Poisson distributions (see Example 4.58), it follows that $X_1 + X_2$ is a Poisson variable with parameter $\lambda_1 + \lambda_2$.

It follows that

$$\begin{aligned} P(X_1 = x) &= \frac{e^{-\lambda_1} \lambda_1^x}{x!} \quad \text{for } x = 0, 1, 2, \dots \\ P(X_2 = y) &= \frac{e^{-\lambda_2} \lambda_2^y}{y!} \quad \text{for } y = 0, 1, 2, \dots \\ P(X_1 + X_2 = z) &= \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^z}{z!} \quad \text{for } z = 0, 1, 2, \dots \end{aligned}$$

Fix any value of $X_1 + X_2 = z$, where $z = 0, 1, 2, \dots$

Then it is clear that X_1 can take values only from $x = 0, 1, 2, \dots, z$.

By definition, the conditional distribution of X_1 given $X_1 + X_2$ is given by

$$f_{X_1|X_1+X_2}(x|z) = \frac{P(X_1 = x, X_1 + X_2 = z)}{P(X_1 + X_2 = z)} = \frac{P(X_1 = x, X_2 = z - x)}{P(X_1 + X_2 = z)} \quad \text{for } x = 0, 1, 2, \dots, z$$

Since X_1 and X_2 are independent, we have

$$f_{X_1|X_1+X_2}(x|z) = \frac{P(X_1 = x) \cdot P(X_2 = z - x)}{P(X_1 + X_2 = z)} \quad \text{for } x = 0, 1, 2, \dots, z$$

Substituting the values for $P(X_1 = x)$, $P(X_2 = z - x)$ and $P(X_1 + X_2 = z)$, we have

$$f_{X_1|X_1+X_2}(x|z) = \frac{e^{-\lambda_1} \lambda_1^x}{x!} \frac{e^{-\lambda_2} \lambda_2^{z-x}}{(z-x)!} \frac{z!}{e^{-(\lambda_1+\lambda_2)} (\lambda_1 + \lambda_2)^z}$$

Simplifying, we have

$$\begin{aligned} f_{X_1|X_1+X_2}(x|z) &= \binom{z}{x} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^x \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{z-x} \\ &= \binom{z}{x} p^x q^{z-x} \quad \text{for } x = 0, 1, 2, \dots, z \end{aligned}$$

where

$$p \triangleq \frac{\lambda_1}{\lambda_1 + \lambda_2} \quad \text{and} \quad q \triangleq 1 - p = \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

Hence, it follows that the conditional distribution of X_1 given $X_1 + X_2$ is a binomial distribution with parameters (z, p) , where z is the value assumed by $X_1 + X_2$ and p is equal to $\frac{\lambda_1}{\lambda_1 + \lambda_2}$. $\blacklozenge$

EXAMPLE 4.61. If X_1, X_2 are independent random variables each having geometric distribution $q^k p$, $k = 0, 1, 2, \dots$, show that the conditional distribution of X_1 given $X_1 + X_2$ is uniform.

(Anna, Nov. 2006)

Solution. Let X and Y be independent random variables each having the geometric distribution $q^k p$, $k = 0, 1, 2, \dots$. Hence, it follows that

$$P(X = x) = q^x p \quad \text{for } x = 0, 1, 2, \dots$$

and

$$P(Y = y) = q^y p \quad \text{for } y = 0, 1, 2, \dots$$

Suppose that $X + Y = n$, where $n = 0, 1, 2, \dots$

Then it is clear that X takes the values $x = 0, 1, 2, \dots, n$.

Hence the conditional distribution of X given $X + Y = n$ is given by

$$f_{X|X+Y}(x|n) = \frac{P(X = x, X + Y = n)}{P(X + Y = n)} = \frac{P(X = x, Y = n - x)}{\sum_{s=0}^n P(X = s, Y = n - s)} \quad \text{for } x = 0, 1, 2, \dots, n$$

Since X and Y are independent, we have

$$f_{X|X+Y}(x|n) = \frac{P(X=x)P(Y=n-x)}{\sum_{s=0}^n P(X=s)P(Y=n-s)}$$

Thus,

$$\begin{aligned} f_{X|X+Y}(x|n) &= \frac{q^x p q^{(n-x)} p}{\sum_{s=0}^n q^s p q^{n-s} p} = \frac{q^n p^2}{\sum_{s=0}^n q^n p^2} = \frac{q^n p^2}{(n+1)q^n p^2} \\ &= \frac{1}{n+1} \quad \text{for } x = 0, 1, 2, \dots, n \end{aligned}$$

which is a discrete uniform distribution. ◆

Theorem 4.15. Let X and Y be independent continuous random variables with marginal probability density functions f_X and f_Y , respectively. Then the probability density function of $X + Y$ is given by

$$f_{X+Y}(\xi) = \int_{y=-\infty}^{\infty} f_X(\xi - y) f_Y(y) dy$$

Proof. Since X and Y are independent random variables, the joint probability density function of X and Y is given by

$$f(x, y) = f_X(x) f_Y(y) \quad \text{for all } (x, y) \in \mathbb{R}^2$$

We obtain the probability density function of $X + Y$ by distribution function technique.

First, we obtain the cumulative distribution function of $X + Y$. We note that

$$F_{X+Y}(\xi) = P(X + Y \leq \xi) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{\xi-y} f_X(x) f_Y(y) dx dy$$

From the observation

$$F_X(\xi - y) = \int_{x=-\infty}^{\xi-y} f_X(x) dx$$

it follows immediately that

$$F_{X+Y}(\xi) = \int_{y=-\infty}^{\infty} F_X(\xi - y) f_Y(y) dy$$

Hence, the probability density function of $X + Y$ is obtained by

$$f_{X+Y}(\xi) = \frac{d}{d\xi} F_{X+Y}(\xi) = \int_{y=-\infty}^{\infty} \frac{d}{d\xi} F_X(\xi - y) f_Y(y) dy$$

Thus, we have

$$f_{X+Y}(\xi) = \int_{y=-\infty}^{\infty} f_X(\xi - y) f_Y(y) dy$$
◆

EXAMPLE 4.62. Let X and Y be independent uniform random variables over $(0, 1)$. Find the probability density function of $Z = X + Y$.

(Anna, Nov. 2005)

Solution. Suppose that X and Y are independent random variables, both are uniformly over the interval $(0, 1)$. Then the probability densities of X and Y are defined by

$$f_X(\xi) = f_Y(\xi) = \begin{cases} 1 & \text{if } 0 < \xi < 1 \\ 0 & \text{otherwise} \end{cases}$$

By Theorem 4.15, the probability density function of $X + Y$ is given by

$$f_{X+Y}(\xi) = \int_{y=-\infty}^{\infty} f_X(\xi - y) f_Y(y) dy = \int_{y=0}^1 f_X(\xi - y) dy$$

If $0 \leq \xi \leq 1$, then we have

$$f_{X+Y}(\xi) = \int_{y=0}^{\xi} dy = \xi$$

If $1 < \xi < 2$, then we have

$$f_{X+Y}(\xi) = \int_{y=\xi-1}^1 dy = 2 - \xi$$

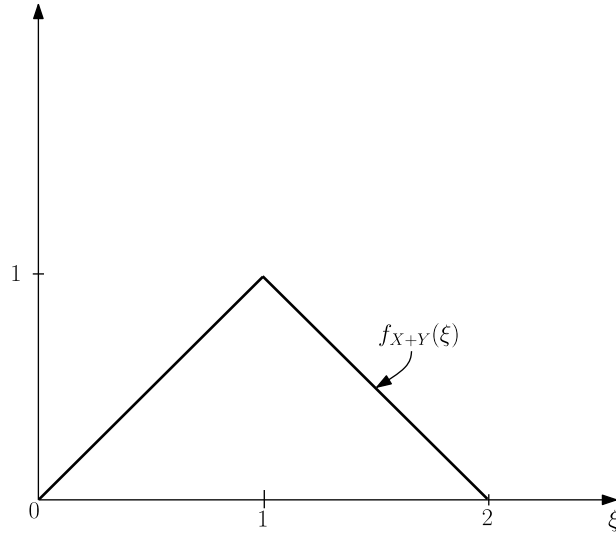


Figure 4.18: Triangular density function.

Thus, the probability density function of $X + Y$ is given by

$$f_{X+Y}(\xi) = \begin{cases} \xi & \text{if } 0 \leq \xi \leq 1 \\ 2 - \xi & \text{if } 1 < \xi < 2 \\ 0 & \text{otherwise} \end{cases}$$

which is illustrated in Figure 4.18. It follows that $X + Y$ has the triangular density function. $\blacklozenge$

EXAMPLE 4.63. If X and Y are independent gamma random variables with parameters (α, λ) and (β, λ) respectively, show that $X + Y$ is a gamma random variable with parameters $(\alpha + \beta, \lambda)$.

Solution. Suppose that X and Y are independent gamma variables with parameters (α, λ) and (β, λ) respectively. By Example 3.73, we know that the moment generating function of X is given by

$$M_X(t) = \left(1 - \frac{t}{\lambda}\right)^{-\alpha}, \text{ where } \left|\frac{t}{\lambda}\right| < 1$$

Similarly, the MGF of Y is given by

$$M_Y(t) = \left(1 - \frac{t}{\lambda}\right)^{-\beta}, \text{ where } \left|\frac{t}{\lambda}\right| < 1$$

Since X and Y are independent, the MGF of $X + Y$ is given by

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t) = \left(1 - \frac{t}{\lambda}\right)^{-\alpha} \left(1 - \frac{t}{\lambda}\right)^{-\beta}$$

Simplifying, we have

$$M_{X+Y}(t) = \left(1 - \frac{t}{\lambda}\right)^{-(\alpha+\beta)}$$

which is the standard form of the MGF of the gamma random variable with parameters $(\alpha + \beta, \lambda)$. By the uniqueness theorem of moment generating functions, it follows that $X + Y$ is a gamma random variable with parameters $(\alpha + \beta, \lambda)$. $\blacklozenge$

EXAMPLE 4.64. If X and Y are independent exponential random variables with the same parameter λ , show that $X + Y$ is a gamma random variable with parameters $(2, \lambda)$.

Solution. Suppose that X and Y are independent exponential random variables with the same parameter λ . By Example 3.62, we know that the MGF of X is given by

$$M_X(t) = \frac{\lambda}{\lambda - t} = \left[1 - \frac{t}{\lambda}\right]^{-1} \text{ where } \frac{t}{\lambda} < 1$$

Since Y has the same parameter λ , the moment generating function of Y is given by

$$M_Y(t) = \frac{\lambda}{\lambda - t} = \left[1 - \frac{t}{\lambda}\right]^{-1} \text{ where } \frac{t}{\lambda} < 1$$

Since X and Y are independent, the MGF of $X + Y$ is given by

$$M_{X+Y}(t) = M_X(t) \cdot M_Y(t) = \left[1 - \frac{t}{\lambda}\right]^{-2} \text{ (where } \frac{t}{\lambda} < 1)$$

which is the standard form of the MGF of the gamma random variable with parameters $(2, \lambda)$. By the uniqueness theorem of moment generating functions, it follows that $X + Y$ is a gamma random variable with parameters $(2, \lambda)$. $\blacklozenge$

EXAMPLE 4.65. $X_1, X_2, \dots, X_n$ are independent, identically distributed exponential random variables each with parameter λ . Prove that $X_1 + X_2 + \dots + X_n$ is a gamma variable with parameters n and λ .

(Anna, Model 2003)

Solution. Suppose that $X_1, X_2, \dots, X_n$ are independent, identically distributed exponential random variables each with parameter λ . Then for each i , we have

$$M_{X_i}(t) = \left[1 - \frac{t}{\lambda}\right]^{-1} \quad \text{where } \frac{t}{\lambda} < 1$$

Since $X_1, X_2, \dots, X_n$ are independent, the MGF of $Z \triangleq X_1 + X_2 + \dots + X_n$ is given by

$$M_Z(t) = \prod_{i=1}^n M_{X_i}(t) = \left[1 - \frac{t}{\lambda}\right]^{-n} \quad (\text{where } \frac{t}{\lambda} < 1)$$

which is the standard form of the MGF of the gamma random variable with parameters (n, λ) . By the uniqueness theorem of moment generating functions, it follows that $Z = X_1 + X_2 + \dots + X_n$ is a gamma random variable with parameters (n, λ) . $\blacklozenge$

EXAMPLE 4.66. If $X_i, i = 1, 2, \dots, n$ are independent normal random variables with parameters (μ_i, σ_i^2) , $i = 1, 2, \dots, n$, respectively, show that $X = X_1 + X_2 + \dots + X_n$ is a normal random variable with parameters (μ, σ^2) , where $\mu = \sum_{i=1}^n \mu_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$. (Additive Property of Normal Distribution)

Solution. Suppose that $X_i, i = 1, 2, \dots, n$ are independent normal random variables with parameters (μ_i, σ_i^2) , $i = 1, 2, \dots, n$, respectively. Define:

$$X \triangleq X_1 + X_2 + \dots + X_n$$

By Example 3.85, we know that the MGF of X_i is given by

$$M_{X_i}(t) = \exp\left(\mu_i t + \frac{\sigma_i^2 t^2}{2}\right), \quad i = 1, 2, \dots, n$$

Since $X_1, X_2, \dots, X_n$ are independent, the MGF of X is given by

$$M_X(t) = \prod_{i=1}^n M_{X_i}(t) = \prod_{i=1}^n \exp\left(\mu_i t + \frac{\sigma_i^2 t^2}{2}\right) = \exp\left(\sum_{i=1}^n \left[\mu_i t + \frac{\sigma_i^2 t^2}{2}\right]\right)$$

Simplifying, we have

$$M_X(t) = \exp\left(\left[\sum_{i=1}^n \mu_i\right] t + \frac{\left[\sum_{i=1}^n \sigma_i^2\right] t^2}{2}\right)$$

If we define

$$\mu = \sum_{i=1}^n \mu_i \quad \text{and} \quad \sigma^2 = \sum_{i=1}^n \sigma_i^2$$

then we have

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$$

which is the standard form of the MGF of a normal random variable with parameters (μ, σ^2) . By the uniqueness theorem of moment generating functions, it follows that X is a normal random variable with parameters (μ, σ^2) . $\blacklozenge$

EXAMPLE 4.67. If X and Y are independent normal random variables each following $N(0, 2)$, find the PDF of $Z = 2X + 3Y$.

(Anna, April 2003)

Solution. Since X follows $N(0, 2)$, the MGF of X is given by

$$M_X(t) = e^{\frac{2^2 t^2}{2}} = e^{2t^2}$$

Similarly, the MGF of Y is given by

$$M_Y(t) = e^{\frac{2^2 t^2}{2}} = e^{2t^2}$$

Since X and Y are independent, the MGF of $Z = 2X + 3Y$ is given by

$$M_Z(t) = E[e^{tZ}] = E[e^{2tX} \cdot e^{3tY}] = E[e^{2tX}] E[e^{3tY}]$$

Thus,

$$\begin{aligned} M_Z(t) &= M_X(2t)M_Y(3t) = e^{2(2t)^2} e^{2(3t)^2} = e^{8t^2} e^{18t^2} = e^{26t^2} \\ &= e^{0t + \frac{\sqrt{26}^2 t^2}{2}} \end{aligned}$$

which is the standard form of the MGF of the normal random variable with parameters $(0, \sigma^2) = (0, 26)$. By the uniqueness theorem of moment generating functions, it follows that Z follows a normal distribution with mean $\mu = 0$ and variance, $\sigma^2 = 26$.

Hence, the probability density function of Z is given by

$$\begin{aligned} f_Z(z) &= \frac{1}{\sqrt{26}\sqrt{2\pi}} e^{-\frac{z^2}{2 \times 26}} \quad \text{for } -\infty < z < \infty \\ &= \frac{1}{\sqrt{52\pi}} e^{-\frac{z^2}{52}} \quad \text{for } -\infty < z < \infty \end{aligned}$$

$\blacklozenge$

PROBLEM SET 4.5

1. If X and Y are independent binomial random variables with parameters $(4, \frac{1}{2})$ and $(6, \frac{1}{2})$, respectively, find $P(X + Y = 4)$.

2. If $X_1, X_2, \dots, X_k$ are k independent binomial random variables with parameters (n_1, p) , (n_2, p) , $\dots$, (n_k, p) , respectively, then show that $X_1 + X_2 + \dots + X_k$ is a binomial random variable with parameters $(n_1 + n_2 + \dots + n_k, p)$.
3. If X and Y are independent Poisson random variables with parameters 3 and 5, respectively, find (a) $P(X + Y < 3)$; (b) $P(X + Y \geq 4)$.
4. If $X_1, X_2, \dots, X_k$ are k independent gamma random variables with parameters (n_i, λ) , where $i = 1, 2, \dots, k$, then show that $X_1 + X_2 + \dots + X_k$ also is a gamma random variable with parameters $(n_1 + n_2 + \dots + n_k, \lambda)$.
5. If X and Y are independent normal random variables with parameters $(0, 9)$ and $(0, 16)$, respectively, find the probability density function of $Z = X + Y$.

4.6 FUNCTIONS OF RANDOM VARIABLES

In this section, we find the probability density function for functions of two-dimensional random variables of the continuous type. We now state a theorem (without proof), which is fundamental for the calculations in this section.

Theorem 4.16. Let (X, Y) be a two-dimensional continuous random variable with joint probability density function $f_{X,Y}(x, y)$. Suppose that $Z = g_1(X, Y)$ and $W = g_2(X, Y)$, where the functions g_1 and g_2 are assumed to satisfy the following conditions:

- (i) The equations

$$\begin{aligned} z &= g_1(x, y) \\ w &= g_2(x, y) \end{aligned}$$

can be uniquely solved for x and y in terms of z and w with solutions given by, say,

$$\begin{aligned} x &= h_1(z, w) \\ y &= h_2(z, w) \end{aligned}$$

- (ii) The functions g_1 and g_2 have continuous partial derivatives at all $(x, y) \in \mathbb{R}^2$ and are such that the following Jacobian determinant

$$J \triangleq \frac{\partial(x, y)}{\partial(z, w)} = \begin{vmatrix} \frac{\partial h_1}{\partial z} & \frac{\partial h_1}{\partial w} \\ \frac{\partial h_2}{\partial z} & \frac{\partial h_2}{\partial w} \end{vmatrix} \neq 0$$

at all points $(x, y) \in \mathbb{R}^2$.

Then the random variables Z and W are jointly continuous with the joint probability density function given by

$$f_{Z,W}(z, w) = f_{X,Y}(x, y) |J|$$

where $x = h_1(z, w)$ and $y = h_2(z, w)$.

Remark 4.8. Given two jointly distributed continuous random variables (X, Y) with joint PDF $f_{X,Y}(x, y)$ and a function $Z = g_1(X, Y)$, we can apply Theorem 4.16 to find the PDF $f_Z(z)$ for Z . The idea is to construct a suitable auxiliary random variable $W = g_2(X, Y)$ in such a way that Conditions (i) and (ii) of Theorem 4.16 are satisfied. Using Theorem 4.16, we can derive the joint probability density function $f_{Z,W}(z, w)$ of the random variables (Z, W) . Then the marginal probability density function $f_Z(z)$ of Z can be derived from $f_{Z,W}(z, w)$ in the usual manner.

Using Theorem 4.16, we can easily determine the probability density functions of sum, product and quotient of the two-dimensional random variable (X, Y) . These are illustrated in the following examples.

EXAMPLE 4.68. (*Sum of Two Random Variables*) Let (X, Y) be a two-dimensional continuous random variable with joint probability density function $f_{X,Y}(x, y)$. Let $Z = X + Y$. Find the probability density and the distribution functions of Z . What do you get when X and Y are independent?

Solution. We shall use the transformation

$$Z = X + Y \quad \text{and} \quad X = X$$

Then it follows that

$$X = X \quad \text{and} \quad Y = Z - X$$

Note that the Jacobian determinant J is given by

$$J = \frac{\partial(x, y)}{\partial(z, x)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial x} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial x} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1 \neq 0$$

By Theorem 4.16, the joint probability density function of (Z, X) is given by

$$f_{Z,X}(z, x) = f_{X,Y}(x, y) |J| = f_{X,Y}(x, z - x)$$

The density of Z is obtained as the marginal density of the two-dimensional random variable (Z, X) . Hence, we have

$$f_Z(z) = \int_{x=-\infty}^{\infty} f_{Z,X}(z, x) dx = \int_{x=-\infty}^{\infty} f_{X,Y}(x, z - x) dx$$

Also, the distribution function of Z is given by

$$F_Z(z) = \int_{t=-\infty}^z f_Z(t) dt = \int_{t=-\infty}^z \int_{x=-\infty}^{\infty} f_{X,Y}(x, z - x) dx dt$$

If X and Y are independent, then

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \quad \text{for all } (x, y) \in \mathbb{R}^2$$

In this case, the probability density function of Z is given by

$$f_Z(z) = \int_{x=-\infty}^{\infty} f_X(x) f_Y(z - x) dx$$

and the distribution function of Z is given by

$$F_Z(z) = \int_{t=-\infty}^z \int_{x=-\infty}^{\infty} f_X(x) f_Y(z - x) dx dt$$

◆

EXAMPLE 4.69. (*Product of Two Random Variables*) Let (X, Y) be a two-dimensional continuous random variable with joint probability density function $f_{X,Y}(x, y)$. Let $Z = XY$. Find the probability density and the distribution functions of Z . What do you get when X and Y are independent?

Solution. We shall use the transformation

$$Z = XY \quad \text{and} \quad X = X$$

Then it follows that

$$X = X \quad \text{and} \quad Y = \frac{Z}{X}$$

Note that the Jacobian determinant J is given by

$$J = \frac{\partial(x, y)}{\partial(z, x)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial x} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial x} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ \frac{1}{x} & -\frac{z}{x^2} \end{vmatrix} = -\frac{1}{x}$$

By Theorem 4.16, the joint probability density function of (Z, X) is given by

$$f_{Z,X}(z, x) = f_{X,Y}(x, y) |J| = f_{X,Y}\left(x, \frac{z}{x}\right) \frac{1}{|x|}$$

The density of Z is obtained as the marginal density of the two-dimensional random variable (Z, X) . Hence, we have

$$f_Z(z) = \int_{x=-\infty}^{\infty} f_{Z,X}(z, x) dx = \int_{x=-\infty}^{\infty} f_{X,Y}\left(x, \frac{z}{x}\right) \frac{1}{|x|} dx$$

Also, the distribution function of Z is given by

$$F_Z(z) = \int_{t=-\infty}^z f_Z(t) dt = \int_{t=-\infty}^z \int_{x=-\infty}^{\infty} f_{X,Y}\left(x, \frac{t}{x}\right) \frac{1}{|x|} dx dt$$

If X and Y are independent, then

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \quad \text{for all } (x, y) \in \mathbb{R}^2$$

In this case, the PDF of Z is given by

$$f_Z(z) = \int_{x=-\infty}^{\infty} f_X(x) f_Y\left(\frac{z}{x}\right) \frac{1}{|x|} dx$$

and the distribution function of Z is given by

$$F_Z(z) = \int_{t=-\infty}^z \int_{x=-\infty}^{\infty} f_X(x) f_Y\left(\frac{t}{x}\right) \frac{1}{|x|} dx dt$$

◆

EXAMPLE 4.70. (*Quotient of Two Random Variables*) Let (X, Y) be a two-dimensional continuous random variable with joint probability density function $f_{X,Y}(x, y)$. Let $Z = \frac{X}{Y}$. Find the probability density and the distribution functions of Z . What do you get when X and Y are independent?

Solution. We shall use the transformation

$$Z = \frac{X}{Y} \quad \text{and} \quad Y = Y$$

Then, it follows that

$$X = YZ \quad \text{and} \quad Y = Y$$

Note that the Jacobian determinant J is given by

$$J = \frac{\partial(x,y)}{\partial(z,y)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial y} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial y} \end{vmatrix} = \begin{vmatrix} y & z \\ 0 & 1 \end{vmatrix} = y$$

By Theorem 4.16, the joint probability density function of (Z, Y) is given by

$$f_{Z,Y}(z,y) = f_{X,Y}(x,y) |J| = f_{X,Y}(yz,y) |y|$$

The density of Z is obtained as the marginal density of the two-dimensional random variable (Z, Y) . Hence, we have

$$f_Z(z) = \int_{y=-\infty}^{\infty} f_{Z,Y}(z,y) dy = \int_{y=-\infty}^{\infty} f_{X,Y}(yz,y) |y| dy$$

Also, the distribution function of Z is given by

$$F_Z(z) = \int_{t=-\infty}^z f_Z(t) dt = \int_{t=-\infty}^z \int_{x=-\infty}^{\infty} f_{X,Y}(yz,y) |y| dy dt$$

If X and Y are independent, then

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \quad \text{for all } (x,y) \in \mathbb{R}^2$$

In this case, the PDF of Z is given by

$$f_Z(z) = \int_{y=-\infty}^{\infty} f_X(yz) f_Y(y) |y| dy$$

and the distribution function of Z is given by

$$F_Z(z) = \int_{t=-\infty}^z \int_{y=-\infty}^{\infty} f_X(yz) f_Y(y) |y| dy dt$$

◆

EXAMPLE 4.71. Let X and Y be positive independent random variables with the identical probability density functions e^{-x} for $x > 0$. Find the joint probability density function of $U = X + Y$ and $V = \frac{X}{Y}$.
(Anna, April 2003)

Solution. Since X and Y are independent exponential random variables with parameter 1, the joint probability density function of X and Y given by

$$f(x,y) = f_X(x) f_Y(y) = \begin{cases} e^{-(x+y)} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

We shall find the joint probability density function of the random variables

$$U = X + Y \quad \text{and} \quad V = \frac{X}{Y}$$

Then, we have

$$X = VY \Rightarrow U = X + Y = VY + Y = (V + 1)Y$$

Thus, solving for X and Y , we have

$$X = \frac{UV}{V+1} \quad \text{and} \quad Y = \frac{U}{V+1}$$

Note also that $X > 0$ and $Y > 0$ imply that $U = X + Y > 0$ and $V = \frac{X}{Y} > 0$.

Next, we find that

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{v}{v+1} & \frac{u}{(v+1)^2} \\ \frac{1}{v+1} & \frac{-u}{(v+1)^2} \end{vmatrix} = -\frac{u}{(v+1)^2}$$

Hence, the joint probability density function of U and V is given by

$$\begin{aligned} f_{U,V}(u,v) &= f(x(u,v), y(u,v)) |J| \quad \text{for } u > 0, v > 0 \\ &= e^{-u} \frac{u}{(v+1)^2} \quad \text{for } u > 0, v > 0 \end{aligned}$$

Thus,

$$f_{U,V}(u,v) = \begin{cases} \frac{ue^{-u}}{(v+1)^2} & \text{for } u > 0, v > 0 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 4.72. If the joint probability density function of two random variables, X and Y , is given by

$$f_{X,Y}(x,y) = \begin{cases} x+y & \text{if } 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

find the PDF of XY .

(Anna, Model, 2003; Nov. 2006; Nov. 2007)

Solution. We are asked to find the PDF of $U = XY$. We assume an auxiliary random variable $V = X$. Thus, we consider the transformation

$$U = XY \text{ and } V = X$$

Solving for X and Y , we have

$$X = V \text{ and } Y = \frac{U}{V}$$

Since $0 \leq X \leq 1$ and $0 \leq Y \leq 1$, it follows that the range of (U, V) is given by

$$A = \{(u, v) \in \mathbb{R}^2 : 0 \leq u \leq v \leq 1\}$$

which is illustrated in Figure 4.19.

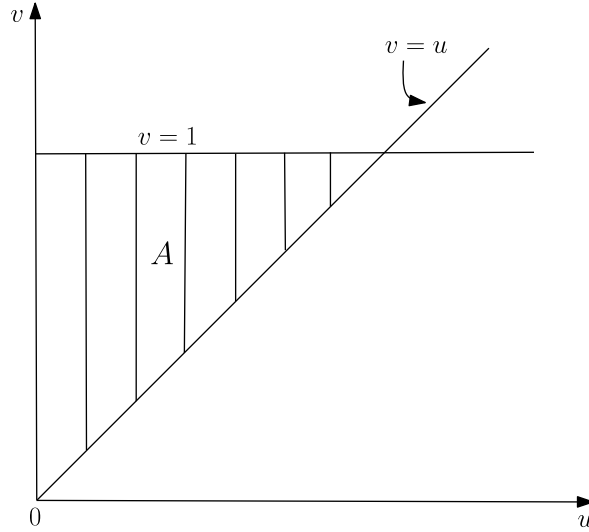


Figure 4.19: Region A.

Next, we find that

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ \frac{1}{v} & -\frac{u}{v^2} \end{vmatrix} = -\frac{1}{v}$$

Hence, the joint probability density function of U and V is given by

$$f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v)) |J| = \left[v + \frac{u}{v} \right] \frac{1}{v} \text{ for } 0 \leq u \leq v \leq 1$$

Simplifying, we have

$$f_{U,V}(u, v) = \begin{cases} 1 + \frac{u}{v^2} & \text{for } 0 \leq u \leq v \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Hence, the marginal density function of U is given by

$$f_U(u) = \int_{v=u}^1 f_{U,V}(u,v) dv = \int v = u^1 \left[1 + \frac{u}{v^2} \right] dv \quad \text{for } 0 \leq u \leq 1$$

[See Figure 4.19].

Integrating, we have

$$\begin{aligned} f_U(u) &= \left[v - \frac{u}{v} \right]_{v=u}^1 = [(1-u) - (u-1)] \\ &= 2(1-u) \quad \text{for } 0 \leq u \leq 1 \end{aligned}$$

Hence, the probability density function of $U = XY$ is given by

$$f_U(u) = \begin{cases} 2(1-u) & \text{for } 0 \leq u \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 4.73. If X and Y are independent exponential distributions with parameter 1, find the PDF of $U = X - Y$.

(Anna, Model 2003; May 2007)

Solution. Since X and Y are independent exponential random variables with parameter 1, the joint probability distribution of (X, Y) is given by

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) = \begin{cases} e^{-(x+y)} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Define the auxiliary variable $V = X + Y$. Thus, we consider the transformation

$$U = X - Y \quad \text{and} \quad V = X + Y$$

Solving for X and Y , we have

$$X = \frac{U+V}{2} \quad \text{and} \quad Y = \frac{V-U}{2}$$

Since $X > 0$ and $Y > 0$, the range space of (U, V) is given by

$$A = \{(u, v) \in \mathbb{R}^2 : v+u > 0, v-u > 0\}$$

which is illustrated in Figure 4.20.

Next, we find that

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2}$$

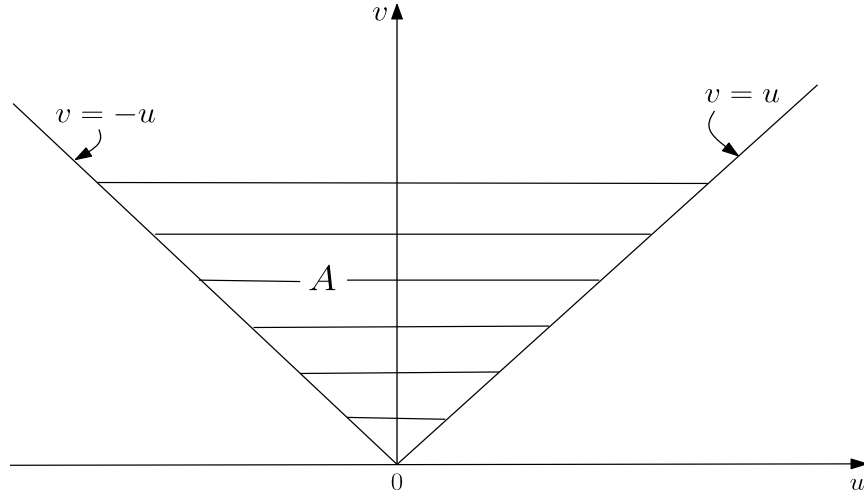


Figure 4.20: Region A.

Thus, the joint probability density function of U and V is given by

$$f_{U,V}(u,v) = f_{X,Y}(x(u,v),y(u,v)) |J| = e^{-v} \frac{1}{2} \quad \text{for } v+u > 0, v-u > 0$$

$$= \begin{cases} \frac{1}{2} e^{-v} & \text{for } v+u > 0, v-u > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we calculate the marginal density function of $U = X - Y$.

We have two cases to consider, namely, (i) $u < 0$ and (ii) $u > 0$.

If $u < 0$, then the marginal density of U is given by

$$f_U(u) = \int_{v=-u}^{\infty} f_{U,V}(u,v) dv = \int_{v=-u}^{\infty} \frac{1}{2} e^{-v} dv$$

[see Figure 4.20].

Integrating, we have

$$f_U(u) = \frac{1}{2} [-e^{-v}]_{v=-u}^{\infty} = \frac{1}{2} e^u$$

If $u > 0$, then the marginal density of U is given by

$$f_U(u) = \int_{v=u}^{\infty} f_{U,V}(u,v) dv = \int_{v=u}^{\infty} \frac{1}{2} e^{-v} dv$$

Integrating, we have

$$f_U(u) = \frac{1}{2} [-e^{-v}]_{v=u}^{\infty} = \frac{1}{2} e^{-u}$$

Combining the two cases, the marginal density function of $U = X - Y$ is given by

$$f_U(u) = \frac{1}{2} e^{-|u|} \quad \text{for } -\infty < u < \infty$$

which is the *double-exponential* or the *Laplace distribution*. ◆

EXAMPLE 4.74. Let X and Y be independent standard normal random variables. Find the PDF of $Z = \frac{X}{Y}$.

(Anna, Model 2003; Nov. 2005)

Solution. Since X is a standard normal random variable, the PDF of X is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for } -\infty < x < \infty$$

Since Y is a standard normal random variable, the PDF of Y is given by

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \quad \text{for } -\infty < y < \infty$$

Since X and Y are independent, the joint probability density function of (X, Y) is given by

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) = \frac{1}{2\pi} e^{-\frac{(x^2+y^2)}{2}} \quad \text{for } -\infty < x,y < \infty$$

Given that $Z = \frac{X}{Y}$. We define an auxiliary variable, $W = Y$.

Thus, we consider the transformation

$$Z = \frac{X}{Y} \quad \text{and} \quad W = Y$$

Solving for X and Y , we have

$$X = ZW \quad \text{and} \quad Y = W$$

Since the range of (X, Y) is the whole of $\mathbb{R}^2$, it follows that the range of (Z, W) is also the whole of $\mathbb{R}^2$.

Next, we find that

$$J = \frac{\partial(x,y)}{\partial(z,w)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{vmatrix} = \begin{vmatrix} w & z \\ 0 & 1 \end{vmatrix} = w$$

The joint probability density function of Z and W is given by

$$f_{Z,W}(z,w) = f_{X,Y}(x(z,w), y(z,w)) |J| \quad \text{for all } -\infty < z, w < \infty$$

Substituting, we have

$$f_{Z,W}(z,w) = \frac{1}{2\pi} e^{-\frac{1}{2}w^2(z^2+1)} |w| \quad \text{for all } -\infty < z, w < \infty$$

Hence, the marginal density function of Z is given by

$$f_Z(z) = \int_{w=-\infty}^{\infty} f_{Z,W}(z,w) dw = \int_{w=-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{1}{2}w^2(z^2+1)} |w| dw$$

Since the integrand is an even function of w , it follows that

$$f_Z(z) = 2 \int_{w=0}^{\infty} \frac{1}{2\pi} e^{-\frac{1}{2}w^2(z^2+1)} w dw = \frac{1}{\pi} \int_{w=0}^{\infty} e^{-\frac{1}{2}w^2(z^2+1)} w dw$$

Substitute $\theta = \frac{1}{2}w^2(z^2+1)$. Then we have

$$d\theta = w(z^2+1) dw \Rightarrow w dw = \frac{d\theta}{z^2+1}$$

Hence, we have

$$f_Z(z) = \frac{1}{\pi} \int_{\theta=0}^{\infty} e^{-\theta} \frac{d\theta}{z^2+1}$$

Integrating,

$$f_Z(z) = \frac{1}{\pi} \frac{1}{z^2+1} \left[-e^{-\theta} \right]_{\theta=0}^{\infty} = \frac{1}{\pi} \frac{1}{z^2+1}$$

Hence, the marginal density function of $Z = \frac{X}{Y}$ is given by

$$f_Z(z) = \frac{1}{\pi} \frac{1}{z^2+1} \quad \text{for } -\infty < z < \infty$$

which is the *Cauchy distribution*. ♦

EXAMPLE 4.75. The joint probability density function of X and Y is given by

$$f(x,y) = e^{-(x+y)}, \quad x > 0, y > 0$$

Find the PDF of $U = \frac{X+Y}{2}$. Are X and Y independent?

(Anna, April 2003; Nov. 2007)

Solution. Define an auxiliary variable, $V = Y$.

Thus, we consider the transformation

$$U = \frac{X+Y}{2} \quad \text{and} \quad V = Y$$

Solving for X and Y , we have

$$X = 2U - V \quad \text{and} \quad Y = V$$

Since $X > 0$ and $Y > 0$, the range of (U, V) is given by

$$A = \{(u, v) \in \mathbb{R}^2 : 2u - v > 0, v > 0\}$$

which is illustrated in Figure 4.21.

Next, we find that

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ 0 & 1 \end{vmatrix} = 2$$

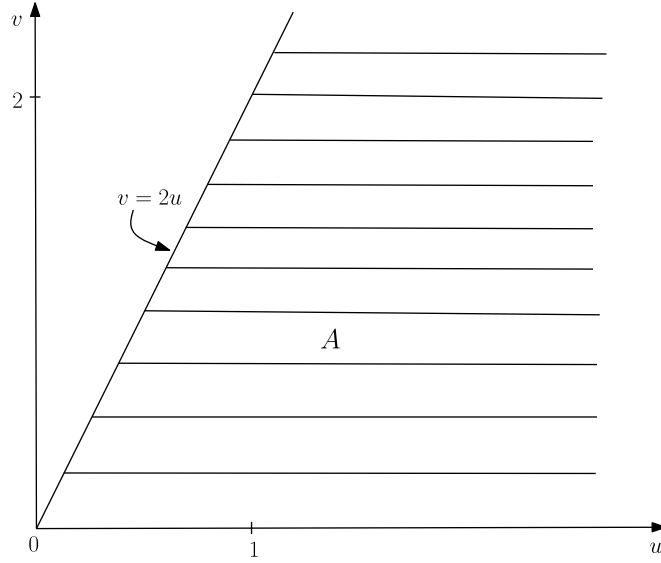


Figure 4.21: Region A.

Hence, the joint probability density function of U and V is given by

$$f_{U,V}(u, v) = f(x(u, v), y(u, v)) |J| = e^{-2u} 2 \text{ for } 2u - v > 0, v > 0$$

Thus, we have

$$f_{u,v}(u, v) = \begin{cases} 2e^{-2u} & \text{for } 2u - v > 0, v > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we compute the marginal density of U .

If $u > 0$, then we have

$$f_U(u) = \int_{v=0}^{2u} f_{U,V}(u, v) dv = \int_{v=0}^{2u} 2e^{-2u} dv$$

(see Figure 4.21).

Integrating, we have

$$f_U(u) = 2e^{-2u} [v]_0^{2u} = 2e^{-2u} [2u] = 4ue^{-2u}$$

Hence, the marginal density of $U = \frac{X+Y}{2}$ is given by

$$f_U(u) = \begin{cases} 4ue^{-2u} & \text{for } u > 0 \\ 0 & \text{elsewhere} \end{cases}$$

which is a *gamma distribution* with parameters $\alpha = 2$ and $\lambda = 2$.

Also, it is easy to see that the marginal densities of X and Y are given by

$$f_X(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and

$$f_Y(y) = \begin{cases} e^{-y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Since $f(x, y) = f_X(x) f_Y(y)$ for all x and y , it is immediate that X and Y are independent random variables. $\blacklozenge$

EXAMPLE 4.76. Let X and Y be two positive independent continuous random variables with the probability density functions $f_1(x)$ and $f_2(y)$, respectively. Find the probability density function of $U = \frac{X}{Y}$.

(Anna, April 2004)

Solution. First, note that the joint probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} f_1(x) f_2(y) & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Now, we are given that $U = \frac{X}{Y}$. Consider the auxiliary variable, $V = Y$. Thus, we consider the transformation

$$U = \frac{X}{Y} \text{ and } V = Y$$

Solving for X and Y , we have

$$X = UV \text{ and } Y = V$$

Since $X > 0$ and $Y > 0$, it follows that $U > 0$ and $V > 0$.

Next, we find that

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ 0 & 1 \end{vmatrix} = v$$

Hence, the joint probability density function of U and V is given by

$$f_{U,V}(u, v) = f(x(u, v), y(u, v)) |J| = f_1(uv) f_2(v) |v| \text{ for } u > 0, v > 0$$

Thus, we have

$$f_{U,V}(u, v) = \begin{cases} v f_1(uv) f_2(v) & \text{for } u > 0, v > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Hence, the marginal density of U is given by

$$f_U(u) = \int_{v=0}^{\infty} f_{U,V}(u, v) dv = \int_{v=0}^{\infty} v f_1(uv) f_2(v) dv \text{ for } u > 0$$

Since $V = Y$, the marginal density of $U = \frac{X}{Y}$ can be written as

$$f_U(u) = \int_{y=0}^{\infty} y f_1(uy) f_2(y) dy \text{ for } u > 0$$

$\blacklozenge$

EXAMPLE 4.77. If the joint density of X_1 and X_2 is given by

$$f(x_1, x_2) = \begin{cases} 6e^{-3x_1-2x_2} & \text{for } x_1 > 0, x_2 > 0 \\ 0 & \text{otherwise} \end{cases}$$

find the probability density of $Y = X_1 + X_2$ and its mean.

(Anna, April 2004; Nov. 2006)

Solution. Define an auxiliary variable $Z = X_1$. Thus, we consider the transformation

$$Y = X_1 + X_2 \text{ and } Z = X_1$$

Solving for X_1 and X_2 , we have

$$X_1 = Z \text{ and } X_2 = Y - Z$$

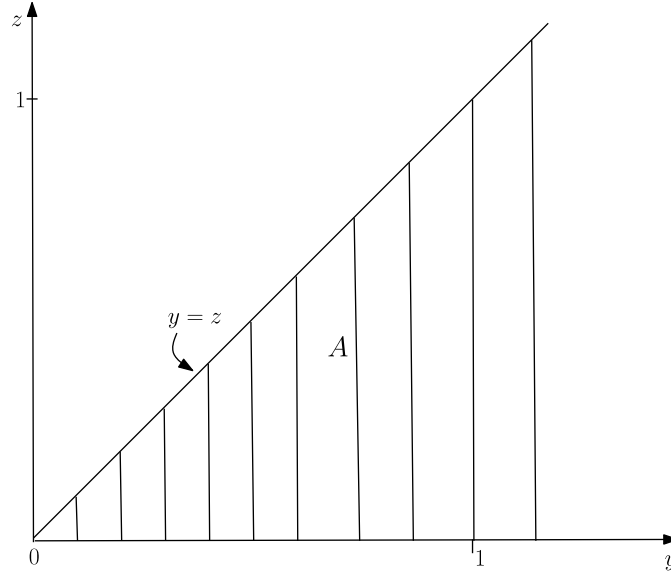


Figure 4.22: Region A.

Since $X_1 > 0$ and $X_2 > 0$, the range of (Y, Z) is given by

$$A = \{(y, z) \in \mathbb{R}^2 : z > 0, y > z\}$$

which is illustrated in Figure 4.22.

Next, we find that

$$J = \frac{\partial(x_1, x_2)}{\partial(y, z)} = \begin{vmatrix} \frac{\partial x_1}{\partial y} & \frac{\partial x_1}{\partial z} \\ \frac{\partial x_2}{\partial y} & \frac{\partial x_2}{\partial z} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$$

Hence, the joint probability density function of Y and Z is given by

$$f_{Y,Z}(y,z) = f(x_1(y,z), x_2(y,z))|J| = 6e^{-2y-z} \text{ for } z > 0, y > z$$

It follows that the marginal density function of Y is given by

$$f_Y(y) = \int_{z=0}^y f_{Y,Z}(y,z) dz = \int_{z=0}^y 6e^{-2y-z} dz = \int_{z=0}^y 6e^{-2y} e^{-z} dz \text{ for } y > 0$$

(see Figure 4.22).

Integrating, we have

$$f_Y(y) = 6e^{-2y} [-e^{-z}]_{z=0}^y = 6e^{-2y} [1 - e^{-y}]$$

Hence, the marginal density of $Y = X_1 + X_2$ is given by

$$f_Y(y) = \begin{cases} 6[e^{-2y} - e^{-3y}] & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

The mean of Y is given by

$$E(Y) = \int_{y=0}^{\infty} y f_Y(y) dy = \int_{y=0}^{\infty} 6y [e^{-2y} - e^{-3y}] dy$$

Thus, we have

$$E(Y) = \int_{y=0}^{\infty} 6ye^{-2y} dy - \int_{y=0}^{\infty} 6ye^{-3y} dy$$

Substituting $s = 2y$ in the first integral and $t = 3y$ in the second integral, we have

$$E(Y) = \frac{3}{2} \int_{s=0}^{\infty} se^{-s} ds - \frac{2}{3} \int_{t=0}^{\infty} te^{-t} dt = \frac{3}{2} \Gamma(2) - \frac{2}{3} \Gamma(2)$$

Since $\Gamma(2) = 1! = 1$, it is immediate that

$$E(Y) = \frac{3}{2} - \frac{2}{3} = \frac{5}{6}$$

◆

EXAMPLE 4.78. If X and Y are independent random variables each normally distributed with mean zero and variance σ^2 , find the density functions of $R = \sqrt{X^2 + Y^2}$ and $\Theta = \tan^{-1}(\frac{Y}{X})$.
(Anna, Nov. 2003)

Solution. Since X and Y are independent with $X \sim N(0, \sigma^2)$ and $Y \sim N(0, \sigma^2)$, it follows that the joint probability density function of X and Y is given by

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) = \frac{1}{\sigma^2 2\pi} e^{-\frac{x^2+y^2}{2\sigma^2}} \text{ for } -\infty < x, y < \infty$$

We consider the polar transformation

$$R = \sqrt{X^2 + Y^2} \text{ and } \Theta = \tan^{-1} \left(\frac{Y}{X} \right)$$

The range space of (r, θ) is clearly $0 < r < \infty$ and $0 < \theta < 2\pi$.

The inverse transformation is given by

$$X = R \cos \Theta \text{ and } Y = R \sin \Theta$$

Note that

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r (\cos^2 \theta + \sin^2 \theta) = r$$

Thus, the joint probability density function of R and Θ is given by

$$f_{R, \Theta}(r, \theta) = f_{X, Y}(x, y) |J| = \frac{1}{\sigma^2 2\pi} r e^{-\frac{r^2}{2\sigma^2}} \text{ for } 0 < r < \infty, 0 < \theta < 2\pi$$

Hence, the marginal density function of R is given by

$$f_R(r) = \int_{\theta=0}^{2\pi} f_{R, \Theta}(r, \theta) d\theta = \int_{\theta=0}^{2\pi} \frac{1}{\sigma^2 2\pi} r e^{-\frac{r^2}{2\sigma^2}} d\theta \text{ for } 0 < r < \infty$$

Integrating, we have

$$f_R(r) = \frac{1}{\sigma^2 2\pi} r e^{-\frac{r^2}{2\sigma^2}} [\theta]_0^{2\pi} = \frac{1}{\sigma^2 2\pi} r e^{-\frac{r^2}{2\sigma^2}} 2\pi$$

Simplifying, the density function of $R = \sqrt{X^2 + Y^2}$ is given by

$$f_R(r) = \frac{1}{\sigma^2} r e^{-\frac{r^2}{2\sigma^2}} \text{ for } 0 < r < \infty$$

which is the *Rayleigh distribution* with parameter σ .

Next, the marginal density function of Θ is given by

$$f_{\Theta}(\theta) = \int_{r=0}^{\infty} f_{R, \Theta}(r, \theta) dr = \int_{r=0}^{\infty} \frac{1}{\sigma^2 2\pi} r e^{-\frac{r^2}{2\sigma^2}} d\theta \text{ for } 0 < \theta < 2\pi$$

Integrating, we have

$$f_{\Theta}(\theta) = \frac{1}{2\pi} \left[-e^{-\frac{r^2}{2\sigma^2}} \right]_0^{\infty} = \frac{1}{2\pi} [0 + 1]$$

Simplifying, the density function of $\Theta = \tan^{-1} \left(\frac{Y}{X} \right)$ is given by

$$f_{\Theta}(\theta) = \frac{1}{2\pi} \text{ for } 0 < \theta < 2\pi$$

which is the *uniform distribution* over the interval $(0, 2\pi)$. ◆

EXAMPLE 4.79. Let X and Y be two independent uniform random variables over $(0, 1)$. Show that the random variables

$$U = \cos(2\pi X)\sqrt{-2\ln Y} \quad \text{and} \quad V = \sin(2\pi X)\sqrt{-2\ln Y}$$

are independent standard normal random variables.

(Anna, Nov. 2004)

Solution. Since X and Y are independent uniform random variables over $(0, 1)$, it follows that the joint probability density function of (X, Y) is given by

$$f(x, y) = f_X(x) f_Y(y) = \begin{cases} 1 & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

The inverse transformation is easily obtained as

$$X = \frac{1}{2\pi} \tan^{-1} \left(\frac{V}{U} \right) \quad \text{and} \quad Y = e^{-\frac{(U^2+V^2)}{2}}$$

Since $0 < X < 1$ and $0 < Y < 1$, it follows that $-\infty < U, V < \infty$.

Next, we find the Jacobian determinant defined by

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

Note that

$$\frac{\partial x}{\partial u} = \frac{1}{2\pi} \frac{1}{1 + \left(\frac{v}{u}\right)^2} \left(-\frac{v}{u^2}\right) = \frac{1}{2\pi} \left[-\frac{v}{u^2 + v^2}\right]$$

$$\frac{\partial x}{\partial v} = \frac{1}{2\pi} \frac{1}{1 + \left(\frac{v}{u}\right)^2} \left(\frac{1}{u}\right) = \frac{1}{2\pi} \left[\frac{u}{u^2 + v^2}\right]$$

$$\frac{\partial y}{\partial u} = e^{-\frac{(u^2+v^2)}{2}} (-u) = -u e^{-\frac{(u^2+v^2)}{2}}$$

$$\frac{\partial y}{\partial v} = e^{-\frac{(u^2+v^2)}{2}} (-v) = -v e^{-\frac{(u^2+v^2)}{2}}$$

Substituting, we have

$$J = \begin{vmatrix} \frac{1}{2\pi} \left[-\frac{v}{u^2+v^2}\right] & \frac{1}{2\pi} \left[\frac{u}{u^2+v^2}\right] \\ -u e^{-\frac{(u^2+v^2)}{2}} & -v e^{-\frac{(u^2+v^2)}{2}} \end{vmatrix}.$$

Using the properties of determinants, J can be simplified as

$$\begin{aligned} J &= \frac{1}{2\pi} \frac{1}{u^2 + v^2} e^{-\frac{(u^2+v^2)}{2}} \begin{vmatrix} -v & u \\ -u & -v \end{vmatrix} = \frac{1}{2\pi} \frac{1}{u^2 + v^2} e^{-\frac{(u^2+v^2)}{2}} (u^2 + v^2) \\ &= \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}} \end{aligned}$$

Thus, the joint probability density function of U and V is given by

$$f_{U,V}(u,v) = f(x,y)|J| = \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}} \quad \text{for } -\infty < u, v < \infty$$

Hence, the marginal density function of U is obtained as

$$\begin{aligned} f_U(u) &= \int_{v=-\infty}^{\infty} f_{U,V}(u,v) dv \quad \text{for } -\infty < u < \infty \\ &= \int_{v=-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}} dv \end{aligned}$$

Since the integrand is an even function of v , we have

$$\begin{aligned} f_U(u) &= 2 \int_{v=0}^{\infty} \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}} dv \\ &= \frac{1}{\pi} e^{-\frac{u^2}{2}} \int_{v=0}^{\infty} e^{-\frac{v^2}{2}} dv \end{aligned}$$

Substituting $z^2 = \frac{v^2}{2}$ in the integral, we have

$$f_U(u) = \frac{1}{\pi} e^{-\frac{u^2}{2}} \sqrt{2} \int_{z=0}^{\infty} e^{-z^2} dz = \frac{1}{\pi} e^{-\frac{u^2}{2}} \sqrt{2} \frac{\sqrt{\pi}}{2}$$

Simplifying, the marginal density function of U is given by

$$f_U(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \quad \text{for } -\infty < u < \infty$$

which is the standard normal density function.

Similarly, the marginal density function of V is obtained as

$$\begin{aligned} f_V(v) &= \int_{u=-\infty}^{\infty} f_{U,V}(u,v) du \quad \text{for } -\infty < v < \infty \\ &= \int_{u=-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}} du \end{aligned}$$

Since the integrand is an even function of u , we have

$$\begin{aligned} f_V(v) &= 2 \int_{u=0}^{\infty} \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}} du \\ &= \frac{1}{\pi} e^{-\frac{v^2}{2}} \int_{u=0}^{\infty} e^{-\frac{u^2}{2}} du \end{aligned}$$

Substituting $z^2 = \frac{v^2}{2}$ in the integral, we have

$$f_V(v) = \frac{1}{\pi} e^{-\frac{v^2}{2}} \sqrt{2} \int_{z=0}^{\infty} e^{-z^2} dz = \frac{1}{\pi} e^{-\frac{v^2}{2}} \sqrt{2} \frac{\sqrt{\pi}}{2}$$

Simplifying, the marginal density function of V is given by

$$f_V(v) = \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} \quad \text{for } -\infty < v < \infty$$

which is the standard normal density function.

Finally, we note that

$$f_U(u) \cdot f_V(v) = \frac{1}{2\pi} e^{-\frac{(u^2+v^2)}{2}} = f_{U,V}(u,v) \quad \text{for } -\infty < u, v < \infty$$

Hence, we conclude that U and V are independent standard normal random variables. $\blacklozenge$

EXAMPLE 4.80. Suppose X and Y are two random variables having the joint probability function

$$f(x,y) = \begin{cases} 4xye^{-(x^2+y^2)}, & x, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Find the probability density function of $Z = \sqrt{X^2 + Y^2}$.

(Anna, April 2005; Nov. 2005; May 2006)

Solution. Define an auxiliary variable, $\Theta = \tan^{-1}\left(\frac{Y}{X}\right)$. Thus, we consider the transformation

$$Z = \sqrt{X^2 + Y^2} \quad \text{and} \quad \Theta = \tan^{-1}\left(\frac{Y}{X}\right)$$

The inverse coordinate transformation is obtained as

$$X = Z \cos \Theta \quad \text{and} \quad Y = Z \sin \Theta$$

Since $X, Y \geq 0$, it follows that $Z \geq 0$ and $0 < \Theta < \frac{\pi}{2}$.

The Jacobian determinant is defined by

$$J = \frac{\partial(x,y)}{\partial(z,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -z \sin \theta \\ \sin \theta & z \cos \theta \end{vmatrix} = z(\cos^2 \theta + \sin^2 \theta) = z$$

Thus, the joint probability density function of (Z, Θ) is given by

$$\begin{aligned} f_{Z,\Theta}(z, \theta) &= f(x,y) |J| = 4z^2 \cos \theta \sin \theta z e^{-z^2} \quad \text{for } z \geq 0, 0 < \theta < \frac{\pi}{2} \\ &= 4z^3 \cos \theta \sin \theta e^{-z^2} \quad \text{for } z \geq 0, 0 < \theta < \frac{\pi}{2} \end{aligned}$$

The marginal density function of Z is given by

$$f_Z(z) = \int_{\theta=0}^{\frac{\pi}{2}} f_{Z,\Theta}(z, \theta) d\theta = \int_{\theta=0}^{\frac{\pi}{2}} 4z^3 \cos \theta \sin \theta e^{-z^2} d\theta$$

Simplifying, we have

$$f_Z(z) = 2z^3 e^{-z^2} \int_{\theta=0}^{\frac{\pi}{2}} \sin 2\theta d\theta = 2z^3 e^{-z^2} \left[-\frac{\cos 2\theta}{2} \right]_{\theta=0}^{\frac{\pi}{2}} = 2z^3 e^{-z^2} \left[\frac{1}{2} + \frac{1}{2} \right]$$

Hence, the density function of $Z = \sqrt{X^2 + Y^2}$ is given by

$$f_Z(z) = \begin{cases} 2z^3 e^{-z^2} & \text{for } z \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

◆

EXAMPLE 4.81. If X_1 and X_2 are independent uniform variates on $[0, 1]$, find the distribution of $\frac{X_1}{X_2}$ and $X_1 X_2$.

(Anna, May 2006)

Solution. Since X_1 and X_2 are independent uniform variates on $[0, 1]$, the joint probability density function of (X_1, X_2) is given by

$$f(x_1, x_2) = f_1(x_1) f_2(x_2) = \begin{cases} 1 & \text{for } 0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

We consider the transformation

$$Z = \frac{X_1}{X_2} \quad \text{and} \quad W = X_1 X_2$$

The inverse transformation is given by

$$X_1 = \sqrt{ZW} \quad \text{and} \quad X_2 = \sqrt{\frac{W}{Z}}$$

Since $0 \leq X_1 \leq 1$ and $0 \leq X_2 \leq 1$, the range of (Z, W) is given by

$$A = \{(z, w) \in \mathbb{R}^2 : 0 < zw < 1, 0 < w < z\}$$

which is illustrated in Figure 4.23.

Next, we calculate the Jacobian determinant defined by

$$J = \frac{\partial(x_1, x_2)}{\partial(z, w)} = \begin{vmatrix} \frac{\partial x_1}{\partial z} & \frac{\partial x_1}{\partial w} \\ \frac{\partial x_2}{\partial z} & \frac{\partial x_2}{\partial w} \end{vmatrix}$$

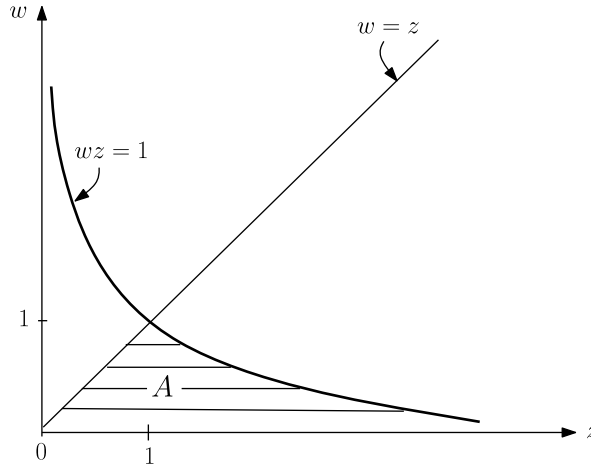


Figure 4.23: Region A.

Note that

$$\frac{\partial x_1}{\partial z} = \frac{1}{2\sqrt{zw}} \quad w = \frac{1}{2} \frac{\sqrt{w}}{\sqrt{z}}$$

$$\frac{\partial x_1}{\partial w} = \frac{1}{2\sqrt{zw}} \quad z = \frac{1}{2} \frac{\sqrt{z}}{\sqrt{w}}$$

$$\frac{\partial x_2}{\partial z} = \frac{1}{2\sqrt{\frac{w}{z}}} \left(-\frac{w}{z^2} \right) = \frac{1}{2} \frac{\sqrt{z}}{\sqrt{w}} \left(-\frac{w}{z^2} \right) = -\frac{\sqrt{w}}{2\sqrt[3]{z}}$$

$$\frac{\partial x_2}{\partial w} = \frac{1}{2\sqrt{\frac{w}{z}}} \left(\frac{1}{z} \right) = \frac{1}{2} \frac{\sqrt{z}}{\sqrt{w}} \left(\frac{1}{z} \right) = \frac{1}{2} \frac{1}{\sqrt{zw}}$$

Substituting these values in the Jacobian determinant J , we have

$$J = \begin{vmatrix} \frac{1}{2} \frac{\sqrt{w}}{\sqrt{z}} & \frac{1}{2} \frac{\sqrt{z}}{\sqrt{w}} \\ -\frac{\sqrt{w}}{2\sqrt[3]{z}} & \frac{1}{2} \frac{1}{\sqrt{zw}} \end{vmatrix} = \frac{1}{4} \left[\frac{\sqrt{w}}{\sqrt{z}} \frac{1}{\sqrt{zw}} + \frac{\sqrt{z}}{\sqrt{w}} \frac{\sqrt{w}}{\sqrt[3]{z}} \right]$$

$$J = \frac{1}{4} \left[\frac{1}{z} + \frac{1}{z} \right] = \frac{1}{4} \left[\frac{2}{z} \right] = \frac{1}{2z}$$

Hence, the joint probability density function of $Z = \frac{X_1}{X_2}$ and $W = X_1 X_2$ is given by

$$f_{Z,W}(z, w) = f(x_1, x_2) |J| = \frac{1}{2z} \quad \text{for } 0 < zw < 1, 0 < w < z$$

◆

EXAMPLE 4.82. If X and Y are independent random variables with probability density functions e^{-x} , $x \geq 0$ and e^{-y} , $y \geq 0$, find the probability density functions of $U = \frac{X}{X+Y}$ and $V = X + Y$. Are they independent?

(Anna, Nov. 2006)

Solution. First, we note that the joint probability density function of X and Y is given by

$$f(x, y) = f_X(x) f_Y(y) = \begin{cases} e^{-(x+y)} & \text{for } x \geq 0, y \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

We consider the transformation

$$U = \frac{X}{X+Y} \quad \text{and} \quad V = X+Y$$

Solving for X and Y , we have

$$X = UV \quad \text{and} \quad Y = V(1-U)$$

Since $X \geq 0$ and $Y \geq 0$, it is clear that

$$0 \leq U = \frac{X}{X+Y} \leq 1 \quad \text{and} \quad V = X+Y \geq 0$$

Next, we find the Jacobian determinant

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ -v & 1-u \end{vmatrix} = v(1-u) + uv = v$$

Hence, the joint probability density function of U and V is given by

$$f_{U,V}(u, v) = f(x, y)|J| = e^{-v}|v| = ve^{-v} \quad \text{for } 0 \leq u \leq 1, v \geq 0$$

and the marginal density function of $U = \frac{X}{X+Y}$ is given by

$$f_U(u) = \int_{v=0}^{\infty} f_{U,V}(u, v) dv = \int_{v=0}^{\infty} ve^{-v} dv = \Gamma(2) = 1 \quad \text{for } 0 \leq u \leq 1$$

Thus, U is uniformly distributed over the interval $[0, 1]$.

Next, the marginal density function of $V = X+Y$ is given by

$$f_V(v) = \int_{u=0}^{\infty} f_{U,V}(u, v) du = \int_{u=0}^1 ve^{-v} du \quad \text{for } v \geq 0$$

Integrating, we have

$$f_V(v) = ve^{-v} \quad \text{for } v \geq 0$$

which is the gamma distribution with parameters $\alpha = 2$ and $\lambda = 1$.

Note that

$$f_U(u) f_V(v) = ve^{-v} = f_{U,V}(u, v) \quad \text{for } 0 \leq u \leq 1, v \geq 0$$

which shows that U and V are independent random variables. ◆

EXAMPLE 4.83. Let X and Y be independent gamma random variables with parameters (α, λ) and (β, λ) respectively. Let $Z = X + Y$ and $W = \frac{X}{X+Y}$.

- Find the joint probability density function of Z and W .
- Find the marginal density functions of Z and W .
- Are Z and W independent?

Solution. (a) The joint probability density function of X and Y is given by

$$\begin{aligned} f_{X,Y}(x,y) &= f_X(x)f_Y(y) = \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} \frac{\lambda e^{-\lambda y}(\lambda y)^{\beta-1}}{\Gamma(\beta)} \quad \text{for } x > 0, y > 0 \\ &= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\lambda(x+y)} x^{\alpha-1} y^{\beta-1} \quad \text{for } x > 0, y > 0 \end{aligned}$$

We consider the transformation

$$Z = X + Y \quad \text{and} \quad W = \frac{X}{X+Y}$$

Solving for X and Y , we have

$$X = ZW \quad \text{and} \quad Y = Z(1-W)$$

Since $X > 0$ and $Y > 0$, it is clear that

$$Z = X + Y > 0 \quad \text{and} \quad 0 < W = \frac{X}{X+Y} < 1$$

Next, we find the Jacobian determinant

$$J = \frac{\partial(x,y)}{\partial(z,w)} = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{vmatrix} = \begin{vmatrix} w & z \\ 1-w & -z \end{vmatrix} = -z$$

Hence, the joint probability density function of Z and W is given by

$$f_{Z,W}(z,w) = f(x(z,w), y(z,w)) |J| \quad \text{for } z > 0, 0 < w < 1$$

Substituting, we have

$$f_{Z,W}(z,w) = \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\lambda z} (zw)^{\alpha-1} [z(1-w)]^{\beta-1} \quad \text{for } z > 0, 0 < w < 1$$

which may be rearranged as

$$f_{Z,W}(z,w) = \frac{\lambda e^{-\lambda z}}{\Gamma(\alpha+\beta)} (\lambda z)^{\alpha+\beta-1} \frac{w^{\alpha-1}(1-w)^{\beta-1}\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \quad \text{for } z > 0, 0 < w < 1$$

Also, using the fact that $\beta(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$, we have

$$f_{Z,W}(z,w) = \frac{\lambda e^{-\lambda z}}{\Gamma(\alpha+\beta)} (\lambda z)^{\alpha+\beta-1} \frac{w^{\alpha-1}(1-w)^{\beta-1}}{\beta(\alpha, \beta)} \quad \text{for } z > 0, 0 < w < 1$$

- (b) From (a), it is easy to see that

$$f_{Z,W}(z, w) = f_Z(z) f_W(w)$$

where

$$f_Z(z) = \frac{\lambda e^{-\lambda z}}{\Gamma(\alpha + \beta)} (\lambda z)^{\alpha + \beta - 1} \text{ for } z > 0$$

and

$$f_W(w) = \frac{w^{\alpha-1}(1-w)^{\beta-1}}{\beta(\alpha, \beta)} \text{ for } 0 < w < 1$$

Thus, it is clear that Z has a gamma distribution with parameters $(\alpha + \beta, \lambda)$ and W has a beta distribution with parameters (α, β) .

- (c) Since the joint probability density function of Z and W is the product of their marginal density functions, we conclude that Z and W are independent. ◆

PROBLEM SET 4.6

1. If X and Y are independent and identically distributed uniform variables over the interval $(0, 1)$, find the probability density function of $Z = X - Y$.
2. If X and Y are independent and identically distributed exponential random variables with parameter 1, find the probability density function of $Z = X - Y$.
3. If X and Y are independent exponential random variables with parameters 1 and 2, respectively, find the probability density function of $Z = \frac{X}{Y}$.
4. If X and Y are continuous random variables with the joint probability density function

$$f(x, y) = \begin{cases} 2e^{-(x+y)} & \text{for } 0 < x < y < \infty \\ 0 & \text{elsewhere} \end{cases}$$

find the probability density function of $Z = X + Y$.

5. If X and Y are continuous random variables with the joint probability density function

$$f(x, y) = \begin{cases} 24xy & \text{for } 0 < x < 1, 0 < y < 1, x + y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the joint probability density function of $Z = X + Y$ and $W = X$.

6. If X and Y are continuous random variables with the joint probability density function

$$f(x, y) = \begin{cases} 4xye^{-(x^2+y^2)} & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the probability density of $R = \sqrt{X^2 + Y^2}$.

7. Let X and Y be continuous random variables with the joint probability density function

$$f(x, y) = \begin{cases} 2 & \text{for } x > 0, y > 0, x + y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find:

- (a) The joint probability density function of $Z = X + Y$ and $W = X - Y$.
 - (b) The marginal density functions of Z and W .
8. Let X and Y be independent and identically distributed uniform variables over $(0, 1)$. Find
- (a) The joint probability density function of $Z = X + Y$ and $W = Y$.
 - (b) The marginal density function of Z .
9. Let X and Y be independent and identically distributed uniform variables over $(0, 1)$. Find
- (a) The joint probability density function of $Z = X + Y$ and $W = X - Y$.
 - (b) The marginal density functions of Z and W .
10. Let X and Y be independent and identically distributed standard normal variables. Find
- (a) The joint probability density function of $Z = X + Y$ and $W = X - Y$.
 - (b) The marginal density functions of Z and W .
 - (c) Are Z and W independent?

4.7 COVARIANCE

Let (X, Y) be a two-dimensional random variable.

Definition 4.14. A *moment of order $i + j$* of (X, Y) for non-negative integers i and j is denoted by μ'_{ij} and is defined as

$$\mu'_{ij} = E(X^i Y^j)$$

provided that $E|X^i Y^j| < \infty$.

If the moments of order 1 exist for (X, Y) , then we have

$$\mu'_{10} = E(X) \text{ and } \mu'_{01} = E(Y)$$

If the moments of order 2 exist for (X, Y) , then we have

$$\mu'_{20} = E(X^2), \mu'_{11} = E(XY) \text{ and } \mu'_{02} = E(Y^2)$$

Definition 4.15. A *central moment of order $i + j$* of (X, Y) for non-negative integers i and j is denoted by μ_{ij} and is defined as

$$\mu_{ij} = E[(X - E(X))^i (Y - E(Y))^j] = E[(X - \mu_X)^i (Y - \mu_Y)^j]$$

provided that $E[(X - \mu_X)^i (Y - \mu_Y)^j] < \infty$.

Note that the first order central moments are clearly zero.

$$\mu_{10} = E[X - \mu_X] = \mu_X - \mu_X = 0$$

and

$$\mu_{01} = E[Y - \mu_Y] = \mu_Y - \mu_Y = 0$$

Note that the second order central moments are given by

$$\mu_{20} = E[(X - \mu_X)^2] = \text{Var}(X)$$

$$\mu_{02} = E[(Y - \mu_Y)^2] = \text{Var}(Y)$$

and

$$\mu_{11} = E[(X - \mu_X)(Y - \mu_Y)] = \text{Cov}(X, Y)$$

Definition 4.16. Let (X, Y) be a two-dimensional random variable. Then the *covariance* between X and Y is denoted by $\text{Cov}(X, Y)$ and is defined as

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where $\mu_X = E(X)$ is the mean of X and $\mu_Y = E(Y)$ is the mean of Y .

EXAMPLE 4.84. Let (X, Y) be a two-dimensional random variable. Show that

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = E(XY) - \mu_X\mu_Y$$

Solution. By definition, we have

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

Thus, we have

$$\text{Cov}(X, Y) = E[XY - E(X)Y - E(Y)X + E(X)E(Y)]$$

Since $E(X)$ and $E(Y)$ are constants, it follows that

$$\begin{aligned} \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y) \\ &= E(XY) - E(X)E(Y) = E(XY) - \mu_X\mu_Y \end{aligned}$$

◆

Theorem 4.17. If X and Y are independent random variables, then $\text{Cov}(X, Y) = 0$.

Proof. By definition, we have

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Since X and Y are independent, it follows that

$$\text{Cov}(X, Y) = E[X - \mu_X] E[Y - \mu_Y] = [\mu_X - \mu_X] [\mu_Y - \mu_Y] = 0$$

◆

Definition 4.17. Two random variables, X and Y , are called *uncorrelated* if $\text{Cov}(X, Y) = 0$.

If X and Y are independent, then, by Theorem 4.17, it follows that $\text{Cov}(X, Y) = 0$. Thus, if X and Y are independent, then they are uncorrelated. However, the converse is not true. That is, if X and Y are uncorrelated, then they need not be independent. This is illustrated by Example 4.85.

EXAMPLE 4.85. Let the random variable X be uniformly distributed over $(-1, 1)$ and $Y = X^2$. Show that X and Y are uncorrelated, but they are dependent.

(Anna, Model 2003)

Solution. First, note that the probability density of X is given by

$$f(x) = \begin{cases} \frac{1}{2} & \text{for } -1 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we calculate $E(X)$, $E(Y)$ and $E(XY)$.

The mean of X is given by

$$\mu_X = E(X) = \int_{x=-1}^1 x f(x) dx = \int_{x=-1}^1 x \left[\frac{1}{2} \right] dx = 0$$

since the integrand is an odd function of x .

The mean of Y is given by

$$\mu_Y = E(Y) = E(X^2) = \int_{x=-1}^1 x^2 f(x) dx = \int_{x=-1}^1 x^2 \left[\frac{1}{2} \right] dx$$

Since the integrand is an even function of x , we can simplify the above integral as

$$\mu_Y = 2 \int_{x=0}^1 x^2 \left[\frac{1}{2} \right] dx = \int_{x=0}^1 x^2 dx = \left[\frac{x^3}{3} \right]_{x=0}^1 = \frac{1}{3}$$

We also find that

$$E(XY) = E(X^3) = \int_{x=-1}^1 x^3 f(x) dx = \int_{x=-1}^1 x^3 \left[\frac{1}{2} \right] dx = 0$$

since the integrand is an odd function of x .

It follows that

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0 - 0 = 0$$

Hence, X and Y are uncorrelated. However, X and Y are not independent, because $Y = X^2$. ◆

EXAMPLE 4.86. Prove that $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$.

(Anna, Model 2003)

Solution. We find that

$$\text{Cov}(aX, bY) = E[(aX - E(aX))(bY - E(bY))]$$

Since a and b are constants, it follows that

$$\text{Cov}(aX, bY) = E[a(X - E(X)) b(Y - E(Y))] = abE[(X - E(X))(Y - E(Y))]$$

Hence, it is immediate that

$$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y) \quad \spadesuit$$

Theorem 4.18. Let $X_1, X_2, \dots, X_n$ be n random variables. Then

$$\text{Var} \left(\sum_{i=1}^n a_i X_i \right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)$$

where $a_1, a_2, \dots, a_n$ are any constants.

Proof. Let $Z = a_1 X_1 + a_2 X_2 + \dots + a_n X_n$. Then we have

$$E(Z) = a_1 E(X_1) + a_2 E(X_2) + \dots + a_n E(X_n)$$

Hence, it follows that

$$Z - E(Z) = a_1 (X_1 - E(X_1)) + a_2 (X_2 - E(X_2)) + \dots + a_n (X_n - E(X_n))$$

Hence, we have

$$\text{Var}(Z) = (Z - E(Z))^2 = \sum_{i=1}^n a_i^2 (X_i - E(X_i))^2 + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j E[(X_i - E(X_i))(X_j - E(X_j))]$$

It follows that

$$\text{Var}(Z) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j)$$

◆

EXAMPLE 4.87. Show that for random variables X, Y, Z and W and constants a, b, c and d ,

$$\text{Cov}(aX + bY, cZ + dW) = ac\text{Cov}(X, Z) + bc\text{Cov}(Y, Z) + ad\text{Cov}(X, W) + bd\text{Cov}(Y, W)$$

(Anna, April 2003)

Solution. Define

$$U = aX + bY \quad \text{and} \quad V = cZ + dW$$

By definition, we have

$$\text{Cov}(aX + bY, cZ + dW) = \text{Cov}(U, V) = E[(U - \mu_U)(V - \mu_V)]$$

Note that

$$U - \mu_U = (aX + bY) - (a\mu_X + b\mu_Y) = a(X - \mu_X) + b(Y - \mu_Y)$$

and

$$V - \mu_V = (cZ + dW) - (c\mu_Z + d\mu_W) = c(Z - \mu_Z) + d(W - \mu_W)$$

Hence, it follows that

$$\begin{aligned} (U - \mu_U)(V - \mu_V) &= ac(X - \mu_X)(Z - \mu_Z) + bc(Y - \mu_Y)(Z - \mu_Z) \\ &\quad + ad(X - \mu_X)(W - \mu_W) + bd(Y - \mu_Y)(W - \mu_W) \end{aligned}$$

Taking expectation on both sides of the above equation, it follows that

$$\text{Cov}(U, V) = ac\text{Cov}(X, Z) + bc\text{Cov}(Y, Z) + ad\text{Cov}(X, W) + bd\text{Cov}(Y, W)$$

◆

EXAMPLE 4.88. Show that $\text{Cov}^2(X, Y) \leq \text{Var}(X) \text{Var}(Y)$.

(Anna, Nov. 2004)

Solution. By Cauchy-Schwartz inequality (see Theorem 4.14), it follows that

$$[E(UV)]^2 \leq E[U^2] E[V^2]$$

for any two random variables U and V . Taking $U = X - \mu_X$ and $V = Y - \mu_Y$, where μ_X and μ_Y are the means of X and Y respectively, it follows that

$$\{E[(X - \mu_X)(Y - \mu_Y)]\}^2 \leq E[(X - \mu_X)]^2 E[(Y - \mu_Y)]^2$$

or

$$\text{Cov}^2(X, Y) \leq \text{Var}(X) \text{Var}(Y)$$

◆

EXAMPLE 4.89. Let X and Y be two random variables each taking three values $-1, 0$ and 1 and having the joint probability distribution:

X \ Y	-1	0	1	Total
-1	0	.1	.1	.2
0	.2	.2	.2	.6
1	0	.1	.1	.2
Total	.2	.4	.4	1.0

Prove that X and Y have different expectations. Also prove that X and Y are uncorrelated and find $\text{Var } X$ and $\text{Var } Y$.

(Anna, Nov. 2005)

Solution. First, note that the marginal distribution of X is given by

$$f_X(x) = P(X = x) = \sum_{y=-1}^1 f(x, y) \text{ for } x = -1, 0, 1$$

Hence, we have

x	-1	0	1
$f_X(x)$	0.2	0.6	0.2

Thus, the expectation of X is given by

$$\mu_X = E(X) = \sum_x x f_X(x) = (-0.2) + 0 + 0.2 = 0$$

Next, note that the marginal distribution of Y is given by

$$f_Y(y) = P(Y = y) = \sum_{x=-1}^1 f(x, y) \text{ for } y = -1, 0, 1$$

Hence, we have

y	-1	0	1
$f_Y(y)$	0.2	0.4	0.4

Thus, the expectation of Y is given by

$$\mu_Y = E(Y) = \sum_y y f_Y(y) = (-0.2) + 0 + 0.4 = 0.2$$

Since $\mu_X = 0$ and $\mu_Y = 0.2$, we have indeed shown that X and Y have different expectations. Next, we note that

$$E(XY) = \sum_x \sum_y xy f(x, y) = 0 + 0 + (-0.1) + 0 + 0 + 0 + 0 + 0 + 0.1 = 0$$

Hence, it is immediate that

$$\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y = 0 - (0)(0.2) = 0 - 0 = 0$$

Thus, X and Y are uncorrelated.

Finally, we find $\text{Var } X$ and $\text{Var } Y$.

Note that

$$E(X^2) = \sum_x x^2 f_X(x) = (1)(0.2) + (0)(0.6) + (1)(0.2) = 0.4$$

Hence, we have

$$\text{Var } X = E(X^2) - \mu_X^2 = 0.4 - 0 = 0.4$$

Also, note that

$$E(Y^2) = \sum_y y^2 f_Y(y) = (1)(0.2) + (0)(0.4) + (1)(0.4) = 0.6$$

and so

$$\text{Var } Y = E(Y^2) - \mu_Y^2 = 0.6 - (0.2)^2 = 0.6 - 0.04 = 0.56$$

◆

EXAMPLE 4.90. Two independent random variables, X and Y , have probability density functions defined by

$$f_X(x) = \begin{cases} 4ax, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad f_Y(y) = \begin{cases} 4by, & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Show that $U = X + Y$ and $V = X - Y$ are uncorrelated.

(Anna, April 2003)

Solution. First, we need to find the values of the constants a and b .

Since f_X is a probability density function defined on $[0, 1]$, we must have

$$\int_0^1 f(x) dx = 1 \Rightarrow \int_0^1 4ax dx = 1$$

Integrating, we have

$$2a \left[x^2 \right]_{x=0}^1 = 1 \Rightarrow 2a = 1$$

Hence, $a = \frac{1}{2}$.

Since the density functions of X and Y are identical, it also follows that $b = \frac{1}{2}$.

Thus, X and Y are independent and identically distributed (i.i.d.) random variables with the common density function

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

We wish to show that the random variables U and V are uncorrelated, where

$$U = X + Y \text{ and } V = X - Y$$

That is, we wish to show that $\text{Cov}(U, V) = E(UV) - E(U)E(V) = 0$.

Note that

$$E(UV) = E[(X + Y)(X - Y)] = E[X^2 - Y^2] = E[X^2] - E[Y^2] = 0$$

and

$$E(U) = E(X) + E(Y)$$

and

$$E(V) = E(X) - E(Y) = 0$$

since X and Y are identically distributed.

Hence, it follows that

$$\text{Cov}(U, V) = E(UV) - E(U)E(V) = 0 - 0 = 0$$

Thus, U and V are uncorrelated. ♦

EXAMPLE 4.91. The fraction X of male runners and the fraction Y of female runners who complete marathon races can be described by the joint density function:

$$f(x, y) = \begin{cases} 8xy, & 0 \leq x \leq 1, 0 \leq y \leq x \\ 0, & \text{otherwise} \end{cases}$$

Find the covariance of X and Y .

(Anna, Model 2003)

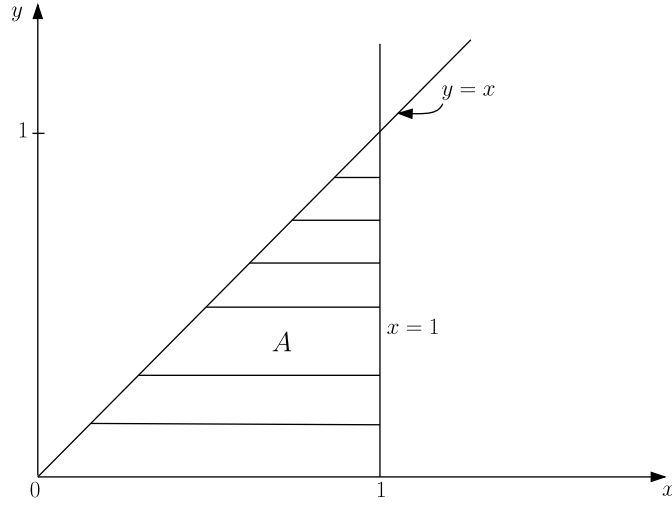
Solution. The joint density function f takes values in the region A defined by

$$A = \{(x, y) \in \mathbb{R}^2 : 0 \leq y \leq x \leq 1\}$$

which is illustrated in Figure 4.24.

The marginal density function of X is easily obtained as

$$f_X(x) = \begin{cases} 4x^3 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

**Figure 4.24:** Region A.

and the marginal density function of Y is easily obtained as

$$f_Y(y) = \begin{cases} 4y(1-y^2) & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Next, the mean of X is given by

$$\mu_X = E(X) = \int_{x=0}^1 x f_X(x) dx = \int_{x=0}^1 x [4x^3] dx = 4 \int_{x=0}^1 x^4 dx$$

Integrating, we have

$$\mu_X = 4 \left[\frac{x^5}{5} \right]_{x=0}^1 = 4 \left[\frac{1}{5} \right] = \frac{4}{5}$$

Next, the mean of Y is given by

$$\mu_Y = E(Y) = \int_{y=0}^1 y f_Y(y) dy = \int_{y=0}^1 y [4y(1-y^2)] dy = 4 \int_{y=0}^1 [y^2 - y^4] dy$$

Integrating, we have

$$\mu_Y = 4 \left[\frac{y^3}{3} - \frac{y^5}{5} \right]_{y=0}^1 = 4 \left[\frac{1}{3} - \frac{1}{5} \right] = \frac{8}{15}$$

Next, we find $E(XY)$.

$$E(XY) = \int_{x=0}^1 \int_{y=0}^x xy f(x,y) dy dx = \int_{x=0}^1 \int_{y=0}^x xy [8xy] dy dx$$

Integrating y , we have

$$E(XY) = 8 \int_{x=0}^1 x^2 \left[\frac{y^3}{3} \right]_{y=0}^x dx = \frac{8}{3} \int_{x=0}^1 x^5 dx$$

Integrating x , we have

$$E(XY) = \frac{8}{3} \left[\frac{x^6}{6} \right]_{x=0}^1 = \frac{8}{3} \times \frac{1}{6} = \frac{4}{9}$$

Hence, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y = \frac{4}{9} - \frac{4}{5} \times \frac{8}{15} = \frac{4}{9} - \frac{32}{75} = \frac{4}{225}$$

◆

EXAMPLE 4.92. The joint probability density function of random variables, X and Y , is given by

$$f(x, y) = \begin{cases} 3(x+y), & 0 \leq x \leq 1, 0 \leq y \leq 1, x+y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find

- (i) the marginal probability density function of X .
- (ii) $P(X+Y < \frac{1}{2})$.
- (iii) $\text{Cov}(X, Y)$.

(Anna, April 2005)

Solution. (i) Clearly, the marginal probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{3}{2} (1-x^2) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(ii) The required probability is given by

$$P\left(X+Y < \frac{1}{2}\right) = \int_{x=0}^{\frac{1}{2}} \int_{y=0}^{\frac{1}{2}} f(x, y) dy dx = \int_{x=0}^{\frac{1}{2}} \int_{y=0}^{\frac{1}{2}} 3(x+y) dy dx$$

Integrating y , we have

$$\begin{aligned} P\left(X+Y < \frac{1}{2}\right) &= 3 \int_{x=0}^{\frac{1}{2}} \left[xy + \frac{y^2}{2} \right]_{y=0}^{\frac{1}{2}} dx \\ &= \frac{3}{2} \int_{x=0}^{\frac{1}{2}} \left(\frac{1}{4} - x^2 \right) dx \end{aligned}$$

Integrating x , we have

$$\begin{aligned} P\left(X+Y < \frac{1}{2}\right) &= \frac{3}{2} \left[\frac{x}{4} - \frac{x^3}{3} \right]_{x=0}^{\frac{1}{2}} \\ &= \frac{3}{2} \left[\frac{1}{8} - \frac{1}{24} \right] = \frac{3}{2} \times \frac{1}{12} = \frac{1}{8} \end{aligned}$$

(iii) We know that $\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y$.

Since we have computed the marginal PDF of X in (i), it follows readily that

$$\begin{aligned} \mu_X &= \int_{x=0}^1 x f_X(x) dx = \int_{x=0}^1 x \left[\frac{3}{2} (1-x^2) \right] dx \\ &= \frac{3}{2} \int_{x=0}^1 (x-x^3) dx \end{aligned}$$

Integrating x , we have

$$\begin{aligned} \mu_X &= \frac{3}{2} \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_{x=0}^1 = \frac{3}{2} \left[\frac{1}{2} - \frac{1}{4} \right] \\ &= \frac{3}{2} \times \frac{1}{4} = \frac{3}{8} \end{aligned}$$

Since the given distribution is symmetric with respect to X and Y , it follows that the marginal PDF of Y is given by

$$f_Y(y) = \begin{cases} \frac{3}{2} (1-y^2) & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

It is evident from this that the mean of Y is the same as the mean of X .

Hence, we have $\mu_Y = \mu_X = \frac{3}{8}$.

Next, we note that

$$\begin{aligned} E(XY) &= \int_{x=0}^1 \int_{y=0}^{1-x} xy f(x,y) dy dx = \int_{x=0}^1 \int_{y=0}^{1-x} xy 3(x+y) dy dx \\ &= 3 \int_{x=0}^1 \int_{y=0}^{1-x} x(xy+y^2) dy dx \end{aligned}$$

Integrating y , we have

$$E(XY) = 3 \int_{x=0}^1 x \left[\frac{xy^2}{2} + \frac{y^3}{3} \right]_{y=0}^{1-x} dx$$

$$= 3 \int_{x=0}^1 x(1-x)^2 \left[\frac{x}{2} + \frac{1-x}{3} \right] dx$$

or

$$\begin{aligned} E(XY) &= \frac{3}{6} \int_{x=0}^1 x(x^2 - 2x + 1)(x + 2) dx \\ &= \frac{1}{2} \int_{x=0}^1 [x^4 - 3x^2 + 2x] dx \end{aligned}$$

Integrating, we have

$$E(XY) = \frac{1}{2} \left[\frac{x^5}{5} - x^3 + x^2 \right]_{x=0}^1 = \frac{1}{10}$$

Hence, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y = \frac{1}{10} - \left[\frac{3}{8} \right]^2 = -\frac{13}{320}$$

◆

EXAMPLE 4.93. Two random variables X and Y have the following joint probability density function

$$f(x, y) = \begin{cases} 2 - x - y; & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find marginal probability density functions of X and Y . Also find the conditional density functions and covariance between X and Y .

(Anna, Nov. 2005; Nov. 2007)

Solution. The marginal probability density functions of X and Y are given, respectively, by

$$f_X(x) = \begin{cases} \frac{3}{2} - x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{3}{2} - y & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

To find $f_{X|Y}(x|y)$, fix any value of y in the range $0 \leq y \leq 1$. Then we have

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{2 - x - y}{\frac{3}{2} - y} \quad \text{for } 0 \leq x \leq 1$$

To find $f_{Y|X}(y|x)$, fix any value of x in the range $0 \leq x \leq 1$. Then we have

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{2 - x - y}{\frac{3}{2} - x} \quad \text{for } 0 \leq y \leq 1$$

Next, we find the covariance between X and Y . By definition, we have

$$\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y$$

First, we find μ_X from the marginal PDF of X .

$$\mu_X = \int_{x=0}^1 x f_X(x) dx = \int_{x=0}^1 \left[\frac{3x}{2} - x^2 \right] dx$$

Integrating, we have

$$\mu_X = \left[\frac{3x^2}{4} - \frac{x^3}{3} \right]_{x=0}^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$$

Next, we find μ_Y from the marginal PDF of Y . We have

$$\mu_Y = \int_{y=0}^1 y f_Y(y) dy = \int_{y=0}^1 \left[\frac{3y}{2} - y^2 \right] dy$$

Integrating,

$$\mu_Y = \left[\frac{3y^2}{4} - \frac{y^3}{3} \right]_{y=0}^1 = \frac{3}{4} - \frac{1}{3} = \frac{5}{12}$$

We also find that

$$E(XY) = \int_{x=0}^1 \int_{y=0}^1 xy f(x, y) dy dx = \int_{x=0}^1 x \int_{y=0}^1 [(2-x)y - y^2] dy dx$$

Integrating y , we have

$$E(XY) = \int_{x=0}^1 x \left[\frac{(2-x)y^2}{2} - \frac{y^3}{3} \right]_{y=0}^1 dx$$

Thus,

$$\begin{aligned} E(XY) &= \int_{x=0}^1 x \left[\frac{2-x}{2} - \frac{1}{3} \right] dx = \frac{1}{6} \int_{x=0}^1 [4x - 3x^2] dx \\ &= \frac{1}{6} [2x^2 - x^3]_{x=0}^1 = \frac{1}{6} \end{aligned}$$

Hence, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y = \frac{1}{6} - \left(\frac{5}{12} \right)^2 = -\frac{1}{144}$$

◆

EXAMPLE 4.94. Two random variables, X and Y have the joint probability density function

$$f(x, y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Find $E(X)$, $E(Y)$, $E(XY)$, $E(2X + 3Y)$, $V(X)$, $V(Y)$, $\text{Cov}(X, Y)$. What can you infer from $\text{Cov}(X, Y)$?
(Anna, May 2006)

Solution. Clearly, the marginal probability density functions of X and Y are given, respectively,

$$f_X(x) = \begin{cases} \frac{x}{8} & \text{if } 0 < x < 4 \\ 0 & \text{otherwise} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{y}{12} & \text{if } 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Hence, we see that

$$f_X(x) f_Y(y) = \begin{cases} \frac{xy}{96} & \text{if } 0 < x < 4, 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Since $f(x, y) = f_X(x) f_Y(y)$ for all $(x, y) \in \mathbb{R}^2$, it follows that X and Y are independent. Hence, it is immediate that

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = E(X)E(Y) - E(X)E(Y) = 0$$

From this, we infer that X and Y are *uncorrelated*.

Next, we find $E(X)$, $E(Y)$, $E(XY)$.

First, we find that

$$\begin{aligned} E(X) &= \int_{x=0}^4 x f_X(x) dx = \int_{x=0}^4 x \left[\frac{x}{8} \right] dx = \frac{1}{8} \left[\frac{x^3}{3} \right]_{x=0}^4 \\ &= \frac{1}{24} [64 - 0] = \frac{8}{3} \end{aligned}$$

Next, we find that

$$\begin{aligned} E(Y) &= \int_{y=1}^5 y f_Y(y) dy = \int_{y=1}^5 y \left[\frac{y}{12} \right] dy = \frac{1}{12} \left[\frac{y^3}{3} \right]_{y=1}^5 \\ E(Y) &= \frac{1}{36} [125 - 1] = \frac{124}{36} = \frac{31}{9} \end{aligned}$$

Since X and Y are independent, it is immediate that

$$E(XY) = E(X)E(Y) = \frac{248}{27}$$

We also find that

$$E(2X + 3Y) = 2E(X) + 3E(Y) = 2 \left[\frac{8}{3} \right] + 3 \left[\frac{31}{9} \right] = \frac{47}{3}$$

Next, we find $\text{Var}(X)$ and $\text{Var}(Y)$.

First, we note that

$$E(X^2) = \int_{x=0}^4 x^2 f_X(x) dx = \int_{x=0}^4 x^2 \left[\frac{x}{8} \right] dx$$

Integrating, we have

$$E(X^2) = \frac{1}{8} \left[\frac{x^4}{4} \right]_0^4 = \frac{1}{32} [256] = 8$$

Next, we note that

$$E(Y^2) = \int_{y=1}^5 y^2 f_Y(y) dy = \int_{y=1}^5 y^2 \left[\frac{y}{12} \right] dy$$

Integrating, we have

$$E(Y^2) = \frac{1}{12} \left[\frac{y^4}{4} \right]_1^5 = \frac{1}{48} [624] = 13$$

Hence, it follows that

$$\text{Var}(X) = E(X^2) - \mu_X^2 = 8 - \left(\frac{8}{3} \right)^2 = \frac{8}{9}$$

and

$$\text{Var}(Y) = E(Y^2) - \mu_Y^2 = 13 - \left(\frac{31}{9} \right)^2 = \frac{92}{81}$$

◆

PROBLEM SET 4.7

1. Let X and Y be two random variables taking three values $-1, 0$ and 1 , and having the joint probability distribution:

X \ Y	Y		
	-1	0	1
-1	$\frac{2}{16}$	$\frac{1}{16}$	$\frac{2}{16}$
0	$\frac{2}{16}$	$\frac{2}{16}$	$\frac{2}{16}$
1	$\frac{2}{16}$	$\frac{1}{16}$	$\frac{2}{16}$

Show that X and Y are not independent and are uncorrelated.

2. If the joint probability density function of X and Y is given by

$$f(x, y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & \text{for } x > 0, y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

show that X and Y are uncorrelated.

3. If X and Y are random variables with the joint probability density function

$$f(x,y) = \frac{x^2 + y^2}{4\pi} e^{-\frac{(x^2+y^2)}{2}} \quad \text{for } -\infty < x < \infty, -\infty < y < \infty$$

show that X and Y are not independent but are uncorrelated.

4. If X and Y are random variables with the joint probability density function

$$f(x,y) = \begin{cases} \frac{1}{3}(x+y) & \text{for } 0 < x < 2, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find $E(X), E(Y), \text{Var}(X), \text{Var}(Y)$ and $\text{Cov}(X, Y)$.

5. If X and Y are random variables with the joint probability density function

$$f(x,y) = \begin{cases} \frac{1}{2} & \text{for } 0 < y < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

find the covariance of X and Y .

6. If X and Y are random variables with the joint probability density function

$$f(x,y) = \begin{cases} x^2 + \frac{xy}{3} & \text{for } 0 < x < 1, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

find the covariance of X and Y .

7. If X and Y are random variables with the joint probability density function

$$f(x,y) = \begin{cases} \frac{2}{3}(x+2y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the covariance of X and Y .

8. If X and Y are random variables with the joint probability density function

$$f(x,y) = \begin{cases} \frac{9}{4} - x - y & \text{for } 0 < x < 2, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

find the covariance of X and Y .

4.8 CONDITIONAL EXPECTATION

In Section 4.3, you have seen that the conditional density function $f_{X|Y}(x|y)$ is a probability density function when considered as a function of the values of X . Hence, we can easily define the expected value of a function of the random variable X given the value of the random variable Y . This is called as *conditional expectation* and is defined as follows.

Definition 4.18. Let (X, Y) be a two-dimensional random variable. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Then the *conditional expectation* of $h(X)$, given $Y = y$, is denoted by $E(h(X)|Y = y)$ and is defined as

$$E(h(X)|Y = y) = \sum_x h(x) f_{X|Y}(x|y)$$

if (X, Y) is a discrete random variable and $f_Y(y) = P(Y = y) > 0$, and

$$E(h(X)|Y = y) = \int_{x=-\infty}^{\infty} h(x) f_{X|Y}(x|y) dx$$

if (X, Y) is a continuous random variable and $f_Y(y) > 0$. (A similar definition can be given for the conditional expectation $E(h(Y)|X = x)$.)

Definition 4.19. Let (X, Y) be a two-dimensional random variable. Then the *conditional mean* of X , given $Y = y$, is defined as the conditional expectation $E(X|Y = y)$, and the conditional moments of X , given $Y = y$ are defined as $E(X^n|Y = y)$, where n is any non-negative integer. The *conditional variance* of X , given $Y = y$, is denoted by $\text{Var}(X|Y = y)$ and is defined as

$$\text{Var}(X|Y = y) = E[X - E(X|Y = y)]^2$$

which can be simplified to get

$$\text{Var}(X|Y = y) = E[X^2|Y = y] - [E(X|Y = y)]^2$$

The conditional mean, conditional moments and conditional variance are similarly defined for the conditional density function $f_{Y|X}(y|x)$.

The following theorem lists some important properties of the conditional expectation, and is stated without proof.

Theorem 4.19. Let (X, Y) be a two-dimensional random variable. Then the following properties hold:

- (i) $E(c|Y = y) = c$ for any constant c .
- (ii) $E[\alpha g(X) + \beta h(X)|Y = y] = \alpha E[g(X)|Y = y] + \beta E[h(X)|Y = y]$, where α and β are constants and $g, h : \mathbb{R} \rightarrow \mathbb{R}$ are continuous functions.

The conditional moments, $E(X^n|Y = y)$, are constant quantities as long as the value of Y is fixed. But $E(X^n|Y)$ is a random variable as Y takes different values. Hence, we may define the expectation of $E(X^n|Y)$. The following theorem asserts that this expectation equals $E(X^n)$.

Theorem 4.20. If $E(X^n)$ exists, then

$$E(X^n) = E[E(X^n|Y)]$$

Proof. Let (X, Y) be a discrete random variable. Fix any value of Y , say $Y = y$.

By definition,

$$\begin{aligned} E(X^n|Y) &= \sum_x x^n f_{X|Y}(x|y) = \sum_x x^n \left[\frac{P(X=x, Y=y)}{P(Y=y)} \right] \\ &= \frac{\sum_x x^n P(X=x, Y=y)}{P(Y=y)} \end{aligned} \quad (4.7)$$

As Y assumes various values, $Y = y_j (j = 1, 2, \dots)$, $E(X^n|Y)$ is a function of Y , and hence, its expectation can be defined. Thus, we have

$$E[E(X^n|Y)] = \sum_y E[X^n|Y=y] P(Y=y) \quad (4.8)$$

From Eqs. (4.7) and (4.8), we have

$$E[E(X^n|Y)] = \sum_x \sum_y x^n P(X=x, Y=y) = \sum_x x^n P(X=x) = E(X^n)$$

since $\sum_y P(X=x, Y=y) = P(X=x)$.

Next, let (X, Y) be a continuous random variable. Fix any value of Y , say $Y = y$.

By definition, we have

$$\begin{aligned} E(X^n|Y) &= \int_{x=-\infty}^{\infty} x^n f_{X|Y}(x|y) dx = \int_{x=-\infty}^{\infty} x^n \left[\frac{f(x, y)}{f_Y(y)} \right] dx \\ &= \frac{\int_{x=-\infty}^{\infty} x^n f(x, y) dx}{f_Y(y)} \end{aligned} \quad (4.9)$$

As Y assumes various values, $Y = y$, where $-\infty < y < \infty$, $E(X^n|Y)$ is a function of Y , and hence, its expectation can be defined. Thus, we have

$$E[E(X^n|Y)] = \int_{y=-\infty}^{\infty} E[X^n|Y=y] f_Y(y) dy \quad (4.10)$$

From Eqs. (4.7) and (4.8), we have

$$E[E(X^n|Y)] = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} x^n f(x, y) dy dx = \int_{x=-\infty}^{\infty} x^n f_X(x) dx = E(X^n)$$

since $\int_{y=-\infty}^{\infty} f(x, y) dy = f_X(x)$.

◆

Theorem 4.21. If X and Y are independent random variables, then

(i) $E(X^n|Y=y) = E(X^n)$.

(ii) $E(Y^n|X = x) = E(Y^n)$.

Proof. Let X and Y be independent random variables of the discrete type. Then

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

for all values of x and y within their range.

(i) By definition, we have

$$E(X^n|Y = y) = \sum_x x^n f_{X|Y}(x|y)$$

If X and Y are independent, then we know that $f_{X|Y}(x|y) = P(X = x)$. Indeed,

$$f_{X|Y}(x|y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{P(X = x)P(Y = y)}{P(Y = y)} = P(X = x)$$

Hence, it follows that

$$E(X^n|Y = y) = \sum_x x^n P(X = x) = E(X^n)$$

(ii) By definition, we have

$$E(Y^n|X = x) = \sum_y y^n f_{Y|X}(y|x)$$

If X and Y are independent, then we know that $f_{Y|X}(y|x) = P(Y = y)$. Indeed,

$$f_{Y|X}(y|x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{P(X = x)P(Y = y)}{P(X = x)} = P(Y = y)$$

Hence, it follows that

$$E(Y^n|X = x) = \sum_y y^n P(Y = y) = E(Y^n)$$

The proof for the continuous case is similar.

◆

Theorem 4.22. $\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]$.

Proof. By definition, we have

$$\text{Var}(X|Y = y) = E[X^2|Y = y] - [E(X|Y = y)]^2$$

It follows that

$$E[\text{Var}(X|Y)] = EE[X^2|Y] - E[E(X|Y = y)]^2$$

By Theorem 4.20, we have

$$E[\text{Var}(X|Y)] = E(X^2) - E[E(X|Y = y)]^2$$

which can be written as

$$E[\text{Var}(X|Y)] = E(X^2) - [E(X)]^2 - E[E(X|Y = y)]^2 + [E(X)]^2$$

Thus, we have

$$E[\text{Var}(X|Y)] = \text{Var}(X) - E[E(X|Y=y)]^2 + E[E(X|Y)]^2 = \text{Var}(X) - \text{Var}[E(X|Y)]$$

Hence, it follows that

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]$$

◆

EXAMPLE 4.95. Suppose the PDF $f(x, y)$ of (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Obtain the marginal PDF of X , the conditional PDF of Y given $X = 0.8$ and then $E(Y|X = 0.8)$.

(Anna, April 2005)

Solution. First, we obtain the marginal probability density function of X .

If $0 \leq x \leq 1$, then we have

$$f_X(x) = \int_{y=0}^1 f(x, y) dy = \int_{y=0}^1 \frac{6}{5}(x + y^2) dy$$

Integrating, we have

$$f_X(x) = \frac{6}{5} \left[xy + \frac{y^3}{3} \right]_{y=0}^1 = \frac{6}{5} \left[x + \frac{1}{3} \right]$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{6}{5} \left(x + \frac{1}{3} \right) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Next, we fix the value of X as $X = 0.8$.

Then, the conditional density function of Y given $X = 0.8$ is obtained as

$$\begin{aligned} f_{Y|X}(y|0.8) &= \frac{f(0.8, y)}{f_X(0.8)} = \frac{\frac{6}{5}(0.8 + y^2)}{\frac{6}{5}(0.8 + \frac{1}{3})} \quad \text{for } 0 \leq y \leq 1 \\ &= \frac{4/5 + y^2}{17/15} = \frac{3}{17} (5y^2 + 4) \end{aligned}$$

Hence, the conditional density function of Y given $X = 0.8$ is

$$f_{Y|X}(y|0.8) = \begin{cases} \frac{3}{17} (5y^2 + 4) & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Hence, we have

$$\begin{aligned} E(Y|X = 0.8) &= \int_{y=0}^1 y f_{Y|X}(y|0.8) dy = \int_{y=0}^1 y \left[\frac{3}{17} (5y^2 + 4) \right] dy \\ &= \frac{3}{17} \int_{y=0}^1 (5y^3 + 4y) dy \end{aligned}$$

Integrating, we have

$$\begin{aligned} E(Y|X = 0.8) &= \frac{3}{17} \left[\frac{5y^4}{4} + 2y^2 \right]_{y=0}^1 = \frac{3}{17} \left[\frac{5}{4} + 2 \right] = \frac{3}{17} \times \frac{13}{4} \\ &= \frac{39}{68} = 0.5735 \end{aligned}$$

◆

EXAMPLE 4.96. If the joint probability density function of random variables X and Y is

$$f(x, y) = \begin{cases} xe^{-x(1+y)}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

find $f(y|x)$ and $E(Y|X = x)$.

(Anna, Nov. 2005)

Solution. First, we compute the marginal density function of X .

If $x > 0$, then we have

$$f_X(x) = \int_{y=0}^{\infty} f(x, y) dy = \int_{y=0}^{\infty} xe^{-x(1+y)} dy = \int_{y=0}^{\infty} xe^{-x} e^{-xy} dy$$

Integrating, we have

$$f_X(x) = e^{-x} [-e^{-xy}]_{y=0}^{\infty} = e^{-x}$$

Hence, the marginal density function of X is given by

$$f_X(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Next, we find the conditional density function of Y given $X = x$.

Fix any value of X in the range $x > 0$. By definition, we have

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f(x, y)}{f_X(x)} = \frac{xe^{-x}e^{-xy}}{e^{-x}} \text{ for } y > 0 \\ &= xe^{-xy} \text{ for } y > 0 \end{aligned}$$

We also find the conditional mean of Y given $X = x$ as

$$E(Y|X = x) = \int_{y=0}^{\infty} y f_{Y|X}(y|x) dy = \int_{y=0}^{\infty} y [xe^{-xy}] dy$$

Using integration by parts, we have

$$\begin{aligned} E(Y|X = x) &= \int_{y=0}^{\infty} y d[-e^{-xy}] = [y(-e^{-xy})]_{y=0}^{\infty} + \int_{y=0}^{\infty} e^{-xy} dy \\ &= 0 + \left[-\frac{e^{-xy}}{x} \right]_{y=0}^{\infty} = 0 + \frac{1}{x} = \frac{1}{x} \end{aligned}$$

Hence, $E(Y|X = x) = \frac{1}{x}$



EXAMPLE 4.97. The joint PDF of a bivariate random variable (X, Y) is given by

$$f(x, y) = \begin{cases} 2, & 0 < y \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Compute $E(Y|X = x)$ and $E(X|Y = y)$.

(Anna, May 2007)

Solution. The marginal density function of X is given by

$$f_X(x) = \begin{cases} 2x & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 2(1-y) & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Next, we calculate the conditional density functions, $f_{Y|X}$ and $f_{X|Y}$.

First, fix any value x of X in the interval $0 < x < 1$. Then the conditional density of Y , given $X = x$, is obtained as

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x} \quad \text{for } 0 < y < x$$

Next, fix any value y of Y in the interval $0 < y < 1$. Then the conditional density of X given $Y = y$ is obtained as

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{2}{2(1-y)} = \frac{1}{1-y} \quad \text{for } y < x < 1$$

Next, we compute $E(Y|X = x)$ and $E(X|Y = y)$.

By the definition of conditional expectation, we have

$$E(Y|X=x) = \int_{y=0}^x y f_{Y|X}(y|x) dy = \int_{y=0}^x y \frac{1}{x} dy$$

Integrating, we have

$$E(Y|X=x) = \frac{1}{x} \left[\frac{y^2}{2} \right]_{y=0}^x = \frac{x}{2}$$

Similarly, we find that

$$E(X|Y=y) = \int_{x=y}^1 x f_{X|Y}(x|y) dx = \int_{x=y}^1 x \frac{1}{1-y} dx$$

Integrating, we have

$$E(X|Y=y) = \frac{1}{1-y} \left[\frac{x^2}{2} \right]_{x=y}^1 = \frac{1-y^2}{2(1-y)} = \frac{1+y}{2}$$



PROBLEM SET 4.8

1. The joint probability distribution of a bivariate random variable (X, Y) is given by

$X \backslash Y$	-1	0	1
-1	0.1	0.1	0.1
0	0.2	0	0.2
1	0.1	0.1	0.1

Find the marginal distribution of X , the conditional distribution of Y given $X = 0$ and then $E(Y|X = 0)$.

2. The joint probability distribution of X and Y is given below:

$X \backslash Y$	1	2	3
0	0.2	0.1	0.1
1	0	0.1	0.2
2	0.1	0.1	0.1

Find:

- The marginal distributions of X and Y .
- The conditional distribution of X given $Y = 1$ and that of Y given $X = 1$.
- $E(X|Y = 1)$ and $E(Y|X = 1)$.

3. If the joint probability density function of a bivariate random variable (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{2}{5}(x + 4y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find $f_{Y|X}(y|x)$ and $E(Y|X = x)$.

4. If the joint probability density function of a bivariate random variable (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{1}{3}(x + y) & \text{for } 0 < x < 2, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find $f_{X|Y}(x|y)$ and $E(X|Y = y)$.

5. If X and Y are random variables with the joint probability density function

$$f(x, y) = \begin{cases} 8xy & \text{for } 0 < x < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

compute $E(Y|X = x)$ and $E(X|Y = y)$.

6. If X and Y are random variables with the joint probability density function

$$f(x, y) = \begin{cases} x^2 + \frac{xy}{3} & \text{for } 0 < x < 1, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

compute $E(Y|X = x)$ and $E(X|Y = y)$.

7. If X and Y are random variables with the joint probability density function

$$f(x, y) = \begin{cases} 2 & \text{for } 0 < x + y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

compute $E(Y|X = x)$ and $E(X|Y = y)$.

4.9 CORRELATION AND REGRESSION

We start this section with a derivation of *regression curves*.

Let (X, Y) be a two-dimensional random variable. Suppose that X and Y are dependent, i.e. there exists a functional relationship, say $Y = g(X)$, connecting X and Y . Using the principle of least squares, the function g can be determined. Thus, we seek the function $Y = g(X)$ which minimizes the “error” defined by

$$S = E[(Y - g(X))^2]$$

Let (X, Y) be a continuous random variable with joint PDF f . Then we have

$$E[(Y - g(X))^2] = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} [y - g(x)]^2 f(x, y) dx dy$$

Since $f(x, y) = f_X(x) f_{Y|X}(y|x)$, it follows that

$$E[(Y - g(X))^2] = \int_{x=-\infty}^{\infty} f_X(x) \left[\int_{y=-\infty}^{\infty} [y - g(x)]^2 f_{Y|X}(y|x) dy \right] dx$$

Hence, $E[(Y - g(X))^2]$ is minimized if

$$\int_{y=-\infty}^{\infty} [y - g(x)]^2 f_{Y|X}(y|x) dy \quad (4.11)$$

is minimized.

Note that the expression on Eq. (4.11) is equal to

$$E[(Y - g(X))^2 | X = x] = E[(Y - E(Y|X = x) + E(Y|X = x) - g(X))^2 | X = x]$$

which may be simplified as

$$E[(Y - g(X))^2 | X = x] = E[(Y - E(Y|X = x))^2 + [g(X) - E(Y|X = x)]^2 + 2E[(Y - E(Y|X = x)][E(Y|X = x) - g(X)] | X = x] \quad (4.12)$$

Note that

$$E[(Y - E(Y|X = x))^2] = \text{Var}(Y|X = x)$$

and $E[(Y - E(Y|X = x)][E(Y|X = x) - g(X)] | X = x]$ is equal to

$$[E(Y|X = x) - g(X)] E[(Y - E(Y|X = x)) | X = x] = 0$$

since

$$E[(Y - E(Y|X = x)) | X = x] = E(Y|X = x) - E(Y|X = x) = 0$$

Hence, Eq. (4.12) simplifies to

$$E[(Y - g(X))^2 | X = x] = \text{Var}(Y|X = x) + [g(X) - E(Y|X = x)]^2$$

from which follows that

$$E[(Y - g(X))^2 | X = x] \geq \text{Var}(Y|X = x)$$

and equality holds if and only if $g(X) = E(Y|X = x)$.

The discrete case is very similar, and the corresponding calculations using the principle of least squares is left as an exercise for the reader.

To summarize, the principle of least squares yields that the best approximation of Y by the dependency relationship $Y = g(X)$ is obtained when $g(X) = E(Y|X = x)$. Interchanging the roles of X and Y , it can be similarly shown that the best approximation of X by the dependency relationship $X = h(Y)$ is obtained when $h(Y) = E(X|Y = y)$.

Definition 4.20. The equation $y = E(Y|X = x)$ is called the *regression curve of Y on X* and the equation $x = E(X|Y = y)$ is called the *regression curve of X on Y* .

Remark 4.9. If X and Y are independent random variables, then the regression curve of Y on X is given by

$$y = E(Y|X = x) = E(Y) = \mu_Y$$

which is a straight line parallel to the x -axis, and the regression curve of X on Y is given by

$$x = E(X|Y = y) = E(X) = \mu_X$$

which is a straight line parallel to the y -axis. These two regression curves (straight lines) intersect orthogonally at the point $(x, y) = (\mu_X, \mu_Y)$.

The regression curves defined by

$$y = E(Y|X = x) \text{ and } x = E(X|Y = y)$$

are non-linear, in general. For practical applications, we are often interested in obtaining linear regression curves, i.e. we wish to approximate Y by a linear relationship of the form $g(X) = a + bX$, where the constants a and b can be determined by the principle of least squares. We seek the linear curve $g(X) = a + bX$ which minimizes the “error” defined by

$$S = E[Y - g(X)]^2 = E[Y - a - bX]^2$$

which may be simplified as

$$S = E[Y^2] + a^2 + b^2E[X^2] - 2bE(XY) - 2aE(Y) + 2abE(X)$$

By Calculus, a necessary condition for a and b to minimize S is given by

$$\frac{\partial S}{\partial a} = 0 \text{ and } \frac{\partial S}{\partial b} = 0$$

which are known as the *normal equations*.

Thus, the normal equations are

$$\begin{aligned} \frac{\partial S}{\partial a} &= 2a - 2E(Y) + 2bE(X) &= 0 \\ \frac{\partial S}{\partial b} &= 2bE(X^2) - 2E(XY) + 2aE(X) &= 0 \end{aligned}$$

which may be simplified and rewritten as

$$\begin{aligned} a + bE(X) &= E(Y) \\ aE(X) + bE(X^2) &= E(XY) \end{aligned} \tag{4.13}$$

Next, we solve the normal equations, Eq. (4.13), for a and b .

Eliminating a between the two equations, we obtain

$$b = \frac{E(XY) - E(X)E(Y)}{E(X^2) - [E(X)]^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

Substituting the value of b in the first equation in Eq. (4.13), we get

$$a = E(Y) - bE(X) = E(Y) - \frac{\text{Cov}(X, Y)}{\text{Var}(X)} E(X)$$

Hence, the best fitting linear curve in the least-squares sense is obtained as

$$y = E(Y) - \frac{\text{Cov}(X, Y)}{\text{Var}(X)} E(X) - \frac{\text{Cov}(X, Y)}{\text{Var}(X)} X$$

i.e.

$$y - E(Y) = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} [X - E(X)]$$

which is called the *regression line of Y on X*. Similarly, we can show that the *regression line of X on Y* is given by

$$x - E(X) = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)} [Y - E(Y)]$$

The above derivations lead to the following definition.

Definition 4.21. Let (X, Y) be a two-dimensional random variable. The *line of regression of Y on X* is defined as

$$y - E(Y) = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} [X - E(X)] \quad (4.14)$$

and the *line of regression of X on Y* is defined as

$$x - E(X) = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)} [Y - E(Y)] \quad (4.15)$$

Remark 4.10. The point $(E(X), E(Y))$ satisfies both Eqs. (4.14) and (4.15). Thus, it follows that the two regression lines intersect at the point $(E(X), E(Y))$ or (μ_X, μ_Y) .

Next, we define the *regression coefficients* for X and Y .

Definition 4.22. If $E(X^2)$ and $E(Y^2)$ exist, then the *regression coefficient of Y on X* is denoted by $\beta_{Y|X}$ and is defined as

$$\beta_{YX} = \frac{\text{Cov}(X, Y)}{\text{Var}(Y)}$$

and the *regression coefficient of X on Y* is denoted by $\beta_{X|Y}$ and is defined as

$$\beta_{XY} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

Remark 4.11. Using the definitions of β_{YX} and β_{XY} , we can simplify the equations of the two regression lines. The regression line of Y on X is simplified as

$$y - \mu_Y = \beta_{YX} [X - \mu_X]$$

and the regression line of X on Y is simplified as

$$x - \mu_X = \beta_{XY} [Y - \mu_Y]$$

Next, we define the *correlation coefficient* between X and Y .

Definition 4.23. Let (X, Y) be a two-dimensional random variable. If $E(X^2)$ and $E(Y^2)$ exist, then the correlation coefficient between X and Y is denoted by ρ or $\rho(X, Y)$, and is defined as

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E(XY) - \mu_X \mu_Y}{\sqrt{E(X^2) - \mu_X^2} \sqrt{E(Y^2) - \mu_Y^2}}$$

Remark 4.12. Using the definition of the correlation coefficient ρ between X and Y , the equation of the regression line of Y on X can be expressed as

$$y - \mu_Y = \frac{\rho \sigma_Y}{\sigma_X} [x - \mu_X]$$

and the equation of the regression line of X on Y can be expressed as

$$x - \mu_X = \frac{\rho \sigma_X}{\sigma_Y} [y - \mu_Y]$$

Remark 4.13. The following properties of ρ can be easily deduced.

- (i) $\beta_{YX} = \frac{\rho \sigma_Y}{\sigma_X}$ and $\beta_{XY} = \frac{\rho \sigma_X}{\sigma_Y}$.
- (ii) $\rho^2 = \beta_{YX} \beta_{XY}$.
- (iii) $\text{sign}(\rho) = \text{sign of } \text{Cov}(X, Y)$.

Recall that two random variables, X and Y , are said to be *uncorrelated* if

$$\text{Cov}(X, Y) = 0 \text{ or } \rho = 0$$

As proved earlier, if two random variables, X and Y , are independent, then

$$\text{Cov}(X, Y) = 0 \text{ or } \rho = 0$$

so that X and Y are uncorrelated. However, as shown earlier, the converse need not be true. That is, if $\rho = 0$, then X and Y need not be independent (see Example 4.85).

Using Cauchy-Schwarz inequality, next we will obtain the limits for the correlation coefficient, ρ .

Theorem 4.23. The correlation coefficient, ρ , is bounded by 1, i.e.

$$|\rho| \leq 1$$

(Anna, May 2007)

Proof. By Cauchy-Schwarz inequality (see Theorem 4.14), we know that

$$[E(UV)]^2 \leq E[U^2] E[V^2]$$

for any two random variables, U and V .

Taking $U = X - \mu_X$ and $V = Y - \mu_Y$, we have

$$[E[(X - \mu_X)(Y - \mu_Y)]]^2 \leq E[(X - \mu_X)^2] E[(Y - \mu_Y)^2]$$

i.e.

$$[\text{Cov}(X, Y)]^2 \leq \sigma_X^2 \sigma_Y^2$$

i.e.

$$\rho^2 = \frac{[\text{Cov}(X, Y)]^2}{\sigma_X^2 \sigma_Y^2} \leq 1$$

i.e.

$$|\rho| = \frac{|\text{Cov}(X, Y)|}{\sigma_X \sigma_Y} \leq 1$$

◆

Theorem 4.24. Correlation coefficient is independent of the change of origin and scale.

Proof. Let $U = \frac{X-a}{h}$ and $V = \frac{Y-b}{k}$, where a, b, h, k are any real constants with $h > 0$ and $k > 0$.

We assert that $\rho(X, Y) = \rho(U, V)$, i.e. the correlation coefficient, ρ , is independent of the change of origin and scale.

First, we note that

$$X = a + hU \quad \text{and} \quad Y = b + kV$$

Then, it follows that

$$\mu_X = a + h\mu_U \quad \text{and} \quad \mu_Y = b + k\mu_V$$

Hence, we have

$$X - \mu_X = h(U - \mu_U) \quad \text{and} \quad Y - \mu_Y = k(V - \mu_V)$$

We find that

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = hkE[(U - \mu_U)(V - \mu_V)] = hk\text{Cov}(U, V)$$

Next, we find that

$$\sigma_X^2 = E[(X - \mu_X)^2] = h^2E[(U - \mu_U)^2] = h^2\sigma_U^2$$

and

$$\sigma_Y^2 = E[(Y - \mu_Y)^2] = k^2E[(V - \mu_V)^2] = k^2\sigma_V^2$$

It follows that

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{hk\text{Cov}(U, V)}{h\sigma_U k\sigma_V} = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \rho(U, V)$$

◆

Theorem 4.25. The angle between the two regression lines is

$$\theta = \tan^{-1} \left[\frac{1 - \rho^2}{\rho} \frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \right]$$

where ρ is the coefficient of correlation between X and Y .

Proof. The line of regression of Y on X is given by

$$y - \mu_Y = \frac{\rho \sigma_Y}{\sigma_X} [x - \mu_X]$$

Hence, it has the slope, $m_1 = \frac{\rho \sigma_Y}{\sigma_X}$.

The line of regression of X on Y is given by

$$x - \mu_X = \frac{\rho \sigma_X}{\sigma_Y} [y - \mu_Y]$$

Hence, it has the slope, $m_2 = \frac{1}{\rho} \frac{\sigma_Y}{\sigma_X}$.

Thus, the acute angle θ between the two lines of regression is given by

$$\begin{aligned} \tan \theta &= \frac{m_1 \sim m_2}{1 + m_1 m_2} = \frac{\frac{\rho \sigma_Y}{\sigma_X} \sim \frac{1}{\rho} \frac{\sigma_Y}{\sigma_X}}{1 + \frac{\rho \sigma_Y}{\sigma_X} \frac{1}{\rho} \frac{\sigma_Y}{\sigma_X}} \\ &= \frac{\rho \sim \frac{1}{\rho} \sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \end{aligned}$$

Since $|\rho| \leq 1$, it follows that

$$\tan \theta = \frac{1 - \rho^2}{\rho} \frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2}$$

or

$$\theta = \tan^{-1} \left[\frac{1 - \rho^2}{\rho} \frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2} \right] \quad \blacklozenge$$

Corollary 4.1. When X and Y are uncorrelated, i.e. $\rho = 0$, the regression lines are perpendicular to each other.

Proof. When $\rho = 0$, $\theta = \tan^{-1}(\infty) = \frac{\pi}{2}$. $\blacklozenge$

Corollary 4.2. When X and Y are perfectly correlated, i.e. $\rho = \pm 1$, the regression lines coincide.

Proof. When $|\rho| = 1$, $\theta = \tan^{-1}(0) = 0$. $\blacklozenge$

EXAMPLE 4.98. The tangent of the angle between the lines of regression of Y on X and X on Y is 0.6 and $\sigma_X = \frac{1}{2} \sigma_Y$. Find the correlation coefficient.

(Anna, April 2005)

Solution. By Theorem 4.25, the angle θ between the two lines of regression is given by the formula

$$\tan \theta = \frac{1 - \rho^2}{\rho} \frac{\sigma_X \sigma_Y}{\sigma_X^2 + \sigma_Y^2}$$

where ρ is the correlation coefficient between X and Y , and σ_X, σ_Y are the standard deviations of X and Y , respectively.

Given that $\tan \theta = 0.6$ and $\sigma_Y = 2\sigma_X$. Substituting, we get

$$0.6 = \frac{1 - \rho^2}{\rho} \frac{\sigma_X (2\sigma_X)}{\sigma_X^2 + 4\sigma_X^2}$$

Simplifying, we get

$$\frac{1 - \rho^2}{\rho} \frac{2}{5} = 0.6 \Rightarrow \frac{1 - \rho^2}{\rho} = \frac{0.6}{0.4} = 1.5$$

Cross-multiplying and rearranging the terms, we get the quadratic equation

$$\rho^2 + 1.5\rho - 1 = 0$$

which has the two roots $\rho = -2, \rho = 0.5$.

The root $\rho = -2$ is inadmissible since $|\rho| \leq 1$. Hence, $\rho = 0.5$. ◆

EXAMPLE 4.99. Let X and Y be jointly distributed with the correlation coefficient $\rho(X, Y) = \frac{1}{2}$, $\sigma_X = 2$, $\sigma_Y = 3$. Find $\text{Var}(2X - 4Y + 3)$. (Anna, April 2004)

Solution. By definition, we have

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Substituting the given data, it follows that

$$\frac{1}{2} = \frac{\text{Cov}(X, Y)}{2 \times 3}$$

This shows that

$$\text{Cov}(X, Y) = 3$$

By Theorem 4.18, it follows that

$$\text{Var}(2X - 4Y + 3) = 4\text{Var}(X) + 16\text{Var}(Y) - 8\text{Cov}(X, Y)$$

Substituting the values, we have

$$\text{Var}(2X - 4Y + 3) = 4(4) + 16(9) - 8(3) = 16 + 144 - 24 = 136$$
◆

EXAMPLE 4.100. Show that the coefficient of correlation ρ between the random variables, X and Y is given by

$$\rho = \frac{\sigma_X^2 + \sigma_Y^2 - \sigma_{X-Y}^2}{2\sigma_X \sigma_Y}$$

Solution. We find that

$$\sigma_{X-Y}^2 = \text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$$

Hence, it follows that

$$\text{Cov}(X, Y) = \frac{1}{2} [\sigma_X^2 + \sigma_Y^2 - \sigma_{X-Y}^2]$$

By definition, we have

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Substituting for $\text{Cov}(X, Y)$, it follows that

$$\rho = \frac{\sigma_X^2 + \sigma_Y^2 - \sigma_{X-Y}^2}{2\sigma_X \sigma_Y}$$
◆

EXAMPLE 4.101. Let X_1 and X_2 be two independent random variables with means 5 and 10 and standard deviations 2 and 3, respectively. Obtain the correlation coefficient between $U = 3X_1 + 4X_2$ and $V = 3X_1 - X_2$.

(Anna, Nov. 2007)

Solution. Let $\mu_1 = E(X_1)$, $\mu_2 = E(X_2)$ and $\sigma_1^2 = \text{Var}(X_1)$ and $\sigma_2^2 = \text{Var}(X_2)$. Given that $\mu_1 = 5$, $\mu_2 = 10$, $\sigma_1 = 2$ and $\sigma_2 = 3$. Since X_1 and X_2 are independent, it follows that $\text{Cov}(X_1, X_2) = 0$.

Note that

$$U - \mu_U = 3(X_1 - \mu_1) + 4(X_2 - \mu_2) \quad \text{and} \quad V - \mu_V = 3(X_1 - \mu_1) - (X_2 - \mu_2)$$

Hence, it is immediate that

$$\text{Var}(U) = 9\text{Var}(X_1) + 4\text{Var}(X_2) + 24\text{Cov}(X_1, X_2)$$

and

$$\text{Var}(V) = 9\text{Var}(X_1) + \text{Var}(X_2) - 6\text{Cov}(X_1, X_2)$$

Substituting the given values,

$$\text{Var}(U) = \sigma_U^2 = 180 \quad \text{and} \quad \text{Var}(V) = \sigma_V^2 = 45$$

Next, we also note that

$$\text{Cov}(U, V) = E[(U - \mu_U)(V - \mu_V)] = 9\text{Var}(X_1) - 4\text{Var}(X_2) + 9\text{Cov}(X_1, X_2)$$

Substituting the given values,

$$\text{Cov}(U, V) = 36 - 4 + 0 = 32$$

Thus, the correlation coefficient between U and V is calculated as

$$\rho(U, V) = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{32}{\sqrt{180}\sqrt{45}} = \frac{32}{90} = \frac{16}{45} = 0.3556$$

◆

EXAMPLE 4.102. The joint probability mass function of X and Y is given below:

X \ Y	Y	
	-1	+1
0	$\frac{1}{8}$	$\frac{3}{8}$
1	$\frac{2}{8}$	$\frac{2}{8}$

Find the correlation coefficient of (X, Y) .

(Anna, Nov. 2004)

Solution. First, we find the marginal density function of X .

By definition, we have

$$f_X(x) = P(X = x) = \sum_y f(x, y) = \sum_y P(X = x, Y = y)$$

Thus, we have

x	0	1
$f_X(x)$	$\frac{1}{2}$	$\frac{1}{2}$

Next, we find the marginal density function of Y .

By definition, we have

$$f_Y(y) = P(Y = y) = \sum_x f(x, y) = \sum_x P(X = x, Y = y)$$

Thus, we have

y	-1	1
$f_Y(y)$	$\frac{3}{8}$	$\frac{5}{8}$

Next, we find the standard deviations of X and Y .

First, we note that

$$E(X) = \sum_x x f_X(x) = 0 + \frac{1}{2} = \frac{1}{2}$$

and

$$E(X^2) = \sum_x x^2 f_X(x) = 0 + \frac{1}{2} = \frac{1}{2}$$

Thus, the variance of X is given by

$$\sigma_X^2 = E(X^2) - [E(X)]^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

and, the standard deviation of X is given by

$$\sigma_X = \frac{1}{2}$$

Next, we note that

$$E(Y) = \sum_y y f_Y(y) = -\frac{3}{8} + \frac{5}{8} = \frac{2}{8} = \frac{1}{4}$$

and

$$E(Y^2) = \sum_y y^2 f_Y(y) = \frac{3}{8} + \frac{5}{8} = 1$$

Hence, the variance of Y is given by

$$\sigma_Y^2 = E(Y^2) - [E(Y)]^2 = 1 - \frac{1}{16} = \frac{15}{16}$$

and, the standard deviation of Y is given by

$$\sigma_Y = \frac{\sqrt{15}}{4} = 0.9682$$

We also note that

$$E(XY) = \sum_x \sum_y xyf(x,y) = 0 + 0 - \frac{2}{8} + \frac{2}{8} = 0$$

Hence, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0 - \frac{1}{2} \times \frac{1}{4} = -\frac{1}{8}$$

and the correlation coefficient between X and Y is given by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = -\frac{0.1250}{0.5 \times 0.9682} = -0.2582$$

◆

EXAMPLE 4.103. If the equations of the two lines of regression of Y on X and X on Y are, respectively, $7x - 16y + 9 = 0$ and $5y - 4x - 3 = 0$, calculate the coefficient of correlation, $E(X)$ and $E(Y)$.

(Anna, May 1999)

Solution. Given that the regression line of Y on X is

$$7x - 16y = -9$$

and the regression line of X on Y is

$$4x - 5y = -3$$

We know that the two regression lines intersect at (μ_X, μ_Y) . Hence, we have

$$\begin{aligned} 7\mu_X - 16\mu_Y &= -9 \\ 4\mu_X - 5\mu_Y &= -3 \end{aligned}$$

Note that

$$\Delta = \begin{vmatrix} 7 & -16 \\ 4 & -5 \end{vmatrix} = -35 + 64 = 29$$

$$\Delta_1 = \begin{vmatrix} -9 & -16 \\ -3 & -5 \end{vmatrix} = 45 - 48 = -3$$

and

$$\Delta_2 = \begin{vmatrix} 7 & -9 \\ 4 & -3 \end{vmatrix} = -21 + 36 = 15$$

By Cramer's rule, it follows that

$$\mu_X = \frac{\Delta_1}{\Delta} = -\frac{3}{29} \quad \text{and} \quad \mu_Y = \frac{\Delta_2}{\Delta} = \frac{15}{29}$$

The line of regression of Y on X can be rewritten as

$$y = \frac{7}{16}x - \frac{9}{16}$$

from which it follows that $\beta_{YX} = \frac{\rho\sigma_Y}{\sigma_X} = \frac{7}{16}$.

The line of regression of X on Y can be rewritten as

$$x = \frac{5}{4}y - \frac{3}{4}$$

from which it follows that $\beta_{XY} = \frac{\rho\sigma_X}{\sigma_Y} = \frac{5}{4}$.

Hence, it follows that

$$\rho = \pm \sqrt{\beta_{YX} \beta_{XY}} = \pm \sqrt{\frac{7}{16} \times \frac{5}{4}} = \pm \frac{\sqrt{35}}{8} = \pm 0.7395$$

Since ρ has the same sign as that of β_{YX} and β_{XY} , it follows that

$$\rho = 0.7395$$

◆

Remark 4.14. In some numerical problems, we may be given two regression lines but without a specification on which is the regression line of Y on X and which is the regression line of X on Y . This can be easily resolved as follows. Take any regression line as the regression line of Y on X and change it into the standard form

$$y = \beta_{YX}x + c$$

Take the other regression line as the regression line of X on Y and change it into the standard form

$$x = \beta_{XY}y + c'$$

We know that the product of the regression coefficients β_{YX} and β_{XY} equals the square of the correlation coefficient, ρ , i.e.

$$\beta_{YX}\beta_{XY} = \rho^2 \leq 1$$

If the product $\beta_{YX}\beta_{XY} \leq 1$, then our assumption about the regression lines is correct. If, on the other hand, $\beta_{YX}\beta_{XY} > 1$, then our assumption about the regression lines is wrong. This means that the regression line which we assumed as the regression line of Y on X should be taken as the regression line of X on Y and vice versa.

EXAMPLE 4.104. The regression lines between the two random variables, X and Y , are given by

$$3X + Y = 10 \text{ and } 3X + 4Y = 12$$

Find the coefficient of correlation between X and Y .

(Anna, April 2004)

Solution. Using the property that $\beta_{YX}\beta_{XY} = \rho^2 \leq 1$, we can easily determine the regression line

Thus, the regression line of Y on X is given by

$$3x + 4y = 12 \text{ or } y = -\frac{3}{4}x + 3$$

from which it follows that $\beta_{YX} = -\frac{3}{4}$.

Also, the regression line of X on Y is given by

$$3x + y = 10 \text{ or } x = -\frac{1}{3}y + \frac{10}{3}$$

from which it follows that $\beta_{XY} = -\frac{1}{3}$.

Since $\rho^2 = \beta_{YX}\beta_{XY}$, we have

$$\rho^2 = \left[-\frac{3}{4}\right] \left[-\frac{1}{3}\right] = \frac{1}{4}$$

Hence, $\rho = \pm \frac{1}{2}$. Since ρ has the same sign as that of β_{YX} and β_{XY} , it follows that the correlation coefficient ρ is given by

$$\rho = -\frac{1}{2}$$

◆

EXAMPLE 4.105. If $y = 2x - 3$ and $y = 5x + 7$ are the two regression lines, find (i) the mean values of X and Y , (ii) the correlation coefficient between X and Y , and (iii) find an estimate of X when $Y = 1$.

(Anna, May 2006)

Solution. (i) First, we find the mean values of X and Y . Since the two regression lines intersect at (μ_X, μ_Y) , it follows that

$$\mu_Y = 2\mu_X - 3$$

$$\mu_Y = 5\mu_X + 7$$

Solving the two equations, we get

$$2\mu_X - 3 = 5\mu_X + 7 \Rightarrow 3\mu_X = -10$$

$$\mu_X = -\frac{10}{3}$$

Hence,

$$\mu_Y = 2\left(-\frac{10}{3}\right) - 3 = -\frac{29}{3}$$

Thus, the mean values of X and Y are given by

$$\mu_X = -\frac{10}{3} \text{ and } \mu_Y = -\frac{29}{3}$$

(ii) Using the property that $\beta_{YX}\beta_{XY} \leq 1$, we can easily determine which regression line is on which. Thus, the regression line of Y on X is given by

$$y = 2x - 3$$

from which we have $\beta_{YX} = 2$.

The regression line of X on Y is given by

$$y = 5x + 7 \text{ or } x = \frac{1}{5}y - \frac{7}{5}$$

from which we have $\beta_{XY} = \frac{1}{5}$.

Thus, it follows that

$$\rho^2 = \beta_{YX}\beta_{XY} = 2 \frac{1}{5} = 0.4$$

$$d\rho = \pm\sqrt{0.4} = \pm 0.6325$$

Since ρ has the same sign as β_{YX} and β_{XY} , it is immediate that

$$\rho = \sqrt{0.4} = 0.6325$$

(iii) Finally, we find an estimate of X when $Y = 1$. The regression line of X on Y is

$$y = 5x + 7 \quad \text{or} \quad x = \frac{y}{5} - \frac{7}{5}$$

When $Y = 1$, we obtain

$$X = \frac{1}{5} - \frac{7}{5} = -\frac{6}{5} = -1.2$$

◆

EXAMPLE 4.106. The regression equations of two random variables, X and Y , are

$$x - \frac{5}{4}y + \frac{33}{5} = 0 \quad \text{and} \quad y - \frac{20}{9}x + \frac{107}{9} = 0$$

The standard deviation of X is 3. Find the standard deviation of Y .

(Anna, May 2000)

Solution. Using the property that $\beta_{YX}\beta_{XY} = \rho^2 \leq 1$, it is easy to see that the regression line of Y on X is given by

$$x - \frac{5}{4}y + \frac{33}{5} = 0 \quad \text{or} \quad y = 0.8x + 5.28$$

from which it follows that $\beta_{YX} = 0.8$

Also, the regression line of X on Y is given by

$$y - \frac{20}{9}x + \frac{107}{9} = 0 \quad \text{or} \quad x = 0.45y + 5.35$$

from which it follows that $\beta_{XY} = 0.45$.

Hence, $\rho^2 = \beta_{YX}\beta_{XY} = 0.8 \times 0.45 = 0.36$.

So, $\rho = \pm 0.6$. Since ρ has the same sign as that of β_{YX} and β_{XY} , it follows that $\rho = 0.6$.

Next, we note that

$$\beta_{YX} = \frac{\rho\sigma_Y}{\sigma_X} = 0.8$$

from which it follows that

$$\frac{\sigma_Y}{\sigma_X} = \frac{0.8}{\rho} = \frac{0.8}{0.6} = \frac{4}{3}$$

Since it is given that $\sigma_X = 3$,

$$\sigma_Y = \frac{4}{3}\sigma_X = \frac{4}{3} \times 3 = 4$$

◆

EXAMPLE 4.107. X and Y are two random variables with variances σ_X^2 and σ_Y^2 , respectively and r is the coefficient of correlation between them. If $U = X + kY$ and $V = X + \frac{\sigma_X}{\sigma_Y} Y$, find the value of k so that U and V are uncorrelated.

(Anna, Nov. 2004)

Solution. Since r is the correlation coefficient between X and Y , we know that

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

from which we have

$$\text{Cov}(X, Y) = r \sigma_X \sigma_Y$$

The given random variables are

$$U = X + kY \quad \text{and} \quad V = X + \frac{\sigma_X}{\sigma_Y} Y$$

Then, we have

$$\mu_U = \mu_X + k\mu_Y \quad \text{and} \quad \mu_V = \mu_X + \frac{\sigma_X}{\sigma_Y} \mu_Y$$

Hence,

$$U - \mu_U = (X - \mu_X) + k(Y - \mu_Y) \quad \text{and} \quad V - \mu_V = (X - \mu_X) + \frac{\sigma_X}{\sigma_Y} (Y - \mu_Y)$$

By definition,

$$\begin{aligned} \text{Cov}(U, V) &= E[(U - \mu_U)(V - \mu_V)] \\ &= E\left\{[(X - \mu_X) + k(Y - \mu_Y)] \left[(X - \mu_X) + \frac{\sigma_X}{\sigma_Y} (Y - \mu_Y)\right]\right\} \end{aligned}$$

Simplifying, we have

$$\text{Cov}(U, V) = \sigma_X^2 + k \frac{\sigma_X}{\sigma_Y} \sigma_Y^2 + \left[k + \frac{\sigma_X}{\sigma_Y}\right] \text{Cov}(X, Y)$$

Since $\text{Cov}(X, Y) = r \sigma_X \sigma_Y$, it follows that

$$\text{Cov}(U, V) = \sigma_X^2 + k \sigma_X \sigma_Y + \left[k + \frac{\sigma_X}{\sigma_Y}\right] r \sigma_X \sigma_Y = \sigma_X^2(1+r) + k \sigma_X \sigma_Y(1+r)$$

The random variables, U and V , are uncorrelated if and only if $\text{Cov}(U, V) = 0$, i.e. we must have

$$\sigma_X^2(1+r) + k \sigma_X \sigma_Y(1+r) = 0$$

or

$$\sigma_X + k \sigma_Y = 0$$

or

$$k = -\frac{\sigma_X}{\sigma_Y}$$

◆

EXAMPLE 4.108. If X, Y are standardized random variables, and coefficient of correlation between $aX + bY$, $bX + aY$ is $\frac{1+2ab}{a^2+b^2}$, then find the coefficient of correlation between X and Y .

(Anna, Nov. 2006)

Solution. Since X and Y are standardized random variables, it follows that

$$E(X) = E(Y) = 0 \quad \text{and} \quad \text{Var}(X) = \text{Var}(Y) = 1$$

Thus,

$$E(X^2) = E(Y^2) = 1 \quad \text{and} \quad \text{Cov}(X, Y) = E(XY)$$

Now, define the random variables

$$U = aX + bY \quad \text{and} \quad V = bX + aY$$

We are given that the correlation coefficient between U and V is

$$\rho(U, V) = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{1 + 2ab}{a^2 + b^2} \quad (4.16)$$

Note that

$$\mu_U = a\mu_X + b\mu_Y = 0 \quad \text{and} \quad \mu_V = b\mu_X + a\mu_Y = 0$$

Hence, it follows that

$$\begin{aligned} \text{Cov}(U, V) &= E[(U - \mu_U)(V - \mu_V)] = E[UV] = E[(aX + bY)(bX + aY)] \\ &= ab[E(X^2) + E(Y^2)] + (a^2 + b^2)E(XY) \\ &= 2ab + (a^2 + b^2)\text{Cov}(X, Y) = 2ab + (a^2 + b^2)\rho(X, Y) \end{aligned}$$

(Note that $\rho(X, Y) = \text{Cov}(X, Y)$ because $\sigma_X = \sigma_Y = 1$.)

We also find that

$$\begin{aligned} \sigma_U^2 &= E[(U - \mu_U)^2] = E[U^2] = E[(aX + bY)^2] \\ &= a^2E(X^2) + b^2E(Y^2) + 2ab\text{Cov}(X, Y) = a^2 + b^2 + 2ab\rho(X, Y) \end{aligned}$$

Hence, we have

$$\sigma_U = \sqrt{a^2 + b^2 + 2ab\rho(X, Y)}$$

Next, we find that

$$\begin{aligned} \sigma_V^2 &= E[(V - \mu_V)^2] = E[V^2] = E[(bX + aY)^2] \\ &= b^2E(X^2) + a^2E(Y^2) + 2ab\text{Cov}(X, Y) = a^2 + b^2 + 2ab\rho(X, Y) \end{aligned}$$

Hence, we have

$$\sigma_V = \sqrt{a^2 + b^2 + 2ab\rho(X, Y)}$$

Substituting the values of $\text{Cov}(U, V)$, σ_U and σ_V in Eq. (4.16), we have

$$\frac{2ab + (a^2 + b^2)\rho(X, Y)}{a^2 + b^2 + 2ab\rho(X, Y)} = \frac{1 + 2ab}{a^2 + b^2}$$

Cross-multiplying, we have

$$(a^2 + b^2)^2 \rho(X, Y) + 2ab(a^2 + b^2) = (1 + 2ab) [a^2 + b^2 + 2ab\rho(X, Y)]$$

Simplifying,

$$[(a^2 + b^2)^2 - 2ab(1 + 2ab)] \rho(X, Y) = a^2 + b^2$$

Using the fact that $(a^2 + b^2)^2 - 4a^2b^2 = (a^2 - b^2)^2$, it follows that

$$\rho(X, Y) = \frac{a^2 + b^2}{(a^2 - b^2)^2 - 2ab}$$

◆

EXAMPLE 4.109. If X and Y are independent random variables with means zero, and variances σ_1^2 and σ_2^2 , respectively, and U and V are defined by

$$U = X \cos \theta + Y \sin \theta \quad \text{and} \quad V = Y \cos \theta - X \sin \theta$$

show that the coefficient of correlation ρ between U and V is given by

$$\rho = \frac{(\sigma_2^2 - \sigma_1^2) \sin 2\theta}{\sqrt{(\sigma_2^2 - \sigma_1^2) \sin^2 2\theta + 4\sigma_1^2 \sigma_2^2}}$$

Solution. Given that $E(X) = E(Y) = 0$ and $\text{Var}(X) = \sigma_1^2$, $\text{Var}(Y) = \sigma_2^2$.

Thus, it follows that $E(X^2) = \sigma_1^2$ and $E(Y^2) = \sigma_2^2$.

Since X and Y are independent, $\text{Cov}(X, Y) = 0$.

Since X and Y have zero means, $E(XY) = 0$.

Since $U = X \cos \theta + Y \sin \theta$, it follows that

$$E(U) = E(X) \cos \theta + E(Y) \sin \theta = 0$$

Since $V = Y \cos \theta - X \sin \theta$, it follows that

$$E(V) = E(Y) \cos \theta - E(X) \sin \theta = 0$$

Hence,

$$\begin{aligned} \text{Cov}(U, V) &= E(UV) = E[(X \cos \theta + Y \sin \theta)(Y \cos \theta - X \sin \theta)] \\ &= -\sin \theta \cos \theta E(X^2) + \sin \theta \cos \theta E(Y^2) + (\cos^2 \theta - \sin^2 \theta) E(XY) \end{aligned}$$

Since $E(X^2) = \sigma_1^2$, $E(Y^2) = \sigma_2^2$ and $E(XY) = 0$, we have

$$\text{Cov}(U, V) = (\sigma_2^2 - \sigma_1^2) \sin \theta \cos \theta = \frac{1}{2} (\sigma_2^2 - \sigma_1^2) \sin 2\theta$$

Next, we find that

$$\begin{aligned}\sigma_U^2 &= E(U^2) = E[(X \cos \theta + Y \sin \theta)^2] \\ \sigma_U^2 &= \cos^2 \theta E(X^2) + \sin^2 \theta E(Y^2) + 2 \sin \theta \cos \theta E(XY) \\ &= \sigma_1^2 \cos^2 \theta + \sigma_2^2 \sin^2 \theta\end{aligned}$$

We also find that

$$\begin{aligned}\sigma_V^2 &= E(V^2) = E[(Y \cos \theta - X \sin \theta)^2] \\ &= \cos^2 \theta E(Y^2) + \sin^2 \theta E(X^2) - 2 \sin \theta \cos \theta E(XY) \\ &= \sigma_2^2 \cos^2 \theta + \sigma_1^2 \sin^2 \theta\end{aligned}$$

Thus, we find that

$$\begin{aligned}\sigma_U^2 \sigma_V^2 &= (\sigma_1^2 \cos^2 \theta + \sigma_2^2 \sin^2 \theta) (\sigma_2^2 \cos^2 \theta + \sigma_1^2 \sin^2 \theta) \\ &= (\sigma_1^4 + \sigma_2^4) \sin^2 \theta \cos^2 \theta + \sigma_1^2 \sigma_2^2 (\cos^4 \theta + \sin^4 \theta)\end{aligned}$$

Note that

$$\cos^4 \theta + \sin^4 \theta = (\cos^2 \theta + \sin^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$$

Thus, we have

$$\begin{aligned}\sigma_U^2 \sigma_V^2 &= (\sigma_1^4 + \sigma_2^4) \sin^2 \theta \cos^2 \theta + \sigma_1^2 \sigma_2^2 [1 - 2 \sin^2 \theta \cos^2 \theta] \\ &= (\sigma_1^4 - 2 \sigma_1^2 \sigma_2^2 + \sigma_2^4) \sin^2 \theta \cos^2 \theta + \sigma_1^2 \sigma_2^2 = (\sigma_1^2 - \sigma_2^2)^2 \sin^2 \theta \cos^2 \theta + \sigma_1^2 \sigma_2^2\end{aligned}$$

Since $\sin 2\theta = 2 \sin \theta \cos \theta$, it follows that

$$\sigma_U^2 \sigma_V^2 = \frac{1}{4} (\sigma_1^2 - \sigma_2^2)^2 \sin^2 2\theta + \sigma_1^2 \sigma_2^2.$$

Hence,

$$\sigma_U \sigma_V = \frac{1}{2} \sqrt{(\sigma_1^2 - \sigma_2^2)^2 \sin^2 2\theta + 4 \sigma_1^2 \sigma_2^2}$$

Thus, the correlation coefficient between U and V is given by

$$\rho = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{1/2 (\sigma_2^2 - \sigma_1^2) \sin 2\theta}{1/2 \sqrt{(\sigma_1^2 - \sigma_2^2)^2 \sin^2 2\theta + 4 \sigma_1^2 \sigma_2^2}}$$

Simplifying, we have

$$\rho = \frac{(\sigma_2^2 - \sigma_1^2) \sin 2\theta}{\sqrt{(\sigma_1^2 - \sigma_2^2)^2 \sin^2 2\theta + 4 \sigma_1^2 \sigma_2^2}}$$

◆

EXAMPLE 4.110. Let X and Y be a pair of correlated random variables. Let U and V be obtained by rotating X and Y about the axes clockwise through an angle α , i.e.

$$U = X \cos \alpha + Y \sin \alpha$$

$$V = X \sin \alpha - Y \cos \alpha$$

Show that U and V will be uncorrelated if α is given by

$$\tan 2\alpha = \frac{2\rho\sigma_X\sigma_Y}{\sigma_X^2 - \sigma_Y^2}$$

where ρ is the coefficient of correlation between X and Y .

Solution. Note that

$$\mu_U = E(U) = \mu_X \cos \alpha + \mu_Y \sin \alpha$$

and

$$\mu_V = E(V) = \mu_X \sin \alpha - \mu_Y \cos \alpha$$

Hence, it follows that

$$U - \mu_U = (X - \mu_X) \cos \alpha + (Y - \mu_Y) \sin \alpha$$

and

$$V - \mu_V = (X - \mu_X) \sin \alpha - (Y - \mu_Y) \cos \alpha$$

Thus, the covariance between X and Y is obtained as

$$\begin{aligned} \text{Cov}(U, V) &= E[(U - \mu_U)(V - \mu_V)] \\ &= E[(X - \mu_X) \cos \alpha + (Y - \mu_Y) \sin \alpha][(X - \mu_X) \sin \alpha - (Y - \mu_Y) \cos \alpha] \end{aligned}$$

Simplifying, we have

$$\begin{aligned} \text{Cov}(U, V) &= \sin \alpha \cos \alpha [E[(X - \mu_X)^2] - E[(Y - \mu_Y)^2]] \\ &\quad - (\cos^2 \alpha - \sin^2 \alpha) E[(X - \mu_X)(Y - \mu_Y)] \\ &= \frac{1}{2} \sin 2\alpha [\sigma_X^2 - \sigma_Y^2] - \cos 2\alpha \text{Cov}(X, Y) \end{aligned}$$

Since the correlation coefficient between X and Y is given by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

it follows that $\text{Cov}(X, Y) = \rho \sigma_X \sigma_Y$. Hence,

$$\text{Cov}(U, V) = \frac{1}{2} \sin 2\alpha [\sigma_X^2 - \sigma_Y^2] - \cos 2\alpha \rho \sigma_X \sigma_Y.$$

The random variables U and V will be uncorrelated if and only if $\text{Cov}(U, V) = 0$ or

$$\frac{1}{2} \sin 2\alpha [\sigma_X^2 - \sigma_Y^2] - \cos 2\alpha \rho \sigma_X \sigma_Y = 0$$

or

$$\sin 2\alpha [\sigma_X^2 - \sigma_Y^2] = 2 \cos 2\alpha \rho \sigma_X \sigma_Y$$

or

$$\tan 2\alpha = \frac{2\rho \sigma_X \sigma_Y}{\sigma_X^2 - \sigma_Y^2}$$

◆

EXAMPLE 4.111. If X and Y are two correlated random variables with the same variance and if r is the correlation coefficient between X and Y , show that the correlation coefficient between X and $X + Y$ is $\sqrt{\frac{1+r}{2}}$.

Solution. Given that $\sigma_X^2 = \sigma_Y^2 = \sigma^2$ and $r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sigma^2}$.

We find that

$$\text{Cov}(X, X + Y) = E[(X - \mu_X)((X - \mu_X) + (Y - \mu_Y))] = \sigma^2 + \text{Cov}(X, Y)$$

Since $\text{Cov}(X, Y) = r\sigma^2$, it follows that

$$\text{Cov}(X, X + Y) = \sigma^2 + r\sigma^2 = \sigma^2(1 + r)$$

Next, we note that

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

which implies that

$$\sigma_{X+Y}^2 = \sigma^2 + \sigma^2 + 2r\sigma^2 = 2\sigma^2(1 + r)$$

Hence, the correlation coefficient between X and $X + Y$ is given by

$$\begin{aligned} r(X, X + Y) &= \frac{\text{Cov}(X, X + Y)}{\sigma_X \sigma_{X+Y}} = \frac{\sigma^2(1 + r)}{\sigma \sqrt{2}\sigma \sqrt{1 + r}} \\ &= \sqrt{\frac{1 + r}{2}} \end{aligned}$$

◆

EXAMPLE 4.112. Let $U = aX + bY, V = cX + dY$, where X and Y have zero means and r is the correlation coefficient between X and Y . If U and V are uncorrelated, show that

$$\sigma_U \sigma_V = (ad - bc) \sigma_X \sigma_Y \sqrt{1 - r^2}$$

Solution. Since $\mu_X = \mu_Y = 0$, it is easy to see that $\mu_U = 0$ and $\mu_V = 0$.

Also, since U and V are uncorrelated, it follows that

$$\text{Cov}(U, V) = E(UV) - \mu_U \mu_V = E(UV) = 0$$

Note that

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Thus, X and Y are obtained as

$$X = \frac{1}{ad-bc} [dU - bV] \text{ and } Y = \frac{1}{ad-bc} [-cU + aV]$$

Thus, it is immediate that

$$\sigma_X^2 = \frac{1}{(ad-bc)^2} [d^2 \sigma_U^2 + b^2 \sigma_V^2]$$

Similarly, it follows that

$$\sigma_Y^2 = \frac{1}{(ad-bc)^2} [c^2 \sigma_U^2 + a^2 \sigma_V^2]$$

Now, we have

$$\begin{aligned} \text{Cov}(X, Y) &= E(XY) - \mu_X \mu_Y = E(XY) = \frac{1}{(ad-bc)^2} E[(dU - bV)(-cU + aV)] \\ &= \frac{1}{(ad-bc)^2} [-cd \sigma_U^2 - ab \sigma_V^2] \end{aligned}$$

Next, we find that

$$(1-r^2) \sigma_X^2 \sigma_Y^2 = \sigma_X^2 \sigma_Y^2 - r^2 \sigma_X^2 \sigma_Y^2 = \sigma_X^2 \sigma_Y^2 - [\text{Cov}(X, Y)]^2$$

Thus, it follows that

$$\begin{aligned} (1-r^2) \sigma_X^2 \sigma_Y^2 &= \frac{1}{(ad-bc)^4} [(d^2 \sigma_U^2 + b^2 \sigma_V^2) (c^2 \sigma_U^2 + a^2 \sigma_V^2) - (cd \sigma_U^2 + ab \sigma_V^2)^2] \\ &= \frac{1}{(ad-bc)^4} [a^2 d^2 + b^2 c^2 - 2abcd] \sigma_U^2 \sigma_V^2 \\ &= \frac{(ad-bc)^2 \sigma_U^2 \sigma_V^2}{(ad-bc)^4} = \frac{\sigma_U^2 \sigma_V^2}{(ad-bc)^2} \end{aligned}$$

Hence, it is immediate that

$$\sigma_U^2 \sigma_V^2 = (ad-bc)^2 \sigma_X^2 \sigma_Y^2 (1-r^2)$$

◆

EXAMPLE 4.113. Let (X, Y) be a two-dimensional random variable with joint PDF

$$f(x, y) = \begin{cases} 2, & 0 < x < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the correlation coefficient between X and Y .

(Anna, Model 2003)

Solution. It is easy to see that the marginal density function of X is given by

$$f_X(x) = \begin{cases} 2(1-x) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 2y & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

We find that

$$\mu'_1(X) = E(X) = \int_0^1 x f_X(x) dx = \int_0^1 x [2(1-x)] dx$$

Integrating, we have

$$\mu'_1(X) = 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 2 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{1}{3}$$

Next, we find that

$$\mu'_2(X) = E(X^2) = \int_0^1 x^2 f_X(x) dx = \int_0^1 x^2 [2(1-x)] dx$$

Integrating, we have

$$\mu'_2(X) = 2 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 2 \left[\frac{1}{3} - \frac{1}{4} \right] = \frac{1}{6}$$

Hence, the variance of X is given by

$$\text{Var}(X) = \sigma_X^2 = E(X^2) - [E(X)]^2 = \frac{1}{6} - \frac{1}{9} = \frac{1}{18}$$

We also note that

$$\mu'_1(Y) = E(Y) = \int_0^1 y f_Y(y) dy = \int_0^1 y [2y] dy$$

Integrating, we have

$$\mu'_1(Y) = 2 \left[\frac{y^3}{3} \right]_0^1 = \frac{2}{3}$$

Next, we find that

$$\mu'_2(Y) = E(Y^2) = \int_0^1 y^2 f_Y(y) dy = \int_0^1 y^2 [2y] dy$$

Integrating, we have

$$\mu'_2(Y) = 2 \left[\frac{y^4}{4} \right]_0^1 = \frac{1}{2}$$

Hence, the variance of Y is given by

$$\text{Var}(Y) = \sigma_Y^2 = E(Y^2) - [E(Y)]^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

We also find that

$$E(XY) = \int_{x=0}^1 \int_{y=x}^1 xy f(x,y) dy dx = \int_{x=0}^1 \int_{y=x}^1 xy [2] dy dx$$

Integrating y , we have

$$E(XY) = \int_{x=0}^1 x [y^2]_{y=x}^1 dx = \int_{x=0}^1 x [1 - x^2] dx$$

Integrating x , we have

$$E(XY) = \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

Thus, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{4} - \frac{1}{3} \times \frac{2}{3} = \frac{1}{36}$$

Hence, the correlation coefficient between X and Y is given by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{1/36}{1/\sqrt{18} \times 1/\sqrt{18}} = \frac{1}{36} \times 18 = \frac{1}{2}$$

◆

EXAMPLE 4.114. If X, Y and Z are uncorrelated random variables with zero, mean and standard deviation 5, 12 and 9, respectively, and $U = X + Y$ and $V = Y + Z$, find the correlation coefficient between U and V .

(Anna, Model 2003)

Solution. From the given data, we have

$$E(X) = E(Y) = E(Z) = 0, \sigma_X = 5, \sigma_Y = 12, \sigma_Z = 9$$

Since X, Y and Z are uncorrelated, it follows that

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = E(XY) = 0$$

$$\text{Cov}(Y, Z) = E(YZ) - E(Y)E(Z) = E(YZ) = 0$$

$$\text{Cov}(X, Z) = E(XZ) - E(X)E(Z) = E(XZ) = 0$$

Note that

$$E(U) = E(X + Y) = E(X) + E(Y) = 0 + 0 = 0$$

and

$$E(V) = E(Y + Z) = E(Y) + E(Z) = 0 + 0 = 0$$

Hence, it follows that

$$\begin{aligned}\text{Cov}(U, V) &= E(UV) - E(U)E(V) = E(UV) = E[(X+Y)(Y+Z)] \\ &= E(XY) + E(XZ) + E(Y^2) + E(YZ) = 0 + 0 + \sigma_Y^2 + 0 = 144\end{aligned}$$

Next, we find that

$$\begin{aligned}\sigma_U^2 &= E(U^2) = E[(X+Y)^2] = E(X^2 + 2XY + Y^2) \\ &= E(X^2) + 2E(XY) + E(Y^2) = \sigma_X^2 + 0 + \sigma_Y^2 = 25 + 0 + 144 = 169\end{aligned}$$

Hence, it follows that

$$\sigma_U = \sqrt{169} = 13$$

We also find that

$$\begin{aligned}\sigma_V^2 &= E(V^2) = E[(Y+Z)^2] = E(Y^2 + 2YZ + Z^2) \\ &= E(Y^2) + 2E(YZ) + E(Z^2) = 144 + 0 + 81 = 225\end{aligned}$$

Hence,

$$\sigma_V = \sqrt{225} = 15$$

Thus, the correlation coefficient between U and V is given by

$$\rho(U, V) = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{144}{13 \times 15} = \frac{48}{65}$$

◆

EXAMPLE 4.115. (X, Y) is a two-dimensional random variable uniformly distributed over the triangular region R bounded by $y = 0$, $x = 3$ and $y = \frac{4}{3}x$. Find the correlation coefficient $r_{x,y}$.
(Anna, April 2003)

Solution. The region is as sketched in Figure 4.25.

Since (X, Y) is uniformly distributed in the region R , it takes a constant value c , which is equal to the reciprocal of the area of the triangle R .

The area of R is readily obtained as

$$\Delta = \frac{1}{2}bh = \frac{1}{2} \times 3 \times 4 = 6$$

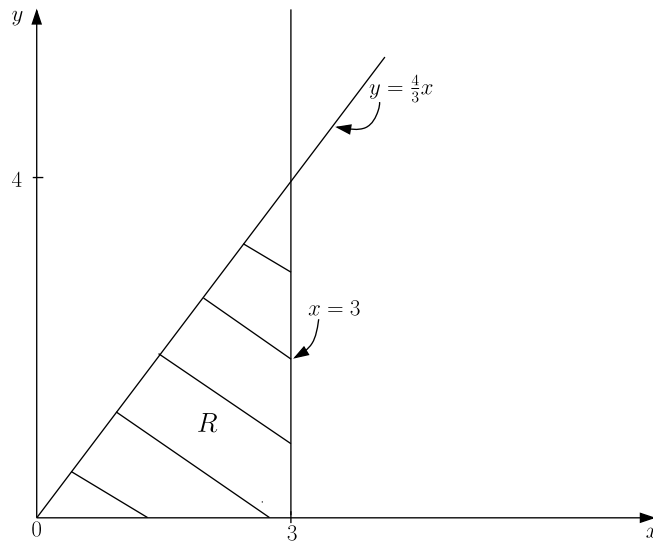
Hence, the probability density function of (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{1}{6} & \text{if } (x, y) \in R \\ 0 & \text{otherwise} \end{cases}$$

Next, we find the marginal density functions of X and Y .

If $0 < x < 3$, then we have

$$f_X(x) = \int_{y=0}^{\frac{4}{3}x} f(x, y) dy = \int_{y=0}^{\frac{4}{3}x} \frac{1}{6} dy$$

Figure 4.25: The region R .

Integrating, we have

$$f_X(x) = \frac{1}{6} \times \frac{4}{3}x = \frac{2x}{9}$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{2x}{9} & \text{if } 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

If $0 < y < 4$, then

$$f_Y(y) = \int_{x=\frac{3}{4}y}^3 f(x,y) dx = \int_{x=\frac{3}{4}y}^3 \frac{1}{6} dx$$

Integrating, we have

$$f_Y(y) = \frac{1}{6} \left[3 - \frac{3}{4}y \right] = \frac{1}{2} \left[1 - \frac{y}{4} \right]$$

Thus, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{2} \left[1 - \frac{y}{4} \right] & \text{if } 0 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

Next, we find the mean and variance of X .

The mean of X is given by

$$\mu'_1(X) = E(X) = \int_0^3 x f_X(x) dx = \int_0^3 x \left[\frac{2x}{9} \right] dx = \frac{2}{9} \int_0^3 x^2 dx$$

Integrating, we have

$$\mu'_1(X) = \frac{2}{9} \left[\frac{x^3}{3} \right]_0^3 = \frac{2}{9} [9] = 2$$

The second raw moment of X is given by

$$\mu'_2(X) = E(X^2) = \int_0^3 x^2 f_X(x) dx = \int_0^3 x^2 \left[\frac{2x}{9} \right] dx = \frac{2}{9} \int_0^3 x^3 dx$$

Integrating, we have

$$\mu'_2(X) = \frac{2}{9} \left[\frac{x^4}{4} \right]_0^3 = \frac{2}{9} \left[\frac{81}{4} \right] = \frac{9}{2}$$

Hence, the variance of X is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{9}{2} - 4 = \frac{1}{2}$$

Next, we find the mean and variance of Y .

The mean of Y is given by

$$\mu'_1(Y) = E(Y) = \int_0^4 y f_Y(y) dy = \int_0^4 y \left[\frac{1}{2} \left(1 - \frac{y}{4} \right) \right] dy$$

Integrating, we have

$$\mu'_1(Y) = \frac{1}{2} \left[\frac{y^2}{2} - \frac{y^3}{12} \right]_0^4 = \frac{4}{3}$$

The second raw moment of Y is given by

$$\mu'_2(Y) = E(Y^2) = \int_0^4 y^2 f_Y(y) dy = \int_0^4 y^2 \left[\frac{1}{2} \left(1 - \frac{y}{4} \right) \right] dy$$

Integrating, we have

$$\mu'_2(Y) = \frac{1}{2} \left[\frac{y^3}{3} - \frac{y^4}{16} \right]_0^4 = \frac{8}{3}$$

Hence, the variance of Y is given by

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{8}{3} - \frac{16}{9} = \frac{8}{9}$$

Next, we calculate the covariance between X and Y .

Note that

$$E(XY) = \int_{x=0}^3 \int_{y=0}^{4/3x} xy f(x,y) dy dx = \int_{x=0}^3 \int_{y=0}^{4/3x} xy \left[\frac{1}{6} \right] dy dx.$$

Integrating y , we have

$$\begin{aligned} E(XY) &= \frac{1}{6} \int_{x=0}^3 x \left[\frac{y^2}{2} \right]_0^{4/3x} dx \\ &= \frac{1}{12} \int_{x=0}^3 x \left[\frac{16}{9} x^2 \right] dx = \frac{4}{27} \int_{x=0}^3 x^3 dx \end{aligned}$$

Integrating, we have

$$E(XY) = \frac{4}{27} \left[\frac{x^4}{4} \right]_{x=0}^3 = \frac{4}{27} \times \frac{81}{4} = 3$$

Thus,

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 3 - 2 \left[\frac{4}{3} \right] = \frac{1}{3}$$

Hence, the correlation coefficient between X and Y is given by

$$r_{x,y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{1/3}{1/\sqrt{2} \times 2\sqrt{2}/3} = \frac{1}{3} \times \frac{3}{2} = \frac{1}{2}$$

◆

EXAMPLE 4.116. Let the random variables X and Y have the joint probability density function

$$f(x, y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Compute the correlation coefficient between X and Y .

(Anna, April 2003; Nov. 2003; Nov. 2004; May 2006; May 2007)

Solution. Clearly, the marginal density function of X is given by

$$f_X(x) = \begin{cases} x + \frac{1}{2} & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} y + \frac{1}{2} & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the mean and variance of X .

The mean of X is given by

$$\mu'_1(X) = E(X) = \int_{x=0}^1 x f_X(x) dx = \int_{x=0}^1 x \left[x + \frac{1}{2} \right] dx$$

Integrating, we have

$$\mu'_1(X) = \left[\frac{x^3}{3} + \frac{x^2}{4} \right]_{x=0}^1 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

The second raw moment of X is given by

$$\mu'_2(X) = E(X^2) = \int_{x=0}^1 x^2 f_X(x) dx = \int_{x=0}^1 x^2 \left[x + \frac{1}{2} \right] dx$$

Integrating, we have

$$\mu'_2(X) = \left[\frac{x^4}{4} + \frac{x^3}{6} \right]_{x=0}^1 = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

Hence, the variance of X is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{5}{12} - \left[\frac{7}{12} \right]^2 = \frac{11}{144}$$

By symmetry, the mean of Y

$$E(Y) = E(X) = \frac{7}{12}$$

and the variance of Y

$$\text{Var}(Y) = \text{Var}(X) = \frac{11}{144}$$

We also find that

$$E(XY) = \int_{x=0}^1 \int_{y=0}^1 xy f(x, y) dy dx = \int_{x=0}^1 \int_{y=0}^1 xy(x+y) dy dx$$

Integrating y , we have

$$\begin{aligned} E(XY) &= \int_{x=0}^1 x \left[\frac{xy^2}{2} + \frac{y^3}{3} \right]_{y=0}^1 dx = \int_{x=0}^1 x \left[\frac{x}{2} + \frac{1}{3} \right] dx \\ &= \int_{x=0}^1 \left[\frac{x^2}{2} + \frac{x}{3} \right] dx \end{aligned}$$

Integrating, we have

$$E(XY) = \left[\frac{x^3}{6} + \frac{x^2}{6} \right]_{x=0}^1 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Hence, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{3} - \left[\frac{7}{12} \right]^2 = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$$

Hence, the correlation coefficient between X and Y is given by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-1/144}{\sqrt{11/144} \times \sqrt{11/144}} = -\frac{1}{144} \times \frac{144}{11} = -\frac{1}{11}$$

◆

EXAMPLE 4.117. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} e^{-x}, & 0 < y < x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

- (i) Find the marginal probability density functions of X and Y .
- (ii) Determine the correlation coefficient between X and Y .

(Anna, April 2004)

Solution. The joint probability density function takes values in the region A defined by

$$A = \{(x, y) \in \mathbb{R}^2 : 0 < y < x < \infty\}$$

which is illustrated in Figure 4.26.

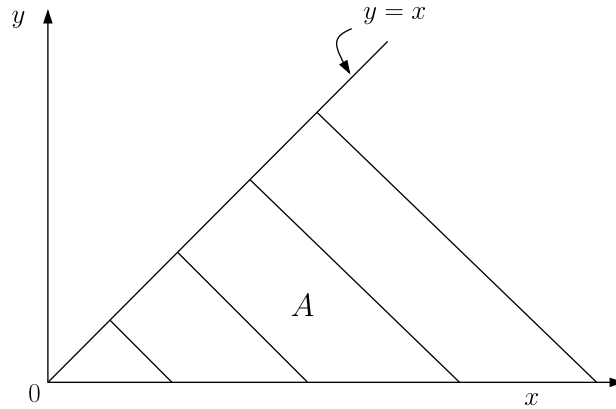


Figure 4.26: Region A .

- (i) First, we find the marginal probability density function of X .

If $0 < x < \infty$, then

$$f_X(x) = \int_{y=0}^x f(x, y) dy = \int_{y=0}^x e^{-x} dy$$

Integrating, we have

$$f_X(x) = e^{-x} [y]_0^x = xe^{-x}$$

Thus, the marginal probability density function of X is given by

$$f_X(x) = \begin{cases} xe^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

If $0 < y < \infty$, then

$$f_Y(y) = \int_{x=y}^{\infty} f(x,y) dx = \int_{x=y}^{\infty} e^{-x} dx$$

Integrating, we have

$$f_Y(y) = [-e^{-x}]_{x=y}^{\infty} = e^{-y}$$

Thus, the marginal probability density function of Y is given by

$$f_Y(y) = \begin{cases} e^{-y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(ii) First, we find the mean and variance of X .

The mean of X is given by

$$\mu'_1(X) = E(X) = \int_0^{\infty} x f_X(x) dx = \int_0^{\infty} x [xe^{-x}] dx$$

Thus, we find that $\mu'_1(X) = \Gamma(3) = 2! = 2$.

Next, the second raw moment of X is given by

$$\mu'_2(X) = E(X^2) = \int_0^{\infty} x^2 f_X(x) dx = \int_0^{\infty} x^2 [xe^{-x}] dx$$

Thus, we find that $\mu'_2(X) = \Gamma(4) = 3! = 6$.

Hence, the variance of X is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 6 - 4 = 2$$

Next, we find the mean and variance of Y .

Since Y follows the exponential distribution with parameter $\lambda = 1$, it is immediate that

$$\text{Mean}(Y) = \text{Var}(Y) = \lambda = 1$$

Next, we find $E(XY)$. We have

$$E(XY) = \int_{x=0}^{\infty} \int_{y=0}^x xy f(x,y) dy dx = \int_{x=0}^{\infty} \int_{y=0}^x xy e^{-x} dy dx$$

Integrating y , we have

$$\begin{aligned} E(XY) &= \int_{x=0}^{\infty} xe^{-x} \left[\frac{y^2}{2} \right]_0^x dx = \frac{1}{2} \int_{x=0}^{\infty} x^3 e^{-x} dx \\ &= \frac{1}{2} \Gamma(4) = \frac{3!}{2} = 3 \end{aligned}$$

Thus, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 3 - (2)(1) = 1$$

Hence, the correlation coefficient between X and Y is given by

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{1}{\sqrt{2} \times 1} = \frac{1}{\sqrt{2}}$$

◆

EXAMPLE 4.118. Let the random variable X have the marginal density

$$f_1(x) = 1, -\frac{1}{2} < x < \frac{1}{2}$$

and the conditional density of Y be

$$f(y|x) = \begin{cases} 1, & x < y < x+1, -\frac{1}{2} < x < 0 \\ 1, & -x < y < 1-x, 0 < x < \frac{1}{2} \end{cases}$$

Prove that the variables X and Y are uncorrelated.

(Anna, Nov. 2003)

Solution. First, we find the mean of X . Note that

$$E(X) = \int_{-\frac{1}{2}}^{\frac{1}{2}} x f_1(x) dx = \int_{-\frac{1}{2}}^{\frac{1}{2}} x[1] dx = 0$$

since the integrand is an odd function of x .

Thus, it is immediate that

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = E(XY) - 0 = E(XY)$$

Hence, to show that X and Y are uncorrelated, it suffices to show that $E(XY) = 0$.

Next, note that the joint probability density function of X and Y is given by

$$f(x, y) = f_1(x)f(y|x) = \begin{cases} 1, & x < y < x+1, -\frac{1}{2} < x < 0 \\ 1, & -x < y < 1-x, 0 < x < \frac{1}{2} \end{cases}$$

Hence,

$$E(XY) = \int_{x=-\frac{1}{2}}^0 \int_{y=x}^{x+1} xy dy dx + \int_{x=0}^{\frac{1}{2}} \int_{y=-x}^{1-x} xy dy dx$$

Substituting $\xi = -x$ in the second double integral on the R.H.S., we obtain

$$E(XY) = \int_{x=-\frac{1}{2}}^0 \int_{y=x}^{x+1} xy dy dx + \int_{\xi=0}^{-\frac{1}{2}} \int_{y=\xi}^{1+\xi} (-\xi)y dy (-d\xi)$$

Replacing ξ by x in the second double integral on the R.H.S., and simplifying, we get

$$E(XY) = \int_{x=-\frac{1}{2}}^0 \int_{y=x}^{x+1} xy \, dy \, dx - \int_{x=-\frac{1}{2}}^0 \int_{y=x}^{1+x} xy \, dy \, dx = 0$$

Hence, X and Y are uncorrelated. ◆

EXAMPLE 4.119. Two random variables, X and Y , are defined as $Y = 4X + 9$. Find the coefficient of correlation between X and Y . (Anna, Nov. 2005)

Solution. Since $Y = 4X + 9$, it is clear that $\mu_Y = 4\mu_X + 9$.

Hence, $Y - \mu_Y = 4(X - \mu_X)$.

Thus, it follows that

$$\sigma_Y^2 = E[(Y - \mu_Y)^2] = 16E[(X - \mu_X)^2] = 16\sigma_X^2$$

Hence, we see that $\sigma_Y = 4\sigma_X$.

We also find that

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = 4E[(X - \mu_X)^2] = 4\sigma_X^2$$

Hence, the coefficient of correlation between X and Y is given by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{4\sigma_X^2}{\sigma_X 4\sigma_X} = 1$$

Thus, X and Y are perfectly correlated with $\rho = 1$. ◆

EXAMPLE 4.120. If the independent random variables, X and Y , have the variances 36 and 16, respectively, find the correlation coefficient between $X + Y$ and $X - Y$. (Anna, Nov. 2006)

Solution. Define $U = X + Y$ and $V = X - Y$.

Since X and Y are independent,

$$\text{Var}(U) = \text{Var}(X) + \text{Var}(Y) = 36 + 16 = 52$$

Hence, it follows that $\sigma_U = \sqrt{52}$.

Since X and Y are independent,

$$\text{Var}(V) = \text{Var}(X) + \text{Var}(Y) = 36 + 16 = 52$$

Hence, it follows that $\sigma_V = \sqrt{52}$.

Next, we find that

$$\begin{aligned} \text{Cov}(U, V) &= E[(U - \mu_U)(V - \mu_V)] = E[((X - \mu_X) + (Y - \mu_Y))((X - \mu_X) - (Y - \mu_Y))] \\ &= E[(X - \mu_X)^2] - E[(Y - \mu_Y)^2] = \sigma_X^2 - \sigma_Y^2 = 36 - 16 = 20 \end{aligned}$$

Hence, the correlation coefficient between U and V is given by

$$\rho = \frac{\text{Cov}(U, V)}{\sigma_U \sigma_V} = \frac{20}{\sqrt{52}\sqrt{52}} = \frac{20}{52} = \frac{5}{13}$$
 ◆

EXAMPLE 4.121. Let X and Y be random variables having joint density function

$$f(x, y) = \begin{cases} \frac{3}{2} (x^2 + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the correlation coefficient, r_{XY} .

(Anna, Nov. 2006)

Solution. The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{3}{2} (x^2 + \frac{1}{3}) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{3}{2} (y^2 + \frac{1}{3}) & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Next, we find the mean and variance of X .

The mean of X is given by

$$\mu'_1(X) = E(X) = \int_{x=0}^1 x f_X(x) dx = \int_{x=0}^1 x \left[\frac{3}{2} \left(x^2 + \frac{1}{3} \right) \right] dx = \frac{3}{2} \int_{x=0}^1 \left[x^3 + \frac{x}{3} \right] dx$$

Integrating, we have

$$\mu'_1(X) = \frac{3}{2} \left[\frac{x^4}{4} + \frac{x^2}{6} \right]_{x=0}^1 = \frac{3}{2} \times \frac{5}{12} = \frac{5}{8}$$

The second raw moment of X is given by

$$\mu'_2(X) = E(X^2) = \int_{x=0}^1 x^2 f_X(x) dx = \int_{x=0}^1 x^2 \left[\frac{3}{2} \left(x^2 + \frac{1}{3} \right) \right] dx = \frac{3}{2} \int_{x=0}^1 \left[x^4 + \frac{x^2}{3} \right] dx$$

Integrating, we have

$$\mu'_2(X) = \frac{3}{2} \left[\frac{x^5}{5} + \frac{x^3}{9} \right]_{x=0}^1 = \frac{3}{2} \times \frac{14}{45} = \frac{7}{15}$$

Thus, the variance of X is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{7}{15} - \frac{25}{64} = \frac{73}{960}$$

Hence, $\sigma_X = \sqrt{\frac{73}{960}}$.

Next, we find the mean and variance of Y .

The mean of Y is given by

$$\mu'_1(Y) = E(Y) = \int_{y=0}^1 y f_Y(y) dy = \int_{y=0}^1 y \left[\frac{3}{2} \left(y^2 + \frac{1}{3} \right) \right] dy = \frac{3}{2} \int_{y=0}^1 \left[y^3 + \frac{y}{3} \right] dy$$

Integrating, we have

$$\mu'_1(Y) = \frac{3}{2} \left[\frac{y^4}{4} + \frac{y^2}{6} \right]_{y=0}^1 = \frac{3}{2} \times \frac{5}{12} = \frac{5}{8}$$

The second raw moment of Y is given by

$$\mu'_2(Y) = E(Y^2) = \int_{y=0}^1 y^2 f_Y(y) dy = \int_{y=0}^1 y^2 \left[\frac{3}{2} \left(y^2 + \frac{1}{3} \right) \right] dy = \frac{3}{2} \int_{y=0}^1 \left[y^4 + \frac{y^2}{3} \right] dy$$

Integrating, we have

$$\mu'_2(Y) = \frac{3}{2} \left[\frac{y^5}{5} + \frac{y^3}{9} \right]_{y=0}^1 = \frac{3}{2} \times \frac{14}{45} = \frac{7}{15}$$

Thus, the variance of Y is given by

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{7}{15} - \frac{25}{64} = \frac{73}{960}$$

Hence, $\sigma_Y = \sqrt{\frac{73}{960}}$.

Next, we find that

$$E(XY) = \int_{x=0}^1 \int_{y=0}^1 xy f(x,y) dy dx = \int_{x=0}^1 \int_{y=0}^1 xy \left[\frac{3}{2} (x^2 + y^2) \right] dy dx$$

Therefore,

$$E(XY) = \frac{3}{2} \int_{x=0}^1 x \left[\int_{y=0}^1 [x^2 y + y^3] dy \right] dx$$

Integrating y , we get

$$\begin{aligned} E(XY) &= \frac{3}{2} \int_{x=0}^1 x \left[\frac{x^2}{y^2} 2 + \frac{y^4}{4} \right]_{y=0}^1 dx = \frac{3}{2} \int_{x=0}^1 x \left[\frac{x^2}{2} + \frac{1}{4} \right] dx \\ &= \frac{3}{2} \int_{x=0}^1 \left[\frac{x^3}{2} + \frac{x}{4} \right] dx \end{aligned}$$

Integrating x , we get

$$E(XY) = \frac{3}{2} \left[\frac{x^4}{8} + \frac{x^2}{8} \right]_{x=0}^1 = \frac{3}{2} \times \frac{1}{4} = \frac{3}{8}$$

Hence, the covariance between X and Y

$$\text{Cov}(X, Y) = E(XY) - \mu_X \mu_Y = \frac{3}{8} - \frac{25}{64} = -\frac{1}{64}$$

and, the correlation coefficient between X and Y

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-\frac{1}{64}}{\sqrt{\frac{73}{960}} \sqrt{\frac{73}{960}}} = -\frac{1}{64} \times \frac{960}{73} = -\frac{15}{73}$$

◆

EXAMPLE 4.122. Let the joint probability density function of X and Y be given by

$$f(x, y) = \begin{cases} 6(1-x-y), & x > 0, y > 0, x+y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find $f_X(x)$, $f_Y(y)$ and $\text{Cov}(X, Y)$. Are X and Y independent? Obtain the regression curves for the means.

(Anna, April 2003)

Solution. As calculated in Example 4.30, the marginal density function of X is given by

$$f_X(x) = \begin{cases} 3(1-x)^2 & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 3(1-y)^2 & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Note that

$$f_X(x)f_Y(y) = \begin{cases} 9(1-x)^2(1-y)^2 & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Since $f(x, y) \neq f_X(x)f_Y(y)$, we conclude that X and Y are not independent.

Next, we find the mean and variance of X .

The mean of X is given by

$$E(X) = \int_{x=0}^1 x f_X(x) dx = \int_{x=0}^1 x [3(1-x)^2] dx = 3 \int_{x=0}^1 [x^3 - 2x^2 + x] dx$$

Integrating, we get

$$E(X) = 3 \left[\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} \right]_{x=0}^1 = 3 \left[\frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right] = \frac{3}{12} = \frac{1}{4}$$

The mean of Y is given by

$$E(Y) = \int_{y=0}^1 y f_Y(y) dy = \int_{y=0}^1 y [3(1-y)^2] dy = 3 \int_{y=0}^1 [y^3 - 2y^2 + y] dy$$

Integrating, we get

$$E(Y) = 3 \left[\frac{y^4}{4} - \frac{2y^3}{3} + \frac{y^2}{2} \right]_{y=0}^1 = 3 \left[\frac{1}{4} - \frac{2}{3} + \frac{1}{2} \right] = \frac{3}{12} = \frac{1}{4}$$

We also find that

$$E(XY) = \int_{x=0}^1 \int_{y=0}^1 y = 0^{1-x} xy f(x, y) dy dx = \int_{x=0}^1 \int_{y=0}^1 y = 0^{1-x} xy [6(1-x-y)] dy dx$$

Integrating y , we have

$$\begin{aligned} E(XY) &= 6 \int_{x=0}^1 x \left[\frac{(1-x)y^2}{2} - \frac{y^3}{3} \right]_{y=0}^{1-x} dx = 6 \int_{x=0}^1 x \left[\frac{(1-x)^3}{2} - \frac{(1-x)^3}{3} \right] dx \\ &= \int_{x=0}^1 x(1-x)^3 dx = \int_{x=0}^1 x(1-3x+3x^2-x^3) dx \end{aligned}$$

Integrating x , we have

$$E(XY) = \left[\frac{x^2}{2} - \frac{x^3}{3} + \frac{3x^4}{4} - \frac{x^5}{5} \right]_{x=0}^1 = \frac{1}{2} - 1 + \frac{3}{4} - \frac{1}{5} = \frac{1}{20}$$

Hence, the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{20} - \left[\frac{1}{4} \right]^2 = -\frac{1}{80}$$

For finding the regression curves for the means, we need to find the two-conditional density functions.

First, we find $f_{Y|X}(y|x)$. Fix any value of X in the range $0 < x < 1$.

Then, by definition, we have

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{6(1-x-y)}{3(1-x)^2} \quad \text{for } 0 < y < 1-x$$

Thus, the conditional density of Y given $X = x$ is obtained as

$$f_{Y|X}(y|x) = \begin{cases} \frac{2(1-x-y)}{(1-x)^2} & \text{for } 0 < y < 1-x \\ 0 & \text{elsewhere} \end{cases}$$

Hence, it follows that

$$\begin{aligned} E(Y|X=x) &= \int_{y=0}^{1-x} y f_{Y|X}(y|x) dy = \int_{y=0}^{1-x} y \left[\frac{2(1-x-y)}{(1-x)^2} \right] dy \\ &= \frac{2}{(1-x)^2} \int_{y=0}^{1-x} [(1-x)y - y^2] dy \end{aligned}$$

Integrating, we have

$$\begin{aligned} E(Y|X=x) &= \frac{2}{(1-x)^2} \left[\frac{(1-x)y^2}{2} - \frac{y^3}{3} \right]_{y=0}^{1-x} = \frac{2}{(1-x)^2} \left[\frac{(1-x)^3}{3} - \frac{(1-x)^3}{3} \right] \\ &= \frac{2}{(1-x)^2} \times \frac{(1-x)^3}{6} = \frac{1-x}{3} \end{aligned}$$

Similarly, it can be shown that the conditional density of X given $Y = y$ is

$$f_{X|Y}(x|y) = \begin{cases} \frac{2(1-x-y)}{(1-y)^2} & \text{for } 0 < x < 1-y \\ 0 & \text{elsewhere} \end{cases}$$

and that

$$E(X|Y=y) = \frac{1-y}{3}$$

Thus, the regression curve of Y on X is given by

$$y = E(Y|X=x) = \frac{1-x}{3}$$

which is linear.

Also, the regression curve of X on Y is given by

$$x = E(X|Y=y) = \frac{1-y}{3}$$

which also is linear. ♦

EXAMPLE 4.123. If the joint density of X and Y is given by

$$f(x,y) = \begin{cases} \frac{x+y}{3} & \text{for } 0 < x < 1, 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

obtain the regression curves of Y on X and of X on Y .

(Anna, Nov. 2003)

Solution. If $0 < x < 1$, then

$$f_X(x) = \int_{y=0}^2 f(x,y) dy = \int_{y=0}^2 \left[\frac{x+y}{3} \right] dy$$

Integrating, we have

$$f_X(x) = \frac{1}{3} \left[xy + \frac{y^2}{2} \right]_{y=0}^2 = \frac{1}{3} [2x + 2] = \frac{2}{3} [x + 1]$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{2}{3}(x+1) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

If $0 < y < 2$, then

$$f_Y(y) = \int_{x=0}^1 f(x,y) dx = \int_{x=0}^1 \left[\frac{x+y}{3} \right] dx$$

Integrating, we have

$$f_Y(y) = \frac{1}{3} \left[\frac{x^2}{2} + xy \right]_{x=0}^1 = \frac{1}{3} \left[\frac{1}{2} + y \right]$$

Thus, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{3} \left(y + \frac{1}{2} \right) & \text{for } 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the conditional density function of Y given $X = x$.

Fix any value of X in the interval $0 < x < 1$.

Then, by definition,

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{\frac{1}{3}(x+y)}{\frac{2}{3}(x+1)} = \frac{(x+y)}{2(x+1)} \quad \text{for } 0 < y < 2$$

Next, we find the conditional density function of X given $Y = y$.

Fix any value of Y in the interval $0 < y < 2$.

Then, by definition,

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{\frac{1}{3}(x+y)}{\frac{1}{3}(y+\frac{1}{2})} = \frac{(x+y)}{(y+\frac{1}{2})} \quad \text{for } 0 < x < 1$$

Next, we find the conditional means $E(Y|X = x)$ and $E(X|Y = y)$.

By definition, we have

$$\begin{aligned} E(Y|X=x) &= \int_{y=0}^2 y f_{Y|X}(y|x) dy = \int_{y=0}^2 y \left[\frac{(x+y)}{2(x+1)} \right] dy \\ &= \frac{1}{2(x+1)} \int_{y=0}^2 [xy + y^2] dy \end{aligned}$$

Integrating,

$$\begin{aligned} E(Y|X=x) &= \frac{1}{2(x+1)} \left[\frac{xy^2}{2} + \frac{y^3}{3} \right]_{y=0}^2 = \frac{1}{2(x+1)} \left[2x + \frac{8}{3} \right] = \frac{1}{x+1} \left[x + \frac{4}{3} \right] \\ &= \frac{1}{3} \left[\frac{3x+4}{x+1} \right] \end{aligned}$$

Next, by definition, we have

$$\begin{aligned} E(X|Y=y) &= \int_{x=0}^1 x f_{X|Y}(x|y) dx = \int_{x=0}^1 x \left[\frac{(x+y)}{(y+\frac{1}{2})} \right] dx \\ &= \frac{1}{y+\frac{1}{2}} \int_{x=0}^1 [x^2 + xy] dx \end{aligned}$$

Integrating, we have

$$\begin{aligned} E(X|Y=y) &= \frac{1}{y+\frac{1}{2}} \left[\frac{x^3}{3} + \frac{x^2 y}{2} \right]_{x=0}^1 = \frac{1}{y+\frac{1}{2}} \left[\frac{1}{3} + \frac{y}{2} \right] \\ &= \frac{1}{3} \left[\frac{3y+2}{2y+1} \right] \end{aligned}$$

Thus, the regression curve of Y on X is given by

$$y = E(Y|X=x) = \frac{1}{3} \left[\frac{3x+4}{x+1} \right]$$

and the regression curve of X on Y is given by

$$x = E(X|Y=y) = \frac{1}{3} \left[\frac{3y+2}{2y+1} \right]$$

◆

EXAMPLE 4.124. Let (X, Y) have the joint probability density function given by:

$$f(x, y) = \begin{cases} 1 & \text{if } |y| < x, 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Show that the regression of Y on X is linear, but the regression of X on Y is not linear.

(Anna, Nov. 2004)

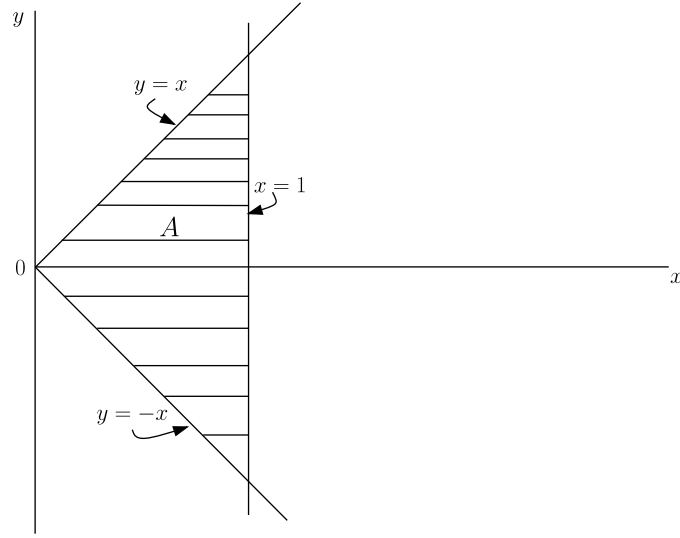


Figure 4.27: Region A.

Solution. The given joint density function f takes values in the region A defined by

$$A = \{(x, y) \in \mathbb{R}^2 : -x < y < x, 0 < x < 1\}$$

which is illustrated in Figure 4.27.

First, we find the marginal density functions of X and Y .

If $0 < x < 1$, then

$$f_X(x) = \int_{y=-x}^x f(x, y) dy = \int_{y=-x}^x 1 dy = 2 \int_{y=0}^x dy = 2x$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

If $-1 < y < 0$, then

$$f_Y(y) = \int_{x=-y}^1 f(x, y) dx = \int_{x=-y}^1 1 dx = 1 + y$$

If $0 < y < 1$, then

$$f_Y(y) = \int_{x=y}^1 f(x, y) dx = \int_{x=y}^1 1 dx = 1 - y$$

Combining the two cases, the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 1 - |y| & \text{for } |y| < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Next, we find the two conditional density functions.

First, we find the conditional density function, $f_{Y|X}(y|x)$.

Fix any value of X in the interval $0 < x < 1$.

Then, by definition, we have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{1}{2x} \quad \text{for } -x < y < x.$$

Next, we find the conditional density function, $f_{X|Y}(x|y)$.

Fix any value of Y in the interval $|y| < 1$.

Then, by definition,

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{1}{1-|y|} \quad \text{for } |y| < x < 1$$

Next, we find the conditional means, $E(Y|X = x)$ and $E(X|Y = y)$.

By definition, we have

$$E(Y|X = x) = \int_{y=-x}^x y f_{Y|X}(y|x) dy = \int_{y=-x}^x y \left[\frac{1}{2x} \right] dy = 0$$

since the integrand is an odd function of y .

Next, by definition,

$$E(X|Y = y) = \int_{x=|y|}^1 x f_{X|Y}(x|y) dx = \int_{x=|y|}^1 x \left[\frac{1}{1-|y|} \right] dx$$

Integrating, we have

$$E(X|Y = y) = \frac{1}{1-|y|} \left[\frac{x^2}{2} \right]_{x=|y|}^1 = \frac{1}{2(1-|y|)} [1 - |y|^2] = \frac{1+|y|}{2}$$

Thus, the regression of Y on X is given by

$$y = E(Y|X = x) = 0$$

which is linear. In fact, this represents the x -axis in the (x,y) -plane.

But the regression of X on Y is given by

$$x = E(X|Y = y) = \frac{1+|y|}{2}$$

which is non-linear. ◆

EXAMPLE 4.125. If the random variables X and Y have the joint probability density function given by

$$f(x,y) = \begin{cases} xe^{-x(y+1)} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

find the regression curve of Y on X .

Solution. First, we find the marginal density function of X .

If $x \geq 0$, then

$$f_X(x) = \int_{y=0}^{\infty} f(x, y) dy = \int_{y=0}^{\infty} [xe^{-x(y+1)}] dy = xe^{-x} \int_{y=0}^{\infty} e^{-xy} dy$$

Integrating, we have

$$f_X(x) = xe^{-x} \left[-\frac{e^{-xy}}{x} \right]_{y=0}^{\infty} = xe^{-x} \left[\frac{1}{x} \right] = e^{-x}$$

Thus, the marginal density function of X is given by

$$f_X(x) = \begin{cases} e^{-x} & \text{if } x \geq 0 \end{cases}$$

Next, we find the conditional density function, $f_{Y|X}(y|x)$.

Fix any value of X in the interval $[0, \infty)$.

Then, by definition, we have

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{xe^{-x(y+1)}}{e^{-x}} = xe^{-xy} \text{ for } y \geq 0$$

Thus, the conditional mean, $E(Y|X = x)$, is obtained as

$$E(Y|X = x) = \int_{y=0}^{\infty} y f_{Y|X}(y|x) dy = \int_{y=0}^{\infty} y [xe^{-xy}] dy = \int_{y=0}^{\infty} y d[-e^{-xy}]$$

Using integration by parts,

$$E(Y|X = x) = [y(-e^{-xy})]_{y=0}^{\infty} - \int_{y=0}^{\infty} (-e^{-xy}) dy = 0 - \int_{y=0}^{\infty} (-e^{-xy}) dy$$

Integrating, we have

$$E(Y|X = x) = 0 - \left[\frac{e^{-xy}}{x} \right]_{y=0}^{\infty} = - \left[0 - \frac{1}{x} \right] = \frac{1}{x}$$

Thus, the regression curve of Y on X is given by

$$y = E(Y|X = x) = \frac{1}{x}$$

which can be expressed as

$$xy = 1.$$

Thus, the regression curve of Y on X is non-linear, which geometrically represents a *rectangular hyperbola*. ◆

EXAMPLE 4.126. Find the correlation coefficient for the following data:

X	10	14	18	22	26	30
Y	18	12	24	6	30	36

(Anna, Nov. 2007)

Solution. We form the following table:

X	X^2	Y	Y^2	XY
10	100	18	324	180
14	196	12	144	168
18	324	24	576	432
22	484	6	36	132
26	676	30	900	780
30	900	36	1296	1080
120	2680	126	3276	2772

From the table, we find that

$$\mu_X = \frac{\sum X}{n} = \frac{120}{6} = 20$$

and

$$\mu_Y = \frac{\sum Y}{n} = \frac{126}{6} = 21$$

The standard deviation of X is

$$\sigma_X = \sqrt{\frac{1}{n} \sum X^2 - \left(\frac{\sum X}{n}\right)^2} = \sqrt{\frac{1}{6}(2680) - (20)^2} = 6.8313$$

Similarly, the standard deviation of Y

$$\sigma_Y = \sqrt{\frac{1}{n} \sum Y^2 - \left(\frac{\sum Y}{n}\right)^2} = \sqrt{\frac{1}{6}(3276) - (21)^2} = 10.2470$$

The covariance between X and Y is

$$\text{Cov}(X, Y) = \frac{1}{n} \sum XY - \left(\frac{\sum X}{n}\right) \left(\frac{\sum Y}{n}\right) = \frac{1}{6}(2772) - (20)(21) = 42$$

Thus, the correlation coefficient between X and Y is given by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{42}{6.8313 \times 10.2470} = \frac{42}{70} = 0.6$$

◆

EXAMPLE 4.127. Find the coefficient of correlation and obtain the lines of regression from the given data:

X	62	64	65	69	70	71	72	74
Y	126	125	139	145	165	152	180	208

(Anna, Nov. 2003; May 2006)

Solution. Let $\xi = X - 69$ and $\eta = Y - 152$.

X	$\xi = X - 69$	ξ^2	Y	$\eta = Y - 152$	η^2	$\xi\eta$
62	-7	49	126	-26	676	182
64	-5	25	125	-27	729	135
65	-4	16	139	-13	169	52
69	0	0	145	-7	49	0
70	1	1	165	13	169	13
71	2	4	152	0	0	0
72	3	9	180	28	784	84
74	5	25	208	56	3136	280
	-5	129		24	5712	746

From the above table, we find that

$$\mu_X = A + \frac{\sum \xi}{n} = 69 - \frac{5}{8} = 68.3750$$

and

$$\mu_Y = B + \frac{\sum \eta}{n} = 152 + \frac{24}{8} = 155$$

The standard deviation of X is

$$\begin{aligned}\sigma_X = \sigma_\xi &= \sqrt{\frac{1}{n} \sum \xi^2 - \left(\frac{\sum \xi}{n}\right)^2} = \sqrt{\frac{1}{8}(129) - \left(-\frac{5}{8}\right)^2} \\ &= \sqrt{\frac{129}{8} - \frac{25}{64}} = \sqrt{\frac{1007}{64}} = 3.9667\end{aligned}$$

Similarly, the standard deviation of Y is

$$\begin{aligned}\sigma_Y = \sigma_\eta &= \sqrt{\frac{1}{n} \sum \eta^2 - \left(\frac{\sum \eta}{n}\right)^2} = \sqrt{\frac{1}{8}(5712) - \left(\frac{24}{8}\right)^2} \\ &= \sqrt{\frac{5712}{8} - 9} = \sqrt{705} = 26.5518\end{aligned}$$

The covariance between X and Y is

$$\text{Cov}(X, Y) = \text{Cov}(\xi, \eta) = \frac{1}{n} \sum \xi\eta - \left(\frac{\sum \xi}{n}\right) \left(\frac{\sum \eta}{n}\right)$$

Substituting from the table, we have

$$\text{Cov}(X, Y) = \frac{746}{8} - \left(\frac{-5}{8}\right) \left(\frac{24}{8}\right) = \frac{746}{8} + \frac{15}{8} = \frac{761}{8} = 95.1250$$

Thus, the correlation coefficient between X and Y is given by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{95.1250}{3.9667 \times 26.5518} = \frac{95.1250}{105.3230} = 0.9032$$

Thus, the regression coefficient of Y on X

$$\beta_{YX} = \frac{\rho \sigma_Y}{\sigma_X} = \frac{0.9032 \times 26.5518}{3.9667} = 6.0457$$

and the regression coefficient of X on Y is given by

$$\beta_{XY} = \frac{\rho \sigma_X}{\sigma_Y} = \frac{0.9032 \times 3.9667}{26.5518} = 0.1349$$

Hence, the regression line of Y on X

$$y - \mu_Y = \beta_{YX} (x - \mu_X)$$

i.e.

$$y - 155 = 6.0457 (x - 68.3750)$$

i.e.

$$y = 6.0457x - 258.3747$$

Also, the regression line of X on Y is given by

$$x - \mu_X = \beta_{XY} (y - \mu_Y)$$

i.e.

$$x - 68.3750 = 0.1349 (y - 155)$$

i.e.

$$x = 0.1349y + 47.4655$$



EXAMPLE 4.128. Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their sons (Y).

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

(Anna, Nov. 2004; May 2007)

Solution. Let $\xi = X - 67$ and $\eta = Y - 68$.

X	$\xi = X - 67$	ξ^2	Y	$\eta = Y - 68$	η^2	$\xi\eta$
65	-2	4	67	-1	1	2
66	-1	1	68	0	0	0
67	0	0	65	-3	9	0
67	0	0	68	0	0	0
68	1	1	72	4	16	4
69	2	4	72	4	16	8
70	3	9	69	1	1	3
72	5	25	71	3	9	15
	8	44		8	52	32

The standard deviation of X is

$$\begin{aligned}\sigma_X = \sigma_\xi &= \sqrt{\frac{1}{n} \sum \xi^2 - \left(\frac{\sum \xi}{n}\right)^2} = \sqrt{\frac{1}{8}(44) - \left(\frac{8}{8}\right)^2} \\ &= \sqrt{\frac{11}{2} - 1} = \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} = 2.1213\end{aligned}$$

Similarly, the standard deviation of Y is

$$\begin{aligned}\sigma_Y = \sigma_\eta &= \sqrt{\frac{1}{n} \sum \eta^2 - \left(\frac{\sum \eta}{n}\right)^2} = \sqrt{\frac{1}{8}(44) - \left(\frac{8}{8}\right)^2} \\ &= \sqrt{\frac{13}{2} - 1} = \sqrt{\frac{11}{2}} = \sqrt{5.5} = 2.3452\end{aligned}$$

The covariance between X and Y is

$$\text{Cov}(X, Y) = \text{Cov}(\xi, \eta) = \frac{1}{n} \sum \xi \eta - \left(\frac{\sum \xi}{n}\right) \left(\frac{\sum \eta}{n}\right)$$

Substituting from the table, we have

$$\text{Cov}(X, Y) = \frac{32}{8} - \left(\frac{8}{8}\right) \left(\frac{8}{8}\right) = 4 - 1 = 3$$

Thus, the correlation coefficient between X and Y is given by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{3}{2.1213 \times 2.3452} = \frac{3}{4.9749} = 0.6030$$

◆

EXAMPLE 4.129. From the given data, find

- (i) the two regression equations.
- (ii) the coefficient of correlation between the marks in Economics and Statistics.
- (iii) the most likely marks in Statistics when the marks in Economics are 30.

Marks in Economics	25	28	35	32	31	36	29	38	34	32
Marks in Statistics	43	46	49	41	36	32	31	30	33	39

(Anna, May 2006; May 2007)

Solution. Let X and Y denote the marks in Economics and Statistics, respectively. Let $\xi = X - 31$ and $\eta = Y - 41$.

X	$\xi = X - 31$	ξ^2	Y	$\eta = Y - 41$	η^2	$\xi\eta$
25	-6	36	43	2	4	-12
28	-3	9	46	5	25	-15
35	4	16	49	8	64	32
32	1	1	41	0	0	0
31	0	0	36	-5	25	0
36	5	25	32	-9	81	-45
29	-2	4	31	-10	100	20
38	7	49	30	-11	121	-77
34	3	9	33	-8	64	-24
32	1	1	39	-2	4	-2
	10	150		-30	488	-123

(i) Here, we obtain the two regression equations.

From the table, we find that

$$\mu_X = A + \frac{\sum \xi}{n} = 31 + \frac{10}{10} = 31 + 1 = 32$$

and

$$\mu_Y = B + \frac{\sum \eta}{n} = 41 - \frac{30}{10} = 41 - 3 = 38$$

The standard deviation of X is given by

$$\begin{aligned}\sigma_X = \sigma_\xi &= \sqrt{\frac{1}{n} \sum \xi^2 - \left(\frac{\sum \xi}{n}\right)^2} = \sqrt{\frac{1}{10} (150) - \left(\frac{10}{10}\right)^2} \\ &= \sqrt{15 - 1} = \sqrt{14} = 3.7417\end{aligned}$$

Similarly, the standard deviation of Y is

$$\begin{aligned}\sigma_Y = \sigma_\eta &= \sqrt{\frac{1}{n} \sum \eta^2 - \left(\frac{\sum \eta}{n}\right)^2} = \sqrt{\frac{1}{10} (488) - \left(\frac{-30}{10}\right)^2} \\ &= \sqrt{48.8 - 9} = \sqrt{39.8} = 6.3087\end{aligned}$$

The covariance between X and Y

$$\text{Cov}(X, Y) = \text{Cov}(\xi, \eta) = \frac{1}{n} \sum \xi\eta - \left(\frac{\sum \xi}{n}\right) \left(\frac{\sum \eta}{n}\right)$$

Substituting from the table, we have

$$\text{Cov}(X, Y) = -\frac{123}{10} - \left(\frac{10}{10}\right) \left(-\frac{30}{10}\right) = -12.3 + 3 = -9.3$$

Thus, the regression coefficient of Y on X is

$$\beta_{YX} = \frac{\text{Cov}(X, Y)}{\sigma_X^2} = -\frac{9.3}{14} = -0.6643$$

and the regression coefficient of X on Y is

$$\beta_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_Y^2} = -\frac{9.3}{39.8} = -0.2337$$

Thus, the regression line of Y on X is given by

$$y - \mu_Y = \beta_{YX} (x - \mu_X)$$

i.e.

$$y - 38 = -0.6643(x - 32)$$

i.e.

$$y = -0.6643x + 59.2576$$

and the regression line of X on Y is given by

$$x - \mu_X = \beta_{XY} (y - \mu_Y)$$

i.e.

$$x - 32 = -0.2337(y - 38)$$

i.e.

$$x = -0.2337y + 40.8806$$

(ii) The correlation coefficient between X and Y is given by

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = -\frac{9.3}{3.7417 \times 6.3087} = -\frac{9.3}{23.6053} = -0.3940$$

(iii) The most likely marks in Statistics (Y) when marks in Economics (X) are 30 are obtained by considering the regression line of Y on X , i.e.

$$y = -0.6643x + 59.2576$$

and substituting $X = 30$ in it. Thus, we obtain the corresponding most likely marks in Statistics as

$$Y = -0.6643(30) + 59.2576 = -19.9290 + 59.2576 = 39.3286 \approx 39$$



PROBLEM SET 4.9

1. In partially destroyed records, the following data are available: $\sigma_X = 4$. Regression equation of X on Y is $5x - 2y = 17$ and the regression equation of Y on X is $4x - 3y = 8$. Find μ_X, μ_Y , the correlation coefficient ρ and σ_Y .

2. If the equations of the two lines of regression of Y on X and X on Y are, respectively, $4x - 3y = 10$ and $5x - y = 18$, and the second moment of X about the origin is 20, find
- The correlation coefficient, ρ .
 - The regression coefficients, β_{XY} and β_{YX} .
 - The standard deviation of Y .
3. Two random variables X and Y have the regression lines

$$2x + y = 13 \quad \text{and} \quad 3x + 2y = 21$$

- Find μ_X and μ_Y .
 - Find ρ , the correlation coefficient.
 - Determine the ratio of the two standard deviations. In particular, if $\sigma_X = 2$, obtain the value of σ_Y .
4. The joint probability distribution of X and Y is given below:

X \ Y	0	1
0	0.4	0.1
1	0.2	0.3

Find the correlation coefficient of (X, Y) .

5. The joint probability distribution of X and Y is given below:

X \ Y	0	1	2
0	0.1	0.1	0.2
1	0.2	0	0.1
2	0.1	0.1	0.1

- Find the correlation coefficient of (X, Y) .
 - Obtain the regression lines of X and Y .
6. Two random variables X and Y have the joint probability density function

$$f(x, y) = \begin{cases} \frac{1}{8}(x+y) & \text{for } 0 < x < 2, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the following:

- The correlation coefficient, ρ .
 - The regression curves of X and Y .
7. If (X, Y) is a two-dimensional random variable with the joint probability density function

$$f(x, y) = \begin{cases} 8xy & \text{for } 0 < y < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the following:

- (a) The correlation coefficient, ρ .
- (b) The regression curves of X and Y .

8. The following marks have been obtained by a random sample of students in a class in Calculus.

Calculus I	50	75	80	72	60	85	55	90
Calculus II	45	70	60	70	70	90	60	80

Find the coefficient of correlation and obtain the regression lines of X and Y .

9. The following table gives the heights (in inches) of mothers (X) and their daughters (Y).

X	60	65	62	64	65	69	68	70
Y	64	64	68	69	70	65	70	72

Find the correlation coefficient and the regression lines of X and Y .

10. The heights (in inches) and weights (in kg) of a random sample of 8 male students in a University are shown in the following data:

Height	65	62	60	64	68	70	68	65
Weight	75	70	65	72	75	80	72	64

Find the following:

- (a) The correlation coefficient, ρ
- (b) The regression lines of X and Y
- (c) The most likely weight of the student if his height is 6 feet.

4.10 CENTRAL LIMIT THEOREM

Central Limit Theorem (CLT) is an important result in Statistics, which states that the normal distribution is the limiting distribution of the sum of independent random variables with finite variance as the number of these random variables gets indefinitely large. Since many real processes yield distributions with finite variance, this theorem has a wide range of applications in Sampling Theory and other areas of Statistics.

The first version of CLT was stated by the English mathematician Abraham de Moivre (1667–1754), which was generalized by Laplace in 1812. In 1901, the Russian mathematician A.M. Lyapunov (1857–1918) postulated the Central Limit Theorem in general terms and provided a rigorous mathematical proof for the same.

We state without proof the statement of the general Central Limit Theorem. (We provide a proof for only an important particular case of this famous theorem.)

Theorem 4.26. (*Central Limit Theorem*) If $X_i (i = 1, 2, \dots, n)$ be independently distributed random variables such that

$$E(X_i) = \mu_i \text{ and } \text{Var}(X_i) = \sigma_i^2 \text{ for } i = 1, 2, \dots, n$$

then, as $n \rightarrow \infty$, the distribution of the sum of these random variables, namely,

$$S_n = X_1 + X_2 + \dots + X_n$$

tends to the normal distribution with mean μ and variance σ^2 , where

$$\mu = \sum_{i=1}^n \mu_i \text{ and } \sigma^2 = \sum_{i=1}^n \sigma_i^2$$

We prove a special case of Theorem 4.26, known as *Lindeberg-Levy Central Limit Theorem*, which can be stated as follows.

Theorem 4.27. (*Lindeberg-Levy Central Limit Theorem*) If $X_1, X_2, \dots, X_n$ are independently and identically distributed random variables with

$$E(X_i) = \mu \text{ and } \text{Var}(X_i) = \sigma^2 \text{ for } i = 1, 2, \dots, n$$

then as $n \rightarrow \infty$, the distribution of the sum of these random variables

$$S_n = X_1 + X_2 + \dots + X_n$$

tends to the normal distribution with mean $n\mu$ and variance $n\sigma^2$.

(Anna, Nov. 2007)

Proof. By the addition theorem of expectation, we have

$$E(S_n) = E(X_1) + E(X_2) + \dots + E(X_n) = n\mu$$

Since the random variables X_i ($i = 1, 2, \dots, n$) are independent, it follows that

$$\text{Var}(S_n) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n) = n\sigma^2$$

Then, we define the standardized random variable

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma}$$

We claim that as $n \rightarrow \infty$, Z_n approaches the standard normal distribution with mean 0 and variance 1.

Let $M_1(t)$ denote the moment generating distribution of the deviation $Y_1 = X_1 - \mu$.

The first two raw moments of Y_1 are given by

$$\mu'_1 = E(Y_1) = E(X_1 - \mu) = E(X_1) - \mu = 0$$

and

$$\mu'_2 = E(Y_1^2) = E((X_1 - \mu)^2) = \sigma^2$$

Hence, it follows that

$$M_1(t) = E(e^{tY_1}) = E(e^{t(X_1 - \mu)}) = 1 + \mu'_1 t + \mu'_2 \frac{t^2}{2} + \mu'_3 \frac{t^3}{3!} + \dots$$

Substituting the values of μ'_1 and μ'_2 in the above, we get

$$M_1(t) = 1 + \frac{\sigma^2 t^2}{2} + O(t^3) \quad (4.17)$$

where $O(t^3)$ contains terms with t^3 and higher powers of t .

Note that

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma} = \frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sqrt{n}\sigma} = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sqrt{n}\sigma} \right)$$

Since the random variables X_i ($i = 1, 2, \dots, n$) are independent and identically distributed, it follows that the moment generating function (MGF) of Z_n is given by

$$M_{Z_n}(t) = E(e^{tZ_n}) = \prod_{i=1}^n E\left(e^{t\left(\frac{X_i - \mu}{\sqrt{n}\sigma}\right)}\right) = \left[E\left(e^{t\left(\frac{X_1 - \mu}{\sqrt{n}\sigma}\right)}\right)\right]^n = \left[M_1\left(\frac{t}{\sqrt{n}\sigma}\right)\right]^n \quad (4.18)$$

where $M_1(t)$ is the (MGF) of the deviation $Y_1 = X_1 - \mu$.

By Eq. (4.17), it follows that

$$M_1\left(\frac{t}{\sqrt{n}\sigma}\right) = 1 + \frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right) \quad (4.19)$$

From the Eqs. (4.18) and (4.19), we have

$$M_{Z_n}(t) = \left[1 + \frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right)\right]^n$$

For every fixed t , the terms $O\left(n^{-\frac{3}{2}}\right) \rightarrow 0$ as $n \rightarrow \infty$. Hence, it follows that

$$\lim_{n \rightarrow \infty} M_{Z_n}(t) = \lim_{n \rightarrow \infty} \left[1 + \frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right)\right]^n = e^{\frac{t^2}{2}},$$

which is the MGF of the standard normal distribution.

Hence, by the uniqueness theorem of moment generating functions, it follows that $Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma}$ is asymptotically normal with mean 0 and variance 1. Equivalently, the sum S_n is asymptotically normal with mean $n\mu$ and variance $n\sigma^2$. $\blacklozenge$

Corollary 4.3. If $X_1, X_2, \dots, X_n$ are independently and identically distributed random variables with

$$E(X_i) = \mu \text{ and } \text{Var}(X_i) = \sigma^2 \text{ for } i = 1, 2, \dots, n$$

then as $n \rightarrow \infty$, the distribution of the sample mean of these random variables

$$\bar{X} = \frac{S_n}{n} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

tends to the normal distribution with mean μ and variance $\frac{\sigma^2}{n}$.

Proof. By CLT, it follows that S_n is asymptotically normal with mean $n\mu$ and variance $n\sigma^2$. Hence, it is immediate that $\bar{X}$ also is asymptotically normal with

$$\text{Mean}(\bar{X}) = E(\bar{X}) = E\left(\frac{S_n}{n}\right) = \frac{E(S_n)}{n} = \frac{n\mu}{n} = \mu$$

and

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{S_n}{n}\right) = \frac{1}{n^2} \text{Var}(S_n) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n} \quad \blacklozenge$$

EXAMPLE 4.130. State and prove the central limit theorem for a sequence of independent and identically distributed random variables.

(Anna, May 2006)

Solution. See the statement and proof of Theorem 4.27. ◆

EXAMPLE 4.131. A random sample of size 100 is taken from a population whose mean is 60 and variance is 400. Using central limit theorem, with what probability can we assert that the mean of the sample will not differ from $\mu = 60$ by more than 4?

(Anna, April 2003; May 2007)

Solution. Let n be the size of the sample.

Given that $n = 100$, $E(X_i) = \mu_i = 60$ and $\text{Var}(X_i) = \sigma_i^2 = 400$ for $i = 1, 2, \dots, 100$

Let $\bar{X}$ denote the sample mean, i.e.

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

By Central Limit Theorem, we know that, in the limits, $\bar{X} \sim N(\mu, \sigma^2)$, where

$$\mu = \frac{n\mu_i}{n} = \mu_i \text{ and } \sigma^2 = \frac{n\sigma_i^2}{n^2} = \frac{\sigma_i^2}{n}$$

Substituting the values, it follows that

$$\mu = 60 \text{ and } \sigma^2 = \frac{\sigma_i^2}{100} = \frac{400}{100} = 4$$

Thus, $\mu = 60$ and $\sigma = 2$. The standard normal variate is defined by

$$Z = \frac{\bar{X} - \mu}{\sigma} = \frac{\bar{X} - 60}{2}$$

The required probability is

$$P(|\bar{X} - \mu| \leq 4) = P\left(\frac{|\bar{X} - \mu|}{2} \leq 2\right) = P(|Z| \leq 2)$$

Since the standard normal distribution $Z \sim N(0, 1)$ is symmetrical about the line $z = 0$, it is immediate that

$$P(|\bar{X} - \mu| \leq 4) = 2P(0 \leq Z \leq 2) = 2 \times 0.4772 = 0.9544 \quad \text{◆}$$

EXAMPLE 4.132. A random sample of size 100 is taken from a population whose mean is 80 and variance is 400. Using CLT, with what probability can we assert that the mean of the sample will not differ from $\mu = 80$ by more than 6?

(Madras, Oct. 2000)

Solution. Let n be the size of the sample.

Given that $n = 100$, $E(X_i) = \mu_i = 80$ and $\text{Var}(X_i) = \sigma_i^2 = 400$ for $i = 1, 2, \dots, 100$
Let $\bar{X}$ denote the sample mean, i.e.

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

By Central Limit Theorem, we know that, in the limits, $\bar{X} \sim N(\mu, \sigma^2)$, where

$$\mu = \frac{n\mu_i}{n} = \mu_i \text{ and } \sigma^2 = \frac{n\sigma_i^2}{n^2} = \frac{\sigma_i^2}{n}$$

Substituting the values, it follows that

$$\mu = 80 \text{ and } \sigma^2 = \frac{\sigma_i^2}{100} = \frac{400}{100} = 4$$

Thus, $\mu = 80$ and $\sigma = 2$. The standard normal variate is defined by

$$Z = \frac{\bar{X} - \mu}{\sigma} = \frac{\bar{X} - 80}{2}$$

The required probability is

$$P(|\bar{X} - \mu| \leq 6) = P\left(\frac{|\bar{X} - \mu|}{2} \leq 3\right) = P(|Z| \leq 3)$$

Since the standard normal distribution $Z \sim N(0, 1)$ is symmetrical about the line $z = 0$, it is immediate that

$$P(|\bar{X} - \mu| \leq 6) = 2P(0 \leq Z \leq 3) = 2 \times 0.49865 = 0.9973 \quad \blacklozenge$$

EXAMPLE 4.133. If $X_1, X_2, \dots, X_n$ are Poisson variables with parameter $\lambda = 2$, use the Central Limit Theorem to estimate $P(120 \leq S_n \leq 160)$, where $S_n = X_1 + X_2 + \dots + X_n$, $n = 75$.

(Madras, April 2002; Anna, Nov. 2007)

Solution. Recall that for a Poisson variable, the mean and variance are equal to the parameter of the distribution.

Since $X_1, X_2, \dots, X_n$ are identically distributed Poisson variables with parameter λ , it follows by Central Limit Theorem that S_n is approximately normal with

$$E(S_n) = \mu = n\lambda = 75 \times 2 = 150 \text{ and } \text{Var}(S_n) = \sigma^2 = n\lambda = 75 \times 2 = 150$$

The standard normal variate is defined by

$$Z = \frac{S_n - \mu}{\sigma} = \frac{S_n - 150}{\sqrt{150}}$$

When $S_n = 120$, $Z = \frac{120-150}{\sqrt{150}} = -2.4495$.

When $S_n = 160$, $Z = \frac{160-150}{\sqrt{150}} = 0.8165$.

Hence, the required probability is equal to

$$P(120 \leq S_n \leq 160) = P(-2.4495 \leq Z \leq 0.8165) = P(-2.4495 \leq Z \leq 0) + P(0 \leq Z \leq 0.8165)$$

Since the standard normal distribution is symmetrical about the line $z = 0$, it follows that

$$\begin{aligned} P(120 \leq S_n \leq 160) &= P(0 \leq Z \leq 2.4495) + P(0 \leq Z \leq 0.8165) \\ &= 0.4929 + 0.2939 = 0.7868 \end{aligned}$$

◆

EXAMPLE 4.134. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with mean 3 and variance $\frac{1}{2}$. Use Central Limit Theorem to estimate $P(340 \leq S_n \leq 370)$, where $S_n = X_1 + X_2 + \dots + X_n$ and $n = 120$.

Solution. By Central Limit Theorem, it follows that S_n is approximately normal with

$$E(S_n) = \mu = n\mu_i = 120 \times 3 = 360 \quad \text{and} \quad \text{Var}(S_n) = \sigma^2 = n\sigma_i^2 = 120 \times \frac{1}{2} = 60$$

The standard normal variate is defined by

$$Z = \frac{S_n - \mu}{\sigma} = \frac{S_n - 360}{\sqrt{60}}$$

$$\text{When } S_n = 340, Z = \frac{340 - 360}{\sqrt{60}} = -2.582.$$

$$\text{When } S_n = 370, Z = \frac{370 - 360}{\sqrt{60}} = 1.2910.$$

Hence, the required probability is equal to

$$P(340 \leq S_n \leq 370) = P(-2.582 \leq Z \leq 1.291) = P(-2.582 \leq Z \leq 0) + P(0 \leq Z \leq 1.291)$$

Since the standard normal distribution is symmetrical about the line $z = 0$, it follows that

$$\begin{aligned} P(320 \leq S_n \leq 450) &= P(0 \leq Z \leq 2.582) + P(0 \leq Z \leq 1.291) \\ &= 0.4951 + 0.4015 = 0.8966 \end{aligned}$$

◆

EXAMPLE 4.135. The lifetime of a certain type of electric bulb may be considered as an exponential random variable with mean 50 hours. Using Central Limit Theorem, find the approximate probability that 100 of these electric bulbs will provide a total of more than 6000 hours of burning time.

Solution. Let X_i be the lifetime of the i th lamp, where $i = 1, 2, \dots, 100$. Given that X_i is an exponential random variable with $\text{Mean}(X_i) = \mu_i = 50$. Thus, by Remark 3.13, we know that $\text{Var}(X_i) = \sigma_i^2 = 50^2 = 2500$.

Define $S_n = X_1 + X_2 + \dots + X_n$ ($n = 100$), where S_n denotes the total lifetime of the given type of 100 electric bulbs.

By Central Limit Theorem, S_n is approximately normal with

$$\text{Mean}(S_n) = \mu = n\mu_i = 100 \times 50 = 5000 \quad \text{and} \quad \text{Var}(S_n) = \sigma^2 = n\sigma_i^2 = 100 \times 2500 = 250000$$

We are asked to estimate $P(S_n \geq 6000)$.

$$Z = \frac{S_n - \mu}{\sigma} = \frac{S_n - 5000}{\sqrt{250000}} = \frac{S_n - 5000}{500}$$

Thus, the required probability is equal to

$$P(S_n > 6000) = P(Z > 2) = 0.5 - P(0 < Z < 2) = 0.5 - 0.4772 = 0.0228$$

Solution. Let p be the probability of getting head in a single trial. Then we know that $p = \frac{1}{2}, q = \frac{1}{2}$ and the number of heads in 200 trials follows a binomial distribution with mean

$$\mu = np = 200 \times \frac{1}{2} = 100$$

$$\sigma^2 = npq = 200 \times \frac{1}{2} \times \frac{1}{2} = 50$$

We define the standard normal variate as

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 100}{\sqrt{50}}$$

When $X = 120$, $Z = \frac{120-100}{\sqrt{50}} = 2.8284$

The required probability is hence calculated using Central Limit Theorem as

$$P(80 \leq X \leq 120) = P(-2.8284 \leq Z \leq 2.8284) = 2P(0 \leq Z \leq 2.8284) = 2 \times 0.4977 = 0.9954$$

(Anna, May 2006)

[illegible]

A simple calculation shows that

$$E(X_i) = \mu_i = \frac{7}{2} \text{ and } \text{Var}(X_i) = \sigma_i^2 = \frac{35}{12}$$

Given that $n = 20$.

Since $X_1, X_2, \dots, X_n$ are identically distributed with mean μ_i and variance σ_i^2 , it follows by Central Limit Theorem that $S_n = X_1 + X_2 + \dots + X_n$ is approximately normal with

$$E(S_n) = \mu = n\mu_i = 20 \times \frac{7}{2} = 70 \text{ and } \text{Var}(S_n) = \sigma^2 = n\sigma_i^2 = 20 \times \frac{35}{12} = \frac{175}{3}$$

The standard normal variate is defined by

$$Z = \frac{S_n - \mu}{\sigma} = \frac{S_n - 70}{\sqrt{\frac{175}{3}}} = \frac{S_n - 70}{7.6376}$$

When $S_n = 65$, $Z = \frac{65-70}{7.6376} = -0.6547$.

When $S_n = 75$, $Z = \frac{75-70}{7.6376} = 0.6547$.

Hence, the required probability is equal to

$$P(65 \leq S_n \leq 75) = P(-0.6547 \leq Z \leq 0.6547)$$

Since the standard normal distribution is symmetrical about the line $z = 0$, it follows that

$$P(65 \leq S_n \leq 75) = 2P(0 \leq Z \leq 0.6547) = 2 \times 0.2422 = 0.4844$$

◆

EXAMPLE 4.138. A distribution with unknown mean μ has variance equal to 1.5. Use Central Limit Theorem to find how large a sample should be taken from the distribution in order that the probability will be at least 0.95 that the sample mean will be within 0.5 of the population mean.

(Madras, Oct. 1999; Anna, April 2003, Nov. 2004)

Solution. Let n be the size of the sample, and a typical member of the sample be X_i .

Given that $E(X_i) = \mu$ and $\text{Var}(X_i) = 1.5$.

Let $\bar{X}$ denote the sample mean. That is,

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Since the variables X_i are independent, it follows that

$$\text{Mean}(\bar{X}) = \frac{n\mu}{n} = \mu \text{ and } \text{Var}(\bar{X}) = \sigma^2 = \frac{n\text{Var}(X_i)}{n^2} = \frac{1.5}{n}$$

By Central Limit Theorem, it follows that as $n \rightarrow \infty$, $\bar{X}$ is normal with mean μ and variance σ^2 . Define the standard normal variable

$$Z = \frac{\bar{X} - \mu}{\sigma} = \frac{\bar{X} - \mu}{\sqrt{\frac{1.5}{n}}}$$

We have to find the sample size n such that

$$P(|\bar{X} - \mu| < 0.5) \geq 0.95$$

or equivalently

$$P\left(|Z| < \frac{0.5}{\sqrt{\frac{1.5}{n}}}\right) \geq 0.95$$

or equivalently

$$P(|Z| < 0.4082\sqrt{n}) \geq 0.95$$

or equivalently

$$2P(0 < Z < 0.4082\sqrt{n}) \geq 0.95$$

or equivalently

$$P(0 < Z < 0.4082\sqrt{n}) \geq 0.475$$

Using the tables, it follows that

$$0.4082\sqrt{n} = 1.96$$

or equivalently

$$n = \left[\frac{1.96}{0.4082} \right]^2 = [4.8016]^2 = 23.0554$$

Hence, the size of the sample should be at least 24.

◆

PROBLEM SET 4.10

1. A random sample of size 200 is taken from a population whose mean is 50 and variance is 600. Using Central Limit Theorem, find the probability that the mean of the sample will not differ from $\mu = 50$ by more than 5.
2. A random sample of size 150 is taken from a population whose mean is 70 and variance is 900. Using Central Limit Theorem, find the probability that the mean of the sample will not differ from $\mu = 70$ by more than 4.
3. The life time of a certain brand of an electric gadget may be considered a random variable with mean 1000 hours and standard deviation 400 hours. Using Central Limit Theorem, find the probability that the average life-time of 50 gadgets exceeds 1100 hours.
4. If $X_1, X_2, \dots, X_n$ are Poisson variables with parameter $\lambda = 3$, use the Central Limit Theorem to estimate $P(260 \leq S_n \leq 320)$, where $S_n = X_1 + X_2 + \dots + X_n$, $n = 100$.
5. A coin is tossed 300 times. Using Central Limit Theorem, find the approximate probability that heads will appear more than 140 times but less than 170 times.
6. 30 dice are thrown. Using Central Limit Theorem, find the approximate probability that the sum obtained is between 90 and 120.

Chapter 5

Random Processes

Numerous problems in the world of science and engineering deal with time waveforms or signals. These signals may be divided into two classes - *deterministic* and *random* signals. The deterministic signals are usually described by mathematical functions with time t as the independent variable. However, a random signal is not purely given by the value of the time t because the signal is often accompanied by an undesired random waveform, for example, the noise. The performance of the communication system is limited whenever the noise interferes with the signals communicated. The *random process* is basically a probabilistic model that is used to characterize random signals. The subject of random processes facilitates the description and analysis of the random signals in a probabilistic sense.

5.1 DEFINITION AND DESCRIPTION OF RANDOM PROCESSES

We begin with an example to motivate the concept of the random process.

EXAMPLE 5.1. Consider the random experiment of tossing a fair coin. It has two outcomes H and T , with equal probability. We can define a random variable, X , by defining X to be the number of heads obtained. Thus, X takes two values given by

$$\begin{aligned} X(s) &= 1 & \text{if } s = H \\ X(s) &= 0 & \text{if } s = T \end{aligned}$$

Thus, a *random variable* X assigns a *real number* $X(s)$ to each outcome s of the experiment. On the other hand, a *random process* X assigns a *real function of time* t to each outcome s of the experiment, where this function is denoted by $X(t, s)$. For instance, a random process X for the given experiment is defined by

$$\begin{aligned} X(t, s) &= X_1(t) = t & \text{if } s = H \\ X(t, s) &= X_2(t) = \sin(2t) & \text{if } s = T \end{aligned}$$

Now, we formally define the random processes as follows.

Definition 5.1. Consider a random experiment with sample space S . If a time function $X(t, s)$ is assigned to each outcome $s \in S$ and where $t \in T$ (the parameter set or the index set), then the family of all such functions, denoted by $X(t, s)$, where $s \in S$ and $t \in T$, is called a *random process*. For a fixed t , say $t = \tau$, $X(\tau, s)$ is a random variable as s varies over the sample space S . On the other hand, for a fixed s , say $s = s_i$, $X(t, s_i) = X_i(t)$ is a single function of time t , called a *sample function* or *ensemble member* or a *realization* of the random process. The totality of all sample functions is called an *ensemble*. Also, if both t and s are fixed, then $X(t, s)$ just represents a real number.

Remark 5.1. For a random process $\{X(t, s) \mid t \in T, s \in S\}$, the dependence on the sample point s is clearly understood. Hence, hereafter, we shall represent the random process simply as $\{X(t) \mid t \in T\}$.

5.1.1 Classification of Random Processes

Consider a random process $\{X(t), t \in T\}$, where T is the index set or parameter set. The values assumed by $X(t)$ are called the *states*, and the set of all possible values of the states forms the *state space* E of the random process.

If the index set T of a random process is discrete, then the process is called a *discrete-parameter* or *discrete-time process*. A discrete-parameter process is also called a *random sequence* and is denoted by $\{X_n, n = 1, 2, \dots\}$. If T and E are both discrete, then the random process is called a *discrete random sequence*. For example, if X_n represents the number of heads obtained in the n th toss of two fair coins, then $\{X_n, n \geq 1\}$ is a discrete random sequence, since $T = \{1, 2, \dots\}$ and $E = \{0, 1, 2\}$. If T is discrete, and E is continuous, then the random process is called a *continuous random sequence*. For example, if X_n represents the temperature at the end of the n th hour of a day, then the states, $X_n, (n = 1, 2, \dots, 24)$ can take any value in an interval and hence are continuous.

If the state space E of a random process is discrete, then the random process is called a *discrete random process* or *discrete-state process*. A discrete random process is also called a *chain*. In this case, the state space E is often assumed to be $\{0, 1, 2, \dots\}$. For example, if $X(t)$ represents the number of SMS messages received in a cellphone in the interval $(0, t)$, then $\{X(t)\}$ is a discrete random process, since $E = \{0, 1, 2, \dots\}$. If the state space E of a random process is continuous, then the random process is called a *continuous random process* or *continuous-state process*. For example, if $X(t)$ is the maximum temperature recorded in a city in the interval $(0, t)$, then $\{X(t)\}$ is a continuous random process.

EXAMPLE 5.2. State the four types of stochastic processes.

(Anna, April 2004)

Solution. The four types of stochastic (or random) processes are

- (i) Discrete time, discrete state random processes (or discrete random sequences).
- (ii) Discrete time, continuous state random processes (or continuous random sequences).
- (iii) Continuous time, discrete state random processes (or discrete random processes).
- (iv) Continuous time, continuous state random processes (or continuous random processes).

◆

In addition to the above four types of random processes, we have two classifications of random processes based on the form of their sample functions.

Definition 5.2. (*Deterministic and Nondeterministic Random Processes*) A random process is called a *deterministic random process* if future values of any sample function can be predicted exactly from past observations or past values. For example

$$X(t) = A \cos \omega_0 t$$

where A is a uniform random variable and ω_0 is a constant.

A random process is called a *nondeterministic random process* if it is not deterministic, i.e. if future values of any sample function cannot be predicted exactly from past observations or past values.

5.1.2 Description of Random Processes

As stated in the definition, a random process becomes a random variable when the time t is fixed at some particular time instant. Thus, the random variable $X(t)$ is characterized by the probability density function and hence it is possible to describe the statistical properties of $X(t)$ such as the mean value, moments, and variance. If two random variables are obtained for the process for two time instants, say t_1 and t_2 , then we can use the joint probability density function of $X(t_1)$ and $X(t_2)$ to describe their statistical properties such as their means, variances, and joint moments. More generally, n random variables will possess statistical properties described by their n -dimensional joint probability density function.

Consider a random process $\{X(t), t \in T\}$, where T is the parameter set. For a fixed time t_1 , $X(t_1) = X_1$ is a random variable and its cumulative distribution function will be denoted by $F_X(x_1; t_1)$ and defined as

$$F_X(x_1; t_1) = P\{X(t_1) \leq x_1\}$$

$F_X(x_1; t_1)$ is called the *first-order distribution* of the random process $X(t)$.

Similarly, given t_1 and t_2 , $X(t_1) = X_1$ and $X(t_2) = X_2$ are two random variables. Their joint cumulative distribution function is called the *second-order distribution* of the random process $X(t)$ and is defined as

$$F_X(x_1, x_2; t_1, t_2) = P\{X(t_1) \leq x_1, X(t_2) \leq x_2\}$$

We can extend this to n random variables $X(t_1) = X_1, \dots, X(t_n) = X_n$. Thus, the *nth order distribution* of the random process $X(t)$ is defined as

$$F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = P\{X(t_1) \leq x_1, X(t_2) \leq x_2, \dots, X(t_n) \leq x_n\}$$

If $X(t)$ is a discrete-time random process, i.e. a random sequence, then $X(t)$ is specified by the joint probability mass function defined as follows:

$$p_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = P(X(t_1) = x_1, X(t_2) = x_2, \dots, X(t_n) = x_n)$$

If $X(t)$ is a continuous-time random process, then $X(t)$ is specified by the joint probability density function defined as follows:

$$f_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = \frac{\partial^n F_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n)}{dt_1 dt_2 \cdots dt_n}$$

5.1.3 Mean, Correlation and Covariance Functions

For a fixed value of t , $X(t)$ is a random variable. Thus, the *mean* of $X(t)$ is denoted by $\mu_X(t)$ and defined as

$$\mu_X(t) = E[X(t)]$$

In general, $\mu_X(t)$ is a function of time, and it is often called the *ensemble average* of $X(t)$.

Given two time instants t_1 and t_2 , $X(t_1) = X_1$ and $X(t_2) = X_2$ are random variables, and the correlation between them is denoted by $R_{XX}(t_1, t_2)$ and defined as

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

$R_{XX}(t_1, t_2)$ is called the *autocorrelation function* of the process $X(t)$, which serves as a measure of the dependence among the random variables of $X(t)$.

Next, the covariance between $X(t_1) = X_1$ and $X(t_2) = X_2$ is denoted by $C_{XX}(t_1, t_2)$ and defined as

$$C_{XX}(t_1, t_2) = \text{Cov}(X(t_1), X(t_2)) = E\{[X(t_1) - \mu_X(t_1)][X(t_2) - \mu_X(t_2)]\}$$

$C_{XX}(t_1, t_2)$ is called the *autocovariance function* of the process $X(t)$.

We note also that the *variance* of $X(t)$ is denoted by $\text{Var}(X(t))$ or $\sigma_X^2(t)$ and defined as

$$\text{Var}(X(t)) = \sigma_X^2(t) = C_{XX}(t, t) = E\{[X(t) - \mu_X(t)]^2\}$$

Also, the *correlation coefficient* of $X(t)$ is denoted by $\rho_{XX}(t_1, t_2)$ and defined as

$$\rho_{XX}(t_1, t_2) = \rho(X(t_1), X(t_2)) = \frac{C_{XX}(t_1, t_2)}{\sqrt{\sigma_X^2(t_1)\sigma_X^2(t_2)}}$$

If $X(t)$ is a complex random process, then the *autocorrelation function* $R_{XX}(t_1, t_2)$ and the *autocovariance function* $C_{XX}(t_1, t_2)$ are defined, respectively, by

$$R_{XX}(t_1, t_2) = E[X(t_1)X^*(t_2)]$$

and

$$C_{XX}(t_1, t_2) = E\{[X(t_1) - \mu_X(t_1)][X(t_2) - \mu_X(t_2)]^*\}$$

where $\star$ denotes the complex conjugate.

The following theorem gives some important properties of the various functions defined above.

Theorem 5.1. For any random process $X(t)$ with the autocorrelation function $R_{XX}(t_1, t_2)$ and autocovariance function $S_{XX}(t_1, t_2)$, the following properties hold:

- (i) $R_{XX}(t_1, t_2) = R_{XX}(t_2, t_1)$.
- (ii) $R_{XX}(t_1, t_1) = E[X^2(t_1)]$.
- (iii) $C_{XX}(t_1, t_2) = C_{XX}(t_2, t_1)$.
- (iv) $C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2)$.
- (v) If $\mu_X(t) = 0$ for all t , then $C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2)$ for all t_1, t_2 .
- (vi) $|\rho_{XX}(t_1, t_2)| \leq 1$.

Proof. The properties stated in the theorem are proved as follows:

(i) $R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = E[X(t_2)X(t_1)] = R_{XX}(t_2, t_1)$

(ii) $R_{XX}(t_1, t_1) = E[X(t_1)X(t_1)] = E[X^2(t_1)]$

(iii) Note that

$$\begin{aligned} C_{XX}(t_1, t_2) &= E\{[X(t_1) - \mu_X(t_1)][X(t_2) - \mu_X(t_2)]\} \\ &= E\{[X(t_2) - \mu_X(t_2)][X(t_1) - \mu_X(t_1)]\} \\ &= C_{XX}(t_2, t_1) \end{aligned}$$

(iv) Note that

$$\begin{aligned} C_{XX}(t_1, t_2) &= E\{[X(t_1) - \mu_X(t_1)][X(t_2) - \mu_X(t_2)]\} \\ &= E[X(t_1)X(t_2)] - \mu_X(t_1)E[X(t_2)] - \mu_X(t_2)E[X(t_1)] + \mu_X(t_1)\mu_X(t_2) \\ &= R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) - \mu_X(t_2)\mu_X(t_1) + \mu_X(t_1)\mu_X(t_2) \\ &= R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) \end{aligned}$$

(v) If $\mu_X(t) = 0$ for all t , then it follows that

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) = R_{XX}(t_1, t_2) - 0 = R_{XX}(t_1, t_2)$$

(vi) This is an immediate consequence of Theorem 4.23. ♦

In dealing with two or more random processes, we may use the joint probability distribution functions or averages to describe the relationship between them. For this purpose, we define the following functions for the random processes $X(t)$ and $Y(t)$ of the same type.

The *cross correlation function* of the random processes $X(t)$ and $Y(t)$ is denoted by $R_{XY}(t_1, t_2)$ and defined as

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$$

where $X(t_1)$ and $Y(t_2)$ are random variables.

The *cross covariance function* of the random processes $X(t)$ and $Y(t)$ is denoted by $C_{XY}(t_1, t_2)$ and defined as

$$C_{XY}(t_1, t_2) = \text{Cov}(X(t_1), Y(t_2)) = E\{[X(t_1) - \mu_X(t_1)][Y(t_2) - \mu_Y(t_2)]\}$$

The *cross correlation coefficient* of the random processes $X(t)$ and $Y(t)$ is denoted by $\rho_{XY}(t_1, t_2)$ and defined as

$$\rho_{XY}(t_1, t_2) = \frac{C_{XY}(t_1, t_2)}{\sqrt{\sigma_X^2(t_1)\sigma_Y^2(t_2)}}$$

Two random processes $X(t)$ and $Y(t)$ are called *uncorrelated* if the cross correlation between them is zero for all values of time, i.e.

$$\rho_{XY}(t_1, t_2) = 0 \quad \text{for all } t_1, t_2$$

The following theorem gives some important properties of the various functions defined above:

Theorem 5.2. If $X(t)$ and $Y(t)$ are two random processes, then the following properties hold:

(i) $R_{XY}(t_1, t_2) = R_{YX}(t_2, t_1).$

(ii) $C_{XY}(t_1, t_2) = C_{YX}(t_2, t_1).$

- (iii) $C_{XY}(t_1, t_2) = R_{XY}(t_1, t_2) - \mu_X(t_1)\mu_Y(t_2)$.
- (iv) If $\mu_X(t) = 0$ or $\mu_Y(t) = 0$ for each t , then $C_{XY}(t_1, t_2) = R_{XY}(t_1, t_2)$ for all t_1, t_2 .
- (v) $|\rho_{XY}(t_1, t_2)| \leq 1$.

Proof. The properties stated in the theorem are proved as follows:

- (i) $R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)] = E[Y(t_2)X(t_1)] = R_{YX}(t_2, t_1)$
- (ii) Note that

$$\begin{aligned} C_{XY}(t_1, t_2) &= E\{[X(t_1) - \mu_X(t_1)][Y(t_2) - \mu_Y(t_2)]\} \\ &= E\{[Y(t_2) - \mu_Y(t_2)][X(t_1) - \mu_X(t_1)]\} \\ &= C_{YX}(t_2, t_1) \end{aligned}$$

- (iii) Note that

$$\begin{aligned} C_{XY}(t_1, t_2) &= E\{[X(t_1) - \mu_X(t_1)][Y(t_2) - \mu_Y(t_2)]\} \\ &= E[X(t_1)Y(t_2)] - \mu_X(t_1)E[Y(t_2)] - \mu_Y(t_2)E[X(t_1)] + \mu_X(t_1)\mu_Y(t_2) \\ &= R_{XY}(t_1, t_2) - \mu_X(t_1)\mu_Y(t_2) - \mu_Y(t_2)\mu_X(t_1) + \mu_X(t_1)\mu_Y(t_2) \\ &= R_{XY}(t_1, t_2) - \mu_X(t_1)\mu_Y(t_2) \end{aligned}$$

- (iv) If $\mu_X(t) = 0$ or $\mu_Y(t) = 0$ for each t , then it follows that

$$C_{XY}(t_1, t_2) = R_{XY}(t_1, t_2) - \mu_X(t_1)\mu_Y(t_2) = R_{XY}(t_1, t_2) - 0 = R_{XY}(t_1, t_2)$$

- (v) This is an immediate consequence of Theorem 4.23. ◆

EXAMPLE 5.3. Consider a random process $X(t)$ defined by

$$X(t) = A \cos(\omega t + \theta)$$

where A and θ are independent and uniform random variables over $(-k, k)$ and $(-\pi, \pi)$, respectively. Find the

- (i) Mean of $X(t)$.
- (ii) Autocorrelation function $R_{XX}(t_1, t_2)$ of $X(t)$.
- (iii) Autocovariance function $C_{XX}(t_1, t_2)$ of $X(t)$.
- (iv) Variance of $X(t)$.

Solution. We are given that A and θ are independent random variables. Since A is uniform over $(-k, k)$, the probability density function of A is given by

$$f_A(a) = \begin{cases} \frac{1}{2k} & \text{if } -k < a < k \\ 0 & \text{otherwise} \end{cases}$$

Since θ is uniform over $(-\pi, \pi)$, the probability density function of θ is given by

$$f_\theta(\theta) = \begin{cases} \frac{1}{2\pi} & \text{if } -\pi < \theta < \pi \\ 0 & \text{otherwise} \end{cases}$$

(i) The mean of $X(t)$ is given by

$$\mu_X(t) = E[A \cos(\omega t + \theta)] = E(A)E[\cos(\omega t + \theta)]$$

since A and θ are independent random variables. Since A is uniform over $(-k, k)$, we know that

$$E(A) = \frac{k + (-k)}{2} = 0$$

from which it follows that

$$\mu_X(t) = E(A)E[\cos(\omega t + \theta)] = 0$$

(ii) The autocorrelation function of $X(t)$ is given by

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = E[A \cos(\omega t_1 + \theta) \cdot A \cos(\omega t_2 + \theta)] = E[A^2 g(\theta)] = E[A^2]E[g(\theta)]$$

since A and θ are independent, where $g(\theta) = \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta)$.

Since A is uniform over $(-k, k)$, we know that

$$E[A^2] = \frac{1}{3} [k^2 + k(-k) + (-k)^2] = \frac{1}{3} [k^2 - k^2 + k^2] = \frac{1}{3} k^2$$

We also find that

$$E[g(\theta)] = \int_{-\pi}^{\pi} g(\theta) f_{\theta}(\theta) d\theta = \int_{-\pi}^{\pi} \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) \left[\frac{1}{2\pi} \right] d\theta$$

which can be written as

$$E[g(\theta)] = \frac{1}{4\pi} \int_{-\pi}^{\pi} 2 \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) d\theta$$

Using the formula $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$, we have

$$E[g(\theta)] = \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ \cos[\omega(t_1 + t_2) + 2\theta] + \cos[\omega(t_1 - t_2)] \} d\theta$$

Integrating, we have

$$E[g(\theta)] = \frac{1}{4\pi} \left\{ \frac{\sin[\omega(t_1 + t_2) + 2\theta]}{2} + \theta \cos[\omega(t_1 - t_2)] \right\}_{-\pi}^{\pi} = \frac{1}{2} \cos[\omega(t_1 - t_2)]$$

Hence, it follows that

$$R_{XX}(t_1, t_2) = E[A^2]E[g(\theta)] = \frac{1}{6} k^2 \cos[\omega(t_1 - t_2)]$$

(iii) The autocovariance function of $X(t)$ is given by

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) = R_{XX}(t_1, t_2) = \frac{1}{6} k^2 \cos[\omega(t_1 - t_2)]$$

since $\mu_X(t_1) = \mu_X(t_2) = 0$.

(iv) The variance of $X(t)$ is given by

$$\sigma_X^2(t) = C_{XX}(t, t) = \frac{1}{6} k^2$$

◆

EXAMPLE 5.4. Consider a random process $X(t) = B\cos(50t + \phi)$, where B and ϕ are independent random variables. B is a random variable with mean 0 and variance 1. ϕ is uniformly distributed in the interval $[-\pi, \pi]$. Find the mean and autocorrelation of the process.

(Anna, April 2004)

Solution. Since B is a random variable with mean 0 and variance 1, it follows that

$$E(B) = 0 \text{ and } E[B^2] = \text{Var}(B) + [E(B)]^2 = 1 + 0 = 1$$

Since ϕ is uniformly distributed in the interval $[-\pi, \pi]$, it has the probability density function

$$f(\phi) = \begin{cases} \frac{1}{2\pi} & \text{for } \phi \in [-\pi, \pi] \\ 0 & \text{elsewhere} \end{cases}$$

First, we find the mean of $X(t) = B\cos(50t + \phi)$. By definition, we have

$$\mu_X(t) = E[X(t)] = E[B\cos(50t + \phi)] = E(B)E[\cos(50t + \phi)] = 0$$

where we have used the fact that B and ϕ are independent random variables and also that $E(B) = 0$.

Next, we find the autocorrelation function of $X(t)$. By definition, we have

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = E\{B\cos(50t_1 + \phi) \cdot B\cos(50t_2 + \phi)\}$$

Since B and ϕ are independent random variables, it follows that

$$R_{XX}(t_1, t_2) = E[B^2] E[\cos(50t_1 + \phi)\cos(50t_2 + \phi)] = E[\cos(50t_1 + \phi)\cos(50t_2 + \phi)]$$

since $E[B^2] = 1$. Thus, we have

$$\begin{aligned} R_{XX}(t_1, t_2) &= \int_{\phi=-\pi}^{\pi} \cos(50t_1 + \phi)\cos(50t_2 + \phi) \cdot \frac{1}{2\pi} d\phi \\ &= \frac{1}{4\pi} \int_{\phi=-\pi}^{\pi} \{\cos[50(t_1 + t_2) + 2\phi] + \cos[50(t_1 - t_2)]\} d\phi \end{aligned}$$

Integrating, we have

$$R_{XX}(t_1, t_2) = \frac{1}{4\pi} \left\{ \frac{\sin[50(t_1 + t_2) + 2\phi]}{2} + \cos[50(t_1 - t_2)] \right\} \phi \Big|_{\phi=-\pi}^{\pi} = \frac{1}{4\pi} \{0 + \cos[50(t_1 - t_2)] \cdot 2\pi\}$$

Thus, the autocorrelation function of $X(t)$ is given by

$$R_{XX}(t_1, t_2) = \frac{1}{2} \cos[50(t_1 - t_2)]$$

◆

EXAMPLE 5.5. Suppose that $X(t)$ is a random process with mean $\mu(t) = 3$ and autocorrelation

$$R(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}$$

Determine the mean, variance and the covariance of the random variables $Z = X(5)$ and $W = X(8)$.

(Bharathiar, Nov. 96)

Solution. Given that $\mu(t) = E[X(t)] = 3$ for all t . Hence, the means of $Z = X(5)$ and $W = X(8)$ are given by

$$\mu_Z = E(Z) = E[X(5)] = \mu(3) = 3$$

$$\mu_W = E(W) = E[X(8)] = \mu(8) = 3$$

Next, we find the variances of $Z = X(5)$ and $W = X(8)$. Note that

$$R(5, 5) = E[X^2(5)] = E[Z^2] = 9 + 4e^{-0.2|5-5|} = 9 + 4(1) = 13$$

Thus, the variance of Z is given by

$$\text{Var}(Z) = \sigma_Z^2 = E[Z^2] - \mu_Z^2 = 13 - 9 = 4$$

Similarly, we find that

$$R(8, 8) = E[X^2(8)] = E[W^2] = 9 + 4e^{-0.2|8-8|} = 9 + 4(1) = 13$$

Thus, the variance of W is given by

$$\text{Var}(W) = \sigma_W^2 = E[W^2] - \mu_W^2 = 13 - 9 = 4$$

Next, we find the covariance of Z and W . Note that

$$R(5, 8) = E[X(5)X(8)] = E[ZW] = 9 + 4e^{-0.2|5-8|} = 9 + 4e^{-0.6}$$

Hence, the covariance of Z and W is given by

$$\text{Cov}(Z, W) = E[ZW] - \mu_Z\mu_W = 9 + 4e^{-0.6} - 9 = 4e^{-0.6} \approx 2.1952$$

◆

EXAMPLE 5.6. Let $X(t) = A\cos\lambda t + B\sin\lambda t$, where A and B are independent normally distributed random variables, $N(0, \sigma^2)$. Obtain the covariance function of the $\{X(t); -\infty < t < \infty\}$ process.

(Anna, Nov. 2003)

Solution. Given that A and B are i.i.d. normal random variables with mean zero and variance, σ^2 . Thus, it follows that

$$E(A) = E(B) = 0, E(AB) = E(A)E(B) = 0 \text{ and } E(A^2) = E(B^2) = \sigma^2$$

$$\mu_X(t) = E[X(t)] = E[A \cos \lambda t + B \sin \lambda t] = \cos \lambda t E(A) + \sin \lambda t E(B) = 0$$

Hence, the autocovariance function of $X(t)$ is given by

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) = R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

Thus, we have

$$\begin{aligned} C_{XX}(t_1, t_2) &= E[(A \cos \lambda t_1 + B \sin \lambda t_1) \cdot (A \cos \lambda t_2 + B \sin \lambda t_2)] \\ &= \cos \lambda t_1 \cos \lambda t_2 E(A^2) + \cos \lambda t_1 \sin \lambda t_2 E(AB) \\ &\quad + \sin \lambda t_1 \cos \lambda t_2 E(AB) + \sin \lambda t_1 \sin \lambda t_2 E(B^2) \\ &= \cos \lambda t_1 \cos \lambda t_2 \sigma^2 + 0 + 0 + \sin \lambda t_1 \sin \lambda t_2 \sigma^2 \end{aligned}$$

$$\text{Simplifying, } C_{XX}(t_1, t_2) = \sigma^2 \sin[\lambda(t_1 - t_2)]$$

◆

EXAMPLE 5.7. For the process $\{X(t) : t \geq 0\}$, $X(t)$ is given by

$$X(t) = a \cos(\theta t) + b \sin(\theta t)$$

Here, a and b are two independent normal variables with $E(a) = E(b) = 0$ and $\text{Var}(a) = \text{Var}(b) = \sigma^2$ and θ is a constant. Obtain the mean, variance, correlation, and the first order and second order densities of $\{X(t)\}$ process.

(Anna, April 2005)

Solution. We are given that a and b are two independent normal variables with $E(a) = E(b) = 0$ and $\text{Var}(a) = \text{Var}(b) = \sigma^2$. Since a and b are independent, it follows that

$$E(ab) = E(a)E(b) = 0 \times 0 = 0$$

First, we find the mean of $X(t)$, where $t \geq 0$. By definition, we have

$$\mu_X(t) = E[X(t)] = E[a \cos \theta t + b \sin \theta t] = E(a) \cos \theta t + E(b) \sin \theta t = 0 + 0 = 0$$

Next, we find the variance of $X(t)$. Note that

$$\begin{aligned} E[X^2(t)] &= E[a^2 \cos^2 \theta t + 2ab \cos \theta t \sin \theta t + b^2 \sin^2 \theta t] \\ &= E(a^2) \cos^2 \theta t + 2E(ab) \cos \theta t \sin \theta t + E(b^2) \sin^2 \theta t \\ &= \sigma^2 \cos^2 \theta t + 0 + \sigma^2 \sin^2 \theta t = \sigma^2 \end{aligned}$$

because $\cos^2 \theta t + \sin^2 \theta t = 1$.

Hence, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = E[X^2(t)] - \mu_X^2(t) = \sigma^2 - 0 = \sigma^2$$

Next, we find the autocorrelation function of $X(t)$. Note that

$$\begin{aligned} R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E\{[a \cos \theta t_1 + b \sin \theta t_1][a \cos \theta t_2 + b \sin \theta t_2]\} \\ &= E[a^2 \cos \theta t_1 \cos \theta t_2 + ab \sin \theta t_1 \cos \theta t_2 + ab \cos \theta t_1 \sin \theta t_2 + b^2 \sin \theta t_1 \sin \theta t_2] \\ &= E(a^2) \cos \theta t_1 \cos \theta t_2 + E(ab)[\sin \theta t_1 \cos \theta t_2 + \cos \theta t_1 \sin \theta t_2] + E(b^2) \sin \theta t_1 \sin \theta t_2 \\ &= \sigma^2 \cos \theta t_1 \cos \theta t_2 + 0 + \sigma^2 \sin \theta t_1 \sin \theta t_2 = \sigma^2 [\cos \theta t_1 \cos \theta t_2 + \sin \theta t_1 \sin \theta t_2] \\ &= \sigma^2 \cos \theta(t_1 - t_2) \end{aligned}$$

We also find the autocovariance function of $X(t)$, which will be useful in the derivation of the second order probability density function of $X(t)$. Note that

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2) = \sigma^2 \cos \theta(t_1 - t_2) - 0 = \sigma^2 \cos \theta(t_1 - t_2)$$

Next, we find the first order probability density function of $X(t)$, where $t \geq 0$.

Since a and b are independent and identically distributed normal random variables with mean zero and variance σ^2 , it follows that $X(t)$ also follows a normal distribution with mean zero and variance σ^2 , i.e. for any fixed time $t_1 \geq 0$, $X_1 = X(t_1)$ has the probability density function

$$f_X(x_1; t_1) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x_1^2}{2\sigma^2}} \quad \text{for } -\infty < x_1 < \infty$$

Next, we find the second order probability density function of $X(t)$, where $t \geq 0$. Fix any times $t_1, t_2 \geq 0$ and set $X_1 = X(t_1), X_2 = X(t_2)$.

Then, we have

$$f_X(x_1, x_2; t_1, t_2) = \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{(x_1^2 - 2\rho x_1 x_2 + x_2^2)}{2(1-\rho^2)\sigma^2}} \quad \text{for } -\infty < x_1, x_2 < \infty$$

where

$$\begin{aligned} \rho &= \rho_{XX}(t_1, t_2) = \text{Correlation Coefficient between } X(t_1) \text{ and } X(t_2) \\ &= \frac{C_{XX}(t_1, t_2)}{\sigma_X(t_1)\sigma_X(t_2)} \\ &= \frac{\sigma^2 \cos \theta(t_1 - t_2)}{\sigma \cdot \sigma} = \cos \theta(t_1 - t_2) \end{aligned}$$

◆

PROBLEM SET 5.1

1. Establish the identity

$$E\{[X(t_1) - X(t_2)]^2\} = E[X^2(t_1)] + E[X^2(t_2)] - 2R_{XX}(t_1, t_2)$$

2. For a random experiment of tossing a fair coin, consider a random process $X(t)$ defined as follows:

$$X(t) = \begin{cases} t & \text{if H is the outcome} \\ \sin 2t & \text{if T is the outcome} \end{cases}$$

Find the:

- mean of $X(t)$.
 - autocorrelation function $R_{XX}(t_1, t_2)$ of $X(t)$.
 - autocovariance function $C_{XX}(t_1, t_2)$ of $X(t)$.
 - variance of $X(t)$.
3. Consider a random process $X(t) = A \cos \omega t, t \geq 0$, where A is a uniform random variable over $(0, 1)$. Find the
- mean of $X(t)$.
 - autocorrelation function $R_{XX}(t_1, t_2)$ of $X(t)$.
 - autocovariance function $C_{XX}(t_1, t_2)$ of $X(t)$.
 - variance of $X(t)$.
4. Consider a random process $X(t) = \cos(\omega t + \theta), t \geq 0$, where θ is a uniform random variable over $(-\pi, \pi)$. Show that the mean and variance of $X(t)$ are independent of time.
5. If $X(t)$ is a random process with mean $\mu(t) = 2$ and autocorrelation

$$R(t_1, t_2) = 5 + 3e^{-0.1|t_1 - t_2|}$$

find the mean, variance and the covariance of the random variables $Z = X(4)$ and $W = X(6)$.

6. Consider a discrete random sequence $X(n) = \{X_n, n \geq 1\}$, where X_n are identically and independently distributed discrete random variables with common probability mass function $p_X(x)$, mean μ and variance σ^2 . Find the
- joint probability mass function of $X_1, X_2, \dots, X_n$.
 - mean of $X(n)$.
 - autocorrelation function $R_{XX}(m, n)$ of $X(n)$.
 - autocovariance function $C_{XX}(m, n)$ of $X(n)$.

5.2 STATIONARY RANDOM PROCESSES

A *stationary process* or *strictly stationary process* is a random process whose probability distribution at a fixed time is the same for all times. As a result, statistical parameters such as mean, variance and the moments, if they exist, do not change over time for stationary processes.

5.2.1 First-Order Stationary Processes

Definition 5.3. A random process is called *stationary to order one* if its first-order probability density function does not change with a shift in time margin. In other words, the condition for $X(t)$ to be a first-order stationary process is given by

$$f_X(x_1; t_1) = f_X(x_1; t_1 + \varepsilon) \quad (5.1)$$

for any time instant t_1 and any time shift ε , where f_X is the first-order probability density function of the random process $X(t)$.

EXAMPLE 5.8. Show that the mean and the variance of a first-order stationary process are constants.

Solution. Let $X(t)$ be a first-order stationary process. Then the first-order probability density function of $X(t)$ satisfies Eq. (5.1) for all t_1 and ε . Now, consider any two time instants, t_1 and t_2 , and define the random variables $X_1 = X(t_1)$ and $X_2 = X(t_2)$. By definition, the mean values of X_1 and X_2 are given by

$$E(X_1) = E[X(t_1)] = \int_{x_1=-\infty}^{\infty} x_1 f_X(x_1; t_1) dx_1 \quad (5.2)$$

$$E(X_2) = E[X(t_2)] = \int_{x_2=-\infty}^{\infty} x_2 f_X(x_2; t_2) dx_2 \quad (5.3)$$

Now, by letting $t_2 = t_1 + \varepsilon$, Eq. (5.3) becomes

$$E[X(t_1 + \varepsilon)] = \int_{x_2=-\infty}^{\infty} x_2 f_X(x_2; t_1 + \varepsilon) dx_2 \quad (5.4)$$

Using the property, Eq. (5.1), of first-order stationary processes, Eq. (5.4) becomes

$$E[X(t_1 + \varepsilon)] = \int_{x_2=-\infty}^{\infty} x_2 f_X(x_2; t_1) dx_2 = \int_{x_1=-\infty}^{\infty} x_1 f_X(x_1; t_1) dx_1 = E[X(t_1)]$$

Since we have shown that

$$E[X(t_1)] = E[X(t_1 + \varepsilon)]$$

it follows that $\mu_X(t) = E[X(t)]$ must be a constant because t_1 and ε are arbitrary. Denote the constant value of $\mu_X(t)$ simply as μ .

Next, we show that the variance of $X(t)$ is also a constant. By definition, we have

$$\text{Var}(X_1) = E\{[X(t_1) - \mu]^2\} = \int_{x_1=-\infty}^{\infty} (x_1 - \mu)^2 f_X(x_1; t_1) dx_1 \quad (5.5)$$

$$\text{Var}(X_2) = E\{[X(t_2) - \mu]^2\} = \int_{x_2=-\infty}^{\infty} (x_2 - \mu)^2 f_X(x_2; t_2) dx_2 \quad (5.6)$$

Now, by letting $t_2 = t_1 + \varepsilon$, Eq. (5.6) becomes

$$\text{Var}[X(t_1 + \varepsilon)] = \int_{x_2=-\infty}^{\infty} (x_2 - \mu)^2 f_X(x_2; t_1 + \varepsilon) dx_2 \quad (5.7)$$

Using the property, Eq. (5.1), of first-order stationary processes, Eq. (5.7) becomes

$$\begin{aligned}\text{Var}[X(t_1 + \varepsilon)] &= \int_{x_2=-\infty}^{\infty} (x_2 - \mu)^2 f_X(x_2 : t_1) dx_2 \\ &= \int_{x_1=-\infty}^{\infty} (x_1 - \mu)^2 f_X(x_1 : t_1) dx_1 = \text{Var}[X(t_1)]\end{aligned}$$

Since we have shown that

$$\text{Var}[X(t_1)] = \text{Var}[X(t_1 + \varepsilon)]$$

it follows that $\text{Var}[X(t)] = \sigma_X^2(t)$ must be a constant because t_1 and ε are arbitrary. ♦

5.2.2 Second-Order Stationary Processes

Definition 5.4. A random process is called *stationary to order two* if its second-order probability density function does not change with a shift in time origin. In other words, the condition for $X(t)$ to be a second-order stationary process is given by

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; t_1 + \varepsilon, t_2 + \varepsilon) \quad (5.8)$$

for all time instants t_1, t_2 and time shift ε .

Remark 5.2. A second-order stationary process is also first-order stationary because the second-order probability density function determines the first-order probability density function. Note that

$$f_X(x_1; t_1) = \int_{x_2=-\infty}^{\infty} f_X(x_1, x_2; t_1, t_2) dx_2$$

Thus, if the second-order probability density function does not change with a shift in time origin, it follows that the first-order probability density function also does not change with a shift in time origin.

Remark 5.3. If $X(t)$ is a second-order process, then its second-order probability density function is a function only of time differences. To see this, substitute $\varepsilon = -t_1$ in the property (5.8). Then

$$f_X(x_1, x_2; t_1, t_2) = f_X(x_1, x_2; 0, t_2 - t_1)$$

Thus, it follows that the autocorrelation function for a second-order stationary process is a function only of time differences, i.e.

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = R_{XX}(t_1 - t_2)$$

Next, we define *wide-sense stationary* (WSS) processes. The motivation for WSS processes stems from the fact that the property Eq. (5.8) for second-order stationary processes is often very restrictive for many of the practical problems in engineering which deal mainly with the autocorrelation function and mean value of a random process. Thus, a weak form of stationary is required, and we define the WSS processes as follows.

Definition 5.5. A random process $X(t)$ is called *wide-sense stationary* if the following conditions hold:

- (i) $\mu_X(t) = E[X(t)] = \mu$ a constant.
- (ii) $R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = R_{XX}(t_1 - t_2)$.

Remark 5.4. By Remarks 5.2 and 5.3, it follows that a second-order stationary process is necessarily a wide-sense stationary process, but the converse is, in general, not true.

Definition 5.6. *Jointly Wide-Sense Processes* Two random processes $X(t)$ and $Y(t)$ are called *jointly wide-sense stationary* if the following conditions hold:

- (i) $X(t)$ is a WSS process.
- (ii) $Y(t)$ is a WSS process.
- (iii) $R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)] = R_{XY}(t_1 - t_2)$.

EXAMPLE 5.9. Show that if a random process $X(t)$ is WSS then it must also be covariance stationary. (Anna, Model 2003)

Solution. If $X(t)$ is WSS, then $E[X(t)] = \mu$ a constant and the autocorrelation function $R_{XX}(t_1, t_2)$ satisfies

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = R_{XX}(t_1 - t_2)$$

Now, we note that the autocovariance function is given by

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - E[X(t_1)X(t_2)] = R_{XX}(t_1 - t_2) - \mu^2$$

which depends only on the time difference. Hence, $X(t)$ is covariance stationary. ♦

Theorem 5.3. If a random process $X(t)$ is WSS with constant mean μ_X , then the autocorrelation function $R_{XX}(t_1, t_2)$ is a function only of the time-difference $t_1 - t_2$. Hence, it may be also represented as

$$R_{XX}(\tau) = E[X(t)X(t + \tau)] \quad (5.9)$$

and the random process $X(t)$ has the following properties:

- (i) $E[X^2(t)] = R_{XX}(0)$ for all t .
- (ii) $R_{XX}(0) \geq 0$.
- (iii) $\text{Var}[X(t)] = R_{XX}(0) - \mu_X^2$, i.e. the variance of $X(t)$ is a constant for all t .

Proof. (i) Substituting $\tau = 0$ in Eq. (5.9), we get

$$R_{XX}(0) = E[X(t)X(t)] = E[X^2(t)] \quad \text{for all } t$$

(ii) Since $X^2(t) \geq 0$ for all t and $R_{XX}(0) = E[X^2(t)]$, it is immediate that $R_{XX}(0) \geq 0$.

(iii) By definition, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = E[X^2(t)] - \mu_X^2$$

Using (i), we have

$$\text{Var}[X(t)] = R_{XX}(0) - \mu_X^2$$

which is a constant. ♦

5.2.3 Order n and Strict-Sense Stationary Processes

Definition 5.7. A random process is called *stationary to order n* if its n th order probability density function does not change with a shift in time origin. In other words, the condition for $X(t)$ to be a n th order stationary process is given by

$$f_X(x_1, \dots, x_n; t_1, \dots, t_n) = f_X(x_1, \dots, x_n; t_1 + \varepsilon, \dots, t_n + \varepsilon) \quad (5.10)$$

for all time instants $t_1, \dots, t_n$ and time shift ε . A random process is called *strict-sense stationary* if it is stationary to all order $n = 1, 2, \dots$.

Remark 5.5. If a random process $X(t)$ is stationary to order n , then it follows that $X(t)$ is stationary to order k for all $k \leq n$.

Definition 5.8. A random process, which is not stationary in any sense, is called *evolutionary*.

EXAMPLE 5.10. Consider a random process defined by $X(t) = A_1 + A_2 t$, where A_1 and A_2 are independent random variables with $E(A_i) = a_i$ and $\text{Var}(A_i) = \sigma_i^2$ for $i = 1, 2$. Prove that the process $X(t)$ is evolutionary.

Solution. Note that

$$\mu_X(t) = E[X(t)] = E[A_1 + A_2 t] = E[A_1] + tE[A_2] = a_1 + ta_2$$

which is not a constant. Thus, the process $X(t)$ is evolutionary. $\blacklozenge$

Definition 5.9. Independent Random Processes A random process $X(t)$ is called *independent* if $X(t_i)$ for $i = 1, 2, \dots, n$ are independent random variables so that for $n = 2, 3, \dots$

$$f_X(x_1, x_2, \dots, x_n; t_1, t_2, \dots, t_n) = \prod_{i=1}^n f_X(x_i; t_i)$$

Thus, for an independent random process, the first-order probability density function is enough to characterize the process.

Definition 5.10. Processes with Stationary Independent Increments A random process $X(t)$ ($t \geq 0$) is said to have *independent increments* if whenever $0 < t_1 < t_2 < \dots < t_n$,

$$X(0), X(t_1) - X(0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$$

are independent random variables. If $X(t)$ ($t \geq 0$) has independent increments and $X(t_2) - X(t_1)$ has the same distribution as $X(t_2 + \varepsilon) - X(t_1 + \varepsilon)$ for all $t_1, t_2, \varepsilon \geq 0, t_1 < t_2$, then the process $X(t)$ is said to have *stationary independent increments*.

EXAMPLE 5.11. If $X(t)$, where $t \geq 0$, is a random process with stationary independent increments with $X(0) = 0$ and $\mu_1 = E[X(1)]$, show that

$$E[X(t)] = \mu_1 t \quad \text{for all } t \geq 0$$

Is $X(t)$ wide-sense stationary?

Solution. Define

$$\phi(t) = E[X(t)] = E[X(t) - X(0)]$$

Then for any t_1 and t_2 , by the property of the stationary independent increments, we have

$$\begin{aligned}\phi(t_1 + t_2) &= E[X(t_1 + t_2) - X(0)] \\ &= E[X(t_1 + t_2) - X(t_2) + X(t_2) - X(0)] \\ &= E[X(t_1 + t_2) - X(t_2)] + E[X(t_2) - X(0)] \\ &= E[X(t_1) - X(0)] + E[X(t_2) - X(0)] \\ &= \phi(t_1) + \phi(t_2)\end{aligned}$$

The only solution to the functional equation $\phi(t_1 + t_2) = \phi(t_1) + \phi(t_2)$ is

$$\phi(t) = ct$$

where c is a constant. Since $\phi(1) = c = E[X(1)] = \mu_1$, it follows that

$$\phi(t) = E[X(t)] = \mu_1 t \quad \text{for all } t \geq 0$$

Since $\mu_X(t) = E[X(t)] = \mu_1 t$ is not a constant, $X(t)$ is not a WSS process. ♦

EXAMPLE 5.12. If $X(t)$, where $t \geq 0$, is a random process with stationary independent increments with $X(0) = 0$ and $\sigma_1^2 = \text{Var}[X(1)]$, prove the following:

- (i) $\text{Var}[X(t)] = \sigma_1^2 t$ for all $t \geq 0$.
- (ii) If $t_2 > t_1$, then $\text{Var}[X(t_2) - X(t_1)] = \sigma_1^2(t_2 - t_1)$.
- (iii) $C_{XX}(t_1, t_2) = \text{Cov}[X(t_1), X(t_2)] = \sigma_1^2 \min(t_1, t_2)$.

Solution. Recall that for any two random variables U and V ,

$$\text{Var}(U + V) = \text{Var}(U) + \text{Var}(V) + 2\text{Cov}(U, V)$$

If U and V are independent, then $\text{Cov}(U, V) = 0$ and

$$\text{Var}(U + V) = \text{Var}(U) + \text{Var}(V)$$

- (i) Define $\psi(t) = \text{Var}[X(t)] = \text{Var}[X(t) - X(0)]$. Then, for any t_1 and t_2 , by the property of the stationary independent increments, we have

$$\begin{aligned}\psi(t_1 + t_2) &= \text{Var}[X(t_1 + t_2) - X(0)] \\ &= \text{Var}[X(t_1 + t_2) - X(t_2) + X(t_2) - X(0)] \\ &= \text{Var}[X(t_1 + t_2) - X(t_2)] + \text{Var}[X(t_2) - X(0)] \\ &= \text{Var}[X(t_1) - X(0)] + \text{Var}[X(t_2) - X(0)] \\ &= \psi(t_1) + \psi(t_2)\end{aligned}$$

The only solution to the functional equation $\psi(t_1 + t_2) = \psi(t_1) + \psi(t_2)$ is

$$\psi(t) = kt$$

where k is a constant. Since $\psi(1) = k = \text{Var}[X(1)] = \sigma_1^2$, it follows that

$$\psi(t) = \text{Var}[X(t)] = \sigma_1^2 t, \quad t \geq 0$$

(ii) Let $t_2 > t_1$. Then by the property of stationary independent increments, we have

$$\begin{aligned}\text{Var}[X(t_2)] &= \text{Var}[X(t_2) - X(t_1) + X(t_1) - X(0)] \\ &= \text{Var}[X(t_2) - X(t_1)] + \text{Var}[X(t_1) - X(0)] \\ &= \text{Var}[X(t_2) - X(t_1)] + \text{Var}[X(t_1)]\end{aligned}$$

Thus,

$$\text{Var}[X(t_2) - X(t_1)] = \text{Var}[X(t_2)] - \text{Var}[X(t_1)] = \sigma_1^2 t_2 - \sigma_1^2 t_1 = \sigma_1^2 (t_2 - t_1)$$

(iii) For any t_1 and t_2 , we know that

$$\text{Var}[X(t_2) - X(t_1)] = \text{Var}[X(t_2)] + \text{Var}[X(t_1)] - 2\text{Cov}[X(t_1)X(t_2)]$$

from which it follows that

$$C_{XX}(t_1, t_2) = \text{Cov}[X(t_1), X(t_2)] = \frac{1}{2} [\text{Var}[X(t_2)] + \text{Var}[X(t_1)] - \text{Var}[X(t_2) - X(t_1)]]$$

Using the results in (i) and (ii), it follows that

$$C_{XX}(t_1, t_2) = \begin{cases} \frac{1}{2} \sigma_1^2 [t_1 + t_2 - (t_2 - t_1)] = \sigma_1^2 t_1 & \text{if } t_2 \geq t_1 \\ \frac{1}{2} \sigma_1^2 [t_1 + t_2 - (t_1 - t_2)] = \sigma_1^2 t_2 & \text{if } t_1 > t_2 \end{cases}$$

Combining the two cases, we conclude that

$$C_{XX}(t_1, t_2) = \sigma_1^2 \min(t_1, t_2)$$

◆

EXAMPLE 5.13. Consider the random process $X(t) = \cos(t + \phi)$, where ϕ is a random variable with density function

$$f(\phi) = \frac{1}{\pi}, \text{ where } -\frac{\pi}{2} < \phi < \frac{\pi}{2}$$

Check whether or not the process is stationary.

(Anna, May 2000)

Solution. We find that

$$\mu_X(t) = E[X(t)] = E[\cos(t + \phi)] = \int_{\phi=-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(t + \phi) \frac{1}{\pi} d\phi$$

Integrating, we have

$$\begin{aligned}\mu_X(t) &= \frac{1}{\pi} [\sin(t + \phi)]_{\phi=-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{\pi} \left[\sin\left(\frac{\pi}{2} + t\right) + \sin\left(\frac{\pi}{2} - t\right) \right] \\ &= \frac{1}{\pi} [\cos t + \cos t] = \frac{2}{\pi} \cos t\end{aligned}$$

Since $\mu_X(t)$ is a function of t , the random process $X(t)$ is not stationary.

◆

EXAMPLE 5.14. Examine whether the Poisson process $\{X(t)\}$ given by the law

$$P[X(t) = r] = \frac{e^{-\lambda t} (\lambda t)^r}{r!}, \quad r = 0, 1, 2, \dots$$

is covariance stationary.

(Anna, May 2006)

Solution. Since the probability distribution of $X(t)$ is a Poisson distribution with parameter λt , it follows by Theorem 3.7 that the mean of $X(t)$ is λt . Thus,

$$\mu_X(t) = E[X(t)] = \lambda t$$

which depends on t . Hence, the Poisson process is not covariance stationary. $\blacklozenge$

EXAMPLE 5.15. Show that the random process $X(t) = A \cos(\omega t + \theta)$ is WSS if A and ω are constants, and θ is a uniformly distributed random variable in $(0, 2\pi)$.

(Anna, Nov. 2003; Nov. 2005; May 2007)

Solution. Since θ is uniformly distributed in $(0, 2\pi)$, it has the probability density function

$$f_\theta(\theta) = \begin{cases} \frac{1}{2\pi} & \text{if } 0 < \theta < 2\pi \\ 0 & \text{elsewhere} \end{cases}$$

First, we find the mean of $X(t)$. By definition, we have

$$\mu_X(t) = E[X(t)] = E[A \cos(\omega t + \theta)] = \int_{\theta=0}^{2\pi} A \cos(\omega t + \theta) \frac{1}{2\pi} d\theta$$

Integrating, we have

$$\mu_X(t) = \frac{A}{2\pi} [\sin(\omega t + \theta)]_{\theta=0}^{2\pi} = 0$$

Thus, $\mu_X(t) \equiv 0$, which is a constant.

Next, we find the autocorrelation function of $X(t)$. By definition, we have

$$\begin{aligned} R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E[A \cos(\omega t_1 + \theta) \cdot A \cos(\omega t_2 + \theta)] \\ &= \int_{\theta=0}^{2\pi} A^2 \cos(\omega t_1 + \theta) \cos(\omega t_2 + \theta) \frac{1}{2\pi} d\theta \end{aligned}$$

Using the identity $2 \cos U \cos V = \cos(U + V) + \cos(U - V)$, we have

$$R_{XX}(t_1, t_2) = \frac{A^2}{4\pi} \int_{\theta=0}^{2\pi} \{\cos[\omega(t_1 + t_2) + 2\theta] + \cos[\omega(t_1 - t_2)]\} d\theta$$

Integrating, we have

$$R_{XX}(t_1, t_2) = \frac{A^2}{4\pi} \left[\frac{\sin[\omega(t_1 + t_2) + 2\theta]}{2} + \theta \cos[\omega(t_1 - t_2)] \right]_{\theta=0}^{2\pi} = \frac{A^2}{4\pi} [0 + 2\pi \cos[\omega(t_1 - t_2)]]$$

Thus, $R_{XX}(t_1, t_2) = \frac{A^2}{2} \cos[\omega(t_1 - t_2)]$, which is a function of the time-difference, $t_1 - t_2$.

Hence, $X(t)$ is a wide-sense stationary process. $\blacklozenge$

EXAMPLE 5.16. The process $\{X(t)\}$, whose probability distribution under certain condition is given by,

$$P\{X(t) = n\} = \begin{cases} \frac{(at)^{n-1}}{(1+at)^{n+1}} & \text{if } n = 1, 2, \dots \\ \frac{at}{1+at} & \text{if } n = 0 \end{cases}$$

Show that it is not stationary.

(Anna, Nov. 2003; May 2006; Nov. 2006; Nov. 2007)

Solution. The probability distribution of $X(t)$ is given by

$X(t) = n$	0	1	2	3	...
P_n	$\frac{at}{1+at}$	$\frac{1}{(1+at)^2}$	$\frac{at}{(1+at)^3}$	$\frac{(at)^2}{(1+at)^4}$	...

The mean of $X(n)$ is given by

$$\mu_X(t) = E[X(t)] = \sum_{n=0}^{\infty} nP_n = \frac{1}{(1+at)^2} + 2 \cdot \frac{at}{(1+at)^3} + 3 \cdot \frac{(at)^2}{(1+at)^4} + \dots$$

which may be rearranged as

$$\mu_X(t) = \frac{1}{(1+at)^2} \left[1 + 2 \left(\frac{at}{1+at} \right) + 3 \left(\frac{at}{1+at} \right)^2 + \dots \right]$$

Using the formula $1 + 2\theta + 3\theta^2 + \dots = (1-\theta)^{-2}$, where $|\theta| < 1$, we have

$$\mu_X(t) = \frac{1}{(1+at)^2} \left[1 - \frac{at}{1+at} \right]^{-2} = \frac{1}{(1+at)^2} \cdot \frac{1}{(1+at)^{-2}} = 1$$

Thus, $\mu_X(t) \equiv 1$, which is a constant.

Next, we find that

$$E[X^2(t)] = \sum_{n=0}^{\infty} n^2 P_n = \sum_{n=1}^{\infty} n^2 \frac{(at)^{n-1}}{(1+at)^{n+1}}$$

which may be rewritten as

$$E[X^2(t)] = \frac{1}{(1+at)^2} \left[\sum_{n=1}^{\infty} n(n+1) \left(\frac{at}{1+at} \right)^{n-1} - \sum_{n=1}^{\infty} n \left(\frac{at}{1+at} \right)^{n-1} \right]$$

Using the standard formulae, we have

$$\begin{aligned} E[X^2(t)] &= \frac{1}{(1+at)^2} \left\{ 2 \left[1 - \frac{at}{1+at} \right]^{-3} - \left[1 - \frac{at}{1+at} \right]^{-2} \right\} \\ &= \frac{1}{(1+at)^2} [2(1+at)^3 - (1+at)^2] = 2(1+at) - 1 = 1 + 2at \end{aligned}$$

Hence, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = E[X^2(t)] - \mu_X^2(t) = 1 + 2at - 1 = 2at$$

which depends on t . Thus, $X(t)$ is not a stationary process. ◆

EXAMPLE 5.17. For the sine wave process $X(t) = Y \cos \omega_0 t$, $-\infty < t < \infty$, ω_0 a constant, the amplitude Y is a random variable with uniform distribution in the interval $0 < t < 1$. Check whether the process is stationary or not.

(Anna, April 2004; Nov. 2006)

Solution. Since Y is a uniform random variable over $(0, 1)$, by Theorem 3.13, it follows that

$$E(Y) = \mu'_1(Y) = \frac{1}{2} \quad \text{and} \quad E(Y^2) = \mu'_2(Y) = \frac{1}{3}$$

To check the stationarity of $X(t)$, we first calculate the mean of $X(t)$. We find that

$$\mu_X(t) = E[X(t)] = E[Y \cos \omega_0 t] = \cos \omega_0 t E(Y) = \cos \omega_0 t \left(\frac{1}{2} \right) = \frac{\cos \omega_0 t}{2}$$

which depends on t .

Thus, $X(t)$ is not a stationary process. ♦

EXAMPLE 5.18. Given a random variable Y with characteristic function $\phi(\omega) = E[e^{i\omega Y}]$ and a random process defined by $X(t) = \cos(\lambda t + Y)$, show that $\{X(t)\}$ is stationary in the wide-sense if $\phi(1) = \phi(2) = 0$.

(Anna, April 2003; April 2005; Nov. 2005)

Solution. Since

$$\phi(k) = E[e^{ikY}] = E[\cos kY + i \sin kY] = E[\cos kY] + iE[\sin kY]$$

it follows that

$$\phi(1) = 0 \iff E[\cos Y] = 0 \quad \text{and} \quad E[\sin Y] = 0$$

$$\phi(2) = 0 \iff E[\cos 2Y] = 0 \quad \text{and} \quad E[\sin 2Y] = 0$$

Now, we suppose that $\phi(1) = 0$ and $\phi(2) = 0$ and establish that $\{X(t)\}$ is a WSS process.

First, we show that $\mu_X(t)$ is a constant. Note that

$$\mu_X(t) = E[X(t)] = E[\cos(\lambda t + Y)] = E[\cos \lambda t \cos Y - \sin \lambda t \sin Y]$$

and so, we have

$$\mu_X(t) = \cos \lambda t E[\cos Y] - \sin \lambda t E[\sin Y] = 0 \quad (\text{since } E[\cos Y] = E[\sin Y] = 0)$$

Thus, $\mu_X(t) \equiv 0$, a constant.

Next, we find the autocorrelation function of the process. We have

$$\begin{aligned} R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E[\cos(\lambda t_1 + Y) \cdot \cos(\lambda t_2 + Y)] \\ &= E[(\cos \lambda t_1 \cos Y - \sin \lambda t_1 \sin Y)(\cos \lambda t_2 \cos Y - \sin \lambda t_2 \sin Y)] \\ &= \cos \lambda t_1 \cos \lambda t_2 E[\cos^2 Y] - \sin \lambda t_1 \cos \lambda t_2 E[\sin Y \cos Y] \\ &\quad - \sin \lambda t_2 \cos \lambda t_1 E[\sin Y \cos Y] + \sin \lambda t_1 \sin \lambda t_2 E[\sin^2 Y] \\ &= \cos \lambda t_1 \cos \lambda t_2 E\left[\frac{1+\cos 2Y}{2}\right] - \sin \lambda t_1 \cos \lambda t_2 E\left[\frac{\sin 2Y}{2}\right] \\ &\quad - \sin \lambda t_2 \cos \lambda t_1 E\left[\frac{\sin 2Y}{2}\right] + \sin \lambda t_1 \sin \lambda t_2 E\left[\frac{1-\cos 2Y}{2}\right] \end{aligned}$$

Since $E[\cos 2Y] = E[\sin 2Y] = 0$, it follows that

$$R_{XX}(t_1, t_2) = \frac{1}{2} \cos \lambda t_1 \cos \lambda t_2 + \frac{1}{2} \sin \lambda t_1 \sin \lambda t_2 = \frac{1}{2} \cos[\lambda(t_1 - t_2)]$$

which is a function of $t_1 - t_2$.

Hence, $\{X(t)\}$ is a WSS process. ♦

EXAMPLE 5.19. Consider a random process $Y(t) = X(t) \cos(\omega_0 t + \theta)$, where $X(t)$ is wide-sense stationary, θ is a uniform random variable independent of $X(t)$ and is distributed uniformly in $(-\pi, \pi)$ and ω_0 is a constant. Prove that $Y(t)$ is wide-sense stationary.

(Anna, April 2003)

Solution. Given that $X(t)$ is a wide-sense stationary. Hence, it follows that

- (i) $\mu_X(t)$ is a constant, say, μ_X .
- (ii) $R_{XX}(t_1, t_2) = R_{XX}(t_1 - t_2)$.

Given also that θ is independent of $X(t)$, and is uniformly distributed in $(-\pi, \pi)$, i.e.

$$f_\theta(\theta) = \begin{cases} \frac{1}{2\pi} & \text{if } -\pi < \theta < \pi \\ 0 & \text{otherwise} \end{cases}$$

To show that $Y(t) = X(t) \cos(\omega_0 t + \theta)$ is a WSS process.

First, we compute the mean of $Y(t)$. We have

$$\mu_Y(t) = E[Y(t)] = E[X(t) \cos(\omega_0 t + \theta)] = E[X(t)]E[\cos(\omega_0 t + \theta)] \quad (5.11)$$

since $X(t)$ and θ are independent random variables.

Note that

$$E[\cos(\omega_0 t + \theta)] = \int_{\theta=-\pi}^{\pi} \cos(\omega_0 t + \theta) \left[\frac{1}{2\pi} \right] d\theta$$

Integrating, we have

$$E[\cos(\omega_0 t + \theta)] = \frac{1}{2\pi} [\sin(\omega_0 t + \theta)]_{\theta=-\pi}^{\pi} = 0$$

Hence, Eq. (5.11) reduces to

$$\mu_Y(t) = \mu_X \cdot 0 = 0, \text{ a constant}$$

Next, we find the autocorrelation function of $Y(t)$.

$$R_{YY}(t_1, t_2) = E[Y(t_1)Y(t_2)] = E[X(t_1) \cos(\omega_0 t_1 + \theta) \cdot X(t_2) \cos(\omega_0 t_2 + \theta)]$$

Since $X(t)$ and θ are independent, it follows that

$$R_{YY}(t_1, t_2) = E[X(t_1)X(t_2)]E[\cos(\omega_0 t_1 + \theta) \cos(\omega_0 t_2 + \theta)]$$

Using (ii), we have

$$R_{YY}(t_1, t_2) = R_{XX}(t_1 - t_2)E[g(\theta)] \quad (5.12)$$

where $g(\theta) = \cos(\omega_0 t_1 + \theta) \cos(\omega_0 t_2 + \theta)$.

Now, note that

$$\begin{aligned} E[g(\theta)] &= \int_{\theta=-\pi}^{\pi} \cos(\omega_0 t_1 + \theta) \cos(\omega_0 t_2 + \theta) \left[\frac{1}{2\pi} \right] d\theta \\ &= \frac{1}{4\pi} \int_{\theta=-\pi}^{\pi} \{ \cos[\omega_0(t_1 + t_2) + 2\theta] + \cos[\omega_0(t_1 - t_2)] \} d\theta \end{aligned}$$

Integrating, we have

$$\begin{aligned} E[g(\theta)] &= \frac{1}{4\pi} \left\{ \frac{\sin[\omega_0(t_1 + t_2) + 2\theta]}{2} + \theta \cos[\omega_0(t_1 - t_2)] \right\}_{\theta=-\pi}^{\pi} \\ &= \frac{1}{4\pi} \{ 0 + 2\pi \cos[\omega_0(t_1 - t_2)] \} \\ E[g(\theta)] &= \frac{1}{2} \cos[\omega_0(t_1 - t_2)] \end{aligned}$$

Substituting the above in Eq. (5.12), we get

$$R_{YY}(t_1, t_2) = R_{XX}(t_1, t_2) \cdot \frac{1}{2} \cos[\omega_0(t_1 - t_2)]$$

which is a function of $t_1 - t_2$.

Hence, we conclude that $Y(t)$ is a WSS process. ♦

EXAMPLE 5.20. If $\{X(t)\}$ is a WSS process with autocorrelation function $R_{XX}(\tau)$ and if $Y(t) = X(t+a) - X(t-a)$, show that

$$R_{YY}(\tau) = 2R_{XX}(\tau) - R_{XX}(\tau+2a) - R_{XX}(\tau-2a)$$

(Anna, April 2003; Nov. 2006)

Solution. Since $\{X(t)\}$ is a WSS process with autocorrelation function $R_{XX}(\tau)$, we know that

$$R_{XX}(\tau) = E[X(t+\tau)X(t)]$$

is a function only of the difference of time.

Since $R_{YY}(\tau) = E[Y(t+\tau)Y(t)]$, it follows that

$$\begin{aligned} R_{YY}(\tau) &= E\{[X(t+\tau+a) - X(t+\tau-a)][X(t+a) - X(t-a)]\} \\ &= E[X(t+\tau+a)X(t+a)] - E[X(t+\tau+a)X(t-a)] \\ &\quad - E[X(t+\tau-a)X(t+a)] + E[X(t+\tau-a)X(t-a)] \\ &= R_{XX}(\tau) - R_{XX}(\tau+2a) - R_{XX}(\tau-2a) + R_{XX}(\tau) \\ &= 2R_{XX}(\tau) - R_{XX}(\tau+2a) - R_{XX}(\tau-2a) \end{aligned}$$

♦

EXAMPLE 5.21. If $\{X(t) = A \cos \lambda t + B \sin \lambda t ; t \geq 0\}$ is a random process, where A and B are independent $N(0, \sigma^2)$ random variables, examine the stationarity of $X(t)$.

(Anna, April 2004)

Solution. Since A and B are independent $N(0, \sigma^2)$ random variables, it follows that

$$E(A) = E(B) = 0 \text{ and } E(A^2) = E(B^2) = \text{Var}(A) = \text{Var}(B) = \sigma^2$$

and

$$E(AB) = E(A)E(B) = 0$$

To examine the stationarity of $X(t)$, we first calculate the mean of $X(t)$. We find that

$$\mu_X(t) = E[X(t)] = E(A \cos \lambda t + B \sin \lambda t) = \cos \lambda t E(A) + \sin \lambda t E(B) = 0$$

Thus, $\mu_X(t) \equiv 0$, which is a constant.

Next, we calculate the autocorrelation function of $X(t)$. We find that

$$\begin{aligned} R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E[(A \cos \lambda t_1 + B \sin \lambda t_1) \cdot (A \cos \lambda t_2 + B \sin \lambda t_2)] \\ &= E(A^2) \cos \lambda t_1 \cos \lambda t_2 + E(AB) [\cos \lambda t_1 \sin \lambda t_2 + \sin \lambda t_1 \cos \lambda t_2] + E(B^2) \sin \lambda t_1 \sin \lambda t_2 \end{aligned}$$

Using the given data, we have

$$\begin{aligned} R_{XX}(t_1, t_2) &= \sigma^2 \cos \lambda t_1 \cos \lambda t_2 + 0 + \sigma^2 \sin \lambda t_1 \sin \lambda t_2 = \sigma^2 (\cos \lambda t_1 \cos \lambda t_2 + \sin \lambda t_1 \sin \lambda t_2) \\ &= \sigma^2 \cos \lambda (t_1 - t_2) \end{aligned}$$

which is a function of $t_1 - t_2$.

Hence, we conclude that $X(t)$ is a WSS process. ◆

EXAMPLE 5.22. Show that the process $X(t) = A \cos \lambda t + B \sin \lambda t$ (where A and B are random variables) is wide-sense stationary, if (i) $E(A) = E(B) = 0$, (ii) $E(A^2) = E(B^2)$ and (iii) $E(AB) = 0$.

(Anna, Nov. 2004)

Solution. First, we find the mean of $X(t)$. Note that

$$\mu_X(t) = E[X(t)] = E[A \cos \lambda t + B \sin \lambda t] = E(A) \cos \lambda t + E(B) \sin \lambda t = 0$$

using Condition (i). Thus, the mean of $X(t)$ is a constant.

Next, we find the autocorrelation function of $X(t)$. Note that

$$\begin{aligned} R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E\{[A \cos \lambda t_1 + B \sin \lambda t_1][A \cos \lambda t_2 + B \sin \lambda t_2]\} \\ &= E(A^2) \cos \lambda t_1 \cos \lambda t_2 + E(AB) [\cos \lambda t_1 \sin \lambda t_2 + \sin \lambda t_1 \cos \lambda t_2] + E(B^2) \sin \lambda t_1 \sin \lambda t_2 \end{aligned}$$

using Conditions (ii) and (iii), we have

$$R_{XX}(t_1, t_2) = E(A^2) [\cos \lambda t_1 \cos \lambda t_2 + \sin \lambda t_1 \sin \lambda t_2] + 0 = E(A^2) \cos \lambda (t_1 - t_2)$$

which is a function of $t_1 - t_2$.

Thus, $X(t)$ is a wide-sense stationary process. ◆

EXAMPLE 5.23. Consider a random process $X(t)$ defined by $X(t) = U \cos t + V \sin t$, where U and V are independent random variables each of which assumes the values -2 and 1 with probabilities $\frac{1}{3}$ and $\frac{2}{3}$, respectively. Show that $X(t)$ is wide-sense stationary and not strict-sense stationary.

(Anna, April 2005)

Solution. First, we calculate the first and second raw moments of the random variables U and V which are identically and independently distributed taking the values -2 and 1 with probabilities $\frac{1}{3}$ and $\frac{2}{3}$, respectively.

By definition, we have

$$E(U) = E(V) = \left(-2 \times \frac{1}{3}\right) + \left(1 \times \frac{2}{3}\right) = -\frac{2}{3} + \frac{2}{3} = 0$$

and

$$E(U^2) = E(V^2) = \left(4 \times \frac{1}{3}\right) + \left(1 \times \frac{2}{3}\right) = \frac{4}{3} + \frac{2}{3} = 2$$

and

$$E(U^3) = E(V^3) = \left(-8 \times \frac{1}{3}\right) + \left(1 \times \frac{2}{3}\right) = -\frac{8}{3} + \frac{2}{3} = -2$$

Since U and V are independent random variables, it also follows that

$$E(UV) = E(U)E(V) = 0 \times 0 = 0$$

$$E(U^2V) = E(U^2)E(V) = 2 \times 0 = 0$$

and

$$E(UV^2) = E(U)E(V^2) = 0 \times 2 = 0$$

Next, we show that the random process $X(t) = U \cos t + V \sin t$ is wide-sense stationary. For this purpose, we first calculate the mean of $X(t)$. We find that

$$\mu_X(t) = E[X(t)] = E[U \cos t + V \sin t] = E(U) \cos t + E(V) \sin t = 0 + 0 = 0$$

which is a constant.

Next, we calculate the autocorrelation function of $X(t)$. We find that

$$\begin{aligned} R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E\{[U \cos t_1 + V \sin t_1][U \cos t_2 + V \sin t_2]\} \\ &= E[U^2 \cos t_1 \cos t_2 + UV \sin t_1 \cos t_2 + UV \cos t_1 \sin t_2 + V^2 \sin t_1 \sin t_2] \\ &= E(U^2) \cos t_1 \cos t_2 + E(UV)(\sin t_1 \cos t_2 + \cos t_1 \sin t_2) + E(V^2) \sin t_1 \sin t_2 \\ &= 2 \cos t_1 \cos t_2 + 0 + 2 \sin t_1 \sin t_2 = 2 \cos(t_1 - t_2) \end{aligned}$$

which is a function of $t_1 - t_2$.

Hence, $X(t)$ is a wide-sense stationary process.

Next, we show that $X(t)$ is not strict-sense stationary by showing that $E[X^3(t)]$ is not a constant. (Note that for a strict sense stationary process, all the moments must be independent of time.)

We find that

$$\begin{aligned}
 E[X^3(t)] &= E[(U \cos t + V \sin t)^3] \\
 &= E[U^3 \cos^3 t + 3U^2V \cos^2 t \sin t + 3UV^2 \cos t \sin^2 t + V^3 \sin^3 t] \\
 &= E(U^3) \cos^3 t + 3E(U^2V) \cos^2 t \sin t + 3E(UV^2) \cos t \sin^2 t + E(V^3) \sin^3 t \\
 &= -2 \cos^3 t + 0 + 0 - 2 \sin^3 t = -2(\cos^3 t + \sin^3 t)
 \end{aligned}$$

which depends on t .

Hence, $X(t)$ is not a strict-sense stationary process. ◆

EXAMPLE 5.24. Two random processes $X(t)$ and $Y(t)$ are defined by

$$X(t) = A \cos \omega t + B \sin \omega t \quad \text{and} \quad Y(t) = B \cos \omega t - A \sin \omega t$$

Show that $X(t)$ and $Y(t)$ are jointly wide-sense stationary if A and B are uncorrelated random variables with zero means and the same variances, and ω is a constant.

(Anna, April 2003)

Solution. We are given that A and B are uncorrelated random variables with zero means and the same variances. Then, it follows that

$$E(A) = E(B) = 0$$

and

$$\text{Var}(A) = \text{Var}(B) = \sigma^2 \quad (\text{say})$$

Thus, it follows that

$$E(A^2) = E(B^2) = \text{Var}(A) + [E(A)]^2 = \sigma^2 + 0 = \sigma^2$$

Since A and B are uncorrelated,

$$\text{Cov}(A, B) = E(AB) - E(A)E(B) = 0 \Rightarrow E(AB) - 0 = 0 \Rightarrow E(AB) = 0$$

To show that $X(t)$ and $Y(t)$ are jointly wide-sense stationary, we must establish the following:

- (i) $X(t)$ is a WSS process.
- (ii) $Y(t)$ is a WSS process.
- (iii) The cross-correlation function $R_{XY}(t_1, t_2)$ is a function of $t_1 - t_2$.

First, we show that $X(t)$ is a WSS process.

We find that the mean of $X(t)$ is given by

$$\begin{aligned}
 \mu_X(t) &= E[X(t)] = E[A \cos \omega t + B \sin \omega t] \\
 &= E(A) \cos \omega t + E(B) \sin \omega t = 0 \cos \omega t + 0 \sin \omega t \\
 &= 0 + 0 = 0
 \end{aligned}$$

which is a constant.

Next, we find the auto-correlation function of the process $X(t)$.

By definition, we have

$$\begin{aligned}
 R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E\{[A \cos \omega t_1 + B \sin \omega t_1][A \cos \omega t_2 + B \sin \omega t_2]\} \\
 &= E[A^2 \cos \omega t_1 \cos \omega t_2 + AB \sin \omega t_1 \cos \omega t_2 + AB \cos \omega t_1 \sin \omega t_2 + B^2 \sin \omega t_1 \sin \omega t_2] \\
 &= E(A^2) \cos \omega t_1 \cos \omega t_2 + E(AB) [\sin \omega t_1 \cos \omega t_2 + \cos \omega t_1 \sin \omega t_2] + E(B^2) \sin \omega t_1 \sin \omega t_2 \\
 &= \sigma^2 \cos \omega t_1 \cos \omega t_2 + 0 + \sigma^2 \sin \omega t_1 \sin \omega t_2 \\
 &= \sigma^2 \cos \omega(t_1 - t_2)
 \end{aligned}$$

which is a function of $t_1 - t_2$.

Thus, we have shown that $X(t)$ is a WSS process.

Next, we show that $Y(t)$ is a WSS process.

We find that the mean of $Y(t)$ is given by

$$\begin{aligned}
 \mu_Y(t) &= E[Y(t)] = E[B \cos \omega t - A \sin \omega t] \\
 &= E(B) \cos \omega t - E(A) \sin \omega t = 0 \cos \omega t - 0 \sin \omega t \\
 &= 0 - 0 = 0
 \end{aligned}$$

which is a constant.

Next, we find the auto-correlation function of the process $Y(t)$.

By definition, we have

$$\begin{aligned}
 R_{YY}(t_1, t_2) &= E[Y(t_1)Y(t_2)] = E\{[B \cos \omega t_1 - A \sin \omega t_1][B \cos \omega t_2 - A \sin \omega t_2]\} \\
 &= E[B^2 \cos \omega t_1 \cos \omega t_2 - AB \sin \omega t_1 \cos \omega t_2 - AB \cos \omega t_1 \sin \omega t_2 + A^2 \sin \omega t_1 \sin \omega t_2] \\
 &= E(B^2) \cos \omega t_1 \cos \omega t_2 - E(AB) [\sin \omega t_1 \cos \omega t_2 + \cos \omega t_1 \sin \omega t_2] + E(A^2) \sin \omega t_1 \sin \omega t_2 \\
 &= \sigma^2 \cos \omega t_1 \cos \omega t_2 + 0 + \sigma^2 \sin \omega t_1 \sin \omega t_2 \\
 &= \sigma^2 \cos \omega(t_1 - t_2)
 \end{aligned}$$

which is a function of $t_1 - t_2$.

Thus, we have shown that $Y(t)$ is a WSS process.

Next, we show that the cross-correlation function $R_{XY}(t_1, t_2)$ is a function of $t_1 - t_2$.

By definition, we have

$$\begin{aligned}
 R_{XY}(t_1, t_2) &= E[X(t_1)Y(t_2)] = E\{[A \cos \omega t_1 + B \sin \omega t_1][B \cos \omega t_2 - A \sin \omega t_2]\} \\
 &= E[AB \cos \omega t_1 \cos \omega t_2 + B^2 \sin \omega t_1 \cos \omega t_2 - A^2 \cos \omega t_1 \sin \omega t_2 - AB \sin \omega t_1 \sin \omega t_2] \\
 &= E(AB) [\cos \omega t_1 \cos \omega t_2 - \sin \omega t_1 \sin \omega t_2] + E(B^2) \sin \omega t_1 \cos \omega t_2 - E(A^2) \cos \omega t_1 \sin \omega t_2 \\
 &= 0 + \sigma^2 \sin \omega t_1 \cos \omega t_2 - \sigma^2 \cos \omega t_1 \sin \omega t_2 = \sigma^2 [\sin \omega t_1 \cos \omega t_2 - \cos \omega t_1 \sin \omega t_2] \\
 &= \sigma^2 \sin \omega(t_1 - t_2)
 \end{aligned}$$

which is a function of $t_1 - t_2$.

Hence, we conclude that $X(t)$ and $Y(t)$ are jointly wide-sense stationary. ◆

PROBLEM SET 5.2

1. Consider the random process $X(t) = \cos(\omega_0 t + \theta)$, where θ is uniformly distributed in the interval $(-\pi, \pi)$. Check whether $\{X(t)\}$ is stationary or not.
2. For a random process $X(t) = Y \sin \omega t$, Y is a uniform random variable in the interval $(-1, 1)$ and ω is a constant. Check whether the process is wide-sense stationary or not.
3. Consider the process $X(t) = A \cos \omega t + B \sin \omega t$, where A and B are uncorrelated random variables each with mean 0 and variance 1, and ω is a positive constant. Show that the process $X(t)$ is covariance stationary. (*Anna, Nov. 2006*)
4. If $X(t) = Y \cos \omega t + Z \sin \omega t$, where Y and Z are two independent normal random variables with $E(Y) = E(Z) = 0$, $E(Y^2) = E(Z^2) = \sigma^2$ and ω is a constant, prove that $\{X(t)\}$ is a strict-sense stationary process of order 2.

(*Anna, May 2007; Nov. 2007*)

5. Consider the process

$$X(t) = (A + 1) \cos \omega t + B \sin \omega t, \quad -\infty < t < \infty$$

where A and B are independent random variables with $E(A) = E(B) = 0$ and $E(A^2) = 1$, $E(B^2) = 2$ and ω is a constant.

- (a) Find the autocorrelation function of $X(t)$.
- (b) Check if $X(t)$ is wide-sense stationary.

6. Suppose that $X(t)$ is a WSS process with mean $\mu \neq 0$ and that $Y(t)$ is defined by

$$Y(t) = X(t + \tau) - X(t)$$

where $\tau > 0$ is a constant.

- (a) Show that the mean of $Y(t)$ is zero for all values of t .
- (b) Show that the variance of $Y(t)$ is given by

$$\sigma_Y^2 = 2[R_{XX}(0) - R_{XX}(\tau)]$$

- (c) Check if $Y(t)$ is wide-sense stationary.

7. Suppose that $X(t)$ is a process with mean $\mu(t) = 3$ and autocorrelation $R(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}$. Determine the mean, variance and covariance of the random variables $Z = X(5)$ and $W = X(8)$.

(*Bharatidasan, Nov. 1996*)

8. If $X(t) = a \cos(\omega t + \theta)$ and $Y(t) = b \sin(\omega t + \theta)$, where a, b, ω are constants and θ is a random variable uniformly distributed in $(0, 2\pi)$, prove that the random processes $X(t)$ and $Y(t)$ are jointly wide-sense stationary.
9. If $X(t) = A \cos t + B \sin t$ and $Y(t) = B \cos t + A \sin t$, where A and B are independent random variables with mean 0 and variance σ^2 , show that $X(t)$ and $Y(t)$ are individually wide-sense stationary, but not jointly wide-sense stationary.
10. Consider the complex process $X(t) = Ae^{i\omega t}$, where A is a random variable and ω is a real constant. Show that $X(t)$ is wide-sense stationary if and only if $E(A) = 0$.

5.3 AUTOCORRELATION AND CROSS-CORRELATION FUNCTIONS

Definition 5.11. The *autocorrelation function* of a random process $X(t)$ is defined as

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] \quad (5.13)$$

If $t_1 = t$ and $t_2 = t + \tau$, where τ is a real number, then Eq. (5.13) becomes

$$R_{XX}(t, t + \tau) = E[X(t)X(t + \tau)]$$

If $X(t)$ is at least wide-sense stationary, then we know by Definition 5.5 that $R_{XX}(t, t + \tau)$ is a function only of the time difference, τ . Thus, for a wide-sense stationary process, we can represent the autocorrelation function simply as

$$R_{XX}(\tau) = E[X(t)X(t + \tau)]$$

Next, we establish some properties of the autocorrelation function $R_{XX}(\tau)$ in the following theorem.

Theorem 5.4. Let $X(t)$ be a wide-sense stationary process with constant mean μ_X and autocorrelation function $R_{XX}(\tau)$. Then the following properties hold:

- (a) $R_{XX}(\tau)$ is an even function of τ , i.e. $R_{XX}(-\tau) = R_{XX}(\tau)$.
- (b) $R_{XX}(0) = E[X^2(t)]$.
- (c) $|R_{XX}(\tau)| \leq R_{XX}(0)$, i.e. the maximum value of $|R_{XX}(\tau)|$ occurs at $\tau = 0$.
- (d) If $Y(t) = X(t) + a$, where a is a constant, then

$$E[Y(t)] = \mu_X + a \quad \text{and} \quad R_{YY}(\tau) = C_{XX}(\tau) + (\mu_X + a)^2$$

where $C_{XX}(\tau) = R_{XX}(\tau) - \mu_X^2$ is the *autocovariance function* of $X(t)$.

- (e) If $R_{XX}(\tau)$ is continuous at $\tau = 0$, then $R_{XX}(\tau)$ is continuous for all τ .
- (f) If there exists a constant $T > 0$ such that $R_{XX}(T) = R_{XX}(0)$, then $R_{XX}(\tau)$ is a periodic function and $X(t)$ is called a *periodic WSS process*.
- (g) If $X(t)$ has no periodic component, then

$$\lim_{\tau \rightarrow \infty} R_{XX}(\tau) = \mu_X^2$$

provided the limit exists.

Proof. (a) By definition, $R_{XX}(\tau) = E[X(t)X(t + \tau)]$. Thus, it follows that

$$R_{XX}(-\tau) = E[X(t)X(t - \tau)] = E[X(t - \tau)X(t)] = R_{XX}(\tau)$$

i.e. $R_{XX}(\tau)$ is an even function of τ .

- (b) By Theorem 5.3, we know that

$$R_{XX}(0) = E[X^2(t)] \quad \text{for all } t$$

and that $R_{XX}(0) \geq 0$.

(c) By Cauchy-Schwarz Inequality (Theorem 4.14), we know that

$$[E(XY)]^2 \leq E(X^2) E(Y^2)$$

Taking $X = X(t)$ and $Y = X(t + \tau)$, the above inequality becomes

$$\{E[X(t)X(t + \tau)]\}^2 \leq E(X^2(t)) E(X^2(t + \tau))$$

i.e.

$$[R_{XX}(\tau)]^2 \leq [R_{XX}(0)]^2$$

Taking square roots on both sides of the above inequality and noting that $R_{XX}(0) \geq 0$, we have

$$|R_{XX}(\tau)| \leq R_{XX}(0)$$

i.e. the function $|R_{XX}(\tau)|$ attains its maximum value at $\tau = 0$.

(d) Let $Y(t) = X(t) + a$, where a is a constant. Then

$$\mu_Y(t) = E[Y(t)] = E[X(t) + a] = E[X(t)] + a = \mu_X + a$$

which is a constant.

We also find that

$$\begin{aligned} R_{YY}(\tau) &= E[Y(t)Y(t + \tau)] = E\{[X(t) + a][X(t + \tau) + a]\} \\ &= E\{X(t)X(t + \tau) + a[X(t) + X(t + \tau)] + a^2\} \\ &= E[X(t)X(t + \tau)] + a\{E[X(t)] + E[X(t + \tau)]\} + a^2 \\ &= R_{XX}(\tau) + a[\mu_X + \mu_X] + a^2 = R_{XX}(\tau) + 2a\mu_X + a^2 \\ &= [R_{XX}(\tau) - \mu_X^2] + [\mu_X^2 + 2a\mu_X + a^2] \\ &= C_{XX}(\tau) + (\mu_X + a)^2 \end{aligned}$$

(e) Suppose that $R_{XX}(\tau)$ is continuous at $\tau = 0$. Now, we note that

$$\begin{aligned} E\{[X(t) - X(t + \tau)]^2\} &= E[X^2(t)] + E[X^2(t + \tau)] - 2E[X(t)X(t + \tau)] \\ &= R_{XX}(0) + R_{XX}(0) - 2R_{XX}(\tau) \\ &= 2[R_{XX}(0) - R_{XX}(\tau)] \end{aligned}$$

Since $R_{XX}(\tau)$ is continuous at $\tau = 0$, it follows that

$$\lim_{\tau \rightarrow 0} R_{XX}(\tau) = R_{XX}(0)$$

Hence, it follows that

$$\begin{aligned} 0 &= \lim_{\tau \rightarrow 0} 2[R_{XX}(0) - R_{XX}(\tau)] \\ &= \lim_{\tau \rightarrow 0} E\{[X(t) - X(t + \tau)]^2\} = 0 \\ &= \lim_{\tau \rightarrow 0} X(t + \tau) = X(t) \end{aligned}$$

which shows that $X(t)$ is continuous for all t .

Next, we note that

$$R_{XX}(\tau + h) - R_{XX}(\tau) = E[X(t)X(t + \tau + h)] - E[X(t)X(t + \tau)] = E\{X(t)[X(t + \tau + h) - X(t + \tau)]\}$$

Since $X(t)$ is continuous for all t , it follows that

$$\lim_{h \rightarrow 0} X(t + \tau + h) = X(t + \tau)$$

Hence, it is immediate that

$$\begin{aligned} \lim_{h \rightarrow 0} [R_{XX}(\tau + h) - R_{XX}(\tau)] &= \lim_{h \rightarrow 0} E\{X(t)[X(t + \tau + h) - X(t + \tau)]\} \\ &= E\left\{X(t) \lim_{h \rightarrow 0} [X(t + \tau + h) - X(t + \tau)]\right\} = 0 \end{aligned}$$

Thus, $R_{XX}(\tau)$ is continuous for all τ .

(f) By Cauchy-Schwarz Inequality (Theorem 4.14), we know that

$$[E(XY)]^2 \leq E(X^2)E(Y^2)$$

Taking $X = X(t)$ and $Y = X(t + \tau + T) - X(t + \tau)$, we have

$$\{E[X(t)[X(t + \tau + T) - X(t + \tau)]\}^2 \leq E[X^2(t)]E\{[X(t + \tau + T) - X(t + \tau)]^2\} \quad (5.14)$$

The L.H.S. of inequality, Eq. (5.14), is equal to

$$\{E[X(t)X(t + \tau + T)] - E[X(t)X(t + \tau)]\}^2 = [R_{XX}(\tau + T) - R_{XX}(\tau)]^2$$

which is always non-negative.

Note that

$$\begin{aligned} E\{[X(t + \tau + T) - X(t + \tau)]^2\} &= E\{X^2(t + \tau + T) + X^2(t + \tau) - 2X(t + \tau + T)X(t + \tau)\} \\ &= E[X^2(t + \tau + T)] + E[X^2(t + \tau)] - 2E[X(t + \tau + T)X(t + \tau)] \\ &= R_{XX}(0) + R_{XX}(0) - 2R_{XX}(T) = 2[R_{XX}(0) - R_{XX}(T)] \end{aligned}$$

Thus, the R.H.S. of inequality, Eq. (5.14), is equal to

$$R_{XX}(0)\{2[R_{XX}(0) - R_{XX}(T)]\} = 2R_{XX}(0)[R_{XX}(0) - R_{XX}(T)]$$

Hence, the inequality, Eq. (5.14) simplifies to

$$[R_{XX}(\tau + T) - R_{XX}(\tau)]^2 \leq 2R_{XX}(0)[R_{XX}(0) - R_{XX}(T)] \quad (5.15)$$

Thus, if $R_{XX}(T) = R_{XX}(0)$ for some $T > 0$, then it follows from the above inequality that

$$0 \leq [R_{XX}(\tau + T) - R_{XX}(\tau)]^2 \leq 0$$

which implies that

$$R_{XX}(\tau + T) - R_{XX}(\tau) = 0 \text{ for all } \tau$$

i.e.

$$R_{XX}(\tau + T) = R_{XX}(\tau) \text{ for all } \tau$$

i.e. $R_{XX}(\tau)$ is a periodic function.

(g) By definition, we have

$$R_{XX}(\tau) = E[X(t)X(t + \tau)]$$

Suppose that $X(t)$ does not have a periodic component. Under this assumption, it follows that as τ is very large, $X(t)$ and $X(t + \tau)$ tend to become independent. Hence, it follows that

$$\begin{aligned} \lim_{\tau \rightarrow \infty} R_{XX}(\tau) &= \lim_{\tau \rightarrow \infty} E[X(t)X(t + \tau)] \\ &= \lim_{\tau \rightarrow \infty} E[X(t)]E[X(t + \tau)] \\ &= \lim_{\tau \rightarrow \infty} \mu_X \mu_X = \mu_X^2 \end{aligned}$$

This completes the proof. ◆

Next, we consider the *cross-correlation function* of jointly wide-sense stationary processes, and study its properties.

Definition 5.12. The *cross-correlation function* of two random processes $X(t)$ and $Y(t)$ is defined as

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)] \quad (5.16)$$

If $t_1 = t$ and $t_2 = t + \tau$, where τ is a real number, then Eq. (5.16) becomes

$$R_{XY}(t, t + \tau) = E[X(t)Y(t + \tau)]$$

If $X(t)$ and $Y(t)$ are at least jointly wide-sense stationary, then we know by Definition 5.6 that $R_{XY}(t, t + \tau)$ is a function only of the time difference, τ . Thus, for jointly wide-sense stationary processes, we can represent their cross-correlation function simply as

$$R_{XY}(\tau) = E[X(t)Y(t + \tau)]$$

Next, we establish some properties of the cross-correlation function $R_{XY}(\tau)$ in the following theorem.

Theorem 5.5. Let $X(t)$ and $Y(t)$ be jointly wide-sense stationary processes with constant means μ_X and μ_Y respectively, and cross-correlation function $R_{XY}(\tau)$. Then the following properties hold:

- (a) $R_{XY}(-\tau) = R_{YX}(\tau)$.
- (b) $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$.
- (c) $|R_{XY}(\tau)| \leq \frac{1}{2} [R_{XX}(0) + R_{YY}(0)]$.
- (d) If $X(t)$ and $Y(t)$ are *orthogonal* processes, then $R_{XY}(\tau) = 0$.
- (e) If $X(t)$ and $Y(t)$ are *statistically independent*, then $R_{XY}(\tau) = \mu_X \mu_Y$.

(f) If $Z(t) = X(t) + Y(t)$, then

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau) + R_{XY}(\tau) + R_{YX}(\tau)$$

If $X(t)$ and $Y(t)$ are orthogonal, then

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau)$$

Proof. (a) By definition, $R_{XY}(\tau) = E[X(t)Y(t + \tau)]$. Hence, it follows that

$$R_{XY}(-\tau) = E[X(t)Y(t - \tau)] = E[Y(t - \tau)X(t)] = R_{YX}(\tau)$$

(b) By Cauchy-Schwarz Inequality (Theorem 4.14), we know that

$$[E(XY)]^2 \leq E(X^2) E(Y^2)$$

Taking $X = X(t)$ and $Y = Y(t + \tau)$ in the above inequality, we have

$$\{E[X(t)Y(t + \tau)]\}^2 \leq E[X^2(t)] E[Y^2(t + \tau)]$$

i.e.

$$|R_{XY}(\tau)|^2 \leq R_{XX}(0)R_{YY}(0)$$

Taking square root on both sides of the above inequality (noting that $R_{XX}(0)$ and $R_{YY}(0)$ are both non-negative), we have

$$|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)} \quad (5.17)$$

(c) If a and b are any two non-negative real numbers, then we know that the geometric mean of a and b cannot exceed their arithmetic mean, i.e.

$$\sqrt{ab} \leq \frac{1}{2} (a + b)$$

Taking $a = R_{XX}(0)$ and $b = R_{YY}(0)$, we have

$$\sqrt{R_{XX}(0)R_{YY}(0)} \leq \frac{1}{2} [R_{XX}(0) + R_{YY}(0)] \quad (5.18)$$

Combining the inequalities, Eqs. (5.17) and (5.18), we have

$$|R_{XY}(\tau)| \leq \frac{1}{2} [R_{XX}(0) + R_{YY}(0)]$$

(d) If X and Y are orthogonal processes, then

$$R_{XY}(\tau) = E[X(t)Y(t + \tau)] = 0$$

(e) If X and Y are statistically independent processes, then

$$R_{XY}(\tau) = E[X(t)Y(t + \tau)] = E[X(t)]E[Y(t + \tau)] = \mu_X \mu_Y$$

(f) Let $Z(t) = X(t) + Y(t)$. Then we have

$$\begin{aligned} R_{ZZ}(\tau) &= E[Z(t)Z(t+\tau)] = E\{[X(t) + Y(t)][X(t+\tau) + Y(t+\tau)]\} \\ &= E[X(t)X(t+\tau) + X(t)Y(t+\tau) + Y(t)X(t+\tau) + Y(t)Y(t+\tau)] \\ &= E[X(t)X(t+\tau)] + E[X(t)Y(t+\tau)] + E[Y(t)X(t+\tau)] + E[Y(t)Y(t+\tau)] \\ &= R_{XX}(\tau) + R_{XY}(\tau) + R_{YX}(\tau) + R_{YY}(\tau) \end{aligned}$$

If $X(t)$ and $Y(t)$ are orthogonal, then

$$R_{XY}(\tau) = E[X(t)Y(t+\tau)] = 0 \text{ and } R_{YX}(\tau) = E[Y(t)X(t+\tau)] = 0$$

Thus, in this case, it follows that

$$R_{ZZ}(\tau) = R_{XX}(\tau) + 0 + 0 + R_{YY}(\tau) = R_{XX}(\tau) + R_{YY}(\tau)$$

This completes the proof. ◆

EXAMPLE 5.25. A stationary random process $X = \{X(t)\}$ with mean 3 has auto-correlation function $R(\tau) = 16 + 9e^{-|\tau|}$. Find the standard deviation of the process.

(Anna, Model 2003)

Solution. Since $X(t)$ is a stationary random process, we know that

$$\mu_X^2 = \lim_{\tau \rightarrow \infty} R(\tau) = \lim_{\tau \rightarrow \infty} [16 + 9e^{-|\tau|}] = 16$$

and that

$$E[X^2(t)] = R(0) = 16 + 9 = 25$$

Hence, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = \sigma_X^2 = E[X^2(t)] - \mu_X^2 = 25 - 16 = 9$$

Thus, the standard deviation of $X(t)$ is given by

$$\sigma_X = +\sqrt{9} = 3$$

◆

EXAMPLE 5.26. The autocorrelation function for a stationary process $X(t)$ is given by

$$R_{XX}(\tau) = 9 + 2e^{-|\tau|}$$

Find the mean value of the random variable $Y = \int_{\tau=0}^2 X(t)dt$ and the variance of $X(t)$.

(Anna, April 2003)

Solution. Since $X(t)$ is a stationary process, we know that

$$\mu_X^2 = \lim_{\tau \rightarrow \infty} R_{XX}(\tau) = \lim_{\tau \rightarrow \infty} 9 + 2e^{-|\tau|} = 9$$

Thus, $\mu_X = 3$.

Next, we note that

$$R_{XX}(0) = E[X^2(t)] = 9 + 2 = 11$$

Hence, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = E[X^2(t)] - \mu_X^2 = 11 - 9 = 2$$

Next, we find the mean value of the random variable $Y = \int_0^2 X(t) dt$. We note that

$$E(Y) = \int_0^2 E[X(t)] dt = \int_0^2 3 dt = [3t]_0^2 = 6$$

◆

EXAMPLE 5.27. Given that the autocorrelation function for a stationary process with no periodic components is $R(z) = 25 + \frac{4}{1+6z^2}$. Find the mean and variance of the process $\{X(t)\}$.

(Madras, May 1999; Anna, Nov. 2004, May 2007)

Solution. Denote the given stationary process by $X(t)$. Then, we know that

$$\mu_X^2 = \lim_{z \rightarrow \infty} R(z) = \lim_{z \rightarrow \infty} \left[25 + \frac{4}{1+6z^2} \right] = 25$$

Hence, the mean of $X(t)$ is $\mu_X = 5$. Next, we find that

$$R(0) = E[X^2(t)] = 25 + 4 = 29$$

Hence, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = \sigma_X^2 = E[X^2(t)] - \mu_X^2 = 29 - 25 = 4$$

◆

EXAMPLE 5.28. Consider two random processes $X(t) = 3\cos(\omega t + \theta)$ and $Y(t) = 2\cos\left(\omega t + \theta - \frac{\pi}{2}\right)$, where θ is a random variable uniformly distributed in $(0, 2\pi)$. Prove that $\sqrt{R_{XX}(0)R_{YY}(0)} \geq |R_{XY}(\tau)|$.

(Anna, April 2003)

Solution. We are given that $X(t) = 3\cos(\omega t + \theta)$ and

$$Y(t) = 2\cos\left(\omega t + \theta - \frac{\pi}{2}\right) = 2\cos\left(\frac{\pi}{2} - \omega t - \theta\right) = 2\sin(\omega t + \theta)$$

where θ is a uniform random variable in $(0, 2\pi)$.

By the solution of Problem 8, Section 5.2, it follows that $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes with correlation functions

$$R_{XX}(t_1, t_2) = \frac{9}{2} \cos(\omega(t_1 - t_2))$$

$$R_{YY}(t_1, t_2) = 2 \cos(\omega(t_1 - t_2))$$

$$R_{XY}(t_1, t_2) = 3 \sin(\omega(t_1 - t_2))$$

Substituting $t_1 = t$ and $t_2 = t + \tau$, we have

$$R_{XX}(\tau) = \frac{9}{2} \cos \omega \tau$$

$$R_{YY}(\tau) = 2 \cos \omega \tau$$

$$R_{XY}(\tau) = -3 \sin \omega \tau$$

Thus, it follows that

$$\sqrt{R_{XX}(0)R_{YY}(0)} = \sqrt{\frac{9}{2} \times 2} = \sqrt{9} = 3$$

Hence, we find that

$$|R_{XY}(\tau)| = |-3 \sin \omega \tau| = 3 |\sin \omega \tau| \leq 3 = \sqrt{R_{XX}(0)R_{YY}(0)}$$

◆

EXAMPLE 5.29. If $X(t) = 4 \sin(\omega t + \phi)$ and $Y(t) = 2 \cos(\omega t + \theta)$, where ω is a constant, $\theta - \phi = \frac{\pi}{2}$ and ϕ is a random variable uniformly distributed in $(0, 2\pi)$, show that $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes. Compute $R_{XX}(\tau)$, $R_{YY}(\tau)$, $R_{XY}(\tau)$ and $R_{YX}(\tau)$. Verify two properties of the autocorrelation function, $R_{XX}(\tau)$ and the cross-correlation function $R_{XY}(\tau)$.

Solution. It can be easily shown that $X(t)$ and $Y(t)$ are individually wide-sense stationary processes with

$$\mu_X = 0, R_{XX}(\tau) = 8 \cos \omega \tau, \mu_Y = 0 \text{ and } R_{YY}(\tau) = 2 \cos \omega \tau$$

Also, the cross-correlation between $X(t)$ and $Y(t)$ can be easily obtained as

$$R_{XY}(\tau) = -4 \cos \omega \tau$$

which is purely a function of τ . Thus, $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes.

Next, we verify two properties of the auto-correlation function for $R_{XX}(\tau)$.

- (a) We show that $R_{XX}(-\tau) = R_{XX}(\tau)$, i.e. $R_{XX}(\tau)$ is an even function of τ . Note that

$$R_{XX}(-\tau) = 8 \cos -\omega \tau = 8 \cos \omega \tau = R_{XX}(\tau)$$

- (b) We show that $|R_{XX}(\tau)| \leq R_{XX}(0)$. Note that $R_{XX}(0) = 8$ and

$$|R_{XX}(\tau)| = |8 \cos \omega \tau| = 8 |\cos \omega \tau| \leq 8 = R_{XX}(0)$$

since $|\cos \theta| \leq 1$ for all θ .

Next, we verify two properties of the cross-correlation function, $R_{XY}(\tau)$.

- (a) We show that $R_{XY}(-\tau) = R_{YX}(\tau)$. Note that

$$R_{XY}(-\tau) = -4 \cos -\omega\tau = -4 \cos \omega\tau = R_{YX}(\tau)$$

- (b) We show that $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$. Note that $R_{XX}(0) = 8$ and $R_{YY}(0) = 2$. Thus, it follows that

$$\sqrt{R_{XX}(0)R_{YY}(0)} = \sqrt{8 \times 2} = \sqrt{16} = 4$$

and

$$|R_{XY}(\tau)| = |-4 \cos \omega\tau| = 4|\cos \omega\tau| \leq 4$$

Hence, it is immediate that $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$.

◆

PROBLEM SET 5.3

1. Given the autocorrelation function for the random process $X(t)$ as

$$R_{XX}(\tau) = 16 + \frac{2}{1 + 3\tau^2}$$

find the mean and variance of the process.

(Madras, May 1999)

2. A stationary process has an autocorrelation function given by

$$R_{XX}(\tau) = \frac{55\tau^2 + 90}{2.2\tau^2 + 3}$$

Find the mean and variance of the process.

3. A stationary process has an autocorrelation given by

$$R_{XX}(\tau) = 2 + 6e^{-3|\tau|}$$

Find the mean and variance of the process.

4. If $X(t)$ is a wide-sense stationary process with autocorrelation function

$$R_{XX}(\tau) = 1 + 4e^{-3|\tau|}$$

find the mean and variance of the process. Also, find the mean of $S = \int_0^1 X(t) dt$.

5. If $X(t) = a \sin(\omega t + \theta)$, where a and ω are constants, and θ is uniformly distributed in the interval $(0, 2\pi)$, show that $X(t)$ is wide-sense stationary and compute its autocorrelation function $R_{XX}(\tau)$. Is $X(t)$ also strictly stationary?

6. Let $X(t)$ be a random process with four sample functions

$$x(t, s_1) = \cos t, x(t, s_2) = \sin t, x(t, s_3) = -\cos t \text{ and } x(t, s_4) = -\sin t$$

which are equally likely. Show that $X(t)$ is wide-sense stationary and compute its autocorrelation function $R_{XX}(\tau)$.

7. If $X(t)$ is a wide-sense stationary process with autocorrelation $R_{XX}(\tau)$, show that

$$P\{|X(t+\tau) - X(t)| \geq a\} \leq \frac{2[R_{XX}(0) - R_{XX}(\tau)]}{a^2}$$

8. If $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes with cross-correlation function $R_{XY}(\tau)$, show that

$$2|R_{XY}(\tau)| \leq R_{XX}(0)R_{YY}(0)$$

9. If $X(t)$ is a WSS process and $\dot{X}(t)$ is the derivative of $X(t)$, then show that the cross-correlation between $X(t)$ and $\dot{X}(t)$ is

$$R_{X\dot{X}}(\tau) = \frac{dR_{XX}(\tau)}{d\tau}$$

10. If $X(t)$ is a WSS process, then show that the autocorrelation of $\dot{X}(t)$ is

$$R_{\dot{X}\dot{X}}(\tau) = -\frac{d^2 R_{XX}(\tau)}{d\tau^2}$$

5.4 ERGODIC PROCESS

In probability theory, a *stationary ergodic process* is a random process that exhibits both *stationarity* and *ergodicity*. We recall that a stationary process or a strictly stationary process is a random process whose probability density functions are invariant under any time shift. Also, a weakly stationary process or a wide-sense stationary process has a constant mean and its autocorrelation function depends only on the time difference. As a result, parameters such as mean and variance do not change over time. An ergodic process is a random process, which conforms to *ergodic theorem* (Birkhoff, 1931). The ergodic theorem allows the time averages of a conforming process to equal the ensemble averages of the random process.

The ensemble averages of a random process $X(t)$ are the mean value $\mu_X(t)$ defined by

$$\mu_X(t) = \int_{-\infty}^{\infty} x f_X(x; t) dx$$

and the autocorrelation function $R_{XX}(\tau)$ defined by

$$R_{XX}(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau) f_X(x; t) dx$$

Next, we define the *time average* of a quantity over an interval $(-T, T)$, where $T > 0$.

Definition 5.13. For any quantity u , the *time average* of u over the interval $(-T, T)$ is defined as

$$A(u) = \frac{1}{2T} \int_{-T}^T u dt$$

where A is used to denote the time average similar to the use of E for statistical expectation. We note that the time average of u may be also denoted by $\langle u \rangle$.

Definition 5.14. Let $X(t)$ be any wide-sense stationary random process. Consider any realization or sample function $x(t)$ of the process. Then the *time average* of the process $X(t)$ over the interval $(-T, T)$ is denoted by $\bar{X}_T$ and defined as

$$\bar{X}_T = A[X(t)] = \frac{1}{2T} \int_{-T}^T X(t) dt \quad (5.19)$$

and the *time correlation function* of the process $X(t)$ over the interval $(-T, T)$ is denoted by $\hat{Z}_T$ and defined as

$$\hat{Z}_T = A[X(t)X(t+\tau)] = \frac{1}{2T} \int_{-T}^T X(t)X(t+\tau) dt \quad (5.20)$$

Taking expectation on both sides of Eqs. (5.19) and (5.20) and simplifying, we get

$$E[\bar{X}_T] = \frac{1}{2T} \int_{-T}^T \mu_X dt = \mu_X$$

$$E[\hat{Z}_T] = \frac{1}{2T} \int_{-T}^T R_{XX}(\tau) dt = R_{XX}(\tau)$$

Next, we define the *ergodicity* of a random process.

Definition 5.15. Let $X(t)$ be any wide-sense stationary random process with constant mean μ_X and autocorrelation function $R_{XX}(\tau) = E[X(t)X(t+\tau)]$. Then the process $X(t)$ is called *ergodic* if the time averages of the random process are equal to the corresponding statistical or ensemble averages in the limits.

(a) The process $X(t)$ is called *mean-ergodic* (or *ergodic in the mean*) if

$$\lim_{T \rightarrow \infty} \bar{X}_T = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = \mu_X$$

(b) The process $X(t)$ is called *correlation ergodic* (or *ergodic in the correlation*) if

$$\lim_{T \rightarrow \infty} \hat{Z}_T = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)X(t+\tau) dt = R_{XX}(\tau)$$

(c) The process $X(t)$ is called *mean-square ergodic* (or *power ergodic*) if

$$\lim_{T \rightarrow \infty} \hat{W}_T = \lim_{T \rightarrow \infty} \left[\frac{1}{2T} \int_{-T}^T X^2(t) dt \right] = R_{XX}(0)$$

(d) The process $X(t)$ is called *wide-sense ergodic* if it is mean ergodic and correlation ergodic.

Remark 5.6. From Definition 5.15, it is clear that if a process is correlation-ergodic, then it must be mean-square ergodic. Thus, mean-square ergodic processes form a special class of correlation-ergodic processes.

Definition 5.16. Let $X(t)$ and $Y(t)$ be jointly wide-sense stationary random processes with cross-correlation function $R_{XY}(\tau) = E[X(t)Y(t+\tau)]$. Then $X(t)$ and $Y(t)$ are called *jointly ergodic* if they are individually ergodic processes and also the *time cross-correlation function* of $X(t)$ and $Y(t)$ is equal to the statistical cross-correlation function in the limits, i.e.

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t)Y(t+\tau) dt = R_{XY}(\tau)$$

Remark 5.7. We note that ergodicity is a stronger condition than stationarity, i.e. not all stationary processes are ergodic. In this section, we shall derive conditions under which a wide-sense stationary process is mean ergodic and correlation ergodic. When these conditions are fulfilled, stationary processes will be ergodic with respect to mean, correlation, etc. In real-world applications involving stationary processes, we often assume that the process is ergodic to simplify the problems.

Next, we establish the *Mean Ergodic Theorem*, which provides an important characterization for a random process $X(t)$ to be mean ergodic.

Theorem 5.6. (Mean Ergodic Theorem) Let $X(t)$ be a random process with constant mean μ_X . Let

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt$$

denote the time average of the process and let

$$\bar{\sigma}_T^2 = \text{Var}(\bar{X}_T)$$

denote the variance of $\bar{X}_T$. Then the process $X(t)$ is mean ergodic if

$$\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = 0 \quad (5.21)$$

Proof. The time average of the process is given by

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt$$

Thus, the mean of $\bar{X}_T$ is given by

$$E[\bar{X}_T] = \frac{1}{2T} \int_{-T}^T E[X(t)] dt = \frac{1}{2T} \int_{-T}^T \mu_X dt = \mu_X$$

By Chebyshev's inequality (Corollary 2.2), we know that

$$P\{|\bar{X}_T - E(\bar{X}_T)| \leq \varepsilon\} \geq 1 - \frac{\text{Var}[\bar{X}_T]}{\varepsilon^2}$$

where $\varepsilon > 0$ is a constant. Using the fact that $E[\bar{X}_T] = \mu_X$, we have

$$P\{|\bar{X}_T - \mu_X| \leq \varepsilon\} \geq 1 - \frac{\bar{\sigma}_T^2}{\varepsilon^2}$$

Taking limits as $T \rightarrow \infty$, we get

$$P\left\{\lim_{T \rightarrow \infty} |\bar{X}_T - \mu_X| \leq \varepsilon\right\} \geq 1 - \frac{\lim_{T \rightarrow \infty} \bar{\sigma}_T^2}{\varepsilon^2}$$

Using Eq. (5.21), we have

$$P\left\{\lim_{T \rightarrow \infty} |\bar{X}_T - \mu_X| \leq \varepsilon\right\} \geq 1 - 0 = 1$$

Since the probability of any event cannot exceed 1,

$$P\left\{\lim_{T \rightarrow \infty} |\bar{X}_T - \mu_X| \leq \varepsilon\right\} = 1$$

i.e.

$$\lim_{T \rightarrow \infty} \bar{X}_T = \mu_X \text{ with probability } 1$$

Thus, we conclude that $X(t)$ is a mean-ergodic process. $\blacklozenge$

EXAMPLE 5.30. Consider a constant random process $X(t) = C$, where C is a random variable with mean μ and variance σ^2 . Examine whether $X(t)$ is mean ergodic.

Solution. First, we calculate the time average of the process. By definition, we have

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt = \frac{1}{2T} \int_{-T}^T C dt = \frac{1}{2T} C[2T] = C$$

Note that

$$\lim_{T \rightarrow \infty} \bar{X}_T = \lim_{T \rightarrow \infty} C = C$$

Note that

$$\bar{\sigma}_T^2 = \text{Var}[\bar{X}_T] = \text{Var}(C) = \sigma^2$$

showing that $\sigma_T = \sigma$, the standard deviation of the random variable C .

Thus, we note that

$$\lim_{T \rightarrow \infty} \sigma_T = \lim_{T \rightarrow \infty} \sigma = \sigma \neq 0$$

Hence, by Mean Ergodic Theorem (Theorem 5.6), it follows that $X(t)$ is not mean ergodic. $\blacklozenge$

EXAMPLE 5.31. Let $X(t)$ be a random process with constant mean μ_X . Show that $X(t)$ is mean ergodic if

$$\lim_{T \rightarrow \infty} \left[\frac{1}{4T^2} \int_{-T}^T \int_{-T}^T C_{XX}(t_1, t_2) dt_1 dt_2 \right] = 0 \quad (5.22)$$

Solution. The time average of the process $X(t)$ is given by

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt$$

We find that

$$E(\bar{X}_T) = \frac{1}{2T} \int_{-T}^T E[X(t)] dt = \frac{1}{2T} \int_{-T}^T \mu_X dt = \mu_X$$

Next, we note that

$$\bar{X}_T^2 = \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T X(t_1)X(t_2) dt_1 dt_2$$

Thus, it follows that

$$E[\bar{X}_T^2] = \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T E[X(t_1)X(t_2)] dt_1 dt_2 = \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) dt_1 dt_2$$

Hence, it follows that

$$\text{Var}[\bar{X}_T] = \bar{\sigma}_T^2 = E[\bar{X}_T^2] - [E(\bar{X}_T)]^2$$

i.e.,

$$\bar{\sigma}_T^2 = \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T R_{XX}(t_1, t_2) dt_1 dt_2 - \mu_X^2$$

i.e.,

$$\bar{\sigma}_T^2 = \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T [R_{XX}(t_1, t_2) - \mu_X^2] dt_1 dt_2 = \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T C_{XX}(t_1, t_2) dt_1 dt_2$$

Thus, if we assume that Eq. (5.22) holds, then it is immediate that

$$\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = 0$$

Hence, by Mean Ergodic Theorem (Theorem 5.6), it follows that the random process $X(t)$ is mean ergodic. $\blacklozenge$

Theorem 5.7. Let $X(t)$ be a wide-sense stationary process with mean μ_X and auto-covariance function $C_{XX}(\tau)$. Let $\bar{X}_T$ denote the time average of the process, i.e.

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt$$

and $\bar{\sigma}_T^2 = \text{Var}[\bar{X}_T]$ be the variance of $\bar{X}_T$.

(a) The variance of $\bar{X}_T$ is given by

$$\bar{\sigma}_T^2 = \frac{1}{2T} \int_{-2T}^{2T} C_{XX}(\tau) \left[1 - \frac{|\tau|}{2T}\right] d\tau = \frac{1}{T} \int_0^{2T} C_{XX}(\tau) \left[1 - \frac{\tau}{2T}\right] d\tau$$

(b) A necessary and sufficient condition for the process $X(t)$ to be mean ergodic is

$$\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = \lim_{T \rightarrow \infty} \left\{ \frac{1}{2T} \int_{-2T}^{2T} C_{XX}(\tau) \left[1 - \frac{|\tau|}{2T}\right] d\tau \right\} = \lim_{T \rightarrow \infty} \left\{ \frac{1}{T} \int_0^{2T} C_{XX}(\tau) \left[1 - \frac{\tau}{2T}\right] d\tau \right\} = 0$$

(c) A necessary and sufficient condition for the process $X(t)$ to be mean ergodic is

$$\int_{-\infty}^{\infty} |C_{XX}(\tau)| d\tau < \infty$$

Proof. (a) The mean of the time average $\bar{X}_T$ is given by

$$E(\bar{X}_T) = \frac{1}{2T} \int_{-T}^T E[X(t)] dt = \frac{1}{2T} \int_{-T}^T \mu_X dt = \mu_X$$

Hence, the variance of the time average $\bar{X}_T$ is given by

$$\begin{aligned} \bar{\sigma}_T^2 &= \text{Var}(\bar{X}_T) = E[(\bar{X}_T - \mu_X)^2] = E \left\{ \left[\frac{1}{2T} \int_{-T}^T [X(t) - \mu_X] dt \right]^2 \right\} \\ &= E \left\{ \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T [X(t) - \mu_X][X(s) - \mu_X] dt ds \right\} \\ &= \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T E\{[X(t) - \mu_X][X(s) - \mu_X]\} dt ds \\ &= \frac{1}{4T^2} \int_{-T}^T \int_{-T}^T C_{XX}(t, s) dt ds \end{aligned} \tag{5.23}$$

We use the substitution

$$\tau = s - t \text{ and } u = t$$

Solving for s and t , we get

$$s = \tau + u \text{ and } t = u$$

We compute the Jacobian of the transformation, viz.

$$J = \frac{\partial(s,t)}{\partial(\tau,u)} = \begin{vmatrix} \frac{\partial s}{\partial \tau} & \frac{\partial s}{\partial u} \\ \frac{\partial t}{\partial \tau} & \frac{\partial t}{\partial u} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$$

Since the process $X(t)$ is wide-sense stationary, the autocovariance function $C_{XX}(s,t)$ is a function only of the time difference $s - t$. Hence, Eq. (5.23) becomes

$$\bar{\sigma}_T^2 = \frac{1}{4T^2} \iint C_{XX}(\tau) d\tau du \quad (5.24)$$

where the integration has to be carried out over the parallelogram shown in Figure 5.1 in regions A and B.

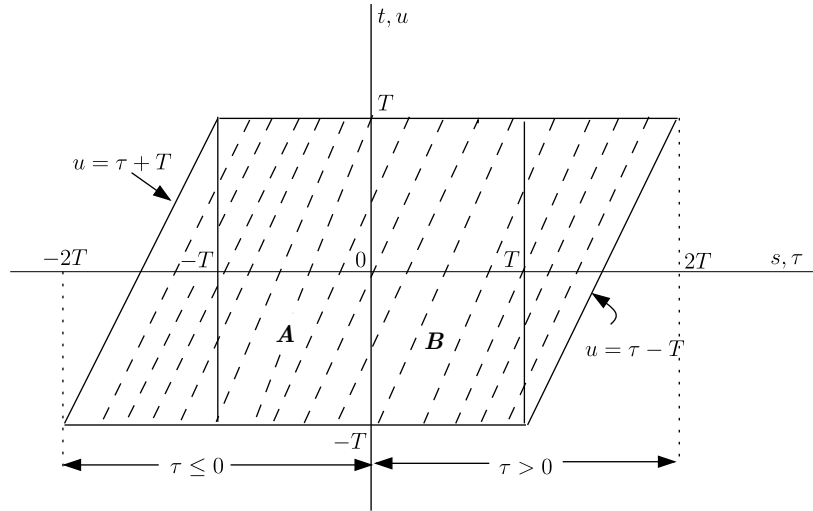


Figure 5.1: The regions A and B.

Region A: $\tau \leq 0$: τ varies from $-2T$ to 0 , while u varies from $-T$ to $T + \tau$.

Region B: $\tau > 0$: τ varies from 0 to $2T$, while u varies from $-T + \tau$ to T .

Substituting the above limits in Eq. (5.24), we obtain the variance $\bar{\sigma}_T^2$ as follows:

$$\bar{\sigma}_T^2 = \frac{1}{4T^2} \left\{ \int_{-T+T}^T \left[\int_0^{2T} C_{XX}(\tau) d\tau \right] du + \int_{-T}^{T+T} \left[\int_{-2T}^0 C_{XX}(\tau) d\tau \right] du \right\}$$

$$\begin{aligned}
&= \frac{1}{4T^2} \left\{ \int_0^{2T} C_{XX}(\tau)[2T - \tau] d\tau + \int_{-2T}^0 C_{XX}(\tau)[2T + \tau] d\tau \right\} \\
&= \frac{1}{4T^2} \left\{ \int_{-2T}^{2T} C_{XX}(\tau)[2T - |\tau|] d\tau \right\}
\end{aligned}$$

i.e.

$$\bar{\sigma}_T^2 = \frac{1}{2T} \int_{-2T}^{2T} C_{XX}(\tau) \left[1 - \frac{|\tau|}{2T} \right] d\tau = \frac{1}{T} \int_0^{2T} C_{XX}(\tau) \left[1 - \frac{\tau}{2T} \right] d\tau \quad (5.25)$$

where we have used the fact that $C_{XX}(\tau) = R_{XX}(\tau) - \mu_X^2$ is an even function of τ .

- (b) By Mean Ergodic Theorem (Theorem 5.6), we know that a necessary and sufficient condition for the process $X(t)$ to be mean ergodic is that

$$\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = 0$$

Using Eq. (5.25), it is immediate that a necessary and sufficient condition for the process $X(t)$ to be mean ergodic is that

$$\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} C_{XX}(\tau) \left[1 - \frac{|\tau|}{2T} \right] d\tau = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{2T} C_{XX}(\tau) \left[1 - \frac{\tau}{2T} \right] d\tau = 0$$

- (c) We note that

$$\left| \int_{-2T}^{2T} C_{XX}(\tau) \left[1 - \frac{|\tau|}{2T} \right] d\tau \right| \leq \int_{-2T}^{2T} |C_{XX}(\tau)| \left[1 - \frac{|\tau|}{2T} \right] d\tau \leq \int_{-2T}^{2T} |C_{XX}(\tau)| d\tau \quad (5.26)$$

From the inequality, Eq. (5.26), and Eq. (5.25), it is immediate that if

$$\int_{-\infty}^{\infty} |C_{XX}(\tau)| d\tau < \infty \quad (5.27)$$

then $\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = 0$. Hence, a necessary and sufficient condition for the random process $X(t)$ to be mean-ergodic is given by the inequality Eq. (5.27). This completes the proof. $\blacklozenge$

We state the following result without proof. (For a proof of this result, see (Papoulis, 1999, p. 430).

Theorem 5.8. (Slutsky's Theorem) Let $X(t)$ be a wide-sense stationary process with autocovariance function $C_{XX}(\tau)$. Then the process $X(t)$ is mean-ergodic if and only if

$$\lim_{T \rightarrow \infty} \left[\frac{1}{T} \int_0^T C_{XX}(\tau) d\tau \right] = 0$$

Remark 5.8. Theorems 5.7 and 5.8 provide some necessary and sufficient conditions for a wide-sense stationary process $X(t)$ to be mean ergodic. However, we cannot always use these results for practical applications because the theorem involves prior knowledge of the auto-covariance function $C_{XX}(\tau)$ of the random process $X(t)$. However, even if we are given some partial knowledge of the auto-covariance function of the process such as $|C_{XX}(\tau)| \rightarrow 0$ as $\tau \rightarrow \infty$, we can conclude that the process is mean ergodic. We also note that ergodic processes must be stationary, but the converse is not true, i.e. stationary processes need not be ergodic. For problems in applications, it is very difficult to show that any arbitrary random process is ergodic, so we usually make an assumption that the process that is stationary also is ergodic.

Next, we derive a characterization of correlation-ergodic processes.

Theorem 5.9. (Correlation Ergodic Theorem) Let $X(t)$ be a wide-sense stationary process with constant mean μ_X and auto-correlation function $R_{XX}(\tau) = E[X(t)X(t+\tau)]$. Let

$$\hat{Z}_T = \frac{1}{2T} \int_{-T}^T X(t)X(t+\tau)dt$$

denote the time auto-correlation function of the process and

$$\hat{\sigma}_T^2 = \text{Var}(\hat{Z}_T)$$

denote the variance of $\hat{Z}_T$. Then the process $X(t)$ is correlation ergodic if

$$\lim_{T \rightarrow \infty} \hat{\sigma}_T = 0 \quad (5.28)$$

Proof. The time auto-correlation of the process is given by

$$\hat{Z}_T = \frac{1}{2T} \int_{-T}^T X(t)X(t+\tau)dt$$

Thus, the mean of $\hat{Z}_T$ is given by

$$E[\hat{Z}_T] = \frac{1}{2T} \int_{-T}^T E[X(t)X(t+\tau)] dt = \frac{1}{2T} \int_{-T}^T R_{XX}(\tau) dt = R_{XX}(\tau)$$

By Chebyshev's inequality (Corollary 2.2), we know that

$$P\{|\hat{Z}_T - E(\hat{Z}_T)| \leq \varepsilon\} \geq 1 - \frac{\text{Var}[\hat{Z}_T]}{\varepsilon^2}$$

where $\varepsilon > 0$ is a constant. Using the fact that $E[\hat{Z}_T] = R_{XX}(\tau)$, we have

$$P\{|\hat{Z}_T - R_{XX}(\tau)| \leq \varepsilon\} \geq 1 - \frac{\hat{\sigma}_T^2}{\varepsilon^2}$$

Taking limits as $T \rightarrow \infty$, we get

$$P \left\{ \left| \lim_{T \rightarrow \infty} \bar{Z}_T - R_{XX}(\tau) \right| \leq \varepsilon \right\} \geq 1 - \frac{\lim_{T \rightarrow \infty} \hat{\sigma}_T^2}{\varepsilon^2}$$

Using Eq. (5.28), we have

$$P \left\{ \left| \lim_{T \rightarrow \infty} \hat{Z}_T - R_{XX}(\tau) \right| \leq \varepsilon \right\} \geq 1 - 0 = 1$$

Since the probability of any event cannot exceed 1, we must have

$$P \left\{ \left| \lim_{T \rightarrow \infty} \hat{Z}_T - R_{XX}(\tau) \right| \leq \varepsilon \right\} = 1$$

i.e.

$$\lim_{T \rightarrow \infty} \hat{Z}_T = R_{XX}(\tau) \text{ with probability } 1$$

Thus, we conclude that $X(t)$ is a correlation-ergodic process. $\blacklozenge$

EXAMPLE 5.32. A random process is defined as $X(t) = A \cos \omega t + B \sin \omega t$, where A and B are random variables with $E(A) = E(B) = 0$, $E(A^2) = E(B^2)$ and $E(AB) = 0$. Show that the process $X(t)$ is mean ergodic.

Solution. First, we compute the mean (ensemble average) of the process $X(t)$. By definition, we have

$$\mu_X(t) = E[X(t)] = E[A \cos \omega t + B \sin \omega t] = E(A) \cos \omega t + E(B) \sin \omega t$$

Since $E(A) = E(B) = 0$, it is immediate that

$$\mu_X(t) = 0 \cos \omega t + 0 \sin \omega t = 0$$

which is a constant.

Next, we compute the time average of the process $X(t)$. By definition, we have

$$\bar{X}_T = \langle X(t) \rangle = \frac{1}{2T} \int_{-T}^T X(t) dt = \frac{1}{2T} \int_{-T}^T [A \cos \omega t + B \sin \omega t] dt$$

Integrating, we have

$$\begin{aligned} \bar{X}_T &= \frac{1}{2T} \left[A \frac{\sin \omega t}{\omega} - B \frac{\cos \omega t}{\omega} \right]_{-T}^T = \frac{1}{2T} \left\{ \left[A \frac{\sin \omega T}{\omega} - B \frac{\cos \omega T}{\omega} \right] - \left[-A \frac{\sin \omega T}{\omega} - B \frac{\cos \omega T}{\omega} \right] \right\} \\ &= \frac{1}{2T} \left[2A \frac{\sin \omega T}{\omega} \right] = \frac{A \sin \omega T}{\omega T} \end{aligned}$$

Thus, it follows that

$$\lim_{T \rightarrow \infty} \bar{X}_T = \lim_{T \rightarrow \infty} \frac{A \sin \omega T}{\omega T} = \frac{A}{\omega} \lim_{T \rightarrow \infty} \frac{\sin \omega T}{T} = 0$$

because $|\sin \omega T| \leq 1$ for all T and $\frac{1}{T} \rightarrow 0$ as $T \rightarrow \infty$.

Since $\lim_{T \rightarrow \infty} \bar{X}_T = 0 = \mu_X$, we conclude that the given process is mean ergodic. $\blacklozenge$

EXAMPLE 5.33. Let $X(t)$ be a wide-sense stationary process with autocorrelation function

$$R_{XX}(\tau) = e^{-a|\tau|}$$

where $a > 0$ is a constant.

- (a) Find the mean and variance of $X(t)$.
- (b) Find the mean and variance of the time average of $X(t)$ over $(-T, T)$.
- (c) Determine if $X(t)$ is mean ergodic.

Solution. (a) First, we calculate the mean of $X(t)$. We note that

$$\mu_X^2 = \lim_{\tau \rightarrow \infty} R_{XX}(\tau) = \lim_{\tau \rightarrow \infty} e^{-a\tau} = 0$$

Thus, it follows that $\mu_X = 0$.

Next, we note that

$$R_{XX}(0) = E[X^2(t)] = 1$$

Thus, the variance of $X(t)$ is given by

$$\sigma_X^2 = \text{Var}[X(t)] = E[X^2(t)] - \mu_X^2 = 1 - 0 = 1$$

- (b) The time average of $X(t)$ over the interval $(-T, T)$ is given by

$$\bar{X}_T = \langle X(t) \rangle = \frac{1}{2T} \int_{-T}^T X(t) dt$$

Thus, the mean of the time average of $X(t)$ is given by

$$E[\bar{X}_T] = \frac{1}{2T} \int_{-T}^T \mu_X dt = \mu_X = 0$$

By Theorem 5.7, we know that the variance of the time average of $X(t)$ is given by

$$\bar{\sigma}_T^2 = \text{Var}[\bar{X}_T] = \frac{1}{T} \int_0^{2T} C_{XX}(\tau) \left(1 - \frac{\tau}{2T}\right) d\tau$$

where $C_{XX}(\tau)$ is the auto-covariance function of $X(t)$ given by

$$C_{XX}(\tau) = R_{XX}(\tau) - \mu_X^2 = e^{-a|\tau|} - 0 = e^{-a|\tau|}$$

Substituting, we get

$$\bar{\sigma}_T^2 = \frac{1}{T} \int_0^{2T} e^{-a\tau} \left(1 - \frac{\tau}{2T}\right) d\tau = \frac{1}{aT} \int_0^{2T} \left(1 - \frac{\tau}{2T}\right) d(-e^{-a\tau})$$

Integrating by parts, we get

$$\begin{aligned}\bar{\sigma}_T^2 &= \frac{1}{aT} \left\{ \left[\left(1 - \frac{\tau}{2T}\right) (-e^{-a\tau}) \right]_0^{2T} - \int_0^{2T} (-e^{-a\tau}) \left(-\frac{1}{2T}\right) d\tau \right\} \\ &= \frac{1}{aT} \left\{ [0+1] + \frac{1}{2aT} \int_0^{2T} (-ae^{-a\tau}) d\tau \right\}\end{aligned}$$

Integrating, we get

$$\bar{\sigma}_T^2 = \frac{1}{aT} \left\{ 1 + \frac{1}{2aT} [e^{-a\tau}]_0^{2T} \right\} = \frac{1}{aT} \left\{ 1 + \frac{1}{2aT} [e^{-2aT} - 1] \right\}$$

i.e.

$$\bar{\sigma}_T^2 = \frac{1}{aT} - \frac{1}{2a^2T^2} [1 - e^{-2aT}]$$

(c) From the result obtained in (b), it follows that

$$\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = \lim_{T \rightarrow \infty} \left\{ \frac{1}{aT} - \frac{1}{2a^2T^2} [1 - e^{-2aT}] \right\} = 0$$

Hence, by Mean Ergodic Theorem (Theorem 5.6), it follows that the process $X(t)$ is mean ergodic. ◆

EXAMPLE 5.34. Show that the random process $X(t) = \cos(t + \phi)$, where ϕ is a random variable uniformly distributed in $(0, 2\pi)$ is

- (i) first order stationary
- (ii) stationary in the wide sense, and
- (iii) ergodic (based on first order or second order averages).

(Anna, May 2006)

Solution. (i) Using the extended formula of the transformation technique (Remark 3.14), it can be easily shown that the first-order probability density function of $X(t)$ is given by

$$f_X(x) = \frac{1}{\pi} \frac{1}{\sqrt{1-x^2}} \quad \text{for } -1 \leq x \leq 1$$

which is independent of t . Hence, the process $X(t)$ is first order stationary.

- (ii) The mean of the process $X(t)$ is given by

$$\mu_X(t) = E[X(t)] = E[\cos(t + \phi)]$$

Since ϕ has the uniform density function

$$f_{\phi}(\phi) = \begin{cases} \frac{1}{2\pi} & \text{if } 0 < \phi < 2\pi \\ 0 & \text{otherwise} \end{cases}$$

it follows that

$$\mu_X(t) = \int_{\phi=0}^{2\pi} \cos(t+\phi) \frac{1}{2\pi} d\phi = \frac{1}{2\pi} \int_{\phi=0}^{2\pi} \cos(t+\phi) d\phi$$

Integrating, we get

$$\mu_X(t) = \frac{1}{2\pi} [\sin(t+\phi)]_{\phi=0}^{2\pi} = \frac{1}{2\pi} [\sin t - \sin t] = 0$$

which is a constant.

Next, we find that the autocorrelation of the process $X(t)$ is given by

$$\begin{aligned} R_{XX}(t_1, t_2) &= E[X(t_1)X(t_2)] = E[\cos(t_1 + \phi) \cos(t_2 + \phi)] \\ &= \int_{\phi=0}^{2\pi} \cos(t_1 + \phi) \cos(t_2 + \phi) \frac{1}{2\pi} d\phi \\ &= \frac{1}{4\pi} \int_{\phi=0}^{2\pi} [\cos(t_1 + t_2 + 2\phi) + \cos(t_1 - t_2)] d\phi \end{aligned}$$

Integrating, we get

$$\begin{aligned} R_{XX}(t_1, t_2) &= \frac{1}{4\pi} \left[\frac{\sin(t_1 + t_2 + 2\phi)}{2} + \phi \cos(t_1 - t_2) \right]_{\phi=0}^{2\pi} = \frac{1}{4\pi} [0 + 2\pi \cos(t_1 - t_2)] \\ &= \frac{1}{2} \cos(t_1 - t_2) \end{aligned}$$

which is a function of the time difference $t_1 - t_2$.

Hence, $X(t)$ is a wide-sense stationary process with the autocorrelation function

$$R_{XX}(\tau) = E[X(t)X(t+\tau)] = \frac{1}{2} \cos \tau$$

(iii) The time average of the process $X(t)$ over the interval $(-T, T)$ is given by

$$\bar{X}_T = \langle X(t) \rangle = \frac{1}{2T} \int_{-T}^T X(t) dt$$

By Theorem 5.7, we know that the variance of the time average of $X(t)$ is given by

$$\bar{\sigma}_T^2 = \text{Var}[\bar{X}_T] = \frac{1}{T} \int_0^{2T} C_{XX}(\tau) \left(1 - \frac{\tau}{2T}\right) d\tau$$

where $C_{XX}(\tau)$ is the auto-covariance function given by

$$C_{XX}(\tau) = R_{XX}(\tau) - \mu_X^2 = \frac{1}{2} \cos \tau - 0 = \frac{1}{2} \cos \tau$$

Thus, it follows that

$$\bar{\sigma}_T^2 = \frac{1}{T} \int_0^{2T} \frac{1}{2} \cos \tau \left(1 - \frac{\tau}{2T}\right) d\tau = \frac{1}{2T} \int_0^{2T} \left(1 - \frac{\tau}{2T}\right) d \sin \tau$$

Integrating by parts, we get

$$\bar{\sigma}_T^2 = \frac{1}{2T} \left\{ \left[\left(1 - \frac{\tau}{2T}\right) \sin \tau \right]_0^{2T} - \int_0^{2T} \sin \tau \left(-\frac{1}{2T}\right) d\tau \right\} = \frac{1}{2T} \left\{ 0 + \frac{1}{2T} \int_0^{2\pi} \sin \tau d\tau \right\}$$

Integrating, we get

$$\bar{\sigma}_T^2 = \frac{1}{4T^2} [-\cos \tau]_0^{2T} = \frac{1 - \cos 2T}{4T^2}$$

Thus, it follows that

$$\lim_{T \rightarrow \infty} \bar{\sigma}_T^2 = \lim_{T \rightarrow \infty} \left[\frac{1 - \cos 2T}{4T^2} \right] = 0$$

because $|1 - \cos 2T| \leq 1$ for all $T > 0$ and $\frac{1}{T^2} \rightarrow 0$ as $T \rightarrow \infty$.

Hence, by Mean Ergodic Theorem (Theorem 5.6), we conclude that $X(t)$ is a mean-ergodic process. ♦

EXAMPLE 5.35. If $X(t)$ is a wide-sense stationary process given by $X(t) = 10 \cos(100t + \theta)$, where θ is uniformly distributed in $(-\pi, \pi)$, prove that the process $X(t)$ is correlation ergodic.

(Anna, Model Nov. 2003; May 2006)

Solution. The autocorrelation of the process $X(t)$ is defined by

$$R_{XX}(\tau) = E[X(t)X(t + \tau)] = E\{10 \cos(100t + \theta) \cdot 10 \cos[100(t + \tau) + \theta]\}$$

Since θ is uniformly distributed in $(-\pi, \pi)$, it has the probability density function

$$f_\theta(\theta) = \begin{cases} \frac{1}{2\pi} & -\pi < \theta < \pi \\ 0 & \text{otherwise} \end{cases}$$

Thus, it follows that

$$\begin{aligned} R_{XX}(\tau) &= \int_{\theta=-\pi}^{\pi} 100 \cos(100t + \theta) \cos[100(t + \tau) + \theta] \frac{1}{2\pi} d\theta \\ &= \frac{25}{\pi} \int_{\theta=-\pi}^{\pi} [\cos(200t + 100\tau + 2\theta) + \cos(100\tau)] d\theta \end{aligned}$$

Integrating, we get

$$\begin{aligned} R_{XX}(\tau) &= \frac{25}{\pi} \left[\frac{\sin(200t + 100\tau + 2\theta)}{2} + \theta \cos(100\tau) \right]_{\theta=-\pi}^{\pi} = \frac{25}{\pi} [0 + 2\pi \cos(100\tau)] \\ &= 50 \cos(100\tau) \end{aligned}$$

Next, we compute the time auto-correlation function of the process $X(t)$ over $(-T, T)$. By definition, we have

$$\begin{aligned} \bar{Z}_T &= \frac{1}{2T} \int_{t=-T}^T X(t)X(t + \tau) dt = \frac{1}{2T} \int_{t=-T}^T 100 \cos(100t + \theta) \cos[100(t + \tau) + \theta] dt \\ &= \frac{25}{T} \int_{t=-T}^T [\cos(200t + 100\tau + \theta) + \cos(100\tau)] dt \end{aligned}$$

Integrating, we get

$$\bar{Z}_T = \frac{25}{T} \left[\frac{\sin(200t + 100\tau + \theta)}{200} + t \cos(100\tau) \right]_{t=-T}^T = \frac{1}{8T} [\sin(200t + 100\tau + \theta)]_{t=-T}^T + 50 \cos(100\tau)$$

Taking limits as $T \rightarrow \infty$, we get

$$\lim_{T \rightarrow \infty} \bar{Z}_T = 0 + 50 \cos(100\tau) = 50 \cos(100\tau) = R_{XX}(\tau)$$

since $|\sin y| \leq 1$ for all y and $\frac{1}{T} \rightarrow 0$ as $T \rightarrow \infty$.

Thus, we conclude that the process $X(t)$ is correlation ergodic. ◆

PROBLEM SET 5.4

1. If $Y(t) = aX(t)$, where $X(t)$ is a mean ergodic process with constant mean $\mu_X \neq 0$ and independent of the random variable a with $E(a) = 0$, show that $Y(t)$ is not mean ergodic.
2. If the auto-covariance function $C_{XX}(\tau)$ of a wide-sense stationary process $X(t)$ satisfies the condition

$$C_{XX}(\tau) \rightarrow 0 \text{ as } \tau \rightarrow \infty$$

show that the process $X(t)$ is mean ergodic.

3. Determine whether or not the process $X(t) = A\cos t + B\sin t$ is mean ergodic if A and B are normally distributed random variables with zero means and unit variances.
4. If $X(t) = A\cos(\omega t + \theta)$, where A, ω are constants and θ is uniformly distributed in $(-\pi, \pi)$, prove that $X(t)$ is correlation ergodic.
5. If $X(t) = a\cos(\omega t + \theta)$ and $Y(t) = b\sin(\omega t + \theta)$, where a, b, ω are constants and θ is a random variable uniformly distributed in $(0, 2\pi)$, prove that the random processes $X(t)$ and $Y(t)$ are jointly correlation ergodic.

5.5 MARKOV PROCESS

In probability theory, a *Markov process* is a random process with the *Markov property*, which is also known as *absence of memory property* or *memoryless property*, i.e. the future behaviour of the process depends only on the current state, and not on the states in the past. Mathematically, the Markov process is defined as follows.

Definition 5.17. A random process $\{X(t), t \in T\}$ is called a *Markov process* if

$$P[X(t_{n+1}) \leq x_{n+1} \mid X(t_n) = x_n, X(t_{n-1}) = x_{n-1}, \dots, X(t_0) = x_0] = P[X(t_{n+1}) \leq x_{n+1} \mid X(t_n) = x_n] \quad (5.29)$$

whenever $t_0 < t_1 < \dots < t_n < t_{n+1}$. $(x_0, x_1, x_2, \dots, x_n, \dots)$ are called the *states* of the process. Eq. (5.29) is called the *Markov property* of the process $X(t)$, which may be interpreted as follows: If the random process $X(t)$ at time t_n is in the state x_n , the future state of the random process $X(t)$ at time t_{n+1} depends only on the present state x_n and not on the past states $x_{n-1}, x_{n-2}, \dots, x_1, x_0$.

EXAMPLE 5.36. Give an example of a Markov process.

(Anna, Nov. 2003)

Solution. Some examples of Markov processes are described below:

- (a) Any random process with independent increments (see Definition 5.10).
- (b) Board games played with dice like Monopoly, snakes and ladders, etc.
- (c) Weather prediction models.



Definition 5.18. A discrete-state Markov process is called a *Markov chain*. Thus, a discrete-parameter Markov chain is defined as a set of random variables $\{X_n, n \geq 0\}$ with the Markov property, that, given the present state, the future and past states are independent, i.e.

$$P(X_{n+1} = y \mid X_n = x, X_{n-1} = x_{n-1}, \dots, X_0 = x_0) = P(X_{n+1} = y \mid X_n = x) \quad (5.30)$$

The set of all possible values of the random variables X_i form a countable set S called the *state space* of the process.

Definition 5.19. Let $\{X_n, n \geq 0\}$ be a Markov chain that takes on a finite or countable number of possible values. Without loss of generality, this set of possible values of the process will be denoted by the set of nonnegative integers $\{0, 1, 2, \dots\}$. If $X_n = i$, we say that the random process is in state i at time n . The conditional probability

$$P\{X_{n+1} = j \mid X_n = i\}$$

is called the *one-step transition probability* from state i to state j at the $(n+1)$ th step and is denoted by $p_{ij}(n, n+1)$.

If the one-step transition probability is independent of n , i.e.

$$p_{ij}(n, n+1) = p_{ij}(m, m+1)$$

the Markov chain is said to have *stationary transition probabilities* and the process is called as a *homogeneous* Markov chain. Otherwise, the process is known as a *nonhomogeneous* Markov chain.

When the Markov chain is homogeneous, the one-step transition probability is denoted by p_{ij} . The matrix $P = [p_{ij}]$ is called the *transition probability matrix*.

Also, for a homogeneous Markov chain, the conditional probability

$$P\{X_n = j \mid X_0 = i\}$$

i.e. the probability that the process is in state j at step n , given that it was in state i at step 0, is called the *n -step transition probability* and is denoted by $p_{ij}^{(n)}$, i.e.

$$p_{ij}^{(n)} = P\{X_n = j \mid X_0 = i\}$$

(Note that $p_{ij}^{(1)} = p_{ij}$, the one-step transition probability defined earlier.)

5.5.1 Markov Chain

Definition 5.20. Let $\{X_n, n \geq 0\}$ be a homogeneous Markov chain with a discrete state space $S = \{0, 1, 2, \dots\}$. Then the one-step transition probability from state i to state j is defined by

$$p_{ij} = P\{X_{n+1} = j \mid X_n = i\} \quad i \geq 0, j \geq 0 \quad (5.31)$$

which is the same for all values of n (as the Markov chain is homogeneous).

The *transition probability matrix (TPM)* of the process $\{X_n, n \geq 0\}$ is defined by

$$P = [p_{ij}] = \begin{bmatrix} p_{00} & p_{01} & p_{02} & \cdots \\ p_{10} & p_{11} & p_{12} & \cdots \\ p_{20} & p_{21} & p_{22} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

where the transition probabilities (elements of P) satisfy

$$p_{ij} \geq 0, \quad \sum_{j=0}^{\infty} p_{ij} = 1 \quad \text{for } i = 0, 1, 2, \dots \quad (5.32)$$

If the state space S is finite and is equal to $\{1, 2, \dots, m\}$, then P is a square matrix of order m , i.e.

$$P = [p_{ij}] = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix}$$

where the transition probabilities (elements of P) satisfy

$$p_{ij} \geq 0, \quad \sum_{j=1}^m p_{ij} = 1 \quad \text{for } i = 1, 2, \dots, m \quad (5.33)$$

We note that a square matrix P whose elements satisfy Eq. (5.32) or (5.33) is called a *Markov matrix* or *stochastic matrix*.

EXAMPLE 5.37. Show that P is a Markov matrix, then P^n is a Markov matrix for any positive integer n .

Solution. We consider any Markov matrix P of order m given by

$$P = [p_{ij}] = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix}$$

Let

$$v = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Then, by the property, Eq. (5.33), we have

$$Pv = [p_{ij}] = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = v$$

i.e.

$$Pv = v \quad (5.34)$$

Premultiplying both sides of Eq. (5.34) by P , we get

$$P(Pv) = P^2v = Pv = v$$

[using Eq. (5.34)]. Thus, P^2 is a Markov matrix.

Thus, the assertion that P^n is a Markov matrix is true for $n = 1, 2$.

Induction Hypothesis: Suppose that P^k is a Markov matrix for some positive integer k , i.e.

$$P^k v = v \quad (5.35)$$

Premultiplying both sides of Eq. (5.35) by P , we get

$$P(P^k v) = P^{k+1}v = Pv = v$$

[using Eq. (5.34)]. Thus, P^{k+1} is a Markov matrix.

This completes the inductive step of the proof. Hence, by the principle of mathematical induction, P^n is a Markov matrix for any positive integer n . $\blacklozenge$

EXAMPLE 5.38. (*Simple Weather Model*) We consider a simple weather model in which a day is either “rainy” (represented by the state 0) or “sunny” (represented by the state 1), and the probabilities of weather conditions, given the weather on the preceding day, can be represented by a transition probability matrix P given below:

$$\begin{array}{cc} & \text{States of } X_{n+1} \\ & \begin{array}{cc} 0 & 1 \end{array} \\ \text{States of } X_n & \begin{array}{cc} 0 & \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} \\ 1 & \begin{bmatrix} 0.1 & 0.9 \end{bmatrix} \end{array} \end{array}$$

The matrix P represents a simple weather model in which a rainy day is 50% likely to be followed by another rainy day, and a sunny day is 90% likely to be followed by another sunny day. We note that the (i, j) th entry of the transition probability matrix P , viz. p_{ij} , represents the probability that if a given day is of type i , it will be followed by a day of type j . We note also that the rows of the matrix P sum to 1 because P is a Markov matrix.

5.5.2 Probability Distribution of a Markov Chain

Definition 5.21. Let $\{X_n, n \geq 0\}$ be a homogeneous Markov chain. Let $p_i(n) = P\{X_n = i\}$, i.e. let $p_i(n)$ denote the probability that the process is in state i in the n th step, and let

$$p(n) = [p_0(n) \quad p_1(n) \quad p_2(n) \quad \cdots]$$

where

$$\sum_{k=0}^{\infty} p_k(n) = 1$$

Then $p_i(0) = P\{X_0 = i\}$, $i \geq 0$ are called the *initial-state probabilities*, and the vector

$$p(0) = [p_0(0) \quad p_1(0) \quad p_2(0) \quad \cdots]$$

is called the *initial-state probability vector* or the *initial probability distribution*. Also,

$$p(n) = [p_0(n) \quad p_1(n) \quad p_2(n) \quad \cdots]$$

is called the *state probability vector after n steps* or the *probability distribution* of the Markov chain.

The probability distribution of any homogeneous Markov chain is completely determined by the transition probability matrix and the initial probability distribution. This is illustrated by Example 5.39.

EXAMPLE 5.39. Consider the simple weather model discussed in Example 5.38. For this model, there are two states, “rainy” (denoted by 0) and “sunny” (denoted by 1), and the transition probability matrix is given by

$$P = \begin{array}{cc} & \begin{array}{cc} 0 & 1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{pmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{pmatrix} \end{array}$$

Suppose that initially, the weather is known to be “rainy”, i.e. $X_0 = 0$. Then the initial probability distribution is given by

$$p(0) = [1 \quad 0]$$

Then the weather on day 1 can be predicted as follows:

$$p(1) = p(0) P = [1 \quad 0] \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix} = [0.5 \quad 0.5]$$

Thus, there is a 50% chance that day 1 will also be rainy.

Next, the weather on day 2 can be predicted in a similar manner:

$$p(2) = p(1) P = [0.5 \quad 0.5] \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix} = [0.3 \quad 0.7]$$

Thus, there is a 30% chance that day 2 will be rainy.

Next, the weather on day 3 can be predicted in a similar manner:

$$p(3) = p(2) P = [0.3 \quad 0.7] \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix} = [0.22 \quad 0.78]$$

Thus, there is a 22% chance that day 3 will be rainy.

Note that in general, the weather on day n can be predicted using the relation

$$p(n) = p(n-1)P$$

Thus, if $p(0)$ and P are available, then the probability distribution $p(n)$ can be calculated for any positive integer n .

Next, we establish an important theorem for homogeneous Markov chains.

5.5.3 Chapman-Kolmogorov Theorem

Theorem 5.10. (*Chapman-Kolmogorov Theorem*) Let $\{X_n, n \geq 0\}$ be a homogeneous Markov chain with transition probability matrix $P = [p_{ij}]$ and n -step transition probability matrix $P^{(n)} = [p_{ij}^{(n)}]$, where

$$p_{ij}^{(n)} = P\{X_n = j | X_0 = i\} \quad \text{and} \quad p_{ij}^{(1)} = p_{ij}$$

Then the following properties hold:

- (a) $P^{(n+m)} = P^{(n)} P^{(m)}$.
- (b) $P^{(n)} = P^n$, i.e. the n -step transition probability matrix $P^{(n)}$ is equal to the n th power of the (one-step) transition probability matrix, P .

Proof. (a) The (i, j) th entry of the matrix $P^{(n+m)}$ is given by

$$\begin{aligned} p_{ij}^{(n+m)} &= P\{X_{n+m} = j | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j, X_n = k | X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_{n+m} = j | X_n = k, X_0 = i\} P\{X_n = k | X_0 = i\} \end{aligned}$$

Using the Markov property, Eq. (5.30), and the fact that the given Markov chain is homogeneous, we have

$$\begin{aligned} p_{ij}^{(n+m)} &= \sum_{k=0}^{\infty} P\{X_{n+m} = j \mid X_n = k\} P\{X_n = k \mid X_0 = i\} \\ &= \sum_{k=0}^{\infty} P\{X_m = j \mid X_0 = k\} P\{X_n = k \mid X_0 = i\} \\ &= \sum_{k=0}^{\infty} p_{kj}^{(m)} p_{ik}^{(n)} = \sum_{k=0}^{\infty} p_{ik}^{(n)} p_{kj}^{(m)} \end{aligned}$$

which is the (i, j) th entry of the matrix product $P^{(n)}P^{(m)}$. Thus, we have shown that

$$P^{(n+m)} = P^{(n)}P^{(m)}$$

(b) We prove this result by induction. By definition, we have

$$P = P^{(1)}$$

Thus, the assertion that $P^{(n)} = P^n$ is true for $n = 1$.

Induction Hypothesis: Suppose that $P^{(m)} = P^m$ for some positive integer m . Then, using the result in (a), we have

$$P^{(m+1)} = P^{(1)}P^{(m)} = P \cdot P^m = P^{m+1}$$

This completes the inductive step of the proof. Hence, by the principle of mathematical induction, $P^{(n)} = P^n$ for any positive integer n . ◆

EXAMPLE 5.40. The transition probability matrix of a Markov chain $\{X_n, n \geq 0\}$ having three states 1, 2 and 3 is

$$P = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.3 & 0.3 \end{bmatrix}$$

and the initial probability distribution is $p(0) = [0.5 \quad 0.3 \quad 0.2]$. Find the following:

- (a) $P\{X_2 = 2\}$.
- (b) $P\{X_3 = 3, X_2 = 2, X_1 = 1, X_0 = 3\}$.

Solution. Since $p(0) = [0.5 \quad 0.3 \quad 0.2]$, it follows that

$$P(X_0 = 1) = 0.5, \quad P(X_0 = 2) = 0.3 \quad \text{and} \quad P(X_0 = 3) = 0.2$$

By Chapman-Kolmogorov Theorem (Theorem 5.10), it follows that

$$P^{(2)} = P^2 = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.3 & 0.3 \end{bmatrix} = \begin{bmatrix} 0.27 & 0.39 & 0.34 \\ 0.20 & 0.48 & 0.32 \\ 0.23 & 0.39 & 0.83 \end{bmatrix}$$

(a) We find that

$$\begin{aligned}
 P\{X_2 = 2\} &= \sum_{i=1}^3 P\{X_2 = 2|X_0 = i\} P\{X_0 = i\} \\
 &= p_{13}^{(2)} P\{X_0 = 1\} + p_{23}^{(2)} P\{X_0 = 2\} + p_{33}^{(2)} P\{X_0 = 3\} \\
 &= (0.34 \times 0.5) + (0.32 \times 0.3) + (0.38 \times 0.2) = 0.17 + 0.096 + 0.076 \\
 &= 0.342
 \end{aligned}$$

(b) First, we find that

$$P\{X_1 = 1|X_0 = 3\} = p_{31} = 0.4$$

Thus, we have

$$P\{X_1 = 1, X_0 = 3\} = P\{X_1 = 1|X_0 = 3\} P\{X_0 = 3\} = 0.4 \times 0.2 = 0.08 \quad (5.36)$$

Next, we find that

$$P\{X_2 = 2, X_1 = 1, X_0 = 3\} = P\{X_2 = 2|X_1 = 1, X_0 = 3\} P\{X_1 = 1, X_0 = 3\}$$

Using the Markov property, Eq. (5.30) and Eq. (5.36), we have

$$P\{X_2 = 2, X_1 = 1, X_0 = 3\} = P\{X_2 = 2|X_1 = 1\} \times 0.08 = p_{12} \times 0.08$$

Since $p_{12} = 0.3$,

$$P\{X_2 = 2, X_1 = 1, X_0 = 3\} = 0.3 \times 0.08 = 0.024 \quad (5.37)$$

Finally, we find that

$$\begin{aligned}
 P\{X_3 = 3, X_2 = 2, X_1 = 1, X_0 = 3\} &= P\{X_3 = 3 | X_2 = 2, X_1 = 1, X_0 = 3\} \\
 &\quad P\{X_2 = 2, X_1 = 1, X_0 = 3\}
 \end{aligned}$$

Using the Markov property, Eq. (5.30), and Eq. (5.37), we have

$$P\{X_3 = 3, X_2 = 2, X_1 = 1, X_0 = 3\} = P\{X_3 = 3|X_2 = 2\} \times 0.024 = p_{23} \times 0.24$$

Thus, we have

$$P\{X_3 = 3, X_2 = 2, X_1 = 1, X_0 = 3\} = 0.3 \times 0.24 = 0.072$$

◆

EXAMPLE 5.41. A message transmission system is found to be Markovian with the transition probability of current message to next message as given by the matrix

$$P = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.2 & 0.7 \\ 0.6 & 0.3 & 0.1 \end{bmatrix}$$

The initial probabilities of the states are $p_1(0) = 0.4, p_2(0) = 0.3, p_3(0) = 0.3$. Find the probabilities of the next message.

(Anna, April 2005)

Solution. The initial probability distribution is

$$p(0) = [0.4 \quad 0.3 \quad 0.3]$$

Thus, the probability distribution after one step is

$$p(1) = p(0)P = [0.4 \quad 0.3 \quad 0.3] \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.2 & 0.7 \\ 0.6 & 0.3 & 0.1 \end{bmatrix} = [0.29 \quad 0.27 \quad 0.44]$$

Thus, the probabilities of the next message are given by

$$p_1(1) = 0.29, \quad p_2(1) = 0.27 \quad \text{and} \quad p_3(1) = 0.44$$

◆

EXAMPLE 5.42. The transition probability matrix of a Markov chain $\{X_n, n = 1, 2, 3, \}$ with three states 0, 1 and 2 is

$$P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

with initial distribution $p(0) = [\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}]$. Find $P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2)$.

(Anna, Nov. 2006)

Solution. Since $p(0) = [\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}]$, it follows that

$$P(X_0 = 0) = P(X_0 = 1) = P(X_0 = 2) = \frac{1}{3}$$

We find that

$$P(X_1 = 1, X_0 = 2) = P(X_0 = 2)P(X_1 = 1|X_0 = 2) = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$$

Thus, we find that

$$P(X_2 = 2, X_1 = 1, X_0 = 2) = P(X_1 = 1, X_0 = 2)P(X_2 = 2|X_1 = 1, X_0 = 2)$$

Using the Markov property Eq. (5.30), we have

$$P(X_2 = 2, X_1 = 1, X_0 = 2) = P(X_1 = 1, X_0 = 2)P(X_2 = 2|X_1 = 1) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

Thus, the required probability is given by

$$P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2) = P(X_2 = 2, X_1 = 1, X_0 = 2)P(X_3 = 1|X_2 = 2, X_1 = 1, X_0 = 2)$$

Using the Markov property Eq. (5.30), we have

$$P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2) = P(X_2 = 2, X_1 = 1, X_0 = 2)P(X_3 = 1|X_2 = 2) = \frac{1}{16} \times \frac{3}{4} = \frac{3}{64}$$

◆

EXAMPLE 5.43. A man tosses a fair coin until 3 heads occur in a row. Let X_n denote the longest string of heads ending at the n th trial, i.e. $X_n = k$ if at the n th trial, the last tail occurred at the $(n - k)$ th trial. Find the transition probability matrix.

(Anna, May 2006)

Solution. The given Markov chain has state space $S = \{0, 1, 2, 3\}$, since the coin is tossed until 3 heads occur in a row. Hence, the transition probability matrix is obtained as follows.

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

◆

EXAMPLE 5.44. At an intersection, a working traffic light will be out of order the next day with probability 0.07, and an out-of-order traffic light will be working the next day with probability 0.88. Let $X_n = 1$ if on day n the traffic light will work; $X_n = 0$ if on day n the traffic light will not work. Is $\{X_n; n = 0, 1, 2, \dots\}$ a Markov chain? If so, write the transition probability matrix.

(Anna, Nov. 2007)

Solution. Yes, $\{X_n; n = 0, 1, 2, \dots\}$ is a Markov chain because X_n depends only on X_{n-1} and not on $X_{n-2}, \dots, X_0$. The transition probability matrix for the given Markov chain is obtained as

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 0.12 & 0.88 \\ 0.07 & 0.93 \end{pmatrix} \end{matrix}$$

◆

5.5.4 Stationary Distribution for a Markov Chain

Definition 5.22. Let $\{X_n, n \geq 0\}$ be a homogeneous Markov chain with transition probability matrix P . If there exists a probability vector π such that

$$\pi P = \pi \tag{5.38}$$

then π is called a *stationary distribution* or *steady-state distribution* for the Markov chain.

Remark 5.9. From Eq. (5.38), it follows that a stationary distribution π for a homogeneous Markov chain is a *left eigenvector* of the transition probability matrix P with corresponding eigenvalue 1. If stationary distribution π exists for a Markov chain, then any scalar multiple of π also is a left eigenvector of P with eigenvalue 1, i.e. $\alpha\pi$ also satisfies Eq. (5.38) for any $\alpha \in \mathbb{R}$. However, the stationary distribution π , if it exists, is unique because it is a probability vector, i.e. its components must sum to unity.

Next, we provide some sufficient condition for the existence of a stationary distribution for a homogeneous Markov chain. For this purpose, we define a *regular* Markov chain.

Definition 5.23. Any stochastic or Markov matrix P for which P^n has all positive entries for some positive integer n is called a *regular* matrix.

A homogeneous Markov chain is called *regular* if its transition probability matrix P is regular, i.e. if there is a finite positive integer n such that P^n , the n -step transition probability matrix, has all positive entries.

For a regular Markov chain, we have the following theorem which we state without proof.

Theorem 5.11. Let $\{X_n, n \geq 0\}$ be a regular homogeneous finite-state Markov chain with transition probability matrix P . Then

$$\lim_{n \rightarrow \infty} P^n = Q$$

where Q is a matrix whose rows are identical and equal to the stationary distribution π for the Markov chain.

EXAMPLE 5.45. Consider the simple weather model discussed in Example 5.38. For this model, there are two states, “rainy” (denoted by 0) and “sunny” (denoted by 1), and the transition probability matrix is given by

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix}$$

Since all the entries of P are positive, P is a regular matrix and hence the given Markov chain is regular. Thus, by Theorem 5.11, it follows that the given Markov chain has a stationary distribution $\pi = [\pi_0 \ \pi_1]$, which is determined by the properties that

$$\pi P = \pi \text{ and } \pi_0 + \pi_1 = 1$$

We note that $\pi P = \pi$ is equivalent to the linear system of equations

$$\begin{aligned} 0.5\pi_0 + 0.1\pi_1 &= \pi_0 & \text{or} & & -0.5\pi_0 + 0.1\pi_1 &= 0 \\ 0.5\pi_0 + 0.9\pi_1 &= \pi_1 & \text{or} & & 0.5\pi_0 - 0.1\pi_1 &= 0 \end{aligned}$$

which reduce to the single homogeneous equation

$$0.5\pi_0 - 0.1\pi_1 = 0 \text{ or } \pi_1 = 5\pi_0$$

Since π is a probability vector, $\pi_0 + \pi_1 = 1$. Thus, we must have

$$\pi_0 + 5\pi_0 = 1 \text{ or } \pi_0 = \frac{1}{6}$$

Thus, $\pi_1 = 5\pi_0 = \frac{5}{6}$.

Thus, the stationary distribution π is given by $\pi = [\frac{1}{6} \ \frac{5}{6}]$.

EXAMPLE 5.46. Consider a two-state Markov chain with transition probability matrix

$$P = \begin{bmatrix} 1-\alpha & \beta \\ \beta & 1-\beta \end{bmatrix} \text{ where } 0 < \alpha < 1, 0 < \beta < 1$$

Find

(a) The stationary distribution.

- (b) The n -step transition probability matrix P^n .
 (c) $\lim_{n \rightarrow \infty} P^n$. What do you conclude?

Solution. (a) Since all the entries of P are positive, P is a regular matrix. Since the given Markov chain is regular, it follows by Theorem 5.11 that the Markov chain has a stationary distribution $\pi = [\pi_1 \ \pi_2]$, which can be determined from the equations

$$\pi P = \pi \text{ and } \pi_1 + \pi_2 = 1$$

We note that $\pi P = \pi$ is equivalent to the following system of linear equations:

$$\begin{aligned} (1 - \alpha)\pi_1 + \beta\pi_2 &= \pi_1 & \text{or} & & -\alpha\pi_1 + \beta\pi_2 &= 0 \\ \alpha\pi_1 + (1 - \beta)\pi_2 &= \pi_2 & \text{or} & & \alpha\pi_1 - \beta\pi_2 &= 0 \end{aligned}$$

which reduce to the single linear homogeneous equation

$$\alpha\pi_1 - \beta\pi_2 = 0 \text{ or } \pi_1 = \frac{\beta}{\alpha} \pi_2$$

Since π is a probability vector, $\pi_1 + \pi_2 = 1$. Thus, we must have

$$\frac{\beta}{\alpha} \pi_2 + \pi_2 = 1 \text{ or } \left(1 + \frac{\beta}{\alpha}\right) \pi_2 = \left(\frac{\alpha + \beta}{\alpha}\right) \pi_2 = 1$$

i.e.

$$\pi_2 = \frac{\alpha}{\alpha + \beta}$$

and so

$$\pi_1 = \frac{\beta}{\alpha} \pi_2 = \frac{\beta}{\alpha + \beta}$$

Thus, the stationary distribution π for the Markov chain is given by

$$\pi = [\pi_1 \ \pi_2] = \left[\frac{\beta}{\alpha + \beta} \quad \frac{\alpha}{\alpha + \beta} \right] = \frac{1}{\alpha + \beta} [\beta \ \alpha]$$

- (b) From linear algebra, we know that the characteristic equation of P is given by

$$\phi(\lambda) = |\lambda I - P| = \begin{vmatrix} \lambda - (1 - \alpha) & -\alpha \\ -\beta & \lambda - (1 - \beta) \end{vmatrix} = 0$$

Simplifying, we get

$$\phi(\lambda) = \lambda^2 + (\alpha + \beta - 2)\lambda + (1 - \alpha - \beta) = 0$$

which can be easily factorized as

$$\phi(\lambda) = (\lambda - 1)[\lambda - (1 - \alpha - \beta)] = 0$$

Thus, the eigenvalues of the transition probability matrix P are

$$\lambda_1 = 1 \text{ and } \lambda_2 = 1 - \alpha - \beta$$

which are distinct because $0 < \alpha < 1$ and $0 < \beta < 1$.

Thus, using the spectral decomposition method, P^n can be obtained as

$$P^n = \lambda_1^n E_1 + \lambda_2^n E_2 \quad (5.39)$$

where E_1 and E_2 are constituent matrices of P given by the expressions

$$E_1 = \frac{1}{\lambda_1 - \lambda_2} [P - \lambda_2 I] \text{ and } E_2 = \frac{1}{\lambda_2 - \lambda_1} [P - \lambda_1 I]$$

A simple calculation using the substitution $\lambda_1 = 1$ and $\lambda_2 = 1 - \alpha - \beta$ yields

$$E_1 = \frac{1}{\alpha + \beta} \begin{bmatrix} \beta & \alpha \\ \beta & \alpha \end{bmatrix} \text{ and } E_2 = \frac{1}{\alpha + \beta} \begin{bmatrix} \alpha & -\alpha \\ -\beta & \beta \end{bmatrix}$$

Thus, using Eq. (5.39), we obtain

$$P^n = E_1 + (1 - \alpha - \beta)^n E_2 = \frac{1}{\alpha + \beta} \left\{ \begin{bmatrix} \beta & \alpha \\ \beta & \alpha \end{bmatrix} + (1 - \alpha - \beta)^n \begin{bmatrix} \alpha & -\alpha \\ -\beta & \beta \end{bmatrix} \right\} \quad (5.40)$$

(c) Taking limits as $n \rightarrow \infty$, we know that

$$\lambda_1^n = 1^n \rightarrow 1 \text{ and } \lambda_2^n = (1 - \alpha - \beta)^n \rightarrow 0$$

because $|1 - \alpha - \beta| < 1$ as $0 < \alpha < 1$ and $0 < \beta < 1$.

Thus, taking limits as $n \rightarrow \infty$ in Eq. (5.40), we obtain

$$\lim_{n \rightarrow \infty} P^n = \frac{1}{\alpha + \beta} \left\{ \begin{bmatrix} \beta & \alpha \\ \beta & \alpha \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\} = \frac{1}{\alpha + \beta} \begin{bmatrix} \beta & \alpha \\ \beta & \alpha \end{bmatrix}$$

Thus, we conclude that

$$\lim_{n \rightarrow \infty} P^n = \begin{bmatrix} \pi \\ \pi \end{bmatrix} = Q$$

where Q is the matrix whose rows are identical and equal to the stationary distribution π . Thus, we have verified Theorem 5.11. ◆

EXAMPLE 5.47. If the transition probability matrix of a Markov chain is

$$P = \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

find the steady-state distribution of the chain.

(Anna, Nov. 2003)

Solution. The steady-state distribution $\pi = [\pi_1 \ \pi_2]$ of the given Markov chain is obtained by solving the equations

$$\pi P = \pi \text{ and } \pi_1 + \pi_2 = 1$$

We note that $\pi P = \pi$ is equivalent to the linear system of equations

$$\begin{aligned} \frac{1}{2}\pi_2 &= \pi_1 \quad \text{or} \quad 2\pi_1 - \pi_2 = 0 \\ \pi_1 + \frac{1}{2}\pi_2 &= \pi_2 \quad \text{or} \quad 2\pi_1 - \pi_2 = 0 \end{aligned}$$

which reduces to the single linear homogeneous equation

$$2\pi_1 - \pi_2 = 0 \text{ or } \pi_2 = 2\pi_1$$

Since $\pi_1 + \pi_2 = 1$, it follows that

$$\pi_1 + 2\pi_1 = 1 \text{ or } 3\pi_1 = 1 \text{ or } \pi_1 = \frac{1}{3}$$

which implies that $\pi_2 = 2\pi_1 = \frac{2}{3}$.

Hence, the steady-state distribution of the Markov chain is given by

$$\pi = [\pi_1 \ \pi_2] = \left[\frac{1}{3} \ \frac{2}{3}\right]$$

◆

EXAMPLE 5.48. Consider a Markov chain with transition probability matrix

$$P = \begin{bmatrix} p & q \\ q & p \end{bmatrix}, \text{ where } 0 < p < 1, p + q = 1$$

Suppose that the initial probability distribution is $p(0) = [\alpha \ 1 - \alpha]$. Find the

- Probability distribution after one step.
- Probability distribution after two steps.
- Probability distribution after infinite number of steps.

Solution. (a) The probability distribution after one step is given by

$$\begin{aligned} p(1) &= p(0)P = [\alpha \ 1 - \alpha] \begin{bmatrix} p & q \\ q & p \end{bmatrix} = [p\alpha + (1 - \alpha)q \quad q\alpha + (1 - \alpha)p] \\ &= [(p - q)\alpha + q \quad (q - p)\alpha + p] \end{aligned}$$

(b) Note that

$$P^2 = PP = \begin{bmatrix} p & q \\ q & p \end{bmatrix} \begin{bmatrix} p & q \\ q & p \end{bmatrix} = \begin{bmatrix} p^2 + q^2 & 2pq \\ 2pq & p^2 + q^2 \end{bmatrix}$$

Thus, the probability distribution after two steps is given by

$$\begin{aligned} p(2) &= p(0)P^2 = [\alpha \ 1 - \alpha] \begin{bmatrix} p^2 + q^2 & 2pq \\ 2pq & p^2 + q^2 \end{bmatrix} \\ &= [(p^2 + q^2)\alpha + 2pq(1 - \alpha) \quad 2pq\alpha + (p^2 + q^2)(1 - \alpha)] \\ &= [(p - q)^2\alpha + 2pq \quad p^2 + q^2 - (p - q)^2\alpha] \end{aligned}$$

- (c) Since all the entries of P are positive, P is a regular stochastic matrix, and so the given Markov chain is a regular chain. By Theorem 5.11, it follows that the given Markov chain has a steady-state probability distribution $\pi = [\pi_1 \ \pi_2]$, which can be determined from the equations

$$\pi P = \pi \text{ and } \pi_1 + \pi_2 = 1$$

From the equation $\pi P = \pi$, we have

$$[\pi_1 \ \pi_2] \begin{bmatrix} p & q \\ q & p \end{bmatrix} = [\pi_1 \ \pi_2]$$

which leads to the system of linear equations

$$\begin{aligned} p\pi_1 + q\pi_2 &= \pi_1 & \text{or} & & -q\pi_1 + q\pi_2 &= 0 \\ q\pi_1 + p\pi_2 &= \pi_2 & \text{or} & & q\pi_1 - q\pi_2 &= 0 \end{aligned}$$

where we have used the fact that $p + q = 1$ or $p = 1 - q$.

Simplifying, we have the single linear homogeneous equation

$$\pi_1 - \pi_2 = 0 \text{ or } \pi_1 = \pi_2$$

Since $\pi_1 + \pi_2 = 1$, it is easy to see that $\pi_1 = \pi_2 = \frac{1}{2}$.

Thus, the probability distribution after infinite number of steps is given by

$$p(\infty) = \pi = [\pi_1 \ \pi_2] = \left[\frac{1}{2} \ \frac{1}{2}\right]$$

◆

EXAMPLE 5.49. A welding process is considered to be a two-state Markov chain, where the state 0 denotes that the process is running in the manufacturing firm and the state 1 denotes that the process is not running in the firm. Suppose that the transition probability matrix for this Markov chain is given by

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix} \end{matrix}$$

- Find the probability that the welding process will be run on the third day from today given that the welding process is run today.
- Find the probability that the welding process will be run on the third day from today if the initial probabilities of states 0 and 1 are equally likely.

Solution. First, we note that

$$P^{(2)} = P^2 = PP = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} = \begin{bmatrix} 0.70 & 0.30 \\ 0.45 & 0.55 \end{bmatrix}$$

and

$$P^{(3)} = P^3 = P^2P = \begin{bmatrix} 0.70 & 0.30 \\ 0.45 & 0.55 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} = \begin{bmatrix} 0.650 & 0.350 \\ 0.525 & 0.475 \end{bmatrix}$$

- (a) We are given that the welding process is run today, which implies that the initial probability distribution is

$$p(0) = [1 \quad 0]$$

Thus, it follows that

$$p(3) = p(0)P^3 = [1 \quad 0] \begin{bmatrix} 0.650 & 0.350 \\ 0.525 & 0.475 \end{bmatrix} = [0.65 \quad 0.35]$$

Hence, the probability that the welding process is run on the third day is given by

$$P(X_3 = 0) = 0.65$$

- (b) We are given that the initial probability distribution is

$$p(0) = [0.5 \quad 0.5]$$

Thus, it follows that

$$p(3) = p(0)P^3 = [0.5 \quad 0.5] \begin{bmatrix} 0.650 & 0.350 \\ 0.525 & 0.475 \end{bmatrix} = [0.5875 \quad 0.4125]$$

Hence, the probability that the welding process is run on the third day is given by

$$P(X_3 = 0) = 0.5875$$



EXAMPLE 5.50. A man either drives a car or catches a train or a bus to go to office every day. Suppose that the transition probability matrix for this Markov chain is given by

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{pmatrix} 0.5 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.1 \\ 0.2 & 0.6 & 0.2 \end{pmatrix} \end{matrix}$$

where the states 0, 1 and 2 denote a car, train and bus, respectively. Suppose that on the first day of a week, the man tossed a fair coin and took a train to work if a head appeared or a bus to work if a tail appeared. Find the probability that:

- (a) He will take a train on the third day of the week.
 (b) He will take a bus to work on the fourth day of the week.

Solution. (a) From the given data, the initial probability distribution (on the first day) is

$$p(1) = [0 \quad 0.5 \quad 0.5]$$

Thus, the probability distribution on the second day is given by

$$p(2) = p(1)P = [0 \quad 0.5 \quad 0.5] \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.1 \\ 0.2 & 0.6 & 0.2 \end{bmatrix} = [0.30 \quad 0.55 \quad 0.15]$$

and the probability distribution on the third day is given by

$$p(3) = p(2)P = [0.30 \quad 0.55 \quad 0.15] \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.1 \\ 0.2 & 0.6 & 0.2 \end{bmatrix} = [0.400 \quad 0.425 \quad 0.175]$$

Thus, the probability that he will take a train to work on the third day is given by

$$P(X_3 = 1) = 0.425$$

which is the probability corresponding to state 1 in $p(3)$.

(b) The probability distribution on the fourth day is given by

$$p(4) = p(3)P = [0.400 \quad 0.425 \quad 0.175] \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.4 & 0.5 & 0.1 \\ 0.2 & 0.6 & 0.2 \end{bmatrix} = [0.4050 \quad 0.3975 \quad 0.1975]$$

Thus, the probability that he will take a bus to work on the fourth day is given by

$$P(X_4 = 2) = 0.1975$$

which is the probability corresponding to state 2 in $p(4)$.

◆

EXAMPLE 5.51. Assume that a computer system is in any one of the three states: busy, idle and under repair, respectively, denoted by 0, 1, 2. Observing its state at 2 P.M. each day, we get the transition probability matrix as

$$P = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.1 & 0.8 & 0.1 \\ 0.6 & 0 & 0.4 \end{bmatrix}$$

Find out the third step transition probability matrix. Determine the limiting probabilities.

(Anna, May 2007)

Solution. First, we compute the second step transition probability matrix as

$$p^{(2)} = P^2 = PP = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.1 & 0.8 & 0.1 \\ 0.6 & 0 & 0.4 \end{bmatrix} \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.1 & 0.8 & 0.1 \\ 0.6 & 0 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.50 & 0.28 & 0.22 \\ 0.20 & 0.66 & 0.14 \\ 0.60 & 0.12 & 0.28 \end{bmatrix}$$

Thus, it follows that the third step transition probability matrix is given by

$$p^{(3)} = P^3 = P^2P = \begin{bmatrix} 0.50 & 0.28 & 0.22 \\ 0.20 & 0.66 & 0.14 \\ 0.60 & 0.12 & 0.28 \end{bmatrix} \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.1 & 0.8 & 0.1 \\ 0.6 & 0 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.460 & 0.324 & 0.216 \\ 0.270 & 0.568 & 0.162 \\ 0.540 & 0.216 & 0.244 \end{bmatrix}$$

Since $p_{ij}^{(2)} > 0$ for all i and j , it follows that P is a regular stochastic matrix. Thus, the given Markov chain is regular. Hence, by Theorem 5.11, the given Markov chain has a steady-state distribution $\pi = [\pi_0 \quad \pi_1 \quad \pi_2]$, which is obtained by solving the equations

$$\pi P = \pi \quad \text{and} \quad \pi_0 + \pi_1 + \pi_2 = 1$$

We note that $\pi P = \pi$ leads to the system of linear equations

$$\begin{array}{lll} 0.6\pi_0 + 0.1\pi_1 + 0.6\pi_2 = \pi_0 & \text{or} & 4\pi_0 - \pi_1 - 6\pi_2 = 0 \\ 0.2\pi_0 + 0.8\pi_1 & = \pi_1 & \text{or} & \pi_0 - \pi_1 = 0 \\ 0.2\pi_0 + 0.1\pi_1 + 0.4\pi_2 = \pi_2 & \text{or} & 2\pi_0 + \pi_1 - 6\pi_2 = 0 \end{array}$$

From the second equation, we have $\pi_0 = \pi_1$. Substituting $\pi_1 = \pi_0$ in the first and third equations, we obtain the same linear homogeneous equation

$$3\pi_0 - 6\pi_2 = 0 \quad \text{or} \quad \pi_0 = 2\pi_2$$

Since $\pi_0 + \pi_1 + \pi_2 = 1$, we must have

$$2\pi_2 + 2\pi_2 + \pi_2 = 1 \quad \text{or} \quad 5\pi_2 = 1$$

Thus, $\pi_2 = \frac{1}{5}$ and hence $\pi_0 = \pi_1 = 2\pi_2 = \frac{2}{5}$.

Hence, the limiting probabilities are

$$\pi_0 = \frac{2}{5}, \quad \pi_1 = \frac{2}{5} \quad \text{and} \quad \pi_2 = \frac{1}{5}$$

◆

EXAMPLE 5.52. An engineer analyzing a series of digital signals generated by a testing system observes that only 1 out of 15 highly distorted signals follows a highly distorted signal, with no recognizable signal between, whereas 20 out of 23 recognizable signals follow recognizable signals with no highly distorted signal between. Given that only highly distorted signals are not recognizable, find the fraction of signals that are highly distorted.

(Anna, Nov. 2007)

Solution. For $n = 1, 2, 3, \dots$, let X_n be the n th signal generated by the testing system. Then we define

$$X_n = \begin{cases} 1 & \text{if the signal is highly distorted} \\ 0 & \text{if the signal is recognizable} \end{cases}$$

Since for any value of $n \geq 1$, X_n depends only on X_{n-1} , and not $X_{n-2}, X_{n-3}, \dots, X_1$, it follows that $\{X_n\}$ is a Markov chain with state space $S = \{0, 1\}$. From the given data, the transition probability matrix is

$$P = \begin{bmatrix} \frac{20}{23} & \frac{3}{23} \\ \frac{14}{15} & \frac{1}{15} \end{bmatrix} = \begin{bmatrix} 0.8696 & 0.1304 \\ 0.9333 & 0.0667 \end{bmatrix}$$

Since all the entries of P are positive, it follows that P is a regular stochastic matrix. Thus, the given Markov chain is regular. Hence, by Theorem 5.11, the given Markov chain has a steady-state distribution $\pi = [\pi_0 \quad \pi_1]$, which is obtained by solving the equations

$$\pi P = \pi \quad \text{and} \quad \pi_0 + \pi_1 = 1$$

We note that $\pi P = \pi$ is equivalent to the system of linear equations

$$\begin{array}{lll} 0.8696\pi_0 + 0.9333\pi_1 = \pi_0 & \text{or} & -0.1304\pi_0 + 0.9333\pi_1 = 0 \\ 0.1304\pi_0 + 0.0667\pi_1 = \pi_1 & \text{or} & 0.1304\pi_0 - 0.9333\pi_1 = 0 \end{array}$$

which reduce to the single linear homogeneous equation

$$0.1304\pi_0 - 0.9333\pi_1 = 0 \text{ or } \pi_0 = 7.1572\pi_1$$

Since $\pi_0 + \pi_1 = 1$, we must have

$$7.1572\pi_1 + \pi_1 = 1 \text{ or } 8.1572\pi_1 = 1 \text{ or } \pi_1 = 0.1226$$

It follows that $\pi_0 = 1 - \pi_1 = 0.8774$.

Hence, the steady-state probability distribution is given by

$$\pi = [\pi_0 \quad \pi_1] = [0.8774 \quad 0.1226]$$

Thus, the fraction of signals that are highly distorted (in the steady-state) is 12.26%. $\blacklozenge$

EXAMPLE 5.53. Using limiting behaviour of homogeneous chain, find the steady-state probabilities of the chain given by the transition probability matrix

$$P = \begin{bmatrix} 0.1 & 0.6 & 0.3 \\ 0.5 & 0.1 & 0.4 \\ 0.1 & 0.2 & 0.7 \end{bmatrix}$$

(Anna, April 2004)

Solution. Since all the entries of P are positive, it follows that P is a regular stochastic matrix. Thus, the Markov chain is regular. Hence, by Theorem 5.11, the given Markov chain has a stationary distribution $\pi = [\pi_1 \quad \pi_2 \quad \pi_3]$, which is obtained by solving the equations

$$\pi P = \pi \text{ and } \pi_1 + \pi_2 + \pi_3 = 1$$

We note that $\pi P = \pi$ is equivalent to the linear system of equations

$$\begin{aligned} 0.1\pi_1 + 0.5\pi_2 + 0.1\pi_3 &= \pi_1 & \text{or} & & 9\pi_1 - 5\pi_2 - \pi_3 &= 0 \\ 0.6\pi_1 + 0.1\pi_2 + 0.2\pi_3 &= \pi_2 & \text{or} & & 6\pi_1 - 9\pi_2 + 2\pi_3 &= 0 \\ 0.3\pi_1 + 0.4\pi_2 + 0.7\pi_3 &= \pi_3 & \text{or} & & 3\pi_1 + 4\pi_2 - 3\pi_3 &= 0 \end{aligned}$$

Clearly, the first equation in the above linear system of equations is redundant because it is just the sum of the second and third equations. Hence, we consider the last two equations in the above linear system together with the condition that

$$\pi_1 + \pi_2 + \pi_3 = 1$$

Thus, we obtain the linear system of equations

$$\begin{bmatrix} 1 & 1 & 1 \\ 6 & -9 & 2 \\ 3 & 4 & -3 \end{bmatrix} \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Solving the linear system, we get $\pi_1 = 0.2021$, $\pi_2 = 0.2553$ and $\pi_3 = 0.5426$. $\blacklozenge$

EXAMPLE 5.54. In a hypothetical market, there are only two brands A and B . A customer buys brand A with probability 0.7 if his last purchase was A and buys brand B with probability 0.4 if his last purchase was B . Assuming MC model, obtain (i) one-step transition probability matrix P (say), (ii) n -step transition probability matrix P^n , and (iii) the stationary distribution. Hence, highlight the proportion of customers who would buy brands A and B in the long run.

(Anna, Nov. 2005)

Solution. (i) The one-step transition probability matrix P is obtained as

$$P = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{pmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{pmatrix} \end{matrix}$$

(ii) In Example 5.46, we showed that if the transition probability matrix P has the general form

$$P = \begin{bmatrix} 1-\alpha & \beta \\ \beta & 1-\beta \end{bmatrix} \text{ where } 0 < \alpha < 1, 0 < \beta < 1$$

then

$$P^n = \frac{1}{\alpha + \beta} \left\{ \begin{bmatrix} \beta & \alpha \\ \beta & \alpha \end{bmatrix} + (1 - \alpha - \beta)^n \begin{bmatrix} \alpha & -\alpha \\ -\beta & \beta \end{bmatrix} \right\} \quad (5.41)$$

Taking $\alpha = 0.3$ and $\beta = 0.6$ in Eq. (5.41), we obtain the n -step transition probability matrix as

$$P^n = \frac{1}{0.9} \left\{ \begin{bmatrix} 0.6 & 0.3 \\ 0.6 & 0.3 \end{bmatrix} + (0.1)^n \begin{bmatrix} 0.3 & -0.3 \\ -0.6 & 0.6 \end{bmatrix} \right\} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} + (0.1)^n \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{bmatrix} \quad (5.42)$$

(iii) The stationary distribution is obtained by taking limits as $n \rightarrow \infty$ in Eq. (5.42).

$$Q = \lim_{n \rightarrow \infty} P^n = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \pi \\ \pi \end{bmatrix}$$

where

$$\pi = [\pi_1 \quad \pi_2] = \left[\frac{2}{3} \quad \frac{1}{3} \right] = [0.6667 \quad 0.3333]$$

is the steady-state probability distribution. Hence, in the long-run, 66.67% of customers would buy brand A and 33.33% of customers would buy brand B .

◆

EXAMPLE 5.55. Describe a simple random walk with two reflecting barriers at the states 0 and 3 and obtain the stationary distribution.

Solution. Consider a model in which a man is at an integral part of the x -axis between the states $x = 0$ and $x = 3$. He takes a unit step to the right or left with equal probability 0.5, unless he is at the origin ($x = 0$), when he takes a step to the right to reach the state $x = 1$ or he is at the state $x = 3$, when he takes a step to the left to reach the state $x = 2$. This Markov chain model is called “random walk with

two reflecting barriers". The state space for the chain is $S = \{0, 1, 2, 3\}$ and the transition probability matrix is given below:

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

The stationary distribution $\pi = [\pi_0 \ \pi_1 \ \pi_2 \ \pi_3]$, if it exists, is obtained by solving the equations $\pi P = \pi$ and $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$.

We note that $\pi P = \pi$ is equivalent to the linear system of equations

$$\begin{aligned} 0.5\pi_1 &= \pi_0 & \text{or} & & 2\pi_0 - \pi_1 &= 0 \\ \pi_0 + 0.5\pi_2 &= \pi_1 & \text{or} & & 2\pi_0 - 2\pi_1 + \pi_2 &= 0 \\ 0.5\pi_1 + \pi_3 &= \pi_2 & \text{or} & & \pi_1 - 2\pi_2 + 2\pi_3 &= 0 \\ 0.5\pi_2 &= \pi_3 & \text{or} & & \pi_2 - 2\pi_3 &= 0 \end{aligned} \tag{5.43}$$

From the first and last equations in (5.43), we have

$$\pi_1 = 2\pi_0 \text{ and } \pi_2 = 2\pi_3$$

Simplifying the second and third equations in Eq. (5.43), we obtain the linear equation

$$-2\pi_0 + 2\pi_3 = 0 \text{ or } \pi_0 = \pi_3$$

Since $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$, we must have

$$\pi_3 + 2\pi_3 + 2\pi_3 + \pi_3 = 1 \text{ or } 6\pi_3 = 1$$

Thus, $\pi_3 = \frac{1}{6}$ and so it follows that

$$\pi_0 = \pi_3 = \frac{1}{6}, \quad \pi_1 = 2\pi_0 = \frac{1}{3} \text{ and } \pi_2 = 2\pi_3 = \frac{1}{3}$$

Thus, the stationary distribution is given by

$$\pi = [\pi_0 \ \pi_1 \ \pi_2 \ \pi_3] = \left[\frac{1}{6} \ \frac{1}{3} \ \frac{1}{3} \ \frac{1}{6} \right]$$

◆

EXAMPLE 5.56. A housewife buys 3 kinds of cereals: A , B and C . She never buys the same cereal in successive weeks. If she buys cereal A , the next week she buys cereal B . However, if she buys B or C , the next week she is 3 times as likely to buy A as the other cereal. In the long run, how often she buys each of the three cereals?

(Anna, Nov. 2005)

Solution. We denote the state space as $S = \{A, B, C\}$, where the states correspond to the three cereals, A , B and C . From the given data, the transition probability matrix is obtained as

$$P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{pmatrix} 0 & 1 & 0 \\ \frac{3}{4} & 0 & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{4} & 0 \end{pmatrix} \end{matrix}$$

The steady-state probability distribution $\pi = [\pi_1 \quad \pi_2 \quad \pi_3]$ is obtained by solving the equations

$$\pi P = \pi \text{ and } \pi_1 + \pi_2 + \pi_3 = 1$$

We note that $\pi P = \pi$ is equivalent to the linear system of equations

$$\begin{aligned} \frac{3}{4}\pi_2 + \frac{3}{4}\pi_3 &= \pi_1 & \text{or} & & 4\pi_1 - 3\pi_2 - 3\pi_3 &= 0 \\ \pi_1 + \frac{1}{4}\pi_3 &= \pi_2 & \text{or} & & 4\pi_1 - 4\pi_2 + \pi_3 &= 0 \\ \frac{1}{4}\pi_2 &= \pi_3 & \text{or} & & \pi_2 &= 4\pi_3 \end{aligned}$$

Using the last equation $\pi_2 = 4\pi_3$, the first and second equations in the above linear system may be simplified as

$$4\pi_1 - 15\pi_3 = 0 \text{ or } \pi_1 = \frac{15}{4}\pi_3$$

Since $\pi_1 + \pi_2 + \pi_3 = 1$, we must have

$$\frac{15}{4}\pi_3 + 4\pi_3 + \pi_3 = 1 \text{ or } \pi_3 = \frac{4}{35}$$

Thus, it follows that $\pi_1 = \frac{15}{4}\pi_3 = \frac{15}{35}$ and $\pi_2 = 4\pi_3 = \frac{16}{35}$.

Hence, the steady-state probability distribution is given by

$$\pi = \begin{pmatrix} A & B & C \\ \frac{15}{35} & \frac{16}{35} & \frac{4}{35} \end{pmatrix} = \begin{pmatrix} A & B & C \\ 0.4286 & 0.4571 & 0.1143 \end{pmatrix}$$

Thus, we conclude that in the long run, the housewife will buy cereal A in 42.9% of the weeks, cereal B in 45.7% of the weeks and cereal B in 11.4% of the weeks. $\blacklozenge$

5.5.5 Classification of States of a Markov Chain

Let $\{X_n, n \geq 0\}$ be a homogeneous Markov chain with the state space $S = \{0, 1, 2, \dots\}$. The states of the Markov chain are classified now.

Definition 5.24. We say that the state j is *accessible* from state i if $p_{ij}^{(n)} > 0$ for some $n \geq 0$, and we write $i \rightarrow j$. If two states, i and j , are accessible to each other, we say that they *communicate*, and we write $i \leftrightarrow j$. If all states communicate with each other, then we say that the Markov chain is *irreducible*. Otherwise, we say that the Markov chain is *reducible*.

Definition 5.25. We define the *period* of state j to be

$$d(j) = \gcd\{n \geq 1 \mid p_{jj}^{(n)} > 0\}$$

where gcd stands for greatest common divisor. If $d(j) > 1$, then we say that the state j is *periodic* with period $d(j)$. If $d(j) = 1$, then we say that the state j is *aperiodic*. We note that whenever $p_{jj} > 0$, the state j is aperiodic.

Definition 5.26. We say that the state j is an *absorbing state* if $p_{jj} = 1$. (Note that if $p_{jj} = 1$, then it follows that once the state j is reached, it is never left.)

Definition 5.27. Let T_j be the time or the number of steps of the first visit to state j after time zero. If the state j is never visited, we set $T_j = \infty$. Then we note that T_j is a discrete random variable taking values $1, 2, \dots, \infty$.

For any states i and j , we define $f_{ij}^{(n)}$ to be the probability that the chain, having started from state i , reaches state j at the n th step (or after n transitions). Formally, we define

$$\begin{aligned} f_{ij}^{(0)} &= 0, \\ f_{ij}^{(1)} &= P\{T_j = 1 | X_0 = i\} = P\{X_1 = j | X_0 = i\} = p_{ij} \\ f_{ij}^{(n)} &= P\{T_j = n | X_0 = i\} = P\{X_n = j, X_k \neq j, k = 1, 2, \dots, n-1 | X_0 = i\} \end{aligned}$$

Note that

$$f_{ij}^{(n)} = \sum_{k \neq j} p_{ik} f_{kj}^{(n-1)}, \text{ where } n = 2, 3, \dots$$

The probability of visiting state j in finite time, starting from state i , is given by

$$f_{ij} = \sum_{n=0}^{\infty} f_{ij}^{(n)} = P\{T_j < \infty | X_0 = i\}$$

Now, we say that the state j is *recurrent* or *persistent* if

$$f_{jj} = P\{T_j < \infty | X_0 = j\} = 1$$

i.e. starting from state j , the probability of eventual return to state a_j is one (or certain). If $f_{jj} < 1$, we say that the state j is *transient*.

We say that a recurrent state a_j is *positive recurrent* if

$$E\{T_j | X_0 = j\} < \infty$$

where

$$E\{T_j | X_0 = j\} = \sum_{n=0}^{\infty} n f_{jj}^{(n)}$$

We say that the state j is *null recurrent* if

$$E\{T_j | X_0 = j\} = \infty$$

Finally, we say that a positive persistent and aperiodic state j is *ergodic*. A Markov chain is called *ergodic* if all its states are ergodic, i.e. all its states are positive recurrent and aperiodic.

We state the following basic theorems without proof (see (Medhi, pp. 98–100, 1994) for a proof).

Theorem 5.12. If a Markov chain is irreducible, then all its states are of the same type. They are either all transient, all null recurrent or all positive recurrent. All its states are either aperiodic or periodic with the same period.

Theorem 5.13. If a Markov chain has finite states and is irreducible, then all its states are positive recurrent.

EXAMPLE 5.57. A gambler has Rs. 2 and he plays a betting game where he wins Re. 1 if a head shows up and loses Re. 1 if a tail shows up in the tossing of a fair coin. He stops playing this game if he wins Rs. 2 or loses Rs. 2. Find the

- Transition probability matrix of the given Markov chain.
- Probability that the gambler has lost his money at the end of 5 plays.
- Probability that the gambler ends the game in 6 plays.

Solution. Let X_n be the amount with the player at the end of n plays. If S denotes the state space of X_n , then it is clear that $S = \{0, 1, 2, 3, 4\}$ because the game ends if the gambler loses all the money ($X_n = 0$) or wins Rs. 4 ($X_n = 4$). [Note that the gambler has Rs. 2 at the commencement of the game, and he stops playing the game if he wins Rs. 2 or loses Rs. 2.]

- The transition probability matrix of the given Markov chain is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

(The given Markov chain is called a *random walk with absorbing barriers at states 0 and 6* because the chain cannot leave the states 0 and 6 once it reaches them, i.e. the game ends when the chain reaches any of the two states, 0 and 6.)

- Since the gambler has initially Rs. 2, the initial probability distribution is given by

$$p(0) = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \end{pmatrix} \end{matrix}$$

The probability distribution after one play is given by

$$p(1) = p(0)P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{pmatrix} 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix} \end{matrix}$$

The probability distribution after 2 plays is given by

$$p(2) = p(1)P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{pmatrix} \frac{1}{4} & 0 & \frac{1}{2} & 0 & \frac{1}{4} \end{pmatrix} \end{matrix}$$

The probability distribution after 3 plays is given by

$$p(3) = p(2)P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} & \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \end{pmatrix} \end{matrix}$$

The probability distribution after 4 plays is given by

$$p(4) = p(3)P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} & \begin{pmatrix} \frac{3}{8} & 0 & \frac{1}{4} & 0 & \frac{3}{8} \end{pmatrix} \end{matrix}$$

The probability distribution after 5 plays is given by

$$p(5) = p(4)P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} & \begin{pmatrix} \frac{3}{8} & \frac{1}{8} & 0 & \frac{1}{8} & \frac{3}{8} \end{pmatrix} \end{matrix}$$

Thus, the probability that the gambler has lost his money at the end of 5 plays is given by

$$P(X_5 = 0) = \frac{3}{8}$$

(c) The probability distribution after 6 plays is given by

$$p(6) = p(5)P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} & \begin{pmatrix} \frac{7}{16} & 0 & \frac{1}{8} & 0 & \frac{7}{16} \end{pmatrix} \end{matrix}$$

Thus, the probability that the game ends in 6 plays is given by

$$P(X_6 = 0 \text{ or } X_6 = 4) = P(X_6 = 0) + P(X_6 = 4) = \frac{7}{16} + \frac{7}{16} = \frac{7}{8}$$

◆

EXAMPLE 5.58. Three boys A , B and C , are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C , but C is as likely to throw the ball to B as to A . Show that the process is Markovian. Find the transition matrix and classify the states.

(Madurai, Nov. 1996; Anna, Nov. 2003, Nov. 2005, Nov. 2007)

Solution. We denote the boys, A , B and C , by states 1, 2 and 3, respectively. Let X_n denote the player holding the ball at the n th stage, where $n = 0, 1, 2, \dots$. Then it is clear that the states of X_{n+1} depend only on X_n , but not on the states of $X_{n-1}, X_{n-2}, \dots, X_0$. Thus, it follows that $\{X_n, n \geq 0\}$ is a Markov chain.

The transition probability matrix of the Markov chain is:

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} \end{matrix}$$

Now, we find that

$$P^2 = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \text{and} \quad P^3 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

Note that $p_{11}^{(3)} > 0, p_{12}^{(1)} > 0, p_{13}^{(2)} > 0, p_{21}^{(2)} > 0, p_{22}^{(3)} > 0, p_{23}^{(1)} > 0, p_{31}^{(1)} > 0, p_{32}^{(1)} > 0$ and $p_{33}^{(2)} > 0$.

Since $p_{ij}^{(n)} > 0$ for all i and j for some positive integer n , it follows that the given Markov chain is irreducible.

Next, we find that

$$P^4 = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}, \quad P^5 = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{3}{8} & \frac{1}{2} \end{bmatrix} \quad \text{and} \quad P^6 = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{3}{8} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{8} & \frac{3}{8} \end{bmatrix}$$

and so on.

Thus, we note that $p_{ii}^{(2)}, p_{ii}^{(3)}, p_{ii}^{(4)}, p_{ii}^{(5)}, p_{ii}^{(6)} > 0$ for $i = 2, 3$. Hence, it follows that the periods of the states 2 and 3 are given by

$$d(2) = \gcd \{ n \geq 1 \mid p_{22}^{(n)} > 0 \} = \gcd \{ 2, 3, 4, 5, 6, \dots \} = 1$$

and

$$d(3) = \gcd \{ n \geq 1 \mid p_{33}^{(n)} > 0 \} = \gcd \{ 2, 3, 4, 5, 6, \dots \} = 1$$

We also note that $p_{11}^{(3)}, p_{11}^{(5)}, p_{11}^{(6)} > 0$ and so the period of the state 1 is given by

$$d(1) = \gcd \{ n \geq 1 \mid p_{11}^{(n)} > 0 \} = \gcd \{ 3, 5, 6, \dots \} = 1$$

Since the states 1, 2 and 3 have period 1, they are aperiodic.

Since the Markov chain is finite and irreducible, by Theorem 5.13, it follows that all its states are positive recurrent. Since we have already shown that all the states are aperiodic as well, it follows that all states of the Markov chain are positive recurrent and aperiodic, i.e. they are ergodic. Hence, the given Markov chain is ergodic. $\blacklozenge$

EXAMPLE 5.59. Let $\{X_n; n = 1, 2, 3, \dots\}$ be a Markov chain with state space $S = \{0, 1, 2\}$ and one-step transition probability matrix

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix}$$

(i) Is the chain ergodic? Explain.

(ii) Find the invariant probabilities.

(Anna, Nov. 2005)

Solution. (i) We find that

$$P^2 = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{3}{4} & \frac{1}{8} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$$

Since $p_{ij}^{(2)} > 0$ for all i and j , it follows that the given Markov chain is irreducible. Since $p_{ii} > 0$ for $i = 1, 2, 3$, it follows that the three states 0, 1 and 2 are aperiodic. Since the Markov chain has finite states and is irreducible, by Theorem 5.13, it follows that all its states are positive recurrent. Hence, all the states of the given Markov chain are both positive recurrent and aperiodic, i.e. they are ergodic. Thus, we conclude that the given Markov chain is ergodic.

- (ii) Since $p_{ij}^{(2)} > 0$ for all i and j , it follows that the transition probability matrix P is a regular stochastic matrix, and so the given Markov chain is regular. Hence, by Theorem 5.11, the given Markov chain has an invariant probability distribution $\pi = [\pi_0 \ \pi_1 \ \pi_2]$, which is obtained by solving the equations

$$\pi P = \pi \text{ and } \pi_0 + \pi_1 + \pi_2 = 1$$

Note that $\pi P = \pi$ is equivalent to the linear system of equations

$$[\pi_0 \ \pi_1 \ \pi_2] \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix} = [\pi_0 \ \pi_1 \ \pi_2]$$

or

$$\begin{aligned} \frac{1}{4}\pi_1 &= \pi_0 \\ \pi_0 + \frac{1}{2}\pi_1 + \pi_2 &= \pi_1 \\ \frac{1}{4}\pi_1 &= \pi_2 \end{aligned}$$

Thus, $\pi_0 = \frac{1}{4}\pi_1$ and $\pi_2 = \frac{1}{4}\pi_1$. Since $\pi_0 + \pi_1 + \pi_2 = 1$, it follows that

$$\frac{1}{4}\pi_1 + \pi_1 + \frac{1}{4}\pi_1 = 1 \text{ or } \frac{3}{2}\pi_1 = 1$$

or $\pi_1 = \frac{2}{3}$.

Thus, it follows that $\pi_0 = \frac{1}{4}\pi_1 = \frac{1}{6}$ and $\pi_2 = \frac{1}{4}\pi_1 = \frac{1}{6}$.

Hence, the invariant probabilities are given by

$$\pi_0 = \frac{1}{6}, \ \pi_1 = \frac{2}{3} \text{ and } \pi_2 = \frac{1}{6}$$

◆

Next, we define the *state transition diagram* of a finite-state Markov chain.

Definition 5.28. A *state transition diagram* of a finite-state homogeneous Markov chain with transition probability matrix $P = [p_{ij}]$ is a line diagram with a vertex corresponding to each state of the Markov chain and a directed line segment between two vertices i and j if $p_{ij} > 0$. In such a diagram, if one can move from the state i to the state j by a path following the arrows, then the state j is *accessible* from state i and we write $i \rightarrow j$.

We note that the state transition diagram of a finite-state homogeneous Markov chain is very useful to determine whether the chain is irreducible or not as well as to determine the period of the states.

EXAMPLE 5.60. The one-step transition probability matrix of a Markov chain with states $\{0, 1\}$ is given as

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- (a) Draw a transition diagram.
 (b) Is it irreducible Markov chain?

Solution. (a) The state transition diagram of the given Markov chain is illustrated in Figure 5.2.

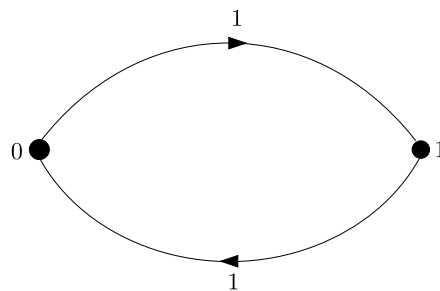


Figure 5.2: State Transition Diagram.

- (b) From Figure 5.2, it is clear that $0 \leftrightarrow 1$. Thus, the given Markov chain is irreducible. ♦

EXAMPLE 5.61. Draw the state transition diagram and classify the states of the Markov chain, whose transition probability matrix is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix} \end{matrix}$$

Solution. The state transition diagram for the given Markov chain is illustrated in Figure 5.3. From the figure, it is clear that

$$0 \rightarrow 3 \rightarrow 1 \rightarrow 0 \text{ and } 0 \rightarrow 3 \rightarrow 2 \rightarrow 0$$

Thus, it is immediate that $0 \leftrightarrow 1 \leftrightarrow 2 \leftrightarrow 3$. Hence, the given Markov chain is irreducible. Since the Markov chain has finite states and it is irreducible, by Theorem 5.13, it follows that all its states are positive recurrent.

Next, we calculate the period of the states 0, 1 and 2. Note that starting from state 0, we can get back to state 0 in only one of the following paths:

- (a) $0 \rightarrow 3 \rightarrow 1 \rightarrow 0$ (in three steps)

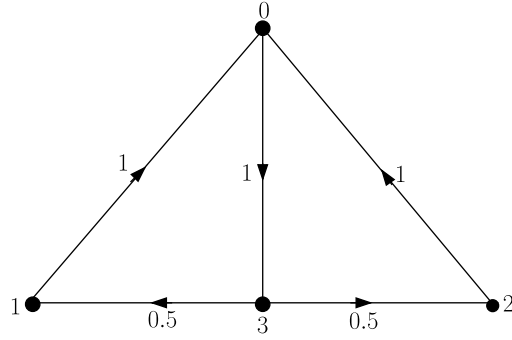


Figure 5.3: State Transition Diagram.

(b) $0 \rightarrow 3 \rightarrow 2 \rightarrow 0$ (in three steps)

Thus, it follows that the period of the state 0 is 3.

Note that starting from state 1, we can get back to state 1 in any of the following paths:

- (a) $1 \rightarrow 0 \rightarrow 3 \rightarrow 1$ (in 3 steps);
- (b) $1 \rightarrow 0 \rightarrow 3 \rightarrow 2 \rightarrow 0 \rightarrow 3 \rightarrow 1$ (in 6 steps);
- (c) $1 \rightarrow 0 \rightarrow 3 \rightarrow 2 \rightarrow 0 \rightarrow 3 \rightarrow 2 \rightarrow 0 \rightarrow 3 \rightarrow 1$ (in 9 steps); and so on.

Thus, the period of the state 1 is given by

$$d(1) = \gcd\{3, 6, 9, \dots\} = 3$$

Thus, the state 1 has period 3.

Note that starting from state 2, we can get back to state 2 in any of the following paths:

- (a) $2 \rightarrow 0 \rightarrow 3 \rightarrow 2$ (in 3 steps);
- (b) $2 \rightarrow 0 \rightarrow 3 \rightarrow 1 \rightarrow 0 \rightarrow 3 \rightarrow 2$ (in 6 steps);
- (c) $2 \rightarrow 0 \rightarrow 3 \rightarrow 1 \rightarrow 0 \rightarrow 3 \rightarrow 1 \rightarrow 0 \rightarrow 3 \rightarrow 2$ (in 9 steps); and so on.

Thus, the period of the state 2 is given by

$$d(2) = \gcd\{3, 6, 9, \dots\} = 3$$

Thus, the state 2 has period 3. Similarly, we can show that the state 3 also has period 3.

Since all the states have period 3, it follows that the given Markov chain is not ergodic. (For an ergodic chain, all states are positive recurrent and aperiodic.) ◆

EXAMPLE 5.62. Consider a Markov chain with state space $\{0, 1\}$ and transition probability matrix

$$P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

- (i) Draw a transition diagram.
- (ii) Show that the state 0 is recurrent.
- (iii) Show that the state 1 is transient.

- (iv) Is the state 1 periodic? If so, what is the period?
- (v) Is the chain irreducible?
- (vi) Is the chain ergodic? Explain.

(Anna, Model, April 2005)

Solution. (i) The state transition diagram for the given Markov chain is illustrated in Figure 5.4.

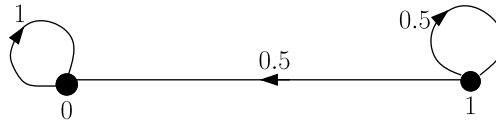


Figure 5.4: State Transition Diagram.

- (ii) We find that

$$\begin{aligned}
 f_{00}^{(1)} &= p_{00} = 1 \\
 f_{10}^{(1)} &= p_{10} = \frac{1}{2} \\
 f_{00}^{(2)} &= p_{01} f_{10}^{(1)} = (0) \frac{1}{2} = 0 \\
 f_{00}^{(n)} &= 0 \text{ for } n \geq 2
 \end{aligned}$$

Thus,

$$f_{00} = P\{T_0 < \infty \mid X_0 = 0\} = \sum_{n=0}^{\infty} f_{00}^{(n)} = 1 + 0 + 0 + \cdots = 1$$

Hence, by Definition 5.27, the state 0 is recurrent.

- (iii) We find that

$$\begin{aligned}
 f_{11}^{(1)} &= p_{11} = \frac{1}{2} \\
 f_{01}^{(1)} &= p_{01} = 0 \\
 f_{11}^{(2)} &= p_{10} f_{01}^{(1)} = \left(\frac{1}{2}\right) 0 = 0 \\
 f_{11}^{(3)} &= 0 \text{ for } n \geq 2
 \end{aligned}$$

Thus,

$$f_{11} = P\{T_1 < \infty \mid X_0 = 1\} = \sum_{n=0}^{\infty} f_{11}^{(n)} = \frac{1}{2} + 0 + 0 + \cdots < 1$$

Hence, by Definition 5.27, the state 1 is transient.

- (iv) Starting from the state 1, we can return to the state 1 by using any of the following paths:

- (a) $1 \rightarrow 1$ (using 1 step);
- (b) $1 \rightarrow 1 \rightarrow 1$ (using 2 steps);
- (c) $1 \rightarrow 1 \rightarrow 1 \rightarrow 1$ (using 3 steps); and so on.

Thus, it is clear that the period of the state 1 is 1, i.e. the state 1 is aperiodic.

- (v) The given Markov chain is not irreducible because the state 1 is not accessible from the state 0, i.e. there is no path from the state 0 to the state 1.
- (vi) The given Markov chain is not ergodic because the state 1 is transient. (For an ergodic chain, all states are aperiodic and positive recurrent.)



PROBLEM SET 5.5

1. The transition probability matrix of a Markov chain with two states 0 and 1 is given by

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

and suppose that the states 0 and 1 are initially equally likely. Find

- (a) $p(2)$, i.e. the probability distribution after 2 steps.
- (b) $p(4)$, i.e. the probability distribution after 4 steps.

2. The transition probability matrix of a Markov chain with states 0, 1 and 2 is given by

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

Show that $P^{2n} = P^2$ for any positive integer n .

3. A fair die is tossed repeatedly. If X_n denotes the maximum of the numbers occurring in the first n tosses, find the transition probability matrix P of the Markov chain $\{X_n\}$. Find also P^2 and $P(X_2 = 3)$.
1. There are three brands of beauty products A , B and C in a shop. It has been found that a person when he purchases a beauty product of a particular brand, he will continue to buy the same brand or switch over to another brand during his next purchase of a beauty product in the shop. The transition probability matrix associated with the three brands is given below:

$$P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{pmatrix} 0.4 & 0.3 & 0.3 \\ 0.6 & 0.1 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix} \end{matrix}$$

If the initial distribution of purchase of the brands A , B and C is $[0.4 \quad 0.4 \quad 0.2]$, determine the distribution of the brands after two purchases.

5. The transition probability matrix of a Markov chain with three states 0, 1, 2 is

$$P = \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.2 & 0.4 \\ 0.1 & 0.6 & 0.3 \end{bmatrix}$$

and the initial probability distribution is $p(0) = [0.3 \quad 0.4 \quad 0.3]$. Find

- (a) $P(X_2 = 2)$.

(b) $P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 0)$.

6. Find the stationary distribution for the Markov chain $\{X_n; n \geq 1\}$ with state space $S = \{0, 1\}$ and one-step transition probability matrix

$$P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

7. A gambler has Rs. 2. He bets 1 rupee at a time and wins 1 rupee with probability $\frac{1}{2}$. He stops fixing if he loses Rs. 2 or wins Rs. 4. Find the
- Transition probability matrix of the related Markov chain.
 - Probability that the gambler has lost his money at the end of 4 fixings.
 - Probability that the game lasts more than 5 fixings.
8. In a tropical city, suppose that the probability of a sunny day (state 0) following a rainy day (state 1) is 0.5 and that the probability of a rainy day following a sunny day is 0.3. We are given that the New Year day is a sunny day. Let X_n denote the weather on the n th day.
- Find the transition probability matrix of the Markov chain.
 - Find the probability that January 3rd is a sunny day.
 - Find the probability that January 4th is a rainy day.
 - In the long run, what will be the proportion of sunny days and rainy days in the given city?
9. There are 2 white balls in bag A and 3 red balls in bag B. At each step of the process, a ball is selected from each bag and the 2 balls selected are interchanged. Let the state a_i of the system be the number of red balls in A after i changes. What is the probability that there are 2 red balls in A after 3 steps? In the long run, what is the probability that there are 2 red balls in bag A?
10. Consider a Markov chain with state space, $S = \{0, 1, 2\}$ and transition probability matrix

$$P = \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

- Draw the state transition diagram.
 - Is the Markov chain irreducible?
 - Is the Markov chain ergodic? Explain.
 - Find the stationary distribution of the chain.
11. Consider a Markov chain with state space, $S = \{0, 1, 2\}$ and transition probability matrix

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & 0 & 0 \end{bmatrix}$$

- Draw the state transition diagram.

- (b) Is the Markov chain irreducible?
- (c) Determine the periods of the states.
- (d) Is the Markov chain ergodic? Explain.

12. The transition probability matrix of a Markov chain with states 0, 1, 2 and 3 is given by

$$P = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0.2 & 0.3 & 0.3 & 0.2 \end{bmatrix}$$

- (a) Draw the state transition diagram.
- (b) Is the Markov chain irreducible?
- (c) Find the periods of the states.
- (d) Is the Markov chain ergodic? Explain.

5.6 BINOMIAL, POISSON AND NORMAL PROCESSES

In Section 1.5, we defined Bernoulli trials (see Definition 1.19) as repeated statistically independent and identically distributed trials, where each trial consists of only two outcomes, *success* and *failure*, with constant probabilities p and q , respectively ($p + q = 1$).

Definition 5.29. Associated with a sequence of Bernoulli trials, we can define a countably infinite sequence of random variables $\{X_n, n = 1, 2, 3, \dots\}$ by

$$X_n = \begin{cases} 1 & \text{if the } n\text{th Bernoulli trial yields a success} \\ 0 & \text{if the } n\text{th Bernoulli trial yields a failure} \end{cases}$$

with probabilities

$$P(X_n = 1) = p \text{ and } P(X_n = 0) = q, \text{ where } 0 < p < 1, p + q = 1$$

Then, we define $\{X_n, n \geq 1\}$ as a *Bernoulli process*.

EXAMPLE 5.63. Consider the experiment of tossing a fair coin repeatedly. Each trial has only two outcomes, viz. H and T , which may be considered as *success* and *failure*, respectively. Note that $p = P(H) = \frac{1}{2}$ and $q = 1 - p = \frac{1}{2}$. Since the trials are independent, and the probability p for getting H remains a constant throughout all the trials, the given trials constitute Bernoulli trials. Associated with these Bernoulli trials, we have the *Bernoulli process* defined by

$$X_n = \begin{cases} 1 & \text{if the } n\text{th toss yields a head} \\ 0 & \text{if the } n\text{th Bernoulli trial yields a tail} \end{cases}$$

The probability distribution for X_n is given by

$$P(X_n = 1) = p = \frac{1}{2} \text{ and } P(X_n = 0) = q = \frac{1}{2}$$

Theorem 5.14. Let $\{X_n, n = 1, 2, 3, \dots\}$ be a Bernoulli process. Then

(a) The mean of the process is

$$\mu_X = E[X_n] = p \quad \text{for all } n$$

(b) The variance of the process is

$$\sigma_X^2 = \text{Var}(X_n) = pq$$

(c) The autocorrelation function of the process is

$$R_{XX}(m, n) = \begin{cases} p & \text{if } m = n \\ p^2 & \text{if } m \neq n \end{cases}$$

(d) The autocovariance function of the process is

$$C_{XX}(m, n) = \begin{cases} pq & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

Proof. (a) The mean of the process is given by

$$\mu_X = E[X_n] = 1 \cdot P(X_n = 1) + 0 \cdot P(X_n = 0) = 1 \cdot p + 0 \cdot q = p$$

(b) The variance of the process is given by

$$\sigma_X^2 = \text{Var}[X_n] = E[X_n^2] - \mu_X^2 = E[X_n^2] - p^2$$

Note that

$$E[X_n^2] = 1^2 \cdot P(X_n = 1) + 0^2 \cdot P(X_n = 0) = 1 \cdot p + 0 \cdot q = p$$

Thus, it follows that

$$\sigma_X^2 = p - p^2 = p(1 - p) = pq$$

(c) The autocorrelation function of the process is given by

$$R_{XX}(m, n) = E[X_m X_n]$$

We have two cases to consider: (i) $m = n$ and (ii) $m \neq n$.

If $m = n$, then we have

$$R_{XX}(m, n) = E[X_m^2] = p$$

and if $m \neq n$, then X_m and X_n are independent random variables and we have

$$R_{XX}(m, n) = E[X_m X_n] = E[X_m]E[X_n] = p \cdot p = p^2$$

Combining both cases, result (c) follows.

(d) The autocovariance function of the process is given by

$$C_{XX}(m, n) = E[(X_m - \mu_X)(X_n - \mu_X)]$$

We have two cases to consider: (i) $m = n$ and (ii) $m \neq n$.

If $m = n$, then we have

$$C_{XX}(m, n) = E[(X_m - \mu_X)^2] = \text{Var}[X_m] = \sigma_X^2 = pq$$

If $m \neq n$, then X_m and X_n are independent random variables and we have

$$C_{XX}(m, n) = E[X_m - \mu_X]E[X_n - \mu_X] = \{E[X_m] - \mu_X\}\{E[X_n] - \mu_X\} = (p - p)(p - p) = 0$$

Combining both cases, result (d) follows. ◆

Definition 5.30. (*Binomial Process*) Let $\{Z_n, n = 1, 2, 3, \dots\}$ be a Bernoulli process and X_n denote the number of the successes in the first n Bernoulli trials, i.e.

$$X_n = Z_1 + Z_2 + \dots + Z_n$$

Then we say that $\{X_n, n = 1, 2, 3, \dots\}$ is a binomial process. Clearly, the probability distribution of X_n is given by

$$P(X_n = k) = \binom{n}{k} p^k q^{n-k} \quad \text{for } k = 0, 1, 2, \dots, n$$

Theorem 5.15. Let $\{X_n, n = 1, 2, 3, \dots\}$ be a binomial process. Then

(a) The mean of the process is

$$E[X_n] = np \quad \text{for all } n$$

(b) The variance of the process is

$$\text{Var}(X_n) = npq$$

(c) The autocorrelation function of the process is

$$R_{XX}(m, n) = pq \min(m, n) + mnp^2$$

(d) The autocovariance function of the process is

$$C_{XX}(m, n) = pq \min(m, n)$$

Proof. (a) The mean of the process $\{X_n\}$ is given by

$$E[X_n] = E[Z_1 + Z_2 + \dots + Z_n] = E[Z_1] + E[Z_2] + \dots + E[Z_n] = p + p + \dots + p = np$$

- (b) Since $Z_1, Z_2, \dots, Z_n$ are independent random variables, it follows that the variance of the process $\{X_n\}$ is given by

$$\begin{aligned}\text{Var}[X_n] &= \text{Var}[Z_1 + Z_2 + \dots + Z_n] = \text{Var}[Z_1] + \text{Var}[Z_2] + \dots + \text{Var}[Z_n] \\ &= pq + pq + \dots + pq = npq\end{aligned}$$

- (c) The autocorrelation function of the process $\{X_n\}$ is given by

$$R_{XX}(m, n) = E[X_m X_n] = E \left[\sum_{i=1}^m \sum_{j=1}^n Z_i Z_j \right]$$

We have two cases to consider: (i) $m \leq n$ and (ii) $n \leq m$.

If $m \leq n$, then we have

$$R_{XX}(m, n) = \sum_{i=1}^m Z_i^2 + \sum_{j=1}^m \sum_{\substack{i=1 \\ i \neq j}}^m E[Z_i Z_j] + \sum_{j=1}^{n-m} \sum_{\substack{i=1 \\ i \neq j}}^m E[Z_i Z_j]$$

Using the results in Theorem 5.14, we have

$$R_{XX}(m, n) = mp + m(m-1)p^2 + m(n-m)p^2 = mpq + mnp^2$$

Similarly, if $n \leq m$, then

$$R_{XX}(m, n) = npq + nmp^2$$

Combining the results for cases (i) and (ii), we have

$$R_{XX}(m, n) = \min(m, n) pq + mnp^2$$

- (d) The covariance of the process $\{X_n\}$ is given by

$$C_{XX}(m, n) = R_{XX}(m, n) - E[X_m]E[X_n] = \min(m, n) pq + mnp^2 - (mp)(np) = \min(m, n) pq$$

◆

Remark 5.10. For a binomial process $\{X_n\}$, we showed that

$$\text{Mean}[X_n] = np \quad \text{and} \quad \text{Var}[X_n] = npq$$

which depend on the time n . Hence, it is immediate that the binomial process is not stationary in any sense.

Definition 5.31. (*Counting Process*) A random process $\{X(t), t \geq 0\}$ is called a *counting process* if $X(t)$ represents the total number of events that have occurred in the interval $(0, t)$, i.e. $X(t)$ must satisfy the following conditions:

- (a) $X(t) \geq 0$ and $X(0) = 0$.
- (b) $X(t)$ takes only integer values.

- (c) $X(t)$ is monotone nondecreasing, i.e. $X(s) \leq X(t)$ if $s < t$.
- (d) $X(t) - X(s)$ is equal to the number of events that have occurred in the interval (s, t) .

A counting process $X(t)$ is said to possess *independent increments* if the number of events that occur in disjoint time intervals are independent. Thus, for a counting process $X(t)$ with independent increments, the number of events occurring up to time s is independent of the number of events occurring between times s and t , i.e. $X(s)$ is independent of $X(t) - X(s)$, where $s < t$. $X(t)$ is said to possess *stationary increments* if $X(t+h) - X(s+h)$ has the same distribution as $X(t) - X(s)$ for all $s < t$ and $h > 0$.

A very important counting process is the *Poisson process*, named after the French mathematician Siméon-Denis Poisson (1781–1840). The Poisson process is used for modelling random events in time that occur to a large extent independently of one another.

Some of the typical applications of Poisson process may be listed now:

- (i) The number of web page requests arriving at a server.
- (ii) The number of telephone calls arriving at a switchboard.
- (iii) The number of particles emitted via radioactive decay.
- (iv) The number of raindrops falling over a wide spatial area.
- (v) The arrival of customers in simple queuing systems.

For defining a Poisson process, we require small- o notation.

Definition 5.32. (*small- o notation*) A function $f(\cdot)$ is called $o(\varepsilon)$ if

$$\lim_{\varepsilon \rightarrow 0} \frac{f(\varepsilon)}{\varepsilon} = 0$$

EXAMPLE 5.64. (a) The function $f(x) = x^2$ is $o(\varepsilon)$ because

$$\lim_{\varepsilon \rightarrow 0} \frac{f(\varepsilon)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{\varepsilon^2}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \varepsilon = 0$$

(b) The function $g(x) = cx$ ($c \neq 0$) is not $o(\varepsilon)$ because

$$\lim_{\varepsilon \rightarrow 0} \frac{g(\varepsilon)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{c\varepsilon}{\varepsilon} = c$$

Definition 5.33. A counting process $X(t)$, which represents the number of occurrences of a certain event in the interval $(0, t)$, is called a *Poisson process* with rate or intensity λ if it satisfies the following conditions known as *Poisson postulates*:

- (a) $X(0) = 0$
- (b) $X(t)$ has independent and stationary increments.
- (c) $P[X(t + \Delta t) - X(t) = 1] = \lambda \Delta t + o(\Delta t)$
- (d) $P[X(t + \Delta t) - X(t) \geq 2] = o(\Delta t)$

Remark 5.11. We note that in Definition 5.33 of the Poisson process, the rate or intensity λ is assumed to be a constant. Such a Poisson process is also known as a *homogeneous Poisson process*. The Poisson process, for which λ is a function of time, is known as a *non-homogeneous Poisson process*. In this chapter, we deal only with homogeneous Poisson processes, i.e. the rate λ will be always assumed to be a constant.

Theorem 5.16. Let $\{X(t), t \geq 0\}$ be a Poisson process. If $p_n(t) = P[X(t) = n]$, then $p_n(t)$ is Poisson distributed with mean λt , i.e.

$$p_n(t) = P[X(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \quad \text{for } n = 0, 1, 2, \dots$$

Proof. First, we derive a differential equation for $p_0(t) = P[X(t) = 0]$. Since $X(t)$ has independent increments, we have

$$\begin{aligned} p_0(t + \Delta t) &= P[X(t + \Delta t) = 0] = P[X(t) = 0, X(t + \Delta t) - X(t) = 0] \\ &= P[X(t) = 0]P[X(t + \Delta t) - X(t) = 0] \end{aligned}$$

Thus, we have

$$p_0(t + \Delta t) = p_0(t) P[X(t + \Delta t) - X(t) = 0] \quad (5.44)$$

Next, we shall show that

$$P[X(t + \Delta t) - X(t) = 0] = 1 - \lambda \Delta t + o(\Delta t) \quad (5.45)$$

Since $X(t)$ is monotone nondecreasing and it can take only nonnegative integer values, it follows that $X(t + \Delta t) - X(t)$ can also take only nonnegative integer values. Thus, it follows that

$$\sum_{k=0}^{\infty} P[X(t + \Delta t) - X(t) = k] = P[X(t + \Delta t) = 0] + P[X(t + \Delta t) = 1] + P[X(t + \Delta t) \geq 2] = 1 \quad (5.46)$$

Using conditions (c) and (d) in Definition 5.33, Eq. (5.46) reduces to

$$P[X(t + \Delta t) = 0] = 1 - \lambda \Delta t + o(\Delta t)$$

which proves Eq. (5.45).

Substituting Eq. (5.45) into Eq. (5.44), we have

$$p_0(t + \Delta t) = p_0(t)[1 - \lambda \Delta t + o(\Delta t)]$$

or

$$\frac{p_0(t + \Delta t) - p_0(t)}{\Delta t} = -\lambda p_0(t) + \frac{o(\Delta t)}{\Delta t} \quad (5.47)$$

Taking limits on both sides of Eq. (5.47) as $\Delta t \rightarrow 0$, we have

$$p_0'(t) = -\lambda p_0(t) \quad \text{or} \quad \frac{dp_0}{dt} = -\lambda p_0$$

Separating the variables, we have

$$\frac{dp_0}{p_0} = -\lambda dt$$

Integrating, we get

$$\ln p_0(t) = -\lambda t + c$$

where c is an integration constant. Thus, it follows that

$$p_0(t) = e^{-\lambda t + c} = Ke^{-\lambda t}$$

Since $p_0(0) = P[X(0) = 0] = 1$ (by condition (a)), it follows that $K = 1$. Thus, we have

$$p_0(t) = e^{-\lambda t} \quad (5.48)$$

In a similar manner, for $n > 0$, we have

$$\begin{aligned} p_n(t + \Delta t) &= P[X(t + \Delta t) = n] \\ &= P[X(t) = n, X(t + \Delta t) - X(t) = 0] + P[X(t) = n - 1, X(t + \Delta t) - X(t) = 1] \\ &\quad + \sum_{k=2}^n P[X(t) = n - k, X(t + \Delta t) - X(t) = k] \end{aligned} \quad (5.49)$$

Using condition (b) in Definition 5.33 and Eq. (5.45), we have

$$\begin{aligned} P[X(t) = n, X(t + \Delta t) - X(t) = 0] &= P[X(t) = n]P[X(t + \Delta t) - X(t) = 0] \\ &= p_n(t)[1 - \lambda \Delta t + o(\Delta t)] \end{aligned} \quad (5.50)$$

Using conditions (b) and (c) in Definition 5.33, we have

$$\begin{aligned} P[X(t) = n - 1, X(t + \Delta t) - X(t) = 1] &= P[X(t) = n - 1]P[X(t + \Delta t) - X(t) = 1] \\ &= p_{n-1}(t)[\lambda \Delta t + o(\Delta t)] \end{aligned} \quad (5.51)$$

For $k \geq 2$, using conditions (b) and (d) in Definition 5.33, we have

$$\begin{aligned} P[X(t) = n - k, X(t + \Delta t) - X(t) = k] &= P[X(t) = n - k]P[X(t + \Delta t) - X(t) = k] \\ &= p_{n-k}(t)o(\Delta t) = o(\Delta t) \end{aligned} \quad (5.52)$$

Substituting Eqs. (5.50), (5.51) and (5.52) into Eq. (5.49), we get

$$p_n(t + \Delta t) = p_n(t)[1 - \lambda \Delta t + o(\Delta t)] + p_{n-1}(t)[\lambda \Delta t + o(\Delta t)] + o(\Delta t)$$

Thus, it follows that

$$\frac{p_n(t + \Delta t) - p_n(t)}{\Delta t} = -\lambda p_n(t) + \lambda p_{n-1}(t) + \frac{o(\Delta t)}{\Delta t} \quad (5.53)$$

Taking limits on both sides of Eq. (5.53) as $\Delta t \rightarrow 0$, we have

$$p'_n(t) = -\lambda p_n(t) + \lambda p_{n-1}(t)$$

or

$$p'_n(t) + \lambda p_n(t) = \lambda p_{n-1}(t)$$

Multiplying both sides by $e^{\lambda t}$, we have

$$e^{\lambda t} [p'_n(t) + \lambda p_n(t)] = \lambda e^{\lambda t} p_{n-1}(t)$$

i.e.

$$\frac{d}{dt} [e^{\lambda t} p_n(t)] = \lambda e^{\lambda t} p_{n-1}(t) \quad (5.54)$$

From Eqs. (5.54) (with $n = 1$) and (5.48), we have

$$\frac{d}{dt} [e^{\lambda t} p_1(t)] = \lambda$$

Integrating, we get

$$e^{\lambda t} p_1(t) = \lambda t + A$$

where A is an integration constant.

Since $p_1(0) = P[X(0) = 1] = 0$, $A = 0$. Thus, it follows that

$$e^{\lambda t} p_1(t) = \lambda t$$

or

$$p_1(t) = e^{-\lambda t} \lambda t \quad (5.55)$$

We prove that

$$p_n(t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \quad (5.56)$$

for any non-negative integer n using mathematical induction.

We note that Eq. (5.56) is true for $n = 0, 1$ by Eqs. (5.48) and (5.55).

Induction Hypothesis: Suppose that Eq. (5.56) is true for any positive integer m , i.e.

$$p_m(t) = e^{-\lambda t} \frac{(\lambda t)^m}{m!} \quad (5.57)$$

From Eqs. (5.54) (with $n = m + 1$) and (5.57), we have

$$\frac{d}{dt} [e^{\lambda t} p_{m+1}(t)] = \lambda e^{\lambda t} p_m(t) = \frac{\lambda^{m+1}}{m!} t^m$$

Integrating, we get

$$e^{\lambda t} p_{m+1}(t) = \frac{(\lambda t)^{m+1}}{(m+1)!} + B$$

where B is an integration constant.

Since $p_{m+1}(0) = P[X(0) = m + 1] = 0$, $B = 0$. Thus, it follows that

$$e^{\lambda t} p_{m+1}(t) = \frac{(\lambda t)^{m+1}}{(m+1)!}$$

or

$$p_{m+1}(t) = e^{-\lambda t} \frac{(\lambda t)^{m+1}}{(m+1)!}$$

Thus, Eq. (5.56) is also true for $n = m + 1$. This completes the inductive step of the proof. Hence, by the principle of mathematical induction, Eq. (5.56) is true for any non-negative integer n .

This completes the proof. $\blacklozenge$

Theorem 5.17. If $X(t)$ is a Poisson process with rate λ , then the number of events in any interval of length t is Poisson distributed with mean λt , i.e. for all $s, t \geq 0$

$$P[X(t+s) - X(s) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \text{ for } n = 0, 1, 2, \dots$$

Proof. By Theorem 5.16, we know that

$$p_n(t) = P[X(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \text{ for } n = 0, 1, 2, \dots \quad (5.58)$$

Since a Poisson process has stationary increments, it follows that $X(t+s) - X(s)$ and $X(t) - X(0)$ have the same probability distribution. Since $X(0) = 0$, it is immediate that $X(t+s) - X(s)$ and $X(t)$ have the same probability distribution. Thus, from Eq. (5.58), it follows that

$$P[X(t+s) - X(s) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \text{ for } n = 0, 1, 2, \dots \quad \blacklozenge$$

Next, we derive the second order probability distribution of a Poisson process with rate λ . If $t_2 > t_1$, then we have

$$\begin{aligned} P[X(t_1) = n_1, X(t_2) = n_2] &= P[X(t_1) = n_1]P[X(t_2) = n_2 \mid X(t_1) = n_1] \\ &= P[X(t_1) = n_1]P[\text{the event occurs } n_2 - n_1 \text{ times in } (t_1, t_2)] \\ &= e^{-\lambda t_1} \frac{(\lambda t_1)^{n_1}}{n_1!} e^{-\lambda(t_2-t_1)} \frac{[\lambda(t_2-t_1)]^{n_2-n_1}}{(n_2-n_1)!} \text{ if } n_2 \geq n_1 \end{aligned}$$

Thus, we have

$$P[X(t_1) = n_1, X(t_2) = n_2] = \begin{cases} e^{-\lambda t_2} \frac{\lambda^{n_2} t_1^{n_1}}{n_1!} \frac{(t_2-t_1)^{n_2-n_1}}{(n_2-n_1)!} & \text{if } n_2 \geq n_1 \\ 0 & \text{otherwise} \end{cases} \quad (5.59)$$

Similarly, we can derive the third order probability distribution of a Poisson process with rate λ as

$$P[X(t_1) = n_1, X(t_2) = n_2, P(X_3) = n_3] = \begin{cases} e^{-\lambda t_3} \frac{\lambda^{n_3} t_1^{n_1}}{n_1!} \frac{(t_2-t_1)^{n_2-n_1}}{(n_2-n_1)!} \frac{(t_3-t_2)^{n_3-n_2}}{(n_3-n_2)!} & \text{if } n_3 \geq n_2 \geq n_1 \\ 0 & \text{otherwise} \end{cases} \quad (5.60)$$

Theorem 5.18. Let $X(t)$ be a Poisson process with rate λ .

(a) The mean and the variance of the process are given by

$$\mu_X(t) = E[X(t)] = \lambda t \quad \text{and} \quad \text{Var}[X(t)] = \sigma_X^2(t) = \lambda t$$

(b) The autocovariance function of the process is given by

$$C_{XX}(t_1, t_2) = \lambda \min(t_1, t_2)$$

(c) The autocorrelation function of the process is given by

$$R_{XX}(t_1, t_2) = \lambda \min(t_1, t_2) + \lambda^2 t_1 t_2$$

(d) The autocorrelation coefficient of the process is given by

$$\rho_{XX}(t_1, t_2) = \frac{\min(t_1, t_2)}{\sqrt{t_1} \sqrt{t_2}}$$

Proof. (a) Since $X(t)$ has a Poisson distribution with parameter λt , it follows by Theorem 3.7 that

$$\mu_X(t) = E[X(t)] = \lambda t \quad \text{and} \quad \text{Var}[X(t)] = \sigma_X^2(t) = \lambda t$$

(b) By definition, a Poisson process has stationary and independent increments. Hence, by Example 5.12, the autocovariance function of $X(t)$ is given by

$$C_{XX}(t_1, t_2) = \sigma_1^2 \min(t_1, t_2) = \lambda \min(t_1, t_2)$$

since $\sigma_1^2 = \text{Var}[X(1)] = \lambda \cdot 1 = \lambda$.

(c) By definition, the autocorrelation function of $X(t)$ is given by

$$R_{XX}(t_1, t_2) = C_{XX}(t_1, t_2) + \mu_X(t_1)\mu_X(t_2) = \lambda \min(t_1, t_2) + \lambda^2 t_1 t_2$$

(d) By definition, the autocorrelation coefficient of $X(t)$ is given by

$$\rho_{XX}(t_1, t_2) = \frac{C_{XX}(t_1, t_2)}{\sigma_X(t_1)\sigma_X(t_2)} = \frac{\lambda \min(t_1, t_2)}{\sqrt{\lambda t_1} \sqrt{\lambda t_2}} = \frac{\min(t_1, t_2)}{\sqrt{t_1} \sqrt{t_2}} \quad \blacklozenge$$

EXAMPLE 5.65. Derive the mean, autocorrelation and autocovariance of a Poisson process.

(Anna, April 2004)

Solution. See the proof of Theorem 5.18. $\blacklozenge$

EXAMPLE 5.66. Define Poisson random process. Is it a stationary process? Justify the answer.

(Anna, Model 2003)

Solution. The definition of a Poisson process is given in Definition 5.33. For a random process to be stationary in any sense, its mean must be a constant. By Theorem 5.18, we know that the mean of a Poisson process with rate λ is given by

$$\mu_X(t) = E[X(t)] = \lambda t$$

which depends on the time t . This shows that the Poisson process is not a stationary process. $\blacklozenge$

EXAMPLE 5.67. If $\{N(t)\}$ is a Poisson process, then prove that the autocorrelation coefficient between $N(t)$ and $N(t+s)$ is $\sqrt{\frac{t}{t+s}}$.

(Anna, Nov. 2004)

Solution. By Theorem 5.18, we know that the autocorrelation coefficient between $N(t)$ and $N(t+s)$ is given by

$$\rho_{XX}(t, t+s) = \frac{\min(t, t+s)}{\sqrt{t}\sqrt{t+s}} = \frac{t}{\sqrt{t}\sqrt{t+s}} = \sqrt{\frac{t}{t+s}}$$

$\blacklozenge$

Next, we derive some important properties of the Poisson process.

Theorem 5.19. Let $X(t)$ be a Poisson process with rate λ . Then the following properties hold:

- (a) The Poisson process is a Markov process.
- (b) The sum of two independent Poisson processes is a Poisson process

(Additive Property) (Anna, Nov. 2004; Nov. 2006)

- (c) The difference of two independent Poisson processes is not a Poisson process.

(Anna, Nov. 2003; May 2007)

Proof. (a) Given the times $t_1 < t_2 < t_3$, we note that

$$P[X(t_3) = n_3 \mid X(t_2) = n_2, X(t_1) = n_1] = \frac{P[X(t_1) = n_1, X(t_2) = n_2, X(t_3) = n_3]}{P[X(t_1) = n_1, X(t_2) = n_2]}$$

Substituting from Eqs. (5.59) and (5.60) and simplifying, it follows that

$$\begin{aligned} P[X(t_3) = n_3 \mid X(t_2) = n_2, X(t_1) = n_1] &= e^{-\lambda(t_3-t_2)} \frac{\lambda^{n_3-n_2} (t_3-t_2)^{n_3-n_2}}{(n_3-n_2)!} \\ &= P[X(t_3) = n_3 \mid X(t_2) = n_2] \end{aligned}$$

This shows that the conditional probability distribution of $X(t_3)$ given all the past values of $X(t_1) = n_1$ and $X(t_2) = n_2$ depends only on the most recent value $X(t_2) = n_2$.

Thus, the Poisson process satisfies the Markov property, Eq. (5.30), i.e. it is a Markov process.

- (b) By Theorem 3.9, we know that the characteristic function of a Poisson distribution with parameter λ is given by

$$\phi_X(\omega) = e^{\lambda(e^{i\omega} - 1)}$$

Suppose that $X_1(t)$ and $X_2(t)$ are independent Poisson processes with rates λ_1 and λ_2 respectively. By Theorem 5.18, it follows that they have the means $\lambda_1 t$ and $\lambda_2 t$, respectively. Thus, it follows that the characteristic functions of $X_1(t)$ and $X_2(t)$ are given by

$$\phi_{X_1(t)}(\omega) = e^{\lambda_1 t(e^{i\omega} - 1)} \quad \text{and} \quad \phi_{X_2(t)}(\omega) = e^{\lambda_2 t(e^{i\omega} - 1)}$$

Since $X_1(t)$ and $X_2(t)$ are independent, we know that the characteristic function of $X_1(t) + X_2(t)$ is given by

$$\begin{aligned} \phi_{X_1(t) + X_2(t)}(\omega) &= \phi_{X_1(t)}(\omega) \phi_{X_2(t)}(\omega) = e^{\lambda_1 t(e^{i\omega} - 1)} e^{\lambda_2 t(e^{i\omega} - 1)} \\ &= e^{(\lambda_1 + \lambda_2)t(e^{i\omega} - 1)} \end{aligned}$$

which is the characteristic function of the Poisson process with rate $\lambda_1 + \lambda_2$.

Hence, by the uniqueness of characteristic functions of random variables, it follows that $X_1(t) + X_2(t)$ is Poisson distributed with mean $(\lambda_1 + \lambda_2)t$. Thus, it is immediate that $X_1(t) + X_2(t)$ is a Poisson process with rate $\lambda_1 + \lambda_2$.

- (c) Let $X(t) = X_1(t) - X_2(t)$, where $X_1(t)$ and $X_2(t)$ are independent Poisson processes with rates λ_1 and λ_2 , respectively. Then, we find that the mean of $X(t)$ is given by

$$\mu_X(t) = E[X(t)] = E[X_1(t)] - E[X_2(t)] = \lambda_1 t - \lambda_2 t = (\lambda_1 - \lambda_2)t$$

Next, we find that the second raw moment of $X(t)$ is given by

$$E[X^2(t)] = E\{[X_1(t) - X_2(t)]^2\} = E\{X_1^2(t) + X_2^2(t) - 2X_1(t)X_2(t)\}$$

Using the independence of $X_1(t)$ and $X_2(t)$, we have

$$\begin{aligned} E[X^2(t)] &= E\{X_1^2(t)\} + E\{X_2^2(t)\} - 2E[X_1(t)]E[X_2(t)] \\ &= (\lambda_1^2 t^2 + \lambda_1 t) + (\lambda_2^2 t^2 + \lambda_2 t) - 2\lambda_1 \lambda_2 t^2 \\ &= (\lambda_1 + \lambda_2)t + (\lambda_1^2 + \lambda_2^2 - 2\lambda_1 \lambda_2)t^2 = (\lambda_1 + \lambda_2)t + (\lambda_1 - \lambda_2)^2 t^2 \end{aligned}$$

Thus, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = E[X^2(t)] - [\mu_X(t)]^2 = (\lambda_1 + \lambda_2)t \neq (\lambda_1 - \lambda_2)t = \mu_X(t)$$

This shows that $X(t)$ is not a Poisson process, i.e. the difference of two independent Poisson processes is not a Poisson process. ◆

EXAMPLE 5.68. If $X_1(t)$ and $X_2(t)$ are two independent Poisson processes, show that the conditional distribution of $X_1(t)$ given $X_1(t) + X_2(t)$ is binomial.

(Anna, Nov. 2006)

Solution. Suppose that $X_1(t)$ and $X_2(t)$ are independent Poisson processes with rates λ_1 and λ_2 , respectively. By Theorem 5.19, we know that $X_1(t) + X_2(t)$ is a Poisson process with rate $\lambda_1 + \lambda_2$. Thus, we have

$$P[X_1(t) = k] = e^{-\lambda_1 t} \frac{(\lambda_1 t)^k}{k!}$$

$$P[X_2(t) = l] = e^{-\lambda_2 t} \frac{(\lambda_2 t)^l}{l!}$$

and

$$P[X_1(t) + X_2(t) = n] = e^{-(\lambda_1 + \lambda_2)t} \frac{[(\lambda_1 + \lambda_2)t]^n}{n!}$$

It follows that

$$P[X_1(t) = k \mid X_1(t) + X_2(t) = n] = \frac{P[X_1(t) = k, X_1(t) + X_2(t) = n]}{P[X_1(t) + X_2(t) = n]} = \frac{P[X_1(t) = k, X_2(t) = n - k]}{P[X_1(t) + X_2(t) = n]}$$

Since $X_1(t)$ and $X_2(t)$ are independent,

$$P[X_1(t) = k \mid X_1(t) + X_2(t) = n] = \frac{P[X_1(t) = k]P[X_2(t) = n - k]}{P[X_1(t) + X_2(t) = n]} = \frac{e^{-\lambda_1 t} \frac{(\lambda_1 t)^k}{k!} e^{-\lambda_2 t} \frac{(\lambda_2 t)^{n-k}}{(n-k)!}}{e^{-(\lambda_1 + \lambda_2)t} \frac{[(\lambda_1 + \lambda_2)t]^n}{n!}}$$

Simplifying, we have

$$P[X_1(t) = k \mid X_1(t) + X_2(t) = n] = \frac{n!}{k!(n-k)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k} = \binom{n}{k} p^k q^{n-k} \quad (5.61)$$

where $p = \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)$ and $q = \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)$.

Thus, we have shown that the conditional distribution of $X_1(t)$ given $X_1(t) + X_2(t)$ is a binomial distribution. $\blacklozenge$

Theorem 5.20. The inter-arrival time of a Poisson process with intensity λ obeys an exponential law with mean $\frac{1}{\lambda}$.

(Anna, Model; Nov. 2003; Nov. 2004; April 2005; May 2006)

Proof. We consider two successive occurrences of the event: E_i and E_{i+1} .

Let E_i take place at time t_i and E_{i+1} take place at time $t_i + T$. Thus, the inter-arrival time between the two successive events is T , which is a continuous random variable. We find that

$$\begin{aligned} P(T > t) &= P[E_{i+1} \text{ did not occur in } (t_i, t_i + t)] \\ &= P[\text{No event occurs in an interval of length } t] \\ &= P[X(t) = 0] \\ &= e^{-\lambda t} \end{aligned}$$

Thus, the cumulative distribution function of T is given by

$$F(t) = P(T \leq t) = 1 - P(T > t) = 1 - e^{-\lambda t} \text{ for } t \geq 0$$

Hence, the probability density function of T is given by

$$f(t) = \frac{dF}{dt} = \lambda e^{-\lambda t} \text{ for } t \geq 0$$

Hence, the inter-arrival time of a Poisson process with intensity λ follows an exponential distribution with mean $\frac{1}{\lambda}$. ◆

EXAMPLE 5.69. Let $\{X(t) : t \geq 0\}$ be a Poisson process with parameter λ . Suppose that each arrival is registered with probability p independent of other arrivals. Let $\{Y(t) : t \geq 0\}$ be the process of registered arrivals. Prove that $Y(t)$ is a Poisson process with parameter λp . (Anna, Nov. 2005)

Solution. We find that

$$P[Y(t) = n] = \sum_{r=0}^{\infty} P[\text{There are } n+r \text{ arrivals in } (0, t) \text{ and } n \text{ of them are registered}]$$

Thus, it follows that

$$\begin{aligned} P[Y(t) = n] &= \sum_{r=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^{n+r}}{(n+r)!} \binom{n+r}{n} p^n q^r = \sum_{r=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^{n+r}}{(n+r)!} \frac{(n+r)!}{n!r!} p^n q^r \\ &= e^{-\lambda t} \frac{(\lambda p t)^n}{n!} \sum_{r=0}^{\infty} \frac{(\lambda q t)^r}{r!} = e^{-\lambda t} \frac{(\lambda p t)^n}{n!} e^{\lambda q t} \\ &= e^{-\lambda t(1-q)} \frac{(\lambda p t)^n}{n!} = e^{-\lambda p t} \frac{(\lambda p t)^n}{n!} \end{aligned}$$

Hence, we conclude that $Y(t)$ is a Poisson process with rate or intensity λp . ◆

EXAMPLE 5.70. Suppose that customers arrive at a bank according to a Poisson process with mean rate of 3 per minute. Find the probability that during a time interval of 2 minutes (i) exactly 4 customers arrive and (ii) more than 4 customers arrive. (Madurai, April 1996)

Solution. Since the Poisson process $X(t)$ has a rate of 3 per minute, its mean is $\mu_X(t) = \lambda t = 3t$. Thus, we have

$$P[X(t) = k] = e^{-3t} \frac{(3t)^k}{k!} \text{ for } k = 0, 1, 2, \dots$$

(i) The probability that during a time interval of 2 minutes, exactly 4 customers arrive, is given by

$$P[X(2) = 4] = e^{-6} \frac{6^4}{4!} = e^{-6} \times 54 = 0.1339$$

(ii) The probability that during a time interval of 2 minutes, more than 4 customers arrive, is given by

$$\begin{aligned} P[X(2) > 4] &= 1 - \{P[X(2) = 0] + P[X(2) = 1] + P[X(2) = 2] + P[X(2) = 3] + P[X(2) = 4]\} \\ &= 1 - (0.0025 + 0.0149 + 0.0446 + 0.0892 + 0.1339) = 0.7149 \end{aligned}$$
◆

EXAMPLE 5.71. The number of particles emitted by a radioactive source is Poisson distributed. The source emits particles at a rate of 6 per minute. Each emitted particle has a probability of 0.7 of being counted. Find the probability that 11 particles are counted in 4 minutes.

(Anna, April 2005)

Solution. We note that the number of emitted particles $N(t)$ follows a Poisson process with intensity $\lambda^* = \lambda p$, where λ is the rate of the particles and p is the probability of being counted. Since $\lambda = 6$ and $p = 0.7$, it follows that

$$\lambda^* = \lambda p = 6 \times 0.7 = 4.2$$

Thus, the probability distribution of $N(t)$ is given by

$$P[N(t) = k] = e^{-4.2t} \frac{(4.2t)^k}{k!} \text{ for } k = 0, 1, 2, \dots$$

Hence, the probability that 11 particles are counted in 4 minutes is given by

$$P[N(4) = 11] = e^{-16.8} \frac{(16.8)^{11}}{11!} = 0.0381$$

◆

EXAMPLE 5.72. A machine goes out of order, whenever a component fails. The failure of this part follows a Poisson process with a mean rate of 1 per week. Find the probability that 2 weeks have elapsed since last failure. If there are 5 spare parts of this component in an inventory and that the next supply is not due in 10 weeks, find the probability that the machine will not be out of order in the next 10 weeks.

(Anna, Nov. 2003)

Solution. We are given that the mean failure rate is $\lambda = 1$. Thus, the number of failures in t weeks is a Poisson process $X(t)$ described by

$$P[X(t) = k] = e^{-t} \frac{t^k}{k!} \text{ for } k = 0, 1, 2, \dots$$

The probability that 2 weeks have elapsed since last failure is given by

$$P[\text{No failure in the 2 weeks}] = P[X(2) = 0] = e^{-2} = 0.1353$$

We are given that there are 5 spare parts of the component in the inventory. Hence, the probability that the machine will not be out of order in the next 10 weeks is given by

$$P[X(10) \leq 5] = \sum_{k=0}^5 e^{-10} \frac{10^k}{k!} = e^{-10} \left[1 + 10 + 50 + \frac{500}{3} + \frac{1250}{3} + \frac{2500}{3} \right] = 0.0671$$

◆

EXAMPLE 5.73. VLSI chips, essential to the running of a computer system, fail in accordance with a Poisson distribution with the rate of one chip in about 5 weeks. If there are two spare chips on hand, and if a new supply will arrive in 8 weeks, what is the probability that during the next 8 weeks the system will be down for a week or more, owing to the lack of chips?

(Anna, Nov. 2007)

Solution. Let $X(t)$ be the number of failures of the VLSI chips measured in t weeks with rate $\lambda = \frac{1}{5} = 0.2$. Since $X(t)$ is a Poisson process, we know that

$$P[X(t) = k] = e^{-0.2t} \frac{(0.2t)^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

Since there are only two spare VLSI chips on hand, the probability that the system will be down for at least one week before new supply in 8 weeks is given by

$$\begin{aligned} P[X(7) > 2] &= 1 - \{P[X(7) = 0] + P[X(7) = 1] + P[X(7) = 2]\} \\ &= 1 - e^{-1.4} [1 + 1.4 + 0.98] = 1 - e^{-1.4} 3.38 \\ &= 0.1665 \end{aligned}$$

◆

EXAMPLE 5.74. Queries presented in a computer database are following a Poisson process of rate $\lambda = 6$ queries per minute. An experiment consists of monitoring the database for m minutes and recording $N(m)$ the number of queries presented. What is the probability of

- (i) No queries in a one-minute interval?
- (ii) Exactly 6 queries arriving in a one-minute interval?
- (iii) Less than 3 queries arriving in a half-minute interval?

(Anna, May 2007)

Solution. Let $N(t)$ be the number of queries presented in t minutes. Since $N(t)$ follows a Poisson process of rate $\lambda = 6$, we know that

$$P[N(t) = k] = e^{-6t} \frac{(6t)^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

- (i) The probability of no queries in a one-minute interval is given by

$$P[N(1) = 0] = e^{-6} = 0.0025$$

- (ii) The probability of exactly 6 queries arriving in a one-minute interval is given by

$$P[N(1) = 6] = e^{-6} \frac{6^6}{6!} = 0.1606$$

- (iii) The probability of less than 3 queries arriving in a half-minute interval is given by

$$P[N(0.5) < 3] = e^{-3} \left[1 + 3 + \frac{3^2}{2!}\right] = 0.4232$$

◆

EXAMPLE 5.75. The number of telephone calls arriving at a certain switch board within a time interval of length (measured in minutes) is a Poisson process $X(t)$ with parameter $\lambda = 2$. Find the probability of

- (i) No telephone calls arriving at this switch board during a 5 minute period.
- (ii) More than one telephone call arriving at this switch board during a given $\frac{1}{2}$ minute period.

(Anna, Model)

Solution. Since the Poisson process $X(t)$ has parameter $\lambda = 2$, we know that

$$P[X(t) = k] = e^{-2t} \frac{(2t)^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

- (i) The required probability is given by

$$P[X(5) = 0] = e^{-10} = 4.54 \times 10^{-5}$$

- (ii) The required probability is given by

$$P[X(0.5) \geq 1] = 1 - P[X(0.5) = 0] = 1 - e^{-1} = 0.6321$$

◆

EXAMPLE 5.76. A fisherman catches fish at a Poisson rate of 2 per hour from a large pond with lots of fish. If he starts fishing at 10:00 A.M., what is the probability that he catches one fish by 10:30 A.M. and three fish by noon?

(Anna, April 2004)

Solution. The given Poisson process $X(t)$ has the rate of $\lambda = 2$ per hour. Thus, we have

$$P[X(t) = k] = e^{-2t} \frac{(2t)^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

It is given that the fisherman starts fishing at 10:00 A.M. Thus, the probability that he catches one fish by 10:30 A.M. (in half-an-hour period) is given by

$$P[X(0.5) = 1] = e^{-1} 1 = e^{-1} = 0.3679$$

Similarly, the probability that the fisherman catches three fish by noon (in 4-hours period) is given by

$$P[X(4) = 3] = e^{-8} \frac{8^3}{3!} = 0.0286$$

◆

EXAMPLE 5.77. If customers arrive at a counter in accordance with a Poisson process with a mean rate of 2 per minute, find the probability that the interval between two consecutive arrivals is

- (i) more than 1 minute
- (ii) between 1 minute and 2 minutes
- (iii) less than or equal to 4 minutes

(Anna, Nov. 2005)

Solution. By Theorem 5.20, we know that the inter-arrival time T of a Poisson process with rate λ follows an exponential distribution with mean $\frac{1}{\lambda}$. The given Poisson process has rate $\lambda = 2$. Thus, the inter-arrival time T follows an exponential distribution with mean $\frac{1}{2}$, i.e. the probability density function of T is given by

$$f(t) = 2e^{-2t} \text{ for } t \geq 0$$

(i) The required probability is given by

$$P(T > 1) = \int_{t=1}^{\infty} f(t)dt = \int_{t=1}^{\infty} 2e^{-2t} dt = [-e^{-2t}]_1^{\infty} = e^{-2} = 0.1353$$

(ii) The required probability is given by

$$P(1 < T < 2) = \int_{t=1}^2 f(t)dt = \int_{t=1}^2 2e^{-2t} dt = [-e^{-2t}]_1^2 = e^{-2} - e^{-4} = 0.1170$$

(iii) The required probability is given by

$$P(0 < T < 4) = \int_{t=0}^4 f(t)dt = \int_{t=0}^4 2e^{-2t} dt = [-e^{-2t}]_0^4 = 1 - e^{-8} = 0.9997$$

◆

Next, we define a *normal* or *Gaussian process*.

Definition 5.34. A random process $\{X(t), t \in T\}$ is said to be a *normal* or *Gaussian process* if for any integer n and any subset $\{t_1, t_2, \dots, t_n\}$ of T , the n random variables $X(t_1), X(t_2), \dots, X(t_n)$ are jointly normally distributed.

The n th order probability density of a Gaussian process is given by

$$f(x_1, \dots, x_n; t_1, \dots, t_n) = \frac{1}{(2\pi)^{n/2} |\Lambda|^{1/2}} \exp \left\{ - \left[\frac{1}{2|\Lambda|} \sum_{i=1}^n \sum_{j=1}^n |\Lambda|_{ij} (x_i - \mu_i)(x_j - \mu_j) \right] \right\} \quad (5.62)$$

where

$$\mu_i = E[X(t_i)]$$

$$\Lambda = [\lambda_{ij}] \text{ is an } n \times n \text{ matrix with } \lambda_{ij} = \text{Cov}[X(t_i), X(t_j)]$$

$$|\Lambda|_{ij} = \text{Cofactor of } \lambda_{ij} \text{ in } |\Lambda|$$

The first order probability density of a Gaussian process may be calculated as follows:

$$\Lambda = [\lambda_{11}], \text{ where } \lambda_{11} = \text{Cov}[X(t_1), X(t_1)] = \text{Var}[X(t_1)] = \sigma_1^2$$

Thus, we have

$$|\Lambda| = \sigma_1^2 \text{ and } |\Lambda|_{11} = \sigma_1^2$$

Hence, from Eq. (5.62), the first order PDF of a Gaussian process is given by

$$f(x_1; t_1) = \frac{1}{\sigma_1 \sqrt{2\pi}} e^{-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2}}$$

The second order probability density of a Gaussian process may be calculated as follows:

$$\Lambda = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \rho_{12} \sigma_1 \sigma_2 \\ \rho_{21} \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}$$

Thus, we have

$$|\Lambda| = \sigma_1^2 \sigma_2^2 (1 - \rho^2)$$

where $\rho_{12} = \rho_{21} = \rho$, the correlation coefficient of $X(t_1)$ and $X(t_2)$.

We also find that

$$\begin{aligned} |\Lambda|_{11} &= \text{Cofactor of } \lambda_{11} = \sigma_2^2 \\ |\Lambda|_{12} &= \text{Cofactor of } \lambda_{12} = -\rho \sigma_1 \sigma_2 \\ |\Lambda|_{21} &= \text{Cofactor of } \lambda_{21} = -\rho \sigma_1 \sigma_2 \\ |\Lambda|_{22} &= \text{Cofactor of } \lambda_{22} = \sigma_1^2 \end{aligned}$$

Hence, from Eq. (5.62), the second order PDF of a Gaussian process is given by

$$f(x_1, x_2; t_1, t_2) = \frac{1}{2\pi \sigma_1 \sigma_2 \sqrt{1 - \rho^2}} e^{-\frac{1}{2\sigma_1^2 \sigma_2^2 (1 - \rho^2)} [\sigma_2^2 (x_1 - \mu_1)^2 - 2\rho \sigma_1 \sigma_2 (x_1 - \mu_1)(x_2 - \mu_2) + \sigma_1^2 (x_2 - \mu_2)^2]}$$

which may be also expressed as

$$\begin{aligned} f(x_1, x_2; t_1, t_2) &= \frac{1}{2\pi \sigma_1 \sigma_2 \sqrt{1 - \rho^2}} \exp \left\{ -\frac{1}{2(1 - \rho^2)} \left\{ \frac{(x_1 - \mu_1)^2}{\sigma_1^2} - \frac{2\rho(x_1 - \mu_1)(x_2 - \mu_2)}{\sigma_1 \sigma_2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} \right\} \right\} \end{aligned}$$

Next, we establish some properties of the Gaussian process.

First, we note that from Eq. (5.62), the n th order probability density function of a Gaussian process is completely characterized by the second-order distributions. Thus, if a Gaussian process is wide-sense stationary, then it is also strictly stationary. This is formally established in the following theorem.

Theorem 5.21. If a Gaussian process $\{X(t), t \in T\}$ is wide-sense stationary, then it is also strict-sense stationary.

(Madurai, April 1996)

Proof. The n th order probability density function of the Gaussian process $X(t)$ is given by

$$f(x_1, \dots, x_n; t_1, \dots, t_n) = \frac{1}{(2\pi)^{n/2} |\Lambda|^{1/2}} \exp \left\{ -\left[\frac{1}{2|\Lambda|} \sum_{i=1}^n \sum_{j=1}^n |\Lambda|_{ij} (x_i - \mu_i)(x_j - \mu_j) \right] \right\}$$

where $\mu_i = E[X(t_i)]$, $\Lambda = [\lambda_{ij}]$ with $\lambda_{ij} = \text{Cov}[X(t_i), X(t_j)]$ and $|\Lambda|_{ij}$ is the cofactor of λ_{ij} in $|\Lambda|$.

If the Gaussian process $X(t)$ is wide-sense stationary, then it follows that $\lambda_{ij} = \text{Cov}[X(t_i), X(t_j)]$ is a function only of the time difference $t_i - t_j$ for all i and j .

Hence, it is immediate that the n th order probability density functions of $\{X(t_1), X(t_2), \dots, X(t_n)\}$ and $\{X(t_1 + h), X(t_2 + h), \dots, X(t_n + h)\}$ are equal.

Hence, we conclude that the Gaussian process is a strict-sense stationary process. $\blacklozenge$

Theorem 5.22. If the member functions of a Gaussian process $\{X(t), t \in T\}$ are uncorrelated, then they are independent random variables.

Proof. We consider any n member functions $X(t_1), X(t_2), \dots, X(t_n)$ of the Gaussian process $\{X(t)\}$. Since the member functions are uncorrelated, it follows that

$$\lambda_{ij} = \text{Cov}[X(t_i), X(t_j)] = \begin{cases} 0 & \text{for } i \neq j \\ \sigma_i^2 & \text{for } i = j \end{cases}$$

Thus, it follows that Λ is a diagonal matrix given by

$$\Lambda = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{bmatrix}$$

Therefore, we have

$$\begin{aligned} |\Lambda| &= \sigma_1^2 \sigma_2^2 \cdots \sigma_n^2 \\ |\Lambda|_{ii} &= \sigma_1^2 \sigma_2^2 \cdots \sigma_{i-1}^2 \sigma_{i+1}^2 \cdots \sigma_n^2 \\ |\Lambda|_{ij} &= 0 \text{ for } i \neq j \end{aligned}$$

Hence, by Eq. (5.62), the n th order PDF of $X(t_1), X(t_2), \dots, X(t_n)$ are given by

$$\begin{aligned} f(x_1, \dots, x_n; t_1, \dots, t_n) &= \frac{1}{(2\pi)^{n/2} \sigma_1 \sigma_2 \cdots \sigma_n} \exp \left[-\frac{1}{2} \sum_{j=1}^n \frac{(x_j - \mu_j)^2}{\sigma_j^2} \right] \\ &= \left[\frac{1}{\sigma_1 \sqrt{2\pi}} e^{-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2}} \right] \cdots \left[\frac{1}{\sigma_n \sqrt{2\pi}} e^{-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}} \right] \\ &= f(x_1; t_1) \cdots f(x_n; t_n) \end{aligned}$$

This shows that $X(t_1), X(t_2), \dots, X(t_n)$ are independent random variables. $\blacklozenge$

EXAMPLE 5.78. If $\{X(t)\}$ is a Gaussian process with $\mu(t) = 10$ and $C(t_1, t_2) = 16e^{-|t_1 - t_2|}$, find the probability that (i) $X(10) \leq 8$, (ii) $|X(10) - X(6)| \leq 4$.

(Anna, April 2003)

Solution. Since $X(t)$ is a Gaussian process, any member of the process is a normal random variable with mean $\mu = 10$ and variance $\sigma^2 = C(t, t) = 16$.

- (i) Here, we find $P[X(10) \leq 8]$.

The standard normal random variable Z is defined by

$$Z = \frac{X(10) - \mu_X}{\sigma_X} = \frac{X(10) - 10}{4}$$

Thus, when $X(10) = 8$, $Z = -\frac{2}{4} = -0.5$. Hence, we have

$$\begin{aligned} P[X(10) \leq 8] &= P[Z \leq -0.5] = P[Z \geq 0.5] = 0.5 - P(0 < Z < 0.5) \\ &= 0.5 - 0.1915 = 0.3085 \end{aligned}$$

- (ii) Here, we find $P[|X(10) - X(6)| \leq 4]$. For this purpose, we define

$$U = X(10) - X(6)$$

Then, we find that the mean of U is given by

$$\mu_U = \mu_X(10) - \mu_X(6) = 10 - 10 = 0$$

and that the variance of U is given by

$$\begin{aligned} \sigma_U^2 &= \sigma_X^2(10) + \sigma_X^2(6) - 2\text{Cov}[X(10), X(6)] \\ &= C(10, 10) + C(6, 6) - 2C(10, 6) = 16 + 16 - 32e^{-4} \\ &= 32 - 32e^{-4} = 32[1 - e^{-4}] \\ &= 31.4139 \end{aligned}$$

Thus, we have $\sigma_U = \sqrt{31.4139} = 5.6048$.

Next, we form the standard normal variable as

$$Z = \frac{U - \mu_U}{\sigma_U} = \frac{U - 0}{5.6048} = \frac{U}{5.6048}$$

When $U = 4$, we have $Z = \frac{4}{5.6048} = 0.7137$.

Thus, the required probability is given by

$$\begin{aligned} P[|X(10) - X(6)| \leq 4] &= P[|U| \leq 4] = P[|Z| \leq 0.7137] \\ &= 2P(0 < Z < 0.7137) = 2(0.2611) = 0.5222 \end{aligned}$$

◆

PROBLEM SET 5.6

1. Let $N(t)$ be a Poisson process with rate λ . For $s < t$, show that

$$P\{N(s) = k \mid N(t) = n\} = \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k} \text{ where } s < t$$

(Anna, Model, 2003)

2. Let $X(t)$ be a Poisson process with rate λ . Find
 - (a) $E[X^2(t)]$.
 - (b) $E\{[X(t) - X(s)]^2\}$ for $t > s$.
3. The emission of particles of a radar is Poisson distributed at a rate of 5 per minute. Each emitted particle has a probability of 0.6 of being recorded. Find the probability that 10 particles are emitted in a 4-minute period.
4. A radioactive source emits particles at the rate of 6 per minute in a Poisson process. Each emitted particle has a probability of 0.5 of being recorded. Find the probability that at least 4 particles are recorded in a 5 minute period.
5. The number of telephone calls arriving at a certain switch board within a time interval of length (measured in minutes) is a Poisson process $X(t)$ with parameter $\lambda = 6$. Find the probability of
 - (1) Two telephone calls arriving at this switch board during a 4 minute period.
 - (2) At least two telephone calls arriving at this switch board during a given $\frac{1}{2}$ minute period.
6. A telephone exchange receives an average of 3 emergency calls in a 10-minute interval. Find the probability that there are at the most 4 emergency calls in a 10-minute interval.
7. A hunter hunts an animal at a Poisson rate of 1 per hour in a dense forest. If he starts hunting at 8:00 A.M., Find the probability that he hunts (a) one animal by 9.00 A.M., and (b) four animals by noon.
8. If customers arrive at a medical shop according to a Poisson process at a rate of 8 per hour, find the conditional probability that in 4 hours 20 customers arrived given that in 8 hours 60 customers arrived.
9. Suppose that IC chips fail in accordance with a Poisson distribution with the rate of one chip in about 4 weeks. If there are 3 spare IC chips on hand, and the next supply is not due in 6 weeks, what is the probability that the system will not be out of order in the next 6 weeks?
10. If the arrival of customers at a teller counter in a bank follows a Poisson process at a mean rate of 3 per minute, find the probability that the interval between two consecutive customer arrivals is (i) more than 2 minutes, (ii) between 2 and 5 minutes and (iii) less than or equal to 6 minutes.
11. Suppose that $\{X(t)\}$ is a normal process with $\mu(t) = 3$ and $C(t_1, t_2) = 4e^{-0.2|t_1 - t_2|}$. Find the probability that (i) $X(5) \leq 2$, and (ii) $|X(8) - X(5)| \leq 1$.

(Bharatidasan, April 1996)

12. Let $Y_n = \sum_{k=1}^n X_k$, where X_k 's are a set of independent random variables each normally distributed with mean μ and variance σ^2 . Show that $\{Y_n : n = 1, 2, \dots\}$ is a Gaussian process.

(Bharatidasan, Nov. 1996)

5.7 SINE WAVE PROCESS

Definition 5.35. A sine wave process is a random process of the form

$$X(t) = A \sin(\omega t + \theta)$$

where the amplitude A or frequency ω or phase θ or any combination of these may be random.

EXAMPLE 5.79. If $X(t) = A\cos(\omega t + \theta)$ is a random process, then $X(t)$ is a sine wave process because it can be represented as

$$X(t) = A\sin\left[\frac{\pi}{2} - (\omega t + \theta)\right] = A\sin(\omega_1 t + \theta_1)$$

with $\omega_1 = -\omega$ and $\theta_1 = \frac{\pi}{2} - \theta$.

EXAMPLE 5.80. Consider the sine wave process $X(t) = A\sin(\omega t + \theta)$, where A is a uniform random variable over $(0, 2\pi)$ and ω, θ are constants. Check whether the process $X(t)$ is wide-sense stationary or not.

Solution. It is easy to show that the mean of $X(t)$ is

$$\mu_X(t) = \pi \sin(\omega t + \theta)$$

which depends on t . This shows that $X(t)$ is not wide-sense stationary. ◆

EXAMPLE 5.81. Consider the sine wave process $X(t) = A\sin(\omega t + \theta)$, where ω is a uniform random variable over $(0, 2\pi)$ and A, θ are constants. Check whether the process $X(t)$ is wide-sense stationary or not.

Solution. It is easy to show that the mean of $X(t)$ is

$$\mu_X(t) = \frac{A}{\pi t} \sin(\pi t + \theta) \sin(\pi t)$$

which depends on t . This shows that $X(t)$ is not wide-sense stationary. ◆

EXAMPLE 5.82. Consider the sine wave process $X(t) = A\sin(\omega t + \theta)$, where θ is a uniform random variable over $(0, 2\pi)$ and A, ω are constants. Check whether the process $X(t)$ is wide-sense stationary or not.

Solution. It is easy to show that $\mu_X(t) = 0$, which is a constant, and $R_{XX}(\tau) = \frac{A^2}{2} \cos \omega \tau$, which is purely a function of τ . Thus, $X(t)$ is wide-sense stationary. ◆

5.8 BIRTH AND DEATH PROCESS

An important and useful class of Markov processes for analyzing queuing systems are *birth and death processes*. Several types of biological and industrial processes are also modelled as birth and death processes. The birth and death processes are designed after population models which gain new members through births and lose members through deaths. For example, the arrival and departure of flights in an airport can be modelled as a birth and death process. Here, the arrival of a flight can be taken as a *birth* and the departure of a flight can be taken as a *death*.

First, we discuss a special case of birth and death processes, namely pure birth processes.

5.8.1 Pure Birth Process

A pure birth process is a continuous-time Markov process $X(t)$ that takes discrete state values $0, 1, 2, \dots$ and increases by 1 at the discontinuity points t_i , which are called as *birth times* or *epochs*. The birth process $X(t)$ represents the size of a population of individuals at birth time t . So, a birth process $X(t)$ consists of a family of increasing staircase functions as shown in Figure 5.5.

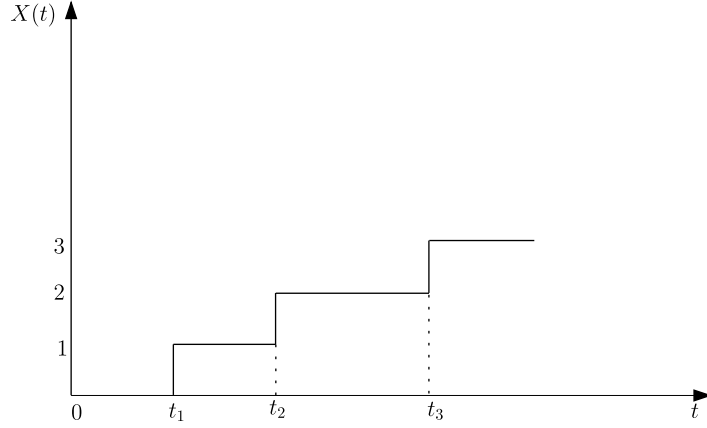


Figure 5.5: Pure Birth Process.

Definition 5.36. A Markov process $X(t)$ with state space $S = \{0, 1, 2, 3, \dots\}$ is called a *birth process* if the following condition is satisfied:

$$P\{X(t+h) - X(t) = k \mid X(t) = n\} = \begin{cases} \lambda_n h + o(h) & \text{if } k = 1, n \geq 0 \\ 1 - \lambda_n h + o(h) & \text{if } k = 0, n \geq 0 \\ o(h) & \text{if } k \geq 2 \end{cases} \quad (5.63)$$

where $\lambda_0, \lambda_1, \dots$ are positive constants called *birth rates* and

$$\lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$$

If $X(t)$ is the size of a population of individuals at time t , then the conditional probability given in Eq. (5.63) provides the probability of a birth during the time interval $(t, t + \Delta t)$. The state diagram for a pure birth process is as illustrated in Figure 5.6. When a birth occurs, the process goes from state n to state $n + 1$ at the birth rate λ_n .

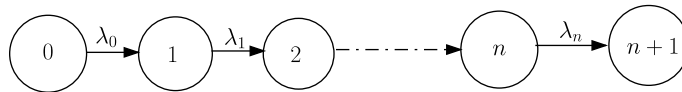


Figure 5.6: State Diagram for a Pure Birth Process.

Let $p_n(t) = P[X(t) = n]$. Then $p_n(t+h) = P[X(t+h) = n]$.

If $n \geq 1$, we note that the event $X(t+h) = n$ can materialize in two different ways:

- (a) At time t , $X(t) = n$ and no birth takes place in the interval $(t, t+h)$.

(b) At time t , $X(t) = n - 1$ and a birth takes place in the interval $(t, t + h)$.

Since the cases (a) and (b) are mutually exclusive, $p_n(t + h)$ is calculated as

$$\begin{aligned} p_n(t + h) &= P[X(t) = n] P[X(t + h) - X(t) = 0 | X(t) = n] \\ &\quad + P[X(t) = n - 1] P[X(t + h) - X(t) = 1 | X(t) = n - 1] \\ &= p_n(t) P[X(t + h) - X(t) = 0 | X(t) = n] + p_{n-1}(t) P[X(t + h) - X(t) = 1 | X(t) = n - 1] \end{aligned}$$

Using Eq. (5.63), we obtain

$$p_n(t + h) = p_n(t)[1 - \lambda_n h + o(h)] + p_{n-1}(t)[\lambda_{n-1} h + o(h)]$$

Thus, for $n \geq 1$, we have

$$\frac{p_n(t + h) - p_n(t)}{h} = -\lambda_n p_n(t) + \lambda_{n-1} p_{n-1}(t) + \frac{o(h)}{h}$$

Taking limits as $h \rightarrow 0$, we have

$$p'_n(t) = -\lambda_n p_n(t) + \lambda_{n-1} p_{n-1}(t) \quad (5.64)$$

which holds for $n \geq 1$.

If $n = 0$, then we have

$$p_0(t + h) = P[X(t + h) = 0] = P[X(t) = 0] P[X(t + h) - X(t) = 0 | X(t) = 0]$$

Using Eq. (5.63), we obtain

$$p_0(t + h) = p_0(t)[1 - \lambda_0 h + o(h)]$$

Therefore, we have

$$\frac{p_0(t + h) - p_0(t)}{h} = -\lambda_0 p_0(t) + \frac{o(h)}{h}$$

Taking limits as $h \rightarrow 0$, we have

$$p'_0(t) = -\lambda_0 p_0(t) \quad (5.65)$$

Equations (5.64) and (5.65) together describe the probabilities for a pure birth process.

5.8.2 Poisson Process

This is a special case of the pure birth process derived under the condition that the birth rate λ_n is a constant for all n , i.e.

$$\lambda_n = \lambda \quad \text{for all } n \geq 0$$

From Eqs. (5.64) and (5.65), we have

$$p'_0(t) = -\lambda p_0(t) \quad (5.66)$$

and, for $n \geq 1$

$$p'_n(t) = -\lambda p_n(t) + \lambda p_{n-1}(t) \quad (5.67)$$

We assume the initial conditions

$$p_0(0) = 1 \text{ and } p_n(0) = 0 \text{ for } n \geq 1$$

Next, we solve Eq. (5.66). Separating the variables, we have

$$\frac{dp_0}{p_0} = -\lambda dt$$

Integrating, we get

$$\ln p_0(t) = -\lambda t + \ln C$$

where C is an integration constant. Thus, we have

$$p_0(t) = Ce^{-\lambda t}$$

Since $p_0(0) = 1$, we have $C = 1$. Thus, it follows that

$$p_0(t) = e^{-\lambda t}$$

Next, from Eq. (5.67) with $n = 1$, we have

$$p_1'(t) = -\lambda p_1(t) + \lambda p_0(t) = -\lambda p_1(t) + \lambda e^{-\lambda t}$$

i.e.

$$p_1'(t) + \lambda p_1(t) = \lambda e^{-\lambda t} \quad (5.68)$$

The above differential equation is of the form

$$\frac{dy}{dt} + Py = Q$$

The integrating factor (I.F.) for this linear differential equation is given by

$$\text{I.F.} = e^{\int P dt} = e^{\int \lambda dt} = e^{\lambda t}$$

Thus, the solution of Eq. (5.68) is given by

$$(\text{I.F.}) p_1(t) = \int Q (\text{I.F.}) dt + C$$

i.e.

$$e^{\lambda t} p_1(t) = \int \lambda e^{-\lambda t} e^{\lambda t} dt + C = \lambda t + C$$

Since $p_1(0) = 0$, it follows that $C = 0$. Thus, we have

$$p_1(t) = e^{-\lambda t} t$$

We claim that

$$p_n(t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \text{ for } n = 0, 1, 2, \dots$$

This can be easily proved by mathematical induction. Clearly, the assertion holds for $n = 0, 1$.

Next, we suppose that the assertion holds for any $n = m$, i.e.

$$p_m(t) = e^{-\lambda t} \frac{(\lambda t)^m}{m!}$$

By Eq. (5.67), we have

$$p'_{m+1}(t) + \lambda p_{m+1}(t) = \lambda p_m(t) \quad (5.69)$$

The integrating factor for this equation is

$$\text{I.F.} = e^{\int \lambda dt} = e^{\lambda t}$$

Hence, the solution of Eq. (5.69) is given by

$$e^{\lambda t} p_{m+1}(t) = \int \lambda e^{-\lambda t} \frac{(\lambda t)^m}{m!} e^{\lambda t} dt + C = \frac{(\lambda t)^{m+1}}{(m+1)!} + C$$

Since $p_{m+1}(0) = 0$, it follows that $C = 0$.

Thus,

$$p_{m+1}(t) = e^{-\lambda t} \frac{(\lambda t)^{m+1}}{(m+1)!}$$

Hence, the assertion is also true for $n = m + 1$. This completes the inductive step of the proof. Thus, by the principle of mathematical induction, we have

$$p_n(t) = P[X(t) = n] = e^{-\lambda t} \frac{(\lambda t)^n}{n!} \quad \text{for } n = 0, 1, 2, \dots$$

which shows that $X(t)$ is a Poisson process.

Yule-Furry process: This is another a special case of the pure birth process derived under the condition that

$$\lambda_n = n\lambda \quad \text{for } n \geq 0 \quad (5.70)$$

Equation (5.70) may be interpreted as follows. If $X(t) = n$ corresponds to the existence of n individuals in the population in time t , then each individual, independent of others, has the same probability λh of producing one offspring in the time interval $(t, t + h)$. For this reason, Yule-Furry process is also called as *simple birth process*.

From Eqs. (5.64) and (5.65), we have

$$p'_0(t) = 0 \quad (5.71)$$

and, for $n \geq 1$

$$p'_n(t) = -n\lambda p_n(t) + (n-1)\lambda p_{n-1}(t) \quad (5.72)$$

We assume the initial conditions

$$p_0(0) = 0, p_1(0) = 1 \quad \text{and} \quad p_n(0) = 0 \quad \text{for } n \geq 1$$

Solving Eq. (5.71) with $p_0(0) = 0$, we obtain

$$p_0(t) = 0$$

Putting $n = 1$ in Eq. (5.72), we obtain

$$p_1'(t) = -\lambda p_1(t)$$

solving which we get

$$p_1'(t) = Ce^{-\lambda t}$$

where C is an integration constant.

Since $p_1(0) = 1$, we must have $C = 1$. Thus, we have

$$p_1(t) = e^{-\lambda t}$$

Putting $n = 2$ in Eq. (5.72), we obtain

$$p_2'(t) = -2\lambda p_2(t) + \lambda p_1(t)$$

i.e.

$$p_2'(t) + 2\lambda p_2(t) = \lambda p_1(t) = \lambda e^{-\lambda t} \quad (5.73)$$

This differential equation is of the form

$$\frac{dy}{dt} + Py = Q$$

The integrating factor for this equation is given by

$$\text{I.F.} = e^{\int P dt} = e^{\int 2\lambda dt} = e^{2\lambda t}$$

Thus, the solution of Eq. (5.73) is given by

$$(\text{I.F.}) p_2(t) = \int p_2(t)(\text{I.F.})dt + C$$

i.e.

$$e^{2\lambda t} p_2(t) = \int \lambda e^{-\lambda t} e^{2\lambda t} dt + C = e^{\lambda t} + C$$

where C is an integration constant.

Since $p_2(0) = 0$, we have

$$0 = 1 + C \text{ or } C = -1$$

Thus, it follows that

$$p_2(t) = e^{-2\lambda t} (e^{\lambda t} - 1) = e^{-\lambda t} (1 - e^{-\lambda t})$$

Similarly, it can be shown that

$$p_3(t) = e^{-\lambda t} (1 - e^{-\lambda t})^2$$

$$p_4(t) = e^{-\lambda t} (1 - e^{-\lambda t})^3$$

and so on.

Using the principle of mathematical induction, it can be easily established that

$$p_n(t) = e^{-\lambda t} \left(1 - e^{-\lambda t}\right)^{n-1} \text{ for } n \geq 1$$

The mean of the Yule-Furry process is calculated as

$$\mu_X(t) = E[X(t)] = e^{\lambda t}$$

and the variance of the Yule-Furry process is calculated as

$$\sigma_X^2(t) = \text{Var}[X(t)] = e^{\lambda t} (e^{\lambda t} - 1)$$

5.8.3 Birth and Death Process

Definition 5.37. A continuous-time Markov chain $X(t)$ with state space $S = \{0, 1, 2, \dots\}$ is called a *birth and death process* if the following condition is satisfied:

$$P[X(t+h) - X(t) = k | X(t) = n] = \begin{cases} \lambda_n h + o(h) & \text{if } k = 1, n \geq 0 \\ \mu_n h + o(h) & \text{if } k = -1, n \geq 1 \\ 1 - (\lambda_n + \mu_n)h + o(h) & \text{if } k = 0, n \geq 1 \\ 0 & \text{otherwise} \end{cases} \quad (5.74)$$

where $\lambda_0, \lambda_1, \lambda_2, \dots$ are positive constants called *birth rates*, $\mu_1, \mu_2, \dots$ are positive constants called *death rates* and

$$\lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$$

If $X(t)$ is the size of a population of individuals at time t , then the conditional probability given in Eq. (5.74) provides the probability that the number of individuals at time $t+h$ is $n+k$ given that the number of individuals at time t is n . The state diagram for a birth and death process is as illustrated in Figure 5.7. When a birth occurs, the process goes from state n to state $n+1$ at the birth rate λ_n and when a death occurs, the process goes from state $n+1$ to state n at the death rate μ_{n+1} .

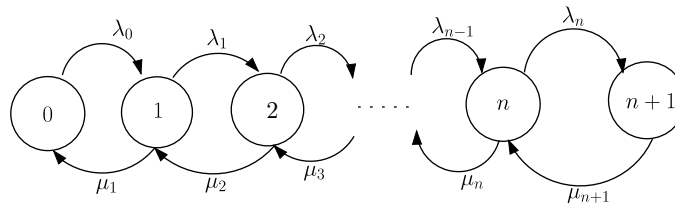


Figure 5.7: State Diagram for a Birth and Death Process.

Let $p_n(t) = P[X(t) = n]$. Then $p_n(t)$ is the probability that the number of individuals at time t is equal to n .

Next, for $n \geq 1$, we calculate $p_n(t+h) = P[X(t+h) = n]$, i.e. the probability that the number of individuals at time $t+h$ is equal to n . For this purpose, we divide the interval $(0, t+h)$ into two subintervals $(0, t)$ and $[t, t+h)$. Then the event $\{X(t+h) = n, n \geq 1\}$ can occur in a number of mutually exclusive ways. These would include events involving more than one birth and/or more than one death in $(t, t+h)$. By Definition 5.37, the probability of such an event is $o(h)$ and we know that

$$\lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$$

Thus, it suffices to consider the following four events:

A_{ij} : $(n-i+j)$ individuals in $(0, t)$, i birth and j death in $[t, t+h)$, where $i, j = 0, 1$

Next, we find the probabilities of the four events A_{ij} , where $i, j = 0, 1$

Since A_{00} represents the event that there are n individuals in $(0, t)$, 0 birth and 0 death in $[t, t+h)$, it follows from Eq. (5.74) that

$$\begin{aligned} P(A_{00}) &= p_n(t) [1 - \lambda_n h + o(h)] [1 - \mu_n h + o(h)] \\ &= p_n(t) [1 - (\lambda_n + \mu_n)h] + o(h) \end{aligned}$$

Since A_{10} represents the event that there are $(n-1)$ individuals in $(0, t)$, 1 birth and 0 death in $[t, t+h)$, it follows from Eq. (5.74) that

$$\begin{aligned} P(A_{10}) &= p_{n-1}(t) [\lambda_{n-1} h + o(h)] [1 - \mu_{n-1} h + o(h)] \\ &= p_{n-1}(t) \lambda_{n-1} h + o(h) \end{aligned}$$

Since A_{01} represents the event that there are $n+1$ individuals in $(0, t)$, 0 birth and 1 death in $[t, t+h)$, it follows from Eq. (5.74) that

$$\begin{aligned} P(A_{01}) &= p_{n+1}(t) [1 - \lambda_{n+1} h + o(h)] [\mu_{n+1} h + o(h)] \\ &= p_{n+1}(t) \mu_{n+1} h + o(h) \end{aligned}$$

Since A_{11} represents the event that there are n individuals in $(0, t)$, 1 birth and 1 death in $[t, t+h)$, it follows from Eq. (5.74) that

$$P(A_{11}) = p_n(t) [\lambda_n h + o(h)] [\mu_n h + o(h)] = o(h)$$

Since the events A_{00}, A_{10}, A_{01} and A_{11} are mutually exclusive, it follows that

$$p_n(t+h) = P(A_{00}) + P(A_{10}) + P(A_{01}) + P(A_{11})$$

Thus

$$p_n(t+h) = p_n(t)[1 - (\lambda_n + \mu_n)h] + p_{n-1}(t)\lambda_{n-1}h + p_{n+1}(t)\mu_{n+1}h + o(h)$$

Hence, we have

$$\frac{p_n(t+h) - p_n(t)}{h} = -(\lambda_n + \mu_n)p_n(t) + \lambda_{n-1}p_{n-1}(t) + \mu_{n+1}p_{n+1}(t) + \frac{o(h)}{h}$$

Taking limits as $h \rightarrow 0$, we obtain

$$p'_n(t) = -(\lambda_n + \mu_n)p_n(t) + \lambda_{n-1}p_{n-1}(t) + \mu_{n+1}p_{n+1}(t) \text{ for } n \geq 1 \quad (5.75)$$

If $n = 0$, then we have

$$\begin{aligned} p_0(t+h) &= p_0(t)[1 - \lambda_0 h + o(h)] + p_1(t)[1 - \lambda_0 h + o(h)][\mu_1 h + o(h)] \\ &= p_0(t)[1 - \lambda_0 h] + p_1(t)\mu_1 h + o(h) \end{aligned}$$

Thus, it follows that

$$\frac{p_0(t+h) - p_0(t)}{h} = -\lambda_0 p_0(t) + \mu_1 p_1(t) + \frac{o(h)}{h}$$

Taking limits as $h \rightarrow 0$, we obtain

$$p'_0(t) = -\lambda_0 p_0(t) + \mu_1 p_1(t) \quad (5.76)$$

If at time $t = 0$ there were i individuals, where $i \geq 0$, then the initial condition for the birth and death process is

$$p_n(0) = 0 \text{ for } n \neq i \text{ and } p_i(0) = 1$$

The differential Eqs. (5.75) and (5.76) are called the *balance equations* of the birth and death process.

Remark 5.12. Some important applications of the birth and death processes are on the following:

- (i) Queuing Theory
- (ii) Reliability Theory
- (iii) Biology
- (iv) Ecology
- (v) Economics

EXAMPLE 5.83. Draw the state diagram of a birth-death process and obtain the balance equations. Hence, find the limiting distribution of the process.

(Anna, May 2006)

Solution. As we have already discussed the state diagram of the birth-death process and derived the balance equations, we find the limiting distribution of the birth-death process.

Suppose that as $t \rightarrow \infty$, $p_n(t) \rightarrow p_n$ (limit) for $n \geq 1$ and $p_0(t) \rightarrow p_0$ (limit). Then it is immediate that for large values of t

$$p'_n(t) = 0 \text{ for } n \geq 1 \text{ and } p'_0(t) = 0$$

From $p'_0(t) = 0$, it follows by Eq. (5.76) that

$$-\lambda_0 p_0 + \mu_1 p_1 = 0 \Rightarrow p_1 = \frac{\lambda_0}{\mu_1} p_0$$

From $p'_1(t) = 0$, it follows by Eq. (5.75) that

$$-(\lambda_1 + \mu_1)p_1 + \lambda_0 p_0 + \mu_2 p_2 = 0 \Rightarrow p_2 = \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} p_0$$

We use the second principle of mathematical induction to show that

$$p_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} p_0 \quad (5.77)$$

where n is any positive integer.

Clearly, Eq. (5.77) is true for $n = 1$ and $n = 2$.

Suppose that Eq. (5.77) is true for $n \leq m$, where m is any positive integer. Taking $n = m$ in Eq. (5.75) and using the assumptions that $p_n(t) \rightarrow p_n$ for large t and all n , we have

$$-(\lambda_m + \mu_m)p_m + \lambda_{m-1}p_{m-1} + \mu_{m+1}p_{m+1} = 0$$

which gives

$$p_{m+1} = \frac{\lambda_m + \mu_m}{\mu_{m+1}} p_m - \frac{\lambda_{m-1}}{\mu_{m+1}} p_{m-1}$$

Using the induction hypothesis, we have

$$\begin{aligned} p_{m+1} &= \left[\frac{\lambda_m + \mu_m}{\mu_{m+1}} \right] \frac{\lambda_0 \lambda_1 \cdots \lambda_{m-1}}{\mu_1 \mu_2 \cdots \mu_m} p_0 - \left[\frac{\lambda_{m-1}}{\mu_{m+1}} \right] \frac{\lambda_0 \lambda_1 \cdots \lambda_{m-2}}{\mu_1 \mu_2 \cdots \mu_{m-1}} p_0 \\ &= \frac{\lambda_0 \lambda_1 \cdots \lambda_m}{\mu_1 \mu_2 \cdots \mu_{m+1}} p_0 \end{aligned}$$

which shows that Eq. (5.77) is also true for $n = m + 1$.

Hence, by the second principle of mathematical induction, it follows that Eq. (5.77) is true for all positive integers n .

We also note that the probability p_0 is determined by the fact that

$$p_0 + p_1 + \cdots + p_n + \cdots = 1$$

which gives

$$p_0 = \frac{1}{1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} + \cdots} = \frac{1}{1 + \sum_{k=1}^{\infty} \prod_{i=0}^{k-1} \frac{\lambda_i}{\mu_{i+1}}} \quad (5.78)$$

From the Eqs. (5.77) and (5.78), we have

$$p_n = \frac{\prod_{i=0}^{n-1} \frac{\lambda_i}{\mu_{i+1}}}{1 + \sum_{k=1}^{\infty} \prod_{i=0}^{k-1} \frac{\lambda_i}{\mu_{i+1}}}$$

◆

Chapter 6

Spectral Analysis of Random Processes

6.1 AUTOCORRELATION AND CROSS-CORRELATION FUNCTIONS

In this section, we review the definition and properties of autocorrelation and cross-correlation functions of random processes studied in Section 5.3.

We recall that the *autocorrelation function* of a random process $X(t)$ is defined as

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

If $X(t)$ is at least wide-sense stationary (WSS), then we know by Definition 5.5 that $R_{XX}(t, t + \tau) = E[X(t)X(t + \tau)]$ is a function only of the time difference, τ . Thus, for a WSS process, we can represent the autocorrelation function simply as

$$R_{XX}(\tau) = E[X(t)X(t + \tau)]$$

Some important properties of the autocorrelation function $R_{XX}(\tau)$ are summarized Theorem 6.1.

Theorem 6.1. Let $X(t)$ be a WSS process with constant mean μ_X and autocorrelation function $R_{XX}(\tau)$. Then the following properties hold:

- (a) $R_{XX}(\tau)$ is an even function of τ , i.e. $R_{XX}(-\tau) = R_{XX}(\tau)$.
- (b) $|R_{XX}(\tau)| \leq R_{XX}(0)$, i.e. the maximum value of $|R_{XX}(\tau)|$ occurs at $\tau = 0$.
- (c) $R_{XX}(0) = E[X^2(t)]$.
- (d) If $Y(t) = X(t) + a$, where a is a constant, then

$$E[Y(t)] = \mu_X + a \text{ and } R_{YY}(\tau) = C_{XX}(\tau) + (\mu_X + a)^2$$

where $C_{XX}(\tau) = R_{XX}(\tau) - \mu_X^2$ is the *autocovariance function* of $X(t)$.

- (e) If $R_{XX}(\tau)$ is continuous at $\tau = 0$, then $R_{XX}(\tau)$ is continuous for all τ .

- (f) If there exists a constant $T > 0$ such that $R_{XX}(T) = R_{XX}(0)$, then $R_{XX}(\tau)$ is a periodic function and $X(t)$ is called a *periodic WSS process*.
- (g) If $X(t)$ has no periodic component, then

$$\lim_{\tau \rightarrow \infty} R_{XX}(\tau) = \mu_X^2$$

provided the limit exists.

Remark 6.1. If $X(t)$ is a voltage waveform across a $1\text{-}\Omega$ resistor, then $E[X^2(t)]$ is the average value of power delivered to the $1\text{-}\Omega$ resistor by the process $X(t)$. For this reason, $E[X^2(t)]$ is often called the *average power* of the random process $X(t)$. For a WSS process $X(t)$, the average power is given by property (c) of Theorem 6.1 as

$$R_{XX}(0) = E[X^2(t)]$$

We recall that the *cross-correlation function* of two random processes $X(t)$ and $Y(t)$ is defined as

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$$

If $X(t)$ and $Y(t)$ are at least jointly wide-sense stationary, then we know by Definition 5.6 that $R_{XY}(t, t + \tau) = E[X(t)Y(t + \tau)]$ is a function only of the time difference, τ . Thus, for jointly WSS processes, we can represent their cross-correlation function simply as

$$R_{XY}(\tau) = E[X(t)Y(t + \tau)]$$

Next, we summarize some important properties of the cross-correlation function $R_{XY}(\tau)$ in the following theorem.

Theorem 6.2. Let $X(t)$ and $Y(t)$ be jointly WSS processes with constant means μ_X and μ_Y respectively, and cross-correlation function $R_{XY}(\tau)$. Then the following properties hold:

- (a) $R_{XY}(-\tau) = R_{YX}(\tau)$.
- (b) $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$.
- (c) $|R_{XY}(\tau)| \leq \frac{1}{2}[R_{XX}(0) + R_{YY}(0)]$.
- (d) If $X(t)$ and $Y(t)$ are *orthogonal* processes, then $R_{XY}(\tau) = 0$.
- (e) If $X(t)$ and $Y(t)$ are *statistically independent*, then $R_{XY}(\tau) = \mu_X \mu_Y$.
- (f) If $Z(t) = X(t) + Y(t)$, then

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau) + R_{XY}(\tau) + R_{YX}(\tau)$$

If $X(t)$ and $Y(t)$ are orthogonal, then

$$R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau)$$

EXAMPLE 6.1. Check whether the following functions can be autocorrelation functions for a WSS random process:

- (a) $f(\tau) = 2 \sin a\tau$.
- (b) $g(\tau) = \frac{1}{1+2\tau^2}$.

Solution. (a) If $f(\tau)$ is the autocorrelation function $R_{XX}(\tau)$ for a WSS random process $X(t)$, then it must satisfy all the properties for $R_{XX}(\tau)$ stated in Theorem 6.1. In particular, it must satisfy property (a), namely $R_{XX}(-\tau) = R_{XX}(\tau)$. But we find that

$$f(-\tau) = 2 \sin a(-\tau) = -2 \sin a\tau = -f(\tau)$$

which shows that $f(\tau)$ cannot be the autocorrelation function for a WSS random process.

(b) If $g(\tau)$ is the autocorrelation function $R_{XX}(\tau)$ for a WSS random process $X(t)$, then it must satisfy all the properties for $R_{XX}(\tau)$ stated in Theorem 6.1. It is easy to see that

$$g(-\tau) = \frac{1}{1+2(-\tau)^2} = \frac{1}{1+2\tau^2} = g(\tau)$$

which shows that $g(\tau)$ is an even function. We also find that

$$0 < g(\tau) = \frac{1}{1+2\tau^2} < 1 \text{ for all } \tau$$

and $\lim_{\tau \rightarrow \infty} g(\tau) = 0 = g(0)$. Thus, the maximum value of $g(\tau)$ is attained at $\tau = 0$.

Thus, we conclude that $g(\tau)$ is a valid autocorrelation function for a random process $X(t)$. ♦

EXAMPLE 6.2. The autocorrelation function for a stationary process $X(t)$ is given by

$$R_{XX}(\tau) = 9 + 2e^{-|\tau|}$$

Find the mean of the random variable $Y = \int_0^2 X(t)dt$ and the variance of $X(t)$.

(Anna, April 2003)

Solution. First, we find the mean of $X(t)$. We know that

$$\mu_X^2 = \lim_{\tau \rightarrow \infty} R_{XX}(\tau) = \lim_{\tau \rightarrow \infty} [9 + 2e^{-|\tau|}] = 9$$

Thus, the mean of $X(t)$ is $\mu_X = \sqrt{9} = 3$.

Next, the average power of $X(t)$ is given by

$$E[X^2(t)] = R_{XX}(0) = 9 + 2 = 11$$

Thus, the variance of $X(t)$ is given by

$$\text{Var}[X(t)] = \sigma_X^2 = E[X^2(t)] - \mu_X^2 = 11 - 9 = 2$$

Next, we find the mean of Y as

$$E[Y] = E\left[\int_0^2 X(t)dt\right] = \int_0^2 E[X(t)] dt = \int_0^2 3 dt = 6$$

♦

EXAMPLE 6.3. If $X(t)$ is a random process defined as $X(t) = A$ for $0 \leq t \leq 1$, where A is a random variable uniformly distributed over $(-\theta, \theta)$, prove that the autocorrelation function of $X(t)$ is $R_{XX}(\tau) = \frac{\theta^2}{3}$.

Solution. Since A is a uniform random variable over $(-\theta, \theta)$, it has the probability density function

$$f_A(a) = \begin{cases} \frac{1}{2\theta} & \text{for } -\theta < a < \theta \\ 0 & \text{elsewhere} \end{cases}$$

Thus, the autocorrelation function of $X(t)$ is given by

$$R_{XX}(\tau) = E[X(t)X(t+\tau)] = E[A \cdot A] = E[A^2] = \int_{a=-\theta}^{\theta} a^2 \left[\frac{1}{2\theta} \right] da$$

Integrating, we have

$$R_{XX}(\tau) = \frac{1}{2\theta} \left[\frac{a^3}{3} \right]_{a=-\theta}^{\theta} = \frac{1}{2\theta} \times \frac{2\theta^3}{3} = \frac{\theta^2}{3}$$

◆

EXAMPLE 6.4. Two random processes $X(t)$ and $Y(t)$ are defined by

$$X(t) = A \cos(\omega t + \theta) \quad \text{and} \quad Y(t) = A \sin(\omega t + \theta)$$

where A and ω are constants and θ is a uniform random variable over $(-\pi, \pi)$.

- Show that $X(t)$ and $Y(t)$ are individually wide-sense stationary processes.
- Find the cross-correlation function $R_{XY}(\tau)$ of $X(t)$ and $Y(t)$, and show that the processes are jointly wide-sense stationary.
- Verify any two properties of $R_{XY}(\tau)$.

Solution. Since θ is a uniform random variable over $(-\pi, \pi)$, it has the probability density function

$$f(\theta) = \begin{cases} \frac{1}{2\pi} & \text{if } -\pi < \theta < \pi \\ 0 & \text{otherwise} \end{cases}$$

- It is easy to show that $\mu_X(t) = 0$, which is a constant, and $R_{XX}(\tau) = \frac{A^2}{2} \cos \omega \tau$, which is purely a function of τ . Thus, $X(t)$ is a wide-sense stationary process. Similarly, it is easy to show that $\mu_Y(t) = 0$, which is a constant and $R_{YY}(\tau) = \frac{A^2}{2} \cos \omega \tau$, which is purely a function of τ . Thus, $Y(t)$ also is a wide-sense stationary process.
- It is easy to show that the cross-correlation function between X and Y is

$$R_{XY}(\tau) = \frac{A^2}{2} \sin \omega \tau$$

which is purely a function of τ . As $X(t)$ and $Y(t)$ are also individually wide-sense stationary processes, we conclude that they are jointly wide-sense stationary.

- (c) We shall verify the following properties of $R_{XY}(\tau)$, viz. (i) $R_{XY}(-\tau) = R_{YX}(\tau)$ and (ii) $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$

(i) We find that

$$R_{XY}(-\tau) = \frac{A^2}{2} \sin \omega(-\tau) = -\frac{A^2}{2} \sin \omega\tau = R_{YX}(\tau)$$

(ii) We find that

$$|R_{XY}(\tau)| = \left| \frac{A^2}{2} \sin \omega\tau \right| \leq \frac{A^2}{2}$$

since $|\sin u| \leq 1$ for all values of u .

We also find that

$$\sqrt{R_{XX}(0)R_{YY}(0)} = \sqrt{\frac{A^2}{2} \cdot \frac{A^2}{2}} = \frac{A^2}{2}$$

Thus, it is immediate that

$$|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$$

◆

PROBLEM SET 6.1

1. Check whether the following functions are valid autocorrelation functions:

- (a) $f(\tau) = A \cos(\omega\tau)$, where A and ω are positive constants.
- (b) $g(\tau) = A \sin(\omega\tau)$, where A and ω are positive constants.
- (c) $h(\tau) = 1 + \frac{|\tau|}{T}$, where T is a positive constant.

2. If $X(t)$ is a WSS process with autocorrelation function

$$R_{XX}(\tau) = 4e^{-2|\tau|} + \frac{9}{4\tau^2 + 3}$$

find the mean and variance of $Y = X(4) - X(2)$.

3. Let $X(t)$ and $Y(t)$ be two independent stationary processes with autocorrelation functions

$$R_{XX}(\tau) = 5e^{-|\tau|} \quad \text{and} \quad R_{YY}(\tau) = \frac{4\tau^2 + 27}{2\tau^2 + 3}$$

Let $Z(t) = 4X(t)Y(t)$. Find the mean and variance of $X(t)$, $Y(t)$ and $Z(t)$.

4. The autocorrelation function of a stationary process $X(t)$ is given by

$$R_{XX}(\tau) = 16 + 3e^{-|\tau|}$$

Find the mean and variance of $S = \int_0^1 X(t) dt$.

5. Two random processes $X(t)$ and $Y(t)$ are defined by

$$X(t) = A \cos(\omega t + \theta) \quad \text{and} \quad Y(t) = A \cos(\omega t + \theta + \phi)$$

where A , ω and ϕ are constants and θ is a random variable uniformly distributed in $(0, 2\pi)$.

- (a) Show that $X(t)$ and $Y(t)$ are individually wide-sense stationary processes.
- (b) Find the cross-correlation function $R_{XY}(\tau)$ and show that $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes.
- (c) For what values of ϕ are $X(t)$ and $Y(t)$ orthogonal?

6.2 POWER SPECTRAL DENSITY AND CROSS-SPECTRAL DENSITY FUNCTIONS

In the study of deterministic signals and systems, frequency domain techniques (using Fourier transforms, etc.) are very useful to gain significant insights into a variety of problems in applications. In this section, we develop frequency domain tools for studying random processes. The results developed in this section will be useful for the study of random processes in linear systems in the next section.

The spectral properties of a deterministic signal $x(t)$ are contained in its Fourier transform $X(\omega)$ given by

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$$

The function $X(\omega)$ in the frequency domain is sometimes simply called the *spectrum* of $x(t)$. If $X(\omega)$ is known, then $x(t)$ can be recovered by taking inverse Fourier transform of $X(\omega)$, i.e.

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$$

Hence, it is clear that $X(\omega)$ provides a complete description of $x(t)$ and vice versa.

Next, we study the spectral properties of random processes.

Definition 6.1. The *power spectral density* or *power spectrum* $S_{XX}(\omega)$ of a continuous-time random process $X(t)$ is defined as the Fourier transform of $R_{XX}(\tau)$:

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau \quad (6.1)$$

Taking the inverse Fourier transform of $S_{XX}(\omega)$, we obtain

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{i\omega\tau} d\omega \quad (6.2)$$

Equations (6.1) and (6.2) are called the *Wiener-Khinchin relations*.

Theorem 6.3. If $X(t)$ is a continuous-time wide-sense stationary random process (real or complex) with power spectral density function $S_{XX}(\omega)$, then the following properties hold:

- (a) The power spectral density at zero frequency gives the area under the graph of the autocorrelation function, i.e.

$$S_{XX}(0) = \int_{-\infty}^{\infty} R_{XX}(\tau) d\tau$$

- (b) $S_{XX}(\omega)$ is real and $S_{XX}(\omega) \geq 0$.

- (c) $S_{XX}(\omega)$ is an even function of ω , i.e. $S_{XX}(-\omega) = S_{XX}(\omega)$.

(d) $E[X^2(t)] = R_{XX}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega.$

(e) If $X(t)$ is a real WSS process, then the spectral density function $S_{XX}(\omega)$ and the autocorrelation function $R_{XX}(\tau)$ form a Fourier cosine transform pair.

Proof. (a) By definition, we have

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau$$

Substituting $\omega = 0$, we have

$$S_{XX}(0) = \int_{-\infty}^{\infty} R_{XX}(\tau) d\tau$$

(b) By definition, the autocorrelation function of a real or complex random process $X(t)$ is given by

$$R_{XX}(\tau) = E[X(t)X^*(t+\tau)]$$

Thus, we have

$$R_{XX}^*(\tau) = E[X^*(t)X(t+\tau)] = E[X(t+\tau)X^*(t)]$$

It follows that

$$R_{XX}^*(\tau) = R_{XX}(-\tau) \quad (6.3)$$

Now, by definition, we have

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau$$

Hence,

$$S_{XX}^*(\omega) = \int_{-\infty}^{\infty} R_{XX}^*(\tau) e^{i\omega\tau} d\tau$$

Using Eq. (6.3), the above equation becomes

$$S_{XX}^*(\omega) = \int_{-\infty}^{\infty} R_{XX}(-\tau) e^{i\omega\tau} d\tau$$

Putting $u = -\tau$ in the integral on the R.H.S. of the above equation, we get

$$S_{XX}^*(\omega) = \int_{-\infty}^{\infty} R_{XX}(u) e^{-i\omega u} du = S_{XX}(\omega)$$

Hence, it is immediate that $S_{XX}(\omega)$ is a real function of ω . (It will be established later that $S_{XX}(\omega) \geq 0$.)

(c) By definition, we have

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau$$

Putting $-\omega$ in place of ω , we get

$$S_{XX}(-\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{i\omega\tau} d\tau$$

Substituting $u = -\tau$ in the integral on the R.H.S. of the above equation, we get

$$S_{XX}(-\omega) = \int_{-\infty}^{\infty} R_{XX}(-u) e^{-i\omega u} du$$

Since $X(t)$ is a WSS process, $R_{XX}(u)$ is an even function of u [by Theorem 6.1 (a)]. Thus, $R_{XX}(-u) = R_{XX}(u)$ and we have

$$S_{XX}(-\omega) = \int_{-\infty}^{\infty} R_{XX}(u) e^{-i\omega u} du = S_{XX}(\omega)$$

Hence, $S_{XX}(\omega)$ is an even function of ω .

(d) Putting $\tau = 0$ in the second Wiener-Khinchin relation, Eq. (6.2), we have

$$R_{XX}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega \quad (6.4)$$

By Theorem 6.1 (c), we know that

$$E[X^2(t)] = R_{XX}(0) \quad (6.5)$$

Combining Eqs. (6.4) and (6.5), we have

$$E[X^2(t)] = R_{XX}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) d\omega$$

(e) Suppose that $X(t)$ is a real WSS process. Then $R_{XX}(\tau) = E[X(t)X(t+\tau)]$ is a real function depending only on the time difference τ . Then, we have

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau = \int_{-\infty}^{\infty} R_{XX}(\tau) [\cos \omega\tau - i \sin \omega\tau] d\tau \quad (6.6)$$

By Theorem 6.1 (a), we know that $R_{XX}(\tau)$ is an even function of τ . Thus, $R_{XX}(\tau) \cos \omega\tau$ is an even function of τ and $R_{XX}(\tau) \sin \omega\tau$ is an odd function of τ .

Thus, Eq. (6.6) simplifies to

$$S_{XX}(\omega) = 2 \int_{-\infty}^{\infty} R_{XX}(\tau) \cos \omega \tau d\tau$$

which is the Fourier cosine transform of $[2R_{XX}(\tau)]$.

Similarly, we have

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) [\cos \omega \tau + i \sin \omega \tau] d\omega \quad (6.7)$$

Since $S_{XX}(\omega)$ is an even function of ω [by property (c)], it is immediate that $S_{XX}(\omega) \cos \omega \tau$ is an even function of ω and $S_{XX}(\omega) \sin \omega \tau$ is an odd function of ω .

Hence, Eq. (6.7) simplifies to

$$R_{XX}(\tau) = \frac{1}{\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) \cos \omega \tau d\omega$$

which is the Fourier inverse cosine transform of $[\frac{1}{2} S_{XX}(\omega)]$.

Hence, we conclude that the spectral power density and the autocorrelation function form a Fourier cosine transform pair.

This completes the proof. ◆

Next, we establish an important theorem, known as *Wiener-Khinchin Theorem*, which provides an useful formula for computing the power spectrum of a random process.

Theorem 6.4. (Wiener-Khinchin Theorem) If $X(t)$ is a real wide-sense stationary random process with autocorrelation function $R_{XX}(\tau)$ and power spectral density function $S_{XX}(\omega)$, and if $X_T(\omega)$ is the Fourier transform of the truncated random process $X_T(t)$ defined as

$$X_T(t) = \begin{cases} X(t) & \text{for } |t| \leq T \\ 0 & \text{for } |t| > T \end{cases}$$

then

$$S_{XX}(\omega) = \lim_{T \rightarrow \infty} \left[\frac{1}{2T} E \{ |X_T(\omega)|^2 \} \right]$$

(Anna, May 2006; Nov. 2006; Nov. 2007)

Proof. Since $X_T(\omega)$ is the Fourier transform of $X_T(t)$, we have

$$X_T(\omega) = \int_{-\infty}^{\infty} X_T(t) e^{-i\omega t} dt \quad (6.8)$$

Using the definition of $X_T(t)$, we can simplify Eq. (6.8) as

$$X_T(\omega) = \int_{-T}^T X(t) e^{-i\omega t} dt \quad (6.9)$$

Since $X(t)$ is real, we find that

$$X_T^*(\omega) = \int_{-T}^T X(t) e^{i\omega t} dt = X_T(-\omega)$$

and thus it follows that

$$|X_T(\omega)|^2 = X_T(\omega)X_T^*(\omega) = X_T(\omega)X_T(-\omega)$$

Using Eq. (6.9), we have

$$\begin{aligned} |X_T(\omega)|^2 &= \left[\int_{t_1=-T}^T X(t_1) e^{-i\omega t_1} dt_1 \right] \left[\int_{t_2=-T}^T X(t_2) e^{i\omega t_2} dt_2 \right] \\ &= \int_{t_2=-T}^T \int_{t_1=-T}^T X(t_1)X(t_2) e^{-i\omega(t_1-t_2)} dt_1 dt_2 \end{aligned} \quad (6.10)$$

Taking expectation on both sides of Eq. (6.10) and dividing by $\frac{1}{2T}$, we have

$$\frac{1}{2T} E \{ |X_T(\omega)|^2 \} = \frac{1}{2T} \int_{t_2=-T}^T \int_{t_1=-T}^T E[X(t_1)X(t_2)] e^{-i\omega(t_1-t_2)} dt_1 dt_2 \quad (6.11)$$

We use the substitution

$$t_1 = t + \tau \text{ and } t_2 = t$$

in the integral on the R.H.S. of Eq. (6.11).

The Jacobian of the above change of variables is given by

$$J = \frac{\partial(t_1, t_2)}{\partial(t, \tau)} = \begin{vmatrix} \frac{\partial t_1}{\partial t} & \frac{\partial t_1}{\partial \tau} \\ \frac{\partial t_2}{\partial t} & \frac{\partial t_2}{\partial \tau} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

Thus, Eq. (6.11) becomes

$$\frac{1}{2T} E \{ |X_T(\omega)|^2 \} = \frac{1}{2T} \int_{t=-T}^T \int_{\tau=-T-t}^{T-t} E[X(t+\tau)X(t)] e^{-i\omega\tau} |J| dt d\tau \quad (6.12)$$

Since $X(t)$ is a real WSS process, $R_{XX}(\tau) = E[X(t)X(t+\tau)]$, which is purely a function of τ . Thus, Eq. (6.12) simplifies to

$$\frac{1}{2T} E \{ |X_T(\omega)|^2 \} = \frac{1}{2T} \int_{t=-T}^T \int_{\tau=-T-t}^{T-t} R_{XX}(\tau) e^{-i\omega\tau} dt d\tau \quad (6.13)$$

Taking limits as $T \rightarrow \infty$ in Eq. (6.13), we have

$$\lim_{T \rightarrow \infty} \left[\frac{1}{2T} E \{ |X_T(\omega)|^2 \} \right] = \int_{\tau=-\infty}^{\infty} \left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{t=-T}^T R_{XX}(\tau) dt \right] e^{-i\omega\tau} d\tau \quad (6.14)$$

Since $R_{XX}(\tau)$ is purely a function of τ , it follows that

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{t=-T}^T R_{XX}(\tau) dt = \lim_{T \rightarrow \infty} R_{XX}(\tau) \frac{1}{2T} (2T) = R_{XX}(\tau) \quad (6.15)$$

Substituting Eq. (6.15) into Eq. (6.14), we have

$$\lim_{T \rightarrow \infty} \left[\frac{1}{2T} E \{ |X_T(\omega)|^2 \} \right] = \int_{\tau=-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} d\tau = S_{XX}(\omega)$$

This completes the proof. ◆

Next, we define the *cross power spectral density* of two random processes $X(t)$ and $Y(t)$.

Definition 6.2. The *cross power spectral density* or *cross power spectrum* $S_{XY}(\omega)$ of two continuous-time random processes $X(t)$ and $Y(t)$ is defined as the Fourier transform of $R_{XY}(\tau)$:

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau$$

Hence, taking the inverse Fourier transform of $S_{XY}(\omega)$, we get

$$R_{XY}(\tau) = \frac{1}{2\pi} S_{XY}(\omega) e^{i\omega\tau} d\omega$$

We remark that unlike the power spectral density $S_{XX}(\omega)$ of a random process $X(t)$, the cross power spectral density $S_{XY}(\omega)$ of two random processes $X(t)$ and $Y(t)$ will be, in general, a complex-valued function.

Theorem 6.5. If $S_{XY}(\omega)$ is the cross power spectral density of two random processes $X(t)$ and $Y(t)$, then the following properties hold:

- (a) $S_{XY}(\omega) = S_{YX}(-\omega)$.
- (b) $S_{XY}(-\omega) = S_{XY}^*(\omega)$.
- (c) If $X(t)$ and $Y(t)$ are orthogonal, then $S_{XY}(\omega) = 0$.

Proof. (a) By definition, we have

$$S_{YX}(\omega) = \int_{-\infty}^{\infty} R_{YX}(\tau) e^{-i\omega\tau} d\tau$$

Replacing ω by $-\omega$ in the above equation, we get

$$S_{YX}(-\omega) = \int_{-\infty}^{\infty} R_{YX}(\tau) e^{i\omega\tau} d\tau$$

Substituting $u = -\tau$ in the integral on the R.H.S. of the above equation, we get

$$S_{YX}(-\omega) = \int_{-\infty}^{\infty} R_{YX}(-u) e^{-i\omega u} du$$

By Theorem 6.2 (a), $R_{YX}(-u) = R_{XY}(u)$. Thus, the above equation simplifies to

$$S_{YX}(-\omega) = \int_{-\infty}^{\infty} R_{XY}(u) e^{-i\omega u} du = S_{XY}(\omega)$$

(b) By definition, we have

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau$$

Thus, we have

$$S_{XY}^*(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{i\omega\tau} d\tau = S_{XY}(-\omega)$$

(c) Suppose that $X(t)$ and $Y(t)$ are orthogonal. Then it follows that

$$R_{XY}(\tau) = E[X(t)Y(t+\tau)] = 0 \text{ for all } \tau$$

Thus, it is immediate that

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau = 0$$

◆

Next, we establish an important theorem, which provides an useful formula for computing the cross power spectrum of two random processes.

Theorem 6.6. Let $X(t)$ and $Y(t)$ be jointly wide-sense stationary random processes with cross-correlation function $R_{XY}(\tau)$ and cross power spectral density function $S_{XY}(\omega)$. Let $X_T(\omega)$ and $Y_T(\omega)$ be the Fourier transforms of the truncated random processes $X_T(t)$ and $Y_T(t)$ defined, respectively, as

$$X_T(t) = \begin{cases} X(t) & \text{for } |t| \leq T \\ 0 & \text{for } |t| > T \end{cases}$$

and

$$Y_T(t) = \begin{cases} Y(t) & \text{for } |t| \leq T \\ 0 & \text{for } |t| > T \end{cases}$$

Then

$$S_{XY}(\omega) = \lim_{T \rightarrow \infty} \left[\frac{1}{2T} E \{X_T^*(\omega) Y_T(\omega)\} \right]$$

Proof. Since $X_T(\omega)$ is the Fourier transform of $X_T(t)$, we have

$$X_T(\omega) = \int_{-\infty}^{\infty} X_T(t) e^{-i\omega t} dt \quad (6.16)$$

Using the definition of $X_T(t)$, we can simplify Eq. (6.16) as

$$X_T(\omega) = \int_{-T}^T X(t) e^{-i\omega t} dt \quad (6.17)$$

We note that $Y_T(\omega)$ may be obtained similarly as

$$Y_T(\omega) = \int_{-T}^T Y(t) e^{-i\omega t} dt \quad (6.18)$$

Since $X(t)$ is real, we find from Eq. (6.17) that

$$X_T^*(\omega) = \int_{-T}^T X(t) e^{i\omega t} dt$$

It follows that

$$X_T^*(\omega) Y_T(\omega) = \left[\int_{-T}^T X(t_2) e^{i\omega t_2} dt_2 \right] \left[\int_{-T}^T Y(t_1) e^{-i\omega t_1} dt_1 \right]$$

Thus, we have

$$X_T^*(\omega) Y_T(\omega) = \int_{t_2=-T}^T \int_{t_1=-T}^T X(t_2) Y(t_1) e^{-i\omega(t_1-t_2)} dt_1 dt_2 \quad (6.19)$$

Taking expectation on both sides of Eq. (6.19) and dividing by $\frac{1}{2T}$, we have

$$\frac{1}{2T} E \{X_T^*(\omega) Y_T(\omega)\} = \frac{1}{2T} \int_{t_2=-T}^T \int_{t_1=-T}^T E[X(t_2) Y(t_1)] e^{-i\omega(t_1-t_2)} dt_1 dt_2 \quad (6.20)$$

We use the substitution

$$t_1 = t + \tau \text{ and } t_2 = t$$

in the integral on the R.H.S. of Eq. (6.20).

The Jacobian of the above change of variables is given by

$$J = \frac{\partial(t_1, t_2)}{\partial(t, \tau)} = \begin{vmatrix} \frac{\partial t_1}{\partial t} & \frac{\partial t_1}{\partial \tau} \\ \frac{\partial t_2}{\partial t} & \frac{\partial t_2}{\partial \tau} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

Thus, Eq. (6.20) becomes

$$\frac{1}{2T} E \{X_T^*(\omega) Y(\omega)\} = \frac{1}{2T} \int_{t=-T}^T \int_{\tau=-T-t}^{T-t} E[X(t)Y(t+\tau)] e^{-i\omega\tau} |J| dt d\tau \quad (6.21)$$

Since $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes, $E[X(t)Y(t+\tau)] = R_{XY}(\tau)$ is purely a function of τ . Thus, Eq. (6.21) simplifies to

$$\frac{1}{2T} E \{X_T^*(\omega) Y(\omega)\} = \frac{1}{2T} \int_{t=-T}^T \int_{\tau=-T-t}^{T-t} R_{XY}(\tau) e^{-i\omega\tau} dt d\tau \quad (6.22)$$

Taking limits as $T \rightarrow \infty$ in Eq. (6.22), we have

$$\lim_{T \rightarrow \infty} \left[\frac{1}{2T} E \{X_T^*(\omega) Y(\omega)\} \right] = \int_{\tau=-\infty}^{\infty} \left[\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{t=-T}^T R_{XY}(\tau) dt \right] e^{-i\omega\tau} d\tau \quad (6.23)$$

Since $R_{XY}(\tau)$ is purely a function of τ , it follows that

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{t=-T}^T R_{XY}(\tau) dt = \lim_{T \rightarrow \infty} R_{XY}(\tau) \frac{1}{2T} (2T) = R_{XY}(\tau) \quad (6.24)$$

Substituting Eq. (6.24) into Eq. (6.23), we have

$$\lim_{T \rightarrow \infty} \left[\frac{1}{2T} E \{X_T^*(\omega) Y(\omega)\} \right] = \int_{\tau=-\infty}^{\infty} R_{XY}(\tau) e^{-i\omega\tau} d\tau = S_{XY}(\omega)$$

This completes the proof. ◆

Next, we define the *white noise*, which is a random process with flat power spectral density.

Definition 6.3. A continuous-time random process $\{W(t), t \in \mathbb{R}\}$ is called a *white noise process* if and only if its mean and autocorrelation function satisfy the following:

- (a) The process $W(t)$ has zero mean, i.e. $\mu_W(t) = E[W(t)] = 0$.

(b) The process $W(t)$ has the autocorrelation function defined by

$$R_{WW}(\tau) = \frac{N_0}{2} \delta(\tau) \quad (6.25)$$

where $N_0 > 0$ and $\delta(\tau)$ is a *unit impulse delta function* (or *Dirac δ function*) defined by

$$\int_{-\infty}^{\infty} \delta(\tau) \phi(\tau) d\tau = \phi(0)$$

where $\phi(\tau)$ is any function continuous at $\tau = 0$.

Remark 6.2. If we take the Fourier transform of Eq. (6.25), we obtain (see Table 6.1)

$$S_{WW}(\omega) = \frac{N_0}{2} \mathcal{F}[\delta(\tau)] = \frac{N_0}{2} (1) = \frac{N_0}{2}$$

This shows that $W(t)$ has a constant power spectral density and this explains the name white noise. We also note that the average power of $W(t)$ is not finite because

$$E[W^2(t)] = R_{WW}(0) = \frac{N_0}{2} \delta(0) = \infty$$

EXAMPLE 6.5. The autocorrelation function of a wide-sense stationary random process is given by

$$R(\tau) = \alpha^2 e^{-2\lambda|\tau|}$$

Determine the power spectral density of the process.

(Anna, Model 2003)

Solution. The power spectral density of the process is given by

$$\begin{aligned} S(\omega) &= \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau = \int_{-\infty}^{\infty} \alpha^2 e^{-2\lambda|\tau|} e^{-i\omega\tau} d\tau \\ &= \int_{-\infty}^0 \alpha^2 e^{2\lambda\tau} e^{-i\omega\tau} d\tau + \int_0^{\infty} \alpha^2 e^{-2\lambda\tau} e^{-i\omega\tau} d\tau \\ &= \alpha^2 \int_{-\infty}^0 e^{(2\lambda-i\omega)\tau} d\tau + \alpha^2 \int_0^{\infty} e^{-(2\lambda+i\omega)\tau} d\tau \end{aligned}$$

Integrating, we have

$$\begin{aligned} S(\omega) &= \alpha^2 \left[\frac{e^{(2\lambda-i\omega)\tau}}{2\lambda-i\omega} \right]_{-\infty}^0 + \alpha^2 \left[\frac{e^{-(2\lambda+i\omega)\tau}}{-(2\lambda+i\omega)} \right]_0^{\infty} = \alpha^2 \left[\frac{1}{2\lambda-i\omega} - 0 \right] + \alpha^2 \left[0 + \frac{1}{2\lambda+i\omega} \right] \\ &= \alpha^2 \left[\frac{1}{2\lambda-i\omega} + \frac{1}{2\lambda+i\omega} \right] = \alpha^2 \left[\frac{(2\lambda+i\omega) + (2\lambda-i\omega)}{(2\lambda-i\omega)(2\lambda+i\omega)} \right] \end{aligned}$$

Table 6.1: List of Fourier Transform Pairs and their definitions

Function $f(t)$	Fourier Transform $F(\omega)$
Definition: $f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega$	Definition: $F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$
$f(t - t_0)$	$F(\omega) e^{-i\omega t_0}$
$f(t) e^{i\omega_0 t}$	$F(\omega - \omega_0)$
$f(at)$	$\frac{1}{ a } F\left(\frac{\omega}{a}\right)$
$\frac{d^n f(t)}{dt^n}$	$(i\omega)^n F(\omega)$
$(-it)^n f(t)$	$\frac{d^n}{d\omega^n} [F(\omega)]$
$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(\omega)}{i\omega} + \pi F(0) \delta(\omega)$
$\delta(t)$	1
1	$2\pi \delta(\omega)$
$e^{i\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$
$u(t)$	$\pi \delta(\omega) + \frac{1}{i\omega}$
$e^{-\alpha t }$	$\frac{2\alpha}{\alpha^2 + \omega^2}$
$e^{-\frac{t^2}{2\sigma^2}}$	$\sigma \sqrt{2\pi} e^{-\frac{\sigma^2 \omega^2}{2}}$
$u(t) e^{-\alpha t}$	$\frac{1}{\alpha + i\omega}$
$u(t) t e^{-\alpha t}$	$\frac{1}{(\alpha + i\omega)^2}$
$\cos \omega_0 t$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$\frac{\pi}{i} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$u(t) \cos \omega_0 t$	$\frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{i\omega}{\omega_0^2 - \omega^2}$
$u(t) \sin \omega_0 t$	$\frac{\pi}{2i} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + \frac{\omega^2}{\omega_0^2 - \omega^2}$

Thus,

$$S(\omega) = \frac{4\lambda \alpha^2}{4\lambda^2 + \omega^2}$$

◆

EXAMPLE 6.6. The power spectral density function of a wide-sense stationary random process is given by

$$S(\omega) = \begin{cases} 1 & \text{if } |\omega| < \omega_0 \\ 0 & \text{otherwise} \end{cases}$$

Find the autocorrelation function of the process.

(Anna, Model, 2003)

Solution. The autocorrelation function of the process is given by

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} (1) e^{i\omega\tau} d\omega$$

Integrating, we have

$$\begin{aligned} R(\tau) &= \frac{1}{2\pi} \left[\frac{e^{i\omega\tau}}{i\tau} \right]_{-\omega_0}^{\omega_0} = \frac{1}{2i\pi\tau} [e^{i\omega_0\tau} - e^{-i\omega_0\tau}] \\ &= \frac{1}{2i\pi\tau} [2i \operatorname{Im}(e^{i\omega_0\tau})] = \frac{1}{2i\pi\tau} [2i \sin \omega_0\tau] \end{aligned}$$

(Note that $e^{i\omega_0\tau} = \cos \omega_0\tau + i \sin \omega_0\tau$.)

Simplifying, we have

$$R(\tau) = \frac{\sin \omega_0\tau}{\pi\tau}$$

◆

EXAMPLE 6.7. The power spectrum of a wide-sense stationary process $X = \{X(t)\}$ is given by

$$S(\omega) = \frac{1}{(1 + \omega^2)^2}$$

Find the autocorrelation function and average power of the process.

(Anna, Model 2003)

Solution. The autocorrelation function of the process is given by

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{1}{(1 + \omega^2)^2} \right] e^{i\omega\tau} d\omega \quad (6.26)$$

We find that

$$\frac{1}{(1 + \omega^2)^2} = \left[\frac{1}{(1 + i\omega)(1 - i\omega)} \right]^2 = \left[\frac{(1 + i\omega) + (1 - i\omega)}{2(1 + i\omega)(1 - i\omega)} \right]^2 = \frac{1}{4} \left[\frac{1}{1 - i\omega} + \frac{1}{1 + i\omega} \right]^2$$

Thus, we have

$$\frac{1}{(1 + \omega^2)^2} = \frac{1}{4} \left[\frac{1}{(1 - i\omega)^2} + \frac{1}{(1 + i\omega)^2} + \frac{2}{1 + \omega^2} \right] \quad (6.27)$$

Substituting Eq. (6.27) into Eq. (6.26), we have

$$R(\tau) = \frac{1}{4} \int_{-\infty}^{\infty} \left[\frac{e^{i\omega\tau}}{(1 - i\omega)^2} + \frac{e^{i\omega\tau}}{(1 + i\omega)^2} + \frac{2e^{i\omega\tau}}{1 + \omega^2} \right] d\omega$$

Integrating, we have

$$R(\tau) = \frac{1}{4} \left[u(\tau) \cdot \tau e^{\tau} + u(\tau) \cdot \tau e^{-\tau} + 2e^{-|\tau|} \right]$$

where $u(\tau)$ is the unit step function.

The average power of the process is given by

$$E[X^2(t)] = R(0) = \frac{1}{4} [0 + 0 + 2] = \frac{1}{2} = 0.5$$

◆

EXAMPLE 6.8. The power spectral density of a wide-sense stationary process is given by

$$S(\omega) = \begin{cases} \frac{b}{a} (a - |\omega|) & \text{if } |\omega| \leq a \\ 0 & \text{if } |\omega| > a \end{cases}$$

Find the autocorrelation function of the process.

(Anna, Model; Nov. 2006; Nov. 2007)

Solution. The autocorrelation function of the process is given by

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-a}^a \left[\frac{b}{a} (a - |\omega|) \right] e^{i\omega\tau} d\omega$$

Since $e^{i\omega\tau} = \cos \omega\tau + i \sin \omega\tau$, we have

$$R(\tau) = \frac{b}{2a\pi} \int_{-a}^a (a - |\omega|) (\cos \omega\tau + i \sin \omega\tau) d\omega$$

Since $\cos \omega\tau$ is an even function of ω , it follows that $(a - |\omega|) \cos \omega\tau$ is an even function of ω . Since $\sin \omega\tau$ is an odd function of ω , it follows that $(a - |\omega|) \sin \omega\tau$ is an odd function of ω . Thus,

$$R(\tau) = \frac{b}{2a\pi} \times 2 \int_0^a (a - \omega) \cos \omega\tau d\omega = \frac{b}{a\pi} \int_0^a (a - \omega) \cos \omega\tau d\omega = \frac{b}{a\tau\pi} \int_0^a (a - \omega) d(\sin \omega\tau)$$

Integrating by parts, we get

$$\begin{aligned} R(\tau) &= \frac{b}{a\tau\pi} \left\{ [(a - \omega) \sin \omega\tau]_0^a - \int_0^a \sin \omega\tau (-1) d\omega \right\} \\ &= \frac{b}{a\tau\pi} \left\{ 0 - \frac{1}{\tau} (\cos \omega\tau)_0^a \right\} = \frac{b}{a\tau\pi} \left\{ -\frac{1}{\tau} (-\cos a\tau + 1) \right\} \end{aligned}$$

Thus, we have

$$R(\tau) = \frac{b}{a\tau^2\pi} (1 - \cos a\tau) = \frac{2b}{a\tau^2\pi} \sin^2 \frac{a\tau}{2}$$

◆

EXAMPLE 6.9. The power spectral density of a zero-mean wide-sense stationary process $X(t)$ is given by

$$S(\omega) = \begin{cases} k & \text{if } |\omega| < \omega_0 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant. Show that $X(t)$ and $X\left(t + \frac{\pi}{\omega_0}\right)$ are uncorrelated.

(Anna, Model 2003)

Solution. First, we find the autocorrelation function $R(\tau)$ of the process. By definition, we have

$$\begin{aligned} R(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-\omega_0}^{\omega_0} k e^{i\omega\tau} d\omega \\ &= \frac{k}{2\pi} \int_{-\omega_0}^{\omega_0} (\cos \omega\tau + i \sin \omega\tau) d\omega \end{aligned}$$

Since $\cos \omega\tau$ is an even function of ω , and $\sin \omega\tau$ is an odd function of ω , it follows that

$$\begin{aligned} R(\tau) &= \frac{k}{\pi} \int_0^{\omega_0} \cos \omega\tau d\omega = \frac{k}{\pi} \left[\frac{\sin \omega\tau}{\tau} \right]_0^{\omega_0} \\ &= \frac{k \sin \omega_0 \tau}{\pi \tau} \end{aligned}$$

Hence,

$$E \left[X(t) X \left(t + \frac{\pi}{\omega_0} \right) \right] = R \left(\frac{\pi}{\omega_0} \right) = \frac{k \sin \pi}{\pi \tau} = 0$$

Since the mean of the process $X(t)$ is zero, it is immediate that

$$\text{Cov} \left[X(t), X \left(t + \frac{\pi}{\omega_0} \right) \right] = R \left(\frac{\pi}{\omega_0} \right) - \mu_X^2 = 0 - 0 = 0$$

Thus, we have shown that $X(t)$ and $X\left(t + \frac{\pi}{\omega_0}\right)$ are uncorrelated. ♦

EXAMPLE 6.10. Let $\{X(t); -\infty < t < \infty\}$ be a process with covariance function $R(\tau)$ of the following form:

$$R(\tau) = ce^{-\alpha|\tau|}, \quad c > 0, \alpha > 0$$

Obtain the spectral density of the $\{X(t)\}$ process.

(Anna, Nov. 2003)

Solution. The spectral density of the $\{X(t)\}$ process is given by

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau = \int_{-\infty}^{\infty} ce^{-\alpha|\tau|} e^{-i\omega\tau} d\tau$$

Thus,

$$S(\omega) = c \int_{-\infty}^0 e^{\alpha\tau} e^{-i\omega\tau} d\tau + c \int_0^{\infty} e^{-\alpha\tau} e^{-i\omega\tau} d\tau = c \int_{-\infty}^0 e^{(\alpha-i\omega)\tau} d\tau + c \int_0^{\infty} e^{-(\alpha+i\omega)\tau} d\tau$$

Integrating, we have

$$\begin{aligned} S(\omega) &= c \left[\frac{e^{(\alpha-i\omega)\tau}}{\alpha-i\omega} \right]_{-\infty}^0 + c \left[\frac{e^{-(\alpha+i\omega)\tau}}{-(\alpha+i\omega)} \right]_0^{\infty} = c \left[\frac{1}{\alpha-i\omega} + \frac{1}{\alpha+i\omega} \right] \\ &= c \left[\frac{(\alpha+i\omega) + (\alpha-i\omega)}{(\alpha-i\omega)(\alpha+i\omega)} \right] = \frac{2c\alpha}{\alpha^2 + \omega^2} \end{aligned}$$

◆

EXAMPLE 6.11. Find the power spectral density of a wide-sense stationary process with autocorrelation function $R(\tau) = e^{-\alpha\tau^2}$, where α is a constant.

(Anna, Model 2003)

Solution. The power spectral density function of the process is given by

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau = \int_{-\infty}^{\infty} e^{-\alpha\tau^2} e^{-i\omega\tau} d\tau$$

We can express $S(\omega)$ as

$$\begin{aligned} S(\omega) &= \int_{-\infty}^{\infty} e^{-(\alpha\tau^2 + i\omega\tau - \frac{\omega^2}{4\alpha})} e^{-\frac{\omega^2}{4\alpha}} d\tau \\ &= e^{-\frac{\omega^2}{4\alpha}} \int_{-\infty}^{\infty} e^{-\left(\sqrt{\alpha}\tau + \frac{i\omega}{2\sqrt{\alpha}}\right)^2} d\omega \end{aligned}$$

Substituting $u = \sqrt{\alpha}\tau + \frac{i\omega}{2\sqrt{\alpha}}$, we get

$$S(\omega) = \frac{1}{\sqrt{\alpha}} e^{-\frac{\omega^2}{4\alpha}} \int_{-\infty}^{\infty} e^{-u^2} du = \frac{2}{\sqrt{\alpha}} e^{-\frac{\omega^2}{4\alpha}} \int_0^{\infty} e^{-u^2} du$$

Substituting $v = u^2$, we get

$$S(\omega) = \frac{2}{\sqrt{\alpha}} e^{-\frac{\omega^2}{4\alpha}} \int_0^{\infty} e^{-v} \frac{dv}{2\sqrt{v}} = \frac{1}{\sqrt{\alpha}} e^{-\frac{\omega^2}{4\alpha}} \Gamma\left(\frac{1}{2}\right)$$

Since $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, it is immediate that

$$S(\omega) = \sqrt{\frac{\pi}{\alpha}} e^{-\frac{\omega^2}{4\alpha}}$$

◆

EXAMPLE 6.12. State any two uses of a spectral density.

(Anna, April 2005)

Solution. (i) The power spectrum of a signal has important applications in *Electronic Communication Systems* (Radio and Microwave Communications, Radars and other systems).

(ii) The power spectral density has important applications in *colorimetry*. It is useful in analyzing the colour characteristics of a particular light source.

◆

EXAMPLE 6.13. Given the power spectral density

$$S_{XX}(\omega) = \frac{1}{4 + \omega^2}$$

find the average power of the process.

(Anna, May 2006)

Solution. The autocorrelation function of the process is given by

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{i\tau\omega} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{1}{4 + \omega^2} \right] e^{i\tau\omega} d\omega$$

Hence, the average power of the process is given by

$$E[X^2(t)] = R(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega}{4 + \omega^2} = \frac{1}{2\pi} \times 2 \int_0^{\infty} \frac{d\omega}{2^2 + \omega^2}$$

Thus, we have

$$E[X^2(t)] = \frac{1}{\pi} \left[\frac{1}{2} \tan^{-1} \left(\frac{\omega}{2} \right) \right]_0^{\infty} = \frac{1}{\pi} \left[\frac{\pi}{4} - 0 \right] = \frac{1}{4}$$

◆

EXAMPLE 6.14. Given the power spectral density of a continuous process as

$$S_{XX}(\omega) = \frac{\omega^2 + 9}{\omega^4 + 5\omega^2 + 4}$$

find the autocorrelation function and mean square value of the process.

(Anna, May 2006)

Solution. The autocorrelation function of the process is given by

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{i\omega\tau} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{\omega^2 + 9}{\omega^4 + 5\omega^2 + 4} \right] e^{i\omega\tau} d\omega \quad (6.28)$$

We note that

$$\frac{\omega^2 + 9}{\omega^4 + 5\omega^2 + 4} = \frac{(\omega^2 + 4) + 5}{(\omega^2 + 4)(\omega^2 + 1)} = \frac{1}{\omega^2 + 1} + \frac{5}{(\omega^2 + 4)(\omega^2 + 1)}$$

and

$$\frac{5}{(\omega^2 + 4)(\omega^2 + 1)} = \frac{5}{3} \left[\frac{(\omega^2 + 4) - (\omega^2 + 1)}{(\omega^2 + 4)(\omega^2 + 1)} \right] = \frac{5}{3} \left[\frac{1}{\omega^2 + 1} - \frac{1}{\omega^2 + 4} \right]$$

Thus, we have

$$\frac{\omega^2 + 9}{\omega^4 + 5\omega^2 + 4} = \frac{1}{\omega^2 + 1} + \frac{5}{3} \left[\frac{1}{\omega^2 + 1} - \frac{1}{\omega^2 + 4} \right] = \frac{8}{3} \frac{1}{\omega^2 + 1} - \frac{5}{3} \frac{1}{\omega^2 + 4} \quad (6.29)$$

Substituting Eq. (6.29) into Eq. (6.28), we have

$$\begin{aligned} R_{XX}(\tau) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{8}{3} \frac{1}{\omega^2 + 1} - \frac{5}{3} \frac{1}{\omega^2 + 4} \right] e^{i\omega\tau} d\omega \\ &= \frac{8}{3} \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + 1} e^{i\omega\tau} d\omega - \frac{5}{3} \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + 4} e^{i\omega\tau} d\omega \end{aligned}$$

Integrating, we get

$$R_{XX}(\tau) = \frac{8}{3} \left(\frac{1}{2} e^{-|\tau|} \right) - \frac{5}{3} \left(\frac{1}{4} e^{-2|\tau|} \right) = \frac{16e^{-|\tau|} - 5e^{-2|\tau|}}{12}$$

Thus, the mean square value of the process is given by

$$E[X^2(t)] = R_{XX}(0) = \frac{16 - 5}{12} = \frac{11}{12}$$

◆

EXAMPLE 6.15. Given that a process $X(t)$ has the autocorrelation

$$R_{XX}(\tau) = Ae^{-\alpha|\tau|} \cos(\omega_0 \tau)$$

where $A > 0$, $\alpha > 0$ and ω_0 are real constants, find the power spectrum of $X(t)$.

(Anna, Nov. 2003)

Solution. For the solution to the given problem, we shall use the formula

$$\int e^{\alpha t} \cos \beta t \, dt = \frac{e^{\alpha t}}{\alpha^2 + \beta^2} [\alpha \cos \beta t + \beta \sin \beta t] \quad (6.30)$$

The power spectrum of $X(t)$ is given by

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau) e^{-i\omega\tau} \, d\tau = \int_{-\infty}^{\infty} A e^{-\alpha|\tau|} \cos(\omega_0\tau) e^{-i\omega\tau} \, d\tau$$

i.e.,

$$\begin{aligned} S_{XX}(\omega) &= A \int_{-\infty}^0 e^{\alpha\tau} \cos(\omega_0\tau) e^{-i\omega\tau} \, d\tau + A \int_0^{\infty} e^{-\alpha\tau} \cos(\omega_0\tau) e^{-i\omega\tau} \, d\tau \\ &= A \int_{-\infty}^0 e^{(\alpha-i\omega)\tau} \cos(\omega_0\tau) \, d\tau + A \int_0^{\infty} e^{-(\alpha+i\omega)\tau} \cos(\omega_0\tau) \, d\tau \end{aligned}$$

Using Eq. (6.30), we have

$$\begin{aligned} S_{XX}(\omega) &= \left\{ \frac{A e^{(\alpha-i\omega)\tau}}{(\alpha-i\omega)^2 + \omega_0^2} [(\alpha-i\omega) \cos \omega_0\tau + \omega_0 \sin \omega_0\tau] \right\}_{-\infty}^0 \\ &\quad + \left\{ \frac{A e^{-(\alpha+i\omega)\tau}}{(\alpha+i\omega)^2 + \omega_0^2} [-(\alpha+i\omega) \cos \omega_0\tau + \omega_0 \sin \omega_0\tau] \right\}_0^{\infty} \\ &= \frac{A(\alpha-i\omega)}{(\alpha-i\omega)^2 + \omega_0^2} + \frac{A(\alpha+i\omega)}{(\alpha+i\omega)^2 + \omega_0^2} \\ &= 2A \operatorname{Re} \left[\frac{\alpha-i\omega}{(\alpha-i\omega)^2 + \omega_0^2} \right] \end{aligned}$$

Note that

$$\frac{\alpha-i\omega}{(\alpha-i\omega)^2 + \omega_0^2} = \frac{\alpha-i\omega}{(\alpha^2 + \omega_0^2 - \omega^2) - 2i\alpha\omega} \times \frac{(\alpha^2 + \omega_0^2 - \omega^2) + 2i\alpha\omega}{(\alpha^2 + \omega_0^2 - \omega^2) + 2i\alpha\omega}$$

Simplifying, we have

$$\operatorname{Re} \left[\frac{\alpha-i\omega}{(\alpha-i\omega)^2 + \omega_0^2} \right] = \frac{\alpha(\alpha^2 + \omega_0^2 - \omega^2) + 2\alpha\omega^2}{(\alpha^2 + \omega_0^2 - \omega^2)^2 + 4\alpha^2\omega^2} = \frac{\alpha(\alpha^2 + \omega_0^2 + \omega^2)}{(\alpha^2 + \omega_0^2 - \omega^2)^2 + 4\alpha^2\omega^2}$$

Hence,

$$S_{XX}(\omega) = \frac{2A\alpha(\alpha^2 + \omega_0^2 + \omega^2)}{(\alpha^2 + \omega_0^2 - \omega^2)^2 + 4\alpha^2\omega^2}$$

◆

EXAMPLE 6.16. The autocorrelation function of the Poisson increment process is given by

$$R(\tau) = \begin{cases} \lambda^2 & \text{for } |\tau| > \varepsilon \\ \lambda^2 + \frac{\lambda}{\varepsilon} \left[1 - \frac{|\tau|}{\varepsilon} \right] & \text{for } |\tau| \leq \varepsilon \end{cases}$$

Prove that its spectral density is given by

$$S(\omega) = 2\pi\lambda^2\delta(\omega) + \frac{4\lambda \sin^2\left(\frac{\omega\varepsilon}{2}\right)}{\varepsilon^2\omega^2} \quad (\text{Anna, May 2007})$$

Solution. The spectral density of the process is given by

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau$$

Using the definition of $R(\tau)$, we have

$$S(\omega) = \int_{-\infty}^{\varepsilon} \lambda^2 e^{-i\omega\tau} d\tau + \int_{-\varepsilon}^{\varepsilon} \left[\lambda^2 + \frac{\lambda}{\varepsilon} \left(1 - \frac{|\tau|}{\varepsilon} \right) \right] e^{-i\omega\tau} d\tau + \int_{\varepsilon}^{\infty} \lambda^2 e^{-i\omega\tau} d\tau$$

which can be easily simplified as

$$\begin{aligned} S(\omega) &= \int_{-\infty}^{\infty} \lambda^2 e^{-i\omega\tau} d\tau + \int_{-\varepsilon}^{\varepsilon} \frac{\lambda}{\varepsilon} \left(1 - \frac{|\tau|}{\varepsilon} \right) e^{-i\omega\tau} d\tau \\ &= \lambda^2 \mathcal{F}(1) + \frac{2\lambda}{\varepsilon} \int_0^{\varepsilon} \left(1 - \frac{\tau}{\varepsilon} \right) \cos \omega\tau d\tau \end{aligned}$$

where

$$\mathcal{F}(1) = \text{Fourier transform of } 1 = 2\pi\delta(\omega) \quad [\text{see Table 6.1}]$$

Thus, we have

$$S(\omega) = 2\pi\lambda^2\delta(\omega) + \frac{2\lambda}{\varepsilon\omega} \int_0^{\varepsilon} \left(1 - \frac{\tau}{\varepsilon} \right) d(\sin \omega\tau)$$

Integrating by parts, we have

$$\begin{aligned} S(\omega) &= 2\pi\lambda^2\delta(\omega) + \frac{2\lambda}{\varepsilon\omega} \left\{ \left[\left(1 - \frac{\tau}{\varepsilon} \right) \sin \omega\tau \right]_0^{\varepsilon} - \int_0^{\varepsilon} \sin \omega\tau \left(\frac{-1}{\varepsilon} \right) d\tau \right\} \\ &= 2\pi\lambda^2\delta(\omega) + \frac{2\lambda}{\varepsilon\omega} \left\{ (0 - 0) + \frac{1}{\varepsilon} \int_0^{\varepsilon} \sin \omega\tau d\tau \right\} \end{aligned}$$

Integrating, we have

$$S(\omega) = 2\pi\lambda^2\delta(\omega) + \frac{2\lambda}{\varepsilon^2\omega^2} [-\cos\omega\tau]_0^\varepsilon = 2\pi\lambda^2\delta(\omega) + \frac{2\lambda}{\varepsilon^2\omega^2} [1 - \cos\omega\varepsilon]$$

Using the identity $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$, we have

$$S(\omega) = 2\pi\lambda^2\delta(\omega) + \frac{4\lambda\sin^2\frac{\omega^2\varepsilon^2}{2}}{\varepsilon^2\omega^2}$$

◆

EXAMPLE 6.17. Find the power spectral density of a wide-sense stationary random process $X(t)$, whose autocorrelation function is given by

$$R(\tau) = \begin{cases} 1 - \frac{|\tau|}{T} & \text{for } |\tau| \leq T \\ 0 & \text{for } |\tau| > T \end{cases}$$

Solution. The power spectral density of $X(t)$ is given by

$$\begin{aligned} S(\omega) &= \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau = \int_{-T}^T \left[1 - \frac{|\tau|}{T}\right] e^{-i\omega\tau} d\tau = \int_{-T}^T \left[1 - \frac{|\tau|}{T}\right] [\cos\omega\tau - i\sin\omega\tau] d\tau \\ &= \int_{-T}^T \left[1 - \frac{|\tau|}{T}\right] \cos\omega\tau d\tau - i \int_{-T}^T \left[1 - \frac{|\tau|}{T}\right] \sin\omega\tau d\tau = 2 \int_0^T \left[1 - \frac{\tau}{T}\right] \cos\omega\tau d\tau - i(0) \end{aligned}$$

since $\cos\omega\tau$ is an even function of τ and $\sin\omega\tau$ is an odd function of τ .

Simplifying, we have

$$S(\omega) = 2 \int_0^T \left[1 - \frac{\tau}{T}\right] \cos\omega\tau d\tau = \frac{2}{\omega} \int_0^T \left[1 - \frac{\tau}{T}\right] d(\sin\omega\tau)$$

Integrating by parts, we get

$$S(\omega) = \frac{2}{\omega} \left\{ \left[\left(1 - \frac{\tau}{T}\right) \sin\omega\tau \right]_0^T - \int_0^T \sin\omega\tau \left(-\frac{1}{T}\right) d\tau \right\} = \frac{2}{\omega} \left\{ (0 - 0) + \frac{1}{T} \int_0^T \sin\omega\tau d\tau \right\}$$

Integrating, we get

$$S(\omega) = \frac{2}{\omega^2 T} [-\cos\omega\tau]_0^T = \frac{2}{\omega^2 T} [1 - \cos\omega T]$$

◆

EXAMPLE 6.18. If $X(t) = A\cos(\omega_0 t + \theta)$, where A and ω_0 are constants, and θ is uniformly distributed over $(0, 2\pi)$, find the autocorrelation function and power spectral density of the process.

Solution. Since θ is a uniform random variable defined over $(0, 2\pi)$, it has the probability density function

$$f_{\theta}(\theta) = \begin{cases} \frac{1}{2\pi} & \text{if } 0 < \theta < 2\pi \\ 0 & \text{otherwise} \end{cases}$$

The autocorrelation function of the process is given by

$$R(\tau) = E[X(t)X(t+\tau)] = E\{A \cos(\omega_0 t + \theta) \cdot A \cos[\omega_0(t+\tau) + \theta]\}$$

i.e.

$$R(\tau) = \int_0^{2\pi} A^2 \cos(\omega_0 t + \theta) \cos[\omega_0(t+\tau) + \theta] \left[\frac{1}{2\pi} \right] d\theta$$

which can be rewritten as

$$R(\tau) = \frac{A^2}{4\pi} \int_0^{2\pi} \{\cos[\omega_0(2t+\tau) + 2\theta] + \cos \omega_0 \tau\} d\theta$$

Integrating, we get the autocorrelation function as

$$R(\tau) = \frac{A^2}{4\pi} \left[\frac{\sin[\omega_0(2t+\tau) + 2\theta]}{2} + \theta \cos \omega_0 \tau \right]_0^{2\pi} = \frac{A^2}{4\pi} [0 + 2\pi \cos \omega_0 \tau]$$

Thus,

$$R(\tau) = \frac{A^2}{2} \cos \omega_0 \tau$$

Hence, the power spectral density function of the process is given by

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau = \int_{-\infty}^{\infty} \frac{A^2}{2} \cos \omega_0 \tau e^{-i\omega\tau} d\tau$$

It follows that

$$S(\omega) = \frac{A^2}{2} \int_{-\infty}^{\infty} \cos \omega_0 \tau e^{-i\omega\tau} d\tau = \frac{A^2}{2} \mathcal{F}(\cos \omega_0 \tau)$$

From Table 6.1, we find that

$$\mathcal{F}(\cos \omega_0 \tau) = \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

Thus, we have

$$S(\omega) = \frac{\pi A^2}{2} [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)] \quad \blacklozenge$$

EXAMPLE 6.19. If $Y(t) = A \cos(\omega_0 t + \theta) + N(t)$, where A is a constant, θ is a random variable with a uniform distribution in $(-\pi, \pi)$ and $\{N(t)\}$ is a band limited Gaussian white noise with a power spectral density

$$S_{NN}(\omega) = \begin{cases} \frac{N_0}{2} & \text{for } |\omega - \omega_0| < \omega_B \\ 0 & \text{elsewhere} \end{cases}$$

find the power spectral density of $\{Y(t)\}$. Assume that $N(t)$ and θ are independent. (Anna, May 2007)

Solution. Let $Y(t) = M(t) + N(t)$, where $M(t) = A\cos(\omega_0 t + \theta)$ and $N(t)$ is as given in the problem. We note that

$$\begin{aligned} Y(t_1)Y(t_2) &= [M(t_1) + N(t_1)][M(t_2) + N(t_2)] \\ &= M(t_1)M(t_2) + M(t_1)N(t_2) + M(t_2)N(t_1) + N(t_1)N(t_2) \end{aligned}$$

Thus, the autocorrelation function of $Y(t)$ is given by

$$\begin{aligned} R_{YY}(t_1, t_2) &= E[Y(t_1)Y(t_2)] \\ &= E[M(t_1)M(t_2)] + E[M(t_1)N(t_2)] + E[N(t_1)M(t_2)] + E[N(t_1)N(t_2)] \\ &= R_{MM}(t_1, t_2) + R_{MN}(t_1, t_2) + R_{NM}(t_1, t_2) + R_{NN}(t_1, t_2) \end{aligned}$$

Since $N(t)$ and θ are independent, it follows that $N(t)$ and $M(t)$ are independent random processes. Hence, we find that

$$R_{MN}(t_1, t_2) = E[M(t_1)N(t_2)] = E[M(t_1)]E[N(t_2)] = \mu_M(t_1)\mu_N(t_2)$$

and

$$R_{NM}(t_1, t_2) = E[N(t_1)M(t_2)] = E[N(t_1)]E[M(t_2)] = \mu_N(t_1)\mu_M(t_2)$$

Thus, it follows that

$$R_{YY}(t_1, t_2) = R_{MM}(t_1, t_2) + \mu_M(t_1)\mu_N(t_2) + \mu_N(t_1)\mu_M(t_2) + R_{NN}(t_1, t_2) \quad (6.31)$$

The mean of $M(t)$ is given by

$$\mu_M(t) = E[M(t)] = E[A\cos(\omega_0 t + \theta)] = AE[\cos(\omega_0 t + \theta)]$$

Since θ is uniformly distributed over $(-\pi, \pi)$, we have

$$\mu_M(t) = A \int_{-\pi}^{\pi} \cos(\omega_0 t + \theta) \left[\frac{1}{2\pi} \right] d\theta$$

Integrating, we get

$$\mu_M(t) = \frac{A}{2\pi} [\sin(\omega_0 t + \theta)]_{-\pi}^{\pi} = \frac{A}{2\pi} [-\sin(\omega_0 t) + \sin(\omega_0 t)] = 0$$

Substituting the above in Eq. (6.31), we get

$$R_{YY}(t_1, t_2) = R_{MM}(t_1, t_2) + 0 + 0 + R_{NN}(t_1, t_2) = R_{MM}(t_1, t_2) + R_{NN}(t_1, t_2)$$

Thus, it follows that

$$R_{YY}(\tau) = R_{MM}(\tau) + R_{NN}(\tau) \quad (6.32)$$

Next, we find that

$$R_{MM}(\tau) = E[M(t)M(t + \tau)] = E\{A\cos(\omega_0 t + \theta) \cdot A\cos[\omega_0(t + \tau) + \theta]\}$$

i.e.

$$R_{MM}(\tau) = \int_{-\pi}^{\pi} A^2 \cos[\omega_0(t + \tau) + \theta] \cos(\omega_0 t + \theta) \left[\frac{1}{2\pi} \right] d\theta$$

which can be rewritten as

$$R_{MM}(\tau) = \frac{A^2}{4\pi} \int_{-\pi}^{\pi} \{ \cos[\omega_0(2t + \tau) + 2\theta] + \cos \omega \tau \} d\theta$$

Integrating, we get

$$R_{MM}(\tau) = \frac{A^2}{4\pi} \left\{ \frac{\sin[\omega_0(2t + \tau) + 2\theta]}{2} + \theta \cos \omega_0 \tau \right\}_{-\pi}^{\pi} = \frac{A^2}{4\pi} [(0 - 0) + 2\pi \cos \omega_0 \tau]$$

Thus, we have

$$R_{MM}(\tau) = \frac{A^2}{2} \cos \omega_0 \tau$$

Substituting the above in Eq. (6.32), we get

$$R_{YY}(\tau) = \frac{A^2}{2} \cos \omega_0 \tau$$

Thus, the power spectral density function of $Y(t)$ is given by

$$\begin{aligned} S_{YY}(\omega) &= \mathcal{F}[R_{YY}(\tau)] = \frac{A^2}{2} \mathcal{F}(\cos \omega_0 \tau) + S_{NN}(\omega) \\ &= \frac{\pi A^2}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + S_{NN}(\omega) \end{aligned}$$

where $S_{NN}(\omega)$ is as given in the problem. ◆

EXAMPLE 6.20. If $X(t)$ and $Y(t)$ are uncorrelated random processes, then find the power spectral density of Z if $Z(t) = X(t) + Y(t)$. Also find the cross-spectral density $S_{XZ}(\omega)$ and $S_{YZ}(\omega)$.
(Anna, Nov. 2006)

Solution. The autocorrelation function of $Z(t)$ is given by

$$\begin{aligned} R_{ZZ}(\tau) &= E[Z(t)Z(t + \tau)] = E\{[X(t) + Y(t)][X(t + \tau) + Y(t + \tau)]\} \\ &= E[X(t)X(t + \tau)] + E[X(t)Y(t + \tau)] + E[Y(t)X(t + \tau)] + E[Y(t)Y(t + \tau)] \\ &= R_{XX}(\tau) + R_{XY}(\tau) + R_{YX}(\tau) + R_{YY}(\tau) \end{aligned}$$

Taking Fourier transform, it follows that the power spectral density of $Z(t)$ is given by

$$S_{ZZ}(\omega) = S_{XX}(\omega) + S_{XY}(\omega) + S_{YX}(\omega) + S_{YY}(\omega)$$

If $X(t)$ and $Y(t)$ are uncorrelated, then it is immediate that

$$S_{XY}(\omega) = S_{YX}(\omega)$$

and the power spectral density of $Z(t)$ simplifies to

$$S_{ZZ}(\omega) = S_{XX}(\omega) + S_{YY}(\omega) + 2S_{XY}(\omega)$$

The cross-correlation function of $X(t)$ and $Z(t)$ is given by

$$\begin{aligned} R_{XZ}(\tau) &= E[X(t)Z(t+\tau)] = E\{X(t)[X(t+\tau) + Y(t+\tau)]\} \\ &= E[X(t)X(t+\tau)] + E[X(t)Y(t+\tau)] \\ &= R_{XX}(\tau) + R_{XY}(\tau) \end{aligned}$$

Taking Fourier transform, it follows that the cross-spectral density of $X(t)$ and $Z(t)$ is given by

$$S_{XZ}(\omega) = S_{XX}(\omega) + S_{XY}(\omega)$$

Similarly, the cross-spectral density of $Y(t)$ and $Z(t)$ is given by

$$S_{YZ}(\omega) = S_{YX}(\omega) + S_{YY}(\omega)$$

Since $S_{YX}(\omega) = S_{XY}(\omega)$, we can also express $S_{YZ}(\omega)$ as

$$S_{YZ}(\omega) = S_{YY}(\omega) + S_{XY}(\omega)$$

◆

EXAMPLE 6.21. Given a cross-power spectrum defined as

$$S_{XY}(\omega) = \begin{cases} a + \frac{ib}{W} \omega & \text{for } -W < \omega < W \\ 0 & \text{elsewhere} \end{cases}$$

Find the cross-correlation function.

(Madras, April 1999; Oct. 2002)

Solution. By definition, the cross-correlation function of $X(t)$ and $Y(t)$ is given by

$$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega) e^{i\omega\tau} d\omega$$

Using the definition of $S_{XY}(\omega)$, we obtain $R_{XY}(\tau)$ as

$$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-W}^W \left[a + \frac{ib\omega}{W} \right] e^{i\omega\tau} d\omega$$

which may be simplified as

$$R_{XY}(\tau) = \frac{a}{2\pi} \int_{-W}^W e^{i\omega\tau} d\omega + \frac{ib}{2\pi W} \int_{-W}^W \omega e^{i\omega\tau} d\omega = I_1(\tau) + I_2(\tau) \quad [\text{say}] \quad (6.33)$$

We find that

$$I_1(\tau) = \frac{a}{2\pi} \int_{-W}^W e^{i\omega\tau} d\omega = \frac{a}{2\pi} \left[\frac{e^{i\omega\tau}}{i\tau} \right]_{-W}^W$$

Simplifying, we have

$$I_1(\tau) = \frac{a}{2\pi} \left[\frac{e^{i\tau W} - e^{-i\tau W}}{i\tau} \right] = \frac{a}{2\pi} \left[\frac{2i \sin W\tau}{i\tau} \right] = \frac{a}{\pi\tau} \sin W\tau \quad (6.34)$$

Next, we find that

$$I_2(\tau) = \frac{ib}{2\pi W} \int_{-W}^W \omega e^{i\omega\tau} d\omega = \frac{b}{2\pi W\tau} \int_{-W}^W \omega d(e^{i\omega\tau})$$

Integrating by parts, we get

$$I_2(\tau) = \frac{b}{2\pi W\tau} \left\{ [\omega e^{i\omega\tau}]_{-W}^W - \int_{-W}^W e^{i\omega\tau} d\omega \right\} = \frac{b}{2\pi W\tau} \left\{ [W e^{iW\tau} + W e^{-iW\tau}] - \left[\frac{e^{i\omega\tau}}{i\tau} \right]_{-W}^W \right\}$$

Simplifying, we get

$$I_2(\tau) = \frac{b}{2\pi W\tau} \left[2W \cos W\tau - \left(\frac{e^{iW\tau} - e^{-iW\tau}}{i\tau} \right) \right] = \frac{b}{2\pi W\tau} \left[2W \cos W\tau - \left(\frac{2i \sin W\tau}{i\tau} \right) \right]$$

Thus, it follows that

$$I_2(\tau) = \frac{b}{\pi W\tau^2} [W\tau \cos W\tau - \sin W\tau] \quad (6.35)$$

Substituting the values of $I_1(\tau)$ and $I_2(\tau)$ from Eqs. (6.34) and (6.35) into Eq. (6.33), we get

$$R_{XY}(\tau) = \frac{a}{\pi\tau} \sin W\tau + \frac{b}{\pi W\tau^2} [W\tau \cos W\tau - \sin W\tau]$$

which may be simplified as

$$R_{XY}(\tau) = \frac{1}{\pi W\tau^2} [(aW\tau - b) \sin W\tau + bW\tau \cos W\tau]$$

◆

EXAMPLE 6.22. If $\{X(t)\}$ is a band limited process such that $S_{XX}(\omega) = 0$ when $|\omega| > \sigma$, prove that

$$2[R_{XX}(0) - R_{XX}(\tau)] \leq \sigma^2 \tau^2 R_{XX}(0)$$

(Anna, May 2007)

Solution. First, we note that

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{i\omega\tau} d\omega$$

which can be written as

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) [\cos \omega \tau + \sin \omega \tau] d\omega$$

Noting that $S_{XX}(\omega)$ and $\cos \omega \tau$ are even functions of ω and $\sin \omega$ is an odd function of ω , the integral in the above equation simplifies to

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) \cos \omega \tau d\omega$$

Since $X(t)$ is a band limited process, it follows that

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\sigma}^{\sigma} S_{XX}(\omega) \cos \omega \tau d\omega$$

Thus, it follows that

$$R_{XX}(0) - R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\sigma}^{\sigma} S_{XX}(\omega) [1 - \cos \omega \tau] d\omega$$

which may be written as

$$R_{XX}(0) - R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\sigma}^{\sigma} S_{XX}(\omega) \left[2 \sin^2 \frac{\omega \tau}{2} \right] d\omega \quad (6.36)$$

From Trigonometry, we know that $|\sin \theta| \leq \theta$ and so $\sin^2 \theta \leq \theta^2$. Thus, Eq. (6.36) can be simplified as

$$R_{XX}(0) - R_{XX}(\tau) \leq \frac{1}{2\pi} \int_{-\sigma}^{\sigma} S_{XX}(\omega) \left[\frac{2\omega^2 \tau^2}{4} \right] d\omega$$

i.e.

$$R_{XX}(0) - R_{XX}(\tau) \leq \frac{1}{2\pi} \frac{\sigma^2 \tau^2}{2} \int_{-\sigma}^{\sigma} S_{XX}(\omega) d\omega \quad (6.37)$$

We also note that

$$R_{XX}(0) = \frac{1}{2\pi} \int_{-\sigma}^{\sigma} S_{XX}(\omega) d\omega \quad (6.38)$$

Substituting Eq. (6.38) into Eq. (6.37), we have

$$R_{XX}(0) - R_{XX}(\tau) \leq \frac{\sigma^2 \tau^2}{2} R_{XX}(0)$$

or equivalently

$$2[R_{XX}(0) - R_{XX}(\tau)] \leq \sigma^2 \tau^2 R_{XX}(0)$$

This completes the proof. ◆

PROBLEM SET 6.2

1. If the power density function of a white noise process is $S(\omega) = \frac{N_0}{2}$, where N_0 is a positive real constant, find the autocorrelation function of the process.
2. Find the autocorrelation function and mean-square value of the wide-sense stationary process $X(t)$, whose power density spectrum is given by

$$S(\omega) = \frac{4}{4 + \omega^2}$$

3. Check whether the following are valid power spectral density functions for WSS processes:

$$(a) f(\omega) = \frac{\omega}{\omega^2 + 4}.$$

$$(b) g(\omega) = \frac{4}{(1 + \omega^2)^2}.$$

$$(c) h(\omega) = \begin{cases} -1 & \text{if } |\omega| < \pi \\ 0 & \text{otherwise} \end{cases}$$

4. Find the power spectral density function of a WSS process $X(t)$ with the autocorrelation function

$$R(\tau) = A + Be^{-c|\tau|}$$

where A, B and c are constants.

5. Let $X(t)$ be a random process given by

$$X(t) = A \cos \beta t + B \sin \beta t$$

where A and B are independent random variables with $\mu_A = \mu_B = 0$ and $\sigma_A = \sigma_B = \sigma$. Find the autocorrelation function and power spectral density of the process $X(t)$.

6. If $X(t)$ is a WSS process with autocorrelation function

$$R(\tau) = e^{-\alpha|\tau|}(1 + \alpha|\tau|)$$

find the power spectral density of the process.

7. If $X(t)$ is a WSS process with autocorrelation function

$$R(\tau) = e^{-\tau^2}$$

find the power spectral density of the process.

8. If $X(t)$ is a WSS process with power spectral density

$$S(\omega) = \begin{cases} \frac{\pi}{2} & \text{for } -\pi < \omega < \pi \\ 0 & \text{elsewhere} \end{cases}$$

find the autocorrelation function of the process.

9. Given that the power spectral density of a continuous WSS process $X(t)$ as

$$S(\omega) = \frac{10}{\omega^2 + 16}$$

find the autocorrelation function and mean-square value of the process.

10. Find the cross-correlation function corresponding to the cross power density spectrum

$$S_{XY}(\omega) = \frac{4}{(a + i\omega)^3}$$

6.3 LINEAR SYSTEMS WITH RANDOM INPUTS

A *control system* or simply *system* is a mathematical model of a physical process which relates the *input* (or *excitation*) signal x to the *output* (or *response*) signal y . Thus, we think of the system as an operator that transforms the input $x(t)$ to yield the output $y(t)$ (see Figure 6.1) and write

$$y(t) = T[x(t)] \quad (6.39)$$

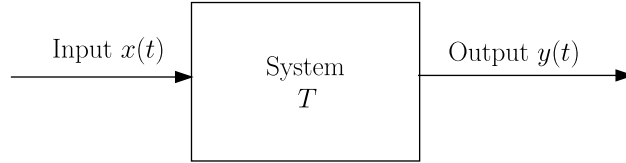


Figure 6.1: System.

Next, we define a linear system.

Definition 6.4. An operator as defined in Eq. (6.39) is called a *linear operator* or *linear system* if the following conditions are satisfied:

(a) If $y_1(t) = T[x_1(t)]$ and $y_2(t) = T[x_2(t)]$, then

$$y_1(t) + y_2(t) = T[x_1(t)] + T[x_2(t)] = T[x_1(t) + x_2(t)] \quad (\text{Additivity})$$

(b) If $y(t) = T[x(t)]$ and α is any scalar, then

$$\alpha y(t) = \alpha T[x(t)] = T[\alpha x(t)] \quad (\text{Homogeneity})$$

Next, we define time-invariant and time-varying linear systems.

Definition 6.5. A linear system T is called *time-invariant* if a timeshift in the input signal $x(t)$ causes the same timeshift in the output signal $y(t)$, i.e. we have

$$y(t - t_0) = T[x(t - t_0)] \quad (6.40)$$

where $t_0 \in \mathbb{R}$ is arbitrary. A linear system T that does not satisfy the condition (6.40) for some input $x(t)$ and timeshift t_0 is called *time-varying*.

For a continuous-time linear time-invariant system T , Eq. (6.39), can be expressed as

$$y(t) = \int_{-\infty}^{\infty} h(\xi) x(t - \xi) d\xi \quad (6.41)$$

where $h(t)$ is the *impulse response* corresponding to the input, $\delta(t)$, the unit-impulse function, i.e.

$$h(t) = T[\delta(t)] \quad (6.42)$$

We note that Eq. (6.41) can also be expressed as

$$y(t) = h(t) \star x(t)$$

i.e. the output signal $y(t)$ of a continuous-time linear time-invariant system is the *convolution integral* of the impulse response $h(t)$ and the input signal $x(t)$.

Next, we define *causal* and *stable* linear time-invariant systems.

Definition 6.6. A linear time-invariant system is called *causal* if the system does not respond prior to the application of an input signal, i.e.

$$y(t) = 0 \text{ for } t < t_0 \text{ if } x(t) = 0 \text{ for } t < t_0 \quad (6.43)$$

where t_0 is any real constant. Using the relation, Eq. (6.41), we can also express the condition Eq. (6.43), for causality as

$$h(t) = 0 \text{ for } t < 0 \quad (6.44)$$

All physically realizable systems are causal systems, i.e. they must satisfy the property Eq. (6.44).

Definition 6.7. A linear time-invariant system is called *stable* if its response to any bounded input is bounded, i.e. if $|x(t)| < M$ for some positive constant M and for all $t \geq 0$, then $|y(t)| < N$ for another positive constant N and for all $t \geq 0$.

By the relation Eq. (6.41), it follows that a necessary and sufficient condition for a linear time-invariant system to be stable is

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty \quad (6.45)$$

When the input to a continuous-time linear system $y(t) = T[x(t)]$ is a random process $\{X(t)\}$, then the output will also be a random process $\{Y(t)\}$, i.e.

$$Y(t) = T[X(t)]$$

For a linear time-invariant system, we can express $Y(t)$ as

$$Y(t) = \int_{-\infty}^{\infty} h(\xi) X(t - \xi) d\xi \quad (6.46)$$

Thus, we may view the system T as a linear system accepting the random process $X(t)$ as its inputs and resulting in the output process $Y(t)$ given by Eq. (6.46).

Next, we prove an important result which asserts that if the input for a linear time-invariant system is a WSS process, then its output also is a WSS process.

Theorem 6.7. For a linear time-invariant system with input process $\{X(t)\}$ that is assumed to be wide-sense stationary and output process $\{Y(t)\}$, the following properties hold:

- (a) If the mean of $X(t)$ is denoted by μ_X (a constant), then the mean of $Y(t)$ is a constant given by

$$\mu_Y = E[Y(t)] = \mu_X \int_{-\infty}^{\infty} h(\xi) d\xi$$

where $h(t)$ is the impulse response of the linear system.

- (b) If the autocorrelation function of $X(t)$ is $R_{XX}(\tau) = E[X(t)X(t + \tau)]$, then the autocorrelation function of $Y(t)$ is given by

$$R_{YY}(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\xi) h(\eta) R_{XX}(\tau + \xi - \eta) d\xi d\eta$$

- (c) $Y(t)$ is a wide-sense stationary process.
 (d) $R_{YY}(\tau) = R_{XX}(\tau) \star h(-\tau) \star h(\tau)$.

Proof. (a) We know that

$$Y(t) = \int_{-\infty}^{\infty} h(\xi) X(t - \xi) d\xi$$

Thus, it follows that the mean of $Y(t)$ is given by

$$\mu_Y(t) = E[Y(t)] = \int_{-\infty}^{\infty} h(\xi) E[X(t - \xi)] d\xi = \int_{-\infty}^{\infty} h(\xi) \mu_X d\xi$$

Hence, we find that the mean of $Y(t)$ is a constant given by

$$\mu_Y = \mu_X \int_{-\infty}^{\infty} h(\xi) d\xi$$

- (b) We note that

$$Y(t)Y(t + \tau) = \int_{-\infty}^{\infty} h(\xi)h(\eta)X(t - \xi)X(t + \tau - \eta) d\xi d\eta$$

Thus, it follows that

$$R_{YY}(\tau) = E[Y(t)Y(t + \tau)] = \int_{-\infty}^{\infty} h(\xi)h(\eta)E[X(t - \xi)X(t + \tau - \eta)] d\xi d\eta$$

Since $X(t)$ is a wide-sense stationary process, its autocorrelation function $R_{XX}(\tau) = E[X(t)X(t + \tau)]$ depends only on the time difference τ . Hence, it follows that

$$E[X(t - \xi)X(t + \tau - \eta)] = R_{XX}(\tau + \xi - \eta)$$

and we have

$$R_{YY}(\tau) = \int_{-\infty}^{\infty} h(\xi)h(\eta)R_{XX}(\tau + \xi - \eta)d\xi d\eta \quad (6.47)$$

which is purely a function of τ .

- (c) This is an immediate consequence of the results (a) and (b).
 (d) This is an immediate consequence of Eq. (6.47). ◆

Next, we prove a result that gives the properties of cross-correlation functions of input and output random processes of a linear time-invariant system.

Theorem 6.8. Consider a linear time-invariant system with input process $\{X(t)\}$ and output process $\{Y(t)\}$. Let $h(t)$ be the impulse response of the linear system. If the input process $\{X(t)\}$ is wide-sense stationary, then the following properties hold:

(a) The cross-correlation function between $X(t)$ and $Y(t)$ is given by

$$R_{XY}(\tau) = R_{XX}(\tau) \star h(\tau)$$

(b) The cross-correlation function between $Y(t)$ and $X(t)$ is given by

$$R_{YX}(\tau) = R_{XX}(\tau) \star h(-\tau)$$

(c) The autocorrelation function of $Y(t)$ is given by

$$R_{YY}(\tau) = R_{XY}(\tau) \star h(-\tau) = R_{YX}(\tau) \star h(\tau)$$

(Anna, Model 2003)

Proof. (a) We know that

$$Y(t) = \int_{-\infty}^{\infty} h(\xi) X(t - \xi) d\xi$$

Thus, it follows that

$$X(t)Y(t + \tau) = \int_{-\infty}^{\infty} h(\xi) X(t) X(t + \tau - \xi) d\xi$$

Hence, the cross-correlation function between $X(t)$ and $Y(t)$ is given by

$$R_{XY}(\tau) = E[X(t)Y(t + \tau)] = \int_{-\infty}^{\infty} h(\xi) E[X(t)X(t + \tau - \xi)] d\xi \quad (6.48)$$

Since $\{X(t)\}$ is a wide-sense stationary process, it follows that

$$E[X(t)X(t + \tau - \xi)] = R_{XX}(\tau - \xi) \quad (6.49)$$

Substituting Eq. (6.49) into Eq. (6.48), we get

$$R_{XY}(\tau) = \int_{-\infty}^{\infty} h(\xi) R_{XX}(\tau - \xi) d\xi = R_{XX}(\tau) \star h(\tau)$$

(b) We find that

$$Y(t)X(t + \tau) = \int_{-\infty}^{\infty} h(\xi) X(t - \xi) X(t + \tau) d\xi$$

Thus, the cross-correlation function between $Y(t)$ and $X(t)$ is given by

$$R_{YX}(\tau) = E[Y(t)X(t + \tau)] = \int_{-\infty}^{\infty} h(\xi) E[X(t - \xi)X(t + \tau)] d\xi \quad (6.50)$$

Since $\{X(t)\}$ is a wide-sense stationary process, it follows that

$$E[X(t - \xi)X(t + \tau)] = R_{XX}(\tau + \xi) \quad (6.51)$$

Substituting Eq. (6.51) into Eq. (6.50), we get

$$R_{YX}(\tau) = \int_{-\infty}^{\infty} h(\xi) R_{XX}(\tau + \xi) d\xi = \int_{-\infty}^{\infty} h(-u) R_{XX}(\tau - u) du = R_{XX}(\tau) \star h(-\tau)$$

- (c) To start with, we note that properties (a) and (b) imply that the processes $\{X(t)\}$ and $\{Y(t)\}$ are jointly wide-sense stationary. Next, we note that

$$Y(t) = \int_{-\infty}^{\infty} h(\xi) X(t - \xi) d\xi$$

It follows that

$$Y(t)Y(t + \tau) = \int_{-\infty}^{\infty} h(\xi) X(t - \xi) Y(t + \tau) d\xi$$

Thus, the autocorrelation function of the output process $Y(t)$ is given by

$$R_{YY}(\tau) = E[Y(t)Y(t + \tau)] = \int_{-\infty}^{\infty} h(\xi) E[X(t - \xi)Y(t + \tau)] d\xi \quad (6.52)$$

Since $X(t)$ and $Y(t)$ are jointly WSS processes, we have

$$E[X(t - \xi)Y(t + \tau)] = R_{XY}(\tau + \xi) \quad (6.53)$$

Substituting Eq. (6.53) into Eq. (6.52), we get

$$R_{YY}(\tau) = \int_{-\infty}^{\infty} h(\xi) R_{XY}(\tau + \xi) d\xi = \int_{-\infty}^{\infty} h(-u) R_{XY}(\tau - u) du$$

i.e.

$$R_{YY}(\tau) = R_{XY}(\tau) \star h(-\tau) \quad (6.54)$$

Next, we note that

$$Y(t + \tau) = \int_{-\infty}^{\infty} h(\xi) X(t + \tau - \xi) d\xi$$

Thus, we have

$$Y(t)Y(t + \tau) = \int_{-\infty}^{\infty} h(\xi) Y(t) X(t + \tau - \xi) d\xi$$

Hence, the autocorrelation function of the output process $Y(t)$ is given by

$$R_{YY}(\tau) = E[Y(t)Y(t+\tau)] = \int_{-\infty}^{\infty} h(\xi) E[Y(t)X(t+\tau-\xi)] d\xi \quad (6.55)$$

Since the processes $\{Y(t)\}$ and $\{X(t)\}$ are jointly wide-sense stationary, it follows that

$$E[Y(t)X(t+\tau-\xi)] = R_{YX}(\tau-\xi) \quad (6.56)$$

Substituting Eq. (6.56) into Eq. (6.55), we get

$$R_{YY}(\tau) = \int_{-\infty}^{\infty} h(\xi) R_{YX}(\tau-\xi) d\xi = R_{YX}(\tau) \star h(\tau) \quad (6.57)$$

Combining Eqs. (6.54) and (6.57), the result (c) follows. ◆

Definition 6.8. The *transfer function* of a time-invariant linear system defined by the equation

$$Y(t) = \int_{-\infty}^{\infty} h(\xi) X(t-\xi) d\xi \quad (6.58)$$

is the Fourier transform of the impulse response $h(t)$ of the linear system and is denoted by $H(\omega)$, i.e.

$$H(\omega) = \int_{-\infty}^{\infty} h(t) e^{-i\omega t} dt \quad (6.59)$$

Note that the system Eq. (6.58) can also be expressed as

$$Y(t) = \int_{-\infty}^{\infty} X(\xi) h(t-\xi) d\xi$$

[This can be easily seen by substituting $u = t - \xi$ in the integral on the R.H.S. of Eq. (6.58)]

The following theorem shows that the transfer function of a time-invariant linear system is the ratio of the Fourier transform of the output process to the Fourier transform of the input process.

Theorem 6.9. If $X(\omega)$, $Y(\omega)$ and $H(\omega)$ are the Fourier transforms of $X(t)$, $Y(t)$ and $h(t)$, respectively, then

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

or equivalently

$$Y(\omega) = H(\omega) X(\omega)$$

i.e. the Fourier transform of the output process of the linear system is the product of the Fourier transforms of the impulse response function and the input process.

Proof. First, we note that

$$Y(\omega) = \int_{-\infty}^{\infty} Y(t) e^{-i\omega t} dt = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} X(\xi) h(t - \xi) d\xi \right] e^{-i\omega t} dt$$

Changing the order of integration, we get

$$\begin{aligned} Y(\omega) &= \int_{-\infty}^{\infty} X(\xi) \left[\int_{-\infty}^{\infty} h(t - \xi) e^{-i\omega(t - \xi)} dt \right] e^{-i\omega\xi} d\xi \\ &= \int_{-\infty}^{\infty} X(\xi) \left[\int_{-\infty}^{\infty} h(u) e^{-i\omega u} du \right] e^{-i\omega\xi} d\xi = \int_{-\infty}^{\infty} X(\xi) H(\omega) e^{-i\omega\xi} d\xi \end{aligned}$$

Hence, it is immediate that

$$Y(\omega) = H(\omega) \int_{-\infty}^{\infty} X(\xi) e^{-i\omega\xi} d\xi = H(\omega) X(\omega)$$

◆

The following result gives a formula for the power spectral density of the output process in terms of the power spectral density of the input process and the system transfer function.

Theorem 6.10. If $S_{XX}(\omega)$ and $S_{YY}(\omega)$ are the power spectral density functions of the wide-sense stationary input process $X(t)$, and output process $Y(t)$, respectively, and $H(\omega)$ is the system transfer function, then

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

(Anna, Model April 2003; April 2005)

Proof. First, we note that

$$Y(t)Y(t + \tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\xi)h(\eta) X(t - \xi)X(t + \tau - \eta) d\xi d\eta$$

Thus, it follows that

$$R_{YY}(\tau) = E[Y(t)Y(t + \tau)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\xi)h(\eta) E[X(t - \xi)X(t + \tau - \eta)] d\xi d\eta \quad (6.60)$$

Since the input process $X(t)$ is wide-sense stationary, we know that

$$E[X(t - \xi)X(t + \tau - \eta)] = R_{XX}(\tau + \xi - \eta) \quad (6.61)$$

Substituting Eq. (6.61) into Eq. (6.60), we get

$$R_{YY}(\tau) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\xi)h(\eta) R_{XX}(\tau + \xi - \eta) d\xi d\eta \quad (6.62)$$

which also shows that the output process $Y(t)$ is a WSS process.

Taking the Fourier transform of Eq. (6.62), we obtain

$$S_{YY}(\omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\xi)h(\eta)R_{XX}(\tau + \xi - \eta)e^{-i\tau\omega} d\xi d\eta d\tau$$

Letting $\lambda = \tau + \xi - \eta$, we get

$$\begin{aligned} S_{YY}(\omega) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(\xi)h(\eta)R_{XX}(\lambda)e^{-i(\lambda-\xi+\eta)\omega} d\xi d\eta d\lambda \\ &= \int_{-\infty}^{\infty} h(\xi)e^{i\omega\xi} d\xi \int_{-\infty}^{\infty} h(\eta)e^{-i\omega\eta} d\eta \int_{-\infty}^{\infty} R_{XX}(\lambda)e^{-i\lambda\omega} d\lambda \end{aligned}$$

Hence, it is immediate that

$$\begin{aligned} S_{YY}(\omega) &= H(-\omega)H(\omega)S_{XX}(\omega) = H^*(\omega)H(\omega)S_{XX}(\omega) \\ &= |H(\omega)|^2 S_{XX}(\omega) \end{aligned}$$

This completes the proof. ◆

Remark 6.3. Theorem 6.10 is also called the *Fundamental Theorem on the power spectrum of the output of a linear system*.

EXAMPLE 6.23. Examine whether the following systems are linear:

- (a) $y(t) = \alpha x(t)$, where α is a scalar.
- (b) $y(t) = x^2(t)$
- (c) $y(t) = tx(t)$

Solution. (a) Let $y_1(t)$ and $y_2(t)$ be the output signals corresponding to the input signals $x_1(t)$ and $x_2(t)$, respectively, i.e.

$$y_1(t) = \alpha x_1(t) \text{ and } y_2(t) = \alpha x_2(t)$$

For any scalars c_1 and c_2 , the output signal for the input signal $x(t) = c_1x_1(t) + c_2x_2(t)$ is given by

$$\begin{aligned} y(t) &= \alpha x(t) = \alpha[c_1x_1(t) + c_2x_2(t)] = c_1[\alpha x_1(t)] + c_2[\alpha x_2(t)] \\ &= c_1y_1(t) + c_2y_2(t) \end{aligned}$$

which shows that the given system is linear.

- (b) Let $y_1(t)$ and $y_2(t)$ be the output signals corresponding to the input signals $x_1(t)$ and $x_2(t)$, respectively, i.e.

$$y_1(t) = x_1^2(t) \text{ and } y_2(t) = x_2^2(t)$$

For any scalars c_1 and c_2 , the output signal for the input signal $x(t) = c_1x_1(t) + c_2x_2(t)$ is given by

$$y(t) = x^2(t) = [c_1x_1(t) + c_2x_2(t)]^2 = c_1^2x_1^2(t) + c_2^2x_2^2(t) + 2c_1c_2x_1(t)x_2(t)$$

But

$$c_1y_1(t) + c_2y_2(t) = c_1x_1^2(t) + c_2x_2^2(t)$$

Since $y(t) \neq c_1y_1(t) + c_2y_2(t)$, we conclude that the given system is non-linear.

- (c) Let $y_1(t)$ and $y_2(t)$ be the output signals corresponding to the input signals $x_1(t)$ and $x_2(t)$, respectively, i.e.

$$y_1(t) = tx_1(t) \quad \text{and} \quad y_2(t) = tx_2(t)$$

For any scalars c_1 and c_2 , the output signal for the input signal $x(t) = c_1x_1(t) + c_2x_2(t)$ is given by

$$\begin{aligned} y(t) &= tx(t) = t[c_1x_1(t) + c_2x_2(t)] = c_1[tx_1(t)] + c_2[tx_2(t)] \\ &= c_1y_1(t) + c_2y_2(t) \end{aligned}$$

which shows that the given system is linear. ♦

EXAMPLE 6.24. Examine whether the following systems are time-invariant:

- (a) $y(t) = \alpha x(t)$.
- (b) $y(t) = x(t) - x(t - a)$.
- (c) $y(t) = tx(t)$.

Solution. (a) If $y(t)$ is the output corresponding to the input $x(t)$, i.e. $y(t) = \alpha x(t)$, then the output corresponding to the input $x(t - t_0)$ is given by

$$\alpha x(t - t_0) = y(t - t_0)$$

which shows that the given system is time-invariant.

- (b) If $y(t)$ is the output corresponding to the input $x(t)$, i.e. $y(t) = x(t) - x(t - a)$, then the output corresponding to the input $x(t - t_0)$ is given by

$$x(t - t_0) - x(t - t_0 - a) = y(t - t_0)$$

which shows that the given system is time-invariant.

- (c) If $y(t)$ is the output corresponding to the input $x(t)$, i.e. $y(t) = tx(t)$, then the output corresponding to the input $x(t - t_0)$ is given by

$$tx(t - t_0) \neq y(t - t_0) = (t - t_0)x(t - t_0)$$

which shows that the given system is time-varying. ♦

EXAMPLE 6.25. Examine whether the following systems are causal:

- (a) $y(t) = x(t) - x(t - a)$.
- (b) $y(t) = tx(t)$.
- (c) $y(t) = x(t + 2)$.

Solution. (a) and (b) The given systems are causal because the present value of $y(t)$ depends only on the present or previous values of the input $x(t)$.

(c) The given system is not causal because the present value of $y(t)$ depends on the future values of the input $x(t)$. ♦

EXAMPLE 6.26. For a linear time-invariant (LTI) system with a WSS process $X(t)$ as the input, show that the mean value of the output is given by

$$\mu_Y = \mu_X H(0)$$

where $H(\omega)$ is the Fourier transform of the impulse response of the system.

Solution. By definition, we have

$$Y(t) = \int_{-\infty}^{\infty} h(\xi) X(t - \xi) d\xi$$

Thus, it follows that

$$\mu_Y(t) = E[Y(t)] = \int_{-\infty}^{\infty} h(\xi) E[X(t - \xi)] d\xi$$

Since $X(t)$ is a WSS process, its mean value is a constant, denoted by μ_X . For an LTI system, the output of a WSS process is a WSS process. Hence, $Y(t)$ also is a WSS process with constant mean given by

$$\mu_Y = \int_{-\infty}^{\infty} h(\xi) \mu_X d\xi = \mu_X \int_{-\infty}^{\infty} h(\xi) d\xi \quad (6.63)$$

By definition, the Fourier transform of the impulse response $h(t)$ is given by

$$H(\omega) = \int_{-\infty}^{\infty} h(\xi) e^{-i\omega\xi} d\xi \quad (6.64)$$

Taking $\omega = 0$ in Eq. (6.64), we have

$$H(0) = \int_{-\infty}^{\infty} h(\xi) d\xi \quad (6.65)$$

Combining Eqs. (6.63) and (6.65), we have

$$\mu_Y = \mu_X H(0) \quad \blacklozenge$$

EXAMPLE 6.27. Consider a linear time-invariant system with impulse response $h(t) = 4e^{-2t}u(t)$ and suppose that a WSS random process $X(t)$ with mean $\mu_X = 2$ is used as the input of the system. Find the mean value of the output of the system.

Solution. First, we find the Fourier transform of the impulse response as

$$H(\omega) = \mathcal{F}[h(t)] = \mathcal{F}[4e^{-2t}u(t)] = 4\mathcal{F}[e^{-2t}u(t)]$$

Using Table 6.1, we find that

$$H(\omega) = 4 \times \frac{1}{2 + i\omega} = \frac{4}{2 + i\omega}$$

It follows that

$$H(0) = \frac{1}{4 + 2(0)} = 2$$

Thus, the mean value of the output of the system is given by

$$\mu_Y = \mu_X H(0) = 2 \times 2 = 4$$

◆

EXAMPLE 6.28. Show that the system represented by

$$Y(t) = \int_{-\infty}^{\infty} h(\xi)X(t - \xi) d\xi$$

is a linear time-invariant system.

Solution. The given system can be represented as $Y(t) = T[X(t)]$, where T is the integral transform as defined in the problem.

First, we show that the given system T is linear.

Let $Y_1(t) = T[X_1(t)]$ and $Y_2(t) = T[X_2(t)]$. We define

$$X(t) = \alpha_1 X_1(t) + \alpha_2 X_2(t)$$

where α_1 and α_2 are any scalars.

Then, it follows that

$$Y(t) = T[X(t)] = \int_{-\infty}^{\infty} h(\xi) [\alpha_1 X_1(t - \xi) + \alpha_2 X_2(t - \xi)] d\xi$$

Thus, we have

$$\begin{aligned} Y(t) &= \alpha_1 \int_{-\infty}^{\infty} h(\xi) X_1(t - \xi) d\xi + \alpha_2 \int_{-\infty}^{\infty} h(\xi) X_2(t - \xi) d\xi \\ &= \alpha_1 Y_1(t) + \alpha_2 Y_2(t) \end{aligned}$$

which shows that the system T is linear.

Next, if we replace $X(t)$ by $X(t - t_0)$ in the definition of T , we have

$$T[X(t - t_0)] = \int_{-\infty}^{\infty} h(\xi) X(t - t_0 - \xi) d\xi = Y(t - t_0)$$

which shows that the system T is time-invariant. $\blacklozenge$

EXAMPLE 6.29. Obtain the spectral density of $\{Y(t), t \geq 0\}$, $Y(t) = AX(t)$, when A is independent of $X(t)$ such that $P(A = 1) = P(A = -1) = \frac{1}{2}$.

(Anna, April 2005)

Solution. First, we find the mean and variance of A .

The mean of A is given by

$$\mu_A = E[A] = \left(1 \times \frac{1}{2}\right) + \left(-1 \times \frac{1}{2}\right) = \frac{1}{2} - \frac{1}{2} = 0$$

Since $\mu_A = 0$, the variance of A is given by

$$\text{Var}(A) = E[A^2] = \left(1 \times \frac{1}{2}\right) + \left(1 \times \frac{1}{2}\right) = \frac{1}{2} + \frac{1}{2} = 1$$

Thus, the mean of Y is given by

$$\mu_Y(t) = E[Y(t)] = E[AX(t)] = E(A)E[X(t)] = 0 \times \mu_X(t) = 0$$

since A is independent of $X(t)$ and $E(A) = 0$.

Next, the autocorrelation function of $Y(t)$ is given by

$$R_{YY}(t, t + \tau) = E[Y(t)Y(t + \tau)] = E[AX(t) \cdot AX(t + \tau)] = E(A^2) E[X(t)X(t + \tau)]$$

since A is independent of $X(t)$.

Since $E(A^2) = 1$, the autocorrelation function of $Y(t)$ is

$$R_{YY}(\tau) = 1 R_{XX}(\tau) = R_{XX}(\tau)$$

(assuming that $X(t)$ is a WSS process).

Thus, the power spectral density of $Y(t)$ is given by

$$S_{YY}(\omega) = \mathcal{F}[R_{YY}(\tau)] = \mathcal{F}[R_{XX}(\tau)] = S_{XX}(\omega)$$

Hence, we conclude that the power spectral density of $Y(t)$ is the same as the power spectral density of $X(t)$. $\blacklozenge$

EXAMPLE 6.30. The impulse response of a linear system is given by

$$h(t) = \begin{cases} \frac{1}{T} & \text{for } 0 \leq t \leq T \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate $S_{YY}(\omega)$ in terms of $S_{XX}(\omega)$.

Solution. The Fourier transform of the impulse response is given by

$$H(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-i\omega\tau} d\tau$$

Using the definition of $h(t)$, we have

$$H(\omega) = \int_0^T \left[\frac{1}{T} \right] e^{-i\omega\tau} d\tau = \frac{1}{T} \int_0^T e^{-i\omega\tau} d\tau$$

Integrating, we have

$$H(\omega) = \frac{1}{T} \left[\frac{e^{-i\omega T}}{-i\omega} \right]_0^T = \frac{1}{i\omega T} [1 - e^{-i\omega T}]$$

which can be written as

$$\begin{aligned} H(\omega) &= \frac{e^{-i\frac{\omega T}{2}}}{i\omega T} \left[e^{i\frac{\omega T}{2}} - e^{-i\frac{\omega T}{2}} \right] = \frac{e^{-i\frac{\omega T}{2}}}{i\omega T} \left[2i \sin \frac{\omega T}{2} \right] \\ &= e^{-i\frac{\omega T}{2}} \frac{\sin \left(\frac{\omega T}{2} \right)}{\frac{\omega T}{2}} \end{aligned}$$

Thus, it follows that

$$|H(\omega)|^2 = \frac{\sin^2 \left(\frac{\omega T}{2} \right)}{\left(\frac{\omega T}{2} \right)^2}$$

By Theorem 6.10, we know that

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

Thus, we have

$$S_{YY}(\omega) = \frac{\sin^2 \left(\frac{\omega T}{2} \right)}{\left(\frac{\omega T}{2} \right)^2} S_{XX}(\omega)$$

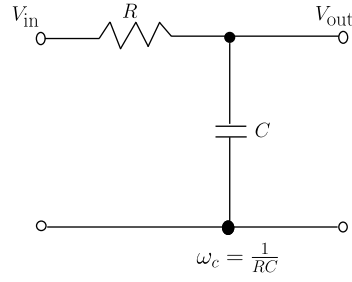
◆

EXAMPLE 6.31. An RC low pass filter as illustrated in Figure 6.2 has a cut-off frequency $\omega_c = \frac{1}{RC}$ and transfer function

$$H(\omega) = \frac{1}{1 + i\frac{\omega}{\omega_c}}$$

If a white noise of spectral density $\frac{N_0}{2}$ is applied to the low pass filter, find the input/output autocorrelation functions and the output spectral density function.

(Madras, Oct. 1999; April 2001; Oct. 2002)

**Figure 6.2:** RC Low Pass Filter.

Solution. The input autocorrelation function is given by

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega) e^{i\omega\tau} d\omega$$

We are given that the input process is a white noise of spectral density $\frac{N_0}{2}$, i.e.

$$S_{XX}(\omega) = \frac{N_0}{2}$$

Thus, it follows that

$$R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{N_0}{2} \right] e^{i\omega\tau} d\omega$$

Using Table 6.1, we have

$$R_{XX}(\tau) = \frac{N_0}{2} \mathcal{F}^{-1}(1) = \frac{N_0}{2} \delta(\tau)$$

By Theorem 6.10, we know that

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

Given that

$$H(\omega) = \frac{1}{1 + i\omega\omega_c^{-1}} = \frac{1}{1 + i\omega RC}$$

Thus, we have

$$H^*(\omega) = \frac{1}{1 - i\omega RC}$$

It follows that

$$|H(\omega)|^2 = H(\omega)H^*(\omega) = \frac{1}{1 + \omega^2 R^2 C^2}$$

Hence, the output spectral density function is

$$\begin{aligned} S_{YY}(\omega) &= S_{XX}(\omega) |H(\omega)|^2 = \frac{N_0}{2} \frac{1}{1 + \omega^2 R^2 C^2} = \frac{N_0}{2} \frac{1}{1 + \frac{\omega^2}{\beta^2}} \\ &= \frac{N_0}{2} \frac{\beta^2}{\beta^2 + \omega^2} = \frac{N_0 \beta}{4} \frac{2\beta}{\beta^2 + \omega^2} \end{aligned}$$

Thus, the output autocorrelation function is given by

$$R_{YY}(\tau) = \mathcal{F}^{-1}[S_{YY}(\omega)] = \frac{N_0\beta}{4} \mathcal{F}^{-1}\left[\frac{2\beta}{\beta^2 + \omega^2}\right] = \frac{N_0\beta}{4} e^{-\beta|\tau|}$$

i.e.

$$R_{YY}(\tau) = \frac{N_0}{4RC} e^{-\frac{|\tau|}{RC}}$$

◆

EXAMPLE 6.32. Consider a circuit with input voltage $X(t)$ and output voltage $Y(t)$. If $X(t)$ is a stationary random process with zero mean and autocorrelation function

$$R_{XX}(\tau) = 3e^{-2|\tau|}$$

and if the system transfer function is

$$H(\omega) = \frac{1}{1 + i\omega}$$

find the following:

- (a) The mean of $Y(t)$.
- (b) The input power spectral density function $S_{XX}(\omega)$.
- (c) The output power spectral density function $S_{YY}(\omega)$.

Solution. (a) The mean of $Y(t)$ is given by

$$\mu_Y = H(0)\mu_X = 1 \times 0 = 0$$

- (b) The input power spectral density is

$$S_{XX}(\omega) = \mathcal{F}[R_{XX}(\tau)] = \mathcal{F}[3e^{-2|\tau|}] = 3\mathcal{F}[e^{-2|\tau|}] = \frac{12}{4 + \omega^2}$$

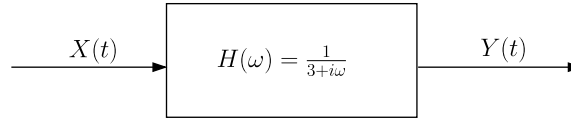
- (c) The output power spectral density is given by

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega) = \frac{1}{1 + \omega^2} \times \frac{12}{4 + \omega^2} = \frac{12}{(1 + \omega^2)(4 + \omega^2)}$$

◆

EXAMPLE 6.33. Consider a linear system as shown in Figure 6.3, where $X(t)$ is the input and $Y(t)$ is the output of the system. The autocorrelation of the input signal is $R_{XX}(\tau) = 2\delta(\tau)$. Find the following:

- (a) The power spectral density of $Y(t)$.
- (b) The autocorrelation function of $Y(t)$.
- (c) The mean square value of the output $Y(t)$.

**Figure 6.3:** Linear system.

Solution. (a) The power spectral density of the output $Y(t)$ is given by

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$$

We find that

$$S_{XX}(\omega) = \mathcal{F}[R_{XX}(\tau)] = \mathcal{F}[2\delta(\tau)] = 2$$

We also find that

$$|H(\omega)|^2 = H^*(\omega)H(\omega) = \frac{1}{3-i\omega} \frac{1}{3+i\omega} = \frac{1}{9+\omega^2}$$

Thus, it follows that

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega) = \frac{2}{9+\omega^2}$$

(b) The autocorrelation function of the output $Y(t)$ is given by

$$R_{YY}(\tau) = \mathcal{F}^{-1}[S_{YY}(\omega)] = \mathcal{F}^{-1}\left[\frac{2}{9+\omega^2}\right] = \frac{1}{3} \mathcal{F}^{-1}\left[\frac{2(3)}{3^2+\omega^2}\right] = \frac{1}{3} e^{-3|\tau|}$$

(c) The mean square value of the output $Y(t)$ is given by

$$E[Y^2(t)] = R_{YY}(0) = \frac{1}{3}$$

◆

PROBLEM SET 6.3

1. Examine whether the following systems are linear:

- (a) $y(t) = x^3(t)$
- (b) $y(t) = \alpha x(t) + \beta$, where α and β are scalars.
- (c) $y(t) = x(2t)$

2. Examine whether the following systems are time-invariant:

- (a) $y(t) = e^{-t}x(t)$
- (b) $y(t) = x(t+a)$, where a is a scalar
- (c) $y(t) = x^2(t)$

3. Examine whether the following systems are causal:

- (a) $y(t) = ax(t) - bx(t-c)$, where a, b and c are scalars
- (b) $y(t) = x(2t)$
- (c) $y(t) = x^2(t)$

4. Consider a linear time-invariant system with impulse response $h(t) = 4e^{-t} \cos 2t u(t)$ and suppose that a WSS random process $X(t)$ with mean $\mu_X = 3$ is used as the input of the system. Find the mean value of the output of the system.
5. Consider a linear time-invariant system with impulse response $h(t) = 3e^{-2t} \sin 2t u(t)$ and suppose that a WSS random process $X(t)$ with mean $\mu_X = 4$ is used as the input of the system. Find the mean value of the output of the system.
6. For a circuit with input voltage $X(t)$ and output voltage $Y(t)$, it is given that $X(t)$ is a stationary random process with zero mean and autocorrelation function

$$R_{XX}(\tau) = 4e^{-3|\tau|}$$

and that the system transfer function is

$$H(\omega) = \frac{1}{2 + i\omega}$$

Then, find the following:

- (a) The mean of $Y(t)$.
 - (b) The input power spectral density function $S_{XX}(\omega)$.
 - (c) The output power spectral density function $S_{YY}(\omega)$.
7. Consider a linear system as shown in Figure 6.4, where $X(t)$ is the input and $Y(t)$ is the output of the system. The autocorrelation of the input signal is $R_{XX}(\tau) = 4\delta(\tau)$. Find the following:
 - (a) The power spectral density of $Y(t)$.
 - (b) The autocorrelation function of $Y(t)$.
 - (c) The mean-square value of the output $Y(t)$.

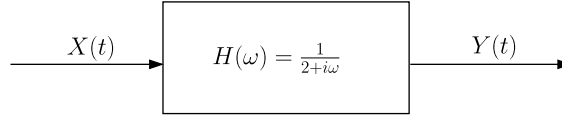


Figure 6.4: Linear System, Exercise 6.7.

8. Consider a linear system with transfer function $H(\omega) = \frac{1}{1 + i\omega}$. If an input signal with autocorrelation function $R(\tau) = c\delta(\tau)$ is applied as input $X(t)$ to the system, find the following:
 - (a) The power spectral density of $Y(t)$
 - (b) The autocorrelation function of $Y(t)$
 - (c) The mean-square value of the output $Y(t)$
9. Consider a RL -circuit with transfer function

$$H(\omega) = \frac{R}{R + i\omega L}$$

and with an input signal of autocorrelation function $R_{XX}(\tau) = e^{-\beta|\tau|}$. Find the following:

- (a) The mean of $Y(t)$
- (b) The power spectral density of $Y(t)$
- (c) The autocorrelation function of $Y(t)$
- (d) The mean-square value of the output $Y(t)$

- 10.** Consider a linear system with input power density spectrum

$$S_{XX}(\omega) = 98\pi\delta(\omega) + \frac{12}{9 + \omega^2}$$

and system transfer function

$$H(\omega) = \frac{6}{\omega^2 + 49}$$

Find the mean value of the output process.

Chapter 7

Queueing Theory

7.1 BASIC CHARACTERISTICS OF QUEUEING MODELS

Queueing theory is the mathematical study of waiting lines or queues. Queueing theory is generally considered as a branch of Operations Research (OR) as the Queueing models are often used in business for making decisions about the resources needed for providing service to customers. Queueing theory had its origin in 1909 when Agner Krarup Erlang (1878–1929), a Danish engineer who worked for the Copenhagen Telephone Exchange, published the first paper (Erlang, 1909) on Queueing theory relating to the study of congestion in telephone traffic. Queueing theory has applications in intelligent transportation systems, telecommunications, networks and traffic flow.

The fundamental goal of Queueing theory is to derive an analytical or mathematical model of customers needing service and use that model to predict queue lengths and waiting times. A queue or waiting line is formed when units (customers or clients) requiring some kind of service arrive at a service counter or service channel. A simple Queueing model is shown in Figure 7.1.

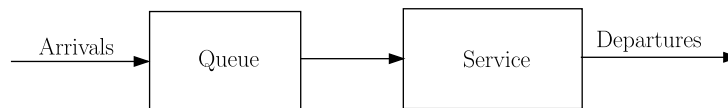


Figure 7.1: A Simple Queueing Model.

The basic characteristics of a Queueing system are: (i) The input or arrival pattern, (ii) The service mechanism (or service pattern), and (iii) The queue discipline, and (iv) The system capacity. A brief explanation of these characteristics are given now.

1. Input or Arrival Pattern: The input describes the manner in which customers (or units) arrive and join the Queueing system. It is not possible to observe or detect the actual amount of customers arriving at the queue for service. Hence, the arrival rate (i.e. the number of arrivals in one time period) or the interval between two successive arrivals cannot be treated as a constant, but only as a random variable. Thus, we express the arrival pattern of customers by the probability distribution of the number of arrivals per unit of time or of the inter-arrival time.

In this chapter, we consider only those Queueing systems in which the number of arrivals per unit of time is a Poisson random variable with mean λ . We know that in this case, the time between

consecutive arrivals, i.e. the *inter-arrival time* of the Poisson process, has an exponential distribution with mean $\frac{1}{\lambda}$ (see Theorem 5.20).

The number of customers may be from finite or infinite sources. Also, the input process should indicate the number of queues that are allowed to form, the maximum queue-length, the maximum number of customers requiring service, etc.

2. Service Mechanism: It can be described by a service rate (i.e. the number of customers serviced in one time period) or by the inter-service time (i.e. the time required to complete the service for a customer).

In this chapter, we consider only those Queueing systems in which the number of customers serviced per unit of time has a Poisson distribution with mean μ or, equivalently, the inter-service time has an exponential distribution with mean $\frac{1}{\mu}$.

The service pattern should also indicate the number of servers and whether the service counters are arranged in parallel or in series, etc.

3. Queue Discipline: In this procedure, the customers are selected for service when a queue is formed. The various types of queue disciplines are tabulated in the following table.

No.	Queue Discipline	Description
1	FIFO or FCFS	<i>First In First Out or First Come First Served.</i> This is the most commonly used procedure in servicing customers.
2	LIFO or LCFS	<i>Last In First Out or Last Come First Served.</i> This procedure is used in inventory systems.
3	SIRO	<i>Selection for service In Random Order.</i>
4	PIR	<i>Priority in Selection</i> , i.e. customers are prioritized upon arrival. This procedure is used in manual transmission messaging systems.

In this chapter, we shall assume that service is provided by the FCFS or FIFO procedure, i.e. on a first come first served basis.

4. System Capacity: The maximum number of customers in the Queueing system can be either finite or infinite. In some Queueing models, only limited customers or units are allowed in the system. When the limiting value is reached, no further customers are allowed to enter the Queueing system.

7.1.1 Transient and Steady-States

A Queueing system is said to be in *transient state* when the operating characteristics of the system depend on time. A Queueing system is said to be in *steady-state* when the operating characteristics of the system are independent of time. For example, if $P_n(t)$ is the probability that there are n customers in the system at time t (where $n > 0$), finding $P_n(t)$ is quite a difficult task even for a simple case. Thus, we are most interested in the steady-state analysis of the system, i.e. in determining $P_n(t)$ in the long-run, i.e. as $t \rightarrow \infty$. Thus, we look for steady-state probabilities, i.e. $P_n(t) \rightarrow P_n$ (a constant) as $t \rightarrow \infty$.

7.1.2 Kendall's Notation of a Queueing System

Kendall's notation (Kendall, 1951) provides a very convenient description of Queueing systems, and is universally accepted and used by statisticians. This notation of a Queueing system has the form

$$(a/b/c) : (d/e)$$

where

a = Inter-arrival distribution

b = Service time distribution

c = Number of channels or servers

d = System capacity

e = Queue discipline

In Kendall's notation, a and b usually take one of the following symbols:

M : for Markovian or Exponential distribution

G : for arbitrary or general distribution

D : for fixed or deterministic distribution

The four important Queueing systems described in Kendall's notation are as follows:

(i) $(M/M/1) : (\infty/\text{FIFO})$

(ii) $(M/M/s) : (\infty/\text{FIFO})$

(iii) $(M/M/1) : (k/\text{FIFO})$

(iv) $(M/M/s) : (k/\text{FIFO})$

We shall study about those models in the following sections. Next, we define the *line length* and *queue length* for a queue.

Definition 7.1. For a queue, the *line length* or *queue size* is defined as the number of customers in the Queueing system. Also, the *queue length* is defined as the difference between the line length and the number of customers being served, i.e.

$$\text{Queue length} = \text{Line length} - \text{Number of customers being served}$$

Terminology:

n	=	Number of customers (units) in the system
$N(t)$	=	Number of customers (units) in the system at time t
$P_n(t)$	=	The probability that there are exactly n customers at time t , i.e. $P[N(t) = n]$
P_n	=	The steady state probability that exactly n customers are in the system
λ_n	=	Mean arrival rate when there are n customers in the system
μ_n	=	Mean service rate when there are n customers in the system
λ	=	Mean arrival rate when λ_n is constant for all n
μ	=	Mean service rate when μ_n is constant for all n
$\rho = \frac{\lambda}{\mu}$	=	Utilization factor or traffic intensity

- $f_s(w)$ = The probability density function of waiting time in the system
 $f_q(w)$ = The probability density function of waiting time in the queue
 L_s = The expected number of customers in the system or the average line length
 L_q = The expected number of customers in the queue or the average queue length
 L_w = The expected number of customers in non-empty queues
 W_s = The expected waiting time of a customer in the system
 W_q = The expected waiting time of a customer in the queue

7.1.3 Transient State Probabilities for Poisson Queue Systems

We now derive the differential equations for the transient state probabilities for the Poisson queue systems, which are also known as *birth-death process* or *immigration-emigration process*.

Let $N(t)$ be the number of customers (or units) in the system at time t and $P_n(t)$ be the probability that there are n customers in the system at time t , where $n \geq 1$, i.e.

$$P_n(t) = P[N(t) = n]$$

First, we derive the differential equation satisfied by $P_n(t)$, and then derive the difference equation satisfied by P_n (in the steady-state).

As we have a Poisson queue model with mean arrival rate λ_n and mean service rate μ_n , we make the following assumptions:

- (i) When $N(t) = n$, the probability of an arrival in $(t, t + \Delta t)$ is $\lambda_n \Delta t + o(\Delta t)$.
- (ii) When $N(t) = n$, the probability of a departure in $(t, t + \Delta t)$ is $\mu_n \Delta t + o(\Delta t)$.
- (iii) When $N(t) = n$, the probability of more than one arrival and/or more than one departure in $(t, t + \Delta t)$ is $o(\Delta t)$.

Next, we find the differential equation satisfied by $P_n(t)$. For this purpose, we first find a formula for $P_n(t + \Delta t)$, i.e. $P[N(t + \Delta t) = n]$.

The event $\{N(t + \Delta t) = n\}$ can happen in a number of mutually exclusive ways. By assumption (iii), the events involving more than one arrival and/or more than one departure in $(t, t + \Delta t)$ is $o(\Delta t)$. There will remain four mutually exclusive events facilitating $\{N(t + \Delta t) = n\}$, which are described as follows:

A_{00} : $N(t) = n$ and no arrival or departure in $(t, t + \Delta t)$.

A_{10} : $N(t) = n - 1$ and one arrival and no departure in $(t, t + \Delta t)$.

A_{01} : $N(t) = n + 1$ and no arrival and one departure in $(t, t + \Delta t)$.

A_{11} : $N(t) = n$ and one arrival and one departure in $(t, t + \Delta t)$.

We find that

$$\begin{aligned}
 P(A_{00}) &= P_n(t) [1 - \lambda_n \Delta t + o(\Delta t)] [1 - \mu_n \Delta t + o(\Delta t)] \\
 &= P_n(t) [1 - (\lambda_n + \mu_n) \Delta t + o(\Delta t)] \\
 P(A_{10}) &= P_{n-1}(t) [\lambda_{n-1} \Delta t + o(\Delta t)] [1 - \mu_{n-1} \Delta t + o(\Delta t)] \\
 &= P_{n-1}(t) \lambda_{n-1} \Delta t + o(\Delta t) \\
 P(A_{01}) &= P_{n+1}(t) [1 - \lambda_{n+1} \Delta t + o(\Delta t)] [\mu_{n+1} \Delta t + o(\Delta t)]
 \end{aligned}$$

$$\begin{aligned}
&= P_{n+1}(t)\mu_{n+1}\Delta t + o(\Delta t) \\
P(A_{11}) &= P_n(t) [\lambda_n\Delta t + o(\Delta t)] [\mu_n\Delta t + o(\Delta t)] \\
&= o(\Delta t)
\end{aligned}$$

Thus, for $n \geq 1$, we have

$$\begin{aligned}
P_n(t + \Delta t) &= P[N(t + \Delta t) = n] = P(A_{00}) + P(A_{10}) + P(A_{01}) + P(A_{11}) \\
P_n(t + \Delta t) &= P_n(t)[1 - (\lambda_n + \mu_n)\Delta t] + \lambda_{n-1}P_{n-1}(t)\Delta t + \mu_{n+1}P_{n+1}(t)\Delta t + o(\Delta t)
\end{aligned}$$

Thus,

$$\frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} = -(\lambda_n + \mu_n)P_n(t) + \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t) + \frac{o(\Delta t)}{\Delta t}$$

Taking limits as $\Delta \rightarrow 0$, it follows that

$$P'_n(t) = -(\lambda_n + \mu_n)P_n(t) + \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t) \quad (\text{where } n \geq 1) \quad (7.1)$$

For $n = 0$, we have

$$\begin{aligned}
P_0(t + \Delta t) &= P_0(t)[1 - \lambda_0\Delta t + o(\Delta t)] + P_1(t)[1 - \lambda_1\Delta t + o(\Delta t)][\mu_1\Delta t + o(\Delta t)] \\
&= P_0(t)[1 - \lambda_0\Delta t] + P_1(t)\mu_1\Delta t + o(\Delta t)
\end{aligned}$$

It follows that

$$\frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = -\lambda_0P_0(t) + \mu_1P_1(t) + \frac{o(\Delta t)}{\Delta t}$$

Taking limits as $\Delta t \rightarrow 0$, we get

$$P'_0(t) = -\lambda_0P_0(t) + \mu_1P_1(t) \quad (7.2)$$

Equations (7.1) and (7.2) together constitute a system of differential equations. The solution of these differential equations yields the transient state probability, $P_n(t)$, for $n \geq 0$, but this is exceedingly difficult even for simple problems.

7.1.4 Steady State Probabilities for Poisson Queue Systems

The steady state probabilities for Poisson queue systems are derived by assuming that $P_n(t) \rightarrow P_n$, independent of t , as $t \rightarrow \infty$. The equations of steady state probabilities, P_n , can be obtained by putting $P'_n(t) = 0$ and replacing $P_n(t)$ by P_n in Eqs. (7.1) and (7.2).

Thus, we obtain

$$0 = -(\lambda_n + \mu_n)P_n + \lambda_{n-1}P_{n-1} + \mu_{n+1}P_{n+1} \quad (7.3)$$

and

$$0 = -\lambda_0P_0 + \mu_1P_1 \quad (7.4)$$

Equations (7.3) and (7.4) are called *balance equations* or *equilibrium equations* of Poisson queue systems.

Next, we obtain the values of the steady state probabilities, P_0 and P_n , for Poisson queue systems. From Eq. (7.4), we have

$$P_1 = \frac{\lambda_0}{\mu_1} P_0 \quad (7.5)$$

Putting $n = 1$ in Eq. (7.3) and using Eq. (7.5), we have

$$\begin{aligned} \mu_2 P_2 &= (\lambda_1 + \mu_1) P_1 - \lambda_0 P_0 \\ &= (\lambda_1 + \mu_1) \frac{\lambda_0}{\mu_1} P_0 - \lambda_0 P_0 \\ &= \frac{\lambda_0 \lambda_1}{\mu_1} P_0 \end{aligned}$$

Thus, we have

$$P_2 = \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} P_0 \quad (7.6)$$

Next, putting $n = 2$ in (7.3) and using (7.5) and (7.6), we have

$$\begin{aligned} \mu_3 P_3 &= (\lambda_2 + \mu_2) P_2 - \lambda_1 P_1 \\ &= (\lambda_2 + \mu_2) \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} P_0 - \frac{\lambda_0 \lambda_1}{\mu_1} P_0 \\ &= \frac{\lambda_0 \lambda_1 \lambda_2}{\mu_1 \mu_2} P_0 \end{aligned}$$

Thus, we have

$$P_3 = \frac{\lambda_0 \lambda_1 \lambda_2}{\mu_1 \mu_2 \mu_3} P_0 \quad (7.7)$$

Using the principle of mathematical induction, it can be easily established that

$$P_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} P_0 \quad (7.8)$$

Since the number of customers in the system can be $0, 1, 2, \dots, \infty$, it follows that

$$P_0 + P_1 + \cdots + P_n + \cdots = P_0 + \sum_{n=1}^{\infty} P_n = 1$$

Thus, it follows that

$$P_0 + \sum_{n=1}^{\infty} \left[\frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right] P_0 = 1$$

Hence, we find that

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \left[\frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right]} \quad (7.9)$$

Equations (7.8) and (7.9) play an important role in the derivation of the important characteristics of various Poisson queue models.

7.2 MODEL I—(M/M/1): (∞ /FIFO) SINGLE SERVER WITH INFINITE CAPACITY

For this model, we make the following assumptions:

- (A1) The mean arrival rate is constant, i.e. $\lambda_n = \lambda$ for all n .
- (A2) The mean service rate is constant, i.e. $\mu_n = \mu$ for all n .
- (A3) The mean arrival rate is less than the mean service rate, i.e. $\lambda < \mu$ or equivalently that

$$\rho = \frac{\lambda}{\mu} < 1$$

The assumption that the traffic intensity $\rho < 1$ ensures that an infinite queue will not build up.

Under assumption (A3), Eq. (7.9) simplifies to

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \left[\frac{\lambda}{\mu}\right]^n} = \frac{1}{1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \dots} = \frac{1}{\left[1 - \frac{\lambda}{\mu}\right]^{-1}} = 1 - \frac{\lambda}{\mu} = 1 - \rho \quad (7.10)$$

Substituting Eq. (7.10) into Eq. (7.8), we have

$$P_n = \left[\frac{\lambda}{\mu}\right]^n P_0 = \rho^n (1 - \rho) \quad (7.11)$$

7.2.1 Characteristics of Model I

1. Average or expected number of customers in the system (L_s): If N denotes the number of customers in the Queueing system, then N is a discrete random variable taking values $N = 0, 1, 2, \dots$ and we know that

$$P_n = P(N = n) = \rho^n (1 - \rho) \text{ for } n = 0, 1, 2, \dots$$

Thus, the expected number of customers in the system is given by

$$L_s = E(N) = \sum_{n=0}^{\infty} n P_n = \sum_{n=0}^{\infty} n \rho^n (1 - \rho) = (1 - \rho) \sum_{n=1}^{\infty} n \rho^n$$

We note that

$$L_s = \rho(1 - \rho) \sum_{n=1}^{\infty} n \rho^{n-1} = \rho(1 - \rho) [1 + 2\rho + 3\rho^2 + \dots] = \rho(1 - \rho) (1 - \rho)^{-2} = \rho(1 - \rho)^{-1}$$

Thus, we have

$$L_s = \frac{\rho}{1 - \rho} = \frac{\frac{\lambda}{\mu}}{1 - \frac{\lambda}{\mu}} = \frac{\lambda}{\mu - \lambda} \quad (7.12)$$

2. Average or expected number of customers in the queue or average length of the queue (L_q): By definition, the queue length is given by

$$\text{Queue length} = \text{Line length} - \text{Number of customers being served}$$

Thus, if the number of customers in the system is n , then the number of customers in the queue or the queue length is $n - 1$.

Hence, the expected length of the queue is

$$L_q = E(N - 1) = \sum_{n=1}^{\infty} (n - 1)P_n = \sum_{n=1}^{\infty} (n - 1)\rho^n(1 - \rho) = \sum_{n=2}^{\infty} (n - 1)\rho^n(1 - \rho) = \sum_{m=1}^{\infty} m\rho^{m+1}(1 - \rho)$$

by substituting $m = n - 1$ in the summation.

Thus, it follows that

$$L_q = \rho^2(1 - \rho) \sum_{m=1}^{\infty} m\rho^{m-1} = \rho^2(1 - \rho)(1 - \rho)^{-2} = \rho^2(1 - \rho)^{-1}$$

Hence, we have

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{\frac{\lambda^2}{\mu^2}}{1 - \frac{\lambda}{\mu}} = \frac{\lambda^2}{\mu(\mu - \lambda)} \quad (7.13)$$

3. Average or expected number of customers in nonempty queue (L_w): It is given by

$$L_w = E\{N - 1 | N - 1 > 0\} = \frac{E(N - 1)}{P(N - 1 > 0)} \quad (7.14)$$

We know that

$$E(N - 1) = L_q = \frac{\rho^2}{1 - \rho}$$

and

$$P(N - 1 > 0) = 1 - P_0 - P_1 = 1 - (1 - \rho) - [\rho(1 - \rho)] = \rho - \rho(1 - \rho) = \rho^2$$

Substituting the above in Eq. (7.14), we get

$$L_w = \frac{\frac{\rho^2}{1 - \rho}}{\rho^2} = \frac{1}{1 - \rho} = \frac{1}{1 - \frac{\lambda}{\mu}} = \frac{\mu}{\mu - \lambda} \quad (7.15)$$

4. The probability that the number of customers in the system exceeds k : It is given by

$$P(N > k) = \sum_{n=k+1}^{\infty} P_n = \sum_{n=k+1}^{\infty} \rho^n(1 - \rho) = \rho^{k+1}(1 - \rho) \sum_{n=k+1}^{\infty} \rho^{n-k-1} = \rho^{k+1}(1 - \rho) \sum_{m=0}^{\infty} \rho^m$$

by substituting $m = n - k - 1$ in the summation.

Thus, it follows that

$$P(N > k) = \rho^{k+1}(1 - \rho)(1 - \rho)^{-1} = \rho^{k+1} = \left(\frac{\lambda}{\mu}\right)^{k+1} \quad (7.16)$$

5. Probability density function of the waiting time of a customer in the system ($f_s(w)$): Let T_s be the random variable that represents the waiting time of a customer in the system, $f_s(w)$ be the probability density function of T_s , and $f_s(w|n)$ be the probability density function of T_s given that there are already n customers in the system when the customer arrives.

Then it is clear that

$$f_s(w) = \sum_{n=0}^{\infty} f_s(w|n)P_n \quad (7.17)$$

where $P_n = P(N(t) = n)$, i.e. P_n represents the probability that there are n customers in the system.

Since the underlying Queueing model has a single server, it is clear that

$$f_s(w|n) = \text{PDF of sum of } (n+1) \text{ service times}$$

which follows because $f_s(w|n)$ is the probability density function of n complete service times and the service time of the customer who is being served.

Since each service time is an exponential random variable with parameter μ and the service times are independent of each other, it follows that $f_s(w|n)$ is the PDF of sum of $(n+1)$ identically and independently distributed exponential random variables with parameter μ , i.e.

$$f_s(w|n) = \frac{\mu^{n+1}}{n!} e^{-\mu w} w^n \quad (\text{where } w > 0) \quad (7.18)$$

which is the probability density function of the gamma distribution with parameters $n+1$ and μ . (By Example 4.65, we know that if $X_1, X_2, \dots, X_n$ are independent and identically distributed exponential random variables each with parameter μ , then $X_1 + X_2 + \dots + X_n$ is a gamma variable with parameters n and μ .)

Substituting Eqs. (7.11) and (7.18) into Eq. (7.17), we obtain

$$f_s(w) = \sum_{n=0}^{\infty} \frac{\mu^{n+1}}{n!} e^{-\mu w} w^n \left[\frac{\lambda}{\mu} \right]^n \left[1 - \frac{\lambda}{\mu} \right] = \sum_{n=0}^{\infty} \frac{\mu^{n+1}}{n!} e^{-\mu w} w^n \frac{\lambda^n}{\mu^n} \frac{(\mu - \lambda)}{\mu}$$

Simplifying, we have

$$f_s(w) = (\mu - \lambda) e^{-\mu w} \sum_{n=0}^{\infty} \frac{(\lambda w)^n}{n!} = (\mu - \lambda) e^{-\mu w} e^{\lambda w} \quad \text{where } w > 0$$

Thus, the probability density function of the waiting time of the customer in the system is

$$f_s(w) = (\mu - \lambda) e^{-(\mu - \lambda)w}, \quad \text{where } w > 0 \quad (7.19)$$

which is the probability density function of the exponential random variable with parameter $\mu - \lambda$.

The distribution function of T_s is given by

$$F_s(w) = P(T_s \leq w) = \begin{cases} 0 & \text{if } w < 0 \\ 1 - e^{-(\mu - \lambda)w} & \text{if } w \geq 0 \end{cases}$$

6. Average or expected waiting time of a customer in the system (W_s): The waiting time of a customer in the system is the random variable T_s , which is exponentially distributed with parameter $\mu - \lambda$ as given in Eq. (7.19). Thus, it is immediate that the average waiting time of the customer in the system is given by

$$W_s = E[T_s] = \frac{1}{\mu - \lambda}$$

7. The probability that the waiting time of a customer in the system exceeds t :

$$P(T_s > t) = \int_{w=t}^{\infty} f_s(w) dw = \int_{w=t}^{\infty} (\mu - \lambda) e^{-(\mu-\lambda)w} dw = \int_{w=t}^{\infty} d \left[-e^{-(\mu-\lambda)w} \right]$$

Integrating, we have

$$P(T_s > t) = \left[-e^{-(\mu-\lambda)w} \right]_{w=t}^{\infty} = e^{-(\mu-\lambda)t}$$

8. Probability density function of the waiting time of a customer in the queue ($f_q(w)$): Let T_q be the random variable that represents the waiting time of a customer in the queue, $f_q(w)$ be the probability density function of T_q , and $f_q(w|n)$ be the probability density function of T_q given that there are already n customers in the system when the customer arrives [or equivalently that one customer is being served and $n - 1$ customers are waiting in the queue when the customer arrives].

Then it is clear that

$$f_q(w) = \sum_{n=1}^{\infty} f_q(w|n) P_n \quad (7.20)$$

where $P_n = P(N(t) = n)$, i.e. P_n represents the probability that there are n customers in the system.

Since the underlying Queueing model has a single server, it is clear that

$$f_q(w|n) = \text{PDF of sum of } n \text{ service times}$$

which follows because $f_q(w|n)$ is the probability density function of $n - 1$ completed service times and the service time of the customer who is being served.

Since each service time is an exponential random variable with parameter μ and the service times are independent of each other, it follows that $f_q(w|n)$ is the P.D.F. of sum of n identically and independently distributed exponential random variables with parameter μ , i.e.

$$f_q(w|n) = \frac{\mu^n}{(n-1)!} e^{-\mu w} w^{n-1} \quad (\text{where } w > 0) \quad (7.21)$$

which is the probability density function of the gamma distribution with parameters n and μ . (By Example 4.65, we know that if $X_1, X_2, \dots, X_n$ are independent and identically distributed exponential random variables each with parameter μ , then $X_1 + X_2 + \dots + X_n$ is a gamma variable with parameters n and μ .)

Substituting Eqs. (7.11) and (7.21) into Eq. (7.20), we obtain

$$f_q(w) = \sum_{n=1}^{\infty} \frac{\mu^n}{(n-1)!} e^{-\mu w} w^{n-1} \left[\frac{\lambda}{\mu} \right]^n \left[1 - \frac{\lambda}{\mu} \right] = \sum_{n=1}^{\infty} \frac{\mu^{n+1}}{n!} e^{-\mu w} w^n \frac{\lambda^n}{\mu^n} \frac{(\mu - \lambda)}{\mu}$$

Simplifying, we have

$$f_q(w) = \frac{\lambda}{\mu} (\mu - \lambda) e^{-\mu w} \sum_{n=1}^{\infty} \frac{(\lambda w)^{n-1}}{(n-1)!} = \frac{\lambda}{\mu} (\mu - \lambda) e^{-\mu w} e^{\lambda w} = \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu-\lambda)w} \quad \text{where } w > 0$$

It should be noted carefully that T_q is not an exponential random variable. In fact, T_q is not even a continuous random variable because it has a non-zero probability at $w = 0$. Indeed,

$$P(T_q = 0) = P(\text{No customer in the queue}) = P_0 = 1 - \rho = 1 - \frac{\lambda}{\mu}$$

Thus, the probability density function of the waiting time of the customer in the queue is

$$f_q(w) = \begin{cases} \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu - \lambda)w} & \text{for } w > 0 \\ 1 - \frac{\lambda}{\mu} & \text{for } w = 0 \\ 0 & \text{for } w < 0 \end{cases} \quad (7.22)$$

The distribution function of the waiting time of the customer in the queue is

$$F_q(w) = P(T_q \leq w) = \begin{cases} 0 & \text{for } w < 0 \\ 1 - \frac{\lambda}{\mu} & \text{for } w = 0 \\ 1 - \frac{\lambda}{\mu} + \frac{\lambda}{\mu} e^{-(\mu - \lambda)w} & \text{for } w > 0 \end{cases}$$

9. The probability that the waiting time of a customer in the queue exceeds t :

$$P(T_q > t) = \int_t^{\infty} f_q(w) dw = \int_t^{\infty} \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu - \lambda)w} dw = \frac{\lambda}{\mu} \int_t^{\infty} d[-e^{-(\mu - \lambda)w}]$$

Integrating, we have

$$P(T_q > t) = \frac{\lambda}{\mu} [-e^{-(\mu - \lambda)w}]_t^{\infty} = \frac{\lambda}{\mu} [0 + e^{-(\mu - \lambda)t}] = \frac{\lambda}{\mu} e^{-(\mu - \lambda)t}$$

10. The average waiting time of a customer in the queue (W_q):

$$W_q = E(T_q) = 0 \times f_q(0) + \int_0^{\infty} w f_q(w) dw = \int_0^{\infty} w \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu - \lambda)w} dw$$

Thus, we have

$$W_q = \frac{\lambda}{\mu} \int_0^{\infty} w d[-e^{-(\mu - \lambda)w}]$$

Integrating by parts,

$$\begin{aligned} W_q &= \frac{\lambda}{\mu} \left\{ w [-e^{-(\mu - \lambda)w}] \Big|_{w=0}^{\infty} - \int_0^{\infty} [-e^{-(\mu - \lambda)w}] dw \right\} = \frac{\lambda}{\mu} \int_0^{\infty} e^{-(\mu - \lambda)w} dw \\ &= \frac{\lambda}{\mu} \left[\frac{-e^{-(\mu - \lambda)w}}{\mu - \lambda} \right]_0^{\infty} = \frac{\lambda}{\mu} \left[0 + \frac{1}{\mu - \lambda} \right] = \frac{\lambda}{\mu(\mu - \lambda)} \end{aligned}$$

11. The average waiting time of a customer in the queue, if he/she has to wait: It is given by

$$E[T_q | T_q > 0] = \frac{E(T_q)}{P(T_q > 0)} = \frac{W_q}{1 - P(T_q = 0)}$$

Since $W_q = \frac{\lambda}{\mu(\mu-\lambda)}$ and $P(T_q = 0) = 1 - \frac{\lambda}{\mu}$, we have

$$E[T_q | T_q > 0] = \frac{\frac{\lambda}{\mu(\mu-\lambda)}}{\frac{\lambda}{\mu}} = \frac{1}{\mu - \lambda}$$

Theorem 7.1. (Little's Formulae) For the $M/M/1 : \infty/\text{FIFO}$ model, the relations between L_s , L_q , W_s and W_q are given as follows:

- (i) $L_s = \lambda W_s$
- (ii) $L_q = \lambda W_q$
- (iii) $W_s = W_q + \frac{1}{\mu}$
- (iv) $L_s = L_q + \frac{\lambda}{\mu}$

Proof. From the characteristics of the $M/M/1 : \infty/\text{FIFO}$ model, we know that

$$L_s = \frac{\lambda}{\mu - \lambda}, L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}, W_s = \frac{1}{\mu - \lambda} \text{ and } W_q = \frac{\lambda}{\mu(\mu - \lambda)} \quad (7.23)$$

- (i) This is immediate from Eq. (7.23).
- (ii) This is immediate from Eq. (7.23).
- (iii) We note that

$$W_q + \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)} + \frac{1}{\mu} = \frac{\lambda + (\mu - \lambda)}{\mu(\mu - \lambda)} = \frac{1}{\mu - \lambda} = W_s$$

- (iv) We note that

$$L_q + \frac{\lambda}{\mu} = \frac{\lambda^2}{\mu(\mu - \lambda)} + \frac{\lambda}{\mu} = \frac{\lambda^2 + \lambda(\mu - \lambda)}{\mu(\mu - \lambda)} = \frac{\lambda}{\mu - \lambda} = L_s \quad \blacklozenge$$

Remark 7.1. Little's formulae are very useful in the study of the $M/M/1 : \infty/\text{FIFO}$ model because if any of the quantities L_s, L_q, W_s and W_q are available, then the other three quantities can be readily obtained by Little's formulae given in Theorem 7.1.

EXAMPLE 7.1. What is the probability that a customer has to wait more than 15 minutes to get his service completed in $(M/M/1) : (\infty/\text{FIFO})$ queue system if $\lambda = 6$ per hour and $\mu = 10$ per hour?
(Anna, Nov. 2003; Nov. 2004)

Solution. From the given data, we find that

$$\lambda = 6 \text{ (per hour) and } \mu = 10 \text{ (per hour)}$$

The probability that the waiting time of a customer in the system exceeds t is given by

$$P(T_s > t) = e^{-(\mu - \lambda)t} = e^{-4t}$$

Thus, the probability that a customer has to wait more than 15 minutes or one-quarter of an hour is given by

$$P(T_s > 0.25) = e^{-4 \times 0.25} = e^{-1} = \frac{1}{e} \quad \blacklozenge$$

EXAMPLE 7.2. Consider an $M/M/1$ Queueing system. Find the probability that there are at least n customers in the system.

(Anna, Model 2003)

Solution. The probability that there are at least n customers in the system is given by

$$P(N \geq n) = \sum_{k=n}^{\infty} P_k = \sum_{k=n}^{\infty} \left(\frac{\lambda}{\mu}\right)^k \left(1 - \frac{\lambda}{\mu}\right) = \sum_{k=n}^{\infty} \rho^k (1 - \rho)$$

where $\rho = \frac{\lambda}{\mu}$ is the traffic intensity.

Thus, we have

$$P(N \geq n) = (1 - \rho) \rho^n \sum_{k=n}^{\infty} \rho^{k-n} = (1 - \rho) \rho^n \left[\frac{1}{1 - \rho} \right] = \rho^n = \left(\frac{\lambda}{\mu}\right)^n \quad \blacklozenge$$

EXAMPLE 7.3. Consider an $M/M/1$ Queueing system. If $\lambda = 6$ and $\mu = 8$, find the probability of at least 10 customers in the system.

(Anna, April 2005)

Solution. The traffic intensity of the system is given by

$$\rho = \frac{\lambda}{\mu} = \frac{6}{8} = \frac{3}{4} = 0.75$$

Thus, the probability of at least 10 customers in the system is given by

$$P(N \geq 10) = \rho^{10} = (0.75)^{10} = 0.0563 \quad \blacklozenge$$

EXAMPLE 7.4. In a given $M/M/1/infty/FCFS$ queue, $\rho = 0.6$. What is the probability that the queue contains 5 or more customers?

(Anna, Nov. 2005)

Solution. The probability that the queue contains 5 or more customers is given by

$$P(N \geq 5) = \rho^5 = (0.6)^5 = 0.0778 \quad \blacklozenge$$

EXAMPLE 7.5. In the usual notation of an $M/M/1$ Queueing system, if $\lambda = 12$ per hour and $\mu = 24$ per hour, find the average number of customers in the system.

(Anna, May 2007)

Solution. The average number of customers in the system is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{12}{24 - 12} = \frac{12}{12} = 1 \quad \blacklozenge$$

EXAMPLE 7.6. Suppose that customers arrive at a Poisson rate of one per every 12 minutes and that the service time is exponential at a rate of one service per 8 minutes. What is

- (a) The average number of customers in the system.
- (b) The average time a customer spends in the system.

(Anna, Nov. 2007)

Solution. From the given data, we have

$$\lambda = \frac{1}{12}/\text{minute} \text{ and } \mu = \frac{1}{8}/\text{minute}$$

- (a) The average number of customers in the system is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{1/12}{1/8 - 1/12} = \frac{1/12}{1/24} = 2$$

- (b) The average time a customer spends in the system is given by

$$W_s = \frac{1}{\lambda} L_s = 12 \times 2 = 24 \text{ minutes}$$



EXAMPLE 7.7. A supermarket has a single cashier. During peak hours, customers arrive at a rate of 20 per hour. The average number of customers that can be processed by the cashier is 24 per hour. Calculate:

- (i) The probability that the cashier is idle.
- (ii) The average number of customers in the Queueing system.
- (iii) The average time a customer spends in the system.
- (iv) The average number of customers in the queue.
- (v) The average time a customer spends in the queue waiting for service.

(C.A., Nov. 1998)

Solution. From the given data, we find:

Mean arrival rate = $\lambda = 20$ customers per hour

Mean service rate = $\mu = 24$ customers per hour

- (i) The probability that the cashier is idle:

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{20}{24} = 1 - \frac{5}{6} = \frac{1}{6} \text{ or } 0.1667$$

- (ii) The average number of customers in the system:

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{20}{24 - 20} = \frac{20}{4} = 5$$

- (iii) The average time a customer spends in the system:

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{24 - 20} = \frac{1}{4} \text{ hour or } 15 \text{ minutes}$$

(iv) The average number of customers waiting in the queue:

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{(20)^2}{24(24 - 20)} = \frac{400}{24 \times 4} = 4.1667$$

(v) The average time a customer spends in the queue:

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{20}{24(24 - 20)} = \frac{20}{24 \times 4} = \frac{5}{24} \text{ hour or } 12.5 \text{ minutes}$$

◆

EXAMPLE 7.8. Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer.

Solution. From the given data, we find:

$$\text{Mean arrival rate} = \lambda = 20 \text{ customers per hour}$$

$$\text{Mean service rate} = \mu = \frac{60 \times 60}{100} = 36 \text{ customers per hour}$$

The average waiting time of a customer in the queue is given by

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{20}{36(36 - 20)} = \frac{5}{144} \text{ hours or } 125 \text{ seconds}$$

The average waiting time of a customer in the system is given by

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{36 - 20} = \frac{1}{16} \text{ hour or } 225 \text{ seconds}$$

◆

EXAMPLE 7.9. Customers arrive at a watch repair shop according to a Poisson process at a rate of one per every 10 minutes and the service time is exponential with mean 8 minutes. Find the average number of customers in the shop (L_s), average waiting time a customer spends in the shop (W_s), and the average time a customer spends in the queue for service W_q .

(Anna, Model 2003)

Solution. From the given data, we find:

$$\text{Mean arrival rate} = \lambda = 6 \text{ customers per hour}$$

$$\text{Mean service rate} = \mu = \frac{60}{8} = 7.5 \text{ customers per hour}$$

$$\text{Traffic intensity} = \rho = \frac{\lambda}{\mu} = \frac{6}{7.5} = 0.8$$

The average number of customers in the shop is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{6}{7.5 - 6} = \frac{6}{1.5} = 4 \text{ per hour}$$

The average waiting time a customer spends in the shop is given by

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{7.5 - 6} = \frac{1}{1.5} = \frac{2}{3} \text{ hour or 40 minutes}$$

The average waiting time a customer spends in the queue is given by

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{6}{7.5(7.5 - 6)} = \frac{8}{15} \text{ hour or 32 minutes}$$



EXAMPLE 7.10. In a public telephone booth having just one phone, the arrivals are considered to be Poisson with the average of 15 per hour. The length of a phone call is assumed to be distributed exponentially with mean 3 minutes. Find the

- (i) Average number of customers waiting in the system.
- (ii) Average number of customers waiting in the queue.
- (iii) Probability that a person arriving at the booth will have to wait in the queue.
- (iv) Expected waiting time of a customer in the system.
- (v) Expected waiting time of a customer in the queue.
- (vi) Percentage of time that the telephone booth will be idle.

Solution. From the given data, we find:

$$\text{Mean arrival rate} = \lambda = 15 \text{ customers per hour}$$

$$\text{Mean service rate} = \mu = \frac{60}{3} = 20 \text{ customers per hour}$$

$$\text{Traffic intensity} = \rho = \frac{\lambda}{\mu} = \frac{15}{20} = \frac{3}{4}$$

- (i) The average number of customers waiting in the system is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{15}{20 - 15} = \frac{15}{5} = 3$$

- (ii) The average number of customers waiting in the queue is given by

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{(15)^2}{20(20 - 15)} = \frac{45}{20} = 2.25$$

- (iii) The probability that a person arriving at the booth will have to wait in the queue is

$$P(W > 0) = 1 - P(W = 0) = 1 - P(\text{no customer in the system}) = 1 - P_0 = \rho = 0.75$$

- (iv) The expected waiting time of a customer in the system is given by

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{20 - 15} = \frac{1}{5} \text{ hour or 12 minutes}$$

- (v) The expected waiting time of a customer in the queue is given by

$$W_q = \frac{\lambda}{\lambda(\mu - \lambda)} = \frac{15}{20(20 - 15)} = \frac{15}{20 \times 5} = \frac{3}{20} \text{ hour or 9 minutes}$$

- (vi) The probability the telephone booth will be idle is given by

$$P(N = 0) = P_0 = 1 - \rho = 1 - 0.75 = 0.25$$

Thus, the telephone booth will be idle in 25% of the time.



EXAMPLE 7.11. In a given $M/M/1$ Queueing system, the average arrivals is 4 customers per minute and $\rho = 0.7$. What are: (i) mean number of customers L_s in the system, (ii) mean number of customers, L_q in the queue, (iii) probability that the server is idle, and (iv) mean waiting time, W_s , in the system?

(Anna, Nov. 2005)

Solution. Since the mean arrival rate $\lambda = 4$ customers per minute and traffic intensity $\rho = 0.7$, we have

$$\text{mean service rate} = \mu = \frac{\lambda}{\rho} = \frac{4}{0.7} = 5.7143$$

- (i) The mean number of customers in the system is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{4}{5.7143 - 4} = 2.3333$$

- (ii) The mean number of customers in the queue is given by

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{4^2}{5.7143(5.7143 - 4)} = 1.6333$$

- (iii) The probability that the server is idle is given by

$$P_0 = 1 - \rho = 1 - 0.7 = 0.3$$

- (iv) The mean waiting time in the system is given by

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{5.7143 - 4} = 0.5833$$



EXAMPLE 7.12. Automatic car wash facility operates with only one bay. Cars arrive according to a Poisson process, with mean of 4 cars per hour and may wait in the facility's parking lot if the bay is busy. If the service time for all cars is constant and equal to 10 minutes, determine L_s, L_q, W_s and W_q .

(Anna, Nov. 2005)

Solution. From the given data, we find that

$$\text{mean arrival rate} = \lambda = 4 \text{ cars per hour}$$

and

$$\text{mean service rate} = \mu = 6 \text{ cars per hour}$$

The mean number of cars in the system is

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{4}{6 - 4} = \frac{4}{2} = 2 \text{ cars}$$

The mean number of cars in the queue is

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{4^2}{6(6 - 4)} = \frac{4}{3} = 1.3333 \text{ cars}$$

The mean waiting time of a car in the system is

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{6 - 4} = \frac{1}{2} \text{ hour or 30 minutes}$$

The mean waiting time of a car in the queue is

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{4}{6(6 - 4)} = \frac{1}{3} \text{ hour or 20 minutes}$$

◆

EXAMPLE 7.13. In a city airport, flights arrive at a rate of 24 flights per day. It is known that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average of 30 minutes. Find the following:

- (i) The probability that the system will be idle.
- (ii) The mean queue size.
- (iii) The average number of flights in the queue.
- (iv) The probability that the queue size exceeds 7.
- (v) If the input of flights increases to an average 30 flights per day, what will be the changes in (i)-(iv)?

Solution. From the given data,

$$\text{mean arrival rate} = \lambda = \frac{24}{24} = 1 \text{ flight per hour}$$

and

$$\text{mean service rate} = \mu = \frac{60}{30} = 2 \text{ flights per hour}$$

Thus, the traffic intensity is given by

$$\rho = \frac{\lambda}{\mu} = \frac{1}{2} = 0.5$$

- (i) The probability that the system will be idle is given by

$$P(\text{no customers in the system}) = P_0 = 1 - \rho = 0.5$$

- (ii) The mean queue size or line length is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{1}{2 - 1} = 1 \text{ flight}$$

- (iii) The average number of flights in the queue is given by

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{1}{2(2 - 1)} = 0.5 \text{ flight}$$

- (iv) The probability that the queue size exceeds 7 is given by

$$P(N > 7) = \rho^8 = 0.5^8 = 0.0039$$

- (v) If the input of flights increases to an average 30 flights per day, then the mean arrival rate λ increases to

$$\lambda = \frac{30}{24} = 1.25 \text{ flights per hour}$$

and the traffic intensity becomes

$$\rho = \frac{\lambda}{\mu} = \frac{1.25}{2} = 0.625$$

In this case, the probability that the system will be idle is given by

$$P(\text{no customers in the system}) = P_0 = 1 - \rho = 0.375$$

In this case, the mean queue size or line length is given by

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{1.25}{2 - 1.25} = \frac{1.25}{0.75} = 1.6667 \text{ flights}$$

Also, the average number of flights in the queue is given by

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{1.25^2}{2(2 - 1.25)} = 1.0417 \text{ flights}$$

The probability that the queue size exceeds 7 is given by

$$P(N > 7) = \rho^8 = 0.625^8 = 0.0233$$



EXAMPLE 7.14. A Xerox machine is maintained in an office and operated by a secretary who does other jobs also. The service rate is Poisson-distributed with a mean service rate of 10 jobs per hour. Generally, the requirements for use are random over the entire 8-hour working day but arrive at a rate of 5 jobs per hour. Several people have noted that a waiting line develops occasionally and have questioned the office policy of maintaining only one Xerox machine. If the time of a secretary is valued at Rs. 10 per hour, find the following:

- (i) Utilization of the Xerox machine.
- (ii) The probability that an arrival has to wait.
- (iii) The mean number of jobs of the system.
- (iv) The average waiting time of a job in the system.
- (v) The average cost per day due to waiting and operating the machine.

Solution. From the given data, we find:

$$\text{Mean arrival rate} = \lambda = 5 \text{ jobs per hour}$$

$$\text{Mean service rate} = \mu = 10 \text{ jobs per hour}$$

- (i) The utilization of the Xerox machine is

$$\rho = \frac{\lambda}{\mu} = \frac{5}{10} = \frac{1}{2} = 0.5$$

Thus, the Xerox machine is in use 50% of the time.

- (ii) The probability that an arrival has to wait is given by

$$P(N \geq 1) = \rho = 0.5$$

- (iii) The mean number of jobs of the system or the line length is

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{5}{10 - 5} = \frac{5}{5} = 1$$

- (iv) The average waiting time of a job in the system is given by

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{10 - 5} = \frac{1}{5} \text{ hour or } 12 \text{ minutes}$$

- (v) We find that

$$\begin{aligned} \text{Average cost per day} &= \text{Average cost per job} \times \text{No. of jobs processed per day} \\ &= 8 \times \text{Rs. } 10 \times L_s \\ &= \text{Rs. } 80 \end{aligned}$$



EXAMPLE 7.15. The arrival rates of telephone calls at telephone booth are according to Poisson distribution with an average time of 12 minutes between arrival of two consecutive calls. The length of telephone call is assumed to be exponentially distributed with mean 4 minutes.

- (i) Determine the probability that the person arriving at the booth will have to wait.
- (ii) Find the average queue length that is formed from time to time.
- (iii) The telephone company will install the second booth when convinced that an arrival would expect to wait at least 5 minutes for the phone. Find the increase in flows of arrivals which will justify the second booth.
- (iv) What is the probability that an arrival will have to wait for more than 15 minutes before the phone is free?

(Anna, Nov. 2006)

Solution. First, we note that the mean inter-arrival time is given by

$$\frac{1}{\lambda} = 12 \text{ minutes}$$

and so the mean arrival rate is given by

$$\lambda = \frac{1}{12} \text{ per minute}$$

Also, we note that the mean inter-service time is given by

$$\frac{1}{\mu} = 4 \text{ minutes}$$

and so the mean service rate is given by

$$\mu = \frac{1}{4} \text{ per minute}$$

Thus, the traffic intensity of the system is

$$\rho = \frac{\lambda}{\mu} = \frac{1/12}{1/4} = \frac{1}{3}$$

- (i) The probability that person arriving at the booth will have to wait is given by

$$P(N \geq 1) = \rho = \frac{1}{3} = 0.3333$$

- (ii) The average queue length that is formed from time to time is given by

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{1/144}{1/4(1/4 - 1/12)} = \frac{1}{144} \times 4 \times 6 = \frac{1}{6}$$

- (iii) The telephone company will install a second booth is $W_q > 5$, i.e.

$$W_q = \frac{\lambda_R}{\mu(\mu - \lambda_R)} > 5$$

where λ_R is the required arrival rate. Thus, we have

$$\frac{\lambda_R}{\frac{1}{4}(\frac{1}{4} - \lambda_R)} > 5 \text{ or } \lambda_R > \frac{5}{4} \left(\frac{1}{4} - \lambda_R \right)$$

i.e.

$$\left(1 + \frac{5}{4} \right) \lambda_R > \frac{5}{16} \text{ or } \frac{9}{4} \lambda_R > \frac{5}{16}$$

i.e.

$$\lambda_R > \frac{5}{36}$$

Hence, the arrival rate should increase by $\frac{5}{36} - \frac{1}{12} = \frac{1}{18}$ per minute to justify the second telephone booth.

(iv) The probability that an arrival will have to wait for more than 15 minutes is given by

$$e^{-(\mu - \lambda)15} = e^{-(\frac{1}{4} - \frac{1}{12})15} = e^{-2.5} = 0.0821$$

◆

PROBLEM SET 7.1

- Find the probability that a customer has to wait more than 20 minutes to get his service completed in (M/M/1) : (∞ /FIFO) queue system if $\lambda = 9$ per hour and $\mu = 15$ per hour.
- For an M/M/1 Queueing system, it is given that $\lambda = 6$ and $\mu = 10$. Find the probability that there are at least 8 customers in the system.
- Suppose that the customers arrive at a Poisson rate of one per every 10 minutes, and that the service time is exponential at a rate of one service per 5 minutes. Find:
 - Average number of customers in the system.
 - Average time a customer spends in the system.
 - Average number of customers in the queue.
 - Average waiting time of a customer in the queue.
- In a given M/M/1 Queueing system, the average arrival is 6 customers per minute and $\rho = 0.8$. Find:
 - The probability that the server is idle.
 - The mean number of customers in the system.
 - The mean number of customers in the queue.
 - The mean waiting time in the system.
 - The mean waiting time in the queue.
- In a given M/M/1 Queueing system, the average service rate is 8 customers per minute and $\rho = 0.5$. Find:
 - The probability that the server is busy.
 - The expected number of customers in the system.
 - The expected number of customers in the queue.
 - The probability that there are at least 6 customers in the system.

- (e) The average waiting time in the system.
 - (f) The average waiting time in the queue.
6. A fast-food restaurant with one food counter is modelled as an $M/M/1$ Queueing system with customer arrival rate of 3 per minute. If the owner wishes to have fewer than 6 customers line up 98% of the time, find out how fast the service rate should be.
 7. The customers arrive at the ATM machine of a bank according to a Poisson process at an average rate of 15 per hour. It is known that the average time taken by each customer is an exponential random variable with mean 2 minutes. Find the following:
 - (a) Probability that an arriving customer will find the ATM machine occupied.
 - (b) Average number of customers in the system.
 - (c) Average number of customers in the queue.
 - (d) Average waiting time in the system.
 - (e) Average waiting time in the queue.
 - (f) The bank has the policy of installing additional ATM machines if customers wait at an average of 3 or more minutes in the queue. Find the average arrival rate of customers required to justify an additional ATM machine.
 8. Suppose that the customers arrive at a box-office window at a Poisson rate of one per every 7 minutes and that the service time is exponentially distributed at a rate of one service per 6 minutes: Find:
 - (a) Traffic intensity, ρ .
 - (b) Average number of customers in the system, L_s .
 - (c) Average waiting time of a customer in the system, W_s .
 - (d) Corresponding changes in ρ , L_s and W_s if the arrival rate increases by 10%. Interpret your results.
 9. A railway reservation centre has a single booking counter. During the rush hours, customers arrive at the rate of 30 customers per hour. The average number of customers that can be attended by the booking operator is 40 per hour. Assume that the arrivals are Poisson and the service time is exponentially distributed. Find:
 - (a) Probability that the operator is idle.
 - (b) Average number of customers in the Queueing system.
 - (c) Average time a customer spends in the system?
 - (d) Average number of customers in the queue?
 - (e) Average time a customer spends in the queue waiting for service?
 10. A grocery store has a single cashier. It is known that the customers arrive at the rate of 20 customers per hour. The average number of customers that can be processed by the cashier is 25 per hour. Assume that the arrivals are Poisson and the service time is exponentially distributed. Find:
 - (a) Probability that the cashier is idle.
 - (b) Average number of customers in the system.
 - (c) Average number of customers in the queue.
 - (d) Average time a customer spends in the system.
 - (e) Average time a customer spends in the queue waiting for service.

11. Customers arrive at a reservation counter of a movie theatre at a rate of 20 per hour. There is one clerk in the counter serving the customers at a rate of 30 per hour. Suppose that the arrivals are Poisson distributed and that the service time is exponentially distributed. Find:
- Probability that the booking counter is idle.
 - Probability that there are more than 5 customers at the counter.
 - Average number of customers in the system.
 - Average number of customers in the queue.
 - Expected waiting time of a customer in the system.
 - Expected waiting time of a customer in the queue.
12. The arrivals for a service centre are Poisson-distributed with a mean arrival rate of 4 units per hour. The mean service time has been shown to be exponentially distributed with a mean service time of 10 minutes per service of the unit. Find:
- The probability that the centre will be idle.
 - The probability of at least 6 units in the centre.
 - The expected number of units in the centre.
 - The expected number of units in the queue.
 - The expected waiting time of a unit in the centre.
 - The expected waiting time of a unit in the queue.
13. The arrival of customers in the only teller counter of a bank is Poisson-distributed at the rate of 25 customers per hour. The teller takes on an average 2 minutes to cash cheque. The service time has been shown to be exponentially distributed. Find:
- Percentage of time the teller is busy.
 - Average number of customers in the bank.
 - Probability of at least 10 customers in the bank.
 - Average number of customers in the queue.
 - Expected waiting time of a customer in the bank.
 - Expected waiting time of a customer in the queue.

7.3 MODEL II—(M/M/s): (∞ /FIFO), MULTIPLE SERVER WITH INFINITE CAPACITY

For the Poisson queue system,

$$P_n = \frac{\lambda_0 \lambda_1 \lambda_2 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \mu_3 \cdots \mu_n} P_0 \quad \text{where } n \geq 1 \quad (7.24)$$

where

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \left[\frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right]} \quad (7.25)$$

If there is a single server, the mean service rate $\mu_n = \mu$ for all n .

But for the given model, there are s servers working independently of each other. When there are n customers in the system, the mean service rate, μ_n , can be calculated in two different situations:

- (i) If $n < s$, only n of the s servers will be busy and the others will be idle. Hence, the mean service rate will be $n\mu$.
- (ii) If $n \geq s$, all the servers will be busy. Hence, the mean service rate will be $s\mu$.

Thus, we assume the following:

(A1) The mean arrival time $\lambda_n = \lambda$ for all n .

(A2) The mean service time μ_n is given by

$$\mu_n = \begin{cases} n\mu, & \text{if } 0 \leq n < s \\ s\mu, & \text{if } n \geq s \end{cases} \quad (7.26)$$

(A3) The mean arrival rate is less than $s\mu$, i.e. $\lambda < s\mu$.

If $0 \leq n < s$, then substituting Eq. (7.26) in Eq. (7.24), we get

$$P_n = \frac{\lambda}{1\mu \cdot 2\mu \cdot 3\mu \cdots n\mu} P_0 = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n P_0 \quad (7.27)$$

if $n \geq s$, then substituting (7.26) in (7.24), we get

$$P_n = \frac{\lambda^n}{[1\mu \cdot 2\mu \cdots (s-1)\mu][s\mu \cdot s\mu \cdots (n-s+1)\text{ times}]}$$

i.e.

$$P_n = \frac{\lambda^n}{(s-1)!\mu^{s-1}(s\mu)^{n-s+1}} = \frac{1}{s!s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n P_0 \quad (7.28)$$

To find the value of P_0 , we use the fact that

$$\sum_{n=0}^{\infty} P_n = 1$$

i.e.

$$\left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \sum_{n=s}^{\infty} \frac{1}{s!s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n \right] P_0 = 1$$

i.e.

$$\left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{s^s}{s!} \sum_{n=s}^{\infty} \left(\frac{\lambda}{\mu s} \right)^n \right] P_0 = 1$$

i.e.

$$\left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{s^s}{s!} \left(\frac{\lambda}{\mu s} \right)^s \frac{1}{1 - \frac{\lambda}{\mu s}} \right] P_0 = 1$$

where we have used the assumption (A3) that $\frac{\lambda}{\mu s} < 1$ and the expansion that

$$\sum_{n=s}^{\infty} \left(\frac{\lambda}{\mu s} \right)^n = \left(\frac{\lambda}{\mu s} \right)^s \left[1 + \frac{\lambda}{\mu s} + \left(\frac{\lambda}{\mu s} \right)^2 + \cdots \right] = \left(\frac{\lambda}{\mu s} \right)^s \frac{1}{1 - \frac{\lambda}{\mu s}}$$

Thus, it follows that

$$\left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s! \left(1 - \frac{\lambda}{\mu s} \right)} \left(\frac{\lambda}{\mu} \right)^s \right] P_0 = 1$$

Hence, we have

$$P_0 = \frac{1}{\left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s! \left(1 - \frac{\lambda}{\mu s} \right)} \left(\frac{\lambda}{\mu} \right)^s \right]} \quad (7.29)$$

7.3.1 Characteristics of Model II

1. Average or expected number of customers in the queue or queue length (L_q): As Model II has a total of s servers, the expected queue length is given by

$$L_q = E(N_q) = E(N - s) = \sum_{n=s}^{\infty} (n - s) P_n = \sum_{x=0}^{\infty} x P_{x+s} \quad (\text{by putting } x = n - s)$$

Thus, we have

$$L_q = \sum_{x=0}^{\infty} x \frac{1}{s! s^x} \left(\frac{\lambda}{\mu} \right)^{s+x} P_0 = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s P_0 \sum_{x=1}^{\infty} x \left(\frac{\lambda}{\mu s} \right)^x$$

It follows that

$$L_q = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s P_0 \frac{\lambda}{\mu s} \frac{1}{\left[1 - \frac{\lambda}{\mu s} \right]^2}$$

where we have used the fact that

$$\sum_{x=1}^{\infty} x \left(\frac{\lambda}{\mu s} \right)^x = \frac{\lambda}{\mu s} \sum_{x=1}^{\infty} x \left(\frac{\lambda}{\mu s} \right)^{x-1} = \frac{\lambda}{\mu s} \left[1 - \frac{\lambda}{\mu s} \right]^{-2}$$

Hence, we deduce that

$$L_q = \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 \quad (7.30)$$

2. Average or expected number of customers in the system (L_s): By Little's formula (iv) (Theorem 7.1) and Eq. (7.30), we have

$$L_s = L_q + \frac{\lambda}{\mu} = \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 + \frac{\lambda}{\mu} \quad (7.31)$$

We remark that we could have also derived Eq. (7.31) directly by using the definition that

$$L_s = E(N) = \sum_{n=0}^{\infty} n P_n$$

3. The average waiting time of a customer in the system (W_s): By Little's formula (i) (Theorem 7.1) and Eq. (7.31), we have

$$W_s = \frac{1}{\lambda} L_s = \frac{1}{\lambda} \left[\frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 + \frac{\lambda}{\mu} \right]$$

Thus,

$$W_s = \frac{1}{\mu} + \frac{1}{\mu} \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^s}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 \quad (7.32)$$

4. The average waiting time of a customer in the queue (W_q): By Little's formula (ii) (Theorem 7.1) and Eq. (7.30), we have

$$W_q = \frac{1}{\lambda} L_q = \frac{1}{\lambda} \left[\frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 \right]$$

Thus,

$$W_q = \frac{1}{\mu} \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^s}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 \quad (7.33)$$

5. The probability that an arrival has to wait for service: Note that an arrival has to wait for service if and only if $T_s > 0$, where T_s denotes the waiting time of a customer in the system, i.e. if and only if there are s or more customers in the system. Thus, the required probability is equal to

$$P(T_s > 0) = P(N \geq s) = \sum_{n=s}^{\infty} P_n = \sum_{n=s}^{\infty} \frac{1}{s! s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n P_0 = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s P_0 \sum_{n=s}^{\infty} \left(\frac{\lambda}{\mu s} \right)^{n-s}$$

Thus, it is immediate that

$$P(T_s > 0) = P(N \geq s) = \frac{\left(\frac{\lambda}{\mu} \right)^s P_0}{s! \left(1 - \frac{\lambda}{\mu s} \right)} \quad (7.34)$$

6. The probability that an arrival enters the service without waiting: It is given by

$$1 - P(T_s > 0) = 1 - \frac{\left(\frac{\lambda}{\mu} \right)^s P_0}{s! \left(1 - \frac{\lambda}{\mu s} \right)}$$

7. The mean waiting time in the queue for those who need to wait:

$$E[T_q | T_s > 0] = \frac{E[T_q]}{P(T_s > 0)} = \frac{W_q}{P(T_s > 0)}$$

Using Eqs. (7.33) and (7.34), we have

$$E[T_q|T_s > 0] = \frac{1}{\mu} \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu}\right]^s}{\left[1 - \frac{\lambda}{\mu s}\right]^2} P_0 \times \frac{s! \left(1 - \frac{\lambda}{\mu s}\right)}{\left(\frac{\lambda}{\mu}\right)^s P_0}$$

Simplifying, we have

$$E[T_q|T_s > 0] = \frac{1}{\mu s \left[1 - \frac{\lambda}{\mu s}\right]} = \frac{1}{\mu s - \lambda} \quad (7.35)$$

8. The average or expected number of customers in non-empty queues (L_n): If N denotes the number of customers in the system and N_q denotes the number of customers in the queue, then L_n is the conditional expectation defined by

$$L_n = E[N_q|N_q \geq 1] = \frac{E[N_q]}{P(N \geq s)} = \frac{L_q}{P(N \geq s)} \quad (7.36)$$

(since there are s servers in Model II)

Substituting from Eqs. (7.30) and (7.34) into Eq. (7.36), we get

$$L_n = \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu}\right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s}\right]^2} P_0 \times \frac{s! \left(1 - \frac{\lambda}{\mu s}\right)}{\left(\frac{\lambda}{\mu}\right)^s P_0}$$

Simplifying, we get

$$L_n = \frac{\left(\frac{\lambda}{\mu s}\right)}{1 - \frac{\lambda}{\mu s}} \quad (7.37)$$

9. The probability that there will be someone waiting: As there are s servers, the required probability is given by

$$P(N \geq s+1) = P(N \geq s) - P(N = s) = P(N \geq s) - P_s$$

Using Eqs. (7.34) and (7.28), we have

$$P(N \geq s+1) = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s! \left(1 - \frac{\lambda}{\mu s}\right)} - \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s P_0 = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0 - \left(\frac{\lambda}{\mu}\right)^s \left(1 - \frac{\lambda}{\mu s}\right) P_0}{s! \left(1 - \frac{\lambda}{\mu s}\right)}$$

Simplifying, we have

$$P(N \geq s+1) = \frac{\left(\frac{\lambda}{\mu}\right)^s \left(\frac{\lambda}{\mu s}\right) P_0}{s! \left(1 - \frac{\lambda}{\mu s}\right)} \quad (7.38)$$

EXAMPLE 7.16. A telephone exchange has two long-distance operators. It has been found that long telephone calls arrive according to a Poisson distribution at an average rate of 15 per hour. The length of service on these calls has been shown to be exponentially distributed with mean length of 2 minutes. Find the:

- (i) Probability that a customer will have to wait for his long distance call.
- (ii) Expected number of customers in the system.
- (iii) Expected number of customers in the queue.
- (iv) Expected time a customer spends in the system.
- (v) Expected waiting time for a customer in the queue.

Solution. From the given data, we find that

$$s = 2, \lambda = \frac{15}{60} = \frac{1}{4} \text{ /minute and } \mu = \frac{1}{2} \text{ /minute}$$

Thus, it follows that

$$\frac{\lambda}{\mu} = \frac{1/4}{1/2} = \frac{1}{2}$$

The probability of no customer in the system is

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s! \left[1 - \frac{\lambda}{\mu s} \right]} \left(\frac{\lambda}{\mu} \right)^s \right]^{-1}$$

Substituting the values, we have

$$\begin{aligned} P_0 &= \left[\sum_{n=0}^1 \frac{1}{n!} \left(\frac{1}{2} \right)^n + \frac{1}{2! \left[1 - \frac{1}{4} \right]} \left(\frac{1}{2} \right)^2 \right]^{-1} = \left[1 + \frac{1}{2} + \frac{1}{3/2} \times \frac{1}{4} \right]^{-1} \\ &= \left[1 + \frac{1}{2} + \frac{1}{6} \right]^{-1} = \left[\frac{5}{3} \right]^{-1} = \frac{3}{5} \end{aligned}$$

- (i) The probability that a customer will have to wait is given by

$$P(N \geq s) = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \frac{1}{\left[1 - \frac{\lambda}{\mu s} \right]} P_0$$

Substituting the values, we have

$$P(N \geq 2) = \frac{1}{2!} \left(\frac{1}{2} \right)^2 \frac{1}{\left[1 - \frac{1}{4} \right]} \frac{3}{5} = \frac{1}{2} \times \frac{1}{4} \times \frac{4}{3} \times \frac{3}{5} = \frac{1}{10}$$

- (ii) The expected number of customers in the system is given by

$$L_s = L_q + \frac{\lambda}{\mu} = \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 + \frac{\lambda}{\mu}$$

Substituting the values, we have

$$L_s = \frac{1}{2 \cdot 2!} \frac{\left[\frac{1}{2}\right]^3}{\left[1 - \frac{1}{4}\right]^2} \frac{3}{5} + \frac{1}{2} = \left[\frac{1}{4} \times \frac{1/8}{9/16} \times \frac{3}{5}\right] + \frac{1}{2} = \left[\frac{1}{4} \times \frac{2}{9} \times \frac{3}{5}\right] + \frac{1}{2}$$

$$L_s = \frac{8}{15} = 0.5333$$

(iii) The expected number of customers in the queue is given by

$$L_q = L_s - \frac{\lambda}{\mu} = 0.5333 - 0.5 = 0.0333$$

(iv) The expected time a customer spends in the system is given by

$$W_s = W_q + \frac{1}{\mu} = \frac{L_q}{\lambda} + \frac{1}{\mu} = \frac{8/15}{1/4} + \frac{1}{1/2} = \frac{32}{15} + 2$$

$$= \frac{62}{15} = 4.1333 \text{ minutes}$$

(v) The expected waiting time for a customer in the queue is given by

$$W_q = \frac{L_q}{\lambda} = \frac{32}{15} = 2.1333 \text{ minutes}$$

◆

EXAMPLE 7.17. A travel centre has three service counters to receive people who visit to book air tickets. The customers arrive in a Poisson distribution with the average arrival of 100 persons in a 10-hour service day. It has been estimated that the service time follows an exponential distribution. The average service time is 15 minutes. Find the:

- (i) Expected number of customers in the system.
- (ii) Expected number of customers in the queue.
- (iii) Expected time a customer spends in the system.
- (iv) Expected waiting time for a customer in the queue.
- (v) Probability that a customer must wait before he gets service.

Solution. From the given data, we have

$$\lambda = 10/\text{hour}, \mu = 4/\text{hour} \text{ and } s = 3$$

The probability of no customer in the system is

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s! \left[1 - \frac{\lambda}{\mu s} \right]} \left(\frac{\lambda}{\mu} \right)^s \right]^{-1}$$

Substituting the values, we have

$$P_0 = \left[\sum_{n=0}^2 \frac{1}{n!} \left(\frac{10}{4} \right)^n + \frac{1}{3! \left[1 - \frac{10}{12} \right]} \left(\frac{10}{4} \right)^3 \right]^{-1}$$

$$= \left[1 + \left(\frac{10}{4} \right) + \frac{1}{2} \left(\frac{10}{4} \right)^2 + \frac{1}{\left(6 \times \frac{2}{12} \right)} \left(\frac{10}{4} \right)^3 \right]^{-1}$$

Thus,

$$P_0 = [1 + 2.5 + 3.125 + 15.625]^{-1} = [22.25]^{-1} = 0.0449$$

(i) The expected number of customers in the system is given by

$$L_s = L_q + \frac{\lambda}{\mu} = \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0 + \frac{\lambda}{\mu}$$

Substituting the values, we have

$$L_s = \frac{1}{2 \cdot 2!} \frac{\left[\frac{10}{4} \right]^3}{\left[1 - \frac{10}{12} \right]^2} (0.0449) + \left(\frac{10}{4} \right)$$

$$= \left[\frac{1}{4} \times \frac{15.6250}{0.0278} \times 0.0449 \right] + 2.5 = 6.3090 + 2.5 = 8.8090 \text{ customers}$$

(ii) The expected number of customers in the queue is given by

$$L_q = L_s - \frac{\lambda}{\mu} = 8.8090 - 2.5 = 6.3090 \text{ customers}$$

(iii) The expected time a customer spends in the system is given by

$$W_s = W_q + \frac{1}{\mu} = \frac{L_q}{\lambda} + \frac{1}{\mu} = \frac{6.3090}{10} + \frac{1}{4}$$

$$= 0.6309 + 0.25 = 0.8809 \text{ hour}$$

(iv) The expected waiting time for a customer is given by

$$W_q = \frac{L_q}{\lambda} = \frac{6.3090}{10} = 0.6309 \text{ hour}$$

(v) The probability that a customer must wait before he gets service is given by

$$P(N \geq s) = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \frac{1}{\left[1 - \frac{\lambda}{s\mu} \right]} P_0$$

Substituting the values, we have

$$P(N \geq 3) = \frac{1}{3!} \left(\frac{10}{4} \right)^3 \frac{1}{\left[1 - \frac{10}{12} \right]} 0.0449 = \frac{1}{6} \times 15.6250 \times 6 \times 0.0449 = 0.7016$$

◆

EXAMPLE 7.18. A road transport company has two reservation clerks serving the customers. The customers arrive in a Poisson fashion at the rate of 8 per hour. The service time for each customer is exponentially distributed with mean 10 minutes. Find the:

- (i) Probability that a customer has to wait for service.
- (ii) Average number of customers in the queue.
- (iii) Average number of customers in the system.
- (iv) Expected waiting time of a customer in the queue.
- (v) Expected time a customer spends in the system.

Solution. From the given data, we find that

$$\lambda = 8/\text{hour}, \mu = 6/\text{hour} \text{ and } s = 2$$

The probability of no customer in the system is

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s! \left[1 - \frac{\lambda}{s\mu} \right]} \left(\frac{\lambda}{\mu} \right)^s \right]^{-1}$$

Substituting the values, we have

$$\begin{aligned} P_0 &= \left[\sum_{n=0}^1 \frac{1}{n!} \left(\frac{8}{6} \right)^n + \frac{1}{2! \left[1 - \frac{8}{12} \right]} \left(\frac{8}{6} \right)^2 \right]^{-1} \\ &= \left[\sum_{n=0}^1 \frac{1}{n!} \left(\frac{4}{3} \right)^n + \frac{1}{2! \left[1 - \frac{2}{3} \right]} \left(\frac{4}{3} \right)^2 \right]^{-1} \end{aligned}$$

Simplifying, we have

$$P_0 = \left[1 + \frac{4}{3} + \left(\frac{3}{2} \times \frac{16}{9} \right) \right]^{-1} = \left[1 + \frac{4}{3} + \frac{8}{3} \right]^{-1} = [5]^{-1} = \frac{1}{5}$$

- (i) The probability that a customer has to wait for service is given by

$$P(N \geq s) = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \frac{1}{\left[1 - \frac{\lambda}{s\mu} \right]} P_0$$

Substituting the values, we have

$$\begin{aligned} P(N \geq 2) &= \frac{1}{2!} \left(\frac{8}{6} \right)^2 \frac{1}{\left[1 - \frac{8}{12} \right]} \frac{1}{5} = \frac{1}{2!} \left(\frac{4}{3} \right)^2 \frac{1}{\left[1 - \frac{2}{3} \right]} \frac{1}{5} \\ &= \frac{1}{2} \times 16 \times 9 \times \frac{3}{1} \times \frac{1}{5} = \frac{8}{15} \approx 0.5333 \end{aligned}$$

(ii) The average number of customers in the queue is given by

$$\begin{aligned} L_q &= \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu}\right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s}\right]^2} P_0 = \frac{1}{2 \cdot 2!} \frac{\left[\frac{8}{6}\right]^3}{\left[1 - \frac{8}{12}\right]^2} \frac{1}{5} = \frac{1}{4} \frac{\left[\frac{4}{3}\right]^3}{\left[1 - \frac{2}{3}\right]^2} \frac{1}{5} \\ &= \frac{1}{4} \times \frac{64/27}{1/9} \times \frac{1}{5} = \frac{16}{15} = 1.0667 \end{aligned}$$

(iii) The average number of customers in the system

$$L_s = L_q + \frac{\lambda}{\mu} = \frac{16}{15} + \frac{8}{6} = \frac{16}{15} + \frac{4}{3} = \frac{12}{5} = 2.4$$

(iv) The expected waiting time of a customer in the queue

$$W_q = \frac{1}{\lambda} L_q = \frac{1}{8} \frac{16}{15} = \frac{2}{15} \text{ hour} = 8 \text{ minutes}$$

(v) The expected time a customer spends in the system

$$W_s = \frac{1}{\lambda} L_s = \frac{1}{8} \frac{12}{5} = \frac{3}{10} \text{ hour} = 18 \text{ minutes}$$

◆

EXAMPLE 7.19. A petrol pump station has 4 pumps. The service time follows an exponential distribution with a mean of 6 minutes and cars arrive for service in a Poisson process at the rate of 30 cars per hour.

- (i) What is the probability that an arrival will have to wait in the line?
- (ii) Find the average waiting time in the queue, average time spent in the system and the average number of cars in the system.
- (iii) For what percentage of time would the pumps be idle on an average?

(Anna, Nov. 2006)

Solution. From the given data, we have

$$\lambda = 30/\text{hour}, \mu = 10/\text{hour} \text{ and } s = 4$$

The probability of no customer in the system is

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n + \frac{1}{s! \left[1 - \frac{\lambda}{\mu s}\right]} \left(\frac{\lambda}{\mu}\right)^s \right]^{-1}$$

Substituting the values, we get

$$\begin{aligned} P_0 &= \left[\sum_{n=0}^3 \frac{1}{n!} \left(\frac{30}{10}\right)^n + \frac{1}{4! \left[1 - \frac{30}{40}\right]} \left(\frac{30}{10}\right)^4 \right]^{-1} = \left[\sum_{n=0}^3 \frac{3^n}{n!} + \frac{3^4}{6} \right]^{-1} \\ &= \left[1 + 3 + \frac{9}{2} + \frac{27}{6} + \frac{81}{6} \right]^{-1} = \left[\frac{53}{2} \right]^{-1} = \frac{2}{53} = 0.0377 \end{aligned}$$

(i) The probability that an arrival will have to wait in the line is given by

$$P(N \geq s) = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \frac{1}{\left[1 - \frac{\lambda}{s\mu} \right]} P_0$$

Substituting the values, we get

$$\begin{aligned} P(N \geq 4) &= \frac{1}{4!} \left(\frac{30}{10} \right)^4 \frac{1}{\left[1 - \frac{30}{40} \right]} \frac{2}{53} = \frac{3^4}{24} \frac{1}{1/4} \frac{2}{53} \\ &= \frac{81}{6} \frac{2}{53} = \frac{27}{53} = 0.5094 \end{aligned}$$

(ii) The average waiting time in the queue is given by

$$W_q = \frac{1}{\mu} \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^s}{\left[1 - \frac{\lambda}{s\mu} \right]^2} P_0 = \frac{1}{10} \frac{1}{4 \cdot 4!} \frac{\left[\frac{30}{10} \right]^4}{\left[1 - \frac{30}{40} \right]^2} \frac{2}{53}$$

Simplifying, we have

$$W_q = \frac{1}{10 \times 96} \times \frac{81}{1/16} \times \frac{2}{53} = \frac{27}{530} = 0.0509 \text{ hour or } 3.0566 \text{ minutes}$$

The average time spent in the system is

$$W_s = \frac{1}{\mu} + W_q = 0.1 + 0.0509 = 0.1509 \text{ hour or } 9.0540 \text{ minutes}$$

The average number of cars in the system is

$$L_s = \lambda \times W_s = 30 \times 0.1509 = 4.5270 \text{ cars}$$

(iii) The probability that the pumps are busy is given by

$$P(N \geq s) = P(N \geq 4) = 0.5094$$

Thus, the probability that the pumps would be idle on average is

$$1 - P(N \geq 4) = 1 - 0.5094 = 0.4946$$

◆

EXAMPLE 7.20. On every Sunday morning, a dental hospital renders free dental service to the patients. As per the hospital rules, 3 dentists who are equally qualified and experienced will be on duty then. It takes on an average 10 minutes for a patient to get treatment and the actual time taken is known to vary approximately exponentially around this average. The patients arrive according to the Poisson distribution with an average of 12 per hour. The hospital management wants to investigate the following:

- (i) The expected number of patients waiting in the queue.
- (ii) The average time that a patient spends at the hospital.

(Anna, Nov. 2007)

Solution. From the given data, we have

$$\lambda = 12/\text{hour}, \mu = 6/\text{hour} \text{ and } s = 3$$

The traffic intensity is

$$\rho = \frac{\lambda}{\mu s} = \frac{12}{6 \times 3} = \frac{12}{18} = \frac{2}{3} < 1$$

The probability of no customer in the system is

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s! \left[1 - \frac{\lambda}{\mu s} \right]} \left(\frac{\lambda}{\mu} \right)^s \right]^{-1} = \left[\sum_{n=0}^2 \frac{1}{n!} \left(\frac{12}{6} \right)^n + \frac{1}{3! \left[1 - \frac{12}{18} \right]} \left(\frac{12}{6} \right)^3 \right]^{-1}$$

i.e.

$$\begin{aligned} P_0 &= \left[\sum_{n=0}^2 \frac{2^n}{n!} + \frac{2^3}{6 \times [1 - 2/3]} \right]^{-1} = \left[1 + 2 + \frac{2^2}{2} + \frac{8}{6 \times 1/3} \right]^{-1} \\ &= [1 + 2 + 2 + 4]^{-1} = [9]^{-1} = \frac{1}{9} \end{aligned}$$

- (i) The average number of customers in the system is

$$L_q = \frac{1}{s \cdot s!} \frac{\left[\frac{\lambda}{\mu} \right]^{s+1}}{\left[1 - \frac{\lambda}{\mu s} \right]^2} P_0$$

Substituting the values, we get

$$L_q = \frac{1}{3 \cdot 3!} \frac{\left[\frac{12}{6} \right]^4}{\left[1 - \frac{12}{18} \right]^2} \frac{1}{9} = \frac{1}{3 \times 6} \frac{2^4}{1/9} \frac{1}{9} = \frac{16}{18} = \frac{8}{9} = 0.8889$$

- (ii) The average waiting time of a customer at the queue is

$$W_q = \frac{1}{\lambda} L_q = \frac{1}{12} \times \frac{8}{9} = \frac{2}{27}$$

The average waiting time of a customer at the system is given by

$$W_s = W_q + \frac{1}{\mu} = \frac{2}{27} + \frac{1}{6} = \frac{13}{54} = 0.2407 \text{ hour or } 14.4444 \text{ minutes}$$

◆

PROBLEM SET 7.2

1. A bank has three teller counters. The customers to the bank choose a teller counter at random. It is given that the arrivals to the teller counters are Poisson-distributed at the average rate λ and that mean service rate at the teller counters is $\frac{\lambda}{2}$. Find the steady average queue at each counter.
2. A watch company in a small town has two service counters. Suppose that customers arrive in a Poisson fashion at the counter at the rate of 16 per hour and that the service time for each customer is exponential with mean 6 minutes. Find:
 - (a) Fraction of time both service counters are busy.
 - (b) Average number of customers in the system.
 - (c) Average number of customers waiting in the queue.
 - (d) Average waiting time a customer spends in the queue.
 - (e) Expected time a customer spends in the system.
 - (f) Traffic intensity of the system.
3. In the (M/M/s) : (∞ /FIFO) model, if $s = 3$ and $\frac{\lambda}{s\mu} = \frac{2}{3}$, find the following:
 - (a) The probability of no customers in the system.
 - (b) The average number of customers in the system.
 - (c) The average number of customers waiting in the queue.
 - (d) The probability that a customer has to wait for service.
4. Telephone calls arrive at the general enquiry section of a government office with two operators. If both operators are busy, the calls are automatically held on queue to be answered on a first-come-first-served basis. It is known that the inter-service time is exponentially distributed with mean of 6 minutes. The calls have exponentially distributed inter-arrival times and the calls occur at a rate of 15 per hour. Find:
 - (a) Fraction of time both operators are busy.
 - (b) Average number of customers waiting in the queue.
 - (c) Average number of customers in the system.
 - (d) Average waiting time a customer spends in the queue.
 - (e) Expected time a customer spends in the system.
 - (f) Traffic intensity of the system.
5. A post office has three clerks serving at the service counters. Customers arrive at the service counters at the average of 18 per hour. The average service time for each customer is 5 minutes. Suppose that the customers arrive in a Poisson fashion and that the inter-service time is exponentially distributed. Find:
 - (a) Probability that all the clerks will be busy.
 - (b) Average number of customers waiting in the queue.
 - (c) Average number of customers in the system.
 - (d) Average waiting time a customer spends in the queue.
 - (v) Find the expected time a customer spends in the system.
 - (e) Find the traffic intensity of the system.
6. A grocery store has 3 service counters. Customers to the store are found to arrive in a Poisson fashion at the rate of 15 per hour. It is estimated that the length of service on the customers is exponentially distributed with mean length 10 minutes. Find:

- (a) Fraction of time all the service counters are busy.
 - (b) Average number of customers waiting in the queue.
 - (c) Average number of customers in the system.
 - (d) Average waiting time a customer spends in the queue.
 - (e) Expected time a customer spends in the system.
 - (f) Utilization rate of the system.
7. A petrol pump station has 3 pumps. The service times follow an exponential distribution with mean of 5 minutes and the cars arrive for service in a Poisson process at the rate of 8 cars per hour. Find the probability that a customer has to wait for service. Find:
- (a) Probability that a customer has to wait for service.
 - (b) Average number of customers waiting in the queue.
 - (c) Average number of cars in the system.
 - (d) Average waiting time a customer spends in the queue.
 - (e) Expected time a customer spends in the system.
 - (f) Traffic intensity of the system.
8. A local hospital has 3 doctors for treating patients. The patients arrive at the hospital according to Poisson distribution with an average of 8 per hour. On an average, a doctor takes about 15 minutes to treat each patient and the actual time taken is known to vary approximately exponentially around this average. Find
- (a) Traffic intensity of the system.
 - (b) Probability that a customer has to wait for service.
 - (c) Average number of customers waiting in the queue.
 - (d) Average number of customers in the system.
 - (e) Average waiting time a customer spends in the queue.
 - (f) Expected time a customer spends in the system.

7.4 MODEL III—(M/M/1): (k/FIFO) SINGLE SERVER WITH FINITE CAPACITY

Model III represents the scenario in which the system can permit or accommodate only a finite number k of arrivals. If a customer arrives and the queue is full, then the customer has to leave and go for service elsewhere without joining the queue.

Thus, we make the following assumptions:

(A1) The mean arrival rate is

$$\lambda_n = \begin{cases} \lambda & \text{for } n = 0, 1, 2, \dots, k-1 \\ 0 & \text{for } n = k, k+1, k+2, \dots \end{cases}$$

(A2) The mean service rate is

$$\mu_n = \mu \quad \text{for } n = 1, 2, 3, \dots$$

For a Poisson queue system, the steady-state probabilities $P_n = P(N = n)$ are given by the difference equations

$$0 = -(\lambda_n + \mu_n)P_n + \lambda_{n-1}P_{n-1} + \mu_{n+1}P_{n+1} \quad (7.39)$$

and

$$0 = -\lambda_0P_0 + \mu_1P_1 \quad (7.40)$$

As the system can accommodate only k arrivals, it is clear that

$$P_n = 0 \text{ for } n > k \quad (7.41)$$

Using the values of λ_n and μ_n in the difference Eqs. (7.39) and (7.40), we get

$$\mu P_{n+1} = (\lambda + \mu)P_n - \lambda P_{n-1} \text{ for } 1 \leq n \leq k-1 \quad (7.42)$$

and

$$\mu P_1 = \lambda P_0 \quad (7.43)$$

Taking $n = k$ in Eq. (7.39) and noting that $\lambda_k = 0$ and using Eq. (7.41), we also have

$$\mu P_k = \lambda P_{k-1} \quad (7.44)$$

From Eq. (7.43), we have

$$P_1 = \frac{\lambda}{\mu} P_0 \quad (7.45)$$

Taking $n = 1$ in Eq. (7.42) and using (7.45), we get

$$\mu P_2 = (\lambda + \mu)P_1 - \lambda P_0 = (\lambda + \mu) \frac{\lambda}{\mu} P_0 - \lambda P_0 = \frac{\lambda^2}{\mu} P_0$$

Thus, we have

$$P_2 = \left(\frac{\lambda}{\mu}\right)^2 P_0 \quad (7.46)$$

Similarly, taking $n = 2$ in Eq. (7.42) and using (7.46) and (7.45), we get

$$\mu P_3 = (\lambda + \mu)P_2 - \lambda P_1 = (\lambda + \mu) \frac{\lambda^2}{\mu^2} P_0 - \frac{\lambda^2}{\mu} P_0 = \frac{\lambda^3}{\mu^2} P_0$$

Thus, we have

$$P_3 = \left(\frac{\lambda}{\mu}\right)^3 P_0$$

Proceeding in this manner, we have

$$P_n = \left(\frac{\lambda}{\mu}\right)^n P_0 \text{ for } 1 \leq n \leq k-1 \quad (7.47)$$

From Eq. (7.44), we also have

$$P_k = \frac{\lambda}{\mu} P_{k-1} = \frac{\lambda}{\mu} \left(\frac{\lambda}{\mu}\right)^{k-1} P_0$$

i.e.

$$P_k = \left(\frac{\lambda}{\mu}\right)^k P_0 \quad (7.48)$$

Since the total probability is 1, we have

$$\sum_{n=0}^{\infty} P_n = 1$$

Using Eq. (7.41), we have

$$\sum_{n=0}^k P_n = 1$$

i.e.

$$\sum_{n=0}^k \left(\frac{\lambda}{\mu}\right)^n P_0 = 1$$

Thus, it follows that

$$P_0 \frac{\left[1 - \left(\frac{\lambda}{\mu}\right)^{k+1}\right]}{1 - \frac{\lambda}{\mu}} = 1 \quad (7.49)$$

which holds whenever $\lambda \neq \mu$.

Hence, when $\lambda \neq \mu$, we have

$$P_0 = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{k+1}} = \frac{1 - \rho}{1 - \rho^{k+1}}$$

where $\rho = \frac{\lambda}{\mu}$ is the traffic intensity.

When $\lambda = \mu$, we have

$$P_0 = \lim_{\rho \rightarrow 1} \left[\frac{1 - \rho}{1 - \rho^{k+1}} \right]$$

Using L'Hospital's rule, it follows that

$$P_0 = \lim_{\rho \rightarrow 1} \left[\frac{-1}{-(k+1)\rho^k} \right] = \frac{1}{k+1}$$

Combining the two cases, we have

$$P_0 = \begin{cases} \frac{1-\rho}{1-\rho^{k+1}} & \text{if } \rho \neq 1 \text{ or } \lambda \neq \mu \\ \frac{1}{k+1} & \text{if } \rho = 1 \text{ or } \lambda = \mu \end{cases} \quad (7.50)$$

From Eqs. (7.47) and (7.48), it follows then that for all values of $n = 0, 1, 2, \dots, k$, we have

$$P_n = \begin{cases} \frac{(1-\rho)\rho^n}{1-\rho^{k+1}} & \text{if } \rho \neq 1 \text{ or } \lambda \neq \mu \\ \frac{1}{k+1} & \text{if } \rho = 1 \text{ or } \lambda = \mu \end{cases} \quad (7.51)$$

Remark 7.2. For Model III, we need not assume that $\lambda < \mu$ since the queue cannot build up unbounded. We note that the steady-state probabilities, P_n , exist even for $\rho \geq 1$.

7.4.1 Characteristics of Model III

1. Average or expected number of customers in the system (L_s): We have two cases to consider: (i) $\rho \neq 1$ and (ii) $\rho = 1$. When $\rho \neq 1$ (i.e. when $\lambda \neq \mu$), we have

$$L_s = \sum_{n=0}^k nP_n = \sum_{n=0}^k n \frac{(1-\rho)\rho^n}{1-\rho^{k+1}} = \frac{1-\rho}{1-\rho^{k+1}} \sum_{n=0}^k n\rho^n = \frac{(1-\rho)\rho}{1-\rho^{k+1}} \sum_{n=0}^k n\rho^{n-1}$$

Thus,

$$L_s = \frac{(1-\rho)\rho}{1-\rho^{k+1}} \sum_{n=0}^k \frac{d}{d\rho} \rho^n = \frac{(1-\rho)\rho}{1-\rho^{k+1}} \frac{d}{d\rho} \left[\frac{1-\rho^{k+1}}{1-\rho} \right] \quad (7.52)$$

We note that

$$\frac{d}{d\rho} \left[\frac{1-\rho^{k+1}}{1-\rho} \right] = \frac{(1-\rho) [-(k+1)\rho^k] - (1-\rho^{k+1})(-1)}{(1-\rho)^2} = \frac{-(k+1)\rho^k + (k+1)\rho^{k+1} + 1 - \rho^{k+1}}{(1-\rho)^2}$$

Simplifying, we have

$$\frac{d}{d\rho} \left[\frac{1-\rho^{k+1}}{1-\rho} \right] = \frac{(1-\rho^{k+1}) - (k+1)\rho^k(1-\rho)}{(1-\rho)^2} \quad (7.53)$$

Substituting Eq. (7.53) into Eq. (7.52), we get

$$L_s = \frac{(1-\rho)\rho}{1-\rho^{k+1}} \left[\frac{(1-\rho^{k+1}) - (k+1)\rho^k(1-\rho)}{(1-\rho)^2} \right]$$

Simplifying,

$$L_s = \frac{\rho}{1-\rho} - \frac{(k+1)\rho^{k+1}}{1-\rho^{k+1}} \quad (7.54)$$

(if $\rho \neq 1$ or $\lambda \neq \mu$)

If $\rho = 1$ or $\lambda = \mu$, then we note that

$$L_s = \sum_{n=0}^k nP_n = \sum_{n=0}^k n \frac{1}{k+1} = \frac{1}{k+1} \sum_{n=0}^k n = \frac{1}{k+1} \frac{k(k+1)}{2} = \frac{k}{2} \quad (7.55)$$

Combining the two cases from Eqs. (7.54) and (7.55), we conclude that

$$L_s = \begin{cases} \frac{\rho}{1-\rho} - \frac{(k+1)\rho^{k+1}}{1-\rho^{k+1}} & \text{if } \rho \neq 1 \\ \frac{k}{2} & \text{if } \rho = 1 \end{cases} \quad (7.56)$$

2. Average or expected number of customers in the queue (L_q):

$$L_q = E(N-1) = \sum_{n=1}^k (n-1)P_n = \sum_{n=1}^k nP_n - \sum_{n=1}^k P_n$$

i.e.

$$L_q = \sum_{n=0}^k nP_n - \sum_{n=1}^k P_n = L_s - (1 - P_0) \quad (7.57)$$

From Little's formula (iv), we know that

$$L_q = L_s - \frac{\lambda}{\mu}$$

which is true when the mean arrival rate is λ throughout. However, from Eq. (7.57), $1 - P_0 \neq \frac{\lambda}{\mu}$ because the mean arrival rate is λ as long as there is a vacancy in the queue and is zero when the system is full.

This motivates us to define the *effective arrival rate*, denoted by λ' , which is defined by using Eq. (7.57) and Little's formula (iv) as

$$\frac{\lambda'}{\mu} = 1 - P_0 \quad \text{or} \quad \lambda' = \mu(1 - P_0)$$

Thus, Eq. (7.57) can be rewritten as

$$L_q = L_s - \frac{\lambda'}{\mu}$$

which is the modified Little's formula for Model III.

3. Average waiting time of a customer in the system (W_s): By the modified Little's formula (i), we know that

$$W_s = \frac{1}{\lambda'} L_s \quad (7.58)$$

where $\lambda' = \mu(1 - P_0)$ is the effective arrival rate and L_s is defined by Eq. (7.56).

4. Average waiting time of a customer in the queue (W_q): By the modified Little's formula (ii), we know that

$$W_q = \frac{1}{\lambda'} L_q \quad (7.59)$$

where $\lambda' = \mu(1 - P_0)$ is the effective arrival rate and L_q is given in §2.

EXAMPLE 7.21. Patients arrive at a doctor's clinic according to Poisson distribution at a rate of 30 patients per hour. The waiting room does not accommodate more than 9 patients. Examination time per patient is exponential with mean rate of 20 per hour. Find the:

- (i) Probability that an arriving patient will not wait.
- (ii) Effective arrival rate.
- (iii) Average number of patients in the clinic.
- (iv) Expected time a patient spends in the clinic.
- (v) Average number of patients in the queue.
- (vi) Expected waiting time of a patient in the queue.

Solution. From the given data, we have

$$\lambda = 30/\text{hour}, \mu = 20/\text{hour} \text{ and } k = 9 + 1 = 10$$

(Note that the maximum capacity k of the system equals 10 because the waiting capacity is 9 and that one patient is treated (served) at a time.)

Then, the traffic intensity is given by

$$\rho = \frac{\lambda}{\mu} = \frac{30}{20} = \frac{3}{2} = 1.5$$

(Note that $\rho \neq 1$.)

(i) The probability that an arriving patient will not wait is given by

$$P(\text{no customer in the system}) = P_0 = \frac{1 - \rho}{1 - \rho^{k+1}} = \frac{\rho - 1}{\rho^{k+1} - 1}$$

Substituting the values of ρ and k , we have

$$P_0 = \frac{1.5 - 1}{1.5^{10} - 1} = \frac{0.5}{56.6650} = 0.0088$$

(ii) The effective arrival rate is given by

$$\lambda' = \mu(1 - P_0) = 20(1 - 0.0088) = 20 \times 0.9912 = 19.8235$$

(iii) The average number of patients in the clinic is given by

$$L_s = \frac{\rho}{1 - \rho} - \frac{(k+1)\rho^{k+1}}{1 - \rho^{k+1}} = \frac{1.5}{1 - 1.5} - \frac{10 \times 1.5^{10}}{-56.6650}$$

i.e.

$$L_s = -\frac{1.5}{0.5} + \frac{576.6504}{56.6650} = -3 + 10.1765 = 7.1765$$

(iv) The expected time a patient spends in the clinic is given by

$$W_s = \frac{1}{\lambda'} L_s = \frac{1}{19.8235} \times 7.1765 = 0.3620 \text{ hour or } 21.72 \text{ minutes}$$

(v) The average number of patients in the queue is given by

$$L_q = L_s - \frac{\lambda'}{\mu} = 7.1765 - \frac{19.8235}{20} = 7.1765 - 0.9912 = 6.1853$$

(vi) The expected waiting time of a patient in the queue is given by

$$W_q = \frac{1}{\lambda'} L_q = \frac{1}{19.8235} \times 6.1853 = 0.3120 \text{ hour or } 18.72 \text{ minutes}$$



EXAMPLE 7.22. The arrival of customers at a one window drive-in bank follows a Poisson distribution with mean 18 per hour. Service time per customer is exponential with mean 10 minutes. The space in front of the window including that for serviced car can accommodate a maximum of 3 cars (including the car being served) and others can wait outside this space. Find:

- (i) probability that an arriving customer will not wait.
- (ii) Effective arrival rate.
- (iii) Expected number of customers in the drive-in bank.
- (iv) Expected time a customer spends in the drive-in bank.
- (v) Expected number of customers in the queue.
- (vi) Expected waiting time of a customer in the queue.

Solution. From the given data, we have $\lambda = 18/\text{hour}$, $\mu = 6/\text{hour}$ and $k = 3$.

The traffic intensity of the system is given by

$$\rho = \frac{\lambda}{\mu} = \frac{18}{6} = 3$$

(Note that $\rho \neq 1$.)

- (i) The probability that an arriving customer will not wait is

$$P_0 = \frac{1 - \rho}{1 - \rho^{k+1}} = \frac{\rho - 1}{\rho^4 - 1} = \frac{2}{80} = \frac{1}{40} = 0.025$$

- (ii) The effective arrival rate is

$$\lambda' = \mu(1 - P_0) = 6 \times \frac{39}{40} = \frac{117}{20} = 5.85$$

- (iii) The expected number of customers in the drive-in bank is

$$L_s = \frac{\rho}{1 - \rho} - \frac{(k+1)\rho^{k+1}}{1 - \rho^{k+1}} = \frac{\rho}{1 - \rho} - \frac{4\rho^4}{1 - \rho^4}$$

Substituting the values, we get

$$L_s = \frac{3}{1 - 3} - \frac{4 \times 81}{(-80)} = -1.5 + 4.05 = 2.55$$

- (iv) The expected time a customer spends in the drive-in bank is

$$W_s = \frac{1}{\lambda'} L_s = \frac{1}{5.85} \times 2.55 = 0.4359 \text{ hour or } 26.1538 \text{ minutes}$$

- (v) The expected number of customers in the queue is

$$L_q = L_s - \frac{\lambda'}{\mu} = 2.55 - \frac{5.85}{6} = 2.55 - 0.9750 = 1.5750$$

(vi) The expected waiting time of a customer in the queue is

$$W_q = \frac{1}{\lambda'} L_q = \frac{1}{5.85} \times 1.5750 = 0.2692 \text{ hour or } 16.1538 \text{ minutes}$$

◆

EXAMPLE 7.23. A city has a one-person barber shop which can accommodate a maximum of 5 people at a time (4 waiting and 1 getting haircut). On average, customers arrive at the rate of 8 per hour and the barber takes 6 minutes for serving each customer. It is estimated that the arrival process is Poisson and the service time is an exponential random variable. Find:

- (i) Percentage of time the barber is idle.
- (ii) Fraction of potential customers who will be turned away.
- (iii) Effective arrival rate of customers for the shop.
- (iv) Expected number of customers in the barber shop.
- (v) Expected time a person spends in the barber shop.
- (vi) Expected number of customers waiting for a hair-cut.
- (vii) Expected waiting time of a customer in the queue.

Solution. From the given data, we have $\lambda = 8/\text{hour}$, $\mu = 10/\text{hour}$ and $k = 5$

Thus, the traffic intensity of the system is given by

$$\rho = \frac{\lambda}{\mu} = \frac{8}{10} = \frac{4}{5} = 0.8$$

(i) The probability that the barber is idle is given by

$$P_0 = \frac{1 - \rho}{1 - \rho^{k+1}} = \frac{1 - \rho}{1 - \rho^6}$$

Note that

$$1 - \rho^6 = 1 - \left(\frac{4}{5}\right)^6 = 1 - (0.8)^6 = 0.7379$$

Then, we have

$$P_0 = \frac{0.2}{0.7379} = 0.2710$$

Thus, the barber is idle during 27.1% of the time.

(ii) The probability that a potential customer will be turned away is given by

$$\begin{aligned} P(N \geq 5) &= \frac{(1 - \rho)\rho^5}{1 - \rho^{k+1}} = \frac{0.2 \times (0.8)^5}{1 - \rho^6} \\ &= \frac{0.2 \times 0.3277}{0.7379} = \frac{0.0655}{0.7379} = 0.0888 \end{aligned}$$

Thus, 8.88% of the potential customers will be turned away due to limited accommodation in the barber's shop.

(iii) The effective arrival rate of customers for the shop is given by

$$\lambda' = \mu(1 - P_0) = 10(1 - 0.2710) = 10 \times 0.7290 = 7.29$$

(iv) The expected number of customers in the barber shop is given by

$$L_s = \frac{\rho}{1 - \rho} - \frac{(k+1)\rho^{k+1}}{1 - \rho^{k+1}} = \frac{\rho}{1 - \rho} - \frac{6\rho^6}{1 - \rho^6}$$

Substituting the values, we get

$$L_s = \frac{0.8}{0.2} - \frac{6 \times (0.8)^6}{0.7379} = 4 - \frac{1.5729}{0.7379} = 4 - 2.1316 = 1.8684$$

(v) The expected time a person spends in the barber shop is given by

$$W_s = \frac{1}{\lambda'} L_s = \frac{1}{7.29} \times 1.8684 = 0.2563 \text{ hour or } 15.3778 \text{ minutes}$$

(vi) The expected number of customers waiting in the queue is given by

$$L_q = L_s - \frac{\lambda'}{\mu} = 1.8684 - \frac{7.29}{10} = 1.8684 - 0.729 = 1.1394$$

(vii) The expected waiting time of a customer in the queue is given by

$$W_q = \frac{1}{\lambda'} L_q = \frac{1}{7.29} \times 1.1394 = 0.1563 \text{ hour or } 9.3778 \text{ minutes}$$



PROBLEM SET 7.3

1. For the $(M/M/1) : k/\text{FIFO}$ Queueing system, prove the following:
 - (a) $L_q = L_s - (1 - P_0)$
 - (b) $W_q = W_s - \frac{1}{\mu}$
2. If $\lambda = 3$ per hour, $\mu = 6$ per hour and maximum capacity $k = 4$ in a $(M/M/1) : (k/\text{FIFO})$ system, find the following:
 - (a) The probability of no customer in the system.
 - (b) The probability of n customers in the system, where $n = 1, 2, 3$.
 - (c) The expected number of customers in the system.
 - (d) The expected number of customers in the queue.
 - (e) The expected time a customer spends in the system.
 - (f) The expected waiting time of a customer in the queue.
3. If $\lambda = 4$ per hour, $\mu = 4$ per hour and maximum capacity $k = 7$ in a $(M/M/1) : (k/\text{FIFO})$ system, find the following:
 - (a) The probability of no customer in the system.

- (b) The probability of n customers in the system, where $n = 1, 2, 3$.
 - (c) The expected number of customers in the system.
 - (d) The expected number of customers in the queue.
 - (e) The expected time a customer spends in the system.
 - (f) The expected waiting time of a customer in the queue.
4. Customers arrive at a beauty parlour with a single beautician according to Poisson distribution at a rate of 20 customers per hour. The waiting room does not accommodate more than 5 people. The service time per customer is exponential with mean rate of 15 per hour. Find
- (a) Probability that an arriving customer has to wait.
 - (b) Effective arrival rate.
 - (c) Average number of customers in the beauty parlour.
 - (d) Expected time a customer spends in the beauty parlour.
 - (e) Average number of customers in the queue.
 - (f) Expected waiting time of a customer in the queue.
5. In a single server Queueing system with Poisson input and exponential service times, it is estimated that the mean arrival rate is 5 calling units per hour, the inter-service time is exponentially distributed with mean 0.2 hour and the maximum possible number of calling units in the system is 3. Find:
- (a) Probability that an arriving customer has to wait.
 - (b) Effective arrival rate.
 - (c) Probability of n customers in the system, where $n = 1, 2, 3$
 - (d) Average number of customers in the system.
 - (e) Expected time a customer spends in the system.
 - (f) Average number of customers in the queue?
 - (g) Expected waiting time of a customer in the queue?
6. A petrol-filling station on a highway has one petrol pump for cars only and has a capacity for four cars (including one at the pump). On average, the cars arrive at the rate of 12 per hour, and each car takes 10 minutes for service. It is estimated that the arrival process is Poisson and the service time is an exponential random variable. Find:
- (a) Probability that an arriving vehicle need not wait.
 - (b) Effective arrival rate.
 - (c) Probability of n cars in the petrol station, where $n = 1, 2, 3$.
 - (d) Average number of cars in the petrol station.
 - (e) Expected time a car spends in the petrol station.
 - (f) Average number of cars in the queue.
 - (g) Expected waiting time of a car in the queue.
7. A local one-person barber shop can accommodate a maximum of 4 people at a time (3 waiting and 1 getting hair-cut). On average, customers arrive at the rate of 6 per hour and the barber takes 15 minutes for each person's hair-cut. It is estimated that the arrival process is Poisson and the service time is an exponential random variable. Find:
- (a) Percentage of time is the barber idle.
 - (b) Fraction of potential customers who will be turned away.

- (c) Effective arrival rate of customers for the shop.
 - (d) Expected number of customers in the barber shop.
 - (e) Expected time a person spends in the barber shop.
 - (f) Expected number of customers waiting for a hair-cut.
 - (g) Expected waiting time of a customer in the queue.
8. Customers arrive at a one window drive-in bank according to Poisson distribution with mean 10 per hour. Service time per customer is exponential with mean 5 minutes. The space in front of the window including that for serviced car can accommodate a maximum of 3 cars and others can wait outside this space. Find:
- (a) The probability that an arriving customer can drive directly to the space in front of the window.
 - (b) The probability that an arriving customer will have to wait outside the indicated space.
 - (c) How long the arriving customer is expected to wait before starting service.
9. Customers arrive at a clinic with only one doctor according to Poisson distribution with mean 15 patients per hour. The waiting room does not accommodate more than 7 patients. Service time per patient is exponential with mean 6 minutes. Find:
- (i) Probability that an arriving patient will not wait.
 - (ii) Effective arrival rate at the clinic.
 - (iii) Expected number of patients at the clinic.
 - (iv) Expected time a patient spends at the clinic.
 - (v) Expected number of patients at the queue.
 - (vi) Expected waiting time of a patient at the queue.
10. Customers arrive at a tailor shop with only one tailor. On average, customers arrive at the rate of 8 per hour and the tailor takes 10 minutes for serving each customer by taking measurements and the tailoring needs. The waiting room does not accommodate more than 3 customers. The arrival process is Poisson and the service time is an exponential random variable. Find
- (a) Probability that an arriving customer will not wait.
 - (b) Effective arrival rate at the tailor shop.
 - (c) Expected number of patients at the tailor shop.
 - (d) Expected time a patient spends at the tailor shop.
 - (e) Expected number of patients at the queue.
 - (f) Expected waiting time of a patient at the queue.

7.5 MODEL IV—(M/M/s): (k/FIFO) MULTIPLE SERVER WITH FINITE CAPACITY

Model IV represents the scenario where the system can accommodate only k arrivals and s service counters, where $s > 1$, serve the customers. Thus, we make the following assumptions:

(A1) The mean arrival rate is given by

$$\lambda_n = \begin{cases} \lambda & \text{for } n = 0, 1, 2, \dots, k-1 \\ 0 & \text{for } n = k, k+1, \dots \end{cases}$$

(A2) The mean service rate is given by

$$\mu_n = \begin{cases} n\mu & \text{for } n = 0, 1, 2, \dots, s-1 \\ s\mu & \text{for } n = s, s+1, \dots \end{cases}$$

(A3) The maximum number of arrivals k that the system can accommodate and the number of service counters s are related by the condition that

$$1 < s < k$$

From the difference Eqs. (7.8) and (7.9) giving steady-state probabilities P_n and P_0 , we have

$$P_n = \frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} P_0 \quad (7.60)$$

where

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \left[\frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right]}$$

or equivalently that

$$P_0^{-1} = 1 + \sum_{n=1}^{\infty} \left[\frac{\lambda_0 \lambda_1 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \right] \quad (7.61)$$

Substituting the values of λ_n and μ_n in Eq. (7.61) and noting that $1 < s < k$, we obtain

$$P_0^{-1} = \left[1 + \frac{\lambda}{1!\mu} + \frac{\lambda^2}{2!\mu^2} + \cdots + \frac{\lambda^{s-1}}{(s-1)!\mu^{s-1}} \right] + \left[\frac{\lambda^s}{(s-1)!\mu^{s-1} \cdot \mu s} + \cdots + \frac{\lambda^k}{(s-1)!\mu^{s-1} \cdot (\mu s)^{k-s+1}} \right]$$

Simplifying, we have

$$\begin{aligned} P_0^{-1} &= \sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{\lambda^s}{s! \mu^s} \left[1 + \frac{\lambda}{\mu s} + \left(\frac{\lambda}{\mu s} \right)^2 + \cdots + \left(\frac{\lambda}{\mu s} \right)^{k-s} \right] \\ &= \sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \sum_{n=s}^k \left(\frac{\lambda}{\mu s} \right)^{n-s} \end{aligned}$$

or equivalently that

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \sum_{n=s}^k \left(\frac{\lambda}{\mu s} \right)^{n-s} \right]^{-1} \quad (7.62)$$

Substituting the values of λ_n and μ_n in (7.60), we also have

$$P_n = \begin{cases} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n P_0 & \text{for } n \leq s \\ \frac{1}{s! s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n P_0 & \text{for } s < n \leq k \\ 0 & \text{for } n > k \end{cases} \quad (7.63)$$

where P_0 is given by Eq. (7.62).

7.5.1 Characteristics of Model IV

1. The average or expected number of customers in the queue or average queue length (L_q): As the given model has s servers and can accommodate only k arrivals, the expected queue length is given by

$$L_q = E[N - s] = \sum_{n=s}^k (n-s)P_n = \sum_{n=s}^k (n-s) \frac{1}{s! s^{n-s}} \left(\frac{\lambda}{\mu}\right)^n P_0$$

i.e.

$$L_q = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s!} \sum_{n=s}^k (n-s) \left(\frac{\lambda}{\mu s}\right)^{n-s} = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s!} \sum_{n=s}^k (n-s) \rho^{n-s} = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0}{s!} \sum_{x=0}^{k-s} x \rho^x$$

where $\rho = \frac{\lambda}{\mu s}$.

Thus, we have

$$L_q = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0 \rho}{s!} \sum_{x=0}^{k-s} x \rho^{x-1} = \frac{\left(\frac{\lambda}{\mu}\right)^s P_0 \rho}{s!} \frac{d}{d\rho} \left[\frac{1 - \rho^{k-s+1}}{1 - \rho} \right] \quad (7.64)$$

We find that

$$\begin{aligned} \frac{d}{d\rho} \left[\frac{1 - \rho^{k-s+1}}{1 - \rho} \right] &= \frac{(1 - \rho) [-(k-s+1)\rho^{k-s}] - (1 - \rho^{k-s+1})(-1)}{(1 - \rho)^2} \\ &= \frac{-(k-s)(1 - \rho)\rho^{k-s} - (1 - \rho)\rho^{k-s} + (1 - \rho^{k-s+1})}{(1 - \rho)^2} \\ &= \frac{-(k-s)(1 - \rho)\rho^{k-s} + 1 - \rho^{k-s}(1 - \rho + \rho)}{(1 - \rho)^2} \end{aligned}$$

Simplifying, we have

$$\frac{d}{d\rho} \left[\frac{1 - \rho^{k-s+1}}{1 - \rho} \right] = \frac{1 - \rho^{k-s} - (k-s)(1 - \rho)\rho^{k-s}}{(1 - \rho)^2} \quad (7.65)$$

Substituting Eq. (7.65) into Eq. (7.64), we get

$$L_q = P_0 \left(\frac{\lambda}{\mu}\right)^s \frac{\rho}{s!(1 - \rho)^2} \left[1 - \rho^{k-s} - (k-s)(1 - \rho)\rho^{k-s} \right] \quad (7.66)$$

where

$$\rho = \frac{\lambda}{\mu s}$$

2. The average or expected number of customers in the system (L_s):

$$\begin{aligned} L_s = E(N) &= \sum_{n=0}^k nP_n = \sum_{n=0}^{s-1} nP_n + \sum_{n=s}^k nP_n \\ &= \sum_{n=0}^{s-1} nP_n + \sum_{n=s}^k (n-s)P_n + \sum_{n=s}^k sP_n \end{aligned}$$

i.e.

$$L_s = \sum_{n=0}^{s-1} nP_n + L_q + s \left[\sum_{n=0}^k P_n - \sum_{n=0}^{s-1} P_n \right] = L_q + \sum_{n=0}^{s-1} nP_n + s \left[1 - \sum_{n=0}^{s-1} P_n \right]$$

i.e.

$$L_s = L_q + \left[s - \sum_{n=0}^{s-1} (s-n)P_n \right] \quad (7.67)$$

Since we know that

$$s - \sum_{n=0}^{s-1} (s-n)P_n \neq \frac{\lambda}{\mu}$$

it is easy to see that Eq. (7.67) does not represent Little's formula. In order that Eq. (7.67) takes the form of Little's formula, we define the *overall effective arrival rate* λ' as follows:

$$\frac{\lambda'}{\mu} = s - \sum_{n=0}^{s-1} (s-n)P_n$$

or equivalently that

$$\lambda' = \mu \left[s - \sum_{n=0}^{s-1} (s-n)P_n \right] \quad (7.68)$$

Using Eq. (7.68), we can express Eq. (7.67) as

$$L_s = L_q + \frac{\lambda'}{\mu} \quad (7.69)$$

which is the modified Little's formula.

3. The average waiting time of a customer in the queue (W_q): From Little's modified formula, the average waiting time of a customer in the queue is given by

$$W_q = \frac{1}{\lambda'} L_q$$

where λ' 's is as defined in Eq. (7.68) and L_q is given by §1.

4. The average waiting time of a customer in the system (W_s): From Little's modified formula, the average waiting time of a customer in the system is given by

$$W_s = \frac{1}{\lambda'} L_s$$

where λ' 's is as defined in Eq. (7.68) and L_s is given by §2.

EXAMPLE 7.24. A dispensary has two doctors and four chairs in the waiting room. The patients who arrive at the dispensary leave when all four chairs in the waiting room of the dispensary are occupied. It is known that patients arrive at the average rate of 8 per hour and spend an average of 10 minutes for their check-up and medical consultation. The arrival process is Poisson and the service time is an exponential random variable. Find:

- (i) Probability that an arriving patient will not wait.

- (ii) Effective arrival rate at the dispensary.
- (iii) Expected number of patients at the queue.
- (iv) Expected waiting time of a patient at the queue.
- (v) Expected number of patients at the dispensary.
- (vi) Expected time a patient spends at the dispensary.

Solution. From the given data, we have $\lambda = 8/\text{hour}$, $\mu = 6/\text{hour}$, $s = 2$, and $k = 2 + 4 = 6$.

We also note that the traffic intensity of the system is

$$\rho = \frac{\lambda}{\mu s} = \frac{8}{6 \times 2} = \frac{8}{12} = \frac{3}{4} = 0.75 < 1$$

- (i) The probability of no customer in the system is given by

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \sum_{n=s}^k \left(\frac{\lambda}{\mu s} \right)^{n-s} \right]^{-1}$$

Substituting the values, we get

$$P_0 = \left[\sum_{n=0}^1 \frac{1}{n!} \left(\frac{8}{6} \right)^n + \frac{1}{2!} \left(\frac{8}{6} \right)^2 \sum_{n=2}^6 \rho^{n-2} \right]^{-1} = \left[\sum_{n=0}^1 \frac{1}{n!} \left(\frac{4}{3} \right)^n + \frac{1}{2!} \left(\frac{4}{3} \right)^2 \sum_{n=2}^6 \rho^{n-2} \right]^{-1}$$

Simplifying,

$$P_0 = \left[1 + \frac{4}{3} + \frac{8}{9} (1 + \rho + \rho^2 + \rho^3 + \rho^4) \right]^{-1} = \left[1 + \frac{4}{3} + \left(\frac{8}{9} \times \frac{1 - \rho^5}{1 - \rho} \right) \right]^{-1}$$

Thus, we have

$$P_0 = \left[1 + \frac{4}{3} + \left(\frac{8}{9} \times \frac{0.7627}{0.25} \right) \right]^{-1} = [1 + 1.3333 + 2.7118]^{-1} = [5.0451]^{-1} = 0.1982$$

- (ii) The effective arrival rate at the dispensary is given by

$$\lambda' = \mu \left[s - \sum_{n=0}^{s-1} (s-n)P_n \right] = 6 \left[2 - \sum_{n=0}^1 (2-n)P_n \right] = 6[2 - 2P_0 - P_1]$$

By (i), $P_0 = 0.1982$. Note that P_1 is calculated as

$$P_1 = \frac{1}{1!} \left(\frac{\lambda}{\mu} \right)^1 P_0 = \frac{8}{6} \times P_0 = \frac{4}{3} \times 0.1982 = 0.2643$$

Thus, we have

$$\lambda' = 6[2 - 0.3964 - 0.2643] = 6 \times 1.3393 = 8.0358$$

(iii) The expected number of patients at the queue is given by

$$L_q = P_0 \left(\frac{\lambda}{\mu} \right)^s \frac{\rho}{s!(1-\rho)^2} \left[1 - \rho^{k-s} - (k-s)(1-\rho)\rho^{k-s} \right]$$

Substituting the values, we get

$$L_q = 0.1982 \times \left(\frac{4}{3} \right)^2 \times \frac{0.75}{2!(1-0.75)^2} \times [1 - (0.75)^4 - 4(0.25)(0.75)^4]$$

Simplifying, we have

$$L_q = 0.1982 \times 1.7778 \times 6 \times 0.3672 = 0.7763$$

(iv) The expected waiting time of a patient at the queue is given by

$$W_q = \frac{1}{\lambda'} L_q = \frac{1}{8.0358} \times 0.7763 = 0.0966 \text{ hour or } 5.7963 \text{ minutes}$$

(v) The expected number of patients at the dispensary is given by

$$L_s = L_q + \frac{\lambda'}{\mu} = 0.7763 + \frac{8.0358}{6} = 0.7763 + 1.3393 = 2.1156$$

(vi) The expected time a patient spends at the dispensary is given by

$$W_s = \frac{1}{\lambda'} L_s = \frac{1}{8.0358} \times 2.1156 = 0.2633 \text{ hour or } 15.7963 \text{ minutes}$$

◆

EXAMPLE 7.25. A beauty salon has 2 barbers and 6 chairs to accommodate waiting customers. Potential customers, who arrive when all the 6 chairs are full, leave the salon immediately. Customers arrive at the salon at the average rate of 10 per hour and spend an average of 10 minutes in the barber's chair. The arrival process is Poisson and the service time is an exponential random variable. Find:

- (i) Probability of no customer in the beauty salon.
- (ii) Effective arrival rate at the beauty salon.
- (iii) Expected number of customers at the queue.
- (iv) Expected waiting time of a customer at the queue.
- (v) Expected number of customers at the beauty salon.
- (vi) Expected time a customer spends at the beauty salon.

Solution. From the given data, we have $\lambda = 10/\text{hour}$, $\mu = 6/\text{hour}$, $s = 2$ and $k = 2 + 6 = 8$

The traffic intensity of the system is given by

$$\rho = \frac{\lambda}{\mu s} = \frac{10}{6 \times 2} = \frac{10}{12} = \frac{5}{6} < 1$$

(i) The probability of no customer in the beauty salon is given by

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \sum_{n=s}^k \left(\frac{\lambda}{\mu s} \right)^{n-s} \right]^{-1}$$

Substituting the values, we get

$$P_0 = \left[\sum_{n=0}^1 \frac{1}{n!} \left(\frac{10}{6} \right)^n + \frac{1}{2!} \left(\frac{10}{6} \right)^2 \sum_{n=2}^8 \rho^{n-2} \right]^{-1} = \left[\sum_{n=0}^1 \frac{1}{n!} \left(\frac{5}{3} \right)^n + \frac{1}{2!} \left(\frac{5}{3} \right)^2 \sum_{m=0}^6 \rho^m \right]^{-1}$$

Simplifying, we have

$$P_0 = \left[1 + \frac{5}{3} + \left(\frac{1}{2} \times \frac{25}{9} \times \frac{1-\rho^7}{1-\rho} \right) \right]^{-1} = \left[1 + \frac{5}{3} + \frac{18.0230}{3} \right]^{-1} = [8.6744]^{-1} = 0.1153$$

(ii) The effective arrival rate at the beauty salon is given by

$$\lambda' = \mu \left[s - \sum_{n=0}^{s-1} (s-n)P_n \right] = 6 \left[2 - \sum_{n=0}^1 (2-n)P_n \right] = 6[2 - 2P_0 - P_1]$$

By (i), $P_0 = 0.1153$. Note that P_1 is calculated as

$$P_1 = \frac{1}{1!} \left(\frac{\lambda}{\mu} \right)^1 P_0 = \frac{8}{6} \times P_0 = \frac{4}{3} \times 0.1982 = 0.2643$$

Thus, we have

$$\lambda' = 6[2 - 0.3964 - 0.2643] = 6 \times 1.3393 = 8.0358$$

(iii) The expected number of customers at the queue is given by

$$L_q = P_0 \left(\frac{\lambda}{\mu} \right)^s \frac{\rho}{s!(1-\rho)^2} \left[1 - \rho^{k-s} - (k-s)(1-\rho)\rho^{k-s} \right]$$

Substituting the values, we get

$$\begin{aligned} L_q &= 0.1153 \times \left(\frac{10}{6} \right)^2 \times \frac{5/6}{2!(1-5/6)^2} \times [1 - \rho^6 - 6(1-\rho)\rho^6] \\ &= 0.1153 \times 2.7778 \times 14.9874 \times 0.3302 = 1.5850 \end{aligned}$$

(iv) The expected waiting time of a customer at the queue is given by

$$W_q = \frac{1}{\lambda'} L_q = \frac{1}{9.4638} \times 1.5850 = 0.1675 \text{ hour or } 10.0488 \text{ minutes}$$

- (v) The expected number of customers at the beauty salon is given by

$$L_s = L_q + \frac{\lambda'}{\mu} = 1.5850 + \frac{9.4638}{6} = 1.5850 + 1.5773 = 3.1623$$

- (vi) The expected time a customer spends at the beauty salon is given by

$$W_s = \frac{1}{\lambda'} L_s = \frac{1}{9.4638} \times 3.1623 = 0.3341 \text{ hour or } 20.0488 \text{ minutes}$$



PROBLEM SET 7.4

1. A barber shop with 2 barbers has 5 chairs to accommodate waiting customers. Potential customers, who arrive when all the 5 chairs are full, leave the salon immediately. Customers arrive at the salon at the average rate of 12 per hour and spend an average of 6 minutes in the barber's chair. The arrival process is Poisson and the service time is an exponential random variable. Find:
 - (a) P_0, P_1 and P_7 .
 - (b) The effective arrival rate at the barber shop.
 - (c) The expected number of customers at the queue.
 - (d) The expected waiting time of a customer at the queue.
 - (e) The expected number of customers at the barber shop.
 - (f) The expected time a customer spends at the barber shop.
2. A petrol-filling station on a national highway has 2 petrol pumps which can fill 2 cars simultaneously. Due to space limitation, only 6 cars can wait in the queue for filling, and other cars will leave the station without entering the queue. The cars arrive at the station at the average rate of 20 per hour and spend an average of 4 minutes in the petrol pump. The arrival process is Poisson and the service time is an exponential random variable. Find:
 - (i) P_0, P_1 and P_8 .
 - (ii) The effective arrival rate at the petrol station.
 - (iii) The expected number of cars at the queue.
 - (iv) The expected waiting time of a car at the queue.
 - (v) The expected number of cars at the petrol station.
 - (vi) The expected time a car spends at the petrol station.
3. A car service centre has 2 bays, which are operated simultaneously to service the cars. Due to space limitation, only 4 cars can wait in the queue for getting service, and other cars leave the centre without entering the queue. The cars arrive at the service centre at the average rate of 18 per day. The service time in both the bays is exponentially distributed with the average service rate $\mu = 12$ cars per day per bay. Find:
 - (i) P_0, P_1 and P_8 .
 - (ii) The effective arrival rate of cars at the car service centre.
 - (iii) The expected number of cars at the queue.
 - (iv) The expected waiting time of a car at the queue.
 - (v) The expected number of cars at the car service centre.
 - (vi) The expected time a car spends at the car service centre.

4. A small town bank has 3 tellers and 7 chairs to accommodate waiting customers. Potential customers, who arrive when all the chairs in the waiting hall are full, leave the bank immediately. Customers arrive at the bank at the average rate of 30 per hour and spend an average of 4 minutes in the teller counter. The arrival process is Poisson and the service time is an exponential random variable. Find:
 - (i) P_0, P_1, P_2 and P_{10} .
 - (ii) The effective arrival rate at the bank.
 - (iii) The expected number of customers at the queue.
 - (iv) The expected waiting time of a customer at the queue.
 - (v) The expected number of customers at the bank.
 - (vi) The expected time a customer spends at the bank.
5. A three person barber shop has 6 chairs to accommodate waiting customers. Potential customers who arrive when all the 6 chairs are full, leave without entering the barber shop. Customers arrive at the average rate of 8 per hour and spend an average of 12 minutes in the barber's chair. The arrival process is Poisson and the service time is an exponential random variable. Find:
 - (i) P_0, P_1, P_2 and P_9 .
 - (ii) The effective arrival rate at the barber shop.
 - (iii) The expected number of customers at the queue.
 - (iv) The expected waiting time of a customer at the queue.
 - (v) The expected number of customers at the barber shop.
 - (vi) The expected time a customer spends at the barber shop.

7.6 THE $(M/G/1)$ QUEUEING SYSTEM

The Models I–IV studied in the preceding sections are all Markovian models, whose steady-state probabilities are described by birth and death processes. In this section, we consider a non-Markovian Queueing model, namely the $(M/G/1)$ Queueing system, i.e. we consider the Queueing model with Poisson input, general output, single server with infinite capacity and general service discipline. We note that for this non-Markovian model, the distribution of the service time does not possess the memoryless property, i.e. it is not exponential. Thus, the process $\{N(t)\}$ which provides the number of customers at time t (or the state of the system or system size at time t) will be no longer Markovian. However, the analysis of the process can be carried out on an associated process, which is Markovian. Kendall used the concept of regeneration point by suitable choice of regeneration points and extracts, from the process $\{N(t)\}$, Markov chains in discrete time at those points. This is called the *technique of imbedded Markov chains*.

We shall carry out our analysis of $M/G/1$ Queueing system by using Kendall's method. Here, we consider a single-server Queueing system with infinite capacity, Poisson arrivals and general service discipline. The model has arbitrary service time, and it is not necessary to be memoryless.

We assume that the input process is Poisson with mean-arrival rate λ and that the service times are independent and identically distributed (i.i.d.) random variables with an arbitrary (general) distribution with mean $\frac{1}{\mu}$. We denote the service time by v , its cumulative distribution function by $B(t)$, and its probability density function (when it exists) by

$$b(t) = B'(t) = \frac{d}{dt} [B(t)] \quad (7.70)$$

Also, the Laplace transform of the density function $b(t)$ is denoted by $B^*(s)$, i.e.

$$B^*(s) = \int_{t=0}^{\infty} e^{-st} b(t) dt \quad (7.71)$$

Let $t_n, n = 1, 2, \dots$ ($t_0 = 0$) be the n th departure epoch, i.e. the instant at which the n th unit or customer completes his/her service and leaves the system. These points t_n are called the *regeneration points* of the process $\{N(t)\}$, where $N(t)$ represents the number of customers in the system at time t .

Let t_n be the time of departure of the n th customer, where $n = 1, 2, \dots$, and X_n be the number of customers in the system at time t_n . Thus, we have

$$X_n = N(t_n)$$

Let A_n be the number of customers arriving during the service time of the n th customer. Then we have

$$X_{n+1} = \begin{cases} X_n - 1 + A_{n+1} & \text{if } X_n \geq 1 \\ A_{n+1} & \text{if } X_n = 0 \end{cases} \quad (7.72)$$

Since the service times of all the units have the same distribution, we have

$$A_n \equiv A \text{ for } n = 1, 2, \dots$$

Thus, we have

$$P\{A = r \mid \text{service time of a unit is } t\} = \frac{e^{-\lambda t} (\lambda t)^r}{r!}$$

Next, we find the probability distribution of A , the number of arrivals during the service time of a unit. We define

$$\alpha_r = P[A = r], \text{ where } r = 0, 1, 2, \dots$$

We find that

$$\alpha_r = P[A = r] = \int_0^{\infty} P\{A = r \mid \text{service time of a unit is } t\} b(t) dt$$

i.e.

$$\alpha_r = P[A = r] = \int_0^{\infty} \frac{e^{-\lambda t} (\lambda t)^r}{r!} b(t) dt \quad (7.73)$$

The probabilities

$$p_{ij} = P[X_{n+1} = j \mid X_n = i]$$

are given by

$$\begin{aligned} p_{ij} &= \alpha_{j-i+1} & \text{for } i \geq 1, j \geq i-1 \\ &= 0 & \text{for } i \geq 1, j < i-1 \\ p_{0j} &= p_{ij} = \alpha_j & \text{for } j \geq 0 \end{aligned} \quad (7.74)$$

From Eq. (7.74), it is clear that $\{X_n, n \geq 0\}$ is a Markov chain having the transition probability matrix

$$P = [p_{ij}] = \begin{bmatrix} \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ \alpha_0 & \alpha_1 & \alpha_2 & \cdots \\ 0 & \alpha_0 & \alpha_1 & \cdots \\ 0 & 0 & \alpha_0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \quad (7.75)$$

As every state can be reached from every other state, the Markov chain $\{X_n\}$ is irreducible. As $p_{ij} \geq 0$, the chain is aperiodic. It can also be shown that when the traffic intensity

$$\rho = \frac{\lambda}{\mu} < 1$$

the Markov chain is persistent, non-null and hence ergodic. Hence, for the Markov chain $\{X_n\}$, we can apply the following ergodic theorem.

Theorem 7.2. (Ergodic Theorem of Markov Chain) For an irreducible ergodic Markov chain, the limiting probabilities

$$v_j = \lim_{n \rightarrow \infty} p_{ij}^{(n)}, \quad j = 0, 1, 2, \dots$$

exist and are independent of the initial state i . The probabilities

$$V = [v_0 \quad v_1 \quad v_2 \quad \cdots], \quad \sum_j v_j = 1$$

are given as the unique solutions of

$$V = VP$$

7.6.1 Pollaczek-Khinchine Formula

Let $K(s) = \sum_j \alpha_j s^j$ and $V(s) = \sum_j v_j s^j$ denote the probability generating functions of $\{\alpha_j\}$ and $\{v_j\}$, respectively. We find that

$$K(s) = \sum_{j=0}^{\infty} \alpha_j s^j = \sum_{j=0}^{\infty} s^j \left\{ \int_0^{\infty} \frac{e^{-\lambda t} (\lambda t)^j}{j!} b(t) dt \right\} = \int_0^{\infty} e^{-\lambda t} \left[\sum_{j=0}^{\infty} \frac{(\lambda s t)^j}{j!} \right] b(t) dt$$

i.e.

$$K(s) = \int_0^{\infty} e^{-\lambda t} e^{\lambda s t} b(t) dt = \int_0^{\infty} e^{-(\lambda - \lambda s)t} b(t) dt = B^*(\lambda - \lambda s) \quad (7.76)$$

where $B^*(s)$ is the Laplace transform of $b(t)$, the probability density function of the service time.

From Eq. (7.76), it follows that

$$K'(s) = -\lambda B^{*'}(\lambda - \lambda s)$$

where $B^{*'}(s)$ denotes the derivative of $B^*(s)$ with respect to s , i.e.

$$B^{*'}(s) = \int_0^{\infty} -te^{-st}b(t) dt$$

Thus, we deduce that

$$K'(1) = -\lambda B^{*'}(0) = -\lambda \int_0^{\infty} -tb(t) dt = \lambda E[v] = \lambda \frac{1}{\mu} = \frac{\lambda}{\mu} = \rho$$

Hence, it follows that

$$E(A) = K'(1) = \frac{\lambda}{\mu} = \rho \quad (7.77)$$

Now, $V = VP$ gives an infinite system of equations, namely

$$\begin{aligned} v_0 &= \alpha_0 v_0 + \alpha_0 v_1 \\ v_1 &= \alpha_1 v_0 + \alpha_1 v_1 + \alpha_0 v_2 \\ v_2 &= \alpha_2 v_0 + \alpha_2 v_1 + \alpha_1 v_2 + \alpha_0 v_3 \\ &\vdots \quad \vdots \quad \vdots \end{aligned} \quad (7.78)$$

Multiplying the $(k+1)$ st Eq. (7.78) by s^k , $k = 0, 1, 2, \dots$ and adding over k , we get

$$v_0 s^0 + v_1 s^1 + \dots = s^0(\alpha_0 v_0 + \alpha_0 v_1) + s^1(\alpha_1 v_0 + \alpha_1 v_1 + \alpha_0 v_2) + s^2(\alpha_2 v_0 + \alpha_2 v_1 + \alpha_1 v_2 + \alpha_0 v_3) + \dots$$

i.e.

$$\sum_{k=0}^{\infty} v_k s^k = v_0 \sum_{k=0}^{\infty} \alpha_k s^k + v_1 \sum_{k=0}^{\infty} \alpha_k s^k + s v_2 \sum_{k=0}^{\infty} \alpha_k s^k + s^2 v_3 \sum_{k=0}^{\infty} \alpha_k s^k + \dots$$

i.e.

$$V(s) = v_0 K(s) + v_1 K(s) + s v_2 K(s) + s^2 v_3 K(s) + \dots$$

i.e.

$$V(s) = [v_0 + v_1 + s v_2 + s^2 v_3 + \dots] K(s) \quad (7.79)$$

Since $V(s) = v_0 + v_1 s + v_2 s^2 + v_3 s^3 + \dots$, it follows that

$$\frac{V(s) - v_0}{s} = v_1 + v_2 s + v_3 s^2 + \dots \quad (7.80)$$

Substituting Eq. (7.80) into Eq. (7.79), we get

$$V(s) = \left[v_0 + \frac{V(s) - v_0}{s} \right] K(s)$$

i.e.

$$V(s) \left[\frac{K(s)}{s} - 1 \right] = \left[\frac{v_0}{s} - v_0 \right] K(s)$$

i.e.

$$V(s)[K(s) - s] = v_0(1 - s)K(s)$$

Thus, it follows that

$$V(s) = \frac{v_0(1 - s)K(s)}{K(s) - s} \quad (7.81)$$

We find the value of v_0 using L'Hospital's rule.

First, note that

$$V(1) = \sum_{k=0}^{\infty} v_k = 1 = K(1)$$

Taking limits as $s \rightarrow 1$ on both sides of (7.81), we have

$$1 = V(1) = \lim_{s \rightarrow 1} \left[\frac{v_0(1 - s)K(s)}{K(s) - s} \right]$$

Using L'Hospital's rule, we have

$$1 = v_0 \lim_{s \rightarrow 1} \left[\frac{-K(s) + (1 - s)K'(s)}{K'(s) - 1} \right] = v_0 \left[\frac{-1 + 0}{K'(1) - 1} \right] = \frac{v_0}{1 - K'(1)}$$

Hence,

$$v_0 = 1 - K'(1) \quad (7.82)$$

Substituting the value of v_0 from Eq. (7.82) into Eq. (7.81), we get

$$V(s) = \frac{[1 - K'(1)](1 - s)K(s)}{K(s) - s} \quad (7.83)$$

Putting $K'(1) = \rho$, we get

$$V(s) = \frac{(1 - \rho)(1 - s)K(s)}{K(s) - s}$$

Using Eq. (7.76), we have

$$V(s) = \frac{(1 - \rho)(1 - s)B^*(\lambda - \lambda s)}{B^*(\lambda - \lambda s) - s} \quad (7.84)$$

Eq. (7.84) is known as *Pollaczek-Khinchine transform formula*.

The $M/G/1$ Queueing system has the following important characteristics:

1. The average or expected number of customers in the queue or the average queue length (L_q)

$$L_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1 - \rho)}$$

where σ^2 is the variance of service time.

2. The average or expected number of customers in the system or the average line length (L_s)

$$L_s = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1 - \rho)} + \rho$$

3. The average waiting time of a customer in the queue (W_q)

$$W_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2\lambda(1-\rho)}$$

4. The average waiting time of a customer in the system (W_s)

$$W_s = \frac{\lambda^2 \sigma^2 + \rho^2}{2\lambda(1-\rho)} + \frac{1}{\mu}$$

(this is another form of the Pollaczek-Khinchine formula).

EXAMPLE 7.26. Write down Pollaczek-Khinchine formula.

(Anna, Nov. 2003)

Solution. The Pollaczek-Khinchine formula is given as follows:

- (i) The average number of customers in the system is given by

$$L_s = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)} + \rho$$

- (ii) The average waiting time of a customer in the system is given by

$$W_s = \frac{\lambda^2 \sigma^2 + \rho^2}{2\lambda(1-\rho)} + \frac{1}{\mu}$$

◆

EXAMPLE 7.27. Derive Pollaczek-Khinchine formula for the average number of customers in the M/G/1 Queueing system.

(Anna, May 2006)

Solution. In the M/G/1 Queueing system under study, we consider a single-server Queueing system with infinite capacity, Poisson arrivals and general service discipline. The model has arbitrary service time, and it is not necessary to be memoryless.

We assume that the input process is Poisson with mean-arrival rate λ and that the service times are independent and identically distributed (i.i.d.) random variables with an arbitrary (general) distribution with mean $\frac{1}{\mu}$ and variance σ^2 . We also define the *traffic intensity* as

$$\rho = \frac{\lambda}{\mu}$$

Let N and N' be the number of customers in the system at times t and $t + T$, respectively, when two consecutive customers have just left the system and T is the random service time.

We also note that $E(T) = \frac{1}{\mu}$ and $\text{Var}(T) = \sigma^2$. Thus, it follows that

$$E(T^2) = \text{Var}(T) + E^2(T) = \sigma^2 + \frac{1}{\mu^2}$$

Let M be the number of customers who are arriving to the system during the random service time T . Then it follows that

$$N' = \begin{cases} M & \text{if } N = 0 \\ N - 1 + M & \text{if } N > 0 \end{cases} \quad (7.85)$$

We can express Eq. (7.85) in the form

$$N' = N - 1 + M + \delta \quad (7.86)$$

where δ is defined as

$$\delta = \begin{cases} 1 & \text{if } N = 0 \\ 0 & \text{if } N > 0 \end{cases} \quad (7.87)$$

From Eq. (7.85), we have

$$E(N') = E(N) - 1 + E(M) + E(\delta)$$

In steady-state, $E(N') = E(N)$ and $E(N^2) = E(N'^2)$. Thus, it follows that

$$E(\delta) = 1 - E(M) \quad (7.88)$$

Next, we find that

$$\begin{aligned} N'^2 &= (N - 1 + M + \delta)(N - 1 + M + \delta) = (N + M - 1 + \delta)(N + M - 1 + \delta) \\ &= N^2 + (M - 1)^2 + \delta^2 + 2N(M - 1) + 2N\delta + 2(M - 1)\delta \end{aligned}$$

Since $\delta^2 = \delta$ and $N\delta = 0$, it follows that

$$\begin{aligned} N'^2 &= N^2 + (M - 1)^2 + \delta + 2N(M - 1) + 0 + (2M - 2)\delta \\ &= N^2 + M^2 - 2M + 1 + 2N(M - 1) + (2M - 1)\delta \end{aligned}$$

Thus, it follows that

$$2N(1 - M) = N^2 - N'^2 + M^2 + (2M - 1)\delta - 2M + 1$$

Taking expectation on both sides and noting that $E(N^2) = E(N'^2)$, we have

$$2E[N(1 - M)] = E(M^2) + E[(2M - 1)\delta] - 2E(M) + 1$$

Since N and M , and δ and M are independent, we have

$$2E(N)E(1 - M) = E(M^2) + E(2M - 1)E(\delta) - 2E(M) + 1 \quad (7.89)$$

By Eq. (7.88), $E(\delta) = 1 - E(M)$. Thus, Eq. (7.89) becomes

$$\begin{aligned} 2E(N)[1 - E(M)] &= E(M^2) + E(2M - 1)[1 - E(M)] - 2E(M) + 1 \\ &= E(M^2) + [2E(M) - 1][1 - E(M)] - 2E(M) + 1 \\ &= E(M^2) - 2E^2(M) + 3E(M) - 1 - 2E(M) + 1 \\ &= E(M^2) - 2E^2(M) + E(M) \end{aligned}$$

Thus, it follows that

$$E(N) = \frac{E(M^2) - 2E^2(M) + E(M)}{2[1 - E(M)]} \quad (7.90)$$

Since the arrival process is Poisson with mean rate λ , it follows that

$$E(M) = \lambda T \quad \text{and} \quad \text{Var}(M) = \lambda T$$

Thus,

$$E(M^2) = \text{Var}(M) + E^2(M) = \lambda T + (\lambda T)^2$$

It follows that

$$E(M) = E[E(M/T)] = \lambda E(T) = \lambda \times \frac{1}{\mu} = \frac{\lambda}{\mu} = \rho$$

and

$$\begin{aligned} E(M^2) &= E[E(M^2/T)] = E[\lambda T + (\lambda T)^2] = \lambda E(T) + \lambda^2 E(T^2) \\ &= \lambda \left[\frac{1}{\mu} \right] + \lambda^2 \left[\sigma^2 + \frac{1}{\mu^2} \right] = \rho + \lambda^2 \sigma^2 + \rho^2 \end{aligned}$$

Substituting the above in Eq. (7.90), we get

$$L_s = E(N) = \frac{(\rho + \lambda^2 \sigma^2 + \rho^2) - 2\rho^2 + \rho}{2(1 - \rho)} = \frac{\lambda^2 \sigma^2 + \rho^2 + 2\rho(1 - \rho)}{2(1 - \rho)}$$

Simplifying, we have

$$L_s = E(N) = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1 - \rho)} + \rho$$

which is the Pollaczek-Khinchine formula for the average number of customers in the M/G/1 Queueing system. $\blacklozenge$

EXAMPLE 7.28. Automatic car wash facility operates with only one bay. Cars arrive according to a Poisson distribution with a mean of 4 cars per hour and may wait in the facility's parking lot if the bay is busy. If the service time for all cars is constant and equal to 10 minutes, determine L_s, L_q, W_s and W_q .
(Anna, April 2004)

Solution. The given problem is in (M/G/1) : ∞ /FIFO model. From the given data, the arrival process is Poisson with mean arrival rate given by $\lambda = 4$ cars/hour.

The service time T is constant with mean $E(T) = \frac{1}{\mu} = 10$ minutes. Thus, it follows that $\text{Var}(T) = \sigma^2 = 0$ and $\mu = 6$ cars/hour.

Thus, the traffic intensity of the system is

$$\rho = \frac{\lambda}{\mu} = \frac{4}{6} = \frac{2}{3}$$

Using Pollaczek-Khintchine formula, we have

$$L_s = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1 - \rho)} + \rho = \frac{0 + 4/9}{2(1 - 2/3)} + \frac{2}{3} = \frac{4/9}{2/3} + \frac{2}{3} = \frac{4}{3} + \frac{2}{3} = \frac{4}{3} = 1.333 \text{ cars}$$

Using Little's formulae, we have

$$L_q = L_s - \frac{\lambda}{\mu} = L_s - \rho = \frac{4}{3} - \frac{2}{3} = \frac{2}{3} = 0.6667 \text{ cars}$$

$$W_s = \frac{1}{\lambda} L_s = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3} \text{ hour or 20 minutes}$$

$$W_q = \frac{1}{\lambda} L_q = \frac{1}{4} \times \frac{2}{3} = \frac{1}{6} \text{ hour or 10 minutes}$$

◆

EXAMPLE 7.29. A car-manufacturing plant uses one big crane for loading cars into a truck. Cars arrive for loading by the crane according to a Poisson distribution with a mean of 5 cars per hour. Given that the service time for all cars is constant and equal to 6 minutes, determine L_s, L_q, W_s and W_q .

Solution. The given problem is in $(M/G/1) : \infty/\text{FIFO}$ model. From the given data, the arrival process is Poisson with mean arrival rate given by $\lambda = 5$ cars/hour.

The service time T is constant with mean $E(T) = \frac{1}{\mu} = 6$ minutes. Thus, it follows that $\text{Var}(T) = \sigma^2 = 0$ and $\mu = 10$ cars/hour.

Thus, the traffic intensity of the system is

$$\rho = \frac{\lambda}{\mu} = \frac{5}{10} = \frac{1}{2}$$

Using Pollaczek-Khintchine formula, we have

$$L_s = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)} + \rho = \frac{0 + 1/4}{2(1-1/2)} + \frac{1}{2} = \frac{1/4}{1} + \frac{1}{2} = \frac{3}{4} = 0.75 \text{ car}$$

Using Little's formulae, we have

$$L_q = L_s - \frac{\lambda}{\mu} = L_s - \rho = \frac{3}{4} - \frac{1}{2} = \frac{1}{4} = 0.25 \text{ car}$$

$$W_s = \frac{1}{\lambda} L_s = \frac{1}{5} \times \frac{3}{4} = \frac{3}{20} \text{ hour or 9 minutes}$$

$$W_q = \frac{1}{\lambda} L_q = \frac{1}{5} \times \frac{1}{4} = \frac{1}{20} \text{ hour or 3 minutes}$$

◆

PROBLEM SET 7.5

1. Automatic car wash facility operates with only one bay. Cars arrive according to a Poisson distribution with a mean of 4 cars per hour and may wait in the facility's parking lot if the bay is busy. If the service time for all cars is constant and equal to 5 minutes, determine L_q, L_s, W_q and W_s .
2. A car, manufacturing plant uses one big crane for loading cars into a truck. Cars arrive for loading by the crane according to a Poisson distribution with a mean of 5 cars per hour. Given that the service time for all cars is constant and equal to 10 minutes, determine L_q, L_s, W_q and W_s .

3. Customers arrive at the automatic teller machine in a bank according to a Poisson distribution with the average arrival rate of 10 per hour. It is known that the service time follows a random distribution with the mean service time of 4 minutes and standard deviation equal to 10 minutes. Find:
 - (i) Average number of customers in the queue.
 - (ii) Average number of customers in the system.
 - (iii) Average waiting time of a customer at the queue.
 - (iv) Average time a customer spends at the system.
4. Customers arrive at a one-man barber shop in a remote village according to a Poisson distribution with the average arrival rate of 8 per hour. It is estimated that the service time follows a random distribution with the mean service time of 6 minutes and standard deviation equal to 15 minutes. Find the
 - (i) Average number of customers in the queue.
 - (ii) Average number of customers in the barber shop.
 - (iii) Average waiting time of a customer at the queue.
 - (iv) Average time a customer spends at the barber shop.
5. Patients arrive at a dentist's clinic according to a Poisson distribution with the average arrival rate of 5 per hour. It is estimated that the service time follows a random distribution with the mean service time of 10 minutes and standard deviation equal to 6 minutes. Find:
 - (i) Average number of patients in the queue.
 - (ii) Average number of patients in the clinic.
 - (iii) Average waiting time of a patient at the queue.
 - (iv) Average time a patient spends at the clinic.

Bibliography

- Baisnab, A. and Jas, M. (1993), *Elements of Probability and Statistics*, Tata McGraw-Hill, New Delhi.
- Birkhoff, G. (1931), ‘Proof of the ergodic theorem’, *Proceedings of the National Academy of Sciences, USA* **17**, 656–660.
- Blake, I. (1979), *An Introduction to Applied Probability*, John Wiley and Sons, New York, USA.
- Chung, K. (1968), *A Course in Probability Theory*, Harcourt Brace and World, New York, USA.
- , (1983), *Elementary Probability Theory with Stochastic Processes*, Narosa Publishing House, New Delhi.
- Erlang, A. (1909), ‘The Theory of Probabilities and Telephone Conversations’, *Nyt Tidsskrift for Matematik B*, Vol. 20, pp. 131–137.
- Ferguson, T. (1967), *Mathematical Statistics*, Academic Press, New York, USA.
- Freund, J.E. (2000), *Mathematical Statistics*, 5th ed., Prentice-Hall of India, New Delhi.
- Gnedenko, B. (1982), *The Theory of Probability*, MIR Publishers, Moscow.
- Gupta, S. and Kapoor, V.M. (2002), *Fundamentals of Mathematical Statistics*, 11th ed., Sultan Chand and Sons, New Delhi.
- Jr., P.P. (2000), *Probability, Random Variables and Random Signal Principles*, 4th ed., McGraw Hill, New York, USA.
- Kendall, D.G. (1951), *The Advanced Theory of Statistics*, Vol. 2, Hafner Press, New York, USA.
- Kolmogorov, A. (1956), *Foundations of the Theory of Probability*, Chelsea Publishing Company, New York, USA.
- Krishnan, V. (2006), *Probability and Random Processes*, John Wiley and Sons, New York, USA.
- Loève, M. (1967), *Probability Theory*, 4th ed., Springer Verlag, New York, USA.
- Medhi, J. (1994), *Stochastic Processes*, New Age Publishers, New Delhi.

- Ochi, M.K. (1990), *Applied Probability and Stochastic Processes*, Wiley Interscience, New York, USA.
- Papoulis, A. (1991), *Probability, Random Variables and Stochastic Processes*, 3rd ed., McGraw Hill, New York, USA.
- Parthasarathy, K. (1980), *Introduction to Probability and Measure*, Macmillan, New Delhi.
- Rohatgi, V.K. and Saleh, A.M.E. (2002), *An Introduction to Probability and Statistics*, 2nd ed., John Wiley and Sons, New York, USA.
- Ross, S.M. (2002), *A First Course in Probability*, 6th edn, Pearson Education, New Delhi.
- (2004a), *Introduction to Probability Models*, 8th ed, Academic Press, San Diego, USA.
- (2004b), *Stochastic Processes*, 2nd ed., John Wiley and Sons, New Delhi.
- Stark, H. and Woods, J.W. (2002), *Probability and Random Processes with Applications to Signal Processing*, 3rd ed., Prentice Hall, New Jersey, USA.
- Wilks, S. (1962), *Mathematical Statistics*, Wiley and Sons, New York, USA.

Answers to Problems

CHAPTER 1

Problem Set 1.1

1. $P(A) = \frac{1}{3}$
2. $P(A) = \frac{9}{38}$
3. $P(A) = \frac{6}{70}$
4. $P(A) = \frac{3}{8}$
5. $P(A) = \frac{11}{36}$
6. (i) $P(A) = \frac{1}{6}$, (ii) $P(B) = \frac{1}{18}$
7. $P(A) = \frac{11}{30}$
8. (i) $P(A) = \frac{1}{2}$, (ii) $P(B) = \frac{5}{18}$
9. (i) $P(A) = \frac{4}{17}$, (ii) $P(B) = \frac{25}{102}$, (iii) $P(C) = \frac{1}{221}$
10. $P(A) = \frac{7}{15}$
11. (i) $P(A) = \frac{14}{99}$, (ii) $P(B) = \frac{56}{165}$
12. (i) $P(A) = \frac{27}{220}$, (ii) $P(B) = \frac{3}{11}$, (iii) $P(C) = \frac{21}{55}$, (iv) $P(D) = \frac{31}{55}$
13. $P(A) = \frac{3}{7}$
14. Clearly, $n(S) = {}^{52}C_{13}$. Next, we note that the number of ways in which 8 cards of a suit can come out of the 13 cards is ${}^{13}C_8$ and the number of ways in which the remaining 4 cards in the hand can come out of a balance of 39 cards of other suits is ${}^{39}C_4$. It is clear that $n(A) = {}^{13}C_8 \times {}^{39}C_4$. Hence, the required probability is

$$P(A) = \frac{n(A)}{n(S)} = \frac{{}^{13}C_8 \times {}^{39}C_4}{{}^{52}C_{13}}$$

15. Since 4 letters can be put in 4 envelopes in $4!$ different ways, it follows that $n(S) = 4! = 24$. We also note that there is exactly one way in which all the 4 letters can be put in the right envelopes. Hence, the probability that all the 4 letters are put in the right envelopes is given by

$$P(A) = \frac{1}{4!} = \frac{1}{24}$$

Thus, the required probability that all letters are not dispatched in the right envelopes is given by

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{24} = \frac{23}{24}$$

Problem Set 1.2

1. (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = p_1 + p_2 - p_3$
 (b) $P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - p_3$
 (c) $P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - p_1 - p_2 + p_3$
 (d) $P(A \cap \bar{B}) = P(A) - P(A \cap B) = p_1 - p_3$
 (e) $P(\bar{A} \cap B) = P(B) - P(A \cap B) = p_2 - p_3$
 (f) $P(\bar{A} \cup B) = P(\bar{A}) + P(A \cap B) = 1 - p_1 + p_3$
 (g) $P(A \cup \bar{B}) = P(\bar{B}) + P(A \cap B) = 1 - p_2 + p_3$
2. (a) $P(\bar{A} \cap B) = \frac{1}{4}$, (b) $P(A \cap \bar{B}) = \frac{1}{12}$, (c) $P(\bar{A} \cap \bar{B}) = \frac{5}{12}$, (d) $P(\bar{A} \cup \bar{B}) = \frac{3}{4}$
3. (a) $P(A) = \frac{3}{10}$, (b) $P(A \cap \bar{B}) = \frac{1}{4}$
4. $P(A \cup B) = \frac{17}{20}$ and $P(\bar{A} \cap \bar{B}) = \frac{3}{20}$
5. $P(A \cup B \cup C) = 0.6s$
6. Since $A - B = A \cap \bar{B}$ and $B - A = B \cap \bar{A}$ are mutually exclusive events, $P(A - B) = P(A) - P(A \cap B)$ and $P(B - A) = P(B) - P(A \cap B)$, it follows that

$$P[(A - B) \cup (B - A)] = P(A - B) + P(B - A) = P(A) + P(B) - 2P(A \cap B)$$

7. Given that $P(A) = 2P(B)$ and $P(B) = 2P(C)$. Clearly, the events A , B and C are mutually exclusive and exhaustive. Thus, it follows that

$$1 = P(A \cup B \cup C) = P(A) + P(B) + P(C) = 4P(C) + 2P(C) + P(C) = 7P(C)$$

Hence, $P(C) = \frac{1}{7}$ and it follows that $P(A) = 4P(C) = \frac{4}{7}$ and $P(B) = 2P(C) = \frac{2}{7}$

8. (a) $P(A) = \frac{1}{12}$, (b) $P(B) = \frac{1}{4}$, (c) $P(C) = \frac{1}{4}$
9. (a) $P(A) = \frac{24}{91}$, (b) $P(B) = \frac{8}{91}$, (c) $P(C) = \frac{36}{455}$
10. (a) $P(A) = \frac{39}{850}$, (b) $P(B) = \frac{33}{2210}$, (c) $P(C) = \frac{169}{1700}$

Problem Set 1.3

1. $P(\bar{B}|\bar{A}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{A})} = \frac{1 - P(A \cup B)}{1 - P(B)} = \frac{37}{45}$
2. If A and B denote the events of the first card being a face card and the second card being a face card respectively, then $P(A) = \frac{12}{52} = \frac{3}{13}$ and $P(B|A) = \frac{11}{51}$. Thus, $P(A \cap B) = P(A) \cdot P(B|A) = \frac{11}{221}$.
3. (a) $P(\bar{A}|B) = \frac{1}{4}$, (b) $P(A|\bar{B}) = \frac{1}{4}$, (c) $P(\bar{A}|\bar{B}) = \frac{3}{4}$
4. (a) $P(\bar{A}|B) = 0.3$, (b) $P(A|\bar{B}) = 0.2$
5. (a) $P(A|B) = \frac{5}{8}$, (b) $P(B|A) = \frac{5}{6}$, (c) $P(A \cap \bar{B}) = \frac{1}{12}$, (d) $P(\bar{A} \cap B) = \frac{1}{4}$, (e) $P(A|\bar{B}) = \frac{1}{4}$, (f) $P(B|\bar{A}) = \frac{1}{2}$, (g) $P(\bar{A} \cap \bar{B}) = \frac{1}{4}$
6. Given that $P(A) = \frac{2}{3}$ and $P(B) = \frac{3}{4}$, and A and B are independent. And, $P(A \cap B) = P(A)P(B) = \frac{1}{2}$. Thus, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{11}{12}$.
7. Given that $P(A) = \frac{2}{5}$, $P(B) = \frac{3}{8}$ and that A and B are independent. Hence, $P(A \cap B) = P(A)P(B) = \frac{3}{20}$. Thus, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{5}{8}$.
8. Given that $P(A) = 0.72$, $P(B) = 0.63$ and $P(A \cap B) = 0.23$. Thus, $P(B|A) = \frac{0.23}{0.72} = 0.3194$.
9. (a) $P(B|A) = 0.4$, (b) $P(A \cup B) = 0.35$

10. (a) $P(A \cap \bar{B}) = 0.16$, (b) $P(\bar{A} \cap B) = 0.36$, (c) $P(\bar{A} \cap \bar{B}) = 0.24$.
 11. (a) $P(A \cup B) = 0.8$, (b) $P[(A - B) \cup (B - A)] = 0.5$, (c) $P(\bar{A} \cap \bar{B}) = 0.28$.
 12. $P(B) = \frac{1}{4}$
 13. If x is the required number of trials, then a simple calculation gives that $1 - q^x = 1 - (0.4)^x \geq 0.8$, which implies that $x \geq 1.7564$. So, $x = 2$ trials.
 14. (a) $P(A \cup B) = 0.85$, (b) $P[(A - B) \cup (B - A)] = 0.5$, (c) $P(A \cap B) = 0.35$.
 15. $P(A \cup B) = \frac{5}{8}$.
 16. $P(A) = \frac{2}{5}$ and $P(B) = \frac{5}{8}$. Since A and B are independent, it follows that $P(A \cap B) = P(A)P(B) = \frac{1}{4}$. Thus, the probability that A and B contradict in a statement is equal to

$$P(A \cap \bar{B}) + P(\bar{A} \cap B) = P(A) + P(B) - 2P(A \cap B) = \frac{21}{40}$$

17. (a) $P(\bar{A} \cap \bar{B} \cap \bar{C}) = \frac{1}{12}$, (b) $P(A \cup B \cup C) = \frac{11}{12}$, (c) $P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) = \frac{3}{8}$.
 18. Given that $P(A) = \frac{4}{7}$, $P(B) = \frac{3}{7}$ and $P(C) = \frac{5}{7}$. The probability that a majority of review will be favourable is equal to $P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) + P(A \cap B \cap C) = \frac{209}{343} = 0.6093$.
 19. By hypotheses, we have (i) $P(A \cap B) = P(A)P(B)$, (ii) $P[A \cap (B \cup C)] = P(A)P(B \cup C)$, (iii) $P[A \cap (B \cap C)] = P(A)P(B \cap C)$. By (ii), we have $P[(A \cap B) \cup (A \cap C)] = P(A)P(B \cup C)$ from which it follows that

$$P(A \cap B) + P(A \cap C) - P(A \cap B \cap C) = P(A)[P(B) + P(C) - P(B \cap C)]$$

Using (i) and (iii), it follows that $P(A \cap C) = P(A)P(C)$.

20. Out of n men, $A_1, A_2, \dots, A_n$, the probability that none dies in a year is equal to $(1 - p)^n$, and hence the probability that, at least one man dies in that year is equal to $1 - (1 - p)^n$. Also, the probability that, out of n men, A is the first to die is $\frac{1}{n}$. Since they are independent, the required probability is $\frac{1}{n} [1 - (1 - p)^n]$.
 21. $P\left(\bigcup_{i=1}^n A_i\right) = 1 - P\left(\bigcap_{i=1}^n \bar{A}_i\right) = 1 - \prod_{i=1}^n P(\bar{A}_i) = 1 - (1 - p_1)(1 - p_2) \cdots (1 - p_n)$.
 22. (a) $P(X = 3) = \frac{5}{16}$, (b) $P(X \geq 3) = \frac{21}{32}$.
 23. $P(X \geq 8) = P(X = 8) + P(X = 9) + \cdots + P(X = 12) = \frac{397}{2048}$.
 24. $r = 12, m = 4$. $n(S) = 6^m = 6^4 = 1296$. Note that

$$n(A) = \text{Coefficient of } x^{r-m} \text{ in } (1 - x^6)^m (1 - x)^{-m} = 435$$

$$\text{Hence, } P(A) = \frac{n(A)}{n(S)} = \frac{435}{1296} = \frac{145}{432}$$

25. $r = 15, m = 5$. $n(S) = 6^m = 6^5 = 7776$. Note that

$$n(A) = \text{Coefficient of } x^{r-m} \text{ in } (1 - x^6)^m (1 - x)^{-m} = 2373$$

$$\text{Hence, } P(A) = \frac{n(A)}{n(S)} = \frac{2373}{7776} = \frac{791}{2592}$$

Problem Set 1.4

1. $P(W) = P(A)P(W|A) + P(B)P(W|B) = \frac{9}{20} = 0.45$.
 2. $P(W) = P(A)P(C|A) + P(B)P(C|B) = \frac{17}{20}$.
 3. Let A = correct diagnosis of X , B = faulty diagnosis of X and let D denote the event that the patient with disease X dies. Then $P(D) = P(A)P(D|A) + P(B)P(D|B) = \frac{45}{100} = 45\%$.

4. Given that $P(A) = \frac{70}{100}$, $P(B) = \frac{30}{100}$, $P(X|A) = \frac{70}{100}$ and $P(X|B) = \frac{40}{100}$. Then $P(X) = P(A)P(X|A) + P(B)P(X|B) = \frac{61}{100} = 61\%$.
5. $P(W) = P(A)P(W|A) + P(B)P(W|B) + P(C)P(W|C) = \frac{4}{9}$
6. $P(C) = P(X)P(C|X) + P(Y)P(C|Y) + P(Z)P(C|Z) = \frac{33}{50} = 0.66$
7. $P(D) = P(X)P(D|X) + P(Y)P(D|Y) + P(Z)P(D|Z) = \frac{28}{200} = 14\%$

Problem Set 1.5

1. $P(\bar{A}|F) = 0.7447$
2. $P(A|E) = 0.9032$
3. (i) $P(A|E) = 0.0845$, (ii) $P(B|E) = 0.2113$, (iii) $P(C|E) = 0.7042$.
4. (i) $P(E|A) = \frac{3}{14}$, (ii) $P(E|B) = \frac{1}{3}$, (iii) $P(E|C) = \frac{2}{7}$.
5. (i) $P(A|D) = 0.8235$, (ii) $P(B|A) = 0.1765$
6. (i) $P(A|G) = 0.3998$, (ii) $P(B|G) = 0.3422$, (iii) $P(C|G) = 0.2580$.
7. (i) $P(A|E) = 0.3585$, (ii) $P(B|E) = 0.5660$, (iii) $P(C|E) = 0.0755$.
8. $P(H) = 0.3$ and $P(B|H) = \frac{P(B)P(H|B)}{P(H)} = 0.2222$.
9. $P(A_3|B) = \frac{P(A_3)P(B|A_3)}{P(B)} = \frac{3}{5} = 0.6$
10. $P(B) = \frac{3}{5}$ and $P(A|B) = \frac{5}{9}$.
11. $P(W) = \frac{7}{15}$ and $P(A|W) = \frac{4}{7}$.
12. $P(E) = 0.2767$ and $P(A|E) = 0.3012$.
13. $P(B) = 0.0880$ and $P(A|B) = 0.5455$.
14. (i) $P(M) = 0.0410$, (ii) $P(X|M) = 0.6098$, (iii) $P(Y|M) = 0.2927$, iv) $P(Z|M) = 0.0976$.

CHAPTER 2**Problem Set 2.1**

1. It is easy to show that $1 - I_A$ is the indicator function of $\bar{A}$. Since the indicator function of any event of the sample space is a random variable, it follows immediately that $1 - I_A$ is a random variable.
2. Here, $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$. By definition, we have $X(HHH) = 3$, $X(HHT) = 2$, etc. Note that

$$X^{-1}(-\infty, x] = \begin{cases} \emptyset & \text{if } x < 0 \\ \{TTT\} & \text{if } 0 \leq x < 1 \\ \{HTT, THT, TTH\} & \text{if } 1 \leq x < 2 \\ \{HHT, HTH, THH\} & \text{if } 2 \leq x < 3 \\ \{HHH\} & \text{if } x \geq 3 \end{cases}$$

Hence, it follows that X is a random variable.

3. (a) $A = \{(1, 3), (2, 2), (3, 1)\}$
- (b) $B = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$
- (c) $C = \{X = 2 \text{ or } X = 3\} = \{(1, 1), (1, 2), (2, 1)\}$
- (d) $D = \{X = 5 \text{ or } X = 6\}$ and so $D = \{(1, 4), (2, 3), (3, 2), (4, 1), (1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$

(e) $E = \{X = 10 \text{ or } X = 11 \text{ or } X = 12\}$ and so $E = \{(4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)\}$

Problem Set 2.2

1. The distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{9} & \text{for } 0 \leq x < 1 \\ \frac{5}{9} & \text{for } 1 \leq x < 2 \\ 1 & \text{for } x \geq 2 \end{cases}$$

2. The distribution function of X is easily obtained as

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{27} & \text{for } 0 \leq x < 1 \\ \frac{7}{27} & \text{for } 1 \leq x < 2 \\ \frac{19}{27} & \text{for } 2 \leq x < 3 \\ 1 & \text{for } x \geq 3 \end{cases}$$

3. Here, $S = \{WW, WR, RW, RR\}$. We find that $P(WW) = \frac{1}{3}$, $P(WR) = \frac{4}{15}$, $P(RW) = \frac{4}{15}$ and $P(RR) = \frac{2}{15}$. Note that X takes the values $x = 0, 1, 2$. It is easy to see that $P[X = 0] = P(RR) = \frac{2}{15}$, $P[X = 1] = P\{WR, RW\} = P(WR) + P(RW) = \frac{8}{15}$, $P[X = 2] = P(WW) = \frac{1}{3}$. Hence, the distribution function of X is obtained as

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{2}{15} & \text{if } 0 \leq x < 1 \\ \frac{2}{3} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

4. The distribution function of X is easily obtained as

$$F(x) = \begin{cases} 0 & \text{if } x < 1 \\ \frac{1}{10} & \text{if } 1 \leq x < 2 \\ \frac{3}{10} & \text{if } 2 \leq x < 3 \\ \frac{3}{5} & \text{if } 3 \leq x < 4 \\ 1 & \text{if } x \geq 4 \end{cases}$$

5. The problem is solved by verifying the properties (i), (ii) and (iii) of the distribution function F stated in Theorem 2.4.

- (a) No, because $F(2) < F(1)$ (Note that F is non-decreasing).
- (b) No, because $F(0) < 1$ (Note that F takes only non-negative values).
- (c) Yes
- (d) No, because $F(4) > 1$ (Note that $F(+\infty) = 1$).
- (e) No, because $F(4) < 1$ (Note that $F(+\infty) = 1$).

6. (a) $F_Y(y) = P(Y \leq y) = P(X - \varepsilon \leq y) = P(X \leq y + \varepsilon) = F_X(y + \varepsilon)$
 (b) $F_Z(z) = P(Z \leq z) = P(\alpha X \leq z) = P(X \leq \frac{z}{\alpha}) = F_X(\frac{z}{\alpha})$
 (c) $F_W(w) = P(W \leq w) = P(\alpha X + \beta \leq w) = P(X \leq \frac{w - \beta}{\alpha}) = F_X(\frac{w - \beta}{\alpha})$
7. From the description of F , it is clear that X takes the values 0, 1, 3, 5 with the following probabilities:

x	0	1	3	5
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Hence, we obtain the following:

- (a) $P(X = 3) = \frac{1}{4}$
 (b) $P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{3}{4}$
 (c) $P(0 < X < 3) = P(X = 1) + P(X = 2) = \frac{1}{2}$
 (d) $P(0 < X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) = \frac{3}{4}$
 (e) $P(1 < X < 5) = P(X = 2) + P(X = 3) = \frac{1}{2}$

Problem Set 2.3

1. The probability distribution function of X is given by

x	0	1	2
$f(x)$	$\frac{1}{9}$	$\frac{4}{9}$	$\frac{4}{9}$

2. The probability distribution function of X is given by

x	0	1	2	3
$f(x)$	$\frac{1}{27}$	$\frac{6}{27}$	$\frac{12}{27}$	$\frac{8}{27}$

3. The probability distribution function of X can be tabulated as

x	0	1	2
$f(x)$	$\frac{14}{30}$	$\frac{7}{15}$	$\frac{1}{15}$

4. (a) We find that probability distribution function of X is given by

x	0	1	2	3
$f(x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

- (b) $P(X < 2) = P(X = 0) + P(X = 1) = \frac{2}{3}$
 (c) $P(0 < X < 3) = P(X = 1) + P(X = 2) = \frac{4}{5}$
5. (a) Using the property that $f(x) = F(x) - F(x - 0)$ for any $x \in \mathbf{R}$, the values of the probability mass function f may be tabulated as follows:

x	-2	0	2	4
$f(x)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$

- (b) Note that

$$P(X \leq 1) = P(X = -2) + P(X = 0) = \frac{1}{4}$$

$$P(-1 < X \leq 1) = P(X = 0) = \frac{1}{8}$$

$$P(0 < X < 3) = P(X = 2) = \frac{1}{4}$$

and

$$P(1 < X \leq 4) = P(X = 2) + P(X = 4) = \frac{3}{4}$$

6. (a) Since the total probability is equal to 1, it follows that $6c^2 + c = 1$. Solving for c , we get $c = -\frac{1}{2}, \frac{1}{3}$. The value $c = -\frac{1}{2}$ is inadmissible since $f(0) = c \geq 0$. Hence, $c = \frac{1}{3}$.
 (b) It will be useful to tabulate the probability distribution of X as follows:

x	0	1	2	3
$f(x)$	$\frac{1}{3}$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{1}{3}$

$$P(X < 2) = P(X = 0) + P(X = 1) = \frac{5}{9}$$

$$P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0) = \frac{2}{3}$$

$$P(1 < X \leq 3) = P(X = 2) + P(X = 3) = \frac{4}{9}$$

- (c) The distribution function of X is easily obtained as follows:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{3} & \text{if } 0 \leq x < 1 \\ \frac{5}{9} & \text{if } 1 \leq x < 2 \\ \frac{2}{3} & \text{if } 2 \leq x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$$

- (d) The largest value of x such that $F(x) < \frac{1}{2}$ is $x = 1$.

7. (a) $P(X = 1 \text{ or } 2) = P(X = 1) + P(X = 2) = \frac{3}{10}$
 (b) We find that

$$P\left[\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right] = \frac{P\left[\left(\frac{1}{2} < X < \frac{5}{2}\right) \text{ and } (X > 1)\right]}{P(X > 1)} = \frac{P\left[1 < X < \frac{5}{2}\right]}{P(X > 1)} = \frac{2}{9}$$

8. (a) $c = \frac{1}{16}$.

- (b) It will be useful to tabulate the probability distribution of X as follows:

x	0	1	2	3	4	5	6
f	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{4}$	$\frac{3}{16}$	$\frac{1}{8}$	$\frac{1}{16}$

We find that

$$P(X < 2) = P(X = 0) + P(X = 1) = \frac{3}{16}$$

$$P(X \geq 2) = 1 - P(X < 2) = \frac{13}{16}$$

and

$$P(0 < X < 4) = P(X = 1) + P(X = 2) + P(X = 3) = \frac{9}{16}$$

(c) We find that

$$P(0 < X < 4 | X \geq 2) = \frac{P[(0 < X < 4) \text{ and } (X \geq 2)]}{P(X \geq 2)} = \frac{P(2 \leq X < 4)}{P(X \geq 2)} = \frac{7}{13}$$

(d) The smallest value of x such that $P(X \leq x) > \frac{1}{2}$ is $x = 3$.

(e) The distribution function of X is easily obtained as

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{16} & \text{if } 0 \leq x < 1 \\ \frac{3}{16} & \text{if } 1 \leq x < 2 \\ \frac{3}{8} & \text{if } 2 \leq x < 3 \\ \frac{5}{8} & \text{if } 3 \leq x < 4 \\ \frac{13}{16} & \text{if } 4 \leq x < 5 \\ \frac{15}{16} & \text{if } 5 \leq x < 6 \\ 1 & \text{if } x \geq 6 \end{cases}$$

9. Note that for each value of $x = 0, 1, 2, \dots, n$, we have

$$f(x) = \frac{2x}{n(n+1)} \geq 0$$

and also that

$$\sum_{x=0}^n f(x) = \sum_{x=0}^n \frac{2x}{n(n+1)} = \frac{2}{n(n+1)} [1 + 2 + \dots + n] = 1$$

Thus, by Theorem 2.9, it follows that f serves as the probability mass function of some discrete random variable X taking the values $x = 0, 1, 2, \dots, n$.

10. By Theorem 2.9, the conditions for f to be the probability mass function of some discrete random variable X taking the values $x = 0, 1, 2, \dots$ are (i) $f(x) \geq 0$ for $x = 0, 1, 2, \dots$ and $\sum_x f(x) = 1$. By (i), we must have $1 - \alpha > 0$ or $\alpha < 1$ and $\alpha > 0$. Thus, $0 < \alpha < 1$. Note also that

$$\sum_{x=0}^{\infty} f(x) = \sum_{x=0}^{\infty} (1 - \alpha)\alpha^x = (1 - \alpha) [1 + \alpha + \alpha^2 + \dots] = 1 \text{ if } 0 < \alpha < 1$$

Hence, the required range of values for α is $0 < \alpha < 1$.

Problem Set 2.4

1. (a) $k = \frac{6}{5}$.

(b) The cumulative distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x^2}{5} (3 + 2x) & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$

$$(c) P\left(X \leq \frac{1}{2} \mid \frac{1}{4} < X < \frac{3}{4}\right) = \frac{P\left(\frac{1}{4} < X \leq \frac{1}{2}\right)}{P\left(\frac{1}{4} < X < \frac{3}{4}\right)} = \frac{5/32}{37/80} = \frac{25}{74}.$$

2. (a) $k = \frac{1}{2}$.
 (b) The cumulative distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x - \frac{x^2}{4} & \text{if } 0 \leq x \leq 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

$$(c) P\left(\frac{1}{2} < X < \frac{3}{2} \mid X \geq 1\right) = \frac{P(1 \leq X < \frac{3}{2})}{P(X \geq 1)} = \frac{3/16}{1/4} = \frac{3}{4}.$$

3. Since $P(X > a) = 1 - P(X \leq a)$, it is clear that we should find a such that $P(X \leq a) = \frac{1}{2}$. A simple calculation gives $P(X \leq a) = a^4 = \frac{1}{2}$, which implies that $a = (0.5)^{\frac{1}{4}} = 0.8409$.
 4. (a) It is easy to note that $f(x) \geq 0$ for all values of x and that

$$\int_{x=1}^3 f(x) dx = \int_1^3 \frac{x}{4} dx = \left[\frac{x^2}{8} \right]_{x=1}^3 = 1$$

Hence, f is a probability density function for some continuous random variable X .

$$(b) P(1 < X < 2) = \int_{x=1}^2 f(x) dx = \left[\frac{x^2}{8} \right]_{x=1}^2 = \frac{3}{8}.$$

$$(c) P\left[X \geq \frac{3}{2} \mid 1 < X < 2\right] = \frac{P[\frac{3}{2} \leq X < 2]}{P(1 < X < 2)} = \frac{7/32}{3/8} = \frac{7}{12}.$$

$$5. (a) F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x^3}{8} & \text{if } 0 \leq x \leq 2 \\ 1 & \text{if } x > 2 \end{cases}$$

$$(b) P\left(\frac{1}{2} < X < \frac{3}{2}\right) = F\left(\frac{3}{2}\right) - F\left(\frac{1}{2}\right) = \frac{27}{64} - \frac{1}{64} = \frac{13}{32}.$$

$$(c) P(X < 1) = F(1) = 1.$$

6. (a) $C = \frac{1}{2}$.

$$(b) F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - \left(1 + x + \frac{x^2}{2}\right)e^{-x} & \text{if } x > 0 \end{cases}$$

$$7. (a) F(x) = \begin{cases} 0 & \text{if } x \leq 1 \\ \frac{2}{21} \left(x + \frac{x^2}{2} - \frac{3}{2}\right) & \text{if } 1 < x < 4 \\ 1 & \text{if } x \geq 4 \end{cases}$$

$$(b) P(2 < X < 3) = F(3) - F(2) = \frac{4}{7} - \frac{5}{21} = \frac{1}{3}.$$

$$8. (a) f(x) = F'(x) = \begin{cases} \frac{3x^2}{16} & \text{if } -2 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$(b) P(0 < X < 1) = F(1) - F(0) = \frac{1}{16}.$$

$$9. (a) F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 3x^2 - 2x^3 & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$

- (b) $\alpha = \frac{1}{2}$.
10. (a) $k = \frac{1}{6}$.
- (b) $F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{\sqrt{x}}{3} & \text{if } 0 < x < 9 \\ 1 & \text{if } x \geq 9 \end{cases}$
- (c) $P(2 < X < 4) = F(4) - F(2) = 0.1953$.
- (d) $P(0 < X < 4 | X > 2) = \frac{P(2 < X < 4)}{P(X > 2)} = 0.3697$.
11. (a) $f(x) = F'(x) = \begin{cases} \frac{8}{x^3} & \text{if } x > 2 \\ 0 & \text{otherwise} \end{cases}$
- (b) $P(3 < X < 5) = F(5) - F(3) = \frac{64}{225}$.
- (c) $P(X > 8) = 1 - P(X \leq 8) = 1 - F(8) = \frac{4}{64} = \frac{1}{16}$.
12. (a) $f(x) = F'(x) = \begin{cases} \frac{1}{2} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
- (b) $P(-\frac{1}{2} < X < \frac{1}{2}) = F(\frac{1}{2}) - F(-\frac{1}{2}) = \frac{1}{2}$
- (c) $P(X > \frac{2}{3}) = 1 - F(\frac{2}{3}) = \frac{1}{6}$

Problem Set 2.5

1. As $E(X) = \sum_{x=1}^{\infty} xf(x) = \sum_{x=1}^{\infty} \frac{1}{x+1} = \infty$, we conclude that $E(X)$ does not exist.
2. $E(X) = \int_{x=0}^{\infty} xf(x) dx = \int_{x=0}^{\infty} \frac{x}{(x+1)^2} dx$. Since $x = (x+1) - 1$, we may simplify the integral as

$$E(X) = \int_{x=0}^{\infty} \frac{1}{x+1} dx - \int_{x=0}^{\infty} \frac{1}{(x+1)^2} dx = [\ln(x+1)]_{x=0}^{\infty} + \left[\frac{1}{x+1} \right]_{x=0}^{\infty} = \infty$$

Hence, $E(X)$ does not exist.

3. The probability distribution function of X is given by

x	0	1	2	3
$f(x)$	$\frac{1}{27}$	$\frac{6}{27}$	$\frac{12}{27}$	$\frac{8}{27}$

Hence, the expectation of X is given by

$$E(X) = \sum_{x=0}^3 xf(x) = \frac{1}{27} [0 + 6 + 24 + 24] = \frac{54}{27} = 2$$

4. The expectations of A and B winning the stake are Rs. 3000 and Rs. 2500, respectively.
5. (a) L.H.S. = $\text{Var}(X) = E[X^2] - [E(X)]^2$ and R.H.S. = $E[X(X-1)] - E(X)[E(X)-1] = E[X^2] - [E(X)]^2$.
- (b) By (a), we have $\text{Var}(X) = E[X(X-1)] - E(X)[E(X)-1] = 4$ and we also find that $\text{Var}(4-5X) = 25\text{Var}(X) = 100$.
6. Since $E(Y) = 12a - b$ and $\text{Var}(Y) = 36a^2$, we obtain $a = \frac{1}{6}$ and $b = 2$.

7. $\mu = E(X) = \frac{7}{8}$ and $\text{Var}(X) = \frac{183}{64}$.
 8. (a) $k = 0.05$, (b) $P(X < 1) = 0.45$, (c) $\mu = E(X) = -0.1$, $\text{Var}(X) = 3.79$.
 9. $\mu = E(X) = 5.37$ and $\text{Var}(X) = 3.7531$
 10. If X is the amount the man gets out of his drawing, then the probability distribution of X is

x	30	40	50	60
f	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$

from which we obtain $E(X) = 42$ and $\text{Var}(X) = 56$.

11. (a) $\alpha = \frac{1}{\ln 3}$, (b) $\mu = E(X) = 1.8205$, $\text{Var}(X) = 0.3268$ (c) $M = e^{\frac{1}{2} \ln 3} = \sqrt{3}$.
 12. (a) $f(x) = F'(x) = \begin{cases} \frac{x}{\alpha^2} e^{-\frac{x^2}{2\alpha^2}} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
 (b) $E(X) = \alpha$, $\mu'_2 = 2\alpha^2$ and $\text{Var}(X) = \alpha^2$
 (c) $P(X > \alpha) = \int_{\alpha}^{\infty} \frac{x}{\alpha^2} e^{-\frac{x^2}{2\alpha^2}} dx = \left[-e^{-\frac{x^2}{2\alpha^2}} \right]_{x=\alpha}^{\infty} = e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}$
 13. $\mu = E(X) = \frac{1}{2}$.
 14. (a) $k = \frac{\pi}{8}$, (b) $E(X) = \mu = 2$, $\text{Var}(X) = 32 \left[\frac{1}{8} - \frac{1}{\pi^2} \right]$, (c) $Q_1 = \frac{4}{3}$, $Q_3 = \frac{8}{3}$, (d) $M = 2$, (e) $c = 2$.
 15. (a) $k = \frac{1}{6}$, (b) $\mu = E(X) = 4$, $\text{Var}(X) = 4$, (c) $H = 3$.

Problem Set 2.6

1. (a) By Chebychev's inequality, $P[|X - 1| \geq 2] \leq \frac{7}{16}$.
 (b) By direct computation, $P[|X - 1| \geq 2] = P(X < -1) + P(X > 3) = \frac{1}{8}$.
 2. By Chebychev's inequality, $P[|X - 7| \geq 4] \leq \frac{\sigma^2}{C^2} = \frac{35/6}{16} = \frac{35}{96}$ and the actual probability is $P[|X - 7| \geq 4] = \frac{1}{6}$.
 3. Note that $\mu = 2$ and $\sigma^2 = 2$. By Chebychev's inequality, $P[|X - 2| \geq 2] \leq \frac{\sigma^2}{C^2} = \frac{2}{4} = \frac{1}{2}$.
 The actual probability is $P[|X - 2| \geq 2] = P[X \leq 0] + P[X \geq 4] = 0 + \int_{x=4}^{\infty} \frac{1}{2} e^{-\frac{x}{2}} dx = e^{-2}$.
 4. (a) By Chebychev's inequality, $P(-2 < X < 14) = P[|X - \mu| < 8] > 1 - \frac{\sigma^2}{C^2} = \frac{15}{16}$.
 (b) By Chebychev's inequality, $P[|X - 6| \geq 6] \leq \frac{\sigma^2}{C^2} = \frac{4}{36} = \frac{1}{9}$.
 5. By Chebychev's inequality, $P[|X - \mu| \leq 2000] \geq 1 - \frac{\sigma^2}{C^2} = 1 - \frac{3000}{2000^2} = 0.9992$.
 6. By Chebychev's inequality, $P[|X - \mu| \leq 20] \geq 1 - \frac{\sigma^2}{C^2} = 1 - \frac{50}{400} = \frac{7}{8}$.
 7. (a) $K = \sqrt{10}$, (b) $K = \sqrt{20}$, (c) $K = 10$.

Problem Set 2.7

1. Note that $\mu = E(X) = E[\overline{X - 2} - 1] = E[X - 2] + 2 = 1 + 2 = 3$.
 The variance is given by $\text{Var}(X) = E[\overline{X - 2} - 1]^2 = E[(X - 2)^2] - 2E[X - 2] + 1 = 9$. Similarly, it can be shown that $\mu_3 = 0$ and $\mu_4 = 5$.
 2. (a) $E[X^r] = \int_{x=0}^2 x^r f(x) dx = \int_{x=0}^2 x^r \left(1 - \frac{x}{2}\right) dx = \int_{x=0}^2 x^r dx - \frac{1}{2} \int_{x=0}^2 x^{r+1} dx = \frac{2^{r+1}}{(r+1)(r+2)}$

- (b) By (a), it is immediate that $E(X) = \frac{2}{3}$ and $E(X^2) = \frac{2}{3}$. Thus, $E[(3X+2)^2] = E[9X^2 + 12X + 4] = 18$.
3. (a) $K = 6$, (b) $\mu = \frac{1}{2}$, $\text{Var}(X) = \frac{1}{20}$ (c) $\mu'_r = \frac{6}{(r+2)(r+3)}$
4. (a) $K = \frac{3}{2}$
 (b) $\mu = 0$ and $\text{Var}(X) = \frac{3}{5}$
 (c) Note that
- $$\mu_{2n+1} = \int_{x=-1}^1 x^{2n+1} f(x) dx = \int_{x=-1}^1 x \frac{3}{2} x^{2n+3} dx = 0$$
- and
- $$\mu_{2n} = \int_{x=-1}^1 x^{2n} f(x) dx = \int_{x=-1}^1 x \frac{3}{2} x^{2n+2} dx = 3 \int_{x=0}^3 x^{2n+2} dx = \frac{3}{2n+3}$$
- (d) The first four central moments are $\mu_1 = 0, \mu_2 = \frac{3}{5}, \mu_3 = 0, \mu_4 = \frac{3}{7}$ and $\beta_1 = 0, \beta_2 = \frac{25}{21}$.
5. (a) $K = \frac{1}{2}$
 (b) $\mu = 0$ and $\text{Var}(X) = \frac{1}{3}$
 (c) Note that
- $$\mu_{2n+1} = \int_{x=-1}^1 x^{2n+1} f(x) dx = \int_{x=-1}^1 x^{2n+1} \left[\frac{1}{2} \right] dx = 0$$
- and
- $$\mu_{2n} = \int_{x=-1}^1 x^{2n} f(x) dx = \int_{x=-1}^1 x^{2n} \left[\frac{1}{2} \right] dx = \int_{x=0}^1 x^{2n} dx = \frac{1}{2n+1}$$
- (d) Using (c), we have $\mu_3 = 0$ and $\mu_4 = \frac{1}{5}$. Thus, it follows that $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0$ and $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{9}{5}$.
6. (a) $\mu = E(X) = 2$ and $\text{Var}(X) = 2$.
 (b) $\mu_1 = 0, \mu_2 = 2, \mu_3 = 4$ and $\mu_4 = 24$.
 (c) The mode of f is $x = 1$.
7. (a) $\mu = E(X) = \frac{1}{\ln 2}$ and $\text{Var}(X) = \frac{3}{2\ln 2} - \frac{1}{(\ln 2)^2} = 0.0827$.
 (b) $\mu'_r = \frac{1}{\ln 2} \left[\frac{2^r - 1}{r} \right]$.
 (c) The median, $M = \sqrt{2}$.

Problem Set 2.8

1. $M_X(t) = \frac{1}{16} [1 + 4e^t + 6e^{2t} + 4e^{3t} + e^{4t}]$.
 We find that $\mu'_1 = M'_X(0) = 2$ and $\mu'_2 = M''_X(0) = 5$. Thus, $\mu = E(X) = \mu'_1 = 2$ and $\text{Var}(X) = \mu'_2 - \mu'^2_1 = 1$.
2. $M_X(t) = \frac{2e^t}{3-e^t}$.
 We find that $\mu'_1 = M'_X(0) = \frac{3}{2}$ and $\mu'_2 = M''_X(0) = 3$. Thus, $\mu = E(X) = \mu'_1 = \frac{3}{2}$ and $\text{Var}(X) = \mu'_2 - \mu'^2_1 = \frac{3}{4}$.

3. $M_X(t) = \frac{e^{bt}-e^{at}}{t(b-a)}$ To find μ'_1 and μ'_2 , we simplify $M_X(t)$ and get

$$M_X(t) = 1 + \left[\frac{b+a}{2} \right] t + \left[\frac{b^2+ab+a^2}{3} \right] \frac{t^2}{2!} + \dots$$

Thus, we find that

$$\mu'_1 = \text{Coefficient of } t \text{ in } M_X(t) = \frac{b+a}{2}$$

and

$$\mu'_2 = \text{Coefficient of } \frac{t^2}{2!} \text{ in } M_X(t) = \frac{b^2+ab+a^2}{3}$$

Hence, the mean of X is $\mu = \mu'_1 = \frac{b+a}{2}$ and the variance of X is $\text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{(b-a)^2}{12}$.

4. $M_X(t) = \frac{1}{(t-1)^2}$ if $-1 < t < 1$ (using integration by parts).

We find that $\mu'_1 = M'_X(0) = 2$ and $\mu'_2 = M''_X(0) = 6$.

Hence, the mean of X is $\mu = \mu'_1 = 2$ and the variance of X is $\text{Var}(X) = \mu'_2 - \mu_1'^2 = 2$.

5. $M_X(t) = \frac{1}{(1-t)^3}$ if $-1 < t < 1$

We find that $\mu'_1 = M'_X(0) = 3$ and $\mu'_2 = M''_X(0) = 12$.

Hence, the mean of X is $\mu = \mu'_1 = 3$ and the variance of X is $\text{Var}(X) = \mu'_2 - \mu_1'^2 = 3$.

6. $M_X(t) = \frac{e^{\alpha t}}{1-\alpha^2 t^2}$ if $-\alpha < t < \alpha$ We note that

$$M_X(t) = e^{\alpha t} [1 - \alpha^2 t^2]^{-1} = \left[1 + \alpha t + \frac{\alpha^2 t^2}{2!} + \dots \right] [1 + \alpha^2 t^2 + \alpha^4 t^4 + \dots]$$

Simplifying, $M_X(t) = 1 + \alpha t + \frac{3\alpha^2 t^2}{2!} + \dots$. Thus,

$$\mu'_1 = \text{Coeff. of } t \text{ in } M_X(t) = \alpha \text{ and } \mu'_2 = \text{Coeff. of } \frac{t^2}{2!} \text{ in } M_X(t) = 3\alpha^2$$

Hence, the mean of X is $\mu = \mu'_1 = \alpha$ and the variance of X is $\text{Var}(X) = \mu'_2 - \mu_1'^2 = 2\alpha^2$.

7. $M_Y(t) = M_X(3t) = \frac{1}{1-9t^2}$ where $-\frac{1}{3} < t < \frac{1}{3}$.

$$8. M_Y(t) = e^{-\frac{4t}{2}} M_X\left(\frac{t}{2}\right) = \frac{1}{4} e^{-2t} \left(1 + e^{\frac{t}{2}}\right)^2.$$

$$9. M_Y(t) = e^{-\frac{6t}{2}} M_X\left(\frac{t}{2}\right) = \frac{1}{t} e^{-3t} [e^t - 1]$$

$$10. M_Y(t) = e^{-\frac{4t}{4}} M_X\left(\frac{t}{4}\right) = \frac{16e^{-t}}{16-t^2} \text{ where } -4 < t < 4.$$

Problem Set 2.9

1. $\phi_X(t) = \frac{1}{2}(1 + \cos t)$. We find that $\mu'_1 = \frac{\phi'_X(0)}{i} = 0$ and $\mu'_2 = \frac{\phi''_X(0)}{i^2} = \frac{1}{2}$.

Hence, $\mu = \mu'_1 = 0$ and $\text{Var}(X) = \mu'_2 - \mu_1'^2 = \frac{1}{2}$.

2. $\phi_X(t) = (q + pe^{it})^n$. We find that $\mu'_1 = \frac{\phi'_X(0)}{i} = np$ and $\mu'_2 = \frac{\phi''_X(0)}{i^2} = n(n-1)p^2 + np$.

Hence, $\mu = \mu'_1 = np$ and $\text{Var}(X) = \mu'_2 - \mu_1'^2 = npq$

3. $\phi_X(t) = e^{\lambda(e^{it}-1)}$. We find that $\mu'_1 = \frac{\phi'_X(0)}{i} = \lambda$ and $\mu'_2 = \frac{\phi''_X(0)}{i^2} = \lambda^2 + \lambda$

Hence, $\mu = \mu'_1 = \lambda$ and $\text{Var}(X) = \mu'_2 - \mu_1'^2 = \lambda$.

4. $\phi_X(t) = \frac{1}{(1-it)^\lambda}$. We find that $\mu'_1 = \frac{\phi'_X(0)}{i} = \lambda$ and $\mu'_2 = \frac{\phi''_X(0)}{i^2} = \lambda(\lambda+1)$.

Hence, $\mu = \mu'_1 = \lambda$ and $\text{Var}(X) = \mu'_2 - \mu'^2_1 = \lambda$.

5. By the inversion theorem, we have

$$f(x) = \frac{1}{2\pi} \int_{t=-\infty}^{\infty} e^{-itx} \phi(t) dt$$

Using the definition of ϕ , we have

$$f(x) = \frac{1}{2\pi} \int_{t=-1}^0 (1+t)e^{-itx} dt + \frac{1}{2\pi} \int_{t=0}^1 (1-t)e^{-itx} dt$$

Evaluating the above integrals using integration by parts, we get

$$f(x) = \frac{1}{2\pi} \left[-\frac{1}{ix} - \frac{1}{(ix)^2} (1 - e^{ix}) + \frac{1}{ix} - \frac{1}{(ix)^2} (e^{-ix} - 1) \right]$$

Simplifying, we have

$$f(x) = \frac{1}{2\pi x^2} [2 - e^{ix} - e^{-ix}] = \frac{1}{\pi x^2} \left[1 - \frac{e^{ix} + e^{-ix}}{2} \right] = \frac{1}{\pi} \left[\frac{1 - \cos x}{x^2} \right]$$

CHAPTER 3

Problem Set 3.1

- (a) $f(0) = P(X=0) = q^{18} = 6.7664 \times 10^{-4}$ and $f(1) = P(X=1) = 18pq^{17} = 0.0061$.
(b) $P(X=4) = {}^{18}C_4 p^4 q^{14} = 3060 p^4 q^{14} = 0.1294$.
- By the uniqueness theorem of M.G.F.s, X is a binomial variable with the parameters $n=3$ and $p=\frac{1}{4}$. Hence,

$$f(x) = P(X=x) = {}^3C_x p^x q^{3-x} = {}^3C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{3-x} \quad \text{for } x=0, 1, 2, 3$$

- $P(X \leq 2) = 1 - P(X=3) = 1 - p^3 = \frac{63}{64}$.
 - Since $\text{Var}(X) = \sigma^2 = npq = \frac{9}{16}$, S.D. = $\sigma = \frac{3}{4}$.
 - $M_Y(t) = e^{3t} M_X(2t) = \frac{e^{3t}}{64} (3 + e^{2t})^3$.
- (a) $p = \frac{1}{4}$, (b) $\text{Mean}(X) = 1$ and $\text{Var}(X) = \frac{3}{4}$.
 - The probability distribution of X is obtained as follows:

x	0	1	2	3	4	5
$f(x)$	$\frac{1}{1024}$	$\frac{15}{1024}$	$\frac{90}{1024}$	$\frac{270}{1024}$	$\frac{405}{1024}$	$\frac{243}{1024}$

- $P(X \geq 3) = P(X=3) + P(X=4) + P(X=5) = \frac{270}{1024} + \frac{405}{1024} + \frac{243}{1024} = \frac{459}{512}$.
 - $P(X \leq 3) = 1 - [P(X=4) + P(X=5)] = 1 - \frac{81}{128} = \frac{47}{128}$.
 - $P(X=2) = \frac{90}{1024} = \frac{45}{512}$.
- $f(x) = P(X=x) = {}^{10}C_x p^x q^{10-x} = {}^{10}C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{10-x} \quad \text{for } x=0, 1, 2, \dots, 10$

$$(a) P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] = 1 - \frac{174}{463} = \frac{289}{463}.$$

$$(b) P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{345}{509}.$$

$$(c) P(X = 10) = p^{10} = \frac{1}{9765625}.$$

$$6. f(x) = P(X = x) = {}^5C_x p^x q^{5-x} = {}^5C_x \left(\frac{1}{10}\right)^x \left(\frac{9}{10}\right)^{5-x} \text{ for } x = 0, 1, 2, \dots, 10.$$

The required probability is given by

$$P(X \geq 1) = 1 - P(X = 0) = 1 - q^{10} = \frac{764}{1173}$$

$$7. f(x) = P(X = x) = {}^{20}C_x p^x q^{20-x} = {}^{20}C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{20-x} \text{ for } x = 0, 1, 2, \dots, 20$$

$$(a) P(X = 2) = 190 p^2 q^{18} = \frac{1453}{7701}.$$

$$(b) P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] = 1 - \frac{1468}{1995} = \frac{527}{1995}.$$

$$(c) P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{1433}{1550}.$$

$$(d) P(X = 0) = q^{20} = \frac{1449}{4042}.$$

$$8. f(x) = P(X = x) = {}^9C_x p^x q^{9-x} = {}^9C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{9-x} \text{ for } x = 0, 1, 2, \dots, 9$$

$$(a) \text{Mean}(X) = np = 3 \text{ and } \text{Var}(X) = npq = 2$$

$$(b) P(X = 2) = 36 p^2 q^7 = \frac{512}{2187} = 0.2341$$

$$(c) P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{671}{1779} = 0.3772$$

$$(d) P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] = \frac{581}{678} = 0.8569$$

$$9. f(x) = P(X = x) = {}^5C_x p^x q^{5-x} = {}^5C_x \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{5-x} \text{ for } x = 0, 1, 2, \dots, 5$$

$$(a) P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)] = 1 - \frac{856}{1065} = \frac{209}{1065} = 0.1962$$

$$(b) P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = 0.9645$$

$$(c) P(X = 3) = 10 p^3 q^2 = 0.0322$$

$$10. P(X = 3) = {}^{10}C_3 p^3 q^7 = 120 p^3 q^7 = 0.0425.$$

11. A simple calculation yields the following table:

No. of Successes	Observed Frequencies	Theoretical Frequencies
0	4	3.1250
1	11	12.5000
2	21	18.7500
3	9	12.5000
4	5	3.1250

12. A simple calculation yields the following table:

No. of Successes	Observed Frequencies	Theoretical Frequencies
0	7	3.1250
1	18	15.6250
2	26	31.2500
3	24	31.2500
4	17	15.6250
5	8	3.1250

Problem Set 3.2

1. (a) $\text{Var}(X) = 1$, (b) $P(X = 0) = e^{-1}$.
2. $\text{Mean}(X) = \text{Var}(X) = \mu = 2$.
3. $P(X = 4) = e^{-3} \frac{81}{24} = 0.1680$.
4. $P(X = 1) = 0.3219$, $P(X = 2) = 0.2590$ and $P(X = 3) = 0.1390$.
5. $P(X = 4 \text{ or } X = 5) = P(X = 4) + P(X = 5) = 2P(X = 4) = 2 \times 0.1755 = 0.3510$.
6. $P(X \leq 2) = e^{-1.5} \left[1 + (1.5) + \frac{(1.5)^2}{2!} \right] = 0.8088$.
7. $P(X \leq 2) = e^{-0.4} \left[1 + (0.4) + \frac{(0.4)^2}{2!} \right] = 0.9921$.
8. $P(X \leq 3) = e^{-8} \left[1 + (8) + \frac{8^2}{2!} + \frac{8^3}{3!} \right] = 0.0424$.
9. (a) $P(X = 0) = 0.3012$, (b) $P(X = 1) = 0.3614$, (c) $P(X \leq 2) = 0.8795$.
10. (a) $P(X = 0) = 0.0025$, (b) $P(X \leq 2) = 0.0620$, (c) $P(X \geq 3) = 0.9380$.
11. (a) $NP(X = 0) \approx 905$, (b) $NP(X = 1) \approx 90$, (c) $NP(X \leq 2) \approx 1000$, (d) $NP(X \geq 2) \approx 5$.
12. (a) $NP(X = 0) \approx 68$, (b) $NP(X \geq 2) \approx 338$, (c) $NP(X \geq 2) \approx 297$.
13. (a) $P(X = 0) = e^{-8} = 3.3546 \times 10^{-4}$, (b) $P(X = 2) = 0.0107$, (c) $P(X > 2) = 0.9862$.
14. A simple calculation results in the following table:

No. of Deaths	Observed Frequencies	Theoretical Frequencies
0	127	121.3061
1	52	60.6531
2	16	15.1633
3	4	2.5272
4	1	0.3159

15. A simple calculation results in the following table:

Mistakes per day	Observed Frequencies	Theoretical Frequencies
0	167	134.7987
1	70	107.8390
2	35	43.1356
3	17	11.5028
4	7	2.3006
5	3	0.3681
6	1	0.0491

Problem Set 3.3

1. Mode = 1.
2. $f(3) + f(6) + f(9) + \cdots = q^3 p + q^6 p + q^9 p + \cdots = \frac{q^3 p}{1 - q^3} = \frac{q^3}{q^2 + q + 1}$.
3. $P(X \text{ is odd}) = \frac{1}{1+q} = \frac{1}{1+0.2} = \frac{1}{1.2} = \frac{5}{6} = 0.8333$.
4. $P(X = 3) = q^2 p = \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = \frac{2}{27}$.
5. By Chebyshev's inequality, $P(|X - 1.5| \leq 2) \geq 1 - \frac{\sigma^2}{C^2} = 1 - \frac{3/4}{4} = \frac{13}{16}$. The actual probability is calculated as $P(|X - 1.5| \leq 2) = P(-0.5 \leq X \leq 3.5) = P(X = 1) + P(X = 2) + P(X = 3) = \frac{26}{27}$.

Problem Set 3.4

1. $P(X = 8) = \binom{7}{3} p^4 q^4 = 35(0.3)^3(0.7)^4 = 0.0681.$
2. $P(X = 10) = \binom{9}{2} p^3 q^7 = 36(0.6)^3(0.4)^7 = 0.0127.$
3. $P(X = 7) = \binom{6}{3} p^4 q^3 = 20(0.7)^4(0.3)^3 = 0.1297.$
4. $P(X = 8) = \binom{7}{2} (0.5)^8 = 21(0.5)^8 = 0.0820.$
5. $P(X \geq 6) = 1 - P(X < 6) = 1 - [P(X = 4) + P(X = 5)] = 1 - [3p^3q + 6p^3q^2] = 0.9998.$

Problem Set 3.5

1. (a) $f(x) = F'(x) = \begin{cases} \frac{1}{2\pi} & \text{if } -\pi < x < \pi \\ 0 & \text{otherwise} \end{cases}$

Thus, it is immediate that X has a uniform density in $[-\pi, \pi]$

- (b) $P(-\frac{\pi}{2} < X < \frac{\pi}{2}) = F(\frac{\pi}{2}) - F(-\frac{\pi}{2}) = \frac{1}{2}$
2. $f(x) = \begin{cases} \frac{1}{2} & \text{if } -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$
 $\text{Mean}(X) = \frac{a+b}{2} = 0$ and $\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{1}{3}.$
3. (a) $P(70 < X < 80) = \frac{1}{5},$ (b) $P(X > 75) = \frac{1}{2},$ (c) $P(X = 80) = 0.$
4. (a) $\mu = a$ and $\text{Var}(X) = \frac{a^2}{12},$ (b) Median, $M = a,$ (c) $Q_1 = \frac{a}{2}$ and $Q_3 = \frac{3a}{2},$ (d) The mode of X is any point in the interval $(0, 2a).$
5. $\phi_X(t) = \frac{e^{i\alpha t}}{2\hbar i t} [2i \sinh t] = \frac{e^{i\alpha t}}{\hbar t} \sinh t.$

Problem Set 3.6

1. (a) $P(X > 2) = e^{-1}.$
 (b) $M_X(t) = \frac{1}{1-2t},$ where $|t| < \frac{1}{2}$
 Note that $M_X(t) = 1 + 2t + (2t)^2 + (2t)^3 + \dots$ and so it follows that

$$\mu'_1 = \text{Coeff. of } t \text{ in } M_X(t) = 2 \text{ and } \mu'_2 = \text{Coeff. of } \frac{t^2}{2!} \text{ in } M_X(t) = 8$$

Thus, $\mu = \mu'_1 = 2$ and $\text{Var}(X) = \mu'_2 - \mu_1'^2 = 4.$

2. By the memoryless property of exponential distribution, the required probability is given by

$$P(X > 240 | X > 120) = P(X > 120) = 1 - P(X \leq 120) = 1 - F(120) = e^{-4}$$

3. By the memoryless property of exponential distribution, the required probability is

$$P(X > 12 | X > 6) = P(X > 6) = 1 - F(6) = e^{-2.4} = 0.0907$$

4. (a) $P(X > 3) = e^{-1},$ (b) $P(X > 8 | X > 2) = P(X > 6) = e^{-2}.$

5. By the memoryless property of exponential distribution, the required probability is

$$P(X \leq 300) = F(300) = 1 - e^{-\frac{300}{500}} = 1 - e^{-0.6} = 0.4512$$

6. $P(X \leq 2) = F(2) = 1 - e^{-\frac{2}{3}} = 0.4866$.

7. (a) $P(X > 8) = 0.1353$, (b) $P(3 < X < 6) = 0.2492$, (c) $P(X < 2) = 0.3935$.

Problem Set 3.7

1. $M_X(t) = (1-t)^{-2} = 1 + 2t + 3t^2 + \dots$ where $|t| < 1$.

Thus,

$$\mu'_1 = \text{Coeff. of } t \text{ in } M_X(t) = 2 \text{ and } \mu'_2 = \text{Coeff. of } \frac{t^2}{2!} \text{ in } M_X(t) = 6$$

Thus, $\mu = \mu'_1 = 2$ and $\text{Var}(X) = \mu'_2 - \mu_1'^2 = 2$.

2. The moment-generating function of X is given by

$$M_X(t) = \sum_{r=0}^{\infty} \mu'_r \frac{t^r}{r!} = \sum_{r=0}^{\infty} \frac{(k+r)!}{k!} \frac{t^r}{r!}$$

and thus it follows that

$$M_X(t) = \sum_{r=0}^{\infty} \binom{k+r}{r} t^r = \left(1 - \frac{t}{k}\right)^{-1}$$

By the uniqueness of MGFs, it follows that X has the probability density function

$$f(x) = \begin{cases} \frac{x^k}{k!} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

3. The probability density function of X is given by

$$f(x) = \frac{\lambda^\alpha e^{-\lambda x}}{\Gamma(\alpha)} x^{\alpha-1} \text{ for } x > 0$$

from which it follows that

$$f'(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} e^{-\lambda x} x^{\alpha-2} [\alpha - 1 - \lambda x]$$

Setting $f'(x) = 0$ and noting that $x > 0$, it follows that the equation has only one root namely

$$x^* = \frac{\alpha - 1}{\lambda}$$

Note that $x^* > 0$ since $\alpha > 1$. Also, it is easy to verify that $f''(x^*) < 0$. Thus, X has a unique mode at x^* .

4. $P(X \geq 2) = 0.0174$.

5. Since the city has a daily stock of 40,000 gallons, the probability that the stock is insufficient on a particular day is given by

$$P(X > 40,000) = P(Y > 10,000) = \int_{y=10,000}^{\infty} \frac{y}{(20,000)^2} e^{-\frac{y}{20,000}} dy \int_{z=0.5}^{\infty} z e^{-z} dz = 0.9098$$

Problem Set 3.8

1. $\mu = 3840$ hours.
2. (a) $\mu = 937.5$ hours, (b) $P(X > 200) = 0.2222$, (c) $P(X < 300) = 0.8108$.
3. $\alpha = 6.1972 \times 10^{-5}$.
4. The probability that a tube will not be replaced during the first 3 months is given by

$$P\left(X > \frac{3}{12}\right) = P\left(X > \frac{1}{4}\right) = 1 - F\left(\frac{1}{4}\right) = e^{-10\left(\frac{1}{4}\right)^3} = 0.8553$$

Hence, the probability that none of the six tubes will be replaced during the first 2 months is given by

$$[0.8553]^6 = 0.3915 \quad [\text{using independence}]$$

Problem Set 3.9

1. $\mu = 20$ and $\sigma = 3$.
2. $\mu = 8$ and $\sigma = 6.6667$.
3. (a) $P(X \geq 35) = 0.0014$, (b) $P(X \leq 30) = 0.9772$, (c) $P(10 \leq X \leq 40) = 0.9772$, (d) $P[|X - 20| > 10] = 0.0456$.
4. (a) $P(13 < X < 43) = 0.9759$, (b) $P(X < 43 | X > 13) = 0.9987$.
5. $\mu = 50.1519$ and $\sigma = 19.3370$.
6. $\mu = 56.3383$ and $\sigma = 22.3048$.
7. (a) $P(X > 500) = 0.0228$, (b) $P(250 < X < 450) = 0.8399$, (c) $P(X < 300) = 0.0228$.
8. (a) $P(X > 2600) = 0.0010$, (b) $P(800 < X < 2000) = 0.0010$, (c) $P(-0.4 < Z < 2) = 0.6326$, (d) $P(X < 500) = 0.0228$.
9. (a) 8.08% students scored marks above 80, (b) 56.7% students scored marks between 30 and 70, (c) 34.46% students scored marks below 35.
10. (a) $P(X < 150) = 0.3372$, (b) $P(X > 180) = 0.2033$, (c) $P(150 < X < 170) = 0.3256$.
11. (a) $NP(40 < X < 70) \approx 533$, (b) $NP(X > 80) \approx 67$, (c) $NP(X < 30) \approx 159$.
12. (a) $NP(X > 3000) \approx 16$, (b) $NP(X < 1500) \approx 264$, (c) $NP(1800 < X < 2300) \approx 1162$.
13. (a) $NP(X > 90) \approx 110$, (b) $NP(X < 35) \approx 12$, (c) $NP(40 < X < 80) \approx 871$.
14. (a) $NP(X > 800) \approx 23$, (b) $NP(X < 300) \approx 92$, (c) $NP(400 < X < 800) \approx 657$.
15. (a) $NP(X > 72) \approx 121$, (b) $NP(X < 60) \approx 154$, (c) $NP(60 < X < 70) \approx 192$.

Problem Set 3.10

1. (a) The distribution function of Y is given by

$$G(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ y^4 & \text{if } 0 < y < 1 \\ 1 & \text{if } y \geq 1 \end{cases}$$

- (b) The probability density function g of Y is obtained as

$$g(y) = G'(y) = \begin{cases} 4y^3 & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

2. (a) If G is the distribution function of Y , then

$$G(y) = \begin{cases} 0 & \text{if } y \leq 0 \\ \sqrt{y} & \text{if } 0 < y < 1 \\ 1 & \text{if } y \geq 1 \end{cases}$$

- (b) If g is the probability density function of Y , then

$$g(y) = G'(y) = \begin{cases} \frac{1}{2\sqrt{y}} & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

3. By Example 3.110, we know that the probability density function of $Y = aX + b$ is given by

$$g(y) = \frac{1}{|a|} f\left(\frac{y-b}{a}\right)$$

where f is the probability density function of X . Hence, it follows that the probability density function of $Y = 2X + 3$ is given by

$$g(y) = \begin{cases} e^{-(y-3)} & \text{if } y > 3 \\ 0 & \text{otherwise} \end{cases}$$

4. By Example 3.110, the probability density function of $Y = X^2$ is given by the formula

$$g(y) = \begin{cases} \frac{1}{2\sqrt{y}} [f(\sqrt{y}) + f(-\sqrt{y})] & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

Since X has the Laplace density function, it is easy to see that the PDF of Y is given by

$$g(y) = \begin{cases} \frac{1}{2\sqrt{y}} e^{-\sqrt{y}} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

5. First, note that the cumulative distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-\theta e^x} & \text{if } x > 0 \end{cases}$$

- (a) The distribution function of Y

$$G(y) = \begin{cases} 1 - e^{-\theta y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (b) The probability density function of Y

$$g(y) = G'(y) = \begin{cases} \theta e^{-\theta y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

6. (a) It is easy to see that the distribution function of Y

$$G(y) = \begin{cases} 1 - e^{-y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(b) The probability density function of Y

$$g(y) = G'(y) = \begin{cases} e^{-y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

7. The probability distribution of $Y = 3X - 2$ is given by

y	-2	1	4	7
$g(y)$	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

8. The probability distribution of $Y = X^2$ is given by

y	0	1	4
$g(y)$	$\frac{1}{9}$	$\frac{4}{9}$	$\frac{4}{9}$

9. The probability mass function of $Y = X^2$ is given by

$$g(y) = \frac{2}{3} \left(\frac{1}{3} \right)^{\sqrt{y}-1}, \quad y = 1, 4, 9, 25, \dots$$

where $\sqrt{y}$ denotes the positive square root of each perfect square y .

10. The probability distribution of Y is obtained as

y	0	1	4
$g(y)$	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{1}{8}$

11. The probability density function of Y

$$g(y) = \begin{cases} e^y & \text{if } y < 0 \\ 0 & \text{otherwise} \end{cases}$$

12. The probability density function of $Y = 8X^3$ is obtained as

$$g(y) = \begin{cases} \frac{1}{3} y^{-\frac{2}{3}} e^{-y^{\frac{1}{3}}} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

13. The probability density function of $Y = \cos X$ is obtained as

$$g(y) = \begin{cases} \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}} & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

It is also easy to show that $\text{Mean}(Y) = \frac{2}{\pi}$ and $\text{Var}(Y) = \frac{1}{2} - \frac{4}{\pi^2}$.

14. The probability density function of $Y = \sqrt{X}$ is obtained as

$$g(y) = \begin{cases} \frac{1}{2} & \text{if } 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

15. The probability density function of $Y = 2X - 3$ is obtained as

$$g(y) = \begin{cases} \frac{y+3}{8} & \text{if } -3 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

CHAPTER 4**Problem Set 4.1**

1. The joint probability distribution of (X, Y) is easily obtained as follows:

$X \backslash Y$	0	1	2
0	$\frac{1}{56}$	$\frac{6}{56}$	$\frac{3}{56}$
1	$\frac{9}{56}$	$\frac{18}{56}$	$\frac{3}{56}$
2	$\frac{9}{56}$	$\frac{6}{56}$	0
3	$\frac{1}{56}$	0	0

2. $F(x, y) = \begin{cases} (1 - e^{-x^2})(1 - e^{-y^2}) & \text{if } x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$
3. $K = 2$
4. $K = \frac{9}{2}$
5. $K = \frac{1}{4}$
6. (a) $K = \frac{1}{8}$, (b) $P(X + Y < 1) = \frac{1}{24}$
7. $P(X + Y < 1) = (1 - \frac{1}{e})^2$
8. $P(Y > X) = \frac{2}{5}$
9. $P(X + Y > 1) = 1 - P(X + Y < 1) = 1 - \frac{5}{48} = \frac{43}{48}$
10. (a) $P(1 < X < 3, 2 < Y < 4) = \frac{(e^6 - 1)(e^8 - 1)}{e^{25}}$, (b) $P(X < 2, Y < 2) = \frac{(e^6 - 1)(e^8 - 1)}{e^{14}}$, (c) $P(X + Y < 1) = 1 - \frac{4}{e^3} + \frac{3}{e^4}$
11. (a) $P(X < \frac{1}{2}, Y > \frac{1}{2}) = \frac{1}{4}$, (b) $P(X + Y < \frac{1}{2}) = \frac{1}{8}$
12. (a) $P(X < \frac{1}{2}, Y < \frac{1}{2}) = \frac{1}{2}$, (b) $P(X + Y < \frac{1}{2}) = \frac{1}{4}$, (c) $P(X > 2Y) = \frac{1}{3}$

Problem Set 4.2

1. (a) The marginal distribution of X is given by

x	0	1
$f_X(x)$	$\frac{1}{2}$	$\frac{1}{2}$

and the marginal distribution of Y is given by

y	0	1
$f_Y(y)$	$\frac{1}{2}$	$\frac{1}{2}$

- (b) $\mu_X = \frac{1}{2}$ and $\mu_Y = \frac{1}{2}$.

(c) Yes.

2. (a) The marginal distribution of X is given by

x	0	1
$f_X(x)$	$\frac{1}{3}$	$\frac{2}{3}$

and the marginal distribution of Y is given by

y	0	1
$f_Y(y)$	$\frac{1}{3}$	$\frac{2}{3}$

(b) $\mu_X = \frac{2}{3}$ and $\mu_Y = \frac{2}{3}$.

(c) No.

3. (a) The marginal distribution of X is given by

x	0	1	2
$f_X(x)$	$\frac{4}{21}$	$\frac{1}{3}$	$\frac{10}{21}$

and the marginal distribution of Y is given by

y	0	1	2
$f_Y(y)$	$\frac{1}{7}$	$\frac{1}{3}$	$\frac{11}{21}$

(b) $\mu_X = \frac{9}{7}$ and $\mu_Y = \frac{29}{21}$.

(c) No.

4. (a) The marginal distribution of X is given by

x	1	2	3
$f_X(x)$	$\frac{5}{18}$	$\frac{1}{3}$	$\frac{7}{18}$

and the marginal distribution of Y is given by

y	1	2	3
$f_Y(y)$	$\frac{2}{9}$	$\frac{1}{3}$	$\frac{4}{9}$

(b) $\mu_X = \frac{19}{9}$ and $\mu_Y = \frac{20}{9}$.

(c) No.

5. (a) $K = \frac{1}{45}$.

(b) The marginal distribution of X is given by

x	0	1	2
$f_X(x)$	$\frac{2}{15}$	$\frac{1}{3}$	$\frac{8}{15}$

and the marginal distribution of Y is given by

y	1	2	3
$f_Y(y)$	$\frac{4}{15}$	$\frac{1}{3}$	$\frac{2}{5}$

(c) $\mu_X = \frac{7}{5}$ and $\mu_Y = \frac{32}{15}$.

(d) X and Y are not independent.

(e) $P(X \leq 1, Y \leq 2) = \frac{4}{15}$.

6. (a) $K = \frac{1}{16}$.

(b) $P(X \geq 1, Y \leq 2) = \frac{3}{16}$.

- (c) The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{x}{2} & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{y}{8} & \text{if } 0 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

- (d) X and Y are independent random variables.

7. (a) $K = \frac{3}{2}$.

- (b) The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{1}{2} (3x^2 + 1) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{2} (3y^2 + 1) & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (c) X and Y are not independent random variables.

- (d) $P(0.2 < X < 0.4) = 0.1120$ and $P(0.3 < Y < 0.6) = 0.2130$.

8. (a) The marginal density function of X is given by

$$f_X(x) = \begin{cases} 6x(1-x) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 3y^2 & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (b) X and Y are not independent random variables.

- (c) $P(X + Y > 1) = \frac{3}{4}$.

9. (a) The marginal density function of X is given by

$$f_X(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} 2y & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (b) X and Y are not independent random variables.

- (c) $\mu_X = \frac{1}{2}$ and $\mu_Y = \frac{2}{3}$.

10. (a) $K = \frac{3}{8}$.

(b) The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{3}{4} (1+x^2) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{2} y & \text{if } 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

(c) X and Y are independent random variables.

(d) $P(X+Y < 1) = \frac{7}{64}$.

11. (a) $K = \frac{1}{2}$.

(b) The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{x}{2} & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} \frac{1}{2} (2-y) & \text{if } 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

(c) X and Y are not independent random variables.

(d) $\mu_X = \frac{4}{3}$ and $\mu_Y = \frac{2}{3}$.

12. (a) $K = \frac{2}{3}$.

(b) The marginal density function of X is given by

$$f_X(x) = \begin{cases} \frac{2}{3} (x+1) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is given by

$$f_Y(y) = \begin{cases} e^{-y} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(c) X and Y are independent random variables.

(d) $P(0 < X < \frac{1}{2}, Y > 1) = \frac{5}{12} e^{-1} = 0.1533$.

Problem Set 4.3

1. (a) The marginal distribution of X is

x	0	1	2
$f_X(x)$	$\frac{3}{10}$	$\frac{2}{5}$	$\frac{3}{10}$

and the marginal distribution of Y is

y	0	1	2
$f_Y(y)$	$\frac{1}{5}$	$\frac{1}{2}$	$\frac{3}{10}$

(b) The conditional distribution of Y given $X = 2$ is

y	0	1	2
$f_{Y X}(y 2)$	$\frac{1}{3}$	$\frac{2}{3}$	0

(c) No.

2. (a) The marginal distribution of X is

x	1	2	3
$f_X(x)$	$\frac{1}{3}$	$\frac{7}{24}$	$\frac{3}{8}$

and the marginal distribution of Y is

y	1	2	3
$f_Y(y)$	$\frac{7}{24}$	$\frac{1}{3}$	$\frac{3}{8}$

(b) The conditional distribution of X given $Y = 1$ is

x	1	2	3
$f_{X Y}(x 1)$	$\frac{4}{7}$	$\frac{1}{7}$	$\frac{2}{7}$

(c) No.

3. (a) The marginal distribution of X is

x	1	2
$f_X(x)$	$\frac{4}{9}$	$\frac{5}{9}$

and the marginal distribution of Y is

y	1	2	3
$f_Y(y)$	$\frac{2}{9}$	$\frac{1}{3}$	$\frac{4}{9}$

(b) The conditional distribution of X given $Y = 1$ is

x	1	2
$f_{X Y}(x 1)$	$\frac{5}{12}$	$\frac{7}{12}$

The conditional distribution of X given $Y = 2$ is

x	1	2
$f_{X Y}(x 2)$	$\frac{4}{9}$	$\frac{5}{9}$

and the conditional distribution of X given $Y = 3$ is

x	1	2
$f_{X Y}(x 3)$	$\frac{11}{24}$	$\frac{13}{24}$

4. (a) The marginal distribution of X is

x	0	1	2
$f_X(x)$	$\frac{4}{15}$	$\frac{1}{3}$	$\frac{2}{5}$

and the marginal distribution of Y is

y	0	1	2
$f_Y(y)$	$\frac{1}{15}$	$\frac{1}{3}$	$\frac{3}{5}$

- (b) The conditional distribution of Y given $X = 0$ is

y	0	1	2
$f_{Y X}(y 0)$	0	$\frac{1}{3}$	$\frac{2}{3}$

The conditional distribution of Y given $X = 1$ is

y	0	1	2
$f_{Y X}(y 1)$	$\frac{1}{15}$	$\frac{1}{3}$	$\frac{3}{5}$

and the conditional distribution of Y given $X = 2$ is

y	0	1	2
$f_{Y X}(y 2)$	$\frac{1}{9}$	$\frac{3}{9}$	$\frac{5}{9}$

5. (a) The marginal density function of X is

$$f_X(x) = \begin{cases} \frac{1}{3}(4x+1) & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is

$$f_Y(y) = \begin{cases} \frac{2}{3}(y+1) & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (b) For any fixed value of y in the interval $0 < y < 1$, the conditional density of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{2x+y}{y+1} \text{ for } 0 < x < 1$$

- (c) For any fixed value of x in the interval $0 < x < 1$, the conditional density of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{2(2x+y)}{4x+1} \text{ for } 0 < y < 1$$

- (d) No.

6. (a) The marginal density function of X is

$$f_X(x) = \begin{cases} \frac{5}{2} - 2x & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is

$$f_Y(y) = \begin{cases} \frac{5}{2} - 2y & \text{if } 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (b) For any fixed value of y in the interval $0 < y < 2$, the conditional density function of X given $Y = y$ is

$$f_{X|Y}(x|y) = \begin{cases} \frac{9-4x-4y}{2(5-4y)} & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (c) For any fixed value of x in the interval $0 < x < 2$, the conditional density function of Y given $X = x$ is

$$f_{X|Y}(x|y) = \begin{cases} \frac{9-4x-4y}{2(5-4x)} & \text{if } 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (d) No.

7. (a) $K = 2$.

- (b) The marginal density function of X is

$$f_X(x) = \begin{cases} \frac{1}{(1+x)^2} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is

$$f_Y(y) = \begin{cases} \frac{1}{(1+y)^2} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (c) For any fixed value of $y > 0$, the conditional density function of X given $Y = y$ is

$$f_{X|Y}(x|y) = \begin{cases} \frac{2(1+y)^2}{(1+x+y)^3} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (d) For any fixed value of $x > 0$, the conditional density function of Y given $X = x$ is

$$f_{Y|X}(y|x) = \begin{cases} \frac{2(1+x)^2}{(1+x+y)^3} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (e) When $Y = 1$, the conditional density of X given $Y = 1$ becomes

$$f_{X|Y}(x|1) = \frac{8}{(x+2)^3} \quad \text{for } x > 0$$

Thus, we obtain

$$P(X < 4|Y = 1) = \int_{x=0}^4 f_{X|Y}(x|1) dx = \frac{8}{9}$$

8. (a) The marginal density function of X is

$$f_X(x) = \begin{cases} \frac{3(3+2x)}{4(1+x)^4} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is

$$f_Y(y) = \begin{cases} \frac{3(3+2y)}{4(1+y)^4} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (b) For any fixed value of $y > 0$, the conditional density of X given $Y = y$ is

$$f_{X|Y}(x|y) = \begin{cases} \frac{6(1+x+y)}{(1+x)^4(3+2y)} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (c) For any fixed value of $x > 0$, the conditional density of Y given $X = x$ is

$$f_{Y|X}(y|x) = \begin{cases} \frac{6(1+x+y)}{(1+y)^4(3+2x)} & \text{if } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (d) No.

9. (a) $K = 2$.

- (b) The marginal density function of X is

$$f_X(x) = \begin{cases} 2x & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is

$$f_Y(y) = \begin{cases} 2(1-y) & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (c) For any fixed value of y in the interval $0 < y < 1$, the conditional density function of X given $Y = y$ is

$$f_{X|Y}(x|y) = \begin{cases} \frac{1}{1-y} & \text{if } y < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (d) For any fixed value of x in the interval $0 < x < 1$, the conditional density function of Y given $X = x$ is

$$f_{Y|X}(y|x) = \begin{cases} \frac{1}{x} & \text{if } 0 < y < x \\ 0 & \text{otherwise} \end{cases}$$

- (e) No.

10. (a) $K = \frac{2}{3}$.

- (b) The marginal density function of X is

$$f_X(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

and the marginal density function of Y is

$$f_Y(y) = \begin{cases} \frac{2}{3}(y+1) & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (c) For any fixed value of y in the interval $0 < y < 1$, the conditional density function of X given $Y = y$ is

$$f_{X|Y}(x|y) = f_X(x) = \begin{cases} e^{-x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (d) For any fixed value of x in the infinite interval $x > 0$, the conditional density function of Y given $X = x$ is

$$f_{Y|X}(y|x) = f_Y(y) = \begin{cases} \frac{2}{3}(y+1) & \text{if } 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

- (e) Yes.

Problem Set 4.4

1. (a) The marginal distribution of X is

x	0	1
$f_X(x)$	$\frac{1}{4}$	$\frac{3}{4}$

and the marginal distribution of Y is given by

y	0	1	2
$f_Y(y)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$

- (b) Since $E(X+Y) = \frac{25}{12}$, $E(X) = \frac{3}{4}$ and $E(Y) = \frac{4}{3}$, it follows that

$$E(X+Y) = \frac{25}{12} = E(X) + E(Y)$$

- (c) $E(XY) = \frac{11}{12}$.

2. (a) The marginal distribution of X is

x	-1	0	1
$f_X(x)$	$\frac{7}{20}$	$\frac{2}{5}$	$\frac{1}{4}$

and the marginal distribution of Y is

y	-1	0	1
$f_Y(y)$	$\frac{3}{10}$	$\frac{7}{20}$	$\frac{7}{20}$

- (b) $\mu_X = -\frac{1}{10}$ and $\mu_Y = \frac{1}{20}$.

- (c) $E(X+Y) = -\frac{1}{20} = \mu_X + \mu_Y$.

- (d) $E(XY) = \frac{1}{20}$.

3. (a) The marginal distribution of X is

x	0	1	2
$f_X(x)$	$\frac{7}{20}$	$\frac{1}{4}$	$\frac{2}{5}$

and the marginal distribution of Y is

y	0	1	2
$f_Y(y)$	$\frac{11}{20}$	$\frac{1}{5}$	$\frac{1}{4}$

- (b) $E(X) = \frac{21}{20}$, $E(X^2) = \frac{37}{20}$, $E(Y) = \frac{7}{10}$ and $E(Y^2) = \frac{6}{5}$.

- (c) $E(X+Y) = \frac{7}{4} = E(X) + E(Y)$.

(d) Since $E(XY) = \frac{3}{4}$, Cauchy-Schwartz inequality holds easily. Note that

$$[E(XY)]^2 = \frac{9}{16} < E(X^2) E(Y^2) = \frac{111}{50}$$

Problem Set 4.5

1. $P(X+Y=4) = \binom{10}{4} \left(\frac{1}{2}\right)^{10} = \frac{105}{512}.$

2. Define $Z = X_1 + X_2 + \cdots + X_n$. Then it is easy to see that

$$M_Z(t) = (q + pe^t)^{n_1+n_2+\cdots+n_k}$$

which shows that Z is a binomial random variable with parameters $(n_1 + n_2 + \cdots + n_k, p)$.

3. (a) $P(X+Y < 3) = 0.0138$, (b) $P(X+Y \geq 4) = 0.9576$.

4. Define $Z = X_1 + X_2 + \cdots + X_k$. Since X_i are independent random variables, it follows that

$$M_Z(t) = M_{X_1}(t) \cdot M_{X_2}(t) \cdots M_{X_k}(t) = \left(1 - \frac{t}{\lambda}\right)^{-n_1} \cdot \left(1 - \frac{t}{\lambda}\right)^{-n_2} \cdots \left(1 - \frac{t}{\lambda}\right)^{-n_k}$$

where $\left|\frac{t}{\lambda}\right| < 1$. Hence, it follows that

$$M_Z(t) = \left(1 - \frac{t}{\lambda}\right)^{-(n_1+n_2+\cdots+n_k)}$$

showing that Z is a gamma random variable with parameters $(\sum_{i=1}^k n_i, \lambda)$.

5. By the additive property of normal distributions, it follows that $Z = X + Y \sim N(0, 25)$. Hence, it is easy to see that the probability density function of Z is given by

$$f_Z(z) = \frac{1}{5\sqrt{2\pi}} e^{-\frac{z^2}{50}} \quad \text{for } -\infty < z < \infty$$

Problem Set 4.6

1. The probability density function of $Z = X - Y$ is

$$f_Z(z) = 1 - |z| \quad \text{for } |z| < 1$$

2. The probability density function of $Z = X - Y$ is

$$f_Z(z) = \frac{1}{2} e^{-|z|} \quad \text{for } -\infty < z < \infty$$

which is the Laplace distribution.

3. The probability density function of $Z = \frac{X}{Y}$ is

$$f_Z(z) = \frac{2}{(z+2)^2} \quad \text{for } z > 0$$

4. The probability density function of $Z = X + Y$ is

$$f_Z(z) = ze^{-z} \quad \text{for } z > 0$$

5. The joint probability density function of Z and W is given by

$$f_{Z,W}(z, w) = 24w(z - w) \quad \text{for } 0 < z < 1, 0 < w < 1, z > w$$

6. The probability density function of $R = \sqrt{X^2 + Y^2}$ is

$$f_R(r) = 2r^3 e^{-r^2} \quad \text{for } r > 0$$

7. (a) The joint probability density function of Z and W is

$$f_{Z,W}(z, w) = f(x, y)|J| = 1 \quad \text{for } z + w > 0, z - w > 0, z < 1$$

- (b) The marginal density function of Z is

$$f_Z(z) = 2z \quad \text{for } 0 < z < 1$$

The marginal density function of W is

$$f_W(w) = 1 - |w| \quad \text{for } |w| < 1$$

8. (a) The joint probability density function of Z and W is

$$f_{Z,W}(z, w) = f(x, y)|J| = 1 \quad \text{for } 0 < z - w < 1, 0 < w < 1$$

- (b) The marginal density function of Z is

$$f_Z(z) = \begin{cases} z & \text{for } 0 < z < 1 \\ 2 - z & \text{for } 1 \leq z < 2 \\ 0 & \text{elsewhere} \end{cases}$$

The marginal density function of W is

$$f_W(w) = 1 \quad \text{for } 0 < w < 1$$

which is the uniform distribution over $(0, 1)$.

9. (a) The joint probability density function of Z and W is

$$f_{Z,W}(z, w) = f(x, y)|J| = \frac{1}{2} \quad \text{for } 0 < z + w < 2, 0 < z - w < 2$$

- (b) The marginal density function of Z is

$$f_Z(z) = \begin{cases} z & \text{for } 0 < z < 1 \\ 2 - z & \text{for } 1 \leq z < 2 \\ 0 & \text{elsewhere} \end{cases}$$

The marginal density function of W is given by

$$f_W(w) = 1 - |w| \quad \text{for } |w| < 1$$

10. (a) The joint probability density function of Z and W is

$$f_{Z,W}(z, w) = \frac{1}{4\pi} e^{-\frac{(z^2 + w^2)}{4}}$$

(b) Both Z and W are normally distributed with mean 0 and variance 2.

(c) Since $f_{Z,W}(z,w) = f_Z(z)f_W(w)$ for all values of z and w , it is immediate that Z and W are independent random variables.

Problem Set 4.7

1. The marginal distributions of X and Y are obtained as follows:

x	-1	0	1
$f_X(x)$	$\frac{5}{16}$	$\frac{6}{16}$	$\frac{5}{16}$

y	-1	0	1
$f_Y(y)$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{6}{16}$

Note that $f(x,y) \neq f_X(x)f_Y(y)$ for the mass points (x,y) . For example,

$$f(0,0) = \frac{2}{16} \neq f_X(0)f_Y(0) = \frac{6}{16} \times \frac{4}{16} = \frac{3}{32}$$

Hence, X and Y are not independent. Next, note that $\mu_X = \mu_Y = 0$ and $E(XY) = 0$. Thus, it follows that

$$\text{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = 0 - 0 = 0$$

showing that X and Y are uncorrelated.

2. Note that we can express the joint probability density function f as

$$f(x,y) = f_X(x)f_Y(y)$$

where

$$f_X(x) = \begin{cases} 1 - e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

is the marginal density function of X and

$$f_Y(y) = \begin{cases} 1 - e^{-y} & \text{for } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

is the marginal density function of Y . Thus, X and Y are independent. Hence, it follows that X and Y are uncorrelated.

3. Note that the marginal density function of X is given by

$$f_X(x) = \frac{1}{2\sqrt{2\pi}} (x^2 + 1)e^{-\frac{x^2}{2}} \quad \text{for } -\infty < x < \infty$$

Since $f(x,y)$ is symmetric in x and y , we have

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} (y^2 + 1)e^{-\frac{y^2}{2}} \quad \text{for } -\infty < y < \infty$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, it follows that X and Y are not independent. Next, we note that

$$\mu_X = E(X) = \int_{-\infty}^{\infty} xf_X(x)dx = 0 \quad \text{and} \quad \mu_Y = E(Y) = \int_{-\infty}^{\infty} yf_Y(y)dy = 0$$

since for each integral, the integrand is an odd function. We note also that

$$E(XY) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} xyf(x,y)dy dx = 0$$

Thus, $\text{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = 0 - 0 = 0$ showing that X and Y are uncorrelated.

4. The marginal density functions of X and Y are obtained as

$$f_X(x) = \begin{cases} \frac{1}{6}(2x+1) & \text{for } 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{2}{3}(y+1) & \text{for } 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$E(X) = \frac{11}{9}, \text{Var}(X) = \frac{23}{81}, E(Y) = \frac{5}{9}, \text{Var}(Y) = \frac{13}{162} \text{ and } \text{Cov}(X,Y) = \frac{7}{162}$$

5. $\mu_X = \frac{4}{3}, \mu_Y = \frac{2}{3}, E(XY) = 1$

$$\text{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = 1 - \frac{8}{9} = \frac{1}{9}$$

6. $\mu_X = \frac{13}{18}, \mu_Y = \frac{10}{9}, E(XY) = \frac{43}{54}$

$$\text{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = \frac{43}{54} - \frac{65}{81} = -\frac{1}{162}$$

7. $\mu_X = \frac{5}{9}, \mu_Y = \frac{11}{18}, E(XY) = \frac{1}{3}$

$$\text{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = \frac{1}{3} - \frac{55}{162} = -\frac{1}{162}$$

8. $\mu_X = -\frac{1}{3}, \mu_Y = -\frac{1}{3}, E(XY) = -\frac{5}{3}$

$$\text{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = -\frac{16}{9}$$

Problem Set 4.8

1. The marginal distribution of X is given by

x	-1	0	1
$f_X(x)$	0.3	0.4	0.3

$$f_{Y|X}(y|X=0) = \frac{f(0,y)}{f_X(0)} = \frac{f(0,y)}{0.4} \text{ for } y = -1, 0, 1$$

Thus, we have

y	-1	0	1
$f_{Y X}(y X=0)$	0.5	0	0.5

Hence, it follows that $E(Y|X=0) = \sum_y y f_{Y|X}(y|X=0) = 0$

2. (a) The marginal distribution of X is given by

x	0	1	2
$f_X(x)$	0.4	0.3	0.3

and the marginal distribution of Y is given by

y	1	2	3
$f_Y(y)$	0.3	0.3	0.4

- (b) Note that $f_{X|Y}(x|Y=1) = \frac{f(x,1)}{f_Y(1)} = \frac{f(x,1)}{0.3}$ for $x = 0, 1, 2$

x	0	1	2
$f_{X Y}(x Y=1)$	$\frac{2}{3}$	0	$\frac{1}{3}$

Note that $f_{Y|X}(y|X=1) = \frac{f(1,y)}{f_X(1)} = \frac{f(1,y)}{0.3}$ for $y = 1, 2, 3$

y	1	2	3
$f_{Y X}(y X=1)$	0	$\frac{1}{3}$	$\frac{2}{3}$

- (c) $E(X|Y=1) = \frac{2}{3}$ and $E(Y|X=1) = \frac{8}{3}$

- (a) We find that

$$f_X(x) = \begin{cases} \frac{2}{5}(x+2) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

For any fixed value of x in the interval $(0, 1)$, we have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{x+4y}{x+2} \text{ for } 0 < y < 1$$

Thus, it follows that

$$E(Y|X=x) = \int_{y=0}^1 y f_{Y|X}(y|x) dy = \frac{3x+8}{6(x+2)}$$

3. We find that

$$f_Y(y) = \begin{cases} \frac{2}{3}(y+1) & \text{for } 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

For any fixed value of y in the interval $(0, 1)$, we have

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{x+y}{2(y+1)} \text{ for } 0 < x < 2$$

Thus, it follows that

$$E(X|Y=y) = \int_{x=0}^2 x f_{X|Y}(x|y) dx = \frac{3y+4}{3(y+1)}$$

4. For any fixed value of x in the interval $(0, 1)$, we have

$$f_{Y|X}(y|x) = \frac{2y}{1-x^2} \quad \text{for } x < y < 1$$

and for any fixed value of y in the interval $(0, 1)$, we have

$$f_{X|Y}(x|y) = \frac{2x}{y^2} \quad \text{for } 0 < x < y$$

Thus, it follows that

$$E(Y|X=x) = \int_{y=x}^1 y f_{Y|X}(y|x) dy = \frac{2(1-x^3)}{3(1-x^2)}$$

and that

$$E(X|Y=y) = \int_{x=0}^y x f_{X|Y}(x|y) dx = \frac{2y}{3}$$

5. For any fixed value of x in the interval $(0, 1)$, we have

$$f_{Y|X}(y|x) = \frac{x + \frac{y}{3}}{2(x + \frac{1}{3})} \quad \text{for } 0 < y < 2$$

and for any fixed value of y in the interval $(0, 2)$, we have

$$f_{X|Y}(x|y) = \frac{2x(3x+y)}{y+2} \quad \text{for } 0 < x < 1$$

Thus, it follows that

$$E(Y|X=x) = \int_{y=0}^2 y f_{Y|X}(y|x) dy = \frac{x + \frac{4}{9}}{x + \frac{1}{3}}$$

and that

$$E(X|Y=y) = \int_{x=0}^1 x f_{X|Y}(x|y) dx = \frac{y + \frac{3}{2}}{y+2}$$

6. For any fixed value of x in the interval $(0, 1)$, we have

$$f_{Y|X}(y|x) = \frac{1}{1-x} \quad \text{for } 0 < y < 1-x$$

and for any fixed value of y in the interval $(0, 1)$, we have

$$f_{X|Y}(x|y) = \frac{1}{1-y} \quad \text{for } 0 < x < 1-y$$

Thus, it follows that

$$E(Y|X=x) = \int_{y=0}^{1-x} y f_{Y|X}(y|x) dy = \frac{1+x}{2}$$

and

$$E(X|Y=y) = \int_{x=0}^{1-y} x f_{X|Y}(x|y) dx = \frac{1+y}{2}$$

Problem Set 4.9

1. $\mu_X = 5, \mu_Y = 4, \rho = \sqrt{\frac{8}{15}} = 0.7303$ and $\sigma_Y = 7.3030$
2. (a) $\rho = \frac{2}{\sqrt{15}} = 0.5164$, (b) $\beta_{XY} = \frac{1}{5}, \beta_{YX} = \frac{4}{3}$, (c) $\sigma_Y = 0.2066$
3. (a) $\mu_X = 5, \mu_Y = 3$, (b) $\rho = -\frac{\sqrt{3}}{2} = -0.8660$ (c) $\frac{\sigma_Y}{\sigma_X} = \sqrt{3}$. Thus, if $\sigma_X = 2$, then $\sigma_Y = 2\sqrt{3}$.
4. $\rho = 0.4082$.
5. (a) $\rho = -0.1346$
 (b) The regression line of Y on X is $y = -0.1449x + 1.1304$.
 The regression line of X on Y is $x = -0.1250y + 1.0250$.
6. (a) $\rho = -\frac{1}{11}$
 (b) The regression curve of Y on X is $y = E(Y|X=x) = \frac{x+\frac{4}{3}}{x+1}$.
 The regression curve of X on Y is $x = E(X|Y=y) = \frac{y+\frac{4}{3}}{y+1}$.
7. (a) $\rho = 0.0804$.
 (b) The regression curve of Y on X is $y = \frac{2x}{3}$.
 The regression curve of X on Y is $x = \frac{2(1-y^3)}{3(1-y^2)}$.
8. $\rho = 0.7692$.
 The regression line of Y on X is given by $y = 0.7218x + 16.9673$.
 The regression line of X on Y is given by $x = 0.8198y + 15.0289$.
9. $\rho = 0.4553$.
 The regression line of Y on X is $y = 0.3764x + 43.1453$.
 The regression line of X on Y is $x = 0.5508y + 28.0587$.
10. (a) $\rho = 0.7328$
 (b) The regression line of Y on X is $y = 1.1709x - 4.7780$.
 The regression line of X on Y is given by $x = 0.4586y + 32.4023$.
 (c) $79.5268 \approx 79.5$ kg.

Problem Set 4.10

1. 0.9164
2. 0.8968
3. 0.0384
4. 0.8756
5. 0.8666
6. 0.8904

CHAPTER 5

Problem Set 5.1

1. We find that

$$E \left\{ [X(t_1) - X(t_2)]^2 \right\} = E \left[X^2(t_1) + X^2(t_2) - 2X(t_1)X(t_2) \right]$$

Hence, it follows that

$$E \left\{ [X(t_1) - X(t_2)]^2 \right\} = E[X^2(t_1)] + E[X^2(t_2)] - 2E[X(t_1)X(t_2)] = E[X^2(t_1)]E[X^2(t_2)] - 2R_{XX}(t_1, t_2)$$

2. (a) $\mu_X(t) = \frac{1}{2}(t + \sin 2t)$.
 (b) If $t_1 \neq t_2$, then $R_{XX}(t_1, t_2) = \frac{1}{4}(t_1 + \sin 2t_1)(t_2 + \sin 2t_2)$.
 If $t_1 = t_2 = t$, then $R_{XX}(t, t) = \frac{1}{2}(t^2 + \sin^2 2t)$.
 (c) If $t_1 \neq t_2$, then $C_{XX}(t_1, t_2) = 0$.
 If $t_1 = t_2 = t$, then $C_{XX}(t, t) = \frac{1}{4}[t - \sin 2t]^2$.
 (d) The variance of $X(t)$ is $\sigma_X^2(t) = C_{XX}(t, t) = \frac{1}{4}[t - \sin 2t]^2$.
3. (a) The mean of $X(t)$ is $\mu_X(t) = \frac{1}{2} \cos \omega t$.
 (b) $R_{XX}(t_1, t_2) = \frac{1}{3} \cos \omega t_1 \cos \omega t_2$.
 (c) $C_{XX}(t_1, t_2) = \frac{1}{12} \cos \omega t_1 \cos \omega t_2$.
 (d) The variance of $X(t)$ is $\sigma_X^2(t) = C_{XX}(t, t) = \frac{1}{12} \cos^2 \omega t$.
4. $\mu_X(t) = 0$ and $\sigma_X^2(t) = \frac{1}{2}$, which are independent of time.
5. $\mu_Z = 2$, $\mu_W = 2$, $\sigma_Z^2 = 4$, $\sigma_W^2 = 4$ and $\text{Cov}(Z, W) = 1 + 3e^{-0.2} \approx 3.4562$.
6. (a) $p_X(x_1, x_2, \dots, x_n) = P[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n] = \prod_{i=1}^n P[X_i = x_i] = \prod_{i=1}^n p_X(x_i)$
 (b) The mean of $X(n)$ is $\mu(n) = E[X(n)] = \mu$ for all $n = 1, 2, \dots$
 (c) $R_{XX}(m, n) = \begin{cases} \mu^2 & \text{if } m \neq n \\ \sigma^2 + \mu^2 & \text{if } m = n \end{cases}$
 (d) $C_{XX}(m, n) = \begin{cases} 0 & \text{if } m \neq n \\ \sigma^2 & \text{if } m = n \end{cases}$

Problem Set 5.2

1. $\mu_X(t) = 0$, which is a constant. Also, $R_{XX} = \frac{1}{2} \cos[\omega_0(t_1 - t_2)]$, which is a function of $t_1 - t_2$. Thus, we conclude that $X(t)$ is wide-sense stationary.
2. $\mu_X(t) = 0$, which is a constant. But, $R_{XX}(t_1, t_2) = \frac{1}{3} \sin \omega t_1 \sin \omega t_2$, which is not a function of $t_1 - t_2$. Thus, we conclude that $X(t)$ is not wide-sense stationary.
3. $\mu_X(t) = 0$, which is a constant. Also, $C_{XX}(t_1, t_2) = \cos \omega(t_1 - t_2)$, which is a function of $t_1 - t_2$. Thus, we conclude that $X(t)$ is covariance stationary.
4. As calculated in Example 5.7, $X_1 = X(t_1)$ has the probability density function

$$f_X(x_1; t_1) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x_1^2}{2\sigma^2}} \quad \text{for } -\infty < x_1 < \infty$$

which is independent of time. Hence, the process $X(t)$ is a strict-sense stationary process of order 1.

In Example 5.7, we also calculated that $X_1 = X(t_1)$ and $X_2 = X(t_2)$ have the joint probability density function

$$f_X(x_1, x_2; t_1, t_2) = \frac{1}{2\pi\sigma^2\sqrt{1-\rho^2}} e^{-\frac{(x_1^2 - 2\rho x_1 x_2 + x_2^2)}{2(1-\rho^2)\sigma^2}} \quad \text{for } -\infty < x_1, x_2 < \infty$$

where

$$\rho = \cos \theta(t_1 - t_2)$$

Thus, it is clear that the second-order probability density function $f_X(x_1, x_2; t_1, t_2)$ depends on $t_1 - t_2$ only. Hence, the process $X(t)$ is a strict-sense stationary process of order 2.

5. (a) $R_{XX}(t_1, t_2) = 2 \cos \omega t_1 \cos \omega t_2 + 0 + 2 \sin \omega t_1 \sin \omega t_2 = 2 \cos \omega(t_1 - t_2)$
 (a) $\mu_X(t) = \cos \omega t$, which depends on t . Thus, $X(t)$ is not wide-sense stationary.
6. (a) The mean of $Y(t)$ is given by

$$\mu_Y(t) = E[Y(t)] = E[X(t + \tau) - X(t)] = E[X(t + \tau)] - E[X(t)] = \mu - \mu = 0$$

- (b) Since the mean of $Y(t)$ is zero, its variance is given by

$$\begin{aligned} \text{Var}[Y(t)] &= E[Y^2(t)] = E\{[X(t + \tau) - X(t)]^2\} \\ &= E[X^2(t + \tau) - 2X(t + \tau)X(t) + X^2(t)] = E[X^2(t + \tau)] - 2E[X(t + \tau)X(t)] + E[X^2(t)] \\ &= R_{XX}(0) - 2R_{XX}(\tau) + R_{XX}(0) = 2(R_{XX}(0) - R_{XX}(\tau)) \end{aligned}$$

- (c) By (a), the mean of $Y(t)$ is always zero, which is a constant. Also, we find that

$$\begin{aligned} R_{YY}(t + s, t) &= E[Y(t + s)Y(t)] = E\{[X(t + s + \tau) - X(t + s)][X(t + \tau) - X(t)]\} \\ &= E[X(t + s + \tau)X(t + \tau)] - E[X(t + s)X(t + \tau)] - E[X(t + s + \tau)X(t)] + E[X(t + s)X(t)] \\ &= R_{XX}(s) - R_{XX}(s - \tau) - R_{XX}(s + \tau) + R_{XX}(s) = -R_{XX}(s - \tau) - R_{XX}(s + \tau) \end{aligned}$$

which is a function of s (note that τ is a constant). Hence, we conclude that $Y(t)$ is wide-sense stationary.

7. $\mu_Z = \mu_W = 3$, $\text{Var}(Z) = \text{Var}(W) = 4$ and $\text{Cov}(Z, W) = 4e^{-0.6} \approx 2.1952$.
8. $\mu_X(t) = 0$, which is a constant, and $R_{XX}(t_1, t_2) = \frac{a^2}{2} \cos[\omega(t_1 - t_2)]$, which is a function of $t_1 - t_2$. Thus, $X(t)$ is a wide-sense stationary process.
 $\mu_Y(t) = 0$, which is a constant, and $R_{YY}(t_1, t_2) = \frac{b^2}{2} \cos[\omega(t_1 - t_2)]$, which is a function of $t_1 - t_2$. Thus, $Y(t)$ is a wide-sense stationary process.
 Also, $R_{XY}(t_1, t_2) = -\frac{ab}{2} \sin[\omega(t_1 - t_2)]$, which is a function of $t_1 - t_2$.
 Hence, we conclude that $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes.
9. $\mu_X(t) = 0$, which is a constant. $R_{XX}(t_1, t_2) = \sigma^2 \cos(t_1 - t_2)$, which is a function of $t_1 - t_2$. Thus, $X(t)$ is a wide-sense stationary process.
 $\mu_Y(t) = 0$, which is a constant. $R_{YY}(t_1, t_2) = \sigma^2 \cos(t_1 - t_2)$, which is a function of $t_1 - t_2$. Thus, $Y(t)$ is a wide-sense stationary process.
 However, $R_{XY}(t_1, t_2) = \sigma^2 \sin(t_1 + t_2)$, which is not a function of $t_1 - t_2$.
 Thus, we conclude that $X(t)$ and $Y(t)$ are not jointly wide-sense stationary.
10. First, suppose that $E(A) = 0$. Then $\mu_X(t) = 0$, which is a constant, and $R_{XX}(t_1, t_2) = E(A^2) e^{i\omega(t_1 - t_2)}$, which is a function of $t_1 - t_2$. Thus, $X(t)$ is a WSS process. Conversely, suppose that $X(t)$ is a WSS process. Then $\mu_X(t) = E(A)e^{i\omega t}$. Since $X(t)$ is a WSS process, the mean of $X(t)$ must be a constant. This can happen only when $E(A) = 0$.

Problem Set 5.3

1. $\mu_X = 4$ and $\text{Var}[X(t)] = 2$.
2. $\mu_X = 5$ and $\text{Var}[X(t)] = 5$.
3. $\mu_X = \sqrt{2}$ and $\text{Var}[X(t)] = 4$.
4. $\mu_X = 1$ and $\text{Var}[X(t)] = 4$. Also, $E(S) = 1$.
5. $\mu_X(t) = 0$, which is a constant, and $R_{XX}(t_1, t_2) = \frac{a^2}{2} \cos[\omega(t_1 - t_2)]$, which is a function of $t_1 - t_2$. Thus, $X(t)$ is wide-sense stationary with the autocorrelation function $R_{XX}(\tau) = \frac{a^2}{2} \cos \omega \tau$. Also, by transformation, it can be shown that the first order probability density function of $X(t)$ follows an arcsine distribution:

$$f_X(x; t) = \frac{1}{\pi \sqrt{1 - x^2}}, \quad -1 < x < 1$$

and it is independent of time. Thus, f_X is invariant to a time shift. Thus, $X(t)$ is strictly stationary to the first order. It can be also shown that the n th order probability density function of $X(t)$ is also invariant to a time shift, but the calculation would be very tedious.

6. $\mu_X(t) = 0$, which is a constant, and $R_{XX}(t_1, t_2) = \frac{1}{2} \cos(t_1 - t_2)$, which is a function of $t_1 - t_2$. Thus, $X(t)$ is wide-sense stationary with the autocorrelation function $R_{XX}(\tau) = E[X(t)X(t + \tau)] = \frac{1}{2} \cos \tau$.
7. By Chebychev's inequality (Corollary 2.3), for any random variable Z with mean μ and variance σ^2

$$P[|Z - \mu| \geq C] \leq \frac{\sigma^2}{C^2} \quad [\text{where } C > 0]$$

We take $Z = X(t + \tau) - X(t)$ and $C = a$. Then $E(Z) = \mu = 0$ and $\text{Var}(Z) = \sigma^2 = 2[R_{XX}(0) - R_{XX}(\tau)]$. Hence, the inequality becomes

$$E[|X(t + \tau) - X(t)| \geq a] \leq \frac{2[R_{XX}(0) - R_{XX}(\tau)]}{a^2}$$

8. Note that

$$E\{[X(t) \pm Y(t + \tau)]^2\} \geq 0$$

i.e.

$$E\{X^2(t) + Y^2(t + \tau) \pm 2X(t)Y(t + \tau)\} \geq 0$$

i.e.

$$E[X^2(t)] + E[Y^2(t + \tau)] \pm 2E[X(t)Y(t + \tau)] \geq 0$$

i.e.

$$R_{XX}(0) + R_{YY}(0) \pm 2R_{XY}(\tau) \geq 0$$

from which it follows that

$$2R_{XY}(\tau) \leq R_{XX}(0) + R_{YY}(0)$$

and

$$2R_{XY}(\tau) \geq -R_{XX}(0) - R_{YY}(0)$$

Combining the above two inequalities, we get

$$2|R_{XY}(\tau)| \leq R_{XX}(0) + R_{YY}(0)$$

9. We know that the derivative of $X(t)$ is given by

$$\dot{X}(t) = \lim_{\varepsilon \rightarrow 0} \left\{ \frac{X(t + \varepsilon) - X(t)}{\varepsilon} \right\}$$

Thus, the cross-correlation function between $X(t)$ and $\dot{X}(t)$ is given by

$$R_{X\dot{X}}(\tau) = E[X(t)\dot{X}(t + \tau)] = \lim_{\varepsilon \rightarrow 0} E \left\{ X(t) \left[\frac{X(t + \tau + \varepsilon) - X(t + \tau)}{\varepsilon} \right] \right\}$$

i.e.

$$R_{X\dot{X}}(\tau) = \lim_{\varepsilon \rightarrow 0} E \left\{ \frac{X(t)X(t + \tau + \varepsilon) - X(t)X(t + \tau)}{\varepsilon} \right\} = \lim_{\varepsilon \rightarrow 0} \left\{ \frac{E[X(t)X(t + \tau + \varepsilon)] - E[X(t)X(t + \tau)]}{\varepsilon} \right\}$$

i.e.

$$R_{X\dot{X}}(\tau) = \lim_{\varepsilon \rightarrow 0} \frac{R_{XX}(\tau + \varepsilon) - R_{XX}(\tau)}{\varepsilon} = \frac{dR_{XX}(\tau)}{d\tau}$$

10. The autocorrelation function of $\dot{X}(t)$ is given by

$$R_{\dot{X}\dot{X}}(\tau) = E[\dot{X}(t)\dot{X}(t + \tau)] = \lim_{\varepsilon \rightarrow 0} E \left[\dot{X}(t) \frac{X(t + \tau + \varepsilon) - X(t + \tau)}{\varepsilon} \right]$$

i.e.

$$R_{\dot{X}\dot{X}}(\tau) = \lim_{h \rightarrow 0} \lim_{\varepsilon \rightarrow 0} E \left\{ \left[\frac{X(t + h) - X(t)}{h} \right] \left[\frac{X(t + \tau + \varepsilon) - X(t + \tau)}{\varepsilon} \right] \right\}$$

i.e.

$$R_{\dot{X}\dot{X}}(\tau) = \lim_{h \rightarrow 0} \frac{1}{h} \lim_{\varepsilon \rightarrow 0} E \left\{ X(t + h) \frac{X(t + \tau + \varepsilon) - X(t + \tau)}{\varepsilon} - X(t) \frac{X(t + \tau + \varepsilon) - X(t + \tau)}{\varepsilon} \right\}$$

i.e.

$$R_{\dot{X}\dot{X}}(\tau) = \lim_{h \rightarrow 0} \frac{1}{h} \lim_{\varepsilon \rightarrow 0} \left[\frac{R_{XX}(\tau - h + \varepsilon) - R_{XX}(\tau - h)}{\varepsilon} - \frac{R_{XX}(\tau + \varepsilon) - R_{XX}(\tau)}{\varepsilon} \right]$$

i.e.

$$R_{\dot{X}\dot{X}}(\tau) = \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{dR_{XX}(\tau - h)}{d\tau} - \frac{dR_{XX}(\tau)}{d\tau} \right] = -\frac{d^2 R_{XX}(\tau)}{d\tau^2}$$

Problem Set 5.4

1. We find that $\mu_Y(t) = 0$ and $\bar{Y}_T = a\bar{X}_T$. Thus,

$$\lim_{T \rightarrow \infty} \bar{Y}_T = a\mu_X \neq 0 = \mu_Y$$

Since the ensemble average of $Y(t)$ is not equal to the time average of $Y(t)$, we conclude that $Y(t)$ is not mean-ergodic.

2. Since $C_{XX}(\tau) \rightarrow 0$ as $T \rightarrow \infty$, it follows that for any $\varepsilon > 0$, there exists a constant $T_0 > 0$ such that

$$|C_{XX}(\tau)| < \varepsilon \quad \text{for all } T > T_0$$

Since $X(t)$ is a WSS process, we know that $C_{XX}(\tau) < C_{XX}(0)$ for all τ . Thus, we find that

$$\frac{1}{T} \int_{\tau=0}^T C_{XX}(\tau) d\tau = \frac{1}{T} \int_{\tau=0}^{T_0} C_{XX}(\tau) d\tau + \frac{1}{T} \int_{\tau=T_0}^T C_{XX}(\tau) d\tau < \frac{1}{T} \int_{\tau=0}^{T_0} C_{XX}(0) d\tau + \frac{1}{T} \int_{\tau=T_0}^T \varepsilon d\tau$$

Simplifying, we get

$$\frac{1}{T} \int_{\tau=0}^T C_{XX}(\tau) d\tau < \frac{T_0}{T} C_{XX}(0) + \left[1 - \frac{T_0}{T}\right] \varepsilon$$

Taking limits on both sides as $T \rightarrow \infty$, we get

$$\lim_{T \rightarrow \infty} \left[\frac{1}{T} \int_{\tau=0}^T C_{XX}(\tau) d\tau \right] < 0 + [1 - 0] \varepsilon = \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, it can be made arbitrarily small. Thus, we must have

$$\lim_{T \rightarrow \infty} \left[\frac{1}{T} \int_{\tau=0}^T C_{XX}(\tau) d\tau \right] = 0$$

By Slutsky's Theorem (5.8), it follows that $X(t)$ is mean-ergodic.

3. First, we compute the ensemble average (mean) of the process $X(t)$.

$$\mu_X(t) = E[X(t)] = E[A \cos t + B \sin t] = E(A) \cos t + E(B) \sin t = 0$$

Next, we compute the time average of the process $X(t)$ over $(-T, T)$.

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt = \frac{1}{2T} \int_{-T}^T (A \cos t + B \sin t) dt$$

Integrating, we have

$$\bar{X}_T = \frac{1}{2T} [A \sin t - B \cos t]_{-T}^T = \frac{1}{2T} [2A \sin T - 0] = \frac{A \sin T}{T}$$

Taking limits as $T \rightarrow \infty$, we get

$$\lim_{T \rightarrow \infty} \bar{X}_T = \lim_{T \rightarrow \infty} \left[\frac{A \sin T}{T} \right] = 0$$

because $|\sin T| \leq 1$ and $\frac{1}{T} \rightarrow 0$ as $T \rightarrow \infty$.

Since the ensemble average of $X(t)$ is equal to the time average of $X(t)$, we conclude that $X(t)$ is mean-ergodic.

4. Since θ is uniformly distributed in $(-\pi, \pi)$, it has the probability density function

$$f_\theta(\theta) = \begin{cases} \frac{1}{2\pi} & \text{if } -\pi < \theta < \pi \\ 0 & \text{otherwise} \end{cases}$$

Also, $R_{XX}(\tau) = \frac{A^2}{2} \cos \omega \tau$.

Next, we obtain the time autocorrelation function of the process over $(-T, T)$ as

$$\bar{Z}_T = \frac{A^2}{8\omega T} \{\sin[\omega(2t + \tau) + 2\theta]\}_{t=-T}^T + \frac{A^2}{2} \cos \omega \tau$$

Taking limits as $T \rightarrow \infty$, we get

$$\lim_{T \rightarrow \infty} \bar{Z}_T = 0 + \frac{A^2}{2} \cos \omega \tau = \frac{A^2}{2} \cos \omega \tau = R_{XX}(\tau)$$

because $|\sin u| \leq 1$ for all u and $\frac{1}{T} \rightarrow 0$ as $T \rightarrow \infty$. Since the time autocorrelation function of $X(t)$ is equal to the autocorrelation function $R_{XX}(\tau)$, we conclude that the process $X(t)$ is correlation-ergodic.

5. By the solution to Problem 8 in Problem Set 5.2, we know that $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes with $\mu_X = \mu_Y = 0$ and

$$R_{XX}(\tau) = \frac{a^2}{2} \cos \omega \tau, \quad R_{YY}(\tau) = \frac{b^2}{2} \cos \omega \tau \quad \text{and} \quad R_{XY}(\tau) = \frac{ab}{2} \sin \omega \tau$$

The time cross-correlation function of $X(t)$ over $(-T, T)$ is given by

$$\begin{aligned} \bar{Z}_T &= \frac{1}{2T} \int_{t=-T}^T X(t)X(t+\tau) dt = \frac{1}{2T} \int_{t=-T}^T a^2 \cos[\omega(t+\tau)+\theta] \cos(\omega t+\theta) dt \\ &= \frac{a^2}{4T} \int_{t=-T}^T \{\cos[\omega(2t+\tau)+2\theta] + \cos \omega \tau\} dt \end{aligned}$$

Integrating, we get

$$\bar{Z}_T = \frac{a^2}{4T} \left\{ \frac{\sin[\omega(2t+\tau)+2\theta]}{2\omega} + t \cos \omega \tau \right\}_{t=-T}^T = \frac{a^2}{8\omega T} \{\sin[\omega(2t+\tau)+2\theta]\}_{t=-T}^T + \frac{a^2}{2} \cos \omega \tau$$

Taking limits as $T \rightarrow \infty$, we get

$$\lim_{T \rightarrow \infty} \bar{Z}_T = 0 + \frac{a^2}{2} \cos \omega \tau = R_{XX}(\tau)$$

because $|\sin u| \leq 1$ for all u and $\frac{1}{T} \rightarrow 0$ as $t \rightarrow \infty$. Since the time autocorrelation of $X(t)$ is equal to the autocorrelation function $R_{XX}(\tau)$, the process $X(t)$ is correlation-ergodic. Next, we calculate the time autocorrelation function of $Y(t)$ over $(-T, T)$. By definition, we have

$$\begin{aligned} \bar{W}_T &= \frac{1}{2T} \int_{t=-T}^T Y(t)Y(t+\tau) dt = \frac{1}{2T} \int_{t=-T}^T b^2 \sin[\omega(t+\tau)+\theta] \sin(\omega t+\theta) dt \\ &= \frac{b^2}{4T} \int_{t=-T}^T \{\cos \omega \tau - \cos[\omega(2t+\tau)+2\theta]\} dt \end{aligned}$$

Integrating, we get

$$\bar{W}_T = \frac{b^2}{4T} \left\{ t \cos \omega \tau - \frac{\sin[\omega(2t+\tau)+2\theta]}{2\omega} \right\}_{t=-T}^T = \frac{b^2}{2} \cos \omega \tau - \frac{b^2}{8\omega T} \{\sin[\omega(2t+\tau)+2\theta]\}_{t=-T}^T$$

Taking limits as $T \rightarrow \infty$, we get

$$\lim_{T \rightarrow \infty} \bar{W}_T = \frac{b^2}{2} \cos \omega \tau - 0 = \frac{b^2}{2} \cos \omega \tau = R_{YY}(\tau)$$

because $|\sin u| \leq 1$ for all u and $\frac{1}{T} \rightarrow 0$ as $T \rightarrow \infty$. Since the time autocorrelation function of $Y(t)$ is equal to the autocorrelation function $R_{YY}(\tau)$, the process $Y(t)$ is correlation-ergodic. Next, the time cross-correlation function between $X(t)$ and $Y(t)$ over $(-T, T)$ is given by

$$\bar{G}_T = \frac{1}{2T} \int_{t=-T}^T X(t)Y(t+\tau) dt = \frac{ab}{4T} \int_{t=-T}^T \{\sin[\omega(2t+\tau)+2\theta] + \sin \omega \tau\} dt$$

Integrating, we get

$$\bar{G}_T = \frac{ab}{4T} \left\{ -\frac{\cos[\omega(2t+\tau)+2\theta]}{2\omega} + t \sin \omega\tau \right\}_{t=-T}^T = -\frac{ab}{8\omega T} \{\cos[\omega(2t+\tau)+2\theta]\}_{t=-T}^T + \frac{ab}{2} \sin \omega\tau$$

Taking limits as $T \rightarrow \infty$, we get

$$\lim_{T \rightarrow \infty} \bar{G}_T = 0 + \frac{ab}{2} \sin \omega\tau = \frac{ab}{2} \sin \omega\tau = R_{XY}(\tau)$$

because $|\cos u| \leq 1$ for all u and $\frac{1}{T} \rightarrow 0$ as $T \rightarrow \infty$. Thus, the time cross-correlation between $X(t)$ and $Y(t)$ is equal to their cross-correlation function $R_{XY}(\tau)$. Since all the three conditions are fulfilled, we conclude that the processes $X(t)$ and $Y(t)$ are jointly correlation-ergodic.

Problem Set 5.5

1. Clearly, $p(0) = [0.5 \quad 0.5]$.

$$(a) \quad p(1) = p(0)P = [0.5 \quad 0.5] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = [0.55 \quad 0.45].$$

$$p(2) = p(1)P = [0.55 \quad 0.45] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = [0.565 \quad 0.435].$$

$$(b) \quad p(3) = p(2)P = [0.565 \quad 0.435] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = [0.5695 \quad 0.4305].$$

$$p(4) = p(3)P = [0.5695 \quad 0.4305] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = [0.5709 \quad 0.4291].$$

2. Use induction on n .

$$3. \quad P = \frac{1}{6} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 1 & 1 & 1 \\ 0 & 0 & 0 & 4 & 1 & 1 \\ 0 & 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 & 6 \end{bmatrix}$$

$$P^2 = \frac{1}{36} \begin{bmatrix} 1 & 3 & 5 & 7 & 9 & 11 \\ 0 & 4 & 5 & 7 & 9 & 11 \\ 0 & 0 & 9 & 7 & 9 & 11 \\ 0 & 0 & 0 & 16 & 9 & 11 \\ 0 & 0 & 0 & 0 & 25 & 11 \\ 0 & 0 & 0 & 0 & 0 & 36 \end{bmatrix}$$

Clearly, $p(0) = [\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}]$. Hence, it is immediate that

$$p(2) = p(0)P^2 = [\frac{1}{216} \quad \frac{7}{216} \quad \frac{19}{216} \quad \frac{37}{216} \quad \frac{61}{216} \quad \frac{91}{216}]$$

$$P(X_2 = 3) = [p(2)]_3 = \frac{19}{216}.$$

4. The initial probability distribution is $p(0) = [0.4 \quad 0.4 \quad 0.2]$.

$$p(1) = p(0)P = [0.4 \quad 0.4 \quad 0.2] \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.6 & 0.1 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix} = [0.44 \quad 0.22 \quad 0.34].$$

$$p(2) = p(1)P = [0.44 \quad 0.22 \quad 0.34] \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.6 & 0.1 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix} = [0.376 \quad 0.256 \quad 0.368].$$

5. (a) The initial probability distribution is $p(0) = [0.3 \quad 0.4 \quad 0.3]$.

$$p(1) = p(0)P = [0.3 \quad 0.4 \quad 0.3] \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.2 & 0.4 \\ 0.1 & 0.6 & 0.3 \end{bmatrix} = [0.34 \quad 0.35 \quad 0.31]$$

$$p(2) = p(1)P = [0.34 \quad 0.35 \quad 0.31] \begin{bmatrix} 0.5 & 0.3 & 0.2 \\ 0.4 & 0.2 & 0.4 \\ 0.1 & 0.6 & 0.3 \end{bmatrix} = \begin{matrix} 0 & 1 & 2 \\ (0.341 & 0.358 & 0.301) \end{matrix}$$

Thus, $P(X_2 = 2) = [p(2)]_2 = 0.301$.

- (b) $P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 0) = 0.0216$.

6. $\pi = [1 \quad 0]$.

7. (a) The transition probability matrix of the Markov chain is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

(Note: This is called a random walk with absorbing barriers at 0 and 6 because the chain cannot come out of the states 0 and 6 once it has entered them.)

- (b) The initial probability distribution is

$$p(0) = \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ (0 & 0 & 1 & 0 & 0 & 0 & 0) \end{matrix}$$

since the gambler has initially Rs. 2 to start the game. A simple calculation shows that

$$p(4) = \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ (\frac{3}{8} & 0 & \frac{5}{16} & 0 & \frac{1}{4} & 0 & \frac{1}{16}) \end{matrix}$$

Thus, the probability that the gambler has lost his money at the end of 4 fixings is

$$P(X_4 = 0) = [p(4)]_0 = \frac{3}{8}$$

- (c) The 5-step probability distribution is

$$p(5) = p(4)P = \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ (\frac{3}{8} & \frac{5}{32} & 0 & \frac{9}{32} & 0 & \frac{1}{8} & \frac{1}{16}) \end{matrix}$$

We note that the probability that the game lasts more than 5 fixings is equal to the probability that $X_5 \neq 0$ and $X_5 \neq 6$, i.e.

$$P[X_5 = 1, 2, 3, 4 \text{ or } 5] = \frac{5}{32} + 0 + \frac{9}{32} + 0 + \frac{1}{8} = \frac{9}{16}$$

8. (a) The transition probability matrix of the given Markov chain is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{pmatrix} \end{matrix}$$

- (b) Since the New Year day is a sunny day, the initial distribution is

$$p(1) = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{pmatrix} 1 & 0 \end{pmatrix} \end{matrix}$$

Then, the probability distribution on January 2nd is given by

$$p(2) = p(1)P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{pmatrix} 0.7 & 0.3 \end{pmatrix} \end{matrix}$$

Thus, the probability distribution on January 3rd is given by

$$p(3) = p(2)P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{pmatrix} 0.64 & 0.36 \end{pmatrix} \end{matrix}$$

Hence, the probability that January 3rd is a sunny day is given by

$$P(X_3 = 0) = [p(3)]_0 = 0.64$$

- (c) The probability distribution on January 4th is given by

$$p(4) = p(3)P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{pmatrix} 0.628 & 0.372 \end{pmatrix} \end{matrix}$$

Hence, the probability that January 4th is a rainy day is given by

$$P(X_4 = 1) = [p(4)]_1 = 0.372$$

- (d) Since all the entries of P are positive, P is a regular matrix. Thus, the given Markov chain is a regular chain. Hence, by Theorem 5.11, the given Markov chain has a stationary distribution $\pi = [\pi_1 \ \pi_2]$, which is obtained by solving the equations

$$\pi P = \pi \text{ and } \pi_1 + \pi_2 = 1$$

A simple calculation gives $\pi = [0.625 \ 0.375]$. Thus, in the long run, 62.5% of the days will be sunny and 37.5% of the days will be rainy.

9. Let X_n be the number of red balls in bag A after n steps. The state space of the chain is $S = \{0, 1, 2\}$ as the bag A can contain a maximum of 2 red balls only. The transition probability matrix is obtained as

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{pmatrix} \end{matrix}$$

Note that the initial probability distribution is $p(0) = [1 \ 0 \ 0]$, since initially there are no red balls in bag A. Thus, the 1-step probability distribution is obtained as

$$p(1) = p(0)P = [0 \ 1 \ 0]$$

Then, the 2-step probability distribution is obtained as

$$p(2) = p(1)P = [\frac{1}{6} \ \frac{1}{2} \ \frac{1}{3}]$$

Also, the 3-step probability distribution is obtained as

$$p(3) = p(2)P = \begin{pmatrix} 0 & 1 & 2 \\ \frac{1}{12} & \frac{23}{36} & \frac{5}{18} \end{pmatrix}$$

Thus, the probability that there are 2 red balls in bag A after 3 steps is given by

$$P(X_3 = 2) = [p(3)]_2 = \frac{5}{18}$$

The steady-state probability distribution $\pi = [\pi_0 \ \pi_1 \ \pi_2]$ is obtained by solving the equations

$$\pi P = \pi \text{ and } \pi_0 + \pi_1 + \pi_2 = 1$$

A simple calculation gives

$$\pi = [\pi_0 \ \pi_1 \ \pi_2] = [\frac{1}{10} \ \frac{6}{10} \ \frac{3}{10}]$$

Hence, the probability that in the long run there are 2 red balls in bag A is $\pi_2 = \frac{3}{10} = 0.3$.

10. (a) The state transition diagram of the given Markov chain is illustrated in Figure P.5.1.

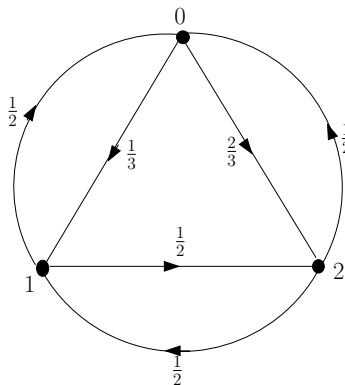


Figure P.5.1: State Transition Diagram.

- (b) From Figure P.5.2, it is clear that $0 \leftrightarrow 1 \leftrightarrow 2$. Thus, the Markov chain is irreducible.
 (c) From Figure P.5.1, it is clear that starting from each state i , we can return to the same state i in 2, 4, 6, 8, ... transitions, where $i = 0, 1, 2$. Thus, the period of each state i is given by

$$d(i) = \gcd\{2, 4, 6, 8, \dots\} = 2 \text{ for } i = 0, 1, 2$$

For a Markov chain to be ergodic, all its states must be positive recurrent and aperiodic. Since the Markov chain has finite states and is irreducible, all its states are positive recurrent by Theorem 5.13. However, the states 0, 1 and 2 are not aperiodic as they have period equal to 2. Thus, we conclude that the given Markov chain is not ergodic.

- (d) The stationary distribution of the Markov chain is $\pi = [\pi_0 \ \pi_1 \ \pi_2]$ which is determined by solving the equations

$$\pi P = \pi \text{ and } \pi_0 + \pi_1 + \pi_2 = 1$$

A simple calculation gives $\pi = [\frac{9}{27} \ \frac{8}{27} \ \frac{10}{27}]$.

11. (a) The state transition diagram of the given Markov chain is illustrated in Figure P.5.2.

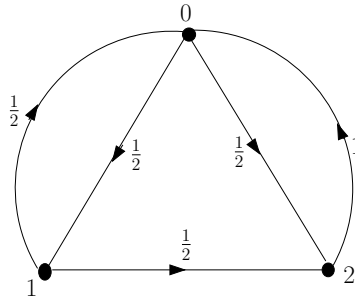


Figure P.5.2: State Transition Diagram.

- (b) From Figure P.5.2, it is clear that $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$. Thus, the given Markov chain is irreducible.
 (c) Starting from state 0, we can return to state 0 in only one of the following paths:

- (i) $0 \rightarrow 1 \rightarrow 0$ (in 2 steps)
- (ii) $0 \rightarrow 2 \rightarrow 0$ (in 2 steps)
- (iii) $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ (in 3 steps)

Thus, the period of the state 0 is given by

$$d(0) = \gcd\{2, 3\} = 1$$

Starting from state 1, we can return to state 1 in any of the following paths:

- (i) $1 \rightarrow 0 \rightarrow 1$ (in 2 steps)
- (ii) $1 \rightarrow 2 \rightarrow 0 \rightarrow 1$ (in 3 steps)
- (iii) $1 \rightarrow 2 \rightarrow 0 \rightarrow 2 \rightarrow 0 \rightarrow 1$ (in 5 steps); and so on.

Thus, the period of the state 1 is given by

$$d(1) = \gcd\{2, 3, 5, \dots\} = 1$$

Starting from state 2, we can return to state 2 in any of the following paths:

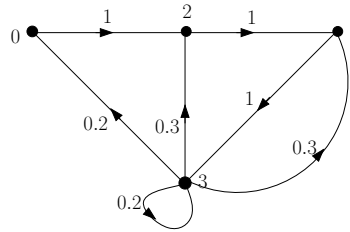
- (i) $2 \rightarrow 0 \rightarrow 2$ (in 2 steps)
- (ii) $2 \rightarrow 0 \rightarrow 1 \rightarrow 2$ (in 3 steps)
- (iii) $2 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 2$ (in 5 steps); and so on.

Thus, the period of the state 1 is given by

$$d(2) = \gcd\{2, 3, 5, \dots\} = 1$$

Thus, we find that all the states of the Markov chain are aperiodic.

- (d) For a Markov chain to be ergodic, all its states must be positive recurrent and aperiodic. Since the Markov chain has finite states and is irreducible, all its states are positive recurrent by Theorem 5.13. Also, by part (c), all states are aperiodic. Thus, we conclude that the given Markov chain is ergodic.

**Figure P.5.3:** State Transition Diagram.

12. (a) The state transition diagram of the given Markov chain is illustrated in Figure P.5.3.
- (b) From Figure P.5.3, it is clear that $0 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 0$. Thus, the given Markov chain is irreducible.
- (c) Starting from state 0, we can return to state 0 in any of the following paths:
- (i) $0 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 0$ (in 4 steps)
 - (ii) $0 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 3 \rightarrow 0$ (in 5 steps)
 - (iii) $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 3 \rightarrow 0$ (in 6 steps); and so on.

Thus, the period of the state 0 is given by

$$d(0) = \gcd\{4, 5, 6, \dots\} = 1$$

Starting from state 1, we can return to state 1 in any of the following paths:

- (i) $1 \rightarrow 3 \rightarrow 1$ (in 2 steps)
- (ii) $1 \rightarrow 3 \rightarrow 3 \rightarrow 1$ (in 3 steps)
- (iii) $1 \rightarrow 3 \rightarrow 3 \rightarrow 3 \rightarrow 1$ (in 5 steps); and so on.

Thus, the period of state 1 is given by

$$d(1) = \gcd\{2, 3, 4, \dots\} = 1$$

Starting from state 2, we can return to state 2 in any of the following paths:

- (i) $2 \rightarrow 1 \rightarrow 3 \rightarrow 2$ (in 3 steps)
- (ii) $2 \rightarrow 1 \rightarrow 3 \rightarrow \rightarrow 2$ (in 4 steps)
- (iii) $2 \rightarrow 1 \rightarrow 3 \rightarrow 3 \rightarrow 3 \rightarrow 2$ (in 5 steps); and so on.

Thus, the period of state 1 is given by

$$d(2) = \gcd\{3, 4, 5, \dots\} = 1$$

Since $3 \rightarrow 3$, it is clear that starting from state 3, we can return to state 3 in any number of steps by the path $3 \rightarrow 3 \rightarrow \dots \rightarrow 3$. Thus, it follows that the period of state 3 is given by

$$d(3) = \gcd\{1, 2, 3, \dots\} = 1$$

Thus, we find that all the states of the Markov chain are aperiodic.

- (d) For a Markov chain to be ergodic, all its states must be positive recurrent and aperiodic. Since the Markov chain has finite states and is irreducible, all its states are positive recurrent by Theorem 5.13. Also, by part (c), all states are aperiodic. Thus, we conclude that the given Markov chain is ergodic.

Problem Set 5.6

1. Since $N(t)$ is a Poisson process with rate λ , we know that

$$P[N(t) = k] = e^{-\lambda t} \frac{(\lambda t)^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

If $s < t$, then we have

$$P\{N(s) = k | N(t) = n\} = \frac{P[N(s) = k, N(t) = n]}{P[N(t) = n]} = \frac{P[N(s) = k, N(t) - N(s) = n - k]}{P[N(t) = n]}$$

Since $N(t)$ is a Poisson process, it has independent increments. Thus, it follows that

$$P\{N(s) = k | N(t) = n\} = \frac{P[N(s) = k]P[N(t) - N(s) = n - k]}{P[N(t) = n]}$$

Since $N(t)$ is a Poisson process, it has stationary increments and $N(0) = 0$. Thus, it follows that

$$P[N(t) - N(s) = n - k] = P[N(t - s) - N(0) = n - k] = P[N(t - s) = n - k] = e^{-\lambda(t-s)} \frac{[\lambda(t-s)]^{n-k}}{(n-k)!}$$

Thus, we have

$$P\{N(s) = k | N(t) = n\} = \frac{e^{-\lambda s} \frac{(\lambda s)^k}{k!} e^{-\lambda(t-s)} \frac{[\lambda(t-s)]^{n-k}}{(n-k)!}}{e^{-\lambda t} \frac{(\lambda t)^n}{n!}} = \frac{n!}{k!(n-k)!} \frac{s^k (t-s)^{n-k}}{t^n}$$

which may be simplified as

$$P\{N(s) = k | N(t) = n\} = \binom{n}{k} \left(\frac{s}{t}\right)^k \left(\frac{t-s}{t}\right)^{n-k} = \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k}$$

2. (a) $E[X^2(t)] = \lambda t + \lambda^2 t^2$.

(b) If $t > s$, then $E\{[X(t) - X(s)]^2\} = \lambda(t-s) + \lambda^2(t-s)^2$.

3. $P[N(4) = 10] = e^{-12} \frac{12^{10}}{10!} = 0.1048$.

4. $P[N(5) \geq 4] = 0.9998$.

5. (a) $P[X(2) = 4] = 0.0053$, (b) $P[X(0.5) \geq 2] = 0.8009$.

6. $P[X(1) \leq 4] = 0.8153$.

7. (a) $P[X(1) = 1] = 0.3679$, (b) $P[X(4) = 4] = 0.1954$.

8. $P[X(4) = 20 | X(8) = 60] = 0.0036$.

9. $P[X(6) \leq 3] = 0.9344$.

10. (i) $P(T > 2) = 0.5134$, (ii) $P(2 < T < 5) = 0.3245$, (iii) $P(0 < T < 6) = 0.8647$.

11. (i) $P[X(5) \leq 2] = 0.3085$, (ii) $P[|X(8) - X(5)| \leq 1] = 0.3544$.

12. Let $Y_n = \sum_{k=1}^n X_k$, where X_k 's are a set of independent random variables, each normally distributed with mean μ and variance σ^2 . By the additive property of characteristic functions, it follows that

$$\phi_{Y_n}(t) = \phi_{X_1}(t) \phi_{X_2}(t) \cdots \phi_{X_n}(t) = e^{\mu i t - \frac{\sigma^2 t^2}{2}} e^{\mu i t - \frac{\sigma^2 t^2}{2}} \cdots e^{\mu i t - \frac{\sigma^2 t^2}{2}} = e^{n \mu i t - \frac{n \sigma^2 t^2}{2}}$$

which is the characteristic function of the normal random variable with mean $n\mu$ and variance $n\sigma^2$. By the uniqueness theorem of characteristic functions (Theorem 2.31), it follows that Y_n is a normally distributed with mean $n\mu$ and variance $n\sigma^2$. Hence, $\{Y_n : n = 1, 2, \dots\}$ is a normal or Gaussian process.

CHAPTER 6

Problem Set 6.1

- (a) Yes, (b) No, (c) Yes
- $\mu_Y = 0$, $\sigma_Y^2 = 12.9062$.
- $\mu_X = 0$, $\sigma_X^2 = 5$, $\mu_Y = \sqrt{2}$, $\sigma_Y^2 = 7$, $\mu_Z = 0$ and $\sigma_Z^2 = 720$.
- $\mu_S = 4$ and $\text{Var}(S) = \frac{6}{e}$.
- $\mu_X(t) = 0$, which is a constant, and $R_{XX}(\tau) = \frac{A^2}{2} \cos \omega \tau$, which is purely a function of τ . Thus, $X(t)$ is a wide-sense stationary process. Similarly, $\mu_Y(t) = 0$, which is a constant, and $R_{YY}(\tau) = \frac{A^2}{2} \cos \omega \tau$, which is purely a function of τ . Thus, $Y(t)$ also is a wide-sense stationary process.
 - $R_{XY}(\tau) = E[X(t)Y(t+\tau)] = \frac{A^2}{2} \cos(\omega \tau + \phi)$, which is purely a function of τ . This shows that $X(t)$ and $Y(t)$ are jointly wide-sense stationary processes.
 - $X(t)$ and $Y(t)$ are orthogonal if $\phi = \frac{(2k+1)\pi}{2} - \omega \tau$, where $k = 0, 1, 2, \dots$

Problem Set 6.2

- $R(\tau) = \frac{N_0}{2} \delta(\tau)$, where $\delta(\tau)$ is the unit impulse function.
- $R(\tau) = e^{-2|\tau|}$ and $E[X^2(t)] = R(0) = 1$.
- (a) No, (b) Yes, (c) No.
- $S(\omega) = 2\pi A \delta(\omega) + \frac{2Bc}{\omega^2 + c^2}$.
- $R(\tau) = \sigma^2 \cos \beta \tau$ and $S(\omega) = \sigma^2 \pi [\delta(\omega - \beta) + \delta(\omega + \beta)]$.
- $S(\omega) = \frac{2\text{Re}[2\alpha^3 + 2\alpha\omega^2 - 2\alpha\omega^2 + i(\dots)]}{(\alpha^2 + \omega^2)^2} = \frac{4\alpha^3}{(\alpha^2 + \omega^2)^2}$
- $S(\omega) = e^{-\frac{\omega^2}{4}} \int_{-\infty}^{\infty} e^{-u^2} du = e^{-\frac{\omega^2}{4}} \sqrt{\pi}$
- $R(\tau) = \frac{\sin \pi \tau}{2\tau}$
- $R(\tau) = \frac{5}{4} e^{-4|\tau|}$ and $E[X^2(t)] = R(0) = \frac{5}{4}$.
- $R_{XY}(\tau) = 2\mathcal{F}^{-1} \left[\frac{2}{(a+i\omega)^3} \right] = 2u(\tau)\tau^2 e^{-a\tau}$.

Problem Set 6.3

- (a) Non-linear, (b) non-linear, (c) linear.
- (a) Time-varying, (b) time-invariant, (c) time-invariant.
- (a) Causal, (b) non-causal, (c) causal.
- $\mu_Y = \mu_X H(0) = 3$.
- $\mu_Y = \mu_X H(0) = 3$.
- $\mu_Y = H(0)\mu_X = 0$.
 - $S_{XX}(\omega) = \frac{24}{9+\omega^2}$.
 - $S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega) = \frac{1}{4+\omega^2} \frac{24}{9+\omega^2} = \frac{24}{(4+\omega^2)(9+\omega^2)}$.
- $S_{YY}(\omega) = \frac{4}{4+\omega^2}$.
 - $R_{YY}(\tau) = e^{-2|\tau|}$.

- (c) $E[Y^2(t)] = R_{YY}(0) = 1.$
8. (a) $S_{YY}(\omega) = \frac{c}{1+\omega^2}.$
 (b) $R_{YY}(\tau) = \frac{c}{2}e^{-|\tau|}.$
 (c) $E[Y^2(t)] = R_{YY}(0) = \frac{c}{2}.$
9. (a) $\mu_Y = H(0)\mu_X = 0.$
 (b) $S_{YY}(\omega) = \frac{2\beta R^2}{L^2} \frac{1}{\left(\frac{R^2}{L^2} + \omega^2\right)(\beta^2 + \omega^2)}.$
 (c) $R_{YY}(\tau) = \frac{R}{R^2 - \beta^2 L^2} \left[Re^{-\beta|\tau|} - \beta L e^{-R/L|\tau|} \right].$
 (d) $E[Y^2(t)] = R_{YY}(0) = \frac{R}{R + \beta L}.$
10. $\mu_Y = H(0)\mu_X = \frac{6}{7}.$

Chapter 7

Problem Set 7.1

- $P(T_s > \frac{1}{3}) = 0.1353.$
- $P(N \geq 8) = 0.0168.$
- (a) $L_s = 1$, (b) $W_s = 10$ minutes, (c) $L_q = \frac{1}{2}$, (d) $W_q = 5$ minutes.
- (a) $P_0 = 0.2$, (b) $L_s = 4$, (c) $L_q = 3.2$, (d) $W_s = 0.6667$, (e) $W_q = 0.5333.$
- (a) $P(N \geq 1) = 0.5$, (b) $L_s = 1$, (c) $L_q = 0.5$, (d) $P(N \geq 6) = 0.0156$, (e) $W_s = 15$ seconds, (f) $W_q = 7.5$ seconds.
- Since the mean arrival rate is $\lambda = 3$, the traffic intensity is given by $\rho = \frac{\lambda}{\mu} = \frac{3}{\mu}$. If the owner wishes to have fewer than 6 customers line up 98% of the time, the probability of 6 or more customers in the system must be less than 2%, i.e. $P(N \geq 6) = \rho^6 = \left(\frac{3}{\mu}\right)^6 \leq 0.02$ or $\mu \geq 5.7582$. Thus, the mean service rate must be at least equal to 5.7582 in order that fewer than 6 customers line up 98% of the time.
- (a) $P(N \geq 1) = 0.5$, (b) $L_s = 1$, (c) $L_q = 0.5$, (d) $W_s = 4$ minutes, (e) $W_q = 2$ minutes, (f) The average arrival rate of customers required to justify an additional ATM machine is $\lambda = 0.3$.
- (a) $\rho = 0.8571$, (b) $L_s = 6$, (c) $W_s = 42$ minutes, (d) If the arrival rate increases by 10%, then $\rho = 0.9429$, $L_s = 16.5$, $W_s = 105$ minutes. (This is an *ill-conditioned* queuing system as a small change in the arrival rate λ has resulted in a significant change in L_s and W_s .)
- (a) $P_0 = 0.25$, (b) $L_s = 3$, (c) $W_s = 6$ minutes, (d) $W_q = 4.5$ minutes.
- (a) $P_0 = 0.2$, (b) $L_s = 4$, (c) $L_q = 3.2$, (d) $W_s = 12$ minutes, (e) $W_q = 9.6$ minutes.
- (a) $P_0 = 0.3333$, (b) $P(N \geq 5) = 0.1317$, (c) $L_s = 2$, (d) $L_q = 1.3333$, (e) $W_s = 6$ minutes, (f) $W_q = 4$ minutes.
- (a) $P_0 = 0.3333$, (b) $P(N \geq 6) = 0.0878$, (c) $L_s = 2$, (d) $L_q = 1.3333$, (e) $W_s = 30$ minutes, (f) $W_q = 20$ minutes.
- (a) $P(N \geq 1) = 0.8333$, (b) $L_s = 5$, (c) $P(N \geq 10) = 0.1615$, (d) $L_q = 4.1667$, (e) $W_s = 12$ minutes, (f) $W_q = 10$ minutes.

Problem Set 7.2

- $L_q = 0.8889.$

2. (a) 71.11%, (b) $L_s = 4.4444$, (c) $L_q = 2.8444$, (d) $W_q = 10.6667$ minutes, (e) $W_s = 16.6667$ minutes, (f) $\rho = 0.8$.
3. (a) $P_0 = \frac{1}{3}$, (b) $L_s = 2.8889$, (c) $L_q = 0.8889$, (d) $P(N \geq s) = 0.4444$.
4. (a) 64.29%, (b) $L_q = 1.9286$, (c) $L_s = 3.4286$, (d) $W_q = 7.7143$ minutes, (e) $W_s = 13.7143$ minutes, (f) $\rho = 0.75$.
5. (a) $P(N \geq s) = 0.2368$, (b) $L_q = 0.2368$, (c) $L_s = 1.7368$, (d) $W_q = 0.7895$ minutes, (e) $W_s = 5.7895$ minutes, (f) $\rho = 0.5$.
6. (a) 70.22%, (b) $L_q = 3.5112$, (c) $L_s = 6.0112$, (d) $W_q = 14.0449$ minutes, (e) $W_s = 24.0449$ minutes, (f) $\rho = 0.8333$.
7. (a) $P(N \geq s) = 0.2368$, (b) $L_q = 0.2368$, (c) $L_s = 1.7368$, (d) $W_q = 1.1820$ minutes, (e) $W_s = 8.6842$ minutes, (f) $\rho = 0.5$.
8. (a) $\rho = 0.5$, (b) $P(N \geq s) = 0.2963$, (c) $L_q = 0.3951$, (d) $L_s = 2.3951$, (e) $W_q = 2.9630$ minutes, (f) $W_s = 17.9360$ minutes.

Problem Set 7.3

1. (a) $L_q = L_s - \frac{\lambda'}{\mu} = L_s - \frac{\mu(1-P_0)}{\mu} = L_s - (1 - P_0)$.
 (b) $W_q = \frac{1}{\lambda'} L_q = \frac{1}{\lambda'} \left[L_s - \frac{\lambda'}{\mu} \right] = \frac{1}{\lambda'} L_s - \frac{1}{\mu} = W_s - \frac{1}{\mu}$.
2. (a) $P_0 = 0.5161$, (b) $P_1 = 0.2581$, $P_2 = 0.1290$, $P_3 = 0.0645$, (c) $L_s = 0.8387$, (d) $L_q = 0.3548$, (e) $W_s = 17.3321$ minutes, (f) $W_q = 7.3321$ minutes.
3. (a) $P_0 = 0.125$, (b) $P_n = 0.125$, (c) $L_s = 3.5$, (d) $L_q = 2.625$, (e) $W_s = 1$ hour, (f) $W_q = 0.75$ hour.
4. (a) $1 - P_0 = 0.9487$, (b) $\lambda' = 14.2298$, (c) $L_s = 4.0784$, (d) $W_s = 17.1966$ minutes, (e) $L_q = 3.1297$, (f) $W_q = 13.1964$ minutes.
5. (a) $1 - P_0 = 0.75$, (b) $\lambda' = 3.75$, (c) $P_n = 0.25$, (d) $L_s = 1.5$, (e) $W_s = 24$ minutes, (f) $L_q = 0.75$, (g) $W_q = 12$ minutes.
6. (a) $P_0 = 0.0323$, (b) $\lambda' = 5.8065$, (c) $P_1 = 0.0645$, $P_2 = 0.1290$, $P_3 = 0.2580$, (d) $L_s = 3.1613$, (e) $W_s = 32.6668$ minutes, (f) $L_q = 2.1936$, (g) $W_q = 22.6672$ minutes.
7. (a) 7.58%, (b) 38.39%, (c) $\lambda' = 3.6968$, (d) $L_s = 2.7583$, (e) $W_s = 44.7679$ minutes, (f) $L_q = 1.8349$, (g) $W_q = 29.7809$ minutes.
8. (a) $P_0 = 0.3219$, (b) $P(N \geq 3) = 0.1864$, (c) $W_q = 4.3932$ minutes.
9. (a) $P_0 = 0.0137$, (b) $\lambda' = 9.863$, (c) $L_s = 6.4939$, (d) $W_s = 39.5046$ minutes, (e) $L_q = 5.5076$, (f) $W_q = 33.5046$ minutes.
10. (a) $P_0 = 0.1037$, (b) $\lambda' = 5.3777$, (c) $L_s = 5.5167$, (d) $W_s = 61.5505$ minutes, (e) $L_q = 4.6204$, (f) $W_q = 51.5505$ minutes.

Problem Set 7.4

1. (a) $P_0 = 0.2554$, $P_1 = 0.3065$, $P_7 = 0.0143$, (b) $\lambda' = 11.8270$, (c) $L_q = 0.5287$, (d) $W_q = 2.6822$ minutes, (e) $L_s = 1.7114$, (f) $W_s = 8.6822$ minutes.
2. (i) $P_0 = 0.2064$, $P_1 = 0.2753$, $P_8 = 0.0161$, (ii) $\lambda' = 19.6785$, (iii) $L_q = 0.8109$, (iv) $W_q = 2.4723$ minutes, (v) $L_s = 2.1228$, (vi) $W_s = 6.4724$ minutes.
3. (i) $P_0 = 0.1686$, $P_1 = 0.2529$, $P_8 = 0.0338$, (ii) $\lambda' = 16.9188$, (iii) $L_q = 1.2635$, (iv) $W_q = 0.8961$ hour, (v) $L_s = 1.6734$, (vi) $W_s = 1.8961$ hour.
4. (i) $P_0 = 0.1131$, $P_1 = 0.2262$, $P_{10} = 0.0088$, (ii) $\lambda' = 29.7315$, (iii) $L_q = 0.7283$, (iv) $W_q = 1.4698$ minutes, (v) $L_s = 2.7104$, (vi) $W_s = 5.4698$ minutes.

5. (i) $P_0 = 0.1878$, $P_1 = 0.3005$, $P_2 = 0.2404$, $P_3 = 0.0029$, (ii) $\lambda' = 7.9763$, (iii) $L_q = 0.5730$, (iv) $W_q = 4.3103$ minutes (v) $L_s = 2.1683$, (vi) $W_s = 16.3103$ minutes.

Problem Set 7.5

1. $L_q = 0.0833$, $L_s = 0.4167$, $W_q = 1.25$ minutes, $W_s = 6.25$ minutes.
2. $L_q = 2.0833$, $L_s = 2.9167$, $W_q = 25$ minutes, $W_s = 35$ minutes.
3. (i) $L_q = 4.8333$, (ii) $L_s = 5.5$, (iii) $W_q = 29$ minutes, (iv) $W_s = 33$ minutes.
4. (i) $L_q = 11.6$, (ii) $L_s = 12.4$, (iii) $W_q = 1.45$ hour, (iv) $W_s = 1.55$ hour.
5. (i) $L_q = 2.8333$, (ii) $L_s = 3.6667$, (iii) $W_q = 35$ minutes, (iv) $W_s = 44$ minutes.

Index

- $M/M/1$: ∞ /FIFO Model, 692
- $(M/M/s)$: (∞ /FIFO) Model, 710
- $M/M/1$: (k /FIFO) Model, 722
- $M/M/s$: (k /FIFO) Model, 733
- $M/G/1$ queueing system, 740
- σ -field, definition, 19

- autocorrelation function, 524
- autocovariance function, 524
- average power, 637

- balance equation of birth-death process, 634
- Bayes' theorem, 81
- Bayes, Thomas, 81
- Bernoulli, J., 2
- Bernoulli distribution, 194
- Bernoulli process, 604
- Bernoulli trials, 68
- binomial distribution, 196
- binomial process, 606
- birth and death process, 632
- birth process, 627
- birth rates, 632
- Bonferroni's inequality, 33
- Boole, George, 32
- Boole's inequality, 32

- Cauchy distribution, 127
- Cauchy-Schwarz inequality, 402
- Central Limit Theorem, 512
- central moments, definition, 164
- Chapman-Kolmogorov Theorem, 577
- characteristic function, 185
- Chebyshev's inequality, 156, 158
- conditional distributions, continuous case, 370
- conditional distributions, discrete case, 369
- conditional expectation, 455, 456
- conditional mean, 456
- conditional variance, 456
- continuous RV, 2-dim, 324
- continuous-state process, 522
- correlation-ergodic, 559
- counting process, 607
- covariance, definition, 441, 442
- cross correlation coefficient, 525
- cross correlation function, 525
- cross covariance function, 525
- cross power spectral density, 646
- cross power spectrum, 646

- de Mere, C., 1
- De Morgan's laws, 19
- death rates, 632
- degenerate distribution, 193
- density function
 - discrete, 105
 - moment generating, 174
 - power spectral, 641
 - probability, 111
 - 2-dim, 324
- density, standard normal, 280
- discrete RV, 2-dim, 322
- discrete RV, PMF, 322
- discrete-state process, 522
- discrete-time process, 522
- distribution function,
 - 2-dim RVs, 319
 - characterization, 99
 - definition, 95
 - discontinuities, 100
 - method, 298
 - properties, 96

- ensemble average, 524
- ergodic,
 - in the correlation, 569
 - in the mean, 559
 - jointly, 560
- events,
 - complementary, 4
 - completely independent, 53
 - compound, 4
 - definition, 4
 - disjoint, 5
 - exhaustive, 4
 - independent, 6
 - mutually exclusive, 5
 - mutually independent, 53
 - pairwise independent, 53
 - simple, 4
- theorem, mean, 560
- expectation,
 - definition, 126
 - properties, 129
- exponential distribution, 255

- functions of 2-dim RVs, 418
- Fundamental Theorem of the power spectrum, 675

- gamma distribution, 265
- Gauss C.F., 276
- Gaussian distribution, 276
- Gaussian process, 621
- geometric distribution, 234

- Hugens, C., 126
- Huyghen's problem, 65

- identically distributed RVs, 347
- Inclusion-Exclusion Principle, 27
- indicator function, 92

- Kendall's method, 740
- Kendall's notation, 688
- Kolmogorov, A.N., 2, 19

- Lindeberg-Levy Central Limit Theorem, 513

- line length, 688
- linear system, 668
 - causal, 669
 - stable, 669
 - time-invariant, 668
 - time-varying, 668
- Little's Formulae, 697

- marginal CMF, 343
- marginal densities, continuous, 342
- marginal distributions, discrete, 342
- Markov chain, 573
 - absorbing state, 594
 - accessible state, 593
 - aperiodic state, 593
 - communicating state, 593
 - ergodic, 594
 - homogeneous, 574
 - irreducible, 593
 - nonhomogeneous, 574
 - null recurrent state, 594
 - period of a state, 593
 - periodic state, 593
 - persistent state, 594
 - recurrent state, 594, 594
 - reducible, 593
 - regular, 582
 - stationary distribution for a, 581
 - steady-state distribution for a, 581
 - technique of imbedded, 740
 - transient state, 594
- Markov process, 573
- Markov's inequality, 155
- mean, arithmetic, 149
- mean, geometric, 149
- mean, harmonic, 149
- mean-ergodic, 559
- mean-square ergodic, 559
- median, definition, 133
- mode, definition, 135
- moments about the mean, definition, 164
- moments about the origin, definition, 163
- moments, 2-dim RVs, 441

- negative binomial distribution, 242
- normal distribution, 276
- normal process, 621

- order parameters, 133

- Pascal distribution, 242
- percentiles, definition, 133
- Poisson distribution, 216
- Poisson postulates, 608
- Poisson process, 608
- Pollaczek-Khinchine formula, 742, 744
- power ergodic, 559
- power spectrum, 641
- probability distribution, 104
- probability mass function, 103
 - 2-dim, 322
- probability,
 - addition rule, 24--26
 - axiomatic definition, 20
 - classical definition, 6
 - conditional, 40
 - frequency definition, 8
 - inverse, 81
 - monotone, 23
 - multiplication rule, 41, 42
 - subtractive, 23
 - total, 74
- pure birth process, 627
- problem of points, 1
- quartiles, definition, 133
- queue discipline, 687
- queueing system, basic characteristics of, 686
- queue length, 688
- random experiment, 2
- random process
 - continuous, 522
 - definition, 522
 - deterministic, 523
 - discrete, 522
 - four types, 522
 - evolutionary, 536
 - first-order distribution, 523
 - independent, 536
 - nondeterministic, 523
 - second-order distribution, 523
 - types, 522
 - variance, 524
- random sequence, 522
 - continuous, 522
 - discrete, 522
- random variable,
 - 2-dim, 319
 - continuous, 111
 - definition, 92
 - discrete, 103
 - functions of, 298
 - properties, 94
 - independent, 344
- raw moments, 164
- regeneration points, 741
- regression coefficients, 466
- regression curves, 464
- regular stochastic matrix, 581
- sample function, 522
- sample space
 - definition, 3
 - types, 3
- simple birth process, 630
- sine wave process, 625
- standard deviation, definition, 131
- standard normal variable, 276
- standard random variable, definition, 133
- stationary process, 532
 - order 1, 532
 - order 2, 534
 - order n , 536
 - strictly, 536
 - wide-sense, 534, 535
- time average, 558, 559
- time correlation function, 559
- total probability theorem, 74
- transfer function, 673
- transient state, 687
- transformation method, 298, 307
- transition probability matrix, 574
- uncorrelated RVs, 442
- uniform distribution, 247
- variance
 - definition, 131
 - properties, 131
- Weibull distribution, 271
- wide-sense ergodic, 559
- Wiener-Khinchin relations, 641
- Wiener-Khinchin theorem, 644
- Yule-Furry process, 630